Namespace Lib9c

Classes

Currencies

SerializeKeys

Class Currencies

Namespace: Lib9c

Assembly: Lib9c.dll

```
public static class Currencies
```

Inheritance

object ← Currencies

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Fields

Crystal

```
public static readonly Currency Crystal
```

Field Value

Currency

DailyRewardRune

```
public static readonly Currency DailyRewardRune
```

Field Value

Currency

FreyaBlessingRune

```
public static readonly Currency FreyaBlessingRune
```

Field Value

Currency

FreyaLiberationRune

```
public static readonly Currency FreyaLiberationRune
```

Field Value

Currency

Garage

```
public static readonly Currency Garage
```

Field Value

Currency

GuildGold

```
public static readonly Currency GuildGold
```

Field Value

Currency

ItemCurrencyTickerPrefix

Ticker prefix shared by every item currency.

```
public const string ItemCurrencyTickerPrefix = "Item_"
```

Field Value

string

Mead

```
public static readonly Currency Mead
```

Field Value

Currency

OdinWeaknessRune

```
public static readonly Currency OdinWeaknessRune
```

Field Value

Currency

OdinWisdomRune

```
public static readonly Currency OdinWisdomRune
```

Field Value

Currency

StakeRune

```
public static readonly Currency StakeRune
```

Field Value

Currency

WrappedCurrencyTickerPrefix

Ticker prefix GetWrappedCurrency(Currency) puts in front of the wrapped ticker.

```
public const string WrappedCurrencyTickerPrefix = "FAV__"
```

Field Value

string

Methods

GetCurrencyByTicker(string)

Gets the appropriate currency object for a given ticker. Uses GetMinterlessCurrency for supported tickers (CRYSTAL, GARAGE, Rune, Soulstone). NCG is not supported as it may have minters and requires special handling.

```
public static Currency GetCurrencyByTicker(string ticker)
```

Parameters

`ticker` string

The currency ticker string

Returns

Currency

The currency object for the given ticker

Exceptions

ArgumentException

Thrown when ticker is NCG or not supported by GetMinterlessCurrency

GetItemCurrency(int, bool)

Returns the currency that stands in for an item, which `ClaimItems` burns back into inventory items.

```
public static Currency GetItemCurrency(int itemId, bool tradable)
```

Parameters

`itemId` int

The item the currency stands in for.

`tradable` bool

Whether the claimed item should be tradable. Note that this only takes effect for material rows — `InventoryExtensions.MintItem` ignores it for every other item type, so a costume claimed through a non tradable ticker is tradable all the same.

Returns

Currency

The currency for the item.

GetMinterlessCurrency(string?)

Covers the reward.CurrencyTicker is following cases: - Currencies.Crystal.Ticker - Currencies.Garage.Ticker - lower case is starting with "rune_" or "runestone_" - lower case is starting with "soulstone_"

```
public static Currency GetMinterlessCurrency(string? ticker)
```

Parameters

`ticker` string

Returns

Currency

Exceptions

ArgumentNullException

ArgumentException

GetRune(string?)

```
public static Currency GetRune(string? ticker)
```

Parameters

`ticker` string

Returns

Currency

GetRunes(RuneSheet?)

```
public static IOrderedEnumerable<Currency> GetRunes(RuneSheet? sheet)
```

Parameters

`sheet` RuneSheet

Returns

IOrderedEnumerable<Currency>

GetRunes(params string?[])

```
public static IOrderedEnumerable<Currency> GetRunes(params string?[] tickers)
```

Parameters

`tickers` string[]

Returns

IOrderedEnumerable<Currency>

GetSoulStone(string?)

```
public static Currency GetSoulStone(string? ticker)
```

Parameters

`ticker` string

Returns

Currency

GetSoulStones(PetSheet?)

```
public static IOrderedEnumerable<Currency> GetSoulStones(PetSheet? sheet)
```

Parameters

`sheet` PetSheet

Returns

IOrderedEnumerable<Currency>

GetSoulStones(params string?[])

```
public static IOrderedEnumerable<Currency> GetSoulStones(params string?[] tickers)
```

Parameters

`tickers` string[]

Returns

IOrderedEnumerable<Currency>

GetUnwrappedCurrency(Currency)

```
public static Currency GetUnwrappedCurrency(Currency currency)
```

Parameters

`currency` Currency

Returns

Currency

GetWrappedCurrency(Currency)

Returns the wrapped counterpart of the currency: a minterless currency whose ticker carries WrappedCurrencyTickerPrefix. GetUnwrappedCurrency(Currency) reverses it.

```
public static Currency GetWrappedCurrency(Currency currency)
```

Parameters

`currency` Currency

The currency to wrap.

Returns

Currency

The wrapped currency.

IsItemCurrencyTicker(string)

Whether the ticker denotes an item wrapped into a currency by GetItemCurrency(int, bool), however many times it was wrapped afterwards. Such a ticker exists per item id, so a policy sheet cannot enumerate them as rows.

```
public static bool IsItemCurrencyTicker(string ticker)
```

Parameters

`ticker` string

The ticker to test.

Returns

bool

IsRuneTicker(string)

```
public static bool IsRuneTicker(string ticker)
```

Parameters

`ticker` string

Returns

bool

IsSoulstoneTicker(string)

```
public static bool IsSoulstoneTicker(string ticker)
```

Parameters

`ticker` string

Returns

bool

IsWrappedCurrency(Currency)

```
public static bool IsWrappedCurrency(Currency currency)
```

Parameters

`currency` Currency

Returns

bool

ParseItemCurrency(Currency)

```
public static (bool tradable, int itemId) ParseItemCurrency(Currency currency)
```

Parameters

`currency` Currency

Returns

(bool tradable, int itemId)

PickAddress(Currency, Address, Address)

pick address by currency ticker

```
public static Address PickAddress(Currency currency, Address agentAddress,
Address avatarAddress)
```

Parameters

`currency` Currency

Libplanet.Types.Assets.Currency

`agentAddress` Address

Libplanet.Crypto.Address

`avatarAddress` Address

Libplanet.Crypto.Address

Returns

Address

agentAddress when ticker is NCG or Crystal or Garage or Mead else avatarAddress

UnwrapTicker(string)

Strips every WrappedCurrencyTickerPrefix the ticker was wrapped with, so that a policy keyed by the plain ticker also covers the wrapped form. A wrapper is not a different asset — `claim_items` unwraps it straight back.

```
public static string UnwrapTicker(string ticker)
```

Parameters

`ticker` string

The ticker to unwrap.

Returns

string

Class SerializeKeys

Namespace: Lib9c

Assembly: Lib9c.dll

```
public static class SerializeKeys
```

Inheritance

object ← SerializeKeys

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Fields

AchievementsKey

```
public const string AchievementsKey = "ach"
```

Field Value

string

ActionPointKey

```
public const string ActionPointKey = "ap"
```

Field Value

string

AdditionalStatValueKey

```
public const string AdditionalStatValueKey = "av"
```

Field Value

string

AddressKey

```
public const string AddressKey = "a"
```

Field Value

string

AgentAddressKey

```
public const string AgentAddressKey = "aga"
```

Field Value

string

AmountKey

```
public const string AmountKey = "am"
```

Field Value

string

ArenaProviderKey

```
public const string ArenaProviderKey = "arp"
```

Field Value

string

AvatarAddressKey

```
public const string AvatarAddressKey = "aa"
```

Field Value

string

BlockIndexKey

```
public const string BlockIndexKey = "bi"
```

Field Value

string

BuffSkillsKey

```
public const string BuffSkillsKey = "bs"
```

Field Value

string

BuyerAgentAddressKey

```
public const string BuyerAgentAddressKey = "bga"
```

Field Value

string

BuyerAvatarAddressKey

```
public const string BuyerAvatarAddressKey = "ba"
```

Field Value

string

ByCustomCraftKey

```
public const string ByCustomCraftKey = "bcc"
```

Field Value

string

CancellableBlockIndexKey

```
public const string CancellableBlockIndexKey = "cbi"
```

Field Value

string

ChampionshipIdKey

```
public const string ChampionshipIdKey = "chi"
```

Field Value

string

ChanceKey

```
public const string ChanceKey = "ch"
```

Field Value

string

CharacterIdKey

```
public const string CharacterIdKey = "ci"
```

Field Value

string

ChargeApKey

```
public const string ChargeApKey = "cha"
```

Field Value

string

ClaimDataKey

```
public const string ClaimDataKey = "cd"
```

Field Value

string

CombatPointKey

```
public const string CombatPointKey = "cp"
```

Field Value

string

CombinationSlotAddressesKey

```
public const string CombinationSlotAddressesKey = "csa"
```

Field Value

string

CompletedQuestIdsKeyDeprecated

```
public const string CompletedQuestIdsKeyDeprecated = "completedQuestIds"
```

Field Value

string

CostumeKey

```
public const string CostumeKey = "c"
```

Field Value

string

CostumesKey

```
public const string CostumesKey = "cs"
```

Field Value

string

CouponWalletKey

```
public const string CouponWalletKey = "coupon_wallet"
```

Field Value

string

CraftWithRandomKey

```
public const string CraftWithRandomKey = "cwr"
```

Field Value

string

DailyRewardReceivedIndexKey

```
public const string DailyRewardReceivedIndexKey = "dri"
```

Field Value

string

EarKey

```
public const string EarKey = "e"
```

Field Value

string

ElementalTypeKey

```
public const string ElementalTypeKey = "et"
```

Field Value

string

EndKey

```
public const string EndKey = "ed"
```

Field Value

string

EnemyAvatarAddressKey

```
public const string EnemyAvatarAddressKey = "eaa"
```

Field Value

string

EquipmentExpKey

```
public const string EquipmentExpKey = "eq_exp"
```

Field Value

string

EquipmentIconIdKey

```
public const string EquipmentIconIdKey = "icon_id"
```

Field Value

string

EquipmentsKey

```
public const string EquipmentsKey = "es"
```

Field Value

string

ErrorCodeKey

```
public const string ErrorCodeKey = "ec"
```

Field Value

string

EventMapKey

```
public const string EventMapKey = "em"
```

Field Value

string

ExpKey

```
public const string ExpKey = "exp"
```

Field Value

string

ExpiredBlockIndexKey

```
public const string ExpiredBlockIndexKey = "ebi"
```

Field Value

string

GradeKey

```
public const string GradeKey = "g"
```

Field Value

string

GrandFinaleIdKey

```
public const string GrandFinaleIdKey = "gfi"
```

Field Value

string

HairKey

```
public const string HairKey = "h"
```

Field Value

string

HasRandomOnlyIconKey

```
public const string HasRandomOnlyIconKey = "hroi"
```

Field Value

string

IdKey

```
public const string IdKey = "id"
```

Field Value

string

ItemCountKey

```
public const string ItemCountKey = "item_count"
```

Field Value

string

ItemIdKey

```
public const string ItemIdKey = "ii"
```

Field Value

string

ItemMapKey

```
public const string ItemMapKey = "im"
```

Field Value

string

ItemSubTypeKey

```
public const string ItemSubTypeKey = "ist"
```

Field Value

string

ItemTypeKey

```
public const string ItemTypeKey = "it"
```

Field Value

string

LegacyActionPointKey

```
public const string LegacyActionPointKey = "actionPoint"
```

Field Value

string

LegacyAdditionalStatValueKey

```
public const string LegacyAdditionalStatValueKey = "additionalValue"
```

Field Value

string

LegacyAddressKey

```
public const string LegacyAddressKey = "address"
```

Field Value

string

LegacyAgentAddressKey

```
public const string LegacyAgentAddressKey = "agentAddress"
```

Field Value

string

LegacyBlockIndexKey

```
public const string LegacyBlockIndexKey = "blockIndex"
```

Field Value

string

LegacyBuffSkillsKey

```
public const string LegacyBuffSkillsKey = "buffSkills"
```

Field Value

string

LegacyChanceKey

```
public const string LegacyChanceKey = "chance"
```

Field Value

string

LegacyCharacterIdKey

```
public const string LegacyCharacterIdKey = "characterId"
```

Field Value

string

LegacyCombinationSlotAddressesKey

```
public const string LegacyCombinationSlotAddressesKey = "combinationSlotAddresses"
```

Field Value

string

LegacyCostumeItemIdKey

```
public const string LegacyCostumeItemIdKey = "item_id"
```

Field Value

string

LegacyCostumeKey

```
public const string LegacyCostumeKey = "costume"
```

Field Value

string

LegacyDailyRewardReceivedIndexKey

```
public const string LegacyDailyRewardReceivedIndexKey = "dailyRewardReceivedIndex"
```

Field Value

string

LegacyEarKey

```
public const string LegacyEarKey = "ear"
```

Field Value

string

LegacyElementalTypeKey

```
public const string LegacyElementalTypeKey = "elemental_type"
```

Field Value

string

LegacyEquippedKey

```
public const string LegacyEquippedKey = "equipped"
```

Field Value

string

LegacyEventMapKey

```
public const string LegacyEventMapKey = "eventMap"
```

Field Value

string

LegacyGradeKey

```
public const string LegacyGradeKey = "grade"
```

Field Value

string

LegacyHairKey

```
public const string LegacyHairKey = "hair"
```

Field Value

string

LegacyInventoryKey

```
public const string LegacyInventoryKey = "inventory"
```

Field Value

string

LegacyItemIdKey

```
public const string LegacyItemIdKey = "itemId"
```

Field Value

string

LegacyItemMapKey

```
public const string LegacyItemMapKey = "itemMap"
```

Field Value

string

LegacyItemSubTypeKey

```
public const string LegacyItemSubTypeKey = "item_sub_type"
```

Field Value

string

LegacyItemTypeKey

```
public const string LegacyItemTypeKey = "item_type"
```

Field Value

string

LegacyItemUsableKey

```
public const string LegacyItemUsableKey = "itemUsable"
```

Field Value

string

LegacyLevelKey

```
public const string LegacyLevelKey = "level"
```

Field Value

string

LegacyMailBoxKey

```
public const string LegacyMailBoxKey = "mailBox"
```

Field Value

string

LegacyMonsterMapKey

```
public const string LegacyMonsterMapKey = "monsterMap"
```

Field Value

string

LegacyNameKey

```
public const string LegacyNameKey = "name"
```

Field Value

string

LegacyNonceKey

```
public const string LegacyNonceKey = "nonce"
```

Field Value

string

LegacyPowerKey

```
public const string LegacyPowerKey = "power"
```

Field Value

string

LegacyPriceKey

```
public const string LegacyPriceKey = "price"
```

Field Value

string

LegacyProductIdKey

```
public const string LegacyProductIdKey = "productId"
```

Field Value

string

LegacyProductsKey

```
public const string LegacyProductsKey = "products"
```

Field Value

string

LegacyQuestListKey

```
public const string LegacyQuestListKey = "questList"
```

Field Value

string

LegacyRankingMapAddressKey

```
public const string LegacyRankingMapAddressKey = "ranking_map_address"
```

Field Value

string

LegacyRequiredBlockIndexKey

```
public const string LegacyRequiredBlockIndexKey = "requiredBlockIndex"
```

Field Value

string

LegacySellerAgentAddressKey

```
public const string LegacySellerAgentAddressKey = "sellerAgentAddress"
```

Field Value

string

LegacySellerAvatarAddressKey

```
public const string LegacySellerAvatarAddressKey = "sellerAvatarAddress"
```

Field Value

string

LegacySetIdKey

```
public const string LegacySetIdKey = "set_id"
```

Field Value

string

LegacySkillRowKey

```
public const string LegacySkillRowKey = "skillRow"
```

Field Value

string

LegacySkillsKey

```
public const string LegacySkillsKey = "skills"
```

Field Value

string

LegacySpineResourcePathKey

```
public const string LegacySpineResourcePathKey = "spine_resource_path"
```

Field Value

string

LegacyStageMapKey

```
public const string LegacyStageMapKey = "stageMap"
```

Field Value

string

LegacyStatKey

```
public const string LegacyStatKey = "stat"
```

Field Value

string

LegacyStatTypeKey

```
public const string LegacyStatTypeKey = "statType"
```

Field Value

string

LegacyStatValueKey

```
public const string LegacyStatValueKey = "value"
```

Field Value

string

LegacyStatsMapKey

```
public const string LegacyStatsMapKey = "statsMap"
```

Field Value

string

LegacyTailKey

```
public const string LegacyTailKey = "tail"
```

Field Value

string

LegacyUpdatedAtKey

```
public const string LegacyUpdatedAtKey = "updatedAt"
```

Field Value

string

LegacyWorldInformationKey

```
public const string LegacyWorldInformationKey = "worldInformation"
```

Field Value

string

LensKey

```
public const string LensKey = "lens"
```

Field Value

string

LevelKey

```
public const string LevelKey = "l"
```

Field Value

string

MadeWithMimisbrunnrRecipeKey

```
public const string MadeWithMimisbrunnrRecipeKey = "mwmr"
```

Field Value

string

MailBoxKey

```
public const string MailBoxKey = "mb"
```

Field Value

string

MemoKey

```
public const string MemoKey = "m"
```

Field Value

string

MonsterCollectionResultKey

```
public const string MonsterCollectionResultKey = "mcr"
```

Field Value

string

MonsterCollectionRoundKey

```
public const string MonsterCollectionRoundKey = "s"
```

Field Value

string

MonsterMapKey

```
public const string MonsterMapKey = "mm"
```

Field Value

string

MyAvatarAddressKey

```
public const string MyAvatarAddressKey = "maa"
```

Field Value

string

NameKey

```
public const string NameKey = "n"
```

Field Value

string

OptionCountFromCombinationKey

```
public const string OptionCountFromCombinationKey = "oc"
```

Field Value

string

OrderDigestListKey

```
public const string OrderDigestListKey = "odl"
```

Field Value

string

OrderIdKey

```
public const string OrderIdKey = "oi"
```

Field Value

string

OrderReceiptListKey

```
public const string OrderReceiptListKey = "orl"
```

Field Value

string

OrderTypeKey

```
public const string OrderTypeKey = "ot"
```

Field Value

string

PowerKey

```
public const string PowerKey = "pw"
```

Field Value

string

PriceKey

```
public const string PriceKey = "p"
```

Field Value

string

ProductIdKey

```
public const string ProductIdKey = "pi"
```

Field Value

string

ProductsKey

```
public const string ProductsKey = "pr"
```

Field Value

string

PurchaseInfosKey

```
public const string PurchaseInfosKey = "pis"
```

Field Value

string

PurchaseResultsKey

```
public const string PurchaseResultsKey = "prs"
```

Field Value

string

QuantityKey

```
public const string QuantityKey = "q"
```

Field Value

string

QuestsKeyDeprecated

```
public const string QuestsKeyDeprecated = "quests"
```

Field Value

string

RankingMapAddressKey

```
public const string RankingMapAddressKey = "rma"
```

Field Value

string

RateKey

```
public const string RateKey = "r"
```

Field Value

string

ReceivedBlockIndexKey

```
public const string ReceivedBlockIndexKey = "rbi2"
```

Field Value

string

RequiredBlockIndexKey

```
public const string RequiredBlockIndexKey = "rbi"
```

Field Value

string

RequiredCharacterLevelKey

```
public const string RequiredCharacterLevelKey = "rc"
```

Field Value

string

RewardLevelKey

```
public const string RewardLevelKey = "rl"
```

Field Value

string

RewardLevelMapKey

```
public const string RewardLevelMapKey = "rlm"
```

Field Value

string

RewardMapKey

```
public const string RewardMapKey = "rm"
```

Field Value

string

RoundKey

```
public const string RoundKey = "rd"
```

Field Value

string

RuneInfos

```
public const string RuneInfos = "ri"
```

Field Value

string

SellerAgentAddressKey

```
public const string SellerAgentAddressKey = "sga"
```

Field Value

string

SellerAvatarAddressKey

```
public const string SellerAvatarAddressKey = "sva"
```

Field Value

string

SellerResultsKey

```
public const string SellerResultsKey = "srs"
```

Field Value

string

SetIdKey

```
public const string SetIdKey = "si"
```

Field Value

string

SkillsKey

```
public const string SkillsKey = "sk"
```

Field Value

string

SpineResourcePathKey

```
public const string SpineResourcePathKey = "srp"
```

Field Value

string

StageMapKey

```
public const string StageMapKey = "sm"
```

Field Value

string

StakeAvatarAddressKey

```
public const string StakeAvatarAddressKey = "saa"
```

Field Value

string

StartedBlockIndexKey

```
public const string StartedBlockIndexKey = "sbi"
```

Field Value

string

StatKey

```
public const string StatKey = "st"
```

Field Value

string

StatTypeKey

```
public const string StatTypeKey = "stt"
```

Field Value

string

StatValueKey

```
public const string StatValueKey = "v"
```

Field Value

string

TailKey

```
public const string TailKey = "t"
```

Field Value

string

TicketKey

```
public const string TicketKey = "tk"
```

Field Value

string

TradableFungibleItemCountKey

```
public const string TradableFungibleItemCountKey = "tradable_fungible_item_count"
```

Field Value

string

TradableFungibleItemKey

```
public const string TradableFungibleItemKey = "tradable_fungible_item"
```

Field Value

string

TradableIdKey

```
public const string TradableIdKey = "ti"
```

Field Value

string

UpdateSellInfoKey

```
public const string UpdateSellInfoKey = "usi"
```

Field Value

string

UpdatedAtKey

```
public const string UpdatedAtKey = "ua"
```

Field Value

string

updateSellOrderIdKey

```
public const string updateSellOrderIdKey = "roi"
```

Field Value

string

Namespace Lib9c.Exceptions

Classes

InvalidStateTypeException

Class InvalidStateTypeException

Namespace: Lib9c.Exceptions

Assembly: Lib9c.dll

```
[Serializable]
public class InvalidStateTypeException : Exception, ISerializable
```

Inheritance

object ← Exception ← InvalidStateTypeException

Implements

ISerializable

Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Constructors

InvalidStateTypeException(string)

```
public InvalidStateTypeException(string message)
```

Parameters

`message` string

Namespace Lib9c.Model.Order

Classes

FungibleOrder

NonFungibleOrder

Order

OrderBase

OrderDigest

OrderDigestListState

OrderFactory

OrderReceipt

Enums

Order.OrderType

Class FungibleOrder

Namespace: Lib9c.Model.Order

Assembly: Lib9c.dll

```
[Serializable]
public class FungibleOrder : Order
```

Inheritance

object ← OrderBase ← Order ← FungibleOrder

Inherited Members

Order.ExpirationInterval , Order.DeriveAddress(Guid) , Order.SellerAgentAddress , Order.SellerAvatarAddress , Order.Price , Order.ItemSubType , Order.GetTax() , Order.Cancel(AvatarState, long) , Order.Equals(Order) , OrderBase.OrderId , OrderBase.TradableId , OrderBase.StartedBlockIndex , OrderBase.ExpiredBlockIndex , OrderBase.Equals(OrderBase) , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

FungibleOrder(Dictionary)

```
public FungibleOrder(Dictionary serialized)
```

Parameters

`serialized` Dictionary

FungibleOrder(Address, Address, Guid, FungibleAssetValue, Guid, long, ItemSubType, int)

```
public FungibleOrder(Address sellerAgentAddress, Address sellerAvatarAddress, Guid
orderId, FungibleAssetValue price, Guid tradableId, long startedBlockIndex,
ItemSubType itemSubType, int itemCount)
```

Parameters

`sellerAgentAddress` Address

`sellerAvatarAddress` Address

`orderId` Guid

`price` FungibleAssetValue

`tradableId` Guid

`startedBlockIndex` long

`itemSubType` ItemSubType

`itemCount` int

Fields

ItemCount

```
public readonly int ItemCount
```

Field Value

int

Properties

Type

```
public override Order.OrderType Type { get; }
```

Property Value

Order.OrderType

Methods

Cancel2(AvatarState, long)

```
[Obsolete("Use Cancel")]
public override ITradableItem Cancel2(AvatarState avatarState, long blockIndex)
```

Parameters

`avatarState` AvatarState

`blockIndex` long

Returns

ITradableItem

Digest(AvatarState, CostumeStatSheet)

```
public override OrderDigest Digest(AvatarState avatarState,
CostumeStatSheet costumeStatSheet)
```

Parameters

`avatarState` AvatarState

`costumeStatSheet` CostumeStatSheet

Returns

OrderDigest

Digest2(AvatarState, CostumeStatSheet)

```
[Obsolete("Use Digest")]
public override OrderDigest Digest2(AvatarState avatarState,
CostumeStatSheet costumeStatSheet)
```

Parameters

`avatarState` AvatarState

`costumeStatSheet` CostumeStatSheet

Returns

OrderDigest

Equals(FungibleOrder)

```
protected bool Equals(FungibleOrder other)
```

Parameters

`other` FungibleOrder

Returns

bool

Equals(object)

Determines whether the specified object is equal to the current object.

```
public override bool Equals(object obj)
```

Parameters

`obj` object

The object to compare with the current object.

Returns

bool

true if the specified object is equal to the current object; otherwise, false.

GetHashCode()

Serves as the default hash function.

```
public override int GetHashCode()
```

Returns

int

A hash code for the current object.

Sell(AvatarState)

```
[Obsolete("Not Used")]
public override ITradableItem Sell(AvatarState avatarState)
```

Parameters

`avatarState` AvatarState

Returns

ITradableItem

Sell2(AvatarState)

```
[Obsolete("Use Sell")]
public override ITradableItem Sell2(AvatarState avatarState)
```

Parameters

`avatarState` AvatarState

Returns

ITradableItem

Sell3(AvatarState)

```
[Obsolete("Use Sell")]
public override ITradableItem Sell3(AvatarState avatarState)
```

Parameters

`avatarState` AvatarState

Returns

ITradableItem

Sell4(AvatarState)

```
[Obsolete("Use Sell")]
public override ITradableItem Sell4(AvatarState avatarState)
```

Parameters

`avatarState` AvatarState

Returns

ITradableItem

Serialize()

```
public override IValue Serialize()
```

Returns

IValue

Transfer(AvatarState, AvatarState, long)

```
public override OrderReceipt Transfer(AvatarState seller, AvatarState buyer,
long blockIndex)
```

Parameters

`seller` AvatarState

`buyer` AvatarState

`blockIndex` long

Returns

OrderReceipt

Transfer2(AvatarState, AvatarState, long)

```
[Obsolete("Use Transfer")]
public override OrderReceipt Transfer2(AvatarState seller, AvatarState buyer,
long blockIndex)
```

Parameters

`seller` AvatarState

`buyer` AvatarState

`blockIndex` long

Returns

OrderReceipt

Transfer3(AvatarState, AvatarState, long)

```
[Obsolete("Use Transfer")]
public override OrderReceipt Transfer3(AvatarState seller, AvatarState buyer,
long blockIndex)
```

Parameters

`seller` AvatarState

`buyer` AvatarState

`blockIndex` long

Returns

OrderReceipt

Validate(AvatarState, int)

```
public override void Validate(AvatarState avatarState, int count)
```

Parameters

`avatarState` AvatarState

`count` int

ValidateCancelOrder(AvatarState, Guid)

```
public override void ValidateCancelOrder(AvatarState avatarState, Guid tradableId)
```

Parameters

`avatarState` AvatarState

`tradableId` Guid

ValidateCancelOrder2(AvatarState, Guid)

```
[Obsolete("Use ValidateCancelOrder")]
public override void ValidateCancelOrder2(AvatarState avatarState, Guid tradableId)
```

Parameters

`avatarState` AvatarState

`tradableId` Guid

ValidateTransfer(AvatarState, Guid, FungibleAssetValue, long)

```
public override int ValidateTransfer(AvatarState avatarState, Guid tradableId,
FungibleAssetValue price, long blockIndex)
```

Parameters

`avatarState` AvatarState

`tradableId` Guid

`price` FungibleAssetValue

`blockIndex` long

Returns

int

ValidateTransfer2(AvatarState, Guid, FungibleAssetValue, long)

```
[Obsolete("Use ValidateTransfer")]
public override int ValidateTransfer2(AvatarState avatarState, Guid tradableId,
FungibleAssetValue price, long blockIndex)
```

Parameters

`avatarState` AvatarState

`tradableId` Guid

`price` FungibleAssetValue

`blockIndex` long

Returns

int

Class NonFungibleOrder

Namespace: Lib9c.Model.Order

Assembly: Lib9c.dll

```
[Serializable]
public class NonFungibleOrder : Order
```

Inheritance

object ← OrderBase ← Order ← NonFungibleOrder

Inherited Members

Order.ExpirationInterval , Order.DeriveAddress(Guid) , Order.SellerAgentAddress , Order.SellerAvatarAddress , Order.Price , Order.ItemSubType , Order.GetTax() , Order.Cancel(AvatarState, long) , Order.Serialize() , Order.Equals(Order) , Order.Equals(object) , Order.GetHashCode() , OrderBase.OrderId , OrderBase.TradableId , OrderBase.StartedBlockIndex , OrderBase.ExpiredBlockIndex , OrderBase.Equals(OrderBase) , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

NonFungibleOrder(Dictionary)

```
public NonFungibleOrder(Dictionary serialized)
```

Parameters

`serialized` Dictionary

NonFungibleOrder(Address, Address, Guid, FungibleAssetValue, Guid, long, ItemSubType)

```
public NonFungibleOrder(Address sellerAgentAddress, Address sellerAvatarAddress,
Guid orderId, FungibleAssetValue price, Guid tradableId, long startedBlockIndex,
```

```
ItemSubType itemSubType)
```

Parameters

`sellerAgentAddress` Address

`sellerAvatarAddress` Address

`orderId` Guid

`price` FungibleAssetValue

`tradableId` Guid

`startedBlockIndex` long

`itemSubType` ItemSubType

Properties

Type

```
public override Order.OrderType Type { get; }
```

Property Value

Order.OrderType

Methods

Cancel2(AvatarState, long)

```
[Obsolete("Use Cancel")]
public override ITradableItem Cancel2(AvatarState avatarState, long blockIndex)
```

Parameters

`avatarState` AvatarState

`blockIndex` long

Returns

ITradableItem

Digest(AvatarState, CostumeStatSheet)

```
public override OrderDigest Digest(AvatarState avatarState,
CostumeStatSheet costumeStatSheet)
```

Parameters

`avatarState` AvatarState

`costumeStatSheet` CostumeStatSheet

Returns

OrderDigest

Digest2(AvatarState, CostumeStatSheet)

```
[Obsolete("Use Digest")]
public override OrderDigest Digest2(AvatarState avatarState,
CostumeStatSheet costumeStatSheet)
```

Parameters

`avatarState` AvatarState

`costumeStatSheet` CostumeStatSheet

Returns

OrderDigest

Sell(AvatarState)

```
public override ITradableItem Sell(AvatarState avatarState)
```

Parameters

`avatarState` AvatarState

Returns

ITradableItem

Sell2(AvatarState)

```
[Obsolete("Use Sell")]
public override ITradableItem Sell2(AvatarState avatarState)
```

Parameters

`avatarState` AvatarState

Returns

ITradableItem

Sell3(AvatarState)

```
[Obsolete("Use Sell")]
public override ITradableItem Sell3(AvatarState avatarState)
```

Parameters

`avatarState` AvatarState

Returns

ITradableItem

Sell4(AvatarState)

```
[Obsolete("Use Sell")]
public override ITradableItem Sell4(AvatarState avatarState)
```

Parameters

`avatarState` AvatarState

Returns

ITradableItem

Transfer(AvatarState, AvatarState, long)

```
public override OrderReceipt Transfer(AvatarState seller, AvatarState buyer,
long blockIndex)
```

Parameters

`seller` AvatarState

`buyer` AvatarState

`blockIndex` long

Returns

OrderReceipt

Transfer2(AvatarState, AvatarState, long)

```
[Obsolete("Use Transfer")]
public override OrderReceipt Transfer2(AvatarState seller, AvatarState buyer,
long blockIndex)
```

Parameters

`seller` AvatarState

`buyer` AvatarState

`blockIndex` long

Returns

OrderReceipt

Transfer3(AvatarState, AvatarState, long)

```
[Obsolete("Use Transfer")]
public override OrderReceipt Transfer3(AvatarState seller, AvatarState buyer,
long blockIndex)
```

Parameters

`seller` AvatarState

`buyer` AvatarState

`blockIndex` long

Returns

OrderReceipt

Validate(AvatarState, int)

```
public override void Validate(AvatarState avatarState, int count)
```

Parameters

`avatarState` AvatarState

`count` int

ValidateCancelOrder(AvatarState, Guid)

```
public override void ValidateCancelOrder(AvatarState avatarState, Guid tradableId)
```

Parameters

`avatarState` AvatarState

`tradableId` Guid

ValidateCancelOrder2(AvatarState, Guid)

```
[Obsolete("Use ValidateCancelOrder")]
public override void ValidateCancelOrder2(AvatarState avatarState, Guid tradableId)
```

Parameters

`avatarState` AvatarState

`tradableId` Guid

ValidateTransfer(AvatarState, Guid, FungibleAssetValue, long)

```
public override int ValidateTransfer(AvatarState avatarState, Guid tradableId,
FungibleAssetValue price, long blockIndex)
```

Parameters

`avatarState` AvatarState

`tradableId` Guid

`price` FungibleAssetValue

`blockIndex` long

Returns

int

ValidateTransfer2(AvatarState, Guid, FungibleAssetValue, long)

```
[Obsolete("Use ValidateTransfer")]
public override int ValidateTransfer2(AvatarState avatarState, Guid tradableId,
FungibleAssetValue price, long blockIndex)
```

Parameters

`avatarState` AvatarState

`tradableId` Guid

`price` FungibleAssetValue

`blockIndex` long

Returns

int

Class Order

Namespace: Lib9c.Model.Order

Assembly: Lib9c.dll

```
[Serializable]
public abstract class Order : OrderBase
```

Inheritance

object ← OrderBase ← Order

Derived

FungibleOrder, NonFungibleOrder

Inherited Members

OrderBase.OrderId , OrderBase.TradableId , OrderBase.StartedBlockIndex , OrderBase.ExpiredBlockIndex , OrderBase.Equals(OrderBase) , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

Order(Dictionary)

```
protected Order(Dictionary serialized)
```

Parameters

`serialized` Dictionary

Order(Address, Address, Guid, FungibleAssetValue, Guid, long, ItemSubType)

```
protected Order(Address sellerAgentAddress, Address sellerAvatarAddress, Guid
orderId, FungibleAssetValue price, Guid tradableId, long startedBlockIndex,
ItemSubType itemSubType)
```

Parameters

`sellerAgentAddress` Address

`sellerAvatarAddress` Address

`orderId` Guid

`price` FungibleAssetValue

`tradableId` Guid

`startedBlockIndex` long

`itemSubType` ItemSubType

Fields

ExpirationInterval

```
public const long ExpirationInterval = 36000
```

Field Value

long

Properties

ItemSubType

```
public ItemSubType ItemSubType { get; }
```

Property Value

ItemSubType

Price

```
public FungibleAssetValue Price { get; }
```

Property Value

FungibleAssetValue

SellerAgentAddress

```
public Address SellerAgentAddress { get; }
```

Property Value

Address

SellerAvatarAddress

```
public Address SellerAvatarAddress { get; }
```

Property Value

Address

Type

```
public abstract Order.OrderType Type { get; }
```

Property Value

Order.OrderType

Methods

Cancel(AvatarState, long)

```
public ITradableItem Cancel(AvatarState avatarState, long blockIndex)
```

Parameters

`avatarState` AvatarState

`blockIndex` long

Returns

ITradableItem

Cancel2(AvatarState, long)

```
[Obsolete("Use Cancel")]
public abstract ITradableItem Cancel2(AvatarState avatarState, long blockIndex)
```

Parameters

`avatarState` AvatarState

`blockIndex` long

Returns

ITradableItem

DeriveAddress(Guid)

```
public static Address DeriveAddress(Guid orderId)
```

Parameters

`orderId` Guid

Returns

Digest(AvatarState, CostumeStatSheet)

```
public abstract OrderDigest Digest(AvatarState avatarState,
CostumeStatSheet costumeStatSheet)
```

Parameters

`avatarState` AvatarState

`costumeStatSheet` CostumeStatSheet

Returns

OrderDigest

Digest2(AvatarState, CostumeStatSheet)

```
[Obsolete("Use Digest")]
public abstract OrderDigest Digest2(AvatarState avatarState,
CostumeStatSheet costumeStatSheet)
```

Parameters

`avatarState` AvatarState

`costumeStatSheet` CostumeStatSheet

Returns

OrderDigest

Equals(Order)

```
protected bool Equals(Order other)
```

Parameters

`other` Order

Returns

bool

Equals(object)

Determines whether the specified object is equal to the current object.

```
public override bool Equals(object obj)
```

Parameters

`obj` object

The object to compare with the current object.

Returns

bool

true if the specified object is equal to the current object; otherwise, false.

GetHashCode()

Serves as the default hash function.

```
public override int GetHashCode()
```

Returns

int

A hash code for the current object.

GetTax()

```
public FungibleAssetValue GetTax()
```

Returns

FungibleAssetValue

Sell(AvatarState)

```
public abstract ITradableItem Sell(AvatarState avatarState)
```

Parameters

`avatarState` AvatarState

Returns

ITradableItem

Sell2(AvatarState)

```
[Obsolete("Use Sell")]
public abstract ITradableItem Sell2(AvatarState avatarState)
```

Parameters

`avatarState` AvatarState

Returns

ITradableItem

Sell3(AvatarState)

```
[Obsolete("Use Sell")]
public abstract ITradableItem Sell3(AvatarState avatarState)
```

Parameters

`avatarState` AvatarState

Returns

ITradableItem

Sell4(AvatarState)

```
[Obsolete("Use Sell")]
public abstract ITradableItem Sell4(AvatarState avatarState)
```

Parameters

`avatarState` AvatarState

Returns

ITradableItem

Serialize()

```
public override IValue Serialize()
```

Returns

IValue

Transfer(AvatarState, AvatarState, long)

```
public abstract OrderReceipt Transfer(AvatarState seller, AvatarState buyer,
long blockIndex)
```

Parameters

`seller` AvatarState

`buyer` AvatarState

`blockIndex` long

Returns

OrderReceipt

Transfer2(AvatarState, AvatarState, long)

```
[Obsolete("Use Transfer")]
public abstract OrderReceipt Transfer2(AvatarState seller, AvatarState buyer,
long blockIndex)
```

Parameters

`seller` AvatarState

`buyer` AvatarState

`blockIndex` long

Returns

OrderReceipt

Transfer3(AvatarState, AvatarState, long)

```
[Obsolete("Use Transfer")]
public abstract OrderReceipt Transfer3(AvatarState seller, AvatarState buyer,
long blockIndex)
```

Parameters

`seller` AvatarState

`buyer` AvatarState

`blockIndex` long

Returns

OrderReceipt

Validate(AvatarState, int)

```
public virtual void Validate(AvatarState avatarState, int count)
```

Parameters

`avatarState` AvatarState

`count` int

ValidateCancelOrder(AvatarState, Guid)

```
public virtual void ValidateCancelOrder(AvatarState avatarState, Guid tradableId)
```

Parameters

`avatarState` AvatarState

`tradableId` Guid

ValidateCancelOrder2(AvatarState, Guid)

```
[Obsolete("Use ValidateCancelOrder")]
public virtual void ValidateCancelOrder2(AvatarState avatarState, Guid tradableId)
```

Parameters

`avatarState` AvatarState

`tradableId` Guid

ValidateTransfer(AvatarState, Guid, FungibleAssetValue, long)

```
public virtual int ValidateTransfer(AvatarState avatarState, Guid tradableId,
FungibleAssetValue price, long blockIndex)
```

Parameters

`avatarState` AvatarState

`tradableId` Guid

`price` FungibleAssetValue

`blockIndex` long

Returns

int

ValidateTransfer2(AvatarState, Guid, FungibleAssetValue, long)

```
[Obsolete("Use ValidateTransfer")]
public virtual int ValidateTransfer2(AvatarState avatarState, Guid tradableId,
FungibleAssetValue price, long blockIndex)
```

Parameters

`avatarState` AvatarState

`tradableId` Guid

`price` FungibleAssetValue

`blockIndex` long

Returns

int

Enum Order.OrderType

Namespace: Lib9c.Model.Order

Assembly: Lib9c.dll

```
public enum Order.OrderType
```

Extension Methods
StateExtensions.Serialize(Enum)

Fields

`Fungible = 0`

`NonFungible = 1`

Class OrderBase

Namespace: Lib9c.Model.Order

Assembly: Lib9c.dll

```
[Serializable]
public class OrderBase
```

Inheritance

object ← OrderBase

Derived

Order, OrderDigest

Inherited Members

object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

OrderBase(Dictionary)

```
public OrderBase(Dictionary serialized)
```

Parameters

`serialized` Dictionary

OrderBase(Guid, Guid, long, long)

```
public OrderBase(Guid orderId, Guid tradableId, long startedBlockIndex,
long expiredBlockIndex)
```

Parameters

`orderId` Guid

`tradableId` Guid

`startedBlockIndex` long

`expiredBlockIndex` long

Fields

OrderId

```
public readonly Guid OrderId
```

Field Value

Guid

TradableId

```
public readonly Guid TradableId
```

Field Value

Guid

Properties

ExpiredBlockIndex

```
public long ExpiredBlockIndex { get; }
```

Property Value

long

StartedBlockIndex

```
public long StartedBlockIndex { get; }
```

Property Value

long

Methods

Equals(OrderBase)

```
protected bool Equals(OrderBase other)
```

Parameters

`other` OrderBase

Returns

bool

Equals(object)

Determines whether the specified object is equal to the current object.

```
public override bool Equals(object obj)
```

Parameters

`obj` object

The object to compare with the current object.

Returns

bool

true if the specified object is equal to the current object; otherwise, false.

GetHashCode()

Serves as the default hash function.

```
public override int GetHashCode()
```

Returns

int

A hash code for the current object.

Serialize()

```
public virtual IValue Serialize()
```

Returns

IValue

Class OrderDigest

Namespace: Lib9c.Model.Order

Assembly: Lib9c.dll

```
[Serializable]
public class OrderDigest : OrderBase
```

Inheritance

object ← OrderBase ← OrderDigest

Inherited Members

OrderBase.OrderId , OrderBase.TradableId , OrderBase.StartedBlockIndex , OrderBase.ExpiredBlockIndex , OrderBase.Equals(OrderBase) , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

OrderDigest(Dictionary)

```
public OrderDigest(Dictionary serialized)
```

Parameters

`serialized` Dictionary

OrderDigest(Address, long, long, Guid, Guid, FungibleAssetValue, long, int, int, int)

```
public OrderDigest(Address sellerAgentAddress, long startedBlockIndex, long
expiredBlockIndex, Guid orderId, Guid tradableId, FungibleAssetValue price, long
combatPoint, int level, int itemId, int itemCount)
```

Parameters

`sellerAgentAddress` Address

`startedBlockIndex` long

`expiredBlockIndex` long

`orderId` Guid

`tradableId` Guid

`price` FungibleAssetValue

`combatPoint` long

`level` int

`itemId` int

`itemCount` int

Fields

CombatPoint

```
public readonly long CombatPoint
```

Field Value

long

ItemCount

```
public readonly int ItemCount
```

Field Value

int

ItemId

```
public readonly int ItemId
```

Field Value

int

Level

```
public readonly int Level
```

Field Value

int

Price

```
public readonly FungibleAssetValue Price
```

Field Value

FungibleAssetValue

SellerAgentAddress

```
public readonly Address SellerAgentAddress
```

Field Value

Address

Methods

Equals(OrderDigest)

```
protected bool Equals(OrderDigest other)
```

Parameters

`other` OrderDigest

Returns

bool

Equals(object)

Determines whether the specified object is equal to the current object.

```
public override bool Equals(object obj)
```

Parameters

`obj` object

The object to compare with the current object.

Returns

bool

true if the specified object is equal to the current object; otherwise, false.

GetHashCode()

Serves as the default hash function.

```
public override int GetHashCode()
```

Returns

int

A hash code for the current object.

Serialize()

```
public override IValue Serialize()
```

Returns

IValue

Class OrderDigestListState

Namespace: Lib9c.Model.Order

Assembly: Lib9c.dll

```
[Serializable]
public class OrderDigestListState
```

Inheritance

object ← OrderDigestListState

Inherited Members

object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

OrderDigestListState(Dictionary)

```
public OrderDigestListState(Dictionary serialized)
```

Parameters

`serialized` Dictionary

OrderDigestListState(Address)

```
public OrderDigestListState(Address address)
```

Parameters

`address` Address

Fields

Address

```
public readonly Address Address
```

Field Value

Address

Properties

OrderDigestList

```
public IReadOnlyList<OrderDigest> OrderDigestList { get; }
```

Property Value

IReadOnlyList<OrderDigest>

Methods

Add(OrderDigest)

```
public void Add(OrderDigest orderDigest)
```

Parameters

`orderDigest` OrderDigest

DeriveAddress(Address)

```
public static Address DeriveAddress(Address avatarAddress)
```

Parameters

`avatarAddress` Address

Returns

Address

Equals(OrderDigestListState)

```
protected bool Equals(OrderDigestListState other)
```

Parameters

`other` OrderDigestListState

Returns

bool

Equals(object)

Determines whether the specified object is equal to the current object.

```
public override bool Equals(object obj)
```

Parameters

`obj` object

The object to compare with the current object.

Returns

bool

true if the specified object is equal to the current object; otherwise, false.

GetHashCode()

Serves as the default hash function.

```
public override int GetHashCode()
```

Returns

int

A hash code for the current object.

Remove(Guid)

```
public void Remove(Guid orderId)
```

Parameters

`orderId` Guid

Serialize()

```
public IValue Serialize()
```

Returns

IValue

Class OrderFactory

Namespace: Lib9c.Model.Order

Assembly: Lib9c.dll

```
public static class OrderFactory
```

Inheritance

object ← OrderFactory

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Methods

Create(Address, Address, Guid, FungibleAssetValue, Guid, long, ItemSubType, int)

```
public static Order Create(Address agentAddress, Address avatarAddress, Guid
orderId, FungibleAssetValue price, Guid tradableId, long startedIndex, ItemSubType
itemSubType, int count)
```

Parameters

`agentAddress` Address

`avatarAddress` Address

`orderId` Guid

`price` FungibleAssetValue

`tradableId` Guid

`startedIndex` long

`itemSubType` ItemSubType

`count` int

Returns

Order

CreateFungibleOrder(Address, Address, Guid, FungibleAssetValue, Guid, long, int, ItemSubType)

```
public static FungibleOrder CreateFungibleOrder(Address sellerAgentAddress, Address
sellerAvatarAddress, Guid orderId, FungibleAssetValue price, Guid itemId, long
startedBlockIndex, int count, ItemSubType itemSubType)
```

Parameters

`sellerAgentAddress` Address

`sellerAvatarAddress` Address

`orderId` Guid

`price` FungibleAssetValue

`itemId` Guid

`startedBlockIndex` long

`count` int

`itemSubType` ItemSubType

Returns

FungibleOrder

CreateNonFungibleOrder(Address, Address, Guid, FungibleAssetValue, Guid, long, ItemSubType)

```
public static NonFungibleOrder CreateNonFungibleOrder(Address sellerAgentAddress,
Address sellerAvatarAddress, Guid orderId, FungibleAssetValue price, Guid itemId,
long startedBlockIndex, ItemSubType itemSubType)
```

Parameters

`sellerAgentAddress` Address

`sellerAvatarAddress` Address

`orderId` Guid

`price` FungibleAssetValue

`itemId` Guid

`startedBlockIndex` long

`itemSubType` ItemSubType

Returns

NonFungibleOrder

Deserialize(Dictionary)

```
public static Order Deserialize(Dictionary dictionary)
```

Parameters

`dictionary` Dictionary

Returns

Order

Class OrderReceipt

Namespace: Lib9c.Model.Order

Assembly: Lib9c.dll

```
[Serializable]
public class OrderReceipt
```

Inheritance

object ← OrderReceipt

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

OrderReceipt(Dictionary)

```
public OrderReceipt(Dictionary serialized)
```

Parameters

`serialized` Dictionary

OrderReceipt(Guid, Address, Address, long)

```
public OrderReceipt(Guid orderId, Address buyerAgentAddress, Address
buyerAvatarAddress, long transferredBlockIndex)
```

Parameters

`orderId` Guid

`buyerAgentAddress` Address

`buyerAvatarAddress` Address

`transferredBlockIndex` long

Fields

BuyerAgentAddress

```
public readonly Address BuyerAgentAddress
```

Field Value

Address

BuyerAvatarAddress

```
public readonly Address BuyerAvatarAddress
```

Field Value

Address

OrderId

```
public readonly Guid OrderId
```

Field Value

Guid

TransferredBlockIndex

```
public readonly long TransferredBlockIndex
```

Field Value

long

Methods

DeriveAddress(Guid)

```
public static Address DeriveAddress(Guid orderId)
```

Parameters

`orderId` Guid

Returns

Address

Serialize()

```
public IValue Serialize()
```

Returns

IValue

Namespace Nekoyume

Classes

ActionObsoleteAttributeExtension

ActionObsoleteConfig

Addresses

CurrencyExtensions

GameConfig

Game-wide constants shared by actions and the client.

GameConfig.MaxEquipmentSlotCount

GameConfig.RequireCharacterLevel

GameConfig.RequireClearedStageLevel

MeadConfig

PlanetExtension

StoreUtils

Structs

AppProtocolVersionExtra

Enums

Planet

Class ActionObsoleteAttributeExtension

Namespace: Nekoyume

Assembly: Lib9c.dll

```
public static class ActionObsoleteAttributeExtension
```

Inheritance

object ← ActionObsoleteAttributeExtension

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Methods

IsObsolete(IEnumerable<ActionObsoleteAttribute>, Planet, long)

```
public static bool IsObsolete(this IEnumerable<ActionObsoleteAttribute> attrs,
Planet planet, long blockIndex)
```

Parameters

`attrs` IEnumerable<ActionObsoleteAttribute>

`planet` Planet

`blockIndex` long

Returns

bool

Class ActionObsoleteConfig

Namespace: Nekoyume

Assembly: Lib9c.dll

```
public static class ActionObsoleteConfig
```

Inheritance

object ← ActionObsoleteConfig

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Fields

V100080ObsoleteIndex

```
public const long V100080ObsoleteIndex = 2448000
```

Field Value

long

V100081ObsoleteIndex

```
public const long V100081ObsoleteIndex = 2550000
```

Field Value

long

V100083ObsoleteIndex

```
public const long V100083ObsoleteIndex = 2680000
```

Field Value

long

V100086ObsoleteIndex

```
public const long V100086ObsoleteIndex = 2800001
```

Field Value

long

V100089ObsoleteIndex

```
public const long V100089ObsoleteIndex = 2908000
```

Field Value

long

V100093ObsoleteIndex

```
public const long V100093ObsoleteIndex = 3050000
```

Field Value

long

V100095ObsoleteIndex

```
public const long V100095ObsoleteIndex = 3317632
```

Field Value

long

V100096ObsoleteIndex

```
public const long V100096ObsoleteIndex = 3317632
```

Field Value

long

V100170ObsoleteIndex

```
public const long V100170ObsoleteIndex = 3810000
```

Field Value

long

V100190ObsoleteIndex

```
public const long V100190ObsoleteIndex = 4204863
```

Field Value

long

V100193ObsoleteIndex

```
public const long V100193ObsoleteIndex = 3975929
```

Field Value

long

V100200ObsoleteIndex

```
public const long V100200ObsoleteIndex = 4246225
```

Field Value

long

V100210ObsoleteIndex

```
public const long V100210ObsoleteIndex = 4366622
```

Field Value

long

V100220ObsoleteIndex

```
public const long V100220ObsoleteIndex = 4390272
```

Field Value

long

V100230ObsoleteIndex

```
public const long V100230ObsoleteIndex = 4558559
```

Field Value

long

V100240ObsoleteIndex

```
public const long V100240ObsoleteIndex = 4578480
```

Field Value

long

V100260ObsoleteIndex

```
public const long V100260ObsoleteIndex = 4596180
```

Field Value

long

V100270ObsoleteIndex

```
public const long V100270ObsoleteIndex = 4841774
```

Field Value

long

V100282ObsoleteIndex

```
public const long V100282ObsoleteIndex = 4835445
```

Field Value

long

V100290ObsoleteIndex

```
public const long V100290ObsoleteIndex = 4913153
```

Field Value

long

V100300ObsoleteIndex

```
public const long V100300ObsoleteIndex = 5150000
```

Field Value

long

V100301ExecutedBlockIndex

```
public const long V100301ExecutedBlockIndex = 5048399
```

Field Value

long

V100310ExecutedBlockIndex

```
public const long V100310ExecutedBlockIndex = 5217577
```

Field Value

long

V100310ObsoleteIndex

```
public const long V100310ObsoleteIndex = 5300000
```

Field Value

long

V100320ObsoleteIndex

```
public const long V100320ObsoleteIndex = 5398001
```

Field Value

long

V100340ObsoleteIndex

```
public const long V100340ObsoleteIndex = 5800000
```

Field Value

long

V100351ObsoleteIndex

```
public const long V100351ObsoleteIndex = 5828750
```

Field Value

long

V100360ObsoleteIndex

```
public const long V100360ObsoleteIndex = 6020000
```

Field Value

long

V200010ObsoleteIndex

```
public const long V200010ObsoleteIndex = 6642114
```

Field Value

long

V200020AccidentObsoleteIndex

```
public const long V200020AccidentObsoleteIndex = 7000000
```

Field Value

long

V200020ObsoleteIndex

```
public const long V200020ObsoleteIndex = 6856587
```

Field Value

long

V200030ObsoleteIndex

```
public const long V200030ObsoleteIndex = 7200000
```

Field Value

long

V200031ObsoleteIndex

```
public const long V200031ObsoleteIndex = 7206000
```

Field Value

long

V200040ObsoleteIndex

```
public const long V200040ObsoleteIndex = 7330000
```

Field Value

long

V200060ObsoleteIndex

```
public const long V200060ObsoleteIndex = 7649999
```

Field Value

long

V200061ObsoleteIndex

```
public const long V200061ObsoleteIndex = 7660000
```

Field Value

long

V200063ObsoleteIndex

```
public const long V200063ObsoleteIndex = 7700000
```

Field Value

long

V200070ObsoleteIndex

```
public const long V200070ObsoleteIndex = 7716400
```

Field Value

long

V200071ObsoleteIndex

```
public const long V200071ObsoleteIndex = 7718878
```

Field Value

long

V200080ObsoleteIndex

```
public const long V200080ObsoleteIndex = 7983895
```

Field Value

long

V200090ObsoleteIndex

```
public const long V200090ObsoleteIndex = 8070865
```

Field Value

long

V200092ObsoleteIndex

```
public const long V200092ObsoleteIndex = 8324909
```

Field Value

long

Class Addresses

Namespace: Nekoyume

Assembly: Lib9c.dll

```
public static class Addresses
```

Inheritance

object ← Addresses

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Fields

AbstainHistory

An address of an account having AbstainHistory.

```
public static readonly Address AbstainHistory
```

Field Value

Address

ActionPoint

```
public static readonly Address ActionPoint
```

Field Value

Address

ActivatedAccount

```
public static readonly Address ActivatedAccount
```

Field Value

Address

Admin

```
public static readonly Address Admin
```

Field Value

Address

AdventureBoss

```
public static readonly Address AdventureBoss
```

Field Value

Address

AdventureCp

```
public static readonly Address AdventureCp
```

Field Value

Address

Agent

```
public static readonly Address Agent
```

Field Value

Address

Arena

```
public static readonly Address Arena
```

Field Value

Address

ArenaCp

```
public static readonly Address ArenaCp
```

Field Value

Address

AssetMinters

```
public static readonly Address AssetMinters
```

Field Value

Address

AuthorizedMiners

```
public static readonly Address AuthorizedMiners
```

Field Value

Address

Avatar

```
public static readonly Address Avatar
```

Field Value

Address

Battle

```
public static readonly Address Battle
```

Field Value

Address

Blacksmith

```
public static readonly Address Blacksmith
```

Field Value

Address

BountyBoard

```
public static readonly Address BountyBoard
```

Field Value

Address

ClaimedGiftIds

```
public static readonly Address ClaimedGiftIds
```

Field Value

Address

Collection

```
public static readonly Address Collection
```

Field Value

Address

CombinationSlot

```
public static readonly Address CombinationSlot
```

Field Value

Address

CommunityPool

An address for community fund.

```
public static readonly Address CommunityPool
```

Field Value

Address

Credits

```
public static readonly Address Credits
```

Field Value

Address

DailyReward

```
public static readonly Address DailyReward
```

Field Value

Address

EmptyAccountAddress

```
public static readonly Address EmptyAccountAddress
```

Field Value

Address

EventDungeon

```
public static readonly Address EventDungeon
```

Field Value

Address

ExploreBoard

```
public static readonly Address ExploreBoard
```

Field Value

Address

ExplorerList

```
public static readonly Address ExplorerList
```

Field Value

Address

GameConfig

```
public static readonly Address GameConfig
```

Field Value

Address

GarageWallet

```
public static readonly Address GarageWallet
```

Field Value

Address

GasPool

```
public static readonly Address GasPool
```

Field Value

Address

GlobalCounter

An account address for the global counters that drive hard world openings.

```
public static readonly Address GlobalCounter
```

Field Value

Address

GoldCurrency

```
public static readonly Address GoldCurrency
```

Field Value

Address

GoldDistribution

```
public static readonly Address GoldDistribution
```

Field Value

Address

Guild

An address of an account having Guild.

```
public static readonly Address Guild
```

Field Value

Address

GuildApplication

An address of an account having Nekoyume.Model.Guild.GuildApplication.

```
public static readonly Address GuildApplication
```

Field Value

Address

GuildBond

An address of an account having Bond.

```
public static readonly Address GuildBond
```

Field Value

Address

GuildDelegateeMetadata

An address of an account having DelegateeMetadata.

```
public static readonly Address GuildDelegateeMetadata
```

Field Value

Address

GuildDelegatorMetadata

An address of an account having DelegatorMetadata.

```
public static readonly Address GuildDelegatorMetadata
```

Field Value

Address

GuildLumpSumRewardsRecord

An address of an account having LumpSumRewardsRecord.

```
public static readonly Address GuildLumpSumRewardsRecord
```

Field Value

Address

GuildMemberCounter

An address of an account having Bencodex.Types.Integer which means the number of the guild.

```
public static readonly Address GuildMemberCounter
```

Field Value

Address

GuildParticipant

An address of an account having GuildParticipant

```
public static readonly Address GuildParticipant
```

Field Value

Address

GuildRebondGrace

An address of an account having RebondGrace.

```
public static readonly Address GuildRebondGrace
```

Field Value

Address

GuildRejoinCooldown

An address of an account having Nekoyume.Model.Guild.GuildRejoinCooldown

```
public static readonly Address GuildRejoinCooldown
```

Field Value

Address

GuildRewardBase

An address of an account having RewardBase.

```
public static readonly Address GuildRewardBase
```

Field Value

Address

GuildUnbondLockIn

An address of an account having UnbondLockIn.

```
public static readonly Address GuildUnbondLockIn
```

Field Value

Address

GuildUnbondingSet

An address of an account having Delegation.UnbondingSet.

```
public static readonly Address GuildUnbondingSet
```

Field Value

Address

InfiniteTowerBoard

Address for the infinite tower board state.

```
public static readonly Address InfiniteTowerBoard
```

Field Value

Address

InfiniteTowerCp

```
public static readonly Address InfiniteTowerCp
```

Field Value

Address

InfiniteTowerInfo

Address for the infinite tower info state.

```
public static readonly Address InfiniteTowerInfo
```

Field Value

Address

Inventory

```
public static readonly Address Inventory
```

Field Value

Address

Market

```
public static readonly Address Market
```

Field Value

Address

MaterialCost

```
public static readonly Address MaterialCost
```

Field Value

Address

Migration

An account address for migration.

```
public static readonly Address Migration
```

Field Value

Address

MortgagePool

```
public static readonly Address MortgagePool
```

Field Value

Address

NonValidatorDelegatee

An address for non validator.

```
[Obsolete("It is not used anymore.")]
public static readonly Address NonValidatorDelegatee
```

Field Value

Address

PatrolReward

```
public static readonly Address PatrolReward
```

Field Value

Address

PayMaster

```
public static readonly Address PayMaster
```

Field Value

Address

PendingActivation

```
public static readonly Address PendingActivation
```

Field Value

Address

QuestList

```
public static readonly Address QuestList
```

Field Value

Address

Raid

```
public static readonly Address Raid
```

Field Value

Address

RaidCp

```
public static readonly Address RaidCp
```

Field Value

Address

Ranking

```
public static readonly Address Ranking
```

Field Value

Address

RedeemCode

```
public static readonly Address RedeemCode
```

Field Value

Address

Relationship

```
public static readonly Address Relationship
```

Field Value

Address

RewardPool

An address for reward pool of validators.

```
public static readonly Address RewardPool
```

Field Value

Address

Rune

```
public static readonly Address Rune
```

Field Value

Address

RuneState

```
public static readonly Address RuneState
```

Field Value

Address

Shop

```
public static readonly Address Shop
```

Field Value

Address

StageRandomBuff

```
public static readonly Address StageRandomBuff
```

Field Value

Address

SuperCraft

```
public static readonly Address SuperCraft
```

Field Value

Address

SwapPool

An address of an account having SwapPool.

```
public static readonly Address SwapPool
```

Field Value

Address

TableSheet

```
public static readonly Address TableSheet
```

Field Value

Address

UnlockEquipmentRecipe

```
public static readonly Address UnlockEquipmentRecipe
```

Field Value

Address

UnlockWorld

```
public static readonly Address UnlockWorld
```

Field Value

Address

ValidatorBond

An address of an account having Bond.

```
public static readonly Address ValidatorBond
```

Field Value

Address

ValidatorDelegatee

An address of an account having ValidatorDelegatee.

```
public static readonly Address ValidatorDelegatee
```

Field Value

Address

ValidatorDelegateeMetadata

An address of an account having DelegateeMetadata.

```
public static readonly Address ValidatorDelegateeMetadata
```

Field Value

Address

ValidatorDelegator

An address of an account having ValidatorDelegator.

```
public static readonly Address ValidatorDelegator
```

Field Value

Address

ValidatorDelegatorMetadata

An address of an account having DelegatorMetadata.

```
public static readonly Address ValidatorDelegatorMetadata
```

Field Value

Address

ValidatorList

An address of an account having ValidatorList.

```
public static readonly Address ValidatorList
```

Field Value

Address

ValidatorLumpSumRewardsRecord

An address of an account having LumpSumRewardsRecord.

```
public static readonly Address ValidatorLumpSumRewardsRecord
```

Field Value

Address

ValidatorRebondGrace

An address of an account having RebondGrace.

```
public static readonly Address ValidatorRebondGrace
```

Field Value

Address

ValidatorRewardBase

An address of an account having RewardBase.

```
public static readonly Address ValidatorRewardBase
```

Field Value

Address

ValidatorUnbondLockIn

An address of an account having UnbondLockIn.

```
public static readonly Address ValidatorUnbondLockIn
```

Field Value

Address

ValidatorUnbondingSet

An address of an account having Delegation.UnbondingSet.

```
public static readonly Address ValidatorUnbondingSet
```

Field Value

Address

WeeklyArena

```
public static readonly Address WeeklyArena
```

Field Value

Address

WorldInformation

```
public static readonly Address WorldInformation
```

Field Value

Address

WorldOpen

An account address for the confirmed hard world openings.

```
public static readonly Address WorldOpen
```

Field Value

Address

Methods

AdventureSeasonAddress(long)

```
public static Address AdventureSeasonAddress(long season)
```

Parameters

`season` long

Returns

Address

CheckAgentHasPermissionOnBalanceAddr(Address, Address)

```
public static bool CheckAgentHasPermissionOnBalanceAddr(Address agentAddr,
Address balanceAddr)
```

Parameters

`agentAddr` Address

`balanceAddr` Address

Returns

bool

CheckAvatarAddrIsContainedInAgent(Address, Address)

```
public static bool CheckAvatarAddrIsContainedInAgent(Address agentAddr,
Address avatarAddr)
```

Parameters

`agentAddr` Address

`avatarAddr` Address

Returns

bool

CheckInventoryAddrIsContainedInAgent(Address, Address)

```
public static bool CheckInventoryAddrIsContainedInAgent(Address agentAddr,
Address inventoryAddr)
```

Parameters

`agentAddr` Address

`inventoryAddr` Address

Returns

bool

GetArenaParticipantAccountAddress(int, int)

Get the account address of an arena participant.

```
public static Address GetArenaParticipantAccountAddress(int championshipId,
int round)
```

Parameters

`championshipId` int

`round` int

Returns

Address

The account address of an arena participant.
"0100000000000000000000000000000000000000" ~
"0199999999999999999999999999999999999999".

GetAvatarAddress(Address, int)

```
public static Address GetAvatarAddress(Address agentAddr, int index)
```

Parameters

`agentAddr` Address

`index` int

Returns

Address

GetCombinationSlotAddress(Address, int)

```
public static Address GetCombinationSlotAddress(Address avatarAddr, int index)
```

Parameters

`avatarAddr` Address

`index` int

Returns

Address

GetCpAccountAddress(BattleType)

Gets the account address for storing CP based on the battle type.

```
public static Address GetCpAccountAddress(BattleType battleType)
```

Parameters

`battleType` BattleType

The type of battle (Adventure, Raid, or Arena)

Returns

Address

The address where CP is stored for the given battle type

Exceptions

ArgumentException

Thrown when an invalid battle type is provided

GetDailyCrystalCostAddress(int)

```
public static Address GetDailyCrystalCostAddress(int index)
```

Parameters

`index` int

Returns

Address

GetGarageAddress(Address, HashDigest<SHA256>)

```
public static Address GetGarageAddress(Address agentAddr,
HashDigest<SHA256> fungibleId)
```

Parameters

`agentAddr` Address

`fungibleId` HashDigest<SHA256>

Returns

Address

GetGarageBalanceAddress(Address)

```
public static Address GetGarageBalanceAddress(Address agentAddr)
```

Parameters

`agentAddr` Address

Returns

Address

GetGuildBanAccountAddress(Address)

Build an Libplanet.Crypto.Address of an Libplanet.Action.State.Account, represented as `agentAddress` ↔ Bencodex.Types.Boolean, indicates whether the `agentAddress` is banned.

```
public static Address GetGuildBanAccountAddress(Address guildAddress)
```

Parameters

`guildAddress` Address

The guild address.

Returns

Address

An account address.

GetHammerPointStateAddress(Address, int)

```
public static Address GetHammerPointStateAddress(Address avatarAddress,
int recipeId)
```

Parameters

`avatarAddress` Address

`recipeId` int

Returns

Address

GetInventoryAddress(Address, int)

```
public static Address GetInventoryAddress(Address agentAddr, int avatarIndex)
```

Parameters

`agentAddr` Address

`avatarIndex` int

Returns

Address

GetItemAddress(Guid)

```
public static Address GetItemAddress(Guid itemId)
```

Parameters

`itemId` Guid

Returns

Address

GetRaiderAddress(Address, int)

```
public static Address GetRaiderAddress(Address avatarAddress, int raidId)
```

Parameters

`avatarAddress` Address

`raidId` int

Returns

Address

GetRaiderListAddress(int)

```
public static Address GetRaiderListAddress(int raidId)
```

Parameters

`raidId` int

Returns

Address

GetSheetAddress(string)

```
public static Address GetSheetAddress(string sheetName)
```

Parameters

`sheetName` string

Returns

Address

GetSheetAddress<T>()

```
public static Address GetSheetAddress<T>() where T : ISheet
```

Returns

Address

Type Parameters

T

GetSkillStateAddressFromAvatarAddress(Address)

```
public static Address GetSkillStateAddressFromAvatarAddress(Address avatarAddress)
```

Parameters

`avatarAddress` Address

Returns

Address

GetWeeklyCrystalCostAddress(int)

```
public static Address GetWeeklyCrystalCostAddress(int index)
```

Parameters

`index` int

Returns

Address

GetWorldBossAddress(int)

```
public static Address GetWorldBossAddress(int raidId)
```

Parameters

`raidId` int

Returns

Address

GetWorldBossKillRewardRecordAddress(Address, int)

```
public static Address GetWorldBossKillRewardRecordAddress(Address avatarAddress,
int raidId)
```

Parameters

`avatarAddress` Address

`raidId` int

Returns

Address

IsArenaParticipantAccountAddress(Address)

Check if the given address is within the range of arena participant account addresses. "0100000000000000000000000000000000000000" ~ "0199999999999999999999999999999999999999".

```
public static bool IsArenaParticipantAccountAddress(Address address)
```

Parameters

`address` Address

Returns

bool

Struct AppProtocolVersionExtra

Namespace: Nekoyume

Assembly: Lib9c.dll

```
public readonly struct AppProtocolVersionExtra
```

Inherited Members
ValueType.Equals(object) , ValueType.GetHashCode() , ValueType.ToString() , object.Equals(object, object) , object.GetType() , object.ReferenceEquals(object, object)

Constructors

AppProtocolVersionExtra(Dictionary)

```
public AppProtocolVersionExtra(Dictionary dictionary)
```

Parameters

`dictionary` Dictionary

AppProtocolVersionExtra(string, string, DateTimeOffset)

```
public AppProtocolVersionExtra(string macOSBinaryUrl, string windowsBinaryUrl,
DateTimeOffset timestamp)
```

Parameters

`macOSBinaryUrl` string

`windowsBinaryUrl` string

`timestamp` DateTimeOffset

Fields

MacOSBinaryUrl

```
public readonly string MacOSBinaryUrl
```

Field Value

string

Timestamp

```
public readonly DateTimeOffset Timestamp
```

Field Value

DateTimeOffset

WindowsBinaryUrl

```
public readonly string WindowsBinaryUrl
```

Field Value

string

Methods

Serialize()

```
public IValue Serialize()
```

Returns

IValue

Class CurrencyExtensions

Namespace: Nekoyume

Assembly: Lib9c.dll

```
public static class CurrencyExtensions
```

Inheritance

object ← CurrencyExtensions

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Methods

Deserialize(Dictionary)

```
public static Currency Deserialize(Dictionary serialized)
```

Parameters

`serialized` Dictionary

Returns

Currency

Serialize(Currency)

```
public static Dictionary Serialize(this Currency currency)
```

Parameters

`currency` Currency

Returns

Dictionary

Class GameConfig

Namespace: Nekoyume

Assembly: Lib9c.dll

Game-wide constants shared by actions and the client.

```
public static class GameConfig
```

Inheritance

object ← GameConfig

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Fields

ActionCostAP

```
public const int ActionCostAP = 5
```

Field Value

int

ArenaChallengeCountMax

```
public const int ArenaChallengeCountMax = 5
```

Field Value

int

ArenaScoreDefault

```
public const int ArenaScoreDefault = 1000
```

Field Value

int

AvatarNickNamePattern

Regular expression pattern that a valid avatar nickname must match (2 to 20 ASCII letters or digits).

```
public const string AvatarNickNamePattern = "^[0-9a-zA-Z]{2,20}$"
```

Field Value

string

DefaultAttackId

```
public const int DefaultAttackId = 100000
```

Field Value

int

DefaultAvatarArmorId

```
public const int DefaultAvatarArmorId = 10200000
```

Field Value

int

DefaultAvatarCharacterId

```
public const int DefaultAvatarCharacterId = 100010
```

Field Value

int

DefaultAvatarWeaponId

```
public const int DefaultAvatarWeaponId = 10100000
```

Field Value

int

EnhanceEquipmentCostAP

```
public const int EnhanceEquipmentCostAP = 0
```

Field Value

int

IsEditor

Whether this build is an editor/development build (`LIB9C_DEV_EXTENSIONS` or `UNITY_EDITOR` defined).

```
public static readonly bool IsEditor
```

Field Value

bool

MaximumProbability

```
public const int MaximumProbability = 10000
```

Field Value

int

MimisbrunnrStartStageId

```
public const int MimisbrunnrStartStageId = 10000001
```

Field Value

int

MimisbrunnrWorldId

```
public const int MimisbrunnrWorldId = 10001
```

Field Value

int

ProgressGatedStageBegin

The first stage that is locked by default and opens only through WorldOpenSheet. Stages below it are always open.

```
public const int ProgressGatedStageBegin = 521
```

Field Value

int

RequiredAppraiseBlock

```
[Obsolete("Use GameConfigState.RequiredAppraiseBlock")]
public const int RequiredAppraiseBlock = 50
```

Field Value

int

SlotCount

The number of avatar slots available per agent (valid avatar indices are 0 to SlotCount - 1).

```
public const int SlotCount = 3
```

Field Value

int

Class GameConfig.MaxEquipmentSlotCount

Namespace: Nekoyume

Assembly: Lib9c.dll

```
public static class GameConfig.MaxEquipmentSlotCount
```

Inheritance

object ← GameConfig.MaxEquipmentSlotCount

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Fields

Armor

```
public const int Armor = 1
```

Field Value

int

Aura

```
public const int Aura = 1
```

Field Value

int

Belt

```
public const int Belt = 1
```

Field Value

int

Grimoire

```
public const int Grimoire = 1
```

Field Value

int

Necklace

```
public const int Necklace = 1
```

Field Value

int

Ring

```
public const int Ring = 2
```

Field Value

int

Weapon

```
public const int Weapon = 1
```

Field Value

int

Class GameConfig.RequireCharacterLevel

Namespace: Nekoyume

Assembly: Lib9c.dll

```
public static class GameConfig.RequireCharacterLevel
```

Inheritance

object ← GameConfig.RequireCharacterLevel

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Fields

CharacterConsumableSlot1

```
[Obsolete("Use GameConfigState.RequireCharacterLevel_ConsumableSlot1")]
public const int CharacterConsumableSlot1 = 1
```

Field Value

int

CharacterConsumableSlot2

```
[Obsolete("Use GameConfigState.RequireCharacterLevel_ConsumableSlot2")]
public const int CharacterConsumableSlot2 = 35
```

Field Value

int

CharacterConsumableSlot3

```
[Obsolete("Use GameConfigState.RequireCharacterLevel_ConsumableSlot3")]
public const int CharacterConsumableSlot3 = 100
```

Field Value

int

CharacterConsumableSlot4

```
[Obsolete("Use GameConfigState.RequireCharacterLevel_ConsumableSlot4")]
public const int CharacterConsumableSlot4 = 200
```

Field Value

int

CharacterConsumableSlot5

```
[Obsolete("Use GameConfigState.RequireCharacterLevel_ConsumableSlot5")]
public const int CharacterConsumableSlot5 = 350
```

Field Value

int

CharacterEarCostumeSlot

```
[Obsolete("Use GameConfigState.RequireCharacterLevel_EarCostumeSlot")]
public const int CharacterEarCostumeSlot = 2
```

Field Value

int

CharacterEquipmentSlotArmor

```
[Obsolete("Use GameConfigState.RequireCharacterLevel_EquipmentSlotArmor")]
public const int CharacterEquipmentSlotArmor = 3
```

Field Value

int

CharacterEquipmentSlotAura

```
[Obsolete("Use GameConfigState.RequireCharacterLevel_EquipmentSlotAura")]
public const int CharacterEquipmentSlotAura = 1
```

Field Value

int

CharacterEquipmentSlotBelt

```
[Obsolete("Use GameConfigState.RequireCharacterLevel_EquipmentSlotBelt")]
public const int CharacterEquipmentSlotBelt = 5
```

Field Value

int

CharacterEquipmentSlotNecklace

```
[Obsolete("Use GameConfigState.RequireCharacterLevel_EquipmentSlotNecklace")]
public const int CharacterEquipmentSlotNecklace = 8
```

Field Value

int

CharacterEquipmentSlotRing1

```
[Obsolete("Use GameConfigState.RequireCharacterLevel_EquipmentSlotRing1")]
public const int CharacterEquipmentSlotRing1 = 13
```

Field Value

int

CharacterEquipmentSlotRing2

```
[Obsolete("Use GameConfigState.RequireCharacterLevel_EquipmentSlotRing2")]
public const int CharacterEquipmentSlotRing2 = 46
```

Field Value

int

CharacterEquipmentSlotWeapon

```
[Obsolete("Use GameConfigState.RequireCharacterLevel_EquipmentSlotWeapon")]
public const int CharacterEquipmentSlotWeapon = 1
```

Field Value

int

CharacterEyeCostumeSlot

```
[Obsolete("Use GameConfigState.RequireCharacterLevel_EyeCostumeSlot")]
public const int CharacterEyeCostumeSlot = 2
```

Field Value

int

CharacterFullCostumeSlot

```
[Obsolete("Use GameConfigState.RequireCharacterLevel_FullCostumeSlot")]
public const int CharacterFullCostumeSlot = 2
```

Field Value

int

CharacterHairCostumeSlot

```
[Obsolete("Use GameConfigState.RequireCharacterLevel_HairCostumeSlot")]
public const int CharacterHairCostumeSlot = 2
```

Field Value

int

CharacterTailCostumeSlot

```
[Obsolete("Use GameConfigState.RequireCharacterLevel_TailCostumeSlot")]
public const int CharacterTailCostumeSlot = 2
```

Field Value

int

CharacterTitleSlot

```
[Obsolete("Use GameConfigState.RequireCharacterLevel_TitleSlot")]
```

```
public const int CharacterTitleSlot = 1
```

Field Value

int

Class GameConfig.RequireClearedStageLevel

Namespace: Nekoyume

Assembly: Lib9c.dll

```
public static class GameConfig.RequireClearedStageLevel
```

Inheritance

object ← GameConfig.RequireClearedStageLevel

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Fields

ActionsInMimisbrunnr

```
[Obsolete("Not used anymore since v200092")]
public const int ActionsInMimisbrunnr = 100
```

Field Value

int

ActionsInRaid

```
[Obsolete("Not used anymore since v200092")]
public const int ActionsInRaid = 50
```

Field Value

int

ActionsInRankingBoard

```
[Obsolete("Not used anymore since v200092")]
public const int ActionsInRankingBoard = 25
```

Field Value

int

ActionsInShop

```
[Obsolete("Not used anymore since v200092")]
public const int ActionsInShop = 17
```

Field Value

int

CombinationConsumableAction

```
[Obsolete("Not used anymore since v200092")]
public const int CombinationConsumableAction = 6
```

Field Value

int

CombinationEquipmentAction

```
[Obsolete("Not used anymore since v200092")]
public const int CombinationEquipmentAction = 3
```

Field Value

int

ItemEnhancementAction

```
[Obsolete("Not used anymore since v200092")]
public const int ItemEnhancementAction = 9
```

Field Value

int

Class MeadConfig

Namespace: Nekoyume

Assembly: Lib9c.dll

```
public static class MeadConfig
```

Inheritance

object ← MeadConfig

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Fields

PatronAddress

```
public static readonly Address PatronAddress
```

Field Value

Address

Enum Planet

Namespace: Nekoyume

Assembly: Lib9c.dll

```
public enum Planet : byte
```

Extension Methods
StateExtensions.Serialize(Enum)

Fields

`Heimdall = 1`

`HeimdallInternal = 5`

`Idun = 2`

`Odin = 0`

`OdinInternal = 4`

`Thor = 3`

`ThorInternal = 6`

Class PlanetExtension

Namespace: Nekoyume

Assembly: Lib9c.dll

```
public static class PlanetExtension
```

Inheritance

object ← PlanetExtension

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Methods

DeterminePlanet(ITransaction)

```
public static Planet? DeterminePlanet(this ITransaction tx)
```

Parameters

`tx` ITransaction

Returns

Planet?

Class StoreUtils

Namespace: Nekoyume

Assembly: Lib9c.dll

```
public static class StoreUtils
```

Inheritance

object ← StoreUtils

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Fields

BackupNameFormat

```
public const string BackupNameFormat = "{0}_{1:yyyyMMdd_HHmmss}"
```

Field Value

string

BackupNamePattern

```
public const string BackupNamePattern = "{0}_*"
```

Field Value

string

Methods

ClearBackupStores(string)

```
public static void ClearBackupStores(string storePath)
```

Parameters

`storePath` string

ResetStore(string)

```
public static void ResetStore(string storePath)
```

Parameters

`storePath` string

Namespace Nekoyume.Action

Classes

ActionBase

ActionContextExtensions

ActionObsoleteAttribute

An attribute on an Libplanet.Action.IAction to indicate that such Libplanet.Action.IAction is obsoleted after a certain index.

Due to a bug introduced earlier and to keep backward compatibility, an Libplanet.Action.IAction with this attribute is obsoleted, i.e. cannot be included, starting from Obsolete Index + 2.

ActionObsoletedException

ActionPointExceededException

ActionUnavailableException

ActivateCollection

ActivatedAccountsDoesNotExistsException

ActivationException

AddRedeemCode

AddressExtension

AdminPermissionException

AgentStateNotContainsAvatarAddressException

AlreadyActivatedException

AlreadyClaimedGiftsException

AlreadyContractedException

AlreadyReceivedException

AlreadyRecipeUnlockedException

AlreadyWorldUnlockedException

AppraiseBlockNotReachedException

ApprovePledge

ArenaNotEndedException

AssetInfo

AttachmentActionResult

AuraSummon

AvatarIndexAlreadyUsedException

AvatarIndexOutOfRangeException

BalanceDoesNotExistsException

BattleArena
Introduce at https://github.com/planetarium/lib9c/pull/2229 Changed at https://github.com/planetarium/lib9c/pull/2242

BurnAsset

Buy
Updated at https://github.com/planetarium/lib9c/pull/1164

Buy7

Buy7.BuyerMultipleResult

Buy7.BuyerResult

Buy7.PurchaseResult

Buy7.SellerMultipleResult

Buy7.SellerResult

BuyMultiple
Introduced at https://github.com/planetarium/lib9c/pull/331 Updated at many pull requests Obsoleted at https://github.com/planetarium/lib9c/pull/487 Updated at many pull requests Updated at https://github.com/planetarium/lib9c/pull/957

BuyMultiple.BuyerResult

BuyMultiple.PurchaseInfo

BuyMultiple.PurchaseResult

BuyMultiple.SellerResult

BuyProduct

CancelProductRegistration

ChargeActionPoint
Hard forked at https://github.com/planetarium/lib9c/pull/430 Updated at https://github.com/planetarium/lib9c/pull/474 Updated at https://github.com/planetarium/lib9c/pull/602 Updated at https://github.com/planetarium/lib9c/pull/861 Updated at https://github.com/planetarium/lib9c/pull/957

ClaimGifts
An action to claim gifts using the Gift Id.
Refer ClaimableGiftsSheet to find gifts data.
It can only claim once at the specified block index.

ClaimItems

ClaimPatrolReward
Claim patrol reward

ClaimRaidReward

ClaimStakeReward
Hard forked at https://github.com/planetarium/lib9c/pull/2097

ClaimStakeReward2

ClaimStakeReward8

ClaimWordBossKillReward

ClaimWorldBossReward
The ClaimWorldBossReward class is an action that allows a player to claim rewards for their contribution in defeating a world boss. This action ensures that only the owner of the avatar can claim the reward and calculates the reward based on the player's contribution.

ClaimableGiftsNotAvailableException

CombinationConsumable

Hard forked at https://github.com/planetarium/lib9c/pull/2195

CombinationConsumable5

CombinationConsumable5.ResultModel

CombinationEquipment
Hard forked at https://github.com/planetarium/lib9c/pull/2195

CombinationEquipment5
Option helpers kept from `CombinationEquipment5` and still used by current crafting.

CombinationSlotResultNullException

CombinationSlotUnlockException

ConsumableSlotOutOfRangeException

ConsumableSlotUnlockException

CostumeSlotUnlockException

CostumeSummon

CreateAvatar
Hard forked at https://github.com/planetarium/lib9c/pull/2195

CreatePendingActivation
Introduced at Initial commit(2e645be18a4e2caea031c347f00777fbad5dbcc6) Updated at https://github.com/planetarium/lib9c/pull/957

CreatePendingActivations
Updated at https://github.com/planetarium/lib9c/pull/746 Updated at https://github.com/planetarium/lib9c/pull/957

CreatePledge
Admin action for pledge contract

DailyReward
Hard forked at https://github.com/planetarium/lib9c/pull/1828 Updated at https://github.com/planetarium/lib9c/pull/1828

DailyReward2

DailyReward2.DailyRewardResult

DuplicateCostumeException

DuplicateEquipmentException

DuplicateMaterialException

DuplicateOrderIdException

EndPledge

EquipmentLevelExceededException

EquipmentSlotUnlockException

EventConsumableItemCrafts

Hard forked at https://github.com/planetarium/lib9c/pull/2195

EventDungeonBattle

Action for participating in an event dungeon battle. This action allows an avatar to battle in a specific stage of an event dungeon, consuming tickets and receiving rewards upon victory. Hard forked at https://github.com/planetarium/lib9c/pull/2195

EventDungeonBattleSweep

Represents a sweep action that automatically completes multiple event dungeon battles without player interaction.

This action allows players to automatically repeat battles on already cleared event dungeon stages, consuming tickets and receiving rewards without manual combat interaction.

Key Features: - Only works on previously cleared stages (no stage clearing) - No ticket purchase functionality (uses existing tickets only) - Automatic reward and experience calculation - Multiple battle execution in a single action

This action modifies the following states: - AvatarState: Updates player's stats, inventory, and experience - EventDungeonInfo: Updates event dungeon progress and ticket usage - RuneState: Updates rune effects and bonuses - ItemSlotState: Updates equipped items - RuneSlotState: Updates equipped runes - CP: Updates character's combat power

Execution Flow: 1. Validate input parameters and avatar state 2. Load and validate game sheets and configurations 3. Validate event dungeon info and ticket availability 4. Check stage clearance requirements 5. Update rune and item slot states 6. Calculate combat power (CP) 7. Generate rewards and experience 8. Update avatar state and return modified world state

Hard forked at https://github.com/planetarium/lib9c/pull/2195

EventMaterialItemCrafts
Hard forked at https://github.com/planetarium/lib9c/pull/2195

FailedLoadSheetException

FailedLoadStateException

FavProductInfo

GameAction

GoldDistribution

GrantItems
Grants items/assets to target avatars from token-like inputs. This action is restricted to a fixed allow-list of signers and the current admin signer. Even if the signer does not have sufficient token balance, the grant will proceed and will burn only the available amount of the token.

Grinding

HackAndSlash
Represents a battle action in the game where a player engages in combat with monsters. This action modifies the following states:

- AvatarState: Updates player's stats, inventory, and experience
- WorldBossKillRewardRecord: Records boss kill rewards
- CollectionState: Updates collection progress
- RuneState: Updates rune effects and bonuses
- ItemSlotState: Updates equipped items
- RuneSlotState: Updates equipped runes
- CP: Updates character's combat power Hard forked at https://github.com/planetarium/lib9c/pull/2195

HackAndSlashRandomBuff
Created at https://github.com/planetarium/lib9c/pull/1031

HackAndSlashSweep
Represents a sweep action that automatically completes multiple battles without player interaction. This action modifies the following states:

- AvatarState: Updates player's stats, inventory, and experience
- CollectionState: Updates collection progress

- RuneState: Updates rune effects and bonuses
- ItemSlotState: Updates equipped items
- RuneSlotState: Updates equipped runes
- CP: Updates character's combat power Hard forked at https://github.com/planetarium/lib9c/pull/2195

InfiniteTowerBattle

Represents a battle action in the infinite tower where players engage in combat with various conditions and restrictions. This action modifies the following states:

- AvatarState: Updates player's stats, inventory, and experience
- InfiniteTowerInfo: Updates tower progress and tickets
- CollectionState: Updates collection progress
- RuneState: Updates rune effects and bonuses
- ItemSlotState: Updates equipped items
- RuneSlotState: Updates equipped runes
- CP: Updates character's combat power

InitializeStates

Introduced at Initial commit(2e645be18a4e2caea031c347f00777fbad5dbcc6) Updated at NCG distribution(7e4515b6e14cc5d6eb716d5ebb587ab04b4246f9) Updated at https://github.com/planetarium/lib9c/pull/19 Updated at https://github.com/planetarium/lib9c/pull/36 Updated at https://github.com/planetarium/lib9c/pull/42 Updated at https://github.com/planetarium/lib9c/pull/55 Updated at https://github.com/planetarium/lib9c/pull/57 Updated at https://github.com/planetarium/lib9c/pull/60 Updated at https://github.com/planetarium/lib9c/pull/102 Updated at https://github.com/planetarium/lib9c/pull/128 Updated at https://github.com/planetarium/lib9c/pull/167 Updated at https://github.com/planetarium/lib9c/pull/422 Updated at https://github.com/planetarium/lib9c/pull/747 Updated at https://github.com/planetarium/lib9c/pull/798 Updated at https://github.com/planetarium/lib9c/pull/957

InvalidAddressException

InvalidClaimException

InvalidCurrencyException

InvalidElementalException

InvalidEquipmentException

InvalidItemCountException

InvalidItemIdException

Represents an exception that is thrown when an invalid item ID is used.

InvalidItemTypeException

InvalidLevelException

InvalidMaterialException

InvalidMinterException

InvalidMonsterCollectionRoundException

InvalidNamePatternException

InvalidPriceException

InvalidProductTypeException

InvalidRecipeIdException

InvalidRepeatPlayException

InvalidShopItemException

InvalidSignatureException

InvalidSlotIndexException

InvalidStageException

InvalidTradableIdException

InvalidTradableItemException

InvalidTransferCurrencyException

InvalidTransferMinterException

InvalidTransferRecipientException

InvalidTransferSignerException

InvalidTransferUnactivatedRecipientException

InvalidWorldException

IssueToken

IssueTokensFromGarage

ItemDoesNotExistException

ItemEnhancement
  Updated at https://github.com/planetarium/lib9c/pull/2195

ItemEnhancement10
  Updated at https://github.com/planetarium/lib9c/pull/1176

ItemEnhancement10.ResultModel

ItemEnhancement11
  Updated at https://github.com/planetarium/lib9c/pull/1164

ItemEnhancement11.ResultModel

ItemEnhancement12
  Updated at https://github.com/planetarium/lib9c/pull/2068

ItemEnhancement12.ResultModel

ItemEnhancement13
  Updated at https://github.com/planetarium/lib9c/pull/2068

ItemEnhancement13.ResultModel

ItemEnhancement7

ItemEnhancement7.ResultModel

ItemEnhancement9

ItemEnhancement9.ResultModel

ItemProductInfo

JoinArena
  Introduced at https://github.com/planetarium/lib9c/pull/2195

JoinArena1
  Introduced at https://github.com/planetarium/lib9c/pull/1495

JoinArena3

Introduced at https://github.com/planetarium/lib9c/pull/1663

MemoLengthOverflowException

MigrateAgentAvatar

MigrateFee

MigrateMonsterCollection

An action to claim remained monster collection rewards and to migrate MonsterCollectionState into LegacyStakeState without cancellation, to keep its staked period.

MigrationActivatedAccountsState

MintAssets

MonsterCollectionExistingClaimableException

MonsterCollectionExistingException

An exception thrown when there is unexpected monster collection.

MonsterCollectionExpiredException

MonsterCollectionLevelException

NotEnoughActionPointException

NotEnoughAvatarLevelException

NotEnoughClearedStageLevelException

NotEnoughCombatPointException

NotEnoughFungibleAssetValueException

NotEnoughHammerPointException

NotEnoughMaterialException

NotEnoughMedalException

NotEnoughRankException

NotEnoughStarException

NotEnoughWeeklyArenaChallengeCountException

OrderIdDoesNotExistException

PatchTableSheet

Introduced at Initial commit(2e645be18a4e2caea031c347f00777fbad5dbcc6) Updated at https://github.com/planetarium/lib9c/pull/42 Updated at https://github.com/planetarium/lib9c/pull/101 Updated at https://github.com/planetarium/lib9c/pull/287 Updated at https://github.com/planetarium/lib9c/pull/315 Updated at https://github.com/planetarium/lib9c/pull/957 Updated at https://github.com/planetarium/lib9c/pull/1560

PatchTableSheetCompressed

Compressed version of PatchTableSheet to reduce transaction size for large CSV files. Uses GZip compression to minimize the size of table data in transactions.

This action is particularly useful for large CSV files like RuneOptionSheet.csv which can be compressed from ~842KB to ~154KB (81% reduction).

Example usage:

```
var compressedAction = new PatchTableSheetCompressed
{
    TableName = "RuneOptionSheet",
    CompressedTableCsv = PatchTableSheetCompressed.CompressCsv(csvData),
};
```

PendingActivationDoesNotExistsException

PermissionDeniedException

PetEnhancement

PetIsLockedException

PlayCountIsZeroException

PolicyExpiredException

PrepareRewardAssets

ProductNotFoundException

PurchaseInfo

PurchaseInfo0

Raid

Represents a raid action where players participate in a world boss battle. This action modifies the following states:

- AvatarState: Updates player's stats, inventory, and experience
- WorldBossKillRewardRecord: Records boss kill rewards
- CollectionState: Updates collection progress
- RuneState: Updates rune effects and bonuses
- ItemSlotState: Updates equipped items
- RuneSlotState: Updates equipped runes
- CP: Updates character's combat power

RankingBattle

Hard forked at https://github.com/planetarium/lib9c/pull/941 Updated at https://github.com/planetarium/lib9c/pull/1135

RankingBattle11

Updated at https://github.com/planetarium/lib9c/pull/1176

RankingExceededException

RapidCombination

Hard forked at https://github.com/planetarium/lib9c/pull/2195

RapidCombination0

RapidCombination0.ResultModel

RapidCombination5

RapidCombination5.ResultModel

ReRegisterProduct

RedeemCode

Hard forked at https://github.com/planetarium/lib9c/pull/602 Updated at https://github.com/planetarium/lib9c/pull/861 Updated at https://github.com/planetarium/lib9c/pull/957

RedeemCode2

RegisterInfo

RegisterProduct

RegisterProduct0

RemoveAddressState
Action to remove state for multiple addresses.

RenewAdminState
Hard forked at https://github.com/planetarium/lib9c/pull/805 Updated at https://github.com/planetarium/lib9c/pull/815 Updated at https://github.com/planetarium/lib9c/pull/957

RequestPledge

RequiredBlockIndexException

RequiredBlockIntervalException

RetrieveAvatarAssets

RewardGold
Introduced at Initial commit(2e645be18a4e2caea031c347f00777fbad5dbcc6) Updated at many pull requests Updated at https://github.com/planetarium/lib9c/pull/1135

RuneEnhancement
Tries to raise a rune's level TryCount times, burning the cost of every try.

RuneSlotInfo
Represents rune slot information for a character.

RuneSummon

SecureMiningReward

Sell
Hard forked at https://github.com/planetarium/lib9c/pull/1640

Sell6

SellCancellation
Hard forked at https://github.com/planetarium/lib9c/pull/602 Updated at https://github.com/planetarium/lib9c/pull/609 Updated at https://github.com/planetarium/lib9c/pull/620 Updated at https://github.com/planetarium/lib9c/pull/861 Updated at https://github.com/planetarium/lib9c/pull/957

SellCancellation.Result

SetAddressState

Action to set state for multiple addresses. Throws an exception if the state already exists for the target address.

SetAddressStateCompressed

Compressed version of SetAddressState to reduce transaction size for large state data. Uses GZip compression to minimize the size of state data in transactions.

This action is particularly useful when setting large state data that would exceed transaction policy size limits. The compression can reduce state data size by 60-80% depending on the content structure and redundancy.

Example usage:

```
var states = new List<(Address, Address, IValue)>
{
    (accountAddress, targetAddress, largeStateValue),
};
var compressedAction = new SetAddressStateCompressed(states);
```

ShopItemExpiredException

SlotAlreadyUnlockedException

StageNotClearedException

StageNotOpenedException

A stage was entered before its hard world segment opened.

Stake

Stake2

StakeExistingClaimableException

Swap

Swap action swaps one currency to another currency.

Synthesize

Synthesize action is a type of action that synthesizes items. MaterialIds: Id list of items as material
ChargeAp: Whether to charge action points with action execution

TableSheetPatchValidation

The per-sheet checks PatchTableSheet and PatchTableSheetCompressed run before a CSV lands on the chain. Both actions call this one helper so the two paths can never disagree about what is accepted.

TotalSupplyDoesNotExistException

TransferAsset

Hard forked at https://github.com/planetarium/lib9c/pull/2143 Updated at https://github.com/planetarium/lib9c/pull/2143

TransferAsset3

TransferAssets

Hard forked at https://github.com/planetarium/lib9c/pull/636 Updated at https://github.com/planetarium/lib9c/pull/957

UnlockCombinationSlot

UnlockEquipmentRecipe

UnlockRuneSlot

UnlockWorld

Unlocks one or more worlds for an avatar and records the result to the legacy `world_ids` state. The required CRYSTAL is calculated from WorldUnlockSheet and transferred to UnlockWorld. If the calculated cost is zero, no asset transfer occurs.

UpdateSell

Hard forked at https://github.com/planetarium/lib9c/pull/1640

UpdateSellInfo

UsageLimitExceedException

ValidatorSetOperate

ValidatorSetOperatorTypeExtensions

WeeklyArenaStateAlreadyEndedException

WeeklyArenaStateNotContainsAvatarAddressException

WorldOpenCounter

Counts player progress for the global counters and confirms world openings whose conditions are met.

WorldOpenGate
: Decides whether a stage may be entered. Stages from ProgressGatedStageBegin are locked until an applied WorldOpenSheet row covers them.

WorldOpenIdentity
: Keeps what each world-open id means fixed across table patches. The first patch that carries an id records its identity; a later patch that gives the id a different identity is rejected. An id may be dropped from the sheet (retired), but its record stays, so it can only come back with the same identity. To fix a wrong rule, drop it and add a new id.

Structs

IssueTokensFromGarage.Spec

MintAssets.MintSpec

RuneEnhancement.LevelUpResult
: The outcome and cost of a run of enhancement tries.

Interfaces

IBuy0
: Common interface used before IBuy5.

IBuy5
: Common interface used after IBuy5.

IClaimItems

IClaimStakeReward

IHackAndSlashV10
: Interface for hack and slash action version 10.

IProductInfo

IPurchaseInfo

IRegisterInfo

ITransferAsset

ITransferAssets

Enums

ItemEnhancement10.EnhancementResult

ItemEnhancement11.EnhancementResult

ItemEnhancement12.EnhancementResult

ItemEnhancement13.EnhancementResult

ItemEnhancement9.EnhancementResult

ShopErrorType

ValidatorSetOperatorType

Class ActionBase

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
public abstract class ActionBase : IAction
```

Inheritance

object ← ActionBase

Implements

IAction

Derived

ApprovePledge, BurnAsset, ClaimPatrolReward, CreatePendingActivation, CreatePendingActivations, CreatePledge, CustomEquipmentCraft, EndPledge, GameAction, BanGuildMember, ClaimGuildReward, ClaimReward, ClaimUnbonded, JoinGuild, MakeGuild, MigrateDelegationHeight, MigratePlanetariumValidator, MigrateStakeAndJoinGuild, MoveGuild, QuitGuild, RemoveGuild, UnbanGuildMember, IssueToken, IssueTokensFromGarage, MigrateAgentAvatar, MigrateFee, MigrateMonsterCollection, MintAssets, PrepareRewardAssets, RemoveAddressState, RequestPledge, RetrieveAvatarAssets, RewardGold, SecureMiningReward, SetAddressState, SetAddressStateCompressed, Swap, TransferAsset, TransferAssets, AllocateGuildReward, AllocateReward, ClaimValidatorRewardSelf, DelegateValidator, Mortgage, PromoteValidator, RecordProposer, Refund, Reward, SetValidatorCommission, SlashValidator, UndelegateValidator, UnjailValidator, UpdateValidators, ValidatorSetOperate

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Properties

PlainValue

```
public abstract IValue PlainValue { get; }
```

Property Value

IValue

Methods

CheckObsolete(long, IActionContext)

```
protected void CheckObsolete(long obsoleteIndex, IActionContext ctx)
```

Parameters

`obsoleteIndex` long

`ctx` IActionContext

CheckPermission(IActionContext)

```
protected void CheckPermission(IActionContext ctx)
```

Parameters

`ctx` IActionContext

Execute(IActionContext)

```
public abstract IWorld Execute(IActionContext context)
```

Parameters

`context` IActionContext

Returns

IWorld

GetSignerAndOtherAddressesHex(IActionContext, params Address[])

returns "[Signer Address, AvatarState Address, ...]"

```
protected string GetSignerAndOtherAddressesHex(IActionContext ctx, params
Address[] addresses)
```

Parameters

`ctx` IActionContext

`addresses` Address[]

Returns

string

LoadPlainValue(IValue)

```
public abstract void LoadPlainValue(IValue plainValue)
```

Parameters

`plainValue` IValue

TryGetAdminState(IActionContext, out AdminState)

```
protected bool TryGetAdminState(IActionContext ctx, out AdminState state)
```

Parameters

`ctx` IActionContext

`state` AdminState

Returns

bool

Class ActionContextExtensions

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
public static class ActionContextExtensions
```

Inheritance

object ← ActionContextExtensions

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Methods

IsMainNet(IActionContext)

```
public static bool IsMainNet(this IActionContext context)
```

Parameters

`context` IActionContext

Returns

bool

Since(IActionContext, long)

```
public static bool Since(this IActionContext context, long blockIndex)
```

Parameters

`context` IActionContext

`blockIndex` long

Returns

bool

Class ActionObsoleteAttribute

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

An attribute on an Libplanet.Action.IAction to indicate that such Libplanet.Action.IAction is obsoleted after a certain index.

Due to a bug introduced earlier and to keep backward compatibility, an Libplanet.Action.IAction with this attribute is obsoleted, i.e. cannot be included, starting from ObsoleteIndex + 2.

```
[AttributeUsage(AttributeTargets.Class, AllowMultiple = true)]
public class ActionObsoleteAttribute : Attribute
```

Inheritance

object ← Attribute ← ActionObsoleteAttribute

Inherited Members

Attribute.Equals(object) , Attribute.GetCustomAttribute(Assembly, Type) ,
Attribute.GetCustomAttribute(Assembly, Type, bool) ,
Attribute.GetCustomAttribute(MemberInfo, Type) ,
Attribute.GetCustomAttribute(MemberInfo, Type, bool) ,
Attribute.GetCustomAttribute(Module, Type) ,
Attribute.GetCustomAttribute(Module, Type, bool) ,
Attribute.GetCustomAttribute(ParameterInfo, Type) ,
Attribute.GetCustomAttribute(ParameterInfo, Type, bool) ,
Attribute.GetCustomAttributes(Assembly) ,
Attribute.GetCustomAttributes(Assembly, bool) ,
Attribute.GetCustomAttributes(Assembly, Type) ,
Attribute.GetCustomAttributes(Assembly, Type, bool) ,
Attribute.GetCustomAttributes(MemberInfo) ,
Attribute.GetCustomAttributes(MemberInfo, bool) ,
Attribute.GetCustomAttributes(MemberInfo, Type) ,
Attribute.GetCustomAttributes(MemberInfo, Type, bool) ,
Attribute.GetCustomAttributes(Module) , Attribute.GetCustomAttributes(Module, bool) ,
Attribute.GetCustomAttributes(Module, Type) ,
Attribute.GetCustomAttributes(Module, Type, bool) ,
Attribute.GetCustomAttributes(ParameterInfo) ,
Attribute.GetCustomAttributes(ParameterInfo, bool) ,

Attribute.GetCustomAttributes(ParameterInfo, Type) ,
Attribute.GetCustomAttributes(ParameterInfo, Type, bool) , Attribute.GetHashCode() ,
Attribute.IsDefaultAttribute() , Attribute.IsDefined(Assembly, Type) ,
Attribute.IsDefined(Assembly, Type, bool) , Attribute.IsDefined(MemberInfo, Type) ,
Attribute.IsDefined(MemberInfo, Type, bool) , Attribute.IsDefined(Module, Type) ,
Attribute.IsDefined(Module, Type, bool) , Attribute.IsDefined(ParameterInfo, Type) ,
Attribute.IsDefined(ParameterInfo, Type, bool) , Attribute.Match(object) ,
Attribute.TypeId , object.Equals(object, object) , object.GetType() ,
object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

ActionObsoleteAttribute(Planet, long)

```
public ActionObsoleteAttribute(Planet planet, long obsoleteIndex)
```

Parameters

`planet` Planet

`obsoleteIndex` long

ActionObsoleteAttribute(long)

```
public ActionObsoleteAttribute(long obsoleteIndex)
```

Parameters

`obsoleteIndex` long

Fields

AffectedPlanet

```
public readonly Planet? AffectedPlanet
```

Field Value

Planet?

ObsoleteIndex

```
public readonly long ObsoleteIndex
```

Field Value

long

Class ActionObsoletedException

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
public class ActionObsoletedException : InvalidOperationException, ISerializable
```

Inheritance

object ← Exception ← SystemException ← InvalidOperationException ← ActionObsoletedException

Implements

ISerializable

Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Constructors

ActionObsoletedException()

```
public ActionObsoletedException()
```

ActionObsoletedException(SerializationInfo, StreamingContext)

```
protected ActionObsoletedException(SerializationInfo info, StreamingContext context)
```

Parameters

`info` SerializationInfo

`context` StreamingContext

ActionObsoletedException(string)

```
public ActionObsoletedException(string s)
```

Parameters

`s` string

Class ActionPointExceededException

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
public class ActionPointExceededException : InvalidOperationException, ISerializable
```

Inheritance

object ← Exception ← SystemException ← InvalidOperationException ← ActionPointExceededException

Implements

ISerializable

Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Constructors

ActionPointExceededException()

```
public ActionPointExceededException()
```

ActionPointExceededException(SerializationInfo, StreamingContext)

```
protected ActionPointExceededException(SerializationInfo info,
StreamingContext context)
```

Parameters

`info` SerializationInfo

`context` StreamingContext

ActionPointExceededException(string)

```
public ActionPointExceededException(string s)
```

Parameters

`s` string

Class ActionUnavailableException

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
public class ActionUnavailableException : InvalidOperationException, ISerializable
```

Inheritance

object ← Exception ← SystemException ← InvalidOperationException ← ActionUnavailableException

Implements

ISerializable

Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Constructors

ActionUnavailableException()

```
public ActionUnavailableException()
```

ActionUnavailableException(SerializationInfo, StreamingContext)

```
protected ActionUnavailableException(SerializationInfo info,
StreamingContext context)
```

Parameters

`info` SerializationInfo

`context` StreamingContext

ActionUnavailableException(string)

```
public ActionUnavailableException(string s)
```

Parameters

`s` string

Class ActivateCollection

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[ActionType("activate_collection")]
public class ActivateCollection : GameAction, IAction
```

Inheritance

object ← ActionBase ← GameAction ← ActivateCollection

Implements

IAction

Inherited Members

GameAction.Id , GameAction.PlainValue , GameAction.LoadPlainValue(IValue) , ActionBase.GetSignerAndOtherAddressesHex(IActionContext, params Address[]) , ActionBase.TryGetAdminState(IActionContext, out AdminState) , ActionBase.CheckPermission(IActionContext) , ActionBase.CheckObsolete(long, IActionContext) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Fields

AvatarAddress

```
public Address AvatarAddress
```

Field Value

Address

CollectionData

```
public List<(int collectionId, List<ICollectionMaterial> materials)> CollectionData
```

Field Value

List<(int collectionId, List<ICollectionMaterial> materials)>

Properties

PlainValueInternal

```
protected override IImmutableDictionary<string, IValue> PlainValueInternal { get; }
```

Property Value

IImmutableDictionary<string, IValue>

Methods

Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

Parameters

`context` IActionContext

Returns

IWorld

LoadPlainValueInternal(IImmutableDictionary<string, IValue>)

```
protected override void LoadPlainValueInternal(IImmutableDictionary<string,
IValue> plainValue)
```

Parameters

`plainValue` IImmutableDictionary<string, IValue>

Class ActivatedAccountsDoesNotExistsException

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
public class ActivatedAccountsDoesNotExistsException : ActivationException,
ISerializable
```

Inheritance

object ← Exception ← ActivationException ← ActivatedAccountsDoesNotExistsException

Implements

ISerializable

Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Constructors

ActivatedAccountsDoesNotExistsException()

```
public ActivatedAccountsDoesNotExistsException()
```

ActivatedAccountsDoesNotExistsException(SerializationInfo, StreamingContext)

```
public ActivatedAccountsDoesNotExistsException(SerializationInfo info,
StreamingContext context)
```

Parameters

`info` SerializationInfo

`context` StreamingContext

ActivatedAccountsDoesNotExistsException(string)

```
public ActivatedAccountsDoesNotExistsException(string message)
```

Parameters

`message` string

Class ActivationException

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
public abstract class ActivationException : Exception, ISerializable
```

Inheritance

object ← Exception ← ActivationException

Implements

ISerializable

Derived

ActivatedAccountsDoesNotExistsException, InvalidSignatureException, PendingActivationDoesNotExistsException

Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Constructors

ActivationException()

```
protected ActivationException()
```

ActivationException(SerializationInfo, StreamingContext)

```
protected ActivationException(SerializationInfo info, StreamingContext context)
```

Parameters

`info` SerializationInfo

`context` StreamingContext

ActivationException(string)

```
protected ActivationException(string message)
```

Parameters

`message` string

Class AddRedeemCode

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
[ActionType("add_redeem_code")]
public class AddRedeemCode : GameAction, IAction, IAddRedeemCodeV1
```

Inheritance

object ← ActionBase ← GameAction ← AddRedeemCode

Implements

IAction, IAddRedeemCodeV1

Inherited Members

GameAction.Id , GameAction.PlainValue , GameAction.LoadPlainValue(IValue) , ActionBase.GetSignerAndOtherAddressesHex(IActionContext, params Address[]) , ActionBase.TryGetAdminState(IActionContext, out AdminState) , ActionBase.CheckPermission(IActionContext) , ActionBase.CheckObsolete(long, IActionContext) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Fields

redeemCsv

```
public string redeemCsv
```

Field Value

string

Properties

PlainValueInternal

```
protected override IImmutableDictionary<string, IValue> PlainValueInternal { get; }
```

Property Value

IImmutableDictionary<string, IValue>

Methods

Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

Parameters

`context` IActionContext

Returns

IWorld

LoadPlainValueInternal(IImmutableDictionary<string, IValue>)

```
protected override void LoadPlainValueInternal(IImmutableDictionary<string,
IValue> plainValue)
```

Parameters

`plainValue` IImmutableDictionary<string, IValue>

Class AddressExtension

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
public static class AddressExtension
```

Inheritance

object ← AddressExtension

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Methods

Derive(Address, byte[])

```
public static Address Derive(this Address address, byte[] key)
```

Parameters

`address` Address

`key` byte[]

Returns

Address

Derive(Address, string)

```
public static Address Derive(this Address address, string key)
```

Parameters

`address` Address

`key` string

Returns

Address

GetPledgeAddress(Address)

```
public static Address GetPledgeAddress(this Address address)
```

Parameters

`address` Address

Returns

Address

Class AdminPermissionException

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
public abstract class AdminPermissionException : Exception, ISerializable
```

Inheritance

object ← Exception ← AdminPermissionException

Implements

ISerializable

Derived

PermissionDeniedException, PolicyExpiredException

Inherited Members

Exception.GetBaseException() , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Constructors

AdminPermissionException(AdminState)

```
protected AdminPermissionException(AdminState policy)
```

Parameters

`policy` AdminState

AdminPermissionException(SerializationInfo, StreamingContext)

```
protected AdminPermissionException(SerializationInfo info, StreamingContext context)
```

Parameters

`info` SerializationInfo

`context` StreamingContext

AdminPermissionException(string)

```
protected AdminPermissionException(string message)
```

Parameters

`message` string

Properties

Policy

```
public AdminState Policy { get; }
```

Property Value

AdminState

Methods

GetObjectData(SerializationInfo, StreamingContext)

When overridden in a derived class, sets the SerializationInfo with information about the exception.

```
public override void GetObjectData(SerializationInfo info, StreamingContext context)
```

Parameters

`info` SerializationInfo

The SerializationInfo that holds the serialized object data about the exception being thrown.

`context` StreamingContext

The StreamingContext that contains contextual information about the source or destination.

Exceptions

ArgumentNullException

The `info` parameter is a null reference (`Nothing` in Visual Basic).

Class AgentStateNotContainsAvatarAddressException

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
public class AgentStateNotContainsAvatarAddressException : Exception, ISerializable
```

Inheritance

object ← Exception ← AgentStateNotContainsAvatarAddressException

Implements

ISerializable

Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Constructors

AgentStateNotContainsAvatarAddressException(SerializationInfo, StreamingContext)

```
public AgentStateNotContainsAvatarAddressException(SerializationInfo info,
StreamingContext context)
```

Parameters

`info` SerializationInfo

`context` StreamingContext

AgentStateNotContainsAvatarAddressException(string)

```
public AgentStateNotContainsAvatarAddressException(string message)
```

Parameters

`message` string

Class AlreadyActivatedException

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
public class AlreadyActivatedException : InvalidOperationException, ISerializable
```

Inheritance

object ← Exception ← SystemException ← InvalidOperationException ← AlreadyActivatedException

Implements

ISerializable

Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Constructors

AlreadyActivatedException(SerializationInfo, StreamingContext)

```
protected AlreadyActivatedException(SerializationInfo info,
StreamingContext context)
```

Parameters

`info` SerializationInfo

`context` StreamingContext

AlreadyActivatedException(string)

```
public AlreadyActivatedException(string s)
```

Parameters

s [string]

Class AlreadyClaimedGiftsException

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
public class AlreadyClaimedGiftsException : Exception, ISerializable
```

Inheritance

object ← Exception ← AlreadyClaimedGiftsException

Implements

ISerializable

Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Constructors

AlreadyClaimedGiftsException()

```
public AlreadyClaimedGiftsException()
```

AlreadyClaimedGiftsException(SerializationInfo, StreamingContext)

```
public AlreadyClaimedGiftsException(SerializationInfo info,
StreamingContext context)
```

Parameters

`info` SerializationInfo

`context` StreamingContext

AlreadyClaimedGiftsException(string)

```
public AlreadyClaimedGiftsException(string msg)
```

Parameters

`msg` string

Class AlreadyContractedException

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
public class AlreadyContractedException : InvalidOperationException, ISerializable
```

Inheritance

object ← Exception ← SystemException ← InvalidOperationException ← AlreadyContractedException

Implements

ISerializable

Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Constructors

AlreadyContractedException(SerializationInfo, StreamingContext)

```
protected AlreadyContractedException(SerializationInfo info,
StreamingContext context)
```

Parameters

`info` SerializationInfo

`context` StreamingContext

AlreadyContractedException(string)

```
public AlreadyContractedException(string s)
```

Parameters

s string

Class AlreadyReceivedException

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
public class AlreadyReceivedException : InvalidOperationException, ISerializable
```

Inheritance

object ← Exception ← SystemException ← InvalidOperationException ← AlreadyReceivedException

Implements

ISerializable

Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Constructors

AlreadyReceivedException(SerializationInfo, StreamingContext)

```
protected AlreadyReceivedException(SerializationInfo info, StreamingContext context)
```

Parameters

`info` SerializationInfo

`context` StreamingContext

AlreadyReceivedException(string)

```
public AlreadyReceivedException(string s)
```

Parameters

s [string]

Class AlreadyRecipeUnlockedException

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
public class AlreadyRecipeUnlockedException : Exception, ISerializable
```

Inheritance

object ← Exception ← AlreadyRecipeUnlockedException

Implements

ISerializable

Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Constructors

AlreadyRecipeUnlockedException()

```
public AlreadyRecipeUnlockedException()
```

AlreadyRecipeUnlockedException(SerializationInfo, StreamingContext)

```
protected AlreadyRecipeUnlockedException(SerializationInfo info,
StreamingContext context)
```

Parameters

`info` SerializationInfo

`context` StreamingContext

AlreadyRecipeUnlockedException(string)

```
public AlreadyRecipeUnlockedException(string msg)
```

Parameters

`msg` string

Class AlreadyWorldUnlockedException

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
public class AlreadyWorldUnlockedException : Exception, ISerializable
```

Inheritance

object ← Exception ← AlreadyWorldUnlockedException

Implements

ISerializable

Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Constructors

AlreadyWorldUnlockedException()

```
public AlreadyWorldUnlockedException()
```

AlreadyWorldUnlockedException(SerializationInfo, StreamingContext)

```
protected AlreadyWorldUnlockedException(SerializationInfo info,
StreamingContext context)
```

Parameters

`info` SerializationInfo

`context` StreamingContext

AlreadyWorldUnlockedException(string)

```
public AlreadyWorldUnlockedException(string msg)
```

Parameters

`msg` string

Class AppraiseBlockNotReachedException

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
public class AppraiseBlockNotReachedException : Exception, ISerializable
```

Inheritance

object ← Exception ← AppraiseBlockNotReachedException

Implements

ISerializable

Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Constructors

AppraiseBlockNotReachedException(SerializationInfo, StreamingContext)

```
protected AppraiseBlockNotReachedException(SerializationInfo info,
StreamingContext context)
```

Parameters

`info` SerializationInfo

`context` StreamingContext

AppraiseBlockNotReachedException(string)

```
public AppraiseBlockNotReachedException(string message)
```

Parameters

`message` string

Class ApprovePledge

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[ActionType("approve_pledge")]
public class ApprovePledge : ActionBase, IAction
```

Inheritance

object ← ActionBase ← ApprovePledge

Implements

IAction

Inherited Members

ActionBase.GetSignerAndOtherAddressesHex(IActionContext, params Address[]) ,
ActionBase.TryGetAdminState(IActionContext, out AdminState) ,
ActionBase.CheckPermission(IActionContext) ,
ActionBase.CheckObsolete(long, IActionContext) , object.Equals(object) ,
object.Equals(object, object) , object.GetHashCode() , object.GetType() ,
object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

ApprovePledge()

```
public ApprovePledge()
```

Fields

PatronAddress

```
public Address PatronAddress
```

Field Value

Address

TypeIdentifier

```
public const string TypeIdentifier = "approve_pledge"
```

Field Value

string

Properties

PlainValue

```
public override IValue PlainValue { get; }
```

Property Value

IValue

Methods

Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

Parameters

`context` IActionContext

Returns

IWorld

LoadPlainValue(IValue)

```
public override void LoadPlainValue(IValue plainValue)
```

Parameters

`plainValue` IValue

Class ArenaNotEndedException

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
public class ArenaNotEndedException : InvalidOperationException, ISerializable
```

Inheritance

object ← Exception ← SystemException ← InvalidOperationException ← ArenaNotEndedException

Implements

ISerializable

Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Constructors

ArenaNotEndedException(SerializationInfo, StreamingContext)

```
protected ArenaNotEndedException(SerializationInfo info, StreamingContext context)
```

Parameters

`info` SerializationInfo

`context` StreamingContext

ArenaNotEndedException(string)

```
public ArenaNotEndedException(string s)
```

Parameters

s string

Class AssetInfo

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
public class AssetInfo : IRegisterInfo
```

Inheritance

object ← AssetInfo

Implements

IRegisterInfo

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

AssetInfo()

```
public AssetInfo()
```

AssetInfo(List)

```
public AssetInfo(List serialized)
```

Parameters

`serialized` List

Properties

Asset

```
public FungibleAssetValue Asset { get; set; }
```

Property Value

FungibleAssetValue

AvatarAddress

```
public Address AvatarAddress { get; set; }
```

Property Value

Address

Price

```
public FungibleAssetValue Price { get; set; }
```

Property Value

FungibleAssetValue

Type

```
public ProductType Type { get; set; }
```

Property Value

ProductType

Methods

Serialize()

```
public IValue Serialize()
```

Returns

IValue

Validate()

```
public void Validate()
```

ValidateAddress(Address)

```
public void ValidateAddress(Address avatarAddress)
```

Parameters

`avatarAddress` Address

ValidatePrice(Currency)

```
public void ValidatePrice(Currency ncg)
```

Parameters

`ncg` Currency

Class AttachmentActionResult

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
public abstract class AttachmentActionResult : IState
```

Inheritance

object ← AttachmentActionResult

Implements

IState

Derived

Buy7.BuyerResult, Buy7.SellerResult, CombinationConsumable5.ResultModel, DailyReward2.DailyRewardResult, ItemEnhancement10.ResultModel, ItemEnhancement11.ResultModel, ItemEnhancement12.ResultModel, ItemEnhancement13.ResultModel, ItemEnhancement7.ResultModel, ItemEnhancement9.ResultModel, RapidCombination5.ResultModel, SellCancellation.Result, MonsterCollectionResult

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

AttachmentActionResult()

```
protected AttachmentActionResult()
```

AttachmentActionResult(Dictionary)

```
protected AttachmentActionResult(Dictionary serialized)
```

Parameters

`serialized` Dictionary

Fields

costume

```
public Costume costume
```

Field Value

Costume

itemUsable

```
public ItemUsable itemUsable
```

Field Value

ItemUsable

tradableFungibleItem

```
public ITradableFungibleItem tradableFungibleItem
```

Field Value

ITradableFungibleItem

tradableFungibleItemCount

```
public int tradableFungibleItemCount
```

Field Value

int

Properties

TypeId

```
protected abstract string TypeId { get; }
```

Property Value

string

Methods

Deserialize(Dictionary)

```
public static AttachmentActionResult Deserialize(Dictionary serialized)
```

Parameters

`serialized` Dictionary

Returns

AttachmentActionResult

Serialize()

```
public virtual IValue Serialize()
```

Returns

IValue

SerializeBackup1()

```
public virtual IValue SerializeBackup1()
```

Returns

IValue

Class AuraSummon

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
[ActionType("aura_summon")]
public class AuraSummon : GameAction, IAction, IAuraSummonV1
```

Inheritance

object ← ActionBase ← GameAction ← AuraSummon

Implements

IAction, IAuraSummonV1

Inherited Members

GameAction.Id , GameAction.PlainValue , GameAction.LoadPlainValue(IValue) , ActionBase.GetSignerAndOtherAddressesHex(IActionContext, params Address[]) , ActionBase.TryGetAdminState(IActionContext, out AdminState) , ActionBase.CheckPermission(IActionContext) , ActionBase.CheckObsolete(long, IActionContext) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

AuraSummon()

```
public AuraSummon()
```

AuraSummon(Address, int, int)

```
public AuraSummon(Address avatarAddress, int groupId, int summonCount)
```

Parameters

`avatarAddress` Address

`groupId` int

`summonCount` int

Fields

AvatarAddress

```
public Address AvatarAddress
```

Field Value

Address

AvatarAddressKey

```
public const string AvatarAddressKey = "aa"
```

Field Value

string

GroupId

```
public int GroupId
```

Field Value

int

GroupIdKey

```
public const string GroupIdKey = "gid"
```

Field Value

string

SummonCount

```
public int SummonCount
```

Field Value

int

SummonCountKey

```
public const string SummonCountKey = "sc"
```

Field Value

string

Properties

PlainValueInternal

```
protected override IImmutableDictionary<string, IValue> PlainValueInternal { get; }
```

Property Value

IImmutableDictionary<string, IValue>

Methods

AddAndUnlockOption(Equipment, IRandom, Row, EquipmentItemOptionSheet, SkillSheet)

```
public static void AddAndUnlockOption(Equipment equipment, IRandom random,
EquipmentItemSubRecipeSheetV2.Row subRecipe, EquipmentItemOptionSheet optionSheet,
SkillSheet skillSheet)
```

Parameters

`equipment` Equipment

`random` IRandom

`subRecipe` EquipmentItemSubRecipeSheetV2.Row

`optionSheet` EquipmentItemOptionSheet

`skillSheet` SkillSheet

Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

Parameters

`context` IActionContext

Returns

IWorld

LoadPlainValueInternal(IImmutableDictionary<string, IValue>)

```
protected override void LoadPlainValueInternal(IImmutableDictionary<string,
IValue> plainValue)
```

Parameters

`plainValue` IImmutableDictionary<string, IValue>

SimulateSummon(string, EquipmentItemRecipeSheet, EquipmentItemSheet, EquipmentItemSubRecipeSheetV2, EquipmentItemOptionSheet, SkillSheet, Row, int, IRandom, long, Row)

Simulates aura equipment summoning with optional grade guarantee system. Applies 10+1 bonus rule and uses grade guarantee settings from SummonSheet.Row when enabled. When grade guarantee is enabled, ensures minimum grade items are obtained based on summon count.

```
public static IEnumerable<(int, Equipment)> SimulateSummon(string addressesHex,
EquipmentItemRecipeSheet recipeSheet, EquipmentItemSheet equipmentItemSheet,
EquipmentItemSubRecipeSheetV2 equipmentItemSubRecipeSheetV2,
EquipmentItemOptionSheet optionSheet, SkillSheet skillSheet, SummonSheet.Row
summonRow, int summonCount, IRandom random, long blockIndex, BoostScheduleSheet.Row
guaranteeBoost = null)
```

Parameters

`addressesHex` string

Addresses hex for error reporting

`recipeSheet` EquipmentItemRecipeSheet

Equipment item recipe sheet

`equipmentItemSheet` EquipmentItemSheet

Equipment item sheet

`equipmentItemSubRecipeSheetV2` EquipmentItemSubRecipeSheetV2

Equipment item sub-recipe sheet

`optionSheet` EquipmentItemOptionSheet

Equipment item option sheet

`skillSheet` SkillSheet

Skill sheet

`summonRow` SummonSheet.Row

Summon configuration row with recipes and guarantee settings

`summonCount` int

Number of items to summon (before 10+1 bonus)

`random` IRandom

Random number generator

`blockIndex` long

Current block index for item creation

`guaranteeBoost` BoostScheduleSheet.Row

The BoostScheduleSheet row adjusting this group's guarantee count, or `null` for none. A caller replaying a summon must pass what the action resolved.

Returns

IEnumerable<(int, Equipment)>

Collection of (recipe ID, equipment) pairs

Class AvatarIndexAlreadyUsedException

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
public class AvatarIndexAlreadyUsedException : InvalidOperationException,
ISerializable
```

Inheritance

object ← Exception ← SystemException ← InvalidOperationException ← AvatarIndexAlreadyUsedException

Implements

ISerializable

Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Constructors

AvatarIndexAlreadyUsedException(SerializationInfo, StreamingContext)

```
protected AvatarIndexAlreadyUsedException(SerializationInfo info,
StreamingContext context)
```

Parameters

`info` SerializationInfo

`context` StreamingContext

AvatarIndexAlreadyUsedException(string)

```
public AvatarIndexAlreadyUsedException(string s)
```

Parameters

s [string]

Class AvatarIndexOutOfRangeException

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
public class AvatarIndexOutOfRangeException : Exception, ISerializable
```

Inheritance

object ← Exception ← AvatarIndexOutOfRangeException

Implements

ISerializable

Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Constructors

AvatarIndexOutOfRangeException(SerializationInfo, StreamingContext)

```
protected AvatarIndexOutOfRangeException(SerializationInfo info,
StreamingContext context)
```

Parameters

`info` SerializationInfo

`context` StreamingContext

AvatarIndexOutOfRangeException(string)

```
public AvatarIndexOutOfRangeException(string message)
```

Parameters

`message` string

Class BalanceDoesNotExistsException

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
public class BalanceDoesNotExistsException : Exception, ISerializable
```

Inheritance

object ← Exception ← BalanceDoesNotExistsException

Implements

ISerializable

Inherited Members

Exception.GetBaseException() , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Constructors

BalanceDoesNotExistsException(Address, Currency)

```
public BalanceDoesNotExistsException(Address address, Currency currency)
```

Parameters

`address` Address

`currency` Currency

BalanceDoesNotExistsException(SerializationInfo, StreamingContext)

```
protected BalanceDoesNotExistsException(SerializationInfo info,
StreamingContext context)
```

Parameters

`info` SerializationInfo

`context` StreamingContext

BalanceDoesNotExistsException(string)

```
public BalanceDoesNotExistsException(string message)
```

Parameters

`message` string

Properties

Address

```
public Address Address { get; }
```

Property Value

Address

Currency

```
public Currency Currency { get; }
```

Property Value

Currency

Methods

GetObjectData(SerializationInfo, StreamingContext)

When overridden in a derived class, sets the SerializationInfo with information about the exception.

```
public override void GetObjectData(SerializationInfo info, StreamingContext context)
```

Parameters

`info` SerializationInfo

The SerializationInfo that holds the serialized object data about the exception being thrown.

`context` StreamingContext

The StreamingContext that contains contextual information about the source or destination.

Exceptions

ArgumentNullException

The `info` parameter is a null reference (`Nothing` in Visual Basic).

Class BattleArena

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

Introduce at https://github.com/planetarium/lib9c/pull/2229 Changed at https://github.com/planetarium/lib9c/pull/2242

```
[Serializable]
[ActionType("battle_arena15")]
public class BattleArena : GameAction, IAction, IBattleArenaV1
```

Inheritance

object ← ActionBase ← GameAction ← BattleArena

Implements

IAction, IBattleArenaV1

Inherited Members

GameAction.Id , GameAction.PlainValue , GameAction.LoadPlainValue(IValue) , ActionBase.GetSignerAndOtherAddressesHex(IActionContext, params Address[]) , ActionBase.TryGetAdminState(IActionContext, out AdminState) , ActionBase.CheckPermission(IActionContext) , ActionBase.CheckObsolete(long, IActionContext) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Fields

HpIncreasingModifier

```
public const int HpIncreasingModifier = 5
```

Field Value

int

PurchasedCountKey

```
public const string PurchasedCountKey = "purchased_count_during_interval"
```

Field Value

string

championshipId

```
public int championshipId
```

Field Value

int

costumes

```
public List<Guid> costumes
```

Field Value

List<Guid>

enemyAvatarAddress

```
public Address enemyAvatarAddress
```

Field Value

Address

equipments

```
public List<Guid> equipments
```

Field Value

List<Guid>

myAvatarAddress

```
public Address myAvatarAddress
```

Field Value

Address

round

```
public int round
```

Field Value

int

runeInfos

```
public List<RuneSlotInfo> runeInfos
```

Field Value

List<RuneSlotInfo>

ticket

```
public int ticket
```

Field Value

int

Properties

PlainValueInternal

```
protected override IImmutableDictionary<string, IValue> PlainValueInternal { get; }
```

Property Value

IImmutableDictionary<string, IValue>

Methods

Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

Parameters

`context` IActionContext

Returns

IWorld

LoadPlainValueInternal(IImmutableDictionary<string, IValue>)

```
protected override void LoadPlainValueInternal(IImmutableDictionary<string,
```

```
IValue> plainValue)
```

Parameters

`plainValue` IImmutableDictionary<string, IValue>

Class BurnAsset

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[ActionType("burn_asset")]
public class BurnAsset : ActionBase, IAction
```

Inheritance

object ← ActionBase ← BurnAsset

Implements

IAction

Inherited Members

ActionBase.GetSignerAndOtherAddressesHex(IActionContext, params Address[]) ,
ActionBase.TryGetAdminState(IActionContext, out AdminState) ,
ActionBase.CheckPermission(IActionContext) ,
ActionBase.CheckObsolete(long, IActionContext) , object.Equals(object) ,
object.Equals(object, object) , object.GetHashCode() , object.GetType() ,
object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

BurnAsset()

```
public BurnAsset()
```

BurnAsset(Address, FungibleAssetValue, string)

```
public BurnAsset(Address owner, FungibleAssetValue amount, string memo)
```

Parameters

`owner` Address

`amount` FungibleAssetValue

`memo` string

Fields

TypeIdentifier

```
public const string TypeIdentifier = "burn_asset"
```

Field Value

string

Properties

Amount

```
public FungibleAssetValue Amount { get; }
```

Property Value

FungibleAssetValue

Memo

```
public string Memo { get; }
```

Property Value

string

Owner

```
public Address Owner { get; }
```

Property Value

Address

PlainValue

```
public override IValue PlainValue { get; }
```

Property Value

IValue

Methods

Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

Parameters

`context` IActionContext

Returns

IWorld

LoadPlainValue(IValue)

```
public override void LoadPlainValue(IValue plainValue)
```

Parameters

`plainValue` IValue

Class Buy

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

Updated at https://github.com/planetarium/lib9c/pull/1164

```
[Serializable]
[ActionObsolete(8324909)]
[ActionType("buy12")]
public class Buy : GameAction, IBuy5, IAction, IBuyV2
```

Inheritance

object ← ActionBase ← GameAction ← Buy

Implements

IBuy5, IAction, IBuyV2

Inherited Members

GameAction.Id , GameAction.PlainValue , GameAction.LoadPlainValue(IValue) , ActionBase.GetSignerAndOtherAddressesHex(IActionContext, params Address[]) , ActionBase.TryGetAdminState(IActionContext, out AdminState) , ActionBase.CheckPermission(IActionContext) , ActionBase.CheckObsolete(long, IActionContext) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Fields

ErrorCodeDuplicateSell

```
public const int ErrorCodeDuplicateSell = 10
```

Field Value

int

ErrorCodeFailedLoadingState

```
public const int ErrorCodeFailedLoadingState = 1
```

Field Value

int

ErrorCodeInsufficientBalance

```
public const int ErrorCodeInsufficientBalance = 4
```

Field Value

int

ErrorCodeInvalidAddress

```
public const int ErrorCodeInvalidAddress = 5
```

Field Value

int

ErrorCodeInvalidItemType

```
public const int ErrorCodeInvalidItemType = 9
```

Field Value

int

ErrorCodeInvalidOrderId

```
public const int ErrorCodeInvalidOrderId = 7
```

Field Value

int

ErrorCodeInvalidPrice

```
public const int ErrorCodeInvalidPrice = 6
```

Field Value

int

ErrorCodeInvalidTradableId

```
public const int ErrorCodeInvalidTradableId = 8
```

Field Value

int

ErrorCodeItemDoesNotExist

```
public const int ErrorCodeItemDoesNotExist = 2
```

Field Value

int

ErrorCodeShopItemExpired

```
public const int ErrorCodeShopItemExpired = 3
```

Field Value

int

TaxRate

```
public const int TaxRate = 8
```

Field Value

int

errors

```
public List<(Guid orderId, int errorCode)> errors
```

Field Value

List<(Guid orderId, int errorCode)>

purchaseInfos

```
public IEnumerable<PurchaseInfo> purchaseInfos
```

Field Value

IEnumerable<PurchaseInfo>

Properties

PlainValueInternal

```
protected override IImmutableDictionary<string, IValue> PlainValueInternal { get; }
```

Property Value

IImmutableDictionary<string, IValue>

buyerAvatarAddress

```
public Address buyerAvatarAddress { get; set; }
```

Property Value

Address

Methods

Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

Parameters

`context` IActionContext

Returns

IWorld

LoadPlainValueInternal(IImmutableDictionary<string, IValue>)

```
protected override void LoadPlainValueInternal(IImmutableDictionary<string,
IValue> plainValue)
```

Parameters

`plainValue` IImmutableDictionary<string, IValue>

Class Buy7

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
[ActionObsolete(7000000)]
[ActionType("buy7")]
public class Buy7 : GameAction, IBuy5, IAction, IBuyV2
```

Inheritance

object ← ActionBase ← GameAction ← Buy7

Implements

IBuy5, IAction, IBuyV2

Inherited Members

GameAction.Id , GameAction.PlainValue , GameAction.LoadPlainValue(IValue) , ActionBase.GetSignerAndOtherAddressesHex(IActionContext, params Address[]) , ActionBase.TryGetAdminState(IActionContext, out AdminState) , ActionBase.CheckPermission(IActionContext) , ActionBase.CheckObsolete(long, IActionContext) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Fields

ErrorCodeFailedLoadingState

```
public const int ErrorCodeFailedLoadingState = 1
```

Field Value

int

ErrorCodeInsufficientBalance

```
public const int ErrorCodeInsufficientBalance = 4
```

Field Value

int

ErrorCodeInvalidAddress

```
public const int ErrorCodeInvalidAddress = 5
```

Field Value

int

ErrorCodeInvalidPrice

```
public const int ErrorCodeInvalidPrice = 6
```

Field Value

int

ErrorCodeItemDoesNotExist

```
public const int ErrorCodeItemDoesNotExist = 2
```

Field Value

int

ErrorCodeShopItemExpired

```
public const int ErrorCodeShopItemExpired = 3
```

Field Value

int

TaxRate

```
public const int TaxRate = 8
```

Field Value

int

buyerMultipleResult

```
public Buy7.BuyerMultipleResult buyerMultipleResult
```

Field Value

Buy7.BuyerMultipleResult

purchaseInfos

```
public IEnumerable<PurchaseInfo0> purchaseInfos
```

Field Value

IEnumerable<PurchaseInfo0>

sellerMultipleResult

```
public Buy7.SellerMultipleResult sellerMultipleResult
```

Field Value

Buy7.SellerMultipleResult

Properties

PlainValueInternal

```
protected override IImmutableDictionary<string, IValue> PlainValueInternal { get; }
```

Property Value

IImmutableDictionary<string, IValue>

buyerAvatarAddress

```
public Address buyerAvatarAddress { get; set; }
```

Property Value

Address

Methods

Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

Parameters

`context` IActionContext

Returns

IWorld

LoadPlainValueInternal(IImmutableDictionary<string, IValue>)

```
protected override void LoadPlainValueInternal(IImmutableDictionary<string,
IValue> plainValue)
```

Parameters

`plainValue` IImmutableDictionary<string, IValue>

Class Buy7.BuyerMultipleResult

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
public class Buy7.BuyerMultipleResult
```

Inheritance

object ← Buy7.BuyerMultipleResult

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

BuyerMultipleResult()

```
public BuyerMultipleResult()
```

BuyerMultipleResult(Dictionary)

```
public BuyerMultipleResult(Dictionary serialized)
```

Parameters

`serialized` Dictionary

Fields

purchaseResults

```
public IEnumerable<Buy7.PurchaseResult> purchaseResults
```

Field Value

IEnumerable<Buy7.PurchaseResult>

Methods

Serialize()

```
public IValue Serialize()
```

Returns

IValue

Class Buy7.BuyerResult

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
public class Buy7.BuyerResult : AttachmentActionResult, IState
```

Inheritance

object ← AttachmentActionResult ← Buy7.BuyerResult

Implements

IState

Derived

Buy7.PurchaseResult, BuyMultiple.PurchaseResult

Inherited Members

AttachmentActionResult.itemUsable , AttachmentActionResult.costume , AttachmentActionResult.tradableFungibleItem , AttachmentActionResult.tradableFungibleItemCount , AttachmentActionResult.SerializeBackup1() , AttachmentActionResult.Deserialize(Dictionary) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

BuyerResult()

```
public BuyerResult()
```

BuyerResult(Dictionary)

```
public BuyerResult(Dictionary serialized)
```

Parameters

`serialized` Dictionary

Fields

id

```
public Guid id
```

Field Value

Guid

shopItem

```
public ShopItem shopItem
```

Field Value

ShopItem

Properties

TypeId

```
protected override string TypeId { get; }
```

Property Value

string

Methods

Serialize()

```
public override IValue Serialize()
```

Returns

IValue

Class Buy7.PurchaseResult

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
public class Buy7.PurchaseResult : Buy7.BuyerResult, IState
```

Inheritance

object ← AttachmentActionResult ← Buy7.BuyerResult ← Buy7.PurchaseResult

Implements

IState

Inherited Members

Buy7.BuyerResult.shopItem , Buy7.BuyerResult.id , Buy7.BuyerResult.TypeId , AttachmentActionResult.itemUsable , AttachmentActionResult.costume , AttachmentActionResult.tradableFungibleItem , AttachmentActionResult.tradableFungibleItemCount , AttachmentActionResult.SerializeBackup1() , AttachmentActionResult.Deserialize(Dictionary) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

PurchaseResult(Dictionary)

```
public PurchaseResult(Dictionary serialized)
```

Parameters

`serialized` Dictionary

PurchaseResult(Guid)

```
public PurchaseResult(Guid shopProductId)
```

Parameters

`shopProductId` Guid

Fields

errorCode

```
public int errorCode
```

Field Value

int

productId

```
public readonly Guid productId
```

Field Value

Guid

Methods

Serialize()

```
public override IValue Serialize()
```

Returns

IValue

Class Buy7.SellerMultipleResult

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
public class Buy7.SellerMultipleResult
```

Inheritance

object ← Buy7.SellerMultipleResult

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

SellerMultipleResult()

```
public SellerMultipleResult()
```

SellerMultipleResult(Dictionary)

```
public SellerMultipleResult(Dictionary serialized)
```

Parameters

`serialized` Dictionary

Fields

sellerResults

```
public IEnumerable<Buy7.SellerResult> sellerResults
```

Field Value

IEnumerable<Buy7.SellerResult>

Methods

Serialize()

```
public IValue Serialize()
```

Returns

IValue

Class Buy7.SellerResult

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
public class Buy7.SellerResult : AttachmentActionResult, IState
```

Inheritance

object ← AttachmentActionResult ← Buy7.SellerResult

Implements

IState

Inherited Members

AttachmentActionResult.itemUsable , AttachmentActionResult.costume , AttachmentActionResult.tradableFungibleItem , AttachmentActionResult.tradableFungibleItemCount , AttachmentActionResult.SerializeBackup1() , AttachmentActionResult.Deserialize(Dictionary) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

SellerResult()

```
public SellerResult()
```

SellerResult(Dictionary)

```
public SellerResult(Dictionary serialized)
```

Parameters

`serialized` Dictionary

Fields

gold

```
public FungibleAssetValue gold
```

Field Value

FungibleAssetValue

id

```
public Guid id
```

Field Value

Guid

shopItem

```
public ShopItem shopItem
```

Field Value

ShopItem

Properties

TypeId

```
protected override string TypeId { get; }
```

Property Value

string

Methods

Serialize()

```
public override IValue Serialize()
```

Returns

IValue

Class BuyMultiple

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

Introduced at https://github.com/planetarium/lib9c/pull/331 Updated at many pull requests Obsoleted at https://github.com/planetarium/lib9c/pull/487 Updated at many pull requests Updated at https://github.com/planetarium/lib9c/pull/957

```
[Serializable]
[ActionObsolete(7000000)]
[ActionType("buy_multiple")]
public class BuyMultiple : GameAction, IAction, IBuyMultipleV1
```

Inheritance

object ← ActionBase ← GameAction ← BuyMultiple

Implements

IAction, IBuyMultipleV1

Inherited Members

GameAction.Id , GameAction.PlainValue , GameAction.LoadPlainValue(IValue) , ActionBase.GetSignerAndOtherAddressesHex(IActionContext, params Address[]) , ActionBase.TryGetAdminState(IActionContext, out AdminState) , ActionBase.CheckPermission(IActionContext) , ActionBase.CheckObsolete(long, IActionContext) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Fields

ERROR_CODE_FAILED_LOADING_STATE

```
public const int ERROR_CODE_FAILED_LOADING_STATE = 1
```

Field Value

int

ERROR_CODE_INSUFFICIENT_BALANCE

```
public const int ERROR_CODE_INSUFFICIENT_BALANCE = 4
```

Field Value

int

ERROR_CODE_ITEM_DOES_NOT_EXIST

```
public const int ERROR_CODE_ITEM_DOES_NOT_EXIST = 2
```

Field Value

int

ERROR_CODE_SHOPITEM_EXPIRED

```
public const int ERROR_CODE_SHOPITEM_EXPIRED = 3
```

Field Value

int

buyerAvatarAddress

```
public Address buyerAvatarAddress
```

Field Value

Address

buyerResult

```
public BuyMultiple.BuyerResult buyerResult
```

Field Value

BuyMultiple.BuyerResult

purchaseInfos

```
public IEnumerable<BuyMultiple.PurchaseInfo> purchaseInfos
```

Field Value

IEnumerable<BuyMultiple.PurchaseInfo>

sellerResult

```
public BuyMultiple.SellerResult sellerResult
```

Field Value

BuyMultiple.SellerResult

Properties

PlainValueInternal

```
protected override IImmutableDictionary<string, IValue> PlainValueInternal { get; }
```

Property Value

IImmutableDictionary<string, IValue>

Methods

Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

Parameters

`context` IActionContext

Returns

IWorld

LoadPlainValueInternal(IImmutableDictionary<string, IValue>)

```
protected override void LoadPlainValueInternal(IImmutableDictionary<string,
IValue> plainValue)
```

Parameters

`plainValue` IImmutableDictionary<string, IValue>

Class BuyMultiple.BuyerResult

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
public class BuyMultiple.BuyerResult
```

Inheritance

object ← BuyMultiple.BuyerResult

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

BuyerResult()

```
public BuyerResult()
```

BuyerResult(Dictionary)

```
public BuyerResult(Dictionary serialized)
```

Parameters

`serialized` Dictionary

Fields

purchaseResults

```
public IEnumerable<BuyMultiple.PurchaseResult> purchaseResults
```

Field Value

IEnumerable<BuyMultiple.PurchaseResult>

Methods

Serialize()

```
public IValue Serialize()
```

Returns

IValue

Class BuyMultiple.PurchaseInfo

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
public class BuyMultiple.PurchaseInfo : IComparable<BuyMultiple.PurchaseInfo>,
IComparable
```

Inheritance

object ← BuyMultiple.PurchaseInfo

Implements

IComparable<BuyMultiple.PurchaseInfo>, IComparable

Inherited Members

object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

PurchaseInfo(Dictionary)

```
public PurchaseInfo(Dictionary serialized)
```

Parameters

`serialized` Dictionary

PurchaseInfo(Guid, Address, Address)

```
public PurchaseInfo(Guid productId, Address sellerAgentAddress,
Address sellerAvatarAddress)
```

Parameters

`productId` Guid

`sellerAgentAddress` Address

`sellerAvatarAddress` Address

Fields

productId

```
public Guid productId
```

Field Value

Guid

sellerAgentAddress

```
public Address sellerAgentAddress
```

Field Value

Address

sellerAvatarAddress

```
public Address sellerAvatarAddress
```

Field Value

Address

Methods

CompareTo(PurchaseInfo)

Compares the current instance with another object of the same type and returns an integer that indicates whether the current instance precedes, follows, or occurs in the same position in the sort order as the other object.

```
public int CompareTo(BuyMultiple.PurchaseInfo other)
```

Parameters

`other` BuyMultiple.PurchaseInfo

An object to compare with this instance.

Returns

int

A value that indicates the relative order of the objects being compared. The return value has these meanings:

Value	Meaning
Less than zero	This instance precedes `other` in the sort order.
Zero	This instance occurs in the same position in the sort order as `other`.
Greater than zero	This instance follows `other` in the sort order.

CompareTo(object)

Compares the current instance with another object of the same type and returns an integer that indicates whether the current instance precedes, follows, or occurs in the same position in the sort order as the other object.

```
public int CompareTo(object obj)
```

Parameters

`obj` object

An object to compare with this instance.

Returns

int

A value that indicates the relative order of the objects being compared. The return value has these meanings:

Value	Meaning
Less than zero	This instance precedes `obj` in the sort order.
Zero	This instance occurs in the same position in the sort order as `obj`.
Greater than zero	This instance follows `obj` in the sort order.

Exceptions

ArgumentException

`obj` is not the same type as this instance.

Equals(PurchaseInfo)

```
protected bool Equals(BuyMultiple.PurchaseInfo other)
```

Parameters

`other` BuyMultiple.PurchaseInfo

Returns

bool

Equals(object)

Determines whether the specified object is equal to the current object.

```
public override bool Equals(object obj)
```

Parameters

`obj` object

The object to compare with the current object.

Returns

bool

true if the specified object is equal to the current object; otherwise, false.

GetHashCode()

Serves as the default hash function.

```
public override int GetHashCode()
```

Returns

int

A hash code for the current object.

Serialize()

```
public IValue Serialize()
```

Returns

IValue

Operators

operator ==(PurchaseInfo, PurchaseInfo)

```
public static bool operator ==(BuyMultiple.PurchaseInfo left,
BuyMultiple.PurchaseInfo right)
```

Parameters

`left` BuyMultiple.PurchaseInfo

`right` BuyMultiple.PurchaseInfo

Returns

bool

operator >(PurchaseInfo, PurchaseInfo)

```
public static bool operator >(BuyMultiple.PurchaseInfo left,
BuyMultiple.PurchaseInfo right)
```

Parameters

`left` BuyMultiple.PurchaseInfo

`right` BuyMultiple.PurchaseInfo

Returns

bool

operator >=(PurchaseInfo, PurchaseInfo)

```
public static bool operator >=(BuyMultiple.PurchaseInfo left,
BuyMultiple.PurchaseInfo right)
```

Parameters

`left` BuyMultiple.PurchaseInfo

`right` BuyMultiple.PurchaseInfo

Returns

bool

operator !=(PurchaseInfo, PurchaseInfo)

```
public static bool operator !=(BuyMultiple.PurchaseInfo left,
BuyMultiple.PurchaseInfo right)
```

Parameters

`left` BuyMultiple.PurchaseInfo

`right` BuyMultiple.PurchaseInfo

Returns

bool

operator <(PurchaseInfo, PurchaseInfo)

```
public static bool operator <(BuyMultiple.PurchaseInfo left,
BuyMultiple.PurchaseInfo right)
```

Parameters

`left` BuyMultiple.PurchaseInfo

`right` BuyMultiple.PurchaseInfo

Returns

bool

operator <=(PurchaseInfo, PurchaseInfo)

```
public static bool operator <=(BuyMultiple.PurchaseInfo left,
BuyMultiple.PurchaseInfo right)
```

Parameters

`left` BuyMultiple.PurchaseInfo

`right` BuyMultiple.PurchaseInfo

Returns

bool

Class BuyMultiple.PurchaseResult

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
public class BuyMultiple.PurchaseResult : Buy7.BuyerResult, IState
```

Inheritance

object ← AttachmentActionResult ← Buy7.BuyerResult ← BuyMultiple.PurchaseResult

Implements

IState

Inherited Members

Buy7.BuyerResult.shopItem , Buy7.BuyerResult.id , Buy7.BuyerResult.TypeId , AttachmentActionResult.itemUsable , AttachmentActionResult.costume , AttachmentActionResult.tradableFungibleItem , AttachmentActionResult.tradableFungibleItemCount , AttachmentActionResult.SerializeBackup1() , AttachmentActionResult.Deserialize(Dictionary) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

PurchaseResult(Dictionary)

```
public PurchaseResult(Dictionary serialized)
```

Parameters

`serialized` Dictionary

PurchaseResult(Guid)

```
public PurchaseResult(Guid shopProductId)
```

Parameters

`shopProductId` Guid

Fields

errorCode

```
public int errorCode
```

Field Value

int

productId

```
public Guid productId
```

Field Value

Guid

Methods

Serialize()

```
public override IValue Serialize()
```

Returns

IValue

Class BuyMultiple.SellerResult

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
public class BuyMultiple.SellerResult
```

Inheritance

object ← BuyMultiple.SellerResult

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

SellerResult()

```
public SellerResult()
```

SellerResult(Dictionary)

```
public SellerResult(Dictionary serialized)
```

Parameters

`serialized` Dictionary

Fields

sellerResults

```
public IEnumerable<Buy7.SellerResult> sellerResults
```

Field Value

IEnumerable<Buy7.SellerResult>

Methods

Serialize()

```
public IValue Serialize()
```

Returns

IValue

Class BuyProduct

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[ActionType("buy_product3")]
public class BuyProduct : GameAction, IAction
```

Inheritance

object ← ActionBase ← GameAction ← BuyProduct

Implements

IAction

Inherited Members

GameAction.Id , GameAction.PlainValue , GameAction.LoadPlainValue(IValue) , ActionBase.GetSignerAndOtherAddressesHex(IActionContext, params Address[]) , ActionBase.TryGetAdminState(IActionContext, out AdminState) , ActionBase.CheckPermission(IActionContext) , ActionBase.CheckObsolete(long, IActionContext) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Fields

AvatarAddress

```
public Address AvatarAddress
```

Field Value

Address

Capacity

```
public const int Capacity = 20
```

Field Value

int

ProductInfos

```
public IEnumerable<IProductInfo> ProductInfos
```

Field Value

IEnumerable<IProductInfo>

Properties

PlainValueInternal

```
protected override IImmutableDictionary<string, IValue> PlainValueInternal { get; }
```

Property Value

IImmutableDictionary<string, IValue>

Methods

Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

Parameters

`context` IActionContext

Returns

IWorld

LoadPlainValueInternal(IImmutableDictionary<string, IValue>)

```
protected override void LoadPlainValueInternal(IImmutableDictionary<string,
IValue> plainValue)
```

Parameters

`plainValue` IImmutableDictionary<string, IValue>

Class CancelProductRegistration

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[ActionType("cancel_product_registration2")]
public class CancelProductRegistration : GameAction, IAction
```

Inheritance

object ← ActionBase ← GameAction ← CancelProductRegistration

Implements

IAction

Inherited Members

GameAction.Id , GameAction.PlainValue , GameAction.LoadPlainValue(IValue) , ActionBase.GetSignerAndOtherAddressesHex(IActionContext, params Address[]) , ActionBase.TryGetAdminState(IActionContext, out AdminState) , ActionBase.CheckPermission(IActionContext) , ActionBase.CheckObsolete(long, IActionContext) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Fields

AvatarAddress

```
public Address AvatarAddress
```

Field Value

Address

Capacity

```
public const int Capacity = 100
```

Field Value

int

ChargeAp

```
public bool ChargeAp
```

Field Value

bool

CostAp

```
public const int CostAp = 5
```

Field Value

int

ProductInfos

```
public List<IProductInfo> ProductInfos
```

Field Value

List<IProductInfo>

Properties

PlainValueInternal

```
protected override IImmutableDictionary<string, IValue> PlainValueInternal { get; }
```

Property Value

IImmutableDictionary<string, IValue>

Methods

Cancel(ProductsState, IProductInfo, IWorld, AvatarState, IActionContext)

```
public static IWorld Cancel(ProductsState productsState, IProductInfo productInfo,
IWorld states, AvatarState avatarState, IActionContext context)
```

Parameters

`productsState` ProductsState

`productInfo` IProductInfo

`states` IWorld

`avatarState` AvatarState

`context` IActionContext

Returns

IWorld

Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

Parameters

`context` IActionContext

Returns

LoadPlainValueInternal(IImmutableDictionary<string, IValue>)

```
protected override void LoadPlainValueInternal(IImmutableDictionary<string,
IValue> plainValue)
```

Parameters

`plainValue` IImmutableDictionary<string, IValue>

Class ChargeActionPoint

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

Hard forked at https://github.com/planetarium/lib9c/pull/430 Updated at https://github.com/planetarium/lib9c/pull/474 Updated at https://github.com/planetarium/lib9c/pull/602 Updated at https://github.com/planetarium/lib9c/pull/861 Updated at https://github.com/planetarium/lib9c/pull/957

```
[Serializable]
[ActionType("charge_action_point3")]
public class ChargeActionPoint : GameAction, IAction, IChargeActionPointV1
```

Inheritance

object ← ActionBase ← GameAction ← ChargeActionPoint

Implements

IAction, IChargeActionPointV1

Inherited Members

GameAction.Id , GameAction.PlainValue , GameAction.LoadPlainValue(IValue) , ActionBase.GetSignerAndOtherAddressesHex(IActionContext, params Address[]) , ActionBase.TryGetAdminState(IActionContext, out AdminState) , ActionBase.CheckPermission(IActionContext) , ActionBase.CheckObsolete(long, IActionContext) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Fields

avatarAddress

```
public Address avatarAddress
```

Field Value

Address

Properties

PlainValueInternal

```
protected override IImmutableDictionary<string, IValue> PlainValueInternal { get; }
```

Property Value

IImmutableDictionary<string, IValue>

Methods

Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

Parameters

`context` IActionContext

Returns

IWorld

LoadPlainValueInternal(IImmutableDictionary<string, IValue>)

```
protected override void LoadPlainValueInternal(IImmutableDictionary<string,
IValue> plainValue)
```

Parameters

`plainValue` IImmutableDictionary<string, IValue>

Class ClaimGifts

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

An action to claim gifts using the Gift Id.
Refer ClaimableGiftsSheet to find gifts data.
It can only claim once at the specified block index.

```
[ActionType("claim_gifts")]
public class ClaimGifts : GameAction, IAction
```

Inheritance

object ← ActionBase ← GameAction ← ClaimGifts

Implements

IAction

Inherited Members

GameAction.Id , GameAction.PlainValue , GameAction.LoadPlainValue(IValue) , ActionBase.GetSignerAndOtherAddressesHex(IActionContext, params Address[]) , ActionBase.TryGetAdminState(IActionContext, out AdminState) , ActionBase.CheckPermission(IActionContext) , ActionBase.CheckObsolete(long, IActionContext) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

ClaimGifts()

```
public ClaimGifts()
```

ClaimGifts(Address, int)

```
public ClaimGifts(Address avatarAddress, int giftId)
```

Parameters

`avatarAddress` Address

`giftId` int

Properties

AvatarAddress

The address of the avatar claiming the gift.

```
public Address AvatarAddress { get; }
```

Property Value

Address

GiftId

The ID of the gift to be claimed. This ID is used in the ClaimableGiftsSheet.

```
public int GiftId { get; }
```

Property Value

int

PlainValueInternal

```
protected override IImmutableDictionary<string, IValue> PlainValueInternal { get; }
```

Property Value

IImmutableDictionary<string, IValue>

Methods

Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

Parameters

`context` IActionContext

Returns

IWorld

Exceptions

FailedLoadStateException

Thrown when the inventory could not be loaded.

SheetRowNotFoundException

Thrown when the gift ID is not found in the ClaimableGiftsSheet.

ClaimableGiftsNotAvailableException

Thrown when the gift is not available at the current block index.

AlreadyClaimedGiftsException

Thrown when the gift has already been claimed.

LoadPlainValueInternal(IImmutableDictionary<string, IValue>)

```
protected override void LoadPlainValueInternal(IImmutableDictionary<string,
IValue> plainValue)
```

Parameters

`plainValue` IImmutableDictionary<string, IValue>

Class ClaimItems

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[ActionType("claim_items")]
public class ClaimItems : GameAction, IAction, IClaimItems
```

Inheritance

object ← ActionBase ← GameAction ← ClaimItems

Implements

IAction, IClaimItems

Inherited Members

GameAction.Id , GameAction.PlainValue , GameAction.LoadPlainValue(IValue) , ActionBase.GetSignerAndOtherAddressesHex(IActionContext, params Address[]) , ActionBase.TryGetAdminState(IActionContext, out AdminState) , ActionBase.CheckPermission(IActionContext) , ActionBase.CheckObsolete(long, IActionContext) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

ClaimItems()

```
public ClaimItems()
```

ClaimItems(IReadOnlyList<(Address, IReadOnlyList<FungibleAssetValue>)>, string)

```
public ClaimItems(IReadOnlyList<(Address, IReadOnlyList<FungibleAssetValue>)>
claimData, string memo = null)
```

Parameters

`claimData` IReadOnlyList<(Address address, IReadOnlyList<FungibleAssetValue> fungibleAssetValues)>

`memo` string

Fields

Memo

```
public string Memo
```

Field Value

string

Properties

ClaimData

```
public IReadOnlyList<(Address address, IReadOnlyList<FungibleAssetValue>
fungibleAssetValues)> ClaimData { get; }
```

Property Value

IReadOnlyList<(Address address, IReadOnlyList<FungibleAssetValue> fungibleAssetValues)>

PlainValueInternal

```
protected override IImmutableDictionary<string, IValue> PlainValueInternal { get; }
```

Property Value

IImmutableDictionary<string, IValue>

Methods

Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

Parameters

`context` IActionContext

Returns

IWorld

LoadPlainValueInternal(IImmutableDictionary<string, IValue>)

```
protected override void LoadPlainValueInternal(IImmutableDictionary<string,
IValue> plainValue)
```

Parameters

`plainValue` IImmutableDictionary<string, IValue>

Class ClaimPatrolReward

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

Claim patrol reward

```
[Serializable]
[ActionType("claim_patrol_reward")]
public class ClaimPatrolReward : ActionBase, IAction
```

Inheritance

object ← ActionBase ← ClaimPatrolReward

Implements

IAction

Inherited Members

ActionBase.GetSignerAndOtherAddressesHex(IActionContext, params Address[]) ,
ActionBase.TryGetAdminState(IActionContext, out AdminState) ,
ActionBase.CheckPermission(IActionContext) ,
ActionBase.CheckObsolete(long, IActionContext) , object.Equals(object) ,
object.Equals(object, object) , object.GetHashCode() , object.GetType() ,
object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

ClaimPatrolReward()

```
public ClaimPatrolReward()
```

ClaimPatrolReward(Address)

```
public ClaimPatrolReward(Address avatarAddress)
```

Parameters

`avatarAddress` Address

Fields

AvatarAddress

The address of the avatar to receive the patrol reward.

```
public Address AvatarAddress
```

Field Value

Address

TypeIdentifier

```
public const string TypeIdentifier = "claim_patrol_reward"
```

Field Value

string

Properties

PlainValue

```
public override IValue PlainValue { get; }
```

Property Value

IValue

Methods

Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

Parameters

`context` IActionContext

Returns

IWorld

LoadPlainValue(IValue)

```
public override void LoadPlainValue(IValue plainValue)
```

Parameters

`plainValue` IValue

Class ClaimRaidReward

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
[ActionType("claim_raid_reward")]
public class ClaimRaidReward : GameAction, IAction, IClaimRaidRewardV1
```

Inheritance

object ← ActionBase ← GameAction ← ClaimRaidReward

Implements

IAction, IClaimRaidRewardV1

Inherited Members

GameAction.Id , GameAction.PlainValue , GameAction.LoadPlainValue(IValue) , ActionBase.GetSignerAndOtherAddressesHex(IActionContext, params Address[]) , ActionBase.TryGetAdminState(IActionContext, out AdminState) , ActionBase.CheckPermission(IActionContext) , ActionBase.CheckObsolete(long, IActionContext) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

ClaimRaidReward()

```
public ClaimRaidReward()
```

ClaimRaidReward(Address)

```
public ClaimRaidReward(Address avatarAddress)
```

Parameters

`avatarAddress` Address

Fields

AvatarAddress

```
public Address AvatarAddress
```

Field Value

Address

Properties

PlainValueInternal

```
protected override IImmutableDictionary<string, IValue> PlainValueInternal { get; }
```

Property Value

IImmutableDictionary<string, IValue>

Methods

Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

Parameters

`context` IActionContext

Returns

IWorld

LoadPlainValueInternal(IImmutableDictionary<string, IValue>)

```
protected override void LoadPlainValueInternal(IImmutableDictionary<string,
IValue> plainValue)
```

Parameters

`plainValue` IImmutableDictionary<string, IValue>

Class ClaimStakeReward

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

Hard forked at https://github.com/planetarium/lib9c/pull/2097

```
[ActionType("claim_stake_reward9")]
public class ClaimStakeReward : GameAction, IClaimStakeReward,
IAction, IClaimStakeRewardV1
```

Inheritance

object ← ActionBase ← GameAction ← ClaimStakeReward

Implements

IClaimStakeReward, IAction, IClaimStakeRewardV1

Inherited Members

GameAction.Id , GameAction.PlainValue , GameAction.LoadPlainValue(IValue) , ActionBase.GetSignerAndOtherAddressesHex(IActionContext, params Address[]) , ActionBase.TryGetAdminState(IActionContext, out AdminState) , ActionBase.CheckPermission(IActionContext) , ActionBase.CheckObsolete(long, IActionContext) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

ClaimStakeReward()

```
public ClaimStakeReward()
```

ClaimStakeReward(Address)

```
public ClaimStakeReward(Address avatarAddress)
```

Parameters

`avatarAddress` Address

Properties

PlainValueInternal

```
protected override IImmutableDictionary<string, IValue> PlainValueInternal { get; }
```

Property Value

IImmutableDictionary<string, IValue>

Methods

Claim(IWorld, IActionContext, Address, Address, AvatarState, StakeState)

```
public static IWorld Claim(IWorld states, IActionContext context, Address
avatarAddress, Address stakeStateAddress, AvatarState avatarState,
StakeState stakeStateV2)
```

Parameters

`states` IWorld

`context` IActionContext

`avatarAddress` Address

`stakeStateAddress` Address

`avatarState` AvatarState

`stakeStateV2` StakeState

Returns

IWorld

Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

Parameters

`context` IActionContext

Returns

IWorld

LoadPlainValueInternal(IImmutableDictionary<string, IValue>)

```
protected override void LoadPlainValueInternal(IImmutableDictionary<string,
IValue> plainValue)
```

Parameters

`plainValue` IImmutableDictionary<string, IValue>

Class ClaimStakeReward2

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
public static class ClaimStakeReward2
```

Inheritance

object ← ClaimStakeReward2

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Fields

ObsoletedIndex

```
public const long ObsoletedIndex = 5549200
```

Field Value

long

Class ClaimStakeReward8

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
public class ClaimStakeReward8
```

Inheritance

object ← ClaimStakeReward8

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Fields

ObsoleteBlockIndex

```
public const long ObsoleteBlockIndex = 7983895
```

Field Value

long

Class ClaimWordBossKillReward

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
[ActionType("claim_world_boss_kill_reward")]
public class ClaimWordBossKillReward : GameAction,
IAction, IClaimWordBossKillRewardV1
```

Inheritance

object ← ActionBase ← GameAction ← ClaimWordBossKillReward

Implements

IAction, IClaimWordBossKillRewardV1

Inherited Members

GameAction.Id , GameAction.PlainValue , GameAction.LoadPlainValue(IValue) , ActionBase.GetSignerAndOtherAddressesHex(IActionContext, params Address[]) , ActionBase.TryGetAdminState(IActionContext, out AdminState) , ActionBase.CheckPermission(IActionContext) , ActionBase.CheckObsolete(long, IActionContext) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Fields

AvatarAddress

```
public Address AvatarAddress
```

Field Value

Address

Properties

PlainValueInternal

```
protected override IImmutableDictionary<string, IValue> PlainValueInternal { get; }
```

Property Value

IImmutableDictionary<string, IValue>

Methods

Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

Parameters

`context` IActionContext

Returns

IWorld

LoadPlainValueInternal(IImmutableDictionary<string, IValue>)

```
protected override void LoadPlainValueInternal(IImmutableDictionary<string,
IValue> plainValue)
```

Parameters

`plainValue` IImmutableDictionary<string, IValue>

Class ClaimWorldBossReward

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

The ClaimWorldBossReward class is an action that allows a player to claim rewards for their contribution in defeating a world boss. This action ensures that only the owner of the avatar can claim the reward and calculates the reward based on the player's contribution.

```
[Serializable]
[ActionType("claim_world_boss_reward")]
public class ClaimWorldBossReward : GameAction, IAction
```

Inheritance

object ← ActionBase ← GameAction ← ClaimWorldBossReward

Implements

IAction

Inherited Members

GameAction.Id , GameAction.PlainValue , GameAction.LoadPlainValue(IValue) , ActionBase.GetSignerAndOtherAddressesHex(IActionContext, params Address[]) , ActionBase.TryGetAdminState(IActionContext, out AdminState) , ActionBase.CheckPermission(IActionContext) , ActionBase.CheckObsolete(long, IActionContext) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

ClaimWorldBossReward()

Initializes a new instance of the ClaimWorldBossReward class.

```
public ClaimWorldBossReward()
```

ClaimWorldBossReward(Address)

Initializes a new instance with the specified avatar address.

```
public ClaimWorldBossReward(Address address)
```

Parameters

`address` Address

The address of the avatar.

Fields

AvatarAddress

The address of the avatar claiming the reward.

```
public Address AvatarAddress
```

Field Value

Address

MinimumContribution

The minimum contribution percentage required to be eligible for a reward.

```
public const decimal MinimumContribution = 0.0001
```

Field Value

decimal

TypeIdentifier

A string identifier for this action type.

```
public const string TypeIdentifier = "claim_world_boss_reward"
```

Field Value

string

Properties

PlainValueInternal

Serializes the action's data into a plain value format.

```
protected override IImmutableDictionary<string, IValue> PlainValueInternal { get; }
```

Property Value

IImmutableDictionary<string, IValue>

Methods

Execute(IActionContext)

Executes the action to claim the world boss reward.

```
public override IWorld Execute(IActionContext context)
```

Parameters

`context` IActionContext

The action context.

Returns

IWorld

The updated world state.

Exceptions

InvalidAddressException

Thrown if the avatar address does not belong to the signer.

AlreadyClaimedException

Thrown if the reward has already been claimed.

InvalidClaimException

Thrown if no valid rewards are calculated.

InvalidOperationException

Thrown if the contribution does not meet the minimum requirement.

LoadPlainValueInternal(IImmutableDictionary<string, IValue>)

Deserializes the plain value back into the action's data.

```
protected override void LoadPlainValueInternal(IImmutableDictionary<string,
IValue> plainValue)
```

Parameters

`plainValue` IImmutableDictionary<string, IValue>

The serialized plain value.

Class ClaimableGiftsNotAvailableException

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
public class ClaimableGiftsNotAvailableException : Exception, ISerializable
```

Inheritance

object ← Exception ← ClaimableGiftsNotAvailableException

Implements

ISerializable

Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Constructors

ClaimableGiftsNotAvailableException()

```
public ClaimableGiftsNotAvailableException()
```

ClaimableGiftsNotAvailableException(SerializationInfo, StreamingContext)

```
public ClaimableGiftsNotAvailableException(SerializationInfo info,
StreamingContext context)
```

Parameters

`info` SerializationInfo

`context` StreamingContext

ClaimableGiftsNotAvailableException(string)

```
public ClaimableGiftsNotAvailableException(string msg)
```

Parameters

`msg` string

Class CombinationConsumable

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

Hard forked at https://github.com/planetarium/lib9c/pull/2195

```
[Serializable]
[ActionType("combination_consumable9")]
public class CombinationConsumable : GameAction, IAction, ICombinationConsumableV1
```

Inheritance

object ← ActionBase ← GameAction ← CombinationConsumable

Implements

IAction, ICombinationConsumableV1

Inherited Members

GameAction.Id , GameAction.PlainValue , GameAction.LoadPlainValue(IValue) , ActionBase.GetSignerAndOtherAddressesHex(IActionContext, params Address[]) , ActionBase.TryGetAdminState(IActionContext, out AdminState) , ActionBase.CheckPermission(IActionContext) , ActionBase.CheckObsolete(long, IActionContext) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Fields

AvatarAddressKey

```
public const string AvatarAddressKey = "a"
```

Field Value

string

RecipeIdKey

```
public const string RecipeIdKey = "r"
```

Field Value

string

SlotIndexKey

```
public const string SlotIndexKey = "s"
```

Field Value

string

avatarAddress

```
public Address avatarAddress
```

Field Value

Address

recipeId

```
public int recipeId
```

Field Value

int

slotIndex

```
public int slotIndex
```

Field Value

int

Properties

PlainValueInternal

```
protected override IImmutableDictionary<string, IValue> PlainValueInternal { get; }
```

Property Value

IImmutableDictionary<string, IValue>

Methods

Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

Parameters

`context` IActionContext

Returns

IWorld

LoadPlainValueInternal(IImmutableDictionary<string, IValue>)

```
protected override void LoadPlainValueInternal(IImmutableDictionary<string,
```

```
IValue> plainValue)
```

Parameters

`plainValue` IImmutableDictionary<string, IValue>

Class CombinationConsumable5

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
[ActionObsolete(7000000)]
[ActionType("combination_consumable5")]
public class CombinationConsumable5 : GameAction, IAction, ICombinationConsumableV1
```

Inheritance

object ← ActionBase ← GameAction ← CombinationConsumable5

Implements

IAction, ICombinationConsumableV1

Inherited Members

GameAction.Id , GameAction.PlainValue , GameAction.LoadPlainValue(IValue) , ActionBase.GetSignerAndOtherAddressesHex(IActionContext, params Address[]) , ActionBase.TryGetAdminState(IActionContext, out AdminState) , ActionBase.CheckPermission(IActionContext) , ActionBase.CheckObsolete(long, IActionContext) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Fields

AvatarAddress

```
public Address AvatarAddress
```

Field Value

Address

recipeId

```
public int recipeId
```

Field Value

int

slotIndex

```
public int slotIndex
```

Field Value

int

Properties

PlainValueInternal

```
protected override IImmutableDictionary<string, IValue> PlainValueInternal { get; }
```

Property Value

IImmutableDictionary<string, IValue>

Methods

Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

Parameters

`context` IActionContext

Returns

IWorld

LoadPlainValueInternal(IImmutableDictionary<string, IValue>)

```
protected override void LoadPlainValueInternal(IImmutableDictionary<string,
IValue> plainValue)
```

Parameters

`plainValue` IImmutableDictionary<string, IValue>

Class CombinationConsumable5.ResultModel

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
public class CombinationConsumable5.ResultModel : AttachmentActionResult, IState
```

Inheritance

object ← AttachmentActionResult ← CombinationConsumable5.ResultModel

Implements

IState

Derived

RapidCombination0.ResultModel

Inherited Members

AttachmentActionResult.itemUsable , AttachmentActionResult.costume , AttachmentActionResult.tradableFungibleItem , AttachmentActionResult.tradableFungibleItemCount , AttachmentActionResult.SerializeBackup1() , AttachmentActionResult.Deserialize(Dictionary) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

ResultModel()

```
public ResultModel()
```

ResultModel(Dictionary)

```
public ResultModel(Dictionary serialized)
```

Parameters

`serialized` Dictionary

Fields

actionPoint

```
public int actionPoint
```

Field Value

int

gold

```
public BigInteger gold
```

Field Value

BigInteger

id

```
public Guid id
```

Field Value

Guid

itemType

```
public ItemType itemType
```

Field Value

ItemType

materials

```
public Dictionary<Material, int> materials
```

Field Value

Dictionary<Material, int>

recipeId

```
public int recipeId
```

Field Value

int

subRecipeId

```
public int? subRecipeId
```

Field Value

int?

Properties

TypeId

```
protected override string TypeId { get; }
```

Property Value

string

Methods

Serialize()

```
public override IValue Serialize()
```

Returns

IValue

Class CombinationEquipment

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

Hard forked at https://github.com/planetarium/lib9c/pull/2195

```
[Serializable]
[ActionType("combination_equipment17")]
public class CombinationEquipment : GameAction, IAction, ICombinationEquipmentV4
```

Inheritance

object ← ActionBase ← GameAction ← CombinationEquipment

Implements

IAction, ICombinationEquipmentV4

Inherited Members

GameAction.Id , GameAction.PlainValue , GameAction.LoadPlainValue(IValue) , ActionBase.GetSignerAndOtherAddressesHex(IActionContext, params Address[]) , ActionBase.TryGetAdminState(IActionContext, out AdminState) , ActionBase.CheckPermission(IActionContext) , ActionBase.CheckObsolete(long, IActionContext) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Fields

AvatarAddressKey

```
public const string AvatarAddressKey = "a"
```

Field Value

string

BasicSubRecipeHammerPoint

```
public const int BasicSubRecipeHammerPoint = 1
```

Field Value

int

PayByCrystalKey

```
public const string PayByCrystalKey = "p"
```

Field Value

string

PetIdKey

```
public const string PetIdKey = "pid"
```

Field Value

string

RecipeIdKey

```
public const string RecipeIdKey = "r"
```

Field Value

string

SlotIndexKey

```
public const string SlotIndexKey = "s"
```

Field Value

string

SpecialSubRecipeHammerPoint

```
public const int SpecialSubRecipeHammerPoint = 2
```

Field Value

int

SubRecipeIdKey

```
public const string SubRecipeIdKey = "i"
```

Field Value

string

UseHammerPointKey

```
public const string UseHammerPointKey = "h"
```

Field Value

string

avatarAddress

```
public Address avatarAddress
```

Field Value

Address

payByCrystal

```
public bool payByCrystal
```

Field Value

bool

petId

```
public int? petId
```

Field Value

int?

recipeId

```
public int recipeId
```

Field Value

int

slotIndex

```
public int slotIndex
```

Field Value

int

subRecipeId

```
public int? subRecipeId
```

Field Value

int?

useHammerPoint

```
public bool useHammerPoint
```

Field Value

bool

Properties

PlainValueInternal

```
protected override IImmutableDictionary<string, IValue> PlainValueInternal { get; }
```

Property Value

IImmutableDictionary<string, IValue>

Methods

AddAndUnlockOption(AgentState, PetState, Equipment, IRandom, Row, EquipmentItemOptionSheet, PetOptionSheet, SkillSheet)

```
public static void AddAndUnlockOption(AgentState agentState, PetState petState,
Equipment equipment, IRandom random, EquipmentItemSubRecipeSheetV2.Row
subRecipe, EquipmentItemOptionSheet optionSheet, PetOptionSheet petOptionSheet,
SkillSheet skillSheet)
```

Parameters

`agentState` AgentState

`petState` PetState

`equipment` Equipment

`random` IRandom

`subRecipe` EquipmentItemSubRecipeSheetV2.Row

`optionSheet` EquipmentItemOptionSheet

`petOptionSheet` PetOptionSheet

`skillSheet` SkillSheet

AddSkillOption(AgentState, Equipment, IRandom, Row, EquipmentItemOptionSheet, SkillSheet)

```
public static void AddSkillOption(AgentState agentState, Equipment equipment,
IRandom random, EquipmentItemSubRecipeSheetV2.Row subRecipe,
EquipmentItemOptionSheet optionSheet, SkillSheet skillSheet)
```

Parameters

`agentState` AgentState

`equipment` Equipment

`random` IRandom

`subRecipe` EquipmentItemSubRecipeSheetV2.Row

`optionSheet` EquipmentItemOptionSheet

`skillSheet` SkillSheet

Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

Parameters

`context` IActionContext

Returns

IWorld

GetSkill(Row, SkillSheet, IRandom)

```
public static Skill GetSkill(EquipmentItemOptionSheet.Row row, SkillSheet
skillSheet, IRandom random)
```

Parameters

`row` EquipmentItemOptionSheet.Row

`skillSheet` SkillSheet

`random` IRandom

Returns

Skill

LoadPlainValueInternal(IImmutableDictionary<string, IValue>)

```
protected override void LoadPlainValueInternal(IImmutableDictionary<string,
IValue> plainValue)
```

Parameters

`plainValue` IImmutableDictionary<string, IValue>

Class CombinationEquipment5

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

Option helpers kept from `CombinationEquipment5` and still used by current crafting.

```
public static class CombinationEquipment5
```

Inheritance

object ← CombinationEquipment5

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Fields

BlacksmithAddress

The address that collects crafting fees.

```
public static readonly Address BlacksmithAddress
```

Field Value

Address

Methods

GetSkill(Row, SkillSheet, IRandom)

```
public static Skill GetSkill(EquipmentItemOptionSheet.Row row, SkillSheet
skillSheet, IRandom random)
```

Parameters

`row` EquipmentItemOptionSheet.Row

`skillSheet` SkillSheet

`random` IRandom

Returns

Skill

GetStat(Row, IRandom)

Draws a stat value in the option's range.

```
public static DecimalStat GetStat(EquipmentItemOptionSheet.Row row, IRandom random)
```

Parameters

`row` EquipmentItemOptionSheet.Row

`random` IRandom

Returns

DecimalStat

SelectOption(EquipmentItemOptionSheet, SkillSheet, Row, IRandom, Equipment)

```
public static HashSet<int> SelectOption(EquipmentItemOptionSheet optionSheet,
SkillSheet skillSheet, EquipmentItemSubRecipeSheet.Row subRecipe, IRandom random,
Equipment equipment)
```

Parameters

`optionSheet` EquipmentItemOptionSheet

`skillSheet` SkillSheet

`subRecipe` EquipmentItemSubRecipeSheet.Row

`random` IRandom

`equipment` Equipment

Returns

HashSet<int>

Class CombinationSlotResultNullException

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
public class CombinationSlotResultNullException : Exception, ISerializable
```

Inheritance

object ← Exception ← CombinationSlotResultNullException

Implements

ISerializable

Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Constructors

CombinationSlotResultNullException(SerializationInfo, StreamingContext)

```
public CombinationSlotResultNullException(SerializationInfo info,
StreamingContext context)
```

Parameters

`info` SerializationInfo

`context` StreamingContext

CombinationSlotResultNullException(string)

```
public CombinationSlotResultNullException(string message)
```

Parameters

`message` string

Class CombinationSlotUnlockException

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
public class CombinationSlotUnlockException : InvalidOperationException,
ISerializable
```

Inheritance

object ← Exception ← SystemException ← InvalidOperationException ← CombinationSlotUnlockException

Implements

ISerializable

Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Constructors

CombinationSlotUnlockException(SerializationInfo, StreamingContext)

```
protected CombinationSlotUnlockException(SerializationInfo info,
StreamingContext context)
```

Parameters

`info` SerializationInfo

`context` StreamingContext

CombinationSlotUnlockException(string)

```
public CombinationSlotUnlockException(string s)
```

Parameters

s [string]

Class ConsumableSlotOutOfRangeException

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
public class ConsumableSlotOutOfRangeException : Exception, ISerializable
```

Inheritance

object ← Exception ← ConsumableSlotOutOfRangeException

Implements

ISerializable

Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Constructors

ConsumableSlotOutOfRangeException()

```
public ConsumableSlotOutOfRangeException()
```

ConsumableSlotOutOfRangeException(SerializationInfo, StreamingContext)

```
protected ConsumableSlotOutOfRangeException(SerializationInfo info,
StreamingContext context)
```

Parameters

`info` SerializationInfo

`context` StreamingContext

ConsumableSlotOutOfRangeException(string)

```
public ConsumableSlotOutOfRangeException(string message)
```

Parameters

`message` string

Class ConsumableSlotUnlockException

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
public class ConsumableSlotUnlockException : InvalidOperationException,
ISerializable
```

Inheritance

object ← Exception ← SystemException ← InvalidOperationException ← ConsumableSlotUnlockException

Implements

ISerializable

Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Constructors

ConsumableSlotUnlockException(SerializationInfo, StreamingContext)

```
protected ConsumableSlotUnlockException(SerializationInfo info,
StreamingContext context)
```

Parameters

`info` SerializationInfo

`context` StreamingContext

ConsumableSlotUnlockException(string)

```
public ConsumableSlotUnlockException(string s)
```

Parameters

s string

Class CostumeSlotUnlockException

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
public class CostumeSlotUnlockException : InvalidOperationException, ISerializable
```

Inheritance

object ← Exception ← SystemException ← InvalidOperationException ← CostumeSlotUnlockException

Implements

ISerializable

Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Constructors

CostumeSlotUnlockException(SerializationInfo, StreamingContext)

```
protected CostumeSlotUnlockException(SerializationInfo info,
StreamingContext context)
```

Parameters

`info` SerializationInfo

`context` StreamingContext

CostumeSlotUnlockException(string)

```
public CostumeSlotUnlockException(string s)
```

Parameters

s string

Class CostumeSummon

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
[ActionType("costume_summon")]
public class CostumeSummon : GameAction, IAction
```

Inheritance

object ← ActionBase ← GameAction ← CostumeSummon

Implements

IAction

Inherited Members

GameAction.Id , GameAction.PlainValue , GameAction.LoadPlainValue(IValue) , ActionBase.GetSignerAndOtherAddressesHex(IActionContext, params Address[]) , ActionBase.TryGetAdminState(IActionContext, out AdminState) , ActionBase.CheckPermission(IActionContext) , ActionBase.CheckObsolete(long, IActionContext) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

CostumeSummon()

```
public CostumeSummon()
```

CostumeSummon(Address, int, int)

```
public CostumeSummon(Address avatarAddress, int groupId, int summonCount)
```

Parameters

`avatarAddress` Address

`groupId` int

`summonCount` int

Fields

AvatarAddress

```
public Address AvatarAddress
```

Field Value

Address

AvatarAddressKey

```
public const string AvatarAddressKey = "aa"
```

Field Value

string

GroupId

```
public int GroupId
```

Field Value

int

GroupIdKey

```
public const string GroupIdKey = "gid"
```

Field Value

string

SummonCount

```
public int SummonCount
```

Field Value

int

SummonCountKey

```
public const string SummonCountKey = "sc"
```

Field Value

string

Properties

PlainValueInternal

```
protected override IImmutableDictionary<string, IValue> PlainValueInternal { get; }
```

Property Value

IImmutableDictionary<string, IValue>

Methods

Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

Parameters

`context` IActionContext

Returns

IWorld

LoadPlainValueInternal(IImmutableDictionary<string, IValue>)

```
protected override void LoadPlainValueInternal(IImmutableDictionary<string,
IValue> plainValue)
```

Parameters

`plainValue` IImmutableDictionary<string, IValue>

SimulateSummon(string, CostumeItemSheet, Row, int, IRandom, Row)

Simulates costume summoning with optional grade guarantee system. Applies 10+1 bonus rule and uses grade guarantee settings from SummonSheet.Row when enabled. When grade guarantee is enabled, ensures minimum grade costumes are obtained based on summon count.

```
public static IEnumerable<Costume> SimulateSummon(string addressesHex,
CostumeItemSheet costumeItemSheet, SummonSheet.Row summonRow, int summonCount,
IRandom random, BoostScheduleSheet.Row guaranteeBoost = null)
```

Parameters

`addressesHex` string

Addresses hex for error reporting

`costumeItemSheet` CostumeItemSheet

Costume item sheet containing costume information

`summonRow` SummonSheet.Row

Summon configuration row with recipes and guarantee settings

`summonCount` int

Number of costumes to summon (before 10+1 bonus)

`random` IRandom

Random number generator

`guaranteeBoost` BoostScheduleSheet.Row

The BoostScheduleSheet row adjusting this group's guarantee count, or `null` for none. A caller replaying a summon must pass what the action resolved.

Returns

IEnumerable<Costume>

Collection of summoned costumes

Class CreateAvatar

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

Hard forked at https://github.com/planetarium/lib9c/pull/2195

```
[Serializable]
[ActionType("create_avatar11")]
public class CreateAvatar : GameAction, IAction, ICreateAvatarV2
```

Inheritance

object ← ActionBase ← GameAction ← CreateAvatar

Implements

IAction, ICreateAvatarV2

Inherited Members

GameAction.Id , GameAction.PlainValue , GameAction.LoadPlainValue(IValue) , ActionBase.GetSignerAndOtherAddressesHex(IActionContext, params Address[]) , ActionBase.TryGetAdminState(IActionContext, out AdminState) , ActionBase.CheckPermission(IActionContext) , ActionBase.CheckObsolete(long, IActionContext) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Fields

DeriveFormat

```
public const string DeriveFormat = "avatar-state-{0}"
```

Field Value

string

ear

```
public int ear
```

Field Value

int

hair

```
public int hair
```

Field Value

int

index

```
public int index
```

Field Value

int

lens

```
public int lens
```

Field Value

int

name

```
public string name
```

Field Value

string

tail

```
public int tail
```

Field Value

int

Properties

PlainValueInternal

```
protected override IImmutableDictionary<string, IValue> PlainValueInternal { get; }
```

Property Value

IImmutableDictionary<string, IValue>

Methods

AddItem(ItemSheet, CreateAvatarItemSheet, AvatarState, IRandom)

```
public static void AddItem(ItemSheet itemSheet, CreateAvatarItemSheet
createAvatarItemSheet, AvatarState avatarState, IRandom random)
```

Parameters

`itemSheet` ItemSheet

`createAvatarItemSheet` CreateAvatarItemSheet

`avatarState` AvatarState

`random` IRandom

CreateAvatarState(string, Address, IActionContext, Address)

```
public static AvatarState CreateAvatarState(string name, Address avatarAddress,
IActionContext ctx, Address rankingMapAddress)
```

Parameters

`name` string

`avatarAddress` Address

`ctx` IActionContext

`rankingMapAddress` Address

Returns

AvatarState

Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

Parameters

`context` IActionContext

Returns

IWorld

LoadPlainValueInternal(IImmutableDictionary<string, IValue>)

```
protected override void LoadPlainValueInternal(IImmutableDictionary<string,
IValue> plainValue)
```

Parameters

`plainValue` IImmutableDictionary<string, IValue>

MintAsset(CreateAvatarFavSheet, AvatarState, IWorld, IActionContext)

```
public static IWorld MintAsset(CreateAvatarFavSheet favSheet, AvatarState
avatarState, IWorld states, IActionContext context)
```

Parameters

`favSheet` CreateAvatarFavSheet

`avatarState` AvatarState

`states` IWorld

`context` IActionContext

Returns

IWorld

Class CreatePendingActivation

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

Introduced at Initial commit(2e645be18a4e2caea031c347f00777fbad5dbcc6) Updated at https://github.com/planetarium/lib9c/pull/957

```
[Serializable]
[ActionType("create_pending_activation")]
[ActionObsolete(7200000)]
public class CreatePendingActivation : ActionBase,
IAction, ICreatePendingActivationV1
```

Inheritance

object ← ActionBase ← CreatePendingActivation

Implements

IAction, ICreatePendingActivationV1

Inherited Members

ActionBase.GetSignerAndOtherAddressesHex(IActionContext, params Address[]) , ActionBase.TryGetAdminState(IActionContext, out AdminState) , ActionBase.CheckPermission(IActionContext) , ActionBase.CheckObsolete(long, IActionContext) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

CreatePendingActivation()

```
public CreatePendingActivation()
```

CreatePendingActivation(PendingActivationState)

```
public CreatePendingActivation(PendingActivationState activationKey)
```

Parameters

`activationKey` PendingActivationState

Properties

PendingActivation

```
public PendingActivationState PendingActivation { get; }
```

Property Value

PendingActivationState

PlainValue

```
public override IValue PlainValue { get; }
```

Property Value

IValue

Methods

Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

Parameters

`context` IActionContext

Returns

IWorld

LoadPlainValue(IValue)

```
public override void LoadPlainValue(IValue plainValue)
```

Parameters

`plainValue` IValue

Class CreatePendingActivations

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

Updated at https://github.com/planetarium/lib9c/pull/746 Updated at https://github.com/planetarium/lib9c/pull/957

```
[Serializable]
[ActionType("create_pending_activations")]
[ActionObsolete(7200000)]
public class CreatePendingActivations : ActionBase,
IAction, ICreatePendingActivationsV1
```

Inheritance

object ← ActionBase ← CreatePendingActivations

Implements

IAction, ICreatePendingActivationsV1

Inherited Members

ActionBase.GetSignerAndOtherAddressesHex(IActionContext, params Address[]) , ActionBase.TryGetAdminState(IActionContext, out AdminState) , ActionBase.CheckPermission(IActionContext) , ActionBase.CheckObsolete(long, IActionContext) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

CreatePendingActivations()

```
public CreatePendingActivations()
```

CreatePendingActivations(IEnumerable<PendingActivationState>)

```
public CreatePendingActivations(IEnumerable<PendingActivationState> states)
```

Parameters

`states` IEnumerable<PendingActivationState>

Properties

PendingActivations

```
public IList<(byte[] Address, byte[] Nonce, byte[] PublicKey)> PendingActivations {
get; }
```

Property Value

IList<(byte[] Address, byte[] Nonce, byte[] PublicKey)>

PlainValue

```
public override IValue PlainValue { get; }
```

Property Value

IValue

Methods

Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

Parameters

`context` IActionContext

Returns

IWorld

LoadPlainValue(IValue)

```
public override void LoadPlainValue(IValue plainValue)
```

Parameters

`plainValue` IValue

Class CreatePledge

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

Admin action for pledge contract

```
[ActionType("create_pledge")]
public class CreatePledge : ActionBase, IAction
```

Inheritance

object ← ActionBase ← CreatePledge

Implements

IAction

Inherited Members

ActionBase.GetSignerAndOtherAddressesHex(IActionContext, params Address[]) , ActionBase.TryGetAdminState(IActionContext, out AdminState) , ActionBase.CheckPermission(IActionContext) , ActionBase.CheckObsolete(long, IActionContext) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Fields

AgentAddresses

```
public IEnumerable<(Address, Address)> AgentAddresses
```

Field Value

IEnumerable<(Address, Address)>

Mead

```
public int Mead
```

Field Value

int

PatronAddress

```
public Address PatronAddress
```

Field Value

Address

TypeIdentifier

```
public const string TypeIdentifier = "create_pledge"
```

Field Value

string

Properties

PlainValue

```
public override IValue PlainValue { get; }
```

Property Value

IValue

Methods

Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

Parameters

`context` IActionContext

Returns

IWorld

LoadPlainValue(IValue)

```
public override void LoadPlainValue(IValue plainValue)
```

Parameters

`plainValue` IValue

Class DailyReward

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

Hard forked at https://github.com/planetarium/lib9c/pull/1828 Updated at https://github.com/planetarium/lib9c/pull/1828

```
[Serializable]
[ActionType("daily_reward7")]
public class DailyReward : GameAction, IAction, IDailyRewardV1
```

Inheritance

object ← ActionBase ← GameAction ← DailyReward

Implements

IAction, IDailyRewardV1

Inherited Members

GameAction.Id , GameAction.PlainValue , GameAction.LoadPlainValue(IValue) , ActionBase.GetSignerAndOtherAddressesHex(IActionContext, params Address[]) , ActionBase.TryGetAdminState(IActionContext, out AdminState) , ActionBase.CheckPermission(IActionContext) , ActionBase.CheckObsolete(long, IActionContext) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Fields

ActionPointMax

```
public const int ActionPointMax = 120
```

Field Value

int

AvatarAddressKey

```
public const string AvatarAddressKey = "a"
```

Field Value

string

DailyRewardInterval

```
public const long DailyRewardInterval = 2550
```

Field Value

long

DailyRuneRewardAmount

```
public const int DailyRuneRewardAmount = 1
```

Field Value

int

avatarAddress

```
public Address avatarAddress
```

Field Value

Address

Properties

PlainValueInternal

```
protected override IImmutableDictionary<string, IValue> PlainValueInternal { get; }
```

Property Value

IImmutableDictionary<string, IValue>

Methods

Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

Parameters

`context` IActionContext

Returns

IWorld

LoadPlainValueInternal(IImmutableDictionary<string, IValue>)

```
protected override void LoadPlainValueInternal(IImmutableDictionary<string,
IValue> plainValue)
```

Parameters

`plainValue` IImmutableDictionary<string, IValue>

Class DailyReward2

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
[ActionObsolete(7000000)]
[ActionType("daily_reward2")]
public class DailyReward2 : GameAction, IAction, IDailyRewardV1
```

Inheritance

object ← ActionBase ← GameAction ← DailyReward2

Implements

IAction, IDailyRewardV1

Inherited Members

GameAction.Id , GameAction.PlainValue , GameAction.LoadPlainValue(IValue) , ActionBase.GetSignerAndOtherAddressesHex(IActionContext, params Address[]) , ActionBase.TryGetAdminState(IActionContext, out AdminState) , ActionBase.CheckPermission(IActionContext) , ActionBase.CheckObsolete(long, IActionContext) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Fields

avatarAddress

```
public Address avatarAddress
```

Field Value

Address

dailyRewardResult

```
public DailyReward2.DailyRewardResult dailyRewardResult
```

Field Value

DailyReward2.DailyRewardResult

Properties

PlainValueInternal

```
protected override IImmutableDictionary<string, IValue> PlainValueInternal { get; }
```

Property Value

IImmutableDictionary<string, IValue>

Methods

Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

Parameters

`context` IActionContext

Returns

IWorld

LoadPlainValueInternal(IImmutableDictionary<string, IValue>)

```
protected override void LoadPlainValueInternal(IImmutableDictionary<string,
```

```
IValue> plainValue)
```

Parameters

`plainValue` IImmutableDictionary<string, IValue>

Class DailyReward2.DailyRewardResult

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
public class DailyReward2.DailyRewardResult : AttachmentActionResult, IState
```

Inheritance

object ← AttachmentActionResult ← DailyReward2.DailyRewardResult

Implements

IState

Inherited Members

AttachmentActionResult.itemUsable , AttachmentActionResult.costume , AttachmentActionResult.tradableFungibleItem , AttachmentActionResult.tradableFungibleItemCount , AttachmentActionResult.SerializeBackup1() , AttachmentActionResult.Deserialize(Dictionary) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

DailyRewardResult()

```
public DailyRewardResult()
```

DailyRewardResult(Dictionary)

```
public DailyRewardResult(Dictionary serialized)
```

Parameters

`serialized` Dictionary

Fields

id

```
public Guid id
```

Field Value

Guid

materials

```
public Dictionary<Material, int> materials
```

Field Value

Dictionary<Material, int>

Properties

TypeId

```
protected override string TypeId { get; }
```

Property Value

string

Methods

Serialize()

```
public override IValue Serialize()
```

Returns

IValue

Class DuplicateCostumeException

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
public class DuplicateCostumeException : InvalidOperationException, ISerializable
```

Inheritance

object ← Exception ← SystemException ← InvalidOperationException ← DuplicateCostumeException

Implements

ISerializable

Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Constructors

DuplicateCostumeException(SerializationInfo, StreamingContext)

```
protected DuplicateCostumeException(SerializationInfo info,
StreamingContext context)
```

Parameters

`info` SerializationInfo

`context` StreamingContext

DuplicateCostumeException(string)

```
public DuplicateCostumeException(string s)
```

Parameters

s string

Class DuplicateEquipmentException

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
public class DuplicateEquipmentException : InvalidOperationException, ISerializable
```

Inheritance

object ← Exception ← SystemException ← InvalidOperationException ← DuplicateEquipmentException

Implements

ISerializable

Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Constructors

DuplicateEquipmentException(SerializationInfo, StreamingContext)

```
protected DuplicateEquipmentException(SerializationInfo info,
StreamingContext context)
```

Parameters

`info` SerializationInfo

`context` StreamingContext

DuplicateEquipmentException(string)

```
public DuplicateEquipmentException(string s)
```

Parameters

s string

Class DuplicateMaterialException

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
public class DuplicateMaterialException : InvalidOperationException, ISerializable
```

Inheritance

object ← Exception ← SystemException ← InvalidOperationException ← DuplicateMaterialException

Implements

ISerializable

Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Constructors

DuplicateMaterialException(SerializationInfo, StreamingContext)

```
protected DuplicateMaterialException(SerializationInfo info,
StreamingContext context)
```

Parameters

`info` SerializationInfo

`context` StreamingContext

DuplicateMaterialException(string)

```
public DuplicateMaterialException(string s)
```

Parameters

s string

Class DuplicateOrderIdException

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
public class DuplicateOrderIdException : InvalidOperationException, ISerializable
```

Inheritance

object ← Exception ← SystemException ← InvalidOperationException ← DuplicateOrderIdException

Implements

ISerializable

Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Constructors

DuplicateOrderIdException(SerializationInfo, StreamingContext)

```
protected DuplicateOrderIdException(SerializationInfo info,
StreamingContext context)
```

Parameters

`info` SerializationInfo

`context` StreamingContext

DuplicateOrderIdException(string)

```
public DuplicateOrderIdException(string s)
```

Parameters

s [string]

Class EndPledge

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[ActionType("end_pledge")]
public class EndPledge : ActionBase, IAction
```

Inheritance

object ← ActionBase ← EndPledge

Implements

IAction

Inherited Members

ActionBase.GetSignerAndOtherAddressesHex(IActionContext, params Address[]) ,
ActionBase.TryGetAdminState(IActionContext, out AdminState) ,
ActionBase.CheckPermission(IActionContext) ,
ActionBase.CheckObsolete(long, IActionContext) , object.Equals(object) ,
object.Equals(object, object) , object.GetHashCode() , object.GetType() ,
object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

EndPledge()

```
public EndPledge()
```

Fields

AgentAddress

```
public Address AgentAddress
```

Field Value

Address

TypeIdentifier

```
public const string TypeIdentifier = "end_pledge"
```

Field Value

string

Properties

PlainValue

```
public override IValue PlainValue { get; }
```

Property Value

IValue

Methods

Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

Parameters

`context` IActionContext

Returns

IWorld

LoadPlainValue(IValue)

```
public override void LoadPlainValue(IValue plainValue)
```

Parameters

`plainValue` IValue

Class EquipmentLevelExceededException

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
public class EquipmentLevelExceededException : InvalidOperationException,
ISerializable
```

Inheritance

object ← Exception ← SystemException ← InvalidOperationException ← EquipmentLevelExceededException

Implements

ISerializable

Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Constructors

EquipmentLevelExceededException(SerializationInfo, StreamingContext)

```
protected EquipmentLevelExceededException(SerializationInfo info,
StreamingContext context)
```

Parameters

`info` SerializationInfo

`context` StreamingContext

EquipmentLevelExceededException(string)

```
public EquipmentLevelExceededException(string s)
```

Parameters

s string

Class EquipmentSlotUnlockException

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
public class EquipmentSlotUnlockException : Exception, ISerializable
```

Inheritance

object ← Exception ← EquipmentSlotUnlockException

Implements

ISerializable

Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Constructors

EquipmentSlotUnlockException()

```
public EquipmentSlotUnlockException()
```

EquipmentSlotUnlockException(SerializationInfo, StreamingContext)

```
protected EquipmentSlotUnlockException(SerializationInfo info,
StreamingContext context)
```

Parameters

`info` SerializationInfo

`context` StreamingContext

EquipmentSlotUnlockException(string)

```
public EquipmentSlotUnlockException(string msg)
```

Parameters

`msg` string

Class EventConsumableItemCrafts

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

Hard forked at https://github.com/planetarium/lib9c/pull/2195

```
[Serializable]
[ActionType("event_consumable_item_crafts2")]
public class EventConsumableItemCrafts : GameAction,
IAction, IEventConsumableItemCraftsV1
```

Inheritance

object ← ActionBase ← GameAction ← EventConsumableItemCrafts

Implements

IAction, IEventConsumableItemCraftsV1

Inherited Members

GameAction.Id , GameAction.PlainValue , GameAction.LoadPlainValue(IValue) , ActionBase.GetSignerAndOtherAddressesHex(IActionContext, params Address[]) , ActionBase.TryGetAdminState(IActionContext, out AdminState) , ActionBase.CheckPermission(IActionContext) , ActionBase.CheckObsolete(long, IActionContext) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Fields

AvatarAddress

```
public Address AvatarAddress
```

Field Value

Address

EventConsumableItemRecipeId

```
public int EventConsumableItemRecipeId
```

Field Value

int

EventScheduleId

```
public int EventScheduleId
```

Field Value

int

SlotIndex

```
public int SlotIndex
```

Field Value

int

Properties

PlainValueInternal

```
protected override IImmutableDictionary<string, IValue> PlainValueInternal { get; }
```

Property Value

IImmutableDictionary<string, IValue>

Methods

Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

Parameters

`context` IActionContext

Returns

IWorld

LoadPlainValueInternal(IImmutableDictionary<string, IValue>)

```
protected override void LoadPlainValueInternal(IImmutableDictionary<string,
IValue> plainValue)
```

Parameters

`plainValue` IImmutableDictionary<string, IValue>

Class EventDungeonBattle

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

Action for participating in an event dungeon battle. This action allows an avatar to battle in a specific stage of an event dungeon, consuming tickets and receiving rewards upon victory. Hard forked at https://github.com/planetarium/lib9c/pull/2195

```
[Serializable]
[ActionType("event_dungeon_battle6")]
public class EventDungeonBattle : GameAction, IAction, IEventDungeonBattleV2
```

Inheritance

object ← ActionBase ← GameAction ← EventDungeonBattle

Implements

IAction, IEventDungeonBattleV2

Inherited Members

GameAction.Id , GameAction.PlainValue , GameAction.LoadPlainValue(IValue) , ActionBase.GetSignerAndOtherAddressesHex(IActionContext, params Address[]) , ActionBase.TryGetAdminState(IActionContext, out AdminState) , ActionBase.CheckPermission(IActionContext) , ActionBase.CheckObsolete(long, IActionContext) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Fields

AvatarAddress

The address of the avatar that will participate in the event dungeon battle.

```
public Address AvatarAddress
```

Field Value

Address

BuyTicketIfNeeded

Whether to automatically purchase a ticket if the avatar doesn't have enough tickets. If true and tickets are insufficient, the action will attempt to buy a ticket using currency.

```
public bool BuyTicketIfNeeded
```

Field Value

bool

Costumes

A list of costume item GUIDs to be equipped for the battle.

```
public List<Guid> Costumes
```

Field Value

List<Guid>

Equipments

A list of equipment item GUIDs to be equipped for the battle.

```
public List<Guid> Equipments
```

Field Value

List<Guid>

EventDungeonId

The ID of the event dungeon to battle in.

```
public int EventDungeonId
```

Field Value

int

EventDungeonStageId

The ID of the specific stage within the event dungeon to battle.

```
public int EventDungeonStageId
```

Field Value

int

EventScheduleId

The ID of the event schedule that defines when the event dungeon is available.

```
public int EventScheduleId
```

Field Value

int

Foods

A list of food item GUIDs to be consumed during the battle.

```
public List<Guid> Foods
```

Field Value

List<Guid>

PlayCount

The number of battles to play in a single action execution. Currently fixed to 1 for this action type.

```
public const int PlayCount = 1
```

Field Value

int

RuneInfos

A list of rune slot information specifying which runes to use in the battle.

```
public List<RuneSlotInfo> RuneInfos
```

Field Value

List<RuneSlotInfo>

Properties

PlainValueInternal

Serializes the action's data into a Bencodex dictionary. The data is stored as a list under the key "l" containing: AvatarAddress, EventScheduleId, EventDungeonId, EventDungeonStageId, Equipments, Costumes, Foods, BuyTicketIfNeeded, and RuneInfos.

```
protected override IImmutableDictionary<string, IValue> PlainValueInternal { get; }
```

Property Value

IImmutableDictionary<string, IValue>

An immutable dictionary containing the serialized action data.

Methods

Execute(IActionContext)

Executes the event dungeon battle action. This method performs the following operations:

1. Validates the avatar state and event dungeon information
2. Checks and consumes event dungeon tickets (or purchases if needed)
3. Validates equipment, costumes, foods, and runes
4. Updates rune and item slot states
5. Simulates the battle using StageSimulator
6. Applies battle results to the avatar state (experience, rewards, etc.)
7. Updates the event dungeon info if the stage is cleared

```
public override IWorld Execute(IActionContext context)
```

Parameters

`context` IActionContext

The action context containing block index, signer, and random seed.

Returns

IWorld

The updated world state after executing the battle.

Exceptions

FailedLoadStateException

Thrown when the avatar state cannot be loaded.

NotEnoughEventDungeonTicketsException

Thrown when the avatar doesn't have enough tickets and BuyTicketIfNeeded is false.

StageNotClearedException

Thrown when attempting to battle a stage without clearing the previous stage.

SheetRowNotFoundException

Thrown when required sheet data cannot be found.

LoadPlainValueInternal(IImmutableDictionary<string, IValue>)

Deserializes the action's data from a Bencodex dictionary. Expects a list under the key "l" containing at least 9 items in the order: AvatarAddress, EventScheduleId, EventDungeonId, EventDungeonStageId, Equipments, Costumes, Foods, BuyTicketIfNeeded, and RuneInfos.

```
protected override void LoadPlainValueInternal(IImmutableDictionary<string,
IValue> plainValue)
```

Parameters

`plainValue` IImmutableDictionary<string, IValue>

The Bencodex dictionary containing the serialized action data.

Exceptions

ArgumentException

Thrown when the plainValue doesn't contain the expected key "l", when "l" is not a Bencodex list, or when the list has fewer than 9 items.

Class EventDungeonBattleSweep

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

Represents a sweep action that automatically completes multiple event dungeon battles without player interaction.

This action allows players to automatically repeat battles on already cleared event dungeon stages, consuming tickets and receiving rewards without manual combat interaction.

Key Features: - Only works on previously cleared stages (no stage clearing) - No ticket purchase functionality (uses existing tickets only) - Automatic reward and experience calculation - Multiple battle execution in a single action

This action modifies the following states: - AvatarState: Updates player's stats, inventory, and experience - EventDungeonInfo: Updates event dungeon progress and ticket usage - RuneState: Updates rune effects and bonuses - ItemSlotState: Updates equipped items - RuneSlotState: Updates equipped runes - CP: Updates character's combat power

Execution Flow: 1. Validate input parameters and avatar state 2. Load and validate game sheets and configurations 3. Validate event dungeon info and ticket availability 4. Check stage clearance requirements 5. Update rune and item slot states 6. Calculate combat power (CP) 7. Generate rewards and experience 8. Update avatar state and return modified world state

Hard forked at https://github.com/planetarium/lib9c/pull/2195

```
[Serializable]
[ActionType("event_dungeon_battle_sweep")]
public class EventDungeonBattleSweep : GameAction, IAction, IEventDungeonBattleSweep
```

Inheritance

object ← ActionBase ← GameAction ← EventDungeonBattleSweep

Implements

IAction, IEventDungeonBattleSweep

Inherited Members

GameAction.Id , GameAction.PlainValue , GameAction.LoadPlainValue(IValue) , ActionBase.GetSignerAndOtherAddressesHex(IActionContext, params Address[]) ,

ActionBase.TryGetAdminState(IActionContext, out AdminState) ,
ActionBase.CheckPermission(IActionContext) ,
ActionBase.CheckObsolete(long, IActionContext) , object.Equals(object) ,
object.Equals(object, object) , object.GetHashCode() , object.GetType() ,
object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Fields

AvatarAddress

The address of the avatar performing the sweep action.

```
public Address AvatarAddress
```

Field Value

Address

Costumes

List of costume item IDs to use during the sweep.

```
public List<Guid> Costumes
```

Field Value

List<Guid>

Equipments

List of equipment item IDs to use during the sweep.

```
public List<Guid> Equipments
```

Field Value

List<Guid>

EventDungeonId

The ID of the event dungeon to sweep.

```
public int EventDungeonId
```

Field Value

int

EventDungeonStageId

The ID of the event dungeon stage to sweep.

```
public int EventDungeonStageId
```

Field Value

int

EventScheduleId

The ID of the event schedule for the dungeon.

```
public int EventScheduleId
```

Field Value

int

Foods

List of food item IDs to use during the sweep.

```
public List<Guid> Foods
```

Field Value

List<Guid>

PlayCount

The number of times to perform the sweep (must be between 1 and MaxPlayCount).

```
public int PlayCount
```

Field Value

int

RuneInfos

List of rune slot information for equipped runes.

```
public List<RuneSlotInfo> RuneInfos
```

Field Value

List<RuneSlotInfo>

Properties

PlainValueInternal

```
protected override IImmutableDictionary<string, IValue> PlainValueInternal { get; }
```

Property Value

IImmutableDictionary<string, IValue>

Methods

Execute(IActionContext)

Executes the event dungeon battle sweep action.

This method performs the following steps: 1. Validates input parameters and loads avatar state 2. Loads and validates all required game sheets 3. Validates event dungeon configuration and ticket availability 4. Checks stage clearance requirements 5. Updates rune and item slot states 6. Calculates combat power (CP) 7. Generates rewards and experience 8. Updates all relevant states and returns the modified world

```
public override IWorld Execute(IActionContext context)
```

Parameters

`context` IActionContext

Action context containing previous state, signer, and block information

Returns

IWorld

Modified world state with updated avatar, dungeon info, and other states

GetRewardItems(IRandom, int, Row, MaterialItemSheet)

Gets reward items for event dungeon sweep based on play count and stage configuration.

This method simulates multiple battle rewards without actually performing combat. It generates items based on the stage's drop configuration and the number of plays requested.

Process: 1. Determine the number of items to drop per play (random between min and max) 2. For each play, generate rewards using the stage's item selector 3. Collect all rewards and sort them by item ID for consistency

```
public static List<ItemBase> GetRewardItems(IRandom random, int playCount,
EventDungeonStageSheet.Row stageRow, MaterialItemSheet materialItemSheet)
```

Parameters

`random` IRandom

Random number generator for deterministic reward generation

`playCount` int

Number of plays to simulate (determines total reward count)

`stageRow` EventDungeonStageSheet.Row

Event dungeon stage configuration containing drop rates and item pools

`materialItemSheet` MaterialItemSheet

Material item sheet for reward generation and validation

Returns

List<ItemBase>

List of reward items sorted by item ID

LoadPlainValueInternal(IImmutableDictionary<string, IValue>)

```
protected override void LoadPlainValueInternal(IImmutableDictionary<string,
IValue> plainValue)
```

Parameters

`plainValue` IImmutableDictionary<string, IValue>

Class EventMaterialItemCrafts

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

Hard forked at https://github.com/planetarium/lib9c/pull/2195

```
[Serializable]
[ActionType("event_material_item_crafts2")]
public class EventMaterialItemCrafts : GameAction,
IAction, IEventMaterialItemCraftsV1
```

Inheritance

object ← ActionBase ← GameAction ← EventMaterialItemCrafts

Implements

IAction, IEventMaterialItemCraftsV1

Inherited Members

GameAction.Id , GameAction.PlainValue , GameAction.LoadPlainValue(IValue) , ActionBase.GetSignerAndOtherAddressesHex(IActionContext, params Address[]) , ActionBase.TryGetAdminState(IActionContext, out AdminState) , ActionBase.CheckPermission(IActionContext) , ActionBase.CheckObsolete(long, IActionContext) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Fields

AvatarAddress

```
public Address AvatarAddress
```

Field Value

Address

EventMaterialItemRecipeId

```
public int EventMaterialItemRecipeId
```

Field Value

int

EventScheduleId

```
public int EventScheduleId
```

Field Value

int

MaterialsToUse

```
public Dictionary<int, int> MaterialsToUse
```

Field Value

Dictionary<int, int>

Properties

PlainValueInternal

```
protected override IImmutableDictionary<string, IValue> PlainValueInternal { get; }
```

Property Value

IImmutableDictionary<string, IValue>

Methods

Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

Parameters

`context` IActionContext

Returns

IWorld

LoadPlainValueInternal(IImmutableDictionary<string, IValue>)

```
protected override void LoadPlainValueInternal(IImmutableDictionary<string,
IValue> plainValue)
```

Parameters

`plainValue` IImmutableDictionary<string, IValue>

Class FailedLoadSheetException

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
public class FailedLoadSheetException : Exception, ISerializable
```

Inheritance

object ← Exception ← FailedLoadSheetException

Implements

ISerializable

Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Constructors

FailedLoadSheetException(SerializationInfo, StreamingContext)

```
protected FailedLoadSheetException(SerializationInfo info, StreamingContext context)
```

Parameters

`info` SerializationInfo

`context` StreamingContext

FailedLoadSheetException(string)

```
public FailedLoadSheetException(string message)
```

Parameters

`message` string

FailedLoadSheetException(Type)

```
public FailedLoadSheetException(Type sheetType)
```

Parameters

`sheetType` Type

Class FailedLoadStateException

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
public class FailedLoadStateException : Exception, ISerializable
```

Inheritance

object ← Exception ← FailedLoadStateException

Implements

ISerializable

Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Constructors

FailedLoadStateException(Address, Type, Exception)

```
public FailedLoadStateException(Address address, Type stateType, Exception
innerException = null)
```

Parameters

`address` Address

`stateType` Type

`innerException` Exception

FailedLoadStateException(SerializationInfo, StreamingContext)

```
protected FailedLoadStateException(SerializationInfo info, StreamingContext context)
```

Parameters

`info` SerializationInfo

`context` StreamingContext

FailedLoadStateException(string)

```
public FailedLoadStateException(string message)
```

Parameters

`message` string

FailedLoadStateException(string, Exception)

```
public FailedLoadStateException(string message, Exception innerException = null)
```

Parameters

`message` string

`innerException` Exception

FailedLoadStateException(string, string, string, Exception)

```
public FailedLoadStateException(string actionType, string addressesHex, string
message, Exception innerException = null)
```

Parameters

`actionType` string

`addressesHex` string

`message` string

`innerException` Exception

FailedLoadStateException(string, string, Type, Address)

```
public FailedLoadStateException(string actionType, string addressesHex, Type
stateType, Address address)
```

Parameters

`actionType` string

`addressesHex` string

`stateType` Type

`address` Address

Class FavProductInfo

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
public class FavProductInfo : IProductInfo
```

Inheritance

object ← FavProductInfo

Implements

IProductInfo

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

FavProductInfo()

```
public FavProductInfo()
```

FavProductInfo(List)

```
public FavProductInfo(List serialized)
```

Parameters

`serialized` List

Properties

AgentAddress

```
public Address AgentAddress { get; set; }
```

Property Value

Address

AvatarAddress

```
public Address AvatarAddress { get; set; }
```

Property Value

Address

Price

```
public FungibleAssetValue Price { get; set; }
```

Property Value

FungibleAssetValue

ProductId

```
public Guid ProductId { get; set; }
```

Property Value

Guid

Type

```
public ProductType Type { get; set; }
```

Property Value

ProductType

Methods

Serialize()

```
public IValue Serialize()
```

Returns

IValue

ValidateType()

```
public void ValidateType()
```

Class GameAction

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
public abstract class GameAction : ActionBase, IAction
```

Inheritance

object ← ActionBase ← GameAction

Implements

IAction

Derived

ActivateCollection, AddRedeemCode, ClaimAdventureBossReward, ExploreAdventureBoss, SweepAdventureBoss, UnlockFloor, Wanted, Battle, AuraSummon, BattleArena, Buy, Buy7, BuyMultiple, BuyProduct, CancelProductRegistration, ChargeActionPoint, ClaimGifts, ClaimItems, ClaimRaidReward, ClaimStakeReward, ClaimWordBossKillReward, ClaimWorldBossReward, CombinationConsumable, CombinationConsumable5, CombinationEquipment, CostumeSummon, IssueCoupons, RedeemCoupon, TransferCoupons, CreateAvatar, DailyReward, DailyReward2, EventConsumableItemCrafts, EventDungeonBattle, EventDungeonBattleSweep, EventMaterialItemCrafts, BulkUnloadFromGarages, DeliverToOthersGarages, LoadIntoMyGarages, UnloadFromMyGarages, GrantItems, Grinding, HackAndSlash, HackAndSlashRandomBuff, HackAndSlashSweep, InfiniteTowerBattle, InitializeStates, ItemEnhancement, ItemEnhancement10, ItemEnhancement11, ItemEnhancement12, ItemEnhancement13, ItemEnhancement7, ItemEnhancement9, JoinArena, JoinArena1, JoinArena3, MigrationActivatedAccountsState, PatchTableSheet, PatchTableSheetCompressed, PetEnhancement, Raid, RankingBattle, RankingBattle11, RapidCombination, RapidCombination0, RapidCombination5, ReRegisterProduct, RedeemCode, RedeemCode2, RegisterProduct, RegisterProduct0, RenewAdminState, RuneEnhancement, RuneSummon, Sell, Sell6, SellCancellation, Stake, Stake2, Synthesize, UnlockCombinationSlot, UnlockEquipmentRecipe, UnlockRuneSlot, UnlockWorld, UpdateSell

Inherited Members

ActionBase.Execute(IActionContext) ,
ActionBase.GetSignerAndOtherAddressesHex(IActionContext, params Address[]) ,
ActionBase.TryGetAdminState(IActionContext, out AdminState) ,
ActionBase.CheckPermission(IActionContext) ,

ActionBase.CheckObsolete(long, IActionContext) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

GameAction()

```
protected GameAction()
```

Properties

Id

```
public Guid Id { get; }
```

Property Value

Guid

PlainValue

```
public override sealed IValue PlainValue { get; }
```

Property Value

IValue

PlainValueInternal

```
protected abstract IImmutableDictionary<string, IValue> PlainValueInternal { get; }
```

Property Value

IImmutableDictionary<string, IValue>

Methods

LoadPlainValue(IValue)

```
public override sealed void LoadPlainValue(IValue plainValue)
```

Parameters

`plainValue` IValue

LoadPlainValueInternal(IImmutableDictionary<string, IValue>)

```
protected abstract void LoadPlainValueInternal(IImmutableDictionary<string,
IValue> plainValue)
```

Parameters

`plainValue` IImmutableDictionary<string, IValue>

Class GoldDistribution

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
public class GoldDistribution : IEquatable<GoldDistribution>
```

Inheritance

object ← GoldDistribution

Implements

IEquatable<GoldDistribution>

Inherited Members

object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

GoldDistribution()

```
public GoldDistribution()
```

GoldDistribution(Dictionary)

```
public GoldDistribution(Dictionary serialized)
```

Parameters

`serialized` Dictionary

GoldDistribution(IValue)

```
public GoldDistribution(IValue serialized)
```

Parameters

`serialized` IValue

Fields

Address

```
[Ignore]
public Address Address
```

Field Value

Address

Properties

AddressString

```
[Index(0, -1)]
public string AddressString { get; set; }
```

Property Value

string

AmountPerBlock

```
[Index(1, -1)]
public BigInteger AmountPerBlock { get; set; }
```

Property Value

BigInteger

EndBlock

```
[Index(3, -1)]
public long EndBlock { get; set; }
```

Property Value

long

StartBlock

```
[Index(2, -1)]
public long StartBlock { get; set; }
```

Property Value

long

Methods

Equals(GoldDistribution)

Indicates whether the current object is equal to another object of the same type.

```
public bool Equals(GoldDistribution other)
```

Parameters

`other` GoldDistribution

An object to compare with this object.

Returns

bool

true if the current object is equal to the `other` parameter; otherwise, false.

Equals(object)

Determines whether the specified object is equal to the current object.

```
public override bool Equals(object obj)
```

Parameters

`obj` object

The object to compare with the current object.

Returns

bool

true if the specified object is equal to the current object; otherwise, false.

GetAmount(long)

```
public BigInteger GetAmount(long blockIndex)
```

Parameters

`blockIndex` long

Returns

BigInteger

GetHashCode()

Serves as the default hash function.

```
public override int GetHashCode()
```

Returns

int

A hash code for the current object.

LoadInDescendingEndBlockOrder(string)

```
public static GoldDistribution[] LoadInDescendingEndBlockOrder(string csvPath)
```

Parameters

`csvPath` string

Returns

GoldDistribution[]

Serialize()

```
public Dictionary Serialize()
```

Returns

Dictionary

Class GrantItems

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

Grants items/assets to target avatars from token-like inputs. This action is restricted to a fixed allow-list of signers and the current admin signer. Even if the signer does not have sufficient token balance, the grant will proceed and will burn only the available amount of the token.

```
[ActionType("grant_items")]
public class GrantItems : GameAction, IAction, IClaimItems
```

Inheritance

object ← ActionBase ← GameAction ← GrantItems

Implements

IAction, IClaimItems

Inherited Members

GameAction.Id , GameAction.PlainValue , GameAction.LoadPlainValue(IValue) , ActionBase.GetSignerAndOtherAddressesHex(IActionContext, params Address[]) , ActionBase.TryGetAdminState(IActionContext, out AdminState) , ActionBase.CheckPermission(IActionContext) , ActionBase.CheckObsolete(long, IActionContext) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

GrantItems()

Creates an empty action instance.

```
public GrantItems()
```

GrantItems(IReadOnlyList<(Address, IReadOnlyList<FungibleAssetValue>)>, string)

Creates an action instance with the given claim data.

```
public GrantItems(IReadOnlyList<(Address, IReadOnlyList<FungibleAssetValue>)>
claimData, string memo = null)
```

Parameters

`claimData` IReadOnlyList<(Address address, IReadOnlyList<FungibleAssetValue> fungibleAssetValues)>

Grant targets and token-like inputs.

`memo` string

Optional memo written to the resulting mail.

Fields

Memo

Optional memo written to the resulting mail.

```
public string Memo
```

Field Value

string

Properties

ClaimData

```
public IReadOnlyList<(Address address, IReadOnlyList<FungibleAssetValue>
fungibleAssetValues)> ClaimData { get; }
```

Property Value

IReadOnlyList<(Address address, IReadOnlyList<FungibleAssetValue> fungibleAssetValues)>

PlainValueInternal

```
protected override IImmutableDictionary<string, IValue> PlainValueInternal { get; }
```

Property Value

IImmutableDictionary<string, IValue>

Methods

Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

Parameters

`context` IActionContext

Returns

IWorld

LoadPlainValueInternal(IImmutableDictionary<string, IValue>)

```
protected override void LoadPlainValueInternal(IImmutableDictionary<string,
IValue> plainValue)
```

Parameters

`plainValue` IImmutableDictionary<string, IValue>

Class Grinding

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[ActionType("grinding2")]
public class Grinding : GameAction, IAction, IGrindingV1
```

Inheritance

object ← ActionBase ← GameAction ← Grinding

Implements

IAction, IGrindingV1

Inherited Members

GameAction.Id , GameAction.PlainValue , GameAction.LoadPlainValue(IValue) , ActionBase.GetSignerAndOtherAddressesHex(IActionContext, params Address[]) , ActionBase.TryGetAdminState(IActionContext, out AdminState) , ActionBase.CheckPermission(IActionContext) , ActionBase.CheckObsolete(long, IActionContext) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Fields

AvatarAddress

```
public Address AvatarAddress
```

Field Value

Address

ChargeAp

```
public bool ChargeAp
```

Field Value

bool

CostAp

```
public const int CostAp = 5
```

Field Value

int

EquipmentIds

```
public List<Guid> EquipmentIds
```

Field Value

List<Guid>

Limit

```
public const int Limit = 50
```

Field Value

int

Properties

PlainValueInternal

```
protected override IImmutableDictionary<string, IValue> PlainValueInternal { get; }
```

Property Value

IImmutableDictionary<string, IValue>

Methods

CalculateMaterialReward(IEnumerable<Equipment>, CrystalEquipmentGrindingSheet, MaterialItemSheet)

```
public static Dictionary<Material, int>
CalculateMaterialReward(IEnumerable<Equipment> equipmentList,
CrystalEquipmentGrindingSheet crystalEquipmentGrindingSheet,
MaterialItemSheet materialItemSheet)
```

Parameters

`equipmentList` IEnumerable<Equipment>

`crystalEquipmentGrindingSheet` CrystalEquipmentGrindingSheet

`materialItemSheet` MaterialItemSheet

Returns

Dictionary<Material, int>

Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

Parameters

`context` IActionContext

Returns

IWorld

LoadPlainValueInternal(IImmutableDictionary<string, IValue>)

```
protected override void LoadPlainValueInternal(IImmutableDictionary<string,
IValue> plainValue)
```

Parameters

`plainValue` IImmutableDictionary<string, IValue>

Class HackAndSlash

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

Represents a battle action in the game where a player engages in combat with monsters. This action modifies the following states:

- AvatarState: Updates player's stats, inventory, and experience
- WorldBossKillRewardRecord: Records boss kill rewards
- CollectionState: Updates collection progress
- RuneState: Updates rune effects and bonuses
- ItemSlotState: Updates equipped items
- RuneSlotState: Updates equipped runes
- CP: Updates character's combat power Hard forked at https://github.com/planetarium/lib9c/pull/2195

```
[Serializable]
[ActionType("hack_and_slash22")]
public class HackAndSlash : GameAction, IAction, IHackAndSlashV10
```

Inheritance

object ← ActionBase ← GameAction ← HackAndSlash

Implements

IAction, IHackAndSlashV10

Inherited Members

GameAction.Id , GameAction.PlainValue , GameAction.LoadPlainValue(IValue) , ActionBase.GetSignerAndOtherAddressesHex(IActionContext, params Address[]) , ActionBase.TryGetAdminState(IActionContext, out AdminState) , ActionBase.CheckPermission(IActionContext) , ActionBase.CheckObsolete(long, IActionContext) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Fields

ApStoneCount

```
public int ApStoneCount
```

Field Value

int

AvatarAddress

```
public Address AvatarAddress
```

Field Value

Address

Costumes

```
public List<Guid> Costumes
```

Field Value

List<Guid>

EntryCostItemCount

Gets or sets the declared total entry cost material item count for this action execution, matching EntryCostItemCount × TotalPlayCount. Required when the stage has an entry material cost; must match the computed value exactly.

```
public int EntryCostItemCount
```

Field Value

int

EntryCostItemId

Gets or sets the declared entry cost material item ID, matching EntryCostItemId. Required when the stage has an entry material cost; must match the sheet value exactly.

```
public int EntryCostItemId
```

Field Value

int

Equipments

```
public List<Guid> Equipments
```

Field Value

List<Guid>

Foods

```
public List<Guid> Foods
```

Field Value

List<Guid>

RuneInfos

```
public List<RuneSlotInfo> RuneInfos
```

Field Value

List<RuneSlotInfo>

StageBuffId

```
public int? StageBuffId
```

Field Value

int?

StageId

```
public int StageId
```

Field Value

int

TotalPlayCount

```
public int TotalPlayCount
```

Field Value

int

UsableApStoneCount

```
public const int UsableApStoneCount = 10
```

Field Value

int

WorldId

```
public int WorldId
```

Field Value

int

Properties

PlainValueInternal

```
protected override IImmutableDictionary<string, IValue> PlainValueInternal { get; }
```

Property Value

IImmutableDictionary<string, IValue>

Methods

Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

Parameters

`context` IActionContext

Returns

IWorld

Execute(IWorld, Address, long, IRandom, IActionContext)

```
public IWorld Execute(IWorld states, Address signer, long blockIndex, IRandom
random, IActionContext context = null)
```

Parameters

`states` IWorld

`signer` Address

`blockIndex` long

`random` IRandom

`context` IActionContext

Returns

IWorld

LoadPlainValueInternal(IImmutableDictionary<string, IValue>)

```
protected override void LoadPlainValueInternal(IImmutableDictionary<string,
IValue> plainValue)
```

Parameters

`plainValue` IImmutableDictionary<string, IValue>

Class HackAndSlashRandomBuff

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

Created at https://github.com/planetarium/lib9c/pull/1031

```
[Serializable]
[ActionType("hack_and_slash_random_buff")]
public class HackAndSlashRandomBuff : GameAction, IAction, IHackAndSlashRandomBuffV1
```

Inheritance

object ← ActionBase ← GameAction ← HackAndSlashRandomBuff

Implements

IAction, IHackAndSlashRandomBuffV1

Inherited Members

GameAction.Id , GameAction.PlainValue , GameAction.LoadPlainValue(IValue) , ActionBase.GetSignerAndOtherAddressesHex(IActionContext, params Address[]) , ActionBase.TryGetAdminState(IActionContext, out AdminState) , ActionBase.CheckPermission(IActionContext) , ActionBase.CheckObsolete(long, IActionContext) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Fields

AdvancedGacha

```
public bool AdvancedGacha
```

Field Value

bool

AdvancedGachaCount

```
public const int AdvancedGachaCount = 10
```

Field Value

int

AvatarAddress

```
public Address AvatarAddress
```

Field Value

Address

NormalGachaCount

```
public const int NormalGachaCount = 5
```

Field Value

int

Properties

PlainValueInternal

```
protected override IImmutableDictionary<string, IValue> PlainValueInternal { get; }
```

Property Value

IImmutableDictionary<string, IValue>

Methods

Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

Parameters

`context` IActionContext

Returns

IWorld

LoadPlainValueInternal(IImmutableDictionary<string, IValue>)

```
protected override void LoadPlainValueInternal(IImmutableDictionary<string,
IValue> plainValue)
```

Parameters

`plainValue` IImmutableDictionary<string, IValue>

Class HackAndSlashSweep

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

Represents a sweep action that automatically completes multiple battles without player interaction. This action modifies the following states:

- AvatarState: Updates player's stats, inventory, and experience
- CollectionState: Updates collection progress
- RuneState: Updates rune effects and bonuses
- ItemSlotState: Updates equipped items
- RuneSlotState: Updates equipped runes
- CP: Updates character's combat power Hard forked at https://github.com/planetarium/lib9c/pull/2195

```
[Serializable]
[ActionType("hack_and_slash_sweep10")]
public class HackAndSlashSweep : GameAction, IAction, IHackAndSlashSweepV3
```

Inheritance

object ← ActionBase ← GameAction ← HackAndSlashSweep

Implements

IAction, IHackAndSlashSweepV3

Inherited Members

GameAction.Id , GameAction.PlainValue , GameAction.LoadPlainValue(IValue) , ActionBase.GetSignerAndOtherAddressesHex(IActionContext, params Address[]) , ActionBase.TryGetAdminState(IActionContext, out AdminState) , ActionBase.CheckPermission(IActionContext) , ActionBase.CheckObsolete(long, IActionContext) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Fields

UsableApStoneCount

```
public const int UsableApStoneCount = 100
```

Field Value

int

actionPoint

```
public int actionPoint
```

Field Value

int

apStoneCount

```
public int apStoneCount
```

Field Value

int

avatarAddress

```
public Address avatarAddress
```

Field Value

Address

costumes

```
public List<Guid> costumes
```

Field Value

List<Guid>

entryCostItemCount

Declared total entry cost material item count for this sweep execution, matching EntryCostItemCount × playCount. Required when the stage has an entry material cost; must match the computed value exactly.

```
public int entryCostItemCount
```

Field Value

int

entryCostItemId

Declared entry cost material item ID, matching EntryCostItemId. Required when the stage has an entry material cost; must match the sheet value exactly.

```
public int entryCostItemId
```

Field Value

int

equipments

```
public List<Guid> equipments
```

Field Value

List<Guid>

runeInfos

```
public List<RuneSlotInfo> runeInfos
```

Field Value

List<RuneSlotInfo>

stageId

```
public int stageId
```

Field Value

int

worldId

```
public int worldId
```

Field Value

int

Properties

PlainValueInternal

```
protected override IImmutableDictionary<string, IValue> PlainValueInternal { get; }
```

Property Value

IImmutableDictionary<string, IValue>

Methods

Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

Parameters

`context` IActionContext

Returns

IWorld

GetRewardItems(IRandom, int, Row, MaterialItemSheet, Row)

Draws the item rewards of `playCount` sweeps of `stageRow`, one object per unit.

```
public static List<ItemBase> GetRewardItems(IRandom random, int playCount,
StageSheet.Row stageRow, MaterialItemSheet materialItemSheet, BoostScheduleSheet.Row
itemRewardBoost = null)
```

Parameters

`random` IRandom

The random number generator.

`playCount` int

The number of clears swept.

`stageRow` StageSheet.Row

The stage being swept.

`materialItemSheet` MaterialItemSheet

The sheet the rewards are created from.

`itemRewardBoost` BoostScheduleSheet.Row

The BoostScheduleSheet row adjusting each clear's items, or `null` for none. Applied per clear exactly as `HackAndSlash` applies it; see ApplyItemRewardBoost(List<ItemBase>, Row, MaterialItemSheet) and GetBoostedItemRewardCount(Row, int). It never changes the random draws. A caller replaying a sweep must pass what the action resolved.

Returns

List<ItemBase>

The items of every clear, one object per unit, ordered by item id. Their count per item id is exactly what the action grants, although the action adds the units a boost adds with one count per item id instead of creating these objects.

GetRewardItems(IRandom, int, Row, MaterialItemSheet, Row, IDictionary<int, int>)

Draws the item rewards of `playCount` sweeps of `stageRow`, counting the units a boost adds instead of creating them.

```
public static List<ItemBase> GetRewardItems(IRandom random, int playCount,
StageSheet.Row stageRow, MaterialItemSheet materialItemSheet, BoostScheduleSheet.Row
itemRewardBoost, IDictionary<int, int> extraCounts)
```

Parameters

`random` IRandom

The random number generator.

`playCount` int

The number of clears swept.

`stageRow` StageSheet.Row

The stage being swept.

`materialItemSheet` MaterialItemSheet

The sheet the rewards are created from.

`itemRewardBoost` BoostScheduleSheet.Row

The BoostScheduleSheet row adjusting each clear's items, or `null` for none. See SplitItemRewardBoost(List<ItemBase>, Row, IDictionary<int, int>).

`extraCounts` IDictionary<int, int>

Accumulates, per item id, the units the boost adds on top of the returned items over every clear. Stays empty without a row.

Returns

List<ItemBase>

The drawn items that remain after the boost, one object per unit, ordered by item id. Without a row, exactly the items the sweep drew.

LoadPlainValueInternal(IImmutableDictionary<string, IValue>)

```
protected override void LoadPlainValueInternal(IImmutableDictionary<string,
IValue> plainValue)
```

Parameters

`plainValue` IImmutableDictionary<string, IValue>

Interface IBuy0

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

Common interface used before IBuy5.

```
public interface IBuy0 : IAction
```

Inherited Members
IAction.LoadPlainValue(IValue) , IAction.Execute(IActionContext) , IAction.PlainValue

Properties

buyerAvatarAddress

```
Address buyerAvatarAddress { get; }
```

Property Value

Address

productId

```
Guid productId { get; }
```

Property Value

Guid

sellerAgentAddress

```
Address sellerAgentAddress { get; }
```

Property Value

Address

sellerAvatarAddress

```
Address sellerAvatarAddress { get; }
```

Property Value

Address

See Also

IBuy5

Interface IBuy5

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

Common interface used after IBuy5.

```
public interface IBuy5 : IAction
```

Inherited Members
IAction.LoadPlainValue(IValue) , IAction.Execute(IActionContext) , IAction.PlainValue

Properties

buyerAvatarAddress

```
Address buyerAvatarAddress { get; }
```

Property Value

Address

purchaseInfos

```
IEnumerable<IPurchaseInfo> purchaseInfos { get; }
```

Property Value

IEnumerable<IPurchaseInfo>

See Also

IBuy5

Interface IClaimItems

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
public interface IClaimItems
```

Properties

ClaimData

```
IReadOnlyList<(Address address, IReadOnlyList<FungibleAssetValue>
fungibleAssetValues)> ClaimData { get; }
```

Property Value

IReadOnlyList<(Address address, IReadOnlyList<FungibleAssetValue> fungibleAssetValues)>

Interface IClaimStakeReward

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
public interface IClaimStakeReward : IAction
```

Inherited Members

IAction.LoadPlainValue(IValue) , IAction.Execute(IActionContext) , IAction.PlainValue

Interface IHackAndSlashV10

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

Interface for hack and slash action version 10.

```
public interface IHackAndSlashV10
```

Properties

ApStoneCount

Gets the number of AP stones to use.

```
int ApStoneCount { get; }
```

Property Value

int

AvatarAddress

Gets the avatar address for the battle.

```
Address AvatarAddress { get; }
```

Property Value

Address

Costumes

Gets the costume IDs to use in battle.

```
IEnumerable<Guid> Costumes { get; }
```

Property Value

IEnumerable<Guid>

Equipments

Gets the equipment IDs to use in battle.

```
IEnumerable<Guid> Equipments { get; }
```

Property Value

IEnumerable<Guid>

Foods

Gets the food IDs to use in battle.

```
IEnumerable<Guid> Foods { get; }
```

Property Value

IEnumerable<Guid>

RuneSlotInfos

Gets the rune slot information for the battle.

```
IEnumerable<IValue> RuneSlotInfos { get; }
```

Property Value

IEnumerable<IValue>

StageBuffId

Gets the stage buff ID to apply.

```
int? StageBuffId { get; }
```

Property Value

int?

StageId

Gets the stage ID to challenge.

```
int StageId { get; }
```

Property Value

int

TotalPlayCount

Gets the total number of plays.

```
int TotalPlayCount { get; }
```

Property Value

int

WorldId

Gets the world ID for the stage.

```
int WorldId { get; }
```

Property Value

int

Interface IProductInfo

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
public interface IProductInfo
```

Properties

AgentAddress

```
Address AgentAddress { get; set; }
```

Property Value

Address

AvatarAddress

```
Address AvatarAddress { get; set; }
```

Property Value

Address

Price

```
FungibleAssetValue Price { get; set; }
```

Property Value

FungibleAssetValue

ProductId

```
Guid ProductId { get; set; }
```

Property Value

Guid

Type

```
ProductType Type { get; set; }
```

Property Value

ProductType

Methods

Serialize()

```
IValue Serialize()
```

Returns

IValue

ValidateType()

```
void ValidateType()
```

Interface IPurchaseInfo

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
public interface IPurchaseInfo
```

Properties

ItemSubType

```
ItemSubType ItemSubType { get; }
```

Property Value

ItemSubType

OrderId

```
Guid? OrderId { get; }
```

Property Value

Guid?

Price

```
FungibleAssetValue Price { get; }
```

Property Value

FungibleAssetValue

SellerAgentAddress

```
Address SellerAgentAddress { get; }
```

Property Value

Address

SellerAvatarAddress

```
Address SellerAvatarAddress { get; }
```

Property Value

Address

Interface IRegisterInfo

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
public interface IRegisterInfo
```

Properties

AvatarAddress

```
Address AvatarAddress { get; set; }
```

Property Value

Address

Price

```
FungibleAssetValue Price { get; set; }
```

Property Value

FungibleAssetValue

Type

```
ProductType Type { get; set; }
```

Property Value

ProductType

Methods

Serialize()

```
IValue Serialize()
```

Returns

IValue

Validate()

```
void Validate()
```

ValidateAddress(Address)

```
void ValidateAddress(Address avatarAddress)
```

Parameters

`avatarAddress` Address

ValidatePrice(Currency)

```
void ValidatePrice(Currency ncg)
```

Parameters

`ncg` Currency

Interface ITransferAsset

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
public interface ITransferAsset
```

Properties

Amount

```
FungibleAssetValue Amount { get; }
```

Property Value

FungibleAssetValue

Memo

```
string? Memo { get; }
```

Property Value

string

Recipient

```
Address Recipient { get; }
```

Property Value

Address

Sender

```
Address Sender { get; }
```

Property Value

Address

Interface ITransferAssets

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
public interface ITransferAssets
```

Properties

Memo

```
string? Memo { get; }
```

Property Value

string

Recipients

```
List<(Address recipient, FungibleAssetValue amount)> Recipients { get; }
```

Property Value

List<(Address recipient, FungibleAssetValue amount)>

Sender

```
Address Sender { get; }
```

Property Value

Address

Class InfiniteTowerBattle

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

Represents a battle action in the infinite tower where players engage in combat with various conditions and restrictions. This action modifies the following states:

- AvatarState: Updates player's stats, inventory, and experience
- InfiniteTowerInfo: Updates tower progress and tickets
- CollectionState: Updates collection progress
- RuneState: Updates rune effects and bonuses
- ItemSlotState: Updates equipped items
- RuneSlotState: Updates equipped runes
- CP: Updates character's combat power

```
[Serializable]
[ActionType("infinite_tower_battle")]
public class InfiniteTowerBattle : GameAction, IAction, IInfiniteTowerBattleV1
```

Inheritance

object ← ActionBase ← GameAction ← InfiniteTowerBattle

Implements

IAction, IInfiniteTowerBattleV1

Inherited Members

GameAction.Id , GameAction.PlainValue , GameAction.LoadPlainValue(IValue) , ActionBase.GetSignerAndOtherAddressesHex(IActionContext, params Address[]) , ActionBase.TryGetAdminState(IActionContext, out AdminState) , ActionBase.CheckPermission(IActionContext) , ActionBase.CheckObsolete(long, IActionContext) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Fields

AvatarAddress

The avatar address for the battle.

```
public Address AvatarAddress
```

Field Value

Address

BuyTicketIfNeeded

Whether to buy a ticket if needed.

```
public bool BuyTicketIfNeeded
```

Field Value

bool

Costumes

The costume IDs to use in battle.

```
public List<Guid> Costumes
```

Field Value

List<Guid>

Equipments

The equipment IDs to use in battle.

```
public List<Guid> Equipments
```

Field Value

List<Guid>

FloorId

The floor ID to challenge.

```
public int FloorId
```

Field Value

int

Foods

The food IDs to use in battle.

```
public List<Guid> Foods
```

Field Value

List<Guid>

InfiniteTowerId

The infinite tower ID.

```
public int InfiniteTowerId
```

Field Value

int

PlayCount

The number of times this action can be played.

```
public const int PlayCount = 1
```

Field Value

int

RuneInfos

The rune slot information for the battle.

```
public List<RuneSlotInfo> RuneInfos
```

Field Value

List<RuneSlotInfo>

UseNcgForTicket

Whether to use NCG for buying tickets.

```
public bool UseNcgForTicket
```

Field Value

bool

Properties

PlainValueInternal

Gets the plain value representation of this action for serialization.

```
protected override IImmutableDictionary<string, IValue> PlainValueInternal { get; }
```

Property Value

IImmutableDictionary<string, IValue>

Methods

Execute(IActionContext)

Executes the infinite tower battle action.

```
public override IWorld Execute(IActionContext context)
```

Parameters

`context` IActionContext

The action context containing state and block information.

Returns

IWorld

The updated world state after the battle.

LoadPlainValueInternal(IImmutableDictionary<string, IValue>)

Loads the action data from a plain value dictionary.

```
protected override void LoadPlainValueInternal(IImmutableDictionary<string,
IValue> plainValue)
```

Parameters

`plainValue` IImmutableDictionary<string, IValue>

The plain value dictionary containing action data.

PerformFullReset(IWorld, Address, int, long)

Helper method to manually trigger full reset for testing purposes.

```
public static void PerformFullReset(IWorld states, Address avatarAddress, int
infiniteTowerId, long currentBlockIndex)
```

Parameters

`states` IWorld

`avatarAddress` Address

`infiniteTowerId` int

`currentBlockIndex` long

RefillDailyTickets(IWorld, Address, int, long)

Helper method to manually trigger daily ticket refill for testing purposes.

```
public static void RefillDailyTickets(IWorld states, Address avatarAddress, int
infiniteTowerId, long currentBlockIndex)
```

Parameters

`states` IWorld

`avatarAddress` Address

`infiniteTowerId` int

`currentBlockIndex` long

Class InitializeStates

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

Introduced at Initial commit(2e645be18a4e2caea031c347f00777fbad5dbcc6) Updated at NCG distribution(7e4515b6e14cc5d6eb716d5ebb587ab04b4246f9) Updated at https://github.com/planetarium/lib9c/pull/19 Updated at https://github.com/planetarium/lib9c/pull/36 Updated at https://github.com/planetarium/lib9c/pull/42 Updated at https://github.com/planetarium/lib9c/pull/55 Updated at https://github.com/planetarium/lib9c/pull/57 Updated at https://github.com/planetarium/lib9c/pull/60 Updated at https://github.com/planetarium/lib9c/pull/102 Updated at https://github.com/planetarium/lib9c/pull/128 Updated at https://github.com/planetarium/lib9c/pull/167 Updated at https://github.com/planetarium/lib9c/pull/422 Updated at https://github.com/planetarium/lib9c/pull/747 Updated at https://github.com/planetarium/lib9c/pull/798 Updated at https://github.com/planetarium/lib9c/pull/957

```
[Serializable]
[ActionType("initialize_states")]
public class InitializeStates : GameAction, IAction, IInitializeStatesV1
```

Inheritance

object ← ActionBase ← GameAction ← InitializeStates

Implements

IAction, IInitializeStatesV1

Inherited Members

GameAction.Id , GameAction.PlainValue , GameAction.LoadPlainValue(IValue) , ActionBase.GetSignerAndOtherAddressesHex(IActionContext, params Address[]) , ActionBase.TryGetAdminState(IActionContext, out AdminState) , ActionBase.CheckPermission(IActionContext) , ActionBase.CheckObsolete(long, IActionContext) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

InitializeStates()

```
public InitializeStates()
```

InitializeStates(ValidatorSet, RankingState0, ShopState, Dictionary<string, string>, GameConfigState, RedeemCodeState, ActivatedAccountsState, GoldCurrencyState, GoldDistribution[], PendingActivationState[], AdminState, AuthorizedMinersState, CreditsState, ISet<Address>)

```
public InitializeStates(ValidatorSet validatorSet, RankingState0 rankingState,
ShopState shopState, Dictionary<string, string> tableSheets, GameConfigState
gameConfigState, RedeemCodeState redeemCodeState, ActivatedAccountsState
activatedAccountsState, GoldCurrencyState goldCurrencyState, GoldDistribution[]
goldDistributions, PendingActivationState[] pendingActivationStates, AdminState
adminAddressState = null, AuthorizedMinersState authorizedMinersState = null,
CreditsState creditsState = null, ISet<Address> assetMinters = null)
```

Parameters

`validatorSet` ValidatorSet

`rankingState` RankingState0

`shopState` ShopState

`tableSheets` Dictionary<string, string>

`gameConfigState` GameConfigState

`redeemCodeState` RedeemCodeState

`activatedAccountsState` ActivatedAccountsState

`goldCurrencyState` GoldCurrencyState

`goldDistributions` GoldDistribution[]

`pendingActivationStates` PendingActivationState[]

`adminAddressState` AdminState

`authorizedMinersState` AuthorizedMinersState

`creditsState` CreditsState

`assetMinters` ISet<Address>

Properties

ActivatedAccounts

```
public Dictionary ActivatedAccounts { get; set; }
```

Property Value

Dictionary

AdminAddressState

```
public Dictionary AdminAddressState { get; set; }
```

Property Value

Dictionary

AssetMinters

```
public ISet<Address> AssetMinters { get; set; }
```

Property Value

ISet<Address>

AuthorizedMiners

```
public Dictionary AuthorizedMiners { get; set; }
```

Property Value

Dictionary

Credits

```
public Dictionary Credits { get; set; }
```

Property Value

Dictionary

GameConfig

```
public Dictionary GameConfig { get; set; }
```

Property Value

Dictionary

GoldCurrency

```
public Dictionary GoldCurrency { get; set; }
```

Property Value

Dictionary

GoldDistributions

```
public List GoldDistributions { get; set; }
```

Property Value

List

PendingActivations

```
public List PendingActivations { get; set; }
```

Property Value

List

PlainValueInternal

```
protected override IImmutableDictionary<string, IValue> PlainValueInternal { get; }
```

Property Value

IImmutableDictionary<string, IValue>

Ranking

```
public Dictionary Ranking { get; set; }
```

Property Value

Dictionary

RedeemCode

```
public Dictionary RedeemCode { get; set; }
```

Property Value

Dictionary

Shop

```
public Dictionary Shop { get; set; }
```

Property Value

Dictionary

TableSheets

```
public Dictionary<string, string> TableSheets { get; set; }
```

Property Value

Dictionary<string, string>

ValidatorSet

```
public IValue ValidatorSet { get; set; }
```

Property Value

IValue

Methods

Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

Parameters

`context` IActionContext

Returns

IWorld

LoadPlainValueInternal(IImmutableDictionary<string, IValue>)

```
protected override void LoadPlainValueInternal(IImmutableDictionary<string,
IValue> plainValue)
```

Parameters

`plainValue` IImmutableDictionary<string, IValue>

Class InvalidAddressException

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
public class InvalidAddressException : InvalidOperationException, ISerializable
```

Inheritance

object ← Exception ← SystemException ← InvalidOperationException ← InvalidAddressException

Implements

ISerializable

Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Constructors

InvalidAddressException()

```
public InvalidAddressException()
```

InvalidAddressException(SerializationInfo, StreamingContext)

```
protected InvalidAddressException(SerializationInfo info, StreamingContext context)
```

Parameters

`info` SerializationInfo

`context` StreamingContext

InvalidAddressException(string)

```
public InvalidAddressException(string msg)
```

Parameters

`msg` string

Class InvalidClaimException

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
public class InvalidClaimException : InvalidOperationException, ISerializable
```

Inheritance

object ← Exception ← SystemException ← InvalidOperationException ← InvalidClaimException

Implements

ISerializable

Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Constructors

InvalidClaimException()

```
public InvalidClaimException()
```

InvalidClaimException(SerializationInfo, StreamingContext)

```
protected InvalidClaimException(SerializationInfo info, StreamingContext context)
```

Parameters

`info` SerializationInfo

`context` StreamingContext

InvalidClaimException(string)

```
public InvalidClaimException(string msg)
```

Parameters

`msg` string

Class InvalidCurrencyException

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
public class InvalidCurrencyException : InvalidOperationException, ISerializable
```

Inheritance

object ← Exception ← SystemException ← InvalidOperationException ← InvalidCurrencyException

Implements

ISerializable

Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Constructors

InvalidCurrencyException(SerializationInfo, StreamingContext)

```
protected InvalidCurrencyException(SerializationInfo info, StreamingContext context)
```

Parameters

`info` SerializationInfo

`context` StreamingContext

InvalidCurrencyException(string)

```
public InvalidCurrencyException(string msg)
```

Parameters

`msg` string

Class InvalidElementalException

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
public class InvalidElementalException : Exception, ISerializable
```

Inheritance

object ← Exception ← InvalidElementalException

Implements

ISerializable

Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Constructors

InvalidElementalException()

```
public InvalidElementalException()
```

InvalidElementalException(SerializationInfo, StreamingContext)

```
protected InvalidElementalException(SerializationInfo info,
StreamingContext context)
```

Parameters

`info` SerializationInfo

`context` StreamingContext

InvalidElementalException(string)

```
public InvalidElementalException(string message)
```

Parameters

`message` string

Class InvalidEquipmentException

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
public class InvalidEquipmentException : Exception, ISerializable
```

Inheritance

object ← Exception ← InvalidEquipmentException

Implements

ISerializable

Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Constructors

InvalidEquipmentException()

```
public InvalidEquipmentException()
```

InvalidEquipmentException(SerializationInfo, StreamingContext)

```
protected InvalidEquipmentException(SerializationInfo info,
StreamingContext context)
```

Parameters

`info` SerializationInfo

`context` StreamingContext

InvalidEquipmentException(string)

```
public InvalidEquipmentException(string msg)
```

Parameters

`msg` string

Class InvalidItemCountException

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
public class InvalidItemCountException : ArgumentOutOfRangeException, ISerializable
```

Inheritance

object ← Exception ← SystemException ← ArgumentException ← ArgumentOutOfRangeException ← InvalidItemCountException

Implements

ISerializable

Inherited Members

ArgumentOutOfRangeException.GetObjectData(SerializationInfo, StreamingContext) , ArgumentOutOfRangeException.ActualValue , ArgumentOutOfRangeException.Message , ArgumentException.ParamName , Exception.GetBaseException() , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Constructors

InvalidItemCountException()

```
public InvalidItemCountException()
```

InvalidItemCountException(SerializationInfo, StreamingContext)

```
protected InvalidItemCountException(SerializationInfo info,
```

```
  StreamingContext context)
```

Parameters

`info` SerializationInfo

`context` StreamingContext

InvalidItemCountException(string)

```
  public InvalidItemCountException(string msg)
```

Parameters

`msg` string

Class InvalidItemIdException

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

Represents an exception that is thrown when an invalid item ID is used.

```
[Serializable]
public class InvalidItemIdException : ArgumentOutOfRangeException, ISerializable
```

Inheritance

object ← Exception ← SystemException ← ArgumentException ← ArgumentOutOfRangeException ← InvalidItemIdException

Implements

ISerializable

Inherited Members

ArgumentOutOfRangeException.GetObjectData(SerializationInfo, StreamingContext) , ArgumentOutOfRangeException.ActualValue , ArgumentOutOfRangeException.Message , ArgumentException.ParamName , Exception.GetBaseException() , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Constructors

InvalidItemIdException()

Initializes a new instance of the InvalidItemIdException class.

```
public InvalidItemIdException()
```

InvalidItemIdException(SerializationInfo, StreamingContext)

Initializes a new instance of the InvalidItemIdException class with a specified error message

```
protected InvalidItemIdException(SerializationInfo info, StreamingContext context)
```

Parameters

`info` SerializationInfo

SerializationInfo

`context` StreamingContext

StreamingContext

InvalidItemIdException(string)

Initializes a new instance of the InvalidItemIdException class with a specified error message.

```
public InvalidItemIdException(string msg)
```

Parameters

`msg` string

The message that describes the error.

Class InvalidItemTypeException

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
public class InvalidItemTypeException : InvalidOperationException, ISerializable
```

Inheritance

object ← Exception ← SystemException ← InvalidOperationException ← InvalidItemTypeException

Implements

ISerializable

Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Constructors

InvalidItemTypeException(SerializationInfo, StreamingContext)

```
protected InvalidItemTypeException(SerializationInfo info, StreamingContext context)
```

Parameters

`info` SerializationInfo

`context` StreamingContext

InvalidItemTypeException(string)

```
public InvalidItemTypeException(string msg)
```

Parameters

`msg` string

Class InvalidLevelException

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
public class InvalidLevelException : InvalidOperationException, ISerializable
```

Inheritance

object ← Exception ← SystemException ← InvalidOperationException ← InvalidLevelException

Implements

ISerializable

Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Constructors

InvalidLevelException(SerializationInfo, StreamingContext)

```
protected InvalidLevelException(SerializationInfo info, StreamingContext context)
```

Parameters

`info` SerializationInfo

`context` StreamingContext

InvalidLevelException(string)

```
public InvalidLevelException(string s)
```

Parameters

s string

Class InvalidMaterialException

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
public class InvalidMaterialException : InvalidOperationException, ISerializable
```

Inheritance

object ← Exception ← SystemException ← InvalidOperationException ← InvalidMaterialException

Implements

ISerializable

Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Constructors

InvalidMaterialException(SerializationInfo, StreamingContext)

```
protected InvalidMaterialException(SerializationInfo info, StreamingContext context)
```

Parameters

`info` SerializationInfo

`context` StreamingContext

InvalidMaterialException(string)

```
public InvalidMaterialException(string s)
```

Parameters

s string

Class InvalidMinterException

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
public class InvalidMinterException : Exception, ISerializable
```

Inheritance

object ← Exception ← InvalidMinterException

Implements

ISerializable

Inherited Members

Exception.GetBaseException() , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Constructors

InvalidMinterException()

```
public InvalidMinterException()
```

InvalidMinterException(Address)

```
public InvalidMinterException(Address signer)
```

Parameters

`signer` Address

InvalidMinterException(SerializationInfo, StreamingContext)

```
protected InvalidMinterException(SerializationInfo info, StreamingContext context)
```

Parameters

`info` SerializationInfo

`context` StreamingContext

InvalidMinterException(string)

```
public InvalidMinterException(string message)
```

Parameters

`message` string

Methods

GetObjectData(SerializationInfo, StreamingContext)

When overridden in a derived class, sets the SerializationInfo with information about the exception.

```
public override void GetObjectData(SerializationInfo info, StreamingContext context)
```

Parameters

`info` SerializationInfo

The SerializationInfo that holds the serialized object data about the exception being thrown.

`context` StreamingContext

The StreamingContext that contains contextual information about the source or destination.

Exceptions

ArgumentNullException

The `info` parameter is a null reference (`Nothing` in Visual Basic).

Class InvalidMonsterCollectionRoundException

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
public class InvalidMonsterCollectionRoundException : InvalidOperationException,
ISerializable
```

Inheritance

object ← Exception ← SystemException ← InvalidOperationException ← InvalidMonsterCollectionRoundException

Implements

ISerializable

Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Constructors

InvalidMonsterCollectionRoundException(SerializationInfo, StreamingContext)

```
protected InvalidMonsterCollectionRoundException(SerializationInfo info,
StreamingContext context)
```

Parameters

`info` SerializationInfo

`context` StreamingContext

InvalidMonsterCollectionRoundException(string)

```
public InvalidMonsterCollectionRoundException(string msg)
```

Parameters

`msg` string

Class InvalidNamePatternException

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
public class InvalidNamePatternException : InvalidOperationException, ISerializable
```

Inheritance

object ← Exception ← SystemException ← InvalidOperationException ← InvalidNamePatternException

Implements

ISerializable

Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Constructors

InvalidNamePatternException(SerializationInfo, StreamingContext)

```
protected InvalidNamePatternException(SerializationInfo info,
StreamingContext context)
```

Parameters

`info` SerializationInfo

`context` StreamingContext

InvalidNamePatternException(string)

```
public InvalidNamePatternException(string s)
```

Parameters

s string

Class InvalidPriceException

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
public class InvalidPriceException : ArgumentOutOfRangeException, ISerializable
```

Inheritance

object ← Exception ← SystemException ← ArgumentException ← ArgumentOutOfRangeException ← InvalidPriceException

Implements

ISerializable

Inherited Members

ArgumentOutOfRangeException.GetObjectData(SerializationInfo, StreamingContext) , ArgumentOutOfRangeException.ActualValue , ArgumentOutOfRangeException.Message , ArgumentException.ParamName , Exception.GetBaseException() , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Constructors

InvalidPriceException(SerializationInfo, StreamingContext)

```
protected InvalidPriceException(SerializationInfo info, StreamingContext context)
```

Parameters

`info` SerializationInfo

`context` StreamingContext

InvalidPriceException(string)

```
public InvalidPriceException(string msg)
```

Parameters

`msg` string

Class InvalidProductTypeException

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
public class InvalidProductTypeException : InvalidOperationException, ISerializable
```

Inheritance

object ← Exception ← SystemException ← InvalidOperationException ← InvalidProductTypeException

Implements

ISerializable

Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Constructors

InvalidProductTypeException(SerializationInfo, StreamingContext)

```
protected InvalidProductTypeException(SerializationInfo info,
StreamingContext context)
```

Parameters

`info` SerializationInfo

`context` StreamingContext

InvalidProductTypeException(string)

```
public InvalidProductTypeException(string msg)
```

Parameters

`msg` string

Class InvalidRecipeIdException

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
public class InvalidRecipeIdException : Exception, ISerializable
```

Inheritance

object ← Exception ← InvalidRecipeIdException

Implements

ISerializable

Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Constructors

InvalidRecipeIdException()

```
public InvalidRecipeIdException()
```

InvalidRecipeIdException(SerializationInfo, StreamingContext)

```
protected InvalidRecipeIdException(SerializationInfo info, StreamingContext context)
```

Parameters

`info` SerializationInfo

`context` StreamingContext

InvalidRecipeIdException(string)

```
public InvalidRecipeIdException(string msg)
```

Parameters

`msg` string

Class InvalidRepeatPlayException

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
public class InvalidRepeatPlayException : Exception, ISerializable
```

Inheritance

object ← Exception ← InvalidRepeatPlayException

Implements

ISerializable

Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Constructors

InvalidRepeatPlayException()

```
public InvalidRepeatPlayException()
```

InvalidRepeatPlayException(SerializationInfo, StreamingContext)

```
protected InvalidRepeatPlayException(SerializationInfo info,
StreamingContext context)
```

Parameters

`info` SerializationInfo

`context` StreamingContext

InvalidRepeatPlayException(string)

```
public InvalidRepeatPlayException(string msg)
```

Parameters

`msg` string

Class InvalidShopItemException

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
public class InvalidShopItemException : InvalidOperationException, ISerializable
```

Inheritance

object ← Exception ← SystemException ← InvalidOperationException ← InvalidShopItemException

Implements

ISerializable

Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Constructors

InvalidShopItemException(SerializationInfo, StreamingContext)

```
protected InvalidShopItemException(SerializationInfo info, StreamingContext context)
```

Parameters

`info` SerializationInfo

`context` StreamingContext

InvalidShopItemException(string)

```
public InvalidShopItemException(string s)
```

Parameters

s [string]

Class InvalidSignatureException

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
public class InvalidSignatureException : ActivationException, ISerializable
```

Inheritance

object ← Exception ← ActivationException ← InvalidSignatureException

Implements

ISerializable

Inherited Members

Exception.GetBaseException() , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Constructors

InvalidSignatureException(PendingActivationState, byte[])

```
public InvalidSignatureException(PendingActivationState pending, byte[] signature)
```

Parameters

`pending` PendingActivationState

`signature` byte[]

InvalidSignatureException(SerializationInfo, StreamingContext)

```
public InvalidSignatureException(SerializationInfo info, StreamingContext context)
```

Parameters

`info` SerializationInfo

`context` StreamingContext

InvalidSignatureException(string)

```
public InvalidSignatureException(string message)
```

Parameters

`message` string

Properties

Pending

```
public PendingActivationState Pending { get; }
```

Property Value

PendingActivationState

Signature

```
public byte[] Signature { get; }
```

Property Value

byte[]

Methods

GetObjectData(SerializationInfo, StreamingContext)

When overridden in a derived class, sets the SerializationInfo with information about the exception.

```
public override void GetObjectData(SerializationInfo info, StreamingContext context)
```

Parameters

`info` SerializationInfo

The SerializationInfo that holds the serialized object data about the exception being thrown.

`context` StreamingContext

The StreamingContext that contains contextual information about the source or destination.

Exceptions

ArgumentNullException

The `info` parameter is a null reference (`Nothing` in Visual Basic).

Class InvalidSlotIndexException

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
public class InvalidSlotIndexException : InvalidOperationException, ISerializable
```

Inheritance

object ← Exception ← SystemException ← InvalidOperationException ← InvalidSlotIndexException

Implements

ISerializable

Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Constructors

InvalidSlotIndexException(SerializationInfo, StreamingContext)

```
protected InvalidSlotIndexException(SerializationInfo info,
StreamingContext context)
```

Parameters

`info` SerializationInfo

`context` StreamingContext

InvalidSlotIndexException(string)

```
public InvalidSlotIndexException(string s)
```

Parameters

s [string]

Class InvalidStageException

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
public class InvalidStageException : Exception, ISerializable
```

Inheritance

object ← Exception ← InvalidStageException

Implements

ISerializable

Derived

StageNotOpenedException

Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Constructors

InvalidStageException()

```
public InvalidStageException()
```

InvalidStageException(SerializationInfo, StreamingContext)

```
protected InvalidStageException(SerializationInfo info, StreamingContext context)
```

Parameters

`info` SerializationInfo

`context` StreamingContext

InvalidStageException(string)

```
public InvalidStageException(string msg)
```

Parameters

`msg` string

Class InvalidTradableIdException

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
public class InvalidTradableIdException : InvalidOperationException, ISerializable
```

Inheritance

object ← Exception ← SystemException ← InvalidOperationException ← InvalidTradableIdException

Implements

ISerializable

Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Constructors

InvalidTradableIdException(SerializationInfo, StreamingContext)

```
protected InvalidTradableIdException(SerializationInfo info,
StreamingContext context)
```

Parameters

`info` SerializationInfo

`context` StreamingContext

InvalidTradableIdException(string)

```
public InvalidTradableIdException(string message)
```

Parameters

`message` string

Class InvalidTradableItemException

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
public class InvalidTradableItemException : InvalidOperationException, ISerializable
```

Inheritance

object ← Exception ← SystemException ← InvalidOperationException ← InvalidTradableItemException

Implements

ISerializable

Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Constructors

InvalidTradableItemException(SerializationInfo, StreamingContext)

```
protected InvalidTradableItemException(SerializationInfo info,
StreamingContext context)
```

Parameters

`info` SerializationInfo

`context` StreamingContext

InvalidTradableItemException(string)

```
public InvalidTradableItemException(string message)
```

Parameters

`message` string

Class InvalidTransferCurrencyException

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
public class InvalidTransferCurrencyException : InvalidOperationException,
ISerializable
```

Inheritance

object ← Exception ← SystemException ← InvalidOperationException ← InvalidTransferCurrencyException

Implements

ISerializable

Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Constructors

InvalidTransferCurrencyException(SerializationInfo, StreamingContext)

```
protected InvalidTransferCurrencyException(SerializationInfo info,
StreamingContext context)
```

Parameters

`info` SerializationInfo

`context` StreamingContext

InvalidTransferCurrencyException(string)

```
public InvalidTransferCurrencyException(string message)
```

Parameters

`message` string

Class InvalidTransferMinterException

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
public class InvalidTransferMinterException : Exception, ISerializable
```

Inheritance

object ← Exception ← InvalidTransferMinterException

Implements

ISerializable

Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Constructors

InvalidTransferMinterException(IEnumerable<Address>, Address, Address)

```
public InvalidTransferMinterException(IEnumerable<Address> minters, Address sender,
Address recipient)
```

Parameters

`minters` IEnumerable<Address>

`sender` Address

`recipient` Address

InvalidTransferMinterException(SerializationInfo, StreamingContext)

```
protected InvalidTransferMinterException(SerializationInfo info,
StreamingContext context)
```

Parameters

`info` SerializationInfo

`context` StreamingContext

InvalidTransferMinterException(string)

```
public InvalidTransferMinterException(string message)
```

Parameters

`message` string

Properties

Minters

```
public HashSet<Address> Minters { get; }
```

Property Value

HashSet<Address>

Recipient

```
public Address Recipient { get; }
```

Property Value

Address

Sender

```
public Address Sender { get; }
```

Property Value

Address

Class InvalidTransferRecipientException

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
public class InvalidTransferRecipientException : Exception, ISerializable
```

Inheritance

object ← Exception ← InvalidTransferRecipientException

Implements

ISerializable

Inherited Members

Exception.GetBaseException() , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Constructors

InvalidTransferRecipientException(Address, Address)

```
public InvalidTransferRecipientException(Address sender, Address recipient)
```

Parameters

`sender` Address

`recipient` Address

InvalidTransferRecipientException(SerializationInfo, StreamingContext)

```
public InvalidTransferRecipientException(SerializationInfo info,
StreamingContext context)
```

Parameters

`info` SerializationInfo

`context` StreamingContext

InvalidTransferRecipientException(string)

```
public InvalidTransferRecipientException(string message)
```

Parameters

`message` string

Properties

Recipient

```
public Address Recipient { get; }
```

Property Value

Address

Sender

```
public Address Sender { get; }
```

Property Value

Address

Methods

GetObjectData(SerializationInfo, StreamingContext)

When overridden in a derived class, sets the SerializationInfo with information about the exception.

```
public override void GetObjectData(SerializationInfo info, StreamingContext context)
```

Parameters

`info` SerializationInfo

The SerializationInfo that holds the serialized object data about the exception being thrown.

`context` StreamingContext

The StreamingContext that contains contextual information about the source or destination.

Exceptions

ArgumentNullException

The `info` parameter is a null reference (`Nothing` in Visual Basic).

Class InvalidTransferSignerException

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
public class InvalidTransferSignerException : Exception, ISerializable
```

Inheritance

object ← Exception ← InvalidTransferSignerException

Implements

ISerializable

Inherited Members

Exception.GetBaseException() , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Constructors

InvalidTransferSignerException(Address, Address, Address)

```
public InvalidTransferSignerException(Address txSigner, Address sender,
Address recipient)
```

Parameters

`txSigner` Address

`sender` Address

`recipient` Address

InvalidTransferSignerException(SerializationInfo, StreamingContext)

```
public InvalidTransferSignerException(SerializationInfo info,
StreamingContext context)
```

Parameters

`info` SerializationInfo

`context` StreamingContext

InvalidTransferSignerException(string)

```
public InvalidTransferSignerException(string message)
```

Parameters

`message` string

Properties

Recipient

```
public Address Recipient { get; }
```

Property Value

Address

Sender

```
public Address Sender { get; }
```

Property Value

Address

TxSigner

```
public Address TxSigner { get; }
```

Property Value

Address

Methods

GetObjectData(SerializationInfo, StreamingContext)

When overridden in a derived class, sets the SerializationInfo with information about the exception.

```
public override void GetObjectData(SerializationInfo info, StreamingContext context)
```

Parameters

`info` SerializationInfo

The SerializationInfo that holds the serialized object data about the exception being thrown.

`context` StreamingContext

The StreamingContext that contains contextual information about the source or destination.

Exceptions

ArgumentNullException

The `info` parameter is a null reference (`Nothing` in Visual Basic).

Class InvalidTransferUnactivatedRecipientException

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
public class InvalidTransferUnactivatedRecipientException : Exception, ISerializable
```

Inheritance

object ← Exception ← InvalidTransferUnactivatedRecipientException

Implements

ISerializable

Inherited Members

Exception.GetBaseException() , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Constructors

InvalidTransferUnactivatedRecipientException(Address, Address)

```
public InvalidTransferUnactivatedRecipientException(Address sender,
Address recipient)
```

Parameters

`sender` Address

`recipient` Address

InvalidTransferUnactivatedRecipientException(SerializationInfo, StreamingContext)

```
public InvalidTransferUnactivatedRecipientException(SerializationInfo info,
StreamingContext context)
```

Parameters

`info` SerializationInfo

`context` StreamingContext

InvalidTransferUnactivatedRecipientException(string)

```
public InvalidTransferUnactivatedRecipientException(string message)
```

Parameters

`message` string

Properties

Recipient

```
public Address Recipient { get; }
```

Property Value

Address

Sender

```
public Address Sender { get; }
```

Property Value

Address

Methods

GetObjectData(SerializationInfo, StreamingContext)

When overridden in a derived class, sets the SerializationInfo with information about the exception.

```
public override void GetObjectData(SerializationInfo info, StreamingContext context)
```

Parameters

`info` SerializationInfo

The SerializationInfo that holds the serialized object data about the exception being thrown.

`context` StreamingContext

The StreamingContext that contains contextual information about the source or destination.

Exceptions

ArgumentNullException

The `info` parameter is a null reference (`Nothing` in Visual Basic).

Class InvalidWorldException

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
public class InvalidWorldException : Exception, ISerializable
```

Inheritance

object ← Exception ← InvalidWorldException

Implements

ISerializable

Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Constructors

InvalidWorldException()

```
public InvalidWorldException()
```

InvalidWorldException(SerializationInfo, StreamingContext)

```
protected InvalidWorldException(SerializationInfo info, StreamingContext context)
```

Parameters

`info` SerializationInfo

`context` StreamingContext

InvalidWorldException(string)

```
public InvalidWorldException(string msg)
```

Parameters

`msg` string

Class IssueToken

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[ActionType("issue_token")]
public class IssueToken : ActionBase, IAction
```

Inheritance

object ← ActionBase ← IssueToken

Implements

IAction

Inherited Members

ActionBase.GetSignerAndOtherAddressesHex(IActionContext, params Address[]) ,
ActionBase.TryGetAdminState(IActionContext, out AdminState) ,
ActionBase.CheckPermission(IActionContext) ,
ActionBase.CheckObsolete(long, IActionContext) , object.Equals(object) ,
object.Equals(object, object) , object.GetHashCode() , object.GetType() ,
object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

IssueToken()

```
public IssueToken()
```

Fields

AvatarAddress

```
public Address AvatarAddress
```

Field Value

Address

FungibleAssetValues

```
public List<FungibleAssetValue> FungibleAssetValues
```

Field Value

List<FungibleAssetValue>

Items

```
public List<(int id, int count, bool tradable)> Items
```

Field Value

List<(int id, int count, bool tradable)>

TypeIdentifier

```
public const string TypeIdentifier = "issue_token"
```

Field Value

string

Properties

PlainValue

```
public override IValue PlainValue { get; }
```

Property Value

IValue

Methods

Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

Parameters

`context` IActionContext

Returns

IWorld

LoadPlainValue(IValue)

```
public override void LoadPlainValue(IValue plainValue)
```

Parameters

`plainValue` IValue

Class IssueTokensFromGarage

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[ActionType("issue_tokens_from_garage")]
public class IssueTokensFromGarage : ActionBase, IAction
```

Inheritance

object ← ActionBase ← IssueTokensFromGarage

Implements

IAction

Inherited Members

ActionBase.GetSignerAndOtherAddressesHex(IActionContext, params Address[]) ,
ActionBase.TryGetAdminState(IActionContext, out AdminState) ,
ActionBase.CheckPermission(IActionContext) ,
ActionBase.CheckObsolete(long, IActionContext) , object.Equals(object) ,
object.Equals(object, object) , object.GetHashCode() , object.GetType() ,
object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

IssueTokensFromGarage()

```
public IssueTokensFromGarage()
```

IssueTokensFromGarage(IEnumerable<Spec>)

```
public IssueTokensFromGarage(IEnumerable<IssueTokensFromGarage.Spec> specs)
```

Parameters

`specs` IEnumerable<IssueTokensFromGarage.Spec>

Fields

TypeIdentifier

```
public const string TypeIdentifier = "issue_tokens_from_garage"
```

Field Value

string

Properties

PlainValue

```
public override IValue PlainValue { get; }
```

Property Value

IValue

Specs

```
public IEnumerable<IssueTokensFromGarage.Spec> Specs { get; }
```

Property Value

IEnumerable<IssueTokensFromGarage.Spec>

Methods

Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

Parameters

`context` IActionContext

Returns

IWorld

LoadPlainValue(IValue)

```
public override void LoadPlainValue(IValue plainValue)
```

Parameters

`plainValue` IValue

Struct IssueTokensFromGarage.Spec

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
public readonly struct IssueTokensFromGarage.Spec
```

Inherited Members
ValueType.Equals(object) , ValueType.GetHashCode() , ValueType.ToString() , object.Equals(object, object) , object.GetType() , object.ReferenceEquals(object, object)

Constructors

Spec(List)

```
public Spec(List bencoded)
```

Parameters

`bencoded` List

Properties

Assets

```
public FungibleAssetValue? Assets { get; }
```

Property Value

FungibleAssetValue?

Items

```
public FungibleItemValue? Items { get; }
```

Property Value

FungibleItemValue?

Methods

FromFungibleAssetValue(FungibleAssetValue)

```
public static IssueTokensFromGarage.Spec FromFungibleAssetValue(FungibleAssetValue
assets)
```

Parameters

`assets` FungibleAssetValue

Returns

IssueTokensFromGarage.Spec

FromFungibleItemValue(FungibleItemValue)

```
public static IssueTokensFromGarage.Spec FromFungibleItemValue(FungibleItemValue
items)
```

Parameters

`items` FungibleItemValue

Returns

IssueTokensFromGarage.Spec

Serialize()

```
public IValue Serialize()
```

Returns

IValue

Class ItemDoesNotExistException

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
public class ItemDoesNotExistException : InvalidOperationException, ISerializable
```

Inheritance

object ← Exception ← SystemException ← InvalidOperationException ← ItemDoesNotExistException

Implements

ISerializable

Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Constructors

ItemDoesNotExistException(SerializationInfo, StreamingContext)

```
public ItemDoesNotExistException(SerializationInfo info, StreamingContext context)
```

Parameters

`info` SerializationInfo

`context` StreamingContext

ItemDoesNotExistException(string)

```
public ItemDoesNotExistException(string msg)
```

Parameters

`msg` string

Class ItemEnhancement

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

Updated at https://github.com/planetarium/lib9c/pull/2195

```
[Serializable]
[ActionType("item_enhancement14")]
public class ItemEnhancement : GameAction, IAction, IItemEnhancementV5
```

Inheritance

object ← ActionBase ← GameAction ← ItemEnhancement

Implements

IAction, IItemEnhancementV5

Inherited Members

GameAction.Id , GameAction.PlainValue , GameAction.LoadPlainValue(IValue) , ActionBase.GetSignerAndOtherAddressesHex(IActionContext, params Address[]) , ActionBase.TryGetAdminState(IActionContext, out AdminState) , ActionBase.CheckPermission(IActionContext) , ActionBase.CheckObsolete(long, IActionContext) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Fields

HammerBannedTypes

```
public static readonly IReadOnlyCollection<ItemSubType> HammerBannedTypes
```

Field Value

IReadOnlyCollection<ItemSubType>

HammerIds

```
public static readonly IReadOnlyCollection<int> HammerIds
```

Field Value

IReadOnlyCollection<int>

MaterialCountLimit

```
public const int MaterialCountLimit = 50
```

Field Value

int

MaxHammerExpBoostFactor

Upper bound of a EnhancementHammerExp row: the adjusted hammer experience never exceeds this many times the configured one, so a mistyped multiplier cannot hand out the top hammer's 500,000,000 experience tenfold.

```
public const long MaxHammerExpBoostFactor = 3
```

Field Value

long

avatarAddress

```
public Address avatarAddress
```

Field Value

Address

hammers

```
public Dictionary<int, int> hammers
```

Field Value

Dictionary<int, int>

itemId

```
public Guid itemId
```

Field Value

Guid

materialIds

```
public List<Guid> materialIds
```

Field Value

List<Guid>

slotIndex

```
public int slotIndex
```

Field Value

int

Properties

PlainValueInternal

```
protected override IImmutableDictionary<string, IValue> PlainValueInternal { get; }
```

Property Value

IImmutableDictionary<string, IValue>

Methods

Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

Parameters

`context` IActionContext

Returns

IWorld

GetEquipmentMaxLevel(Equipment, EnhancementCostSheetV3)

```
public static int GetEquipmentMaxLevel(Equipment equipment,
EnhancementCostSheetV3 sheet)
```

Parameters

`equipment` Equipment

`sheet` EnhancementCostSheetV3

Returns

int

GetHammerExp(EnhancementCostSheetV3, int, Row)

Experience one hammer gives, adjusted by a EnhancementHammerExp row.

```
public static long GetHammerExp(EnhancementCostSheetV3 enhancementCostSheet, int
hammerId, BoostScheduleSheet.Row hammerExpBoost)
```

Parameters

`enhancementCostSheet` EnhancementCostSheetV3

The sheet configuring each hammer's experience.

`hammerId` int

The hammer item id.

`hammerExpBoost` BoostScheduleSheet.Row

The row active for `hammerId` at the block, or `null` for none. A client previewing an enhancement must pass the row for the block it expects, or it shows less experience, a lower level and a lower NCG cost than the action produces.

Returns

long

The configured experience adjusted by the row, rounded down and kept between the configured experience and MaxHammerExpBoostFactor times it.

GetRequiredBlockCount(Equipment, Equipment, EnhancementCostSheetV3)

```
public static int GetRequiredBlockCount(Equipment preEquipment, Equipment
targetEquipment, EnhancementCostSheetV3 sheet)
```

Parameters

`preEquipment` Equipment

`targetEquipment` Equipment

`sheet` EnhancementCostSheetV3

Returns

int

LoadPlainValueInternal(IImmutableDictionary<string, IValue>)

```
protected override void LoadPlainValueInternal(IImmutableDictionary<string,
IValue> plainValue)
```

Parameters

`plainValue` IImmutableDictionary<string, IValue>

TryGetRow(Equipment, EnhancementCostSheetV3, out Row)

```
public static bool TryGetRow(Equipment equipment, EnhancementCostSheetV3 sheet, out
EnhancementCostSheetV3.Row row)
```

Parameters

`equipment` Equipment

`sheet` EnhancementCostSheetV3

`row` EnhancementCostSheetV3.Row

Returns

bool

Class ItemEnhancement10

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

Updated at https://github.com/planetarium/lib9c/pull/1176

```
[Serializable]
[ActionType("item_enhancement10")]
[ActionObsolete(7200000)]
public class ItemEnhancement10 : GameAction, IAction, IItemEnhancementV2
```

Inheritance

object ← ActionBase ← GameAction ← ItemEnhancement10

Implements

IAction, IItemEnhancementV2

Inherited Members

GameAction.Id , GameAction.PlainValue , GameAction.LoadPlainValue(IValue) , ActionBase.GetSignerAndOtherAddressesHex(IActionContext, params Address[]) , ActionBase.TryGetAdminState(IActionContext, out AdminState) , ActionBase.CheckPermission(IActionContext) , ActionBase.CheckObsolete(long, IActionContext) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Fields

avatarAddress

```
public Address avatarAddress
```

Field Value

Address

itemId

```
public Guid itemId
```

Field Value

Guid

materialId

```
public Guid materialId
```

Field Value

Guid

slotIndex

```
public int slotIndex
```

Field Value

int

Properties

PlainValueInternal

```
protected override IImmutableDictionary<string, IValue> PlainValueInternal { get; }
```

Property Value

IImmutableDictionary<string, IValue>

Methods

Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

Parameters

`context` IActionContext

Returns

IWorld

GetEnhancementResult(Row, IRandom)

```
public static ItemEnhancement10.EnhancementResult
GetEnhancementResult(EnhancementCostSheetV2.Row row, IRandom random)
```

Parameters

`row` EnhancementCostSheetV2.Row

`random` IRandom

Returns

ItemEnhancement10.EnhancementResult

GetEquipmentMaxLevel(Equipment, EnhancementCostSheetV2)

```
public static int GetEquipmentMaxLevel(Equipment equipment,
EnhancementCostSheetV2 sheet)
```

Parameters

`equipment` Equipment

`sheet` EnhancementCostSheetV2

Returns

int

GetFeeStoreAddress()

```
public static Address GetFeeStoreAddress()
```

Returns

Address

GetRequiredAp()

```
public static int GetRequiredAp()
```

Returns

int

GetRequiredBlockCount(Row, EnhancementResult)

```
public static int GetRequiredBlockCount(EnhancementCostSheetV2.Row row,
ItemEnhancement10.EnhancementResult result)
```

Parameters

`row` EnhancementCostSheetV2.Row

`result` ItemEnhancement10.EnhancementResult

Returns

int

LoadPlainValueInternal(IImmutableDictionary<string, IValue>)

```
protected override void LoadPlainValueInternal(IImmutableDictionary<string,
IValue> plainValue)
```

Parameters

`plainValue` IImmutableDictionary<string, IValue>

TryGetRow(Equipment, EnhancementCostSheetV2, out Row)

```
public static bool TryGetRow(Equipment equipment, EnhancementCostSheetV2 sheet, out
EnhancementCostSheetV2.Row row)
```

Parameters

`equipment` Equipment

`sheet` EnhancementCostSheetV2

`row` EnhancementCostSheetV2.Row

Returns

bool

Enum ItemEnhancement10.EnhancementResult

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
public enum ItemEnhancement10.EnhancementResult
```

Extension Methods

StateExtensions.Serialize(Enum)

Fields

`Fail = 2`

`GreatSuccess = 0`

`Success = 1`

Class ItemEnhancement10.ResultModel

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
public class ItemEnhancement10.ResultModel : AttachmentActionResult, IState
```

Inheritance

object ← AttachmentActionResult ← ItemEnhancement10.ResultModel

Implements

IState

Inherited Members

AttachmentActionResult.itemUsable , AttachmentActionResult.costume , AttachmentActionResult.tradableFungibleItem , AttachmentActionResult.tradableFungibleItemCount , AttachmentActionResult.SerializeBackup1() , AttachmentActionResult.Deserialize(Dictionary) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

ResultModel()

```
public ResultModel()
```

ResultModel(Dictionary)

```
public ResultModel(Dictionary serialized)
```

Parameters

`serialized` Dictionary

Fields

actionPoint

```
public int actionPoint
```

Field Value

int

enhancementResult

```
public ItemEnhancement10.EnhancementResult enhancementResult
```

Field Value

ItemEnhancement10.EnhancementResult

gold

```
public BigInteger gold
```

Field Value

BigInteger

id

```
public Guid id
```

Field Value

Guid

materialItemIdList

```
public IEnumerable<Guid> materialItemIdList
```

Field Value

IEnumerable<Guid>

preItemUsable

```
public ItemUsable preItemUsable
```

Field Value

ItemUsable

Properties

TypeId

```
protected override string TypeId { get; }
```

Property Value

string

Methods

Serialize()

```
public override IValue Serialize()
```

Returns

IValue

Class ItemEnhancement11

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

Updated at https://github.com/planetarium/lib9c/pull/1164

```
[Serializable]
[ActionType("item_enhancement11")]
public class ItemEnhancement11 : GameAction, IAction, IItemEnhancementV2
```

Inheritance

object ← ActionBase ← GameAction ← ItemEnhancement11

Implements

IAction, IItemEnhancementV2

Inherited Members

GameAction.Id , GameAction.PlainValue , GameAction.LoadPlainValue(IValue) , ActionBase.GetSignerAndOtherAddressesHex(IActionContext, params Address[]) , ActionBase.TryGetAdminState(IActionContext, out AdminState) , ActionBase.CheckPermission(IActionContext) , ActionBase.CheckObsolete(long, IActionContext) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Fields

avatarAddress

```
public Address avatarAddress
```

Field Value

Address

itemId

```
public Guid itemId
```

Field Value

Guid

materialId

```
public Guid materialId
```

Field Value

Guid

slotIndex

```
public int slotIndex
```

Field Value

int

Properties

PlainValueInternal

```
protected override IImmutableDictionary<string, IValue> PlainValueInternal { get; }
```

Property Value

IImmutableDictionary<string, IValue>

Methods

Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

Parameters

`context` IActionContext

Returns

IWorld

GetEnhancementResult(Row, IRandom)

```
public static ItemEnhancement11.EnhancementResult
GetEnhancementResult(EnhancementCostSheetV2.Row row, IRandom random)
```

Parameters

`row` EnhancementCostSheetV2.Row

`random` IRandom

Returns

ItemEnhancement11.EnhancementResult

GetEquipmentMaxLevel(Equipment, EnhancementCostSheetV2)

```
public static int GetEquipmentMaxLevel(Equipment equipment,
EnhancementCostSheetV2 sheet)
```

Parameters

`equipment` Equipment

`sheet` EnhancementCostSheetV2

Returns

int

GetRequiredAp()

```
public static int GetRequiredAp()
```

Returns

int

GetRequiredBlockCount(Row, EnhancementResult)

```
public static int GetRequiredBlockCount(EnhancementCostSheetV2.Row row,
ItemEnhancement11.EnhancementResult result)
```

Parameters

`row` EnhancementCostSheetV2.Row

`result` ItemEnhancement11.EnhancementResult

Returns

int

LoadPlainValueInternal(IImmutableDictionary<string, IValue>)

```
protected override void LoadPlainValueInternal(IImmutableDictionary<string,
IValue> plainValue)
```

Parameters

`plainValue` IImmutableDictionary<string, IValue>

TryGetRow(Equipment, EnhancementCostSheetV2, out Row)

```
public static bool TryGetRow(Equipment equipment, EnhancementCostSheetV2 sheet, out
EnhancementCostSheetV2.Row row)
```

Parameters

`equipment` Equipment

`sheet` EnhancementCostSheetV2

`row` EnhancementCostSheetV2.Row

Returns

bool

Enum ItemEnhancement11.EnhancementResult

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
public enum ItemEnhancement11.EnhancementResult
```

Extension Methods

StateExtensions.Serialize(Enum)

Fields

`Fail = 2`

`GreatSuccess = 0`

`Success = 1`

Class ItemEnhancement11.ResultModel

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
public class ItemEnhancement11.ResultModel : AttachmentActionResult, IState
```

Inheritance

object ← AttachmentActionResult ← ItemEnhancement11.ResultModel

Implements

IState

Inherited Members

AttachmentActionResult.itemUsable , AttachmentActionResult.costume , AttachmentActionResult.tradableFungibleItem , AttachmentActionResult.tradableFungibleItemCount , AttachmentActionResult.SerializeBackup1() , AttachmentActionResult.Deserialize(Dictionary) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

ResultModel()

```
public ResultModel()
```

ResultModel(Dictionary)

```
public ResultModel(Dictionary serialized)
```

Parameters

`serialized` Dictionary

Fields

CRYSTAL

```
public FungibleAssetValue CRYSTAL
```

Field Value

FungibleAssetValue

actionPoint

```
public int actionPoint
```

Field Value

int

enhancementResult

```
public ItemEnhancement11.EnhancementResult enhancementResult
```

Field Value

ItemEnhancement11.EnhancementResult

gold

```
public BigInteger gold
```

Field Value

BigInteger

id

```
public Guid id
```

Field Value

Guid

materialItemIdList

```
public IEnumerable<Guid> materialItemIdList
```

Field Value

IEnumerable<Guid>

preItemUsable

```
public ItemUsable preItemUsable
```

Field Value

ItemUsable

Properties

TypeId

```
protected override string TypeId { get; }
```

Property Value

string

Methods

Serialize()

```
public override IValue Serialize()
```

Returns

IValue

Class ItemEnhancement12

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

Updated at https://github.com/planetarium/lib9c/pull/2068

```
[Serializable]
[ActionObsolete(7718878)]
[ActionType("item_enhancement12")]
public class ItemEnhancement12 : GameAction, IAction, IItemEnhancementV4
```

Inheritance

object ← ActionBase ← GameAction ← ItemEnhancement12

Implements

IAction, IItemEnhancementV4

Inherited Members

GameAction.Id , GameAction.PlainValue , GameAction.LoadPlainValue(IValue) , ActionBase.GetSignerAndOtherAddressesHex(IActionContext, params Address[]) , ActionBase.TryGetAdminState(IActionContext, out AdminState) , ActionBase.CheckPermission(IActionContext) , ActionBase.CheckObsolete(long, IActionContext) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Fields

MaterialCountLimit

```
public const int MaterialCountLimit = 50
```

Field Value

int

avatarAddress

```
public Address avatarAddress
```

Field Value

Address

itemId

```
public Guid itemId
```

Field Value

Guid

materialIds

```
public List<Guid> materialIds
```

Field Value

List<Guid>

slotIndex

```
public int slotIndex
```

Field Value

int

Properties

PlainValueInternal

```
protected override IImmutableDictionary<string, IValue> PlainValueInternal { get; }
```

Property Value

IImmutableDictionary<string, IValue>

Methods

Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

Parameters

`context` IActionContext

Returns

IWorld

GetEquipmentMaxLevel(Equipment, EnhancementCostSheetV3)

```
public static int GetEquipmentMaxLevel(Equipment equipment,
EnhancementCostSheetV3 sheet)
```

Parameters

`equipment` Equipment

`sheet` EnhancementCostSheetV3

Returns

int

GetRequiredAp()

```
public static int GetRequiredAp()
```

Returns

int

GetRequiredBlockCount(Equipment, Equipment, EnhancementCostSheetV3)

```
public static int GetRequiredBlockCount(Equipment preEquipment, Equipment
targetEquipment, EnhancementCostSheetV3 sheet)
```

Parameters

`preEquipment` Equipment

`targetEquipment` Equipment

`sheet` EnhancementCostSheetV3

Returns

int

LoadPlainValueInternal(IImmutableDictionary<string, IValue>)

```
protected override void LoadPlainValueInternal(IImmutableDictionary<string,
IValue> plainValue)
```

Parameters

`plainValue` IImmutableDictionary<string, IValue>

TryGetRow(Equipment, EnhancementCostSheetV3, out Row)

```
public static bool TryGetRow(Equipment equipment, EnhancementCostSheetV3 sheet, out
EnhancementCostSheetV3.Row row)
```

Parameters

`equipment` Equipment

`sheet` EnhancementCostSheetV3

`row` EnhancementCostSheetV3.Row

Returns

bool

Enum ItemEnhancement12.EnhancementResult

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
public enum ItemEnhancement12.EnhancementResult
```

Extension Methods

StateExtensions.Serialize(Enum)

Fields

`Success = 1`

Class ItemEnhancement12.ResultModel

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
public class ItemEnhancement12.ResultModel : AttachmentActionResult, IState
```

Inheritance

object ← AttachmentActionResult ← ItemEnhancement12.ResultModel

Implements

IState

Inherited Members

AttachmentActionResult.itemUsable , AttachmentActionResult.costume , AttachmentActionResult.tradableFungibleItem , AttachmentActionResult.tradableFungibleItemCount , AttachmentActionResult.SerializeBackup1() , AttachmentActionResult.Deserialize(Dictionary) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

ResultModel()

```
public ResultModel()
```

ResultModel(Dictionary)

```
public ResultModel(Dictionary serialized)
```

Parameters

`serialized` Dictionary

Fields

CRYSTAL

```
public FungibleAssetValue CRYSTAL
```

Field Value

FungibleAssetValue

actionPoint

```
public int actionPoint
```

Field Value

int

enhancementResult

```
public ItemEnhancement12.EnhancementResult enhancementResult
```

Field Value

ItemEnhancement12.EnhancementResult

gold

```
public BigInteger gold
```

Field Value

BigInteger

id

```
public Guid id
```

Field Value

Guid

materialItemIdList

```
public IEnumerable<Guid> materialItemIdList
```

Field Value

IEnumerable<Guid>

preItemUsable

```
public ItemUsable preItemUsable
```

Field Value

ItemUsable

Properties

TypeId

```
protected override string TypeId { get; }
```

Property Value

string

Methods

Serialize()

```
public override IValue Serialize()
```

Returns

IValue

Class ItemEnhancement13

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

Updated at https://github.com/planetarium/lib9c/pull/2068

```
[Serializable]
[ActionObsolete(8324909)]
[ActionType("item_enhancement13")]
public class ItemEnhancement13 : GameAction, IAction, IItemEnhancementV4
```

Inheritance

object ← ActionBase ← GameAction ← ItemEnhancement13

Implements

IAction, IItemEnhancementV4

Inherited Members

GameAction.Id , GameAction.PlainValue , GameAction.LoadPlainValue(IValue) , ActionBase.GetSignerAndOtherAddressesHex(IActionContext, params Address[]) , ActionBase.TryGetAdminState(IActionContext, out AdminState) , ActionBase.CheckPermission(IActionContext) , ActionBase.CheckObsolete(long, IActionContext) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Fields

MaterialCountLimit

```
public const int MaterialCountLimit = 50
```

Field Value

int

avatarAddress

```
public Address avatarAddress
```

Field Value

Address

itemId

```
public Guid itemId
```

Field Value

Guid

materialIds

```
public List<Guid> materialIds
```

Field Value

List<Guid>

slotIndex

```
public int slotIndex
```

Field Value

int

Properties

PlainValueInternal

```
protected override IImmutableDictionary<string, IValue> PlainValueInternal { get; }
```

Property Value

IImmutableDictionary<string, IValue>

Methods

Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

Parameters

`context` IActionContext

Returns

IWorld

GetEquipmentMaxLevel(Equipment, EnhancementCostSheetV3)

```
public static int GetEquipmentMaxLevel(Equipment equipment,
EnhancementCostSheetV3 sheet)
```

Parameters

`equipment` Equipment

`sheet` EnhancementCostSheetV3

Returns

int

GetRequiredAp()

```
public static int GetRequiredAp()
```

Returns

int

GetRequiredBlockCount(Equipment, Equipment, EnhancementCostSheetV3)

```
public static int GetRequiredBlockCount(Equipment preEquipment, Equipment
targetEquipment, EnhancementCostSheetV3 sheet)
```

Parameters

`preEquipment` Equipment

`targetEquipment` Equipment

`sheet` EnhancementCostSheetV3

Returns

int

LoadPlainValueInternal(IImmutableDictionary<string, IValue>)

```
protected override void LoadPlainValueInternal(IImmutableDictionary<string,
IValue> plainValue)
```

Parameters

`plainValue` IImmutableDictionary<string, IValue>

TryGetRow(Equipment, EnhancementCostSheetV3, out Row)

```
public static bool TryGetRow(Equipment equipment, EnhancementCostSheetV3 sheet, out
EnhancementCostSheetV3.Row row)
```

Parameters

`equipment` Equipment

`sheet` EnhancementCostSheetV3

`row` EnhancementCostSheetV3.Row

Returns

bool

Enum ItemEnhancement13.EnhancementResult

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
public enum ItemEnhancement13.EnhancementResult
```

Extension Methods

StateExtensions.Serialize(Enum)

Fields

`Success = 1`

Class ItemEnhancement13.ResultModel

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
public class ItemEnhancement13.ResultModel : AttachmentActionResult, IState
```

Inheritance

object ← AttachmentActionResult ← ItemEnhancement13.ResultModel

Implements

IState

Inherited Members

AttachmentActionResult.itemUsable , AttachmentActionResult.costume , AttachmentActionResult.tradableFungibleItem , AttachmentActionResult.tradableFungibleItemCount , AttachmentActionResult.SerializeBackup1() , AttachmentActionResult.Deserialize(Dictionary) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

ResultModel()

```
public ResultModel()
```

ResultModel(Dictionary)

```
public ResultModel(Dictionary serialized)
```

Parameters

`serialized` Dictionary

Fields

CRYSTAL

```
public FungibleAssetValue CRYSTAL
```

Field Value

FungibleAssetValue

actionPoint

```
public int actionPoint
```

Field Value

int

enhancementResult

```
public ItemEnhancement13.EnhancementResult enhancementResult
```

Field Value

ItemEnhancement13.EnhancementResult

gold

```
public BigInteger gold
```

Field Value

BigInteger

id

```
public Guid id
```

Field Value

Guid

materialItemIdList

```
public IEnumerable<Guid> materialItemIdList
```

Field Value

IEnumerable<Guid>

preItemUsable

```
public ItemUsable preItemUsable
```

Field Value

ItemUsable

Properties

TypeId

```
protected override string TypeId { get; }
```

Property Value

string

Methods

Serialize()

```
public override IValue Serialize()
```

Returns

IValue

Class ItemEnhancement7

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
[ActionObsolete(7000000)]
[ActionType("item_enhancement7")]
public class ItemEnhancement7 : GameAction, IAction, IItemEnhancementV2
```

Inheritance

object ← ActionBase ← GameAction ← ItemEnhancement7

Implements

IAction, IItemEnhancementV2

Inherited Members

GameAction.Id , GameAction.PlainValue , GameAction.LoadPlainValue(IValue) , ActionBase.GetSignerAndOtherAddressesHex(IActionContext, params Address[]) , ActionBase.TryGetAdminState(IActionContext, out AdminState) , ActionBase.CheckPermission(IActionContext) , ActionBase.CheckObsolete(long, IActionContext) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Fields

BlacksmithAddress

```
public static readonly Address BlacksmithAddress
```

Field Value

Address

RequiredBlockCount

```
public const int RequiredBlockCount = 1
```

Field Value

int

avatarAddress

```
public Address avatarAddress
```

Field Value

Address

itemId

```
public Guid itemId
```

Field Value

Guid

materialId

```
public Guid materialId
```

Field Value

Guid

slotIndex

```
public int slotIndex
```

Field Value

int

Properties

PlainValueInternal

```
protected override IImmutableDictionary<string, IValue> PlainValueInternal { get; }
```

Property Value

IImmutableDictionary<string, IValue>

Methods

Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

Parameters

`context` IActionContext

Returns

IWorld

GetRequiredAp()

```
public static int GetRequiredAp()
```

Returns

int

GetRequiredNCG(EnhancementCostSheet, int, int)

```
public static BigInteger GetRequiredNCG(EnhancementCostSheet costSheet, int grade,
int level)
```

Parameters

`costSheet` EnhancementCostSheet

`grade` int

`level` int

Returns

BigInteger

LoadPlainValueInternal(IImmutableDictionary<string, IValue>)

```
protected override void LoadPlainValueInternal(IImmutableDictionary<string,
IValue> plainValue)
```

Parameters

`plainValue` IImmutableDictionary<string, IValue>

UpgradeEquipment(Equipment)

```
public static Equipment UpgradeEquipment(Equipment equipment)
```

Parameters

`equipment` Equipment

Returns

Equipment

Class ItemEnhancement7.ResultModel

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
public class ItemEnhancement7.ResultModel : AttachmentActionResult, IState
```

Inheritance

object ← AttachmentActionResult ← ItemEnhancement7.ResultModel

Implements

IState

Inherited Members

AttachmentActionResult.itemUsable , AttachmentActionResult.costume , AttachmentActionResult.tradableFungibleItem , AttachmentActionResult.tradableFungibleItemCount , AttachmentActionResult.SerializeBackup1() , AttachmentActionResult.Deserialize(Dictionary) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

ResultModel()

```
public ResultModel()
```

ResultModel(Dictionary)

```
public ResultModel(Dictionary serialized)
```

Parameters

`serialized` Dictionary

Fields

actionPoint

```
public int actionPoint
```

Field Value

int

gold

```
public BigInteger gold
```

Field Value

BigInteger

id

```
public Guid id
```

Field Value

Guid

materialItemIdList

```
public IEnumerable<Guid> materialItemIdList
```

Field Value

IEnumerable<Guid>

Properties

TypeId

```
protected override string TypeId { get; }
```

Property Value

string

Methods

Serialize()

```
public override IValue Serialize()
```

Returns

IValue

Class ItemEnhancement9

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
[ActionObsolete(7000000)]
[ActionType("item_enhancement9")]
public class ItemEnhancement9 : GameAction, IAction, IItemEnhancementV2
```

Inheritance

object ← ActionBase ← GameAction ← ItemEnhancement9

Implements

IAction, IItemEnhancementV2

Inherited Members

GameAction.Id , GameAction.PlainValue , GameAction.LoadPlainValue(IValue) , ActionBase.GetSignerAndOtherAddressesHex(IActionContext, params Address[]) , ActionBase.TryGetAdminState(IActionContext, out AdminState) , ActionBase.CheckPermission(IActionContext) , ActionBase.CheckObsolete(long, IActionContext) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Fields

BlacksmithAddress

```
public static readonly Address BlacksmithAddress
```

Field Value

Address

avatarAddress

```
public Address avatarAddress
```

Field Value

Address

itemId

```
public Guid itemId
```

Field Value

Guid

materialId

```
public Guid materialId
```

Field Value

Guid

slotIndex

```
public int slotIndex
```

Field Value

int

Properties

PlainValueInternal

```
protected override IImmutableDictionary<string, IValue> PlainValueInternal { get; }
```

Property Value

IImmutableDictionary<string, IValue>

Methods

Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

Parameters

`context` IActionContext

Returns

IWorld

GetEnhancementResult(Row, IRandom)

```
public static ItemEnhancement9.EnhancementResult
GetEnhancementResult(EnhancementCostSheetV2.Row row, IRandom random)
```

Parameters

`row` EnhancementCostSheetV2.Row

`random` IRandom

Returns

ItemEnhancement9.EnhancementResult

GetEquipmentMaxLevel(Equipment, EnhancementCostSheetV2)

```
public static int GetEquipmentMaxLevel(Equipment equipment,
EnhancementCostSheetV2 sheet)
```

Parameters

`equipment` Equipment

`sheet` EnhancementCostSheetV2

Returns

int

GetRequiredAp()

```
public static int GetRequiredAp()
```

Returns

int

GetRequiredBlockCount(Row, EnhancementResult)

```
public static int GetRequiredBlockCount(EnhancementCostSheetV2.Row row,
ItemEnhancement9.EnhancementResult result)
```

Parameters

`row` EnhancementCostSheetV2.Row

`result` ItemEnhancement9.EnhancementResult

Returns

int

LoadPlainValueInternal(IImmutableDictionary<string, IValue>)

```
protected override void LoadPlainValueInternal(IImmutableDictionary<string,
IValue> plainValue)
```

Parameters

`plainValue` IImmutableDictionary<string, IValue>

TryGetRow(Equipment, EnhancementCostSheetV2, out Row)

```
public static bool TryGetRow(Equipment equipment, EnhancementCostSheetV2 sheet, out
EnhancementCostSheetV2.Row row)
```

Parameters

`equipment` Equipment

`sheet` EnhancementCostSheetV2

`row` EnhancementCostSheetV2.Row

Returns

bool

Enum ItemEnhancement9.EnhancementResult

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
public enum ItemEnhancement9.EnhancementResult
```

Extension Methods
StateExtensions.Serialize(Enum)

Fields

`Fail = 2`

`GreatSuccess = 0`

`Success = 1`

Class ItemEnhancement9.ResultModel

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
public class ItemEnhancement9.ResultModel : AttachmentActionResult, IState
```

Inheritance

object ← AttachmentActionResult ← ItemEnhancement9.ResultModel

Implements

IState

Inherited Members

AttachmentActionResult.itemUsable , AttachmentActionResult.costume , AttachmentActionResult.tradableFungibleItem , AttachmentActionResult.tradableFungibleItemCount , AttachmentActionResult.SerializeBackup1() , AttachmentActionResult.Deserialize(Dictionary) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

ResultModel()

```
public ResultModel()
```

ResultModel(Dictionary)

```
public ResultModel(Dictionary serialized)
```

Parameters

`serialized` Dictionary

Fields

actionPoint

```
public int actionPoint
```

Field Value

int

enhancementResult

```
public ItemEnhancement9.EnhancementResult enhancementResult
```

Field Value

ItemEnhancement9.EnhancementResult

gold

```
public BigInteger gold
```

Field Value

BigInteger

id

```
public Guid id
```

Field Value

Guid

materialItemIdList

```
public IEnumerable<Guid> materialItemIdList
```

Field Value

IEnumerable<Guid>

preItemUsable

```
public ItemUsable preItemUsable
```

Field Value

ItemUsable

Properties

TypeId

```
protected override string TypeId { get; }
```

Property Value

string

Methods

Serialize()

```
public override IValue Serialize()
```

Returns

IValue

Class ItemProductInfo

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
public class ItemProductInfo : IProductInfo
```

Inheritance

object ← ItemProductInfo

Implements

IProductInfo

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

ItemProductInfo()

```
public ItemProductInfo()
```

ItemProductInfo(List)

```
public ItemProductInfo(List serialized)
```

Parameters

`serialized` List

Properties

AgentAddress

```
public Address AgentAddress { get; set; }
```

Property Value

Address

AvatarAddress

```
public Address AvatarAddress { get; set; }
```

Property Value

Address

ItemSubType

```
public ItemSubType ItemSubType { get; set; }
```

Property Value

ItemSubType

Legacy

```
public bool Legacy { get; set; }
```

Property Value

bool

Price

```
public FungibleAssetValue Price { get; set; }
```

Property Value

FungibleAssetValue

ProductId

```
public Guid ProductId { get; set; }
```

Property Value

Guid

TradableId

```
public Guid TradableId { get; set; }
```

Property Value

Guid

Type

```
public ProductType Type { get; set; }
```

Property Value

ProductType

Methods

Serialize()

```
public IValue Serialize()
```

Returns

IValue

ValidateType()

```
public void ValidateType()
```

Class JoinArena

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

Introduced at https://github.com/planetarium/lib9c/pull/2195

```
[Serializable]
[ActionType("join_arena4")]
public class JoinArena : GameAction, IAction, IJoinArenaV1
```

Inheritance

object ← ActionBase ← GameAction ← JoinArena

Implements

IAction, IJoinArenaV1

Inherited Members

GameAction.Id , GameAction.PlainValue , GameAction.LoadPlainValue(IValue) , ActionBase.GetSignerAndOtherAddressesHex(IActionContext, params Address[]) , ActionBase.TryGetAdminState(IActionContext, out AdminState) , ActionBase.CheckPermission(IActionContext) , ActionBase.CheckObsolete(long, IActionContext) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Fields

avatarAddress

```
public Address avatarAddress
```

Field Value

Address

championshipId

```
public int championshipId
```

Field Value

int

costumes

```
public List<Guid> costumes
```

Field Value

List<Guid>

equipments

```
public List<Guid> equipments
```

Field Value

List<Guid>

round

```
public int round
```

Field Value

int

runeInfos

```
public List<RuneSlotInfo> runeInfos
```

Field Value

List<RuneSlotInfo>

Properties

PlainValueInternal

```
protected override IImmutableDictionary<string, IValue> PlainValueInternal { get; }
```

Property Value

IImmutableDictionary<string, IValue>

Methods

Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

Parameters

`context` IActionContext

Returns

IWorld

LoadPlainValueInternal(IImmutableDictionary<string, IValue>)

```
protected override void LoadPlainValueInternal(IImmutableDictionary<string,
```

```
IValue> plainValue)
```

Parameters

`plainValue` IImmutableDictionary<string, IValue>

Class JoinArena1

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

Introduced at https://github.com/planetarium/lib9c/pull/1495

```
[Serializable]
[ActionObsolete(7000000)]
[ActionType("join_arena")]
public class JoinArena1 : GameAction, IAction, IJoinArenaV1
```

Inheritance

object ← ActionBase ← GameAction ← JoinArena1

Implements

IAction, IJoinArenaV1

Inherited Members

GameAction.Id , GameAction.PlainValue , GameAction.LoadPlainValue(IValue) , ActionBase.GetSignerAndOtherAddressesHex(IActionContext, params Address[]) , ActionBase.TryGetAdminState(IActionContext, out AdminState) , ActionBase.CheckPermission(IActionContext) , ActionBase.CheckObsolete(long, IActionContext) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Fields

avatarAddress

```
public Address avatarAddress
```

Field Value

Address

championshipId

```
public int championshipId
```

Field Value

int

costumes

```
public List<Guid> costumes
```

Field Value

List<Guid>

equipments

```
public List<Guid> equipments
```

Field Value

List<Guid>

round

```
public int round
```

Field Value

int

Properties

PlainValueInternal

```
protected override IImmutableDictionary<string, IValue> PlainValueInternal { get; }
```

Property Value

IImmutableDictionary<string, IValue>

Methods

Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

Parameters

`context` IActionContext

Returns

IWorld

LoadPlainValueInternal(IImmutableDictionary<string, IValue>)

```
protected override void LoadPlainValueInternal(IImmutableDictionary<string,
IValue> plainValue)
```

Parameters

`plainValue` IImmutableDictionary<string, IValue>

Class JoinArena3

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

Introduced at https://github.com/planetarium/lib9c/pull/1663

```
[Serializable]
[ActionObsolete(8324909)]
[ActionType("join_arena3")]
public class JoinArena3 : GameAction, IAction, IJoinArenaV1
```

Inheritance

object ← ActionBase ← GameAction ← JoinArena3

Implements

IAction, IJoinArenaV1

Inherited Members

GameAction.Id , GameAction.PlainValue , GameAction.LoadPlainValue(IValue) , ActionBase.GetSignerAndOtherAddressesHex(IActionContext, params Address[]) , ActionBase.TryGetAdminState(IActionContext, out AdminState) , ActionBase.CheckPermission(IActionContext) , ActionBase.CheckObsolete(long, IActionContext) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Fields

avatarAddress

```
public Address avatarAddress
```

Field Value

Address

championshipId

```
public int championshipId
```

Field Value

int

costumes

```
public List<Guid> costumes
```

Field Value

List<Guid>

equipments

```
public List<Guid> equipments
```

Field Value

List<Guid>

round

```
public int round
```

Field Value

int

runeInfos

```
public List<RuneSlotInfo> runeInfos
```

Field Value

List<RuneSlotInfo>

Properties

PlainValueInternal

```
protected override IImmutableDictionary<string, IValue> PlainValueInternal { get; }
```

Property Value

IImmutableDictionary<string, IValue>

Methods

Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

Parameters

`context` IActionContext

Returns

IWorld

LoadPlainValueInternal(IImmutableDictionary<string, IValue>)

```
protected override void LoadPlainValueInternal(IImmutableDictionary<string,
```

```
IValue> plainValue)
```

Parameters

`plainValue` IImmutableDictionary<string, IValue>

Class MemoLengthOverflowException

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
public class MemoLengthOverflowException : InvalidOperationException, ISerializable
```

Inheritance

object ← Exception ← SystemException ← InvalidOperationException ← MemoLengthOverflowException

Implements

ISerializable

Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Constructors

MemoLengthOverflowException(SerializationInfo, StreamingContext)

```
protected MemoLengthOverflowException(SerializationInfo info,
StreamingContext context)
```

Parameters

`info` SerializationInfo

`context` StreamingContext

MemoLengthOverflowException(string)

```
public MemoLengthOverflowException(string message)
```

Parameters

`message` string

Class MigrateAgentAvatar

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[ActionType("migrate_agent_avatar")]
public class MigrateAgentAvatar : ActionBase, IAction
```

Inheritance

object ← ActionBase ← MigrateAgentAvatar

Implements

IAction

Inherited Members

ActionBase.GetSignerAndOtherAddressesHex(IActionContext, params Address[]) ,
ActionBase.TryGetAdminState(IActionContext, out AdminState) ,
ActionBase.CheckPermission(IActionContext) ,
ActionBase.CheckObsolete(long, IActionContext) , object.Equals(object) ,
object.Equals(object, object) , object.GetHashCode() , object.GetType() ,
object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

MigrateAgentAvatar()

```
public MigrateAgentAvatar()
```

Fields

AgentAddresses

```
public List<Address> AgentAddresses
```

Field Value

List<Address>

TypeIdentifier

```
public const string TypeIdentifier = "migrate_agent_avatar"
```

Field Value

string

Properties

PlainValue

```
public override IValue PlainValue { get; }
```

Property Value

IValue

Methods

Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

Parameters

`context` IActionContext

Returns

IWorld

LoadPlainValue(IValue)

```
public override void LoadPlainValue(IValue plainValue)
```

Parameters

`plainValue` IValue

Class MigrateFee

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[ActionType("migrate_fee")]
public class MigrateFee : ActionBase, IAction
```

Inheritance

object ← ActionBase ← MigrateFee

Implements

IAction

Inherited Members

ActionBase.GetSignerAndOtherAddressesHex(IActionContext, params Address[]) ,
ActionBase.TryGetAdminState(IActionContext, out AdminState) ,
ActionBase.CheckPermission(IActionContext) ,
ActionBase.CheckObsolete(long, IActionContext) , object.Equals(object) ,
object.Equals(object, object) , object.GetHashCode() , object.GetType() ,
object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

MigrateFee()

```
public MigrateFee()
```

Fields

Memo

```
public string Memo
```

Field Value

string

TransferData

```
public List<(Address sender, Address recipient, BigInteger amount)> TransferData
```

Field Value

List<(Address sender, Address recipient, BigInteger amount)>

TypeIdentifier

```
public const string TypeIdentifier = "migrate_fee"
```

Field Value

string

Properties

PlainValue

```
public override IValue PlainValue { get; }
```

Property Value

IValue

Methods

Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

Parameters

`context` IActionContext

Returns

IWorld

LoadPlainValue(IValue)

```
public override void LoadPlainValue(IValue plainValue)
```

Parameters

`plainValue` IValue

Class MigrateMonsterCollection

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

An action to claim remained monster collection rewards and to migrate MonsterCollectionState into LegacyStakeState without cancellation, to keep its staked period.

```
[ActionType("migrate_monster_collection")]
public class MigrateMonsterCollection : ActionBase,
IAction, IMigrateMonsterCollectionV1
```

Inheritance

object ← ActionBase ← MigrateMonsterCollection

Implements

IAction, IMigrateMonsterCollectionV1

Inherited Members

ActionBase.GetSignerAndOtherAddressesHex(IActionContext, params Address[]) , ActionBase.TryGetAdminState(IActionContext, out AdminState) , ActionBase.CheckPermission(IActionContext) , ActionBase.CheckObsolete(long, IActionContext) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

MigrateMonsterCollection()

```
public MigrateMonsterCollection()
```

MigrateMonsterCollection(Address)

```
public MigrateMonsterCollection(Address avatarAddress)
```

Parameters

`avatarAddress` Address

Properties

AvatarAddress

```
public Address AvatarAddress { get; }
```

Property Value

Address

PlainValue

```
public override IValue PlainValue { get; }
```

Property Value

IValue

Methods

Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

Parameters

`context` IActionContext

Returns

IWorld

LoadPlainValue(IValue)

```
public override void LoadPlainValue(IValue plainValue)
```

Parameters

`plainValue` IValue

Class MigrationActivatedAccountsState

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
[ActionType("migration_activated_accounts_state")]
public class MigrationActivatedAccountsState : GameAction, IAction,
IMigrationActivatedAccountsStateV1
```

Inheritance

object ← ActionBase ← GameAction ← MigrationActivatedAccountsState

Implements

IAction, IMigrationActivatedAccountsStateV1

Inherited Members

GameAction.Id , GameAction.PlainValue , GameAction.LoadPlainValue(IValue) , ActionBase.GetSignerAndOtherAddressesHex(IActionContext, params Address[]) , ActionBase.TryGetAdminState(IActionContext, out AdminState) , ActionBase.CheckPermission(IActionContext) , ActionBase.CheckObsolete(long, IActionContext) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Properties

PlainValueInternal

```
protected override IImmutableDictionary<string, IValue> PlainValueInternal { get; }
```

Property Value

IImmutableDictionary<string, IValue>

Methods

Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

Parameters

`context` IActionContext

Returns

IWorld

LoadPlainValueInternal(IImmutableDictionary<string, IValue>)

```
protected override void LoadPlainValueInternal(IImmutableDictionary<string,
IValue> plainValue)
```

Parameters

`plainValue` IImmutableDictionary<string, IValue>

Class MintAssets

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[ActionType("mint_assets")]
public class MintAssets : ActionBase, IAction
```

Inheritance

object ← ActionBase ← MintAssets

Implements

IAction

Inherited Members

ActionBase.GetSignerAndOtherAddressesHex(IActionContext, params Address[]) ,
ActionBase.TryGetAdminState(IActionContext, out AdminState) ,
ActionBase.CheckPermission(IActionContext) ,
ActionBase.CheckObsolete(long, IActionContext) , object.Equals(object) ,
object.Equals(object, object) , object.GetHashCode() , object.GetType() ,
object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

MintAssets()

```
public MintAssets()
```

MintAssets(IEnumerable<MintSpec>, string?)

```
public MintAssets(IEnumerable<MintAssets.MintSpec> specs, string? memo)
```

Parameters

`specs` IEnumerable<MintAssets.MintSpec>

memo string

Fields

TypeIdentifier

```
public const string TypeIdentifier = "mint_assets"
```

Field Value

string

Properties

Memo

```
public string? Memo { get; }
```

Property Value

string

MintSpecs

```
public List<MintAssets.MintSpec>? MintSpecs { get; }
```

Property Value

List<MintAssets.MintSpec>

PlainValue

```
public override IValue PlainValue { get; }
```

Property Value

IValue

Methods

Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

Parameters

`context` IActionContext

Returns

IWorld

LoadPlainValue(IValue)

```
public override void LoadPlainValue(IValue plainValue)
```

Parameters

`plainValue` IValue

Struct MintAssets.MintSpec

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
public readonly struct MintAssets.MintSpec
```

Inherited Members
ValueType.Equals(object) , ValueType.GetHashCode() , ValueType.ToString() , object.Equals(object, object) , object.GetType() , object.ReferenceEquals(object, object)

Constructors

MintSpec(List)

```
public MintSpec(List bencoded)
```

Parameters

`bencoded` List

MintSpec(Address, FungibleAssetValue?, FungibleItemValue?)

```
public MintSpec(Address recipient, FungibleAssetValue? assets,
FungibleItemValue? items)
```

Parameters

`recipient` Address

`assets` FungibleAssetValue?

`items` FungibleItemValue?

Properties

Assets

```
public FungibleAssetValue? Assets { get; }
```

Property Value

FungibleAssetValue?

Items

```
public FungibleItemValue? Items { get; }
```

Property Value

FungibleItemValue?

Recipient

```
public Address Recipient { get; }
```

Property Value

Address

Methods

Serialize()

```
public IValue Serialize()
```

Returns

IValue

Class MonsterCollectionExistingClaimableException

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
public class MonsterCollectionExistingClaimableException : Exception, ISerializable
```

Inheritance

object ← Exception ← MonsterCollectionExistingClaimableException

Implements

ISerializable

Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Constructors

MonsterCollectionExistingClaimableException()

```
public MonsterCollectionExistingClaimableException()
```

MonsterCollectionExistingClaimableException(SerializationInfo, StreamingContext)

```
protected MonsterCollectionExistingClaimableException(SerializationInfo info,
StreamingContext context)
```

Parameters

`info` SerializationInfo

`context` StreamingContext

MonsterCollectionExistingClaimableException(string)

```
public MonsterCollectionExistingClaimableException(string message)
```

Parameters

`message` string

MonsterCollectionExistingClaimableException(string, Exception)

```
public MonsterCollectionExistingClaimableException(string message,
Exception innerException)
```

Parameters

`message` string

`innerException` Exception

Class MonsterCollectionExistingException

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

An exception thrown when there is unexpected monster collection.

```
[Serializable]
public class MonsterCollectionExistingException : Exception, ISerializable
```

Inheritance

object ← Exception ← MonsterCollectionExistingException

Implements

ISerializable

Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Constructors

MonsterCollectionExistingException()

```
public MonsterCollectionExistingException()
```

MonsterCollectionExistingException(SerializationInfo, StreamingContext)

```
protected MonsterCollectionExistingException(SerializationInfo info,
StreamingContext context)
```

Parameters

`info` SerializationInfo

`context` StreamingContext

MonsterCollectionExistingException(string)

```
public MonsterCollectionExistingException(string message)
```

Parameters

`message` string

MonsterCollectionExistingException(string, Exception)

```
public MonsterCollectionExistingException(string message, Exception innerException)
```

Parameters

`message` string

`innerException` Exception

Class MonsterCollectionExpiredException

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
public class MonsterCollectionExpiredException : InvalidOperationException,
ISerializable
```

Inheritance

object ← Exception ← SystemException ← InvalidOperationException ← MonsterCollectionExpiredException

Implements

ISerializable

Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Constructors

MonsterCollectionExpiredException(SerializationInfo, StreamingContext)

```
protected MonsterCollectionExpiredException(SerializationInfo info,
StreamingContext context)
```

Parameters

`info` SerializationInfo

`context` StreamingContext

MonsterCollectionExpiredException(string)

```
public MonsterCollectionExpiredException(string msg)
```

Parameters

`msg` string

Class MonsterCollectionLevelException

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
public class MonsterCollectionLevelException : InvalidOperationException,
ISerializable
```

Inheritance

object ← Exception ← SystemException ← InvalidOperationException ← MonsterCollectionLevelException

Implements

ISerializable

Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Constructors

MonsterCollectionLevelException()

```
public MonsterCollectionLevelException()
```

MonsterCollectionLevelException(SerializationInfo, StreamingContext)

```
protected MonsterCollectionLevelException(SerializationInfo info,
```

```
    StreamingContext context)
```

Parameters

`info` SerializationInfo

`context` StreamingContext

MonsterCollectionLevelException(string)

```
public MonsterCollectionLevelException(string msg)
```

Parameters

`msg` string

Class NotEnoughActionPointException

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
public class NotEnoughActionPointException : Exception, ISerializable
```

Inheritance

object ← Exception ← NotEnoughActionPointException

Implements

ISerializable

Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Constructors

NotEnoughActionPointException(SerializationInfo, StreamingContext)

```
public NotEnoughActionPointException(SerializationInfo info,
StreamingContext context)
```

Parameters

`info` SerializationInfo

`context` StreamingContext

NotEnoughActionPointException(string)

```
public NotEnoughActionPointException(string msg)
```

Parameters

`msg` string

Class NotEnoughAvatarLevelException

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
public class NotEnoughAvatarLevelException : Exception, ISerializable
```

Inheritance

object ← Exception ← NotEnoughAvatarLevelException

Implements

ISerializable

Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Constructors

NotEnoughAvatarLevelException(int, bool, int, int)

```
public NotEnoughAvatarLevelException(int itemId, bool isMadeWithMimisbrunnrRecipe,
int require, int current)
```

Parameters

`itemId` int

`isMadeWithMimisbrunnrRecipe` bool

`require` int

`current` int

NotEnoughAvatarLevelException(int, int)

```
public NotEnoughAvatarLevelException(int require, int current)
```

Parameters

`require` int

`current` int

NotEnoughAvatarLevelException(SerializationInfo, StreamingContext)

```
public NotEnoughAvatarLevelException(SerializationInfo info,
StreamingContext context)
```

Parameters

`info` SerializationInfo

`context` StreamingContext

NotEnoughAvatarLevelException(string)

```
public NotEnoughAvatarLevelException(string message)
```

Parameters

`message` string

Class NotEnoughClearedStageLevelException

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
public class NotEnoughClearedStageLevelException : Exception, ISerializable
```

Inheritance

object ← Exception ← NotEnoughClearedStageLevelException

Implements

ISerializable

Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Constructors

NotEnoughClearedStageLevelException(SerializationInfo, StreamingContext)

```
public NotEnoughClearedStageLevelException(SerializationInfo info,
StreamingContext context)
```

Parameters

`info` SerializationInfo

`context` StreamingContext

NotEnoughClearedStageLevelException(string)

```
public NotEnoughClearedStageLevelException(string message)
```

Parameters

`message` string

NotEnoughClearedStageLevelException(string, int, int)

```
public NotEnoughClearedStageLevelException(string addressesHex, int require,
int current)
```

Parameters

`addressesHex` string

`require` int

`current` int

NotEnoughClearedStageLevelException(string, string, int, int)

```
public NotEnoughClearedStageLevelException(string actionType, string addressesHex,
int require, int current)
```

Parameters

`actionType` string

`addressesHex` string

`require` int

`current` int

Class NotEnoughCombatPointException

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
public class NotEnoughCombatPointException : InvalidOperationException,
ISerializable
```

Inheritance

object ← Exception ← SystemException ← InvalidOperationException ← NotEnoughCombatPointException

Implements

ISerializable

Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Constructors

NotEnoughCombatPointException(SerializationInfo, StreamingContext)

```
protected NotEnoughCombatPointException(SerializationInfo info,
StreamingContext context)
```

Parameters

`info` SerializationInfo

`context` StreamingContext

NotEnoughCombatPointException(string)

```
public NotEnoughCombatPointException(string s)
```

Parameters

s [string]

Class NotEnoughFungibleAssetValueException

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
public class NotEnoughFungibleAssetValueException : Exception, ISerializable
```

Inheritance

object ← Exception ← NotEnoughFungibleAssetValueException

Implements

ISerializable

Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Constructors

NotEnoughFungibleAssetValueException(SerializationInfo, StreamingContext)

```
public NotEnoughFungibleAssetValueException(SerializationInfo info,
StreamingContext context)
```

Parameters

`info` SerializationInfo

`context` StreamingContext

NotEnoughFungibleAssetValueException(string)

```
public NotEnoughFungibleAssetValueException(string message)
```

Parameters

`message` string

NotEnoughFungibleAssetValueException(string, Exception)

```
public NotEnoughFungibleAssetValueException(string message,
Exception innerException)
```

Parameters

`message` string

`innerException` Exception

NotEnoughFungibleAssetValueException(string, BigInteger, FungibleAssetValue)

```
public NotEnoughFungibleAssetValueException(string addressesHex, BigInteger require,
FungibleAssetValue current)
```

Parameters

`addressesHex` string

`require` BigInteger

`current` FungibleAssetValue

NotEnoughFungibleAssetValueException(string, string, FungibleAssetValue, FungibleAssetValue)

```
public NotEnoughFungibleAssetValueException(string actionType, string addressesHex,
FungibleAssetValue require, FungibleAssetValue current)
```

Parameters

`actionType` string

`addressesHex` string

`require` FungibleAssetValue

`current` FungibleAssetValue

NotEnoughFungibleAssetValueException(string, string, FungibleAssetValue, BigInteger)

```
public NotEnoughFungibleAssetValueException(string actionType, string addressesHex,
FungibleAssetValue require, BigInteger current)
```

Parameters

`actionType` string

`addressesHex` string

`require` FungibleAssetValue

`current` BigInteger

NotEnoughFungibleAssetValueException(string, string, BigInteger, FungibleAssetValue)

```
public NotEnoughFungibleAssetValueException(string actionType, string addressesHex,
BigInteger require, FungibleAssetValue current)
```

Parameters

`actionType` string

`addressesHex` string

`require` BigInteger

`current` FungibleAssetValue

NotEnoughFungibleAssetValueException(string, string, BigInteger, BigInteger)

```
public NotEnoughFungibleAssetValueException(string actionType, string addressesHex,
BigInteger require, BigInteger current)
```

Parameters

`actionType` string

`addressesHex` string

`require` BigInteger

`current` BigInteger

NotEnoughFungibleAssetValueException(string, string, string, Exception)

```
public NotEnoughFungibleAssetValueException(string addressesHex, string require,
string current, Exception innerException = null)
```

Parameters

`addressesHex` string

`require` string

`current` string

`innerException` Exception

NotEnoughFungibleAssetValueException(string, string, string, string, Exception)

```
public NotEnoughFungibleAssetValueException(string actionType, string addressesHex,
string require, string current, Exception innerException = null)
```

Parameters

`actionType` string

`addressesHex` string

`require` string

`current` string

`innerException` Exception

Class NotEnoughHammerPointException

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
public class NotEnoughHammerPointException : Exception, ISerializable
```

Inheritance

object ← Exception ← NotEnoughHammerPointException

Implements

ISerializable

Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Constructors

NotEnoughHammerPointException(SerializationInfo, StreamingContext)

```
protected NotEnoughHammerPointException(SerializationInfo info,
StreamingContext context)
```

Parameters

`info` SerializationInfo

`context` StreamingContext

NotEnoughHammerPointException(string)

```
public NotEnoughHammerPointException(string s)
```

Parameters

s [string]

Class NotEnoughMaterialException

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
public class NotEnoughMaterialException : InvalidOperationException, ISerializable
```

Inheritance

object ← Exception ← SystemException ← InvalidOperationException ← NotEnoughMaterialException

Implements

ISerializable

Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Constructors

NotEnoughMaterialException(SerializationInfo, StreamingContext)

```
protected NotEnoughMaterialException(SerializationInfo info,
StreamingContext context)
```

Parameters

`info` SerializationInfo

`context` StreamingContext

NotEnoughMaterialException(string)

```
public NotEnoughMaterialException(string s)
```

Parameters

s [string](#)

Class NotEnoughMedalException

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
public class NotEnoughMedalException : Exception, ISerializable
```

Inheritance

object ← Exception ← NotEnoughMedalException

Implements

ISerializable

Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Constructors

NotEnoughMedalException(SerializationInfo, StreamingContext)

```
public NotEnoughMedalException(SerializationInfo info, StreamingContext context)
```

Parameters

`info` SerializationInfo

`context` StreamingContext

NotEnoughMedalException(string)

```
public NotEnoughMedalException(string msg)
```

Parameters

`msg` string

Class NotEnoughRankException

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
public class NotEnoughRankException : InvalidOperationException, ISerializable
```

Inheritance

object ← Exception ← SystemException ← InvalidOperationException ← NotEnoughRankException

Implements

ISerializable

Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Constructors

NotEnoughRankException()

```
public NotEnoughRankException()
```

NotEnoughRankException(SerializationInfo, StreamingContext)

```
public NotEnoughRankException(SerializationInfo info, StreamingContext context)
```

Parameters

`info` SerializationInfo

`context` StreamingContext

NotEnoughRankException(string)

```
public NotEnoughRankException(string message)
```

Parameters

`message` string

Class NotEnoughStarException

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
public class NotEnoughStarException : Exception, ISerializable
```

Inheritance

object ← Exception ← NotEnoughStarException

Implements

ISerializable

Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Constructors

NotEnoughStarException(SerializationInfo, StreamingContext)

```
protected NotEnoughStarException(SerializationInfo info, StreamingContext context)
```

Parameters

`info` SerializationInfo

`context` StreamingContext

NotEnoughStarException(string)

```
public NotEnoughStarException(string s)
```

Parameters

s string

Class NotEnoughWeeklyArenaChallengeCountException

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
public class NotEnoughWeeklyArenaChallengeCountException : Exception, ISerializable
```

Inheritance

object ← Exception ← NotEnoughWeeklyArenaChallengeCountException

Implements

ISerializable

Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Constructors

NotEnoughWeeklyArenaChallengeCountException()

```
public NotEnoughWeeklyArenaChallengeCountException()
```

NotEnoughWeeklyArenaChallengeCountException(SerializationInfo, StreamingContext)

```
public NotEnoughWeeklyArenaChallengeCountException(SerializationInfo info,
StreamingContext context)
```

Parameters

`info` SerializationInfo

`context` StreamingContext

NotEnoughWeeklyArenaChallengeCountException(string)

```
public NotEnoughWeeklyArenaChallengeCountException(string message)
```

Parameters

`message` string

Fields

BaseMessage

```
public const string BaseMessage = "Aborted as the arena state reached the
daily limit."
```

Field Value

string

Class OrderIdDoesNotExistException

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
public class OrderIdDoesNotExistException : InvalidOperationException, ISerializable
```

Inheritance

object ← Exception ← SystemException ← InvalidOperationException ← OrderIdDoesNotExistException

Implements

ISerializable

Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Constructors

OrderIdDoesNotExistException(SerializationInfo, StreamingContext)

```
protected OrderIdDoesNotExistException(SerializationInfo info,
StreamingContext context)
```

Parameters

`info` SerializationInfo

`context` StreamingContext

OrderIdDoesNotExistException(string)

```
public OrderIdDoesNotExistException(string msg)
```

Parameters

`msg` string

Class PatchTableSheet

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

Introduced at Initial commit(2e645be18a4e2caea031c347f00777fbad5dbcc6) Updated at https://github.com/planetarium/lib9c/pull/42 Updated at https://github.com/planetarium/lib9c/pull/101 Updated at https://github.com/planetarium/lib9c/pull/287 Updated at https://github.com/planetarium/lib9c/pull/315 Updated at https://github.com/planetarium/lib9c/pull/957 Updated at https://github.com/planetarium/lib9c/pull/1560

```
[Serializable]
[ActionType("patch_table_sheet")]
public class PatchTableSheet : GameAction, IAction, IPatchTableSheetV1
```

Inheritance

object ← ActionBase ← GameAction ← PatchTableSheet

Implements

IAction, IPatchTableSheetV1

Inherited Members

GameAction.Id , GameAction.PlainValue , GameAction.LoadPlainValue(IValue) , ActionBase.GetSignerAndOtherAddressesHex(IActionContext, params Address[]) , ActionBase.TryGetAdminState(IActionContext, out AdminState) , ActionBase.CheckPermission(IActionContext) , ActionBase.CheckObsolete(long, IActionContext) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Fields

Operator

The operator address that has special permission to operation actions. When the action is signed by this operator, permission check is skipped. This is a temporary solution until

v100340, after which this concept should be eliminated or justified differently. For any questions about this account, please consult Nine Chronicles Dev team.

```
public static readonly Address Operator
```

Field Value

Address

TableCsv

```
public string TableCsv
```

Field Value

string

TableName

```
public string TableName
```

Field Value

string

Properties

PlainValueInternal

```
protected override IImmutableDictionary<string, IValue> PlainValueInternal { get; }
```

Property Value

IImmutableDictionary<string, IValue>

Methods

Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

Parameters

`context` IActionContext

Returns

IWorld

LoadPlainValueInternal(IImmutableDictionary<string, IValue>)

```
protected override void LoadPlainValueInternal(IImmutableDictionary<string,
IValue> plainValue)
```

Parameters

`plainValue` IImmutableDictionary<string, IValue>

Class PatchTableSheetCompressed

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

Compressed version of PatchTableSheet to reduce transaction size for large CSV files. Uses GZip compression to minimize the size of table data in transactions.

This action is particularly useful for large CSV files like RuneOptionSheet.csv which can be compressed from ~842KB to ~154KB (81% reduction).

Example usage:

```
var compressedAction = new PatchTableSheetCompressed
{
    TableName = "RuneOptionSheet",
    CompressedTableCsv = PatchTableSheetCompressed.CompressCsv(csvData),
};
```

```
[Serializable]
[ActionType("patch_table_sheet_compressed")]
public class PatchTableSheetCompressed : GameAction,
IAction, IPatchTableSheetCompressedV1
```

Inheritance

object ← ActionBase ← GameAction ← PatchTableSheetCompressed

Implements

IAction, IPatchTableSheetCompressedV1

Inherited Members

GameAction.Id , GameAction.PlainValue , GameAction.LoadPlainValue(IValue) , ActionBase.GetSignerAndOtherAddressesHex(IActionContext, params Address[]) , ActionBase.TryGetAdminState(IActionContext, out AdminState) , ActionBase.CheckPermission(IActionContext) , ActionBase.CheckObsolete(long, IActionContext) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Fields

Operator

The operator address that has special permission to operation actions. When the action is signed by this operator, permission check is skipped. This is a temporary solution until v100340, after which this concept should be eliminated or justified differently. For any questions about this account, please consult Nine Chronicles Dev team.

```
public static readonly Address Operator
```

Field Value

Address

Properties

CompressedTableCsv

Gets or sets the compressed CSV data as a byte array. This should be compressed using the CompressCsv method.

```
public byte[] CompressedTableCsv { get; set; }
```

Property Value

byte[]

Remarks

The compression can reduce file size by 60-80% depending on the CSV content. For example, RuneOptionSheet.csv (842KB) can be compressed to ~154KB.

PlainValueInternal

Serializes the action data for blockchain storage. Contains the table name and compressed CSV data.

```
protected override IImmutableDictionary<string, IValue> PlainValueInternal { get; }
```

Property Value

IImmutableDictionary<string, IValue>

TableName

Gets or sets the name of the table sheet to patch. This should match the name of the CSV file without the .csv extension.

```
public string TableName { get; set; }
```

Property Value

string

Examples

"RuneOptionSheet" for RuneOptionSheet.csv "ItemSheet" for ItemSheet.csv

Methods

CompressCsv(string)

Compresses CSV data using GZip compression.

This method is used to prepare CSV data for the CompressedTableCsv property. GZip compression typically reduces CSV file size by 60-80%.

Example:

```
var csvData = File.ReadAllText("RuneOptionSheet.csv");
var compressedData = PatchTableSheetCompressed.CompressCsv(csvData);
// compressedData can now be used in CompressedTableCsv property
```

```
public static byte[] CompressCsv(string csvData)
```

Parameters

`csvData` string

The CSV data to compress. Must not be null.

Returns

byte[]

Compressed byte array using GZip compression.

Exceptions

ArgumentNullException

Thrown when csvData is null.

DecompressCsv(byte[])

Decompresses CSV data using GZip compression.

This method is used internally to restore the original CSV data from the compressed byte array stored in CompressedTableCsv.

Example:

```
var originalCsv = PatchTableSheetCompressed.DecompressCsv(compressedData);
// originalCsv now contains the decompressed CSV string
```

```
public static string DecompressCsv(byte[] compressedData)
```

Parameters

`compressedData` byte[]

The compressed data to decompress. Must not be null.

Returns

string

Decompressed CSV string in UTF-8 encoding.

Exceptions

ArgumentNullException

Thrown when compressedData is null.

InvalidDataException

Thrown when the compressed data is corrupted or not valid GZip format.

Execute(IActionContext)

Executes the compressed table sheet patching action.

This method:

1. Validates permissions (unless signed by operator)
2. Decompresses the CSV data
3. Updates the table sheet state
4. Handles special cases like GameConfigSheet

The action reduces transaction size significantly compared to PatchTableSheet while maintaining the same functionality.

```
public override IWorld Execute(IActionContext context)
```

Parameters

`context` IActionContext

The action context containing state and transaction information.

Returns

IWorld

The updated world state with the patched table sheet.

Exceptions

InvalidDataException

Thrown when the compressed data is corrupted or invalid.

LoadPlainValueInternal(IImmutableDictionary<string, IValue>)

Deserializes the action data from blockchain storage. Restores the table name and compressed CSV data.

```
protected override void LoadPlainValueInternal(IImmutableDictionary<string,
IValue> plainValue)
```

Parameters

`plainValue` IImmutableDictionary<string, IValue>

The serialized action data.

Class PendingActivationDoesNotExistsException

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
public class PendingActivationDoesNotExistsException : ActivationException,
ISerializable
```

Inheritance

object ← Exception ← ActivationException ← PendingActivationDoesNotExistsException

Implements

ISerializable

Inherited Members

Exception.GetBaseException() , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Constructors

PendingActivationDoesNotExistsException(Address)

```
public PendingActivationDoesNotExistsException(Address pendingAddress)
```

Parameters

`pendingAddress` Address

PendingActivationDoesNotExistsException(SerializationInfo, StreamingContext)

```
public PendingActivationDoesNotExistsException(SerializationInfo info,
StreamingContext context)
```

Parameters

`info` SerializationInfo

`context` StreamingContext

PendingActivationDoesNotExistsException(string)

```
public PendingActivationDoesNotExistsException(string message)
```

Parameters

`message` string

Properties

PendingAddress

```
public Address PendingAddress { get; }
```

Property Value

Address

Methods

GetObjectData(SerializationInfo, StreamingContext)

When overridden in a derived class, sets the SerializationInfo with information about the exception.

```
public override void GetObjectData(SerializationInfo info, StreamingContext context)
```

Parameters

`info` SerializationInfo

The SerializationInfo that holds the serialized object data about the exception being thrown.

`context` StreamingContext

The StreamingContext that contains contextual information about the source or destination.

Exceptions

ArgumentNullException

The `info` parameter is a null reference (`Nothing` in Visual Basic).

Class PermissionDeniedException

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
public class PermissionDeniedException : AdminPermissionException, ISerializable
```

Inheritance

object ← Exception ← AdminPermissionException ← PermissionDeniedException

Implements

ISerializable

Inherited Members

AdminPermissionException.Policy , Exception.GetBaseException() , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Constructors

PermissionDeniedException(AdminState, Address)

```
public PermissionDeniedException(AdminState policy, Address signer)
```

Parameters

`policy` AdminState

`signer` Address

PermissionDeniedException(SerializationInfo, StreamingContext)

```
public PermissionDeniedException(SerializationInfo info, StreamingContext context)
```

Parameters

`info` SerializationInfo

`context` StreamingContext

PermissionDeniedException(string)

```
public PermissionDeniedException(string message)
```

Parameters

`message` string

Properties

Signer

```
public Address Signer { get; }
```

Property Value

Address

Methods

GetObjectData(SerializationInfo, StreamingContext)

When overridden in a derived class, sets the SerializationInfo with information about the exception.

```
public override void GetObjectData(SerializationInfo info, StreamingContext context)
```

Parameters

`info` SerializationInfo

The SerializationInfo that holds the serialized object data about the exception being thrown.

`context` StreamingContext

The StreamingContext that contains contextual information about the source or destination.

Exceptions

ArgumentNullException

The `info` parameter is a null reference (`Nothing` in Visual Basic).

Class PetEnhancement

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
[ActionType("pet_enhancement")]
public class PetEnhancement : GameAction, IAction
```

Inheritance

object ← ActionBase ← GameAction ← PetEnhancement

Implements

IAction

Inherited Members

GameAction.Id , GameAction.PlainValue , GameAction.LoadPlainValue(IValue) , ActionBase.GetSignerAndOtherAddressesHex(IActionContext, params Address[]) , ActionBase.TryGetAdminState(IActionContext, out AdminState) , ActionBase.CheckPermission(IActionContext) , ActionBase.CheckObsolete(long, IActionContext) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Fields

ActionTypeIdentifier

```
public const string ActionTypeIdentifier = "pet_enhancement"
```

Field Value

string

AvatarAddress

```
public Address AvatarAddress
```

Field Value

Address

PetId

```
public int PetId
```

Field Value

int

TargetLevel

```
public int TargetLevel
```

Field Value

int

Properties

PlainValueInternal

```
protected override IImmutableDictionary<string, IValue> PlainValueInternal { get; }
```

Property Value

IImmutableDictionary<string, IValue>

Methods

Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

Parameters

`context` IActionContext

Returns

IWorld

LoadPlainValueInternal(IImmutableDictionary<string, IValue>)

```
protected override void LoadPlainValueInternal(IImmutableDictionary<string,
IValue> plainValue)
```

Parameters

`plainValue` IImmutableDictionary<string, IValue>

Class PetIsLockedException

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
public class PetIsLockedException : Exception, ISerializable
```

Inheritance

object ← Exception ← PetIsLockedException

Implements

ISerializable

Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Constructors

PetIsLockedException(SerializationInfo, StreamingContext)

```
public PetIsLockedException(SerializationInfo info, StreamingContext context)
```

Parameters

`info` SerializationInfo

`context` StreamingContext

PetIsLockedException(string)

```
public PetIsLockedException(string msg)
```

Parameters

`msg` string

Class PlayCountIsZeroException

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
public class PlayCountIsZeroException : Exception, ISerializable
```

Inheritance

object ← Exception ← PlayCountIsZeroException

Implements

ISerializable

Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Constructors

PlayCountIsZeroException(SerializationInfo, StreamingContext)

```
public PlayCountIsZeroException(SerializationInfo info, StreamingContext context)
```

Parameters

`info` SerializationInfo

`context` StreamingContext

PlayCountIsZeroException(string)

```
public PlayCountIsZeroException(string msg)
```

Parameters

`msg` string

Class PolicyExpiredException

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
public class PolicyExpiredException : AdminPermissionException, ISerializable
```

Inheritance

object ← Exception ← AdminPermissionException ← PolicyExpiredException

Implements

ISerializable

Inherited Members

AdminPermissionException.Policy , Exception.GetBaseException() , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Constructors

PolicyExpiredException(AdminState, long)

```
public PolicyExpiredException(AdminState policy, long blockIndex)
```

Parameters

`policy` AdminState

`blockIndex` long

PolicyExpiredException(SerializationInfo, StreamingContext)

```
public PolicyExpiredException(SerializationInfo info, StreamingContext context)
```

Parameters

`info` SerializationInfo

`context` StreamingContext

PolicyExpiredException(string)

```
public PolicyExpiredException(string message)
```

Parameters

`message` string

Properties

BlockIndex

```
public long BlockIndex { get; }
```

Property Value

long

Methods

GetObjectData(SerializationInfo, StreamingContext)

When overridden in a derived class, sets the SerializationInfo with information about the exception.

```
public override void GetObjectData(SerializationInfo info, StreamingContext context)
```

Parameters

`info` SerializationInfo

The SerializationInfo that holds the serialized object data about the exception being thrown.

`context` StreamingContext

The StreamingContext that contains contextual information about the source or destination.

Exceptions

ArgumentNullException

The `info` parameter is a null reference (`Nothing` in Visual Basic).

Class PrepareRewardAssets

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[ActionType("prepare_reward_assets")]
public class PrepareRewardAssets : ActionBase, IAction, IPrepareRewardAssetsV1
```

Inheritance

object ← ActionBase ← PrepareRewardAssets

Implements

IAction, IPrepareRewardAssetsV1

Inherited Members

ActionBase.GetSignerAndOtherAddressesHex(IActionContext, params Address[]) ,
ActionBase.TryGetAdminState(IActionContext, out AdminState) ,
ActionBase.CheckPermission(IActionContext) ,
ActionBase.CheckObsolete(long, IActionContext) , object.Equals(object) ,
object.Equals(object, object) , object.GetHashCode() , object.GetType() ,
object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

PrepareRewardAssets()

```
public PrepareRewardAssets()
```

PrepareRewardAssets(Address, List<FungibleAssetValue>)

```
public PrepareRewardAssets(Address rewardPoolAddress, List<FungibleAssetValue>
assets)
```

Parameters

`rewardPoolAddress` Address

`assets` List<FungibleAssetValue>

Fields

Assets

```
public List<FungibleAssetValue> Assets
```

Field Value

List<FungibleAssetValue>

RewardPoolAddress

```
public Address RewardPoolAddress
```

Field Value

Address

Properties

PlainValue

```
public override IValue PlainValue { get; }
```

Property Value

IValue

Methods

Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

Parameters

`context` IActionContext

Returns

IWorld

LoadPlainValue(IValue)

```
public override void LoadPlainValue(IValue plainValue)
```

Parameters

`plainValue` IValue

Class ProductNotFoundException

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
public class ProductNotFoundException : InvalidOperationException, ISerializable
```

Inheritance

object ← Exception ← SystemException ← InvalidOperationException ← ProductNotFoundException

Implements

ISerializable

Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Constructors

ProductNotFoundException(SerializationInfo, StreamingContext)

```
public ProductNotFoundException(SerializationInfo info, StreamingContext context)
```

Parameters

`info` SerializationInfo

`context` StreamingContext

ProductNotFoundException(string)

```
public ProductNotFoundException(string msg)
```

Parameters

`msg` string

Class PurchaseInfo

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
public class PurchaseInfo : IComparable<PurchaseInfo>, IComparable, IPurchaseInfo
```

Inheritance

object ← PurchaseInfo

Implements

IComparable<PurchaseInfo>, IComparable, IPurchaseInfo

Inherited Members

object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

PurchaseInfo(Dictionary)

```
public PurchaseInfo(Dictionary serialized)
```

Parameters

`serialized` Dictionary

PurchaseInfo(Guid, Guid, Address, Address, ItemSubType, FungibleAssetValue)

```
public PurchaseInfo(Guid orderId, Guid tradableId, Address agentAddress, Address
avatarAddress, ItemSubType type, FungibleAssetValue itemPrice)
```

Parameters

`orderId` Guid

`tradableId` Guid

`agentAddress` Address

`avatarAddress` Address

`type` ItemSubType

`itemPrice` FungibleAssetValue

Fields

OrderId

```
public readonly Guid OrderId
```

Field Value

Guid

TradableId

```
public readonly Guid TradableId
```

Field Value

Guid

Properties

ItemSubType

```
public ItemSubType ItemSubType { get; }
```

Property Value

ItemSubType

Price

```
public FungibleAssetValue Price { get; }
```

Property Value

FungibleAssetValue

SellerAgentAddress

```
public Address SellerAgentAddress { get; }
```

Property Value

Address

SellerAvatarAddress

```
public Address SellerAvatarAddress { get; }
```

Property Value

Address

Methods

CompareTo(PurchaseInfo)

Compares the current instance with another object of the same type and returns an integer that indicates whether the current instance precedes, follows, or occurs in the same position in the sort order as the other object.

```
public int CompareTo(PurchaseInfo other)
```

Parameters

`other` PurchaseInfo

An object to compare with this instance.

Returns

int

A value that indicates the relative order of the objects being compared. The return value has these meanings:

Value	Meaning
Less than zero	This instance precedes `other` in the sort order.
Zero	This instance occurs in the same position in the sort order as `other`.
Greater than zero	This instance follows `other` in the sort order.

CompareTo(object)

Compares the current instance with another object of the same type and returns an integer that indicates whether the current instance precedes, follows, or occurs in the same position in the sort order as the other object.

```
public int CompareTo(object obj)
```

Parameters

`obj` object

An object to compare with this instance.

Returns

int

A value that indicates the relative order of the objects being compared. The return value has these meanings:

Value	Meaning
Less than zero	This instance precedes `obj` in the sort order.
Zero	This instance occurs in the same position in the sort order as `obj`.
Greater than zero	This instance follows `obj` in the sort order.

Exceptions

ArgumentException

`obj` is not the same type as this instance.

Equals(PurchaseInfo)

```
protected bool Equals(PurchaseInfo other)
```

Parameters

`other` PurchaseInfo

Returns

bool

Equals(object)

Determines whether the specified object is equal to the current object.

```
public override bool Equals(object obj)
```

Parameters

`obj` object

The object to compare with the current object.

Returns

bool

true if the specified object is equal to the current object; otherwise, false.

GetHashCode()

Serves as the default hash function.

```
public override int GetHashCode()
```

Returns

int

A hash code for the current object.

Serialize()

```
public IValue Serialize()
```

Returns

IValue

Operators

operator ==(PurchaseInfo, PurchaseInfo)

```
public static bool operator ==(PurchaseInfo left, PurchaseInfo right)
```

Parameters

`left` PurchaseInfo

`right` PurchaseInfo

Returns

bool

operator >(PurchaseInfo, PurchaseInfo)

```
public static bool operator >(PurchaseInfo left, PurchaseInfo right)
```

Parameters

`left` PurchaseInfo

`right` PurchaseInfo

Returns

bool

operator >=(PurchaseInfo, PurchaseInfo)

```
public static bool operator >=(PurchaseInfo left, PurchaseInfo right)
```

Parameters

`left` PurchaseInfo

`right` PurchaseInfo

Returns

bool

operator !=(PurchaseInfo, PurchaseInfo)

```
public static bool operator !=(PurchaseInfo left, PurchaseInfo right)
```

Parameters

`left` PurchaseInfo

`right` PurchaseInfo

Returns

bool

operator <(PurchaseInfo, PurchaseInfo)

```
public static bool operator <(PurchaseInfo left, PurchaseInfo right)
```

Parameters

`left` PurchaseInfo

`right` PurchaseInfo

Returns

bool

operator <=(PurchaseInfo, PurchaseInfo)

```
public static bool operator <=(PurchaseInfo left, PurchaseInfo right)
```

Parameters

`left` PurchaseInfo

`right` PurchaseInfo

Returns

bool

Class PurchaseInfo0

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
public class PurchaseInfo0 : IComparable<PurchaseInfo0>, IComparable, IPurchaseInfo
```

Inheritance

object ← PurchaseInfo0

Implements

IComparable<PurchaseInfo0>, IComparable, IPurchaseInfo

Inherited Members

object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

PurchaseInfo0(Dictionary)

```
public PurchaseInfo0(Dictionary serialized)
```

Parameters

`serialized` Dictionary

PurchaseInfo0(Guid, Address, Address, ItemSubType, FungibleAssetValue)

```
public PurchaseInfo0(Guid id, Address agentAddress, Address avatarAddress,
ItemSubType type, FungibleAssetValue itemPrice = default)
```

Parameters

`id` Guid

`agentAddress` Address

`avatarAddress` Address

`type` ItemSubType

`itemPrice` FungibleAssetValue

Fields

itemSubType

```
public readonly ItemSubType itemSubType
```

Field Value

ItemSubType

price

```
public readonly FungibleAssetValue price
```

Field Value

FungibleAssetValue

productId

```
public readonly Guid productId
```

Field Value

Guid

sellerAgentAddress

```
public readonly Address sellerAgentAddress
```

Field Value

Address

sellerAvatarAddress

```
public readonly Address sellerAvatarAddress
```

Field Value

Address

Methods

CompareTo(PurchaseInfo0)

Compares the current instance with another object of the same type and returns an integer that indicates whether the current instance precedes, follows, or occurs in the same position in the sort order as the other object.

```
public int CompareTo(PurchaseInfo0 other)
```

Parameters

`other` PurchaseInfo0

An object to compare with this instance.

Returns

int

A value that indicates the relative order of the objects being compared. The return value has these meanings:

Value	Meaning
Less than zero	This instance precedes `other` in the sort order.
Zero	This instance occurs in the same position in the sort order as `other`.
Greater than zero	This instance follows `other` in the sort order.

CompareTo(object)

Compares the current instance with another object of the same type and returns an integer that indicates whether the current instance precedes, follows, or occurs in the same position in the sort order as the other object.

```
public int CompareTo(object obj)
```

Parameters

`obj` object

An object to compare with this instance.

Returns

int

A value that indicates the relative order of the objects being compared. The return value has these meanings:

Value	Meaning
Less than zero	This instance precedes `obj` in the sort order.
Zero	This instance occurs in the same position in the sort order as `obj`.

Value	Meaning
Greater than zero	This instance follows `obj` in the sort order.

Exceptions

ArgumentException

`obj` is not the same type as this instance.

Equals(PurchaseInfo0)

```
protected bool Equals(PurchaseInfo0 other)
```

Parameters

`other` PurchaseInfo0

Returns

bool

Equals(object)

Determines whether the specified object is equal to the current object.

```
public override bool Equals(object obj)
```

Parameters

`obj` object

The object to compare with the current object.

Returns

bool

true if the specified object is equal to the current object; otherwise, false.

GetHashCode()

Serves as the default hash function.

```
public override int GetHashCode()
```

Returns

int

A hash code for the current object.

Serialize()

```
public IValue Serialize()
```

Returns

IValue

Operators

operator ==(PurchaseInfo0, PurchaseInfo0)

```
public static bool operator ==(PurchaseInfo0 left, PurchaseInfo0 right)
```

Parameters

`left` PurchaseInfo0

`right` PurchaseInfo0

Returns

bool

operator >(PurchaseInfo0, PurchaseInfo0)

```
public static bool operator >(PurchaseInfo0 left, PurchaseInfo0 right)
```

Parameters

`left` PurchaseInfo0

`right` PurchaseInfo0

Returns

bool

operator >=(PurchaseInfo0, PurchaseInfo0)

```
public static bool operator >=(PurchaseInfo0 left, PurchaseInfo0 right)
```

Parameters

`left` PurchaseInfo0

`right` PurchaseInfo0

Returns

bool

operator !=(PurchaseInfo0, PurchaseInfo0)

```
public static bool operator !=(PurchaseInfo0 left, PurchaseInfo0 right)
```

Parameters

`left` PurchaseInfo0

`right` PurchaseInfo0

Returns

bool

operator <(PurchaseInfo0, PurchaseInfo0)

```
public static bool operator <(PurchaseInfo0 left, PurchaseInfo0 right)
```

Parameters

`left` PurchaseInfo0

`right` PurchaseInfo0

Returns

bool

operator <=(PurchaseInfo0, PurchaseInfo0)

```
public static bool operator <=(PurchaseInfo0 left, PurchaseInfo0 right)
```

Parameters

`left` PurchaseInfo0

`right` PurchaseInfo0

Returns

bool

Class Raid

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

Represents a raid action where players participate in a world boss battle. This action modifies the following states:

- AvatarState: Updates player's stats, inventory, and experience
- WorldBossKillRewardRecord: Records boss kill rewards
- CollectionState: Updates collection progress
- RuneState: Updates rune effects and bonuses
- ItemSlotState: Updates equipped items
- RuneSlotState: Updates equipped runes
- CP: Updates character's combat power

```
[Serializable]
[ActionType("raid7")]
public class Raid : GameAction, IAction, IRaidV2
```

Inheritance

object ← ActionBase ← GameAction ← Raid

Implements

IAction, IRaidV2

Inherited Members

GameAction.Id , GameAction.PlainValue , GameAction.LoadPlainValue(IValue) , ActionBase.GetSignerAndOtherAddressesHex(IActionContext, params Address[]) , ActionBase.TryGetAdminState(IActionContext, out AdminState) , ActionBase.CheckPermission(IActionContext) , ActionBase.CheckObsolete(long, IActionContext) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Fields

AvatarAddress

```
public Address AvatarAddress
```

Field Value

Address

CostumeIds

```
public List<Guid> CostumeIds
```

Field Value

List<Guid>

EquipmentIds

```
public List<Guid> EquipmentIds
```

Field Value

List<Guid>

FoodIds

```
public List<Guid> FoodIds
```

Field Value

List<Guid>

PayNcg

```
public bool PayNcg
```

Field Value

bool

RuneInfos

```
public List<RuneSlotInfo> RuneInfos
```

Field Value

List<RuneSlotInfo>

Properties

PlainValueInternal

```
protected override IImmutableDictionary<string, IValue> PlainValueInternal { get; }
```

Property Value

IImmutableDictionary<string, IValue>

Methods

Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

Parameters

`context` IActionContext

Returns

IWorld

LoadPlainValueInternal(IImmutableDictionary<string, IValue>)

```
protected override void LoadPlainValueInternal(IImmutableDictionary<string,
IValue> plainValue)
```

Parameters

`plainValue` IImmutableDictionary<string, IValue>

Class RankingBattle

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

Hard forked at https://github.com/planetarium/lib9c/pull/941 Updated at https://github.com/planetarium/lib9c/pull/1135

```
[Serializable]
[ActionObsolete(8324909)]
[ActionType("ranking_battle12")]
public class RankingBattle : GameAction, IAction, IRankingBattleV2
```

Inheritance

object ← ActionBase ← GameAction ← RankingBattle

Implements

IAction, IRankingBattleV2

Inherited Members

GameAction.Id , GameAction.PlainValue , GameAction.LoadPlainValue(IValue) , ActionBase.GetSignerAndOtherAddressesHex(IActionContext, params Address[]) , ActionBase.TryGetAdminState(IActionContext, out AdminState) , ActionBase.CheckPermission(IActionContext) , ActionBase.CheckObsolete(long, IActionContext) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Fields

PreviousArenaInfo

```
public ArenaInfo PreviousArenaInfo
```

Field Value

ArenaInfo

PreviousEnemyArenaInfo

```
public ArenaInfo PreviousEnemyArenaInfo
```

Field Value

ArenaInfo

PreviousEnemyPlayerDigest

```
public EnemyPlayerDigest PreviousEnemyPlayerDigest
```

Field Value

EnemyPlayerDigest

StageId

```
public const int StageId = 999999
```

Field Value

int

avatarAddress

```
public Address avatarAddress
```

Field Value

Address

costumeIds

```
public List<Guid> costumeIds
```

Field Value

List<Guid>

enemyAddress

```
public Address enemyAddress
```

Field Value

Address

equipmentIds

```
public List<Guid> equipmentIds
```

Field Value

List<Guid>

weeklyArenaAddress

```
public Address weeklyArenaAddress
```

Field Value

Address

Properties

PlainValueInternal

```
protected override IImmutableDictionary<string, IValue> PlainValueInternal { get; }
```

Property Value

IImmutableDictionary<string, IValue>

Methods

Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

Parameters

`context` IActionContext

Returns

IWorld

LoadPlainValueInternal(IImmutableDictionary<string, IValue>)

```
protected override void LoadPlainValueInternal(IImmutableDictionary<string,
IValue> plainValue)
```

Parameters

`plainValue` IImmutableDictionary<string, IValue>

Class RankingBattle11

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

Updated at https://github.com/planetarium/lib9c/pull/1176

```
[Serializable]
[ActionType("ranking_battle11")]
[ActionObsolete(7200000)]
public class RankingBattle11 : GameAction, IAction, IRankingBattleV2
```

Inheritance

object ← ActionBase ← GameAction ← RankingBattle11

Implements

IAction, IRankingBattleV2

Inherited Members

GameAction.Id , GameAction.PlainValue , GameAction.LoadPlainValue(IValue) , ActionBase.GetSignerAndOtherAddressesHex(IActionContext, params Address[]) , ActionBase.TryGetAdminState(IActionContext, out AdminState) , ActionBase.CheckPermission(IActionContext) , ActionBase.CheckObsolete(long, IActionContext) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Fields

ArenaInfo

```
public ArenaInfo ArenaInfo
```

Field Value

ArenaInfo

EnemyArenaInfo

```
public ArenaInfo EnemyArenaInfo
```

Field Value

ArenaInfo

EnemyPlayerDigest

```
public EnemyPlayerDigest EnemyPlayerDigest
```

Field Value

EnemyPlayerDigest

EntranceFee

```
public static readonly BigInteger EntranceFee
```

Field Value

BigInteger

StageId

```
public const int StageId = 999999
```

Field Value

int

UpdateTargetBlockIndex

```
public const long UpdateTargetBlockIndex = 3808000
```

Field Value

long

UpdateTargetWeeklyArenaIndex

```
public const int UpdateTargetWeeklyArenaIndex = 68
```

Field Value

int

avatarAddress

```
public Address avatarAddress
```

Field Value

Address

costumeIds

```
public List<Guid> costumeIds
```

Field Value

List<Guid>

enemyAddress

```
public Address enemyAddress
```

Field Value

Address

equipmentIds

```
public List<Guid> equipmentIds
```

Field Value

List<Guid>

weeklyArenaAddress

```
public Address weeklyArenaAddress
```

Field Value

Address

Properties

PlainValueInternal

```
protected override IImmutableDictionary<string, IValue> PlainValueInternal { get; }
```

Property Value

IImmutableDictionary<string, IValue>

Methods

Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

Parameters

`context` IActionContext

Returns

IWorld

LoadPlainValueInternal(IImmutableDictionary<string, IValue>)

```
protected override void LoadPlainValueInternal(IImmutableDictionary<string,
IValue> plainValue)
```

Parameters

`plainValue` IImmutableDictionary<string, IValue>

Class RankingExceededException

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
public class RankingExceededException : InvalidOperationException, ISerializable
```

Inheritance

object ← Exception ← SystemException ← InvalidOperationException ← RankingExceededException

Implements

ISerializable

Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Constructors

RankingExceededException()

```
public RankingExceededException()
```

RankingExceededException(SerializationInfo, StreamingContext)

```
protected RankingExceededException(SerializationInfo info, StreamingContext context)
```

Parameters

`info` SerializationInfo

`context` StreamingContext

RankingExceededException(string)

```
public RankingExceededException(string message)
```

Parameters

`message` string

Class RapidCombination

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

Hard forked at https://github.com/planetarium/lib9c/pull/2195

```
[Serializable]
[ActionType("rapid_combination10")]
public class RapidCombination : GameAction, IAction, IRapidCombinationV2
```

Inheritance

object ← ActionBase ← GameAction ← RapidCombination

Implements

IAction, IRapidCombinationV2

Inherited Members

GameAction.Id , GameAction.PlainValue , GameAction.LoadPlainValue(IValue) , ActionBase.GetSignerAndOtherAddressesHex(IActionContext, params Address[]) , ActionBase.TryGetAdminState(IActionContext, out AdminState) , ActionBase.CheckPermission(IActionContext) , ActionBase.CheckObsolete(long, IActionContext) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Fields

avatarAddress

```
public Address avatarAddress
```

Field Value

Address

slotIndexList

```
public List<int> slotIndexList
```

Field Value

List<int>

Properties

PlainValueInternal

```
protected override IImmutableDictionary<string, IValue> PlainValueInternal { get; }
```

Property Value

IImmutableDictionary<string, IValue>

Methods

Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

Parameters

`context` IActionContext

Returns

IWorld

LoadPlainValueInternal(IImmutableDictionary<string, IValue>)

```
protected override void LoadPlainValueInternal(IImmutableDictionary<string,
IValue> plainValue)
```

Parameters

`plainValue` IImmutableDictionary<string, IValue>

Class RapidCombination0

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
[ActionObsolete(7000000)]
[ActionType("rapid_combination")]
public class RapidCombination0 : GameAction, IAction, IRapidCombinationV1
```

Inheritance

object ← ActionBase ← GameAction ← RapidCombination0

Implements

IAction, IRapidCombinationV1

Inherited Members

GameAction.Id , GameAction.PlainValue , GameAction.LoadPlainValue(IValue) , ActionBase.GetSignerAndOtherAddressesHex(IActionContext, params Address[]) , ActionBase.TryGetAdminState(IActionContext, out AdminState) , ActionBase.CheckPermission(IActionContext) , ActionBase.CheckObsolete(long, IActionContext) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Fields

avatarAddress

```
public Address avatarAddress
```

Field Value

Address

slotIndex

```
public int slotIndex
```

Field Value

int

Properties

PlainValueInternal

```
protected override IImmutableDictionary<string, IValue> PlainValueInternal { get; }
```

Property Value

IImmutableDictionary<string, IValue>

Methods

CalculateHourglassCount(GameConfigState, long)

```
public static int CalculateHourglassCount(GameConfigState state, long diff)
```

Parameters

`state` GameConfigState

`diff` long

Returns

int

CalculateHourglassCount(decimal, long)

```
public static int CalculateHourglassCount(decimal hourglassPerBlock, long diff)
```

Parameters

`hourglassPerBlock` decimal

`diff` long

Returns

int

Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

Parameters

`context` IActionContext

Returns

IWorld

LoadPlainValueInternal(IImmutableDictionary<string, IValue>)

```
protected override void LoadPlainValueInternal(IImmutableDictionary<string,
IValue> plainValue)
```

Parameters

`plainValue` IImmutableDictionary<string, IValue>

Class RapidCombination0.ResultModel

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
public class RapidCombination0.ResultModel : CombinationConsumable5.ResultModel,
IState
```

Inheritance

object ← AttachmentActionResult ← CombinationConsumable5.ResultModel ← RapidCombination0.ResultModel

Implements

IState

Inherited Members

CombinationConsumable5.ResultModel.materials , CombinationConsumable5.ResultModel.id , CombinationConsumable5.ResultModel.gold , CombinationConsumable5.ResultModel.actionPoint , CombinationConsumable5.ResultModel.recipeId , CombinationConsumable5.ResultModel.subRecipeId , CombinationConsumable5.ResultModel.itemType , AttachmentActionResult.itemUsable , AttachmentActionResult.costume , AttachmentActionResult.tradableFungibleItem , AttachmentActionResult.tradableFungibleItemCount , AttachmentActionResult.SerializeBackup1() , AttachmentActionResult.Deserialize(Dictionary) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

ResultModel(Dictionary)

```
public ResultModel(Dictionary serialized)
```

Parameters

`serialized` Dictionary

Fields

cost

```
public Dictionary<Material, int> cost
```

Field Value

Dictionary<Material, int>

Properties

TypeId

```
protected override string TypeId { get; }
```

Property Value

string

Methods

Serialize()

```
public override IValue Serialize()
```

Returns

IValue

Class RapidCombination5

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
[ActionObsolete(7000000)]
[ActionType("rapid_combination5")]
public class RapidCombination5 : GameAction, IAction, IRapidCombinationV1
```

Inheritance

object ← ActionBase ← GameAction ← RapidCombination5

Implements

IAction, IRapidCombinationV1

Inherited Members

GameAction.Id , GameAction.PlainValue , GameAction.LoadPlainValue(IValue) , ActionBase.GetSignerAndOtherAddressesHex(IActionContext, params Address[]) , ActionBase.TryGetAdminState(IActionContext, out AdminState) , ActionBase.CheckPermission(IActionContext) , ActionBase.CheckObsolete(long, IActionContext) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Fields

avatarAddress

```
public Address avatarAddress
```

Field Value

Address

slotIndex

```
public int slotIndex
```

Field Value

int

Properties

PlainValueInternal

```
protected override IImmutableDictionary<string, IValue> PlainValueInternal { get; }
```

Property Value

IImmutableDictionary<string, IValue>

Methods

Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

Parameters

`context` IActionContext

Returns

IWorld

LoadPlainValueInternal(IImmutableDictionary<string, IValue>)

```
protected override void LoadPlainValueInternal(IImmutableDictionary<string,
```

```
IValue> plainValue)
```

Parameters

`plainValue` IImmutableDictionary<string, IValue>

Class RapidCombination5.ResultModel

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
public class RapidCombination5.ResultModel : AttachmentActionResult, IState
```

Inheritance

object ← AttachmentActionResult ← RapidCombination5.ResultModel

Implements

IState

Inherited Members

AttachmentActionResult.itemUsable , AttachmentActionResult.costume , AttachmentActionResult.tradableFungibleItem , AttachmentActionResult.tradableFungibleItemCount , AttachmentActionResult.SerializeBackup1() , AttachmentActionResult.Deserialize(Dictionary) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

ResultModel(Dictionary)

```
public ResultModel(Dictionary serialized)
```

Parameters

`serialized` Dictionary

Fields

cost

```
public Dictionary<Material, int> cost
```

Field Value

Dictionary<Material, int>

id

```
public Guid id
```

Field Value

Guid

Properties

TypeId

```
protected override string TypeId { get; }
```

Property Value

string

Methods

Serialize()

```
public override IValue Serialize()
```

Returns

IValue

Class ReRegisterProduct

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[ActionType("re_register_product2")]
public class ReRegisterProduct : GameAction, IAction
```

Inheritance

object ← ActionBase ← GameAction ← ReRegisterProduct

Implements

IAction

Inherited Members

GameAction.Id , GameAction.PlainValue , GameAction.LoadPlainValue(IValue) , ActionBase.GetSignerAndOtherAddressesHex(IActionContext, params Address[]) , ActionBase.TryGetAdminState(IActionContext, out AdminState) , ActionBase.CheckPermission(IActionContext) , ActionBase.CheckObsolete(long, IActionContext) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Fields

AvatarAddress

```
public Address AvatarAddress
```

Field Value

Address

Capacity

```
public const int Capacity = 100
```

Field Value

int

ChargeAp

```
public bool ChargeAp
```

Field Value

bool

CostAp

```
public const int CostAp = 5
```

Field Value

int

ReRegisterInfos

```
public List<(IProductInfo, IRegisterInfo)> ReRegisterInfos
```

Field Value

List<(IProductInfo, IRegisterInfo)>

Properties

PlainValueInternal

```
protected override IImmutableDictionary<string, IValue> PlainValueInternal { get; }
```

Property Value

IImmutableDictionary<string, IValue>

Methods

Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

Parameters

`context` IActionContext

Returns

IWorld

LoadPlainValueInternal(IImmutableDictionary<string, IValue>)

```
protected override void LoadPlainValueInternal(IImmutableDictionary<string,
IValue> plainValue)
```

Parameters

`plainValue` IImmutableDictionary<string, IValue>

Class RedeemCode

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

Hard forked at https://github.com/planetarium/lib9c/pull/602 Updated at https://github.com/planetarium/lib9c/pull/861 Updated at https://github.com/planetarium/lib9c/pull/957

```
[Serializable]
[ActionType("redeem_code3")]
public class RedeemCode : GameAction, IAction, IRedeemCodeV1
```

Inheritance

object ← ActionBase ← GameAction ← RedeemCode

Implements

IAction, IRedeemCodeV1

Inherited Members

GameAction.Id , GameAction.PlainValue , GameAction.LoadPlainValue(IValue) , ActionBase.GetSignerAndOtherAddressesHex(IActionContext, params Address[]) , ActionBase.TryGetAdminState(IActionContext, out AdminState) , ActionBase.CheckPermission(IActionContext) , ActionBase.CheckObsolete(long, IActionContext) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

RedeemCode()

```
public RedeemCode()
```

RedeemCode(string, Address)

```
public RedeemCode(string code, Address avatarAddress)
```

Parameters

`code` string

`avatarAddress` Address

Properties

AvatarAddress

```
public Address AvatarAddress { get; }
```

Property Value

Address

Code

```
public string Code { get; }
```

Property Value

string

PlainValueInternal

```
protected override IImmutableDictionary<string, IValue> PlainValueInternal { get; }
```

Property Value

IImmutableDictionary<string, IValue>

Methods

Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

Parameters

`context` IActionContext

Returns

IWorld

LoadPlainValueInternal(IImmutableDictionary<string, IValue>)

```
protected override void LoadPlainValueInternal(IImmutableDictionary<string,
IValue> plainValue)
```

Parameters

`plainValue` IImmutableDictionary<string, IValue>

Class RedeemCode2

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
[ActionObsolete(7000000)]
[ActionType("redeem_code2")]
public class RedeemCode2 : GameAction, IAction, IRedeemCodeV1
```

Inheritance

object ← ActionBase ← GameAction ← RedeemCode2

Implements

IAction, IRedeemCodeV1

Inherited Members

GameAction.Id , GameAction.PlainValue , GameAction.LoadPlainValue(IValue) , ActionBase.GetSignerAndOtherAddressesHex(IActionContext, params Address[]) , ActionBase.TryGetAdminState(IActionContext, out AdminState) , ActionBase.CheckPermission(IActionContext) , ActionBase.CheckObsolete(long, IActionContext) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

RedeemCode2()

```
public RedeemCode2()
```

RedeemCode2(string, Address)

```
public RedeemCode2(string code, Address avatarAddress)
```

Parameters

`code` string

`avatarAddress` Address

Properties

AvatarAddress

```
public Address AvatarAddress { get; }
```

Property Value

Address

Code

```
public string Code { get; }
```

Property Value

string

PlainValueInternal

```
protected override IImmutableDictionary<string, IValue> PlainValueInternal { get; }
```

Property Value

IImmutableDictionary<string, IValue>

Methods

Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

Parameters

`context` IActionContext

Returns

IWorld

LoadPlainValueInternal(IImmutableDictionary<string, IValue>)

```
protected override void LoadPlainValueInternal(IImmutableDictionary<string,
IValue> plainValue)
```

Parameters

`plainValue` IImmutableDictionary<string, IValue>

Class RegisterInfo

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
public class RegisterInfo : IRegisterInfo
```

Inheritance

object ← RegisterInfo

Implements

IRegisterInfo

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

RegisterInfo()

```
public RegisterInfo()
```

RegisterInfo(List)

```
public RegisterInfo(List serialized)
```

Parameters

`serialized` List

Properties

AvatarAddress

```
public Address AvatarAddress { get; set; }
```

Property Value

Address

ItemCount

```
public int ItemCount { get; set; }
```

Property Value

int

Price

```
public FungibleAssetValue Price { get; set; }
```

Property Value

FungibleAssetValue

TradableId

```
public Guid TradableId { get; set; }
```

Property Value

Guid

Type

```
public ProductType Type { get; set; }
```

Property Value

ProductType

Methods

Serialize()

```
public IValue Serialize()
```

Returns

IValue

Validate()

```
public void Validate()
```

ValidateAddress(Address)

```
public void ValidateAddress(Address avatarAddress)
```

Parameters

avatarAddress Address

ValidatePrice(Currency)

```
public void ValidatePrice(Currency ncg)
```

Parameters

ncg Currency

Class RegisterProduct

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[ActionType("register_product3")]
public class RegisterProduct : GameAction, IAction
```

Inheritance

object ← ActionBase ← GameAction ← RegisterProduct

Implements

IAction

Inherited Members

GameAction.Id , GameAction.PlainValue , GameAction.LoadPlainValue(IValue) , ActionBase.GetSignerAndOtherAddressesHex(IActionContext, params Address[]) , ActionBase.TryGetAdminState(IActionContext, out AdminState) , ActionBase.CheckPermission(IActionContext) , ActionBase.CheckObsolete(long, IActionContext) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Fields

AvatarAddress

```
public Address AvatarAddress
```

Field Value

Address

Capacity

```
public const int Capacity = 100
```

Field Value

int

ChargeAp

```
public bool ChargeAp
```

Field Value

bool

CostAp

```
public const int CostAp = 5
```

Field Value

int

NonTradableTickerCurrencies

```
public static readonly IReadOnlyCollection<Currency> NonTradableTickerCurrencies
```

Field Value

IReadOnlyCollection<Currency>

RegisterInfos

```
public IEnumerable<IRegisterInfo> RegisterInfos
```

Field Value

IEnumerable<IRegisterInfo>

Properties

PlainValueInternal

```
protected override IImmutableDictionary<string, IValue> PlainValueInternal { get; }
```

Property Value

IImmutableDictionary<string, IValue>

Methods

Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

Parameters

`context` IActionContext

Returns

IWorld

LoadPlainValueInternal(IImmutableDictionary<string, IValue>)

```
protected override void LoadPlainValueInternal(IImmutableDictionary<string,
IValue> plainValue)
```

Parameters

`plainValue` IImmutableDictionary<string, IValue>

Register(IActionContext, IRegisterInfo, AvatarState, ProductsState, IWorld, IRandom, RestrictionSheet)

```
public static IWorld Register(IActionContext context, IRegisterInfo info,
AvatarState avatarState, ProductsState productsState, IWorld states, IRandom random,
RestrictionSheet restrictionSheet)
```

Parameters

`context` IActionContext

The action context the registration runs in.

`info` IRegisterInfo

What to register: an item or a fungible asset.

`avatarState` AvatarState

The seller's avatar, whose inventory the item leaves.

`productsState` ProductsState

The seller's product list the new product id joins.

`states` IWorld

The world the product is written into.

`random` IRandom

Source of the product id.

`restrictionSheet` RestrictionSheet

Restrictions on top of the hardcoded ones, or null on a chain that has not been patched with RestrictionSheet yet. The sheet only adds: a patched sheet that forgets an entry must not lift a restriction that code already enforces. Required rather than optional so that a new call site has to decide instead of silently registering without the policy.

Returns

Class RegisterProduct0

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[ActionType("register_product")]
[ActionObsolete(8324909)]
public class RegisterProduct0 : GameAction, IAction
```

Inheritance

object ← ActionBase ← GameAction ← RegisterProduct0

Implements

IAction

Inherited Members

GameAction.Id , GameAction.PlainValue , GameAction.LoadPlainValue(IValue) , ActionBase.GetSignerAndOtherAddressesHex(IActionContext, params Address[]) , ActionBase.TryGetAdminState(IActionContext, out AdminState) , ActionBase.CheckPermission(IActionContext) , ActionBase.CheckObsolete(long, IActionContext) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Fields

AvatarAddress

```
public Address AvatarAddress
```

Field Value

Address

Capacity

```
public const int Capacity = 100
```

Field Value

int

ChargeAp

```
public bool ChargeAp
```

Field Value

bool

CostAp

```
public const int CostAp = 5
```

Field Value

int

RegisterInfos

```
public IEnumerable<IRegisterInfo> RegisterInfos
```

Field Value

IEnumerable<IRegisterInfo>

Properties

PlainValueInternal

```
protected override IImmutableDictionary<string, IValue> PlainValueInternal { get; }
```

Property Value

IImmutableDictionary<string, IValue>

Methods

Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

Parameters

`context` IActionContext

Returns

IWorld

LoadPlainValueInternal(IImmutableDictionary<string, IValue>)

```
protected override void LoadPlainValueInternal(IImmutableDictionary<string,
IValue> plainValue)
```

Parameters

`plainValue` IImmutableDictionary<string, IValue>

Register(IActionContext, IRegisterInfo, AvatarState, ProductsState, IWorld, IRandom)

```
public static IWorld Register(IActionContext context, IRegisterInfo info,
AvatarState avatarState, ProductsState productsState, IWorld states, IRandom random)
```

Parameters

`context` IActionContext

`info` IRegisterInfo

`avatarState` AvatarState

`productsState` ProductsState

`states` IWorld

`random` IRandom

Returns

IWorld

Class RemoveAddressState

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

Action to remove state for multiple addresses.

```
[ActionType("remove_address_state")]
public class RemoveAddressState : ActionBase, IAction
```

Inheritance

object ← ActionBase ← RemoveAddressState

Implements

IAction

Inherited Members

ActionBase.GetSignerAndOtherAddressesHex(IActionContext, params Address[]) ,
ActionBase.TryGetAdminState(IActionContext, out AdminState) ,
ActionBase.CheckPermission(IActionContext) ,
ActionBase.CheckObsolete(long, IActionContext) , object.Equals(object) ,
object.Equals(object, object) , object.GetHashCode() , object.GetType() ,
object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

RemoveAddressState()

Default constructor.

```
public RemoveAddressState()
```

RemoveAddressState(IReadOnlyList<(Address accountAddress, Address targetAddress)>)

Constructor that initializes with a list of removals.

```
public RemoveAddressState(IReadOnlyList<(Address accountAddress, Address
targetAddress)> removals)
```

Parameters

`removals` IReadOnlyList<(Address accountAddress, Address targetAddress)>

List of tuples to remove state for

Fields

TypeId

The type identifier of the action.

```
public const string TypeId = "remove_address_state"
```

Field Value

string

Properties

PlainValue

```
public override IValue PlainValue { get; }
```

Property Value

IValue

Removals

List of tuples (account address, target address) to remove state for.

```
public IReadOnlyList<(Address accountAddress, Address targetAddress)> Removals {
get; }
```

Property Value

IReadOnlyList<(Address accountAddress, Address targetAddress)>

Methods

Execute(IActionContext)

Executes the action to remove state for the specified addresses.

```
public override IWorld Execute(IActionContext context)
```

Parameters

`context` IActionContext

Action context

Returns

IWorld

The updated world state

LoadPlainValue(IValue)

```
public override void LoadPlainValue(IValue plainValue)
```

Parameters

`plainValue` IValue

Class RenewAdminState

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

Hard forked at https://github.com/planetarium/lib9c/pull/805 Updated at https://github.com/planetarium/lib9c/pull/815 Updated at https://github.com/planetarium/lib9c/pull/957

```
[Serializable]
[ActionType("renew_admin_state")]
public class RenewAdminState : GameAction, IAction, IRenewAdminStateV1
```

Inheritance

object ← ActionBase ← GameAction ← RenewAdminState

Implements

IAction, IRenewAdminStateV1

Inherited Members

GameAction.Id , GameAction.PlainValue , GameAction.LoadPlainValue(IValue) , ActionBase.GetSignerAndOtherAddressesHex(IActionContext, params Address[]) , ActionBase.TryGetAdminState(IActionContext, out AdminState) , ActionBase.CheckPermission(IActionContext) , ActionBase.CheckObsolete(long, IActionContext) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

RenewAdminState()

```
public RenewAdminState()
```

RenewAdminState(long)

```
public RenewAdminState(long newValidUntil)
```

Parameters

`newValidUntil` long

Properties

NewValidUntil

```
public long NewValidUntil { get; }
```

Property Value

long

PlainValueInternal

```
protected override IImmutableDictionary<string, IValue> PlainValueInternal { get; }
```

Property Value

IImmutableDictionary<string, IValue>

Methods

Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

Parameters

`context` IActionContext

Returns

IWorld

LoadPlainValueInternal(IImmutableDictionary<string, IValue>)

```
protected override void LoadPlainValueInternal(IImmutableDictionary<string,
IValue> plainValue)
```

Parameters

`plainValue` IImmutableDictionary<string, IValue>

Class RequestPledge

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[ActionType("request_pledge")]
public class RequestPledge : ActionBase, IAction
```

Inheritance

object ← ActionBase ← RequestPledge

Implements

IAction

Inherited Members

ActionBase.GetSignerAndOtherAddressesHex(IActionContext, params Address[]) ,
ActionBase.TryGetAdminState(IActionContext, out AdminState) ,
ActionBase.CheckPermission(IActionContext) ,
ActionBase.CheckObsolete(long, IActionContext) , object.Equals(object) ,
object.Equals(object, object) , object.GetHashCode() , object.GetType() ,
object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

RequestPledge()

```
public RequestPledge()
```

Fields

AgentAddress

```
public Address AgentAddress
```

Field Value

DefaultRefillMead

```
public const int DefaultRefillMead = 4
```

Field Value

int

RefillMead

```
public int RefillMead
```

Field Value

int

TypeIdentifier

```
public const string TypeIdentifier = "request_pledge"
```

Field Value

string

Properties

PlainValue

```
public override IValue PlainValue { get; }
```

Property Value

IValue

Methods

Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

Parameters

`context` IActionContext

Returns

IWorld

LoadPlainValue(IValue)

```
public override void LoadPlainValue(IValue plainValue)
```

Parameters

`plainValue` IValue

Class RequiredBlockIndexException

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
public class RequiredBlockIndexException : Exception, ISerializable
```

Inheritance

object ← Exception ← RequiredBlockIndexException

Implements

ISerializable

Derived

RequiredBlockIntervalException

Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Constructors

RequiredBlockIndexException()

```
public RequiredBlockIndexException()
```

RequiredBlockIndexException(SerializationInfo, StreamingContext)

```
protected RequiredBlockIndexException(SerializationInfo info,
StreamingContext context)
```

Parameters

`info` SerializationInfo

`context` StreamingContext

RequiredBlockIndexException(string)

```
public RequiredBlockIndexException(string msg)
```

Parameters

`msg` string

RequiredBlockIndexException(string, string, long)

```
public RequiredBlockIndexException(string actionType, string addressesHex,
long currentBlockIndex)
```

Parameters

`actionType` string

`addressesHex` string

`currentBlockIndex` long

RequiredBlockIndexException(string, string, long, long)

```
public RequiredBlockIndexException(string actionType, string addressesHex, long
requiredBlockIndex, long currentBlockIndex)
```

Parameters

`actionType` string

`addressesHex` string

`requiredBlockIndex` long

`currentBlockIndex` long

Class RequiredBlockIntervalException

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
public class RequiredBlockIntervalException : RequiredBlockIndexException,
ISerializable
```

Inheritance

object ← Exception ← RequiredBlockIndexException ← RequiredBlockIntervalException

Implements

ISerializable

Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Constructors

RequiredBlockIntervalException()

```
public RequiredBlockIntervalException()
```

RequiredBlockIntervalException(SerializationInfo, StreamingContext)

```
protected RequiredBlockIntervalException(SerializationInfo info,
StreamingContext context)
```

Parameters

`info` SerializationInfo

`context` StreamingContext

RequiredBlockIntervalException(string)

```
public RequiredBlockIntervalException(string msg)
```

Parameters

`msg` string

RequiredBlockIntervalException(string, string, long)

```
public RequiredBlockIntervalException(string actionType, string addressesHex,
long currentBlockIndex)
```

Parameters

`actionType` string

`addressesHex` string

`currentBlockIndex` long

RequiredBlockIntervalException(string, string, long, long)

```
public RequiredBlockIntervalException(string actionType, string addressesHex, long
requiredBlockIndex, long currentBlockIndex)
```

Parameters

`actionType` string

`addressesHex` string

`requiredBlockIndex` long

`currentBlockIndex` long

Class RetrieveAvatarAssets

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
[ActionType("retrieve_avatar_assets")]
public class RetrieveAvatarAssets : ActionBase, IAction
```

Inheritance

object ← ActionBase ← RetrieveAvatarAssets

Implements

IAction

Inherited Members

ActionBase.GetSignerAndOtherAddressesHex(IActionContext, params Address[]) ,
ActionBase.TryGetAdminState(IActionContext, out AdminState) ,
ActionBase.CheckPermission(IActionContext) ,
ActionBase.CheckObsolete(long, IActionContext) , object.Equals(object) ,
object.Equals(object, object) , object.GetHashCode() , object.GetType() ,
object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

RetrieveAvatarAssets()

```
public RetrieveAvatarAssets()
```

RetrieveAvatarAssets(Address)

```
public RetrieveAvatarAssets(Address avatarAddress)
```

Parameters

`avatarAddress` Address

Fields

AvatarAddress

```
public Address AvatarAddress
```

Field Value

Address

TypeIdentifier

```
public const string TypeIdentifier = "retrieve_avatar_assets"
```

Field Value

string

Properties

PlainValue

```
public override IValue PlainValue { get; }
```

Property Value

IValue

Methods

Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

Parameters

`context` IActionContext

Returns

IWorld

LoadPlainValue(IValue)

```
public override void LoadPlainValue(IValue plainValue)
```

Parameters

`plainValue` IValue

Class RewardGold

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

Introduced at Initial commit(2e645be18a4e2caea031c347f00777fbad5dbcc6) Updated at many pull requests Updated at https://github.com/planetarium/lib9c/pull/1135

```
[Serializable]
public class RewardGold : ActionBase, IAction
```

Inheritance

object ← ActionBase ← RewardGold

Implements

IAction

Inherited Members

ActionBase.GetSignerAndOtherAddressesHex(IActionContext, params Address[]) , ActionBase.TryGetAdminState(IActionContext, out AdminState) , ActionBase.CheckPermission(IActionContext) , ActionBase.CheckObsolete(long, IActionContext) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Fields

FilterInactiveArenaInfoBlockIndex

```
public const long FilterInactiveArenaInfoBlockIndex = 3976000
```

Field Value

long

Properties

PlainValue

```
public override IValue PlainValue { get; }
```

Property Value

IValue

Methods

Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

Parameters

`context` IActionContext

Returns

IWorld

GenesisGoldDistribution(IActionContext, IWorld)

```
public IWorld GenesisGoldDistribution(IActionContext ctx, IWorld states)
```

Parameters

`ctx` IActionContext

`states` IWorld

Returns

IWorld

LoadPlainValue(IValue)

```
public override void LoadPlainValue(IValue plainValue)
```

Parameters

`plainValue` IValue

MinerReward(IActionContext, IWorld)

```
public IWorld MinerReward(IActionContext ctx, IWorld states)
```

Parameters

`ctx` IActionContext

`states` IWorld

Returns

IWorld

PrepareNextArena(IActionContext, IWorld)

```
public IWorld PrepareNextArena(IActionContext ctx, IWorld states)
```

Parameters

`ctx` IActionContext

`states` IWorld

Returns

IWorld

ResetChallengeCount(IActionContext, IWorld)

```
public IWorld ResetChallengeCount(IActionContext ctx, IWorld states)
```

Parameters

`ctx` IActionContext

`states` IWorld

Returns

IWorld

WeeklyArenaRankingBoard(IActionContext, IWorld)

```
[Obsolete("Use WeeklyArenaRankingBoard2 for performance.")]
public IWorld WeeklyArenaRankingBoard(IActionContext ctx, IWorld states)
```

Parameters

`ctx` IActionContext

`states` IWorld

Returns

IWorld

WeeklyArenaRankingBoard2(IActionContext, IWorld)

```
public IWorld WeeklyArenaRankingBoard2(IActionContext ctx, IWorld states)
```

Parameters

`ctx` IActionContext

`states` IWorld

Returns

IWorld

Class RuneEnhancement

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

Tries to raise a rune's level TryCount times, burning the cost of every try.

```
[Serializable]
[ActionType("runeEnhancement2")]
public class RuneEnhancement : GameAction, IAction, IRuneEnhancementV1
```

Inheritance

object ← ActionBase ← GameAction ← RuneEnhancement

Implements

IAction, IRuneEnhancementV1

Inherited Members

GameAction.Id , GameAction.PlainValue , GameAction.LoadPlainValue(IValue) , ActionBase.GetSignerAndOtherAddressesHex(IActionContext, params Address[]) , ActionBase.TryGetAdminState(IActionContext, out AdminState) , ActionBase.CheckPermission(IActionContext) , ActionBase.CheckObsolete(long, IActionContext) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Fields

AvatarAddress

The avatar whose rune is enhanced.

```
public Address AvatarAddress
```

Field Value

Address

RuneId

The rune to enhance.

```
public int RuneId
```

Field Value

int

TryCount

How many levels to try, each paying its own cost.

```
public int TryCount
```

Field Value

int

Properties

PlainValueInternal

```
protected override IImmutableDictionary<string, IValue> PlainValueInternal { get; }
```

Property Value

IImmutableDictionary<string, IValue>

Methods

Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

Parameters

`context` IActionContext

Returns

IWorld

LoadPlainValueInternal(IImmutableDictionary<string, IValue>)

```
protected override void LoadPlainValueInternal(IImmutableDictionary<string,
IValue> plainValue)
```

Parameters

`plainValue` IImmutableDictionary<string, IValue>

Struct RuneEnhancement.LevelUpResult

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

The outcome and cost of a run of enhancement tries.

```
public struct RuneEnhancement.LevelUpResult
```

Inherited Members
ValueType.Equals(object) , ValueType.GetHashCode() , object.Equals(object, object) , object.GetType() , object.ReferenceEquals(object, object)

Properties

CrystalCost

Crystal burned over all tries.

```
public long CrystalCost { readonly get; set; }
```

Property Value

long

LevelUpCount

Tries that succeeded.

```
public int LevelUpCount { readonly get; set; }
```

Property Value

int

NcgCost

NCG burned over all tries.

```
public int NcgCost { readonly get; set; }
```

Property Value

int

RuneCost

Rune stones burned over all tries.

```
public int RuneCost { readonly get; set; }
```

Property Value

int

Methods

ToString()

Returns the fully qualified type name of this instance.

```
public override string ToString()
```

Returns

string

The fully qualified type name.

Class RuneSlotInfo

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

Represents rune slot information for a character.

```
[Serializable]
public class RuneSlotInfo
```

Inheritance

object ← RuneSlotInfo

Inherited Members

object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

RuneSlotInfo(List)

Initializes a new instance of the RuneSlotInfo class from serialized data.

```
public RuneSlotInfo(List serialized)
```

Parameters

`serialized` List

The serialized data.

RuneSlotInfo(int, int)

Initializes a new instance of the RuneSlotInfo class.

```
public RuneSlotInfo(int slotIndex, int runeId)
```

Parameters

`slotIndex` int

The slot index.

`runeId` int

The rune ID.

Properties

RuneId

Gets the rune ID equipped in this slot.

```
public int RuneId { get; }
```

Property Value

int

SlotIndex

Gets the slot index for the rune.

```
public int SlotIndex { get; }
```

Property Value

int

Methods

Equals(object)

Determines whether the specified object is equal to the current object.

```
public override bool Equals(object obj)
```

Parameters

`obj` object

The object to compare with the current object.

Returns

bool

True if the specified object is equal to the current object; otherwise, false.

GetHashCode()

Serves as the default hash function.

```
public override int GetHashCode()
```

Returns

int

A hash code for the current object.

Serialize()

Serializes the rune slot info to Bencodex format.

```
public IValue Serialize()
```

Returns

IValue

The serialized data.

Class RuneSummon

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[ActionType("rune_summon")]
public class RuneSummon : GameAction, IAction, IRuneSummonV1
```

Inheritance

object ← ActionBase ← GameAction ← RuneSummon

Implements

IAction, IRuneSummonV1

Inherited Members

GameAction.Id , GameAction.PlainValue , GameAction.LoadPlainValue(IValue) , ActionBase.GetSignerAndOtherAddressesHex(IActionContext, params Address[]) , ActionBase.TryGetAdminState(IActionContext, out AdminState) , ActionBase.CheckPermission(IActionContext) , ActionBase.CheckObsolete(long, IActionContext) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Fields

AvatarAddress

```
public Address AvatarAddress
```

Field Value

Address

AvatarAddressKey

```
public const string AvatarAddressKey = "aa"
```

Field Value

string

GroupId

```
public int GroupId
```

Field Value

int

GroupIdKey

```
public const string GroupIdKey = "gid"
```

Field Value

string

RuneQuantity

Number of runes obtained per summon.

```
public const int RuneQuantity = 10
```

Field Value

int

SummonCount

```
public int SummonCount
```

Field Value

int

SummonCountKey

```
public const string SummonCountKey = "sc"
```

Field Value

string

Properties

PlainValueInternal

```
protected override IImmutableDictionary<string, IValue> PlainValueInternal { get; }
```

Property Value

IImmutableDictionary<string, IValue>

Methods

Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

Parameters

`context` IActionContext

Returns

IWorld

LoadPlainValueInternal(IImmutableDictionary<string, IValue>)

```
protected override void LoadPlainValueInternal(IImmutableDictionary<string,
IValue> plainValue)
```

Parameters

`plainValue` IImmutableDictionary<string, IValue>

SimulateSummon(RuneSheet, Row, int, IRandom, RuneListSheet, Row)

Simulates rune summoning with optional grade guarantee system. Applies 10+1 bonus rule and uses grade guarantee settings from SummonSheet.Row when enabled. When grade guarantee is enabled, ensures minimum grade runes are obtained based on summon count. Each summon yields 10 runes of the selected type.

```
public static Dictionary<Currency, int> SimulateSummon(RuneSheet runeSheet,
SummonSheet.Row summonRow, int summonCount, IRandom random, RuneListSheet
runeListSheet, BoostScheduleSheet.Row guaranteeBoost = null)
```

Parameters

`runeSheet` RuneSheet

Rune sheet containing rune information

`summonRow` SummonSheet.Row

Summon configuration row with recipes and guarantee settings

`summonCount` int

Number of summons to perform (before 10+1 bonus)

`random` IRandom

Random number generator

`runeListSheet` RuneListSheet

Rune list sheet for grade information

`guaranteeBoost` BoostScheduleSheet.Row

The BoostScheduleSheet row adjusting this group's guarantee count, or `null` for none. A caller replaying a summon must pass what the action resolved.

Returns

Dictionary<Currency, int>

Dictionary mapping rune currencies to quantities

Summon(IActionContext, Address, RuneSheet, Row, int, IRandom, IWorld, RuneListSheet, Row)

```
public static IWorld Summon(IActionContext context, Address avatarAddress, RuneSheet runeSheet, SummonSheet.Row summonRow, int summonCount, IRandom random, IWorld states, RuneListSheet runeListSheet, BoostScheduleSheet.Row guaranteeBoost = null)
```

Parameters

`context` IActionContext

`avatarAddress` Address

`runeSheet` RuneSheet

`summonRow` SummonSheet.Row

`summonCount` int

`random` IRandom

`states` IWorld

`runeListSheet` RuneListSheet

`guaranteeBoost` BoostScheduleSheet.Row

Returns

Class SecureMiningReward

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[ActionType("secure_mining_reward")]
public class SecureMiningReward : ActionBase, IAction, ISecureMiningReward
```

Inheritance

object ← ActionBase ← SecureMiningReward

Implements

IAction, ISecureMiningReward

Inherited Members

ActionBase.GetSignerAndOtherAddressesHex(IActionContext, params Address[]) ,
ActionBase.TryGetAdminState(IActionContext, out AdminState) ,
ActionBase.CheckPermission(IActionContext) ,
ActionBase.CheckObsolete(long, IActionContext) , object.Equals(object) ,
object.Equals(object, object) , object.GetHashCode() , object.GetType() ,
object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

SecureMiningReward()

```
public SecureMiningReward()
```

SecureMiningReward(Address)

```
public SecureMiningReward(Address recipient)
```

Parameters

`recipient` Address

Properties

PlainValue

```
public override IValue PlainValue { get; }
```

Property Value

IValue

Recipient

```
public Address Recipient { get; }
```

Property Value

Address

Methods

Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

Parameters

`context` IActionContext

Returns

IWorld

LoadPlainValue(IValue)

```
public override void LoadPlainValue(IValue plainValue)
```

Parameters

`plainValue` IValue

Class Sell

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

Hard forked at https://github.com/planetarium/lib9c/pull/1640

```
[Serializable]
[ActionType("sell12")]
[ActionObsolete(7200000)]
public class Sell : GameAction, IAction, ISellV2
```

Inheritance

object ← ActionBase ← GameAction ← Sell

Implements

IAction, ISellV2

Inherited Members

GameAction.Id , GameAction.PlainValue , GameAction.LoadPlainValue(IValue) , ActionBase.GetSignerAndOtherAddressesHex(IActionContext, params Address[]) , ActionBase.TryGetAdminState(IActionContext, out AdminState) , ActionBase.CheckPermission(IActionContext) , ActionBase.CheckObsolete(long, IActionContext) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Fields

count

```
public int count
```

Field Value

int

itemSubType

```
public ItemSubType itemSubType
```

Field Value

ItemSubType

orderId

```
public Guid orderId
```

Field Value

Guid

price

```
public FungibleAssetValue price
```

Field Value

FungibleAssetValue

sellerAvatarAddress

```
public Address sellerAvatarAddress
```

Field Value

Address

tradableId

```
public Guid tradableId
```

Field Value

Guid

Properties

PlainValueInternal

```
protected override IImmutableDictionary<string, IValue> PlainValueInternal { get; }
```

Property Value

IImmutableDictionary<string, IValue>

Methods

Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

Parameters

`context` IActionContext

Returns

IWorld

LoadPlainValueInternal(IImmutableDictionary<string, IValue>)

```
protected override void LoadPlainValueInternal(IImmutableDictionary<string,
```

```
IValue> plainValue)
```

Parameters

`plainValue` IImmutableDictionary<string, IValue>

Class Sell6

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
[ActionObsolete(7000000)]
[ActionType("sell6")]
public class Sell6 : GameAction, IAction, ISellV2
```

Inheritance

object ← ActionBase ← GameAction ← Sell6

Implements

IAction, ISellV2

Inherited Members

GameAction.Id , GameAction.PlainValue , GameAction.LoadPlainValue(IValue) , ActionBase.GetSignerAndOtherAddressesHex(IActionContext, params Address[]) , ActionBase.TryGetAdminState(IActionContext, out AdminState) , ActionBase.CheckPermission(IActionContext) , ActionBase.CheckObsolete(long, IActionContext) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Fields

ExpiredBlockIndex

```
public const long ExpiredBlockIndex = 16000
```

Field Value

long

count

```
public int count
```

Field Value

int

itemSubType

```
public ItemSubType itemSubType
```

Field Value

ItemSubType

price

```
public FungibleAssetValue price
```

Field Value

FungibleAssetValue

sellerAvatarAddress

```
public Address sellerAvatarAddress
```

Field Value

Address

tradableId

```
public Guid tradableId
```

Field Value

Guid

Properties

PlainValueInternal

```
protected override IImmutableDictionary<string, IValue> PlainValueInternal { get; }
```

Property Value

IImmutableDictionary<string, IValue>

Methods

Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

Parameters

`context` IActionContext

Returns

IWorld

LoadPlainValueInternal(IImmutableDictionary<string, IValue>)

```
protected override void LoadPlainValueInternal(IImmutableDictionary<string,
```

```
IValue> plainValue)
```

Parameters

`plainValue` IImmutableDictionary<string, IValue>

Class SellCancellation

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

Hard forked at https://github.com/planetarium/lib9c/pull/602 Updated at https://github.com/planetarium/lib9c/pull/609 Updated at https://github.com/planetarium/lib9c/pull/620 Updated at https://github.com/planetarium/lib9c/pull/861 Updated at https://github.com/planetarium/lib9c/pull/957

```
[Serializable]
[ActionObsolete(8324909)]
[ActionType("sell_cancellation9")]
public class SellCancellation : GameAction, IAction, ISellCancellationV3
```

Inheritance

object ← ActionBase ← GameAction ← SellCancellation

Implements

IAction, ISellCancellationV3

Inherited Members

GameAction.Id , GameAction.PlainValue , GameAction.LoadPlainValue(IValue) , ActionBase.GetSignerAndOtherAddressesHex(IActionContext, params Address[]) , ActionBase.TryGetAdminState(IActionContext, out AdminState) , ActionBase.CheckPermission(IActionContext) , ActionBase.CheckObsolete(long, IActionContext) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Fields

itemSubType

```
public ItemSubType itemSubType
```

Field Value

ItemSubType

orderId

```
public Guid orderId
```

Field Value

Guid

sellerAvatarAddress

```
public Address sellerAvatarAddress
```

Field Value

Address

tradableId

```
public Guid tradableId
```

Field Value

Guid

Properties

PlainValueInternal

```
protected override IImmutableDictionary<string, IValue> PlainValueInternal { get; }
```

Property Value

IImmutableDictionary<string, IValue>

Methods

Cancel(IActionContext, IWorld, AvatarState, string, Order)

```
public static IWorld Cancel(IActionContext context, IWorld states, AvatarState
avatarState, string addressesHex, Order order)
```

Parameters

`context` IActionContext

`states` IWorld

`avatarState` AvatarState

`addressesHex` string

`order` Order

Returns

IWorld

CancelV2(IActionContext, IWorld, AvatarState, string, Order, Guid, ItemSubType)

```
public static IWorld CancelV2(IActionContext context, IWorld states,
AvatarState avatarState, string addressesHex, Order order, Guid tradableId,
ItemSubType itemSubType)
```

Parameters

`context` IActionContext

`states` IWorld

`avatarState` AvatarState

`addressesHex` string

`order` Order

`tradableId` Guid

`itemSubType` ItemSubType

Returns

IWorld

Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

Parameters

`context` IActionContext

Returns

IWorld

LoadPlainValueInternal(IImmutableDictionary<string, IValue>)

```
protected override void LoadPlainValueInternal(IImmutableDictionary<string,
IValue> plainValue)
```

Parameters

`plainValue` IImmutableDictionary<string, IValue>

Class SellCancellation.Result

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
public class SellCancellation.Result : AttachmentActionResult, IState
```

Inheritance

object ← AttachmentActionResult ← SellCancellation.Result

Implements

IState

Inherited Members

AttachmentActionResult.itemUsable , AttachmentActionResult.costume , AttachmentActionResult.tradableFungibleItem , AttachmentActionResult.tradableFungibleItemCount , AttachmentActionResult.SerializeBackup1() , AttachmentActionResult.Deserialize(Dictionary) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

Result()

```
public Result()
```

Result(Dictionary)

```
public Result(Dictionary serialized)
```

Parameters

`serialized` Dictionary

Fields

id

```
public Guid id
```

Field Value

Guid

shopItem

```
public ShopItem shopItem
```

Field Value

ShopItem

Properties

TypeId

```
protected override string TypeId { get; }
```

Property Value

string

Methods

Serialize()

```
public override IValue Serialize()
```

Returns

IValue

Class SetAddressState

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

Action to set state for multiple addresses. Throws an exception if the state already exists for the target address.

```
[ActionType("set_address_state")]
public class SetAddressState : ActionBase, IAction
```

Inheritance

object ← ActionBase ← SetAddressState

Implements

IAction

Inherited Members

ActionBase.GetSignerAndOtherAddressesHex(IActionContext, params Address[]) , ActionBase.TryGetAdminState(IActionContext, out AdminState) , ActionBase.CheckPermission(IActionContext) , ActionBase.CheckObsolete(long, IActionContext) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

SetAddressState()

Default constructor.

```
public SetAddressState()
```

SetAddressState(IReadOnlyList<(Address accountAddress, Address targetAddress, IValue state)>)

Constructor that initializes with a list of states to set.

```
public SetAddressState(IReadOnlyList<(Address accountAddress, Address targetAddress,
IValue state)> states)
```

Parameters

`states` IReadOnlyList<(Address accountAddress, Address targetAddress, IValue state)>

List of tuples to set state for

Exceptions

ArgumentNullException

Thrown if states is null or any state value is null

Fields

TypeId

The type identifier of the action.

```
public const string TypeId = "set_address_state"
```

Field Value

string

Properties

PlainValue

```
public override IValue PlainValue { get; }
```

Property Value

IValue

States

List of tuples (account address, target address, state value) to set state for.

```
public IReadOnlyList<(Address accountAddress, Address targetAddress, IValue state)>
States { get; }
```

Property Value

IReadOnlyList<(Address accountAddress, Address targetAddress, IValue state)>

Methods

Execute(IActionContext)

Executes the action to set state for the specified addresses. Throws an exception if the state already exists for the target address.

```
public override IWorld Execute(IActionContext context)
```

Parameters

`context` IActionContext

Action context

Returns

IWorld

The updated world state

Exceptions

InvalidOperationException

Thrown if the state already exists for the target address

LoadPlainValue(IValue)

```
public override void LoadPlainValue(IValue plainValue)
```

Parameters

`plainValue` IValue

Class SetAddressStateCompressed

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

Compressed version of SetAddressState to reduce transaction size for large state data. Uses GZip compression to minimize the size of state data in transactions.

This action is particularly useful when setting large state data that would exceed transaction policy size limits. The compression can reduce state data size by 60-80% depending on the content structure and redundancy.

Example usage:

```
var states = new List<(Address, Address, IValue)>
{
    (accountAddress, targetAddress, largeStateValue),
};
var compressedAction = new SetAddressStateCompressed(states);
```

```
[ActionType("set_address_state_compressed")]
public class SetAddressStateCompressed : ActionBase, IAction
```

Inheritance

object ← ActionBase ← SetAddressStateCompressed

Implements

IAction

Inherited Members

ActionBase.GetSignerAndOtherAddressesHex(IActionContext, params Address[]) , ActionBase.TryGetAdminState(IActionContext, out AdminState) , ActionBase.CheckPermission(IActionContext) , ActionBase.CheckObsolete(long, IActionContext) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

SetAddressStateCompressed()

Default constructor.

```
public SetAddressStateCompressed()
```

SetAddressStateCompressed(IReadOnlyList<(Address accountAddress, Address targetAddress, IValue state)>)

Constructor that initializes with a list of states to set. The states will be automatically compressed using GZip compression.

```
public SetAddressStateCompressed(IReadOnlyList<(Address accountAddress, Address
targetAddress, IValue state)> states)
```

Parameters

`states` IReadOnlyList<(Address accountAddress, Address targetAddress, IValue state)>

List of tuples to set state for

Exceptions

ArgumentNullException

Thrown if states is null or any state value is null

ArgumentException

Thrown if states list is empty

SetAddressStateCompressed(IReadOnlyList<(Address accountAddress, Address targetAddress, byte[] compressedState)>)

Constructor that initializes with already compressed state data.

```
public SetAddressStateCompressed(IReadOnlyList<(Address accountAddress, Address
targetAddress, byte[] compressedState)> compressedStates)
```

Parameters

`compressedStates` IReadOnlyList<(Address accountAddress, Address targetAddress, byte[] compressedState)>

List of tuples with compressed state data

Exceptions

ArgumentNullException

Thrown if compressedStates is null or any compressed state is null

ArgumentException

Thrown if compressedStates list is empty

Fields

TypeId

The type identifier of the action.

```
public const string TypeId = "set_address_state_compressed"
```

Field Value

string

Properties

CompressedStates

List of tuples (account address, target address, compressed state data) to set state for. The state data is compressed using GZip compression.

```
public IReadOnlyList<(Address accountAddress, Address targetAddress, byte[]
compressedState)> CompressedStates { get; }
```

Property Value

IReadOnlyList<(Address accountAddress, Address targetAddress, byte[] compressedState)>

PlainValue

```
public override IValue PlainValue { get; }
```

Property Value

IValue

Methods

CompressState(IValue)

Compresses state data using GZip compression.

This method is used to prepare state data for the CompressedStates property. GZip compression typically reduces state data size by 60-80% depending on content.

Example:

```
var state = (Text)"large state data";
var compressedData = SetAddressStateCompressed.CompressState(state);
// compressedData can now be used in CompressedStates property
```

```
public static byte[] CompressState(IValue state)
```

Parameters

`state` IValue

The state data to compress. Must not be null.

Returns

byte[]

Compressed byte array using GZip compression.

Exceptions

ArgumentNullException

Thrown when state is null.

InvalidOperationException

Thrown when compression fails.

DecompressState(byte[])

Decompresses state data using GZip compression.

This method is used internally to restore the original state data from the compressed byte array stored in CompressedStates.

Example:

```
var originalState = SetAddressStateCompressed.DecompressState(compressedData);
// originalState now contains the decompressed IValue
```

```
public static IValue DecompressState(byte[] compressedData)
```

Parameters

`compressedData` byte[]

The compressed data to decompress. Must not be null.

Returns

IValue

Decompressed IValue state data.

Exceptions

ArgumentNullException

Thrown when compressedData is null.

ArgumentException

Thrown when compressedData is empty.

InvalidDataException

Thrown when the compressed data is corrupted or not valid GZip format.

InvalidOperationException

Thrown when decompression fails.

Execute(IActionContext)

Executes the action to set state for the specified addresses. Throws an exception if the state already exists for the target address.

```
public override IWorld Execute(IActionContext context)
```

Parameters

`context` IActionContext

Action context

Returns

IWorld

The updated world state

Exceptions

ArgumentNullException

Thrown if context is null

InvalidOperationException

Thrown if the state already exists for the target address

InvalidDataException

Thrown when the compressed data is corrupted or invalid

UnauthorizedAccessException

Thrown when the signer does not have permission to execute this action

LoadPlainValue(IValue)

```
public override void LoadPlainValue(IValue plainValue)
```

Parameters

`plainValue` IValue

Enum ShopErrorType

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
public enum ShopErrorType
```

Extension Methods
StateExtensions.Serialize(Enum)

Fields

ERROR_CODE_DUPLICATE_SELL = 10

ERROR_CODE_FAILED_LOADING_STATE = 1

ERROR_CODE_INSUFFICIENT_BALANCE = 4

ERROR_CODE_INVALID_ADDRESS = 5

ERROR_CODE_INVALID_ITEMTYPE = 9

ERROR_CODE_INVALID_ORDERID = 7

ERROR_CODE_INVALID_PRICE = 6

ERROR_CODE_INVALID_TRADABLEID = 8

ERROR_CODE_ITEM_DOES_NOT_EXIST = 2

ERROR_CODE_SHOPITEM_EXPIRED = 3

Class ShopItemExpiredException

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
public class ShopItemExpiredException : Exception, ISerializable
```

Inheritance

object ← Exception ← ShopItemExpiredException

Implements

ISerializable

Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Constructors

ShopItemExpiredException()

```
public ShopItemExpiredException()
```

ShopItemExpiredException(SerializationInfo, StreamingContext)

```
protected ShopItemExpiredException(SerializationInfo info, StreamingContext context)
```

Parameters

`info` SerializationInfo

`context` StreamingContext

ShopItemExpiredException(string)

```
public ShopItemExpiredException(string msg)
```

Parameters

`msg` string

Class SlotAlreadyUnlockedException

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
public class SlotAlreadyUnlockedException : InvalidOperationException, ISerializable
```

Inheritance

object ← Exception ← SystemException ← InvalidOperationException ← SlotAlreadyUnlockedException

Implements

ISerializable

Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Constructors

SlotAlreadyUnlockedException(SerializationInfo, StreamingContext)

```
protected SlotAlreadyUnlockedException(SerializationInfo info,
StreamingContext context)
```

Parameters

`info` SerializationInfo

`context` StreamingContext

SlotAlreadyUnlockedException(string)

```
public SlotAlreadyUnlockedException(string s)
```

Parameters

s string

Class StageNotClearedException

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
public class StageNotClearedException : Exception, ISerializable
```

Inheritance

object ← Exception ← StageNotClearedException

Implements

ISerializable

Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Constructors

StageNotClearedException()

```
public StageNotClearedException()
```

StageNotClearedException(SerializationInfo, StreamingContext)

```
protected StageNotClearedException(SerializationInfo info, StreamingContext context)
```

Parameters

`info` SerializationInfo

`context` StreamingContext

StageNotClearedException(string)

```
public StageNotClearedException(string message)
```

Parameters

`message` string

StageNotClearedException(string, Exception)

```
public StageNotClearedException(string message, Exception innerException)
```

Parameters

`message` string

`innerException` Exception

StageNotClearedException(string, string, int, int)

```
public StageNotClearedException(string actionType, string addressesHex, int
requiredToClearedStage, int currentClearedStage)
```

Parameters

`actionType` string

`addressesHex` string

`requiredToClearedStage` int

`currentClearedStage` int

Class StageNotOpenedException

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

A stage was entered before its hard world segment opened.

```
[Serializable]
public class StageNotOpenedException : InvalidStageException, ISerializable
```

Inheritance

object ← Exception ← InvalidStageException ← StageNotOpenedException

Implements

ISerializable

Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Constructors

StageNotOpenedException(int, int)

Creates the exception for `stageId` of `worldId`.

```
public StageNotOpenedException(int stageId, int worldId)
```

Parameters

`stageId` int

`worldId` int

StageNotOpenedException(SerializationInfo, StreamingContext)

Deserialization constructor.

```
protected StageNotOpenedException(SerializationInfo info, StreamingContext context)
```

Parameters

`info` SerializationInfo

`context` StreamingContext

Class Stake

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[ActionType("stake3")]
public class Stake : GameAction, IAction, IStakeV1
```

Inheritance

object ← ActionBase ← GameAction ← Stake

Implements

IAction, IStakeV1

Inherited Members

GameAction.Id , GameAction.PlainValue , GameAction.LoadPlainValue(IValue) , ActionBase.GetSignerAndOtherAddressesHex(IActionContext, params Address[]) , ActionBase.TryGetAdminState(IActionContext, out AdminState) , ActionBase.CheckPermission(IActionContext) , ActionBase.CheckObsolete(long, IActionContext) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

Stake()

```
public Stake()
```

Stake(BigInteger)

```
public Stake(BigInteger amount)
```

Parameters

`amount` BigInteger

Stake(BigInteger, Address)

```
public Stake(BigInteger amount, Address avatarAddress)
```

Parameters

`amount` BigInteger

`avatarAddress` Address

Properties

PlainValueInternal

```
protected override IImmutableDictionary<string, IValue> PlainValueInternal { get; }
```

Property Value

IImmutableDictionary<string, IValue>

Methods

Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

Parameters

`context` IActionContext

Returns

IWorld

LoadPlainValueInternal(IImmutableDictionary<string, IValue>)

```
protected override void LoadPlainValueInternal(IImmutableDictionary<string,
IValue> plainValue)
```

Parameters

`plainValue` IImmutableDictionary<string, IValue>

Class Stake2

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[ActionType("stake2")]
[ActionObsolete(7983895)]
public class Stake2 : GameAction, IAction, IStakeV1
```

Inheritance

object ← ActionBase ← GameAction ← Stake2

Implements

IAction, IStakeV1

Inherited Members

GameAction.Id , GameAction.PlainValue , GameAction.LoadPlainValue(IValue) , ActionBase.GetSignerAndOtherAddressesHex(IActionContext, params Address[]) , ActionBase.TryGetAdminState(IActionContext, out AdminState) , ActionBase.CheckPermission(IActionContext) , ActionBase.CheckObsolete(long, IActionContext) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

Stake2()

```
public Stake2()
```

Stake2(BigInteger)

```
public Stake2(BigInteger amount)
```

Parameters

`amount` BigInteger

Fields

ObsoleteBlockIndex

```
public const long ObsoleteBlockIndex = 7983895
```

Field Value

long

Properties

PlainValueInternal

```
protected override IImmutableDictionary<string, IValue> PlainValueInternal { get; }
```

Property Value

IImmutableDictionary<string, IValue>

Methods

Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

Parameters

`context` IActionContext

Returns

IWorld

LoadPlainValueInternal(IImmutableDictionary<string, IValue>)

```
protected override void LoadPlainValueInternal(IImmutableDictionary<string,
IValue> plainValue)
```

Parameters

`plainValue` IImmutableDictionary<string, IValue>

Class StakeExistingClaimableException

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
public class StakeExistingClaimableException : Exception, ISerializable
```

Inheritance

object ← Exception ← StakeExistingClaimableException

Implements

ISerializable

Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Constructors

StakeExistingClaimableException()

```
public StakeExistingClaimableException()
```

StakeExistingClaimableException(SerializationInfo, StreamingContext)

```
protected StakeExistingClaimableException(SerializationInfo info,
StreamingContext context)
```

Parameters

`info` SerializationInfo

`context` StreamingContext

StakeExistingClaimableException(string)

```
public StakeExistingClaimableException(string message)
```

Parameters

`message` string

StakeExistingClaimableException(string, Exception)

```
public StakeExistingClaimableException(string message, Exception innerException)
```

Parameters

`message` string

`innerException` Exception

Class Swap

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

Swap action swaps one currency to another currency.

```
[Serializable]
[ActionType("swap")]
public class Swap : ActionBase, IAction
```

Inheritance

object ← ActionBase ← Swap

Implements

IAction

Inherited Members

ActionBase.GetSignerAndOtherAddressesHex(IActionContext, params Address[]) ,
ActionBase.TryGetAdminState(IActionContext, out AdminState) ,
ActionBase.CheckPermission(IActionContext) ,
ActionBase.CheckObsolete(long, IActionContext) , object.Equals(object) ,
object.Equals(object, object) , object.GetHashCode() , object.GetType() ,
object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

Swap()

Initializes a new instance of the Swap class.

```
public Swap()
```

Swap(FungibleAssetValue, Currency)

Initializes a new instance of the Swap class.

```
public Swap(FungibleAssetValue from, Currency to)
```

Parameters

`from` FungibleAssetValue

The Libplanet.Types.Assets.FungibleAssetValue to swap from.

`to` Currency

The Libplanet.Types.Assets.Currency to swap to.

Fields

TypeIdentifier

The type identifier for the Swap class.

```
public const string TypeIdentifier = "swap"
```

Field Value

string

Properties

From

The Libplanet.Types.Assets.FungibleAssetValue to swap from.

```
public FungibleAssetValue From { get; }
```

Property Value

FungibleAssetValue

PlainValue

```
public override IValue PlainValue { get; }
```

Property Value

IValue

To

The Libplanet.Types.Assets.Currency to swap to.

```
public Currency To { get; }
```

Property Value

Currency

Methods

Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

Parameters

`context` IActionContext

Returns

IWorld

LoadPlainValue(IValue)

```
public override void LoadPlainValue(IValue plainValue)
```

Parameters

`plainValue` IValue

Class Synthesize

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

Synthesize action is a type of action that synthesizes items. MaterialIds: Id list of items as material
ChargeAp: Whether to charge action points with action execution

```
[Serializable]
[ActionType("synthesize")]
public class Synthesize : GameAction, IAction
```

Inheritance

object ← ActionBase ← GameAction ← Synthesize

Implements

IAction

Inherited Members

GameAction.Id , GameAction.PlainValue , GameAction.LoadPlainValue(IValue) , ActionBase.GetSignerAndOtherAddressesHex(IActionContext, params Address[]) , ActionBase.TryGetAdminState(IActionContext, out AdminState) , ActionBase.CheckPermission(IActionContext) , ActionBase.CheckObsolete(long, IActionContext) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Fields

AvatarAddress

AvatarAddress of the signer.

```
public Address AvatarAddress
```

Field Value

Address

ChargeAp

Whether to charge action points with action execution.

```
public bool ChargeAp
```

Field Value

bool

InvalidMaterialItemId

The list of invalid item ids for material.

```
public static readonly int[] InvalidMaterialItemId
```

Field Value

int[]

MaterialGradeId

MaterialGrade of the material item.

```
public int MaterialGradeId
```

Field Value

int

MaterialIds

Id list of items as material.

```
public List<Guid> MaterialIds
```

Field Value

List<Guid>

MaterialItemSubTypeId

ItemSubType of the material item.

```
public int MaterialItemSubTypeId
```

Field Value

int

Properties

PlainValueInternal

```
protected override IImmutableDictionary<string, IValue> PlainValueInternal { get; }
```

Property Value

IImmutableDictionary<string, IValue>

Methods

Execute(IActionContext)

Execute Synthesize action, Input MaterialIds and AvatarAddress. Depending on the probability, you can get different items. Success: Return new AvatarState with synthesized item in inventory.

```
public override IWorld Execute(IActionContext context)
```

Parameters

`context` IActionContext

Returns

IWorld

LoadPlainValueInternal(IImmutableDictionary<string, IValue>)

```
protected override void LoadPlainValueInternal(IImmutableDictionary<string,
IValue> plainValue)
```

Parameters

`plainValue` IImmutableDictionary<string, IValue>

Class TableSheetPatchValidation

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

The per-sheet checks PatchTableSheet and PatchTableSheetCompressed run before a CSV lands on the chain. Both actions call this one helper so the two paths can never disagree about what is accepted.

```
public static class TableSheetPatchValidation
```

Inheritance

object ← TableSheetPatchValidation

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Methods

Validate(string, string)

Validates `csv` for the sheet `tableName`. Sheets without a validator are accepted as they are.

```
public static void Validate(string tableName, string csv)
```

Parameters

`tableName` string

`csv` string

Remarks

The validated sheets are read leniently at runtime: a row that does not parse is kept but never matches, so a malformed patch would otherwise land and silently misconfigure the game with nothing reporting it. The world-open sheets are also read by every

HackAndSlash, where an empty CSV would make Set(string, bool) throw and fail all of them, so an empty or whitespace-only CSV is rejected for them too.

Exceptions

SheetRowValidateException

The CSV is malformed.

Class TotalSupplyDoesNotExistException

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
public class TotalSupplyDoesNotExistException : Exception, ISerializable
```

Inheritance

object ← Exception ← TotalSupplyDoesNotExistException

Implements

ISerializable

Inherited Members

Exception.GetBaseException() , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Constructors

TotalSupplyDoesNotExistException(Currency)

```
public TotalSupplyDoesNotExistException(Currency currency)
```

Parameters

`currency` Currency

TotalSupplyDoesNotExistException(SerializationInfo, StreamingContext)

```
protected TotalSupplyDoesNotExistException(SerializationInfo info,
StreamingContext context)
```

Parameters

`info` SerializationInfo

`context` StreamingContext

TotalSupplyDoesNotExistException(string)

```
public TotalSupplyDoesNotExistException(string message)
```

Parameters

`message` string

Properties

Currency

```
public Currency Currency { get; }
```

Property Value

Currency

Methods

GetObjectData(SerializationInfo, StreamingContext)

When overridden in a derived class, sets the SerializationInfo with information about the exception.

```
public override void GetObjectData(SerializationInfo info, StreamingContext context)
```

Parameters

`info` SerializationInfo

The SerializationInfo that holds the serialized object data about the exception being thrown.

`context` StreamingContext

The StreamingContext that contains contextual information about the source or destination.

Exceptions

ArgumentNullException

The `info` parameter is a null reference (`Nothing` in Visual Basic).

Class TransferAsset

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

Hard forked at https://github.com/planetarium/lib9c/pull/2143 Updated at https://github.com/planetarium/lib9c/pull/2143

```
[Serializable]
[ActionType("transfer_asset5")]
public class TransferAsset : ActionBase, IAction, ISerializable,
ITransferAsset, ITransferAssetV1
```

Inheritance

object ← ActionBase ← TransferAsset

Implements

IAction, ISerializable, ITransferAsset, ITransferAssetV1

Inherited Members

ActionBase.GetSignerAndOtherAddressesHex(IActionContext, params Address[]) , ActionBase.TryGetAdminState(IActionContext, out AdminState) , ActionBase.CheckPermission(IActionContext) , ActionBase.CheckObsolete(long, IActionContext) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

TransferAsset()

```
public TransferAsset()
```

TransferAsset(Address, Address, FungibleAssetValue, string)

```
public TransferAsset(Address sender, Address recipient, FungibleAssetValue amount,
string memo = null)
```

Parameters

`sender` Address

`recipient` Address

`amount` FungibleAssetValue

`memo` string

TransferAsset(SerializationInfo, StreamingContext)

```
protected TransferAsset(SerializationInfo info, StreamingContext context)
```

Parameters

`info` SerializationInfo

`context` StreamingContext

Fields

TypeIdentifier

```
public const string TypeIdentifier = "transfer_asset5"
```

Field Value

string

Properties

Amount

```
public FungibleAssetValue Amount { get; }
```

Property Value

FungibleAssetValue

Memo

```
public string Memo { get; }
```

Property Value

string

PlainValue

```
public override IValue PlainValue { get; }
```

Property Value

IValue

Recipient

```
public Address Recipient { get; }
```

Property Value

Address

Sender

```
public Address Sender { get; }
```

Property Value

Address

Methods

Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

Parameters

`context` IActionContext

Returns

IWorld

GetObjectData(SerializationInfo, StreamingContext)

Populates a SerializationInfo with the data needed to serialize the target object.

```
public void GetObjectData(SerializationInfo info, StreamingContext context)
```

Parameters

`info` SerializationInfo

The SerializationInfo to populate with data.

`context` StreamingContext

The destination (see StreamingContext) for this serialization.

Exceptions

SecurityException

The caller does not have the required permission.

LoadPlainValue(IValue)

```
public override void LoadPlainValue(IValue plainValue)
```

Parameters

`plainValue` IValue

ThrowIfStakeState(IWorld, Address)

```
public static void ThrowIfStakeState(IWorld state, Address recipient)
```

Parameters

`state` IWorld

`recipient` Address

Class TransferAsset3

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
public static class TransferAsset3
```

Inheritance

object ← TransferAsset3

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Fields

AllowedCrystalTransfers

```
public static readonly IReadOnlyList<Address> AllowedCrystalTransfers
```

Field Value

IReadOnlyList<Address>

CrystalTransferringRestrictionStartIndex

```
public const long CrystalTransferringRestrictionStartIndex = 6220000
```

Field Value

long

Methods

CheckCrystalSender(Currency, long, Address)

```
public static void CheckCrystalSender(Currency currency, long blockIndex,
Address sender)
```

Parameters

`currency` Currency

`blockIndex` long

`sender` Address

Class TransferAssets

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

Hard forked at https://github.com/planetarium/lib9c/pull/636 Updated at https://github.com/planetarium/lib9c/pull/957

```
[Serializable]
[ActionType("transfer_assets3")]
public class TransferAssets : ActionBase, IAction, ISerializable,
ITransferAssets, ITransferAssetsV1
```

Inheritance

object ← ActionBase ← TransferAssets

Implements

IAction, ISerializable, ITransferAssets, ITransferAssetsV1

Inherited Members

ActionBase.GetSignerAndOtherAddressesHex(IActionContext, params Address[]) , ActionBase.TryGetAdminState(IActionContext, out AdminState) , ActionBase.CheckPermission(IActionContext) , ActionBase.CheckObsolete(long, IActionContext) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

TransferAssets()

```
public TransferAssets()
```

TransferAssets(Address, List<(Address, FungibleAssetValue)>, string)

```
public TransferAssets(Address sender, List<(Address, FungibleAssetValue)>
recipients, string memo = null)
```

Parameters

`sender` Address

`recipients` List<(Address recipient, FungibleAssetValue amount)>

`memo` string

TransferAssets(SerializationInfo, StreamingContext)

```
protected TransferAssets(SerializationInfo info, StreamingContext context)
```

Parameters

`info` SerializationInfo

`context` StreamingContext

Fields

RecipientsCapacity

```
public const int RecipientsCapacity = 100
```

Field Value

int

TypeIdentifier

```
public const string TypeIdentifier = "transfer_assets3"
```

Field Value

string

Properties

Memo

```
public string Memo { get; }
```

Property Value

string

PlainValue

```
public override IValue PlainValue { get; }
```

Property Value

IValue

Recipients

```
public List<(Address recipient, FungibleAssetValue amount)> Recipients { get; }
```

Property Value

List<(Address recipient, FungibleAssetValue amount)>

Sender

```
public Address Sender { get; }
```

Property Value

Address

Methods

Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

Parameters

`context` IActionContext

Returns

IWorld

GetObjectData(SerializationInfo, StreamingContext)

Populates a SerializationInfo with the data needed to serialize the target object.

```
public void GetObjectData(SerializationInfo info, StreamingContext context)
```

Parameters

`info` SerializationInfo

The SerializationInfo to populate with data.

`context` StreamingContext

The destination (see StreamingContext) for this serialization.

Exceptions

SecurityException

The caller does not have the required permission.

LoadPlainValue(IValue)

```
public override void LoadPlainValue(IValue plainValue)
```

Parameters

`plainValue` IValue

Class UnlockCombinationSlot

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
[ActionType("unlock_combination_slot")]
public class UnlockCombinationSlot : GameAction, IAction
```

Inheritance

object ← ActionBase ← GameAction ← UnlockCombinationSlot

Implements

IAction

Inherited Members

GameAction.Id , GameAction.PlainValue , GameAction.LoadPlainValue(IValue) , ActionBase.GetSignerAndOtherAddressesHex(IActionContext, params Address[]) , ActionBase.TryGetAdminState(IActionContext, out AdminState) , ActionBase.CheckPermission(IActionContext) , ActionBase.CheckObsolete(long, IActionContext) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Fields

AvatarAddress

```
public Address AvatarAddress
```

Field Value

Address

GoldenDustId

```
public const int GoldenDustId = 600201
```

Field Value

int

RubyDustId

```
public const int RubyDustId = 600202
```

Field Value

int

SlotIndex

```
public int SlotIndex
```

Field Value

int

Properties

PlainValueInternal

```
protected override IImmutableDictionary<string, IValue> PlainValueInternal { get; }
```

Property Value

IImmutableDictionary<string, IValue>

Methods

Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

Parameters

`context` IActionContext

Returns

IWorld

LoadPlainValueInternal(IImmutableDictionary<string, IValue>)

```
protected override void LoadPlainValueInternal(IImmutableDictionary<string,
IValue> plainValue)
```

Parameters

`plainValue` IImmutableDictionary<string, IValue>

Class UnlockEquipmentRecipe

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[ActionType("unlock_equipment_recipe2")]
public class UnlockEquipmentRecipe : GameAction, IAction, IUnlockEquipmentRecipeV1
```

Inheritance

object ← ActionBase ← GameAction ← UnlockEquipmentRecipe

Implements

IAction, IUnlockEquipmentRecipeV1

Inherited Members

GameAction.Id , GameAction.PlainValue , GameAction.LoadPlainValue(IValue) , ActionBase.GetSignerAndOtherAddressesHex(IActionContext, params Address[]) , ActionBase.TryGetAdminState(IActionContext, out AdminState) , ActionBase.CheckPermission(IActionContext) , ActionBase.CheckObsolete(long, IActionContext) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Fields

AvatarAddress

```
public Address AvatarAddress
```

Field Value

Address

RecipeIds

```
public List<int> RecipeIds
```

Field Value

List<int>

Properties

PlainValueInternal

```
protected override IImmutableDictionary<string, IValue> PlainValueInternal { get; }
```

Property Value

IImmutableDictionary<string, IValue>

Methods

Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

Parameters

`context` IActionContext

Returns

IWorld

LoadPlainValueInternal(IImmutableDictionary<string, IValue>)

```
protected override void LoadPlainValueInternal(IImmutableDictionary<string,
IValue> plainValue)
```

Parameters

`plainValue` IImmutableDictionary<string, IValue>

UnlockedIds(IWorld, Address, EquipmentItemRecipeSheet, WorldInformation, List<int>)

```
public static List<int> UnlockedIds(IWorld states, Address unlockedRecipeIdsAddress,
EquipmentItemRecipeSheet equipmentRecipeSheet, WorldInformation worldInformation,
List<int> recipeIds)
```

Parameters

`states` IWorld

`unlockedRecipeIdsAddress` Address

`equipmentRecipeSheet` EquipmentItemRecipeSheet

`worldInformation` WorldInformation

`recipeIds` List<int>

Returns

List<int>

Class UnlockRuneSlot

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
[ActionType("unlock_rune_slot")]
public class UnlockRuneSlot : GameAction, IAction, IUnlockRuneSlotV1
```

Inheritance

object ← ActionBase ← GameAction ← UnlockRuneSlot

Implements

IAction, IUnlockRuneSlotV1

Inherited Members

GameAction.Id , GameAction.PlainValue , GameAction.LoadPlainValue(IValue) , ActionBase.GetSignerAndOtherAddressesHex(IActionContext, params Address[]) , ActionBase.TryGetAdminState(IActionContext, out AdminState) , ActionBase.CheckPermission(IActionContext) , ActionBase.CheckObsolete(long, IActionContext) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Fields

AvatarAddress

```
public Address AvatarAddress
```

Field Value

Address

SlotIndex

```
public int SlotIndex
```

Field Value

int

Properties

PlainValueInternal

```
protected override IImmutableDictionary<string, IValue> PlainValueInternal { get; }
```

Property Value

IImmutableDictionary<string, IValue>

Methods

Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

Parameters

`context` IActionContext

Returns

IWorld

LoadPlainValueInternal(IImmutableDictionary<string, IValue>)

```
protected override void LoadPlainValueInternal(IImmutableDictionary<string,
```

```
IValue> plainValue)
```

Parameters

`plainValue` IImmutableDictionary<string, IValue>

Class UnlockWorld

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

Unlocks one or more worlds for an avatar and records the result to the legacy `world_ids` state. The required CRYSTAL is calculated from WorldUnlockSheet and transferred to UnlockWorld. If the calculated cost is zero, no asset transfer occurs.

```
[ActionType("unlock_world2")]
public class UnlockWorld : GameAction, IAction, IUnlockWorldV1
```

Inheritance

object ← ActionBase ← GameAction ← UnlockWorld

Implements

IAction, IUnlockWorldV1

Inherited Members

GameAction.Id , GameAction.PlainValue , GameAction.LoadPlainValue(IValue) , ActionBase.GetSignerAndOtherAddressesHex(IActionContext, params Address[]) , ActionBase.TryGetAdminState(IActionContext, out AdminState) , ActionBase.CheckPermission(IActionContext) , ActionBase.CheckObsolete(long, IActionContext) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Remarks

Hard-forked at https://github.com/planetarium/lib9c/pull/1309.

Fields

AvatarAddress

The avatar address to apply the unlock to.

```
public Address AvatarAddress
```

Field Value

Address

WorldIds

Target world IDs to unlock.

```
public List<int> WorldIds
```

Field Value

List<int>

Properties

PlainValueInternal

```
protected override IImmutableDictionary<string, IValue> PlainValueInternal { get; }
```

Property Value

IImmutableDictionary<string, IValue>

Methods

Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

Parameters

`context` IActionContext

Returns

LoadPlainValueInternal(IImmutableDictionary<string, IValue>)

```
protected override void LoadPlainValueInternal(IImmutableDictionary<string,
IValue> plainValue)
```

Parameters

`plainValue` IImmutableDictionary<string, IValue>

Class UpdateSell

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

Hard forked at https://github.com/planetarium/lib9c/pull/1640

```
[Serializable]
[ActionType("update_sell5")]
[ActionObsolete(7200000)]
public class UpdateSell : GameAction, IAction, IUpdateSellV2
```

Inheritance

object ← ActionBase ← GameAction ← UpdateSell

Implements

IAction, IUpdateSellV2

Inherited Members

GameAction.Id , GameAction.PlainValue , GameAction.LoadPlainValue(IValue) , ActionBase.GetSignerAndOtherAddressesHex(IActionContext, params Address[]) , ActionBase.TryGetAdminState(IActionContext, out AdminState) , ActionBase.CheckPermission(IActionContext) , ActionBase.CheckObsolete(long, IActionContext) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Fields

sellerAvatarAddress

```
public Address sellerAvatarAddress
```

Field Value

Address

updateSellInfos

```
public IEnumerable<UpdateSellInfo> updateSellInfos
```

Field Value

IEnumerable<UpdateSellInfo>

Properties

PlainValueInternal

```
protected override IImmutableDictionary<string, IValue> PlainValueInternal { get; }
```

Property Value

IImmutableDictionary<string, IValue>

Methods

Cancel(IWorld, UpdateSellInfo, string, AvatarState, OrderDigestListState, IActionContext, Address)

```
public static IWorld Cancel(IWorld states, UpdateSellInfo updateSellInfo, string
addressesHex, AvatarState avatarState, OrderDigestListState digestList,
IActionContext context, Address sellerAvatarAddress)
```

Parameters

`states` IWorld

`updateSellInfo` UpdateSellInfo

`addressesHex` string

`avatarState` AvatarState

`digestList` OrderDigestListState

`context` IActionContext

`sellerAvatarAddress` Address

Returns

IWorld

Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

Parameters

`context` IActionContext

Returns

IWorld

LoadPlainValueInternal(IImmutableDictionary<string, IValue>)

```
protected override void LoadPlainValueInternal(IImmutableDictionary<string,
IValue> plainValue)
```

Parameters

`plainValue` IImmutableDictionary<string, IValue>

Class UpdateSellInfo

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
public class UpdateSellInfo
```

Inheritance

object ← UpdateSellInfo

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

UpdateSellInfo(List)

```
public UpdateSellInfo(List serialized)
```

Parameters

`serialized` List

UpdateSellInfo(Guid, Guid, Guid, ItemSubType, FungibleAssetValue, int)

```
public UpdateSellInfo(Guid orderId, Guid updateSellOrderId, Guid tradableId,
ItemSubType itemSubType, FungibleAssetValue price, int count)
```

Parameters

`orderId` Guid

`updateSellOrderId` Guid

`tradableId` Guid

`itemSubType` ItemSubType

`price` FungibleAssetValue

`count` int

Fields

count

```
public int count
```

Field Value

int

itemSubType

```
public ItemSubType itemSubType
```

Field Value

ItemSubType

orderId

```
public Guid orderId
```

Field Value

Guid

price

```
public FungibleAssetValue price
```

Field Value

FungibleAssetValue

tradableId

```
public Guid tradableId
```

Field Value

Guid

updateSellOrderId

```
public Guid updateSellOrderId
```

Field Value

Guid

Methods

Serialize()

```
public IValue Serialize()
```

Returns

IValue

Class UsageLimitExceedException

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
public class UsageLimitExceedException : InvalidOperationException, ISerializable
```

Inheritance

object ← Exception ← SystemException ← InvalidOperationException ← UsageLimitExceedException

Implements

ISerializable

Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Constructors

UsageLimitExceedException()

```
public UsageLimitExceedException()
```

UsageLimitExceedException(SerializationInfo, StreamingContext)

```
protected UsageLimitExceedException(SerializationInfo info,
StreamingContext context)
```

Parameters

`info` SerializationInfo

`context` StreamingContext

UsageLimitExceedException(string)

```
public UsageLimitExceedException(string message)
```

Parameters

`message` string

Class ValidatorSetOperate

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
[ActionType("op_validator_set")]
public sealed class ValidatorSetOperate : ActionBase, IAction,
IEquatable<ValidatorSetOperate>
```

Inheritance

object ← ActionBase ← ValidatorSetOperate

Implements

IAction, IEquatable<ValidatorSetOperate>

Inherited Members

object.Equals(object, object) , object.GetType() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

ValidatorSetOperate()

```
public ValidatorSetOperate()
```

ValidatorSetOperate(ValidatorSetOperatorType, Validator)

```
public ValidatorSetOperate(ValidatorSetOperatorType @operator, Validator operand)
```

Parameters

`operator` ValidatorSetOperatorType

`operand` Validator

Fields

TypeIdentifier

```
public const string TypeIdentifier = "op_validator_set"
```

Field Value

string

ValidatorSetOperateKey

```
public static readonly string ValidatorSetOperateKey
```

Field Value

string

Properties

Error

```
public string Error { get; }
```

Property Value

string

Operand

```
public Validator Operand { get; }
```

Property Value

Validator

Operator

```
public ValidatorSetOperatorType Operator { get; }
```

Property Value

ValidatorSetOperatorType

PlainValue

```
public override IValue PlainValue { get; }
```

Property Value

IValue

Methods

Append(Validator)

```
public static ValidatorSetOperate Append(Validator operand)
```

Parameters

`operand` Validator

Returns

ValidatorSetOperate

Equals(ValidatorSetOperate)

Indicates whether the current object is equal to another object of the same type.

```
public bool Equals(ValidatorSetOperate other)
```

Parameters

`other` ValidatorSetOperate

An object to compare with this object.

Returns

bool

true if the current object is equal to the `other` parameter; otherwise, false.

Equals(object)

Determines whether the specified object is equal to the current object.

```
public override bool Equals(object obj)
```

Parameters

`obj` object

The object to compare with the current object.

Returns

bool

true if the specified object is equal to the current object; otherwise, false.

Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

Parameters

`context` IActionContext

Returns

IWorld

GetHashCode()

Serves as the default hash function.

```
public override int GetHashCode()
```

Returns

int

A hash code for the current object.

LoadPlainValue(IValue)

```
public override void LoadPlainValue(IValue plainValue)
```

Parameters

`plainValue` IValue

Remove(Validator)

```
public static ValidatorSetOperate Remove(Validator operand)
```

Parameters

`operand` Validator

Returns

ValidatorSetOperate

Update(Validator)

```
public static ValidatorSetOperate Update(Validator operand)
```

Parameters

`operand` Validator

Returns

ValidatorSetOperate

Enum ValidatorSetOperatorType

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
public enum ValidatorSetOperatorType
```

Extension Methods
ValidatorSetOperatorTypeExtensions.ToFunc(ValidatorSetOperatorType) ,
StateExtensions.Serialize(Enum)

Fields

`Append = 0`

Append the validator. no-op if validator public key already exists.

`Remove = 1`

Set validator's power to 0.

`Update = 2`

Update validator. if not exists, append it.

Class ValidatorSetOperatorTypeExtensions

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
public static class ValidatorSetOperatorTypeExtensions
```

Inheritance

object ← ValidatorSetOperatorTypeExtensions

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Methods

ToFunc(ValidatorSetOperatorType)

```
public static Func<ValidatorSet, Validator, Validator> ToFunc(this
ValidatorSetOperatorType @operator)
```

Parameters

`operator` ValidatorSetOperatorType

Returns

Func<ValidatorSet, Validator, Validator>

Class WeeklyArenaStateAlreadyEndedException

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
public class WeeklyArenaStateAlreadyEndedException : Exception, ISerializable
```

Inheritance

object ← Exception ← WeeklyArenaStateAlreadyEndedException

Implements

ISerializable

Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Constructors

WeeklyArenaStateAlreadyEndedException()

```
public WeeklyArenaStateAlreadyEndedException()
```

WeeklyArenaStateAlreadyEndedException(SerializationInfo, StreamingContext)

```
public WeeklyArenaStateAlreadyEndedException(SerializationInfo info,
StreamingContext context)
```

Parameters

`info` SerializationInfo

`context` StreamingContext

WeeklyArenaStateAlreadyEndedException(string)

```
public WeeklyArenaStateAlreadyEndedException(string message)
```

Parameters

`message` string

Fields

BaseMessage

```
public const string BaseMessage = "Aborted as the weekly arena state already ended."
```

Field Value

string

Class WeeklyArenaStateNotContainsAvatarAddressException

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

```
[Serializable]
public class WeeklyArenaStateNotContainsAvatarAddressException :
Exception, ISerializable
```

Inheritance

object ← Exception ← WeeklyArenaStateNotContainsAvatarAddressException

Implements

ISerializable

Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Constructors

WeeklyArenaStateNotContainsAvatarAddressException(SerializationInfo, StreamingContext)

```
public WeeklyArenaStateNotContainsAvatarAddressException(SerializationInfo info,
StreamingContext context)
```

Parameters

`info` SerializationInfo

`context` StreamingContext

WeeklyArenaStateNotContainsAvatarAddressException(string)

```
public WeeklyArenaStateNotContainsAvatarAddressException(string message)
```

Parameters

`message` string

WeeklyArenaStateNotContainsAvatarAddressException(string, Address)

```
public WeeklyArenaStateNotContainsAvatarAddressException(string addressesHex,
Address avatarAddress)
```

Parameters

`addressesHex` string

`avatarAddress` Address

Class WorldOpenCounter

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

Counts player progress for the global counters and confirms world openings whose conditions are met.

```
public static class WorldOpenCounter
```

Inheritance

object ← WorldOpenCounter

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Methods

ConfirmSatisfiedAfterPatch(IWorld, string, long)

Confirms every pending opening whose conditions are already met, after `tableName` was patched. A table patch can lower a threshold below a counter that no longer moves often (a rarely cleared stage, a rarely crafted recipe); confirming here means such an opening does not wait for the next matching play. Only a WorldOpenSheet or WorldOpenConditionSheet patch triggers it; any other sheet, or absent world-open sheets, writes nothing.

```
public static IWorld ConfirmSatisfiedAfterPatch(IWorld states, string tableName,
long blockIndex)
```

Parameters

`states` IWorld

The world after the patched sheet was stored.

`tableName` string

The patched sheet's name.

`blockIndex` long

The block being evaluated; recorded as the confirmed block.

Returns

IWorld

The updated world, or `states` itself when nothing was written.

IsSatisfied(IWorldState, WorldOpenConditionSheet, int)

Whether any condition group of `openId` is fully met (all conditions in a group must hold; any one group suffices). An opening without condition rows is never satisfied.

```
public static bool IsSatisfied(IWorldState state, WorldOpenConditionSheet
conditions, int openId)
```

Parameters

`state` IWorldState

`conditions` WorldOpenConditionSheet

`openId` int

Returns

bool

Tick(IWorld, GlobalCounterType, int, long, int, long)

Adds `delta` to every counter of `type` covering `target` that a pending opening references, then confirms the openings whose conditions are now met. Writes nothing when no sheet defines such a counter, so ordinary actions cost nothing extra.

```
public static IWorld Tick(IWorld states, GlobalCounterType type, int target, long
```

```
  delta, int avatarLevel, long blockIndex)
```

Parameters

`states` IWorld

The world to update.

`type` GlobalCounterType

What happened.

`target` int

The stage or recipe id it happened to.

`delta` long

How much to add; a non-positive delta writes nothing.

`avatarLevel` int

The acting avatar's level, checked against `min_avatar_level`.

`blockIndex` long

The block being evaluated; recorded as the confirmed block.

Returns

IWorld

The updated world, or `states` itself when nothing was written.

Remarks

Matching counters are processed one at a time in ascending id (not CSV order), and whether an opening is still pending is re-read for each counter. So if counter A's tick confirms opening X, a later counter B that only X references is not written in the same call: X is no longer pending by then. A counter shared by several pending openings is written once per call, not once per opening.

Class WorldOpenGate

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

Decides whether a stage may be entered. Stages from ProgressGatedStageBegin are locked until an applied WorldOpenSheet row covers them.

```
public static class WorldOpenGate
```

Inheritance

object ← WorldOpenGate

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Methods

GetOpenedStageEnd(IWorldState, Row, long)

The last enterable stage of `worldRow`; `StageBegin - 1` when all of it is locked.

```
public static int GetOpenedStageEnd(IWorldState state, WorldSheet.Row worldRow,
long blockIndex)
```

Parameters

`state` IWorldState

The world to read the sheet and confirmed openings from.

`worldRow` WorldSheet.Row

The world to check.

`blockIndex` long

The block to evaluate at. A caller predicting the next action must pass `tip + 1`, not the tip (see IsStageOpen(IWorldState, Row, int, long)).

Returns

int

Remarks

Global only: this ignores whether the avatar has unlocked the world or cleared the previous stage. Combine it with the avatar's own progress to know what that avatar can enter.

IsStageOpen(IWorldState, Row, int, long)

Whether `stageId` of `worldRow` is open at `blockIndex`. A missing sheet leaves gated stages locked. Mimisbrunnr is never gated. Precondition: the caller has already validated that `stageId` lies within `StageBegin..StageEnd` of `worldRow`.

```
public static bool IsStageOpen(IWorldState state, WorldSheet.Row worldRow, int
stageId, long blockIndex)
```

Parameters

`state` IWorldState

The world to read the sheet and confirmed openings from.

`worldRow` WorldSheet.Row

The world the stage belongs to.

`stageId` int

The stage to check.

`blockIndex` long

The block the entry is evaluated at. Inside an action this is `context.BlockIndex`; a caller outside the chain predicting whether its next action will pass must pass `tip + 1`, the block that action lands in at the earliest, not the tip itself.

Returns

bool

Class WorldOpenIdentity

Namespace: Nekoyume.Action

Assembly: Lib9c.dll

Keeps what each world-open id means fixed across table patches. The first patch that carries an id records its identity; a later patch that gives the id a different identity is rejected. An id may be dropped from the sheet (retired), but its record stays, so it can only come back with the same identity. To fix a wrong rule, drop it and add a new id.

```
public static class WorldOpenIdentity
```

Inheritance

object ← WorldOpenIdentity

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Remarks

Counter values and confirmations are keyed by id, so changing what an id means would carry them over to a different rule: a counter would keep counts of something else, and a new opening reusing a confirmed id would open without its conditions. The identity is:

- GlobalCounterSheet: every column (type, target range, minimum level).
- WorldOpenSheet: `world_id` and `stage_end`. The block window and the prerequisite stay adjustable; the prerequisite follows the stage_end line, so it has to change when an opening is inserted below.
- WorldOpenConditionSheet: per `open_id`, which counters are grouped together. Group numbers and row order do not matter; thresholds stay adjustable.

The recorded strings are part of the chain state: changing their format (adding a column, renaming a GlobalCounterType, reordering fields) would make every recorded id mismatch and reject every later patch of that sheet. Keep the format, or compare against the old format as well when it has to change. The tests pin it.

Methods

CheckAndRecord(IWorld, string, string)

Rejects `csv` if it changes the identity of an id recorded earlier, and records the identity of every id seen for the first time. Any other sheet is left alone.

```
public static IWorld CheckAndRecord(IWorld states, string tableName, string csv)
```

Parameters

`states` IWorld

The world before the patched sheet is stored.

`tableName` string

The patched sheet's name.

`csv` string

The patched sheet, already validated by TableSheetPatchValidation.

Returns

IWorld

The world with the new identities recorded, or `states` itself when nothing new was seen.

Exceptions

SheetRowValidateException

An id's identity differs from its record.

Namespace Nekoyume.Action.AdventureBoss

Classes

ClaimAdventureBossReward

ExploreAdventureBoss

Represents an action where a player explores and battles through adventure boss stages. This action modifies the following states:

- AvatarState: Updates player's stats, inventory, and experience
- ExploreBoard: Updates exploration progress and points
- Explorer: Updates player's exploration status and achievements
- CollectionState: Updates collection progress
- RuneState: Updates rune effects and bonuses
- ItemSlotState: Updates equipped items
- RuneSlotState: Updates equipped runes
- CP: Updates character's combat power

SweepAdventureBoss

Represents a sweep action that automatically completes multiple adventure boss stages without player interaction. This action modifies the following states:

- AvatarState: Updates player's stats, inventory, and experience
- ExploreBoard: Updates exploration progress and points
- Explorer: Updates player's exploration status and achievements
- CollectionState: Updates collection progress
- RuneState: Updates rune effects and bonuses
- ItemSlotState: Updates equipped items
- RuneSlotState: Updates equipped runes
- CP: Updates character's combat power

UnlockFloor

Wanted

Class ClaimAdventureBossReward

Namespace: Nekoyume.Action.AdventureBoss

Assembly: Lib9c.dll

```
[Serializable]
[ActionType("claim_adventure_boss_reward")]
public class ClaimAdventureBossReward : GameAction, IAction
```

Inheritance

object ← ActionBase ← GameAction ← ClaimAdventureBossReward

Implements

IAction

Inherited Members

GameAction.Id , GameAction.PlainValue , GameAction.LoadPlainValue(IValue) , ActionBase.GetSignerAndOtherAddressesHex(IActionContext, params Address[]) , ActionBase.TryGetAdminState(IActionContext, out AdminState) , ActionBase.CheckPermission(IActionContext) , ActionBase.CheckObsolete(long, IActionContext) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Fields

AvatarAddress

```
public Address AvatarAddress
```

Field Value

Address

TypeIdentifier

```
public const string TypeIdentifier = "claim_adventure_boss_reward"
```

Field Value

string

Properties

PlainValueInternal

```
protected override IImmutableDictionary<string, IValue> PlainValueInternal { get; }
```

Property Value

IImmutableDictionary<string, IValue>

Methods

Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

Parameters

`context` IActionContext

Returns

IWorld

LoadPlainValueInternal(IImmutableDictionary<string, IValue>)

```
protected override void LoadPlainValueInternal(IImmutableDictionary<string,
```

```
IValue> plainValue)
```

Parameters

`plainValue` IImmutableDictionary<string, IValue>

Class ExploreAdventureBoss

Namespace: Nekoyume.Action.AdventureBoss

Assembly: Lib9c.dll

Represents an action where a player explores and battles through adventure boss stages.
This action modifies the following states:

- AvatarState: Updates player's stats, inventory, and experience
- ExploreBoard: Updates exploration progress and points
- Explorer: Updates player's exploration status and achievements
- CollectionState: Updates collection progress
- RuneState: Updates rune effects and bonuses
- ItemSlotState: Updates equipped items
- RuneSlotState: Updates equipped runes
- CP: Updates character's combat power

```
[Serializable]
[ActionType("explore_adventure_boss")]
public class ExploreAdventureBoss : GameAction, IAction
```

Inheritance

object ← ActionBase ← GameAction ← ExploreAdventureBoss

Implements

IAction

Inherited Members

GameAction.Id , GameAction.PlainValue , GameAction.LoadPlainValue(IValue) , ActionBase.GetSignerAndOtherAddressesHex(IActionContext, params Address[]) , ActionBase.TryGetAdminState(IActionContext, out AdminState) , ActionBase.CheckPermission(IActionContext) , ActionBase.CheckObsolete(long, IActionContext) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Fields

AvatarAddress

```
public Address AvatarAddress
```

Field Value

Address

Costumes

```
public List<Guid> Costumes
```

Field Value

List<Guid>

Equipments

```
public List<Guid> Equipments
```

Field Value

List<Guid>

Foods

```
public List<Guid> Foods
```

Field Value

List<Guid>

RuneInfos

```
public List<RuneSlotInfo> RuneInfos
```

Field Value

List<RuneSlotInfo>

Season

```
public int Season
```

Field Value

int

StageBuffId

```
public int? StageBuffId
```

Field Value

int?

TypeIdentifier

```
public const string TypeIdentifier = "explore_adventure_boss"
```

Field Value

string

Properties

PlainValueInternal

```
protected override IImmutableDictionary<string, IValue> PlainValueInternal { get; }
```

Property Value

IImmutableDictionary<string, IValue>

Methods

Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

Parameters

`context` IActionContext

Returns

IWorld

LoadPlainValueInternal(IImmutableDictionary<string, IValue>)

```
protected override void LoadPlainValueInternal(IImmutableDictionary<string,
IValue> plainValue)
```

Parameters

`plainValue` IImmutableDictionary<string, IValue>

Class SweepAdventureBoss

Namespace: Nekoyume.Action.AdventureBoss

Assembly: Lib9c.dll

Represents a sweep action that automatically completes multiple adventure boss stages without player interaction. This action modifies the following states:

- AvatarState: Updates player's stats, inventory, and experience
- ExploreBoard: Updates exploration progress and points
- Explorer: Updates player's exploration status and achievements
- CollectionState: Updates collection progress
- RuneState: Updates rune effects and bonuses
- ItemSlotState: Updates equipped items
- RuneSlotState: Updates equipped runes
- CP: Updates character's combat power

```
[Serializable]
[ActionType("sweep_adventure_boss")]
public class SweepAdventureBoss : GameAction, IAction
```

Inheritance

object ← ActionBase ← GameAction ← SweepAdventureBoss

Implements

IAction

Inherited Members

GameAction.Id , GameAction.PlainValue , GameAction.LoadPlainValue(IValue) , ActionBase.GetSignerAndOtherAddressesHex(IActionContext, params Address[]) , ActionBase.TryGetAdminState(IActionContext, out AdminState) , ActionBase.CheckPermission(IActionContext) , ActionBase.CheckObsolete(long, IActionContext) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Fields

AvatarAddress

```
public Address AvatarAddress
```

Field Value

Address

Costumes

```
public List<Guid> Costumes
```

Field Value

List<Guid>

Equipments

```
public List<Guid> Equipments
```

Field Value

List<Guid>

RuneInfos

```
public List<RuneSlotInfo> RuneInfos
```

Field Value

List<RuneSlotInfo>

Season

```
public int Season
```

Field Value

int

TypeIdentifier

```
public const string TypeIdentifier = "sweep_adventure_boss"
```

Field Value

string

Properties

PlainValueInternal

```
protected override IImmutableDictionary<string, IValue> PlainValueInternal { get; }
```

Property Value

IImmutableDictionary<string, IValue>

Methods

Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

Parameters

`context` IActionContext

Returns

IWorld

LoadPlainValueInternal(IImmutableDictionary<string, IValue>)

```
protected override void LoadPlainValueInternal(IImmutableDictionary<string,
IValue> plainValue)
```

Parameters

`plainValue` IImmutableDictionary<string, IValue>

Class UnlockFloor

Namespace: Nekoyume.Action.AdventureBoss

Assembly: Lib9c.dll

```
[Serializable]
[ActionType("unlock_floor")]
public class UnlockFloor : GameAction, IAction
```

Inheritance

object ← ActionBase ← GameAction ← UnlockFloor

Implements

IAction

Inherited Members

GameAction.Id , GameAction.PlainValue , GameAction.LoadPlainValue(IValue) , ActionBase.GetSignerAndOtherAddressesHex(IActionContext, params Address[]) , ActionBase.TryGetAdminState(IActionContext, out AdminState) , ActionBase.CheckPermission(IActionContext) , ActionBase.CheckObsolete(long, IActionContext) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Fields

AvatarAddress

```
public Address AvatarAddress
```

Field Value

Address

GoldenDustId

```
public const int GoldenDustId = 600201
```

Field Value

int

OpeningFloor

```
public const int OpeningFloor = 5
```

Field Value

int

Season

```
public int Season
```

Field Value

int

TotalFloor

```
public const int TotalFloor = 30
```

Field Value

int

TypeIdentifier

```
public const string TypeIdentifier = "unlock_floor"
```

Field Value

string

UseNcg

```
public bool UseNcg
```

Field Value

bool

Properties

PlainValueInternal

```
protected override IImmutableDictionary<string, IValue> PlainValueInternal { get; }
```

Property Value

IImmutableDictionary<string, IValue>

Methods

Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

Parameters

`context` IActionContext

Returns

IWorld

LoadPlainValueInternal(IImmutableDictionary<string, IValue>)

```
protected override void LoadPlainValueInternal(IImmutableDictionary<string,
IValue> plainValue)
```

Parameters

`plainValue` IImmutableDictionary<string, IValue>

Class Wanted

Namespace: Nekoyume.Action.AdventureBoss

Assembly: Lib9c.dll

```
[Serializable]
[ActionType("wanted")]
public class Wanted : GameAction, IAction
```

Inheritance

object ← ActionBase ← GameAction ← Wanted

Implements

IAction

Inherited Members

GameAction.Id , GameAction.PlainValue , GameAction.LoadPlainValue(IValue) , ActionBase.GetSignerAndOtherAddressesHex(IActionContext, params Address[]) , ActionBase.TryGetAdminState(IActionContext, out AdminState) , ActionBase.CheckPermission(IActionContext) , ActionBase.CheckObsolete(long, IActionContext) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Fields

AvatarAddress

```
public Address AvatarAddress
```

Field Value

Address

Bounty

```
public FungibleAssetValue Bounty
```

Field Value

FungibleAssetValue

Season

```
public int Season
```

Field Value

int

TypeIdentifier

```
public const string TypeIdentifier = "wanted"
```

Field Value

string

Properties

PlainValueInternal

```
protected override IImmutableDictionary<string, IValue> PlainValueInternal { get; }
```

Property Value

IImmutableDictionary<string, IValue>

Methods

Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

Parameters

`context` IActionContext

Returns

IWorld

LoadPlainValueInternal(IImmutableDictionary<string, IValue>)

```
protected override void LoadPlainValueInternal(IImmutableDictionary<string,
IValue> plainValue)
```

Parameters

`plainValue` IImmutableDictionary<string, IValue>

Namespace Nekoyume.Action.Arena

Classes

Battle

Temp

Enums

ArenaProvider

Temp

Enum ArenaProvider

Namespace: Nekoyume.Action.Arena

Assembly: Lib9c.dll

Temp

```
public enum ArenaProvider
```

Extension Methods
StateExtensions.Serialize(Enum)

Fields

`PLANETARIUM = 0`

Class Battle

Namespace: Nekoyume.Action.Arena

Assembly: Lib9c.dll

Temp

```
[Serializable]
[ActionType("battle")]
public class Battle : GameAction, IAction
```

Inheritance

object ← ActionBase ← GameAction ← Battle

Implements

IAction

Inherited Members

GameAction.Id , GameAction.PlainValue , GameAction.LoadPlainValue(IValue) , ActionBase.GetSignerAndOtherAddressesHex(IActionContext, params Address[]) , ActionBase.TryGetAdminState(IActionContext, out AdminState) , ActionBase.CheckPermission(IActionContext) , ActionBase.CheckObsolete(long, IActionContext) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Fields

CostAp

```
public const int CostAp = 5
```

Field Value

int

HpIncreasingModifier

```
public const int HpIncreasingModifier = 5
```

Field Value

int

arenaProvider

```
public ArenaProvider arenaProvider
```

Field Value

ArenaProvider

chargeAp

```
public bool chargeAp
```

Field Value

bool

costumes

```
public List<Guid> costumes
```

Field Value

List<Guid>

enemyAvatarAddress

```
public Address enemyAvatarAddress
```

Field Value

Address

equipments

```
public List<Guid> equipments
```

Field Value

List<Guid>

memo

```
public string memo
```

Field Value

string

myAvatarAddress

```
public Address myAvatarAddress
```

Field Value

Address

runeInfos

```
public List<RuneSlotInfo> runeInfos
```

Field Value

List<RuneSlotInfo>

Properties

PlainValueInternal

```
protected override IImmutableDictionary<string, IValue> PlainValueInternal { get; }
```

Property Value

IImmutableDictionary<string, IValue>

Methods

Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

Parameters

`context` IActionContext

Returns

IWorld

LoadPlainValueInternal(IImmutableDictionary<string, IValue>)

```
protected override void LoadPlainValueInternal(IImmutableDictionary<string,
```

```
IValue> plainValue)
```

Parameters

`plainValue` IImmutableDictionary<string, IValue>

Namespace Nekoyume.Action.Coupons

Classes

IssueCoupons

RedeemCoupon

TransferCoupons

Class IssueCoupons

Namespace: Nekoyume.Action.Coupons

Assembly: Lib9c.dll

```
[Serializable]
[ActionType("issue_coupons")]
public sealed class IssueCoupons : GameAction, IAction
```

Inheritance

object ← ActionBase ← GameAction ← IssueCoupons

Implements

IAction

Inherited Members

GameAction.Id , GameAction.PlainValue , GameAction.LoadPlainValue(IValue) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

IssueCoupons()

```
public IssueCoupons()
```

IssueCoupons(IImmutableDictionary<RewardSet, uint>, Address)

```
public IssueCoupons(IImmutableDictionary<RewardSet, uint> rewards,
Address recipient)
```

Parameters

`rewards` IImmutableDictionary<RewardSet, uint>

`recipient` Address

Properties

PlainValueInternal

```
protected override IImmutableDictionary<string, IValue> PlainValueInternal { get; }
```

Property Value

IImmutableDictionary<string, IValue>

Recipient

```
public Address Recipient { get; }
```

Property Value

Address

Rewards

```
public IImmutableDictionary<RewardSet, uint> Rewards { get; }
```

Property Value

IImmutableDictionary<RewardSet, uint>

Methods

Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

Parameters

`context` IActionContext

Returns

IWorld

LoadPlainValueInternal(IImmutableDictionary<string, IValue>)

```
protected override void LoadPlainValueInternal(IImmutableDictionary<string,
IValue> plainValue)
```

Parameters

`plainValue` IImmutableDictionary<string, IValue>

Class RedeemCoupon

Namespace: Nekoyume.Action.Coupons

Assembly: Lib9c.dll

```
[Serializable]
[ActionType("redeem_coupon")]
public sealed class RedeemCoupon : GameAction, IAction
```

Inheritance

object ← ActionBase ← GameAction ← RedeemCoupon

Implements

IAction

Inherited Members

GameAction.Id , GameAction.PlainValue , GameAction.LoadPlainValue(IValue) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

RedeemCoupon()

```
public RedeemCoupon()
```

RedeemCoupon(Guid, Address)

```
public RedeemCoupon(Guid couponId, Address avatarAddress)
```

Parameters

`couponId` Guid

`avatarAddress` Address

Properties

AvatarAddress

```
public Address AvatarAddress { get; }
```

Property Value

Address

CouponId

```
public Guid CouponId { get; }
```

Property Value

Guid

PlainValueInternal

```
protected override IImmutableDictionary<string, IValue> PlainValueInternal { get; }
```

Property Value

IImmutableDictionary<string, IValue>

Methods

Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

Parameters

`context` IActionContext

Returns

IWorld

LoadPlainValueInternal(IImmutableDictionary<string, IValue>)

```
protected override void LoadPlainValueInternal(IImmutableDictionary<string,
IValue> plainValue)
```

Parameters

`plainValue` IImmutableDictionary<string, IValue>

Class TransferCoupons

Namespace: Nekoyume.Action.Coupons

Assembly: Lib9c.dll

```
[Serializable]
[ActionType("transfer_coupons")]
public sealed class TransferCoupons : GameAction, IAction
```

Inheritance

object ← ActionBase ← GameAction ← TransferCoupons

Implements

IAction

Inherited Members

GameAction.Id , GameAction.PlainValue , GameAction.LoadPlainValue(IValue) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

TransferCoupons()

```
public TransferCoupons()
```

TransferCoupons(IImmutableDictionary<Address, IImmutableSet<Guid>>)

```
public TransferCoupons(IImmutableDictionary<Address, IImmutableSet<Guid>>
couponsPerRecipient)
```

Parameters

`couponsPerRecipient` IImmutableDictionary<Address, IImmutableSet<Guid>>

Properties

CouponsPerRecipient

```
public IImmutableDictionary<Address, IImmutableSet<Guid>> CouponsPerRecipient {
get; }
```

Property Value

IImmutableDictionary<Address, IImmutableSet<Guid>>

PlainValueInternal

```
protected override IImmutableDictionary<string, IValue> PlainValueInternal { get; }
```

Property Value

IImmutableDictionary<string, IValue>

Methods

Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

Parameters

`context` IActionContext

Returns

IWorld

LoadPlainValueInternal(IImmutableDictionary<string, IValue>)

```
protected override void LoadPlainValueInternal(IImmutableDictionary<string,
IValue> plainValue)
```

Parameters

`plainValue` IImmutableDictionary<string, IValue>

Namespace Nekoyume.Action.Custom EquipmentCraft

Classes

CustomEquipmentCraft

Structs

CustomCraftData

Struct CustomCraftData

Namespace: Nekoyume.Action.CustomEquipmentCraft

Assembly: Lib9c.dll

```
[Serializable]
public struct CustomCraftData
```

Inherited Members

ValueType.Equals(object) , ValueType.GetHashCode() , ValueType.ToString() , object.Equals(object, object) , object.GetType() , object.ReferenceEquals(object, object)

Fields

IconId

```
public int IconId
```

Field Value

int

RecipeId

```
public int RecipeId
```

Field Value

int

SlotIndex

```
public int SlotIndex
```

Field Value

int

Methods

Serialize()

```
public IValue Serialize()
```

Returns

IValue

Class CustomEquipmentCraft

Namespace: Nekoyume.Action.CustomEquipmentCraft

Assembly: Lib9c.dll

```
[Serializable]
[ActionType("custom_equipment_craft")]
public class CustomEquipmentCraft : ActionBase, IAction
```

Inheritance

object ← ActionBase ← CustomEquipmentCraft

Implements

IAction

Inherited Members

ActionBase.GetSignerAndOtherAddressesHex(IActionContext, params Address[]) , ActionBase.TryGetAdminState(IActionContext, out AdminState) , ActionBase.CheckPermission(IActionContext) , ActionBase.CheckObsolete(long, IActionContext) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Fields

AvatarAddress

```
public Address AvatarAddress
```

Field Value

Address

CraftList

```
public List<CustomCraftData> CraftList
```

Field Value

List<CustomCraftData>

RandomIconId

```
public const int RandomIconId = 0
```

Field Value

int

TypeIdentifier

```
public const string TypeIdentifier = "custom_equipment_craft"
```

Field Value

string

Properties

PlainValue

```
public override IValue PlainValue { get; }
```

Property Value

IValue

Methods

Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

Parameters

`context` IActionContext

Returns

IWorld

LoadPlainValue(IValue)

```
public override void LoadPlainValue(IValue plainValue)
```

Parameters

`plainValue` IValue

Namespace Nekoyume.Action.Exceptions

Classes

CombatPowerException
: Exception thrown when player's combat power doesn't meet the floor requirements.

DuplicatedCraftSlotIndexException

EmptyRewardException

FloorAlreadyClearedException
: Exception thrown when trying to re-enter a floor that has already been cleared in the current season.

InfiniteTowerScheduleEndedException
: Exception thrown when infinite tower schedule has ended.

InfiniteTowerScheduleNotActiveException
: Exception thrown when infinite tower schedule is not active.

InfiniteTowerScheduleNotStartedException
: Exception thrown when infinite tower schedule has not started yet.

InvalidItemGradeException
: Exception thrown when an item grade doesn't meet the floor requirements.

InvalidItemLevelException
: Exception thrown when an item level doesn't meet the floor requirements.

InvalidSummonCountException

NotEnoughInfiniteTowerTicketsException
: Exception thrown when player doesn't have enough infinite tower tickets.

StageNotClearedException
: Exception thrown when trying to access a stage that hasn't been cleared yet.

Class CombatPowerException

Namespace: Nekoyume.Action.Exceptions

Assembly: Lib9c.dll

Exception thrown when player's combat power doesn't meet the floor requirements.

```
public class CombatPowerException : Exception, ISerializable
```

Inheritance

object ← Exception ← CombatPowerException

Implements

ISerializable

Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Constructors

CombatPowerException(string, string, long, long, bool)

Initializes a new instance of the CombatPowerException class.

```
public CombatPowerException(string actionType, string addressesHex, long requiredCp,
long currentCp, bool isInsufficient = true)
```

Parameters

`actionType` string

The type of action that failed.

`addressesHex` string

The hex representation of addresses involved.

`requiredCp` long

The required combat power.

`currentCp` long

The current combat power.

`isInsufficient` bool

Whether the combat power is insufficient (true) or excessive (false).

Class DuplicatedCraftSlotIndexException

Namespace: Nekoyume.Action.Exceptions

Assembly: Lib9c.dll

```
[Serializable]
public class DuplicatedCraftSlotIndexException : Exception, ISerializable
```

Inheritance

object ← Exception ← DuplicatedCraftSlotIndexException

Implements

ISerializable

Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Constructors

DuplicatedCraftSlotIndexException(SerializationInfo, StreamingContext)

```
protected DuplicatedCraftSlotIndexException(SerializationInfo info,
StreamingContext context)
```

Parameters

`info` SerializationInfo

`context` StreamingContext

DuplicatedCraftSlotIndexException(string)

```
public DuplicatedCraftSlotIndexException(string message)
```

Parameters

`message` string

Class EmptyRewardException

Namespace: Nekoyume.Action.Exceptions

Assembly: Lib9c.dll

```
[Serializable]
public class EmptyRewardException : Exception, ISerializable
```

Inheritance

object ← Exception ← EmptyRewardException

Implements

ISerializable

Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Constructors

EmptyRewardException()

```
public EmptyRewardException()
```

EmptyRewardException(SerializationInfo, StreamingContext)

```
protected EmptyRewardException(SerializationInfo info, StreamingContext context)
```

Parameters

`info` SerializationInfo

`context` StreamingContext

EmptyRewardException(string)

```
public EmptyRewardException(string msg)
```

Parameters

`msg` string

Class FloorAlreadyClearedException

Namespace: Nekoyume.Action.Exceptions

Assembly: Lib9c.dll

Exception thrown when trying to re-enter a floor that has already been cleared in the current season.

```
public class FloorAlreadyClearedException : Exception, ISerializable
```

Inheritance

object ← Exception ← FloorAlreadyClearedException

Implements

ISerializable

Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Constructors

FloorAlreadyClearedException(string, string, int, int)

Initializes a new instance of the FloorAlreadyClearedException class.

```
public FloorAlreadyClearedException(string actionType, string addressesHex, int
floorId, int clearedFloor)
```

Parameters

`actionType` string

The type of action that failed.

`addressesHex` string

The hex representation of addresses involved.

`floorId` int

The floor that was attempted.

`clearedFloor` int

The highest cleared floor.

Class InfiniteTowerScheduleEndedException

Namespace: Nekoyume.Action.Exceptions

Assembly: Lib9c.dll

Exception thrown when infinite tower schedule has ended.

```
public class InfiniteTowerScheduleEndedException : Exception, ISerializable
```

Inheritance

object ← Exception ← InfiniteTowerScheduleEndedException

Implements

ISerializable

Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Constructors

InfiniteTowerScheduleEndedException(string, string, int, long, long)

Initializes a new instance of the InfiniteTowerScheduleEndedException class.

```
public InfiniteTowerScheduleEndedException(string actionType, string addressesHex,
int infiniteTowerId, long currentBlockIndex, long endBlockIndex)
```

Parameters

`actionType` string

The type of action that failed.

`addressesHex` string

The hex representation of addresses involved.

`infiniteTowerId` int

The infinite tower ID.

`currentBlockIndex` long

The current block index.

`endBlockIndex` long

The block index when the schedule ended.

Class InfiniteTowerScheduleNotActiveException

Namespace: Nekoyume.Action.Exceptions

Assembly: Lib9c.dll

Exception thrown when infinite tower schedule is not active.

```
public class InfiniteTowerScheduleNotActiveException : Exception, ISerializable
```

Inheritance

object ← Exception ← InfiniteTowerScheduleNotActiveException

Implements

ISerializable

Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Constructors

InfiniteTowerScheduleNotActiveException(string, string, int, long, long, long)

Initializes a new instance of the InfiniteTowerScheduleNotActiveException class.

```
public InfiniteTowerScheduleNotActiveException(string actionType, string
addressesHex, int infiniteTowerId, long currentBlockIndex, long startBlockIndex,
long endBlockIndex)
```

Parameters

`actionType` string

The type of action that failed.

`addressesHex` string

The hex representation of addresses involved.

`infiniteTowerId` int

The infinite tower ID.

`currentBlockIndex` long

The current block index.

`startBlockIndex` long

The block index when the schedule starts.

`endBlockIndex` long

The block index when the schedule ends.

Class InfiniteTowerScheduleNotStartedException

Namespace: Nekoyume.Action.Exceptions

Assembly: Lib9c.dll

Exception thrown when infinite tower schedule has not started yet.

```
public class InfiniteTowerScheduleNotStartedException : Exception, ISerializable
```

Inheritance

object ← Exception ← InfiniteTowerScheduleNotStartedException

Implements

ISerializable

Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Constructors

InfiniteTowerScheduleNotStartedException(string, string, int, long, long)

Initializes a new instance of the InfiniteTowerScheduleNotStartedException class.

```
public InfiniteTowerScheduleNotStartedException(string actionType, string
addressesHex, int infiniteTowerId, long currentBlockIndex, long startBlockIndex)
```

Parameters

`actionType` string

The type of action that failed.

`addressesHex` string

The hex representation of addresses involved.

`infiniteTowerId` int

The infinite tower ID.

`currentBlockIndex` long

The current block index.

`startBlockIndex` long

The block index when the schedule starts.

Class InvalidItemGradeException

Namespace: Nekoyume.Action.Exceptions

Assembly: Lib9c.dll

Exception thrown when an item grade doesn't meet the floor requirements.

```
public class InvalidItemGradeException : Exception, ISerializable
```

Inheritance

object ← Exception ← InvalidItemGradeException

Implements

ISerializable

Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Constructors

InvalidItemGradeException(string, string, int, int, bool)

Initializes a new instance of the InvalidItemGradeException class.

```
public InvalidItemGradeException(string actionType, string addressesHex, int
itemGrade, int requiredGrade, bool isMinimum = true)
```

Parameters

`actionType` string

The type of action that failed.

`addressesHex` string

The hex representation of addresses involved.

`itemGrade` int

The item grade that failed validation.

`requiredGrade` int

The required grade.

`isMinimum` bool

Whether the requirement is a minimum (true) or maximum (false).

Class InvalidItemLevelException

Namespace: Nekoyume.Action.Exceptions

Assembly: Lib9c.dll

Exception thrown when an item level doesn't meet the floor requirements.

```
public class InvalidItemLevelException : Exception, ISerializable
```

Inheritance

object ← Exception ← InvalidItemLevelException

Implements

ISerializable

Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Constructors

InvalidItemLevelException(string, string, int, int, bool)

Initializes a new instance of the InvalidItemLevelException class.

```
public InvalidItemLevelException(string actionType, string addressesHex, int
itemLevel, int requiredLevel, bool isMinimum = true)
```

Parameters

`actionType` string

The type of action that failed.

`addressesHex` string

The hex representation of addresses involved.

`itemLevel` int

The item level that failed validation.

`requiredLevel` int

The required level.

`isMinimum` bool

Whether the requirement is a minimum (true) or maximum (false).

Class InvalidSummonCountException

Namespace: Nekoyume.Action.Exceptions

Assembly: Lib9c.dll

```
public class InvalidSummonCountException : ArgumentException, ISerializable
```

Inheritance

object ← Exception ← SystemException ← ArgumentException ← InvalidSummonCountException

Implements

ISerializable

Inherited Members

ArgumentException.GetObjectData(SerializationInfo, StreamingContext) , ArgumentException.Message , ArgumentException.ParamName , Exception.GetBaseException() , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Constructors

InvalidSummonCountException()

```
public InvalidSummonCountException()
```

InvalidSummonCountException(SerializationInfo, StreamingContext)

```
protected InvalidSummonCountException(SerializationInfo info,
StreamingContext context)
```

Parameters

`info` SerializationInfo

`context` StreamingContext

InvalidSummonCountException(string)

```
public InvalidSummonCountException(string msg)
```

Parameters

`msg` string

Class NotEnoughInfiniteTowerTicketsException

Namespace: Nekoyume.Action.Exceptions

Assembly: Lib9c.dll

Exception thrown when player doesn't have enough infinite tower tickets.

```
public class NotEnoughInfiniteTowerTicketsException : Exception, ISerializable
```

Inheritance

object ← Exception ← NotEnoughInfiniteTowerTicketsException

Implements

ISerializable

Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Constructors

NotEnoughInfiniteTowerTicketsException(string, string, int, int)

Initializes a new instance of the NotEnoughInfiniteTowerTicketsException class.

```
public NotEnoughInfiniteTowerTicketsException(string actionType, string
addressesHex, int requiredTickets, int availableTickets)
```

Parameters

`actionType` string

The type of action that failed.

`addressesHex` string

The hex representation of addresses involved.

`requiredTickets` int

The number of tickets required.

`availableTickets` int

The number of tickets available.

Class StageNotClearedException

Namespace: Nekoyume.Action.Exceptions

Assembly: Lib9c.dll

Exception thrown when trying to access a stage that hasn't been cleared yet.

```
public class StageNotClearedException : Exception, ISerializable
```

Inheritance

object ← Exception ← StageNotClearedException

Implements

ISerializable

Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Constructors

StageNotClearedException(string, string, int, int)

Initializes a new instance of the StageNotClearedException class.

```
public StageNotClearedException(string actionType, string addressesHex, int
requiredStage, int clearedStage)
```

Parameters

`actionType` string

The type of action that failed.

`addressesHex` string

The hex representation of addresses involved.

`requiredStage` int

The stage that needs to be cleared.

`clearedStage` int

The highest stage that has been cleared.

Namespace Nekoyume.Action.Exceptions.AdventureBoss

Classes

AlreadyClaimedException

ClaimExpiredException

InsufficientStakingException

InvalidAdventureBossSeasonException

InvalidBountyException

MaxInvestmentCountExceededException

PreviousBountyException

SeasonInProgressException

Class AlreadyClaimedException

Namespace: Nekoyume.Action.Exceptions.AdventureBoss

Assembly: Lib9c.dll

```
[Serializable]
public class AlreadyClaimedException : Exception, ISerializable
```

Inheritance

object ← Exception ← AlreadyClaimedException

Implements

ISerializable

Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Constructors

AlreadyClaimedException()

```
public AlreadyClaimedException()
```

AlreadyClaimedException(SerializationInfo, StreamingContext)

```
protected AlreadyClaimedException(SerializationInfo info, StreamingContext context)
```

Parameters

`info` SerializationInfo

`context` StreamingContext

AlreadyClaimedException(string)

```
public AlreadyClaimedException(string msg)
```

Parameters

`msg` string

Class ClaimExpiredException

Namespace: Nekoyume.Action.Exceptions.AdventureBoss

Assembly: Lib9c.dll

```
[Serializable]
public class ClaimExpiredException : Exception, ISerializable
```

Inheritance

object ← Exception ← ClaimExpiredException

Implements

ISerializable

Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Constructors

ClaimExpiredException()

```
public ClaimExpiredException()
```

ClaimExpiredException(SerializationInfo, StreamingContext)

```
protected ClaimExpiredException(SerializationInfo info, StreamingContext context)
```

Parameters

`info` SerializationInfo

`context` StreamingContext

ClaimExpiredException(string)

```
public ClaimExpiredException(string msg)
```

Parameters

`msg` string

Class InsufficientStakingException

Namespace: Nekoyume.Action.Exceptions.AdventureBoss

Assembly: Lib9c.dll

```
[Serializable]
public class InsufficientStakingException : Exception, ISerializable
```

Inheritance

object ← Exception ← InsufficientStakingException

Implements

ISerializable

Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Constructors

InsufficientStakingException()

```
public InsufficientStakingException()
```

InsufficientStakingException(SerializationInfo, StreamingContext)

```
protected InsufficientStakingException(SerializationInfo info,
StreamingContext context)
```

Parameters

`info` SerializationInfo

`context` StreamingContext

InsufficientStakingException(string)

```
public InsufficientStakingException(string msg)
```

Parameters

`msg` string

Class InvalidAdventureBossSeasonException

Namespace: Nekoyume.Action.Exceptions.AdventureBoss

Assembly: Lib9c.dll

```
[Serializable]
public class InvalidAdventureBossSeasonException : Exception, ISerializable
```

Inheritance

object ← Exception ← InvalidAdventureBossSeasonException

Implements

ISerializable

Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Constructors

InvalidAdventureBossSeasonException()

```
public InvalidAdventureBossSeasonException()
```

InvalidAdventureBossSeasonException(SerializationInfo, StreamingContext)

```
protected InvalidAdventureBossSeasonException(SerializationInfo info,
StreamingContext context)
```

Parameters

`info` SerializationInfo

`context` StreamingContext

InvalidAdventureBossSeasonException(string)

```
public InvalidAdventureBossSeasonException(string message)
```

Parameters

`message` string

Class InvalidBountyException

Namespace: Nekoyume.Action.Exceptions.AdventureBoss

Assembly: Lib9c.dll

```
[Serializable]
public class InvalidBountyException : Exception, ISerializable
```

Inheritance

object ← Exception ← InvalidBountyException

Implements

ISerializable

Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Constructors

InvalidBountyException()

```
public InvalidBountyException()
```

InvalidBountyException(SerializationInfo, StreamingContext)

```
protected InvalidBountyException(SerializationInfo info, StreamingContext context)
```

Parameters

`info` SerializationInfo

`context` StreamingContext

InvalidBountyException(string)

```
public InvalidBountyException(string msg)
```

Parameters

`msg` string

Class MaxInvestmentCountExceededException

Namespace: Nekoyume.Action.Exceptions.AdventureBoss

Assembly: Lib9c.dll

```
[Serializable]
public class MaxInvestmentCountExceededException : Exception, ISerializable
```

Inheritance

object ← Exception ← MaxInvestmentCountExceededException

Implements

ISerializable

Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Constructors

MaxInvestmentCountExceededException()

```
public MaxInvestmentCountExceededException()
```

MaxInvestmentCountExceededException(SerializationInfo, StreamingContext)

```
protected MaxInvestmentCountExceededException(SerializationInfo info,
StreamingContext context)
```

Parameters

`info` SerializationInfo

`context` StreamingContext

MaxInvestmentCountExceededException(string)

```
public MaxInvestmentCountExceededException(string message)
```

Parameters

`message` string

Class PreviousBountyException

Namespace: Nekoyume.Action.Exceptions.AdventureBoss

Assembly: Lib9c.dll

```
[Serializable]
public class PreviousBountyException : Exception, ISerializable
```

Inheritance

object ← Exception ← PreviousBountyException

Implements

ISerializable

Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Constructors

PreviousBountyException()

```
public PreviousBountyException()
```

PreviousBountyException(SerializationInfo, StreamingContext)

```
protected PreviousBountyException(SerializationInfo info, StreamingContext context)
```

Parameters

`info` SerializationInfo

`context` StreamingContext

PreviousBountyException(string)

```
public PreviousBountyException(string msg)
```

Parameters

`msg` string

Class SeasonInProgressException

Namespace: Nekoyume.Action.Exceptions.AdventureBoss

Assembly: Lib9c.dll

```
[Serializable]
public class SeasonInProgressException : Exception, ISerializable
```

Inheritance

object ← Exception ← SeasonInProgressException

Implements

ISerializable

Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Constructors

SeasonInProgressException()

```
public SeasonInProgressException()
```

SeasonInProgressException(SerializationInfo, StreamingContext)

```
protected SeasonInProgressException(SerializationInfo info,
StreamingContext context)
```

Parameters

`info` SerializationInfo

`context` StreamingContext

SeasonInProgressException(string)

```
public SeasonInProgressException(string msg)
```

Parameters

`msg` string

Namespace Nekoyume.Action.Exceptions.Arena

Classes

AlreadyJoinedArenaException

Class AlreadyJoinedArenaException

Namespace: Nekoyume.Action.Exceptions.Arena

Assembly: Lib9c.dll

```
[Serializable]
public class AlreadyJoinedArenaException : Exception, ISerializable
```

Inheritance

object ← Exception ← AlreadyJoinedArenaException

Implements

ISerializable

Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Constructors

AlreadyJoinedArenaException(int, int, Address)

```
public AlreadyJoinedArenaException(int championshipId, int round,
Address avatarAddress)
```

Parameters

`championshipId` int

`round` int

`avatarAddress` Address

AlreadyJoinedArenaException(SerializationInfo, StreamingContext)

```
protected AlreadyJoinedArenaException(SerializationInfo info,
StreamingContext context)
```

Parameters

`info` SerializationInfo

`context` StreamingContext

AlreadyJoinedArenaException(string)

```
public AlreadyJoinedArenaException(string msg)
```

Parameters

`msg` string

Namespace Nekoyume.Action.Exceptions.CustomEquipmentCraft

Classes

NotEnoughRelationshipException

RandomOnlyIconException

Class NotEnoughRelationshipException

Namespace: Nekoyume.Action.Exceptions.CustomEquipmentCraft

Assembly: Lib9c.dll

```
[Serializable]
public class NotEnoughRelationshipException : Exception, ISerializable
```

Inheritance

object ← Exception ← NotEnoughRelationshipException

Implements

ISerializable

Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Constructors

NotEnoughRelationshipException(SerializationInfo, StreamingContext)

```
protected NotEnoughRelationshipException(SerializationInfo info,
StreamingContext context)
```

Parameters

`info` SerializationInfo

`context` StreamingContext

NotEnoughRelationshipException(string)

```
public NotEnoughRelationshipException(string s)
```

Parameters

s [string]

Class RandomOnlyIconException

Namespace: Nekoyume.Action.Exceptions.CustomEquipmentCraft

Assembly: Lib9c.dll

```
public class RandomOnlyIconException : Exception, ISerializable
```

Inheritance

object ← Exception ← RandomOnlyIconException

Implements

ISerializable

Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Constructors

RandomOnlyIconException(int)

```
public RandomOnlyIconException(int iconId)
```

Parameters

`iconId` int

RandomOnlyIconException(SerializationInfo, StreamingContext)

```
protected RandomOnlyIconException(SerializationInfo info, StreamingContext context)
```

Parameters

`info` SerializationInfo

`context` StreamingContext

RandomOnlyIconException(string)

```
public RandomOnlyIconException(string s)
```

Parameters

`s` string

Namespace Nekoyume.Action.Factory

Classes

ClaimStakeRewardFactory

Class ClaimStakeRewardFactory

Namespace: Nekoyume.Action.Factory

Assembly: Lib9c.dll

```
public static class ClaimStakeRewardFactory
```

Inheritance

object ← ClaimStakeRewardFactory

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Methods

CreateByBlockIndex(long, Address)

```
public static IClaimStakeReward CreateByBlockIndex(long blockIndex,
Address avatarAddress)
```

Parameters

`blockIndex` long

`avatarAddress` Address

Returns

IClaimStakeReward

CreateByVersion(int, Address)

```
public static IClaimStakeReward CreateByVersion(int version, Address avatarAddress)
```

Parameters

`version` int

`avatarAddress` Address

Returns

IClaimStakeReward

Namespace Nekoyume.Action.Garages

Classes

BulkUnloadFromGarages

DeliverToOthersGarages

GarageUtils

LoadIntoMyGarages

UnloadFromMyGarages

Class BulkUnloadFromGarages

Namespace: Nekoyume.Action.Garages

Assembly: Lib9c.dll

```
[ActionType("bulk_unload_from_garages")]
public class BulkUnloadFromGarages : GameAction, IBulkUnloadFromGaragesV1, IAction
```

Inheritance

object ← ActionBase ← GameAction ← BulkUnloadFromGarages

Implements

IBulkUnloadFromGaragesV1, IAction

Inherited Members

GameAction.Id , GameAction.PlainValue , GameAction.LoadPlainValue(IValue) , ActionBase.GetSignerAndOtherAddressesHex(IActionContext, params Address[]) , ActionBase.TryGetAdminState(IActionContext, out AdminState) , ActionBase.CheckPermission(IActionContext) , ActionBase.CheckObsolete(long, IActionContext) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

BulkUnloadFromGarages()

```
public BulkUnloadFromGarages()
```

BulkUnloadFromGarages(IReadOnlyList<(Address recipientAvatarAddress, IEnumerable<(Address balanceAddress, FungibleAssetValue value)>? fungibleAssetValues, IEnumerable<(HashDigest<SHA256> fungibleId, int count)>? fungibleIdAndCounts, string? memo)>)

```
public BulkUnloadFromGarages(IReadOnlyList<(Address recipientAvatarAddress,
IEnumerable<(Address balanceAddress, FungibleAssetValue value)>?
fungibleAssetValues, IEnumerable<(HashDigest<SHA256> fungibleId, int count)>?
fungibleIdAndCounts, string? memo)> unloadData)
```

Parameters

`unloadData` IReadOnlyList<(Address recipientAvatarAddress, IEnumerable<(Address balanceAddress, FungibleAssetValue value)> fungibleAssetValues, IEnumerable<(HashDigest<SHA256> fungibleId, int count)> fungibleIdAndCounts, string memo)>

Properties

PlainValueInternal

```
protected override IImmutableDictionary<string, IValue> PlainValueInternal { get; }
```

Property Value

IImmutableDictionary<string, IValue>

UnloadData

```
public IReadOnlyList<(Address recipientAvatarAddress, IOrderedEnumerable<(Address
balanceAddress, FungibleAssetValue value)>? fungibleAssetValues,
IOrderedEnumerable<(HashDigest<SHA256> fungibleId, int count)>? fungibleIdAndCounts,
string? memo)> UnloadData { get; }
```

Property Value

IReadOnlyList<(Address recipientAvatarAddress, IOrderedEnumerable<(Address balanceAddress, FungibleAssetValue value)> fungibleAssetValues, IOrdered Enumerable<(HashDigest<SHA256> fungibleId, int count)> fungibleIdAndCounts , string memo)>

Methods

Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

Parameters

`context` IActionContext

Returns

IWorld

LoadPlainValueInternal(IImmutableDictionary<string, IValue>)

```
protected override void LoadPlainValueInternal(IImmutableDictionary<string,
IValue> plainValue)
```

Parameters

`plainValue` IImmutableDictionary<string, IValue>

Class DeliverToOthersGarages

Namespace: Nekoyume.Action.Garages

Assembly: Lib9c.dll

```
[ActionType("deliver_to_others_garages")]
public class DeliverToOthersGarages : GameAction, IDeliverToOthersGaragesV1, IAction
```

Inheritance

object ← ActionBase ← GameAction ← DeliverToOthersGarages

Implements

IDeliverToOthersGaragesV1, IAction

Inherited Members

GameAction.Id , GameAction.PlainValue , GameAction.LoadPlainValue(IValue) , ActionBase.GetSignerAndOtherAddressesHex(IActionContext, params Address[]) , ActionBase.TryGetAdminState(IActionContext, out AdminState) , ActionBase.CheckPermission(IActionContext) , ActionBase.CheckObsolete(long, IActionContext) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

DeliverToOthersGarages()

```
public DeliverToOthersGarages()
```

DeliverToOthersGarages(Address, IEnumerable<FungibleAssetValue>?, IEnumerable<(HashDigest<SHA256> fungibleId, int count)>?, string?)

```
public DeliverToOthersGarages(Address recipientAgentAddr,
IEnumerable<FungibleAssetValue>? fungibleAssetValues,
IEnumerable<(HashDigest<SHA256> fungibleId, int count)>? fungibleIdAndCounts,
string? memo)
```

Parameters

`recipientAgentAddr` Address

`fungibleAssetValues` IEnumerable<FungibleAssetValue>

`fungibleIdAndCounts` IEnumerable<(HashDigest<SHA256> fungibleId, int count)>

`memo` string

Properties

FungibleAssetValues

```
public IOrderedEnumerable<FungibleAssetValue>? FungibleAssetValues { get; }
```

Property Value

IOrderedEnumerable<FungibleAssetValue>

FungibleIdAndCounts

```
public IOrderedEnumerable<(HashDigest<SHA256> fungibleId, int count)>?
FungibleIdAndCounts { get; }
```

Property Value

IOrderedEnumerable<(HashDigest<SHA256> fungibleId, int count)>

Memo

```
public string? Memo { get; }
```

Property Value

string

PlainValueInternal

```
protected override IImmutableDictionary<string, IValue> PlainValueInternal { get; }
```

Property Value

IImmutableDictionary<string, IValue>

RecipientAgentAddr

```
public Address RecipientAgentAddr { get; }
```

Property Value

Address

Methods

Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

Parameters

`context` IActionContext

Returns

LoadPlainValueInternal(IImmutableDictionary<string, IValue>)

```
protected override void LoadPlainValueInternal(IImmutableDictionary<string,
IValue> plainValue)
```

Parameters

`plainValue` IImmutableDictionary<string, IValue>

Class GarageUtils

Namespace: Nekoyume.Action.Garages

Assembly: Lib9c.dll

```
public static class GarageUtils
```

Inheritance

object ← GarageUtils

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Methods

ConvertToFungibleItemGarage(IValue?, Address, HashDigest<SHA256>)

```
public static FungibleItemGarage ConvertToFungibleItemGarage(IValue? garageState,
Address garageAddr, HashDigest<SHA256> fungibleId)
```

Parameters

`garageState` IValue

`garageAddr` Address

`fungibleId` HashDigest<SHA256>

Returns

FungibleItemGarage

MergeAndSort(IEnumerable<(HashDigest<SHA256> fungibleId, int count)>?)

```
public static IOrderedEnumerable<(HashDigest<SHA256> fungibleId, int
count)>? MergeAndSort(IEnumerable<(HashDigest<SHA256> fungibleId, int
count)>? fungibleIdAndCounts)
```

Parameters

`fungibleIdAndCounts` IEnumerable<(HashDigest<SHA256> fungibleId, int count)>

Returns

IOrderedEnumerable<(HashDigest<SHA256> fungibleId, int count)>

MergeAndSort(IEnumerable<(Address balanceAddr, FungibleAssetValue value)>?)

```
public static IOrderedEnumerable<(Address balanceAddr, FungibleAssetValue
value)>? MergeAndSort(IEnumerable<(Address balanceAddr, FungibleAssetValue
value)>? fungibleAssetValues)
```

Parameters

`fungibleAssetValues` IEnumerable<(Address balanceAddr, FungibleAssetValue value)>

Returns

IOrderedEnumerable<(Address balanceAddr, FungibleAssetValue value)>

WithGarageStateTuples(Address, IWorldState, IEnumerable<(HashDigest<SHA256> fungibleId, int count)>)

```
public static IOrderedEnumerable<(HashDigest<SHA256> fungibleId, int count,
Address garageAddr, IValue? garageState)> WithGarageStateTuples(Address
agentAddr, IWorldState states, IEnumerable<(HashDigest<SHA256> fungibleId, int
count)> fungibleIdAndCounts)
```

Parameters

`agentAddr` Address

`states` IWorldState

`fungibleIdAndCounts` IEnumerable<(HashDigest<SHA256> fungibleId, int count)>

Returns

IOrderedEnumerable<(HashDigest<SHA256> fungibleId, int count, Address garageAddr, IValue garageState)>

WithGarageStateTuples(Address, Address, IWorldState, IEnumerable<(HashDigest<SHA256> fungibleId, int count)>)

```
public static IOrderedEnumerable<(HashDigest<SHA256> fungibleId, int count, Address
senderGarageAddr, FungibleItemGarage senderGarage, Address recipientGarageAddr,
IValue? recipientGarageState)> WithGarageStateTuples(Address senderAgentAddr,
Address recipientAgentAddr, IWorldState states, IEnumerable<(HashDigest<SHA256>
fungibleId, int count)> fungibleIdAndCounts)
```

Parameters

`senderAgentAddr` Address

`recipientAgentAddr` Address

`states` IWorldState

`fungibleIdAndCounts` IEnumerable<(HashDigest<SHA256> fungibleId, int count)>

Returns

IOrderedEnumerable<(HashDigest<SHA256> fungibleId, int count, Address senderGarageAddr, FungibleItemGarage senderGarage, Address recipientGarageAddr, IValue recipientGarageState)>

WithGarageTuples(Address, IWorldState, IEnumerable<(HashDigest<SHA256> fungibleId, int count)>)

```
public static IOrderedEnumerable<(HashDigest<SHA256> fungibleId, int count,
Address garageAddr, FungibleItemGarage garage)> WithGarageTuples(Address
agentAddr, IWorldState states, IEnumerable<(HashDigest<SHA256> fungibleId, int
count)> fungibleIdAndCounts)
```

Parameters

`agentAddr` Address

`states` IWorldState

`fungibleIdAndCounts` IEnumerable<(HashDigest<SHA256> fungibleId, int count)>

Returns

IOrderedEnumerable<(HashDigest<SHA256> fungibleId, int count, Address garageAddr, FungibleItemGarage garage)>

Class LoadIntoMyGarages

Namespace: Nekoyume.Action.Garages

Assembly: Lib9c.dll

```
[ActionType("load_into_my_garages")]
public class LoadIntoMyGarages : GameAction, ILoadIntoMyGaragesV1, IAction
```

Inheritance

object ← ActionBase ← GameAction ← LoadIntoMyGarages

Implements

ILoadIntoMyGaragesV1, IAction

Inherited Members

GameAction.Id , GameAction.PlainValue , GameAction.LoadPlainValue(IValue) , ActionBase.GetSignerAndOtherAddressesHex(IActionContext, params Address[]) , ActionBase.TryGetAdminState(IActionContext, out AdminState) , ActionBase.CheckPermission(IActionContext) , ActionBase.CheckObsolete(long, IActionContext) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

LoadIntoMyGarages()

```
public LoadIntoMyGarages()
```

LoadIntoMyGarages(IEnumerable<(Address balanceAddr, FungibleAssetValue value)>?, Address?, IEnumerable<(HashDigest<SHA256> fungibleId, int count)>?, string?)

```
public LoadIntoMyGarages(IEnumerable<(Address balanceAddr, FungibleAssetValue
value)>? fungibleAssetValues, Address? avatarAddr, IEnumerable<(HashDigest<SHA256>
fungibleId, int count)>? fungibleIdAndCounts, string? memo)
```

Parameters

`fungibleAssetValues` IEnumerable<(Address balanceAddr, FungibleAssetValue value)>

`avatarAddr` Address?

`fungibleIdAndCounts` IEnumerable<(HashDigest<SHA256> fungibleId, int count)>

`memo` string

Properties

AvatarAddr

This address should belong to one of the signer's avatars. If the avatar state is v1, there is no separate inventory, so it should be execute another action first to migrate the avatar state to v2. And then, the inventory address will be set.

```
public Address? AvatarAddr { get; }
```

Property Value

Address?

FungibleAssetValues

```
public IOrderedEnumerable<(Address balanceAddr, FungibleAssetValue value)>?
FungibleAssetValues { get; }
```

Property Value

IOrderedEnumerable<(Address balanceAddr, FungibleAssetValue value)>

FungibleIdAndCounts

```
public IOrderedEnumerable<(HashDigest<SHA256> fungibleId, int count)>?
FungibleIdAndCounts { get; }
```

Property Value

IOrderedEnumerable<(HashDigest<SHA256> fungibleId, int count)>

Memo

```
public string? Memo { get; }
```

Property Value

string

PlainValueInternal

```
protected override IImmutableDictionary<string, IValue> PlainValueInternal { get; }
```

Property Value

IImmutableDictionary<string, IValue>

Methods

Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

Parameters

`context` IActionContext

Returns

IWorld

LoadPlainValueInternal(IImmutableDictionary<string, IValue>)

```
protected override void LoadPlainValueInternal(IImmutableDictionary<string,
IValue> plainValue)
```

Parameters

`plainValue` IImmutableDictionary<string, IValue>

Class UnloadFromMyGarages

Namespace: Nekoyume.Action.Garages

Assembly: Lib9c.dll

```
[ActionType("unload_from_my_garages")]
public class UnloadFromMyGarages : GameAction, IUnloadFromMyGaragesV1, IAction
```

Inheritance

object ← ActionBase ← GameAction ← UnloadFromMyGarages

Implements

IUnloadFromMyGaragesV1, IAction

Inherited Members

GameAction.Id , GameAction.PlainValue , GameAction.LoadPlainValue(IValue) , ActionBase.GetSignerAndOtherAddressesHex(IActionContext, params Address[]) , ActionBase.TryGetAdminState(IActionContext, out AdminState) , ActionBase.CheckPermission(IActionContext) , ActionBase.CheckObsolete(long, IActionContext) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

UnloadFromMyGarages()

```
public UnloadFromMyGarages()
```

UnloadFromMyGarages(Address, IEnumerable<(Address balanceAddr, FungibleAssetValue value)>?, IEnumerable<(HashDigest<SHA256> fungibleId, int count)>?, string?)

```
public UnloadFromMyGarages(Address recipientAvatarAddr, IEnumerable<(Address
balanceAddr, FungibleAssetValue value)>? fungibleAssetValues,
IEnumerable<(HashDigest<SHA256> fungibleId, int count)>? fungibleIdAndCounts,
string? memo)
```

Parameters

`recipientAvatarAddr` Address

`fungibleAssetValues` IEnumerable<(Address balanceAddr, FungibleAssetValue value)>

`fungibleIdAndCounts` IEnumerable<(HashDigest<SHA256> fungibleId, int count)>

`memo` string

Properties

FungibleAssetValues

```
public IOrderedEnumerable<(Address balanceAddr, FungibleAssetValue value)>?
FungibleAssetValues { get; }
```

Property Value

IOrderedEnumerable<(Address balanceAddr, FungibleAssetValue value)>

FungibleIdAndCounts

```
public IOrderedEnumerable<(HashDigest<SHA256> fungibleId, int count)>?
FungibleIdAndCounts { get; }
```

Property Value

IOrderedEnumerable<(HashDigest<SHA256> fungibleId, int count)>

Memo

```
public string? Memo { get; }
```

Property Value

string

PlainValueInternal

```
protected override IImmutableDictionary<string, IValue> PlainValueInternal { get; }
```

Property Value

IImmutableDictionary<string, IValue>

RecipientAvatarAddr

If the avatar state is v1, there is no separate inventory, so it should be execute another action first to migrate the avatar state to v2. And then, the inventory address will be set.

```
public Address RecipientAvatarAddr { get; }
```

Property Value

Address

Methods

Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

Parameters

`context` IActionContext

Returns

IWorld

LoadPlainValueInternal(IImmutableDictionary<string, IValue>)

```
protected override void LoadPlainValueInternal(IImmutableDictionary<string,
IValue> plainValue)
```

Parameters

`plainValue` IImmutableDictionary<string, IValue>

Namespace Nekoyume.Action.Guild

Classes

BanGuildMember

ClaimGuildReward

ClaimReward

ClaimUnbonded
: An action to claim unbonded assets. This action can be executed only when the unbonding period is over.

GuildConfig

JoinGuild

MakeGuild

MoveGuild

QuitGuild

RemoveGuild
: An action to remove the guild.

UnbanGuildMember

Class BanGuildMember

Namespace: Nekoyume.Action.Guild

Assembly: Lib9c.dll

```
[ActionType("ban_guild_member")]
public class BanGuildMember : ActionBase, IAction
```

Inheritance

object ← ActionBase ← BanGuildMember

Implements

IAction

Inherited Members

ActionBase.GetSignerAndOtherAddressesHex(IActionContext, params Address[]) ,
ActionBase.TryGetAdminState(IActionContext, out AdminState) ,
ActionBase.CheckPermission(IActionContext) ,
ActionBase.CheckObsolete(long, IActionContext) , object.Equals(object) ,
object.Equals(object, object) , object.GetHashCode() , object.GetType() ,
object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

BanGuildMember()

```
public BanGuildMember()
```

BanGuildMember(AgentAddress)

```
public BanGuildMember(AgentAddress target)
```

Parameters

`target` AgentAddress

Fields

TypeIdentifier

```
public const string TypeIdentifier = "ban_guild_member"
```

Field Value

string

Properties

PlainValue

```
public override IValue PlainValue { get; }
```

Property Value

IValue

Target

```
public AgentAddress Target { get; }
```

Property Value

AgentAddress

Methods

Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

Parameters

`context` IActionContext

Returns

IWorld

LoadPlainValue(IValue)

```
public override void LoadPlainValue(IValue plainValue)
```

Parameters

`plainValue` IValue

Class ClaimGuildReward

Namespace: Nekoyume.Action.Guild

Assembly: Lib9c.dll

```
[ActionType("claim_guild_reward")]
public sealed class ClaimGuildReward : ActionBase, IAction
```

Inheritance

object ← ActionBase ← ClaimGuildReward

Implements

IAction

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

ClaimGuildReward()

```
public ClaimGuildReward()
```

Fields

TypeIdentifier

```
public const string TypeIdentifier = "claim_guild_reward"
```

Field Value

string

Properties

PlainValue

```
public override IValue PlainValue { get; }
```

Property Value

IValue

Methods

Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

Parameters

`context` IActionContext

Returns

IWorld

LoadPlainValue(IValue)

```
public override void LoadPlainValue(IValue plainValue)
```

Parameters

`plainValue` IValue

Class ClaimReward

Namespace: Nekoyume.Action.Guild

Assembly: Lib9c.dll

```
[ActionType("claim_reward")]
public sealed class ClaimReward : ActionBase, IAction
```

Inheritance

object ← ActionBase ← ClaimReward

Implements

IAction

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

ClaimReward()

```
public ClaimReward()
```

Fields

TypeIdentifier

```
public const string TypeIdentifier = "claim_reward"
```

Field Value

string

Properties

PlainValue

```
public override IValue PlainValue { get; }
```

Property Value

IValue

Methods

Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

Parameters

`context` IActionContext

Returns

IWorld

LoadPlainValue(IValue)

```
public override void LoadPlainValue(IValue plainValue)
```

Parameters

`plainValue` IValue

Class ClaimUnbonded

Namespace: Nekoyume.Action.Guild

Assembly: Lib9c.dll

An action to claim unbonded assets. This action can be executed only when the unbonding period is over.

```
[ActionType("claim_unbonded")]
public sealed class ClaimUnbonded : ActionBase, IAction
```

Inheritance

object ← ActionBase ← ClaimUnbonded

Implements

IAction

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

ClaimUnbonded()

```
public ClaimUnbonded()
```

Fields

TypeIdentifier

```
public const string TypeIdentifier = "claim_unbonded"
```

Field Value

string

Properties

PlainValue

```
public override IValue PlainValue { get; }
```

Property Value

IValue

Methods

Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

Parameters

`context` IActionContext

Returns

IWorld

LoadPlainValue(IValue)

```
public override void LoadPlainValue(IValue plainValue)
```

Parameters

`plainValue` IValue

Class GuildConfig

Namespace: Nekoyume.Action.Guild

Assembly: Lib9c.dll

```
public static class GuildConfig
```

Inheritance

object ← GuildConfig

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Fields

PlanetariumGuildOwner

```
public static readonly AgentAddress PlanetariumGuildOwner
```

Field Value

AgentAddress

Class JoinGuild

Namespace: Nekoyume.Action.Guild

Assembly: Lib9c.dll

```
[ActionType("join_guild")]
public class JoinGuild : ActionBase, IAction
```

Inheritance

object ← ActionBase ← JoinGuild

Implements

IAction

Inherited Members

ActionBase.GetSignerAndOtherAddressesHex(IActionContext, params Address[]) ,
ActionBase.TryGetAdminState(IActionContext, out AdminState) ,
ActionBase.CheckPermission(IActionContext) ,
ActionBase.CheckObsolete(long, IActionContext) , object.Equals(object) ,
object.Equals(object, object) , object.GetHashCode() , object.GetType() ,
object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

JoinGuild()

```
public JoinGuild()
```

JoinGuild(GuildAddress)

```
public JoinGuild(GuildAddress guildAddress)
```

Parameters

`guildAddress` GuildAddress

Fields

TypeIdentifier

```
public const string TypeIdentifier = "join_guild"
```

Field Value

string

Properties

GuildAddress

```
public GuildAddress GuildAddress { get; }
```

Property Value

GuildAddress

PlainValue

```
public override IValue PlainValue { get; }
```

Property Value

IValue

Methods

Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

Parameters

`context` IActionContext

Returns

IWorld

LoadPlainValue(IValue)

```
public override void LoadPlainValue(IValue plainValue)
```

Parameters

`plainValue` IValue

Class MakeGuild

Namespace: Nekoyume.Action.Guild

Assembly: Lib9c.dll

```
[ActionType("make_guild")]
public class MakeGuild : ActionBase, IAction
```

Inheritance

object ← ActionBase ← MakeGuild

Implements

IAction

Inherited Members

ActionBase.GetSignerAndOtherAddressesHex(IActionContext, params Address[]) ,
ActionBase.TryGetAdminState(IActionContext, out AdminState) ,
ActionBase.CheckPermission(IActionContext) ,
ActionBase.CheckObsolete(long, IActionContext) , object.Equals(object) ,
object.Equals(object, object) , object.GetHashCode() , object.GetType() ,
object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

MakeGuild()

```
public MakeGuild()
```

MakeGuild(Address)

```
public MakeGuild(Address validatorAddress)
```

Parameters

`validatorAddress` Address

Fields

TypeIdentifier

```
public const string TypeIdentifier = "make_guild"
```

Field Value

string

Properties

PlainValue

```
public override IValue PlainValue { get; }
```

Property Value

IValue

ValidatorAddress

```
public Address ValidatorAddress { get; }
```

Property Value

Address

Methods

Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

Parameters

`context` IActionContext

Returns

IWorld

LoadPlainValue(IValue)

```
public override void LoadPlainValue(IValue plainValue)
```

Parameters

`plainValue` IValue

Class MoveGuild

Namespace: Nekoyume.Action.Guild

Assembly: Lib9c.dll

```
[ActionType("move_guild")]
public class MoveGuild : ActionBase, IAction
```

Inheritance

object ← ActionBase ← MoveGuild

Implements

IAction

Inherited Members

ActionBase.GetSignerAndOtherAddressesHex(IActionContext, params Address[]) ,
ActionBase.TryGetAdminState(IActionContext, out AdminState) ,
ActionBase.CheckPermission(IActionContext) ,
ActionBase.CheckObsolete(long, IActionContext) , object.Equals(object) ,
object.Equals(object, object) , object.GetHashCode() , object.GetType() ,
object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

MoveGuild()

```
public MoveGuild()
```

MoveGuild(GuildAddress)

```
public MoveGuild(GuildAddress guildAddress)
```

Parameters

`guildAddress` GuildAddress

Fields

TypeIdentifier

```
public const string TypeIdentifier = "move_guild"
```

Field Value

string

Properties

GuildAddress

```
public GuildAddress GuildAddress { get; }
```

Property Value

GuildAddress

PlainValue

```
public override IValue PlainValue { get; }
```

Property Value

IValue

Methods

Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

Parameters

`context` IActionContext

Returns

IWorld

LoadPlainValue(IValue)

```
public override void LoadPlainValue(IValue plainValue)
```

Parameters

`plainValue` IValue

Class QuitGuild

Namespace: Nekoyume.Action.Guild

Assembly: Lib9c.dll

```
[ActionType("quit_guild")]
public class QuitGuild : ActionBase, IAction
```

Inheritance

object ← ActionBase ← QuitGuild

Implements

IAction

Inherited Members

ActionBase.GetSignerAndOtherAddressesHex(IActionContext, params Address[]) ,
ActionBase.TryGetAdminState(IActionContext, out AdminState) ,
ActionBase.CheckPermission(IActionContext) ,
ActionBase.CheckObsolete(long, IActionContext) , object.Equals(object) ,
object.Equals(object, object) , object.GetHashCode() , object.GetType() ,
object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Fields

TypeIdentifier

```
public const string TypeIdentifier = "quit_guild"
```

Field Value

string

Properties

PlainValue

```
public override IValue PlainValue { get; }
```

Property Value

IValue

Methods

Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

Parameters

`context` IActionContext

Returns

IWorld

LoadPlainValue(IValue)

```
public override void LoadPlainValue(IValue plainValue)
```

Parameters

`plainValue` IValue

Class RemoveGuild

Namespace: Nekoyume.Action.Guild

Assembly: Lib9c.dll

An action to remove the guild.

```
[ActionType("remove_guild")]
public class RemoveGuild : ActionBase, IAction
```

Inheritance

object ← ActionBase ← RemoveGuild

Implements

IAction

Inherited Members

ActionBase.GetSignerAndOtherAddressesHex(IActionContext, params Address[]) ,
ActionBase.TryGetAdminState(IActionContext, out AdminState) ,
ActionBase.CheckPermission(IActionContext) ,
ActionBase.CheckObsolete(long, IActionContext) , object.Equals(object) ,
object.Equals(object, object) , object.GetHashCode() , object.GetType() ,
object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Fields

TypeIdentifier

```
public const string TypeIdentifier = "remove_guild"
```

Field Value

string

Properties

PlainValue

```
public override IValue PlainValue { get; }
```

Property Value

IValue

Methods

Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

Parameters

`context` IActionContext

Returns

IWorld

LoadPlainValue(IValue)

```
public override void LoadPlainValue(IValue plainValue)
```

Parameters

`plainValue` IValue

Class UnbanGuildMember

Namespace: Nekoyume.Action.Guild

Assembly: Lib9c.dll

```
[ActionType("unban_guild_member")]
public class UnbanGuildMember : ActionBase, IAction
```

Inheritance

object ← ActionBase ← UnbanGuildMember

Implements

IAction

Inherited Members

ActionBase.GetSignerAndOtherAddressesHex(IActionContext, params Address[]) ,
ActionBase.TryGetAdminState(IActionContext, out AdminState) ,
ActionBase.CheckPermission(IActionContext) ,
ActionBase.CheckObsolete(long, IActionContext) , object.Equals(object) ,
object.Equals(object, object) , object.GetHashCode() , object.GetType() ,
object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

UnbanGuildMember()

```
public UnbanGuildMember()
```

UnbanGuildMember(Address)

```
public UnbanGuildMember(Address target)
```

Parameters

`target` Address

Fields

TypeIdentifier

```
public const string TypeIdentifier = "unban_guild_member"
```

Field Value

string

Properties

PlainValue

```
public override IValue PlainValue { get; }
```

Property Value

IValue

Target

```
public Address Target { get; }
```

Property Value

Address

Methods

Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

Parameters

`context` IActionContext

Returns

IWorld

LoadPlainValue(IValue)

```
public override void LoadPlainValue(IValue plainValue)
```

Parameters

`plainValue` IValue

Namespace Nekoyume.Action.Guild.Migration

Classes

MigrateDelegationHeight

An action to migrate the delegation height. This action triggers the migration of the delegation height.

MigratePlanetariumValidator

An action to migrate the planetarium validator. With this migration, planetarium validator is now active, and bonded FAVs are moved to active address.

MigrateStakeAndJoinGuild

An action to migrate guild delegation and join guild. After migration is done, guild participant now have delegation with validator.

Class MigrateDelegationHeight

Namespace: Nekoyume.Action.Guild.Migration

Assembly: Lib9c.dll

An action to migrate the delegation height. This action triggers the migration of the delegation height.

```
[ActionType("migrate_delegation_height")]
public class MigrateDelegationHeight : ActionBase, IAction
```

Inheritance

object ← ActionBase ← MigrateDelegationHeight

Implements

IAction

Inherited Members

ActionBase.GetSignerAndOtherAddressesHex(IActionContext, params Address[]) , ActionBase.TryGetAdminState(IActionContext, out AdminState) , ActionBase.CheckPermission(IActionContext) , ActionBase.CheckObsolete(long, IActionContext) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

MigrateDelegationHeight()

```
public MigrateDelegationHeight()
```

MigrateDelegationHeight(long)

```
public MigrateDelegationHeight(long height)
```

Parameters

`height` long

Fields

TypeIdentifier

```
public const string TypeIdentifier = "migrate_delegation_height"
```

Field Value

string

Properties

Height

```
public long Height { get; }
```

Property Value

long

PlainValue

```
public override IValue PlainValue { get; }
```

Property Value

IValue

Methods

Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

Parameters

`context` IActionContext

Returns

IWorld

LoadPlainValue(IValue)

```
public override void LoadPlainValue(IValue plainValue)
```

Parameters

`plainValue` IValue

Class MigratePlanetariumValidator

Namespace: Nekoyume.Action.Guild.Migration

Assembly: Lib9c.dll

An action to migrate the planetarium validator. With this migration, planetarium validator is now active, and bonded FAVs are moved to active address.

```
[ActionType("migrate_planetarium_validator")]
public class MigratePlanetariumValidator : ActionBase, IAction
```

Inheritance

object ← ActionBase ← MigratePlanetariumValidator

Implements

IAction

Inherited Members

ActionBase.GetSignerAndOtherAddressesHex(IActionContext, params Address[]) , ActionBase.TryGetAdminState(IActionContext, out AdminState) , ActionBase.CheckPermission(IActionContext) , ActionBase.CheckObsolete(long, IActionContext) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

MigratePlanetariumValidator()

```
public MigratePlanetariumValidator()
```

Fields

TypeIdentifier

```
public const string TypeIdentifier = "migrate_planetarium_validator"
```

Field Value

string

Properties

PlainValue

```
public override IValue PlainValue { get; }
```

Property Value

IValue

Methods

Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

Parameters

`context` IActionContext

Returns

IWorld

LoadPlainValue(IValue)

```
public override void LoadPlainValue(IValue plainValue)
```

Parameters

`plainValue` IValue

Class MigrateStakeAndJoinGuild

Namespace: Nekoyume.Action.Guild.Migration

Assembly: Lib9c.dll

An action to migrate guild delegation and join guild. After migration is done, guild participant now have delegation with validator.

```
[ActionType("migrate_stake_and_join_guild")]
public class MigrateStakeAndJoinGuild : ActionBase, IAction
```

Inheritance

object ← ActionBase ← MigrateStakeAndJoinGuild

Implements

IAction

Inherited Members

ActionBase.GetSignerAndOtherAddressesHex(IActionContext, params Address[]) , ActionBase.TryGetAdminState(IActionContext, out AdminState) , ActionBase.CheckPermission(IActionContext) , ActionBase.CheckObsolete(long, IActionContext) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

MigrateStakeAndJoinGuild()

Initializes a new instance of the MigrateStakeAndJoinGuild class.

```
public MigrateStakeAndJoinGuild()
```

MigrateStakeAndJoinGuild(AgentAddress)

Initializes a new instance of the MigrateStakeAndJoinGuild class.

```
public MigrateStakeAndJoinGuild(AgentAddress target)
```

Parameters

`target` AgentAddress

Fields

TypeIdentifier

The type identifier for this action.

```
public const string TypeIdentifier = "migrate_stake_and_join_guild"
```

Field Value

string

Properties

PlainValue

```
public override IValue PlainValue { get; }
```

Property Value

IValue

Target

The target address to migrate.

```
public AgentAddress Target { get; }
```

Property Value

AgentAddress

Methods

Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

Parameters

`context` IActionContext

Returns

IWorld

LoadPlainValue(IValue)

```
public override void LoadPlainValue(IValue plainValue)
```

Parameters

`plainValue` IValue

Namespace Nekoyume.Action.Guild.Migration.LegacyModels

Classes

LegacyGuildParticipant
: The legacy model for GuildParticipant.

MigrationModule
: The module for delegation height for migration.

Class LegacyGuildParticipant

Namespace: Nekoyume.Action.Guild.Migration.LegacyModels

Assembly: Lib9c.dll

The legacy model for GuildParticipant.

```
public class LegacyGuildParticipant : IBencodable,
IEquatable<LegacyGuildParticipant>
```

Inheritance

object ← LegacyGuildParticipant

Implements

IBencodable, IEquatable<LegacyGuildParticipant>

Inherited Members

object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

LegacyGuildParticipant(List)

Constructor of LegacyGuildParticipant.

```
public LegacyGuildParticipant(List list)
```

Parameters

`list` List

The serialized data.

Exceptions

InvalidCastException

Throws when the deserialization failed.

FailedLoadStateException

Throws when the state is un-deserializable.

LegacyGuildParticipant(GuildAddress)

Constructor of LegacyGuildParticipant.

```
public LegacyGuildParticipant(GuildAddress guildAddress)
```

Parameters

`guildAddress` GuildAddress

The guild address.

Fields

GuildAddress

The guild address.

```
public readonly GuildAddress GuildAddress
```

Field Value

GuildAddress

Properties

Bencoded

Serialize the state.

```
public List Bencoded { get; }
```

Property Value

List

Methods

Equals(LegacyGuildParticipant)

Indicates whether the current object is equal to another object of the same type.

```
public bool Equals(LegacyGuildParticipant other)
```

Parameters

`other` LegacyGuildParticipant

An object to compare with this object.

Returns

bool

true if the current object is equal to the `other` parameter; otherwise, false.

Equals(object)

Determines whether the specified object is equal to the current object.

```
public override bool Equals(object obj)
```

Parameters

`obj` object

The object to compare with the current object.

Returns

bool

true if the specified object is equal to the current object; otherwise, false.

GetHashCode()

Serves as the default hash function.

```
public override int GetHashCode()
```

Returns

int

A hash code for the current object.

Class MigrationModule

Namespace: Nekoyume.Action.Guild.Migration.LegacyModels

Assembly: Lib9c.dll

The module for delegation height for migration.

```
public static class MigrationModule
```

Inheritance

object ← MigrationModule

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Fields

DelegationMigrationHeight

An address for delegation height migration.

```
public static readonly Address DelegationMigrationHeight
```

Field Value

Address

Methods

GetDelegationMigrationHeight(IWorldState)

```
public static long? GetDelegationMigrationHeight(this IWorldState worldState)
```

Parameters

`worldState` IWorldState

Returns

long?

SetDelegationMigrationHeight(IWorld, long)

```
public static IWorld SetDelegationMigrationHeight(this IWorld world, long height)
```

Parameters

`world` IWorld

`height` long

Returns

IWorld

Namespace Nekoyume.Action.Loader

Classes

NCActionLoader

An Libplanet.Action.Loader.IActionLoader implementation for Nine Chronicles's BlockChain. This is a simple wrapper around a Libplanet.Action.Loader.TypedActionLoader for loading all Libplanet.Action.IAction classes within the same assembly as the Action Base class, inheriting from the ActionBase class, and has an Libplanet.Action.ActionType Attribute from Bencodex.Types.IValues

Class NCActionLoader

Namespace: Nekoyume.Action.Loader

Assembly: Lib9c.dll

An Libplanet.Action.Loader.IActionLoader implementation for Nine Chronicles's BlockChain. This is a simple wrapper around a Libplanet.Action.Loader.TypedActionLoader for loading all Libplanet.Action.IAction classes within the same assembly as the ActionBase class, inheriting from the ActionBase class, and has an Libplanet.Action.ActionTypeAttribute from Bencodex.Types.IValues

```
public class NCActionLoader : IActionLoader
```

Inheritance

object ← NCActionLoader

Implements

IActionLoader

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

NCActionLoader()

```
public NCActionLoader()
```

Methods

LoadAction(long, IValue)

```
public IAction LoadAction(long index, IValue value)
```

Parameters

`index` long

`value` IValue

Returns

IAction

See Also

TypedActionLoader

Namespace Nekoyume.Action.Validator Delegation

Classes

AllocateGuildReward

AllocateReward

ClaimValidatorRewardSelf

DelegateValidator

Mortgage
An action for mortgage gas fee for a transaction. Should be executed at the beginning of the tx.

PromoteValidator

RecordProposer
An action for recording proposer of the block to use in next block's reward distribution.

Refund
An action for refund gas fee for a transaction. Should be executed at the beginning of the tx.

Reward
An action for reward for a transaction. Should be executed at the beginning of the tx.

SetValidatorCommission
Set the commission percentage of the validator.

SlashValidator

UndelegateValidator

UnjailValidator

UpdateValidators

Class AllocateGuildReward

Namespace: Nekoyume.Action.ValidatorDelegation

Assembly: Lib9c.dll

```
public sealed class AllocateGuildReward : ActionBase, IAction
```

Inheritance

object ← ActionBase ← AllocateGuildReward

Implements

IAction

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

AllocateGuildReward()

```
public AllocateGuildReward()
```

Properties

PlainValue

```
public override IValue PlainValue { get; }
```

Property Value

IValue

Methods

Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

Parameters

`context` IActionContext

Returns

IWorld

LoadPlainValue(IValue)

```
public override void LoadPlainValue(IValue plainValue)
```

Parameters

`plainValue` IValue

Class AllocateReward

Namespace: Nekoyume.Action.ValidatorDelegation

Assembly: Lib9c.dll

```
public sealed class AllocateReward : ActionBase, IAction
```

Inheritance

object ← ActionBase ← AllocateReward

Implements

IAction

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

AllocateReward()

```
public AllocateReward()
```

Properties

PlainValue

```
public override IValue PlainValue { get; }
```

Property Value

IValue

Methods

Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

Parameters

`context` IActionContext

Returns

IWorld

LoadPlainValue(IValue)

```
public override void LoadPlainValue(IValue plainValue)
```

Parameters

`plainValue` IValue

Class ClaimValidatorRewardSelf

Namespace: Nekoyume.Action.ValidatorDelegation

Assembly: Lib9c.dll

```
[ActionType("claim_validator_reward_self")]
public sealed class ClaimValidatorRewardSelf : ActionBase, IAction
```

Inheritance

object ← ActionBase ← ClaimValidatorRewardSelf

Implements

IAction

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

ClaimValidatorRewardSelf()

```
public ClaimValidatorRewardSelf()
```

Fields

TypeIdentifier

```
public const string TypeIdentifier = "claim_validator_reward_self"
```

Field Value

string

Properties

PlainValue

```
public override IValue PlainValue { get; }
```

Property Value

IValue

ValidatorDelegatee

```
public Address ValidatorDelegatee { get; }
```

Property Value

Address

Methods

Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

Parameters

`context` IActionContext

Returns

IWorld

LoadPlainValue(IValue)

```
public override void LoadPlainValue(IValue plainValue)
```

Parameters

`plainValue` IValue

Class DelegateValidator

Namespace: Nekoyume.Action.ValidatorDelegation

Assembly: Lib9c.dll

```
[ActionType("delegate_validator")]
public sealed class DelegateValidator : ActionBase, IAction
```

Inheritance

object ← ActionBase ← DelegateValidator

Implements

IAction

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

DelegateValidator()

```
public DelegateValidator()
```

DelegateValidator(FungibleAssetValue)

```
public DelegateValidator(FungibleAssetValue fav)
```

Parameters

`fav` FungibleAssetValue

Fields

TypeIdentifier

```
public const string TypeIdentifier = "delegate_validator"
```

Field Value

string

Properties

FAV

```
public FungibleAssetValue FAV { get; }
```

Property Value

FungibleAssetValue

PlainValue

```
public override IValue PlainValue { get; }
```

Property Value

IValue

Methods

Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

Parameters

`context` IActionContext

Returns

IWorld

LoadPlainValue(IValue)

```
public override void LoadPlainValue(IValue plainValue)
```

Parameters

`plainValue` IValue

Class Mortgage

Namespace: Nekoyume.Action.ValidatorDelegation

Assembly: Lib9c.dll

An action for mortgage gas fee for a transaction. Should be executed at the beginning of the tx.

```
public sealed class Mortgage : ActionBase, IAction
```

Inheritance

object ← ActionBase ← Mortgage

Implements

IAction

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

Mortgage()

Creates a new instance of Mortgage.

```
public Mortgage()
```

Properties

PlainValue

```
public override IValue PlainValue { get; }
```

Property Value

IValue

Methods

Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

Parameters

`context` IActionContext

Returns

IWorld

LoadPlainValue(IValue)

```
public override void LoadPlainValue(IValue plainValue)
```

Parameters

`plainValue` IValue

Class PromoteValidator

Namespace: Nekoyume.Action.ValidatorDelegation

Assembly: Lib9c.dll

```
[ActionType("promote_validator")]
public sealed class PromoteValidator : ActionBase, IAction
```

Inheritance

object ← ActionBase ← PromoteValidator

Implements

IAction

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

PromoteValidator()

```
public PromoteValidator()
```

PromoteValidator(PublicKey, FungibleAssetValue)

```
public PromoteValidator(PublicKey publicKey, FungibleAssetValue fav)
```

Parameters

`publicKey` PublicKey

`fav` FungibleAssetValue

PromoteValidator(PublicKey, FungibleAssetValue, BigInteger)

```
public PromoteValidator(PublicKey publicKey, FungibleAssetValue fav,
BigInteger commissionPercentage)
```

Parameters

`publicKey` PublicKey

`fav` FungibleAssetValue

`commissionPercentage` BigInteger

Fields

TypeIdentifier

```
public const string TypeIdentifier = "promote_validator"
```

Field Value

string

Properties

CommissionPercentage

```
public BigInteger CommissionPercentage { get; }
```

Property Value

BigInteger

FAV

```
public FungibleAssetValue FAV { get; }
```

Property Value

FungibleAssetValue

PlainValue

```
public override IValue PlainValue { get; }
```

Property Value

IValue

PublicKey

```
public PublicKey PublicKey { get; }
```

Property Value

PublicKey

Methods

Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

Parameters

`context` IActionContext

Returns

IWorld

ExecutePublic(IActionContext)

```
public IWorld ExecutePublic(IActionContext context)
```

Parameters

`context` IActionContext

Returns

IWorld

LoadPlainValue(IValue)

```
public override void LoadPlainValue(IValue plainValue)
```

Parameters

`plainValue` IValue

Class RecordProposer

Namespace: Nekoyume.Action.ValidatorDelegation

Assembly: Lib9c.dll

An action for recording proposer of the block to use in next block's reward distribution.

```
public sealed class RecordProposer : ActionBase, IAction
```

Inheritance

object ← ActionBase ← RecordProposer

Implements

IAction

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

RecordProposer()

Creates a new instance of RecordProposer.

```
public RecordProposer()
```

Properties

PlainValue

```
public override IValue PlainValue { get; }
```

Property Value

IValue

Methods

Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

Parameters

`context` IActionContext

Returns

IWorld

LoadPlainValue(IValue)

```
public override void LoadPlainValue(IValue plainValue)
```

Parameters

`plainValue` IValue

Class Refund

Namespace: Nekoyume.Action.ValidatorDelegation

Assembly: Lib9c.dll

An action for refund gas fee for a transaction. Should be executed at the beginning of the tx.

```
public sealed class Refund : ActionBase, IAction
```

Inheritance

object ← ActionBase ← Refund

Implements

IAction

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

Refund()

Creates a new instance of Refund.

```
public Refund()
```

Properties

PlainValue

```
public override IValue PlainValue { get; }
```

Property Value

IValue

Methods

Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

Parameters

`context` IActionContext

Returns

IWorld

LoadPlainValue(IValue)

```
public override void LoadPlainValue(IValue plainValue)
```

Parameters

`plainValue` IValue

Class Reward

Namespace: Nekoyume.Action.ValidatorDelegation

Assembly: Lib9c.dll

An action for reward for a transaction. Should be executed at the beginning of the tx.

```
public sealed class Reward : ActionBase, IAction
```

Inheritance

object ← ActionBase ← Reward

Implements

IAction

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

Reward()

Creates a new instance of Refund.

```
public Reward()
```

Properties

PlainValue

```
public override IValue PlainValue { get; }
```

Property Value

IValue

Methods

Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

Parameters

`context` IActionContext

Returns

IWorld

LoadPlainValue(IValue)

```
public override void LoadPlainValue(IValue plainValue)
```

Parameters

`plainValue` IValue

Class SetValidatorCommission

Namespace: Nekoyume.Action.ValidatorDelegation

Assembly: Lib9c.dll

Set the commission percentage of the validator.

```
[ActionType("set_validator_commission")]
public sealed class SetValidatorCommission : ActionBase, IAction
```

Inheritance

object ← ActionBase ← SetValidatorCommission

Implements

IAction

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

SetValidatorCommission()

```
public SetValidatorCommission()
```

SetValidatorCommission(BigInteger)

```
public SetValidatorCommission(BigInteger commissionPercentage)
```

Parameters

`commissionPercentage` BigInteger

Fields

TypeIdentifier

```
public const string TypeIdentifier = "set_validator_commission"
```

Field Value

string

Properties

CommissionPercentage

```
public BigInteger CommissionPercentage { get; }
```

Property Value

BigInteger

PlainValue

```
public override IValue PlainValue { get; }
```

Property Value

IValue

Methods

Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

Parameters

`context` IActionContext

Returns

IWorld

LoadPlainValue(IValue)

```
public override void LoadPlainValue(IValue plainValue)
```

Parameters

`plainValue` IValue

Class SlashValidator

Namespace: Nekoyume.Action.ValidatorDelegation

Assembly: Lib9c.dll

```
public sealed class SlashValidator : ActionBase, IAction
```

Inheritance

object ← ActionBase ← SlashValidator

Implements

IAction

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

SlashValidator()

```
public SlashValidator()
```

Properties

AbstainJailTime

```
public static long AbstainJailTime { get; }
```

Property Value

long

DuplicateVoteSlashFactor

```
public static BigInteger DuplicateVoteSlashFactor { get; }
```

Property Value

BigInteger

LivenessSlashFactor

```
public static BigInteger LivenessSlashFactor { get; }
```

Property Value

BigInteger

PlainValue

```
public override IValue PlainValue { get; }
```

Property Value

IValue

Methods

Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

Parameters

`context` IActionContext

Returns

LoadPlainValue(IValue)

```
public override void LoadPlainValue(IValue plainValue)
```

Parameters

`plainValue` IValue

Class UndelegateValidator

Namespace: Nekoyume.Action.ValidatorDelegation

Assembly: Lib9c.dll

```
[ActionType("undelegate_validator")]
public sealed class UndelegateValidator : ActionBase, IAction
```

Inheritance

object ← ActionBase ← UndelegateValidator

Implements

IAction

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

UndelegateValidator()

```
public UndelegateValidator()
```

UndelegateValidator(BigInteger)

```
public UndelegateValidator(BigInteger share)
```

Parameters

`share` BigInteger

Fields

TypeIdentifier

```
public const string TypeIdentifier = "undelegate_validator"
```

Field Value

string

Properties

PlainValue

```
public override IValue PlainValue { get; }
```

Property Value

IValue

Share

```
public BigInteger Share { get; }
```

Property Value

BigInteger

Methods

Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

Parameters

`context` IActionContext

Returns

IWorld

LoadPlainValue(IValue)

```
public override void LoadPlainValue(IValue plainValue)
```

Parameters

`plainValue` IValue

Class UnjailValidator

Namespace: Nekoyume.Action.ValidatorDelegation

Assembly: Lib9c.dll

```
[ActionType("unjail_validator")]
public sealed class UnjailValidator : ActionBase, IAction
```

Inheritance

object ← ActionBase ← UnjailValidator

Implements

IAction

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

UnjailValidator()

```
public UnjailValidator()
```

Fields

TypeIdentifier

```
public const string TypeIdentifier = "unjail_validator"
```

Field Value

string

Properties

PlainValue

```
public override IValue PlainValue { get; }
```

Property Value

IValue

Methods

Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

Parameters

`context` IActionContext

Returns

IWorld

LoadPlainValue(IValue)

```
public override void LoadPlainValue(IValue plainValue)
```

Parameters

`plainValue` IValue

Class UpdateValidators

Namespace: Nekoyume.Action.ValidatorDelegation

Assembly: Lib9c.dll

```
public sealed class UpdateValidators : ActionBase, IAction
```

Inheritance

object ← ActionBase ← UpdateValidators

Implements

IAction

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

UpdateValidators()

```
public UpdateValidators()
```

Properties

PlainValue

```
public override IValue PlainValue { get; }
```

Property Value

IValue

Methods

Execute(IActionContext)

```
public override IWorld Execute(IActionContext context)
```

Parameters

`context` IActionContext

Returns

IWorld

LoadPlainValue(IValue)

```
public override void LoadPlainValue(IValue plainValue)
```

Parameters

`plainValue` IValue

Namespace Nekoyume.Arena

Classes

ArenaHelper

There are only things that don't change

ArenaSimulator

Changed at https://github.com/planetarium/lib9c/pull/2229

ArenaSimulator2

This class is only for previewnet.

Interfaces

IArenaSimulator

Class ArenaHelper

Namespace: Nekoyume.Arena

Assembly: Lib9c.dll

There are only things that don't change

```
public static class ArenaHelper
```

Inheritance

object ← ArenaHelper

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Fields

ScoreLimits

```
public static readonly IReadOnlyDictionary<ArenaType, (int upper, int
lower)> ScoreLimits
```

Field Value

IReadOnlyDictionary<ArenaType, (int, int)>

ScoreLimitsV1

```
[Obsolete("Use `ScoreLimits` instead.")]
public static readonly IReadOnlyDictionary<ArenaType, (int, int)> ScoreLimitsV1
```

Field Value

IReadOnlyDictionary<ArenaType, (int, int)>

ScoreLimitsV2

```
[Obsolete("Use `ScoreLimits` instead.")]
public static readonly IReadOnlyDictionary<ArenaType, (int upper, int
lower)> ScoreLimitsV2
```

Field Value

IReadOnlyDictionary<ArenaType, (int, int)>

ScoreLimitsV3

```
[Obsolete("Use `ScoreLimits` instead.")]
public static readonly IReadOnlyDictionary<ArenaType, (int upper, int
lower)> ScoreLimitsV3
```

Field Value

IReadOnlyDictionary<ArenaType, (int, int)>

Methods

DeriveArenaAddress(int, int)

```
public static Address DeriveArenaAddress(int championshipId, int round)
```

Parameters

`championshipId` int

`round` int

Returns

Address

GetCurrentTicketResetCount(long, long, int)

```
public static int GetCurrentTicketResetCount(long currentBlockIndex, long
roundStartBlockIndex, int interval)
```

Parameters

`currentBlockIndex` long

`roundStartBlockIndex` long

`interval` int

Returns

int

GetEntranceFee(RoundData, long, int)

```
public static FungibleAssetValue GetEntranceFee(ArenaSheet.RoundData roundData, long
currentBlockIndex, int avatarLevel)
```

Parameters

`roundData` ArenaSheet.RoundData

`currentBlockIndex` long

`avatarLevel` int

Returns

FungibleAssetValue

GetMaxPurchasedTicketCount(RoundData)

```
[Obsolete("not use since v100320, battle_arena6. Use
`ArenaSheet.RoundData.MaxPurchaseCount`")]
public static long GetMaxPurchasedTicketCount(ArenaSheet.RoundData roundData)
```

Parameters

`roundData` ArenaSheet.RoundData

Returns

long

GetMedalTotalCount(Row, AvatarState)

```
public static int GetMedalTotalCount(ArenaSheet.Row row, AvatarState avatarState)
```

Parameters

`row` ArenaSheet.Row

`avatarState` AvatarState

Returns

int

GetRewardCount(int)

```
public static int GetRewardCount(int score)
```

Parameters

`score` int

Returns

int

GetScores(int, int)

```
public static (int myWinScore, int myDefeatScore, int enemyDefeatScore)
GetScores(int myScore, int enemyScore)
```

Parameters

`myScore` int

`enemyScore` int

Returns

(int myWinScore, int myDefeatScore, int enemyDefeatScore)

GetScoresV1(int, int)

```
[Obsolete("Use `GetScores()` instead.")]
public static (int myWinScore, int myDefeatScore, int enemyDefeatScore)
GetScoresV1(int myScore, int enemyScore)
```

Parameters

`myScore` int

`enemyScore` int

Returns

(int myWinScore, int myDefeatScore, int enemyDefeatScore)

GetTicketPrice(RoundData, ArenaInformation, Currency)

```
public static FungibleAssetValue GetTicketPrice(ArenaSheet.RoundData roundData,
ArenaInformation arenaInformation, Currency currency)
```

Parameters

`roundData` ArenaSheet.RoundData

`arenaInformation` ArenaInformation

`currency` Currency

Returns

FungibleAssetValue

ValidateScoreDifference(IReadOnlyDictionary<ArenaType, (int, int)>, ArenaType, int, int)

```
public static bool ValidateScoreDifference(IReadOnlyDictionary<ArenaType, (int,
int)> scoreLimits, ArenaType arenaType, int myScore, int enemyScore)
```

Parameters

`scoreLimits` IReadOnlyDictionary<ArenaType, (int, int)>

`arenaType` ArenaType

`myScore` int

`enemyScore` int

Returns

bool

ValidateScoreDifferenceV1(IReadOnlyDictionary<ArenaType, (int, int)>, ArenaType, int, int)

```
[Obsolete("Use `ValidateScoreDifference()` instead.")]
public static bool ValidateScoreDifferenceV1(IReadOnlyDictionary<ArenaType, (int,
int)> scoreLimits, ArenaType arenaType, int myScore, int enemyScore)
```

Parameters

`scoreLimits` IReadOnlyDictionary<ArenaType, (int, int)>

`arenaType` ArenaType

`myScore` int

`enemyScore` int

Returns

bool

ValidateScoreDifferenceV2(IReadOnlyDictionary<ArenaType, (int, int)>, ArenaType, int, int)

```
[Obsolete("Use `ValidateScoreDifference()` instead.")]
public static bool ValidateScoreDifferenceV2(IReadOnlyDictionary<ArenaType, (int, int)> scoreLimits, ArenaType arenaType, int myScore, int enemyScore)
```

Parameters

`scoreLimits` IReadOnlyDictionary<ArenaType, (int, int)>

`arenaType` ArenaType

`myScore` int

`enemyScore` int

Returns

bool

Class ArenaSimulator

Namespace: Nekoyume.Arena

Assembly: Lib9c.dll

Changed at https://github.com/planetarium/lib9c/pull/2229

```
public class ArenaSimulator : IArenaSimulator
```

Inheritance

object ← ArenaSimulator

Implements

IArenaSimulator

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

ArenaSimulator(IRandom, int, long)

```
public ArenaSimulator(IRandom random, int hpModifier = 2, long
shatterStrikeMaxDamage = 400000)
```

Parameters

`random` IRandom

`hpModifier` int

`shatterStrikeMaxDamage` long

Properties

BuffLimitSheet

```
public BuffLimitSheet BuffLimitSheet { get; }
```

Property Value

BuffLimitSheet

BuffLinkSheet

```
public BuffLinkSheet BuffLinkSheet { get; set; }
```

Property Value

BuffLinkSheet

HpModifier

```
public int HpModifier { get; }
```

Property Value

int

Log

```
public ArenaLog Log { get; }
```

Property Value

ArenaLog

Random

```
public IRandom Random { get; }
```

Property Value

IRandom

ShatterStrikeMaxDamage

```
public long ShatterStrikeMaxDamage { get; }
```

Property Value

long

Turn

```
public int Turn { get; }
```

Property Value

int

Methods

Simulate(ArenaPlayerDigest, ArenaPlayerDigest, ArenaSimulatorSheets, List<StatModifier>, List<StatModifier>, BuffLimitSheet, BuffLinkSheet, bool)

```
public ArenaLog Simulate(ArenaPlayerDigest challenger, ArenaPlayerDigest enemy,
ArenaSimulatorSheets sheets, List<StatModifier> challengerCollectionModifiers,
List<StatModifier> enemyCollectionModifiers, BuffLimitSheet buffLimitSheet,
BuffLinkSheet buffLinkSheet, bool setExtraValueBuffBeforeGetBuffs = false)
```

Parameters

`challenger` ArenaPlayerDigest

`enemy` ArenaPlayerDigest

`sheets` ArenaSimulatorSheets

`challengerCollectionModifiers` List<StatModifier>

`enemyCollectionModifiers` List<StatModifier>

`buffLimitSheet` BuffLimitSheet

`buffLinkSheet` BuffLinkSheet

`setExtraValueBuffBeforeGetBuffs` bool

Returns

ArenaLog

Class ArenaSimulator2

Namespace: Nekoyume.Arena

Assembly: Lib9c.dll

This class is only for previewnet.

```
public class ArenaSimulator2 : Simulator, ISimulator
```

Inheritance

object ← Simulator ← ArenaSimulator2

Implements

ISimulator

Inherited Members

Simulator.Random , Simulator.Log , Simulator.Player , Simulator.Result , Simulator.Characters , Simulator.TurnPriority , Simulator.MaterialItemSheet , Simulator.SkillSheet , Simulator.SkillBuffSheet , Simulator.StatBuffSheet , Simulator.SkillActionBuffSheet , Simulator.ActionBuffSheet , Simulator.CharacterSheet , Simulator.CharacterLevelSheet , Simulator.EquipmentItemSetEffectSheet , Simulator.BuffLimitSheet , Simulator.BuffLinkSheet , Simulator.ShatterStrikeMaxDamage , Simulator.MaxTurn , Simulator.TurnNumber , Simulator.WaveNumber , Simulator.WaveTurn , Simulator.LogEvent , Simulator.SetReward(WeightedSelector<StageSheet.RewardData>, int, IRandom, MaterialItemSheet) , Simulator.SetRewardV2(WeightedSelector<StageSheet.RewardData>, int, IRandom, MaterialItemSheet) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

ArenaSimulator2(IRandom, ArenaPlayerDigest, ArenaPlayerDigest, ArenaSimulatorSheetsV1)

```
public ArenaSimulator2(IRandom random, ArenaPlayerDigest myDigest, ArenaPlayerDigest
enemyDigest, ArenaSimulatorSheetsV1 simulatorSheets)
```

Parameters

`random` IRandom

`myDigest` ArenaPlayerDigest

`enemyDigest` ArenaPlayerDigest

`simulatorSheets` ArenaSimulatorSheetsV1

Properties

Reward

do not use it

```
public override IEnumerable<ItemBase> Reward { get; }
```

Property Value

IEnumerable<ItemBase>

Methods

Simulate()

```
public Player Simulate()
```

Returns

Player

Interface IArenaSimulator

Namespace: Nekoyume.Arena

Assembly: Lib9c.dll

```
public interface IArenaSimulator
```

Properties

BuffLimitSheet

```
BuffLimitSheet BuffLimitSheet { get; }
```

Property Value

BuffLimitSheet

BuffLinkSheet

```
BuffLinkSheet BuffLinkSheet { get; set; }
```

Property Value

BuffLinkSheet

Log

```
ArenaLog Log { get; }
```

Property Value

ArenaLog

Random

```
IRandom Random { get; }
```

Property Value

IRandom

ShatterStrikeMaxDamage

```
long ShatterStrikeMaxDamage { get; }
```

Property Value

long

Turn

```
int Turn { get; }
```

Property Value

int

Namespace Nekoyume.Battle

Classes

ArenaScoreHelper

AttackCountHelper

CPHelper

CriticalHelper

DamageHelper

HitHelper

InfiniteTowerLog
: Log for infinite tower battle results. Contains information about the battle outcome and player performance.

InfiniteTowerSimulator
: Simulator for infinite tower battles with conditions and restrictions. Handles turn-based combat with wave progression, special conditions, and reward processing.

InfiniteTowerWave
: Represents a wave in the infinite tower. Contains information about the enemies that will spawn in this wave.

InvalidCountException

ListEmptyException

RaidSimulator

RaidSimulatorV1

RaidSimulatorV2

RankingSimulator

RankingSimulatorV1

RewardSelector

Simulator

StageRewardExpHelper

StageSimulator
: Simulator for stage battles that handles wave-based combat.

StageSimulatorV1

StageSimulatorV2

StageSimulatorV3

TestSimulator

Wave
: Represents a wave of enemies in a battle.

WaveResult
: Result of a single wave battle. Contains information about the wave completion status and battle outcome.

WeightedSelector<T>

Structs

EnemyPlayerDigest

Interfaces

ISimulator

IStageSimulator

Enums

InfiniteTowerLog.InfiniteTowerResult
: Represents the possible results of an infinite tower battle.

Class ArenaScoreHelper

Namespace: Nekoyume.Battle

Assembly: Lib9c.dll

```
public static class ArenaScoreHelper
```

Inheritance

object ← ArenaScoreHelper

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Fields

DifferLowerLimit

```
[Obsolete("Only to used for V1")]
public const int DifferLowerLimit = -1000
```

Field Value

int

DifferUpperLimit

```
[Obsolete("Only to used for V1")]
public const int DifferUpperLimit = 1000
```

Field Value

int

LoseScoreMax

```
[Obsolete("Only to used for V1")]
public const int LoseScoreMax = -30
```

Field Value

int

LoseScoreMin

```
[Obsolete("Only to used for V1")]
public const int LoseScoreMin = -5
```

Field Value

int

WinScoreMax

```
[Obsolete("Only to used for V1")]
public const int WinScoreMax = 60
```

Field Value

int

WinScoreMin

```
[Obsolete("Only to used for V1")]
public const int WinScoreMin = 1
```

Field Value

int

Methods

GetScore(int, int, Result)

```
public static (int challengerScoreDelta, int defenderScoreDelta) GetScore(int
challengerRating, int defenderRating, BattleLog.Result result)
```

Parameters

`challengerRating` int

`defenderRating` int

`result` BattleLog.Result

Returns

(int challengerScoreDelta, int defenderScoreDelta)

GetScoreV1(int, int, Result)

```
[Obsolete("Use GetScore()")]
public static int GetScoreV1(int challengerRating, int defenderRating,
BattleLog.Result result)
```

Parameters

`challengerRating` int

`defenderRating` int

`result` BattleLog.Result

Returns

int

GetScoreV2(int, int, Result)

```
[Obsolete("Use GetScore()")]
public static int GetScoreV2(int challengerRating, int defenderRating,
BattleLog.Result result)
```

Parameters

`challengerRating` int

`defenderRating` int

`result` BattleLog.Result

Returns

int

GetScoreV3(int, int, Result)

```
[Obsolete("Use GetScore()")]
public static (int challengerScore, int defenderScore) GetScoreV3(int
challengerRating, int defenderRating, BattleLog.Result result)
```

Parameters

`challengerRating` int

`defenderRating` int

`result` BattleLog.Result

Returns

(int challengerScoreDelta, int defenderScoreDelta)

GetScoreV4(int, int, Result)

```
[Obsolete("Use GetScore()")]
public static (int challengerScoreDelta, int defenderScoreDelta) GetScoreV4(int
challengerRating, int defenderRating, BattleLog.Result result)
```

Parameters

`challengerRating` int

`defenderRating` int

`result` BattleLog.Result

Returns

(int challengerScoreDelta, int defenderScoreDelta)

Class AttackCountHelper

Namespace: Nekoyume.Battle

Assembly: Lib9c.dll

```
public static class AttackCountHelper
```

Inheritance

object ← AttackCountHelper

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Fields

CountMaxLowerLimit

```
public const int CountMaxLowerLimit = 2
```

Field Value

int

CountMaxUpperLimit

```
public const int CountMaxUpperLimit = 5
```

Field Value

int

Methods

GetAdditionalCriticalChance(int, int)

```
public static int GetAdditionalCriticalChance(int attackCount, int attackCountMax)
```

Parameters

`attackCount` int

`attackCountMax` int

Returns

int

GetCountMax(int)

```
public static int GetCountMax(int level)
```

Parameters

`level` int

Returns

int

GetDamageMultiplier(int, int)

```
public static int GetDamageMultiplier(int attackCount, int attackCountMax)
```

Parameters

`attackCount` int

`attackCountMax` int

Returns

int

Class CPHelper

Namespace: Nekoyume.Battle

Assembly: Lib9c.dll

```
public static class CPHelper
```

Inheritance

object ← CPHelper

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Methods

ConvertCpToStat(StatType, decimal, int)

```
public static decimal ConvertCpToStat(StatType statType, decimal cp,
int characterLevel)
```

Parameters

`statType` StatType

`cp` decimal

`characterLevel` int

Returns

decimal

DecimalToLong(decimal)

```
public static long DecimalToLong(decimal value)
```

Parameters

`value` decimal

Returns

long

GetCP(Costume, CostumeStatSheet)

```
public static long GetCP(Costume costume, CostumeStatSheet sheet)
```

Parameters

`costume` Costume

`sheet` CostumeStatSheet

Returns

long

GetCP(INonFungibleItem, CostumeStatSheet)

```
[Obsolete("Use GetCp")]
public static long GetCP(INonFungibleItem tradableItem, CostumeStatSheet sheet)
```

Parameters

`tradableItem` INonFungibleItem

`sheet` CostumeStatSheet

Returns

long

GetCP(ItemUsable)

```
public static long GetCP(ItemUsable itemUsable)
```

Parameters

`itemUsable` ItemUsable

Returns

long

GetCP(AvatarState, CharacterSheet)

```
[Obsolete("Use TotalCP")]
public static long GetCP(AvatarState avatarState, CharacterSheet characterSheet)
```

Parameters

`avatarState` AvatarState

`characterSheet` CharacterSheet

Returns

long

GetCPOfATK(decimal)

```
public static decimal GetCPOfATK(decimal value)
```

Parameters

`value` decimal

Returns

decimal

GetCPOfArmorPenetration(decimal)

```
public static decimal GetCPOfArmorPenetration(decimal value)
```

Parameters

`value` decimal

Returns

decimal

GetCPOfCDMG(decimal, int)

```
public static decimal GetCPOfCDMG(decimal value, int characterLevel)
```

Parameters

`value` decimal

`characterLevel` int

Returns

decimal

GetCPOfCRI(decimal, int)

```
public static decimal GetCPOfCRI(decimal value, int characterLevel)
```

Parameters

`value` decimal

`characterLevel` int

Returns

decimal

GetCPOfDEF(decimal)

```
public static decimal GetCPOfDEF(decimal value)
```

Parameters

`value` decimal

Returns

decimal

GetCPOfDRR(decimal, int)

```
public static decimal GetCPOfDRR(decimal value, int characterLevel)
```

Parameters

`value` decimal

`characterLevel` int

Returns

decimal

GetCPOfDRV(decimal)

```
public static decimal GetCPOfDRV(decimal value)
```

Parameters

`value` decimal

Returns

decimal

GetCPOfHIT(decimal)

```
public static decimal GetCPOfHIT(decimal value)
```

Parameters

`value` decimal

Returns

decimal

GetCPOfHP(decimal)

```
public static decimal GetCPOfHP(decimal value)
```

Parameters

`value` decimal

Returns

decimal

GetCPOfSPD(decimal)

```
public static decimal GetCPOfSPD(decimal value)
```

Parameters

`value` decimal

Returns

decimal

GetCPOfThorn(decimal)

```
public static decimal GetCPOfThorn(decimal value)
```

Parameters

`value` decimal

Returns

decimal

GetCPV2(AvatarState, CharacterSheet, CostumeStatSheet)

```
[Obsolete("Use TotalCP")]
public static long GetCPV2(AvatarState avatarState, CharacterSheet characterSheet,
CostumeStatSheet costumeStatSheet)
```

Parameters

`avatarState` AvatarState

`characterSheet` CharacterSheet

`costumeStatSheet` CostumeStatSheet

Returns

long

GetSkillsMultiplier(int)

```
public static decimal GetSkillsMultiplier(int skillsCount)
```

Parameters

`skillsCount` int

Returns

decimal

GetStatCP(StatType, decimal, int)

```
public static decimal GetStatCP(StatType statType, decimal statValue, int
characterLevel = 1)
```

Parameters

`statType` StatType

`statValue` decimal

`characterLevel` int

Returns

decimal

GetStatsCP(IStats, int)

```
public static decimal GetStatsCP(IStats stats, int characterLevel = 1)
```

Parameters

`stats` IStats

`characterLevel` int

Returns

decimal

TotalCP(IReadOnlyCollection<Equipment>, IReadOnlyCollection<Costume>, IReadOnlyCollection<RuneOptionInfo>, int, Row, CostumeStatSheet, List<StatModifier>, int)

```
public static long TotalCP(IReadOnlyCollection<Equipment> equipments,
IReadOnlyCollection<Costume> costumes,
IReadOnlyCollection<RuneOptionSheet.Row.RuneOptionInfo> runeOptions, int level,
CharacterSheet.Row row, CostumeStatSheet costumeStatSheet, List<StatModifier>
collectionStatModifiers, int runeLevelBonus)
```

Parameters

`equipments` IReadOnlyCollection<Equipment>

`costumes` IReadOnlyCollection<Costume>

`runeOptions` IReadOnlyCollection<RuneOptionSheet.Row.RuneOptionInfo>

`level` int

`row` CharacterSheet.Row

`costumeStatSheet` CostumeStatSheet

`collectionStatModifiers` List<StatModifier>

`runeLevelBonus` int

Returns

long

Class CriticalHelper

Namespace: Nekoyume.Battle

Assembly: Lib9c.dll

```
public static class CriticalHelper
```

Inheritance

object ← CriticalHelper

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Methods

GetCriticalDamage(CharacterBase, long)

```
public static long GetCriticalDamage(CharacterBase caster, long originalDamage)
```

Parameters

`caster` CharacterBase

`originalDamage` long

Returns

long

GetCriticalDamageForArena(ArenaCharacter, long)

```
public static long GetCriticalDamageForArena(ArenaCharacter caster,
long originalDamage)
```

Parameters

`caster` ArenaCharacter

`originalDamage` long

Returns

long

Class DamageHelper

Namespace: Nekoyume.Battle

Assembly: Lib9c.dll

```
public static class DamageHelper
```

Inheritance

object ← DamageHelper

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Methods

GetDamageReductionRate(long)

calculate damage reduction rate.

```
public static decimal GetDamageReductionRate(long drr)
```

Parameters

`drr` long

enemy DRR stats

Returns

decimal

damage reduction rate.

GetFinalDefense(long, long)

Calculates final defense after applying armor penetration. Uses Math.Clamp to ensure the result is between 0 and long.MaxValue, preventing overflow issues when dealing with large defense values.

```
public static long GetFinalDefense(long targetDefense, long armorPenetration)
```

Parameters

`targetDefense` long

enemy DEF stats

`armorPenetration` long

caster ArmorPenetration stats

Returns

long

calculated final defense.

GetReducedDamage(long, long, long)

Calculates reduced damage after applying damage reduction values (DRV) and damage reduction rate (DRR). Uses NumberConversionHelper.SafeDecimalToInt64 to prevent overflow when dealing with large damage values. Ensures minimum damage of 1 to prevent zero damage attacks.

```
public static long GetReducedDamage(long damage, long drv, long drr)
```

Parameters

`damage` long

damage

`drv` long

enemy DRV stats

`drr` long

enemy DRR stats

Returns

long

calculated damage.

Struct EnemyPlayerDigest

Namespace: Nekoyume.Battle

Assembly: Lib9c.dll

```
public readonly struct EnemyPlayerDigest : IState
```

Implements
IState

Inherited Members
ValueType.Equals(object) , ValueType.GetHashCode() , ValueType.ToString() , object.Equals(object, object) , object.GetType() , object.ReferenceEquals(object, object)

Constructors

EnemyPlayerDigest(List)

```
public EnemyPlayerDigest(List serialized)
```

Parameters

`serialized` List

EnemyPlayerDigest(AvatarState)

```
public EnemyPlayerDigest(AvatarState enemyAvatarState)
```

Parameters

`enemyAvatarState` AvatarState

Fields

CharacterId

```
public readonly int CharacterId
```

Field Value

int

Costumes

```
public readonly IReadOnlyList<Costume> Costumes
```

Field Value

IReadOnlyList<Costume>

EarIndex

```
public readonly int EarIndex
```

Field Value

int

Equipments

```
public readonly IReadOnlyList<Equipment> Equipments
```

Field Value

IReadOnlyList<Equipment>

HairIndex

```
public readonly int HairIndex
```

Field Value

int

LensIndex

```
public readonly int LensIndex
```

Field Value

int

Level

```
public readonly int Level
```

Field Value

int

NameWithHash

```
public readonly string NameWithHash
```

Field Value

string

TailIndex

```
public readonly int TailIndex
```

Field Value

int

Methods

Serialize()

```
public IValue Serialize()
```

Returns

IValue

Class HitHelper

Namespace: Nekoyume.Battle

Assembly: Lib9c.dll

```
public static class HitHelper
```

Inheritance

object ← HitHelper

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Fields

GetHitStep1CorrectionMax

```
public const long GetHitStep1CorrectionMax = 50
```

Field Value

long

GetHitStep1CorrectionMin

```
public const long GetHitStep1CorrectionMin = -5
```

Field Value

long

GetHitStep1LevelDiffMax

```
public const long GetHitStep1LevelDiffMax = 10
```

Field Value

long

GetHitStep1LevelDiffMin

```
public const long GetHitStep1LevelDiffMin = -14
```

Field Value

long

GetHitStep2AdditionalCorrectionMax

```
public const long GetHitStep2AdditionalCorrectionMax = 50
```

Field Value

long

GetHitStep2AdditionalCorrectionMin

```
public const long GetHitStep2AdditionalCorrectionMin = 0
```

Field Value

long

GetHitStep3CorrectionMax

```
public const long GetHitStep3CorrectionMax = 90
```

Field Value

long

GetHitStep3CorrectionMin

```
public const long GetHitStep3CorrectionMin = 10
```

Field Value

long

Methods

GetHitStep1(long, long)

```
public static long GetHitStep1(long attackerLevel, long defenderLevel)
```

Parameters

`attackerLevel` long

`defenderLevel` long

Returns

long

GetHitStep2(long, long)

```
public static long GetHitStep2(long attackerHit, long defenderHit)
```

Parameters

`attackerHit` long

`defenderHit` long

Returns

long

GetHitStep3(long)

```
public static long GetHitStep3(long correction)
```

Parameters

`correction` long

Returns

long

GetHitStep4(long, long)

```
public static bool GetHitStep4(long lowLimitChance, long correction)
```

Parameters

`lowLimitChance` long

`correction` long

Returns

bool

IsHit(long, long, long, long, long)

```
public static bool IsHit(long attackerLevel, long attackerHit, long defenderLevel,
long defenderHit, long lowLimitChance)
```

Parameters

`attackerLevel` long

`attackerHit` long

`defenderLevel` long

`defenderHit` long

`lowLimitChance` long

Returns

bool

IsHitWithoutLevelCorrection(long, long, long, long, long)

```
public static bool IsHitWithoutLevelCorrection(long attackerLevel, long attackerHit,
long defenderLevel, long defenderHit, long lowLimitChance)
```

Parameters

`attackerLevel` long

`attackerHit` long

`defenderLevel` long

`defenderHit` long

`lowLimitChance` long

Returns

bool

Interface ISimulator

Namespace: Nekoyume.Battle

Assembly: Lib9c.dll

```
public interface ISimulator
```

Properties

Characters

```
SimplePriorityQueue<CharacterBase, decimal> Characters { get; }
```

Property Value

SimplePriorityQueue<CharacterBase, decimal>

Log

```
BattleLog Log { get; }
```

Property Value

BattleLog

LogEvent

```
bool LogEvent { get; }
```

Property Value

bool

Player

```
Player Player { get; }
```

Property Value

Player

Reward

```
IEnumerable<ItemBase> Reward { get; }
```

Property Value

IEnumerable<ItemBase>

WaveNumber

```
int WaveNumber { get; }
```

Property Value

int

WaveTurn

```
int WaveTurn { get; }
```

Property Value

int

Interface IStageSimulator

Namespace: Nekoyume.Battle

Assembly: Lib9c.dll

```
public interface IStageSimulator : ISimulator
```

Inherited Members
ISimulator.Player , ISimulator.Log , ISimulator.Characters , ISimulator.Reward , ISimulator.WaveNumber , ISimulator.WaveTurn , ISimulator.LogEvent

Properties

EnemySkillSheet

```
EnemySkillSheet EnemySkillSheet { get; }
```

Property Value

EnemySkillSheet

ItemMap

```
CollectionMap ItemMap { get; }
```

Property Value

CollectionMap

StageId

```
int StageId { get; }
```

Property Value

int

Class InfiniteTowerLog

Namespace: Nekoyume.Battle

Assembly: Lib9c.dll

Log for infinite tower battle results. Contains information about the battle outcome and player performance.

```
public class InfiniteTowerLog
```

Inheritance

object ← InfiniteTowerLog

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Properties

ClearedWaveNumber

Gets or sets the number of waves cleared.

```
public int ClearedWaveNumber { get; set; }
```

Property Value

int

EnemiesDefeated

Gets or sets the number of enemies defeated.

```
public int EnemiesDefeated { get; set; }
```

Property Value

int

IsClear

Gets or sets whether the floor was cleared.

```
public bool IsClear { get; set; }
```

Property Value

bool

PlayerHpRemaining

Gets or sets the player's remaining HP after battle.

```
public int PlayerHpRemaining { get; set; }
```

Property Value

int

Result

Gets or sets the battle result (Win/Lose).

```
public InfiniteTowerLog.InfiniteTowerResult Result { get; set; }
```

Property Value

InfiniteTowerLog.InfiniteTowerResult

Enum InfiniteTowerLog.InfiniteTowerResult

Namespace: Nekoyume.Battle

Assembly: Lib9c.dll

Represents the possible results of an infinite tower battle.

```
public enum InfiniteTowerLog.InfiniteTowerResult
```

Extension Methods

StateExtensions.Serialize(Enum)

Fields

`Lose = 1`

The player lost the battle.

`Win = 0`

The player won the battle.

Class InfiniteTowerSimulator

Namespace: Nekoyume.Battle

Assembly: Lib9c.dll

Simulator for infinite tower battles with conditions and restrictions. Handles turn-based combat with wave progression, special conditions, and reward processing.

```
public class InfiniteTowerSimulator : Simulator, IStageSimulator, ISimulator
```

Inheritance

object ← Simulator ← InfiniteTowerSimulator

Implements

IStageSimulator, ISimulator

Inherited Members

Simulator.Random , Simulator.Log , Simulator.Player , Simulator.Result , Simulator.Characters , Simulator.TurnPriority , Simulator.MaterialItemSheet , Simulator.SkillSheet , Simulator.SkillBuffSheet , Simulator.StatBuffSheet , Simulator.SkillActionBuffSheet , Simulator.ActionBuffSheet , Simulator.CharacterSheet , Simulator.CharacterLevelSheet , Simulator.EquipmentItemSetEffectSheet , Simulator.BuffLimitSheet , Simulator.BuffLinkSheet , Simulator.ShatterStrikeMaxDamage , Simulator.MaxTurn , Simulator.TurnNumber , Simulator.WaveNumber , Simulator.WaveTurn , Simulator.LogEvent , Simulator.SetReward(WeightedSelector<StageSheet.RewardData>, int, IRandom, MaterialItemSheet) , Simulator.SetRewardV2(WeightedSelector<StageSheet.RewardData>, int, IRandom, MaterialItemSheet) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

InfiniteTowerSimulator(IRandom, AvatarState, List<Guid>, AllRuneState, RuneSlotState, int, int, Row, List<WaveData>, bool, int, SimulatorSheets,

EnemySkillSheet, CostumeStatSheet, ItemSheet, List<StatModifier>, BuffLimitSheet, BuffLinkSheet, List<InfiniteTowerCondition>, int, int, bool)

Initializes a new instance of the InfiniteTowerSimulator class.

```
public InfiniteTowerSimulator(IRandom random, AvatarState avatarState, List<Guid>
foods, AllRuneState runeStates, RuneSlotState runeSlotState, int infiniteTowerId,
int floorId, InfiniteTowerFloorSheet.Row floorRow,
List<InfiniteTowerFloorWaveSheet.WaveData> waveRows, bool isCleared, int exp,
SimulatorSheets simulatorSheets, EnemySkillSheet enemySkillSheet, CostumeStatSheet
costumeStatSheet, ItemSheet itemSheet, List<StatModifier> collectionModifiers,
BuffLimitSheet buffLimitSheet, BuffLinkSheet buffLinkSheet,
List<InfiniteTowerCondition> conditions, int shatterStrikeMaxDamage, int turnLimit =
150, bool logEvent = false)
```

Parameters

`random` IRandom

Random number generator for battle calculations.

`avatarState` AvatarState

The player's avatar state.

`foods` List<Guid>

List of food items to be consumed.

`runeStates` AllRuneState

Player's rune states.

`runeSlotState` RuneSlotState

Player's rune slot configuration.

`infiniteTowerId` int

The infinite tower ID.

`floorId` int

The floor ID being challenged.

`floorRow` InfiniteTowerFloorSheet.Row

Floor data containing rewards and requirements.

`waveRows` List<InfiniteTowerFloorWaveSheet.WaveData>

List of wave data for this floor.

`isCleared` bool

Whether this floor has been cleared before.

`exp` int

Experience reward (always 0 for infinite tower).

`simulatorSheets` SimulatorSheets

Game data sheets for simulation.

`enemySkillSheet` EnemySkillSheet

Enemy skill data.

`costumeStatSheet` CostumeStatSheet

Costume stat bonuses.

`itemSheet` ItemSheet

Item data for reward creation.

`collectionModifiers` List<StatModifier>

Collection-based stat modifiers.

`buffLimitSheet` BuffLimitSheet

Buff limit data.

`buffLinkSheet` BuffLinkSheet

Buff link data.

`conditions` List<InfiniteTowerCondition>

Infinite tower conditions for this floor.

`shatterStrikeMaxDamage` int

Maximum damage for shatter strike.

`turnLimit` int

Turn limit for all waves combined (default: 200).

`logEvent` bool

Whether to log battle events (default: true).

Properties

CollectionModifiers

Gets the collection modifiers for stat bonuses.

```
public List<StatModifier> CollectionModifiers { get; }
```

Property Value

List<StatModifier>

Conditions

Gets the infinite tower conditions that apply to this floor.

```
public List<InfiniteTowerCondition> Conditions { get; }
```

Property Value

List<InfiniteTowerCondition>

CostumeStatSheet

Gets the costume stat sheet for costume bonuses.

```
public CostumeStatSheet CostumeStatSheet { get; }
```

Property Value

CostumeStatSheet

EnemySkillSheet

Gets the enemy skill sheet for enemy abilities.

```
public EnemySkillSheet EnemySkillSheet { get; }
```

Property Value

EnemySkillSheet

Exp

Gets the experience reward (always 0 for infinite tower).

```
public int Exp { get; }
```

Property Value

int

FloorId

Gets the floor ID being challenged.

```
public int FloorId { get; }
```

Property Value

int

FloorRow

Gets the floor data containing rewards and requirements.

```
public InfiniteTowerFloorSheet.Row FloorRow { get; }
```

Property Value

InfiniteTowerFloorSheet.Row

Foods

Gets the list of food items used by the player.

```
public List<Guid> Foods { get; }
```

Property Value

List<Guid>

FungibleAssetRewards

Gets the fungible asset rewards (NCG, Crystal, etc.) obtained from this floor.

```
public Dictionary<string, int> FungibleAssetRewards { get; }
```

Property Value

Dictionary<string, int>

InfiniteTowerId

Gets the infinite tower ID.

```
public int InfiniteTowerId { get; }
```

Property Value

int

IsCleared

Gets whether this floor has been cleared before.

```
public bool IsCleared { get; }
```

Property Value

bool

ItemMap

Gets the item map for collection tracking (not used in infinite tower).

```
public CollectionMap ItemMap { get; }
```

Property Value

CollectionMap

ItemSheet

Gets the item sheet for creating reward items.

```
public ItemSheet ItemSheet { get; }
```

Property Value

ItemSheet

Reward

Gets the reward items obtained from this floor. Returns the list of items that were dropped during the battle.

```
public override IEnumerable<ItemBase> Reward { get; }
```

Property Value

IEnumerable<ItemBase>

RewardItems

Gets the list of reward items obtained from this floor.

```
public List<ItemBase> RewardItems { get; }
```

Property Value

List<ItemBase>

RuneSlotState

Gets the rune slot state for the player.

```
public RuneSlotState RuneSlotState { get; }
```

Property Value

RuneSlotState

RuneStates

Gets the rune states for the player.

```
public AllRuneState RuneStates { get; }
```

Property Value

AllRuneState

SimulatorSheets

Gets the simulator sheets containing game data.

```
public SimulatorSheets SimulatorSheets { get; }
```

Property Value

SimulatorSheets

StageId

Gets the stage ID, which is the same as the floor ID for infinite tower.

```
public int StageId { get; }
```

Property Value

int

TurnLimit

Gets the turn limit for all waves combined.

```
public int TurnLimit { get; }
```

Property Value

int

WaveRows

Gets the list of wave data for this floor.

```
public List<InfiniteTowerFloorWaveSheet.WaveData> WaveRows { get; }
```

Property Value

List<InfiniteTowerFloorWaveSheet.WaveData>

Methods

ApplyConditionsToCharacter(CharacterBase, bool)

Applies infinite tower conditions to any character (Player, Enemy, Boss, etc.). This unified method works for all CharacterBase subclasses and applies stat modifiers based on the condition's targeting rules.

```
public void ApplyConditionsToCharacter(CharacterBase character, bool isPlayer)
```

Parameters

`character` CharacterBase

The character to apply conditions to.

`isPlayer` bool

Whether the character is a player (affects targeting rules).

SetConditions(List<InfiniteTowerCondition>)

Sets the infinite tower conditions for this floor. This method allows setting conditions after simulator creation. The conditions will be applied when Simulate() is called.

```
public void SetConditions(List<InfiniteTowerCondition> conditions)
```

Parameters

`conditions` List<InfiniteTowerCondition>

The infinite tower conditions to apply.

Simulate()

Simulates the infinite tower battle with wave progression and turn-based combat. Processes all waves sequentially with integrated turn limit and applies conditions.

```
public void Simulate()
```

Class InfiniteTowerWave

Namespace: Nekoyume.Battle

Assembly: Lib9c.dll

Represents a wave in the infinite tower. Contains information about the enemies that will spawn in this wave.

```
public class InfiniteTowerWave
```

Inheritance

object ← InfiniteTowerWave

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

InfiniteTowerWave(int, int, int)

Initializes a new instance of the InfiniteTowerWave class.

```
public InfiniteTowerWave(int waveNumber, int enemyId, int enemyCount)
```

Parameters

`waveNumber` int

The wave number.

`enemyId` int

The enemy character ID.

`enemyCount` int

The number of enemies to spawn.

Properties

EnemyCount

Gets the number of enemies to spawn in this wave.

```
public int EnemyCount { get; }
```

Property Value

int

EnemyId

Gets the enemy character ID for this wave.

```
public int EnemyId { get; }
```

Property Value

int

WaveNumber

Gets the wave number.

```
public int WaveNumber { get; }
```

Property Value

int

Class InvalidCountException

Namespace: Nekoyume.Battle

Assembly: Lib9c.dll

```
public class InvalidCountException : InvalidOperationException, ISerializable
```

Inheritance

object ← Exception ← SystemException ← InvalidOperationException ← InvalidCountException

Implements

ISerializable

Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Constructors

InvalidCountException()

```
public InvalidCountException()
```

InvalidCountException(string)

```
public InvalidCountException(string message)
```

Parameters

`message` string

Class ListEmptyException

Namespace: Nekoyume.Battle

Assembly: Lib9c.dll

```
public class ListEmptyException : InvalidOperationException, ISerializable
```

Inheritance

object ← Exception ← SystemException ← InvalidOperationException ← ListEmptyException

Implements

ISerializable

Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Constructors

ListEmptyException(string)

```
public ListEmptyException(string message = "list is empty. add value first")
```

Parameters

`message` string

Class RaidSimulator

Namespace: Nekoyume.Battle

Assembly: Lib9c.dll

```
public class RaidSimulator : Simulator, ISimulator
```

Inheritance

object ← Simulator ← RaidSimulator

Implements

ISimulator

Inherited Members

Simulator.Random , Simulator.Log , Simulator.Player , Simulator.Result , Simulator.Characters , Simulator.TurnPriority , Simulator.MaterialItemSheet , Simulator.SkillSheet , Simulator.SkillBuffSheet , Simulator.StatBuffSheet , Simulator.SkillActionBuffSheet , Simulator.ActionBuffSheet , Simulator.CharacterSheet , Simulator.CharacterLevelSheet , Simulator.EquipmentItemSetEffectSheet , Simulator.BuffLimitSheet , Simulator.BuffLinkSheet , Simulator.ShatterStrikeMaxDamage , Simulator.MaxTurn , Simulator.TurnNumber , Simulator.WaveNumber , Simulator.WaveTurn , Simulator.LogEvent , Simulator.SetReward(WeightedSelector<StageSheet.RewardData>, int, IRandom, MaterialItemSheet) , Simulator.SetRewardV2(WeightedSelector<StageSheet.RewardData>, int, IRandom, MaterialItemSheet) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

RaidSimulator(int, IRandom, AvatarState, List<Guid>, AllRuneState, RuneSlotState, RaidSimulatorSheets, CostumeStatSheet, List<StatModifier>, BuffLimitSheet, BuffLinkSheet, long, Row)

```
public RaidSimulator(int bossId, IRandom random, AvatarState avatarState, List<Guid>
foods, AllRuneState runeStates, RuneSlotState runeSlotState, RaidSimulatorSheets
simulatorSheets, CostumeStatSheet costumeStatSheet, List<StatModifier>
collectionModifiers, BuffLimitSheet buffLimitSheet, BuffLinkSheet buffLinkSheet,
long shatterStrikeMaxDamage = 400000, BoostScheduleSheet.Row battleRewardBoost
= null)
```

Parameters

`bossId` int

`random` IRandom

`avatarState` AvatarState

`foods` List<Guid>

`runeStates` AllRuneState

`runeSlotState` RuneSlotState

`simulatorSheets` RaidSimulatorSheets

`costumeStatSheet` CostumeStatSheet

`collectionModifiers` List<StatModifier>

`buffLimitSheet` BuffLimitSheet

`buffLinkSheet` BuffLinkSheet

`shatterStrikeMaxDamage` long

`battleRewardBoost` BoostScheduleSheet.Row

Properties

AssetReward

```
public List<FungibleAssetValue> AssetReward { get; }
```

Property Value

List<FungibleAssetValue>

BossId

```
public int BossId { get; }
```

Property Value

int

DamageDealt

```
public long DamageDealt { get; }
```

Property Value

long

Reward

```
public override IEnumerable<ItemBase> Reward { get; }
```

Property Value

IEnumerable<ItemBase>

Methods

Simulate()

```
public BattleLog Simulate()
```

Returns

BattleLog

SpawnBoss(RaidBoss)

```
public void SpawnBoss(RaidBoss raidBoss)
```

Parameters

`raidBoss` RaidBoss

Class RaidSimulatorV1

Namespace: Nekoyume.Battle

Assembly: Lib9c.dll

```
public class RaidSimulatorV1 : Simulator, ISimulator
```

Inheritance

object ← Simulator ← RaidSimulatorV1

Implements

ISimulator

Inherited Members

Simulator.Random , Simulator.Log , Simulator.Player , Simulator.Result , Simulator.Characters , Simulator.TurnPriority , Simulator.MaterialItemSheet , Simulator.SkillSheet , Simulator.SkillBuffSheet , Simulator.StatBuffSheet , Simulator.SkillActionBuffSheet , Simulator.ActionBuffSheet , Simulator.CharacterSheet , Simulator.CharacterLevelSheet , Simulator.EquipmentItemSetEffectSheet , Simulator.BuffLimitSheet , Simulator.BuffLinkSheet , Simulator.ShatterStrikeMaxDamage , Simulator.MaxTurn , Simulator.TurnNumber , Simulator.WaveNumber , Simulator.WaveTurn , Simulator.LogEvent , Simulator.SetReward(WeightedSelector<StageSheet.RewardData>, int, IRandom, MaterialItemSheet) , Simulator.SetRewardV2(WeightedSelector<StageSheet.RewardData>, int, IRandom, MaterialItemSheet) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

RaidSimulatorV1(int, IRandom, AvatarState, List<Guid>, RaidSimulatorSheetsV1, CostumeStatSheet)

```
public RaidSimulatorV1(int bossId, IRandom random, AvatarState
avatarState, List<Guid> foods, RaidSimulatorSheetsV1 simulatorSheets,
CostumeStatSheet costumeStatSheet)
```

Parameters

`bossId` int

`random` IRandom

`avatarState` AvatarState

`foods` List<Guid>

`simulatorSheets` RaidSimulatorSheetsV1

`costumeStatSheet` CostumeStatSheet

Properties

AssetReward

```
public List<FungibleAssetValue> AssetReward { get; }
```

Property Value

List<FungibleAssetValue>

BossId

```
public int BossId { get; }
```

Property Value

int

DamageDealt

```
public long DamageDealt { get; }
```

Property Value

long

Reward

```
public override IEnumerable<ItemBase> Reward { get; }
```

Property Value

IEnumerable<ItemBase>

Methods

Simulate()

```
public BattleLog Simulate()
```

Returns

BattleLog

SpawnBoss(RaidBoss)

```
public void SpawnBoss(RaidBoss raidBoss)
```

Parameters

`raidBoss` RaidBoss

Class RaidSimulatorV2

Namespace: Nekoyume.Battle

Assembly: Lib9c.dll

```
public class RaidSimulatorV2 : Simulator, ISimulator
```

Inheritance

object ← Simulator ← RaidSimulatorV2

Implements

ISimulator

Inherited Members

Simulator.Random , Simulator.Log , Simulator.Player , Simulator.Result , Simulator.Characters , Simulator.TurnPriority , Simulator.MaterialItemSheet , Simulator.SkillSheet , Simulator.SkillBuffSheet , Simulator.StatBuffSheet , Simulator.SkillActionBuffSheet , Simulator.ActionBuffSheet , Simulator.CharacterSheet , Simulator.CharacterLevelSheet , Simulator.EquipmentItemSetEffectSheet , Simulator.BuffLimitSheet , Simulator.BuffLinkSheet , Simulator.ShatterStrikeMaxDamage , Simulator.MaxTurn , Simulator.TurnNumber , Simulator.WaveNumber , Simulator.WaveTurn , Simulator.LogEvent , Simulator.SetReward(WeightedSelector<StageSheet.RewardData>, int, IRandom, MaterialItemSheet) , Simulator.SetRewardV2(WeightedSelector<StageSheet.RewardData>, int, IRandom, MaterialItemSheet) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

RaidSimulatorV2(int, IRandom, AvatarState, List<Guid>, List<RuneState>, RaidSimulatorSheets, CostumeStatSheet)

```
public RaidSimulatorV2(int bossId, IRandom random, AvatarState avatarState,
List<Guid> foods, List<RuneState> runeStates, RaidSimulatorSheets simulatorSheets,
CostumeStatSheet costumeStatSheet)
```

Parameters

`bossId` int

`random` IRandom

`avatarState` AvatarState

`foods` List<Guid>

`runeStates` List<RuneState>

`simulatorSheets` RaidSimulatorSheets

`costumeStatSheet` CostumeStatSheet

Properties

AssetReward

```
public List<FungibleAssetValue> AssetReward { get; }
```

Property Value

List<FungibleAssetValue>

BossId

```
public int BossId { get; }
```

Property Value

int

DamageDealt

```
public long DamageDealt { get; }
```

Property Value

long

Reward

```
public override IEnumerable<ItemBase> Reward { get; }
```

Property Value

IEnumerable<ItemBase>

Methods

Simulate()

```
public BattleLog Simulate()
```

Returns

BattleLog

SpawnBoss(RaidBoss)

```
public void SpawnBoss(RaidBoss raidBoss)
```

Parameters

`raidBoss` RaidBoss

Class RankingSimulator

Namespace: Nekoyume.Battle

Assembly: Lib9c.dll

```
public class RankingSimulator : Simulator, ISimulator
```

Inheritance

object ← Simulator ← RankingSimulator

Implements

ISimulator

Inherited Members

Simulator.Random , Simulator.Log , Simulator.Player , Simulator.Result , Simulator.Characters , Simulator.TurnPriority , Simulator.MaterialItemSheet , Simulator.SkillSheet , Simulator.SkillBuffSheet , Simulator.StatBuffSheet , Simulator.SkillActionBuffSheet , Simulator.ActionBuffSheet , Simulator.CharacterSheet , Simulator.CharacterLevelSheet , Simulator.EquipmentItemSetEffectSheet , Simulator.BuffLimitSheet , Simulator.BuffLinkSheet , Simulator.ShatterStrikeMaxDamage , Simulator.MaxTurn , Simulator.TurnNumber , Simulator.WaveNumber , Simulator.WaveTurn , Simulator.LogEvent , Simulator.SetReward(WeightedSelector<StageSheet.RewardData>, int, IRandom, MaterialItemSheet) , Simulator.SetRewardV2(WeightedSelector<StageSheet.RewardData>, int, IRandom, MaterialItemSheet) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

RankingSimulator(IRandom, Player, EnemyPlayerDigest, List<Guid>, RankingSimulatorSheetsV1, int, CostumeStatSheet)

```
public RankingSimulator(IRandom random, Player player, EnemyPlayerDigest
enemyPlayerDigest, List<Guid> foods, RankingSimulatorSheetsV1
rankingSimulatorSheets, int stageId, CostumeStatSheet costumeStatSheet)
```

Parameters

`random` IRandom

`player` Player

`enemyPlayerDigest` EnemyPlayerDigest

`foods` List<Guid>

`rankingSimulatorSheets` RankingSimulatorSheetsV1

`stageId` int

`costumeStatSheet` CostumeStatSheet

RankingSimulator(IRandom, AvatarState, AvatarState, List<Guid>, RankingSimulatorSheetsV1, int)

```
public RankingSimulator(IRandom random, AvatarState avatarState, AvatarState
enemyAvatarState, List<Guid> foods, RankingSimulatorSheetsV1 rankingSimulatorSheets,
int stageId)
```

Parameters

`random` IRandom

`avatarState` AvatarState

`enemyAvatarState` AvatarState

`foods` List<Guid>

`rankingSimulatorSheets` RankingSimulatorSheetsV1

`stageId` int

RankingSimulator(IRandom, AvatarState, AvatarState, List<Guid>, RankingSimulatorSheetsV1, int, CostumeStatSheet)

```
public RankingSimulator(IRandom random, AvatarState avatarState, AvatarState
enemyAvatarState, List<Guid> foods, RankingSimulatorSheetsV1 rankingSimulatorSheets,
int stageId, CostumeStatSheet costumeStatSheet)
```

Parameters

`random` IRandom

`avatarState` AvatarState

`enemyAvatarState` AvatarState

`foods` List<Guid>

`rankingSimulatorSheets` RankingSimulatorSheetsV1

`stageId` int

`costumeStatSheet` CostumeStatSheet

Properties

Reward

Use this property after invoke the `PostSimulate(rewards)` function. This property will returns `null` basically.

```
public override IEnumerable<ItemBase> Reward { get; }
```

Property Value

IEnumerable<ItemBase>

Methods

PostSimulate(List<ItemBase>, int, int)

This function should invoke after the `Simulate()` function invoked.

```
public void PostSimulate(List<ItemBase> rewards, int challengerScoreDelta,
int challengerScore)
```

Parameters

`rewards` List<ItemBase>

`challengerScoreDelta` int

`challengerScore` int

Simulate()

```
public Player Simulate()
```

Returns

Player

Class RankingSimulatorV1

Namespace: Nekoyume.Battle

Assembly: Lib9c.dll

```
public class RankingSimulatorV1 : Simulator, ISimulator
```

Inheritance

object ← Simulator ← RankingSimulatorV1

Implements

ISimulator

Inherited Members

Simulator.Random , Simulator.Log , Simulator.Player , Simulator.Result , Simulator.Characters , Simulator.TurnPriority , Simulator.MaterialItemSheet , Simulator.SkillSheet , Simulator.SkillBuffSheet , Simulator.StatBuffSheet , Simulator.SkillActionBuffSheet , Simulator.ActionBuffSheet , Simulator.CharacterSheet , Simulator.CharacterLevelSheet , Simulator.EquipmentItemSetEffectSheet , Simulator.BuffLimitSheet , Simulator.BuffLinkSheet , Simulator.ShatterStrikeMaxDamage , Simulator.MaxTurn , Simulator.TurnNumber , Simulator.WaveNumber , Simulator.WaveTurn , Simulator.LogEvent , Simulator.SetReward(WeightedSelector<StageSheet.RewardData>, int, IRandom, MaterialItemSheet) , Simulator.SetRewardV2(WeightedSelector<StageSheet.RewardData>, int, IRandom, MaterialItemSheet) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

RankingSimulatorV1(IRandom, Player, EnemyPlayerDigest, List<Guid>, RankingSimulatorSheetsV1, int, ArenaInfo, ArenaInfo, CostumeStatSheet)

```
public RankingSimulatorV1(IRandom random, Player player, EnemyPlayerDigest
enemyPlayerDigest, List<Guid> foods, RankingSimulatorSheetsV1
rankingSimulatorSheets, int stageId, ArenaInfo arenaInfo, ArenaInfo enemyInfo,
CostumeStatSheet costumeStatSheet)
```

Parameters

`random` IRandom

`player` Player

`enemyPlayerDigest` EnemyPlayerDigest

`foods` List<Guid>

`rankingSimulatorSheets` RankingSimulatorSheetsV1

`stageId` int

`arenaInfo` ArenaInfo

`enemyInfo` ArenaInfo

`costumeStatSheet` CostumeStatSheet

RankingSimulatorV1(IRandom, AvatarState, AvatarState, List<Guid>, RankingSimulatorSheetsV1, int, ArenaInfo, ArenaInfo)

```
public RankingSimulatorV1(IRandom random, AvatarState avatarState, AvatarState
enemyAvatarState, List<Guid> foods, RankingSimulatorSheetsV1 rankingSimulatorSheets,
int stageId, ArenaInfo arenaInfo, ArenaInfo enemyInfo)
```

Parameters

`random` IRandom

`avatarState` AvatarState

`enemyAvatarState` AvatarState

foods List<Guid>

rankingSimulatorSheets RankingSimulatorSheetsV1

stageId int

arenaInfo ArenaInfo

enemyInfo ArenaInfo

RankingSimulatorV1(IRandom, AvatarState, AvatarState, List<Guid>, RankingSimulatorSheetsV1, int, ArenaInfo, ArenaInfo, CostumeStatSheet)

```
public RankingSimulatorV1(IRandom random, AvatarState avatarState,
AvatarState enemyAvatarState, List<Guid> foods, RankingSimulatorSheetsV1
rankingSimulatorSheets, int stageId, ArenaInfo arenaInfo, ArenaInfo enemyInfo,
CostumeStatSheet costumeStatSheet)
```

Parameters

random IRandom

avatarState AvatarState

enemyAvatarState AvatarState

foods List<Guid>

rankingSimulatorSheets RankingSimulatorSheetsV1

stageId int

arenaInfo ArenaInfo

enemyInfo ArenaInfo

costumeStatSheet CostumeStatSheet

Fields

WeeklyArenaRewardSheet

```
public readonly WeeklyArenaRewardSheet WeeklyArenaRewardSheet
```

Field Value

WeeklyArenaRewardSheet

Properties

Reward

```
public override IEnumerable<ItemBase> Reward { get; }
```

Property Value

IEnumerable<ItemBase>

Methods

Simulate()

```
public Player Simulate()
```

Returns

Player

SimulateV1()

```
[Obsolete("Use Simulate")]
public Player SimulateV1()
```

Returns

Player

SimulateV2()

```
[Obsolete("Use Simulate")]
public Player SimulateV2()
```

Returns

Player

SimulateV3()

```
[Obsolete("Use Simulate")]
public Player SimulateV3()
```

Returns

Player

SimulateV4()

```
[Obsolete("use Simulate")]
public Player SimulateV4()
```

Returns

Player

Class RewardSelector

Namespace: Nekoyume.Battle

Assembly: Lib9c.dll

```
public static class RewardSelector
```

Inheritance

object ← RewardSelector

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Methods

Select(IRandom, WeeklyArenaRewardSheet, MaterialItemSheet, int, int)

```
public static List<ItemBase> Select(IRandom random, WeeklyArenaRewardSheet
weeklyArenaRewardSheet, MaterialItemSheet materialItemSheet, int playerLevel, int
maxCount = 1)
```

Parameters

`random` IRandom

`weeklyArenaRewardSheet` WeeklyArenaRewardSheet

`materialItemSheet` MaterialItemSheet

`playerLevel` int

`maxCount` int

Returns

List<ItemBase>

Class Simulator

Namespace: Nekoyume.Battle

Assembly: Lib9c.dll

```
public abstract class Simulator : ISimulator
```

Inheritance

object ← Simulator

Implements

ISimulator

Derived

ArenaSimulator2, AdventureBossSimulator, InfiniteTowerSimulator, RaidSimulator, RaidSimulatorV1, RaidSimulatorV2, RankingSimulator, RankingSimulatorV1, StageSimulator, StageSimulatorV1, StageSimulatorV2, StageSimulatorV3, TestSimulator

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

Simulator(IRandom, Player, List<Guid>, SimulatorSheets)

```
protected Simulator(IRandom random, Player player, List<Guid> foods,
SimulatorSheets simulatorSheets)
```

Parameters

`random` IRandom

`player` Player

`foods` List<Guid>

`simulatorSheets` SimulatorSheets

Simulator(IRandom, Player, List<Guid>, SimulatorSheetsV1)

```
protected Simulator(IRandom random, Player player, List<Guid> foods,
SimulatorSheetsV1 simulatorSheets)
```

Parameters

`random` IRandom

`player` Player

`foods` List<Guid>

`simulatorSheets` SimulatorSheetsV1

Simulator(IRandom, AvatarState, List<Guid>, SimulatorSheets, bool, long)

```
protected Simulator(IRandom random, AvatarState avatarState, List<Guid> foods,
SimulatorSheets simulatorSheets, bool logEvent = true, long shatterStrikeMaxDamage
= 400000)
```

Parameters

`random` IRandom

`avatarState` AvatarState

`foods` List<Guid>

`simulatorSheets` SimulatorSheets

`logEvent` bool

`shatterStrikeMaxDamage` long

Simulator(IRandom, AvatarState, List<Guid>, SimulatorSheetsV1, bool)

```
protected Simulator(IRandom random, AvatarState avatarState, List<Guid> foods,
SimulatorSheetsV1 simulatorSheets, bool logEvent = true)
```

Parameters

`random` IRandom

`avatarState` AvatarState

`foods` List<Guid>

`simulatorSheets` SimulatorSheetsV1

`logEvent` bool

Fields

ActionBuffSheet

```
public readonly ActionBuffSheet ActionBuffSheet
```

Field Value

ActionBuffSheet

CharacterLevelSheet

```
public readonly CharacterLevelSheet CharacterLevelSheet
```

Field Value

CharacterLevelSheet

CharacterSheet

```
public readonly CharacterSheet CharacterSheet
```

Field Value

CharacterSheet

EquipmentItemSetEffectSheet

```
public readonly EquipmentItemSetEffectSheet EquipmentItemSetEffectSheet
```

Field Value

EquipmentItemSetEffectSheet

MaterialItemSheet

```
public readonly MaterialItemSheet MaterialItemSheet
```

Field Value

MaterialItemSheet

MaxTurn

```
protected const int MaxTurn = 200
```

Field Value

int

Random

```
public readonly IRandom Random
```

Field Value

IRandom

SkillActionBuffSheet

```
public readonly SkillActionBuffSheet SkillActionBuffSheet
```

Field Value

SkillActionBuffSheet

SkillBuffSheet

```
public readonly SkillBuffSheet SkillBuffSheet
```

Field Value

SkillBuffSheet

SkillSheet

```
public readonly SkillSheet SkillSheet
```

Field Value

SkillSheet

StatBuffSheet

```
public readonly StatBuffSheet StatBuffSheet
```

Field Value

StatBuffSheet

TurnNumber

```
public int TurnNumber
```

Field Value

int

TurnPriority

```
public const decimal TurnPriority = 100
```

Field Value

decimal

Properties

BuffLimitSheet

```
public BuffLimitSheet BuffLimitSheet { get; protected set; }
```

Property Value

BuffLimitSheet

BuffLinkSheet

```
public BuffLinkSheet BuffLinkSheet { get; set; }
```

Property Value

BuffLinkSheet

Characters

```
public SimplePriorityQueue<CharacterBase, decimal> Characters { get; set; }
```

Property Value

SimplePriorityQueue<CharacterBase, decimal>

Log

```
public BattleLog Log { get; }
```

Property Value

BattleLog

LogEvent

```
public bool LogEvent { get; protected set; }
```

Property Value

bool

Player

```
public Player Player { get; }
```

Property Value

Player

Result

```
public BattleLog.Result Result { get; protected set; }
```

Property Value

BattleLog.Result

Reward

```
public abstract IEnumerable<ItemBase> Reward { get; }
```

Property Value

IEnumerable<ItemBase>

ShatterStrikeMaxDamage

```
public long ShatterStrikeMaxDamage { get; }
```

Property Value

long

WaveNumber

```
public int WaveNumber { get; protected set; }
```

Property Value

int

WaveTurn

```
public int WaveTurn { get; set; }
```

Property Value

int

Methods

SetReward(WeightedSelector<RewardData>, int, IRandom, MaterialItemSheet)

```
public static List<ItemBase> SetReward(WeightedSelector<StageSheet.RewardData>
itemSelector, int maxCount, IRandom random, MaterialItemSheet materialItemSheet)
```

Parameters

`itemSelector` WeightedSelector<StageSheet.RewardData>

`maxCount` int

`random` IRandom

`materialItemSheet` MaterialItemSheet

Returns

List<ItemBase>

SetRewardV2(WeightedSelector<RewardData>, int, IRandom, MaterialItemSheet)

```
public static List<ItemBase> SetRewardV2(WeightedSelector<StageSheet.RewardData>
itemSelector, int maxCount, IRandom random, MaterialItemSheet materialItemSheet)
```

Parameters

`itemSelector` WeightedSelector<StageSheet.RewardData>

`maxCount` int

`random` IRandom

`materialItemSheet` MaterialItemSheet

Returns

List<ItemBase>

Class StageRewardExpHelper

Namespace: Nekoyume.Battle

Assembly: Lib9c.dll

```
public static class StageRewardExpHelper
```

Inheritance

object ← StageRewardExpHelper

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Fields

CachedExp

key: differ (stage number - character level) value: exp

```
public static readonly IReadOnlyDictionary<int, int> CachedExp
```

Field Value

IReadOnlyDictionary<int, int>

DifferLowerLimit

```
public const int DifferLowerLimit = -15
```

Field Value

int

DifferUpperLimit

```
public const int DifferUpperLimit = 10
```

Field Value

int

RewardExpMax

```
public const int RewardExpMax = 300
```

Field Value

int

RewardExpMin

```
public const int RewardExpMin = 0
```

Field Value

int

Methods

GetExp(int, int)

```
public static int GetExp(int characterLevel, int stageNumber)
```

Parameters

`characterLevel` int

`stageNumber` int

Returns

int

Class StageSimulator

Namespace: Nekoyume.Battle

Assembly: Lib9c.dll

Simulator for stage battles that handles wave-based combat.

```
public class StageSimulator : Simulator, IStageSimulator, ISimulator
```

Inheritance

object ← Simulator ← StageSimulator

Implements

IStageSimulator, ISimulator

Inherited Members

Simulator.Random , Simulator.Log , Simulator.Player , Simulator.Result , Simulator.Characters , Simulator.TurnPriority , Simulator.MaterialItemSheet , Simulator.SkillSheet , Simulator.SkillBuffSheet , Simulator.StatBuffSheet , Simulator.SkillActionBuffSheet , Simulator.ActionBuffSheet , Simulator.CharacterSheet , Simulator.CharacterLevelSheet , Simulator.EquipmentItemSetEffectSheet , Simulator.BuffLimitSheet , Simulator.BuffLinkSheet , Simulator.ShatterStrikeMaxDamage , Simulator.MaxTurn , Simulator.TurnNumber , Simulator.WaveNumber , Simulator.WaveTurn , Simulator.LogEvent , Simulator.SetReward(WeightedSelector<StageSheet.RewardData>, int, IRandom, MaterialItemSheet) , Simulator.SetRewardV2(WeightedSelector<StageSheet.RewardData>, int, IRandom, MaterialItemSheet) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

StageSimulator(IRandom, AvatarState, List<Guid>, AllRuneState, RuneSlotState, List<Skill>, int, int, Row, Row, bool, int, SimulatorSheets, EnemySkillSheet,

CostumeStatSheet, List<ItemBase>, List<StatModifier>, BuffLimitSheet, BuffLinkSheet, bool, long, Row)

Initializes a new instance of the StageSimulator class.

```
public StageSimulator(IRandom random, AvatarState avatarState, List<Guid> foods,
AllRuneState runeStates, RuneSlotState runeSlotState, List<Skill> skillsOnWaveStart,
int worldId, int stageId, StageSheet.Row stageRow, StageWaveSheet.Row stageWaveRow,
bool isCleared, int exp, SimulatorSheets simulatorSheets, EnemySkillSheet
enemySkillSheet, CostumeStatSheet costumeStatSheet, List<ItemBase> waveRewards,
List<StatModifier> collectionModifiers, BuffLimitSheet buffLimitSheet, BuffLinkSheet
buffLinkSheet, bool logEvent = true, long shatterStrikeMaxDamage = 400000,
BoostScheduleSheet.Row favRewardBoost = null)
```

Parameters

`random` IRandom

The random number generator.

`avatarState` AvatarState

The avatar state.

`foods` List<Guid>

The food items to use.

`runeStates` AllRuneState

The rune states.

`runeSlotState` RuneSlotState

The rune slot state.

`skillsOnWaveStart` List<Skill>

The skills to use at wave start.

`worldId` int

The world ID.

`stageId` int

The stage ID.

`stageRow` StageSheet.Row

The stage row data.

`stageWaveRow` StageWaveSheet.Row

The stage wave row data.

`isCleared` bool

Whether the stage is cleared.

`exp` int

The experience points.

`simulatorSheets` SimulatorSheets

The simulator sheets.

`enemySkillSheet` EnemySkillSheet

The enemy skill sheet.

`costumeStatSheet` CostumeStatSheet

The costume stat sheet.

`waveRewards` List<ItemBase>

The item rewards already drawn for this clear.

`collectionModifiers` List<StatModifier>

The stat modifiers from activated collections.

`buffLimitSheet` BuffLimitSheet

The buff limit sheet.

`buffLinkSheet` BuffLinkSheet

The buff link sheet.

`logEvent` bool

Whether to record the battle log.

`shatterStrikeMaxDamage` long

Maximum damage for shatter strike.

`favRewardBoost` BoostScheduleSheet.Row

The BoostScheduleSheet row adjusting this stage's fungible asset rewards, or `null` for none. See ApplyFavRewardBoost(List<(string ticker, int amount)>, Row). A caller replaying a battle must pass what the action resolved.

Fields

MaxItemRewardBoostFactor

Upper bound of the boosted count of each distinct item in one clear, as a multiple of how many times it was drawn in that clear: a BoostScheduleSheet row grants at most this many times the drawn items per clear, whatever its value. It bounds a mistyped value (e.g. `MUL 200` for `MUL 2.00`) to a tenfold over-reward, and the objects ApplyItemRewardBoost(List<ItemBase>, Row, MaterialItemSheet) creates per clear to this many times `DropItemMax`. It is not a per-transaction bound; `HackAndSlashSweep` adds its extra units with one count per item id instead of one object per unit, so its cost does not grow with the value.

```
public const int MaxItemRewardBoostFactor = 10
```

Field Value

int

Properties

EnemySkillSheet

Gets the enemy skill sheet.

```
public EnemySkillSheet EnemySkillSheet { get; }
```

Property Value

EnemySkillSheet

FungibleAssetRewards

Gets the fungible asset rewards (e.g., Crystal) obtained from wave 2 clear.

```
public Dictionary<string, int> FungibleAssetRewards { get; }
```

Property Value

Dictionary<string, int>

ItemMap

Gets the collection map for items.

```
public CollectionMap ItemMap { get; }
```

Property Value

CollectionMap

Reward

Gets the rewards from the battle.

```
public override IEnumerable<ItemBase> Reward { get; }
```

Property Value

IEnumerable<ItemBase>

StageId

Gets the stage ID.

```
public int StageId { get; }
```

Property Value

int

Methods

ApplyFavRewardBoost(List<(string ticker, int amount)>, Row)

Applies a BoostScheduleSheet row to the fungible asset rewards of one clear.

```
public static List<(string ticker, int amount)> ApplyFavRewardBoost(List<(string
ticker, int amount)> rewards, BoostScheduleSheet.Row favRewardBoost)
```

Parameters

`rewards` List<(string ticker, int amount)>

The amount drawn for each ticker.

`favRewardBoost` BoostScheduleSheet.Row

The row to apply, or `null` for none.

Returns

List<(string ticker, int amount)>

`rewards` itself when `favRewardBoost` is `null`; otherwise a new list with each positive amount replaced by Apply(int) of it, in the same order. A ticker whose adjusted amount is 0 is left out so that nothing mints a zero amount.

Remarks

Draws nothing, so the random sequence is the same as without a row. A boost adjusts what was drawn and never creates a reward: a ticker that was not drawn, or drawn as 0, stays as it is.

ApplyItemRewardBoost(List<ItemBase>, Row, MaterialItemSheet)

Applies a BoostScheduleSheet row to the item rewards of one clear, one object per unit.

```
public static List<ItemBase> ApplyItemRewardBoost(List<ItemBase> rewards,
BoostScheduleSheet.Row itemRewardBoost, MaterialItemSheet materialItemSheet)
```

Parameters

`rewards` List<ItemBase>

The items drawn in one clear, one object per unit.

`itemRewardBoost` BoostScheduleSheet.Row

The row to apply, or `null` for none.

`materialItemSheet` MaterialItemSheet

The sheet the rewards were created from.

Returns

List<ItemBase>

`rewards` itself when `itemRewardBoost` is `null`; otherwise a new list, ordered by item id, in which each distinct item drawn `n` times appears GetBoostedItemRewardCount(Row, int) (row, `n`) times, at most MaxItemRewardBoostFactor × `n`.

Remarks

Draws nothing, so the random sequence is the same as without a row. `MUL 2` grants exactly twice each drawn item; `ADD 1` one more of each distinct item; a multiplier below 1 takes items away. A `MUL` above 1 turns an item drawn once into at least 2, so `MUL 1.5` grants 2 of a single drop (and 3 of a double). The count can exceed the stage's `DropItemMax`, which only limits the draws. Added items are created the way

SetRewardV2(WeightedSelector<RewardData>, int, IRandom, MaterialItemSheet) creates them, so a circle is tradable as usual. Equivalent to SplitItemRewardBoost(List<ItemBase>, Row, IDictionary<int, int>) followed by CreateItemRewardBoostExtras(IReadOnlyDictionary<int, int>, MaterialItemSheet).

CreateItemRewardBoostExtras(IReadOnlyDictionary<int, int>, MaterialItemSheet)

Creates the items SplitItemRewardBoost(List<ItemBase>, Row, IDictionary<int, int>) counted, one object per unit, for a caller that needs objects (a battle's drop log, a result view).

```
public static List<ItemBase> CreateItemRewardBoostExtras(IReadOnlyDictionary<int,
int> extraCounts, MaterialItemSheet materialItemSheet)
```

Parameters

`extraCounts` IReadOnlyDictionary<int, int>

Units to create per item id.

`materialItemSheet` MaterialItemSheet

The sheet the rewards were created from.

Returns

List<ItemBase>

The items, ordered by item id.

CreateItemRewardBoostItem(Row)

Creates one unit of a boosted item reward the way SetRewardV2(WeightedSelector<RewardData>, int, IRandom, MaterialItemSheet) creates a drawn one, so a circle is tradable as usual. A fungible item, so one object may stand for any number of units when added with a count.

```
public static ItemBase CreateItemRewardBoostItem(MaterialItemSheet.Row materialRow)
```

Parameters

`materialRow` MaterialItemSheet.Row

The item's row.

Returns

ItemBase

The item.

GetBoostedItemRewardCount(Row, int)

The boosted count of an item drawn `count` times in one clear.

```
public static int GetBoostedItemRewardCount(BoostScheduleSheet.Row itemRewardBoost,
int count)
```

Parameters

`itemRewardBoost` BoostScheduleSheet.Row

The row to apply.

`count` int

How many times the item was drawn in the clear.

Returns

int

Apply(int)(`count`), except that a Mul above 1 turns an item drawn once into at least 2, so that a fractional multiplier still rewards the single drops most stages make. A count of 2 or more keeps the plain floor (`MUL 1.2` on 2 stays 2). Capped at MaxItemRewardBoost Factor × `count`.

GetFavWaveRewards(IRandom, Row, Row)

Draws the fungible asset rewards of one clear of `stageRow`.

```
public static List<(string ticker, int amount)> GetFavWaveRewards(IRandom random,
StageSheet.Row stageRow, BoostScheduleSheet.Row favRewardBoost = null)
```

Parameters

`random` IRandom

The random number generator.

`stageRow` StageSheet.Row

The stage being cleared.

`favRewardBoost` BoostScheduleSheet.Row

The BoostScheduleSheet row adjusting the amounts, or `null` for none. See ApplyFavRewardBoost(List<(string ticker, int amount)>, Row). It never changes the random draws.

Returns

List<(string ticker, int amount)>

The amount drawn for each ticker.

GetWaveRewards(IRandom, Row, MaterialItemSheet, int, Row)

Draws the item rewards of `playCount` clears of `stageRow`.

```
public static List<ItemBase> GetWaveRewards(IRandom random, StageSheet.Row stageRow,
MaterialItemSheet materialItemSheet, int playCount = 1, BoostScheduleSheet.Row
itemRewardBoost = null)
```

Parameters

`random` IRandom

The random number generator.

`stageRow` StageSheet.Row

The stage being cleared.

`materialItemSheet` MaterialItemSheet

The sheet the rewards are created from.

`playCount` int

The number of clears.

`itemRewardBoost` BoostScheduleSheet.Row

The BoostScheduleSheet row adjusting each clear's items, or `null` for none. See ApplyItemRewardBoost(List<ItemBase>, Row, MaterialItemSheet). It never changes the random draws. A caller replaying a battle must pass what the action resolved.

Returns

List<ItemBase>

The items, one object per unit, each clear's ordered by item id.

Simulate()

```
public Player Simulate()
```

Returns

Player

SplitItemRewardBoost(List<ItemBase>, Row, IDictionary<int, int>)

Applies a BoostScheduleSheet row to the item rewards of one clear without creating an object per added unit: units taken away are dropped from the returned list, and units added are only counted into `extraCounts`.

```
public static List<ItemBase> SplitItemRewardBoost(List<ItemBase> rewards,
BoostScheduleSheet.Row itemRewardBoost, IDictionary<int, int> extraCounts)
```

Parameters

`rewards` List<ItemBase>

The items drawn in one clear, one object per unit.

`itemRewardBoost` BoostScheduleSheet.Row

The row to apply, or `null` for none.

`extraCounts` IDictionary<int, int>

Accumulates, per item id, how many units the row adds on top of the returned list. Left untouched when nothing is added, so one dictionary can collect several clears.

Returns

List<ItemBase>

`rewards` itself when `itemRewardBoost` is `null`; otherwise the drawn items that remain, ordered by item id. Together with `extraCounts` they add up to exactly what ApplyItemRewardBoost(List<ItemBase>, Row, MaterialItemSheet) grants for the clear.

Class StageSimulatorV1

Namespace: Nekoyume.Battle

Assembly: Lib9c.dll

```
public class StageSimulatorV1 : Simulator, IStageSimulator, ISimulator
```

Inheritance

object ← Simulator ← StageSimulatorV1

Implements

IStageSimulator, ISimulator

Inherited Members

Simulator.Random , Simulator.Log , Simulator.Player , Simulator.Result , Simulator.Characters , Simulator.TurnPriority , Simulator.MaterialItemSheet , Simulator.SkillSheet , Simulator.SkillBuffSheet , Simulator.StatBuffSheet , Simulator.SkillActionBuffSheet , Simulator.ActionBuffSheet , Simulator.CharacterSheet , Simulator.CharacterLevelSheet , Simulator.EquipmentItemSetEffectSheet , Simulator.BuffLimitSheet , Simulator.BuffLinkSheet , Simulator.ShatterStrikeMaxDamage , Simulator.MaxTurn , Simulator.TurnNumber , Simulator.WaveNumber , Simulator.WaveTurn , Simulator.LogEvent , Simulator.SetReward(WeightedSelector<StageSheet.RewardData>, int, IRandom, MaterialItemSheet) , Simulator.SetRewardV2(WeightedSelector<StageSheet.RewardData>, int, IRandom, MaterialItemSheet) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

StageSimulatorV1(IRandom, AvatarState, List<Guid>, List<Skill>, int, int, StageSimulatorSheetsV1, CostumeStatSheet, int, int)

```
public StageSimulatorV1(IRandom random, AvatarState avatarState, List<Guid> foods,
List<Skill> skillsOnWaveStart, int worldId, int stageId, StageSimulatorSheetsV1
stageSimulatorSheets, CostumeStatSheet costumeStatSheet, int constructorVersion, int
playCount = 1)
```

Parameters

`random` IRandom

`avatarState` AvatarState

`foods` List<Guid>

`skillsOnWaveStart` List<Skill>

`worldId` int

`stageId` int

`stageSimulatorSheets` StageSimulatorSheetsV1

`costumeStatSheet` CostumeStatSheet

`constructorVersion` int

`playCount` int

StageSimulatorV1(IRandom, AvatarState, List<Guid>, int, int, StageSimulatorSheetsV1)

Do not use anymore since v100025.

```
public StageSimulatorV1(IRandom random, AvatarState avatarState, List<Guid> foods,
int worldId, int stageId, StageSimulatorSheetsV1 stageSimulatorSheets)
```

Parameters

`random` IRandom

`avatarState` AvatarState

`foods` List<Guid>

`worldId` int

`stageId` int

`stageSimulatorSheets` StageSimulatorSheetsV1

StageSimulatorV1(IRandom, AvatarState, List<Guid>, int, int, StageSimulatorSheetsV1, Skill)

Do not use anymore since v100025.

```
public StageSimulatorV1(IRandom random, AvatarState avatarState, List<Guid> foods,
int worldId, int stageId, StageSimulatorSheetsV1 stageSimulatorSheets, Skill skill)
```

Parameters

`random` IRandom

`avatarState` AvatarState

`foods` List<Guid>

`worldId` int

`stageId` int

`stageSimulatorSheets` StageSimulatorSheetsV1

`skill` Skill

StageSimulatorV1(IRandom, AvatarState, List<Guid>, int, int, StageSimulatorSheetsV1, Skill, int, int)

```
public StageSimulatorV1(IRandom random, AvatarState avatarState, List<Guid> foods,
int worldId, int stageId, StageSimulatorSheetsV1 stageSimulatorSheets, Skill skill,
int constructorVersion, int playCount)
```

Parameters

`random` IRandom

`avatarState` AvatarState

`foods` List<Guid>

`worldId` int

`stageId` int

`stageSimulatorSheets` StageSimulatorSheetsV1

`skill` Skill

`constructorVersion` int

`playCount` int

StageSimulatorV1(IRandom, AvatarState, List<Guid>, int, int, StageSimulatorSheetsV1, CostumeStatSheet)

Do not use anymore since v100025.

```
public StageSimulatorV1(IRandom random, AvatarState avatarState, List<Guid> foods,
int worldId, int stageId, StageSimulatorSheetsV1 stageSimulatorSheets,
CostumeStatSheet costumeStatSheet)
```

Parameters

`random` IRandom

`avatarState` AvatarState

`foods` List<Guid>

`worldId` int

`stageId` int

`stageSimulatorSheets` StageSimulatorSheetsV1

`costumeStatSheet` CostumeStatSheet

StageSimulatorV1(IRandom, AvatarState, List<Guid>, int, int, StageSimulatorSheetsV1, CostumeStatSheet, int, int)

```
public StageSimulatorV1(IRandom random, AvatarState avatarState, List<Guid> foods,
int worldId, int stageId, StageSimulatorSheetsV1 stageSimulatorSheets,
CostumeStatSheet costumeStatSheet, int constructorVersion, int playCount = 1)
```

Parameters

`random` IRandom

`avatarState` AvatarState

`foods` List<Guid>

`worldId` int

`stageId` int

`stageSimulatorSheets` StageSimulatorSheetsV1

`costumeStatSheet` CostumeStatSheet

`constructorVersion` int

`playCount` int

StageSimulatorV1(IRandom, AvatarState, List<Guid>, int, int, StageSimulatorSheetsV1, int, int)

```
public StageSimulatorV1(IRandom random, AvatarState avatarState, List<Guid> foods,
int worldId, int stageId, StageSimulatorSheetsV1 stageSimulatorSheets, int
constructorVersion, int playCount)
```

Parameters

`random` IRandom

`avatarState` AvatarState

`foods` List<Guid>

`worldId` int

`stageId` int

`stageSimulatorSheets` StageSimulatorSheetsV1

`constructorVersion` int

`playCount` int

Fields

ConstructorVersionDefault

```
public const int ConstructorVersionDefault = 1
```

Field Value

int

ConstructorVersionV100025

```
public const int ConstructorVersionV100025 = 2
```

Field Value

int

ConstructorVersionV100080

```
public const int ConstructorVersionV100080 = 3
```

Field Value

int

Properties

EnemySkillSheet

```
public EnemySkillSheet EnemySkillSheet { get; }
```

Property Value

EnemySkillSheet

ItemMap

```
public CollectionMap ItemMap { get; }
```

Property Value

CollectionMap

Reward

```
public override IEnumerable<ItemBase> Reward { get; }
```

Property Value

IEnumerable<ItemBase>

StageId

```
public int StageId { get; }
```

Property Value

int

Methods

SetItemSelector(Row, IRandom)

```
public static WeightedSelector<StageSheet.RewardData> SetItemSelector(StageSheet.Row
stageRow, IRandom random)
```

Parameters

`stageRow` StageSheet.Row

`random` IRandom

Returns

WeightedSelector<StageSheet.RewardData>

Simulate(int)

```
public Player Simulate(int playCount)
```

Parameters

`playCount` int

Returns

Player

SimulateV1()

```
[Obsolete("Use Simulate")]
```

```
public Player SimulateV1()
```

Returns

Player

SimulateV2()

```
[Obsolete("Use Simulate")]
public Player SimulateV2()
```

Returns

Player

SimulateV3()

```
[Obsolete("Use Simulate")]
public Player SimulateV3()
```

Returns

Player

SimulateV4()

```
[Obsolete("Use Simulate")]
public Player SimulateV4()
```

Returns

Player

SimulateV5(int)

```
[Obsolete("Use Simulate")]
public Player SimulateV5(int playCount)
```

Parameters

`playCount` int

Returns

Player

Class StageSimulatorV2

Namespace: Nekoyume.Battle

Assembly: Lib9c.dll

```
public class StageSimulatorV2 : Simulator, IStageSimulator, ISimulator
```

Inheritance

object ← Simulator ← StageSimulatorV2

Implements

IStageSimulator, ISimulator

Inherited Members

Simulator.Random , Simulator.Log , Simulator.Player , Simulator.Result , Simulator.Characters , Simulator.TurnPriority , Simulator.MaterialItemSheet , Simulator.SkillSheet , Simulator.SkillBuffSheet , Simulator.StatBuffSheet , Simulator.SkillActionBuffSheet , Simulator.ActionBuffSheet , Simulator.CharacterSheet , Simulator.CharacterLevelSheet , Simulator.EquipmentItemSetEffectSheet , Simulator.BuffLimitSheet , Simulator.BuffLinkSheet , Simulator.ShatterStrikeMaxDamage , Simulator.MaxTurn , Simulator.TurnNumber , Simulator.WaveNumber , Simulator.WaveTurn , Simulator.LogEvent , Simulator.SetReward(WeightedSelector<StageSheet.RewardData>, int, IRandom, MaterialItemSheet) , Simulator.SetRewardV2(WeightedSelector<StageSheet.RewardData>, int, IRandom, MaterialItemSheet) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

StageSimulatorV2(IRandom, AvatarState, List<Guid>, List<Skill>, int, int, Row, Row, bool, int, SimulatorSheetsV1, EnemySkillSheet, CostumeStatSheet, List<ItemBase>)

```
public StageSimulatorV2(IRandom random, AvatarState avatarState, List<Guid> foods,
List<Skill> skillsOnWaveStart, int worldId, int stageId, StageSheet.Row stageRow,
StageWaveSheet.Row stageWaveRow, bool isCleared, int exp, SimulatorSheetsV1
simulatorSheets, EnemySkillSheet enemySkillSheet, CostumeStatSheet costumeStatSheet,
List<ItemBase> waveRewards)
```

Parameters

`random` IRandom

`avatarState` AvatarState

`foods` List<Guid>

`skillsOnWaveStart` List<Skill>

`worldId` int

`stageId` int

`stageRow` StageSheet.Row

`stageWaveRow` StageWaveSheet.Row

`isCleared` bool

`exp` int

`simulatorSheets` SimulatorSheetsV1

`enemySkillSheet` EnemySkillSheet

`costumeStatSheet` CostumeStatSheet

`waveRewards` List<ItemBase>

Properties

EnemySkillSheet

```
public EnemySkillSheet EnemySkillSheet { get; }
```

Property Value

EnemySkillSheet

ItemMap

```
public CollectionMap ItemMap { get; }
```

Property Value

CollectionMap

Reward

```
public override IEnumerable<ItemBase> Reward { get; }
```

Property Value

IEnumerable<ItemBase>

StageId

```
public int StageId { get; }
```

Property Value

int

Methods

GetWaveRewards(IRandom, Row, MaterialItemSheet, int)

```
public static List<ItemBase> GetWaveRewards(IRandom random, StageSheet.Row stageRow,
MaterialItemSheet materialItemSheet, int playCount = 1)
```

Parameters

`random` IRandom

`stageRow` StageSheet.Row

`materialItemSheet` MaterialItemSheet

`playCount` int

Returns

List<ItemBase>

Simulate()

```
public Player Simulate()
```

Returns

Player

Class StageSimulatorV3

Namespace: Nekoyume.Battle

Assembly: Lib9c.dll

```
public class StageSimulatorV3 : Simulator, IStageSimulator, ISimulator
```

Inheritance

object ← Simulator ← StageSimulatorV3

Implements

IStageSimulator, ISimulator

Inherited Members

Simulator.Random , Simulator.Log , Simulator.Player , Simulator.Result , Simulator.Characters , Simulator.TurnPriority , Simulator.MaterialItemSheet , Simulator.SkillSheet , Simulator.SkillBuffSheet , Simulator.StatBuffSheet , Simulator.SkillActionBuffSheet , Simulator.ActionBuffSheet , Simulator.CharacterSheet , Simulator.CharacterLevelSheet , Simulator.EquipmentItemSetEffectSheet , Simulator.BuffLimitSheet , Simulator.BuffLinkSheet , Simulator.ShatterStrikeMaxDamage , Simulator.MaxTurn , Simulator.TurnNumber , Simulator.WaveNumber , Simulator.WaveTurn , Simulator.LogEvent , Simulator.SetReward(WeightedSelector<StageSheet.RewardData>, int, IRandom, MaterialItemSheet) , Simulator.SetRewardV2(WeightedSelector<StageSheet.RewardData>, int, IRandom, MaterialItemSheet) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

StageSimulatorV3(IRandom, AvatarState, List<Guid>, List<RuneState>, List<Skill>, int, int, Row, Row, bool, int, SimulatorSheets, EnemySkillSheet, CostumeStatSheet, List<ItemBase>)

```
public StageSimulatorV3(IRandom random, AvatarState avatarState, List<Guid> foods,
List<RuneState> runeStates, List<Skill> skillsOnWaveStart, int worldId, int stageId,
StageSheet.Row stageRow, StageWaveSheet.Row stageWaveRow, bool isCleared, int exp,
SimulatorSheets simulatorSheets, EnemySkillSheet enemySkillSheet, CostumeStatSheet
costumeStatSheet, List<ItemBase> waveRewards)
```

Parameters

`random` IRandom

`avatarState` AvatarState

`foods` List<Guid>

`runeStates` List<RuneState>

`skillsOnWaveStart` List<Skill>

`worldId` int

`stageId` int

`stageRow` StageSheet.Row

`stageWaveRow` StageWaveSheet.Row

`isCleared` bool

`exp` int

`simulatorSheets` SimulatorSheets

`enemySkillSheet` EnemySkillSheet

`costumeStatSheet` CostumeStatSheet

`waveRewards` List<ItemBase>

Properties

EnemySkillSheet

```
public EnemySkillSheet EnemySkillSheet { get; }
```

Property Value

EnemySkillSheet

ItemMap

```
public CollectionMap ItemMap { get; }
```

Property Value

CollectionMap

Reward

```
public override IEnumerable<ItemBase> Reward { get; }
```

Property Value

IEnumerable<ItemBase>

StageId

```
public int StageId { get; }
```

Property Value

int

Methods

GetWaveRewards(IRandom, Row, MaterialItemSheet, int)

```
public static List<ItemBase> GetWaveRewards(IRandom random, StageSheet.Row stageRow,
MaterialItemSheet materialItemSheet, int playCount = 1)
```

Parameters

`random` IRandom

`stageRow` StageSheet.Row

`materialItemSheet` MaterialItemSheet

`playCount` int

Returns

List<ItemBase>

Simulate()

```
public Player Simulate()
```

Returns

Player

Class TestSimulator

Namespace: Nekoyume.Battle

Assembly: Lib9c.dll

```
public class TestSimulator : Simulator, ISimulator
```

Inheritance

object ← Simulator ← TestSimulator

Implements

ISimulator

Inherited Members

Simulator.Random , Simulator.Log , Simulator.Player , Simulator.Result , Simulator.Characters , Simulator.TurnPriority , Simulator.MaterialItemSheet , Simulator.SkillSheet , Simulator.SkillBuffSheet , Simulator.StatBuffSheet , Simulator.SkillActionBuffSheet , Simulator.ActionBuffSheet , Simulator.CharacterSheet , Simulator.CharacterLevelSheet , Simulator.EquipmentItemSetEffectSheet , Simulator.BuffLimitSheet , Simulator.BuffLinkSheet , Simulator.ShatterStrikeMaxDamage , Simulator.MaxTurn , Simulator.TurnNumber , Simulator.WaveNumber , Simulator.WaveTurn , Simulator.LogEvent , Simulator.SetReward(WeightedSelector<StageSheet.RewardData>, int, IRandom, MaterialItemSheet) , Simulator.SetRewardV2(WeightedSelector<StageSheet.RewardData>, int, IRandom, MaterialItemSheet) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

TestSimulator(IRandom, AvatarState, List<Guid>, SimulatorSheets)

```
public TestSimulator(IRandom random, AvatarState avatarState, List<Guid> foods,
SimulatorSheets simulatorSheets)
```

Parameters

`random` IRandom

`avatarState` AvatarState

`foods` List<Guid>

`simulatorSheets` SimulatorSheets

Properties

Reward

```
public override IEnumerable<ItemBase> Reward { get; }
```

Property Value

IEnumerable<ItemBase>

Class Wave

Namespace: Nekoyume.Battle

Assembly: Lib9c.dll

Represents a wave of enemies in a battle.

```
public class Wave
```

Inheritance

object ← Wave

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Fields

HasBoss

Gets or sets whether this wave contains a boss enemy.

```
public bool HasBoss
```

Field Value

bool

Methods

Add(Enemy)

Adds an enemy to this wave.

```
public void Add(Enemy enemy)
```

Parameters

`enemy` Enemy

The enemy to add.

Spawn(ISimulator)

Spawns all enemies in this wave into the simulator.

```
public void Spawn(ISimulator simulator)
```

Parameters

`simulator` ISimulator

The simulator to spawn enemies into.

SpawnV1(ISimulator)

```
[Obsolete("Use Spawn")]
public void SpawnV1(ISimulator simulator)
```

Parameters

`simulator` ISimulator

SpawnV2(ISimulator)

```
[Obsolete("Use Spawn")]
public void SpawnV2(ISimulator simulator)
```

Parameters

`simulator` ISimulator

Class WaveResult

Namespace: Nekoyume.Battle

Assembly: Lib9c.dll

Result of a single wave battle. Contains information about the wave completion status and battle outcome.

```
public class WaveResult
```

Inheritance

object ← WaveResult

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Properties

EnemiesDefeated

Gets or sets the number of enemies defeated in this wave.

```
public int EnemiesDefeated { get; set; }
```

Property Value

int

IsClear

Gets or sets whether the wave was cleared.

```
public bool IsClear { get; set; }
```

Property Value

bool

PlayerHpRemaining

Gets or sets the player's remaining HP after the wave.

```
public int PlayerHpRemaining { get; set; }
```

Property Value

int

Result

Gets or sets the battle result for this wave.

```
public BattleLog.Result Result { get; set; }
```

Property Value

BattleLog.Result

Class WeightedSelector<T>

Namespace: Nekoyume.Battle

Assembly: Lib9c.dll

```
public class WeightedSelector<T>
```

Type Parameters

T

Inheritance

object ← WeightedSelector<T>

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

WeightedSelector(IRandom)

```
public WeightedSelector(IRandom random)
```

Parameters

random IRandom

Properties

Count

```
public int Count { get; }
```

Property Value

int

Methods

Add(T, decimal)

```
public void Add(T item, decimal weight)
```

Parameters

`item` T

`weight` decimal

Clear()

```
public void Clear()
```

Select(int)

```
public IEnumerable<T> Select(int count)
```

Parameters

`count` int

Returns

IEnumerable<T>

SelectV1(int)

```
[Obsolete("Use Select")]
public IEnumerable<T> SelectV1(int count)
```

Parameters

`count` int

Returns

IEnumerable<T>

SelectV2(int)

```
[Obsolete("Use Select")]
public IEnumerable<T> SelectV2(int count)
```

Parameters

`count` int

Returns

IEnumerable<T>

Namespace Nekoyume.Battle.AdventureBoss

Classes

AdventureBossSimulator

Class AdventureBossSimulator

Namespace: Nekoyume.Battle.AdventureBoss

Assembly: Lib9c.dll

```
public class AdventureBossSimulator : Simulator, IStageSimulator, ISimulator
```

Inheritance

object ← Simulator ← AdventureBossSimulator

Implements

IStageSimulator, ISimulator

Inherited Members

Simulator.Random , Simulator.Log , Simulator.Player , Simulator.Result , Simulator.Characters , Simulator.TurnPriority , Simulator.MaterialItemSheet , Simulator.SkillSheet , Simulator.SkillBuffSheet , Simulator.StatBuffSheet , Simulator.SkillActionBuffSheet , Simulator.ActionBuffSheet , Simulator.CharacterSheet , Simulator.CharacterLevelSheet , Simulator.EquipmentItemSetEffectSheet , Simulator.BuffLimitSheet , Simulator.BuffLinkSheet , Simulator.ShatterStrikeMaxDamage , Simulator.MaxTurn , Simulator.TurnNumber , Simulator.WaveNumber , Simulator.WaveTurn , Simulator.LogEvent , Simulator.SetReward(WeightedSelector<StageSheet.RewardData>, int, IRandom, MaterialItemSheet) , Simulator.SetRewardV2(WeightedSelector<StageSheet.RewardData>, int, IRandom, MaterialItemSheet) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

AdventureBossSimulator(int, int, IRandom, AvatarState, SimulatorSheets, bool, long)

```
public AdventureBossSimulator(int bossId, int floorId, IRandom random, AvatarState
avatarState, SimulatorSheets simulatorSheets, bool logEvent = true, long
```

```
  shatterStrikeMaxDamage = 400000)
```

Parameters

`bossId` int

`floorId` int

`random` IRandom

`avatarState` AvatarState

`simulatorSheets` SimulatorSheets

`logEvent` bool

`shatterStrikeMaxDamage` long

AdventureBossSimulator(int, int, IRandom, AvatarState, List<Guid>, AllRuneState, RuneSlotState, Row, Row, SimulatorSheets, EnemySkillSheet, CostumeStatSheet, List<ItemBase>, List<StatModifier>, BuffLimitSheet, BuffLinkSheet, bool, long)

```
  public AdventureBossSimulator(int bossId, int floorId, IRandom random, AvatarState
  avatarState, List<Guid> foods, AllRuneState runeStates, RuneSlotState runeSlotState,
  AdventureBossFloorSheet.Row floorRow, AdventureBossFloorWaveSheet.Row floorWaveRow,
  SimulatorSheets simulatorSheets, EnemySkillSheet enemySkillSheet, CostumeStatSheet
  costumeStatSheet, List<ItemBase> waveRewards, List<StatModifier>
  collectionModifiers, BuffLimitSheet buffLimitSheet, BuffLinkSheet buffLinkSheet,
  bool logEvent = true, long shatterStrikeMaxDamage = 400000)
```

Parameters

`bossId` int

`floorId` int

`random` IRandom

`avatarState` AvatarState

`foods` List<Guid>

`runeStates` AllRuneState

`runeSlotState` RuneSlotState

`floorRow` AdventureBossFloorSheet.Row

`floorWaveRow` AdventureBossFloorWaveSheet.Row

`simulatorSheets` SimulatorSheets

`enemySkillSheet` EnemySkillSheet

`costumeStatSheet` CostumeStatSheet

`waveRewards` List<ItemBase>

`collectionModifiers` List<StatModifier>

`buffLimitSheet` BuffLimitSheet

`buffLinkSheet` BuffLinkSheet

`logEvent` bool

`shatterStrikeMaxDamage` long

Properties

BossId

```
public int BossId { get; }
```

Property Value

int

EnemySkillSheet

```
public EnemySkillSheet EnemySkillSheet { get; }
```

Property Value

EnemySkillSheet

FloorId

```
public int FloorId { get; }
```

Property Value

int

ItemMap

```
public CollectionMap ItemMap { get; }
```

Property Value

CollectionMap

Reward

```
public override IEnumerable<ItemBase> Reward { get; }
```

Property Value

IEnumerable<ItemBase>

StageId

```
public int StageId { get; }
```

Property Value

int

Methods

AddBreakthrough(List<int>, AdventureBossFloorWaveSheet)

```
public void AddBreakthrough(List<int> floorIdList, AdventureBossFloorWaveSheet
adventureBossFloorWaveSheet)
```

Parameters

`floorIdList` List<int>

`adventureBossFloorWaveSheet` AdventureBossFloorWaveSheet

GetWaveRewards(IRandom, Row, MaterialItemSheet, int)

```
public static List<ItemBase> GetWaveRewards(IRandom random,
AdventureBossFloorSheet.Row floorRow, MaterialItemSheet materialItemSheet, int
playCount = 1)
```

Parameters

`random` IRandom

`floorRow` AdventureBossFloorSheet.Row

`materialItemSheet` MaterialItemSheet

`playCount` int

Returns

List<ItemBase>

SetItemSelector(Row, IRandom)

```
public static WeightedSelector<AdventureBossFloorSheet.RewardData>
SetItemSelector(AdventureBossFloorSheet.Row floorRow, IRandom random)
```

Parameters

`floorRow` AdventureBossFloorSheet.Row

`random` IRandom

Returns

WeightedSelector<AdventureBossFloorSheet.RewardData>

SetReward(WeightedSelector<RewardData>, int, IRandom, MaterialItemSheet)

```
public static List<ItemBase>
SetReward(WeightedSelector<AdventureBossFloorSheet.RewardData> itemSelector, int
maxCount, IRandom random, MaterialItemSheet materialItemSheet)
```

Parameters

`itemSelector` WeightedSelector<AdventureBossFloorSheet.RewardData>

`maxCount` int

`random` IRandom

`materialItemSheet` MaterialItemSheet

Returns

List<ItemBase>

Simulate()

```
public Player Simulate()
```

Returns

Player

Namespace Nekoyume.Data

Classes

AdventureBossGameData

Structs

AdventureBossGameData.ClaimableReward

Class AdventureBossGameData

Namespace: Nekoyume.Data

Assembly: Lib9c.dll

```
public static class AdventureBossGameData
```

Inheritance

object ← AdventureBossGameData

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Fields

AdventureBossOperationalAddress

```
public static readonly Address AdventureBossOperationalAddress
```

Field Value

Address

Struct AdventureBossGameData.ClaimableReward

Namespace: Nekoyume.Data

Assembly: Lib9c.dll

```
public struct AdventureBossGameData.ClaimableReward
```

Inherited Members

ValueType.Equals(object) , ValueType.GetHashCode() , ValueType.ToString() , object.Equals(object, object) , object.GetType() , object.ReferenceEquals(object, object)

Fields

FavReward

```
public Dictionary<int, int> FavReward
```

Field Value

Dictionary<int, int>

ItemReward

```
public Dictionary<int, int> ItemReward
```

Field Value

Dictionary<int, int>

NcgReward

```
public FungibleAssetValue? NcgReward
```

Field Value

FungibleAssetValue?

Methods

IsEmpty()

```
public bool IsEmpty()
```

Returns

bool

Namespace Nekoyume.Delegation

Classes

Bond

CurrencyComparer
Represents a comparer for Libplanet.Types.Assets.Currency.

DelegateeMetadata

Delegatee<TRepository, TDelegatee, TDelegator>

DelegationAddress

DelegationRepository<TRepository, TDelegatee, TDelegator>

DelegatorMetadata

Delegator<TRepository, TDelegatee, TDelegator>

LumpSumRewardsRecord

RebondGrace

RewardBase
RewardBase is a class that represents the base of the reward. If it's multiplied by the number of shares, it will be the reward for the period. Also, it holds the significant figure to calculate the reward.

UnbondLockIn

UnbondingEntry

UnbondingEntry.Comparer

UnbondingFactory

UnbondingRef

Interfaces

IDelegatee

IDelegateeMetadata

IDelegationRepository

IDelegator

IDelegatorMetadata

IUnbonding

Enums

UnbondingType

Class Bond

Namespace: Nekoyume.Delegation

Assembly: Lib9c.dll

```
public sealed class Bond : IBencodable, IEquatable<Bond>
```

Inheritance

object ← Bond

Implements

IBencodable, IEquatable<Bond>

Inherited Members

object.Equals(object, object) , object.GetType() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

Bond(Address)

```
public Bond(Address address)
```

Parameters

`address` Address

Bond(Address, IValue)

```
public Bond(Address address, IValue bencoded)
```

Parameters

`address` Address

`bencoded` IValue

Bond(Address, List)

```
public Bond(Address address, List bencoded)
```

Parameters

`address` Address

`bencoded` List

Properties

Address

```
public Address Address { get; }
```

Property Value

Address

Bencoded

```
public List Bencoded { get; }
```

Property Value

List

IsEmpty

```
public bool IsEmpty { get; }
```

Property Value

bool

LastDistributeHeight

```
public long? LastDistributeHeight { get; }
```

Property Value

long?

Share

```
public BigInteger Share { get; }
```

Property Value

BigInteger

Methods

Equals(Bond?)

Indicates whether the current object is equal to another object of the same type.

```
public bool Equals(Bond? other)
```

Parameters

`other` Bond

An object to compare with this object.

Returns

bool

true if the current object is equal to the `other` parameter; otherwise, false.

Equals(object?)

Determines whether the specified object is equal to the current object.

```
public override bool Equals(object? obj)
```

Parameters

`obj` object

The object to compare with the current object.

Returns

bool

true if the specified object is equal to the current object; otherwise, false.

GetHashCode()

Serves as the default hash function.

```
public override int GetHashCode()
```

Returns

int

A hash code for the current object.

Class CurrencyComparer

Namespace: Nekoyume.Delegation

Assembly: Lib9c.dll

Represents a comparer for Libplanet.Types.Assets.Currency.

```
public abstract class CurrencyComparer : IComparer<Currency>
```

Inheritance

object ← CurrencyComparer

Implements

IComparer<Currency>

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Properties

HashBytes

Gets a CurrencyComparer that compares Libplanet.Types.Assets.Currency by their hash byte representation.

```
public static CurrencyComparer HashBytes { get; }
```

Property Value

CurrencyComparer

Methods

Compare(Currency, Currency)

Compares two objects and returns a value indicating whether one is less than, equal to, or greater than the other.

```
public abstract int Compare(Currency x, Currency y)
```

Parameters

x Currency

The first object to compare.

y Currency

The second object to compare.

Returns

int

A signed integer that indicates the relative values of x and y, as shown in the following table.

Value	Meaning
Less than zero	x is less than y.
Zero	x equals y.
Greater than zero	x is greater than y.

Class DelegateeMetadata

Namespace: Nekoyume.Delegation

Assembly: Lib9c.dll

```
public class DelegateeMetadata : IDelegateeMetadata, IBencodable
```

Inheritance

object ← DelegateeMetadata

Implements

IDelegateeMetadata, IBencodable

Inherited Members

object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

DelegateeMetadata(Address, Address, IValue)

```
public DelegateeMetadata(Address delegateeAddress, Address delegateeAccountAddress,
IValue bencoded)
```

Parameters

`delegateeAddress` Address

`delegateeAccountAddress` Address

`bencoded` IValue

DelegateeMetadata(Address, Address, List)

```
public DelegateeMetadata(Address delegateeAddress, Address delegateeAccountAddress,
List bencoded)
```

Parameters

`delegateeAddress` Address

`delegateeAccountAddress` Address

`bencoded` List

DelegateeMetadata(Address, Address, Currency, IEnumerable<Currency>, Address, Address, Address, Address, long, int, int)

Create a new instance of DelegateeMetadata.

```
public DelegateeMetadata(Address delegateeAddress, Address delegateeAccountAddress,
Currency delegationCurrency, IEnumerable<Currency> rewardCurrencies, Address
delegationPoolAddress, Address rewardPoolAddress, Address
rewardRemainderPoolAddress, Address slashedPoolAddress, long unbondingPeriod, int
maxUnbondLockInEntries, int maxRebondGraceEntries)
```

Parameters

`delegateeAddress` Address

The Address of the Delegatee<T, TSelf>.

`delegateeAccountAddress` Address

The Address of the account of the Delegatee<T, TSelf>.

`delegationCurrency` Currency

The Libplanet.Types.Assets.Currency used for delegation.

`rewardCurrencies` IEnumerable<Currency>

The enumerable of Libplanet.Types.Assets.Currencys used for reward.

`delegationPoolAddress` Address

The Address of the delegation pool that stores delegated Libplanet.Types.Assets.FungibleAssetValues.

`rewardPoolAddress` Address

The Address of the reward pool that gathers rewards to be distributed.

`rewardRemainderPoolAddress` Address

The Address of the reward remainder pool to sends the remainder of the rewards to.

`slashedPoolAddress` Address

The Address of the pool that sends the slashed Libplanet.Types.Assets.FungibleAsset Values to.

`unbondingPeriod` long

The period in blocks that the unbonded Libplanet.Types.Assets.FungibleAssetValues can be withdrawn.

`maxUnbondLockInEntries` int

The maximum number of entries that can be locked in for unbonding.

`maxRebondGraceEntries` int

The maximum number of entries that can be locked in for rebonding.

Properties

Address

```
public Address Address { get; }
```

Property Value

Address

Bencoded

```
public List Bencoded { get; }
```

Property Value

List

DelegateeAccountAddress

```
public Address DelegateeAccountAddress { get; }
```

Property Value

Address

DelegateeAddress

```
public Address DelegateeAddress { get; }
```

Property Value

Address

DelegationCurrency

```
public Currency DelegationCurrency { get; }
```

Property Value

Currency

DelegationPoolAddress

```
public Address DelegationPoolAddress { get; }
```

Property Value

Address

Jailed

```
public bool Jailed { get; }
```

Property Value

bool

JailedUntil

```
public long JailedUntil { get; }
```

Property Value

long

MaxRebondGraceEntries

```
public int MaxRebondGraceEntries { get; }
```

Property Value

int

MaxUnbondLockInEntries

```
public int MaxUnbondLockInEntries { get; }
```

Property Value

int

RewardCurrencies

```
public ImmutableSortedSet<Currency> RewardCurrencies { get; }
```

Property Value

ImmutableSortedSet<Currency>

RewardPoolAddress

```
public Address RewardPoolAddress { get; }
```

Property Value

Address

RewardRemainderPoolAddress

```
public Address RewardRemainderPoolAddress { get; }
```

Property Value

Address

SlashedPoolAddress

```
public Address SlashedPoolAddress { get; }
```

Property Value

Address

Tombstoned

```
public bool Tombstoned { get; }
```

Property Value

bool

TotalDelegatedFAV

```
public FungibleAssetValue TotalDelegatedFAV { get; }
```

Property Value

FungibleAssetValue

TotalShares

```
public BigInteger TotalShares { get; }
```

Property Value

BigInteger

UnbondingPeriod

```
public long UnbondingPeriod { get; }
```

Property Value

long

UnbondingRefs

```
public ImmutableSortedSet<UnbondingRef> UnbondingRefs { get; }
```

Property Value

ImmutableSortedSet<UnbondingRef>

Methods

AddDelegatedFAV(FungibleAssetValue)

```
public void AddDelegatedFAV(FungibleAssetValue fav)
```

Parameters

`fav` FungibleAssetValue

AddShare(BigInteger)

```
public void AddShare(BigInteger share)
```

Parameters

`share` BigInteger

AddUnbondingRef(UnbondingRef)

```
public void AddUnbondingRef(UnbondingRef unbondingRef)
```

Parameters

`unbondingRef` UnbondingRef

BondAddress(Address)

```
public Address BondAddress(Address delegatorAddress)
```

Parameters

`delegatorAddress` Address

Returns

Address

CurrentLumpSumRewardsRecordAddress()

Get the Address of the current lump sum rewards record. This will be removed after the migration is done.

```
public virtual Address CurrentLumpSumRewardsRecordAddress()
```

Returns

Address

Address of the current lump sum rewards record.

CurrentRewardBaseAddress()

Get the Address of the current RewardBase.

```
public virtual Address CurrentRewardBaseAddress()
```

Returns

Address

Address of the current RewardBase.

DistributionPoolAddress()

Get the Address of the distribution pool where the rewards are distributed from.

```
public virtual Address DistributionPoolAddress()
```

Returns

Address

Address of the distribution pool.

Equals(IDelegateeMetadata?)

Check if the given IDelegateeMetadata is equal to this instance.

```
public virtual bool Equals(IDelegateeMetadata? other)
```

Parameters

`other` IDelegateeMetadata

The IDelegateeMetadata to compare with this instance.

Returns

bool

true if the given IDelegateeMetadata is equal to this instance;

Equals(object?)

Determines whether the specified object is equal to the current object.

```
public override bool Equals(object? obj)
```

Parameters

`obj` object

The object to compare with the current object.

Returns

bool

true if the specified object is equal to the current object; otherwise, false.

FAVFromShare(BigInteger)

```
public FungibleAssetValue FAVFromShare(BigInteger share)
```

Parameters

`share` BigInteger

Returns

FungibleAssetValue

GetHashCode()

Serves as the default hash function.

```
public override int GetHashCode()
```

Returns

int

A hash code for the current object.

LumpSumRewardsRecordAddress(long)

Get the Address of the lump sum rewards record at the given height. This will be removed after the migration is done.

```
public virtual Address LumpSumRewardsRecordAddress(long height)
```

Parameters

`height` long

The height of the lump sum rewards record.

Returns

Address

Address of the lump sum rewards record at the given height.

RebondGraceAddress(Address)

```
public virtual Address RebondGraceAddress(Address delegatorAddress)
```

Parameters

`delegatorAddress` Address

Returns

Address

RemoveDelegatedFAV(FungibleAssetValue)

```
public void RemoveDelegatedFAV(FungibleAssetValue fav)
```

Parameters

`fav` FungibleAssetValue

RemoveShare(BigInteger)

```
public void RemoveShare(BigInteger share)
```

Parameters

`share` BigInteger

RemoveUnbondingRef(UnbondingRef)

```
public void RemoveUnbondingRef(UnbondingRef unbondingRef)
```

Parameters

`unbondingRef` UnbondingRef

RewardBaseAddress(long)

Get the Address of the RewardBase at the given height.

```
public virtual Address RewardBaseAddress(long height)
```

Parameters

`height` long

Returns

Address

Address of the RewardBase at the given height.

ShareFromFAV(FungibleAssetValue)

```
public BigInteger ShareFromFAV(FungibleAssetValue fav)
```

Parameters

`fav` FungibleAssetValue

Returns

BigInteger

UnbondLockInAddress(Address)

```
public Address UnbondLockInAddress(Address delegatorAddress)
```

Parameters

`delegatorAddress` Address

Returns

Address

UpdateUnbondingPeriod(long)

Update the unbonding period.

```
public void UpdateUnbondingPeriod(long unbondingPeriod)
```

Parameters

`unbondingPeriod` long

The new unbonding period.

Class Delegatee<TRepository, TDelegatee, TDelegator>

Namespace: Nekoyume.Delegation

Assembly: Lib9c.dll

```
public abstract class Delegatee<TRepository, TDelegatee, TDelegator> : IDelegatee
where TRepository : DelegationRepository<TRepository, TDelegatee, TDelegator> where
TDelegatee : Delegatee<TRepository, TDelegatee, TDelegator> where TDelegator :
Delegator<TRepository, TDelegatee, TDelegator>
```

Type Parameters

`TRepository`

`TDelegatee`

`TDelegator`

Inheritance

object ← Delegatee<TRepository, TDelegatee, TDelegator>

Implements

IDelegatee

Derived

GuildDelegatee, ValidatorDelegatee

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

Delegatee(Address, Address, Currency, IEnumerable<Currency>, Address, Address, Address, Address, long, int, int, TRepository)

```
public Delegatee(Address address, Address accountAddress, Currency
delegationCurrency, IEnumerable<Currency> rewardCurrencies, Address
delegationPoolAddress, Address rewardPoolAddress, Address
rewardRemainderPoolAddress, Address slashedPoolAddress, long unbondingPeriod, int
maxUnbondLockInEntries, int maxRebondGraceEntries, TRepository repository)
```

Parameters

`address` Address

`accountAddress` Address

`delegationCurrency` Currency

`rewardCurrencies` IEnumerable<Currency>

`delegationPoolAddress` Address

`rewardPoolAddress` Address

`rewardRemainderPoolAddress` Address

`slashedPoolAddress` Address

`unbondingPeriod` long

`maxUnbondLockInEntries` int

`maxRebondGraceEntries` int

`repository` TRepository

Delegatee(Address, TRepository)

```
public Delegatee(Address address, TRepository repository)
```

Parameters

`address` Address

`repository` TRepository

Properties

AccountAddress

```
public Address AccountAddress { get; }
```

Property Value

Address

Address

```
public Address Address { get; }
```

Property Value

Address

DelegationCurrency

```
public Currency DelegationCurrency { get; }
```

Property Value

Currency

DelegationPoolAddress

```
public Address DelegationPoolAddress { get; }
```

Property Value

Address

Jailed

```
public bool Jailed { get; }
```

Property Value

bool

JailedUntil

```
public long JailedUntil { get; }
```

Property Value

long

MaxRebondGraceEntries

```
public int MaxRebondGraceEntries { get; }
```

Property Value

int

MaxUnbondLockInEntries

```
public int MaxUnbondLockInEntries { get; }
```

Property Value

int

Metadata

```
public DelegateeMetadata Metadata { get; }
```

Property Value

DelegateeMetadata

MetadataAddress

```
public Address MetadataAddress { get; }
```

Property Value

Address

MetadataBencoded

```
public List MetadataBencoded { get; }
```

Property Value

List

Repository

```
public TRepository Repository { get; }
```

Property Value

TRepository

RewardCurrencies

```
public ImmutableSortedSet<Currency> RewardCurrencies { get; }
```

Property Value

ImmutableSortedSet<Currency>

RewardPoolAddress

```
public Address RewardPoolAddress { get; }
```

Property Value

Address

RewardRemainderPoolAddress

```
public Address RewardRemainderPoolAddress { get; }
```

Property Value

Address

SlashedPoolAddress

```
public Address SlashedPoolAddress { get; }
```

Property Value

Address

Tombstoned

```
public bool Tombstoned { get; }
```

Property Value

bool

TotalDelegated

```
public FungibleAssetValue TotalDelegated { get; }
```

Property Value

FungibleAssetValue

TotalShares

```
public BigInteger TotalShares { get; }
```

Property Value

BigInteger

UnbondingPeriod

```
public long UnbondingPeriod { get; }
```

Property Value

long

Methods

Bond(TDelegator, FungibleAssetValue, long)

```
public virtual BigInteger Bond(TDelegator delegator, FungibleAssetValue fav,
long height)
```

Parameters

`delegator` TDelegator

`fav` FungibleAssetValue

`height` long

Returns

BigInteger

BondAddress(Address)

```
public Address BondAddress(Address delegatorAddress)
```

Parameters

`delegatorAddress` Address

Returns

Address

CalculateRewards(BigInteger, RewardBase, RewardBase?)

```
public IEnumerable<FungibleAssetValue> CalculateRewards(BigInteger share, RewardBase
currentRewardBase, RewardBase? lastRewardBase)
```

Parameters

`share` BigInteger

`currentRewardBase` RewardBase

`lastRewardBase` RewardBase

Returns

IEnumerable<FungibleAssetValue>

CollectRewards(long)

```
public void CollectRewards(long height)
```

Parameters

`height` long

CurrentLumpSumRewardsRecordAddress()

```
public Address CurrentLumpSumRewardsRecordAddress()
```

Returns

Address

CurrentRewardBaseAddress()

Get the Address of the current RewardBase.

```
public Address CurrentRewardBaseAddress()
```

Returns

Address

Address of the current RewardBase.

DistributeReward(TDelegator, long)

```
public void DistributeReward(TDelegator delegator, long height)
```

Parameters

`delegator` TDelegator

`height` long

DistributionPoolAddress()

Get the Address of the distribution pool where the rewards are distributed from.

```
public Address DistributionPoolAddress()
```

Returns

Address

Address of the distribution pool.

FAVFromShare(BigInteger)

```
public FungibleAssetValue FAVFromShare(BigInteger share)
```

Parameters

`share` BigInteger

Returns

FungibleAssetValue

Jail(long)

```
public void Jail(long releaseHeight)
```

Parameters

`releaseHeight` long

LumpSumRewardsRecordAddress(long)

```
public Address LumpSumRewardsRecordAddress(long height)
```

Parameters

`height` long

Returns

Address

OnDelegationChanged(long)

```
protected virtual void OnDelegationChanged(long height)
```

Parameters

`height` long

OnEnjailed()

```
protected virtual void OnEnjailed()
```

OnUnjailed()

```
protected virtual void OnUnjailed()
```

RebondGraceAddress(Address)

```
public Address RebondGraceAddress(Address delegatorAddress)
```

Parameters

`delegatorAddress` Address

Returns

Address

RewardBaseAddress(long)

Get the Address of the RewardBase at the given height.

```
public Address RewardBaseAddress(long height)
```

Parameters

`height` long

Returns

Address

Address of the RewardBase at the given height.

ShareFromFAV(FungibleAssetValue)

```
public BigInteger ShareFromFAV(FungibleAssetValue fav)
```

Parameters

`fav` FungibleAssetValue

Returns

BigInteger

Slash(BigInteger, long, long)

```
public virtual void Slash(BigInteger slashFactor, long infractionHeight,
long height)
```

Parameters

`slashFactor` BigInteger

`infractionHeight` long

`height` long

StartNewRewardPeriod(long)

Start a new reward period. It generates a new RewardBase and archives the current one.

```
public void StartNewRewardPeriod(long height)
```

Parameters

`height` long

The height of the block where the new reward period starts.

Tombstone()

```
public void Tombstone()
```

Unbond(TDelegator, BigInteger, long)

```
public FungibleAssetValue Unbond(TDelegator delegator, BigInteger share,
long height)
```

Parameters

`delegator` TDelegator

`share` BigInteger

`height` long

Returns

FungibleAssetValue

UnbondLockInAddress(Address)

```
public Address UnbondLockInAddress(Address delegatorAddress)
```

Parameters

`delegatorAddress` Address

Returns

Address

Unjail(long)

```
public void Unjail(long height)
```

Parameters

`height` long

Class DelegationAddress

Namespace: Nekoyume.Delegation

Assembly: Lib9c.dll

```
public static class DelegationAddress
```

Inheritance

object ← DelegationAddress

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Methods

BondAddress(Address, Address)

```
public static Address BondAddress(Address delegateeMetadataAddress,
Address delegatorAddress)
```

Parameters

`delegateeMetadataAddress` Address

`delegatorAddress` Address

Returns

Address

BondAddress(Address, Address, Address)

```
public static Address BondAddress(Address delegateeAddress, Address
delegateeAccountAddress, Address delegatorAddress)
```

Parameters

`delegateeAddress` Address

`delegateeAccountAddress` Address

`delegatorAddress` Address

Returns

Address

CurrentRewardBaseAddress(Address)

Get the Libplanet.Crypto.Address of the current RewardBase.

```
public static Address CurrentRewardBaseAddress(Address delegateeMetadataAddress)
```

Parameters

`delegateeMetadataAddress` Address

Libplanet.Crypto.Address of the DelegateeMetadata.

Returns

Address

Libplanet.Crypto.Address of the current RewardBase.

CurrentRewardBaseAddress(Address, Address)

Get the Libplanet.Crypto.Address of the current RewardBase.

```
public static Address CurrentRewardBaseAddress(Address delegateeAddress,
Address delegateeAccountAddress)
```

Parameters

`delegateeAddress` Address

Libplanet.Crypto.Address of the Delegatee<T, TSelf>.

`delegateeAccountAddress` Address

Libplanet.Crypto.Address of the account of the Delegatee<T, TSelf>.

Returns

Address

Libplanet.Crypto.Address of the current RewardBase.

DelegateeMetadataAddress(Address, Address)

```
public static Address DelegateeMetadataAddress(Address delegateeAddress,
Address delegateeAccountAddress)
```

Parameters

`delegateeAddress` Address

`delegateeAccountAddress` Address

Returns

Address

DelegationPoolAddress(Address)

```
public static Address DelegationPoolAddress(Address delegateeMetadataAddress)
```

Parameters

`delegateeMetadataAddress` Address

Returns

Address

DelegationPoolAddress(Address, Address)

```
public static Address DelegationPoolAddress(Address delegateeAddress,
Address delegateeAccountAddress)
```

Parameters

`delegateeAddress` Address

`delegateeAccountAddress` Address

Returns

Address

DelegatorMetadataAddress(Address, Address)

```
public static Address DelegatorMetadataAddress(Address delegatorAddress,
Address delegatorAccountAddress)
```

Parameters

`delegatorAddress` Address

`delegatorAccountAddress` Address

Returns

Address

DistributionPoolAddress(Address)

Get the Libplanet.Crypto.Address of the distribution pool where the rewards are distributed from.

```
public static Address DistributionPoolAddress(Address delegateeMetadataAddress)
```

Parameters

`delegateeMetadataAddress` Address

Libplanet.Crypto.Address of the DelegateeMetadata.

Returns

Address

Libplanet.Crypto.Address of the distribution pool.

DistributionPoolAddress(Address, Address)

Get the Libplanet.Crypto.Address of the distribution pool where the rewards are distributed from.

```
public static Address DistributionPoolAddress(Address delegateeAddress,
Address delegateeAccountAddress)
```

Parameters

`delegateeAddress` Address

Libplanet.Crypto.Address of the Delegatee<T, TSelf>.

`delegateeAccountAddress` Address

Libplanet.Crypto.Address of the account of the Delegatee<T, TSelf>.

Returns

Address

Libplanet.Crypto.Address of the distribution pool.

RebondGraceAddress(Address, Address)

```
public static Address RebondGraceAddress(Address delegateeMetadataAddress,
Address delegatorAddress)
```

Parameters

`delegateeMetadataAddress` Address

`delegatorAddress` Address

Returns

Address

RebondGraceAddress(Address, Address, Address)

```
public static Address RebondGraceAddress(Address delegateeAddress, Address
delegateeAccountAddress, Address delegatorAddress)
```

Parameters

`delegateeAddress` Address

`delegateeAccountAddress` Address

`delegatorAddress` Address

Returns

Address

RewardBaseAddress(Address, Address, long)

Get the Libplanet.Crypto.Address of the RewardBase at the given height.

```
public static Address RewardBaseAddress(Address delegateeAddress, Address
delegateeAccountAddress, long height)
```

Parameters

`delegateeAddress` Address

Libplanet.Crypto.Address of the Delegatee<T, TSelf>.

`delegateeAccountAddress` Address

Libplanet.Crypto.Address of the account of the Delegatee<T, TSelf>.

`height` long

The height of the RewardBase.

Returns

Address

Libplanet.Crypto.Address of the RewardBase at the given height.

RewardBaseAddress(Address, long)

Get the Libplanet.Crypto.Address of the RewardBase at the given height.

```
public static Address RewardBaseAddress(Address delegateeMetadataAddress,
long height)
```

Parameters

`delegateeMetadataAddress` Address

Libplanet.Crypto.Address of the DelegateeMetadata.

`height` long

The height of the RewardBase.

Returns

Address

Libplanet.Crypto.Address of the RewardBase at the given height.

RewardPoolAddress(Address)

```
public static Address RewardPoolAddress(Address delegateeMetadataAddress)
```

Parameters

`delegateeMetadataAddress` Address

Returns

Address

RewardPoolAddress(Address, Address)

```
public static Address RewardPoolAddress(Address delegateeAddress,
Address delegateeAccountAddress)
```

Parameters

`delegateeAddress` Address

`delegateeAccountAddress` Address

Returns

Address

UnbondLockInAddress(Address, Address)

```
public static Address UnbondLockInAddress(Address delegateeMetadataAddress,
Address delegatorAddress)
```

Parameters

`delegateeMetadataAddress` Address

`delegatorAddress` Address

Returns

Address

UnbondLockInAddress(Address, Address, Address)

```
public static Address UnbondLockInAddress(Address delegateeAddress, Address
delegateeAccountAddress, Address delegatorAddress)
```

Parameters

`delegateeAddress` Address

`delegateeAccountAddress` Address

`delegatorAddress` Address

Returns

Address

Class DelegationRepository<TRepository, TDelegatee, TDelegator>

Namespace: Nekoyume.Delegation

Assembly: Lib9c.dll

```
public abstract class DelegationRepository<TRepository, TDelegatee, TDelegator> :
IDelegationRepository where TRepository : DelegationRepository<TRepository,
TDelegatee, TDelegator> where TDelegatee : Delegatee<TRepository, TDelegatee,
TDelegator> where TDelegator : Delegator<TRepository, TDelegatee, TDelegator>
```

Type Parameters

`TRepository`

`TDelegatee`

`TDelegator`

Inheritance

object ← DelegationRepository<TRepository, TDelegatee, TDelegator>

Implements

IDelegationRepository

Derived

GuildRepository, ValidatorRepository

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

DelegationRepository(IWorld, IActionContext, Address, Address, Address, Address, Address, Address, Address, Address)

```
public DelegationRepository(IWorld world, IActionContext actionContext, Address
delegateeAccountAddress, Address delegatorAccountAddress, Address
delegateeMetadataAccountAddress, Address delegatorMetadataAccountAddress, Address
bondAccountAddress, Address unbondLockInAccountAddress, Address
rebondGraceAccountAddress, Address unbondingSetAccountAddress, Address
rewardBaseAccountAddress, Address lumpSumRewardRecordAccountAddress)
```

Parameters

`world` IWorld

`actionContext` IActionContext

`delegateeAccountAddress` Address

`delegatorAccountAddress` Address

`delegateeMetadataAccountAddress` Address

`delegatorMetadataAccountAddress` Address

`bondAccountAddress` Address

`unbondLockInAccountAddress` Address

`rebondGraceAccountAddress` Address

`unbondingSetAccountAddress` Address

`rewardBaseAccountAddress` Address

`lumpSumRewardRecordAccountAddress` Address

Fields

bondAccount

```
protected IAccount bondAccount
```

Field Value

IAccount

delegateeAccount

```
protected IAccount delegateeAccount
```

Field Value

IAccount

delegateeMetadataAccount

```
protected IAccount delegateeMetadataAccount
```

Field Value

IAccount

delegatorAccount

```
protected IAccount delegatorAccount
```

Field Value

IAccount

delegatorMetadataAccount

```
protected IAccount delegatorMetadataAccount
```

Field Value

IAccount

lumpSumRewardsRecordAccount

```
protected IAccount lumpSumRewardsRecordAccount
```

Field Value

IAccount

previousWorld

```
protected IWorld previousWorld
```

Field Value

IWorld

rebondGraceAccount

```
protected IAccount rebondGraceAccount
```

Field Value

IAccount

rewardBaseAccount

```
protected IAccount rewardBaseAccount
```

Field Value

IAccount

unbondLockInAccount

```
protected IAccount unbondLockInAccount
```

Field Value

IAccount

unbondingSetAccount

```
protected IAccount unbondingSetAccount
```

Field Value

IAccount

Properties

ActionContext

Libplanet.Action.IActionContext of the current action.

```
public IActionContext ActionContext { get; }
```

Property Value

IActionContext

BondAccountAddress

Libplanet.Crypto.Address of the Bond account.

```
public Address BondAccountAddress { get; }
```

Property Value

Address

DelegateeAccountAddress

```
public Address DelegateeAccountAddress { get; }
```

Property Value

Address

DelegateeMetadataAccountAddress

```
public Address DelegateeMetadataAccountAddress { get; }
```

Property Value

Address

DelegatorAccountAddress

```
public Address DelegatorAccountAddress { get; }
```

Property Value

Address

DelegatorMetadataAccountAddress

```
public Address DelegatorMetadataAccountAddress { get; }
```

Property Value

Address

LumpSumRewardsRecordAccountAddress

Libplanet.Crypto.Address of the LumpSumRewardsRecord account.

```
public Address LumpSumRewardsRecordAccountAddress { get; }
```

Property Value

Address

RebondGraceAccountAddress

Libplanet.Crypto.Address of the RebondGrace account

```
public Address RebondGraceAccountAddress { get; }
```

Property Value

Address

RewardBaseAccountAddress

Libplanet.Crypto.Address of the RewardBase account.

```
public Address RewardBaseAccountAddress { get; }
```

Property Value

Address

UnbondLockInAccountAddress

Libplanet.Crypto.Address of the UnbondLockIn account.

```
public Address UnbondLockInAccountAddress { get; }
```

Property Value

Address

UnbondingSetAccountAddress

Libplanet.Crypto.Address of the UnbondingSet account.

```
public Address UnbondingSetAccountAddress { get; }
```

Property Value

Address

World

```
public virtual IWorld World { get; }
```

Property Value

IWorld

Methods

GetBalance(Address, Currency)

```
public FungibleAssetValue GetBalance(Address address, Currency currency)
```

Parameters

`address` Address

`currency` Currency

Returns

GetBond(TDelegatee, Address)

```
public Bond GetBond(TDelegatee delegatee, Address delegatorAddress)
```

Parameters

`delegatee` TDelegatee

`delegatorAddress` Address

Returns

Bond

GetCurrentLumpSumRewardsRecord(TDelegatee)

```
public LumpSumRewardsRecord? GetCurrentLumpSumRewardsRecord(TDelegatee delegatee)
```

Parameters

`delegatee` TDelegatee

Returns

LumpSumRewardsRecord

GetCurrentRewardBase(TDelegatee)

```
public RewardBase? GetCurrentRewardBase(TDelegatee delegatee)
```

Parameters

`delegatee` TDelegatee

Returns

RewardBase

GetDelegatee(Address)

```
public abstract TDelegatee GetDelegatee(Address address)
```

Parameters

`address` Address

Returns

TDelegatee

GetDelegateeMetadata(Address)

```
public DelegateeMetadata GetDelegateeMetadata(Address delegateeAddress)
```

Parameters

`delegateeAddress` Address

Returns

DelegateeMetadata

GetDelegator(Address)

```
public abstract TDelegator GetDelegator(Address address)
```

Parameters

`address` Address

Returns

TDelegator

GetDelegatorMetadata(Address)

```
public DelegatorMetadata GetDelegatorMetadata(Address delegatorAddress)
```

Parameters

`delegatorAddress` Address

Returns

DelegatorMetadata

GetLumpSumRewardsRecord(TDelegatee, long)

```
public LumpSumRewardsRecord? GetLumpSumRewardsRecord(TDelegatee delegatee,
long height)
```

Parameters

`delegatee` TDelegatee

`height` long

Returns

LumpSumRewardsRecord

GetRebondGrace(TDelegatee, Address)

```
public RebondGrace GetRebondGrace(TDelegatee delegatee, Address delegatorAddress)
```

Parameters

`delegatee` TDelegatee

`delegatorAddress` Address

Returns

RebondGrace

GetRewardBase(TDelegatee, long)

```
public RewardBase? GetRewardBase(TDelegatee delegatee, long height)
```

Parameters

`delegatee` TDelegatee

`height` long

Returns

RewardBase

GetUnbondLockIn(TDelegatee, Address)

```
public UnbondLockIn GetUnbondLockIn(TDelegatee delegatee, Address delegatorAddress)
```

Parameters

`delegatee` TDelegatee

`delegatorAddress` Address

Returns

UnbondLockIn

GetUnlimitedRebondGrace(Address)

```
public RebondGrace GetUnlimitedRebondGrace(Address address)
```

Parameters

`address` Address

Returns

RebondGrace

GetUnlimitedUnbondLockIn(Address)

```
public UnbondLockIn GetUnlimitedUnbondLockIn(Address address)
```

Parameters

`address` Address

Returns

UnbondLockIn

RemoveLumpSumRewardsRecord(LumpSumRewardsRecord)

Remove the LumpSumRewardsRecord from the IDelegatee. This is used when the LumpSumRewardsRecord is no longer needed. This can be removed when the migration for RewardBase is done.

```
public void RemoveLumpSumRewardsRecord(LumpSumRewardsRecord lumpSumRewardsRecord)
```

Parameters

`lumpSumRewardsRecord` LumpSumRewardsRecord

The LumpSumRewardsRecord to remove.

SetBond(Bond)

```
public void SetBond(Bond bond)
```

Parameters

`bond` Bond

SetDelegatee(TDelegatee)

```
public abstract void SetDelegatee(TDelegatee delegatee)
```

Parameters

`delegatee` TDelegatee

SetDelegateeMetadata(DelegateeMetadata)

```
public void SetDelegateeMetadata(DelegateeMetadata delegateeMetadata)
```

Parameters

`delegateeMetadata` DelegateeMetadata

SetDelegator(TDelegator)

```
public abstract void SetDelegator(TDelegator delegator)
```

Parameters

`delegator` TDelegator

SetDelegatorMetadata(DelegatorMetadata)

```
public void SetDelegatorMetadata(DelegatorMetadata delegatorMetadata)
```

Parameters

`delegatorMetadata` DelegatorMetadata

SetLumpSumRewardsRecord(LumpSumRewardsRecord)

```
public void SetLumpSumRewardsRecord(LumpSumRewardsRecord lumpSumRewardsRecord)
```

Parameters

`lumpSumRewardsRecord` LumpSumRewardsRecord

SetRebondGrace(RebondGrace)

```
public void SetRebondGrace(RebondGrace rebondGrace)
```

Parameters

`rebondGrace` RebondGrace

SetRewardBase(RewardBase)

Set the RewardBase of the IDelegatee.

```
public void SetRewardBase(RewardBase rewardBase)
```

Parameters

`rewardBase` RewardBase

The RewardBase to set.

SetUnbondLockIn(UnbondLockIn)

```
public void SetUnbondLockIn(UnbondLockIn unbondLockIn)
```

Parameters

`unbondLockIn` UnbondLockIn

TransferAsset(Address, Address, FungibleAssetValue)

```
public void TransferAsset(Address sender, Address recipient,
FungibleAssetValue value)
```

Parameters

`sender` Address

`recipient` Address

`value` FungibleAssetValue

UpdateWorld(IWorld)

```
public virtual void UpdateWorld(IWorld world)
```

Parameters

`world` IWorld

Class DelegatorMetadata

Namespace: Nekoyume.Delegation

Assembly: Lib9c.dll

```
public class DelegatorMetadata : IDelegatorMetadata, IBencodable
```

Inheritance

object ← DelegatorMetadata

Implements

IDelegatorMetadata, IBencodable

Inherited Members

object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

DelegatorMetadata(Address, Address, IValue)

```
public DelegatorMetadata(Address address, Address accountAddress, IValue bencoded)
```

Parameters

`address` Address

`accountAddress` Address

`bencoded` IValue

DelegatorMetadata(Address, Address, List)

```
public DelegatorMetadata(Address delegatorAddress, Address delegatorAccountAddress,
List bencoded)
```

Parameters

`delegatorAddress` Address

`delegatorAccountAddress` Address

`bencoded` List

DelegatorMetadata(Address, Address, Address, Address)

```
public DelegatorMetadata(Address address, Address accountAddress, Address
delegationPoolAddress, Address rewardAddress)
```

Parameters

`address` Address

`accountAddress` Address

`delegationPoolAddress` Address

`rewardAddress` Address

Properties

Address

```
public Address Address { get; }
```

Property Value

Address

Bencoded

```
public List Bencoded { get; }
```

Property Value

List

Delegatees

```
public ImmutableSortedSet<Address> Delegatees { get; }
```

Property Value

ImmutableSortedSet<Address>

DelegationPoolAddress

```
public Address DelegationPoolAddress { get; }
```

Property Value

Address

DelegatorAccountAddress

```
public Address DelegatorAccountAddress { get; }
```

Property Value

Address

DelegatorAddress

```
public Address DelegatorAddress { get; }
```

Property Value

Address

RewardAddress

```
public Address RewardAddress { get; }
```

Property Value

Address

UnbondingRefs

```
public ImmutableSortedSet<UnbondingRef> UnbondingRefs { get; }
```

Property Value

ImmutableSortedSet<UnbondingRef>

Methods

AddDelegatee(Address)

```
public void AddDelegatee(Address delegatee)
```

Parameters

`delegatee` Address

AddUnbondingRef(UnbondingRef)

```
public void AddUnbondingRef(UnbondingRef unbondingRef)
```

Parameters

`unbondingRef` UnbondingRef

Equals(IDelegatorMetadata?)

Check if the given IDelegatorMetadata is equal to this instance.

```
public virtual bool Equals(IDelegatorMetadata? other)
```

Parameters

`other` IDelegatorMetadata

The IDelegatorMetadata to compare with this instance.

Returns

bool

true if the given IDelegatorMetadata is equal to this instance;

Equals(object?)

Determines whether the specified object is equal to the current object.

```
public override bool Equals(object? obj)
```

Parameters

`obj` object

The object to compare with the current object.

Returns

bool

true if the specified object is equal to the current object; otherwise, false.

GetHashCode()

Serves as the default hash function.

```
public override int GetHashCode()
```

Returns

int

A hash code for the current object.

RemoveDelegatee(Address)

```
public void RemoveDelegatee(Address delegatee)
```

Parameters

`delegatee` Address

RemoveUnbondingRef(UnbondingRef)

```
public void RemoveUnbondingRef(UnbondingRef unbondingRef)
```

Parameters

`unbondingRef` UnbondingRef

Class Delegator<TRepository, TDelegatee, TDelegator>

Namespace: Nekoyume.Delegation

Assembly: Lib9c.dll

```
public abstract class Delegator<TRepository, TDelegatee, TDelegator> : IDelegator
where TRepository : DelegationRepository<TRepository, TDelegatee, TDelegator> where
TDelegatee : Delegatee<TRepository, TDelegatee, TDelegator> where TDelegator :
Delegator<TRepository, TDelegatee, TDelegator>
```

Type Parameters

`TRepository`

`TDelegatee`

`TDelegator`

Inheritance

object ← Delegator<TRepository, TDelegatee, TDelegator>

Implements

IDelegator

Derived

GuildDelegator, ValidatorDelegator

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

Delegator(Address, Address, Address, Address, TRepository)

```
public Delegator(Address address, Address accountAddress, Address
delegationPoolAddress, Address rewardAddress, TRepository repository)
```

Parameters

`address` Address

`accountAddress` Address

`delegationPoolAddress` Address

`rewardAddress` Address

`repository` TRepository

Delegator(Address, TRepository)

```
public Delegator(Address address, TRepository repository)
```

Parameters

`address` Address

`repository` TRepository

Properties

AccountAddress

```
public Address AccountAddress { get; }
```

Property Value

Address

Address

```
public Address Address { get; }
```

Property Value

Address

Delegatees

```
public ImmutableSortedSet<Address> Delegatees { get; }
```

Property Value

ImmutableSortedSet<Address>

DelegationPoolAddress

```
public Address DelegationPoolAddress { get; }
```

Property Value

Address

Metadata

```
public DelegatorMetadata Metadata { get; }
```

Property Value

DelegatorMetadata

MetadataAddress

```
public Address MetadataAddress { get; }
```

Property Value

Address

MetadataBencoded

```
public List MetadataBencoded { get; }
```

Property Value

List

Repository

```
public TRepository Repository { get; }
```

Property Value

TRepository

RewardAddress

```
public Address RewardAddress { get; }
```

Property Value

Address

UnbondingRefs

```
public ImmutableSortedSet<UnbondingRef> UnbondingRefs { get; }
```

Property Value

ImmutableSortedSet<UnbondingRef>

Methods

CancelUndelegate(TDelegatee, FungibleAssetValue, long)

```
public void CancelUndelegate(TDelegatee delegatee, FungibleAssetValue fav,
long height)
```

Parameters

`delegatee` TDelegatee

`fav` FungibleAssetValue

`height` long

ClaimReward(TDelegatee, long)

```
public void ClaimReward(TDelegatee delegatee, long height)
```

Parameters

`delegatee` TDelegatee

`height` long

Delegate(TDelegatee, FungibleAssetValue, long)

```
public virtual void Delegate(TDelegatee delegatee, FungibleAssetValue fav,
long height)
```

Parameters

`delegatee` TDelegatee

`fav` FungibleAssetValue

`height` long

OnUnbondingReleased(long, IUnbonding, FungibleAssetValue?)

```
protected virtual void OnUnbondingReleased(long height, IUnbonding
releasedUnbonding, FungibleAssetValue? releasedFAV)
```

Parameters

`height` long

`releasedUnbonding` IUnbonding

`releasedFAV` FungibleAssetValue?

Redelegate(TDelegatee, TDelegatee, BigInteger, long)

```
public virtual void Redelegate(TDelegatee srcDelegatee, TDelegatee dstDelegatee,
BigInteger share, long height)
```

Parameters

`srcDelegatee` TDelegatee

`dstDelegatee` TDelegatee

`share` BigInteger

`height` long

ReleaseUnbondings(long)

```
public void ReleaseUnbondings(long height)
```

Parameters

`height` long

Undelegate(TDelegatee, BigInteger, long)

```
public virtual void Undelegate(TDelegatee delegatee, BigInteger share, long height)
```

Parameters

`delegatee` TDelegatee

`share` BigInteger

`height` long

Interface IDelegatee

Namespace: Nekoyume.Delegation

Assembly: Lib9c.dll

```
public interface IDelegatee
```

Properties

AccountAddress

```
Address AccountAddress { get; }
```

Property Value

Address

Address

```
Address Address { get; }
```

Property Value

Address

DelegationCurrency

```
Currency DelegationCurrency { get; }
```

Property Value

Currency

DelegationPoolAddress

```
Address DelegationPoolAddress { get; }
```

Property Value

Address

Jailed

```
bool Jailed { get; }
```

Property Value

bool

JailedUntil

```
long JailedUntil { get; }
```

Property Value

long

MaxRebondGraceEntries

```
int MaxRebondGraceEntries { get; }
```

Property Value

int

MaxUnbondLockInEntries

```
int MaxUnbondLockInEntries { get; }
```

Property Value

int

RewardCurrencies

```
ImmutableSortedSet<Currency> RewardCurrencies { get; }
```

Property Value

ImmutableSortedSet<Currency>

RewardPoolAddress

```
Address RewardPoolAddress { get; }
```

Property Value

Address

RewardRemainderPoolAddress

```
Address RewardRemainderPoolAddress { get; }
```

Property Value

Address

Tombstoned

```
bool Tombstoned { get; }
```

Property Value

bool

TotalDelegated

```
FungibleAssetValue TotalDelegated { get; }
```

Property Value

FungibleAssetValue

TotalShares

```
BigInteger TotalShares { get; }
```

Property Value

BigInteger

UnbondingPeriod

```
long UnbondingPeriod { get; }
```

Property Value

long

Methods

Bond(IDelegator, FungibleAssetValue, long)

```
BigInteger Bond(IDelegator delegator, FungibleAssetValue fav, long height)
```

Parameters

`delegator` IDelegator

`fav` FungibleAssetValue

`height` long

Returns

BigInteger

BondAddress(Address)

```
Address BondAddress(Address delegatorAddress)
```

Parameters

`delegatorAddress` Address

Returns

Address

CollectRewards(long)

```
void CollectRewards(long height)
```

Parameters

`height` long

CurrentLumpSumRewardsRecordAddress()

```
Address CurrentLumpSumRewardsRecordAddress()
```

Returns

Address

CurrentRewardBaseAddress()

Get the Address of the current RewardBase.

```
Address CurrentRewardBaseAddress()
```

Returns

Address

The Address of the current RewardBase.

DistributeReward(IDelegator, long)

```
void DistributeReward(IDelegator delegator, long height)
```

Parameters

`delegator` IDelegator

`height` long

FAVFromShare(BigInteger)

```
FungibleAssetValue FAVFromShare(BigInteger share)
```

Parameters

`share` BigInteger

Returns

FungibleAssetValue

Jail(long)

```
void Jail(long releaseHeight)
```

Parameters

`releaseHeight` long

LumpSumRewardsRecordAddress(long)

```
Address LumpSumRewardsRecordAddress(long height)
```

Parameters

`height` long

Returns

Address

RebondGraceAddress(Address)

```
Address RebondGraceAddress(Address delegatorAddress)
```

Parameters

`delegatorAddress` Address

Returns

Address

RewardBaseAddress(long)

Get the Address of the RewardBase at the given height.

```
Address RewardBaseAddress(long height)
```

Parameters

`height` long

The height of the RewardBase.

Returns

Address

The Address of the RewardBase at the given height.

ShareFromFAV(FungibleAssetValue)

```
BigInteger ShareFromFAV(FungibleAssetValue fav)
```

Parameters

`fav` FungibleAssetValue

Returns

BigInteger

Slash(BigInteger, long, long)

```
void Slash(BigInteger slashFactor, long infractionHeight, long height)
```

Parameters

`slashFactor` BigInteger

`infractionHeight` long

`height` long

Tombstone()

```
void Tombstone()
```

Unbond(IDelegator, BigInteger, long)

```
FungibleAssetValue Unbond(IDelegator delegator, BigInteger share, long height)
```

Parameters

`delegator` IDelegator

`share` BigInteger

`height` long

Returns

FungibleAssetValue

UnbondLockInAddress(Address)

```
Address UnbondLockInAddress(Address delegatorAddress)
```

Parameters

`delegatorAddress` Address

Returns

Address

Unjail(long)

```
void Unjail(long height)
```

Parameters

`height` long

Interface IDelegateeMetadata

Namespace: Nekoyume.Delegation

Assembly: Lib9c.dll

```
public interface IDelegateeMetadata : IBencodable
```

Inherited Members
IBencodable.Bencoded

Properties

Address

```
Address Address { get; }
```

Property Value

Address

DelegateeAccountAddress

```
Address DelegateeAccountAddress { get; }
```

Property Value

Address

DelegateeAddress

```
Address DelegateeAddress { get; }
```

Property Value

Address

DelegationCurrency

```
Currency DelegationCurrency { get; }
```

Property Value

Currency

DelegationPoolAddress

```
Address DelegationPoolAddress { get; }
```

Property Value

Address

Jailed

```
bool Jailed { get; }
```

Property Value

bool

JailedUntil

```
long JailedUntil { get; }
```

Property Value

long

MaxRebondGraceEntries

```
int MaxRebondGraceEntries { get; }
```

Property Value

int

MaxUnbondLockInEntries

```
int MaxUnbondLockInEntries { get; }
```

Property Value

int

RewardCurrencies

```
ImmutableSortedSet<Currency> RewardCurrencies { get; }
```

Property Value

ImmutableSortedSet<Currency>

RewardPoolAddress

```
Address RewardPoolAddress { get; }
```

Property Value

Address

RewardRemainderPoolAddress

```
Address RewardRemainderPoolAddress { get; }
```

Property Value

Address

Tombstoned

```
bool Tombstoned { get; }
```

Property Value

bool

TotalDelegatedFAV

```
FungibleAssetValue TotalDelegatedFAV { get; }
```

Property Value

FungibleAssetValue

TotalShares

```
BigInteger TotalShares { get; }
```

Property Value

BigInteger

UnbondingPeriod

```
long UnbondingPeriod { get; }
```

Property Value

long

Methods

FAVFromShare(BigInteger)

```
FungibleAssetValue FAVFromShare(BigInteger share)
```

Parameters

`share` BigInteger

Returns

FungibleAssetValue

ShareFromFAV(FungibleAssetValue)

```
BigInteger ShareFromFAV(FungibleAssetValue fav)
```

Parameters

`fav` FungibleAssetValue

Returns

BigInteger

Interface IDelegationRepository

Namespace: Nekoyume.Delegation

Assembly: Lib9c.dll

```
public interface IDelegationRepository
```

Properties

ActionContext

```
IActionContext ActionContext { get; }
```

Property Value

IActionContext

DelegateeAccountAddress

```
Address DelegateeAccountAddress { get; }
```

Property Value

Address

DelegatorAccountAddress

```
Address DelegatorAccountAddress { get; }
```

Property Value

Address

World

```
IWorld World { get; }
```

Property Value

IWorld

Methods

GetBalance(Address, Currency)

```
FungibleAssetValue GetBalance(Address address, Currency currency)
```

Parameters

`address` Address

`currency` Currency

Returns

FungibleAssetValue

GetBond(IDelegatee, Address)

```
Bond GetBond(IDelegatee delegatee, Address delegatorAddress)
```

Parameters

`delegatee` IDelegatee

`delegatorAddress` Address

Returns

Bond

GetCurrentLumpSumRewardsRecord(IDelegatee)

```
LumpSumRewardsRecord? GetCurrentLumpSumRewardsRecord(IDelegatee delegatee)
```

Parameters

`delegatee` IDelegatee

Returns

LumpSumRewardsRecord

GetCurrentRewardBase(IDelegatee)

Get the current RewardBase of the `delegatee`.

```
RewardBase? GetCurrentRewardBase(IDelegatee delegatee)
```

Parameters

`delegatee` IDelegatee

The IDelegatee to get the current RewardBase.

Returns

RewardBase

The current RewardBase of the `delegatee`.

GetDelegatee(Address)

```
IDelegatee GetDelegatee(Address address)
```

Parameters

`address` Address

Returns

IDelegatee

GetDelegateeMetadata(Address)

```
DelegateeMetadata GetDelegateeMetadata(Address delegateeAddress)
```

Parameters

`delegateeAddress` Address

Returns

DelegateeMetadata

GetDelegator(Address)

```
IDelegator GetDelegator(Address address)
```

Parameters

`address` Address

Returns

IDelegator

GetDelegatorMetadata(Address)

```
DelegatorMetadata GetDelegatorMetadata(Address delegatorAddress)
```

Parameters

`delegatorAddress` Address

Returns

DelegatorMetadata

GetLumpSumRewardsRecord(IDelegatee, long)

```
LumpSumRewardsRecord? GetLumpSumRewardsRecord(IDelegatee delegatee, long height)
```

Parameters

`delegatee` IDelegatee

`height` long

Returns

LumpSumRewardsRecord

GetRebondGrace(IDelegatee, Address)

```
RebondGrace GetRebondGrace(IDelegatee delegatee, Address delegatorAddress)
```

Parameters

`delegatee` IDelegatee

`delegatorAddress` Address

Returns

RebondGrace

GetRewardBase(IDelegatee, long)

Get the RewardBase of the `delegatee` at the given `height`.

```
RewardBase? GetRewardBase(IDelegatee delegatee, long height)
```

Parameters

`delegatee` IDelegatee

The IDelegatee to get the RewardBase of.

`height` long

The height to get the RewardBase at.

Returns

RewardBase

The RewardBase of the `delegatee` at the given `height`.

GetUnbondLockIn(IDelegatee, Address)

```
UnbondLockIn GetUnbondLockIn(IDelegatee delegatee, Address delegatorAddress)
```

Parameters

`delegatee` IDelegatee

`delegatorAddress` Address

Returns

UnbondLockIn

GetUnlimitedRebondGrace(Address)

```
RebondGrace GetUnlimitedRebondGrace(Address address)
```

Parameters

`address` Address

Returns

RebondGrace

GetUnlimitedUnbondLockIn(Address)

```
UnbondLockIn GetUnlimitedUnbondLockIn(Address address)
```

Parameters

`address` Address

Returns

UnbondLockIn

RemoveLumpSumRewardsRecord(LumpSumRewardsRecord)

Remove the LumpSumRewardsRecord from the IDelegatee. This is used when the LumpSumRewardsRecord is no longer needed. This can be removed when the migration for RewardBase is done.

```
void RemoveLumpSumRewardsRecord(LumpSumRewardsRecord lumpSumRewardsRecord)
```

Parameters

`lumpSumRewardsRecord` LumpSumRewardsRecord

The LumpSumRewardsRecord to remove.

SetBond(Bond)

```
void SetBond(Bond bond)
```

Parameters

`bond` Bond

SetDelegatee(IDelegatee)

```
void SetDelegatee(IDelegatee delegatee)
```

Parameters

`delegatee` IDelegatee

SetDelegateeMetadata(DelegateeMetadata)

```
void SetDelegateeMetadata(DelegateeMetadata delegateeMetadata)
```

Parameters

`delegateeMetadata` DelegateeMetadata

SetDelegator(IDelegator)

```
void SetDelegator(IDelegator delegator)
```

Parameters

`delegator` IDelegator

SetDelegatorMetadata(DelegatorMetadata)

```
void SetDelegatorMetadata(DelegatorMetadata delegatorMetadata)
```

Parameters

`delegatorMetadata` DelegatorMetadata

SetLumpSumRewardsRecord(LumpSumRewardsRecord)

```
void SetLumpSumRewardsRecord(LumpSumRewardsRecord lumpSumRewardsRecord)
```

Parameters

`lumpSumRewardsRecord` LumpSumRewardsRecord

SetRebondGrace(RebondGrace)

```
void SetRebondGrace(RebondGrace rebondGrace)
```

Parameters

`rebondGrace` RebondGrace

SetRewardBase(RewardBase)

Set the RewardBase of the IDelegatee.

```
void SetRewardBase(RewardBase rewardBase)
```

Parameters

`rewardBase` RewardBase

The RewardBase to set.

SetUnbondLockIn(UnbondLockIn)

```
void SetUnbondLockIn(UnbondLockIn unbondLockIn)
```

Parameters

`unbondLockIn` UnbondLockIn

TransferAsset(Address, Address, FungibleAssetValue)

```
void TransferAsset(Address sender, Address recipient, FungibleAssetValue value)
```

Parameters

`sender` Address

`recipient` Address

`value` FungibleAssetValue

UpdateWorld(IWorld)

```
void UpdateWorld(IWorld world)
```

Parameters

`world` IWorld

Interface IDelegator

Namespace: Nekoyume.Delegation

Assembly: Lib9c.dll

```
public interface IDelegator
```

Properties

AccountAddress

```
Address AccountAddress { get; }
```

Property Value

Address

Address

```
Address Address { get; }
```

Property Value

Address

Delegatees

```
ImmutableSortedSet<Address> Delegatees { get; }
```

Property Value

ImmutableSortedSet<Address>

DelegationPoolAddress

```
Address DelegationPoolAddress { get; }
```

Property Value

Address

RewardAddress

```
Address RewardAddress { get; }
```

Property Value

Address

Methods

CancelUndelegate(IDelegatee, FungibleAssetValue, long)

```
void CancelUndelegate(IDelegatee delegatee, FungibleAssetValue fav, long height)
```

Parameters

`delegatee` IDelegatee

`fav` FungibleAssetValue

`height` long

ClaimReward(IDelegatee, long)

```
void ClaimReward(IDelegatee delegatee, long height)
```

Parameters

`delegatee` IDelegatee

`height` long

Delegate(IDelegatee, FungibleAssetValue, long)

```
void Delegate(IDelegatee delegatee, FungibleAssetValue fav, long height)
```

Parameters

`delegatee` IDelegatee

`fav` FungibleAssetValue

`height` long

Redelegate(IDelegatee, IDelegatee, BigInteger, long)

```
void Redelegate(IDelegatee srcDelegatee, IDelegatee dstDelegatee, BigInteger share,
long height)
```

Parameters

`srcDelegatee` IDelegatee

`dstDelegatee` IDelegatee

`share` BigInteger

`height` long

Undelegate(IDelegatee, BigInteger, long)

```
void Undelegate(IDelegatee delegatee, BigInteger share, long height)
```

Parameters

`delegatee` IDelegatee

`share` BigInteger

`height` long

Interface IDelegatorMetadata

Namespace: Nekoyume.Delegation

Assembly: Lib9c.dll

```
public interface IDelegatorMetadata : IBencodable
```

Inherited Members
IBencodable.Bencoded

Properties

Address

```
Address Address { get; }
```

Property Value

Address

Delegatees

```
ImmutableSortedSet<Address> Delegatees { get; }
```

Property Value

ImmutableSortedSet<Address>

DelegationPoolAddress

```
Address DelegationPoolAddress { get; }
```

Property Value

Address

DelegatorAccountAddress

```
Address DelegatorAccountAddress { get; }
```

Property Value

Address

DelegatorAddress

```
Address DelegatorAddress { get; }
```

Property Value

Address

Methods

AddDelegatee(Address)

```
void AddDelegatee(Address delegatee)
```

Parameters

`delegatee` Address

RemoveDelegatee(Address)

```
void RemoveDelegatee(Address delegatee)
```

Parameters

`delegatee` Address

Interface IUnbonding

Namespace: Nekoyume.Delegation

Assembly: Lib9c.dll

```
public interface IUnbonding
```

Properties

Address

```
Address Address { get; }
```

Property Value

Address

DelegateeAddress

```
Address DelegateeAddress { get; }
```

Property Value

Address

DelegatorAddress

```
Address DelegatorAddress { get; }
```

Property Value

Address

IsEmpty

```
bool IsEmpty { get; }
```

Property Value

bool

IsFull

```
bool IsFull { get; }
```

Property Value

bool

LowestExpireHeight

```
long LowestExpireHeight { get; }
```

Property Value

long

Methods

Release(long, out FungibleAssetValue?)

```
IUnbonding Release(long height, out FungibleAssetValue? releasedFAV)
```

Parameters

`height` long

`releasedFAV` FungibleAssetValue?

Returns

IUnbonding

Slash(BigInteger, long, long, Address)

```
IUnbonding Slash(BigInteger slashFactor, long infractionHeight, long height,
Address slashedPoolAddress)
```

Parameters

`slashFactor` BigInteger

`infractionHeight` long

`height` long

`slashedPoolAddress` Address

Returns

IUnbonding

Class LumpSumRewardsRecord

Namespace: Nekoyume.Delegation

Assembly: Lib9c.dll

```
public class LumpSumRewardsRecord : IBencodable, IEquatable<LumpSumRewardsRecord>
```

Inheritance

object ← LumpSumRewardsRecord

Implements

IBencodable, IEquatable<LumpSumRewardsRecord>

Inherited Members

object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

LumpSumRewardsRecord(Address, IValue)

```
public LumpSumRewardsRecord(Address address, IValue bencoded)
```

Parameters

`address` Address

`bencoded` IValue

LumpSumRewardsRecord(Address, List)

```
public LumpSumRewardsRecord(Address address, List bencoded)
```

Parameters

`address` Address

`bencoded` List

LumpSumRewardsRecord(Address, long, BigInteger, IEnumerable<Currency>)

```
public LumpSumRewardsRecord(Address address, long startHeight, BigInteger
totalShares, IEnumerable<Currency> currencies)
```

Parameters

`address` Address

`startHeight` long

`totalShares` BigInteger

`currencies` IEnumerable<Currency>

LumpSumRewardsRecord(Address, long, BigInteger, IEnumerable<Currency>, long?)

```
public LumpSumRewardsRecord(Address address, long startHeight, BigInteger
totalShares, IEnumerable<Currency> currencies, long? lastStartHeight)
```

Parameters

`address` Address

`startHeight` long

`totalShares` BigInteger

`currencies` IEnumerable<Currency>

`lastStartHeight` long?

LumpSumRewardsRecord(Address, long, BigInteger, IEnumerable<FungibleAssetValue>, long?)

```
public LumpSumRewardsRecord(Address address, long startHeight, BigInteger
totalShares, IEnumerable<FungibleAssetValue> lumpSumRewards, long? lastStartHeight)
```

Parameters

`address` Address

`startHeight` long

`totalShares` BigInteger

`lumpSumRewards` IEnumerable<FungibleAssetValue>

`lastStartHeight` long?

Properties

Address

```
public Address Address { get; }
```

Property Value

Address

Bencoded

```
public List Bencoded { get; }
```

Property Value

List

LastStartHeight

```
public long? LastStartHeight { get; }
```

Property Value

long?

LumpSumRewards

```
public ImmutableDictionary<Currency, FungibleAssetValue> LumpSumRewards { get; }
```

Property Value

ImmutableDictionary<Currency, FungibleAssetValue>

StartHeight

```
public long StartHeight { get; }
```

Property Value

long

TotalShares

```
public BigInteger TotalShares { get; }
```

Property Value

BigInteger

Methods

AddLumpSumRewards(FungibleAssetValue)

```
public LumpSumRewardsRecord AddLumpSumRewards(FungibleAssetValue rewards)
```

Parameters

`rewards` FungibleAssetValue

Returns

LumpSumRewardsRecord

AddLumpSumRewards(LumpSumRewardsRecord, FungibleAssetValue)

```
public static LumpSumRewardsRecord AddLumpSumRewards(LumpSumRewardsRecord record,
FungibleAssetValue rewards)
```

Parameters

`record` LumpSumRewardsRecord

`rewards` FungibleAssetValue

Returns

LumpSumRewardsRecord

AddLumpSumRewards(IEnumerable<FungibleAssetValue>)

```
public LumpSumRewardsRecord AddLumpSumRewards(IEnumerable<FungibleAssetValue>
rewards)
```

Parameters

`rewards` IEnumerable<FungibleAssetValue>

Returns

LumpSumRewardsRecord

Equals(LumpSumRewardsRecord?)

Indicates whether the current object is equal to another object of the same type.

```
public bool Equals(LumpSumRewardsRecord? other)
```

Parameters

`other` LumpSumRewardsRecord

An object to compare with this object.

Returns

bool

true if the current object is equal to the `other` parameter; otherwise, false.

Equals(object?)

Determines whether the specified object is equal to the current object.

```
public override bool Equals(object? obj)
```

Parameters

`obj` object

The object to compare with the current object.

Returns

bool

true if the specified object is equal to the current object; otherwise, false.

GetHashCode()

Serves as the default hash function.

```
public override int GetHashCode()
```

Returns

int

A hash code for the current object.

MoveAddress(Address)

```
public LumpSumRewardsRecord MoveAddress(Address address)
```

Parameters

`address` Address

Returns

LumpSumRewardsRecord

RewardsDuringPeriod(BigInteger)

```
public ImmutableSortedDictionary<Currency, FungibleAssetValue>
RewardsDuringPeriod(BigInteger share)
```

Parameters

`share` BigInteger

Returns

ImmutableSortedDictionary<Currency, FungibleAssetValue>

RewardsDuringPeriod(BigInteger, Currency)

```
public FungibleAssetValue RewardsDuringPeriod(BigInteger share, Currency currency)
```

Parameters

`share` BigInteger

`currency` Currency

Returns

FungibleAssetValue

Class RebondGrace

Namespace: Nekoyume.Delegation

Assembly: Lib9c.dll

```
public sealed class RebondGrace : IUnbonding, IBencodable, IEquatable<RebondGrace>
```

Inheritance

object ← RebondGrace

Implements

IUnbonding, IBencodable, IEquatable<RebondGrace>

Inherited Members

object.Equals(object, object) , object.GetType() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

RebondGrace(Address, int, IValue, IDelegationRepository?)

```
public RebondGrace(Address address, int maxEntries, IValue bencoded,
IDelegationRepository? repository = null)
```

Parameters

`address` Address

`maxEntries` int

`bencoded` IValue

`repository` IDelegationRepository

RebondGrace(Address, int, List, IDelegationRepository?)

```
public RebondGrace(Address address, int maxEntries, List bencoded,
IDelegationRepository? repository = null)
```

Parameters

`address` Address

`maxEntries` int

`bencoded` List

`repository` IDelegationRepository

RebondGrace(Address, int, Address, Address, IDelegationRepository?)

```
public RebondGrace(Address address, int maxEntries, Address delegateeAddress,
Address delegatorAddress, IDelegationRepository? repository = null)
```

Parameters

`address` Address

`maxEntries` int

`delegateeAddress` Address

`delegatorAddress` Address

`repository` IDelegationRepository

RebondGrace(Address, int, Address, Address, IEnumerable<UnbondingEntry>, IDelegationRepository?)

```
public RebondGrace(Address address, int maxEntries, Address delegateeAddress,
Address delegatorAddress, IEnumerable<UnbondingEntry> entries,
IDelegationRepository? repository = null)
```

Parameters

`address` Address

`maxEntries` int

`delegateeAddress` Address

`delegatorAddress` Address

`entries` IEnumerable<UnbondingEntry>

`repository` IDelegationRepository

Properties

Address

```
public Address Address { get; }
```

Property Value

Address

Bencoded

```
public List Bencoded { get; }
```

Property Value

List

DelegateeAddress

```
public Address DelegateeAddress { get; }
```

Property Value

Address

DelegatorAddress

```
public Address DelegatorAddress { get; }
```

Property Value

Address

Entries

```
public ImmutableSortedDictionary<long, ImmutableList<UnbondingEntry>> Entries {
get; }
```

Property Value

ImmutableSortedDictionary<long, ImmutableList<UnbondingEntry>>

FlattenedEntries

```
public ImmutableArray<UnbondingEntry> FlattenedEntries { get; }
```

Property Value

ImmutableArray<UnbondingEntry>

IsEmpty

```
public bool IsEmpty { get; }
```

Property Value

bool

IsFull

```
public bool IsFull { get; }
```

Property Value

bool

LowestExpireHeight

```
public long LowestExpireHeight { get; }
```

Property Value

long

MaxEntries

```
public int MaxEntries { get; }
```

Property Value

int

Repository

```
public IDelegationRepository? Repository { get; }
```

Property Value

IDelegationRepository

Methods

Equals(RebondGrace?)

Indicates whether the current object is equal to another object of the same type.

```
public bool Equals(RebondGrace? other)
```

Parameters

`other` RebondGrace

An object to compare with this object.

Returns

bool

true if the current object is equal to the `other` parameter; otherwise, false.

Equals(object?)

Determines whether the specified object is equal to the current object.

```
public override bool Equals(object? obj)
```

Parameters

`obj` object

The object to compare with the current object.

Returns

bool

true if the specified object is equal to the current object; otherwise, false.

GetHashCode()

Serves as the default hash function.

```
public override int GetHashCode()
```

Returns

int

A hash code for the current object.

Release(long, out FungibleAssetValue?)

```
public RebondGrace Release(long height, out FungibleAssetValue? releasedFAV)
```

Parameters

`height` long

`releasedFAV` FungibleAssetValue?

Returns

RebondGrace

Slash(BigInteger, long, long, Address)

```
public RebondGrace Slash(BigInteger slashFactor, long infractionHeight, long height,
Address slashedPoolAddress)
```

Parameters

`slashFactor` BigInteger

`infractionHeight` long

`height` long

`slashedPoolAddress` Address

Returns

RebondGrace

Class RewardBase

Namespace: Nekoyume.Delegation

Assembly: Lib9c.dll

RewardBase is a class that represents the base of the reward. If it's multiplied by the number of shares, it will be the reward for the period. Also, it holds the significant figure to calculate the reward.

```
public class RewardBase : IBencodable, IEquatable<RewardBase>
```

Inheritance

object ← RewardBase

Implements

IBencodable, IEquatable<RewardBase>

Inherited Members

object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

RewardBase(Address, IValue)

```
public RewardBase(Address address, IValue bencoded)
```

Parameters

`address` Address

`bencoded` IValue

RewardBase(Address, List)

```
public RewardBase(Address address, List bencoded)
```

Parameters

`address` Address

`bencoded` List

RewardBase(Address, BigInteger, IEnumerable<Currency>)

```
public RewardBase(Address address, BigInteger totalShares,
IEnumerable<Currency> currencies)
```

Parameters

`address` Address

`totalShares` BigInteger

`currencies` IEnumerable<Currency>

Fields

Margin

Margin for significant figure. It's used to calculate the significant figure of the reward base.

```
public const int Margin = 2
```

Field Value

int

Properties

Address

```
public Address Address { get; }
```

Property Value

Address

Bencoded

```
public List Bencoded { get; }
```

Property Value

List

RewardPortion

Cumulative reward portion of RewardBase. When it's multiplied by the number of shares, it will be the reward for the period.

```
public ImmutableSortedDictionary<Currency, BigInteger> RewardPortion { get; }
```

Property Value

ImmutableSortedDictionary<Currency, BigInteger>

SigFig

Significant figure of RewardBase.

```
public int SigFig { get; }
```

Property Value

int

StartHeight

Start height of RewardBase that attached when archived.

```
public long? StartHeight { get; }
```

Property Value

long?

Methods

AddReward(FungibleAssetValue, BigInteger)

```
public RewardBase AddReward(FungibleAssetValue reward, BigInteger totalShares)
```

Parameters

`reward` FungibleAssetValue

`totalShares` BigInteger

Returns

RewardBase

AddRewards(IEnumerable<FungibleAssetValue>, BigInteger)

Add rewards to the RewardBase.

```
public RewardBase AddRewards(IEnumerable<FungibleAssetValue> rewards,
BigInteger totalShares)
```

Parameters

`rewards` IEnumerable<FungibleAssetValue>

Rewards to add.

`totalShares` BigInteger

TotalShares used as denominator of the portion.

Returns

RewardBase

New RewardBase with added rewards.

AttachHeight(Address, long)

```
public RewardBase AttachHeight(Address address, long startHeight)
```

Parameters

`address` Address

`startHeight` long

Returns

RewardBase

CumulativeRewardDuringPeriod(BigInteger)

Calculate the cumulative reward during the period.

```
public ImmutableSortedDictionary<Currency, FungibleAssetValue>
CumulativeRewardDuringPeriod(BigInteger share)
```

Parameters

`share` BigInteger

The number of shares to calculate the reward.

Returns

ImmutableSortedDictionary<Currency, FungibleAssetValue>

Cumulative reward during the period.

CumulativeRewardDuringPeriod(BigInteger, Currency)

Calculate the cumulative reward during the period, for the specific currency.

```
public FungibleAssetValue CumulativeRewardDuringPeriod(BigInteger share,
Currency currency)
```

Parameters

`share` BigInteger

The number of shares to calculate the reward.

`currency` Currency

The currency to calculate the reward.

Returns

FungibleAssetValue

Cumulative reward during the period, for the specific currency.

Exceptions

ArgumentException

Thrown when the `currency` is not in the RewardPortion.

Equals(RewardBase?)

Indicates whether the current object is equal to another object of the same type.

```
public bool Equals(RewardBase? other)
```

Parameters

`other` RewardBase

An object to compare with this object.

Returns

bool

true if the current object is equal to the `other` parameter; otherwise, false.

Equals(object?)

Determines whether the specified object is equal to the current object.

```
public override bool Equals(object? obj)
```

Parameters

`obj` object

The object to compare with the current object.

Returns

bool

true if the specified object is equal to the current object; otherwise, false.

GetHashCode()

Serves as the default hash function.

```
public override int GetHashCode()
```

Returns

int

A hash code for the current object.

RecommendedSigFig(BigInteger)

```
public static int RecommendedSigFig(BigInteger totalShares)
```

Parameters

`totalShares` BigInteger

Returns

int

UpdateSigFig(BigInteger)

```
public RewardBase UpdateSigFig(BigInteger totalShares)
```

Parameters

`totalShares` BigInteger

Returns

RewardBase

Class UnbondLockIn

Namespace: Nekoyume.Delegation

Assembly: Lib9c.dll

```
public sealed class UnbondLockIn : IUnbonding, IBencodable, IEquatable<UnbondLockIn>
```

Inheritance

object ← UnbondLockIn

Implements

IUnbonding, IBencodable, IEquatable<UnbondLockIn>

Inherited Members

object.Equals(object, object) , object.GetType() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

UnbondLockIn(Address, int, IValue, IDelegationRepository?)

```
public UnbondLockIn(Address address, int maxEntries, IValue bencoded,
IDelegationRepository? repository = null)
```

Parameters

`address` Address

`maxEntries` int

`bencoded` IValue

`repository` IDelegationRepository

UnbondLockIn(Address, int, List,

IDelegationRepository?)

```
public UnbondLockIn(Address address, int maxEntries, List bencoded,
IDelegationRepository? repository = null)
```

Parameters

`address` Address

`maxEntries` int

`bencoded` List

`repository` IDelegationRepository

UnbondLockIn(Address, int, Address, Address, IDelegationRepository?)

```
public UnbondLockIn(Address address, int maxEntries, Address delegateeAddress,
Address delegatorAddress, IDelegationRepository? repository)
```

Parameters

`address` Address

`maxEntries` int

`delegateeAddress` Address

`delegatorAddress` Address

`repository` IDelegationRepository

UnbondLockIn(Address, int, Address, Address, IEnumerable<UnbondingEntry>, IDelegationRepository?)

```
public UnbondLockIn(Address address, int maxEntries, Address delegateeAddress,
Address delegatorAddress, IEnumerable<UnbondingEntry> entries,
IDelegationRepository? repository = null)
```

Parameters

`address` Address

`maxEntries` int

`delegateeAddress` Address

`delegatorAddress` Address

`entries` IEnumerable<UnbondingEntry>

`repository` IDelegationRepository

Properties

Address

```
public Address Address { get; }
```

Property Value

Address

Bencoded

```
public List Bencoded { get; }
```

Property Value

List

DelegateeAddress

```
public Address DelegateeAddress { get; }
```

Property Value

Address

DelegatorAddress

```
public Address DelegatorAddress { get; }
```

Property Value

Address

Entries

```
public ImmutableSortedDictionary<long, ImmutableList<UnbondingEntry>> Entries {
get; }
```

Property Value

ImmutableSortedDictionary<long, ImmutableList<UnbondingEntry>>

FlattenedEntries

```
public ImmutableArray<UnbondingEntry> FlattenedEntries { get; }
```

Property Value

ImmutableArray<UnbondingEntry>

IsEmpty

```
public bool IsEmpty { get; }
```

Property Value

bool

IsFull

```
public bool IsFull { get; }
```

Property Value

bool

LowestExpireHeight

```
public long LowestExpireHeight { get; }
```

Property Value

long

MaxEntries

```
public int MaxEntries { get; }
```

Property Value

int

Repository

```
public IDelegationRepository? Repository { get; }
```

Property Value

IDelegationRepository

Methods

Equals(UnbondLockIn?)

Indicates whether the current object is equal to another object of the same type.

```
public bool Equals(UnbondLockIn? other)
```

Parameters

`other` UnbondLockIn

An object to compare with this object.

Returns

bool

true if the current object is equal to the `other` parameter; otherwise, false.

Equals(object?)

Determines whether the specified object is equal to the current object.

```
public override bool Equals(object? obj)
```

Parameters

`obj` object

The object to compare with the current object.

Returns

bool

true if the specified object is equal to the current object; otherwise, false.

GetHashCode()

Serves as the default hash function.

```
public override int GetHashCode()
```

Returns

int

A hash code for the current object.

Release(long, out FungibleAssetValue?)

```
public UnbondLockIn Release(long height, out FungibleAssetValue? releasedFAV)
```

Parameters

`height` long

`releasedFAV` FungibleAssetValue?

Returns

UnbondLockIn

Slash(BigInteger, long, long, Address)

```
public UnbondLockIn Slash(BigInteger slashFactor, long infractionHeight, long
height, Address slashedPoolAddress)
```

Parameters

`slashFactor` BigInteger

`infractionHeight` long

`height` long

`slashedPoolAddress` Address

Returns

UnbondLockIn

Class UnbondingEntry

Namespace: Nekoyume.Delegation

Assembly: Lib9c.dll

```
public class UnbondingEntry : IBencodable, IEquatable<UnbondingEntry>
```

Inheritance

object ← UnbondingEntry

Implements

IBencodable, IEquatable<UnbondingEntry>

Inherited Members

object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

UnbondingEntry(IValue)

```
public UnbondingEntry(IValue bencoded)
```

Parameters

`bencoded` IValue

UnbondingEntry(Address, FungibleAssetValue, FungibleAssetValue, long, long)

```
public UnbondingEntry(Address unbondeeAddress, FungibleAssetValue
initialUnbondingFAV, FungibleAssetValue unbondingFAV, long creationHeight,
long expireHeight)
```

Parameters

`unbondeeAddress` Address

`initialUnbondingFAV` FungibleAssetValue

`unbondingFAV` FungibleAssetValue

`creationHeight` long

`expireHeight` long

UnbondingEntry(Address, FungibleAssetValue, long, long)

```
public UnbondingEntry(Address unbondeeAddress, FungibleAssetValue unbondingFAV, long
creationHeight, long expireHeight)
```

Parameters

`unbondeeAddress` Address

`unbondingFAV` FungibleAssetValue

`creationHeight` long

`expireHeight` long

Properties

Bencoded

```
public List Bencoded { get; }
```

Property Value

List

CreationHeight

```
public long CreationHeight { get; }
```

Property Value

long

ExpireHeight

```
public long ExpireHeight { get; }
```

Property Value

long

InitialUnbondingFAV

```
public FungibleAssetValue InitialUnbondingFAV { get; }
```

Property Value

FungibleAssetValue

UnbondeeAddress

```
public Address UnbondeeAddress { get; }
```

Property Value

Address

UnbondingFAV

```
public FungibleAssetValue UnbondingFAV { get; }
```

Property Value

FungibleAssetValue

Methods

Equals(UnbondingEntry?)

Indicates whether the current object is equal to another object of the same type.

```
public bool Equals(UnbondingEntry? other)
```

Parameters

`other` UnbondingEntry

An object to compare with this object.

Returns

bool

true if the current object is equal to the `other` parameter; otherwise, false.

Equals(object?)

Determines whether the specified object is equal to the current object.

```
public override bool Equals(object? obj)
```

Parameters

`obj` object

The object to compare with the current object.

Returns

bool

true if the specified object is equal to the current object; otherwise, false.

GetHashCode()

Serves as the default hash function.

```
public override int GetHashCode()
```

Returns

int

A hash code for the current object.

Slash(BigInteger, long, out FungibleAssetValue)

```
public UnbondingEntry Slash(BigInteger slashFactor, long infractionHeight, out
FungibleAssetValue slashedFAV)
```

Parameters

`slashFactor` BigInteger

`infractionHeight` long

`slashedFAV` FungibleAssetValue

Returns

UnbondingEntry

Class UnbondingEntry.Comparer

Namespace: Nekoyume.Delegation

Assembly: Lib9c.dll

```
public class UnbondingEntry.Comparer : IComparer<UnbondingEntry>
```

Inheritance

object ← UnbondingEntry.Comparer

Implements

IComparer<UnbondingEntry>

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Methods

Compare(UnbondingEntry?, UnbondingEntry?)

Compares two objects and returns a value indicating whether one is less than, equal to, or greater than the other.

```
public int Compare(UnbondingEntry? x, UnbondingEntry? y)
```

Parameters

x UnbondingEntry

The first object to compare.

y UnbondingEntry

The second object to compare.

Returns

int

A signed integer that indicates the relative values of x and y, as shown in the following table.

Value	Meaning
Less than zero	x is less than y.
Zero	x equals y.
Greater than zero	x is greater than y.

Class UnbondingFactory

Namespace: Nekoyume.Delegation

Assembly: Lib9c.dll

```
public static class UnbondingFactory
```

Inheritance

object ← UnbondingFactory

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Methods

GetUnbondingFromRef(IValue, IDelegationRepository)

```
public static IUnbonding GetUnbondingFromRef(IValue bencoded,
IDelegationRepository repository)
```

Parameters

`bencoded` IValue

`repository` IDelegationRepository

Returns

IUnbonding

GetUnbondingFromRef(UnbondingRef, IDelegationRepository)

```
public static IUnbonding GetUnbondingFromRef(UnbondingRef reference,
IDelegationRepository repository)
```

Parameters

`reference` UnbondingRef

`repository` IDelegationRepository

Returns

IUnbonding

ToReference(IUnbonding)

```
public static UnbondingRef ToReference(IUnbonding unbonding)
```

Parameters

`unbonding` IUnbonding

Returns

UnbondingRef

Class UnbondingRef

Namespace: Nekoyume.Delegation

Assembly: Lib9c.dll

```
public class UnbondingRef : IEquatable<UnbondingRef>, IComparable<UnbondingRef>,
IComparable, IBencodable
```

Inheritance

object ← UnbondingRef

Implements

IEquatable<UnbondingRef>, IComparable<UnbondingRef>, IComparable, IBencodable

Inherited Members

object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

UnbondingRef(IValue)

```
public UnbondingRef(IValue bencoded)
```

Parameters

`bencoded` IValue

UnbondingRef(List)

```
public UnbondingRef(List list)
```

Parameters

`list` List

UnbondingRef(Address, UnbondingType)

```
public UnbondingRef(Address address, UnbondingType unbondingType)
```

Parameters

`address` Address

`unbondingType` UnbondingType

Properties

Address

```
public Address Address { get; }
```

Property Value

Address

Bencoded

```
public List Bencoded { get; }
```

Property Value

List

UnbondingType

```
public UnbondingType UnbondingType { get; }
```

Property Value

UnbondingType

Methods

CompareTo(UnbondingRef?)

Compares the current instance with another object of the same type and returns an integer that indicates whether the current instance precedes, follows, or occurs in the same position in the sort order as the other object.

```
public int CompareTo(UnbondingRef? other)
```

Parameters

`other` UnbondingRef

An object to compare with this instance.

Returns

int

A value that indicates the relative order of the objects being compared. The return value has these meanings:

Value	Meaning
Less than zero	This instance precedes `other` in the sort order.
Zero	This instance occurs in the same position in the sort order as `other`.
Greater than zero	This instance follows `other` in the sort order.

CompareTo(object?)

Compares the current instance with another object of the same type and returns an integer that indicates whether the current instance precedes, follows, or occurs in the same position in the sort order as the other object.

```
public int CompareTo(object? obj)
```

Parameters

`obj` object

An object to compare with this instance.

Returns

int

A value that indicates the relative order of the objects being compared. The return value has these meanings:

Value	Meaning
Less than zero	This instance precedes `obj` in the sort order.
Zero	This instance occurs in the same position in the sort order as `obj`.
Greater than zero	This instance follows `obj` in the sort order.

Exceptions

ArgumentException

`obj` is not the same type as this instance.

Equals(UnbondingRef?)

Indicates whether the current object is equal to another object of the same type.

```
public bool Equals(UnbondingRef? other)
```

Parameters

`other` UnbondingRef

An object to compare with this object.

Returns

bool

true if the current object is equal to the `other` parameter; otherwise, false.

Equals(object?)

Determines whether the specified object is equal to the current object.

```
public override bool Equals(object? obj)
```

Parameters

`obj` object

The object to compare with the current object.

Returns

bool

true if the specified object is equal to the current object; otherwise, false.

GetHashCode()

Serves as the default hash function.

```
public override int GetHashCode()
```

Returns

int

A hash code for the current object.

Operators

operator ==(UnbondingRef?, UnbondingRef?)

```
public static bool operator ==(UnbondingRef? left, UnbondingRef? right)
```

Parameters

`left` UnbondingRef

`right` UnbondingRef

Returns

bool

operator >(UnbondingRef, UnbondingRef)

```
public static bool operator >(UnbondingRef left, UnbondingRef right)
```

Parameters

`left` UnbondingRef

`right` UnbondingRef

Returns

bool

operator >=(UnbondingRef, UnbondingRef)

```
public static bool operator >=(UnbondingRef left, UnbondingRef right)
```

Parameters

`left` UnbondingRef

`right` UnbondingRef

Returns

bool

operator !=(UnbondingRef?, UnbondingRef?)

```
public static bool operator !=(UnbondingRef? left, UnbondingRef? right)
```

Parameters

`left` UnbondingRef

`right` UnbondingRef

Returns

bool

operator <(UnbondingRef, UnbondingRef)

```
public static bool operator <(UnbondingRef left, UnbondingRef right)
```

Parameters

`left` UnbondingRef

`right` UnbondingRef

Returns

bool

operator <=(UnbondingRef, UnbondingRef)

```
public static bool operator <=(UnbondingRef left, UnbondingRef right)
```

Parameters

`left` UnbondingRef

`right` UnbondingRef

Returns

bool

Enum UnbondingType

Namespace: Nekoyume.Delegation

Assembly: Lib9c.dll

```
public enum UnbondingType
```

Extension Methods
StateExtensions.Serialize(Enum)

Fields

`RebondGrace = 1`

`UnbondLockIn = 0`

Namespace Nekoyume.Exceptions

Classes

AlreadyFoughtAvatarException

BencodexTypesExtensions

InvalidActionFieldException

InvalidItemIdException

InvalidMaterialCountException

ItemNotFoundException

MedalIdNotFoundException

NotEnoughEventDungeonTicketsException

NotEnoughItemException

PetCostNotFoundException

StateNullException

UnsupportedStateException

Class AlreadyFoughtAvatarException

Namespace: Nekoyume.Exceptions

Assembly: Lib9c.dll

```
[Serializable]
public class AlreadyFoughtAvatarException : Exception, ISerializable
```

Inheritance

object ← Exception ← AlreadyFoughtAvatarException

Implements

ISerializable

Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Constructors

AlreadyFoughtAvatarException()

```
public AlreadyFoughtAvatarException()
```

AlreadyFoughtAvatarException(string)

```
public AlreadyFoughtAvatarException(string message)
```

Parameters

`message` string

Class BencodexTypesExtensions

Namespace: Nekoyume.Exceptions

Assembly: Lib9c.dll

```
public static class BencodexTypesExtensions
```

Inheritance

object ← BencodexTypesExtensions

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Methods

Replace(List, int, IValue)

Returns a new list with the value replaced at the specified index.

```
public static List Replace(this List list, int index, IValue value)
```

Parameters

`list` List

`index` int

`value` IValue

Returns

List

Exceptions

ArgumentOutOfRangeException

Class InvalidActionFieldException

Namespace: Nekoyume.Exceptions

Assembly: Lib9c.dll

```
[Serializable]
public class InvalidActionFieldException : Exception, ISerializable
```

Inheritance

object ← Exception ← InvalidActionFieldException

Implements

ISerializable

Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Constructors

InvalidActionFieldException(SerializationInfo, StreamingContext)

```
protected InvalidActionFieldException(SerializationInfo info,
StreamingContext context)
```

Parameters

`info` SerializationInfo

`context` StreamingContext

InvalidActionFieldException(string?)

```
public InvalidActionFieldException(string? message = null)
```

Parameters

`message` string

InvalidActionFieldException(string?, Exception?)

```
public InvalidActionFieldException(string? message = null, Exception? innerException
= null)
```

Parameters

`message` string

`innerException` Exception

InvalidActionFieldException(string, string, string, string, Exception?)

```
public InvalidActionFieldException(string actionType, string addressesHex, string
fieldName, string message, Exception? innerException = null)
```

Parameters

`actionType` string

`addressesHex` string

`fieldName` string

`message` string

`innerException` Exception

Class InvalidItemIdException

Namespace: Nekoyume.Exceptions

Assembly: Lib9c.dll

```
[Serializable]
public class InvalidItemIdException : Exception, ISerializable
```

Inheritance

object ← Exception ← InvalidItemIdException

Implements

ISerializable

Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Constructors

InvalidItemIdException()

```
public InvalidItemIdException()
```

InvalidItemIdException(SerializationInfo, StreamingContext)

```
protected InvalidItemIdException(SerializationInfo info, StreamingContext context)
```

Parameters

`info` SerializationInfo

`context` StreamingContext

InvalidItemIdException(string)

```
public InvalidItemIdException(string message)
```

Parameters

`message` string

InvalidItemIdException(string, Exception)

```
public InvalidItemIdException(string message, Exception innerException)
```

Parameters

`message` string

`innerException` Exception

InvalidItemIdException(string, string, int)

```
public InvalidItemIdException(string actionType, string addressesHex, int itemId)
```

Parameters

`actionType` string

`addressesHex` string

`itemId` int

Class InvalidMaterialCountException

Namespace: Nekoyume.Exceptions

Assembly: Lib9c.dll

```
[Serializable]
public class InvalidMaterialCountException : Exception, ISerializable
```

Inheritance

object ← Exception ← InvalidMaterialCountException

Implements

ISerializable

Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Constructors

InvalidMaterialCountException(string)

```
public InvalidMaterialCountException(string message)
```

Parameters

`message` string

InvalidMaterialCountException(string, string, int, int)

```
public InvalidMaterialCountException(string actionType, string addressesHex, int
required, int playerHas)
```

Parameters

`actionType` string

`addressesHex` string

`required` int

`playerHas` int

Class ItemNotFoundException

Namespace: Nekoyume.Exceptions

Assembly: Lib9c.dll

```
[Serializable]
public class ItemNotFoundException : Exception, ISerializable
```

Inheritance

object ← Exception ← ItemNotFoundException

Implements

ISerializable

Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Constructors

ItemNotFoundException()

```
public ItemNotFoundException()
```

ItemNotFoundException(Address, HashDigest<SHA256>, Exception?)

```
public ItemNotFoundException(Address inventoryAddr, HashDigest<SHA256> fungibleId,
Exception? innerException = null)
```

Parameters

`inventoryAddr` Address

`fungibleId` HashDigest<SHA256>

`innerException` Exception

ItemNotFoundException(Address, Guid, Exception?)

```
public ItemNotFoundException(Address inventoryAddr, Guid nonFungibleId, Exception?
innerException = null)
```

Parameters

`inventoryAddr` Address

`nonFungibleId` Guid

`innerException` Exception

ItemNotFoundException(SerializationInfo, StreamingContext)

```
protected ItemNotFoundException(SerializationInfo info, StreamingContext context)
```

Parameters

`info` SerializationInfo

`context` StreamingContext

ItemNotFoundException(string?)

```
public ItemNotFoundException(string? message)
```

Parameters

`message` string

ItemNotFoundException(string?, Exception?)

```
public ItemNotFoundException(string? message, Exception? innerException)
```

Parameters

`message` string

`innerException` Exception

Class MedalIdNotFoundException

Namespace: Nekoyume.Exceptions

Assembly: Lib9c.dll

```
[Serializable]
public class MedalIdNotFoundException : Exception, ISerializable
```

Inheritance

object ← Exception ← MedalIdNotFoundException

Implements

ISerializable

Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Constructors

MedalIdNotFoundException()

```
public MedalIdNotFoundException()
```

MedalIdNotFoundException(SerializationInfo, StreamingContext)

```
protected MedalIdNotFoundException(SerializationInfo info, StreamingContext context)
```

Parameters

`info` SerializationInfo

`context` StreamingContext

MedalIdNotFoundException(string)

```
public MedalIdNotFoundException(string message)
```

Parameters

`message` string

MedalIdNotFoundException(string, Exception)

```
public MedalIdNotFoundException(string message, Exception innerException)
```

Parameters

`message` string

`innerException` Exception

MedalIdNotFoundException(string, string, int, int)

```
public MedalIdNotFoundException(string actionType, string addressesHex, int
requiredAmount, int currentAmount)
```

Parameters

`actionType` string

`addressesHex` string

`requiredAmount` int

`currentAmount` int

Class NotEnoughEventDungeonTicketsException

Namespace: Nekoyume.Exceptions

Assembly: Lib9c.dll

```
[Serializable]
public class NotEnoughEventDungeonTicketsException : Exception, ISerializable
```

Inheritance

object ← Exception ← NotEnoughEventDungeonTicketsException

Implements

ISerializable

Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Constructors

NotEnoughEventDungeonTicketsException()

```
public NotEnoughEventDungeonTicketsException()
```

NotEnoughEventDungeonTicketsException(SerializationInfo, StreamingContext)

```
protected NotEnoughEventDungeonTicketsException(SerializationInfo info,
StreamingContext context)
```

Parameters

`info` SerializationInfo

`context` StreamingContext

NotEnoughEventDungeonTicketsException(string)

```
public NotEnoughEventDungeonTicketsException(string message)
```

Parameters

`message` string

NotEnoughEventDungeonTicketsException(string, Exception)

```
public NotEnoughEventDungeonTicketsException(string message,
Exception innerException)
```

Parameters

`message` string

`innerException` Exception

NotEnoughEventDungeonTicketsException(string, string, int, int)

```
public NotEnoughEventDungeonTicketsException(string actionType, string addressesHex,
int requiredAmount, int currentAmount)
```

Parameters

`actionType` string

`addressesHex` string

`requiredAmount` int

`currentAmount` int

Class NotEnoughItemException

Namespace: Nekoyume.Exceptions

Assembly: Lib9c.dll

```
[Serializable]
public class NotEnoughItemException : Exception, ISerializable
```

Inheritance

object ← Exception ← NotEnoughItemException

Implements

ISerializable

Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Constructors

NotEnoughItemException()

```
public NotEnoughItemException()
```

NotEnoughItemException(Address, HashDigest<SHA256>, int, int, Exception?)

```
public NotEnoughItemException(Address inventoryAddr, HashDigest<SHA256> fungibleId,
int expectedCount, int actualCount, Exception? innerException = null)
```

Parameters

`inventoryAddr` Address

`fungibleId` HashDigest<SHA256>

`expectedCount` int

`actualCount` int

`innerException` Exception

NotEnoughItemException(SerializationInfo, StreamingContext)

```
protected NotEnoughItemException(SerializationInfo info, StreamingContext context)
```

Parameters

`info` SerializationInfo

`context` StreamingContext

NotEnoughItemException(string?)

```
public NotEnoughItemException(string? message)
```

Parameters

`message` string

NotEnoughItemException(string?, Exception?)

```
public NotEnoughItemException(string? message, Exception? innerException)
```

Parameters

`message` string

`innerException` Exception

Class PetCostNotFoundException

Namespace: Nekoyume.Exceptions

Assembly: Lib9c.dll

```
[Serializable]
public class PetCostNotFoundException : Exception, ISerializable
```

Inheritance

object ← Exception ← PetCostNotFoundException

Implements

ISerializable

Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Constructors

PetCostNotFoundException(string)

```
public PetCostNotFoundException(string message)
```

Parameters

`message` string

PetCostNotFoundException(string, Exception)

```
public PetCostNotFoundException(string message, Exception innerException)
```

Parameters

`message` string

`innerException` Exception

PetCostNotFoundException(string, string, string, Exception)

```
public PetCostNotFoundException(string actionType, string addressesHex, string
message, Exception innerException = null)
```

Parameters

`actionType` string

`addressesHex` string

`message` string

`innerException` Exception

Class StateNullException

Namespace: Nekoyume.Exceptions

Assembly: Lib9c.dll

```
[Serializable]
public class StateNullException : Exception, ISerializable
```

Inheritance

object ← Exception ← StateNullException

Implements

ISerializable

Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Constructors

StateNullException()

```
public StateNullException()
```

StateNullException(Address, Address, Exception?)

```
public StateNullException(Address accountAddress, Address address, Exception?
innerException = null)
```

Parameters

`accountAddress` Address

`address` Address

`innerException` Exception

StateNullException(SerializationInfo, StreamingContext)

```
protected StateNullException(SerializationInfo info, StreamingContext context)
```

Parameters

`info` SerializationInfo

`context` StreamingContext

StateNullException(string?)

```
public StateNullException(string? message)
```

Parameters

`message` string

StateNullException(string?, Exception?)

```
public StateNullException(string? message, Exception? innerException)
```

Parameters

`message` string

`innerException` Exception

Class UnsupportedStateException

Namespace: Nekoyume.Exceptions

Assembly: Lib9c.dll

```
[Serializable]
public class UnsupportedStateException : Exception, ISerializable
```

Inheritance

object ← Exception ← UnsupportedStateException

Implements

ISerializable

Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Constructors

UnsupportedStateException(int, int, Exception)

```
public UnsupportedStateException(int expectedVersion, int actualVersion, Exception
innerException = null)
```

Parameters

`expectedVersion` int

`actualVersion` int

`innerException` Exception

UnsupportedStateException(SerializationInfo, StreamingContext)

```
protected UnsupportedStateException(SerializationInfo info,
StreamingContext context)
```

Parameters

`info` SerializationInfo

`context` StreamingContext

UnsupportedStateException(string)

```
public UnsupportedStateException(string msg)
```

Parameters

`msg` string

UnsupportedStateException(string, int, string, int, Exception)

```
public UnsupportedStateException(string expectedName, int expectedVersion, string
actualName, int actualVersion, Exception innerException = null)
```

Parameters

`expectedName` string

`expectedVersion` int

`actualName` string

`actualVersion` int

`innerException` Exception

Namespace Nekoyume.Extensions

Classes

ActionContextExtensions

AgentAddressExtensions

CombinationSlotStateExtensions

EquipmentExtensions

EventDungeonExtensions

EventRecipeExtensions

EventScheduleExtensions

IntegerExtensions

MaterialItemSheetExtensions

RandomExtensions

SheetsExtensions

WorldExtensions

WorldInformationExtensions

Class ActionContextExtensions

Namespace: Nekoyume.Extensions

Assembly: Lib9c.dll

```
public static class ActionContextExtensions
```

Inheritance

object ← ActionContextExtensions

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Methods

GetAgentAddress(IActionContext)

```
public static AgentAddress GetAgentAddress(this IActionContext context)
```

Parameters

`context` IActionContext

Returns

AgentAddress

Class AgentAddressExtensions

Namespace: Nekoyume.Extensions

Assembly: Lib9c.dll

```
public static class AgentAddressExtensions
```

Inheritance

object ← AgentAddressExtensions

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Methods

GetPledgeAddress(AgentAddress)

```
public static PledgeAddress GetPledgeAddress(this AgentAddress address)
```

Parameters

`address` AgentAddress

Returns

PledgeAddress

Class CombinationSlotStateExtensions

Namespace: Nekoyume.Extensions

Assembly: Lib9c.dll

```
public static class CombinationSlotStateExtensions
```

Inheritance

object ← CombinationSlotStateExtensions

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Methods

TryGetMail(CombinationSlotState, long, long, out CombinationMail, out ItemEnhanceMail)

```
public static bool TryGetMail(this CombinationSlotState state, long
blockIndex, long requiredBlockIndex, out CombinationMail combinationMail, out
ItemEnhanceMail itemEnhanceMail)
```

Parameters

`state` CombinationSlotState

`blockIndex` long

`requiredBlockIndex` long

`combinationMail` CombinationMail

`itemEnhanceMail` ItemEnhanceMail

Returns

bool

TryGetResultId(CombinationSlotState, out Guid)

```
public static bool TryGetResultId(this CombinationSlotState state, out
Guid resultId)
```

Parameters

`state` CombinationSlotState

`resultId` Guid

Returns

bool

ValidateFromAction(CombinationSlotState, long, AvatarState, int, string, string)

```
public static void ValidateFromAction(this CombinationSlotState slotState,
long blockIndex, AvatarState avatarState, int slotIndex, string actionTypeText,
string addressesHex)
```

Parameters

`slotState` CombinationSlotState

`blockIndex` long

`avatarState` AvatarState

`slotIndex` int

`actionTypeText` string

`addressesHex` string

Class EquipmentExtensions

Namespace: Nekoyume.Extensions

Assembly: Lib9c.dll

```
public static class EquipmentExtensions
```

Inheritance

object ← EquipmentExtensions

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Methods

GetRequirementLevel(Equipment, ItemRequirementSheet, EquipmentItemRecipeSheet, EquipmentItemSubRecipeSheetV2, EquipmentItemOptionSheet)

```
public static int GetRequirementLevel(this Equipment equipment,
ItemRequirementSheet requirementSheet, EquipmentItemRecipeSheet recipeSheet,
EquipmentItemSubRecipeSheetV2 subRecipeSheet, EquipmentItemOptionSheet
itemOptionSheet)
```

Parameters

`equipment` Equipment

`requirementSheet` ItemRequirementSheet

`recipeSheet` EquipmentItemRecipeSheet

`subRecipeSheet` EquipmentItemSubRecipeSheetV2

`itemOptionSheet` EquipmentItemOptionSheet

Returns

int

IsMadeWithMimisbrunnrRecipe(Equipment, EquipmentItemRecipeSheet, EquipmentItemSubRecipeSheetV2, EquipmentItemOptionSheet)

```
public static bool IsMadeWithMimisbrunnrRecipe(this Equipment equipment,
EquipmentItemRecipeSheet recipeSheet, EquipmentItemSubRecipeSheetV2 subRecipeSheet,
EquipmentItemOptionSheet itemOptionSheet)
```

Parameters

`equipment` Equipment

`recipeSheet` EquipmentItemRecipeSheet

`subRecipeSheet` EquipmentItemSubRecipeSheetV2

`itemOptionSheet` EquipmentItemOptionSheet

Returns

bool

Class EventDungeonExtensions

Namespace: Nekoyume.Extensions

Assembly: Lib9c.dll

```
public static class EventDungeonExtensions
```

Inheritance

object ← EventDungeonExtensions

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Methods

GetRemainingTicketsConsiderReset(EventDungeonInfo, Row, long)

```
public static int GetRemainingTicketsConsiderReset(this EventDungeonInfo info,
EventScheduleSheet.Row eventScheduleRow, long currentBlockIndex)
```

Parameters

`info` EventDungeonInfo

`eventScheduleRow` EventScheduleSheet.Row

`currentBlockIndex` long

Returns

int

GetStageNumber(Row)

```
public static int GetStageNumber(this EventDungeonStageSheet.Row row)
```

Parameters

`row` EventDungeonStageSheet.Row

Returns

int

GetStageRows(EventDungeonStageSheet, int, int)

```
public static List<EventDungeonStageSheet.Row> GetStageRows(this
EventDungeonStageSheet sheet, int beginningStageId, int endStageId)
```

Parameters

`sheet` EventDungeonStageSheet

`beginningStageId` int

`endStageId` int

Returns

List<EventDungeonStageSheet.Row>

GetStageWaveRows(EventDungeonStageWaveSheet, int, int)

```
public static List<EventDungeonStageWaveSheet.Row> GetStageWaveRows(this
EventDungeonStageWaveSheet sheet, int beginningStageId, int endStageId)
```

Parameters

`sheet` EventDungeonStageWaveSheet

`beginningStageId` int

`endStageId` int

Returns

List<EventDungeonStageWaveSheet.Row>

ToEventDungeonStageNumber(int)

```
public static int ToEventDungeonStageNumber(this int eventDungeonStageId)
```

Parameters

`eventDungeonStageId` int

Returns

int

TryGetRowByEventDungeonStageId(EventDungeonSheet, int, out Row)

```
public static bool TryGetRowByEventDungeonStageId(this EventDungeonSheet sheet, int
eventDungeonStageId, out EventDungeonSheet.Row row)
```

Parameters

`sheet` EventDungeonSheet

`eventDungeonStageId` int

`row` EventDungeonSheet.Row

Returns

bool

TryGetRowByEventScheduleId(EventDungeonSheet, int, out Row)

```
public static bool TryGetRowByEventScheduleId(this EventDungeonSheet sheet, int
eventScheduleId, out EventDungeonSheet.Row row)
```

Parameters

`sheet` EventDungeonSheet

`eventScheduleId` int

`row` EventDungeonSheet.Row

Returns

bool

ValidateFromAction(EventDungeonSheet, int, int, string, string)

```
public static EventDungeonSheet.Row ValidateFromAction(this EventDungeonSheet
sheet, int eventDungeonId, int eventDungeonStageId, string actionTypeText,
string addressesHex)
```

Parameters

`sheet` EventDungeonSheet

`eventDungeonId` int

`eventDungeonStageId` int

`actionTypeText` string

`addressesHex` string

Returns

EventDungeonSheet.Row

ValidateFromAction(EventDungeonStageSheet, int, string, string)

```
public static EventDungeonStageSheet.Row ValidateFromAction(this
EventDungeonStageSheet sheet, int eventDungeonStageId, string actionTypeText,
string addressesHex)
```

Parameters

`sheet` EventDungeonStageSheet

`eventDungeonStageId` int

`actionTypeText` string

`addressesHex` string

Returns

EventDungeonStageSheet.Row

Class EventRecipeExtensions

Namespace: Nekoyume.Extensions

Assembly: Lib9c.dll

```
public static class EventRecipeExtensions
```

Inheritance

object ← EventRecipeExtensions

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Methods

GetRecipeRows(EventConsumableItemRecipeSheet, int)

```
public static List<EventConsumableItemRecipeSheet.Row> GetRecipeRows(this
EventConsumableItemRecipeSheet sheet, int eventScheduleId)
```

Parameters

`sheet` EventConsumableItemRecipeSheet

`eventScheduleId` int

Returns

List<EventConsumableItemRecipeSheet.Row>

GetRecipeRows(EventMaterialItemRecipeSheet, int)

```
public static List<EventMaterialItemRecipeSheet.Row> GetRecipeRows(this
EventMaterialItemRecipeSheet sheet, int eventScheduleId)
```

Parameters

`sheet` EventMaterialItemRecipeSheet

`eventScheduleId` int

Returns

List<EventMaterialItemRecipeSheet.Row>

ValidateFromAction(EventConsumableItemRecipeSheet, int, string, string)

```
public static EventConsumableItemRecipeSheet.Row ValidateFromAction(this
EventConsumableItemRecipeSheet sheet, int eventConsumableItemRecipeId, string
actionTypeText, string addressesHex)
```

Parameters

`sheet` EventConsumableItemRecipeSheet

`eventConsumableItemRecipeId` int

`actionTypeText` string

`addressesHex` string

Returns

EventConsumableItemRecipeSheet.Row

ValidateFromAction(EventMaterialItemRecipeSheet, int, string, string)

```
public static EventMaterialItemRecipeSheet.Row ValidateFromAction(this
EventMaterialItemRecipeSheet sheet, int eventMaterialItemRecipeId, string
actionTypeText, string addressesHex)
```

Parameters

`sheet` EventMaterialItemRecipeSheet

`eventMaterialItemRecipeId` int

`actionTypeText` string

`addressesHex` string

Returns

EventMaterialItemRecipeSheet.Row

ValidateFromAction(Row, MaterialItemSheet, Dictionary<int, int>, string, string)

```
public static void ValidateFromAction(this EventMaterialItemRecipeSheet.Row
recipeRow, MaterialItemSheet materialItemSheet, Dictionary<int, int> materialsToUse,
string actionTypeText, string addressesHex)
```

Parameters

`recipeRow` EventMaterialItemRecipeSheet.Row

`materialItemSheet` MaterialItemSheet

`materialsToUse` Dictionary<int, int>

`actionTypeText` string

`addressesHex` string

Class EventScheduleExtensions

Namespace: Nekoyume.Extensions

Assembly: Lib9c.dll

```
public static class EventScheduleExtensions
```

Inheritance

object ← EventScheduleExtensions

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Methods

GetDungeonTicketCost(Row, int, Currency)

```
public static FungibleAssetValue GetDungeonTicketCost(this EventScheduleSheet.Row
row, int numberOfTicketPurchases, Currency currency)
```

Parameters

`row` EventScheduleSheet.Row

`numberOfTicketPurchases` int

`currency` Currency

Returns

FungibleAssetValue

GetDungeonTicketCostV1(Row, int)

```
[Obsolete("Use `GetDungeonTicketCost()` instead.")]
public static long GetDungeonTicketCostV1(this EventScheduleSheet.Row row,
int numberOfTicketPurchases)
```

Parameters

`row` EventScheduleSheet.Row

`numberOfTicketPurchases` int

Returns

long

GetStageExp(Row, int, int)

```
public static int GetStageExp(this EventScheduleSheet.Row row, int stageNumber, int
multiplier = 1)
```

Parameters

`row` EventScheduleSheet.Row

`stageNumber` int

`multiplier` int

Returns

int

ToEventScheduleId(int)

```
public static int ToEventScheduleId(this int eventDungeonOrRecipeId)
```

Parameters

`eventDungeonOrRecipeId` int

Returns

int

TryGetRowForDungeon(EventScheduleSheet, long, out Row)

```
public static bool TryGetRowForDungeon(this EventScheduleSheet sheet, long
blockIndex, out EventScheduleSheet.Row row)
```

Parameters

`sheet` EventScheduleSheet

`blockIndex` long

`row` EventScheduleSheet.Row

Returns

bool

TryGetRowForRecipe(EventScheduleSheet, long, out Row)

```
public static bool TryGetRowForRecipe(this EventScheduleSheet sheet, long
blockIndex, out EventScheduleSheet.Row row)
```

Parameters

`sheet` EventScheduleSheet

`blockIndex` long

`row` EventScheduleSheet.Row

Returns

bool

ValidateFromActionForDungeon(EventScheduleSheet, long, int, int, string, string)

```
public static EventScheduleSheet.Row ValidateFromActionForDungeon(this
EventScheduleSheet scheduleSheet, long blockIndex, int eventScheduleId, int
eventDungeonId, string actionTypeText, string addressesHex)
```

Parameters

`scheduleSheet` EventScheduleSheet

`blockIndex` long

`eventScheduleId` int

`eventDungeonId` int

`actionTypeText` string

`addressesHex` string

Returns

EventScheduleSheet.Row

ValidateFromActionForRecipe(EventScheduleSheet, long, int, int, string, string)

```
public static EventScheduleSheet.Row ValidateFromActionForRecipe(this
EventScheduleSheet scheduleSheet, long blockIndex, int eventScheduleId, int
eventRecipeId, string actionTypeText, string addressesHex)
```

Parameters

`scheduleSheet` EventScheduleSheet

`blockIndex` long

`eventScheduleId` int

`eventRecipeId` int

`actionTypeText` string

`addressesHex` string

Returns

EventScheduleSheet.Row

Class IntegerExtensions

Namespace: Nekoyume.Extensions

Assembly: Lib9c.dll

```
public static class IntegerExtensions
```

Inheritance

object ← IntegerExtensions

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Methods

NormalizeFromTenThousandths(int)

```
public static decimal NormalizeFromTenThousandths(this int value)
```

Parameters

`value` int

Returns

decimal

Class MaterialItemSheetExtensions

Namespace: Nekoyume.Extensions

Assembly: Lib9c.dll

```
public static class MaterialItemSheetExtensions
```

Inheritance

object ← MaterialItemSheetExtensions

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Methods

ValidateFromAction(MaterialItemSheet, List<MaterialInfo>, Dictionary<int, int>, string)

```
public static void ValidateFromAction(this MaterialItemSheet materialItemSheet,
List<EquipmentItemSubRecipeSheet.MaterialInfo> materialInfos, Dictionary<int, int>
requiredFungibleItems, string addressesHex)
```

Parameters

`materialItemSheet` MaterialItemSheet

`materialInfos` List<EquipmentItemSubRecipeSheet.MaterialInfo>

`requiredFungibleItems` Dictionary<int, int>

`addressesHex` string

Class RandomExtensions

Namespace: Nekoyume.Extensions

Assembly: Lib9c.dll

```
public static class RandomExtensions
```

Inheritance

object ← RandomExtensions

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Methods

GenerateAddress(IRandom)

```
public static Address GenerateAddress(this IRandom random)
```

Parameters

`random` IRandom

Returns

Address

NextLong(IRandom, long, long)

Returns a value in `[minValue, maxValue)`, like Next(int, int) but for 64-bit bounds.

```
public static long NextLong(this IRandom random, long minValue, long maxValue)
```

Parameters

`random` IRandom

The random source.

`minValue` long

The inclusive lower bound.

`maxValue` long

The exclusive upper bound.

Returns

long

The drawn value.

Remarks

When both bounds fit in int, this calls Next(int, int) and nothing else, so sheet values that already fit draw exactly what they drew before the sheet columns became long. Only a wider range reads 8 bytes from NextBytes(byte[]); its modulo bias is at most range / 2^64, negligible for sheet-sized ranges.

Exceptions

ArgumentOutOfRangeException

`minValue` is greater than `maxValue`.

Class SheetsExtensions

Namespace: Nekoyume.Extensions

Assembly: Lib9c.dll

```
public static class SheetsExtensions
```

Inheritance

object ← SheetsExtensions

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Methods

FindLevelByStakedAmount(IStakeRewardSheet, Address, FungibleAssetValue)

```
public static int FindLevelByStakedAmount(this IStakeRewardSheet sheet, Address
agentAddress, FungibleAssetValue balance)
```

Parameters

`sheet` IStakeRewardSheet

`agentAddress` Address

`balance` FungibleAssetValue

Returns

int

FindPreviousRaidIdByBlockIndex(WorldBossListSheet, long)

```
public static int FindPreviousRaidIdByBlockIndex(this WorldBossListSheet sheet,
long blockIndex)
```

Parameters

`sheet` WorldBossListSheet

`blockIndex` long

Returns

int

FindPreviousRowByBlockIndex(WorldBossListSheet, long)

```
public static WorldBossListSheet.Row FindPreviousRowByBlockIndex(this
WorldBossListSheet sheet, long blockIndex)
```

Parameters

`sheet` WorldBossListSheet

`blockIndex` long

Returns

WorldBossListSheet.Row

FindRaidIdByBlockIndex(WorldBossListSheet, long)

```
public static int FindRaidIdByBlockIndex(this WorldBossListSheet sheet,
long blockIndex)
```

Parameters

`sheet` WorldBossListSheet

`blockIndex` long

Returns

int

FindRowByBlockIndex(WorldBossListSheet, long)

```
public static WorldBossListSheet.Row FindRowByBlockIndex(this WorldBossListSheet
sheet, long blockIndex)
```

Parameters

`sheet` WorldBossListSheet

`blockIndex` long

Returns

WorldBossListSheet.Row

GetActionPointByStaking(StakeActionPointCoefficientSheet, int, int, int)

```
public static int GetActionPointByStaking(this StakeActionPointCoefficientSheet
sheet, int originAp, int playCount, int level)
```

Parameters

`sheet` StakeActionPointCoefficientSheet

`originAp` int

`playCount` int

`level` int

Returns

int

GetAddress(Dictionary<Type, (Address address, ISheet sheet)>, Type)

```
public static Address GetAddress(this Dictionary<Type, (Address address, ISheet
sheet)> sheets, Type type)
```

Parameters

`sheets` Dictionary<Type, (Address address, ISheet sheet)>

`type` Type

Returns

Address

GetAddress<T>(Dictionary<Type, (Address address, ISheet sheet)>)

```
public static Address GetAddress<T>(this Dictionary<Type, (Address address, ISheet
sheet)> sheets) where T : ISheet
```

Parameters

`sheets` Dictionary<Type, (Address address, ISheet sheet)>

Returns

Address

Type Parameters

T

GetArenaSimulatorSheets(Dictionary<Type, (Address address, ISheet sheet)>)

```
public static ArenaSimulatorSheets GetArenaSimulatorSheets(this Dictionary<Type,
(Address address, ISheet sheet)> sheets)
```

Parameters

`sheets` Dictionary<Type, (Address address, ISheet sheet)>

Returns

ArenaSimulatorSheets

GetArenaSimulatorSheetsV1(Dictionary<Type, (Address address, ISheet sheet)>)

```
public static ArenaSimulatorSheetsV1 GetArenaSimulatorSheetsV1(this Dictionary<Type,
(Address address, ISheet sheet)> sheets)
```

Parameters

`sheets` Dictionary<Type, (Address address, ISheet sheet)>

Returns

ArenaSimulatorSheetsV1

GetArenaSimulatorSheets_v100291(Dictionary<Type, (Address address, ISheet sheet)>)

```
public static ArenaSimulatorSheetsV1 GetArenaSimulatorSheets_v100291(this
Dictionary<Type, (Address address, ISheet sheet)> sheets)
```

Parameters

`sheets` Dictionary<Type, (Address address, ISheet sheet)>

Returns

ArenaSimulatorSheetsV1

GetAvatarSheets(Dictionary<Type, (Address address, ISheet sheet)>)

```
public static AvatarSheets GetAvatarSheets(this Dictionary<Type, (Address address,
ISheet sheet)> sheets)
```

Parameters

`sheets` Dictionary<Type, (Address address, ISheet sheet)>

Returns

AvatarSheets

GetItemSheet(Dictionary<Type, (Address address, ISheet sheet)>)

```
public static ItemSheet GetItemSheet(this Dictionary<Type, (Address address, ISheet
sheet)> sheets)
```

Parameters

`sheets` Dictionary<Type, (Address address, ISheet sheet)>

Returns

ItemSheet

GetQuestSheet(Dictionary<Type, (Address address, ISheet sheet)>)

```
public static QuestSheet GetQuestSheet(this Dictionary<Type, (Address address,
ISheet sheet)> sheets)
```

Parameters

`sheets` Dictionary<Type, (Address address, ISheet sheet)>

Returns

QuestSheet

GetRaidSimulatorSheets(Dictionary<Type, (Address address, ISheet sheet)>)

```
public static RaidSimulatorSheets GetRaidSimulatorSheets(this Dictionary<Type,
(Address address, ISheet sheet)> sheets)
```

Parameters

`sheets` Dictionary<Type, (Address address, ISheet sheet)>

Returns

RaidSimulatorSheets

GetRaidSimulatorSheetsV1(Dictionary<Type, (Address address, ISheet sheet)>)

```
public static RaidSimulatorSheetsV1 GetRaidSimulatorSheetsV1(this Dictionary<Type,
(Address address, ISheet sheet)> sheets)
```

Parameters

`sheets` Dictionary<Type, (Address address, ISheet sheet)>

Returns

RaidSimulatorSheetsV1

GetRankingSimulatorSheets(Dictionary<Type, (Address address, ISheet sheet)>)

```
public static RankingSimulatorSheets GetRankingSimulatorSheets(this Dictionary<Type,
(Address address, ISheet sheet)> sheets)
```

Parameters

`sheets` Dictionary<Type, (Address address, ISheet sheet)>

Returns

RankingSimulatorSheets

GetRankingSimulatorSheetsV1(Dictionary<Type, (Address address, ISheet sheet)>)

```
public static RankingSimulatorSheetsV1 GetRankingSimulatorSheetsV1(this
Dictionary<Type, (Address address, ISheet sheet)> sheets)
```

Parameters

`sheets` Dictionary<Type, (Address address, ISheet sheet)>

Returns

RankingSimulatorSheetsV1

GetRankingSimulatorSheetsV100291(Dictionary<Type, (Address address, ISheet sheet)>)

```
public static RankingSimulatorSheetsV1 GetRankingSimulatorSheetsV100291(this
Dictionary<Type, (Address address, ISheet sheet)> sheets)
```

Parameters

`sheets` Dictionary<Type, (Address address, ISheet sheet)>

Returns

RankingSimulatorSheetsV1

GetSheet(Dictionary<Type, (Address address, ISheet sheet)>, Type)

```
public static ISheet GetSheet(this Dictionary<Type, (Address address, ISheet sheet)>
sheets, Type sheetType)
```

Parameters

`sheets` Dictionary<Type, (Address address, ISheet sheet)>

`sheetType` Type

Returns

ISheet

GetSheet<T>(Dictionary<Type, (Address address, ISheet sheet)>)

```
public static T GetSheet<T>(this Dictionary<Type, (Address address, ISheet sheet)>
sheets) where T : ISheet
```

Parameters

`sheets` Dictionary<Type, (Address address, ISheet sheet)>

Returns

T

Type Parameters

`T`

GetSimulatorSheets(Dictionary<Type, (Address address, ISheet sheet)>)

```
public static SimulatorSheets GetSimulatorSheets(this Dictionary<Type, (Address address, ISheet sheet)> sheets)
```

Parameters

`sheets` Dictionary<Type, (Address address, ISheet sheet)>

Returns

SimulatorSheets

GetSimulatorSheetsV1(Dictionary<Type, (Address address, ISheet sheet)>)

```
public static SimulatorSheetsV1 GetSimulatorSheetsV1(this Dictionary<Type, (Address address, ISheet sheet)> sheets)
```

Parameters

`sheets` Dictionary<Type, (Address address, ISheet sheet)>

Returns

SimulatorSheetsV1

GetSimulatorSheetsV100291(Dictionary<Type, (Address address, ISheet sheet)>)

```
public static SimulatorSheetsV1 GetSimulatorSheetsV100291(this Dictionary<Type,
(Address address, ISheet sheet)> sheets)
```

Parameters

`sheets` Dictionary<Type, (Address address, ISheet sheet)>

Returns

SimulatorSheetsV1

GetStageSimulatorSheets(Dictionary<Type, (Address address, ISheet sheet)>)

```
public static StageSimulatorSheets GetStageSimulatorSheets(this Dictionary<Type,
(Address address, ISheet sheet)> sheets)
```

Parameters

`sheets` Dictionary<Type, (Address address, ISheet sheet)>

Returns

StageSimulatorSheets

GetStageSimulatorSheetsV1(Dictionary<Type, (Address address, ISheet sheet)>)

```
public static StageSimulatorSheetsV1 GetStageSimulatorSheetsV1(this Dictionary<Type,
(Address address, ISheet sheet)> sheets)
```

Parameters

`sheets` Dictionary<Type, (Address address, ISheet sheet)>

Returns

StageSimulatorSheetsV1

GetStageSimulatorSheetsV100291(Dictionary<Type, (Address address, ISheet sheet)>)

```
public static StageSimulatorSheetsV1 GetStageSimulatorSheetsV100291(this
Dictionary<Type, (Address address, ISheet sheet)> sheets)
```

Parameters

`sheets` Dictionary<Type, (Address address, ISheet sheet)>

Returns

StageSimulatorSheetsV1

TryGetAddress(Dictionary<Type, (Address address, ISheet sheet)>, Type, out Address)

```
public static bool TryGetAddress(this Dictionary<Type, (Address address, ISheet
sheet)> sheets, Type type, out Address address)
```

Parameters

`sheets` Dictionary<Type, (Address address, ISheet sheet)>

`type` Type

`address` Address

Returns

bool

TryGetAddress<T>(Dictionary<Type, (Address address, ISheet sheet)>, out Address)

```
public static bool TryGetAddress<T>(this Dictionary<Type, (Address address, ISheet sheet)> sheets, out Address address) where T : ISheet
```

Parameters

`sheets` Dictionary<Type, (Address address, ISheet sheet)>

`address` Address

Returns

bool

Type Parameters

`T`

TryGetSheet(Dictionary<Type, (Address address, ISheet sheet)>, Type, out ISheet)

```
public static bool TryGetSheet(this Dictionary<Type, (Address address, ISheet sheet)> sheets, Type type, out ISheet sheet)
```

Parameters

`sheets` Dictionary<Type, (Address address, ISheet sheet)>

`type` Type

`sheet` ISheet

Returns

bool

TryGetSheet<T>(Dictionary<Type, (Address address, ISheet sheet)>, out T)

```
public static bool TryGetSheet<T>(this Dictionary<Type, (Address address, ISheet sheet)> sheets, out T sheet) where T : ISheet
```

Parameters

`sheets` Dictionary<Type, (Address address, ISheet sheet)>

`sheet` T

Returns

bool

Type Parameters

`T`

Class WorldExtensions

Namespace: Nekoyume.Extensions

Assembly: Lib9c.dll

```
public static class WorldExtensions
```

Inheritance

object ← WorldExtensions

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Methods

GetFeeAddress(IWorld, long)

Retrieves the fee address based on the block index and migration height.

```
public static Address GetFeeAddress(this IWorld world, long blockIndex)
```

Parameters

`world` IWorld

The current world instance.

`blockIndex` long

The block index for which the fee address is required.

Returns

Address

The fee address, which may vary based on migration height and arena data.

MutateAccount(IWorld, Address, Func<IAccount, IAccount>)

```
public static IWorld MutateAccount(this IWorld world, Address accountAddress,
Func<IAccount, IAccount> mutateFn)
```

Parameters

`world` IWorld

`accountAddress` Address

`mutateFn` Func<IAccount, IAccount>

Returns

IWorld

Class WorldInformationExtensions

Namespace: Nekoyume.Extensions

Assembly: Lib9c.dll

```
public static class WorldInformationExtensions
```

Inheritance

object ← WorldInformationExtensions

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Methods

ValidateFromAction(WorldInformation, int, string, string)

```
public static void ValidateFromAction(this WorldInformation worldInformation, int
stageId, string actionTypeText, string addressesHex)
```

Parameters

`worldInformation` WorldInformation

`stageId` int

`actionTypeText` string

`addressesHex` string

Namespace Nekoyume.Helper

Classes

AdventureBossHelper

AvatarStateExtensions

CrystalCalculator

CustomCraftHelper

Helpers for `CustomEquipmentCraft`.

InventoryExtensions

ItemOptionHelper

ItemOptionInfo

NumberConversionHelper

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PetHelper

RuneEnhancementCostBoost

A BoostScheduleSheet already checked to apply to the enhancement cost of one rune, RuneId: an enhancement cost row was active and the rune was summonable when it was resolved. Only ResolveEnhancementCostBoost(BoostScheduleSheet?, int, long, Func<RuneSummonSheet?>) creates one, and TryEnhancement(int, Row, IRandom, int, out LevelUpResult, RuneEnhancementCostBoost?, long) and ApplyEnhancementCostBoost(RuneCostData, int, int, RuneEnhancementCostBoost?, long) ignore it for any other rune, so a boost only ever discounts the rune it was resolved for — neither handing over the raw patched sheet nor reusing a boost after switching runes can skip the check.

RuneHelper

SummonHelper

SynthesizeSimulator

A class that simulates the synthesis of items. SynthesizeSimulator.InputData

WorldBossHelper

World boss (raid) rules shared by the raid actions and clients: ranks, tickets, rewards and their BoostScheduleSheet adjustments.

Structs

SynthesizeResult

Represents the result of the synthesis.

SynthesizeSimulator.InputData

Simulate the synthesis of items.

Class AdventureBossHelper

Namespace: Nekoyume.Helper

Assembly: Lib9c.dll

```
public static class AdventureBossHelper
```

Inheritance

object ← AdventureBossHelper

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Fields

FixedRewardRatio

```
public const decimal FixedRewardRatio = 0.7
```

Field Value

decimal

MaxAdventureBossRewardBoostFactor

Upper bound of a AdventureBossFloorReward or AdventureBossFirstClearReward row: a boosted reward amount never exceeds this many times the amount drawn, whatever the row's `op` and value. It bounds a mistyped value to a fivefold over-reward and keeps boosted amounts added to one inventory slot from overflowing.

```
public const long MaxAdventureBossRewardBoostFactor = 5
```

Field Value

long

RaffleRewardPercent

Share of a season's total bounty, in percent, paid as the raffle reward.

```
public const int RaffleRewardPercent = 5
```

Field Value

int

RandomRewardRatio

```
public const decimal RandomRewardRatio = 0.3
```

Field Value

decimal

TotalRewardMultiplier

```
public const decimal TotalRewardMultiplier = 1.2
```

Field Value

decimal

Methods

AddExploreRewards(IActionContext, IWorld, Address, Inventory, IEnumerable<RewardAmountData>)

```
public static IWorld AddExploreRewards(IActionContext context, IWorld states,
Address avatarAddress, Inventory inventory,
IEnumerable<AdventureBossSheet.RewardAmountData> rewardList)
```

Parameters

`context` IActionContext

`states` IWorld

`avatarAddress` Address

`inventory` Inventory

`rewardList` IEnumerable<AdventureBossSheet.RewardAmountData>

Returns

IWorld

ApplyRewardBoost(RewardAmountData, Row)

Applies a BoostScheduleSheet row to one explore or sweep reward entry.

```
public static AdventureBossSheet.RewardAmountData
ApplyRewardBoost(AdventureBossSheet.RewardAmountData reward,
BoostScheduleSheet.Row rewardBoost)
```

Parameters

`reward` AdventureBossSheet.RewardAmountData

The reward as the sheets and the random draws decided it.

`rewardBoost` BoostScheduleSheet.Row

The row adjusting this floor's reward at the current block — see AdventureBossFloorReward and AdventureBossFirstClearReward — or `null` for none. A caller replaying an action must pass what the action resolved.

Returns

AdventureBossSheet.RewardAmountData

`reward` itself when there is no row or nothing to adjust, otherwise a copy whose Amount is adjusted and kept between 1 and MaxAdventureBossRewardBoostFactor times the drawn amount.

Remarks

Only the granted amount changes: the entry was already drawn, so a boost never adds or reorders random draws. A boost adjusts a reward but never creates one, so an entry with no amount stays as it is; and it never takes a reward below 1, because granting 0 of an item or a currency fails the action. The cap applies to `ADD` too, so `ADD 10` on an amount of 1 grants 5.

CalculateExploreReward(ClaimableReward, BountyBoard, ExploreBoard, Explorer, Address, AdventureBossNcgRewardRatioSheet, decimal, decimal, bool, out FungibleAssetValue)

Calculate reward for adventure boss explorers. This method only calculates reward for given avatar, not actually give rewards.

```
public static AdventureBossGameData.ClaimableReward
CalculateExploreReward(AdventureBossGameData.ClaimableReward reward, BountyBoard
bountyBoard, ExploreBoard exploreBoard, Explorer explorer, Address avatarAddress,
AdventureBossNcgRewardRatioSheet sheet, decimal ncgApRatio, decimal ncgRuneRatio,
bool isReal, out FungibleAssetValue ncgReward)
```

Parameters

`reward` AdventureBossGameData.ClaimableReward

Claimable reward for this avatar so far.

`bountyBoard` BountyBoard

Bounty board for this season. NCG reward is based on totalBounty on this board.

`exploreBoard` ExploreBoard

Explore board for this season. Total reward amount is base on usedApPotion on this board.

`explorer` Explorer

Target explorer to calculate reward

`avatarAddress` Address

Target avatar address to calculate reward.

`sheet` AdventureBossNcgRewardRatioSheet

NCG to reward exchange ratio sheet. Calculate total reward amount based on this sheet.

`ncgApRatio` decimal

Exchange ratio between used AP potion to NCG. Used to set total reward amount.

`ncgRuneRatio` decimal

If a reward is rune, use this fixed ratio, not in sheet.

`isReal` bool

`ncgReward` FungibleAssetValue

out value: calculated NCG reward in this function. We must handle NCG reward separately because NCG reward must be transferred from each season's bounty address.

Returns

AdventureBossGameData.ClaimableReward

Updated Claimable reward after calculation.

CalculateRaffleReward(BountyBoard)

```
public static FungibleAssetValue CalculateRaffleReward(BountyBoard bountyBoard)
```

Parameters

`bountyBoard` BountyBoard

Returns

FungibleAssetValue

CalculateWantedReward(ClaimableReward, BountyBoard, Address, AdventureBossNcgRewardRatioSheet, decimal, out FungibleAssetValue)

Calculate reward for adventure boss operators. This method only calculates reward for given avatar, not actually give rewards.

```
public static AdventureBossGameData.ClaimableReward
CalculateWantedReward(AdventureBossGameData.ClaimableReward reward, BountyBoard
bountyBoard, Address avatarAddress, AdventureBossNcgRewardRatioSheet sheet, decimal
ncgRuneRatio, out FungibleAssetValue ncgReward)
```

Parameters

`reward` AdventureBossGameData.ClaimableReward

Claimable reward for this avatar so far.

`bountyBoard` BountyBoard

Bounty board for this season. All the reward amount is based on totalBounty on this board.

`avatarAddress` Address

Target avatar address to calculate reward.

`sheet` AdventureBossNcgRewardRatioSheet

NCG to reward exchange ratio sheet. Calculate total reward amount based on this sheet.

`ncgRuneRatio` decimal

If a reward is rune, use this fixed ratio, not in sheet.

`ncgReward` FungibleAssetValue

out value: calculated NCG reward in this function. We must handle NCG reward separately because NCG reward must be transferred from each season's bounty address.

Returns

AdventureBossGameData.ClaimableReward

Updated Claimable reward after calculation.

CollectExploreReward(ClaimableReward, GameConfigState, AdventureBossNcgRewardRatioSheet, SeasonInfo, BountyBoard, ExploreBoard, Explorer, long, Address, ref ClaimableReward, out FungibleAssetValue)

```
public static bool CollectExploreReward(AdventureBossGameData.ClaimableReward
reward, GameConfigState gameConfig, AdventureBossNcgRewardRatioSheet
ncgRewardRatioSheet, SeasonInfo seasonInfo, BountyBoard bountyBoard,
ExploreBoard exploreBoard, Explorer explorer, long currentBlockIndex, Address
avatarAddress, ref AdventureBossGameData.ClaimableReward updatedReward, out
FungibleAssetValue ncgReward)
```

Parameters

`reward` AdventureBossGameData.ClaimableReward

`gameConfig` GameConfigState

`ncgRewardRatioSheet` AdventureBossNcgRewardRatioSheet

`seasonInfo` SeasonInfo

`bountyBoard` BountyBoard

`exploreBoard` ExploreBoard

`explorer` Explorer

`currentBlockIndex` long

`avatarAddress` Address

`updatedReward` AdventureBossGameData.ClaimableReward

`ncgReward` FungibleAssetValue

Returns

bool

CollectWantedReward(ClaimableReward, GameConfigState, AdventureBossNcgRewardRatioSheet, SeasonInfo, BountyBoard, Investor, long, Address, ref ClaimableReward)

```
public static bool CollectWantedReward(AdventureBossGameData.ClaimableReward reward,
GameConfigState gameConfig, AdventureBossNcgRewardRatioSheet ncgRewardRatioSheet,
SeasonInfo seasonInfo, BountyBoard bountyBoard, Investor investor, long
currentBlockIndex, Address avatarAddress, ref AdventureBossGameData.ClaimableReward
updatedReward)
```

Parameters

`reward` AdventureBossGameData.ClaimableReward

`gameConfig` GameConfigState

`ncgRewardRatioSheet` AdventureBossNcgRewardRatioSheet

`seasonInfo` SeasonInfo

`bountyBoard` BountyBoard

`investor` Investor

`currentBlockIndex` long

`avatarAddress` Address

`updatedReward` AdventureBossGameData.ClaimableReward

Returns

bool

GetSeasonAsAddressForm(long)

```
public static string GetSeasonAsAddressForm(long season)
```

Parameters

`season` long

Returns

string

PickExploreRaffle(BountyBoard, ExploreBoard, ExplorerList, IRandom)

```
public static ExploreBoard PickExploreRaffle(BountyBoard bountyBoard, ExploreBoard
exploreBoard, ExplorerList explorerList, IRandom random)
```

Parameters

`bountyBoard` BountyBoard

`exploreBoard` ExploreBoard

`explorerList` ExplorerList

`random` IRandom

Returns

ExploreBoard

PickReward(IRandom, IEnumerable<RewardRatioData>)

```
public static (int?, int?) PickReward(IRandom random,
IEnumerable<AdventureBossSheet.RewardRatioData> rewardData)
```

Parameters

`random` IRandom

`rewardData` IEnumerable<AdventureBossSheet.RewardRatioData>

Returns

(int?, int?)

Class AvatarStateExtensions

Namespace: Nekoyume.Helper

Assembly: Lib9c.dll

```
public static class AvatarStateExtensions
```

Inheritance

object ← AvatarStateExtensions

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Methods

GetLevelAndExp(AvatarState, CharacterLevelSheet, int, int)

```
public static (int, long) GetLevelAndExp(this AvatarState avatarState,
CharacterLevelSheet characterLevelSheet, int stageId, int repeatCount)
```

Parameters

`avatarState` AvatarState

`characterLevelSheet` CharacterLevelSheet

`stageId` int

`repeatCount` int

Returns

(int, long)

GetLevelAndExpForEventDungeon(AvatarState, CharacterLevelSheet, Row, int, int)

Calculates level and experience for event dungeon sweep with level-up consideration. This method handles experience accumulation correctly without subtracting experience during level up.

```
public static (int, long) GetLevelAndExpForEventDungeon(this AvatarState
avatarState, CharacterLevelSheet characterLevelSheet, EventScheduleSheet.Row
scheduleRow, int stageNumber, int repeatCount)
```

Parameters

`avatarState` AvatarState

The avatar state to calculate experience for

`characterLevelSheet` CharacterLevelSheet

Character level sheet for level information

`scheduleRow` EventScheduleSheet.Row

Event schedule row containing dungeon experience configuration

`stageNumber` int

Event dungeon stage number

`repeatCount` int

Number of times to repeat the battle

Returns

(int, long)

Tuple containing final level and experience

GetLevelAndExpV1(AvatarState, CharacterLevelSheet, int, int)

```
[Obsolete("Use GetLevelAndExp")]
public static (int, long) GetLevelAndExpV1(this AvatarState avatarState,
CharacterLevelSheet characterLevelSheet, int stageId, int repeatCount)
```

Parameters

`avatarState` AvatarState

`characterLevelSheet` CharacterLevelSheet

`stageId` int

`repeatCount` int

Returns

(int, long)

UpdateExp(AvatarState, int, long)

```
public static void UpdateExp(this AvatarState avatarState, int level, long exp)
```

Parameters

`avatarState` AvatarState

`level` int

`exp` long

UpdateInventory(AvatarState, List<ItemBase>)

```
public static void UpdateInventory(this AvatarState avatarState,
List<ItemBase> rewards)
```

Parameters

`avatarState` AvatarState

`rewards` List<ItemBase>

UpdateMonsterMap(AvatarState, StageWaveSheet, int)

```
public static void UpdateMonsterMap(this AvatarState avatarState, StageWaveSheet
stageWaveSheet, int stageId)
```

Parameters

`avatarState` AvatarState

`stageWaveSheet` StageWaveSheet

`stageId` int

ValidEquipmentAndCostume(AvatarState, IEnumerable<Guid>, List<Guid>, ItemRequirementSheet, EquipmentItemRecipeSheet, EquipmentItemSubRecipeSheetV2, EquipmentItemOptionSheet, long, string)

```
public static void ValidEquipmentAndCostume(this AvatarState avatarState,
IEnumerable<Guid> costumeIds, List<Guid> equipmentIds, ItemRequirementSheet
itemRequirementSheet, EquipmentItemRecipeSheet equipmentItemRecipeSheet,
EquipmentItemSubRecipeSheetV2 equipmentItemSubRecipeSheetV2,
EquipmentItemOptionSheet equipmentItemOptionSheet, long blockIndex,
string addressesHex)
```

Parameters

`avatarState` AvatarState

`costumeIds` IEnumerable<Guid>

`equipmentIds` List<Guid>

`itemRequirementSheet` ItemRequirementSheet

`equipmentItemRecipeSheet` EquipmentItemRecipeSheet

`equipmentItemSubRecipeSheetV2` EquipmentItemSubRecipeSheetV2

`equipmentItemOptionSheet` EquipmentItemOptionSheet

`blockIndex` long

`addressesHex` string

ValidEquipmentAndCostumeV2(AvatarState, IEnumerable<Guid>, List<Guid>, ItemRequirementSheet, EquipmentItemRecipeSheet, EquipmentItemSubRecipeSheetV2, EquipmentItemOptionSheet, long, string, GameConfigState)

```
public static (List<Equipment> equipments, List<Costume> costumes)
ValidEquipmentAndCostumeV2(this AvatarState avatarState, IEnumerable<Guid>
costumeIds, List<Guid> equipmentIds, ItemRequirementSheet itemRequirementSheet,
EquipmentItemRecipeSheet equipmentItemRecipeSheet, EquipmentItemSubRecipeSheetV2
equipmentItemSubRecipeSheetV2, EquipmentItemOptionSheet equipmentItemOptionSheet,
long blockIndex, string addressesHex, GameConfigState gameConfigState)
```

Parameters

`avatarState` AvatarState

`costumeIds` IEnumerable<Guid>

`equipmentIds` List<Guid>

`itemRequirementSheet` ItemRequirementSheet

`equipmentItemRecipeSheet` EquipmentItemRecipeSheet

`equipmentItemSubRecipeSheetV2` EquipmentItemSubRecipeSheetV2

`equipmentItemOptionSheet` EquipmentItemOptionSheet

`blockIndex` long

`addressesHex` string

`gameConfigState` GameConfigState

Returns

(List<Equipment> equipments, List<Costume> costumes)

Class CrystalCalculator

Namespace: Nekoyume.Helper

Assembly: Lib9c.dll

```
public static class CrystalCalculator
```

Inheritance

object ← CrystalCalculator

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Fields

CRYSTAL

The crystal currency.

```
public static readonly Currency CRYSTAL
```

Field Value

Currency

MaxLevelExponent

Enhancement levels above this add no more grinding crystal: the multiplier stops at 2^5 - 1.

```
public const int MaxLevelExponent = 5
```

Field Value

int

Methods

CalculateBuffGachaCost(int, bool, CrystalStageBuffGachaSheet)

```
public static FungibleAssetValue CalculateBuffGachaCost(int stageId, bool
advancedGacha, CrystalStageBuffGachaSheet stageBuffGachaSheet)
```

Parameters

`stageId` int

`advancedGacha` bool

`stageBuffGachaSheet` CrystalStageBuffGachaSheet

Returns

FungibleAssetValue

CalculateCombinationCost(FungibleAssetValue, Row, CrystalCostState, CrystalCostState)

```
public static FungibleAssetValue CalculateCombinationCost(FungibleAssetValue
crystal, CrystalFluctuationSheet.Row row, CrystalCostState prevWeeklyCostState =
null, CrystalCostState beforePrevWeeklyCostState = null)
```

Parameters

`crystal` FungibleAssetValue

`row` CrystalFluctuationSheet.Row

`prevWeeklyCostState` CrystalCostState

`beforePrevWeeklyCostState` CrystalCostState

Returns

FungibleAssetValue

CalculateCrystal(Address, IEnumerable<Equipment>, FungibleAssetValue, bool, CrystalEquipmentGrindingSheet, CrystalMonsterCollectionMultiplierSheet, StakeRegularRewardSheet)

```
public static FungibleAssetValue CalculateCrystal(Address agentAddress,
IEnumerable<Equipment> equipmentList, FungibleAssetValue stakedAmount, bool
enhancementFailed, CrystalEquipmentGrindingSheet crystalEquipmentGrindingSheet,
CrystalMonsterCollectionMultiplierSheet crystalMonsterCollectionMultiplierSheet,
StakeRegularRewardSheet stakeRegularRewardSheet)
```

Parameters

`agentAddress` Address

`equipmentList` IEnumerable<Equipment>

`stakedAmount` FungibleAssetValue

`enhancementFailed` bool

`crystalEquipmentGrindingSheet` CrystalEquipmentGrindingSheet

`crystalMonsterCollectionMultiplierSheet` CrystalMonsterCollectionMultiplierSheet

`stakeRegularRewardSheet` StakeRegularRewardSheet

Returns

FungibleAssetValue

CalculateCrystal(IEnumerable<Equipment>, bool, CrystalEquipmentGrindingSheet,

CrystalMonsterCollectionMultiplierSheet, int)

```
public static FungibleAssetValue CalculateCrystal(IEnumerable<Equipment>
equipmentList, bool enhancementFailed, CrystalEquipmentGrindingSheet
crystalEquipmentGrindingSheet, CrystalMonsterCollectionMultiplierSheet
crystalMonsterCollectionMultiplierSheet, int stakingLevel)
```

Parameters

`equipmentList` IEnumerable<Equipment>

`enhancementFailed` bool

`crystalEquipmentGrindingSheet` CrystalEquipmentGrindingSheet

`crystalMonsterCollectionMultiplierSheet` CrystalMonsterCollectionMultiplierSheet

`stakingLevel` int

Returns

FungibleAssetValue

CalculateEntranceFee(int, BigInteger)

```
public static FungibleAssetValue CalculateEntranceFee(int level,
BigInteger entranceFee)
```

Parameters

`level` int

`entranceFee` BigInteger

Returns

FungibleAssetValue

CalculateMaterialCost(int, int, CrystalMaterialCostSheet)

```
public static FungibleAssetValue CalculateMaterialCost(int materialId, int
materialCount, CrystalMaterialCostSheet crystalMaterialCostSheet)
```

Parameters

`materialId` int

`materialCount` int

`crystalMaterialCostSheet` CrystalMaterialCostSheet

Returns

FungibleAssetValue

CalculateRecipeUnlockCost(IEnumerable<int>, EquipmentItemRecipeSheet)

Sums the crystal needed to unlock the given equipment recipes.

```
public static FungibleAssetValue CalculateRecipeUnlockCost(IEnumerable<int>
recipeIds, EquipmentItemRecipeSheet equipmentItemRecipeSheet)
```

Parameters

`recipeIds` IEnumerable<int>

The recipes to unlock.

`equipmentItemRecipeSheet` EquipmentItemRecipeSheet

The sheet holding each recipe's unlock cost.

Returns

FungibleAssetValue

The total unlock cost.

CalculateWorldUnlockCost(IEnumerable<int>, WorldUnlockSheet)

```
public static FungibleAssetValue CalculateWorldUnlockCost(IEnumerable<int> worldIds,
WorldUnlockSheet worldUnlockSheet)
```

Parameters

`worldIds` IEnumerable<int>

`worldUnlockSheet` WorldUnlockSheet

Returns

FungibleAssetValue

Class CustomCraftHelper

Namespace: Nekoyume.Helper

Assembly: Lib9c.dll

Helpers for `CustomEquipmentCraft`.

```
public static class CustomCraftHelper
```

Inheritance

object ← CustomCraftHelper

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Methods

CalculateAdditionalCost(int, CustomEquipmentCraftRelationshipSheet)

```
public static (BigInteger, IDictionary<int, int>)? CalculateAdditionalCost(int
relationship, CustomEquipmentCraftRelationshipSheet relationshipSheet)
```

Parameters

`relationship` int

`relationshipSheet` CustomEquipmentCraftRelationshipSheet

Returns

(BigInteger, IDictionary<int, int>)?

CalculateCraftCost(int, int, MaterialItemSheet, Row, Row, decimal)

```
public static (BigInteger, IDictionary<int, int>) CalculateCraftCost(int iconId, int
relationship, MaterialItemSheet materialItemSheet,
CustomEquipmentCraftRecipeSheet.Row recipeRow,
CustomEquipmentCraftRelationshipSheet.Row relationshipRow,
decimal iconCostMultiplier)
```

Parameters

`iconId` int

`relationship` int

`materialItemSheet` MaterialItemSheet

`recipeRow` CustomEquipmentCraftRecipeSheet.Row

`relationshipRow` CustomEquipmentCraftRelationshipSheet.Row

`iconCostMultiplier` decimal

Returns

(BigInteger, IDictionary<int, int>)

SelectCp(Row, IRandom)

Picks a CP group by its ratio, then a total CP within it.

```
public static long SelectCp(CustomEquipmentCraftRelationshipSheet.Row
relationshipRow, IRandom random)
```

Parameters

`relationshipRow` CustomEquipmentCraftRelationshipSheet.Row

`random` IRandom

Returns

long

Class InventoryExtensions

Namespace: Nekoyume.Helper

Assembly: Lib9c.dll

```
public static class InventoryExtensions
```

Inheritance

object ← InventoryExtensions

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Methods

GetEquippedFullCostumeOrArmorId(Inventory)

```
public static int GetEquippedFullCostumeOrArmorId(this Inventory inventory)
```

Parameters

`inventory` Inventory

Returns

int

MintItem(Inventory, Row, int, bool, IRandom)

Create and Add item.

```
public static void MintItem(this Inventory inventory, ItemSheet.Row itemRow, int
quantity, bool tradable, IRandom random)
```

Parameters

`inventory` Inventory

Inventory>

`itemRow` ItemSheet.Row

ItemSheet.Row

`quantity` int

create item count

`tradable` bool

item is tradable

`random` IRandom

Libplanet.Action.IRandom>

UseActionPoint(Inventory, long, int, bool, MaterialItemSheet, long)

```
public static long UseActionPoint(this Inventory inventory, long originalAp, int
requiredAp, bool chargeAp, MaterialItemSheet materialItemSheet, long blockIndex)
```

Parameters

`inventory` Inventory

`originalAp` long

`requiredAp` int

`chargeAp` bool

`materialItemSheet` MaterialItemSheet

`blockIndex` long

Returns

long

Class ItemOptionHelper

Namespace: Nekoyume.Helper

Assembly: Lib9c.dll

```
public static class ItemOptionHelper
```

Inheritance

object ← ItemOptionHelper

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Methods

GetStatOptionRows(Row, EquipmentItemOptionSheet, ItemUsable)

```
public static List<EquipmentItemOptionSheet.Row>
GetStatOptionRows(EquipmentItemSubRecipeSheetV2.Row subRecipeRow,
EquipmentItemOptionSheet itemOptionSheet, ItemUsable itemUsable)
```

Parameters

`subRecipeRow` EquipmentItemSubRecipeSheetV2.Row

`itemOptionSheet` EquipmentItemOptionSheet

`itemUsable` ItemUsable

Returns

List<EquipmentItemOptionSheet.Row>

GetStatOptionRows(long, ItemUsable, EquipmentItemSubRecipeSheetV2, EquipmentItemOptionSheet)

```
public static List<EquipmentItemOptionSheet.Row> GetStatOptionRows(long subRecipeId,
ItemUsable itemUsable, EquipmentItemSubRecipeSheetV2 subRecipeSheet,
EquipmentItemOptionSheet itemOptionSheet)
```

Parameters

`subRecipeId` long

`itemUsable` ItemUsable

`subRecipeSheet` EquipmentItemSubRecipeSheetV2

`itemOptionSheet` EquipmentItemOptionSheet

Returns

List<EquipmentItemOptionSheet.Row>

TryGet(ItemUsable, out ItemOptionInfo)

```
public static bool TryGet(ItemUsable itemUsable, out ItemOptionInfo itemOptionInfo)
```

Parameters

`itemUsable` ItemUsable

`itemOptionInfo` ItemOptionInfo

Returns

bool

Class ItemOptionInfo

Namespace: Nekoyume.Helper

Assembly: Lib9c.dll

```
public class ItemOptionInfo
```

Inheritance

object ← ItemOptionInfo

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

ItemOptionInfo(Equipment)

```
public ItemOptionInfo(Equipment equipment)
```

Parameters

`equipment` Equipment

ItemOptionInfo(ItemUsable)

```
public ItemOptionInfo(ItemUsable itemUsable)
```

Parameters

`itemUsable` ItemUsable

Fields

CP

```
public readonly long CP
```

Field Value

long

MainStat

```
public readonly (StatType type, long baseValue, long totalValue) MainStat
```

Field Value

(StatType type, long baseValue, long totalValue)

OptionCountFromCombination

```
public readonly int OptionCountFromCombination
```

Field Value

int

SkillOptions

```
public readonly List<(SkillSheet.Row skillRow, long power, int chance, int
statPowerRatio, StatType refStatType)> SkillOptions
```

Field Value

List<(SkillSheet.Row skillRow, long power, int chance, int statPowerRatio, StatType refStatType)>

StatOptions

```
public readonly List<(StatType type, long value, int count)> StatOptions
```

Field Value

List<(StatType type, long value, int count)>

Class NumberConversionHelper

Namespace: Nekoyume.Helper

Assembly: Lib9c.dll

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```
public static class NumberConversionHelper
```

Inheritance

object ← NumberConversionHelper

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Methods

SafeBigIntegerToInt32(BigInteger)

Safely converts BigInteger to int32, clamping to int32 bounds to prevent overflow.

```
public static int SafeBigIntegerToInt32(BigInteger value)
```

Parameters

`value` BigInteger

The BigInteger value to convert.

Returns

int

The converted int32 value, clamped to int32 bounds.

SafeDecimalToInt32(decimal)

Safely converts decimal to int32, clamping to int32 bounds to prevent overflow.

```
public static int SafeDecimalToInt32(decimal value)
```

Parameters

`value` decimal

The decimal value to convert.

Returns

int

The converted int32 value, clamped to int32 bounds.

SafeDecimalToInt64(decimal)

Safely converts decimal to long, clamping to long bounds to prevent overflow.

```
public static long SafeDecimalToInt64(decimal value)
```

Parameters

`value` decimal

The decimal value to convert.

Returns

long

The converted long value, clamped to long bounds.

Class PetHelper

Namespace: Nekoyume.Helper

Assembly: Lib9c.dll

```
public static class PetHelper
```

Inheritance

object ← PetHelper

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Methods

CalculateDiscountedHourglass(long, int, PetState, PetOptionSheet)

```
public static int CalculateDiscountedHourglass(long diff, int hourglassPerBlock,
PetState petState, PetOptionSheet petOptionSheet)
```

Parameters

`diff` long

`hourglassPerBlock` int

`petState` PetState

`petOptionSheet` PetOptionSheet

Returns

int

CalculateDiscountedMaterialCost(FungibleAssetValue, PetState, PetOptionSheet)

```
public static FungibleAssetValue CalculateDiscountedMaterialCost(FungibleAssetValue
originalCost, PetState petState, PetOptionSheet petOptionSheet)
```

Parameters

`originalCost` FungibleAssetValue

`petState` PetState

`petOptionSheet` PetOptionSheet

Returns

FungibleAssetValue

CalculateEnhancementCost(PetCostSheet, int, int, int)

```
public static (int ncgQuantity, int soulStoneQuantity)
CalculateEnhancementCost(PetCostSheet costSheet, int petId, int currentLevel,
int targetLevel)
```

Parameters

`costSheet` PetCostSheet

`petId` int

`currentLevel` int

`targetLevel` int

Returns

(int challengerScoreDelta, int defenderScoreDelta)

CalculateReducedBlockOnCraft(long, long, PetState, PetOptionSheet)

```
public static long CalculateReducedBlockOnCraft(long originalBlock, long
minimumBlock, PetState petState, PetOptionSheet petOptionSheet)
```

Parameters

`originalBlock` long

`minimumBlock` long

`petState` PetState

`petOptionSheet` PetOptionSheet

Returns

long

GetBonusOptionProbability(int, PetState, PetOptionSheet)

```
public static int GetBonusOptionProbability(int originalRatio, PetState petState,
PetOptionSheet petOptionSheet)
```

Parameters

`originalRatio` int

`petState` PetState

`petOptionSheet` PetOptionSheet

Returns

int

GetSoulstoneCurrency(string)

```
public static Currency GetSoulstoneCurrency(string ticker)
```

Parameters

`ticker` string

Returns

Currency

Class RuneEnhancementCostBoost

Namespace: Nekoyume.Helper

Assembly: Lib9c.dll

A BoostScheduleSheet already checked to apply to the enhancement cost of one rune, RuneId: an enhancement cost row was active and the rune was summonable when it was resolved. Only ResolveEnhancementCostBoost(BoostScheduleSheet?, int, long, Func<RuneSummonSheet?>) creates one, and TryEnhancement(int, Row, IRandom, int, out LevelUpResult, RuneEnhancementCostBoost?, long) and ApplyEnhancementCostBoost(RuneCostData, int, int, RuneEnhancementCostBoost?, long) ignore it for any other rune, so a boost only ever discounts the rune it was resolved for — neither handing over the raw patched sheet nor reusing a boost after switching runes can skip the check.

```
public sealed class RuneEnhancementCostBoost
```

Inheritance

object ← RuneEnhancementCostBoost

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.ReferenceEquals(object, object) , object.ToString()

Properties

RuneId

The rune this boost was resolved for, and the only one it adjusts.

```
public int RuneId { get; }
```

Property Value

int

Class RuneHelper

Namespace: Nekoyume.Helper

Assembly: Lib9c.dll

```
public static class RuneHelper
```

Inheritance

object ← RuneHelper

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Fields

DailyRewardRune

```
public static readonly Currency DailyRewardRune
```

Field Value

Currency

StakeRune

```
public static readonly Currency StakeRune
```

Field Value

Currency

Methods

ApplyEnhancementCostBoost(RuneCostData, int, int, RuneEnhancementCostBoost?, long)

Applies the SummonRuneEnhancementCrystalCost and SummonRuneEnhancementRuneStoneCost rows active at `blockIndex` to the cost of one try that reaches `targetLevel`.

```
public static RuneCostSheet.RuneCostData
ApplyEnhancementCostBoost(RuneCostSheet.RuneCostData cost, int runeId, int
targetLevel, RuneEnhancementCostBoost? boost, long blockIndex)
```

Parameters

`cost` RuneCostSheet.RuneCostData

The sheet's cost for `targetLevel`.

`runeId` int

The rune `cost` belongs to. `boost` applies only when it was resolved for this rune.

`targetLevel` int

The level the try attempts to reach, matched against `target_id`.

`boost` RuneEnhancementCostBoost

What ResolveEnhancementCostBoost(BoostScheduleSheet?, int, long, Func<RuneSummonSheet?>) returned for the rune, or `null` for none. Ignored when resolved for a rune other than `runeId`.

`blockIndex` long

The block being evaluated.

Returns

RuneCostSheet.RuneCostData

`cost` itself when no row applies; otherwise a copy whose crystal and rune stone quantities are adjusted by ApplyEnhancementCostBoostQuantity(Row?, int). NCG quantity and success rate are always copied unchanged.

Remarks

Callers showing a cost — or a maximum try count — should pass every per-level cost through this, with the boost from ResolveEnhancementCostBoost(BoostScheduleSheet?, int, long, Func<RuneSummonSheet?>), so that they agree with RuneEnhancement.

ApplyEnhancementCostBoostQuantity(Row?, int)

Applies one BoostScheduleSheet row to one enhancement cost quantity.

```
public static int ApplyEnhancementCostBoostQuantity(BoostScheduleSheet.Row? boost,
int baseCost)
```

Parameters

`boost` BoostScheduleSheet.Row

The row to apply, or `null` for none.

`baseCost` int

The sheet's quantity.

Returns

int

The adjusted quantity, kept within `[1, baseCost]` when `baseCost` is positive. A non-positive `baseCost` is returned unchanged.

Remarks

Discount only: a row can never raise a cost. Raising one is not an event anyone has asked for, and it would let a mistyped row push the per-try sum in TryEnhancement(int, Row, IRandom, int, out LevelUpResult, RuneEnhancementCostBoost?, long) past its type's maximum, where it wraps negative and the cost is skipped instead of charged.

At least 1: rounding down a small cost (4 stones × 0.2 = 0.8) would otherwise make the try free, turning a discount into a giveaway that the row never declared. A free enhancement has to be configured in RuneCostSheet itself. A cost the sheet already sets to 0 stays 0, so a boost never creates a cost either.

ApplyEnhancementCostBoostQuantity(Row?, long)

ApplyEnhancementCostBoostQuantity(Row?, int) for the 64-bit crystal quantity: the same discount-only, at-least-1 rule.

```
public static long ApplyEnhancementCostBoostQuantity(BoostScheduleSheet.Row? boost,
long baseCost)
```

Parameters

`boost` BoostScheduleSheet.Row

The row to apply, or `null` for none.

`baseCost` long

The sheet's quantity.

Returns

long

The adjusted quantity, kept within `[1, baseCost]` when `baseCost` is positive. A non-positive `baseCost` is returned unchanged.

CalculateRuneLevelBonus(AllRuneState, RuneListSheet, RuneLevelBonusSheet)

```
public static int CalculateRuneLevelBonus(AllRuneState allRuneState, RuneListSheet
runeListSheet, RuneLevelBonusSheet runeLevelBonusSheet)
```

Parameters

`allRuneState` AllRuneState

`runeListSheet` RuneListSheet

`runeLevelBonusSheet` RuneLevelBonusSheet

Returns

int

CalculateStakeReward(FungibleAssetValue, int)

```
public static FungibleAssetValue CalculateStakeReward(FungibleAssetValue
stakeAmount, int rate)
```

Parameters

`stakeAmount` FungibleAssetValue

`rate` int

Returns

FungibleAssetValue

GetRuneOptions(IEnumerable<RuneSlotInfo>, AllRuneState, RuneOptionSheet)

```
public static List<RuneOptionSheet.Row.RuneOptionInfo>
GetRuneOptions(IEnumerable<RuneSlotInfo> runeInfos, AllRuneState runeStates,
RuneOptionSheet runeOptionSheet)
```

Parameters

`runeInfos` IEnumerable<RuneSlotInfo>

`runeStates` AllRuneState

`runeOptionSheet` RuneOptionSheet

Returns

List<RuneOptionSheet.Row.RuneOptionInfo>

GetRuneOptions(IEnumerable<RuneInfo>, AllRuneState, RuneOptionSheet)

```
public static List<RuneOptionSheet.Row.RuneOptionInfo>
GetRuneOptions(IEnumerable<RuneWeightSheet.RuneInfo> runeInfos, AllRuneState
runeStates, RuneOptionSheet runeOptionSheet)
```

Parameters

`runeInfos` IEnumerable<RuneWeightSheet.RuneInfo>

`runeStates` AllRuneState

`runeOptionSheet` RuneOptionSheet

Returns

List<RuneOptionSheet.Row.RuneOptionInfo>

ResolveEnhancementCostBoost(BoostScheduleSheet?, int, long, Func<RuneSummonSheet?>)

Decides whether a rune's enhancement cost may be adjusted at all, and returns what to pass to TryEnhancement(int, Row, IRandom, int, out LevelUpResult, RuneEnhancementCostBoost?, long) and ApplyEnhancementCostBoost(RuneCostData, int, int, RuneEnhancementCostBoost?, long).

```
public static RuneEnhancementCostBoost?
ResolveEnhancementCostBoost(BoostScheduleSheet? boostScheduleSheet, int runeId, long
blockIndex, Func<RuneSummonSheet?> getRuneSummonSheet)
```

Parameters

`boostScheduleSheet` BoostScheduleSheet

The patched BoostScheduleSheet, or `null` when the chain has none.

`runeId` int

The rune being enhanced.

`blockIndex` long

The block being evaluated.

`getRuneSummonSheet` Func<RuneSummonSheet>

Loads the RuneSummonSheet, returning `null` when it is absent or unreadable. Called only when an enhancement cost row is active at `blockIndex`, so that evaluation without one reads no extra state.

Returns

RuneEnhancementCostBoost

A RuneEnhancementCostBoost over `boostScheduleSheet` when a SummonRuneEnhancementCrystalCost or SummonRuneEnhancementRuneStoneCost row is active at `blockIndex` and `runeId` is a recipe id of any RuneSummonSheet group; otherwise `null`, meaning no adjustment.

Remarks

The summon pool is derived from the sheet rather than listed, so the adjustment follows later changes to it. A recipe counts whatever its ratio and whether or not its group is currently sold. A missing RuneSummonSheet means no rune is summonable, never an error.

ToCurrency(Row)

```
public static Currency ToCurrency(RuneSheet.Row runeRow)
```

Parameters

`runeRow` RuneSheet.Row

Returns

Currency

ToFungibleAssetValue(Row, int)

```
public static FungibleAssetValue ToFungibleAssetValue(RuneSheet.Row runeRow,
int quantity)
```

Parameters

`runeRow` RuneSheet.Row

`quantity` int

Returns

FungibleAssetValue

TryEnhancement(int, Row, IRandom, int, out LevelUpResult, RuneEnhancementCostBoost?, long)

Simulates `tryCount` enhancement tries from `startRuneLevel`, summing the cost of every try and drawing one random number per try for its success.

```
public static bool TryEnhancement(int startRuneLevel, RuneCostSheet.Row costRow,
IRandom random, int tryCount, out RuneEnhancement.LevelUpResult levelUpResult,
RuneEnhancementCostBoost? boost = null, long blockIndex = 0)
```

Parameters

`startRuneLevel` int

The rune level before the first try.

`costRow` RuneCostSheet.Row

The rune's cost row.

`random` IRandom

The random source; one draw per try.

`tryCount` int

The number of tries.

`levelUpResult` RuneEnhancement.LevelUpResult

The level-ups and the summed cost.

`boost` RuneEnhancementCostBoost

What ResolveEnhancementCostBoost(BoostScheduleSheet?, int, long, Func<RuneSummonSheet?>) returned for this rune, or `null` for none. Adjusts each try's crystal and rune stone cost; see ApplyEnhancementCostBoost(RuneCostData, int, int, RuneEnhancementCostBoost?, long). Ignored, not rejected, when it was resolved for a rune other than `costRow`'s, so a preview holding on to a stale boost shows the undiscounted cost the chain would charge.

`blockIndex` long

The block being evaluated, against which `boost` rows are matched. Ignored when `boost` is `null`.

Returns

bool

`false` when a try reaches a level `costRow` has no cost for.

Class SummonHelper

Namespace: Nekoyume.Helper

Assembly: Lib9c.dll

```
public static class SummonHelper
```

Inheritance

object ← SummonHelper

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Fields

AllowedSummonCount

Array of allowed summon counts. Only these values are permitted for summon actions.

```
public static readonly int[] AllowedSummonCount
```

Field Value

int[]

Methods

ApplyGuaranteeBoost(Row, int, int)

Applies a BoostScheduleSheet row to a configured guarantee count.

```
public static int ApplyGuaranteeBoost(BoostScheduleSheet.Row guaranteeBoost, int
guaranteeCount, int summonCount)
```

Parameters

`guaranteeBoost` BoostScheduleSheet.Row

The row to apply.

`guaranteeCount` int

The guarantee count the summon sheet configures.

`summonCount` int

The number of summons, bonus included.

Returns

int

The adjusted count, kept below `summonCount` so that at least one draw stays a normal draw — unless the sheet itself already configures more.

Remarks

A boost adjusts a guarantee but never creates one, since a guarantee that exists only during an event reads as one taken away when the event ends. That also covers a group configured with a count of 0: its guaranteed-grade path has never run, so starting it mid-event could surface a grade with no eligible recipe and fail every summon.

CalculateSummonCount(int)

Calculates the actual summon count by applying the 10+1 bonus rule. For every 10 summons, 1 additional summon is granted as a bonus.

```
public static int CalculateSummonCount(int count)
```

Parameters

`count` int

Base summon count

Returns

int

Total summon count including bonus

CheckSummonCountIsValid(int)

Validates if the given summon count is allowed. Only counts in AllowedSummonCount array (1, 10, 100) are permitted.

```
public static bool CheckSummonCountIsValid(int count)
```

Parameters

`count` int

Summon count to validate

Returns

bool

True if the count is allowed, false otherwise

GetGuaranteeSettings(Row, int, Row)

Determines guarantee settings based on summon count and SummonSheet.Row configuration.

```
public static (bool useGuarantee, int minimumGrade, int guaranteeCount)
GetGuaranteeSettings(SummonSheet.Row summonRow, int summonCount,
BoostScheduleSheet.Row guaranteeBoost = null)
```

Parameters

`summonRow` SummonSheet.Row

The summon row containing guarantee settings

`summonCount` int

The number of summons

`guaranteeBoost` BoostScheduleSheet.Row

The BoostScheduleSheet row adjusting this group's guarantee count at the current block, or `null` for none. See ApplyGuaranteeBoost(Row, int, int).

Returns

(bool useGuarantee, int minimumGrade, int guaranteeCount)

A tuple containing (useGuarantee, minimumGrade, guaranteeCount)

GetSummonRecipeIdByRandom(Row, IRandom)

Gets a random recipe ID from the summon row based on weighted ratios. Uses cumulative ratio calculation to select recipes according to their configured probabilities.

```
public static int GetSummonRecipeIdByRandom(SummonSheet.Row summonRow,
IRandom random)
```

Parameters

`summonRow` SummonSheet.Row

Summon sheet row containing recipes and their ratios

`random` IRandom

Random number generator

Returns

int

Selected recipe ID

GetSummonRecipeIdsWithGradeGuarantee(Row, int, IRandom, CostumeItemSheet, EquipmentItemRecipeSheet, Row)

Gets summon recipe IDs with grade guarantee system for costumes. Applies different guarantee settings for 11 summons vs 110 summons. This method processes items one by one to maintain the same random call order as the original implementation. Uses grade guarantee settings from SummonSheet.Row.

```
public static List<int> GetSummonRecipeIdsWithGradeGuarantee(SummonSheet.Row
summonRow, int summonCount, IRandom random, CostumeItemSheet costumeItemSheet,
EquipmentItemRecipeSheet equipmentItemRecipeSheet, BoostScheduleSheet.Row
guaranteeBoost = null)
```

Parameters

`summonRow` SummonSheet.Row

Summon sheet row containing recipes, ratios, and grade guarantee settings

`summonCount` int

Number of items to summon

`random` IRandom

Random number generator

`costumeItemSheet` CostumeItemSheet

Costume item sheet to check grades

`equipmentItemRecipeSheet` EquipmentItemRecipeSheet

Equipment item recipe sheet (not used for costumes, can be null)

`guaranteeBoost` BoostScheduleSheet.Row

Guarantee count adjustment, or `null` for none.

Returns

List<int>

List of recipe IDs with grade guarantee applied

GetSummonRecipeIdsWithGradeGuarantee(Row, int, IRandom, EquipmentItemSheet, EquipmentItemRecipeSheet, Row)

Gets summon recipe IDs with grade guarantee system based on summon count. Applies different guarantee settings for 11 summons vs 110 summons. This method processes items one by one to maintain the same random call order as the original implementation. Uses grade guarantee settings from SummonSheet.Row.

```
public static List<int> GetSummonRecipeIdsWithGradeGuarantee(SummonSheet.Row
summonRow, int summonCount, IRandom random, EquipmentItemSheet equipmentItemSheet,
EquipmentItemRecipeSheet equipmentItemRecipeSheet, BoostScheduleSheet.Row
guaranteeBoost = null)
```

Parameters

`summonRow` SummonSheet.Row

Summon sheet row containing recipes, ratios, and grade guarantee settings

`summonCount` int

Number of items to summon

`random` IRandom

Random number generator

`equipmentItemSheet` EquipmentItemSheet

Equipment item sheet to check grades

`equipmentItemRecipeSheet` EquipmentItemRecipeSheet

Equipment item recipe sheet to map recipe IDs to equipment IDs

`guaranteeBoost` BoostScheduleSheet.Row

Guarantee count adjustment, or `null` for none.

Returns

List<int>

List of recipe IDs with grade guarantee applied

Struct SynthesizeResult

Namespace: Nekoyume.Helper

Assembly: Lib9c.dll

Represents the result of the synthesis.

```
public struct SynthesizeResult
```

Inherited Members
ValueType.Equals(object) , ValueType.GetHashCode() , ValueType.ToString() , object.Equals(object, object) , object.GetType() , object.ReferenceEquals(object, object)

Fields

IsSuccess

```
public bool IsSuccess
```

Field Value

bool

ItemBase

```
public ItemBase ItemBase
```

Field Value

ItemBase

RecipeId

```
public int RecipeId
```

Field Value

int

SubRecipeId

```
public int SubRecipeId
```

Field Value

int

Properties

IsEquipment

```
public bool IsEquipment { get; }
```

Property Value

bool

Class SynthesizeSimulator

Namespace: Nekoyume.Helper

Assembly: Lib9c.dll

A class that simulates the synthesis of items. SynthesizeSimulator.InputData

```
public static class SynthesizeSimulator
```

Inheritance

object ← SynthesizeSimulator

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Methods

GetCostumeMaterialList(List<Guid>, AvatarState, Grade, ItemSubType, string)

```
public static List<ItemBase> GetCostumeMaterialList(List<Guid> materialIds,
AvatarState avatarState, Grade grade, ItemSubType itemSubType, string addressesHex)
```

Parameters

`materialIds` List<Guid>

`avatarState` AvatarState

`grade` Grade

`itemSubType` ItemSubType

`addressesHex` string

Returns

List<ItemBase>

GetEquipmentMaterialList(List<Guid>, AvatarState, long, Grade, ItemSubType, string)

```
public static List<ItemBase> GetEquipmentMaterialList(List<Guid> materialIds,
AvatarState avatarState, long blockIndex, Grade grade, ItemSubType itemSubType,
string addressesHex)
```

Parameters

`materialIds` List<Guid>

`avatarState` AvatarState

`blockIndex` long

`grade` Grade

`itemSubType` ItemSubType

`addressesHex` string

Returns

List<ItemBase>

GetItemGuid(ItemBase)

Get the item id from the item.

```
public static List<Guid> GetItemGuid(ItemBase itemBase)
```

Parameters

`itemBase` ItemBase

item to get id

Returns

List<Guid>

item id

Exceptions

ArgumentException

GetItemGuids(IEnumerable<ItemBase>)

Get the item id from the item.

```
public static List<Guid> GetItemGuids(IEnumerable<ItemBase> itemBases)
```

Parameters

`itemBases` IEnumerable<ItemBase>

items to get id

Returns

List<Guid>

item id list

Exceptions

ArgumentException

GetMaterialList(List<Guid>, AvatarState, long, Grade, ItemSubType, string)

```
public static List<ItemBase> GetMaterialList(List<Guid> materialIds,
AvatarState avatarState, long blockIndex, Grade grade, ItemSubType itemSubType,
string addressesHex)
```

Parameters

`materialIds` List<Guid>

`avatarState` AvatarState

`blockIndex` long

`grade` Grade

`itemSubType` ItemSubType

`addressesHex` string

Returns

List<ItemBase>

GetRequiredCount(int, Row?)

Returns how many materials one synthesis consumes, after a BoostScheduleSheet adjustment.

```
public static int GetRequiredCount(int requiredCount, BoostScheduleSheet.Row?
requiredCountBoost)
```

Parameters

`requiredCount` int

The `required_count` that SynthesizeSheet configures.

`requiredCountBoost` BoostScheduleSheet.Row

The row found for SynthesizeRequiredCount and the material grade at the current block, or `null` for none.

Returns

int

`requiredCount` itself when there is no row or it is not positive; otherwise the adjusted count, rounded down but never below 1.

Remarks

The floor of 1 keeps a discount from making a synthesis free (and the division by the count from failing): `1 × 0.6` stays `1`. A count the sheet configures as 0 or less is left alone so that it fails exactly as it did without the boost.

GetSynthesizeResultPool(Grade, ItemSubType, CostumeItemSheet)

get expected result item pool (Costume)

```
public static HashSet<int> GetSynthesizeResultPool(Grade sourceGrade, ItemSubType subType, CostumeItemSheet sheet)
```

Parameters

`sourceGrade` Grade

grade of material items

`subType` ItemSubType

subtype of material items

`sheet` CostumeItemSheet

costume item sheet to use

Returns

HashSet<int>

expected result item pool of costume

GetSynthesizeResultPool(Grade, ItemSubType, EquipmentItemSheet)

get expected result item pool (Equipment)

```
public static HashSet<int> GetSynthesizeResultPool(Grade sourceGrade, ItemSubType
subType, EquipmentItemSheet sheet)
```

Parameters

`sourceGrade` Grade

grade of material items

`subType` ItemSubType

subtype of material items

`sheet` EquipmentItemSheet

equipment item sheet to use

Returns

HashSet<int>

expected result item pool of equipment

GetSynthesizeResultPool(HashSet<Grade>, ItemSubType, CostumeItemSheet)

Returns a list of items that may come out as a result of that synthesis.

```
public static HashSet<(int, Grade)> GetSynthesizeResultPool(HashSet<Grade>
sourceGrades, ItemSubType subType, CostumeItemSheet sheet)
```

Parameters

`sourceGrades` HashSet<Grade>

grades of material items

`subType` ItemSubType

excepted FullCostume,Title

`sheet` CostumeItemSheet

CostumeItemSheet to use

Returns

HashSet<(int, Grade)>

list of items key(int), grade(Grade) tuple

GetSynthesizeResultPool(HashSet<Grade>, ItemSubType, EquipmentItemSheet)

Returns a list of items that may come out as a result of that synthesis.

```
public static HashSet<(int, Grade)> GetSynthesizeResultPool(HashSet<Grade>
sourceGrades, ItemSubType subType, EquipmentItemSheet sheet)
```

Parameters

`sourceGrades` HashSet<Grade>

grades of material items

`subType` ItemSubType

excepted Grimoire,Aura

`sheet` EquipmentItemSheet

EquipmentItemSheet to use

Returns

HashSet<(int, Grade)>

list of items key(int), grade(Grade) tuple

GetTargetGrade(Grade)

Get the target grade of the item: the grade one step above the given one.

```
public static Grade GetTargetGrade(Grade grade)
```

Parameters

`grade` Grade

grade of the item

Returns

Grade

target grade

Remarks

Since there is no upper bound (see the int overload), the returned value is one step above whatever it is given and so need not be a defined Grade member. Callers that hand it to a sheet lookup are fine — an unlisted grade simply matches no row. Callers that switch on it should keep a default arm.

Exceptions

ArgumentOutOfRangeException

Thrown when `grade` is below the first grade.

GetTargetGrade(int)

Get the target grade of the item: the grade id one step above the given one.

```
public static int GetTargetGrade(int gradeId)
```

Parameters

`gradeId` int

grade id of the item

Returns

int

target grade id

Remarks

There is deliberately no upper bound here. Whether the grade above exists is a question the item sheets answer, and both callers inside this class ask them: the result-pool lookups fall back to the source grade when the pool for the grade above comes back empty. Capping here as well would only mean editing this method every time a grade is added. Callers outside this class have to arrange that fallback themselves.

Exceptions

ArgumentOutOfRangeException

Thrown when `gradeId` is below the first grade, or so high that the grade above it does not fit in an int.

GetUpgradeGrade(Grade, ItemSubType, CostumeItemSheet)

Returns the grade of the item that can be obtained by synthesizing the item. (Costume)

```
public static Grade GetUpgradeGrade(Grade grade, ItemSubType subType,
CostumeItemSheet sheet)
```

Parameters

`grade` Grade

source grade

`subType` ItemSubType

subtype of material items

`sheet` CostumeItemSheet

costume item sheet to use

Returns

Grade

grade of the item that can be obtained by synthesizing the item

GetUpgradeGrade(Grade, ItemSubType, EquipmentItemSheet)

Returns the grade of the item that can be obtained by synthesizing the item. (Equipment)

```
public static Grade GetUpgradeGrade(Grade grade, ItemSubType subType,
EquipmentItemSheet sheet)
```

Parameters

`grade` Grade

source grade

`subType` ItemSubType

subtype of material items

`sheet` EquipmentItemSheet

equipment item sheet to use

Returns

Grade

grade of the item that can be obtained by synthesizing the item

GetWeight(int, SynthesizeWeightSheet)

Returns the weight of the item that can be obtained by synthesizing the item.

```
public static int GetWeight(int itemId, SynthesizeWeightSheet sheet)
```

Parameters

`itemId` int

item id of material item

`sheet` SynthesizeWeightSheet

SynthesizeWeightSheet to use

Returns

int

weight of the item that can be obtained by synthesizing the item

Simulate(InputData)

Simulate the synthesis of items.

```
public static List<SynthesizeResult> Simulate(SynthesizeSimulator.InputData
inputData)
```

Parameters

`inputData` SynthesizeSimulator.InputData

SynthesizeSimulator.InputData

Returns

List<SynthesizeResult>

Exceptions

SheetRowNotFoundException

NotEnoughMaterialException

Struct SynthesizeSimulator.InputData

Namespace: Nekoyume.Helper

Assembly: Lib9c.dll

Simulate the synthesis of items.

```
public struct SynthesizeSimulator.InputData
```

Inherited Members
ValueType.Equals(object) , ValueType.GetHashCode() , ValueType.ToString() , object.Equals(object, object) , object.GetType() , object.ReferenceEquals(object, object)

Fields

BlockIndex

The block index when action is executed.

```
public long BlockIndex
```

Field Value

long

CostumeItemSheet

The sheet that contains the costume item information. CostumeItemSheet

```
public CostumeItemSheet CostumeItemSheet
```

Field Value

CostumeItemSheet

EquipmentItemOptionSheet

The sheet that contains the equipment item option information. EquipmentItemOption Sheet

```
public EquipmentItemOptionSheet EquipmentItemOptionSheet
```

Field Value

EquipmentItemOptionSheet

EquipmentItemRecipeSheet

The sheet that contains the equipment item recipe information. EquipmentItemRecipeSheet

```
public EquipmentItemRecipeSheet EquipmentItemRecipeSheet
```

Field Value

EquipmentItemRecipeSheet

EquipmentItemSheet

The sheet that contains the equipment item information. EquipmentItemSheet

```
public EquipmentItemSheet EquipmentItemSheet
```

Field Value

EquipmentItemSheet

EquipmentItemSubRecipeSheetV2

The sheet that contains the equipment item sub recipe information. EquipmentItemSub RecipeSheetV2

```
public EquipmentItemSubRecipeSheetV2 EquipmentItemSubRecipeSheetV2
```

Field Value

EquipmentItemSubRecipeSheetV2

Grade

The grade of the material item.

```
public Grade Grade
```

Field Value

Grade

ItemSubType

The subtype of the material item.

```
public ItemSubType ItemSubType
```

Field Value

ItemSubType

MaterialCount

The number of materials.

```
public int MaterialCount
```

Field Value

int

RandomObject

The random object to use. Caution: Must have the same seed as when the action is executed

```
public IRandom RandomObject
```

Field Value

IRandom

RequiredCountBoost

The BoostScheduleSheet row adjusting the required material count of Grade at BlockIndex, or `null` for none. See GetRequiredCount(int, Row?). Caution: Must be the row the action resolved, or the material count may not divide.

```
public BoostScheduleSheet.Row? RequiredCountBoost
```

Field Value

BoostScheduleSheet.Row

SkillSheet

The sheet that contains the skill information. SkillSheet

```
public SkillSheet SkillSheet
```

Field Value

SkillSheet

SynthesizeSheet

The sheet that contains the synthesis information. SynthesizeSheet

```
public SynthesizeSheet SynthesizeSheet
```

Field Value

SynthesizeSheet

SynthesizeWeightSheet

The sheet that contains the synthesis weight information. SynthesizeWeightSheet

```
public SynthesizeWeightSheet SynthesizeWeightSheet
```

Field Value

SynthesizeWeightSheet

Class WorldBossHelper

Namespace: Nekoyume.Helper

Assembly: Lib9c.dll

World boss (raid) rules shared by the raid actions and clients: ranks, tickets, rewards and their BoostScheduleSheet adjustments.

```
public static class WorldBossHelper
```

Inheritance

object ← WorldBossHelper

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Fields

MaxChallengeCount

Number of raid tickets a raider holds after a refill.

```
public const int MaxChallengeCount = 3
```

Field Value

int

MaxWorldBossRewardBoostFactor

Upper bound of a WorldBossBattleReward, WorldBossKillReward or WorldBossRankReward row: each boosted amount never exceeds this many times the amount drawn, whatever the row's `op` and value. It bounds a mistyped value (e.g. a block index pasted into the value cell) to a fivefold over-reward and keeps several boosted amounts added to one inventory slot from overflowing.

```
public const long MaxWorldBossRewardBoostFactor = 5
```

Field Value

long

RefillInterval

Legacy ticket refill interval in blocks, used by CanRefillTicketV1(long, long, long).

```
[Obsolete("Use GameConfigState.DailyWorldBossInterval")]
public const long RefillInterval = 7200
```

Field Value

long

Methods

ApplyRewardBoost(Row, int)

Applies a BoostScheduleSheet row to one granted amount of a world boss reward: the count of one drawn rune, the crystal or the circles.

```
public static int ApplyRewardBoost(BoostScheduleSheet.Row rewardBoost, int amount)
```

Parameters

`rewardBoost` BoostScheduleSheet.Row

The row to apply, or `null` for none.

`amount` int

The amount the reward row and the draws grant.

Returns

int

`amount` unchanged when there is no row or nothing is granted; otherwise Apply(int) of it, capped at MaxWorldBossRewardBoostFactor times `amount`. A result of 0 means the amount is not granted at all.

Remarks

A boost adjusts what a reward grants but never adds a kind the row does not grant — a row with no crystal stays without crystal under `ADD` — since a reward that exists only during an event reads as one taken away when the event ends. `ADD` applies to every amount separately, i.e. once per distinct rune drawn; use `MUL` to scale a reward as a whole. The cap applies to `ADD` too, so `ADD 10` on a single drawn rune grants 5, not 11. Lowering is not bounded: a `MUL` below 1 or a negative `ADD` still lowers an amount, down to 0.

CalculateContribution(BigInteger, long)

Calculates the contribution percentage based on total damage and individual damage.

```
public static decimal CalculateContribution(BigInteger totalDamage, long myDamage)
```

Parameters

`totalDamage` BigInteger

The total damage dealt.

`myDamage` long

The damage dealt by the individual.

Returns

decimal

The contribution percentage rounded to four decimal places.

Exceptions

ArgumentOutOfRangeException

Thrown when totalDamage is lower than 0.

CalculateContributionReward(Row, decimal)

Calculates the contribution reward based on the given contribution percentage.

```
public static (List<(int id, int count)>, List<FungibleAssetValue>)
CalculateContributionReward(WorldBossContributionRewardSheet.Row row,
decimal contribution)
```

Parameters

`row` WorldBossContributionRewardSheet.Row

The row from the WorldBossContributionRewardSheet containing reward details.

`contribution` decimal

The contribution percentage of the player.

Returns

(List<(int id, int count)>, List<FungibleAssetValue>)

A tuple containing a list of item rewards and a list of fungible asset values.

CalculateRank(Row, long)

```
public static int CalculateRank(WorldBossCharacterSheet.Row row, long score)
```

Parameters

`row` WorldBossCharacterSheet.Row

`score` long

Returns

int

CalculateReward(int, int, RuneWeightSheet, IWorldBossRewardSheet, RuneSheet, MaterialItemSheet, IRandom, Row)

Calculates a world boss reward — runes drawn by RuneWeightSheet, crystal and circles — for `rank` of `bossId`.

```
public static (List<FungibleAssetValue> assets, Dictionary<TradableMaterial, int>
materials) CalculateReward(int rank, int bossId, RuneWeightSheet sheet,
IWorldBossRewardSheet rewardSheet, RuneSheet runeSheet, MaterialItemSheet
materialSheet, IRandom random, BoostScheduleSheet.Row rewardBoost = null)
```

Parameters

`rank` int

The rank whose reward row is granted.

`bossId` int

The boss id, as in `WorldBossListSheet.boss_id`.

`sheet` RuneWeightSheet

Weights the drawn runes.

`rewardSheet` IWorldBossRewardSheet

The battle, kill or rank reward sheet the reward row is taken from.

`runeSheet` RuneSheet

Maps a drawn rune id to its currency.

`materialSheet` MaterialItemSheet

Supplies the circle material.

`random` IRandom

Draws the rune count, for a ranged row, and each rune.

`rewardBoost` BoostScheduleSheet.Row

The BoostScheduleSheet row adjusting this reward at the current block, or `null` for none. A caller replaying a reward must pass what the action resolved. See ApplyRewardBoost(Row, int).

Returns

(List<FungibleAssetValue> assets, Dictionary<TradableMaterial, int> materials)

The granted assets, runes first and crystal last, and circles.

Remarks

`rewardBoost` scales what was drawn rather than how many draws are made, so `random` is consumed exactly as without it and everything drawn after this reward (e.g. the next kill reward of the same action) stays the same.

CalculateTicketPrice(Row, RaiderState, Currency)

```
public static FungibleAssetValue CalculateTicketPrice(WorldBossListSheet.Row row,
RaiderState raiderState, Currency currency)
```

Parameters

`row` WorldBossListSheet.Row

`raiderState` RaiderState

`currency` Currency

Returns

FungibleAssetValue

CanRefillTicket(long, long, long, int)

```
public static bool CanRefillTicket(long blockIndex, long refilledIndex, long
startedIndex, int refillInterval)
```

Parameters

`blockIndex` long

`refilledIndex` long

`startedIndex` long

`refillInterval` int

Returns

bool

CanRefillTicketV1(long, long, long)

```
public static bool CanRefillTicketV1(long blockIndex, long refilledIndex,
long startedIndex)
```

Parameters

`blockIndex` long

`refilledIndex` long

`startedIndex` long

Returns

bool

Namespace Nekoyume.Model

Classes

ArenaCharacter

ArenaSkills

CharacterBase

CollectionMap

Represents a collection map with key-value pairs. Supports both Dictionary and List serialization formats for backward compatibility.

Serialization Format: - Dictionary (Legacy): Uses key-value pairs for backward compatibility - List (New): Uses ordered list for better performance and smaller size

Field Order (List Format): 1. version - Serialization version number 2. entries - List of key-value pairs ordered by key

Enemy

EnemyPlayer

FailedAddWorldException

FailedToUnlockWorldException

Player

Player.ExpData

RaidBoss

Skills

WorldInformation

Structs

ActivationKey

ArenaPlayerDigest

Introduced at https://github.com/planetarium/lib9c/pull/1156

FungibleItemValue

Interfaces

IArena

IStage

Struct ActivationKey

Namespace: Nekoyume.Model

Assembly: Lib9c.dll

```
public struct ActivationKey
```

Inherited Members
ValueType.Equals(object) , ValueType.GetHashCode() , ValueType.ToString() , object.Equals(object, object) , object.GetType() , object.ReferenceEquals(object, object)

Fields

DeriveKey

```
public const string DeriveKey = "activated"
```

Field Value

string

Properties

PendingAddress

```
public readonly Address PendingAddress { get; }
```

Property Value

Address

PrivateKey

```
public readonly PrivateKey PrivateKey { get; }
```

Property Value

PrivateKey

Methods

Create(PrivateKey, byte[])

```
public static (ActivationKey, PendingActivationState) Create(PrivateKey privateKey,
byte[] nonce)
```

Parameters

`privateKey` PrivateKey

`nonce` byte[]

Returns

(ActivationKey, PendingActivationState)

Decode(string)

```
public static ActivationKey Decode(string hexWithSlash)
```

Parameters

`hexWithSlash` string

Returns

ActivationKey

Encode()

```
public string Encode()
```

Returns

string

Class ArenaCharacter

Namespace: Nekoyume.Model

Assembly: Lib9c.dll

```
public class ArenaCharacter : ICloneable
```

Inheritance

object ← ArenaCharacter

Implements

ICloneable

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

ArenaCharacter(IArenaSimulator, ArenaPlayerDigest, ArenaSimulatorSheets, bool)

```
public ArenaCharacter(IArenaSimulator simulator, ArenaPlayerDigest digest,
ArenaSimulatorSheets sheets, bool isEnemy = false)
```

Parameters

`simulator` IArenaSimulator

`digest` ArenaPlayerDigest

`sheets` ArenaSimulatorSheets

`isEnemy` bool

ArenaCharacter(IArenaSimulator, ArenaPlayerDigest, ArenaSimulatorSheets, long, List<StatModifier>, bool, bool)

```
public ArenaCharacter(IArenaSimulator simulator, ArenaPlayerDigest
digest, ArenaSimulatorSheets sheets, long hpModifier, List<StatModifier>
collectionModifiers, bool isEnemy = false, bool setExtraValueBuffBeforeGetBuffs
= false)
```

Parameters

`simulator` IArenaSimulator

`digest` ArenaPlayerDigest

`sheets` ArenaSimulatorSheets

`hpModifier` long

`collectionModifiers` List<StatModifier>

`isEnemy` bool

`setExtraValueBuffBeforeGetBuffs` bool

Fields

CriticalMultiplier

```
public const decimal CriticalMultiplier = 1.5
```

Field Value

decimal

RuneSkillCooldownMap

```
public readonly Dictionary<int, int> RuneSkillCooldownMap
```

Field Value

Dictionary<int, int>

Simulator

```
public readonly IArenaSimulator Simulator
```

Field Value

IArenaSimulator

_runeSkills

```
public readonly ArenaSkills _runeSkills
```

Field Value

ArenaSkills

usedSkill

```
public ArenaSkill usedSkill
```

Field Value

ArenaSkill

Properties

ATK

```
public long ATK { get; }
```

Property Value

long

ActionBuffs

```
public IEnumerable<ActionBuff> ActionBuffs { get; }
```

Property Value

IEnumerable<ActionBuff>

AdditionalHP

```
public long AdditionalHP { get; }
```

Property Value

long

ArmorPenetration

```
public long ArmorPenetration { get; }
```

Property Value

long

AttackCount

```
public int AttackCount { get; }
```

Property Value

int

AttackRange

```
public float AttackRange { get; }
```

Property Value

float

Buffs

```
public Dictionary<int, Buff> Buffs { get; }
```

Property Value

Dictionary<int, Buff>

CDMG

```
public long CDMG { get; }
```

Property Value

long

CRI

```
public long CRI { get; }
```

Property Value

long

CharacterId

```
public int CharacterId { get; }
```

Property Value

int

CurrentHP

```
public long CurrentHP { get; set; }
```

Property Value

long

DEF

```
public long DEF { get; }
```

Property Value

long

DRR

```
public long DRR { get; }
```

Property Value

long

DRV

```
public long DRV { get; }
```

Property Value

long

DefenseElementalType

```
public ElementalType DefenseElementalType { get; }
```

Property Value

ElementalType

HIT

```
public long HIT { get; }
```

Property Value

long

HP

```
public long HP { get; }
```

Property Value

long

Id

```
public Guid Id { get; }
```

Property Value

Guid

IsDead

```
public bool IsDead { get; }
```

Property Value

bool

IsEnemy

```
public bool IsEnemy { get; }
```

Property Value

bool

Level

```
public int Level { get; set; }
```

Property Value

int

OffensiveElementalType

```
public ElementalType OffensiveElementalType { get; }
```

Property Value

ElementalType

RunSpeed

```
public float RunSpeed { get; }
```

Property Value

float

SPD

```
public long SPD { get; }
```

Property Value

long

SizeType

```
public SizeType SizeType { get; }
```

Property Value

SizeType

SkillLog

```
public ArenaSkill SkillLog { get; }
```

Property Value

ArenaSkill

StatBuffs

```
public IEnumerable<StatBuff> StatBuffs { get; }
```

Property Value

IEnumerable<StatBuff>

Stats

```
public CharacterStats Stats { get; }
```

Property Value

CharacterStats

Thorn

```
public long Thorn { get; }
```

Property Value

long

Methods

AddBuff(Buff, bool)

```
public IEnumerable<Buff> AddBuff(Buff buff, bool updateImmediate = true)
```

Parameters

`buff` Buff

`updateImmediate` bool

Returns

IEnumerable<Buff>

Clone()

Creates a new object that is a copy of the current instance.

```
public object Clone()
```

Returns

object

A new object that is a copy of this instance.

GetDamage(long, bool)

```
public long GetDamage(long damage, bool considerAttackCount = true)
```

Parameters

`damage` long

`considerAttackCount` bool

Returns

long

Heal(long)

```
public void Heal(long heal)
```

Parameters

`heal` long

IsCritical(bool)

```
public bool IsCritical(bool considerAttackCount = true)
```

Parameters

`considerAttackCount` bool

Returns

bool

IsHit(ArenaCharacter)

```
public virtual bool IsHit(ArenaCharacter caster)
```

Parameters

`caster` ArenaCharacter

Returns

bool

OnPostSkill(ArenaSkill)

```
protected virtual void OnPostSkill(ArenaSkill usedSkill)
```

Parameters

`usedSkill` ArenaSkill

OnPreSkill()

```
protected virtual bool OnPreSkill()
```

Returns

bool

RemoveActionBuff(ActionBuff)

```
public void RemoveActionBuff(ActionBuff removedBuff)
```

Parameters

`removedBuff` ActionBuff

RemoveAllStatBuffs()

Removes all positive stat buffs from the character. Only buffs with a positive value are removed; debuffs (negative value) are preserved.

```
public void RemoveAllStatBuffs()
```

RemoveRecentStatBuff()

```
public void RemoveRecentStatBuff()
```

RemoveStatBuff(StatBuff)

```
public void RemoveStatBuff(StatBuff removedBuff)
```

Parameters

`removedBuff` StatBuff

ResetCurrentHP()

```
protected void ResetCurrentHP()
```

SetRuneSkills(IEnumerable<RuneState>, RuneOptionSheet, SkillSheet)

```
public void SetRuneSkills(IEnumerable<RuneState> runes, RuneOptionSheet
runeOptionSheet, SkillSheet skillSheet)
```

Parameters

`runes` IEnumerable<RuneState>

`runeOptionSheet` RuneOptionSheet

`skillSheet` SkillSheet

SetRuneStats(IEnumerable<RuneState>, RuneOptionSheet, int)

```
public void SetRuneStats(IEnumerable<RuneState> runes, RuneOptionSheet
runeOptionSheet, int runeLevelBonus)
```

Parameters

`runes` IEnumerable<RuneState>

`runeOptionSheet` RuneOptionSheet

runeLevelBonus int

SetRuneV1(List<RuneState>, RuneOptionSheet, SkillSheet)

```
[Obsolete("Use SetRune instead.")]
public void SetRuneV1(List<RuneState> runes, RuneOptionSheet runeOptionSheet,
SkillSheet skillSheet)
```

Parameters

runes List<RuneState>

runeOptionSheet RuneOptionSheet

skillSheet SkillSheet

SetRuneV2(List<RuneState>, RuneOptionSheet, SkillSheet)

```
[Obsolete("Use SetRune instead.")]
public void SetRuneV2(List<RuneState> runes, RuneOptionSheet runeOptionSheet,
SkillSheet skillSheet)
```

Parameters

runes List<RuneState>

runeOptionSheet RuneOptionSheet

skillSheet SkillSheet

Spawn(ArenaCharacter)

```
public void Spawn(ArenaCharacter target)
```

Parameters

`target` ArenaCharacter

SpawnV2(ArenaCharacter)

```
[Obsolete("Use Spawn")]
public void SpawnV2(ArenaCharacter target)
```

Parameters

`target` ArenaCharacter

Tick()

```
public void Tick()
```

Struct ArenaPlayerDigest

Namespace: Nekoyume.Model

Assembly: Lib9c.dll

Introduced at https://github.com/planetarium/lib9c/pull/1156

```
public readonly struct ArenaPlayerDigest : IState
```

Implements
IState

Inherited Members
ValueType.Equals(object) , ValueType.GetHashCode() , ValueType.ToString() , object.Equals(object, object) , object.GetType() , object.ReferenceEquals(object, object)

Constructors

ArenaPlayerDigest(List)

```
public ArenaPlayerDigest(List serialized)
```

Parameters

`serialized` List

ArenaPlayerDigest(AvatarState, AllRuneState, RuneSlotState)

```
public ArenaPlayerDigest(AvatarState avatarState, AllRuneState runes,
RuneSlotState runeSlotState)
```

Parameters

`avatarState` AvatarState

`runes` AllRuneState

`runeSlotState` RuneSlotState

ArenaPlayerDigest(AvatarState, ArenaAvatarState)

```
[Obsolete("Do not use it")]
public ArenaPlayerDigest(AvatarState avatarState, ArenaAvatarState arenaAvatarState)
```

Parameters

`avatarState` AvatarState

`arenaAvatarState` ArenaAvatarState

ArenaPlayerDigest(AvatarState, List<Costume>, List<Equipment>, AllRuneState, RuneSlotState)

```
public ArenaPlayerDigest(AvatarState avatarState, List<Costume> costumes,
List<Equipment> equipments, AllRuneState runes, RuneSlotState runeSlotState)
```

Parameters

`avatarState` AvatarState

`costumes` List<Costume>

`equipments` List<Equipment>

`runes` AllRuneState

`runeSlotState` RuneSlotState

ArenaPlayerDigest(AvatarState, List<Guid>, List<Guid>, AllRuneState, RuneSlotState)

```
public ArenaPlayerDigest(AvatarState avatarState, List<Guid> equipments, List<Guid>
costumes, AllRuneState runes, RuneSlotState runeSlotState)
```

Parameters

`avatarState` AvatarState

`equipments` List<Guid>

`costumes` List<Guid>

`runes` AllRuneState

`runeSlotState` RuneSlotState

Fields

CharacterId

```
public readonly int CharacterId
```

Field Value

int

Costumes

```
public readonly List<Costume> Costumes
```

Field Value

List<Costume>

EarIndex

```
public readonly int EarIndex
```

Field Value

int

Equipments

```
public readonly List<Equipment> Equipments
```

Field Value

List<Equipment>

HairIndex

```
public readonly int HairIndex
```

Field Value

int

LensIndex

```
public readonly int LensIndex
```

Field Value

int

Level

```
public readonly int Level
```

Field Value

int

NameWithHash

```
public readonly string NameWithHash
```

Field Value

string

RuneSlotState

```
public readonly RuneSlotState RuneSlotState
```

Field Value

RuneSlotState

Runes

```
public readonly AllRuneState Runes
```

Field Value

AllRuneState

TailIndex

```
public readonly int TailIndex
```

Field Value

int

Methods

Serialize()

```
public IValue Serialize()
```

Returns

IValue

Class ArenaSkills

Namespace: Nekoyume.Model

Assembly: Lib9c.dll

```
[Serializable]
public class ArenaSkills : IEnumerable<ArenaSkill>, IEnumerable
```

Inheritance

object ← ArenaSkills

Implements

IEnumerable<ArenaSkill>, IEnumerable

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Methods

Add(ArenaSkill)

```
public void Add(ArenaSkill skill)
```

Parameters

`skill` ArenaSkill

Clear()

```
public void Clear()
```

GetCooldown(int)

```
public int GetCooldown(int skillId)
```

Parameters

`skillId` int

Returns

int

GetEnumerator()

Returns an enumerator that iterates through the collection.

```
public IEnumerator<ArenaSkill> GetEnumerator()
```

Returns

IEnumerator<ArenaSkill>

An enumerator that can be used to iterate through the collection.

ReduceCooldown()

```
public void ReduceCooldown()
```

Select(IRandom)

```
public ArenaSkill Select(IRandom random)
```

Parameters

`random` IRandom

Returns

ArenaSkill

SelectWithoutDefaultAttack(IRandom)

```
public ArenaSkill SelectWithoutDefaultAttack(IRandom random)
```

Parameters

`random` IRandom

Returns

ArenaSkill

SetCooldown(int, int)

```
public void SetCooldown(int skillId, int cooldown)
```

Parameters

`skillId` int

`cooldown` int

Class CharacterBase

Namespace: Nekoyume.Model

Assembly: Lib9c.dll

```
[Serializable]
public abstract class CharacterBase : ICloneable
```

Inheritance

object ← CharacterBase

Implements

ICloneable

Derived

Enemy, Player

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

CharacterBase(Simulator, CharacterStats, int, ElementalType, Row, SizeType, float, float)

```
protected CharacterBase(Simulator simulator, CharacterStats stat, int characterId,
ElementalType elementalType, CharacterSheet.Row rowData = null, SizeType sizeType =
SizeType.XL, float attackRange = 4, float runSpeed = 0.3)
```

Parameters

`simulator` Simulator

`stat` CharacterStats

`characterId` int

`elementalType` ElementalType

`rowData` CharacterSheet.Row

`sizeType` SizeType

`attackRange` float

`runSpeed` float

CharacterBase(Simulator, CharacterSheet, int, int, IEnumerable<StatModifier>)

```
protected CharacterBase(Simulator simulator, CharacterSheet characterSheet, int
characterId, int level, IEnumerable<StatModifier> optionalStatModifiers = null)
```

Parameters

`simulator` Simulator

`characterSheet` CharacterSheet

`characterId` int

`level` int

`optionalStatModifiers` IEnumerable<StatModifier>

CharacterBase(CharacterBase)

```
protected CharacterBase(CharacterBase value)
```

Parameters

`value` CharacterBase

CharacterBase(Row)

```
protected CharacterBase(CharacterSheet.Row row)
```

Parameters

`row` CharacterSheet.Row

Fields

BuffSkills

```
public readonly Skills BuffSkills
```

Field Value

Skills

Buffs

```
public readonly Dictionary<int, Buff> Buffs
```

Field Value

Dictionary<int, Buff>

CriticalMultiplier

```
public const decimal CriticalMultiplier = 1.5
```

Field Value

decimal

Id

```
public readonly Guid Id
```

Field Value

Guid

Simulator

```
[NonSerialized]
public Simulator Simulator
```

Field Value

Simulator

Skills

```
public readonly Skills Skills
```

Field Value

Skills

Targets

```
public readonly List<CharacterBase> Targets
```

Field Value

List<CharacterBase>

atkElementType

```
public ElementalType atkElementType
```

Field Value

ElementalType

attackRange

```
public float attackRange
```

Field Value

float

defElementType

```
public ElementalType defElementType
```

Field Value

ElementalType

Properties

ATK

```
public long ATK { get; }
```

Property Value

long

ActionBuffs

```
public IEnumerable<ActionBuff> ActionBuffs { get; }
```

Property Value

IEnumerable<ActionBuff>

AdditionalHP

```
public long AdditionalHP { get; }
```

Property Value

long

ArmorPenetration

```
public long ArmorPenetration { get; }
```

Property Value

long

AttackCount

```
public int AttackCount { get; set; }
```

Property Value

int

AttackCountMax

```
public int AttackCountMax { get; protected set; }
```

Property Value

int

CDMG

```
public long CDMG { get; }
```

Property Value

long

CRI

```
public long CRI { get; }
```

Property Value

long

CharacterId

```
public int CharacterId { get; }
```

Property Value

int

CurrentHP

```
public long CurrentHP { get; set; }
```

Property Value

long

DEF

```
public long DEF { get; }
```

Property Value

long

DRR

```
public long DRR { get; }
```

Property Value

long

DRV

```
public long DRV { get; }
```

Property Value

long

HIT

```
public long HIT { get; }
```

Property Value

long

HP

```
public long HP { get; }
```

Property Value

long

IsDead

```
public bool IsDead { get; }
```

Property Value

bool

Level

```
public int Level { get; set; }
```

Property Value

int

RowData

```
public CharacterSheet.Row RowData { get; }
```

Property Value

CharacterSheet.Row

RunSpeed

```
public float RunSpeed { get; }
```

Property Value

float

SPD

```
public long SPD { get; }
```

Property Value

long

SizeType

```
public SizeType SizeType { get; }
```

Property Value

SizeType

StatBuffs

```
public IEnumerable<StatBuff> StatBuffs { get; }
```

Property Value

IEnumerable<StatBuff>

Stats

```
public CharacterStats Stats { get; }
```

Property Value

CharacterStats

Thorn

```
public long Thorn { get; }
```

Property Value

long

usedSkill

```
public Skill usedSkill { get; set; }
```

Property Value

Skill

Methods

AddBuff(Buff, bool)

Add buff/debuff to target; it means buff/debuff is used by caster. When `Dispel` is used, it can remove prev. debuffs on target. All the removed debuffs will be returned and saved in battle log.

```
public IEnumerable<Buff> AddBuff(Buff buff, bool updateImmediate = true)
```

Parameters

`buff` Buff

`updateImmediate` bool

Returns

IEnumerable<Buff>

An enumerable of removed debuffs from target. `null` will be returned if nothing eliminated.

Clone()

Creates a new object that is a copy of the current instance.

```
public abstract object Clone()
```

Returns

object

A new object that is a copy of this instance.

Die()

```
public void Die()
```

EndTurn()

```
protected virtual void EndTurn()
```

GetChance(int)

```
public bool GetChance(int chance)
```

Parameters

`chance` int

Returns

bool

GetDamage(long, bool)

```
public long GetDamage(long damage, bool considerAttackCount = true)
```

Parameters

`damage` long

`considerAttackCount` bool

Returns

long

Heal(long)

```
public void Heal(long heal)
```

Parameters

`heal` long

InitAI()

```
public virtual void InitAI()
```

InitAIV1()

```
[Obsolete("Use InitAI")]
public void InitAIV1()
```

InitAIV2()

```
[Obsolete("Use InitAI")]
public void InitAIV2()
```

IsCritical(bool)

```
public bool IsCritical(bool considerAttackCount = true)
```

Parameters

`considerAttackCount` bool

Returns

bool

IsHit(CharacterBase)

```
public virtual bool IsHit(CharacterBase caster)
```

Parameters

`caster` CharacterBase

Returns

bool

IsHit(ElementalResult)

```
public bool IsHit(ElementalResult result)
```

Parameters

`result` ElementalResult

Returns

bool

OnDead()

```
protected virtual void OnDead()
```

OnPostSkill(Skill)

```
protected virtual void OnPostSkill(Skill usedSkill)
```

Parameters

`usedSkill` Skill

OnPreSkill()

```
protected virtual bool OnPreSkill()
```

Returns

bool

ReduceSkillCooldown()

```
protected virtual void ReduceSkillCooldown()
```

RemoveActionBuff(ActionBuff)

```
public void RemoveActionBuff(ActionBuff removedBuff)
```

Parameters

`removedBuff` ActionBuff

RemoveAllStatBuffs()

Removes all positive stat buffs from the character. Only buffs with a positive value are removed; debuffs (negative value) are preserved.

```
public void RemoveAllStatBuffs()
```

RemoveRecentStatBuff()

```
public void RemoveRecentStatBuff()
```

RemoveStatBuff(StatBuff)

```
public void RemoveStatBuff(StatBuff removedBuff)
```

Parameters

`removedBuff` StatBuff

ResetCurrentHP()

```
public void ResetCurrentHP()
```

SetSkill()

```
protected virtual void SetSkill()
```

Tick()

```
public void Tick()
```

UseSkill()

```
protected virtual Skill UseSkill()
```

Returns

Skill

Class CollectionMap

Namespace: Nekoyume.Model

Assembly: Lib9c.dll

Represents a collection map with key-value pairs. Supports both Dictionary and List serialization formats for backward compatibility.

Serialization Format: - Dictionary (Legacy): Uses key-value pairs for backward compatibility - List (New): Uses ordered list for better performance and smaller size

Field Order (List Format): 1. version - Serialization version number 2. entries - List of key-value pairs ordered by key

```
[Serializable]
public class CollectionMap : IState, IDictionary<int, int>,
ICollection<KeyValuePair<int, int>>, IEnumerable<KeyValuePair<int, int>>,
IEnumerable, ISerializable
```

Inheritance

object ← CollectionMap

Implements

IState, IDictionary<int, int>, ICollection<KeyValuePair<int, int>>, IEnumerable<KeyValuePair<int, int>>, IEnumerable, ISerializable

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Remarks

This class implements dual serialization support to ensure smooth migration from the legacy Dictionary format to the new List format. The List format provides better performance and smaller serialized data size.

Example usage:

```
// Create collection map
var map = new CollectionMap();
map.Add(1, 100);
```

```
map.Add(2, 200);
// Serialize to List format (new)
var serialized = map.Serialize(); // Returns List
// Deserialize from any format
var deserialized = new CollectionMap(serialized); // Supports both Dictionary
and List
```

Constructors

CollectionMap()

```
public CollectionMap()
```

CollectionMap(Dictionary)

Constructor for backward compatibility with Dictionary format.

```
public CollectionMap(Dictionary serialized)
```

Parameters

`serialized` Dictionary

Dictionary containing serialized data

CollectionMap(IValue)

Constructor for deserialization that supports both Dictionary and List formats.

```
public CollectionMap(IValue serialized)
```

Parameters

`serialized` IValue

Serialized data in either Dictionary or List format

Exceptions

ArgumentNullException

Thrown when serialized is null

ArgumentException

Thrown when serialized format is not supported

Properties

Count

Gets the number of elements contained in the ICollection<T>.

```
public int Count { get; }
```

Property Value

int

The number of elements contained in the ICollection<T>.

IsReadOnly

Gets a value indicating whether the ICollection<T> is read-only.

```
public bool IsReadOnly { get; }
```

Property Value

bool

true if the ICollection<T> is read-only; otherwise, false.

this[int]

Gets or sets the element with the specified key.

```
public int this[int key] { get; set; }
```

Parameters

`key` int

The key of the element to get or set.

Property Value

int

The element with the specified key.

Exceptions

ArgumentNullException

`key` is null.

KeyNotFoundException

The property is retrieved and `key` is not found.

NotSupportedException

The property is set and the IDictionary<TKey, TValue> is read-only.

Keys

Gets an ICollection<T> containing the keys of the IDictionary<TKey, TValue>.

```
public ICollection<int> Keys { get; }
```

Property Value

ICollection<int>

An ICollection<T> containing the keys of the object that implements IDictionary<TKey, TValue>.

Values

Gets an ICollection<T> containing the values in the IDictionary<TKey, TValue>.

```
public ICollection<int> Values { get; }
```

Property Value

ICollection<int>

An ICollection<T> containing the values in the object that implements IDictionary<TKey, TValue>.

Methods

Add(KeyValuePair<int, int>)

Adds an item to the ICollection<T>.

```
public void Add(KeyValuePair<int, int> item)
```

Parameters

`item` KeyValuePair<int, int>

The object to add to the ICollection<T>.

Exceptions

NotSupportedException

The ICollection<T> is read-only.

Add(int, int)

Adds an element with the provided key and value to the IDictionary<TKey, TValue>.

```
public void Add(int key, int value)
```

Parameters

`key` int

The object to use as the key of the element to add.

`value` int

The object to use as the value of the element to add.

Exceptions

ArgumentNullException

`key` is null.

ArgumentException

An element with the same key already exists in the IDictionary<TKey, TValue>.

NotSupportedException

The IDictionary<TKey, TValue> is read-only.

Clear()

Removes all items from the ICollection<T>.

```
public void Clear()
```

Exceptions

NotSupportedException

The ICollection<T> is read-only.

Contains(KeyValuePair<int, int>)

Determines whether the ICollection<T> contains a specific value.

```
public bool Contains(KeyValuePair<int, int> item)
```

Parameters

`item` KeyValuePair<int, int>

The object to locate in the ICollection<T>.

Returns

bool

true if `item` is found in the ICollection<T>; otherwise, false.

ContainsKey(int)

Determines whether the IDictionary<TKey, TValue> contains an element with the specified key.

```
public bool ContainsKey(int key)
```

Parameters

`key` int

The key to locate in the IDictionary<TKey, TValue>.

Returns

bool

true if the IDictionary<TKey, TValue> contains an element with the key; otherwise, false.

Exceptions

ArgumentNullException

`key` is null.

CopyTo(KeyValuePair<int, int>[], int)

Copies the elements of the ICollection<T> to an Array, starting at a particular Array index.

```
public void CopyTo(KeyValuePair<int, int>[] array, int arrayIndex)
```

Parameters

`array` KeyValuePair<int, int>[]

The one-dimensional Array that is the destination of the elements copied from ICollection<T>. The Array must have zero-based indexing.

`arrayIndex` int

The zero-based index in `array` at which copying begins.

Exceptions

ArgumentNullException

`array` is null.

ArgumentOutOfRangeException

`arrayIndex` is less than 0.

ArgumentException

The number of elements in the source ICollection<T> is greater than the available space from `arrayIndex` to the end of the destination `array`.

GetEnumerator()

Returns an enumerator that iterates through the collection.

```
public IEnumerator<KeyValuePair<int, int>> GetEnumerator()
```

Returns

IEnumerator<KeyValuePair<int, int>>

An enumerator that can be used to iterate through the collection.

GetObjectData(SerializationInfo, StreamingContext)

Populates a SerializationInfo with the data needed to serialize the target object.

```
public void GetObjectData(SerializationInfo info, StreamingContext context)
```

Parameters

`info` SerializationInfo

The SerializationInfo to populate with data.

`context` StreamingContext

The destination (see StreamingContext) for this serialization.

Exceptions

SecurityException

The caller does not have the required permission.

Remove(KeyValuePair<int, int>)

Removes the first occurrence of a specific object from the ICollection<T>.

```
public bool Remove(KeyValuePair<int, int> item)
```

Parameters

`item` KeyValuePair<int, int>

The object to remove from the ICollection<T>.

Returns

bool

true if `item` was successfully removed from the ICollection<T>; otherwise, false. This method also returns false if `item` is not found in the original ICollection<T>.

Exceptions

NotSupportedException

The ICollection<T> is read-only.

Remove(int)

Removes the element with the specified key from the IDictionary<TKey, TValue>.

```
public bool Remove(int key)
```

Parameters

`key` int

The key of the element to remove.

Returns

bool

true if the element is successfully removed; otherwise, false. This method also returns false if `key` was not found in the original IDictionary<TKey, TValue>.

Exceptions

ArgumentNullException

`key` is null.

NotSupportedException

The IDictionary<TKey, TValue> is read-only.

Serialize()

Serializes the CollectionMap to List format (new format). Order: [version, entries]

```
public IValue Serialize()
```

Returns

IValue

List containing serialized data

TryGetValue(int, out int)

Gets the value associated with the specified key.

```
public bool TryGetValue(int key, out int value)
```

Parameters

`key` int

The key whose value to get.

`value` int

When this method returns, the value associated with the specified key, if the key is found; otherwise, the default value for the type of the `value` parameter. This parameter is passed uninitialized.

Returns

bool

true if the object that implements IDictionary<TKey, TValue> contains an element with the specified key; otherwise, false.

Exceptions

ArgumentNullException

`key` is null.

Class Enemy

Namespace: Nekoyume.Model

Assembly: Lib9c.dll

```
[Serializable]
public class Enemy : CharacterBase, ICloneable
```

Inheritance

object ← CharacterBase ← Enemy

Implements

ICloneable

Derived

RaidBoss

Inherited Members

CharacterBase.CriticalMultiplier , CharacterBase.Id , CharacterBase.Simulator , CharacterBase.atkElementType , CharacterBase.attackRange , CharacterBase.defElementType , CharacterBase.Skills , CharacterBase.BuffSkills , CharacterBase.Buffs , CharacterBase.StatBuffs , CharacterBase.ActionBuffs , CharacterBase.Targets , CharacterBase.RowData , CharacterBase.CharacterId , CharacterBase.SizeType , CharacterBase.RunSpeed , CharacterBase.Stats , CharacterBase.Level , CharacterBase.HP , CharacterBase.AdditionalHP , CharacterBase.ATK , CharacterBase.DEF , CharacterBase.CRI , CharacterBase.HIT , CharacterBase.SPD , CharacterBase.DRV , CharacterBase.DRR , CharacterBase.CDMG , CharacterBase.ArmorPenetration , CharacterBase.Thorn , CharacterBase.CurrentHP , CharacterBase.IsDead , CharacterBase.AttackCount , CharacterBase.AttackCountMax , CharacterBase.usedSkill , CharacterBase.InitAI() , CharacterBase.InitAIV1() , CharacterBase.InitAIV2() , CharacterBase.Tick() , CharacterBase.ReduceSkillCooldown() , CharacterBase.UseSkill() , CharacterBase.EndTurn() , CharacterBase.AddBuff(Buff, bool) , CharacterBase.RemoveActionBuff(ActionBuff) , CharacterBase.RemoveStatBuff(StatBuff) , CharacterBase.RemoveRecentStatBuff() , CharacterBase.RemoveAllStatBuffs() , CharacterBase.ResetCurrentHP() , CharacterBase.IsCritical(bool) , CharacterBase.IsHit(ElementalResult) , CharacterBase.IsHit(CharacterBase) , CharacterBase.GetDamage(long, bool) , CharacterBase.Die() , CharacterBase.Heal(long) , CharacterBase.GetChance(int) , CharacterBase.OnPreSkill() , CharacterBase.OnPostSkill(Skill) , object.Equals(object) , object.Equals(object, object) ,

object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

Enemy(CharacterBase, CharacterStats, Row, ElementalType)

```
public Enemy(CharacterBase player, CharacterStats stat, CharacterSheet.Row rowData,
ElementalType elementalType)
```

Parameters

`player` CharacterBase

`stat` CharacterStats

`rowData` CharacterSheet.Row

`elementalType` ElementalType

Enemy(CharacterBase, CharacterStats, int, ElementalType)

```
public Enemy(CharacterBase player, CharacterStats stat, int characterId,
ElementalType elementalType)
```

Parameters

`player` CharacterBase

`stat` CharacterStats

`characterId` int

`elementalType` ElementalType

Enemy(CharacterBase, Row, int, IEnumerable<StatModifier>)

```
public Enemy(CharacterBase player, CharacterSheet.Row rowData, int monsterLevel,
IEnumerable<StatModifier> optionalStatModifiers = null)
```

Parameters

`player` CharacterBase

`rowData` CharacterSheet.Row

`monsterLevel` int

`optionalStatModifiers` IEnumerable<StatModifier>

Enemy(Enemy)

```
public Enemy(Enemy value)
```

Parameters

`value` Enemy

Enemy(Row)

```
public Enemy(CharacterSheet.Row rowData)
```

Parameters

`rowData` CharacterSheet.Row

Fields

spawnIndex

```
public int spawnIndex
```

Field Value

int

Methods

Clone()

Creates a new object that is a copy of the current instance.

```
public override object Clone()
```

Returns

object

A new object that is a copy of this instance.

OnDead()

```
protected override void OnDead()
```

SetSkill()

```
protected override void SetSkill()
```

Class EnemyPlayer

Namespace: Nekoyume.Model

Assembly: Lib9c.dll

```
[Serializable]
public class EnemyPlayer : Player, ICloneable
```

Inheritance

object ← CharacterBase ← Player ← EnemyPlayer

Implements

ICloneable

Inherited Members

Player.Exp , Player.Inventory , Player.worldInformation , Player.weapon , Player.armor , Player.belt , Player.necklace , Player.ring , Player.aura , Player.Grimoire , Player.monsterMap , Player.eventMap , Player.monsterMapForBeforeV100310 , Player.eventMapForBeforeV100310 , Player.hairIndex , Player.lensIndex , Player.earIndex , Player.tailIndex , Player.characterLevelSheet , Player.costumes , Player.equipments , Player.RuneSkills , Player.RuneSkillCooldownMap , Player.Costumes , Player.Equipments , Player.IsHit(CharacterBase) , Player.RemoveTarget(Enemy) , Player.RemoveTarget(EnemyPlayer) , Player.SetEquipmentStat(EquipmentItemSetEffectSheet) , Player.GetExp(long, bool) , Player.GetExp2(long, bool) , Player.GetExp3(long, bool) , Player.GetRewards(List<ItemBase>) , Player.Use(List<Guid>) , Player.OverrideSkill(Skill) , Player.AddSkill(Skill) , Player.UseSkill() , Player.ReduceSkillCooldown() , Player.SetCostumeStat(CostumeStatSheet) , Player.SetRuneStats(List<RuneState>, RuneOptionSheet, int) , Player.SetRuneSkills(List<RuneState>, RuneOptionSheet, SkillSheet) , Player.SetCollections(IEnumerable<StatModifier>) , Player.ConfigureStats(CostumeStatSheet, List<RuneState>, RuneOptionSheet, int, SkillSheet, List<StatModifier>) , Player.SetRuneV1(List<RuneState>, RuneOptionSheet, SkillSheet) , Player.EndTurn() , CharacterBase.CriticalMultiplier , CharacterBase.Id , CharacterBase.Simulator , CharacterBase.atkElementType , CharacterBase.attackRange , CharacterBase.defElementType , CharacterBase.Skills , CharacterBase.BuffSkills , CharacterBase.Buffs , CharacterBase.StatBuffs , CharacterBase.ActionBuffs , CharacterBase.Targets , CharacterBase.RowData , CharacterBase.CharacterId ,

CharacterBase.SizeType , CharacterBase.RunSpeed , CharacterBase.Stats , CharacterBase.Level , CharacterBase.HP , CharacterBase.AdditionalHP , CharacterBase.ATK , CharacterBase.DEF , CharacterBase.CRI , CharacterBase.HIT , CharacterBase.SPD , CharacterBase.DRV , CharacterBase.DRR , CharacterBase.CDMG , CharacterBase.ArmorPenetration , CharacterBase.Thorn , CharacterBase.CurrentHP , CharacterBase.IsDead , CharacterBase.AttackCount , CharacterBase.AttackCountMax , CharacterBase.usedSkill , CharacterBase.InitAI() , CharacterBase.InitAIV1() , CharacterBase.InitAIV2() , CharacterBase.Tick() , CharacterBase.AddBuff(Buff, bool) , CharacterBase.RemoveActionBuff(ActionBuff) , CharacterBase.RemoveStatBuff(StatBuff) , CharacterBase.RemoveRecentStatBuff() , CharacterBase.RemoveAllStatBuffs() , CharacterBase.ResetCurrentHP() , CharacterBase.IsCritical(bool) , CharacterBase.IsHit(ElementalResult) , CharacterBase.GetDamage(long, bool) , CharacterBase.Die() , CharacterBase.Heal(long) , CharacterBase.SetSkill() , CharacterBase.GetChance(int) , CharacterBase.OnPreSkill() , CharacterBase.OnPostSkill(Skill) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

EnemyPlayer(EnemyPlayerDigest, CharacterSheet, CharacterLevelSheet, EquipmentItemSetEffectSheet)

```
public EnemyPlayer(EnemyPlayerDigest enemyPlayerDigest, CharacterSheet
characterSheet, CharacterLevelSheet levelSheet, EquipmentItemSetEffectSheet
equipmentItemSetEffectSheet)
```

Parameters

`enemyPlayerDigest` EnemyPlayerDigest

`characterSheet` CharacterSheet

`levelSheet` CharacterLevelSheet

`equipmentItemSetEffectSheet` EquipmentItemSetEffectSheet

EnemyPlayer(ArenaPlayerDigest, ArenaSimulatorSheetsV1)

```
public EnemyPlayer(ArenaPlayerDigest arenaPlayerDigest, ArenaSimulatorSheetsV1
simulatorSheets)
```

Parameters

`arenaPlayerDigest` ArenaPlayerDigest

`simulatorSheets` ArenaSimulatorSheetsV1

EnemyPlayer(AvatarState, Simulator)

```
public EnemyPlayer(AvatarState avatarState, Simulator simulator)
```

Parameters

`avatarState` AvatarState

`simulator` Simulator

EnemyPlayer(AvatarState, CharacterSheet, CharacterLevelSheet, EquipmentItemSetEffectSheet)

```
public EnemyPlayer(AvatarState avatarState, CharacterSheet characterSheet,
CharacterLevelSheet characterLevelSheet, EquipmentItemSetEffectSheet
equipmentItemSetEffectSheet)
```

Parameters

`avatarState` AvatarState

`characterSheet` CharacterSheet

`characterLevelSheet` CharacterLevelSheet

`equipmentItemSetEffectSheet` EquipmentItemSetEffectSheet

EnemyPlayer(AvatarState, SimulatorSheetsV1)

```
public EnemyPlayer(AvatarState avatarState, SimulatorSheetsV1 simulatorSheets)
```

Parameters

`avatarState` AvatarState

`simulatorSheets` SimulatorSheetsV1

EnemyPlayer(int, CharacterSheet, CharacterLevelSheet, EquipmentItemSetEffectSheet)

```
public EnemyPlayer(int level, CharacterSheet characterSheet, CharacterLevelSheet
characterLevelSheet, EquipmentItemSetEffectSheet equipmentItemSetEffectSheet)
```

Parameters

`level` int

`characterSheet` CharacterSheet

`characterLevelSheet` CharacterLevelSheet

`equipmentItemSetEffectSheet` EquipmentItemSetEffectSheet

Fields

NameWithHash

```
public readonly string NameWithHash
```

Field Value

string

Methods

Clone()

Creates a new object that is a copy of the current instance.

```
public override object Clone()
```

Returns

object

A new object that is a copy of this instance.

OnDead()

```
protected override void OnDead()
```

Spawn()

```
public override void Spawn()
```

SpawnV1()

```
[Obsolete("Use Spawn")]
public override void SpawnV1()
```

SpawnV2()

```
[Obsolete("Use Spawn")]
```

```
public override void SpawnV2()
```

Class FailedAddWorldException

Namespace: Nekoyume.Model

Assembly: Lib9c.dll

```
[Serializable]
public class FailedAddWorldException : Exception, ISerializable
```

Inheritance

object ← Exception ← FailedAddWorldException

Implements

ISerializable

Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Constructors

FailedAddWorldException(SerializationInfo, StreamingContext)

```
protected FailedAddWorldException(SerializationInfo info, StreamingContext context)
```

Parameters

`info` SerializationInfo

`context` StreamingContext

FailedAddWorldException(string)

```
public FailedAddWorldException(string message)
```

Parameters

`message` string

Class FailedToUnlockWorldException

Namespace: Nekoyume.Model

Assembly: Lib9c.dll

```
[Serializable]
public class FailedToUnlockWorldException : Exception, ISerializable
```

Inheritance

object ← Exception ← FailedToUnlockWorldException

Implements

ISerializable

Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Constructors

FailedToUnlockWorldException(SerializationInfo, StreamingContext)

```
protected FailedToUnlockWorldException(SerializationInfo info,
StreamingContext context)
```

Parameters

`info` SerializationInfo

`context` StreamingContext

FailedToUnlockWorldException(string)

```
public FailedToUnlockWorldException(string message)
```

Parameters

`message` string

Struct FungibleItemValue

Namespace: Nekoyume.Model

Assembly: Lib9c.dll

```
public readonly struct FungibleItemValue
```

Inherited Members
ValueType.Equals(object) , ValueType.GetHashCode() , ValueType.ToString() , object.Equals(object, object) , object.GetType() , object.ReferenceEquals(object, object)

Constructors

FungibleItemValue(List)

```
public FungibleItemValue(List bencoded)
```

Parameters

`bencoded` List

FungibleItemValue(HashDigest<SHA256>, int)

```
public FungibleItemValue(HashDigest<SHA256> id, int count)
```

Parameters

`id` HashDigest<SHA256>

`count` int

Properties

Count

```
public int Count { get; }
```

Property Value

int

Id

```
public HashDigest<SHA256> Id { get; }
```

Property Value

HashDigest<SHA256>

Methods

Serialize()

```
public IValue Serialize()
```

Returns

IValue

Interface IArena

Namespace: Nekoyume.Model

Assembly: Lib9c.dll

```
public interface IArena
```

Methods

CoAreaAttack(ArenaCharacter, IEnumerable<ArenaSkillInfo>, IEnumerable<ArenaSkillInfo>)

```
IEnumerator CoAreaAttack(ArenaCharacter caster,
IEnumerable<ArenaSkill.ArenaSkillInfo> skillInfos,
IEnumerable<ArenaSkill.ArenaSkillInfo> buffInfos)
```

Parameters

`caster` ArenaCharacter

`skillInfos` IEnumerable<ArenaSkill.ArenaSkillInfo>

`buffInfos` IEnumerable<ArenaSkill.ArenaSkillInfo>

Returns

IEnumerator

CoBlowAttack(ArenaCharacter, IEnumerable<ArenaSkillInfo>, IEnumerable<ArenaSkillInfo>)

```
IEnumerator CoBlowAttack(ArenaCharacter caster,
IEnumerable<ArenaSkill.ArenaSkillInfo> skillInfos,
```

```
IEnumerable<ArenaSkill.ArenaSkillInfo> buffInfos)
```

Parameters

`caster` ArenaCharacter

`skillInfos` IEnumerable<ArenaSkill.ArenaSkillInfo>

`buffInfos` IEnumerable<ArenaSkill.ArenaSkillInfo>

Returns

IEnumerator

CoBuff(ArenaCharacter, IEnumerable<ArenaSkillInfo>, IEnumerable<ArenaSkillInfo>)

```
IEnumerator CoBuff(ArenaCharacter caster, IEnumerable<ArenaSkill.ArenaSkillInfo>
skillInfos, IEnumerable<ArenaSkill.ArenaSkillInfo> buffInfos)
```

Parameters

`caster` ArenaCharacter

`skillInfos` IEnumerable<ArenaSkill.ArenaSkillInfo>

`buffInfos` IEnumerable<ArenaSkill.ArenaSkillInfo>

Returns

IEnumerator

CoBuffRemovalAttack(ArenaCharacter, IEnumerable<ArenaSkillInfo>, IEnumerable<ArenaSkillInfo>)

```
IEnumerator CoBuffRemovalAttack(ArenaCharacter caster,
IEnumerable<ArenaSkill.ArenaSkillInfo> skillInfos,
```

```
IEnumerable<ArenaSkill.ArenaSkillInfo> buffInfos)
```

Parameters

`caster` ArenaCharacter

`skillInfos` IEnumerable<ArenaSkill.ArenaSkillInfo>

`buffInfos` IEnumerable<ArenaSkill.ArenaSkillInfo>

Returns

IEnumerator

CoCustomEvent(ArenaCharacter, ArenaEventBase)

```
IEnumerator CoCustomEvent(ArenaCharacter caster, ArenaEventBase eventBase)
```

Parameters

`caster` ArenaCharacter

`eventBase` ArenaEventBase

Returns

IEnumerator

CoDead(ArenaCharacter)

```
IEnumerator CoDead(ArenaCharacter caster)
```

Parameters

`caster` ArenaCharacter

Returns

IEnumerator

CoDoubleAttack(ArenaCharacter, IEnumerable<ArenaSkillInfo>, IEnumerable<ArenaSkillInfo>)

```
IEnumerator CoDoubleAttack(ArenaCharacter caster,
IEnumerable<ArenaSkill.ArenaSkillInfo> skillInfos,
IEnumerable<ArenaSkill.ArenaSkillInfo> buffInfos)
```

Parameters

`caster` ArenaCharacter

`skillInfos` IEnumerable<ArenaSkill.ArenaSkillInfo>

`buffInfos` IEnumerable<ArenaSkill.ArenaSkillInfo>

Returns

IEnumerator

CoDoubleAttackWithCombo(ArenaCharacter, IEnumerable<ArenaSkillInfo>, IEnumerable<ArenaSkillInfo>)

```
IEnumerator CoDoubleAttackWithCombo(ArenaCharacter caster,
IEnumerable<ArenaSkill.ArenaSkillInfo> skillInfos,
IEnumerable<ArenaSkill.ArenaSkillInfo> buffInfos)
```

Parameters

`caster` ArenaCharacter

`skillInfos` IEnumerable<ArenaSkill.ArenaSkillInfo>

`buffInfos` IEnumerable<ArenaSkill.ArenaSkillInfo>

Returns

IEnumerator

CoFullBuffRemovalAttack(ArenaCharacter, IEnumerable<ArenaSkillInfo>, IEnumerable<ArenaSkillInfo>)

Coroutine for full buff removal attack that removes all positive stat buffs.

```
IEnumerator CoFullBuffRemovalAttack(ArenaCharacter caster,
IEnumerable<ArenaSkill.ArenaSkillInfo> skillInfos,
IEnumerable<ArenaSkill.ArenaSkillInfo> buffInfos)
```

Parameters

`caster` ArenaCharacter

`skillInfos` IEnumerable<ArenaSkill.ArenaSkillInfo>

`buffInfos` IEnumerable<ArenaSkill.ArenaSkillInfo>

Returns

IEnumerator

CoHeal(ArenaCharacter, IEnumerable<ArenaSkillInfo>, IEnumerable<ArenaSkillInfo>)

```
IEnumerator CoHeal(ArenaCharacter caster, IEnumerable<ArenaSkill.ArenaSkillInfo>
skillInfos, IEnumerable<ArenaSkill.ArenaSkillInfo> buffInfos)
```

Parameters

`caster` ArenaCharacter

`skillInfos` IEnumerable<ArenaSkill.ArenaSkillInfo>

`buffInfos` IEnumerable<ArenaSkill.ArenaSkillInfo>

Returns

IEnumerator

CoNormalAttack(ArenaCharacter, IEnumerable<ArenaSkillInfo>, IEnumerable<ArenaSkillInfo>)

```
IEnumerator CoNormalAttack(ArenaCharacter caster,
IEnumerable<ArenaSkill.ArenaSkillInfo> skillInfos,
IEnumerable<ArenaSkill.ArenaSkillInfo> buffInfos)
```

Parameters

`caster` ArenaCharacter

`skillInfos` IEnumerable<ArenaSkill.ArenaSkillInfo>

`buffInfos` IEnumerable<ArenaSkill.ArenaSkillInfo>

Returns

IEnumerator

CoRemoveBuffs(ArenaCharacter)

```
IEnumerator CoRemoveBuffs(ArenaCharacter caster)
```

Parameters

`caster` ArenaCharacter

Returns

IEnumerator

CoShatterStrike(ArenaCharacter, IEnumerable<ArenaSkillInfo>, IEnumerable<ArenaSkillInfo>)

```
IEnumerator CoShatterStrike(ArenaCharacter caster,
IEnumerable<ArenaSkill.ArenaSkillInfo> skillInfos,
IEnumerable<ArenaSkill.ArenaSkillInfo> buffInfos)
```

Parameters

`caster` ArenaCharacter

`skillInfos` IEnumerable<ArenaSkill.ArenaSkillInfo>

`buffInfos` IEnumerable<ArenaSkill.ArenaSkillInfo>

Returns

IEnumerator

CoSpawnCharacter(ArenaCharacter)

```
IEnumerator CoSpawnCharacter(ArenaCharacter character)
```

Parameters

`character` ArenaCharacter

Returns

IEnumerator

CoTickDamage(ArenaCharacter, IEnumerable<ArenaSkillInfo>)

```
IEnumerator CoTickDamage(ArenaCharacter affectedCharacter,
```

```
IEnumerable<ArenaSkill.ArenaSkillInfo> skillInfos)
```

Parameters

`affectedCharacter` ArenaCharacter

`skillInfos` IEnumerable<ArenaSkill.ArenaSkillInfo>

Returns

IEnumerator

CoTurnEnd(int)

```
IEnumerator CoTurnEnd(int turnNumber)
```

Parameters

`turnNumber` int

Returns

IEnumerator

Interface IStage

Namespace: Nekoyume.Model

Assembly: Lib9c.dll

```
public interface IStage
```

Methods

CoAreaAttack(CharacterBase, int, IEnumerable<SkillInfo>, IEnumerable<SkillInfo>)

```
IEnumerator CoAreaAttack(CharacterBase caster, int skillId,
IEnumerable<Skill.SkillInfo> skillInfos, IEnumerable<Skill.SkillInfo> buffInfos)
```

Parameters

`caster` CharacterBase

`skillId` int

`skillInfos` IEnumerable<Skill.SkillInfo>

`buffInfos` IEnumerable<Skill.SkillInfo>

Returns

IEnumerator

CoBlowAttack(CharacterBase, int, IEnumerable<SkillInfo>, IEnumerable<SkillInfo>)

```
IEnumerator CoBlowAttack(CharacterBase caster, int skillId,
IEnumerable<Skill.SkillInfo> skillInfos, IEnumerable<Skill.SkillInfo> buffInfos)
```

Parameters

`caster` CharacterBase

`skillId` int

`skillInfos` IEnumerable<Skill.SkillInfo>

`buffInfos` IEnumerable<Skill.SkillInfo>

Returns

IEnumerator

CoBreakthrough(CharacterBase, int, List<MonsterData>)

```
IEnumerator CoBreakthrough(CharacterBase character, int floorId,
List<AdventureBossFloorWaveSheet.MonsterData> monsters)
```

Parameters

`character` CharacterBase

`floorId` int

`monsters` List<AdventureBossFloorWaveSheet.MonsterData>

Returns

IEnumerator

CoBuff(CharacterBase, int, IEnumerable<SkillInfo>, IEnumerable<SkillInfo>)

```
IEnumerator CoBuff(CharacterBase caster, int skillId, IEnumerable<Skill.SkillInfo>
skillInfos, IEnumerable<Skill.SkillInfo> buffInfos)
```

Parameters

`caster` CharacterBase

`skillId` int

`skillInfos` IEnumerable<Skill.SkillInfo>

`buffInfos` IEnumerable<Skill.SkillInfo>

Returns

IEnumerator

CoBuffRemovalAttack(CharacterBase, int, IEnumerable<SkillInfo>, IEnumerable<SkillInfo>)

```
IEnumerator CoBuffRemovalAttack(CharacterBase caster, int skillId,
IEnumerable<Skill.SkillInfo> skillInfos, IEnumerable<Skill.SkillInfo> buffInfos)
```

Parameters

`caster` CharacterBase

`skillId` int

`skillInfos` IEnumerable<Skill.SkillInfo>

`buffInfos` IEnumerable<Skill.SkillInfo>

Returns

IEnumerator

CoCustomEvent(CharacterBase, EventBase)

```
IEnumerator CoCustomEvent(CharacterBase character, EventBase eventBase)
```

Parameters

`character` CharacterBase

`eventBase` EventBase

Returns

IEnumerator

CoDead(CharacterBase)

```
IEnumerator CoDead(CharacterBase character)
```

Parameters

`character` CharacterBase

Returns

IEnumerator

CoDoubleAttack(CharacterBase, int, IEnumerable<SkillInfo>, IEnumerable<SkillInfo>)

```
IEnumerator CoDoubleAttack(CharacterBase caster, int skillId,
IEnumerable<Skill.SkillInfo> skillInfos, IEnumerable<Skill.SkillInfo> buffInfos)
```

Parameters

`caster` CharacterBase

`skillId` int

`skillInfos` IEnumerable<Skill.SkillInfo>

`buffInfos` IEnumerable<Skill.SkillInfo>

Returns

IEnumerator

CoDoubleAttackWithCombo(CharacterBase, int, IEnumerable<SkillInfo>, IEnumerable<SkillInfo>)

```
IEnumerator CoDoubleAttackWithCombo(CharacterBase caster, int skillId,
IEnumerable<Skill.SkillInfo> skillInfos, IEnumerable<Skill.SkillInfo> buffInfos)
```

Parameters

`caster` CharacterBase

`skillId` int

`skillInfos` IEnumerable<Skill.SkillInfo>

`buffInfos` IEnumerable<Skill.SkillInfo>

Returns

IEnumerator

CoDropBox(List<ItemBase>)

```
IEnumerator CoDropBox(List<ItemBase> items)
```

Parameters

`items` List<ItemBase>

Returns

IEnumerator

CoFullBuffRemovalAttack(CharacterBase, int, IEnumerable<SkillInfo>, IEnumerable<SkillInfo>)

Coroutine for full buff removal attack that removes all positive stat buffs.

```
IEnumerator CoFullBuffRemovalAttack(CharacterBase caster, int skillId,
IEnumerable<Skill.SkillInfo> skillInfos, IEnumerable<Skill.SkillInfo> buffInfos)
```

Parameters

`caster` CharacterBase

`skillId` int

`skillInfos` IEnumerable<Skill.SkillInfo>

`buffInfos` IEnumerable<Skill.SkillInfo>

Returns

IEnumerator

CoGetExp(long)

```
IEnumerator CoGetExp(long exp)
```

Parameters

`exp` long

Returns

IEnumerator

CoGetReward(List<ItemBase>, Dictionary<string, int>)

```
IEnumerator CoGetReward(List<ItemBase> rewards, Dictionary<string, int> favRewards)
```

Parameters

`rewards` List<ItemBase>

`favRewards` Dictionary<string, int>

Returns

IEnumerator

CoHeal(CharacterBase, int, IEnumerable<SkillInfo>, IEnumerable<SkillInfo>)

```
IEnumerator CoHeal(CharacterBase caster, int skillId, IEnumerable<Skill.SkillInfo>
skillInfos, IEnumerable<Skill.SkillInfo> buffInfos)
```

Parameters

`caster` CharacterBase

`skillId` int

`skillInfos` IEnumerable<Skill.SkillInfo>

`buffInfos` IEnumerable<Skill.SkillInfo>

Returns

IEnumerator

CoNormalAttack(CharacterBase, int, IEnumerable<SkillInfo>, IEnumerable<SkillInfo>)

```
IEnumerator CoNormalAttack(CharacterBase caster, int skillId,
IEnumerable<Skill.SkillInfo> skillInfos, IEnumerable<Skill.SkillInfo> buffInfos)
```

Parameters

`caster` CharacterBase

`skillId` int

`skillInfos` IEnumerable<Skill.SkillInfo>

`buffInfos` IEnumerable<Skill.SkillInfo>

Returns

IEnumerator

CoRemoveBuffs(CharacterBase)

```
IEnumerator CoRemoveBuffs(CharacterBase caster)
```

Parameters

`caster` CharacterBase

Returns

IEnumerator

CoShatterStrike(CharacterBase, int, IEnumerable<SkillInfo>, IEnumerable<SkillInfo>)

```
IEnumerator CoShatterStrike(CharacterBase caster, int skillId,
IEnumerable<Skill.SkillInfo> skillInfos, IEnumerable<Skill.SkillInfo> buffInfos)
```

Parameters

`caster` CharacterBase

`skillId` int

`skillInfos` IEnumerable<Skill.SkillInfo>

`buffInfos` IEnumerable<Skill.SkillInfo>

Returns

IEnumerator

CoSpawnEnemyPlayer(EnemyPlayer)

```
IEnumerator CoSpawnEnemyPlayer(EnemyPlayer character)
```

Parameters

`character` EnemyPlayer

Returns

IEnumerator

CoSpawnPlayer(Player)

```
IEnumerator CoSpawnPlayer(Player character)
```

Parameters

`character` Player

Returns

IEnumerator

CoSpawnWave(int, int, List<Enemy>, bool)

```
IEnumerator CoSpawnWave(int waveNumber, int waveTurn, List<Enemy> enemies,
bool hasBoss)
```

Parameters

`waveNumber` int

`waveTurn` int

`enemies` List<Enemy>

`hasBoss` bool

Returns

IEnumerator

CoStageBuff(CharacterBase, int, IEnumerable<SkillInfo>, IEnumerable<SkillInfo>)

```
IEnumerator CoStageBuff(CharacterBase affected, int skillId,
IEnumerable<Skill.SkillInfo> skillInfos, IEnumerable<Skill.SkillInfo> buffInfos)
```

Parameters

`affected` CharacterBase

`skillId` int

`skillInfos` IEnumerable<Skill.SkillInfo>

`buffInfos` IEnumerable<Skill.SkillInfo>

Returns

IEnumerator

CoTickDamage(CharacterBase, int, IEnumerable<SkillInfo>)

```
IEnumerator CoTickDamage(CharacterBase affectedCharacter, int skillId,
IEnumerable<Skill.SkillInfo> skillInfos)
```

Parameters

`affectedCharacter` CharacterBase

`skillId` int

`skillInfos` IEnumerable<Skill.SkillInfo>

Returns

IEnumerator

CoWaveTurnEnd(int, int)

```
IEnumerator CoWaveTurnEnd(int turnNumber, int waveTurn)
```

Parameters

`turnNumber` int

`waveTurn` int

Returns

IEnumerator

Class Player

Namespace: Nekoyume.Model

Assembly: Lib9c.dll

```
[Serializable]
public class Player : CharacterBase, ICloneable
```

Inheritance

object ← CharacterBase ← Player

Implements

ICloneable

Derived

EnemyPlayer

Inherited Members

CharacterBase.CriticalMultiplier , CharacterBase.Id , CharacterBase.Simulator , CharacterBase.atkElementType , CharacterBase.attackRange , CharacterBase.defElementType , CharacterBase.Skills , CharacterBase.BuffSkills , CharacterBase.Buffs , CharacterBase.StatBuffs , CharacterBase.ActionBuffs , CharacterBase.Targets , CharacterBase.RowData , CharacterBase.CharacterId , CharacterBase.SizeType , CharacterBase.RunSpeed , CharacterBase.Stats , CharacterBase.Level , CharacterBase.HP , CharacterBase.AdditionalHP , CharacterBase.ATK , CharacterBase.DEF , CharacterBase.CRI , CharacterBase.HIT , CharacterBase.SPD , CharacterBase.DRV , CharacterBase.DRR , CharacterBase.CDMG , CharacterBase.ArmorPenetration , CharacterBase.Thorn , CharacterBase.CurrentHP , CharacterBase.IsDead , CharacterBase.AttackCount , CharacterBase.AttackCountMax , CharacterBase.usedSkill , CharacterBase.InitAI() , CharacterBase.InitAIV1() , CharacterBase.InitAIV2() , CharacterBase.Tick() , CharacterBase.AddBuff(Buff, bool) , CharacterBase.RemoveActionBuff(ActionBuff) , CharacterBase.RemoveStatBuff(StatBuff) , CharacterBase.RemoveRecentStatBuff() , CharacterBase.RemoveAllStatBuffs() , CharacterBase.ResetCurrentHP() , CharacterBase.IsCritical(bool) , CharacterBase.IsHit(ElementalResult) , CharacterBase.GetDamage(long, bool) , CharacterBase.Die() , CharacterBase.Heal(long) , CharacterBase.SetSkill() , CharacterBase.GetChance(int) , CharacterBase.OnPreSkill() , CharacterBase.OnPostSkill(Skill) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

Player(ArenaPlayerDigest, ArenaSimulatorSheetsV1)

```
public Player(ArenaPlayerDigest enemyArenaPlayerDigest, ArenaSimulatorSheetsV1
simulatorSheets)
```

Parameters

`enemyArenaPlayerDigest` ArenaPlayerDigest

`simulatorSheets` ArenaSimulatorSheetsV1

Player(Player)

```
protected Player(Player value)
```

Parameters

`value` Player

Player(AvatarState, Simulator)

```
public Player(AvatarState avatarState, Simulator simulator)
```

Parameters

`avatarState` AvatarState

`simulator` Simulator

Player(AvatarState, CharacterSheet, CharacterLevelSheet, EquipmentItemSetEffectSheet)

```
public Player(AvatarState avatarState, CharacterSheet characterSheet,
CharacterLevelSheet characterLevelSheet, EquipmentItemSetEffectSheet
equipmentItemSetEffectSheet)
```

Parameters

`avatarState` AvatarState

`characterSheet` CharacterSheet

`characterLevelSheet` CharacterLevelSheet

`equipmentItemSetEffectSheet` EquipmentItemSetEffectSheet

Player(AvatarState, SimulatorSheetsV1)

```
public Player(AvatarState avatarState, SimulatorSheetsV1 simulatorSheets)
```

Parameters

`avatarState` AvatarState

`simulatorSheets` SimulatorSheetsV1

Player(int, CharacterSheet, CharacterLevelSheet, EquipmentItemSetEffectSheet)

```
public Player(int level, CharacterSheet characterSheet, CharacterLevelSheet
characterLevelSheet, EquipmentItemSetEffectSheet equipmentItemSetEffectSheet)
```

Parameters

`level` int

`characterSheet` CharacterSheet

`characterLevelSheet` CharacterLevelSheet

`equipmentItemSetEffectSheet` EquipmentItemSetEffectSheet

Fields

Exp

```
public readonly Player.ExpData Exp
```

Field Value

Player.ExpData

Grimoire

```
public Grimoire Grimoire
```

Field Value

Grimoire

Inventory

```
public readonly Inventory Inventory
```

Field Value

Inventory

RuneSkillCooldownMap

```
public readonly Dictionary<int, int> RuneSkillCooldownMap
```

Field Value

Dictionary<int, int>

RuneSkills

```
public readonly Skills RuneSkills
```

Field Value

Skills

armor

```
public Armor armor
```

Field Value

Armor

aura

```
public Aura aura
```

Field Value

Aura

belt

```
public Belt belt
```

Field Value

Belt

characterLevelSheet

```
public CharacterLevelSheet characterLevelSheet
```

Field Value

CharacterLevelSheet

costumes

```
protected List<Costume> costumes
```

Field Value

List<Costume>

earIndex

```
public int earIndex
```

Field Value

int

equipments

```
protected List<Equipment> equipments
```

Field Value

List<Equipment>

eventMap

```
public CollectionMap eventMap
```

Field Value

CollectionMap

eventMapForBeforeV100310

```
public CollectionMap eventMapForBeforeV100310
```

Field Value

CollectionMap

hairIndex

```
public int hairIndex
```

Field Value

int

lensIndex

```
public int lensIndex
```

Field Value

int

monsterMap

```
public CollectionMap monsterMap
```

Field Value

CollectionMap

monsterMapForBeforeV100310

```
public CollectionMap monsterMapForBeforeV100310
```

Field Value

CollectionMap

necklace

```
public Necklace necklace
```

Field Value

Necklace

ring

```
public Ring ring
```

Field Value

Ring

tailIndex

```
public int tailIndex
```

Field Value

int

weapon

```
public Weapon weapon
```

Field Value

Weapon

worldInformation

```
public WorldInformation worldInformation
```

Field Value

WorldInformation

Properties

Costumes

```
public IReadOnlyList<Costume> Costumes { get; }
```

Property Value

IReadOnlyList<Costume>

Equipments

```
public IReadOnlyList<Equipment> Equipments { get; }
```

Property Value

IReadOnlyList<Equipment>

Methods

AddSkill(Skill)

```
public void AddSkill(Skill skill)
```

Parameters

`skill` Skill

Clone()

Creates a new object that is a copy of the current instance.

```
public override object Clone()
```

Returns

object

A new object that is a copy of this instance.

ConfigureStats(CostumeStatSheet, List<RuneState>, RuneOptionSheet, int, SkillSheet, List<StatModifier>)

```
public void ConfigureStats(CostumeStatSheet costumeStatSheet, List<RuneState>
runeStates, RuneOptionSheet runeOptionSheet, int runeLevelBonus, SkillSheet
skillSheet, List<StatModifier> collectionModifiers)
```

Parameters

`costumeStatSheet` CostumeStatSheet

`runeStates` List<RuneState>

`runeOptionSheet` RuneOptionSheet

`runeLevelBonus` int

`skillSheet` SkillSheet

`collectionModifiers` List<StatModifier>

EndTurn()

```
protected override void EndTurn()
```

GetExp(long, bool)

```
public void GetExp(long waveExp, bool log = false)
```

Parameters

`waveExp` long

`log` bool

GetExp2(long, bool)

```
[Obsolete("Use GetExp")]
public void GetExp2(long waveExp, bool log = false)
```

Parameters

`waveExp` long

`log` bool

GetExp3(long, bool)

```
[Obsolete("Use GetExp")]
public void GetExp3(long waveExp, bool log = false)
```

Parameters

`waveExp` long

`log` bool

GetRewards(List<ItemBase>)

```
public CollectionMap GetRewards(List<ItemBase> items)
```

Parameters

`items` List<ItemBase>

Returns

CollectionMap

IsHit(CharacterBase)

```
public override bool IsHit(CharacterBase caster)
```

Parameters

`caster` CharacterBase

Returns

bool

OnDead()

```
protected override void OnDead()
```

OverrideSkill(Skill)

```
public void OverrideSkill(Skill skill)
```

Parameters

`skill` Skill

ReduceSkillCooldown()

```
protected override void ReduceSkillCooldown()
```

RemoveTarget(Enemy)

```
public void RemoveTarget(Enemy enemy)
```

Parameters

`enemy` Enemy

RemoveTarget(EnemyPlayer)

```
public void RemoveTarget(EnemyPlayer enemy)
```

Parameters

`enemy` EnemyPlayer

SetCollections(IEnumerable<StatModifier>)

```
public void SetCollections(IEnumerable<StatModifier> statModifiers)
```

Parameters

`statModifiers` IEnumerable<StatModifier>

SetCostumeStat(CostumeStatSheet)

```
public void SetCostumeStat(CostumeStatSheet costumeStatSheet)
```

Parameters

`costumeStatSheet` CostumeStatSheet

SetEquipmentStat(EquipmentItemSetEffectSheet)

```
protected void SetEquipmentStat(EquipmentItemSetEffectSheet sheet)
```

Parameters

`sheet` EquipmentItemSetEffectSheet

SetRuneSkills(List<RuneState>, RuneOptionSheet, SkillSheet)

Sets the rune skills for the player.

```
public void SetRuneSkills(List<RuneState> runes, RuneOptionSheet runeOptionSheet,
SkillSheet skillSheet)
```

Parameters

`runes` List<RuneState>

The list of rune states.

`runeOptionSheet` RuneOptionSheet

The rune option sheet.

`skillSheet` SkillSheet

The skill sheet.

SetRuneStats(List<RuneState>, RuneOptionSheet, int)

Sets the rune stats for a player character.

```
public void SetRuneStats(List<RuneState> runes, RuneOptionSheet runeOptionSheet,
int runeLevelBonus)
```

Parameters

`runes` List<RuneState>

The AllRuneState for the player character.

`runeOptionSheet` RuneOptionSheet

The rune option sheet that contains information about rune options.

`runeLevelBonus` int

The rune level bonus value from RuneLevelBonusSheet. This enhances equipped rune stats.

SetRuneV1(List<RuneState>, RuneOptionSheet, SkillSheet)

```
[Obsolete("Use SetRune")]
public void SetRuneV1(List<RuneState> runes, RuneOptionSheet runeOptionSheet,
```

```
  SkillSheet skillSheet)
```

Parameters

`runes` List<RuneState>

`runeOptionSheet` RuneOptionSheet

`skillSheet` SkillSheet

Spawn()

```
  public virtual void Spawn()
```

SpawnV1()

```
  [Obsolete("Use Spawn")]
  public virtual void SpawnV1()
```

SpawnV2()

```
  [Obsolete("Use Spawn")]
  public virtual void SpawnV2()
```

Use(List<Guid>)

```
  public void Use(List<Guid> consumableIds)
```

Parameters

`consumableIds` List<Guid>

UseSkill()

```
protected override Skill UseSkill()
```

Returns

Skill

Class Player.ExpData

Namespace: Nekoyume.Model

Assembly: Lib9c.dll

```
[Serializable]
public class Player.ExpData : ICloneable
```

Inheritance

object ← Player.ExpData

Implements

ICloneable

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

ExpData()

```
public ExpData()
```

ExpData(ExpData)

```
protected ExpData(Player.ExpData value)
```

Parameters

`value` Player.ExpData

Properties

Current

```
public long Current { get; set; }
```

Property Value

long

Max

```
public long Max { get; }
```

Property Value

long

Need

```
public long Need { get; }
```

Property Value

long

Methods

Clone()

Creates a new object that is a copy of the current instance.

```
public object Clone()
```

Returns

object

A new object that is a copy of this instance.

Set(Row)

```
public void Set(CharacterLevelSheet.Row row)
```

Parameters

`row` CharacterLevelSheet.Row

Class RaidBoss

Namespace: Nekoyume.Model

Assembly: Lib9c.dll

```
public class RaidBoss : Enemy, ICloneable
```

Inheritance

object ← CharacterBase ← Enemy ← RaidBoss

Implements

ICloneable

Inherited Members

Enemy.spawnIndex , Enemy.OnDead() , CharacterBase.CriticalMultiplier , CharacterBase.Id , CharacterBase.Simulator , CharacterBase.atkElementType , CharacterBase.attackRange , CharacterBase.defElementType , CharacterBase.Skills , CharacterBase.BuffSkills , CharacterBase.Buffs , CharacterBase.StatBuffs , CharacterBase.ActionBuffs , CharacterBase.Targets , CharacterBase.CharacterId , CharacterBase.SizeType , CharacterBase.RunSpeed , CharacterBase.Stats , CharacterBase.Level , CharacterBase.HP , CharacterBase.AdditionalHP , CharacterBase.ATK , CharacterBase.DEF , CharacterBase.CRI , CharacterBase.HIT , CharacterBase.SPD , CharacterBase.DRV , CharacterBase.DRR , CharacterBase.CDMG , CharacterBase.ArmorPenetration , CharacterBase.Thorn , CharacterBase.CurrentHP , CharacterBase.IsDead , CharacterBase.AttackCount , CharacterBase.AttackCountMax , CharacterBase.usedSkill , CharacterBase.InitAI() , CharacterBase.InitAIV1() , CharacterBase.InitAIV2() , CharacterBase.Tick() , CharacterBase.ReduceSkillCooldown() , CharacterBase.AddBuff(Buff, bool) , CharacterBase.RemoveActionBuff(ActionBuff) , CharacterBase.RemoveStatBuff(StatBuff) , CharacterBase.RemoveRecentStatBuff() , CharacterBase.RemoveAllStatBuffs() , CharacterBase.ResetCurrentHP() , CharacterBase.IsCritical(bool) , CharacterBase.IsHit(ElementalResult) , CharacterBase.GetDamage(long, bool) , CharacterBase.Die() , CharacterBase.Heal(long) , CharacterBase.GetChance(int) , CharacterBase.OnPreSkill() , CharacterBase.OnPostSkill(Skill) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

RaidBoss(CharacterBase, Row, Row, WaveStatData, bool)

```
public RaidBoss(CharacterBase player, WorldBossCharacterSheet.Row characterRow,
WorldBossActionPatternSheet.Row patternRow, WorldBossCharacterSheet.WaveStatData
statData, bool ignoreLevelCorrectionOnHit)
```

Parameters

`player` CharacterBase

`characterRow` WorldBossCharacterSheet.Row

`patternRow` WorldBossActionPatternSheet.Row

`statData` WorldBossCharacterSheet.WaveStatData

`ignoreLevelCorrectionOnHit` bool

RaidBoss(RaidBoss)

```
public RaidBoss(RaidBoss value)
```

Parameters

`value` RaidBoss

Properties

Enraged

```
public bool Enraged { get; protected set; }
```

Property Value

bool

IgnoreLevelCorrectionOnHit

```
public bool IgnoreLevelCorrectionOnHit { get; protected set; }
```

Property Value

bool

PatternRowData

```
public WorldBossActionPatternSheet.Row PatternRowData { get; }
```

Property Value

WorldBossActionPatternSheet.Row

RowData

```
public WorldBossCharacterSheet.Row RowData { get; }
```

Property Value

WorldBossCharacterSheet.Row

Methods

Clone()

Creates a new object that is a copy of the current instance.

```
public override object Clone()
```

Returns

object

A new object that is a copy of this instance.

EndTurn()

```
protected override void EndTurn()
```

Enrage()

```
public void Enrage()
```

IsHit(CharacterBase)

```
public override bool IsHit(CharacterBase caster)
```

Parameters

`caster` CharacterBase

Returns

bool

SetSkill()

```
protected override void SetSkill()
```

UseSkill()

```
protected override Skill UseSkill()
```

Returns

Skill

Class Skills

Namespace: Nekoyume.Model

Assembly: Lib9c.dll

```
[Serializable]
public class Skills : IEnumerable<Skill>, IEnumerable
```

Inheritance

object ← Skills

Implements

IEnumerable<Skill>, IEnumerable

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Methods

Add(Skill)

```
public void Add(Skill skill)
```

Parameters

`skill` Skill

Clear()

```
public void Clear()
```

GetCooldown(int)

```
public int GetCooldown(int skillId)
```

Parameters

`skillId` int

Returns

int

GetEnumerator()

Returns an enumerator that iterates through the collection.

```
public IEnumerator<Skill> GetEnumerator()
```

Returns

IEnumerator<Skill>

An enumerator that can be used to iterate through the collection.

ReduceCooldown()

```
public void ReduceCooldown()
```

ReduceCooldownV1()

```
[Obsolete("ReduceCooldown")]
public void ReduceCooldownV1()
```

Select(IRandom)

```
public Skill Select(IRandom random)
```

Parameters

`random` IRandom

Returns

Skill

SelectV1(IRandom)

```
[Obsolete("Use Select")]
public Skill SelectV1(IRandom random)
```

Parameters

`random` IRandom

Returns

Skill

SelectV2(IRandom)

```
[Obsolete("Use Select")]
public Skill SelectV2(IRandom random)
```

Parameters

`random` IRandom

Returns

Skill

SelectWithoutDefaultAttack(IRandom)

```
public Skill SelectWithoutDefaultAttack(IRandom random)
```

Parameters

`random` IRandom

Returns

Skill

SetCooldown(int, int)

```
public void SetCooldown(int skillId, int cooldown)
```

Parameters

`skillId` int

`cooldown` int

Class WorldInformation

Namespace: Nekoyume.Model

Assembly: Lib9c.dll

```
[Serializable]
public class WorldInformation : IState, ISerializable, ICloneable
```

Inheritance

object ← WorldInformation

Implements

IState, ISerializable, ICloneable

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Extension Methods

WorldInformationExtensions.ValidateFromAction(WorldInformation, int, string, string)

Constructors

WorldInformation(Dictionary)

```
public WorldInformation(Dictionary serialized)
```

Parameters

`serialized` Dictionary

WorldInformation(long, WorldSheet, bool, string)

```
public WorldInformation(long blockIndex, WorldSheet worldSheet, bool
openAllOfWorldsAndStages = false, string avatarName = "")
```

Parameters

`blockIndex` long

`worldSheet` WorldSheet

`openAllOfWorldsAndStages` bool

`avatarName` string

WorldInformation(long, WorldSheet, int)

```
public WorldInformation(long blockIndex, WorldSheet worldSheet, int clearStageId
= 0)
```

Parameters

`blockIndex` long

`worldSheet` WorldSheet

`clearStageId` int

Methods

AddAndUnlockMimisbrunnrWorld(Row, long, WorldSheet, WorldUnlockSheet)

```
public void AddAndUnlockMimisbrunnrWorld(WorldSheet.Row worldRow, long unlockedAt,
WorldSheet worldSheet, WorldUnlockSheet worldUnlockSheet)
```

Parameters

`worldRow` WorldSheet.Row

`unlockedAt` long

`worldSheet` WorldSheet

`worldUnlockSheet` WorldUnlockSheet

AddAndUnlockNewWorld(Row, long, WorldSheet)

```
public void AddAndUnlockNewWorld(WorldSheet.Row worldRow, long unlockedAt,
WorldSheet worldSheet)
```

Parameters

`worldRow` WorldSheet.Row

`unlockedAt` long

`worldSheet` WorldSheet

ClearStage(int, int, long, WorldSheet, WorldUnlockSheet)

Clear a specific stage. And consider world unlock.

```
public void ClearStage(int worldId, int stageId, long clearedAt, WorldSheet
worldSheet, WorldUnlockSheet worldUnlockSheet)
```

Parameters

`worldId` int

`stageId` int

`clearedAt` long

`worldSheet` WorldSheet

`worldUnlockSheet` WorldUnlockSheet

Exceptions

FailedToUnlockWorldException

Clone()

Creates a new object that is a copy of the current instance.

```
public object Clone()
```

Returns

object

A new object that is a copy of this instance.

GetObjectData(SerializationInfo, StreamingContext)

Populates a SerializationInfo with the data needed to serialize the target object.

```
public void GetObjectData(SerializationInfo info, StreamingContext context)
```

Parameters

`info` SerializationInfo

The SerializationInfo to populate with data.

`context` StreamingContext

The destination (see StreamingContext) for this serialization.

Exceptions

SecurityException

The caller does not have the required permission.

IsStageCleared(int)

```
public bool IsStageCleared(int stageId)
```

Parameters

`stageId` int

Returns

bool

IsWorldUnlocked(int)

```
public bool IsWorldUnlocked(int worldId)
```

Parameters

`worldId` int

Returns

bool

Serialize()

```
public IValue Serialize()
```

Returns

IValue

TryAddWorld(Row, out World)

```
public bool TryAddWorld(WorldSheet.Row worldRow, out WorldInformation.World world)
```

Parameters

`worldRow` WorldSheet.Row

`world` WorldInformation.World

Returns

bool

TryGetFirstWorld(out World)

```
public bool TryGetFirstWorld(out WorldInformation.World world)
```

Parameters

`world` WorldInformation.World

Returns

bool

TryGetLastClearedMimisbrunnrStageId(out int)

```
public bool TryGetLastClearedMimisbrunnrStageId(out int stageId)
```

Parameters

`stageId` int

Returns

bool

TryGetLastClearedStageId(out int)

Get stage id of the most recent cleared.(ignore mimisbrunnr)

```
public bool TryGetLastClearedStageId(out int stageId)
```

Parameters

`stageId` int

Returns

bool

TryGetUnlockedWorldByStageClearedBlockIndex(out World)

Get `World` object that contains the most recent stage clear.

```
public bool TryGetUnlockedWorldByStageClearedBlockIndex(out
WorldInformation.World world)
```

Parameters

`world` WorldInformation.World

Returns

bool

TryGetWorld(int, out World)

Get `World` object that equals to `worldId` argument.

```
public bool TryGetWorld(int worldId, out WorldInformation.World world)
```

Parameters

`worldId` int

`world` WorldInformation.World

Returns

bool

Exceptions

KeyNotFoundException

TryGetWorldByStageId(int, out World)

Get `World` object that contains `stageId` argument.

```
public bool TryGetWorldByStageId(int stageId, out WorldInformation.World world)
```

Parameters

`stageId` int

`world` WorldInformation.World

Returns

bool

UnlockWorld(int, long, WorldSheet)

Unlock a specific world.

```
public void UnlockWorld(int worldId, long unlockedAt, WorldSheet worldSheet)
```

Parameters

`worldId` int

`unlockedAt` long

`worldSheet` WorldSheet

Exceptions

FailedToUnlockWorldException

UpdateWorld(Row)

```
public void UpdateWorld(WorldSheet.Row worldRow)
```

Parameters

`worldRow` WorldSheet.Row

Struct WorldInformation.World

Namespace: Nekoyume.Model

Assembly: Lib9c.dll

```
[Serializable]
public struct WorldInformation.World : IState
```

Implements
IState

Inherited Members
ValueType.Equals(object) , ValueType.GetHashCode() , ValueType.ToString() , object.Equals(object, object) , object.GetType() , object.ReferenceEquals(object, object)

Constructors

World(Dictionary)

```
public World(Dictionary serialized)
```

Parameters

`serialized` Dictionary

World(World, long)

```
public World(WorldInformation.World world, long unlockedBlockIndex = -1)
```

Parameters

`world` WorldInformation.World

`unlockedBlockIndex` long

World(World, long, int)

```
public World(WorldInformation.World world, long stageClearedBlockIndex,
int stageClearedId)
```

Parameters

`world` WorldInformation.World

`stageClearedBlockIndex` long

`stageClearedId` int

World(Row, long, long, int)

```
public World(WorldSheet.Row worldRow, long unlockedBlockIndex = -1, long
stageClearedBlockIndex = -1, int stageClearedId = -1)
```

Parameters

`worldRow` WorldSheet.Row

`unlockedBlockIndex` long

`stageClearedBlockIndex` long

`stageClearedId` int

Fields

Id

```
public readonly int Id
```

Field Value

int

Name

```
public readonly string Name
```

Field Value

string

StageBegin

```
public readonly int StageBegin
```

Field Value

int

StageClearedBlockIndex

```
public readonly long StageClearedBlockIndex
```

Field Value

long

StageClearedId

```
public readonly int StageClearedId
```

Field Value

int

StageEnd

```
public readonly int StageEnd
```

Field Value

int

UnlockedBlockIndex

```
public readonly long UnlockedBlockIndex
```

Field Value

long

Properties

IsStageCleared

```
public bool IsStageCleared { get; }
```

Property Value

bool

IsUnlocked

```
public bool IsUnlocked { get; }
```

Property Value

bool

Methods

ContainsStageId(int)

```
public bool ContainsStageId(int stageId)
```

Parameters

`stageId` int

Returns

bool

GetNextStageId()

```
public int GetNextStageId()
```

Returns

int

GetNextStageIdForPlay()

```
public int GetNextStageIdForPlay()
```

Returns

int

IsPlayable(int)

```
public bool IsPlayable(int stageId)
```

Parameters

`stageId` int

Returns

bool

Serialize()

```
public IValue Serialize()
```

Returns

IValue

Namespace Nekoyume.Model.Adventure Boss

Classes

BountyBoard

ExploreBoard

Explorer

ExplorerList

Investor

SeasonInfo

Class BountyBoard

Namespace: Nekoyume.Model.AdventureBoss

Assembly: Lib9c.dll

```
public class BountyBoard
```

Inheritance

object ← BountyBoard

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

BountyBoard(List)

```
public BountyBoard(List bencoded)
```

Parameters

`bencoded` List

BountyBoard(long)

```
public BountyBoard(long season)
```

Parameters

`season` long

Fields

FixedRewardFavId

```
public int? FixedRewardFavId
```

Field Value

int?

FixedRewardItemId

```
public int? FixedRewardItemId
```

Field Value

int?

Investors

```
public List<Investor> Investors
```

Field Value

List<Investor>

RandomRewardFavId

```
public int? RandomRewardFavId
```

Field Value

int?

RandomRewardItemId

```
public int? RandomRewardItemId
```

Field Value

int?

Season

```
public long Season
```

Field Value

long

Properties

Bencoded

```
public IValue Bencoded { get; }
```

Property Value

IValue

Methods

AddOrUpdate(Address, string, FungibleAssetValue)

```
public void AddOrUpdate(Address avatarAddress, string name,
FungibleAssetValue price)
```

Parameters

avatarAddress Address

`name` string

`price` FungibleAssetValue

SetReward(Row, IRandom)

```
public void SetReward(AdventureBossWantedRewardSheet.Row rewardInfo, IRandom random)
```

Parameters

`rewardInfo` AdventureBossWantedRewardSheet.Row

`random` IRandom

totalBounty()

```
public FungibleAssetValue totalBounty()
```

Returns

FungibleAssetValue

Class ExploreBoard

Namespace: Nekoyume.Model.AdventureBoss

Assembly: Lib9c.dll

```
public class ExploreBoard
```

Inheritance

object ← ExploreBoard

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

ExploreBoard(List)

```
public ExploreBoard(List bencoded)
```

Parameters

`bencoded` List

ExploreBoard(long)

```
public ExploreBoard(long season)
```

Parameters

`season` long

Fields

ExplorerCount

```
public int ExplorerCount
```

Field Value

int

FixedRewardFavId

```
public int? FixedRewardFavId
```

Field Value

int?

FixedRewardItemId

```
public int? FixedRewardItemId
```

Field Value

int?

RaffleReward

```
public FungibleAssetValue? RaffleReward
```

Field Value

FungibleAssetValue?

RaffleWinner

```
public Address? RaffleWinner
```

Field Value

Address?

RaffleWinnerName

```
public string RaffleWinnerName
```

Field Value

string

Season

```
public long Season
```

Field Value

long

TotalPoint

```
public long TotalPoint
```

Field Value

long

UsedApPotion

```
public long UsedApPotion
```

Field Value

long

UsedGoldenDust

```
public long UsedGoldenDust
```

Field Value

long

UsedNcg

```
public BigInteger UsedNcg
```

Field Value

BigInteger

Methods

Bencoded()

```
public IValue Bencoded()
```

Returns

IValue

SetReward(Row, IRandom)

```
public void SetReward(AdventureBossContributionRewardSheet.Row rewardInfo,
IRandom random)
```

Parameters

`rewardInfo` AdventureBossContributionRewardSheet.Row

`random` IRandom

Class Explorer

Namespace: Nekoyume.Model.AdventureBoss

Assembly: Lib9c.dll

```
public class Explorer : IBencodable
```

Inheritance

object ← Explorer

Implements

IBencodable

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

Explorer(IValue)

```
public Explorer(IValue bencoded)
```

Parameters

`bencoded` IValue

Explorer(Address, string)

```
public Explorer(Address avatarAddress, string name)
```

Parameters

`avatarAddress` Address

name string

Fields

AvatarAddress

```
public Address AvatarAddress
```

Field Value

Address

Claimed

```
public bool Claimed
```

Field Value

bool

Floor

```
public int Floor
```

Field Value

int

MaxFloor

```
public int MaxFloor
```

Field Value

int

Name

```
public string Name
```

Field Value

string

Score

```
public int Score
```

Field Value

int

UsedApPotion

```
public int UsedApPotion
```

Field Value

int

UsedGoldenDust

```
public int UsedGoldenDust
```

Field Value

int

UsedNcg

```
public BigInteger UsedNcg
```

Field Value

BigInteger

Properties

Bencoded

An Bencodex.Types.IValue representation of this object that can be decoded back to instantiate an equal object. The decoded object must be equal to the original in the sense that Equals(T) should be true.

```
public IValue Bencoded { get; }
```

Property Value

IValue

Remarks

Note that the only requirement is that the produced Bencodex.Types.IValue can be decoded back to an equal object. This representation may not be canonical in the sense that additional junk data may be present in an Bencodex.Types.IValue that one may wish to decode and this may be discarded while decoding.

A specific implemnetation may decide to only allow the canonical representation to be decoded.

Class ExplorerList

Namespace: Nekoyume.Model.AdventureBoss

Assembly: Lib9c.dll

```
public class ExplorerList
```

Inheritance

object ← ExplorerList

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

ExplorerList(List)

```
public ExplorerList(List bencoded)
```

Parameters

`bencoded` List

ExplorerList(long)

```
public ExplorerList(long season)
```

Parameters

`season` long

Fields

Explorers

```
public HashSet<(Address, string)> Explorers
```

Field Value

HashSet<(Address, string)>

Season

```
public long Season
```

Field Value

long

Properties

Bencoded

```
public IValue Bencoded { get; }
```

Property Value

IValue

Methods

AddExplorer(Address, string)

```
public void AddExplorer(Address avatarAddress, string name)
```

Parameters

`avatarAddress` Address

`name` string

Class Investor

Namespace: Nekoyume.Model.AdventureBoss

Assembly: Lib9c.dll

```
public class Investor : IBencodable
```

Inheritance

object ← Investor

Implements

IBencodable

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

Investor(IValue)

```
public Investor(IValue value)
```

Parameters

`value` IValue

Investor(Address, string, FungibleAssetValue)

```
public Investor(Address avatarAddress, string name, FungibleAssetValue price)
```

Parameters

`avatarAddress` Address

`name` string

`price` FungibleAssetValue

Fields

AvatarAddress

```
public Address AvatarAddress
```

Field Value

Address

Claimed

```
public bool Claimed
```

Field Value

bool

Count

```
public int Count
```

Field Value

int

MaxInvestmentCount

```
public const int MaxInvestmentCount = 3
```

Field Value

int

Name

```
public string Name
```

Field Value

string

Price

```
public FungibleAssetValue Price
```

Field Value

FungibleAssetValue

Properties

Bencoded

An Bencodex.Types.IValue representation of this object that can be decoded back to instantiate an equal object. The decoded object must be equal to the original in the sense that Equals(T) should be true.

```
public IValue Bencoded { get; }
```

Property Value

IValue

Remarks

Note that the only requirement is that the produced Bencodex.Types.IValue can be decoded back to an equal object. This representation may not be canonical in the sense that additional junk data may be present in an Bencodex.Types.IValue that one may wish to decode and this may be discarded while decoding.

A specific implemnetation may decide to only allow the canonical representation to be decoded.

Class SeasonInfo

Namespace: Nekoyume.Model.AdventureBoss

Assembly: Lib9c.dll

```
public class SeasonInfo
```

Inheritance

object ← SeasonInfo

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

SeasonInfo(List)

```
public SeasonInfo(List serialized)
```

Parameters

`serialized` List

SeasonInfo(long, long, long, long, long?, long?)

```
public SeasonInfo(long season, long startBlockIndex, long activeInterval, long
inactiveInterval, long? endBlockIndex = null, long? nextStartBlockIndex = null)
```

Parameters

`season` long

`startBlockIndex` long

`activeInterval` long

`inactiveInterval` long

`endBlockIndex` long?

`nextStartBlockIndex` long?

Fields

BossId

```
public int BossId
```

Field Value

int

EndBlockIndex

```
public readonly long EndBlockIndex
```

Field Value

long

NextStartBlockIndex

```
public readonly long NextStartBlockIndex
```

Field Value

long

Season

```
public readonly long Season
```

Field Value

long

StartBlockIndex

```
public readonly long StartBlockIndex
```

Field Value

long

Properties

Bencoded

```
public IValue Bencoded { get; }
```

Property Value

IValue

Namespace Nekoyume.Model.Arena

Classes

AddressNotFoundInArenaParticipantsException

ArenaAvatarStateNotFoundException

ArenaInformation
Introduced at https://github.com/planetarium/lib9c/pull/1029

ArenaInformationAlreadyContainsException

ArenaInformationNotFoundException

ArenaParticipant
This class combines the information from ArenaScore and ArenaInformation, and brings together information about the arena participant, including additional information.

ArenaParticipants
Introduced at https://github.com/planetarium/lib9c/pull/1027

ArenaParticipantsNotFoundException

ArenaResult
temp

ArenaScore
Introduced at https://github.com/planetarium/lib9c/pull/1027

ArenaScoreAlreadyContainsException

ArenaScoreNotFoundException

CoolDownBlockException

ExceedPlayCountException

ExceedTicketPurchaseLimitDuringIntervalException

ExceedTicketPurchaseLimitException

InvalidSeasonException

NotEnoughTicketException

RoundNotFoundException

ThisArenaIsClosedException

TicketPurchaseLimitExceedException

ValidateScoreDifferenceException

Class AddressNotFoundInArenaParticipantsException

Namespace: Nekoyume.Model.Arena

Assembly: Lib9c.dll

```
[Serializable]
public class AddressNotFoundInArenaParticipantsException : Exception, ISerializable
```

Inheritance

object ← Exception ← AddressNotFoundInArenaParticipantsException

Implements

ISerializable

Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Constructors

AddressNotFoundInArenaParticipantsException(SerializationInfo, StreamingContext)

```
protected AddressNotFoundInArenaParticipantsException(SerializationInfo info,
StreamingContext context)
```

Parameters

`info` SerializationInfo

`context` StreamingContext

AddressNotFoundInArenaParticipantsException(string)

```
public AddressNotFoundInArenaParticipantsException(string message)
```

Parameters

`message` string

Class ArenaAvatarStateNotFoundException

Namespace: Nekoyume.Model.Arena

Assembly: Lib9c.dll

```
[Serializable]
public class ArenaAvatarStateNotFoundException : Exception, ISerializable
```

Inheritance

object ← Exception ← ArenaAvatarStateNotFoundException

Implements

ISerializable

Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Constructors

ArenaAvatarStateNotFoundException(SerializationInfo, StreamingContext)

```
protected ArenaAvatarStateNotFoundException(SerializationInfo info,
StreamingContext context)
```

Parameters

`info` SerializationInfo

`context` StreamingContext

ArenaAvatarStateNotFoundException(string)

```
public ArenaAvatarStateNotFoundException(string message)
```

Parameters

`message` string

Class ArenaInformation

Namespace: Nekoyume.Model.Arena

Assembly: Lib9c.dll

Introduced at https://github.com/planetarium/lib9c/pull/1029

```
public class ArenaInformation : IState
```

Inheritance

object ← ArenaInformation

Implements

IState

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

ArenaInformation(List)

```
public ArenaInformation(List serialized)
```

Parameters

`serialized` List

ArenaInformation(Address, int, int)

```
public ArenaInformation(Address avatarAddress, int championshipId, int round)
```

Parameters

`avatarAddress` Address

`championshipId` int

`round` int

Fields

Address

```
public Address Address
```

Field Value

Address

MaxTicketCount

```
public const int MaxTicketCount = 8
```

Field Value

int

Properties

Lose

```
public int Lose { get; }
```

Property Value

int

PurchasedTicketCount

```
public int PurchasedTicketCount { get; }
```

Property Value

int

Ticket

```
public int Ticket { get; }
```

Property Value

int

TicketResetCount

```
public int TicketResetCount { get; }
```

Property Value

int

Win

```
public int Win { get; }
```

Property Value

int

Methods

BuyTicket(long)

```
public void BuyTicket(long maxCount)
```

Parameters

`maxCount` long

DeriveAddress(Address, int, int)

```
public static Address DeriveAddress(Address avatarAddress, int championshipId,
int round)
```

Parameters

`avatarAddress` Address

`championshipId` int

`round` int

Returns

Address

ResetTicket(int)

```
public void ResetTicket(int resetCount)
```

Parameters

`resetCount` int

Serialize()

```
public IValue Serialize()
```

Returns

IValue

UpdateRecord(int, int)

```
public void UpdateRecord(int win, int lose)
```

Parameters

`win` int

`lose` int

UseTicket(int)

```
public void UseTicket(int ticketCount)
```

Parameters

`ticketCount` int

Class ArenaInformationAlreadyContainsException

Namespace: Nekoyume.Model.Arena

Assembly: Lib9c.dll

```
[Serializable]
public class ArenaInformationAlreadyContainsException : Exception, ISerializable
```

Inheritance

object ← Exception ← ArenaInformationAlreadyContainsException

Implements

ISerializable

Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Constructors

ArenaInformationAlreadyContainsException(SerializationInfo, StreamingContext)

```
protected ArenaInformationAlreadyContainsException(SerializationInfo info,
StreamingContext context)
```

Parameters

`info` SerializationInfo

`context` StreamingContext

ArenaInformationAlreadyContainsException(string)

```
public ArenaInformationAlreadyContainsException(string message)
```

Parameters

`message` string

Class ArenaInformationNotFoundException

Namespace: Nekoyume.Model.Arena

Assembly: Lib9c.dll

```
[Serializable]
public class ArenaInformationNotFoundException : Exception, ISerializable
```

Inheritance

object ← Exception ← ArenaInformationNotFoundException

Implements

ISerializable

Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Constructors

ArenaInformationNotFoundException(SerializationInfo, StreamingContext)

```
protected ArenaInformationNotFoundException(SerializationInfo info,
StreamingContext context)
```

Parameters

`info` SerializationInfo

`context` StreamingContext

ArenaInformationNotFoundException(string)

```
public ArenaInformationNotFoundException(string message)
```

Parameters

`message` string

Class ArenaParticipant

Namespace: Nekoyume.Model.Arena

Assembly: Lib9c.dll

This class combines the information from ArenaScore and ArenaInformation, and brings together information about the arena participant, including additional information.

```
public class ArenaParticipant : IBencodable, IState
```

Inheritance

object ← ArenaParticipant

Implements

IBencodable, IState

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

ArenaParticipant()

```
public ArenaParticipant()
```

ArenaParticipant(IValue)

```
public ArenaParticipant(IValue bencoded)
```

Parameters

`bencoded` IValue

ArenaParticipant(Address)

```
public ArenaParticipant(Address avatarAddr)
```

Parameters

`avatarAddr` Address

Fields

AvatarAddr

```
public readonly Address AvatarAddr
```

Field Value

Address

Cp

```
public long Cp
```

Field Value

long

DefaultScore

```
public const int DefaultScore = 1000
```

Field Value

int

LastBattleBlockIndex

```
public long LastBattleBlockIndex
```

Field Value

long

Level

```
public int Level
```

Field Value

int

Lose

```
public int Lose
```

Field Value

int

MaxTicketCount

```
public const int MaxTicketCount = 8
```

Field Value

int

Name

If you need to know NameWithHash, check PostConstructor() method of the AvatarState class and you can find the relevant information there. It provides a formatted string that includes the avatar's name and a shortened version of their address.

```
public string Name
```

Field Value

string

Examples

```
$"{name} <size=80%><color=#A68F7E>#{address.ToHex().Substring(0, 4)}
</color></size>";
```

PortraitId

```
public int PortraitId
```

Field Value

int

PurchasedTicketCount

```
public int PurchasedTicketCount
```

Field Value

int

Score

```
public int Score
```

Field Value

int

Ticket

```
public int Ticket
```

Field Value

int

TicketResetCount

```
public int TicketResetCount
```

Field Value

int

Win

```
public int Win
```

Field Value

int

Properties

Bencoded

An Bencodex.Types.IValue representation of this object that can be decoded back to instantiate an equal object. The decoded object must be equal to the original in the sense that Equals(T) should be true.

```
public IValue Bencoded { get; }
```

Property Value

IValue

Remarks

Note that the only requirement is that the produced Bencodex.Types.IValue can be decoded back to an equal object. This representation may not be canonical in the sense that additional junk data may be present in an Bencodex.Types.IValue that one may wish to decode and this may be discarded while decoding.

A specific implemnetation may decide to only allow the canonical representation to be decoded.

Methods

AddScore(int)

```
public void AddScore(int score)
```

Parameters

`score` int

BuyTicket(long)

```
public void BuyTicket(long maxCount)
```

Parameters

`maxCount` long

ResetTicket(int)

```
public void ResetTicket(int resetCount)
```

Parameters

`resetCount` int

Serialize()

```
public IValue Serialize()
```

Returns

IValue

UpdateRecord(int, int)

```
public void UpdateRecord(int win, int lose)
```

Parameters

`win` int

`lose` int

UseTicket(int)

```
public void UseTicket(int ticketCount)
```

Parameters

`ticketCount` int

Class ArenaParticipants

Namespace: Nekoyume.Model.Arena

Assembly: Lib9c.dll

Introduced at https://github.com/planetarium/lib9c/pull/1027

```
public class ArenaParticipants : IState
```

Inheritance

object ← ArenaParticipants

Implements

IState

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

ArenaParticipants(List)

```
public ArenaParticipants(List serialized)
```

Parameters

`serialized` List

ArenaParticipants(int, int)

```
public ArenaParticipants(int championshipId, int round)
```

Parameters

`championshipId` int

`round` int

Fields

Address

```
public Address Address
```

Field Value

Address

AvatarAddresses

```
public readonly List<Address> AvatarAddresses
```

Field Value

List<Address>

Methods

Add(Address)

```
public void Add(Address address)
```

Parameters

`address` Address

DeriveAddress(int, int)

```
public static Address DeriveAddress(int championshipId, int round)
```

Parameters

`championshipId` int

`round` int

Returns

Address

Serialize()

```
public IValue Serialize()
```

Returns

IValue

Class ArenaParticipantsNotFoundException

Namespace: Nekoyume.Model.Arena

Assembly: Lib9c.dll

```
[Serializable]
public class ArenaParticipantsNotFoundException : Exception, ISerializable
```

Inheritance

object ← Exception ← ArenaParticipantsNotFoundException

Implements

ISerializable

Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Constructors

ArenaParticipantsNotFoundException(SerializationInfo, StreamingContext)

```
protected ArenaParticipantsNotFoundException(SerializationInfo info,
StreamingContext context)
```

Parameters

`info` SerializationInfo

`context` StreamingContext

ArenaParticipantsNotFoundException(string)

```
public ArenaParticipantsNotFoundException(string message)
```

Parameters

`message` string

Class ArenaResult

Namespace: Nekoyume.Model.Arena

Assembly: Lib9c.dll

temp

```
public class ArenaResult : IBencodable, IState
```

Inheritance

object ← ArenaResult

Implements

IBencodable, IState

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

ArenaResult(IValue)

```
public ArenaResult(IValue bencoded)
```

Parameters

`bencoded` IValue

ArenaResult(bool, int, int, long)

```
public ArenaResult(bool isVictory, int portraitId, int level, long cp)
```

Parameters

`isVictory` bool

`portraitId` int

`level` int

`cp` long

Fields

Cp

```
public long Cp
```

Field Value

long

IsVictory

```
public bool IsVictory
```

Field Value

bool

Level

```
public int Level
```

Field Value

int

PortraitId

```
public int PortraitId
```

Field Value

int

Properties

Bencoded

An Bencodex.Types.IValue representation of this object that can be decoded back to instantiate an equal object. The decoded object must be equal to the original in the sense that Equals(T) should be true.

```
public IValue Bencoded { get; }
```

Property Value

IValue

Remarks

Note that the only requirement is that the produced Bencodex.Types.IValue can be decoded back to an equal object. This representation may not be canonical in the sense that additional junk data may be present in an Bencodex.Types.IValue that one may wish to decode and this may be discarded while decoding.

A specific implemnetation may decide to only allow the canonical representation to be decoded.

Methods

DeriveAddress(Address, TxId)

```
public static Address DeriveAddress(Address avatarAddress, TxId txId)
```

Parameters

`avatarAddress` Address

`txId` TxId

Returns

Address

Serialize()

```
public IValue Serialize()
```

Returns

IValue

Class ArenaScore

Namespace: Nekoyume.Model.Arena

Assembly: Lib9c.dll

Introduced at https://github.com/planetarium/lib9c/pull/1027

```
public class ArenaScore : IState
```

Inheritance

object ← ArenaScore

Implements

IState

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

ArenaScore(List)

```
public ArenaScore(List serialized)
```

Parameters

`serialized` List

ArenaScore(Address, int, int, int)

```
public ArenaScore(Address avatarAddress, int championshipId, int round, int score
= 1000)
```

Parameters

`avatarAddress` Address

`championshipId` int

`round` int

`score` int

Fields

Address

```
public Address Address
```

Field Value

Address

ArenaScoreDefault

```
public const int ArenaScoreDefault = 1000
```

Field Value

int

Properties

Score

```
public int Score { get; }
```

Property Value

int

Methods

AddScore(int)

```
public void AddScore(int score)
```

Parameters

`score` int

DeriveAddress(Address, int, int)

```
public static Address DeriveAddress(Address avatarAddress, int championshipId,
int round)
```

Parameters

`avatarAddress` Address

`championshipId` int

`round` int

Returns

Address

Serialize()

```
public IValue Serialize()
```

Returns

IValue

Class ArenaScoreAlreadyContainsException

Namespace: Nekoyume.Model.Arena

Assembly: Lib9c.dll

```
[Serializable]
public class ArenaScoreAlreadyContainsException : Exception, ISerializable
```

Inheritance

object ← Exception ← ArenaScoreAlreadyContainsException

Implements

ISerializable

Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Constructors

ArenaScoreAlreadyContainsException(SerializationInfo, StreamingContext)

```
protected ArenaScoreAlreadyContainsException(SerializationInfo info,
StreamingContext context)
```

Parameters

`info` SerializationInfo

`context` StreamingContext

ArenaScoreAlreadyContainsException(string)

```
public ArenaScoreAlreadyContainsException(string message)
```

Parameters

`message` string

Class ArenaScoreNotFoundException

Namespace: Nekoyume.Model.Arena

Assembly: Lib9c.dll

```
[Serializable]
public class ArenaScoreNotFoundException : Exception, ISerializable
```

Inheritance

object ← Exception ← ArenaScoreNotFoundException

Implements

ISerializable

Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Constructors

ArenaScoreNotFoundException(SerializationInfo, StreamingContext)

```
protected ArenaScoreNotFoundException(SerializationInfo info,
StreamingContext context)
```

Parameters

`info` SerializationInfo

`context` StreamingContext

ArenaScoreNotFoundException(string)

```
public ArenaScoreNotFoundException(string message)
```

Parameters

`message` string

Class CoolDownBlockException

Namespace: Nekoyume.Model.Arena

Assembly: Lib9c.dll

```
[Serializable]
public class CoolDownBlockException : Exception, ISerializable
```

Inheritance

object ← Exception ← CoolDownBlockException

Implements

ISerializable

Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Constructors

CoolDownBlockException(SerializationInfo, StreamingContext)

```
protected CoolDownBlockException(SerializationInfo info, StreamingContext context)
```

Parameters

`info` SerializationInfo

`context` StreamingContext

CoolDownBlockException(string)

```
public CoolDownBlockException(string message)
```

Parameters

`message` string

Class ExceedPlayCountException

Namespace: Nekoyume.Model.Arena

Assembly: Lib9c.dll

```
[Serializable]
public class ExceedPlayCountException : Exception, ISerializable
```

Inheritance

object ← Exception ← ExceedPlayCountException

Implements

ISerializable

Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Constructors

ExceedPlayCountException(SerializationInfo, StreamingContext)

```
protected ExceedPlayCountException(SerializationInfo info, StreamingContext context)
```

Parameters

`info` SerializationInfo

`context` StreamingContext

ExceedPlayCountException(string)

```
public ExceedPlayCountException(string message)
```

Parameters

`message` string

Class ExceedTicketPurchaseLimitDuringIntervalException

Namespace: Nekoyume.Model.Arena

Assembly: Lib9c.dll

```
[Serializable]
public class ExceedTicketPurchaseLimitDuringIntervalException :
Exception, ISerializable
```

Inheritance

object ← Exception ← ExceedTicketPurchaseLimitDuringIntervalException

Implements

ISerializable

Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Constructors

ExceedTicketPurchaseLimitDuringIntervalException(SerializationInfo, StreamingContext)

```
protected ExceedTicketPurchaseLimitDuringIntervalException(SerializationInfo info,
StreamingContext context)
```

Parameters

`info` SerializationInfo

`context` StreamingContext

ExceedTicketPurchaseLimitDuringIntervalException(string)

```
public ExceedTicketPurchaseLimitDuringIntervalException(string message)
```

Parameters

`message` string

Class ExceedTicketPurchaseLimitException

Namespace: Nekoyume.Model.Arena

Assembly: Lib9c.dll

```
[Serializable]
public class ExceedTicketPurchaseLimitException : Exception, ISerializable
```

Inheritance

object ← Exception ← ExceedTicketPurchaseLimitException

Implements

ISerializable

Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Constructors

ExceedTicketPurchaseLimitException(SerializationInfo, StreamingContext)

```
protected ExceedTicketPurchaseLimitException(SerializationInfo info,
StreamingContext context)
```

Parameters

`info` SerializationInfo

`context` StreamingContext

ExceedTicketPurchaseLimitException(string)

```
public ExceedTicketPurchaseLimitException(string message)
```

Parameters

`message` string

Class InvalidSeasonException

Namespace: Nekoyume.Model.Arena

Assembly: Lib9c.dll

```
[Serializable]
public class InvalidSeasonException : Exception, ISerializable
```

Inheritance

object ← Exception ← InvalidSeasonException

Implements

ISerializable

Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Constructors

InvalidSeasonException(SerializationInfo, StreamingContext)

```
protected InvalidSeasonException(SerializationInfo info, StreamingContext context)
```

Parameters

`info` SerializationInfo

`context` StreamingContext

InvalidSeasonException(string)

```
public InvalidSeasonException(string message)
```

Parameters

`message` string

Class NotEnoughTicketException

Namespace: Nekoyume.Model.Arena

Assembly: Lib9c.dll

```
[Serializable]
public class NotEnoughTicketException : Exception, ISerializable
```

Inheritance

object ← Exception ← NotEnoughTicketException

Implements

ISerializable

Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Constructors

NotEnoughTicketException(SerializationInfo, StreamingContext)

```
protected NotEnoughTicketException(SerializationInfo info, StreamingContext context)
```

Parameters

`info` SerializationInfo

`context` StreamingContext

NotEnoughTicketException(string)

```
public NotEnoughTicketException(string message)
```

Parameters

`message` string

Class RoundNotFoundException

Namespace: Nekoyume.Model.Arena

Assembly: Lib9c.dll

```
[Serializable]
public class RoundNotFoundException : Exception, ISerializable
```

Inheritance

object ← Exception ← RoundNotFoundException

Implements

ISerializable

Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Constructors

RoundNotFoundException(SerializationInfo, StreamingContext)

```
protected RoundNotFoundException(SerializationInfo info, StreamingContext context)
```

Parameters

`info` SerializationInfo

`context` StreamingContext

RoundNotFoundException(string)

```
public RoundNotFoundException(string message)
```

Parameters

`message` string

Class ThisArenaIsClosedException

Namespace: Nekoyume.Model.Arena

Assembly: Lib9c.dll

```
[Serializable]
public class ThisArenaIsClosedException : Exception, ISerializable
```

Inheritance

object ← Exception ← ThisArenaIsClosedException

Implements

ISerializable

Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Constructors

ThisArenaIsClosedException(SerializationInfo, StreamingContext)

```
protected ThisArenaIsClosedException(SerializationInfo info,
StreamingContext context)
```

Parameters

`info` SerializationInfo

`context` StreamingContext

ThisArenaIsClosedException(string)

```
public ThisArenaIsClosedException(string message)
```

Parameters

`message` string

Class TicketPurchaseLimitExceedException

Namespace: Nekoyume.Model.Arena

Assembly: Lib9c.dll

```
public class TicketPurchaseLimitExceedException : Exception, ISerializable
```

Inheritance

object ← Exception ← TicketPurchaseLimitExceedException

Implements

ISerializable

Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Constructors

TicketPurchaseLimitExceedException(SerializationInfo, StreamingContext)

```
protected TicketPurchaseLimitExceedException(SerializationInfo info,
StreamingContext context)
```

Parameters

`info` SerializationInfo

`context` StreamingContext

TicketPurchaseLimitExceedException(string)

```
public TicketPurchaseLimitExceedException(string message)
```

Parameters

`message` string

Class ValidateScoreDifferenceException

Namespace: Nekoyume.Model.Arena

Assembly: Lib9c.dll

```
[Serializable]
public class ValidateScoreDifferenceException : Exception, ISerializable
```

Inheritance

object ← Exception ← ValidateScoreDifferenceException

Implements

ISerializable

Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Constructors

ValidateScoreDifferenceException(SerializationInfo, StreamingContext)

```
protected ValidateScoreDifferenceException(SerializationInfo info,
StreamingContext context)
```

Parameters

`info` SerializationInfo

`context` StreamingContext

ValidateScoreDifferenceException(string)

```
public ValidateScoreDifferenceException(string message)
```

Parameters

`message` string

Namespace Nekoyume.Model.BattleStatus

Classes

AreaAttack

BattleLog

BlowAttack

Buff

BuffRemovalAttack

Dead

DoubleAttack

DoubleAttackWithCombo

DropBox

EventBase

FullBuffRemovalAttack
: Battle status event for the full buff removal attack skill. Triggers the CoFullBuffRemovalAttack(CharacterBase, int, IEnumerable<SkillInfo>, IEnumerable<SkillInfo>) coroutine for stage rendering.

GetExp

GetReward
: Event that represents a character receiving rewards after battle. Supports both item rewards and fungible asset rewards.

HealSkill

NormalAttack

RemoveBuffs

ShatterStrike

Skill

Skill.SkillInfo

SpawnEnemyPlayer

SpawnPlayer

SpawnWave

Tick

TickDamage

WaveTurnEnd

Enums

BattleLog.Result

Class AreaAttack

Namespace: Nekoyume.Model.BattleStatus

Assembly: Lib9c.dll

```
[Serializable]
public class AreaAttack : Skill
```

Inheritance

object ← EventBase ← Skill ← AreaAttack

Inherited Members

Skill.SkillId , Skill.SkillInfos , Skill.BuffInfos , EventBase.Character , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

AreaAttack(int, CharacterBase, IEnumerable<SkillInfo>, IEnumerable<SkillInfo>)

```
public AreaAttack(int skillId, CharacterBase character, IEnumerable<Skill.SkillInfo>
skillInfos, IEnumerable<Skill.SkillInfo> buffInfos)
```

Parameters

`skillId` int

`character` CharacterBase

`skillInfos` IEnumerable<Skill.SkillInfo>

`buffInfos` IEnumerable<Skill.SkillInfo>

Methods

CoExecute(IStage)

```
public override IEnumerator CoExecute(IStage stage)
```

Parameters

`stage` IStage

Returns

IEnumerator

Class BattleLog

Namespace: Nekoyume.Model.BattleStatus

Assembly: Lib9c.dll

```
[Serializable]
public class BattleLog : IEnumerable<EventBase>, IEnumerable
```

Inheritance

object ← BattleLog

Implements

IEnumerable<EventBase>, IEnumerable

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Fields

clearedWaveNumber

```
public int clearedWaveNumber
```

Field Value

int

diffScore

```
public int diffScore
```

Field Value

int

events

```
public List<EventBase> events
```

Field Value

List<EventBase>

id

```
public Guid id
```

Field Value

Guid

newlyCleared

```
public bool newlyCleared
```

Field Value

bool

result

```
public BattleLog.Result result
```

Field Value

BattleLog.Result

score

```
public int score
```

Field Value

int

stageId

```
public int stageId
```

Field Value

int

waveCount

```
public int waveCount
```

Field Value

int

worldId

```
public int worldId
```

Field Value

int

Properties

Count

```
public int Count { get; }
```

Property Value

int

IsClear

```
public bool IsClear { get; }
```

Property Value

bool

Methods

Add(EventBase)

```
public void Add(EventBase e)
```

Parameters

e EventBase

GetEnumerator()

Returns an enumerator that iterates through the collection.

```
public IEnumerator<EventBase> GetEnumerator()
```

Returns

IEnumerator<EventBase>

An enumerator that can be used to iterate through the collection.

Enum BattleLog.Result

Namespace: Nekoyume.Model.BattleStatus

Assembly: Lib9c.dll

```
public enum BattleLog.Result
```

Extension Methods
StateExtensions.Serialize(Enum)

Fields

`Lose = 1`

`TimeOver = 2`

`Win = 0`

Class BlowAttack

Namespace: Nekoyume.Model.BattleStatus

Assembly: Lib9c.dll

```
[Serializable]
public class BlowAttack : Skill
```

Inheritance

object ← EventBase ← Skill ← BlowAttack

Inherited Members

Skill.SkillId , Skill.SkillInfos , Skill.BuffInfos , EventBase.Character , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

BlowAttack(int, CharacterBase, IEnumerable<SkillInfo>, IEnumerable<SkillInfo>)

```
public BlowAttack(int skillId, CharacterBase character, IEnumerable<Skill.SkillInfo>
skillInfos, IEnumerable<Skill.SkillInfo> buffInfos)
```

Parameters

`skillId` int

`character` CharacterBase

`skillInfos` IEnumerable<Skill.SkillInfo>

`buffInfos` IEnumerable<Skill.SkillInfo>

Methods

CoExecute(IStage)

```
public override IEnumerator CoExecute(IStage stage)
```

Parameters

`stage` IStage

Returns

IEnumerator

Class Buff

Namespace: Nekoyume.Model.BattleStatus

Assembly: Lib9c.dll

```
[Serializable]
public class Buff : Skill
```

Inheritance

object ← EventBase ← Skill ← Buff

Inherited Members

Skill.SkillId , Skill.SkillInfos , Skill.BuffInfos , EventBase.Character , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

Buff(int, CharacterBase, IEnumerable<SkillInfo>)

```
public Buff(int skillId, CharacterBase character, IEnumerable<Skill.SkillInfo>
skillInfos)
```

Parameters

`skillId` int

`character` CharacterBase

`skillInfos` IEnumerable<Skill.SkillInfo>

Methods

CoExecute(IStage)

```
public override IEnumerator CoExecute(IStage stage)
```

Parameters

`stage` IStage

Returns

IEnumerator

Class BuffRemovalAttack

Namespace: Nekoyume.Model.BattleStatus

Assembly: Lib9c.dll

```
[Serializable]
public class BuffRemovalAttack : Skill
```

Inheritance

object ← EventBase ← Skill ← BuffRemovalAttack

Inherited Members

Skill.SkillId , Skill.SkillInfos , Skill.BuffInfos , EventBase.Character , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

BuffRemovalAttack(int, CharacterBase, IEnumerable<SkillInfo>, IEnumerable<SkillInfo>)

```
public BuffRemovalAttack(int skillId, CharacterBase character,
IEnumerable<Skill.SkillInfo> skillInfos, IEnumerable<Skill.SkillInfo> buffInfos)
```

Parameters

`skillId` int

`character` CharacterBase

`skillInfos` IEnumerable<Skill.SkillInfo>

`buffInfos` IEnumerable<Skill.SkillInfo>

Methods

CoExecute(IStage)

```
public override IEnumerator CoExecute(IStage stage)
```

Parameters

`stage` IStage

Returns

IEnumerator

Class Dead

Namespace: Nekoyume.Model.BattleStatus

Assembly: Lib9c.dll

```
[Serializable]
public class Dead : EventBase
```

Inheritance

object ← EventBase ← Dead

Inherited Members

EventBase.Character , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

Dead(CharacterBase)

```
public Dead(CharacterBase character)
```

Parameters

`character` CharacterBase

Methods

CoExecute(IStage)

```
public override IEnumerator CoExecute(IStage stage)
```

Parameters

`stage` IStage

Returns

IEnumerator

Class DoubleAttack

Namespace: Nekoyume.Model.BattleStatus

Assembly: Lib9c.dll

```
[Serializable]
public class DoubleAttack : Skill
```

Inheritance

object ← EventBase ← Skill ← DoubleAttack

Inherited Members

Skill.SkillId , Skill.SkillInfos , Skill.BuffInfos , EventBase.Character , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

DoubleAttack(int, CharacterBase, IEnumerable<SkillInfo>, IEnumerable<SkillInfo>)

```
public DoubleAttack(int skillId, CharacterBase character,
IEnumerable<Skill.SkillInfo> skillInfos, IEnumerable<Skill.SkillInfo> buffInfos)
```

Parameters

`skillId` int

`character` CharacterBase

`skillInfos` IEnumerable<Skill.SkillInfo>

`buffInfos` IEnumerable<Skill.SkillInfo>

Methods

CoExecute(IStage)

```
public override IEnumerator CoExecute(IStage stage)
```

Parameters

`stage` IStage

Returns

IEnumerator

Class DoubleAttackWithCombo

Namespace: Nekoyume.Model.BattleStatus

Assembly: Lib9c.dll

```
[Serializable]
public class DoubleAttackWithCombo : Skill
```

Inheritance

object ← EventBase ← Skill ← DoubleAttackWithCombo

Inherited Members

Skill.SkillId , Skill.SkillInfos , Skill.BuffInfos , EventBase.Character , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

DoubleAttackWithCombo(int, CharacterBase, IEnumerable<SkillInfo>, IEnumerable<SkillInfo>)

```
public DoubleAttackWithCombo(int skillId, CharacterBase character,
IEnumerable<Skill.SkillInfo> skillInfos, IEnumerable<Skill.SkillInfo> buffInfos)
```

Parameters

`skillId` int

`character` CharacterBase

`skillInfos` IEnumerable<Skill.SkillInfo>

`buffInfos` IEnumerable<Skill.SkillInfo>

Methods

CoExecute(IStage)

```
public override IEnumerator CoExecute(IStage stage)
```

Parameters

`stage` IStage

Returns

IEnumerator

Class DropBox

Namespace: Nekoyume.Model.BattleStatus

Assembly: Lib9c.dll

```
[Serializable]
public class DropBox : EventBase
```

Inheritance

object ← EventBase ← DropBox

Inherited Members

EventBase.Character , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

DropBox(CharacterBase, List<ItemBase>)

```
public DropBox(CharacterBase character, List<ItemBase> items)
```

Parameters

`character` CharacterBase

`items` List<ItemBase>

Fields

Items

```
public readonly List<ItemBase> Items
```

Field Value

List<ItemBase>

Methods

CoExecute(IStage)

```
public override IEnumerator CoExecute(IStage stage)
```

Parameters

`stage` IStage

Returns

IEnumerator

Class EventBase

Namespace: Nekoyume.Model.BattleStatus

Assembly: Lib9c.dll

```
[Serializable]
public abstract class EventBase
```

Inheritance

object ← EventBase

Derived

Breakthrough, Dead, DropBox, GetExp, GetReward, RemoveBuffs, Skill, SpawnEnemyPlayer, SpawnPlayer, SpawnWave, WaveTurnEnd

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

EventBase(CharacterBase)

```
protected EventBase(CharacterBase character)
```

Parameters

`character` CharacterBase

Fields

Character

```
public readonly CharacterBase Character
```

Field Value

CharacterBase

Methods

CoExecute(IStage)

```
public abstract IEnumerator CoExecute(IStage stage)
```

Parameters

`stage` IStage

Returns

IEnumerator

Class FullBuffRemovalAttack

Namespace: Nekoyume.Model.BattleStatus

Assembly: Lib9c.dll

Battle status event for the full buff removal attack skill. Triggers the CoFullBuffRemovalAttack(CharacterBase, int, IEnumerable<SkillInfo>, IEnumerable<SkillInfo>) coroutine for stage rendering.

```
[Serializable]
public class FullBuffRemovalAttack : Skill
```

Inheritance

object ← EventBase ← Skill ← FullBuffRemovalAttack

Inherited Members

Skill.SkillId , Skill.SkillInfos , Skill.BuffInfos , EventBase.Character , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

FullBuffRemovalAttack(int, CharacterBase, IEnumerable<SkillInfo>, IEnumerable<SkillInfo>)

```
public FullBuffRemovalAttack(int skillId, CharacterBase character,
IEnumerable<Skill.SkillInfo> skillInfos, IEnumerable<Skill.SkillInfo> buffInfos)
```

Parameters

`skillId` int

`character` CharacterBase

`skillInfos` IEnumerable<Skill.SkillInfo>

`buffInfos` IEnumerable<Skill.SkillInfo>

Methods

CoExecute(IStage)

```
public override IEnumerator CoExecute(IStage stage)
```

Parameters

`stage` IStage

Returns

IEnumerator

Class GetExp

Namespace: Nekoyume.Model.BattleStatus

Assembly: Lib9c.dll

```
[Serializable]
public class GetExp : EventBase
```

Inheritance

object ← EventBase ← GetExp

Inherited Members

EventBase.Character , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

GetExp(CharacterBase, long)

```
public GetExp(CharacterBase character, long exp)
```

Parameters

`character` CharacterBase

`exp` long

Properties

Exp

```
public long Exp { get; }
```

Property Value

long

Methods

CoExecute(IStage)

```
public override IEnumerator CoExecute(IStage stage)
```

Parameters

`stage` IStage

Returns

IEnumerator

Class GetReward

Namespace: Nekoyume.Model.BattleStatus

Assembly: Lib9c.dll

Event that represents a character receiving rewards after battle. Supports both item rewards and fungible asset rewards.

```
[Serializable]
public class GetReward : EventBase
```

Inheritance

object ← EventBase ← GetReward

Inherited Members

EventBase.Character , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

GetReward(CharacterBase, List<ItemBase>)

Initializes a new instance of the GetReward class with item rewards only.

```
public GetReward(CharacterBase character, List<ItemBase> rewards)
```

Parameters

`character` CharacterBase

The character receiving the rewards.

`rewards` List<ItemBase>

List of item rewards.

GetReward(CharacterBase, List<ItemBase>, Dictionary<string, int>)

Initializes a new instance of the GetReward class with both item and fungible asset rewards.

```
public GetReward(CharacterBase character, List<ItemBase> rewards, Dictionary<string,
int> fungibleAssetRewards)
```

Parameters

`character` CharacterBase

The character receiving the rewards.

`rewards` List<ItemBase>

List of item rewards.

`fungibleAssetRewards` Dictionary<string, int>

Dictionary of fungible asset rewards.

Fields

FungibleAssetRewards

Dictionary of fungible asset rewards (e.g., NCG, Crystal) received by the character.

```
public readonly Dictionary<string, int> FungibleAssetRewards
```

Field Value

Dictionary<string, int>

Rewards

List of item rewards received by the character.

```
public readonly List<ItemBase> Rewards
```

Field Value

List<ItemBase>

Methods

CoExecute(IStage)

Executes the reward event by giving rewards to the character.

```
public override IEnumerator CoExecute(IStage stage)
```

Parameters

`stage` IStage

The stage context for reward processing.

Returns

IEnumerator

Coroutine enumerator for reward execution.

Class HealSkill

Namespace: Nekoyume.Model.BattleStatus

Assembly: Lib9c.dll

```
[Serializable]
public class HealSkill : Skill
```

Inheritance

object ← EventBase ← Skill ← HealSkill

Inherited Members

Skill.SkillId , Skill.SkillInfos , Skill.BuffInfos , EventBase.Character , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

HealSkill(int, CharacterBase, IEnumerable<SkillInfo>, IEnumerable<SkillInfo>)

```
public HealSkill(int skillId, CharacterBase character, IEnumerable<Skill.SkillInfo>
skillInfos, IEnumerable<Skill.SkillInfo> buffInfos)
```

Parameters

`skillId` int

`character` CharacterBase

`skillInfos` IEnumerable<Skill.SkillInfo>

`buffInfos` IEnumerable<Skill.SkillInfo>

Methods

CoExecute(IStage)

```
public override IEnumerator CoExecute(IStage stage)
```

Parameters

`stage` IStage

Returns

IEnumerator

Class NormalAttack

Namespace: Nekoyume.Model.BattleStatus

Assembly: Lib9c.dll

```
[Serializable]
public class NormalAttack : Skill
```

Inheritance

object ← EventBase ← Skill ← NormalAttack

Inherited Members

Skill.SkillId , Skill.SkillInfos , Skill.BuffInfos , EventBase.Character , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

NormalAttack(int, CharacterBase, IEnumerable<SkillInfo>, IEnumerable<SkillInfo>)

```
public NormalAttack(int skillId, CharacterBase character,
IEnumerable<Skill.SkillInfo> skillInfos, IEnumerable<Skill.SkillInfo> buffInfos)
```

Parameters

`skillId` int

`character` CharacterBase

`skillInfos` IEnumerable<Skill.SkillInfo>

`buffInfos` IEnumerable<Skill.SkillInfo>

Methods

CoExecute(IStage)

```
public override IEnumerator CoExecute(IStage stage)
```

Parameters

`stage` IStage

Returns

IEnumerator

Class RemoveBuffs

Namespace: Nekoyume.Model.BattleStatus

Assembly: Lib9c.dll

```
[Serializable]
public class RemoveBuffs : EventBase
```

Inheritance

object ← EventBase ← RemoveBuffs

Inherited Members

EventBase.Character , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

RemoveBuffs(CharacterBase)

```
public RemoveBuffs(CharacterBase character)
```

Parameters

`character` CharacterBase

Methods

CoExecute(IStage)

```
public override IEnumerator CoExecute(IStage stage)
```

Parameters

`stage` IStage

Returns

IEnumerator

Class ShatterStrike

Namespace: Nekoyume.Model.BattleStatus

Assembly: Lib9c.dll

```
[Serializable]
public class ShatterStrike : Skill
```

Inheritance

object ← EventBase ← Skill ← ShatterStrike

Inherited Members

Skill.SkillId , Skill.SkillInfos , Skill.BuffInfos , EventBase.Character , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

ShatterStrike(int, CharacterBase, IEnumerable<SkillInfo>, IEnumerable<SkillInfo>)

```
public ShatterStrike(int skillId, CharacterBase character,
IEnumerable<Skill.SkillInfo> skillInfos, IEnumerable<Skill.SkillInfo> buffInfos)
```

Parameters

`skillId` int

`character` CharacterBase

`skillInfos` IEnumerable<Skill.SkillInfo>

`buffInfos` IEnumerable<Skill.SkillInfo>

Methods

CoExecute(IStage)

```
public override IEnumerator CoExecute(IStage stage)
```

Parameters

`stage` IStage

Returns

IEnumerator

Class Skill

Namespace: Nekoyume.Model.BattleStatus

Assembly: Lib9c.dll

```
[Serializable]
public abstract class Skill : EventBase
```

Inheritance

object ← EventBase ← Skill

Derived

StageBuff, AreaAttack, BlowAttack, Buff, BuffRemovalAttack, DoubleAttack, DoubleAttackWithCombo, FullBuffRemovalAttack, HealSkill, NormalAttack, ShatterStrike, Tick, TickDamage

Inherited Members

EventBase.Character , EventBase.CoExecute(IStage) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

Skill(int, CharacterBase, IEnumerable<SkillInfo>, IEnumerable<SkillInfo>)

```
protected Skill(int skillId, CharacterBase character, IEnumerable<Skill.SkillInfo>
skillInfos, IEnumerable<Skill.SkillInfo> buffInfos)
```

Parameters

`skillId` int

`character` CharacterBase

`skillInfos` IEnumerable<Skill.SkillInfo>

`buffInfos` IEnumerable<Skill.SkillInfo>

Fields

BuffInfos

```
public readonly IEnumerable<Skill.SkillInfo>? BuffInfos
```

Field Value

IEnumerable<Skill.SkillInfo>

SkillId

```
public readonly int SkillId
```

Field Value

int

SkillInfos

```
public readonly IEnumerable<Skill.SkillInfo> SkillInfos
```

Field Value

IEnumerable<Skill.SkillInfo>

Class Skill.SkillInfo

Namespace: Nekoyume.Model.BattleStatus

Assembly: Lib9c.dll

```
[Serializable]
public class Skill.SkillInfo
```

Inheritance

object ← Skill.SkillInfo

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

SkillInfo(Guid, bool, long, long, bool, SkillCategory, int, ElementalType, SkillTargetType, Buff?, CharacterBase?, bool, IEnumerable<Buff>?, IceShield?)

```
public SkillInfo(Guid characterId, bool isDead, long thorn, long effect, bool
critical, SkillCategory skillCategory, int waveTurn, ElementalType elementalType =
ElementalType.Normal, SkillTargetType targetType = SkillTargetType.Enemy, Buff? buff
= null, CharacterBase? target = null, bool affected = true, IEnumerable<Buff>?
dispelList = null, IceShield? iceShield = null)
```

Parameters

`characterId` Guid

`isDead` bool

`thorn` long

`effect` long

`critical` bool

`skillCategory` SkillCategory

`waveTurn` int

`elementalType` ElementalType

`targetType` SkillTargetType

`buff` Buff

`target` CharacterBase

`affected` bool

`dispelList` IEnumerable<Buff>

`iceShield` IceShield

Fields

Affected

```
public readonly bool Affected
```

Field Value

bool

Buff

```
public readonly Buff? Buff
```

Field Value

Buff

CharacterId

```
public readonly Guid CharacterId
```

Field Value

Guid

Critical

```
public readonly bool Critical
```

Field Value

bool

DispelList

```
public readonly IEnumerable<Buff>? DispelList
```

Field Value

IEnumerable<Buff>

Effect

```
public readonly long Effect
```

Field Value

long

ElementalType

```
public readonly ElementalType ElementalType
```

Field Value

ElementalType

IceShield

```
public readonly IceShield? IceShield
```

Field Value

IceShield

IsDead

```
public readonly bool IsDead
```

Field Value

bool

SkillCategory

```
public readonly SkillCategory SkillCategory
```

Field Value

SkillCategory

SkillTargetType

```
public readonly SkillTargetType SkillTargetType
```

Field Value

SkillTargetType

Target

```
public readonly CharacterBase? Target
```

Field Value

CharacterBase

Thorn

```
public readonly long Thorn
```

Field Value

long

WaveTurn

```
public readonly int WaveTurn
```

Field Value

int

Class SpawnEnemyPlayer

Namespace: Nekoyume.Model.BattleStatus

Assembly: Lib9c.dll

```
[Serializable]
public class SpawnEnemyPlayer : EventBase
```

Inheritance

object ← EventBase ← SpawnEnemyPlayer

Inherited Members

EventBase.Character , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

SpawnEnemyPlayer(CharacterBase)

```
public SpawnEnemyPlayer(CharacterBase character)
```

Parameters

`character` CharacterBase

Methods

CoExecute(IStage)

```
public override IEnumerator CoExecute(IStage stage)
```

Parameters

`stage` IStage

Returns

IEnumerator

Class SpawnPlayer

Namespace: Nekoyume.Model.BattleStatus

Assembly: Lib9c.dll

```
[Serializable]
public class SpawnPlayer : EventBase
```

Inheritance

object ← EventBase ← SpawnPlayer

Inherited Members

EventBase.Character , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

SpawnPlayer(CharacterBase)

```
public SpawnPlayer(CharacterBase character)
```

Parameters

`character` CharacterBase

Methods

CoExecute(IStage)

```
public override IEnumerator CoExecute(IStage stage)
```

Parameters

`stage` IStage

Returns

IEnumerator

Class SpawnWave

Namespace: Nekoyume.Model.BattleStatus

Assembly: Lib9c.dll

```
[Serializable]
public class SpawnWave : EventBase
```

Inheritance

object ← EventBase ← SpawnWave

Inherited Members

EventBase.Character , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

SpawnWave(CharacterBase, int, int, List<Enemy>, bool)

```
public SpawnWave(CharacterBase character, int waveNumber, int waveTurn, List<Enemy>
enemies, bool hasBoss)
```

Parameters

`character` CharacterBase

`waveNumber` int

`waveTurn` int

`enemies` List<Enemy>

`hasBoss` bool

Fields

Enemies

```
public readonly List<Enemy> Enemies
```

Field Value

List<Enemy>

HasBoss

```
public readonly bool HasBoss
```

Field Value

bool

WaveNumber

```
public readonly int WaveNumber
```

Field Value

int

WaveTurn

```
public readonly int WaveTurn
```

Field Value

int

Methods

CoExecute(IStage)

```
public override IEnumerator CoExecute(IStage stage)
```

Parameters

`stage` IStage

Returns

IEnumerator

Class Tick

Namespace: Nekoyume.Model.BattleStatus

Assembly: Lib9c.dll

```
[Serializable]
public class Tick : Skill
```

Inheritance

object ← EventBase ← Skill ← Tick

Inherited Members

Skill.SkillId , Skill.SkillInfos , Skill.BuffInfos , EventBase.Character , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

Tick(CharacterBase)

```
public Tick(CharacterBase character)
```

Parameters

`character` CharacterBase

Tick(int, CharacterBase, IEnumerable<SkillInfo>, IEnumerable<SkillInfo>)

```
public Tick(int skillId, CharacterBase character, IEnumerable<Skill.SkillInfo>
skillInfos, IEnumerable<Skill.SkillInfo> buffInfos)
```

Parameters

`skillId` int

`character` CharacterBase

`skillInfos` IEnumerable<Skill.SkillInfo>

`buffInfos` IEnumerable<Skill.SkillInfo>

Methods

CoExecute(IStage)

```
public override IEnumerator CoExecute(IStage stage)
```

Parameters

`stage` IStage

Returns

IEnumerator

Class TickDamage

Namespace: Nekoyume.Model.BattleStatus

Assembly: Lib9c.dll

```
[Serializable]
public class TickDamage : Skill
```

Inheritance

object ← EventBase ← Skill ← TickDamage

Inherited Members

Skill.SkillId , Skill.SkillInfos , Skill.BuffInfos , EventBase.Character , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

TickDamage(int, CharacterBase, IEnumerable<SkillInfo>, IEnumerable<SkillInfo>)

```
public TickDamage(int skillId, CharacterBase character, IEnumerable<Skill.SkillInfo>
skillInfos, IEnumerable<Skill.SkillInfo> buffInfos)
```

Parameters

`skillId` int

`character` CharacterBase

`skillInfos` IEnumerable<Skill.SkillInfo>

`buffInfos` IEnumerable<Skill.SkillInfo>

Methods

CoExecute(IStage)

```
public override IEnumerator CoExecute(IStage stage)
```

Parameters

`stage` IStage

Returns

IEnumerator

Class WaveTurnEnd

Namespace: Nekoyume.Model.BattleStatus

Assembly: Lib9c.dll

```
[Serializable]
public class WaveTurnEnd : EventBase
```

Inheritance

object ← EventBase ← WaveTurnEnd

Inherited Members

EventBase.Character , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

WaveTurnEnd(CharacterBase, int, int)

```
public WaveTurnEnd(CharacterBase character, int turnNumber, int waveTurn)
```

Parameters

`character` CharacterBase

`turnNumber` int

`waveTurn` int

Fields

TurnNumber

```
public readonly int TurnNumber
```

Field Value

int

WaveTurn

```
public readonly int WaveTurn
```

Field Value

int

Methods

CoExecute(IStage)

```
public override IEnumerator CoExecute(IStage stage)
```

Parameters

`stage` IStage

Returns

IEnumerator

Namespace Nekoyume.Model.Battle Status.AdventureBoss

Classes

Breakthrough

StageBuff

Class Breakthrough

Namespace: Nekoyume.Model.BattleStatus.AdventureBoss

Assembly: Lib9c.dll

```
[Serializable]
public class Breakthrough : EventBase
```

Inheritance

object ← EventBase ← Breakthrough

Inherited Members

EventBase.Character , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

Breakthrough(CharacterBase, int, List<MonsterData>)

```
public Breakthrough(CharacterBase character, int floorId,
List<AdventureBossFloorWaveSheet.MonsterData> monsters)
```

Parameters

`character` CharacterBase

`floorId` int

`monsters` List<AdventureBossFloorWaveSheet.MonsterData>

Fields

FloorId

```
public readonly int FloorId
```

Field Value

int

Monsters

```
public readonly List<AdventureBossFloorWaveSheet.MonsterData> Monsters
```

Field Value

List<AdventureBossFloorWaveSheet.MonsterData>

Methods

CoExecute(IStage)

```
public override IEnumerator CoExecute(IStage stage)
```

Parameters

`stage` IStage

Returns

IEnumerator

Class StageBuff

Namespace: Nekoyume.Model.BattleStatus.AdventureBoss

Assembly: Lib9c.dll

```
[Serializable]
public class StageBuff : Skill
```

Inheritance

object ← EventBase ← Skill ← StageBuff

Inherited Members

Skill.SkillId , Skill.SkillInfos , Skill.BuffInfos , EventBase.Character , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

StageBuff(int, CharacterBase, IEnumerable<SkillInfo>, IEnumerable<SkillInfo>)

```
public StageBuff(int skillId, CharacterBase character, IEnumerable<Skill.SkillInfo>
skillInfos, IEnumerable<Skill.SkillInfo> buffInfos)
```

Parameters

`skillId` int

`character` CharacterBase

`skillInfos` IEnumerable<Skill.SkillInfo>

`buffInfos` IEnumerable<Skill.SkillInfo>

Methods

CoExecute(IStage)

```
public override IEnumerator CoExecute(IStage stage)
```

Parameters

`stage` IStage

Returns

IEnumerator

Namespace Nekoyume.Model.Battle Status.Arena

Classes

ArenaAreaAttack

ArenaBlowAttack

ArenaBuff

ArenaBuffRemovalAttack

ArenaDead

ArenaDoubleAttack

ArenaDoubleAttackWithCombo

ArenaEventBase

ArenaFullBuffRemovalAttack

Arena battle status event for the full buff removal attack skill. Triggers the CoFullBuffRemovalAttack(ArenaCharacter, IEnumerable<ArenaSkillInfo>, IEnumerable<ArenaSkillInfo>) coroutine for arena rendering.

ArenaHeal

ArenaLog

ArenaNormalAttack

ArenaRemoveBuffs

ArenaShatterStrike

ArenaSkill

ArenaSkill.ArenaSkillInfo

ArenaSpawnCharacter

ArenaTick

ArenaTickDamage

ArenaTurnEnd

Enums

ArenaLog.ArenaResult

Class ArenaAreaAttack

Namespace: Nekoyume.Model.BattleStatus.Arena

Assembly: Lib9c.dll

```
[Serializable]
public class ArenaAreaAttack : ArenaSkill
```

Inheritance

object ← ArenaEventBase ← ArenaSkill ← ArenaAreaAttack

Inherited Members

ArenaSkill.SkillId , ArenaSkill.SkillInfos , ArenaSkill.BuffInfos , ArenaEventBase.Character , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

ArenaAreaAttack(int, ArenaCharacter, IEnumerable<ArenaSkillInfo>, IEnumerable<ArenaSkillInfo>)

```
public ArenaAreaAttack(int skillId, ArenaCharacter character,
IEnumerable<ArenaSkill.ArenaSkillInfo> skillInfos,
IEnumerable<ArenaSkill.ArenaSkillInfo> buffInfos)
```

Parameters

`skillId` int

`character` ArenaCharacter

`skillInfos` IEnumerable<ArenaSkill.ArenaSkillInfo>

`buffInfos` IEnumerable<ArenaSkill.ArenaSkillInfo>

Methods

CoExecute(IArena)

```
public override IEnumerator CoExecute(IArena arena)
```

Parameters

`arena` IArena

Returns

IEnumerator

Class ArenaBlowAttack

Namespace: Nekoyume.Model.BattleStatus.Arena

Assembly: Lib9c.dll

```
[Serializable]
public class ArenaBlowAttack : ArenaSkill
```

Inheritance

object ← ArenaEventBase ← ArenaSkill ← ArenaBlowAttack

Inherited Members

ArenaSkill.SkillId , ArenaSkill.SkillInfos , ArenaSkill.BuffInfos , ArenaEventBase.Character , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

ArenaBlowAttack(int, ArenaCharacter, IEnumerable<ArenaSkillInfo>, IEnumerable<ArenaSkillInfo>)

```
public ArenaBlowAttack(int skillId, ArenaCharacter character,
IEnumerable<ArenaSkill.ArenaSkillInfo> skillInfos,
IEnumerable<ArenaSkill.ArenaSkillInfo> buffInfos)
```

Parameters

`skillId` int

`character` ArenaCharacter

`skillInfos` IEnumerable<ArenaSkill.ArenaSkillInfo>

`buffInfos` IEnumerable<ArenaSkill.ArenaSkillInfo>

Methods

CoExecute(IArena)

```
public override IEnumerator CoExecute(IArena arena)
```

Parameters

`arena` IArena

Returns

IEnumerator

Class ArenaBuff

Namespace: Nekoyume.Model.BattleStatus.Arena

Assembly: Lib9c.dll

```
[Serializable]
public class ArenaBuff : ArenaSkill
```

Inheritance

object ← ArenaEventBase ← ArenaSkill ← ArenaBuff

Inherited Members

ArenaSkill.SkillId , ArenaSkill.SkillInfos , ArenaSkill.BuffInfos , ArenaEventBase.Character , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

ArenaBuff(int, ArenaCharacter, IEnumerable<ArenaSkillInfo>)

```
public ArenaBuff(int skillId, ArenaCharacter character,
IEnumerable<ArenaSkill.ArenaSkillInfo> skillInfos)
```

Parameters

`skillId` int

`character` ArenaCharacter

`skillInfos` IEnumerable<ArenaSkill.ArenaSkillInfo>

Methods

CoExecute(IArena)

```
public override IEnumerator CoExecute(IArena arena)
```

Parameters

`arena` IArena

Returns

IEnumerator

Class ArenaBuffRemovalAttack

Namespace: Nekoyume.Model.BattleStatus.Arena

Assembly: Lib9c.dll

```
[Serializable]
public class ArenaBuffRemovalAttack : ArenaSkill
```

Inheritance

object ← ArenaEventBase ← ArenaSkill ← ArenaBuffRemovalAttack

Inherited Members

ArenaSkill.SkillId , ArenaSkill.SkillInfos , ArenaSkill.BuffInfos , ArenaEventBase.Character , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

ArenaBuffRemovalAttack(int, ArenaCharacter, IEnumerable<ArenaSkillInfo>, IEnumerable<ArenaSkillInfo>)

```
public ArenaBuffRemovalAttack(int skillId, ArenaCharacter character,
IEnumerable<ArenaSkill.ArenaSkillInfo> skillInfos,
IEnumerable<ArenaSkill.ArenaSkillInfo> buffInfos)
```

Parameters

`skillId` int

`character` ArenaCharacter

`skillInfos` IEnumerable<ArenaSkill.ArenaSkillInfo>

`buffInfos` IEnumerable<ArenaSkill.ArenaSkillInfo>

Methods

CoExecute(IArena)

```
public override IEnumerator CoExecute(IArena arena)
```

Parameters

`arena` IArena

Returns

IEnumerator

Class ArenaDead

Namespace: Nekoyume.Model.BattleStatus.Arena

Assembly: Lib9c.dll

```
[Serializable]
public class ArenaDead : ArenaEventBase
```

Inheritance

object ← ArenaEventBase ← ArenaDead

Inherited Members

ArenaEventBase.Character , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

ArenaDead(ArenaCharacter)

```
public ArenaDead(ArenaCharacter character)
```

Parameters

`character` ArenaCharacter

Methods

CoExecute(IArena)

```
public override IEnumerator CoExecute(IArena arena)
```

Parameters

`arena` IArena

Returns

IEnumerator

Class ArenaDoubleAttack

Namespace: Nekoyume.Model.BattleStatus.Arena

Assembly: Lib9c.dll

```
[Serializable]
public class ArenaDoubleAttack : ArenaSkill
```

Inheritance

object ← ArenaEventBase ← ArenaSkill ← ArenaDoubleAttack

Inherited Members

ArenaSkill.SkillId , ArenaSkill.SkillInfos , ArenaSkill.BuffInfos , ArenaEventBase.Character , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

ArenaDoubleAttack(int, ArenaCharacter, IEnumerable<ArenaSkillInfo>, IEnumerable<ArenaSkillInfo>)

```
public ArenaDoubleAttack(int skillId, ArenaCharacter character,
IEnumerable<ArenaSkill.ArenaSkillInfo> skillInfos,
IEnumerable<ArenaSkill.ArenaSkillInfo> buffInfos)
```

Parameters

`skillId` int

`character` ArenaCharacter

`skillInfos` IEnumerable<ArenaSkill.ArenaSkillInfo>

`buffInfos` IEnumerable<ArenaSkill.ArenaSkillInfo>

Methods

CoExecute(IArena)

```
public override IEnumerator CoExecute(IArena arena)
```

Parameters

`arena` IArena

Returns

IEnumerator

Class ArenaDoubleAttackWithCombo

Namespace: Nekoyume.Model.BattleStatus.Arena

Assembly: Lib9c.dll

```
[Serializable]
public class ArenaDoubleAttackWithCombo : ArenaSkill
```

Inheritance

object ← ArenaEventBase ← ArenaSkill ← ArenaDoubleAttackWithCombo

Inherited Members

ArenaSkill.SkillId , ArenaSkill.SkillInfos , ArenaSkill.BuffInfos , ArenaEventBase.Character , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

ArenaDoubleAttackWithCombo(int, ArenaCharacter, IEnumerable<ArenaSkillInfo>, IEnumerable<ArenaSkillInfo>)

```
public ArenaDoubleAttackWithCombo(int skillId, ArenaCharacter character,
IEnumerable<ArenaSkill.ArenaSkillInfo> skillInfos,
IEnumerable<ArenaSkill.ArenaSkillInfo> buffInfos)
```

Parameters

`skillId` int

`character` ArenaCharacter

`skillInfos` IEnumerable<ArenaSkill.ArenaSkillInfo>

`buffInfos` IEnumerable<ArenaSkill.ArenaSkillInfo>

Methods

CoExecute(IArena)

```
public override IEnumerator CoExecute(IArena arena)
```

Parameters

`arena` IArena

Returns

IEnumerator

Class ArenaEventBase

Namespace: Nekoyume.Model.BattleStatus.Arena

Assembly: Lib9c.dll

```
[Serializable]
public abstract class ArenaEventBase
```

Inheritance

object ← ArenaEventBase

Derived

ArenaDead, ArenaRemoveBuffs, ArenaSkill, ArenaSpawnCharacter, ArenaTurnEnd

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

ArenaEventBase(ArenaCharacter)

```
protected ArenaEventBase(ArenaCharacter character)
```

Parameters

`character` ArenaCharacter

Fields

Character

```
public readonly ArenaCharacter Character
```

Field Value

ArenaCharacter

Methods

CoExecute(IArena)

```
public abstract IEnumerator CoExecute(IArena arena)
```

Parameters

`arena` IArena

Returns

IEnumerator

Class ArenaFullBuffRemovalAttack

Namespace: Nekoyume.Model.BattleStatus.Arena

Assembly: Lib9c.dll

Arena battle status event for the full buff removal attack skill. Triggers the CoFullBuffRemovalAttack(ArenaCharacter, IEnumerable<ArenaSkillInfo>, IEnumerable<ArenaSkillInfo>) coroutine for arena rendering.

```
[Serializable]
public class ArenaFullBuffRemovalAttack : ArenaSkill
```

Inheritance

object ← ArenaEventBase ← ArenaSkill ← ArenaFullBuffRemovalAttack

Inherited Members

ArenaSkill.SkillId , ArenaSkill.SkillInfos , ArenaSkill.BuffInfos , ArenaEventBase.Character , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

ArenaFullBuffRemovalAttack(int, ArenaCharacter, IEnumerable<ArenaSkillInfo>, IEnumerable<ArenaSkillInfo>)

```
public ArenaFullBuffRemovalAttack(int skillId, ArenaCharacter character,
IEnumerable<ArenaSkill.ArenaSkillInfo> skillInfos,
IEnumerable<ArenaSkill.ArenaSkillInfo> buffInfos)
```

Parameters

`skillId` int

`character` ArenaCharacter

`skillInfos` IEnumerable<ArenaSkill.ArenaSkillInfo>

`buffInfos` IEnumerable<ArenaSkill.ArenaSkillInfo>

Methods

CoExecute(IArena)

```
public override IEnumerator CoExecute(IArena arena)
```

Parameters

`arena` IArena

Returns

IEnumerator

Class ArenaHeal

Namespace: Nekoyume.Model.BattleStatus.Arena

Assembly: Lib9c.dll

```
[Serializable]
public class ArenaHeal : ArenaSkill
```

Inheritance

object ← ArenaEventBase ← ArenaSkill ← ArenaHeal

Inherited Members

ArenaSkill.SkillId , ArenaSkill.SkillInfos , ArenaSkill.BuffInfos , ArenaEventBase.Character , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

ArenaHeal(int, ArenaCharacter, IEnumerable<ArenaSkillInfo>, IEnumerable<ArenaSkillInfo>)

```
public ArenaHeal(int skillId, ArenaCharacter character,
IEnumerable<ArenaSkill.ArenaSkillInfo> skillInfos,
IEnumerable<ArenaSkill.ArenaSkillInfo> buffInfos)
```

Parameters

`skillId` int

`character` ArenaCharacter

`skillInfos` IEnumerable<ArenaSkill.ArenaSkillInfo>

`buffInfos` IEnumerable<ArenaSkill.ArenaSkillInfo>

Methods

CoExecute(IArena)

```
public override IEnumerator CoExecute(IArena arena)
```

Parameters

`arena` IArena

Returns

IEnumerator

Class ArenaLog

Namespace: Nekoyume.Model.BattleStatus.Arena

Assembly: Lib9c.dll

```
[Serializable]
public class ArenaLog : IEnumerable<ArenaEventBase>, IEnumerable
```

Inheritance

object ← ArenaLog

Implements

IEnumerable<ArenaEventBase>, IEnumerable

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Fields

Events

```
public List<ArenaEventBase> Events
```

Field Value

List<ArenaEventBase>

Id

```
public Guid Id
```

Field Value

Guid

Result

```
public ArenaLog.ArenaResult Result
```

Field Value

ArenaLog.ArenaResult

Score

```
public int Score
```

Field Value

int

Methods

Add(ArenaEventBase)

```
public void Add(ArenaEventBase e)
```

Parameters

e ArenaEventBase

GetEnumerator()

Returns an enumerator that iterates through the collection.

```
public IEnumerator<ArenaEventBase> GetEnumerator()
```

Returns

IEnumerator<ArenaEventBase>

An enumerator that can be used to iterate through the collection.

Enum ArenaLog.ArenaResult

Namespace: Nekoyume.Model.BattleStatus.Arena

Assembly: Lib9c.dll

```
public enum ArenaLog.ArenaResult
```

Extension Methods
StateExtensions.Serialize(Enum)

Fields

`Lose = 1`

`Win = 0`

Class ArenaNormalAttack

Namespace: Nekoyume.Model.BattleStatus.Arena

Assembly: Lib9c.dll

```
[Serializable]
public class ArenaNormalAttack : ArenaSkill
```

Inheritance

object ← ArenaEventBase ← ArenaSkill ← ArenaNormalAttack

Inherited Members

ArenaSkill.SkillId , ArenaSkill.SkillInfos , ArenaSkill.BuffInfos , ArenaEventBase.Character , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

ArenaNormalAttack(int, ArenaCharacter, IEnumerable<ArenaSkillInfo>, IEnumerable<ArenaSkillInfo>)

```
public ArenaNormalAttack(int skillId, ArenaCharacter character,
IEnumerable<ArenaSkill.ArenaSkillInfo> skillInfos,
IEnumerable<ArenaSkill.ArenaSkillInfo> buffInfos)
```

Parameters

`skillId` int

`character` ArenaCharacter

`skillInfos` IEnumerable<ArenaSkill.ArenaSkillInfo>

`buffInfos` IEnumerable<ArenaSkill.ArenaSkillInfo>

Methods

CoExecute(IArena)

```
public override IEnumerator CoExecute(IArena arena)
```

Parameters

`arena` IArena

Returns

IEnumerator

Class ArenaRemoveBuffs

Namespace: Nekoyume.Model.BattleStatus.Arena

Assembly: Lib9c.dll

```
[Serializable]
public class ArenaRemoveBuffs : ArenaEventBase
```

Inheritance

object ← ArenaEventBase ← ArenaRemoveBuffs

Inherited Members

ArenaEventBase.Character , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

ArenaRemoveBuffs(ArenaCharacter)

```
public ArenaRemoveBuffs(ArenaCharacter character)
```

Parameters

`character` ArenaCharacter

Methods

CoExecute(IArena)

```
public override IEnumerator CoExecute(IArena arena)
```

Parameters

`arena` IArena

Returns

IEnumerator

Class ArenaShatterStrike

Namespace: Nekoyume.Model.BattleStatus.Arena

Assembly: Lib9c.dll

```
[Serializable]
public class ArenaShatterStrike : ArenaSkill
```

Inheritance

object ← ArenaEventBase ← ArenaSkill ← ArenaShatterStrike

Inherited Members

ArenaSkill.SkillId , ArenaSkill.SkillInfos , ArenaSkill.BuffInfos , ArenaEventBase.Character , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

ArenaShatterStrike(int, ArenaCharacter, IEnumerable<ArenaSkillInfo>, IEnumerable<ArenaSkillInfo>)

```
public ArenaShatterStrike(int skillId, ArenaCharacter character,
IEnumerable<ArenaSkill.ArenaSkillInfo> skillInfos,
IEnumerable<ArenaSkill.ArenaSkillInfo> buffInfos)
```

Parameters

`skillId` int

`character` ArenaCharacter

`skillInfos` IEnumerable<ArenaSkill.ArenaSkillInfo>

`buffInfos` IEnumerable<ArenaSkill.ArenaSkillInfo>

Methods

CoExecute(IArena)

```
public override IEnumerator CoExecute(IArena arena)
```

Parameters

`arena` IArena

Returns

IEnumerator

Class ArenaSkill

Namespace: Nekoyume.Model.BattleStatus.Arena

Assembly: Lib9c.dll

```
[Serializable]
public abstract class ArenaSkill : ArenaEventBase
```

Inheritance

object ← ArenaEventBase ← ArenaSkill

Derived

ArenaAreaAttack, ArenaBlowAttack, ArenaBuff, ArenaBuffRemovalAttack, ArenaDoubleAttack, ArenaDoubleAttackWithCombo, ArenaFullBuffRemovalAttack, ArenaHeal, ArenaNormalAttack, ArenaShatterStrike, ArenaTick, ArenaTickDamage

Inherited Members

ArenaEventBase.Character , ArenaEventBase.CoExecute(IArena) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

ArenaSkill(int, ArenaCharacter, IEnumerable<ArenaSkillInfo>, IEnumerable<ArenaSkillInfo>)

```
protected ArenaSkill(int skillId, ArenaCharacter character,
IEnumerable<ArenaSkill.ArenaSkillInfo> skillInfos,
IEnumerable<ArenaSkill.ArenaSkillInfo> buffInfos)
```

Parameters

`skillId` int

`character` ArenaCharacter

`skillInfos` IEnumerable<ArenaSkill.ArenaSkillInfo>

`buffInfos` IEnumerable<ArenaSkill.ArenaSkillInfo>

Fields

BuffInfos

```
public readonly IEnumerable<ArenaSkill.ArenaSkillInfo>? BuffInfos
```

Field Value

IEnumerable<ArenaSkill.ArenaSkillInfo>

SkillId

```
public readonly int SkillId
```

Field Value

int

SkillInfos

```
public readonly IEnumerable<ArenaSkill.ArenaSkillInfo> SkillInfos
```

Field Value

IEnumerable<ArenaSkill.ArenaSkillInfo>

Class ArenaSkill.ArenaSkillInfo

Namespace: Nekoyume.Model.BattleStatus.Arena

Assembly: Lib9c.dll

```
[Serializable]
public class ArenaSkill.ArenaSkillInfo
```

Inheritance

object ← ArenaSkill.ArenaSkillInfo

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

ArenaSkillInfo(ArenaCharacter, long, bool, SkillCategory, int, ElementalType, SkillTargetType, Buff?, bool, IEnumerable<Buff>?, IceShield?)

```
public ArenaSkillInfo(ArenaCharacter character, long effect, bool critical,
SkillCategory skillCategory, int turn, ElementalType elementalType =
ElementalType.Normal, SkillTargetType targetType = SkillTargetType.Enemy, Buff? buff
= null, bool affected = true, IEnumerable<Buff>? dispelList = null, IceShield?
iceShield = null)
```

Parameters

`character` ArenaCharacter

`effect` long

`critical` bool

`skillCategory` SkillCategory

`turn` int

`elementalType` ElementalType

`targetType` SkillTargetType

`buff` Buff

`affected` bool

`dispelList` IEnumerable<Buff>

`iceShield` IceShield

Fields

Affected

```
public readonly bool Affected
```

Field Value

bool

Buff

```
public readonly Buff? Buff
```

Field Value

Buff

Critical

```
public readonly bool Critical
```

Field Value

bool

DispelList

```
public readonly IEnumerable<Buff>? DispelList
```

Field Value

IEnumerable<Buff>

Effect

```
public readonly long Effect
```

Field Value

long

ElementalType

```
public readonly ElementalType ElementalType
```

Field Value

ElementalType

IceShield

```
public readonly IceShield? IceShield
```

Field Value

IceShield

SkillCategory

```
public readonly SkillCategory SkillCategory
```

Field Value

SkillCategory

SkillTargetType

```
public readonly SkillTargetType SkillTargetType
```

Field Value

SkillTargetType

Target

```
public readonly ArenaCharacter Target
```

Field Value

ArenaCharacter

Turn

```
public readonly int Turn
```

Field Value

int

Class ArenaSpawnCharacter

Namespace: Nekoyume.Model.BattleStatus.Arena

Assembly: Lib9c.dll

```
public class ArenaSpawnCharacter : ArenaEventBase
```

Inheritance

object ← ArenaEventBase ← ArenaSpawnCharacter

Inherited Members

ArenaEventBase.Character , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

ArenaSpawnCharacter(ArenaCharacter)

```
public ArenaSpawnCharacter(ArenaCharacter character)
```

Parameters

`character` ArenaCharacter

Methods

CoExecute(IArena)

```
public override IEnumerator CoExecute(IArena arena)
```

Parameters

`arena` IArena

Returns

IEnumerator

Class ArenaTick

Namespace: Nekoyume.Model.BattleStatus.Arena

Assembly: Lib9c.dll

```
[Serializable]
public class ArenaTick : ArenaSkill
```

Inheritance

object ← ArenaEventBase ← ArenaSkill ← ArenaTick

Inherited Members

ArenaSkill.SkillId , ArenaSkill.SkillInfos , ArenaSkill.BuffInfos , ArenaEventBase.Character , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

ArenaTick(ArenaCharacter)

```
public ArenaTick(ArenaCharacter character)
```

Parameters

`character` ArenaCharacter

ArenaTick(int, ArenaCharacter, IEnumerable<ArenaSkillInfo>, IEnumerable<ArenaSkillInfo>)

```
public ArenaTick(int skillId, ArenaCharacter character,
IEnumerable<ArenaSkill.ArenaSkillInfo> skillInfos,
IEnumerable<ArenaSkill.ArenaSkillInfo> buffInfos)
```

Parameters

`skillId` int

`character` ArenaCharacter

`skillInfos` IEnumerable<ArenaSkill.ArenaSkillInfo>

`buffInfos` IEnumerable<ArenaSkill.ArenaSkillInfo>

Methods

CoExecute(IArena)

```
public override IEnumerator CoExecute(IArena arena)
```

Parameters

`arena` IArena

Returns

IEnumerator

Class ArenaTickDamage

Namespace: Nekoyume.Model.BattleStatus.Arena

Assembly: Lib9c.dll

```
[Serializable]
public class ArenaTickDamage : ArenaSkill
```

Inheritance

object ← ArenaEventBase ← ArenaSkill ← ArenaTickDamage

Inherited Members

ArenaSkill.SkillId , ArenaSkill.SkillInfos , ArenaSkill.BuffInfos , ArenaEventBase.Character , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

ArenaTickDamage(int, ArenaCharacter, IEnumerable<ArenaSkillInfo>, IEnumerable<ArenaSkillInfo>)

```
public ArenaTickDamage(int skillId, ArenaCharacter character,
IEnumerable<ArenaSkill.ArenaSkillInfo> skillInfos,
IEnumerable<ArenaSkill.ArenaSkillInfo> buffInfos)
```

Parameters

`skillId` int

`character` ArenaCharacter

`skillInfos` IEnumerable<ArenaSkill.ArenaSkillInfo>

`buffInfos` IEnumerable<ArenaSkill.ArenaSkillInfo>

Methods

CoExecute(IArena)

```
public override IEnumerator CoExecute(IArena arena)
```

Parameters

`arena` IArena

Returns

IEnumerator

Class ArenaTurnEnd

Namespace: Nekoyume.Model.BattleStatus.Arena

Assembly: Lib9c.dll

```
[Serializable]
public class ArenaTurnEnd : ArenaEventBase
```

Inheritance

object ← ArenaEventBase ← ArenaTurnEnd

Inherited Members

ArenaEventBase.Character , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

ArenaTurnEnd(ArenaCharacter, int)

```
public ArenaTurnEnd(ArenaCharacter character, int turnNumber)
```

Parameters

`character` ArenaCharacter

`turnNumber` int

Fields

TurnNumber

```
public readonly int TurnNumber
```

Field Value

int

Methods

CoExecute(IArena)

```
public override IEnumerator CoExecute(IArena arena)
```

Parameters

`arena` IArena

Returns

IEnumerator

Namespace Nekoyume.Model.Buff

Classes

ActionBuff

Bleed

Buff

BuffFactory

Dispel

Focus

IceShield

StatBuff

Stun

Vampiric

Structs

BuffInfo

Class ActionBuff

Namespace: Nekoyume.Model.Buff

Assembly: Lib9c.dll

```
[Serializable]
public abstract class ActionBuff : Buff, ICloneable
```

Inheritance

object ← Buff ← ActionBuff

Implements

ICloneable

Derived

Bleed, Dispel, Focus, IceShield, Stun, Vampiric

Inherited Members

Buff.BuffInfo , Buff.OriginalDuration , Buff.RemainedDuration , Buff.GetTarget(CharacterBase) , Buff.Clone() , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

ActionBuff(ActionBuff)

```
protected ActionBuff(ActionBuff value)
```

Parameters

`value` ActionBuff

ActionBuff(SkillCustomField, Row)

```
public ActionBuff(SkillCustomField customField, ActionBuffSheet.Row row)
```

Parameters

`customField` SkillCustomField

`row` ActionBuffSheet.Row

ActionBuff(Row)

```
public ActionBuff(ActionBuffSheet.Row row)
```

Parameters

`row` ActionBuffSheet.Row

Properties

CustomField

```
public SkillCustomField? CustomField { get; }
```

Property Value

SkillCustomField?

RowData

```
public ActionBuffSheet.Row RowData { get; }
```

Property Value

ActionBuffSheet.Row

Methods

IsBuff()

```
public override bool IsBuff()
```

Returns

bool

IsDebuff()

```
public override bool IsDebuff()
```

Returns

bool

Class Bleed

Namespace: Nekoyume.Model.Buff

Assembly: Lib9c.dll

```
[Serializable]
public class Bleed : ActionBuff, ICloneable
```

Inheritance

object ← Buff ← ActionBuff ← Bleed

Implements

ICloneable

Inherited Members

ActionBuff.RowData , ActionBuff.CustomField , ActionBuff.IsBuff() , ActionBuff.IsDebuff() , Buff.BuffInfo , Buff.OriginalDuration , Buff.RemainedDuration , Buff.GetTarget(CharacterBase) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

Bleed(Bleed)

```
protected Bleed(Bleed value)
```

Parameters

`value` Bleed

Bleed(SkillCustomField, Row)

```
public Bleed(SkillCustomField customField, ActionBuffSheet.Row row)
```

Parameters

`customField` SkillCustomField

`row` ActionBuffSheet.Row

Bleed(Row, long)

```
public Bleed(ActionBuffSheet.Row row, long power)
```

Parameters

`row` ActionBuffSheet.Row

`power` long

Properties

Power

```
public long Power { get; }
```

Property Value

long

Methods

Clone()

Creates a new object that is a copy of the current instance.

```
public override object Clone()
```

Returns

object

A new object that is a copy of this instance.

GiveEffect(CharacterBase, int, bool)

```
public Skill GiveEffect(CharacterBase affectedCharacter, int simulatorWaveTurn, bool
copyCharacter = true)
```

Parameters

`affectedCharacter` CharacterBase

`simulatorWaveTurn` int

`copyCharacter` bool

Returns

Skill

GiveEffectForArena(ArenaCharacter, int)

```
public ArenaSkill GiveEffectForArena(ArenaCharacter affectedCharacter,
int simulatorWaveTurn)
```

Parameters

`affectedCharacter` ArenaCharacter

`simulatorWaveTurn` int

Returns

ArenaSkill

Class Buff

Namespace: Nekoyume.Model.Buff

Assembly: Lib9c.dll

```
[Serializable]
public abstract class Buff : ICloneable
```

Inheritance

object ← Buff

Implements

ICloneable

Derived

ActionBuff, StatBuff

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

Buff(Buff)

```
protected Buff(Buff value)
```

Parameters

`value` Buff

Buff(BuffInfo)

```
protected Buff(BuffInfo buffInfo)
```

Parameters

`buffInfo` BuffInfo

Properties

BuffInfo

```
public BuffInfo BuffInfo { get; }
```

Property Value

BuffInfo

OriginalDuration

```
public int OriginalDuration { get; }
```

Property Value

int

RemainedDuration

```
public int RemainedDuration { get; set; }
```

Property Value

int

Methods

Clone()

Creates a new object that is a copy of the current instance.

```
public abstract object Clone()
```

Returns

object

A new object that is a copy of this instance.

GetTarget(CharacterBase)

```
public virtual IEnumerable<CharacterBase> GetTarget(CharacterBase caster)
```

Parameters

`caster` CharacterBase

Returns

IEnumerable<CharacterBase>

IsBuff()

```
public abstract bool IsBuff()
```

Returns

bool

IsDebuff()

```
public abstract bool IsDebuff()
```

Returns

bool

Class BuffFactory

Namespace: Nekoyume.Model.Buff

Assembly: Lib9c.dll

```
public static class BuffFactory
```

Inheritance

object ← BuffFactory

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Methods

GetActionBuff(Stats, Row)

```
public static ActionBuff GetActionBuff(Stats stat, ActionBuffSheet.Row row)
```

Parameters

`stat` Stats

`row` ActionBuffSheet.Row

Returns

ActionBuff

GetBuffs(CharacterStats, ISkill, SkillBuffSheet, StatBuffSheet, SkillActionBuffSheet, ActionBuffSheet, bool)

```
public static IList<Buff> GetBuffs(CharacterStats stats, ISkill skill,
SkillBuffSheet skillBuffSheet, StatBuffSheet statBuffSheet, SkillActionBuffSheet
skillActionBuffSheet, ActionBuffSheet actionBuffSheet, bool hasExtraValueBuff
= false)
```

Parameters

`stats` CharacterStats

`skill` ISkill

`skillBuffSheet` SkillBuffSheet

`statBuffSheet` StatBuffSheet

`skillActionBuffSheet` SkillActionBuffSheet

`actionBuffSheet` ActionBuffSheet

`hasExtraValueBuff` bool

Returns

IList<Buff>

GetCustomActionBuff(SkillCustomField, Row)

```
public static ActionBuff GetCustomActionBuff(SkillCustomField customField,
ActionBuffSheet.Row row)
```

Parameters

`customField` SkillCustomField

`row` ActionBuffSheet.Row

Returns

ActionBuff

GetCustomStatBuff(Row, SkillCustomField)

```
public static StatBuff GetCustomStatBuff(StatBuffSheet.Row row,
SkillCustomField customField)
```

Parameters

`row` StatBuffSheet.Row

`customField` SkillCustomField

Returns

StatBuff

GetStatBuff(Row)

```
public static StatBuff GetStatBuff(StatBuffSheet.Row row)
```

Parameters

`row` StatBuffSheet.Row

Returns

StatBuff

Struct BuffInfo

Namespace: Nekoyume.Model.Buff

Assembly: Lib9c.dll

```
[Serializable]
public struct BuffInfo
```

Inherited Members

ValueType.Equals(object) , ValueType.GetHashCode() , ValueType.ToString() , object.Equals(object, object) , object.GetType() , object.ReferenceEquals(object, object)

Constructors

BuffInfo(int, int, int, int, SkillTargetType)

```
public BuffInfo(int id, int groupId, int chance, int duration,
SkillTargetType skillTargetType)
```

Parameters

`id` int

`groupId` int

`chance` int

`duration` int

`skillTargetType` SkillTargetType

Fields

Chance

```
public int Chance
```

Field Value

int

Duration

```
public int Duration
```

Field Value

int

GroupId

```
public int GroupId
```

Field Value

int

Id

```
public int Id
```

Field Value

int

SkillTargetType

```
public SkillTargetType SkillTargetType
```

Field Value

SkillTargetType

Class Dispel

Namespace: Nekoyume.Model.Buff

Assembly: Lib9c.dll

```
[Serializable]
public class Dispel : ActionBuff, ICloneable
```

Inheritance

object ← Buff ← ActionBuff ← Dispel

Implements

ICloneable

Inherited Members

ActionBuff.RowData , ActionBuff.CustomField , ActionBuff.IsBuff() , ActionBuff.IsDebuff() , Buff.BuffInfo , Buff.OriginalDuration , Buff.RemainedDuration , Buff.GetTarget(CharacterBase) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

Dispel(ActionBuff)

```
protected Dispel(ActionBuff value)
```

Parameters

`value` ActionBuff

Dispel(SkillCustomField, Row)

```
public Dispel(SkillCustomField customField, ActionBuffSheet.Row row)
```

Parameters

`customField` SkillCustomField

`row` ActionBuffSheet.Row

Dispel(Row)

```
public Dispel(ActionBuffSheet.Row row)
```

Parameters

`row` ActionBuffSheet.Row

Methods

Clone()

Creates a new object that is a copy of the current instance.

```
public override object Clone()
```

Returns

object

A new object that is a copy of this instance.

Class Focus

Namespace: Nekoyume.Model.Buff

Assembly: Lib9c.dll

```
[Serializable]
public class Focus : ActionBuff, ICloneable
```

Inheritance

object ← Buff ← ActionBuff ← Focus

Implements

ICloneable

Inherited Members

ActionBuff.RowData , ActionBuff.CustomField , ActionBuff.IsBuff() , ActionBuff.IsDebuff() , Buff.BuffInfo , Buff.OriginalDuration , Buff.RemainedDuration , Buff.GetTarget(CharacterBase) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

Focus(Focus)

```
protected Focus(Focus value)
```

Parameters

`value` Focus

Focus(SkillCustomField, Row)

```
public Focus(SkillCustomField customField, ActionBuffSheet.Row row)
```

Parameters

`customField` SkillCustomField

`row` ActionBuffSheet.Row

Focus(Row)

```
public Focus(ActionBuffSheet.Row row)
```

Parameters

`row` ActionBuffSheet.Row

Methods

Clone()

Creates a new object that is a copy of the current instance.

```
public override object Clone()
```

Returns

object

A new object that is a copy of this instance.

Class IceShield

Namespace: Nekoyume.Model.Buff

Assembly: Lib9c.dll

```
[Serializable]
public class IceShield : ActionBuff, ICloneable
```

Inheritance

object ← Buff ← ActionBuff ← IceShield

Implements

ICloneable

Inherited Members

ActionBuff.RowData , ActionBuff.CustomField , ActionBuff.IsBuff() , ActionBuff.IsDebuff() , Buff.BuffInfo , Buff.OriginalDuration , Buff.RemainedDuration , Buff.GetTarget(CharacterBase) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

IceShield(IceShield)

```
protected IceShield(IceShield value)
```

Parameters

`value` IceShield

IceShield(SkillCustomField, Row)

```
public IceShield(SkillCustomField customField, ActionBuffSheet.Row row)
```

Parameters

`customField` SkillCustomField

`row` ActionBuffSheet.Row

IceShield(Row)

```
public IceShield(ActionBuffSheet.Row row)
```

Parameters

`row` ActionBuffSheet.Row

Methods

Clone()

Creates a new object that is a copy of the current instance.

```
public override object Clone()
```

Returns

object

A new object that is a copy of this instance.

FrostBite(StatBuffSheet, BuffLinkSheet)

```
public StatBuff FrostBite(StatBuffSheet statBuffSheet, BuffLinkSheet buffLinkSheet)
```

Parameters

`statBuffSheet` StatBuffSheet

`buffLinkSheet` BuffLinkSheet

Returns

StatBuff

Class StatBuff

Namespace: Nekoyume.Model.Buff

Assembly: Lib9c.dll

```
[Serializable]
public class StatBuff : Buff, ICloneable
```

Inheritance

object ← Buff ← StatBuff

Implements

ICloneable

Inherited Members

Buff.BuffInfo , Buff.OriginalDuration , Buff.RemainedDuration , Buff.GetTarget(CharacterBase) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

StatBuff(StatBuff)

```
protected StatBuff(StatBuff value)
```

Parameters

`value` StatBuff

StatBuff(SkillCustomField, Row)

```
public StatBuff(SkillCustomField customField, StatBuffSheet.Row row)
```

Parameters

customField SkillCustomField

row StatBuffSheet.Row

StatBuff(Row)

```
public StatBuff(StatBuffSheet.Row row)
```

Parameters

row StatBuffSheet.Row

Properties

CustomField

```
public SkillCustomField? CustomField { get; }
```

Property Value

SkillCustomField?

RowData

```
public StatBuffSheet.Row RowData { get; }
```

Property Value

StatBuffSheet.Row

Stack

```
public int Stack { get; }
```

Property Value

int

Methods

Clone()

Creates a new object that is a copy of the current instance.

```
public override object Clone()
```

Returns

object

A new object that is a copy of this instance.

GetModifier()

```
public StatModifier GetModifier()
```

Returns

StatModifier

IsBuff()

```
public override bool IsBuff()
```

Returns

bool

IsDebuff()

```
public override bool IsDebuff()
```

Returns

bool

SetStack(int)

```
public void SetStack(int stack)
```

Parameters

`stack` int

Class Stun

Namespace: Nekoyume.Model.Buff

Assembly: Lib9c.dll

```
[Serializable]
public class Stun : ActionBuff, ICloneable
```

Inheritance

object ← Buff ← ActionBuff ← Stun

Implements

ICloneable

Inherited Members

ActionBuff.RowData , ActionBuff.CustomField , ActionBuff.IsBuff() , ActionBuff.IsDebuff() , Buff.BuffInfo , Buff.OriginalDuration , Buff.RemainedDuration , Buff.GetTarget(CharacterBase) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

Stun(ActionBuff)

```
protected Stun(ActionBuff value)
```

Parameters

`value` ActionBuff

Stun(SkillCustomField, Row)

```
public Stun(SkillCustomField customField, ActionBuffSheet.Row row)
```

Parameters

`customField` SkillCustomField

`row` ActionBuffSheet.Row

Stun(Row)

```
public Stun(ActionBuffSheet.Row row)
```

Parameters

`row` ActionBuffSheet.Row

Methods

Clone()

Creates a new object that is a copy of the current instance.

```
public override object Clone()
```

Returns

object

A new object that is a copy of this instance.

Class Vampiric

Namespace: Nekoyume.Model.Buff

Assembly: Lib9c.dll

```
[Serializable]
public class Vampiric : ActionBuff, ICloneable
```

Inheritance

object ← Buff ← ActionBuff ← Vampiric

Implements

ICloneable

Inherited Members

ActionBuff.RowData , ActionBuff.CustomField , ActionBuff.IsBuff() , ActionBuff.IsDebuff() , Buff.BuffInfo , Buff.OriginalDuration , Buff.RemainedDuration , Buff.GetTarget(CharacterBase) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

Vampiric(Vampiric)

```
protected Vampiric(Vampiric value)
```

Parameters

`value` Vampiric

Vampiric(SkillCustomField, Row)

```
public Vampiric(SkillCustomField customField, ActionBuffSheet.Row row)
```

Parameters

`customField` SkillCustomField

`row` ActionBuffSheet.Row

Vampiric(Row, long)

```
public Vampiric(ActionBuffSheet.Row row, long basisPoint)
```

Parameters

`row` ActionBuffSheet.Row

`basisPoint` long

Properties

BasisPoint

```
public long BasisPoint { get; }
```

Property Value

long

Methods

Clone()

Creates a new object that is a copy of the current instance.

```
public override object Clone()
```

Returns

object

A new object that is a copy of this instance.

GiveEffect(CharacterBase, SkillInfo, int, bool)

공격 피해(`skillInfo` 의 Effect)에 BasisPoint(만분율)를 곱한 만큼 회복한다. 회복량은 long 이라 int.MaxValue 를 넘어도 잘리지 않는다.

```
public Skill GiveEffect(CharacterBase affectedCharacter, Skill.SkillInfo skillInfo,
int simulatorWaveTurn, bool copyCharacter = true)
```

Parameters

`affectedCharacter` CharacterBase

회복하는 캐릭터.

`skillInfo` Skill.SkillInfo

흡혈을 일으킨 공격의 결과.

`simulatorWaveTurn` int

현재 웨이브 턴.

`copyCharacter` bool

전투 로그용으로 캐릭터를 복제할지.

Returns

Skill

회복을 기록한 틱.

GiveEffectForArena(ArenaCharacter, ArenaSkillInfo, int)

아레나용 GiveEffect(CharacterBase, SkillInfo, int, bool). 공격 피해에 BasisPoint(만분율)를 곱한 만큼 회복한다.

```
public ArenaSkill GiveEffectForArena(ArenaCharacter affectedCharacter,
ArenaSkill.ArenaSkillInfo skillInfo, int simulatorWaveTurn)
```

Parameters

`affectedCharacter` ArenaCharacter

회복하는 캐릭터.

`skillInfo` ArenaSkill.ArenaSkillInfo

흡혈을 일으킨 공격의 결과.

`simulatorWaveTurn` int

현재 웨이브 턴.

Returns

ArenaSkill

회복을 기록한 틱.

Namespace Nekoyume.Model.Character

Enums

SizeType

Enum SizeType

Namespace: Nekoyume.Model.Character

Assembly: Lib9c.dll

```
public enum SizeType
```

Extension Methods
StateExtensions.Serialize(Enum)

Fields

`L = 3`

`M = 2`

`S = 1`

`XL = 4`

`XS = 0`

Namespace Nekoyume.Model.Collection

Classes

CollectionFactory

FungibleCollectionMaterial

NonFungibleCollectionMaterial

Interfaces

ICollectionMaterial

Enums

MaterialType

Class CollectionFactory

Namespace: Nekoyume.Model.Collection

Assembly: Lib9c.dll

```
public static class CollectionFactory
```

Inheritance

object ← CollectionFactory

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Methods

DeserializeMaterial(List)

```
public static ICollectionMaterial DeserializeMaterial(List serialized)
```

Parameters

`serialized` List

Returns

ICollectionMaterial

Class FungibleCollectionMaterial

Namespace: Nekoyume.Model.Collection

Assembly: Lib9c.dll

```
public class FungibleCollectionMaterial : ICollectionMaterial, IBencodable
```

Inheritance

object ← FungibleCollectionMaterial

Implements

ICollectionMaterial, IBencodable

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

FungibleCollectionMaterial()

```
public FungibleCollectionMaterial()
```

FungibleCollectionMaterial(IValue)

```
public FungibleCollectionMaterial(IValue bencoded)
```

Parameters

`bencoded` IValue

FungibleCollectionMaterial(List)

```
public FungibleCollectionMaterial(List serialized)
```

Parameters

`serialized` List

Properties

Bencoded

An Bencodex.Types.IValue representation of this object that can be decoded back to instantiate an equal object. The decoded object must be equal to the original in the sense that Equals(T) should be true.

```
public IValue Bencoded { get; }
```

Property Value

IValue

Remarks

Note that the only requirement is that the produced Bencodex.Types.IValue can be decoded back to an equal object. This representation may not be canonical in the sense that additional junk data may be present in an Bencodex.Types.IValue that one may wish to decode and this may be discarded while decoding.

A specific implemnetation may decide to only allow the canonical representation to be decoded.

ItemCount

```
public int ItemCount { get; set; }
```

Property Value

int

ItemId

```
public int ItemId { get; set; }
```

Property Value

int

Type

```
public MaterialType Type { get; }
```

Property Value

MaterialType

Methods

BurnMaterial(Row, Inventory, long)

Burns the material from the inventory based on the given item row, inventory, and block index.

```
public void BurnMaterial(ItemSheet.Row itemRow, Inventory inventory,
long blockIndex)
```

Parameters

`itemRow` ItemSheet.Row

The item row from the item sheet.

`inventory` Inventory

The inventory object.

`blockIndex` long

The block index.

Interface ICollectionMaterial

Namespace: Nekoyume.Model.Collection

Assembly: Lib9c.dll

```
public interface ICollectionMaterial : IBencodable
```

Inherited Members
IBencodable.Bencoded

Properties

ItemCount

```
int ItemCount { get; set; }
```

Property Value

int

ItemId

```
int ItemId { get; set; }
```

Property Value

int

Type

```
MaterialType Type { get; }
```

Property Value

MaterialType

Enum MaterialType

Namespace: Nekoyume.Model.Collection

Assembly: Lib9c.dll

```
public enum MaterialType
```

Extension Methods
StateExtensions.Serialize(Enum)

Fields

`Fungible = 0`

`NonFungible = 1`

Class NonFungibleCollectionMaterial

Namespace: Nekoyume.Model.Collection

Assembly: Lib9c.dll

```
public class NonFungibleCollectionMaterial : ICollectionMaterial, IBencodable
```

Inheritance

object ← NonFungibleCollectionMaterial

Implements

ICollectionMaterial, IBencodable

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

NonFungibleCollectionMaterial()

```
public NonFungibleCollectionMaterial()
```

NonFungibleCollectionMaterial(List)

```
public NonFungibleCollectionMaterial(List serialized)
```

Parameters

`serialized` List

Properties

Bencoded

An Bencodex.Types.IValue representation of this object that can be decoded back to instantiate an equal object. The decoded object must be equal to the original in the sense that Equals(T) should be true.

```
public IValue Bencoded { get; }
```

Property Value

IValue

Remarks

Note that the only requirement is that the produced Bencodex.Types.IValue can be decoded back to an equal object. This representation may not be canonical in the sense that additional junk data may be present in an Bencodex.Types.IValue that one may wish to decode and this may be discarded while decoding.

A specific implemnetation may decide to only allow the canonical representation to be decoded.

ItemCount

```
public int ItemCount { get; set; }
```

Property Value

int

ItemId

```
public int ItemId { get; set; }
```

Property Value

int

Level

```
public int Level { get; set; }
```

Property Value

int

NonFungibleId

```
public Guid NonFungibleId { get; set; }
```

Property Value

Guid

SkillContains

```
public bool SkillContains { get; set; }
```

Property Value

bool

Type

```
public MaterialType Type { get; }
```

Property Value

MaterialType

Methods

BurnMaterial(Row, Inventory, RequiredMaterial, long)

Burns the specified material from the inventory based on the item type.

```
public void BurnMaterial(ItemSheet.Row itemRow, Inventory inventory,
CollectionSheet.RequiredMaterial materialInfo, long blockIndex)
```

Parameters

`itemRow` ItemSheet.Row

The ItemSheet.Row object representing the item.

`inventory` Inventory

The Inventory object representing the player's inventory.

`materialInfo` CollectionSheet.RequiredMaterial

The CollectionSheet.RequiredMaterial object representing the material info.

`blockIndex` long

The block index where the burn operation is taking place.

Exceptions

ItemDoesNotExistException

Thrown when the material item does not exist in the inventory.

InvalidItemTypeException

Thrown when the item type is not supported by NonFungibleCollectionMaterial.

Namespace Nekoyume.Model.Coupons

Structs

Coupon

RewardSet

RewardSet.Comparer

Struct Coupon

Namespace: Nekoyume.Model.Coupons

Assembly: Lib9c.dll

```
public readonly struct Coupon : IOrderedEnumerable<(int ItemId, uint Quantity)>,
IEnumerable<(int ItemId, uint Quantity)>, IEnumerable, IEquatable<Coupon>
```

Implements
IOrderedEnumerable<(int ItemId, uint Quantity)>, IEnumerable<(int ItemId, uint Quantity)>, IEnumerable, IEquatable<Coupon>

Inherited Members
ValueType.ToString() , object.Equals(object, object) , object.GetType() , object.ReferenceEquals(object, object)

Constructors

Coupon(IValue)

```
public Coupon(IValue serialized)
```

Parameters

`serialized` IValue

Coupon(Guid, in RewardSet)

```
public Coupon(Guid id, in RewardSet rewards)
```

Parameters

`id` Guid

`rewards` RewardSet

Coupon(Guid, IEnumerable<(int ItemId, uint Quantity)>)

```
public Coupon(Guid id, IEnumerable<(int ItemId, uint Quantity)> rewards)
```

Parameters

`id` Guid

`rewards` IEnumerable<(int ItemId, uint Quantity)>

Coupon(Guid, params (int ItemId, uint Quantity)[])

```
public Coupon(Guid id, params (int ItemId, uint Quantity)[] rewards)
```

Parameters

`id` Guid

`rewards` (int ItemId, uint Quantity)[]

Fields

Id

```
public readonly Guid Id
```

Field Value

Guid

Rewards

```
public readonly RewardSet Rewards
```

Field Value

RewardSet

Methods

Equals(Coupon)

Indicates whether the current object is equal to another object of the same type.

```
public bool Equals(Coupon other)
```

Parameters

`other` Coupon

An object to compare with this object.

Returns

bool

true if the current object is equal to the `other` parameter; otherwise, false.

Equals(object?)

Indicates whether this instance and a specified object are equal.

```
public override bool Equals(object? obj)
```

Parameters

`obj` object

The object to compare with the current instance.

Returns

bool

true if `obj` and this instance are the same type and represent the same value; otherwise, false.

GetHashCode()

Returns the hash code for this instance.

```
public override int GetHashCode()
```

Returns

int

A 32-bit signed integer that is the hash code for this instance.

Serialize()

```
public IValue Serialize()
```

Returns

IValue

Struct RewardSet

Namespace: Nekoyume.Model.Coupons

Assembly: Lib9c.dll

```
public readonly struct RewardSet : IImmutableDictionary<int, uint>,
IReadOnlyDictionary<int, uint>, IReadOnlyCollection<KeyValuePair<int, uint>>,
IEnumerable<KeyValuePair<int, uint>>, IEnumerable, IEquatable<RewardSet>
```

Implements

IImmutableDictionary<int, uint>, IReadOnlyDictionary<int, uint>, IReadOnlyCollection<KeyValuePair<int, uint>>, IEnumerable<KeyValuePair<int, uint>>, IEnumerable, IEquatable<RewardSet>

Inherited Members

ValueType.ToString() , object.Equals(object, object) , object.GetType() , object.ReferenceEquals(object, object)

Constructors

RewardSet(Dictionary)

```
public RewardSet(Dictionary serialized)
```

Parameters

`serialized` Dictionary

RewardSet(IEnumerable<(int ItemId, uint Quantity)>)

```
public RewardSet(IEnumerable<(int ItemId, uint Quantity)> rewards)
```

Parameters

`rewards` IEnumerable<(int ItemId, uint Quantity)>

RewardSet(in ImmutableDictionary<int, uint>)

```
public RewardSet(in ImmutableDictionary<int, uint> rewards)
```

Parameters

`rewards` ImmutableDictionary<int, uint>

RewardSet(params (int ItemId, uint Quantity)[])

```
public RewardSet(params (int ItemId, uint Quantity)[] rewards)
```

Parameters

`rewards` (int ItemId, uint Quantity)[]

Properties

Count

Gets the number of elements in the collection.

```
public int Count { get; }
```

Property Value

int

The number of elements in the collection.

this[int]

Gets the element that has the specified key in the read-only dictionary.

```
public uint this[int key] { get; }
```

Parameters

`key` int

The key to locate.

Property Value

uint

The element that has the specified key in the read-only dictionary.

Exceptions

ArgumentNullException

`key` is null.

KeyNotFoundException

The property is retrieved and `key` is not found.

Keys

Gets an enumerable collection that contains the keys in the read-only dictionary.

```
public IEnumerable<int> Keys { get; }
```

Property Value

IEnumerable<int>

An enumerable collection that contains the keys in the read-only dictionary.

Values

Gets an enumerable collection that contains the values in the read-only dictionary.

```
public IEnumerable<uint> Values { get; }
```

Property Value

IEnumerable<uint>

An enumerable collection that contains the values in the read-only dictionary.

Methods

Add(int, uint)

Adds an element with the specified key and value to the dictionary.

```
public IImmutableDictionary<int, uint> Add(int key, uint value)
```

Parameters

`key` int

The key of the element to add.

`value` uint

The value of the element to add.

Returns

IImmutableDictionary<int, uint>

A new immutable dictionary that contains the additional key/value pair.

Exceptions

ArgumentException

The given key already exists in the dictionary but has a different value.

AddRange(IEnumerable<KeyValuePair<int, uint>>)

Adds the specified key/value pairs to the dictionary.

```
public IImmutableDictionary<int, uint> AddRange(IEnumerable<KeyValuePair<int,
uint>> pairs)
```

Parameters

`pairs` IEnumerable<KeyValuePair<int, uint>>

The key/value pairs to add.

Returns

IImmutableDictionary<int, uint>

A new immutable dictionary that contains the additional key/value pairs.

Exceptions

ArgumentException

One of the given keys already exists in the dictionary but has a different value.

Clear()

Retrieves an empty dictionary that has the same ordering and key/value comparison rules as this dictionary instance.

```
public IImmutableDictionary<int, uint> Clear()
```

Returns

IImmutableDictionary<int, uint>

An empty dictionary with equivalent ordering and key/value comparison rules.

Contains(KeyValuePair<int, uint>)

Determines whether the immutable dictionary contains the specified key/value pair.

```
public bool Contains(KeyValuePair<int, uint> pair)
```

Parameters

`pair` KeyValuePair<int, uint>

The key/value pair to locate.

Returns

bool

true if the specified key/value pair is found in the dictionary; otherwise, false.

ContainsKey(int)

Determines whether the read-only dictionary contains an element that has the specified key.

```
public bool ContainsKey(int key)
```

Parameters

`key` int

The key to locate.

Returns

bool

true if the read-only dictionary contains an element that has the specified key; otherwise, false.

Exceptions

ArgumentNullException

`key` is null.

Equals(RewardSet)

Indicates whether the current object is equal to another object of the same type.

```
public bool Equals(RewardSet other)
```

Parameters

`other` RewardSet

An object to compare with this object.

Returns

bool

true if the current object is equal to the `other` parameter; otherwise, false.

Equals(object?)

Indicates whether this instance and a specified object are equal.

```
public override bool Equals(object? obj)
```

Parameters

`obj` object

The object to compare with the current instance.

Returns

bool

true if `obj` and this instance are the same type and represent the same value; otherwise, false.

GetEnumerator()

Returns an enumerator that iterates through the collection.

```
public IEnumerator<KeyValuePair<int, uint>> GetEnumerator()
```

Returns

IEnumerator<KeyValuePair<int, uint>>

An enumerator that can be used to iterate through the collection.

GetHashCode()

Returns the hash code for this instance.

```
public override int GetHashCode()
```

Returns

int

A 32-bit signed integer that is the hash code for this instance.

Remove(int)

Removes the element with the specified key from the immutable dictionary.

```
public IImmutableDictionary<int, uint> Remove(int key)
```

Parameters

`key` int

The key of the element to remove.

Returns

IImmutableDictionary<int, uint>

A new immutable dictionary with the specified element removed; or this instance if the specified key cannot be found in the dictionary.

RemoveRange(IEnumerable<int>)

Removes the elements with the specified keys from the immutable dictionary.

```
public IImmutableDictionary<int, uint> RemoveRange(IEnumerable<int> keys)
```

Parameters

`keys` IEnumerable<int>

The keys of the elements to remove.

Returns

IImmutableDictionary<int, uint>

A new immutable dictionary with the specified keys removed; or this instance if the specified keys cannot be found in the dictionary.

Serialize()

```
public Dictionary Serialize()
```

Returns

Dictionary

SetItem(int, uint)

Sets the specified key and value in the immutable dictionary, possibly overwriting an existing value for the key.

```
public IImmutableDictionary<int, uint> SetItem(int key, uint value)
```

Parameters

`key` int

The key of the entry to add.

`value` uint

The key value to set.

Returns

IImmutableDictionary<int, uint>

A new immutable dictionary that contains the specified key/value pair.

SetItems(IEnumerable<KeyValuePair<int, uint>>)

Sets the specified key/value pairs in the immutable dictionary, possibly overwriting existing values for the keys.

```
public IImmutableDictionary<int, uint> SetItems(IEnumerable<KeyValuePair<int,
uint>> items)
```

Parameters

`items` IEnumerable<KeyValuePair<int, uint>>

The key/value pairs to set in the dictionary. If any of the keys already exist in the dictionary, this method will overwrite their previous values.

Returns

IImmutableDictionary<int, uint>

A new immutable dictionary that contains the specified key/value pairs.

TryGetKey(int, out int)

Determines whether this dictionary contains a specified key.

```
public bool TryGetKey(int equalKey, out int actualKey)
```

Parameters

`equalKey` int

The key to search for.

`actualKey` int

The matching key located in the dictionary if found, or `equalkey` if no match is found.

Returns

bool

true if a match for `equalKey` is found; otherwise, false.

TryGetValue(int, out uint)

Gets the value that is associated with the specified key.

```
public bool TryGetValue(int key, out uint value)
```

Parameters

`key` int

The key to locate.

`value` uint

When this method returns, the value associated with the specified key, if the key is found; otherwise, the default value for the type of the `value` parameter. This parameter is passed uninitialized.

Returns

bool

true if the object that implements the IReadOnlyDictionary<TKey, TValue> interface contains an element that has the specified key; otherwise, false.

Exceptions

ArgumentNullException

`key` is null.

Struct RewardSet.Comparer

Namespace: Nekoyume.Model.Coupons

Assembly: Lib9c.dll

```
public struct RewardSet.Comparer : IComparer<RewardSet>
```

Implements
IComparer<RewardSet>

Inherited Members
ValueType.Equals(object) , ValueType.GetHashCode() , ValueType.ToString() , object.Equals(object, object) , object.GetType() , object.ReferenceEquals(object, object)

Methods

Compare(RewardSet, RewardSet)

Compares two objects and returns a value indicating whether one is less than, equal to, or greater than the other.

```
public int Compare(RewardSet x, RewardSet y)
```

Parameters

x RewardSet

The first object to compare.

y RewardSet

The second object to compare.

Returns

int

A signed integer that indicates the relative values of x and y, as shown in the following table.

Value	Meaning
Less than zero	x is less than y.
Zero	x equals y.
Greater than zero	x is greater than y.

Namespace Nekoyume.Model.Elemental

Classes

ElementalTypeComparer

ElementalTypeExtension

Enums

ElementalResult

ElementalType

Enum ElementalResult

Namespace: Nekoyume.Model.Elemental

Assembly: Lib9c.dll

```
public enum ElementalResult
```

Extension Methods
StateExtensions.Serialize(Enum)

Fields

`Draw = 1`

`Lose = 2`

`Win = 0`

Enum ElementalType

Namespace: Nekoyume.Model.Elemental

Assembly: Lib9c.dll

```
public enum ElementalType
```

Extension Methods
ElementalTypeExtension.GetBattleResult(ElementalType, ElementalType) ,
ElementalTypeExtension.GetDamage(ElementalType, ElementalType, long) ,
ElementalTypeExtension.GetMultiplier(ElementalType, ElementalType) ,
ElementalTypeExtension.TryGetLoseCase(ElementalType, out ElementalType) ,
ElementalTypeExtension.TryGetWinCase(ElementalType, out ElementalType) ,
StateExtensions.Serialize(Enum)

Fields

`Fire = 1`

`Land = 3`

`Normal = 0`

`Water = 2`

`Wind = 4`

Class ElementalTypeComparer

Namespace: Nekoyume.Model.Elemental

Assembly: Lib9c.dll

```
public class ElementalTypeComparer : IEqualityComparer<ElementalType>
```

Inheritance

object ← ElementalTypeComparer

Implements

IEqualityComparer<ElementalType>

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Fields

Instance

```
public static readonly ElementalTypeComparer Instance
```

Field Value

ElementalTypeComparer

Methods

Equals(ElementalType, ElementalType)

Determines whether the specified objects are equal.

```
public bool Equals(ElementalType x, ElementalType y)
```

Parameters

`x` ElementalType

The first object of type `T` to compare.

`y` ElementalType

The second object of type `T` to compare.

Returns

bool

[true] if the specified objects are equal; otherwise, [false].

GetHashCode(ElementalType)

Returns a hash code for the specified object.

```
public int GetHashCode(ElementalType obj)
```

Parameters

`obj` ElementalType

The [object] for which a hash code is to be returned.

Returns

int

A hash code for the specified object.

Exceptions

ArgumentNullException

The type of `obj` is a reference type and `obj` is [null].

Class ElementalTypeExtension

Namespace: Nekoyume.Model.Elemental

Assembly: Lib9c.dll

```
public static class ElementalTypeExtension
```

Inheritance

object ← ElementalTypeExtension

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Fields

WinMultiplier

```
public const decimal WinMultiplier = 1.2
```

Field Value

decimal

Methods

GetAllTypes()

```
public static List<ElementalType> GetAllTypes()
```

Returns

List<ElementalType>

GetBattleResult(ElementalType, ElementalType)

```
public static ElementalResult GetBattleResult(this ElementalType from,
ElementalType to)
```

Parameters

`from` ElementalType

`to` ElementalType

Returns

ElementalResult

1: Win 0: Draw -1: Lose

GetDamage(ElementalType, ElementalType, long)

```
public static long GetDamage(this ElementalType from, ElementalType to, long damage)
```

Parameters

`from` ElementalType

`to` ElementalType

`damage` long

Returns

long

GetMultiplier(ElementalType, ElementalType)

```
public static decimal GetMultiplier(this ElementalType from, ElementalType to)
```

Parameters

`from` ElementalType

`to` ElementalType

Returns

decimal

TryGetLoseCase(ElementalType, out ElementalType)

```
public static bool TryGetLoseCase(this ElementalType lose, out ElementalType win)
```

Parameters

`lose` ElementalType

`win` ElementalType

Returns

bool

TryGetWinCase(ElementalType, out ElementalType)

```
public static bool TryGetWinCase(this ElementalType win, out ElementalType lose)
```

Parameters

`win` ElementalType

`lose` ElementalType

Returns

bool

Namespace Nekoyume.Model.EnumType

Classes

BattleTypeExtensions

Enums

ArenaType

BattleType

CraftType

Grade
Enumeration of item grades.

RuneSlotType

RuneType

RuneUsePlace

StatReferenceType

TradeType

Enum ArenaType

Namespace: Nekoyume.Model.EnumType

Assembly: Lib9c.dll

```
public enum ArenaType
```

Extension Methods
StateExtensions.Serialize(Enum)

Fields

`Championship = 2`

`OffSeason = 0`

`Season = 1`

Enum BattleType

Namespace: Nekoyume.Model.EnumType

Assembly: Lib9c.dll

```
public enum BattleType
```

Extension Methods
BattleTypeExtensions.IsEquippableRune(BattleType, RuneUsePlace) ,
StateExtensions.Serialize(Enum)

Fields

`Adventure = 1`

`Arena = 2`

`End = 5`

`InfiniteTower = 4`

`Raid = 3`

Class BattleTypeExtensions

Namespace: Nekoyume.Model.EnumType

Assembly: Lib9c.dll

```
public static class BattleTypeExtensions
```

Inheritance

object ← BattleTypeExtensions

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Methods

IsEquippableRune(BattleType, RuneUsePlace)

```
public static bool IsEquippableRune(this BattleType battleType,
RuneUsePlace runePlace)
```

Parameters

`battleType` BattleType

`runePlace` RuneUsePlace

Returns

bool

Enum CraftType

Namespace: Nekoyume.Model.EnumType

Assembly: Lib9c.dll

```
public enum CraftType
```

Extension Methods
StateExtensions.Serialize(Enum)

Fields

`Mimisbrunnr = 2`

`Normal = 0`

`Premium = 1`

Enum Grade

Namespace: Nekoyume.Model.EnumType

Assembly: Lib9c.dll

Enumeration of item grades.

```
public enum Grade
```

Extension Methods
StateExtensions.Serialize(Enum)

Fields

`Divinity = 6`

Divinity grade.

`Epic = 3`

Epic grade.

`Legendary = 5`

Legendary grade.

`Mythic = 7`

Mythic grade.

`Normal = 1`

Normal grade.

`Rare = 2`

Rare grade.

`Transcendent = 8`

Transcendent grade.

`Ultimate = 9`

Ultimate grade.

```
Unique = 4
```

Unique grade.

Enum RuneSlotType

Namespace: Nekoyume.Model.EnumType

Assembly: Lib9c.dll

```
public enum RuneSlotType
```

Extension Methods
StateExtensions.Serialize(Enum)

Fields

`Crystal = 4`

`Default = 1`

`Ncg = 2`

`Stake = 3`

Enum RuneType

Namespace: Nekoyume.Model.EnumType

Assembly: Lib9c.dll

```
public enum RuneType
```

Extension Methods
StateExtensions.Serialize(Enum)

Fields

`Skill = 2`

`Stat = 1`

Enum RuneUsePlace

Namespace: Nekoyume.Model.EnumType

Assembly: Lib9c.dll

```
public enum RuneUsePlace
```

Extension Methods
StateExtensions.Serialize(Enum)

Fields

`Adventure = 1`

`AdventureAndArena = 3`

`All = 7`

`Arena = 2`

`Raid = 4`

`RaidAndAdventure = 5`

`RaidAndArena = 6`

Enum StatReferenceType

Namespace: Nekoyume.Model.EnumType

Assembly: Lib9c.dll

```
public enum StatReferenceType
```

Extension Methods
StateExtensions.Serialize(Enum)

Fields

`Caster = 0`

`Target = 1`

Enum TradeType

Namespace: Nekoyume.Model.EnumType

Assembly: Lib9c.dll

```
public enum TradeType
```

Extension Methods
StateExtensions.Serialize(Enum)

Fields

`Buy = 0`

`Sell = 1`

Namespace Nekoyume.Model.Event

Classes

EventDungeonInfo

Class EventDungeonInfo

Namespace: Nekoyume.Model.Event

Assembly: Lib9c.dll

```
public class EventDungeonInfo : IState
```

Inheritance

object ← EventDungeonInfo

Implements

IState

Inherited Members

object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Extension Methods

EventDungeonExtensions.GetRemainingTicketsConsiderReset(EventDungeonInfo, EventScheduleSheet.Row, long)

Constructors

EventDungeonInfo(IValue)

```
public EventDungeonInfo(IValue serialized)
```

Parameters

`serialized` IValue

EventDungeonInfo(List)

```
public EventDungeonInfo(List serialized)
```

Parameters

serialized List

EventDungeonInfo(int, int, int, int)

```
public EventDungeonInfo(int resetTicketsInterval = 0, int remainingTickets = 0, int
numberOfTicketPurchases = 0, int clearedStageId = 0)
```

Parameters

resetTicketsInterval int

remainingTickets int

numberOfTicketPurchases int

clearedStageId int

Properties

ClearedStageId

```
public int ClearedStageId { get; }
```

Property Value

int

NumberOfTicketPurchases

```
public int NumberOfTicketPurchases { get; }
```

Property Value

int

RemainingTickets

```
public int RemainingTickets { get; }
```

Property Value

int

ResetTicketsInterval

```
public int ResetTicketsInterval { get; }
```

Property Value

int

Methods

ClearStage(int)

```
public void ClearStage(int stageId)
```

Parameters

`stageId` int

DeriveAddress(Address, int)

```
public static Address DeriveAddress(Address address, int dungeonId)
```

Parameters

`address` Address

`dungeonId` int

Returns

Address

Equals(EventDungeonInfo)

```
protected bool Equals(EventDungeonInfo other)
```

Parameters

`other` EventDungeonInfo

Returns

bool

Equals(object)

Determines whether the specified object is equal to the current object.

```
public override bool Equals(object obj)
```

Parameters

`obj` object

The object to compare with the current object.

Returns

bool

true if the specified object is equal to the current object; otherwise, false.

GetHashCode()

Serves as the default hash function.

```
public override int GetHashCode()
```

Returns

int

A hash code for the current object.

HasTickets(int)

```
public bool HasTickets(int tickets)
```

Parameters

`tickets` int

Returns

bool

IncreaseNumberOfTicketPurchases()

```
public void IncreaseNumberOfTicketPurchases()
```

IsCleared(int)

```
public bool IsCleared(int stageId)
```

Parameters

`stageId` int

Returns

bool

ResetTickets(int, int)

```
public void ResetTickets(int interval, int tickets)
```

Parameters

`interval` int

`tickets` int

Serialize()

```
public IValue Serialize()
```

Returns

IValue

TryUseTickets(int)

```
public bool TryUseTickets(int tickets)
```

Parameters

`tickets` int

Returns

bool

Namespace Nekoyume.Model.Faucet

Classes

FaucetRuneInfo

Class FaucetRuneInfo

Namespace: Nekoyume.Model.Faucet

Assembly: Lib9c.dll

```
[Serializable]
public class FaucetRuneInfo
```

Inheritance

object ← FaucetRuneInfo

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

FaucetRuneInfo(List)

```
public FaucetRuneInfo(List serialized)
```

Parameters

`serialized` List

FaucetRuneInfo(int, int)

```
public FaucetRuneInfo(int runeId, int amount)
```

Parameters

`runeId` int

`amount` int

Properties

Amount

```
public int Amount { get; }
```

Property Value

int

RuneId

```
public int RuneId { get; }
```

Property Value

int

Methods

Equals(FaucetRuneInfo)

```
protected bool Equals(FaucetRuneInfo other)
```

Parameters

`other` FaucetRuneInfo

Returns

bool

Serialize()

```
public IValue Serialize()
```

Returns

IValue

Namespace Nekoyume.Model.Garages

Classes

FungibleItemGarage

Interfaces

IFungibleItemGarage

IGarage<T1, T2>

Class FungibleItemGarage

Namespace: Nekoyume.Model.Garages

Assembly: Lib9c.dll

```
public class FungibleItemGarage : IFungibleItemGarage, IGarage<IFungibleItemGarage,
int>
```

Inheritance

object ← FungibleItemGarage

Implements

IFungibleItemGarage, IGarage<IFungibleItemGarage, int>

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

FungibleItemGarage(IValue?)

```
public FungibleItemGarage(IValue? serialized)
```

Parameters

`serialized` IValue

FungibleItemGarage(IFungibleItem, int)

```
public FungibleItemGarage(IFungibleItem item, int count)
```

Parameters

`item` IFungibleItem

count int

Properties

Count

```
public int Count { get; }
```

Property Value

int

Item

```
public IFungibleItem Item { get; }
```

Property Value

IFungibleItem

Methods

Deliver(IFungibleItemGarage, int)

```
public void Deliver(IFungibleItemGarage to, int count)
```

Parameters

to IFungibleItemGarage

count int

Load(int)

```
public void Load(int count)
```

Parameters

`count` int

Serialize()

```
public IValue Serialize()
```

Returns

IValue

Unload(int)

```
public void Unload(int count)
```

Parameters

`count` int

Interface IFungibleItemGarage

Namespace: Nekoyume.Model.Garages

Assembly: Lib9c.dll

```
public interface IFungibleItemGarage : IGarage<IFungibleItemGarage, int>
```

Inherited Members
IGarage<IFungibleItemGarage, int>.Load(int) ,
IGarage<IFungibleItemGarage, int>.Deliver(IFungibleItemGarage, int) ,
IGarage<IFungibleItemGarage, int>.Unload(int)

Properties

Count

```
int Count { get; }
```

Property Value

int

Item

```
IFungibleItem Item { get; }
```

Property Value

IFungibleItem

Interface IGarage<T1, T2>

Namespace: Nekoyume.Model.Garages

Assembly: Lib9c.dll

```
public interface IGarage<in T1, in T2> where T1 : IGarage<in T1, in T2>
```

Type Parameters

T1

T2

Methods

Deliver(T1, T2)

```
void Deliver(T1 to, T2 amount)
```

Parameters

to T1

amount T2

Load(T2)

```
void Load(T2 amount)
```

Parameters

amount T2

Unload(T2)

```
void Unload(T2 amount)
```

Parameters

`amount` T2

Namespace Nekoyume.Model.GrandFinale

Classes

GrandFinaleInformation

Class GrandFinaleInformation

Namespace: Nekoyume.Model.GrandFinale

Assembly: Lib9c.dll

```
public class GrandFinaleInformation : IState
```

Inheritance

object ← GrandFinaleInformation

Implements

IState

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

GrandFinaleInformation(List)

```
public GrandFinaleInformation(List serialized)
```

Parameters

`serialized` List

GrandFinaleInformation(Address, int)

```
public GrandFinaleInformation(Address avatarAddress, int grandFinaleId)
```

Parameters

`avatarAddress` Address

`grandFinaleId` int

Fields

Address

```
public Address Address
```

Field Value

Address

Methods

DeriveAddress(Address, int)

```
public static Address DeriveAddress(Address avatarAddress, int grandFinaleId)
```

Parameters

`avatarAddress` Address

`grandFinaleId` int

Returns

Address

GetBattleRecordList()

```
public List<KeyValuePair<Address, bool>> GetBattleRecordList()
```

Returns

List<KeyValuePair<Address, bool>>

Serialize()

```
public IValue Serialize()
```

Returns

IValue

TryGetBattleRecord(Address, out bool)

```
public bool TryGetBattleRecord(Address enemyAddress, out bool win)
```

Parameters

`enemyAddress` Address

`win` bool

Returns

bool

UpdateRecord(Address, bool)

```
public void UpdateRecord(Address enemyAddress, bool win)
```

Parameters

`enemyAddress` Address

`win` bool

Namespace Nekoyume.Model.Guild

Classes

Guild

GuildDelegatee

GuildDelegator

GuildParticipant

GuildRepository

Class Guild

Namespace: Nekoyume.Model.Guild

Assembly: Lib9c.dll

```
public class Guild : IBencodable, IEquatable<Guild>
```

Inheritance

object ← Guild

Implements

IBencodable, IEquatable<Guild>

Inherited Members

object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

Guild(GuildAddress, IValue, GuildRepository)

```
public Guild(GuildAddress address, IValue bencoded, GuildRepository repository)
```

Parameters

`address` GuildAddress

`bencoded` IValue

`repository` GuildRepository

Guild(GuildAddress, AgentAddress, Address, GuildRepository)

```
public Guild(GuildAddress address, AgentAddress guildMasterAddress, Address
validatorAddress, GuildRepository repository)
```

Parameters

`address` GuildAddress

`guildMasterAddress` AgentAddress

`validatorAddress` Address

`repository` GuildRepository

Fields

GuildMasterAddress

```
public readonly AgentAddress GuildMasterAddress
```

Field Value

AgentAddress

ValidatorAddress

```
public readonly Address ValidatorAddress
```

Field Value

Address

Properties

Address

```
public GuildAddress Address { get; }
```

Property Value

GuildAddress

Bencoded

```
public List Bencoded { get; }
```

Property Value

List

Repository

```
public GuildRepository Repository { get; }
```

Property Value

GuildRepository

Methods

ClaimReward(Address, long)

```
public void ClaimReward(Address validatorAddress, long height)
```

Parameters

`validatorAddress` Address

`height` long

Equals(Guild?)

Indicates whether the current object is equal to another object of the same type.

```
public bool Equals(Guild? other)
```

Parameters

`other` Guild

An object to compare with this object.

Returns

bool

true if the current object is equal to the `other` parameter; otherwise, false.

Equals(object?)

Determines whether the specified object is equal to the current object.

```
public override bool Equals(object? obj)
```

Parameters

`obj` object

The object to compare with the current object.

Returns

bool

true if the specified object is equal to the current object; otherwise, false.

GetHashCode()

Serves as the default hash function.

```
public override int GetHashCode()
```

Returns

int

A hash code for the current object.

Class GuildDelegatee

Namespace: Nekoyume.Model.Guild

Assembly: Lib9c.dll

```
public class GuildDelegatee : Delegatee<GuildRepository, GuildDelegatee,
GuildDelegator>, IDelegatee, IEquatable<GuildDelegatee>
```

Inheritance

object ← Delegatee<GuildRepository, GuildDelegatee, GuildDelegator> ← GuildDelegatee

Implements

IDelegatee, IEquatable<GuildDelegatee>

Inherited Members

Delegatee<GuildRepository, GuildDelegatee, GuildDelegator>.Metadata ,
Delegatee<GuildRepository, GuildDelegatee, GuildDelegator>.Repository ,
Delegatee<GuildRepository, GuildDelegatee, GuildDelegator>.Address ,
Delegatee<GuildRepository, GuildDelegatee, GuildDelegator>.AccountAddress ,
Delegatee<GuildRepository, GuildDelegatee, GuildDelegator>.MetadataAddress ,
Delegatee<GuildRepository, GuildDelegatee, GuildDelegator>.DelegationCurrency ,
Delegatee<GuildRepository, GuildDelegatee, GuildDelegator>.RewardCurrencies ,
Delegatee<GuildRepository, GuildDelegatee, GuildDelegator>.DelegationPoolAddress ,
Delegatee<GuildRepository, GuildDelegatee, GuildDelegator>.RewardPoolAddress ,
Delegatee<GuildRepository, GuildDelegatee,
GuildDelegator>.RewardRemainderPoolAddress ,
Delegatee<GuildRepository, GuildDelegatee, GuildDelegator>.SlashedPoolAddress ,
Delegatee<GuildRepository, GuildDelegatee, GuildDelegator>.UnbondingPeriod ,
Delegatee<GuildRepository, GuildDelegatee, GuildDelegator>.MaxUnbondLockInEntries ,
Delegatee<GuildRepository, GuildDelegatee, GuildDelegator>.MaxRebondGraceEntries ,
Delegatee<GuildRepository, GuildDelegatee, GuildDelegator>.TotalDelegated ,
Delegatee<GuildRepository, GuildDelegatee, GuildDelegator>.TotalShares ,
Delegatee<GuildRepository, GuildDelegatee, GuildDelegator>.Jailed ,
Delegatee<GuildRepository, GuildDelegatee, GuildDelegator>.JailedUntil ,
Delegatee<GuildRepository, GuildDelegatee, GuildDelegator>.Tombstoned ,
Delegatee<GuildRepository, GuildDelegatee, GuildDelegator>.MetadataBencoded ,
Delegatee<GuildRepository, GuildDelegatee,
GuildDelegator>.ShareFromFAV(FungibleAssetValue) ,
Delegatee<GuildRepository, GuildDelegatee, GuildDelegator>.FAVFromShare(BigInteger) ,

Delegatee<GuildRepository, GuildDelegatee, GuildDelegator>.Jail(long) ,
Delegatee<GuildRepository, GuildDelegatee, GuildDelegator>.Unjail(long) ,
Delegatee<GuildRepository, GuildDelegatee, GuildDelegator>.Tombstone() ,
Delegatee<GuildRepository, GuildDelegatee, GuildDelegator>.BondAddress(Address) ,
Delegatee<GuildRepository, GuildDelegatee,
GuildDelegator>.UnbondLockInAddress(Address) ,
Delegatee<GuildRepository, GuildDelegatee,
GuildDelegator>.RebondGraceAddress(Address) ,
Delegatee<GuildRepository, GuildDelegatee, GuildDelegator>.DistributionPoolAddress() ,
Delegatee<GuildRepository, GuildDelegatee,
GuildDelegator>.CurrentRewardBaseAddress() ,
Delegatee<GuildRepository, GuildDelegatee, GuildDelegator>.RewardBaseAddress(long) ,
Delegatee<GuildRepository, GuildDelegatee,
GuildDelegator>.CurrentLumpSumRewardsRecordAddress() ,
Delegatee<GuildRepository, GuildDelegatee,
GuildDelegator>.LumpSumRewardsRecordAddress(long) ,
Delegatee<GuildRepository, GuildDelegatee, GuildDelegator>.Bond(GuildDelegator,
FungibleAssetValue, long) ,
Delegatee<GuildRepository, GuildDelegatee, GuildDelegator>.Unbond(GuildDelegator,
BigInteger, long) ,
Delegatee<GuildRepository, GuildDelegatee,
GuildDelegator>.DistributeReward(GuildDelegator, long) ,
Delegatee<GuildRepository, GuildDelegatee, GuildDelegator>.CollectRewards(long) ,
Delegatee<GuildRepository, GuildDelegatee, GuildDelegator>.Slash(BigInteger, long, long)
,
Delegatee<GuildRepository, GuildDelegatee, GuildDelegator>.StartNewRewardPeriod(long)
,
Delegatee<GuildRepository, GuildDelegatee,
GuildDelegator>.CalculateRewards(BigInteger, RewardBase, RewardBase) ,
Delegatee<GuildRepository, GuildDelegatee, GuildDelegator>.OnDelegationChanged(long)
,
Delegatee<GuildRepository, GuildDelegatee, GuildDelegator>.OnEnjailed() ,
Delegatee<GuildRepository, GuildDelegatee, GuildDelegator>.OnUnjailed() ,
object.Equals(object, object) , object.GetType() , object.MemberwiseClone() ,
object.ReferenceEquals(object, object) , object.ToString()

Constructors

GuildDelegatee(Address, GuildRepository)

```
public GuildDelegatee(Address address, GuildRepository repository)
```

Parameters

`address` Address

`repository` GuildRepository

GuildDelegatee(Address, IEnumerable<Currency>, GuildRepository)

```
public GuildDelegatee(Address address, IEnumerable<Currency> rewardCurrencies,
GuildRepository repository)
```

Parameters

`address` Address

`rewardCurrencies` IEnumerable<Currency>

`repository` GuildRepository

Properties

GuildMaxRebondGraceEntries

```
public static int GuildMaxRebondGraceEntries { get; }
```

Property Value

int

GuildMaxUnbondLockInEntries

```
public static int GuildMaxUnbondLockInEntries { get; }
```

Property Value

int

Methods

Activate()

```
public void Activate()
```

Deactivate()

```
public void Deactivate()
```

Equals(GuildDelegatee?)

Indicates whether the current object is equal to another object of the same type.

```
public bool Equals(GuildDelegatee? other)
```

Parameters

`other` GuildDelegatee

An object to compare with this object.

Returns

bool

true if the current object is equal to the `other` parameter; otherwise, false.

Equals(object?)

Determines whether the specified object is equal to the current object.

```
public override bool Equals(object? obj)
```

Parameters

`obj` object

The object to compare with the current object.

Returns

bool

true if the specified object is equal to the current object; otherwise, false.

GetHashCode()

Serves as the default hash function.

```
public override int GetHashCode()
```

Returns

int

A hash code for the current object.

Slash(BigInteger, long, long)

```
public override void Slash(BigInteger slashFactor, long infractionHeight,
long height)
```

Parameters

`slashFactor` BigInteger

`infractionHeight` long

`height` long

Class GuildDelegator

Namespace: Nekoyume.Model.Guild

Assembly: Lib9c.dll

```
public class GuildDelegator : Delegator<GuildRepository, GuildDelegatee,
GuildDelegator>, IDelegator, IEquatable<GuildDelegator>
```

Inheritance

object ← Delegator<GuildRepository, GuildDelegatee, GuildDelegator> ← GuildDelegator

Implements

IDelegator, IEquatable<GuildDelegator>

Inherited Members

Delegator<GuildRepository, GuildDelegatee, GuildDelegator>.Metadata ,
Delegator<GuildRepository, GuildDelegatee, GuildDelegator>.Repository ,
Delegator<GuildRepository, GuildDelegatee, GuildDelegator>.Address ,
Delegator<GuildRepository, GuildDelegatee, GuildDelegator>.AccountAddress ,
Delegator<GuildRepository, GuildDelegatee, GuildDelegator>.MetadataAddress ,
Delegator<GuildRepository, GuildDelegatee, GuildDelegator>.DelegationPoolAddress ,
Delegator<GuildRepository, GuildDelegatee, GuildDelegator>.RewardAddress ,
Delegator<GuildRepository, GuildDelegatee, GuildDelegator>.Delegatees ,
Delegator<GuildRepository, GuildDelegatee, GuildDelegator>.UnbondingRefs ,
Delegator<GuildRepository, GuildDelegatee, GuildDelegator>.MetadataBencoded ,
Delegator<GuildRepository, GuildDelegatee, GuildDelegator>.Delegate(GuildDelegatee, FungibleAssetValue, long) ,
Delegator<GuildRepository, GuildDelegatee, GuildDelegator>.Undelegate(GuildDelegatee, BigInteger, long) ,
Delegator<GuildRepository, GuildDelegatee, GuildDelegator>.Redelegate(GuildDelegatee, GuildDelegatee, BigInteger, long) ,
Delegator<GuildRepository, GuildDelegatee, GuildDelegator>.CancelUndelegate(GuildDelegatee, FungibleAssetValue, long) ,
Delegator<GuildRepository, GuildDelegatee, GuildDelegator>.ClaimReward(GuildDelegatee, long) ,
Delegator<GuildRepository, GuildDelegatee, GuildDelegator>.ReleaseUnbondings(long) ,
Delegator<GuildRepository, GuildDelegatee, GuildDelegator>.OnUnbondingReleased(long, IUnbonding, FungibleAssetValue?) ,

object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

GuildDelegator(Address, Address, GuildRepository)

```
public GuildDelegator(Address address, Address delegationPoolAddress,
GuildRepository repository)
```

Parameters

`address` Address

`delegationPoolAddress` Address

`repository` GuildRepository

GuildDelegator(Address, GuildRepository)

```
public GuildDelegator(Address address, GuildRepository repository)
```

Parameters

`address` Address

`repository` GuildRepository

Methods

Equals(GuildDelegator?)

Indicates whether the current object is equal to another object of the same type.

```
public bool Equals(GuildDelegator? other)
```

Parameters

`other` GuildDelegator

An object to compare with this object.

Returns

bool

true if the current object is equal to the `other` parameter; otherwise, false.

Equals(object?)

Determines whether the specified object is equal to the current object.

```
public override bool Equals(object? obj)
```

Parameters

`obj` object

The object to compare with the current object.

Returns

bool

true if the specified object is equal to the current object; otherwise, false.

GetHashCode()

Serves as the default hash function.

```
public override int GetHashCode()
```

Returns

int

A hash code for the current object.

OnUnbondingReleased(long, IUnbonding, FungibleAssetValue?)

```
protected override void OnUnbondingReleased(long height, IUnbonding
releasedUnbonding, FungibleAssetValue? releasedFAV)
```

Parameters

`height` long

`releasedUnbonding` IUnbonding

`releasedFAV` FungibleAssetValue?

Class GuildParticipant

Namespace: Nekoyume.Model.Guild

Assembly: Lib9c.dll

```
public class GuildParticipant : IBencodable, IEquatable<GuildParticipant>
```

Inheritance

object ← GuildParticipant

Implements

IBencodable, IEquatable<GuildParticipant>

Inherited Members

object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

GuildParticipant(AgentAddress, IValue, GuildRepository)

```
public GuildParticipant(AgentAddress address, IValue bencoded,
GuildRepository repository)
```

Parameters

`address` AgentAddress

`bencoded` IValue

`repository` GuildRepository

GuildParticipant(AgentAddress, GuildAddress, GuildRepository)

```
public GuildParticipant(AgentAddress address, GuildAddress guildAddress,
```

```
  GuildRepository repository)
```

Parameters

`address` AgentAddress

`guildAddress` GuildAddress

`repository` GuildRepository

Fields

GuildAddress

```
  public readonly GuildAddress GuildAddress
```

Field Value

GuildAddress

Properties

Address

```
  public AgentAddress Address { get; }
```

Property Value

AgentAddress

Bencoded

```
  public List Bencoded { get; }
```

Property Value

List

DelegationPoolAddress

```
public Address DelegationPoolAddress { get; }
```

Property Value

Address

Repository

```
public GuildRepository Repository { get; }
```

Property Value

GuildRepository

RewardAddress

```
public Address RewardAddress { get; }
```

Property Value

Address

Methods

ClaimReward(Guild, long)

```
public void ClaimReward(Guild guild, long height)
```

Parameters

`guild` Guild

`height` long

Delegate(Guild, FungibleAssetValue, long)

```
public void Delegate(Guild guild, FungibleAssetValue fav, long height)
```

Parameters

`guild` Guild

`fav` FungibleAssetValue

`height` long

Equals(GuildParticipant?)

Indicates whether the current object is equal to another object of the same type.

```
public bool Equals(GuildParticipant? other)
```

Parameters

`other` GuildParticipant

An object to compare with this object.

Returns

bool

true if the current object is equal to the `other` parameter; otherwise, false.

Equals(object?)

Determines whether the specified object is equal to the current object.

```
public override bool Equals(object? obj)
```

Parameters

`obj` object

The object to compare with the current object.

Returns

bool

true if the specified object is equal to the current object; otherwise, false.

GetHashCode()

Serves as the default hash function.

```
public override int GetHashCode()
```

Returns

int

A hash code for the current object.

Redelegate(Guild, Guild, BigInteger, long)

```
public void Redelegate(Guild srcGuild, Guild dstGuild, BigInteger share,
long height)
```

Parameters

`srcGuild` Guild

`dstGuild` Guild

`share` BigInteger

`height` long

Undelegate(Guild, BigInteger, long)

```
public void Undelegate(Guild guild, BigInteger share, long height)
```

Parameters

`guild` Guild

`share` BigInteger

`height` long

Class GuildRepository

Namespace: Nekoyume.Model.Guild

Assembly: Lib9c.dll

```
public class GuildRepository : DelegationRepository<GuildRepository, GuildDelegatee,
GuildDelegator>, IDelegationRepository
```

Inheritance

object ← DelegationRepository<GuildRepository, GuildDelegatee, GuildDelegator> ← GuildRepository

Implements

IDelegationRepository

Inherited Members

DelegationRepository<GuildRepository, GuildDelegatee, GuildDelegator>.previousWorld ,
DelegationRepository<GuildRepository, GuildDelegatee, GuildDelegator>.delegateeAccount ,
DelegationRepository<GuildRepository, GuildDelegatee, GuildDelegator>.delegatorAccount ,
DelegationRepository<GuildRepository, GuildDelegatee, GuildDelegator>.delegateeMetadataAccount ,
DelegationRepository<GuildRepository, GuildDelegatee, GuildDelegator>.delegatorMetadataAccount ,
DelegationRepository<GuildRepository, GuildDelegatee, GuildDelegator>.bondAccount ,
DelegationRepository<GuildRepository, GuildDelegatee, GuildDelegator>.unbondLockInAccount ,
DelegationRepository<GuildRepository, GuildDelegatee, GuildDelegator>.rebondGraceAccount ,
DelegationRepository<GuildRepository, GuildDelegatee, GuildDelegator>.unbondingSetAccount ,
DelegationRepository<GuildRepository, GuildDelegatee, GuildDelegator>.rewardBaseAccount ,
DelegationRepository<GuildRepository, GuildDelegatee, GuildDelegator>.lumpSumRewardsRecordAccount ,
DelegationRepository<GuildRepository, GuildDelegatee, GuildDelegator>.World ,
DelegationRepository<GuildRepository, GuildDelegatee, GuildDelegator>.ActionContext ,

DelegationRepository<GuildRepository, GuildDelegatee, GuildDelegator>.DelegateeAccountAddress ,
DelegationRepository<GuildRepository, GuildDelegatee, GuildDelegator>.DelegatorAccountAddress ,
DelegationRepository<GuildRepository, GuildDelegatee, GuildDelegator>.DelegateeMetadataAccountAddress ,
DelegationRepository<GuildRepository, GuildDelegatee, GuildDelegator>.DelegatorMetadataAccountAddress ,
DelegationRepository<GuildRepository, GuildDelegatee, GuildDelegator>.BondAccountAddress ,
DelegationRepository<GuildRepository, GuildDelegatee, GuildDelegator>.UnbondLockInAccountAddress ,
DelegationRepository<GuildRepository, GuildDelegatee, GuildDelegator>.RebondGraceAccountAddress ,
DelegationRepository<GuildRepository, GuildDelegatee, GuildDelegator>.UnbondingSetAccountAddress ,
DelegationRepository<GuildRepository, GuildDelegatee, GuildDelegator>.RewardBaseAccountAddress ,
DelegationRepository<GuildRepository, GuildDelegatee, GuildDelegator>.LumpSumRewardsRecordAccountAddress ,
DelegationRepository<GuildRepository, GuildDelegatee, GuildDelegator>.GetDelegatee(Address) ,
DelegationRepository<GuildRepository, GuildDelegatee, GuildDelegator>.GetDelegator(Address) ,
DelegationRepository<GuildRepository, GuildDelegatee, GuildDelegator>.SetDelegatee(GuildDelegatee) ,
DelegationRepository<GuildRepository, GuildDelegatee, GuildDelegator>.SetDelegator(GuildDelegator) ,
DelegationRepository<GuildRepository, GuildDelegatee, GuildDelegator>.GetDelegateeMetadata(Address) ,
DelegationRepository<GuildRepository, GuildDelegatee, GuildDelegator>.GetDelegatorMetadata(Address) ,
DelegationRepository<GuildRepository, GuildDelegatee, GuildDelegator>.GetBond(GuildDelegatee, Address) ,
DelegationRepository<GuildRepository, GuildDelegatee, GuildDelegator>.GetUnbondLockIn(GuildDelegatee, Address) ,
DelegationRepository<GuildRepository, GuildDelegatee, GuildDelegator>.GetUnlimitedUnbondLockIn(Address) ,
DelegationRepository<GuildRepository, GuildDelegatee, GuildDelegator>.GetRebondGrace(GuildDelegatee, Address) ,

DelegationRepository<GuildRepository, GuildDelegatee, GuildDelegator>.GetUnlimitedRebondGrace(Address) ,
DelegationRepository<GuildRepository, GuildDelegatee, GuildDelegator>.GetCurrentRewardBase(GuildDelegatee) ,
DelegationRepository<GuildRepository, GuildDelegatee, GuildDelegator>.GetRewardBase(GuildDelegatee, long) ,
DelegationRepository<GuildRepository, GuildDelegatee, GuildDelegator>.GetLumpSumRewardsRecord(GuildDelegatee, long) ,
DelegationRepository<GuildRepository, GuildDelegatee, GuildDelegator>.GetCurrentLumpSumRewardsRecord(GuildDelegatee) ,
DelegationRepository<GuildRepository, GuildDelegatee, GuildDelegator>.GetBalance(Address, Currency) ,
DelegationRepository<GuildRepository, GuildDelegatee, GuildDelegator>.SetDelegateeMetadata(DelegateeMetadata) ,
DelegationRepository<GuildRepository, GuildDelegatee, GuildDelegator>.SetDelegatorMetadata(DelegatorMetadata) ,
DelegationRepository<GuildRepository, GuildDelegatee, GuildDelegator>.SetBond(Bond) ,
DelegationRepository<GuildRepository, GuildDelegatee, GuildDelegator>.SetUnbondLockIn(UnbondLockIn) ,
DelegationRepository<GuildRepository, GuildDelegatee, GuildDelegator>.SetRebondGrace(RebondGrace) ,
DelegationRepository<GuildRepository, GuildDelegatee, GuildDelegator>.SetRewardBase(RewardBase) ,
DelegationRepository<GuildRepository, GuildDelegatee, GuildDelegator>.SetLumpSumRewardsRecord(LumpSumRewardsRecord) ,
DelegationRepository<GuildRepository, GuildDelegatee, GuildDelegator>.RemoveLumpSumRewardsRecord(LumpSumRewardsRecord) ,
DelegationRepository<GuildRepository, GuildDelegatee, GuildDelegator>.TransferAsset(Address, Address, FungibleAssetValue) ,
DelegationRepository<GuildRepository, GuildDelegatee, GuildDelegator>.UpdateWorld(IWorld) ,
object.Equals(object) , object.Equals(object, object) , object.GetHashCode() ,
object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) ,
object.ToString()

Extension Methods

GuildBanModule.Ban(GuildRepository, AgentAddress, AgentAddress) ,
GuildBanModule.IsBanned(GuildRepository, GuildAddress, Address) ,
GuildBanModule.RemoveBanList(GuildRepository, GuildAddress) ,
GuildBanModule.Unban(GuildRepository, AgentAddress, Address) ,
GuildDelegateeModule.CreateDelegatee(GuildRepository, Address) ,

GuildDelegateeModule.TryGetDelegatee(GuildRepository, Address, out GuildDelegatee?) ,
GuildMemberCounterModule.DecreaseGuildMemberCount(GuildRepository, GuildAddress) ,
GuildMemberCounterModule.GetGuildMemberCount(GuildRepository, GuildAddress) ,
GuildMemberCounterModule.IncreaseGuildMemberCount(GuildRepository, GuildAddress) ,
GuildModule.GetDelegationInfo(GuildRepository, Address) ,
GuildModule.MakeGuild(GuildRepository, GuildAddress, Address) ,
GuildModule.RemoveGuild(GuildRepository) ,
GuildModule.TryGetGuild(GuildRepository, GuildAddress, out Guild?) ,
GuildParticipantModule.GetJoinedGuild(GuildRepository, AgentAddress) ,
GuildParticipantModule.JoinGuild(GuildRepository, GuildAddress, AgentAddress) ,
GuildParticipantModule.LeaveGuild(GuildRepository, AgentAddress) ,
GuildParticipantModule.MoveGuild(GuildRepository, AgentAddress, GuildAddress) ,
GuildParticipantModule.TryGetGuildParticipant(GuildRepository, AgentAddress, out GuildParticipant?)

Constructors

GuildRepository(IWorld, IActionContext)

```
public GuildRepository(IWorld world, IActionContext actionContext)
```

Parameters

`world` IWorld

`actionContext` IActionContext

GuildRepository(IDelegationRepository)

```
public GuildRepository(IDelegationRepository repository)
```

Parameters

`repository` IDelegationRepository

Properties

World

```
public override IWorld World { get; }
```

Property Value

IWorld

Methods

GetDelegatee(Address)

```
public override GuildDelegatee GetDelegatee(Address address)
```

Parameters

`address` Address

Returns

GuildDelegatee

GetDelegator(Address)

```
public override GuildDelegator GetDelegator(Address address)
```

Parameters

`address` Address

Returns

GuildDelegator

GetGuild(Address)

```
public Guild GetGuild(Address address)
```

Parameters

`address` Address

Returns

Guild

GetGuildParticipant(Address)

```
public GuildParticipant GetGuildParticipant(Address address)
```

Parameters

`address` Address

Returns

GuildParticipant

RemoveGuildParticipant(Address)

```
public void RemoveGuildParticipant(Address guildParticipantAddress)
```

Parameters

`guildParticipantAddress` Address

SetDelegatee(GuildDelegatee)

```
public override void SetDelegatee(GuildDelegatee delegatee)
```

Parameters

`delegatee` GuildDelegatee

SetDelegator(GuildDelegator)

```
public override void SetDelegator(GuildDelegator delegator)
```

Parameters

`delegator` GuildDelegator

SetGuild(Guild)

```
public void SetGuild(Guild guild)
```

Parameters

`guild` Guild

SetGuildParticipant(GuildParticipant)

```
public void SetGuildParticipant(GuildParticipant guildParticipant)
```

Parameters

`guildParticipant` GuildParticipant

UpdateWorld(IWorld)

```
public override void UpdateWorld(IWorld world)
```

Parameters

```
world IWorld
```

Namespace Nekoyume.Model.InfiniteTower

Classes

InfiniteTowerBattleCondition

Infinite tower battle condition for managing floor-specific restrictions and requirements. Represents various types of battle conditions that can be applied to infinite tower floors.

InfiniteTowerCondition

Infinite tower condition for managing battle conditions and restrictions. Uses StatType, SkillTargetType, and StatModifier for flexible stat modifications.

InfiniteTowerInfo

Infinite tower info for managing player's tower progress and tickets.

Enums

BattleConditionType

Represents different types of battle conditions that can be applied to infinite tower floors.

Enum BattleConditionType

Namespace: Nekoyume.Model.InfiniteTower

Assembly: Lib9c.dll

Represents different types of battle conditions that can be applied to infinite tower floors.

```
public enum BattleConditionType
```

Extension Methods
StateExtensions.Serialize(Enum)

Fields

`CP = 0`

Combat Power restrictions (RequiredCp, MaxCp)

`ForbiddenItemSubTypes = 5`

Forbidden Item Sub Types restrictions

`ForbiddenRuneTypes = 3`

Forbidden Rune Types restrictions

`ItemGrade = 1`

Item Grade restrictions (MinItemGrade, MaxItemGrade)

`ItemLevel = 2`

Item Level restrictions (MinItemLevel, MaxItemLevel)

`RequiredElementalType = 4`

Required Elemental Type restrictions

Class InfiniteTowerBattleCondition

Namespace: Nekoyume.Model.InfiniteTower

Assembly: Lib9c.dll

Infinite tower battle condition for managing floor-specific restrictions and requirements. Represents various types of battle conditions that can be applied to infinite tower floors.

```
[Serializable]
public class InfiniteTowerBattleCondition
```

Inheritance

object ← InfiniteTowerBattleCondition

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Constructors

InfiniteTowerBattleCondition(List<ElementalType>, bool)

Initializes a new instance of the InfiniteTowerBattleCondition class for Required Elemental Type.

```
public InfiniteTowerBattleCondition(List<ElementalType> requiredElementalTypes,
bool isElementalType)
```

Parameters

`requiredElementalTypes` List<ElementalType>

List of required elemental types

`isElementalType` bool

Must be true to indicate this is for elemental type

InfiniteTowerBattleCondition(List<RuneType>)

Initializes a new instance of the InfiniteTowerBattleCondition class for Forbidden Rune Types.

```
public InfiniteTowerBattleCondition(List<RuneType> forbiddenRuneTypes)
```

Parameters

`forbiddenRuneTypes` List<RuneType>

List of forbidden rune types

InfiniteTowerBattleCondition(List<ItemSubType>, bool)

Initializes a new instance of the InfiniteTowerBattleCondition class for Forbidden Item Sub Types.

```
public InfiniteTowerBattleCondition(List<ItemSubType> forbiddenItemSubTypes,
bool isItemSubTypes)
```

Parameters

`forbiddenItemSubTypes` List<ItemSubType>

List of forbidden item sub types

`isItemSubTypes` bool

Must be true to indicate this is for item sub types

InfiniteTowerBattleCondition(int?, int?)

Initializes a new instance of the InfiniteTowerBattleCondition class for Item Level restrictions.

```
public InfiniteTowerBattleCondition(int? minItemLevel, int? maxItemLevel)
```

Parameters

`minItemLevel` int?

Minimum item level

`maxItemLevel` int?

Maximum item level

InfiniteTowerBattleCondition(int?, int?, bool)

Initializes a new instance of the InfiniteTowerBattleCondition class for Item Grade restrictions.

```
public InfiniteTowerBattleCondition(int? minItemGrade, int? maxItemGrade,
bool isItemGrade)
```

Parameters

`minItemGrade` int?

Minimum item grade

`maxItemGrade` int?

Maximum item grade

`isItemGrade` bool

Must be true to indicate this is for item grade

InfiniteTowerBattleCondition(long?, long?)

Initializes a new instance of the InfiniteTowerBattleCondition class for CP restrictions.

```
public InfiniteTowerBattleCondition(long? requiredCp, long? maxCp)
```

Parameters

`requiredCp` long?

Minimum required combat power

`maxCp` long?

Maximum allowed combat power

Properties

ForbiddenItemSubTypes

Gets the list of forbidden item sub types (for ForbiddenItemSubTypes type).

```
public List<ItemSubType> ForbiddenItemSubTypes { get; }
```

Property Value

List<ItemSubType>

ForbiddenRuneTypes

Gets the list of forbidden rune types (for ForbiddenRuneTypes type).

```
public List<RuneType> ForbiddenRuneTypes { get; }
```

Property Value

List<RuneType>

MaxCp

Gets the maximum allowed combat power (for CP type).

```
public long? MaxCp { get; }
```

Property Value

long?

MaxItemGrade

Gets the maximum item grade (for ItemGrade type).

```
public int? MaxItemGrade { get; }
```

Property Value

int?

MaxItemLevel

Gets the maximum item level (for ItemLevel type).

```
public int? MaxItemLevel { get; }
```

Property Value

int?

MinItemGrade

Gets the minimum item grade (for ItemGrade type).

```
public int? MinItemGrade { get; }
```

Property Value

int?

MinItemLevel

Gets the minimum item level (for ItemLevel type).

```
public int? MinItemLevel { get; }
```

Property Value

int?

RequiredCp

Gets the minimum required combat power (for CP type).

```
public long? RequiredCp { get; }
```

Property Value

long?

RequiredElementalTypes

Gets the list of required elemental types (for RequiredElementalType type).

```
public List<ElementalType> RequiredElementalTypes { get; }
```

Property Value

List<ElementalType>

Type

Gets the type of this battle condition.

```
public BattleConditionType Type { get; }
```

Property Value

BattleConditionType

Methods

HasRestrictions()

Checks if this battle condition has any restrictions set.

```
public bool HasRestrictions()
```

Returns

bool

True if the condition has restrictions, false otherwise

ToString()

Gets a description of this battle condition.

```
public override string ToString()
```

Returns

string

String description of the condition

Class InfiniteTowerCondition

Namespace: Nekoyume.Model.InfiniteTower

Assembly: Lib9c.dll

Infinite tower condition for managing battle conditions and restrictions. Uses StatType, SkillTargetType, and StatModifier for flexible stat modifications.

```
[Serializable]
public class InfiniteTowerCondition
```

Inheritance

object ← InfiniteTowerCondition

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

InfiniteTowerCondition(Row)

Initializes a new instance of the InfiniteTowerCondition class.

```
public InfiniteTowerCondition(InfiniteTowerConditionSheet.Row row)
```

Parameters

`row` InfiniteTowerConditionSheet.Row

The condition sheet row containing the condition data.

Properties

Id

Gets the unique identifier for this condition.

```
public int Id { get; }
```

Property Value

int

OperationType

Gets the operation type for the stat modification.

```
public StatModifier.OperationType OperationType { get; }
```

Property Value

StatModifier.OperationType

StatType

Gets the stat type that this condition affects.

```
public StatType StatType { get; }
```

Property Value

StatType

TargetType

Gets the list of target types for this condition (Self, Enemy, etc.).

```
public List<SkillTargetType> TargetType { get; }
```

Property Value

List<SkillTargetType>

Value

Gets the value for the stat modification.

```
public int Value { get; }
```

Property Value

int

Methods

GetStatModifier()

Creates a StatModifier from this condition.

```
public StatModifier GetStatModifier()
```

Returns

StatModifier

GetValue()

Gets the value of this condition.

```
public int GetValue()
```

Returns

int

The condition value.

IsApplicableToEquipment(Equipment)

Checks if this condition applies to the given equipment for stat modification. This is only used for stat modifiers, not equipment restrictions.

```
public bool IsApplicableToEquipment(Equipment equipment)
```

Parameters

`equipment` Equipment

Returns

bool

IsApplicableToRune(RuneState)

Checks if this condition applies to the given rune for stat modification. This is only used for stat modifiers, not rune restrictions.

```
public bool IsApplicableToRune(RuneState runeState)
```

Parameters

`runeState` RuneState

Returns

bool

Class InfiniteTowerInfo

Namespace: Nekoyume.Model.InfiniteTower

Assembly: Lib9c.dll

Infinite tower info for managing player's tower progress and tickets.

```
[Serializable]
public class InfiniteTowerInfo : IState
```

Inheritance

object ← InfiniteTowerInfo

Implements

IState

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

InfiniteTowerInfo(List)

Initializes a new instance of the InfiniteTowerInfo class from serialized data.

```
public InfiniteTowerInfo(List serialized)
```

Parameters

`serialized` List

The serialized data.

InfiniteTowerInfo(Address, int, int)

Initializes a new instance of the InfiniteTowerInfo class.

```
public InfiniteTowerInfo(Address address, int infiniteTowerId, int initialTickets)
```

Parameters

`address` Address

The avatar address.

`infiniteTowerId` int

The infinite tower ID.

`initialTickets` int

The initial number of tickets to grant on creation.

Properties

Address

Gets the avatar address for this infinite tower info.

```
public Address Address { get; }
```

Property Value

Address

ClearedFloor

Gets the highest floor that has been cleared by the player.

```
public int ClearedFloor { get; }
```

Property Value

int

InfiniteTowerId

Gets the infinite tower ID this info belongs to.

```
public int InfiniteTowerId { get; }
```

Property Value

int

LastResetBlockIndex

Gets the block index when the tower was last reset.

```
public long LastResetBlockIndex { get; }
```

Property Value

long

LastTicketRefillBlockIndex

Gets the block index when tickets were last refilled.

```
public long LastTicketRefillBlockIndex { get; }
```

Property Value

long

NumberOfTicketPurchases

Gets the number of times the player has purchased tickets.

```
public int NumberOfTicketPurchases { get; }
```

Property Value

int

RemainingTickets

Gets the number of remaining tickets for the player.

```
public int RemainingTickets { get; }
```

Property Value

int

TotalTicketsUsed

Gets the total number of tickets used by the player.

```
public int TotalTicketsUsed { get; }
```

Property Value

int

Methods

AddTickets(int)

```
public void AddTickets(int count)
```

Parameters

`count` int

ClearFloor(int)

```
public void ClearFloor(int floor)
```

Parameters

`floor` int

GetScheduleStatus(long, long, long)

Gets the schedule status for the infinite tower.

```
public string GetScheduleStatus(long currentBlockIndex, long startBlockIndex,
long endBlockIndex)
```

Parameters

`currentBlockIndex` long

Current block index

`startBlockIndex` long

Schedule start block index

`endBlockIndex` long

Schedule end block index

Returns

string

Schedule status string

IncreaseNumberOfTicketPurchases()

```
public void IncreaseNumberOfTicketPurchases()
```

IsAccessible(long, long, long)

Checks if the infinite tower is currently accessible based on schedule.

```
public bool IsAccessible(long currentBlockIndex, long startBlockIndex,
long endBlockIndex)
```

Parameters

`currentBlockIndex` long

Current block index

`startBlockIndex` long

Schedule start block index

`endBlockIndex` long

Schedule end block index

Returns

bool

True if accessible, false otherwise

IsCleared(int)

```
public bool IsCleared(int floor)
```

Parameters

`floor` int

Returns

bool

PerformSeasonReset(long, int, int)

Performs a season reset of the infinite tower progress. This is called when a new season starts. Grants exactly 1 ticket on season start.

```
public void PerformSeasonReset(long currentBlockIndex, int dailyFreeTickets,
int maxTickets)
```

Parameters

`currentBlockIndex` long

Current block index

`dailyFreeTickets` int

Ignored for initial grant; kept for backward-compatibility

`maxTickets` int

Ignored; capacity is fixed to 10

ResetTickets(int)

```
public void ResetTickets(int maxTickets)
```

Parameters

`maxTickets` int

Serialize()

```
public IValue Serialize()
```

Returns

IValue

TryRefillDailyTickets(int, int, long, int)

Refills daily free tickets according to elapsed block periods since the last refill. Caps the total number of tickets to a fixed capacity (10).

```
public bool TryRefillDailyTickets(int dailyFreeTickets, int maxTickets, long
currentBlockIndex, int blocksPerDay = 10800)
```

Parameters

`dailyFreeTickets` int

Number of free tickets to give per period (from schedule's DailyFreeTickets)

`maxTickets` int

Maximum number of tickets player can hold (from schedule's MaxTickets)

`currentBlockIndex` long

Current block index

`blocksPerDay` int

Number of blocks per period (from schedule's ResetIntervalBlocks, default: 10800)

Returns

bool

True if tickets were refilled, false otherwise

TryUseTickets(int)

```
public bool TryUseTickets(int count)
```

Parameters

`count` int

Returns

bool

Namespace Nekoyume.Model.Item

Classes

Armor

Represents armor equipment items. Supports both Dictionary and List serialization formats for backward compatibility.

Aura

Represents aura equipment items. Supports both Dictionary and List serialization formats for backward compatibility.

Belt

Represents belt equipment items. Supports both Dictionary and List serialization formats for backward compatibility.

Consumable

Represents consumable items that can be used by characters. Supports both Dictionary and List serialization formats for backward compatibility.

Costume

Represents costume items that can be equipped by characters. Supports both Dictionary and List serialization formats for backward compatibility.

Equipment

Represents equipment items that can be equipped by characters. Supports both Dictionary and List serialization formats for backward compatibility.

Field Order (List Format): Base fields (0~10): 11 fields from ItemUsable Equipment fields (11~22): equipped, level, stat, setId, spineResourcePath, iconId, byCustomCraft, craftWithRandom, hasRandomOnlyIcon, optionCountFromCombination, madeWithMimisbrunnrRecipe, exp

Equipment Properties: - Equipped: Whether the equipment is currently equipped - Level: Equipment enhancement level - Stat: Primary stat of the equipment - SetId: ID for equipment set bonuses - SpineResourcePath: Path to spine animation resource - IconId: Icon identifier for UI display - ByCustomCraft: Whether crafted through custom crafting - CraftWithRandom: Whether random options were applied during crafting - HasRandomOnlyIcon: Whether has random-only icon - OptionCountFromCombination: Number of options from combination - MadeWithMimisbrunnrRecipe: Whether made with Mimisbrunnr recipe - Exp: Experience points for leveling

Grimoire

Represents grimoire equipment items. Supports both Dictionary and List serialization formats for backward compatibility.

Inventory

Inventory.Item

ItemBase

Base class for all items in the game. Supports both Dictionary and List serialization formats for backward compatibility.

Serialization Format: - Dictionary (Legacy): Uses key-value pairs for backward compatibility - List (New): Uses ordered list for better performance and smaller size

Field Order (List Format): 1. version - Serialization version number 2. id - Item ID (Binary format) 3. itemType - Item type as string 4. itemSubType - Item sub-type as string 5. grade - Item grade 6. elementalType - Elemental type as string

ItemFactory

Factory class for creating and deserializing items. Supports both Dictionary and List serialization formats for backward compatibility.

ItemSubTypeComparer

ItemTypeComparer

ItemUsable

Material

Represents a material item that can be used for crafting and other purposes. Supports both Dictionary and List serialization formats for backward compatibility.

Field Order (List Format): Base fields (0~5): version, id, itemType, itemSubType, grade, elementalType Material fields (6): itemId

Material Properties: - ItemId: Unique identifier for the material item

Necklace

Represents necklace equipment items. Supports both Dictionary and List serialization formats for backward compatibility.

Ring

Represents ring equipment items. Supports both Dictionary and List serialization formats for backward compatibility.

ShopItem

TradableMaterial
: Represents a tradable material item that can be traded between players. Supports both Dictionary and List serialization formats for backward compatibility.

Weapon
: Represents weapon equipment items. Supports both Dictionary and List serialization formats for backward compatibility.

Structs

OrderLock

Interfaces

IEquippableItem

IFungibleItem

IItem

ILock

INonFungibleItem

ITradableFungibleItem

ITradableItem

Enums

ItemSubType

ItemType

LockType

Class Armor

Namespace: Nekoyume.Model.Item

Assembly: Lib9c.dll

Represents armor equipment items. Supports both Dictionary and List serialization formats for backward compatibility.

```
[Serializable]
public class Armor : Equipment, INonFungibleItem, IEquippableItem, ITradableItem,
IItem, IState
```

Inheritance

object ← ItemBase ← ItemUsable ← Equipment ← Armor

Implements

INonFungibleItem, IEquippableItem, ITradableItem, IItem, IState

Inherited Members

Equipment.equipped , Equipment.level , Equipment.Exp , Equipment.optionCountFromCombination , Equipment.IconId , Equipment.ByCustomCraft , Equipment.CraftWithRandom , Equipment.HasRandomOnlyIcon , Equipment.Stat , Equipment.SetId , Equipment.SpineResourcePath , Equipment.MadeWithMimisbrunnrRecipe , Equipment.UniqueStatType , Equipment.Equipped , Equipment.GetIncrementAmountOfEnhancement() , Equipment.Serialize() , Equipment.GetFieldNames() , Equipment.Equip() , Equipment.Unequip() , Equipment.LevelUpV1() , Equipment.LevelUp(IRandom, EnhancementCostSheetV2.Row, bool) , Equipment.SetLevel(IRandom, int, EnhancementCostSheetV3) , Equipment.GetRealExp(EquipmentItemSheet, EnhancementCostSheetV3) , Equipment.Equals(Equipment) , Equipment.Equals(object) , Equipment.GetHashCode() , ItemUsable.ITEM_USABLE_FIELD_COUNT , ItemUsable.ItemId , ItemUsable.NonFungibleId , ItemUsable.StatsMap , ItemUsable.Skills , ItemUsable.BuffSkills , ItemUsable.RequiredBlockIndex , ItemUsable.Equals(ItemUsable) , ItemUsable.GetOptionCount() , ItemUsable.Update(long) , ItemBase.SerializationVersion , ItemBase.BaseFieldCount , ItemBase.Codec , ItemBase.Id , ItemBase.Grade , ItemBase.ItemType , ItemBase.ItemSubType , ItemBase.ElementalType , ItemBase.GetObjectData(SerializationInfo, StreamingContext) , ItemBase.Equals(ItemBase) , ItemBase.ToString() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Extension Methods
EquipmentExtensions.GetRequirementLevel(Equipment, ItemRequirementSheet, EquipmentItemRecipeSheet, EquipmentItemSubRecipeSheetV2, EquipmentItemOptionSheet) ,
EquipmentExtensions.IsMadeWithMimisbrunnrRecipe(Equipment, EquipmentItemRecipeSheet, EquipmentItemSubRecipeSheetV2, EquipmentItemOptionSheet)

Constructors

Armor(IValue)

Constructor for deserialization that supports both Dictionary and List formats.

```
public Armor(IValue serialized)
```

Parameters

`serialized` IValue

Serialized data in either Dictionary or List format

Armor(Row, Guid, long, bool)

```
public Armor(EquipmentItemSheet.Row data, Guid id, long requiredBlockIndex, bool
madeWithMimisbrunnrRecipe = false)
```

Parameters

`data` EquipmentItemSheet.Row

`id` Guid

`requiredBlockIndex` long

`madeWithMimisbrunnrRecipe` bool

Armor(SerializationInfo, StreamingContext)

```
protected Armor(SerializationInfo info, StreamingContext _)
```

Parameters

`info` SerializationInfo

`_` StreamingContext

Properties

TradableId

```
public Guid TradableId { get; }
```

Property Value

Guid

Class Aura

Namespace: Nekoyume.Model.Item

Assembly: Lib9c.dll

Represents aura equipment items. Supports both Dictionary and List serialization formats for backward compatibility.

```
[Serializable]
public class Aura : Equipment, INonFungibleItem, IEquippableItem, IItem, IState
```

Inheritance

object ← ItemBase ← ItemUsable ← Equipment ← Aura

Implements

INonFungibleItem, IEquippableItem, IItem, IState

Inherited Members

Equipment.equipped , Equipment.level , Equipment.Exp , Equipment.optionCountFromCombination , Equipment.IconId , Equipment.ByCustomCraft , Equipment.CraftWithRandom , Equipment.HasRandomOnlyIcon , Equipment.Stat , Equipment.SetId , Equipment.SpineResourcePath , Equipment.MadeWithMimisbrunnrRecipe , Equipment.UniqueStatType , Equipment.Equipped , Equipment.GetIncrementAmountOfEnhancement() , Equipment.Serialize() , Equipment.GetFieldNames() , Equipment.Equip() , Equipment.Unequip() , Equipment.LevelUpV1() , Equipment.LevelUp(IRandom, EnhancementCostSheetV2.Row, bool) , Equipment.SetLevel(IRandom, int, EnhancementCostSheetV3) , Equipment.GetRealExp(EquipmentItemSheet, EnhancementCostSheetV3) , Equipment.Equals(Equipment) , Equipment.Equals(object) , Equipment.GetHashCode() , ItemUsable.ITEM_USABLE_FIELD_COUNT , ItemUsable.ItemId , ItemUsable.NonFungibleId , ItemUsable.StatsMap , ItemUsable.Skills , ItemUsable.BuffSkills , ItemUsable.RequiredBlockIndex , ItemUsable.Equals(ItemUsable) , ItemUsable.GetOptionCount() , ItemUsable.Update(long) , ItemBase.SerializationVersion , ItemBase.BaseFieldCount , ItemBase.Codec , ItemBase.Id , ItemBase.Grade , ItemBase.ItemType , ItemBase.ItemSubType , ItemBase.ElementalType , ItemBase.GetObjectData(SerializationInfo, StreamingContext) , ItemBase.Equals(ItemBase) , ItemBase.ToString() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Extension Methods
EquipmentExtensions.GetRequirementLevel(Equipment, ItemRequirementSheet, EquipmentItemRecipeSheet, EquipmentItemSubRecipeSheetV2, EquipmentItemOptionSheet) ,
EquipmentExtensions.IsMadeWithMimisbrunnrRecipe(Equipment, EquipmentItemRecipeSheet, EquipmentItemSubRecipeSheetV2, EquipmentItemOptionSheet)

Constructors

Aura(IValue)

Constructor for deserialization that supports both Dictionary and List formats.

```
public Aura(IValue serialized)
```

Parameters

`serialized` IValue

Serialized data in either Dictionary or List format

Aura(Row, Guid, long, bool)

```
public Aura(EquipmentItemSheet.Row data, Guid id, long requiredBlockIndex, bool
madeWithMimisbrunnrRecipe = false)
```

Parameters

`data` EquipmentItemSheet.Row

`id` Guid

`requiredBlockIndex` long

`madeWithMimisbrunnrRecipe` bool

Aura(SerializationInfo, StreamingContext)

```
protected Aura(SerializationInfo info, StreamingContext _)
```

Parameters

`info` SerializationInfo

`_` StreamingContext

Class Belt

Namespace: Nekoyume.Model.Item

Assembly: Lib9c.dll

Represents belt equipment items. Supports both Dictionary and List serialization formats for backward compatibility.

```
[Serializable]
public class Belt : Equipment, INonFungibleItem, IEquippableItem, ITradableItem,
IItem, IState
```

Inheritance

object ← ItemBase ← ItemUsable ← Equipment ← Belt

Implements

INonFungibleItem, IEquippableItem, ITradableItem, IItem, IState

Inherited Members

Equipment.equipped , Equipment.level , Equipment.Exp , Equipment.optionCountFromCombination , Equipment.IconId , Equipment.ByCustomCraft , Equipment.CraftWithRandom , Equipment.HasRandomOnlyIcon , Equipment.Stat , Equipment.SetId , Equipment.SpineResourcePath , Equipment.MadeWithMimisbrunnrRecipe , Equipment.UniqueStatType , Equipment.Equipped , Equipment.GetIncrementAmountOfEnhancement() , Equipment.Serialize() , Equipment.GetFieldNames() , Equipment.Equip() , Equipment.Unequip() , Equipment.LevelUpV1() , Equipment.LevelUp(IRandom, EnhancementCostSheetV2.Row, bool) , Equipment.SetLevel(IRandom, int, EnhancementCostSheetV3) , Equipment.GetRealExp(EquipmentItemSheet, EnhancementCostSheetV3) , Equipment.Equals(Equipment) , Equipment.Equals(object) , Equipment.GetHashCode() , ItemUsable.ITEM_USABLE_FIELD_COUNT , ItemUsable.ItemId , ItemUsable.NonFungibleId , ItemUsable.StatsMap , ItemUsable.Skills , ItemUsable.BuffSkills , ItemUsable.RequiredBlockIndex , ItemUsable.Equals(ItemUsable) , ItemUsable.GetOptionCount() , ItemUsable.Update(long) , ItemBase.SerializationVersion , ItemBase.BaseFieldCount , ItemBase.Codec , ItemBase.Id , ItemBase.Grade , ItemBase.ItemType , ItemBase.ItemSubType , ItemBase.ElementalType , ItemBase.GetObjectData(SerializationInfo, StreamingContext) , ItemBase.Equals(ItemBase) , ItemBase.ToString() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Extension Methods
EquipmentExtensions.GetRequirementLevel(Equipment, ItemRequirementSheet, EquipmentItemRecipeSheet, EquipmentItemSubRecipeSheetV2, EquipmentItemOptionSheet) ,
EquipmentExtensions.IsMadeWithMimisbrunnrRecipe(Equipment, EquipmentItemRecipeSheet, EquipmentItemSubRecipeSheetV2, EquipmentItemOptionSheet)

Constructors

Belt(IValue)

Constructor for deserialization that supports both Dictionary and List formats.

```
public Belt(IValue serialized)
```

Parameters

`serialized` IValue

Serialized data in either Dictionary or List format

Belt(Row, Guid, long, bool)

```
public Belt(EquipmentItemSheet.Row data, Guid id, long requiredBlockIndex, bool
madeWithMimisbrunnrRecipe = false)
```

Parameters

`data` EquipmentItemSheet.Row

`id` Guid

`requiredBlockIndex` long

`madeWithMimisbrunnrRecipe` bool

Belt(SerializationInfo, StreamingContext)

```
protected Belt(SerializationInfo info, StreamingContext _)
```

Parameters

`info` SerializationInfo

`_` StreamingContext

Properties

TradableId

```
public Guid TradableId { get; }
```

Property Value

Guid

Class Consumable

Namespace: Nekoyume.Model.Item

Assembly: Lib9c.dll

Represents consumable items that can be used by characters. Supports both Dictionary and List serialization formats for backward compatibility.

```
[Serializable]
public class Consumable : ItemUsable, INonFungibleItem, ITradableItem, IItem, IState
```

Inheritance

object ← ItemBase ← ItemUsable ← Consumable

Implements

INonFungibleItem, ITradableItem, IItem, IState

Inherited Members

ItemUsable.ITEM_USABLE_FIELD_COUNT , ItemUsable.ItemId , ItemUsable.NonFungibleId , ItemUsable.StatsMap , ItemUsable.Skills , ItemUsable.BuffSkills , ItemUsable.RequiredBlockIndex , ItemUsable.Equals(ItemUsable) , ItemUsable.Equals(object) , ItemUsable.GetHashCode() , ItemUsable.GetOptionCount() , ItemUsable.Update(long) , ItemUsable.GetFieldNames() , ItemBase.SerializationVersion , ItemBase.BaseFieldCount , ItemBase.Codec , ItemBase.Id , ItemBase.Grade , ItemBase.ItemType , ItemBase.ItemSubType , ItemBase.ElementalType , ItemBase.GetObjectData(SerializationInfo, StreamingContext) , ItemBase.Equals(ItemBase) , ItemBase.ToString() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Constructors

Consumable(IValue)

Constructor for deserialization that supports both Dictionary and List formats.

```
public Consumable(IValue serialized)
```

Parameters

`serialized` IValue

Serialized data in either Dictionary or List format

Consumable(Row, Guid, long)

```
public Consumable(ConsumableItemSheet.Row data, Guid id, long requiredBlockIndex)
```

Parameters

`data` ConsumableItemSheet.Row

`id` Guid

`requiredBlockIndex` long

Consumable(SerializationInfo, StreamingContext)

```
protected Consumable(SerializationInfo info, StreamingContext _)
```

Parameters

`info` SerializationInfo

`_` StreamingContext

Properties

MainStat

```
public StatType MainStat { get; }
```

Property Value

StatType

Stats

```
public List<DecimalStat> Stats { get; set; }
```

Property Value

List<DecimalStat>

TradableId

```
public Guid TradableId { get; }
```

Property Value

Guid

Methods

Serialize()

Serializes the consumable to List format (new format). Order: [baseData..., stats]

```
public override IValue Serialize()
```

Returns

IValue

List containing serialized data

Class Costume

Namespace: Nekoyume.Model.Item

Assembly: Lib9c.dll

Represents costume items that can be equipped by characters. Supports both Dictionary and List serialization formats for backward compatibility.

```
[Serializable]
public class Costume : ItemBase, INonFungibleItem, IEquippableItem, ITradableItem,
IItem, IState
```

Inheritance

object ← ItemBase ← Costume

Implements

INonFungibleItem, IEquippableItem, ITradableItem, IItem, IState

Inherited Members

ItemBase.SerializationVersion , ItemBase.BaseFieldCount , ItemBase.Codec , ItemBase.Id , ItemBase.Grade , ItemBase.ItemType , ItemBase.ItemSubType , ItemBase.ElementalType , ItemBase.GetObjectData(SerializationInfo, StreamingContext) , ItemBase.Equals(ItemBase) , ItemBase.ToString() , ItemBase.GetFieldNames() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Constructors

Costume(IValue)

Constructor for deserialization that supports both Dictionary and List formats.

```
public Costume(IValue serialized)
```

Parameters

`serialized` IValue

Serialized data in either Dictionary or List format

Costume(Row, Guid)

```
public Costume(CostumeItemSheet.Row data, Guid itemId)
```

Parameters

`data` CostumeItemSheet.Row

`itemId` Guid

Costume(SerializationInfo, StreamingContext)

```
protected Costume(SerializationInfo info, StreamingContext _)
```

Parameters

`info` SerializationInfo

`_` StreamingContext

Fields

equipped

```
public bool equipped
```

Field Value

bool

Properties

Equipped

```
public bool Equipped { get; }
```

Property Value

bool

ItemId

```
public Guid ItemId { get; }
```

Property Value

Guid

NonFungibleId

```
public Guid NonFungibleId { get; }
```

Property Value

Guid

RequiredBlockIndex

```
public long RequiredBlockIndex { get; set; }
```

Property Value

long

SpineResourcePath

```
public string SpineResourcePath { get; set; }
```

Property Value

string

TradableId

```
public Guid TradableId { get; }
```

Property Value

Guid

Methods

Equals(Costume)

```
protected bool Equals(Costume other)
```

Parameters

`other` Costume

Returns

bool

Equals(object)

Determines whether the specified object is equal to the current object.

```
public override bool Equals(object obj)
```

Parameters

`obj` object

The object to compare with the current object.

Returns

bool

true if the specified object is equal to the current object; otherwise, false.

Equip()

```
public void Equip()
```

GetHashCode()

Serves as the default hash function.

```
public override int GetHashCode()
```

Returns

int

A hash code for the current object.

Serialize()

Serializes the costume to List format (new format). Order: [baseData..., equipped, spineResourcePath, itemId, requiredBlockIndex]

```
public override IValue Serialize()
```

Returns

IValue

List containing serialized data

Unequip()

```
public void Unequip()
```

Update(long)

```
public void Update(long blockIndex)
```

Parameters

`blockIndex` long

Class Equipment

Namespace: Nekoyume.Model.Item

Assembly: Lib9c.dll

Represents equipment items that can be equipped by characters. Supports both Dictionary and List serialization formats for backward compatibility.

Field Order (List Format): Base fields (0~10): 11 fields from ItemUsable Equipment fields (11~22): equipped, level, stat, setId, spineResourcePath, iconId, byCustomCraft, craftWithRandom, hasRandomOnlyIcon, optionCountFromCombination, madeWithMimisbrunnrRecipe, exp

Equipment Properties: - Equipped: Whether the equipment is currently equipped - Level: Equipment enhancement level - Stat: Primary stat of the equipment - SetId: ID for equipment set bonuses - SpineResourcePath: Path to spine animation resource - IconId: Icon identifier for UI display - ByCustomCraft: Whether crafted through custom crafting - CraftWithRandom: Whether random options were applied during crafting - HasRandomOnlyIcon: Whether has random-only icon - OptionCountFromCombination: Number of options from combination - MadeWithMimisbrunnrRecipe: Whether made with Mimisbrunnr recipe - Exp: Experience points for leveling

```
[Serializable]
public class Equipment : ItemUsable, INonFungibleItem, IEquippableItem,
IItem, IState
```

Inheritance

object ← ItemBase ← ItemUsable ← Equipment

Implements

INonFungibleItem, IEquippableItem, IItem, IState

Derived

Armor, Aura, Belt, Grimoire, Necklace, Ring, Weapon

Inherited Members

ItemUsable.ITEM_USABLE_FIELD_COUNT , ItemUsable.ItemId , ItemUsable.NonFungibleId , ItemUsable.StatsMap , ItemUsable.Skills , ItemUsable.BuffSkills , ItemUsable.RequiredBlockIndex , ItemUsable.Equals(ItemUsable) , ItemUsable.GetOptionCount() , ItemUsable.Update(long) , ItemBase.SerializationVersion , ItemBase.BaseFieldCount , ItemBase.Codec , ItemBase.Id , ItemBase.Grade ,

ItemBase.ItemType , ItemBase.ItemSubType , ItemBase.ElementalType , ItemBase.GetObjectData(SerializationInfo, StreamingContext) , ItemBase.Equals(ItemBase) , ItemBase.ToString() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Extension Methods
EquipmentExtensions.GetRequirementLevel(Equipment, ItemRequirementSheet, EquipmentItemRecipeSheet, EquipmentItemSubRecipeSheetV2, EquipmentItemOptionSheet) , EquipmentExtensions.IsMadeWithMimisbrunnrRecipe(Equipment, EquipmentItemRecipeSheet, EquipmentItemSubRecipeSheetV2, EquipmentItemOptionSheet)

Remarks

Equipment items can be enhanced, equipped, and provide various stats and bonuses. The equipment system supports both regular crafting and custom crafting with random options.

Example usage:

```
// Create equipment
var equipment = new Equipment(equipmentRow, Guid.NewGuid(), 1000L);
// Enhance equipment
equipment.LevelUp();
// Equip/unequip
equipment.Equipped = true;
// Check stats
var statValue = equipment.Stat.StatType;
var setBonus = equipment.SetId;
```

Constructors

Equipment(IValue)

Constructor for deserialization that supports both Dictionary and List formats.

```
public Equipment(IValue serialized)
```

Parameters

`serialized` IValue

Serialized data in either Dictionary or List format

Equipment(Row, Guid, long, bool, int)

```
public Equipment(EquipmentItemSheet.Row data, Guid id, long requiredBlockIndex, bool
madeWithMimisbrunnrRecipe = false, int iconId = 0)
```

Parameters

`data` EquipmentItemSheet.Row

`id` Guid

`requiredBlockIndex` long

`madeWithMimisbrunnrRecipe` bool

`iconId` int

Equipment(SerializationInfo, StreamingContext)

```
protected Equipment(SerializationInfo info, StreamingContext _)
```

Parameters

`info` SerializationInfo

`_` StreamingContext

Fields

ByCustomCraft

```
public bool ByCustomCraft
```

Field Value

bool

CraftWithRandom

```
public bool CraftWithRandom
```

Field Value

bool

Exp

```
public long Exp
```

Field Value

long

HasRandomOnlyIcon

```
public bool HasRandomOnlyIcon
```

Field Value

bool

IconId

```
public int IconId
```

Field Value

int

equipped

```
public bool equipped
```

Field Value

bool

level

```
public int level
```

Field Value

int

optionCountFromCombination

```
public int optionCountFromCombination
```

Field Value

int

Properties

Equipped

```
public bool Equipped { get; }
```

Property Value

bool

MadeWithMimisbrunnrRecipe

```
public bool MadeWithMimisbrunnrRecipe { get; set; }
```

Property Value

bool

SetId

```
public int SetId { get; }
```

Property Value

int

SpineResourcePath

```
public string SpineResourcePath { get; }
```

Property Value

string

Stat

```
public DecimalStat Stat { get; }
```

Property Value

DecimalStat

UniqueStatType

```
public StatType UniqueStatType { get; }
```

Property Value

StatType

Methods

Equals(Equipment)

```
protected bool Equals(Equipment other)
```

Parameters

`other` Equipment

Returns

bool

Equals(object)

Determines whether the specified object is equal to the current object.

```
public override bool Equals(object obj)
```

Parameters

`obj` object

The object to compare with the current object.

Returns

bool

true if the specified object is equal to the current object; otherwise, false.

Equip()

```
public void Equip()
```

GetFieldNames()

Gets the field names for serialization in order.

```
protected override string[] GetFieldNames()
```

Returns

string[]

Array of field names

GetHashCode()

Serves as the default hash function.

```
public override int GetHashCode()
```

Returns

int

A hash code for the current object.

GetIncrementAmountOfEnhancement()

```
public decimal GetIncrementAmountOfEnhancement()
```

Returns

decimal

GetRealExp(EquipmentItemSheet, EnhancementCostSheetV3)

```
public long GetRealExp(EquipmentItemSheet itemSheet, EnhancementCostSheetV3
costSheet)
```

Parameters

`itemSheet` EquipmentItemSheet

`costSheet` EnhancementCostSheetV3

Returns

long

LevelUp(IRandom, Row, bool)

```
[Obsolete("Since ItemEnhancement12, Use `SetLevel` instead.")]
public void LevelUp(IRandom random, EnhancementCostSheetV2.Row row,
bool isGreatSuccess)
```

Parameters

`random` IRandom

`row` EnhancementCostSheetV2.Row

`isGreatSuccess` bool

LevelUpV1()

```
[Obsolete("Use LevelUp")]
public void LevelUpV1()
```

Serialize()

Serializes the equipment to List format (new format). Order: [baseData..., equipped, level, stat, setId, spineResourcePath, iconId, byCustomCraft, craftWithRandom, hasRandomOnlyIcon, optionCountFromCombination, madeWithMimisbrunnrRecipe, exp]

```
public override IValue Serialize()
```

Returns

IValue

List containing serialized data

SetLevel(IRandom, int, EnhancementCostSheetV3)

```
public void SetLevel(IRandom random, int targetLevel, EnhancementCostSheetV3 sheet)
```

Parameters

`random` IRandom

`targetLevel` int

`sheet` EnhancementCostSheetV3

Unequip()

```
public void Unequip()
```

Class Grimoire

Namespace: Nekoyume.Model.Item

Assembly: Lib9c.dll

Represents grimoire equipment items. Supports both Dictionary and List serialization formats for backward compatibility.

```
[Serializable]
public class Grimoire : Equipment, INonFungibleItem, IEquippableItem, IItem, IState
```

Inheritance

object ← ItemBase ← ItemUsable ← Equipment ← Grimoire

Implements

INonFungibleItem, IEquippableItem, IItem, IState

Inherited Members

Equipment.equipped , Equipment.level , Equipment.Exp , Equipment.optionCountFromCombination , Equipment.IconId , Equipment.ByCustomCraft , Equipment.CraftWithRandom , Equipment.HasRandomOnlyIcon , Equipment.Stat , Equipment.SetId , Equipment.SpineResourcePath , Equipment.MadeWithMimisbrunnrRecipe , Equipment.UniqueStatType , Equipment.Equipped , Equipment.GetIncrementAmountOfEnhancement() , Equipment.Serialize() , Equipment.GetFieldNames() , Equipment.Equip() , Equipment.Unequip() , Equipment.LevelUpV1() , Equipment.LevelUp(IRandom, EnhancementCostSheetV2.Row, bool) , Equipment.SetLevel(IRandom, int, EnhancementCostSheetV3) , Equipment.GetRealExp(EquipmentItemSheet, EnhancementCostSheetV3) , Equipment.Equals(Equipment) , Equipment.Equals(object) , Equipment.GetHashCode() , ItemUsable.ITEM_USABLE_FIELD_COUNT , ItemUsable.ItemId , ItemUsable.NonFungibleId , ItemUsable.StatsMap , ItemUsable.Skills , ItemUsable.BuffSkills , ItemUsable.RequiredBlockIndex , ItemUsable.Equals(ItemUsable) , ItemUsable.GetOptionCount() , ItemUsable.Update(long) , ItemBase.SerializationVersion , ItemBase.BaseFieldCount , ItemBase.Codec , ItemBase.Id , ItemBase.Grade , ItemBase.ItemType , ItemBase.ItemSubType , ItemBase.ElementalType , ItemBase.GetObjectData(SerializationInfo, StreamingContext) , ItemBase.Equals(ItemBase) , ItemBase.ToString() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Extension Methods
EquipmentExtensions.GetRequirementLevel(Equipment, ItemRequirementSheet, EquipmentItemRecipeSheet, EquipmentItemSubRecipeSheetV2, EquipmentItemOptionSheet) ,
EquipmentExtensions.IsMadeWithMimisbrunnrRecipe(Equipment, EquipmentItemRecipeSheet, EquipmentItemSubRecipeSheetV2, EquipmentItemOptionSheet)

Constructors

Grimoire(IValue)

Constructor for deserialization that supports both Dictionary and List formats.

```
public Grimoire(IValue serialized)
```

Parameters

`serialized` IValue

Serialized data in either Dictionary or List format

Grimoire(Row, Guid, long, bool)

```
public Grimoire(EquipmentItemSheet.Row data, Guid id, long requiredBlockIndex, bool
madeWithMimisbrunnrRecipe = false)
```

Parameters

`data` EquipmentItemSheet.Row

`id` Guid

`requiredBlockIndex` long

`madeWithMimisbrunnrRecipe` bool

Grimoire(SerializationInfo, StreamingContext)

```
protected Grimoire(SerializationInfo info, StreamingContext _)
```

Parameters

`info` SerializationInfo

`_` StreamingContext

Interface IEquippableItem

Namespace: Nekoyume.Model.Item

Assembly: Lib9c.dll

```
public interface IEquippableItem : IItem, IState
```

Inherited Members
IItem.ItemType , IItem.ItemSubType , IState.Serialize()

Properties

Equipped

```
bool Equipped { get; }
```

Property Value

bool

Methods

Equip()

```
void Equip()
```

Unequip()

```
void Unequip()
```

Interface IFungibleItem

Namespace: Nekoyume.Model.Item

Assembly: Lib9c.dll

```
public interface IFungibleItem : IItem, IState
```

Inherited Members
IItem.ItemType , IItem.ItemSubType , IState.Serialize()

Properties

FungibleId

```
HashDigest<SHA256> FungibleId { get; }
```

Property Value

HashDigest<SHA256>

Interface IItem

Namespace: Nekoyume.Model.Item

Assembly: Lib9c.dll

```
public interface IItem : IState
```

Inherited Members
IState.Serialize()

Properties

ItemSubType

```
ItemSubType ItemSubType { get; }
```

Property Value

ItemSubType

ItemType

```
ItemType ItemType { get; }
```

Property Value

ItemType

Interface ILock

Namespace: Nekoyume.Model.Item

Assembly: Lib9c.dll

```
public interface ILock
```

Properties

Type

```
LockType Type { get; }
```

Property Value

LockType

Methods

Serialize()

```
IValue Serialize()
```

Returns

IValue

Interface INonFungibleItem

Namespace: Nekoyume.Model.Item

Assembly: Lib9c.dll

```
public interface INonFungibleItem : IItem, IState
```

Inherited Members
IItem.ItemType , IItem.ItemSubType , IState.Serialize()

Properties

NonFungibleId

```
Guid NonFungibleId { get; }
```

Property Value

Guid

RequiredBlockIndex

```
long RequiredBlockIndex { get; set; }
```

Property Value

long

Interface ITradableFungibleItem

Namespace: Nekoyume.Model.Item

Assembly: Lib9c.dll

```
public interface ITradableFungibleItem : ITradableItem, IFungibleItem, IItem,
IState, ICloneable
```

Inherited Members

ITradableItem.TradableId , ITradableItem.RequiredBlockIndex , IFungibleItem.FungibleId , IItem.ItemType , IItem.ItemSubType , IState.Serialize() , ICloneable.Clone()

Interface ITradableItem

Namespace: Nekoyume.Model.Item

Assembly: Lib9c.dll

```
public interface ITradableItem : IItem, IState
```

Inherited Members
IItem.ItemType , IItem.ItemSubType , IState.Serialize()

Properties

RequiredBlockIndex

```
long RequiredBlockIndex { get; set; }
```

Property Value

long

TradableId

```
Guid TradableId { get; }
```

Property Value

Guid

Class Inventory

Namespace: Nekoyume.Model.Item

Assembly: Lib9c.dll

```
[Serializable]
public class Inventory : IState, ICloneable
```

Inheritance

object ← Inventory

Implements

IState, ICloneable

Inherited Members

object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Extension Methods

InventoryExtensions.GetEquippedFullCostumeOrArmorId(Inventory) , InventoryExtensions.MintItem(Inventory, ItemSheet.Row, int, bool, IRandom) , InventoryExtensions.UseActionPoint(Inventory, long, int, bool, MaterialItemSheet, long)

Constructors

Inventory()

```
public Inventory()
```

Inventory(List)

```
public Inventory(List serialized)
```

Parameters

`serialized` List

Properties

Consumables

```
public IEnumerable<Consumable> Consumables { get; }
```

Property Value

IEnumerable<Consumable>

Costumes

```
public IEnumerable<Costume> Costumes { get; }
```

Property Value

IEnumerable<Costume>

Equipments

```
public IEnumerable<Equipment> Equipments { get; }
```

Property Value

IEnumerable<Equipment>

Items

```
public IReadOnlyList<Inventory.Item> Items { get; }
```

Property Value

IReadOnlyList<Inventory.Item>

Materials

```
public IEnumerable<Material> Materials { get; }
```

Property Value

IEnumerable<Material>

Methods

AddFungibleItem(ItemBase, int, ILock)

```
public Inventory.Item AddFungibleItem(ItemBase itemBase, int count = 1, ILock iLock
= null)
```

Parameters

`itemBase` ItemBase

`count` int

`iLock` ILock

Returns

Inventory.Item

AddItem(ItemBase, int, ILock)

```
public KeyValuePair<int, int> AddItem(ItemBase itemBase, int count = 1, ILock iLock
= null)
```

Parameters

`itemBase` ItemBase

`count` int

`iLock` ILock

Returns

KeyValuePair<int, int>

AddNonFungibleItem(ItemBase, ILock)

```
public Inventory.Item AddNonFungibleItem(ItemBase itemBase, ILock iLock = null)
```

Parameters

`itemBase` ItemBase

`iLock` ILock

Returns

Inventory.Item

AddTradableFungibleItem(ITradableFungibleItem, int)

Add tradable fungible item. If there is same item which is not locked, add count to it. The same item is item which has same item sheet id, fungible id, tradable id and same required block index. If there is no same item, add new item.

```
public Inventory.Item AddTradableFungibleItem(ITradableFungibleItem
tradableFungibleItem, int count)
```

Parameters

`tradableFungibleItem` ITradableFungibleItem

`count` int

Returns

Inventory.Item

Exceptions

ArgumentNullException

ArgumentOutOfRangeException

Clone()

Creates a new object that is a copy of the current instance.

```
public object Clone()
```

Returns

object

A new object that is a copy of this instance.

Equals(Inventory)

```
protected bool Equals(Inventory other)
```

Parameters

`other` Inventory

Returns

bool

Equals(object)

Determines whether the specified object is equal to the current object.

```
public override bool Equals(object obj)
```

Parameters

`obj` object

The object to compare with the current object.

Returns

bool

true if the specified object is equal to the current object; otherwise, false.

FilterConsumables(int, long)

```
public List<Consumable> FilterConsumables(int id, long blockIndex)
```

Parameters

`id` int

`blockIndex` long

Returns

List<Consumable>

GetHashCode()

Serves as the default hash function.

```
public override int GetHashCode()
```

Returns

int

A hash code for the current object.

HasFungibleItem(HashDigest<SHA256>, long, int)

```
public bool HasFungibleItem(HashDigest<SHA256> fungibleId, long blockIndex, int
count = 1)
```

Parameters

`fungibleId` HashDigest<SHA256>

`blockIndex` long

`count` int

Returns

bool

HasItem(int, int)

```
public bool HasItem(int rowId, int count = 1)
```

Parameters

`rowId` int

`count` int

Returns

bool

HasNonFungibleItem(Guid)

```
public bool HasNonFungibleItem(Guid nonFungibleId)
```

Parameters

`nonFungibleId` Guid

Returns

bool

HasTradableFungibleItem(HashDigest<SHA256>, long?, long?, int)

```
public bool HasTradableFungibleItem(HashDigest<SHA256> fungibleId, long?
requiredBlockIndex, long? blockIndex, int count)
```

Parameters

`fungibleId` HashDigest<SHA256>

`requiredBlockIndex` long?

`blockIndex` long?

`count` int

Returns

bool

HasTradableItem(Guid, long, int)

```
public bool HasTradableItem(Guid tradableId, long blockIndex, int count)
```

Parameters

`tradableId` Guid

`blockIndex` long

`count` int

Returns

bool

IsMaterialRemovable(Item, int)

Checks if the given item is non-tradable material and can be removed from the inventory.

```
public static bool IsMaterialRemovable(Inventory.Item item, int id)
```

Parameters

`item` Inventory.Item

The item to check for removal.

`id` int

The ID of the material to remove.

Returns

bool

Returns a boolean indicating whether the item is removable.

IsMaterialRemovable(Item, int, long)

Checks if the given item is material and can be removed from the inventory.

```
public static bool IsMaterialRemovable(Inventory.Item item, int id, long blockIndex)
```

Parameters

`item` Inventory.Item

The item to check for removal.

`id` int

The ID of the material to remove.

`blockIndex` long

The block index.

Returns

bool

Returns a boolean indicating whether the item is removable.

LockByReferringToDigestList(OrderDigestListState, Guid, long)

```
public void LockByReferringToDigestList(OrderDigestListState digestList, Guid
tradableId, long blockIndex)
```

Parameters

`digestList` OrderDigestListState

`tradableId` Guid

`blockIndex` long

ReconfigureFungibleItem(OrderDigestListState, Guid)

```
public void ReconfigureFungibleItem(OrderDigestListState digestList,
Guid tradableId)
```

Parameters

`digestList` OrderDigestListState

`tradableId` Guid

RemoveConsumable(int, long, int)

```
public bool RemoveConsumable(int id, long blockIndex, int count = 1)
```

Parameters

`id` int

`blockIndex` long

`count` int

Returns

bool

RemoveFungibleItem(HashDigest<SHA256>, long, int, bool)

```
public bool RemoveFungibleItem(HashDigest<SHA256> fungibleId, long blockIndex, int
count = 1, bool onlyTradableItem = false)
```

Parameters

`fungibleId` HashDigest<SHA256>

`blockIndex` long

`count` int

`onlyTradableItem` bool

Returns

bool

RemoveFungibleItem(IFungibleItem, long, int, bool)

```
public bool RemoveFungibleItem(IFungibleItem fungibleItem, long blockIndex, int
count = 1, bool onlyTradableItem = false)
```

Parameters

`fungibleItem` IFungibleItem

`blockIndex` long

`count` int

`onlyTradableItem` bool

Returns

bool

RemoveFungibleItem2(HashDigest<SHA256>, int, bool)

```
[Obsolete("Use RemoveFungibleItem")]
public bool RemoveFungibleItem2(HashDigest<SHA256> fungibleId, int count = 1, bool
onlyTradableItem = false)
```

Parameters

`fungibleId` HashDigest<SHA256>

`count` int

`onlyTradableItem` bool

Returns

bool

RemoveFungibleItem2(IFungibleItem, int, bool)

```
[Obsolete("Use RemoveFungibleItem")]
public bool RemoveFungibleItem2(IFungibleItem fungibleItem, int count = 1, bool
```

```
    onlyTradableItem = false)
```

Parameters

`fungibleItem` IFungibleItem

`count` int

`onlyTradableItem` bool

Returns

bool

RemoveItem(Item)

```
public void RemoveItem(Inventory.Item item)
```

Parameters

`item` Inventory.Item

RemoveMaterial(int, long, int)

Remove a material from the inventory. contains tradable materials.

```
public bool RemoveMaterial(int id, long blockIndex, int count = 1)
```

Parameters

`id` int

The ID of the material item to remove.

`blockIndex` long

The block index.

`count` int

The number of materials to remove. Default value is 1.

Returns

bool

True if the material is successfully removed, false otherwise.

RemoveNonFungibleItem(INonFungibleItem)

```
public bool RemoveNonFungibleItem(INonFungibleItem nonFungibleItem)
```

Parameters

`nonFungibleItem` INonFungibleItem

Returns

bool

RemoveNonFungibleItem(Guid)

```
public bool RemoveNonFungibleItem(Guid nonFungibleId)
```

Parameters

`nonFungibleId` Guid

Returns

bool

RemoveNonFungibleItem2(Guid)

```
[Obsolete("Use RemoveNonFungibleItem(Guid nonFungibleId)")]
public bool RemoveNonFungibleItem2(Guid nonFungibleId)
```

Parameters

`nonFungibleId` Guid

Returns

bool

RemoveNonTradableMaterial(int, int)

Remove a non-tradable material from the inventory.

```
public bool RemoveNonTradableMaterial(int id, int count = 1)
```

Parameters

`id` int

The ID of the material item to remove.

`count` int

The number of materials to remove. Default value is 1.

Returns

bool

True if the material is successfully removed, false otherwise.

RemoveTradableFungibleItem(HashDigest<SHA256>, long?, long?, int)

Remove tradable item from inventory. Use the requiredBlockIndex when you want to remove the item that has same requiredBlockIndex. Use the blockIndex when you want to remove the item that has requiredBlockIndex less than or equal to blockIndex. Do not use both requiredBlockIndex and blockIndex at the same time.

```
public bool RemoveTradableFungibleItem(HashDigest<SHA256> fungibleId, long?
```

```
requiredBlockIndex, long? blockIndex, int count)
```

Parameters

`fungibleId` HashDigest<SHA256>

`requiredBlockIndex` long?

`blockIndex` long?

`count` int

Returns

bool

Exceptions

ArgumentNullException

RemoveTradableItem(Guid, long, int)

```
public bool RemoveTradableItem(Guid tradableId, long blockIndex, int count = 1)
```

Parameters

`tradableId` Guid

`blockIndex` long

`count` int

Returns

bool

RemoveTradableItemV1(ITradableItem, int)

```
[Obsolete("Use RemoveTradableItem()")]
public bool RemoveTradableItemV1(ITradableItem tradableItem, int count = 1)
```

Parameters

`tradableItem` ITradableItem

`count` int

Returns

bool

RemoveTradableItemV1(Guid, long, int)

```
[Obsolete("Use RemoveTradableItem()")]
public bool RemoveTradableItemV1(Guid tradableId, long blockIndex, int count = 1)
```

Parameters

`tradableId` Guid

`blockIndex` long

`count` int

Returns

bool

RemoveTradableMaterial(int, long, int)

Remove a tradable material from the inventory.

```
public bool RemoveTradableMaterial(int materialId, long blockIndex, int count = 1)
```

Parameters

`materialId` int

The ID of the material item to remove.

`blockIndex` long

The block index.

`count` int

The number of materials to remove. Default value is 1.

Returns

bool

True if the material is successfully removed, false otherwise.

SellItem(Guid, long, int)

```
public ITradableItem SellItem(Guid tradableId, long blockIndex, int count)
```

Parameters

`tradableId` Guid

`blockIndex` long

`count` int

Returns

ITradableItem

Serialize()

```
public IValue Serialize()
```

Returns

IValue

TryGetCostume(int, out Costume)

```
[Obsolete("Use public bool TryGetNonFungibleItem<T>(Guid itemId, out
T outNonFungibleItem)")]
public bool TryGetCostume(int rowId, out Costume outCostume)
```

Parameters

`rowId` int

`outCostume` Costume

Returns

bool

TryGetFungibleItems(HashDigest<SHA256>, out List<Item>)

```
public bool TryGetFungibleItems(HashDigest<SHA256> fungibleId, out
List<Inventory.Item> outItems)
```

Parameters

`fungibleId` HashDigest<SHA256>

`outItems` List<Inventory.Item>

Returns

bool

TryGetItem(int, out Item)

```
public bool TryGetItem(int rowId, out Inventory.Item outItem)
```

Parameters

`rowId` int

`outItem` Inventory.Item

Returns

bool

TryGetLockedItem(ILock, out Item)

```
public bool TryGetLockedItem(ILock iLock, out Inventory.Item outItem)
```

Parameters

`iLock` ILock

`outItem` Inventory.Item

Returns

bool

TryGetNonFungibleItem(Guid, out Item)

```
public bool TryGetNonFungibleItem(Guid nonFungibleId, out Inventory.Item outItem)
```

Parameters

`nonFungibleId` Guid

`outItem` Inventory.Item

Returns

bool

TryGetNonFungibleItem<T>(Guid, out T, long)

Tries to get a non-fungible item from the inventory with the specified item ID.

```
public bool TryGetNonFungibleItem<T>(Guid itemId, out T outNonFungibleItem, long
blockIndex = 9223372036854775807) where T : INonFungibleItem
```

Parameters

`itemId` Guid

The ID of the non-fungible item to retrieve.

`outNonFungibleItem` T

When this method returns, contains the non-fungible item with the specified ID, if found; otherwise, the default value for the type.

`blockIndex` long

The block index for the non-fungible item to be considered valid. Defaults to MaxValue.

Returns

bool

`true` if a non-fungible item with the specified ID is found in the inventory; otherwise, `false`.

Type Parameters

`T`

The type of non-fungible item to retrieve.

TryGetNonFungibleItem<T>(T, out T)

```
public bool TryGetNonFungibleItem<T>(T nonFungibleItem, out T outNonFungibleItem)
where T : INonFungibleItem
```

Parameters

`nonFungibleItem` T

`outNonFungibleItem` T

Returns

bool

Type Parameters

`T`

TryGetTradableFungibleItems(HashDigest<SHA256>, long?, long?, out IEnumerable<Item>)

Get ITradableFungibleItem from the inventory. This method check the item is not locked, and the item's required block index is less than or equal to the given block index. Do not use both requiredBlockIndex and blockIndex at the same time.

```
public bool TryGetTradableFungibleItems(HashDigest<SHA256> fungibleId, long?
requiredBlockIndex, long? blockIndex, out IEnumerable<Inventory.Item> outItems)
```

Parameters

`fungibleId` HashDigest<SHA256>

`requiredBlockIndex` long?

`blockIndex` long?

`outItems` IEnumerable<Inventory.Item>

Returns

bool

Exceptions

ArgumentNullException

TryGetTradableItem(Guid, long, int, out Item)

```
public bool TryGetTradableItem(Guid tradeId, long blockIndex, int count, out
Inventory.Item outItem)
```

Parameters

`tradeId` Guid

`blockIndex` long

`count` int

`outItem` Inventory.Item

Returns

bool

TryGetTradableItems(Guid, long, int, out List<Item>)

```
public bool TryGetTradableItems(Guid tradeId, long blockIndex, int count, out
List<Inventory.Item> outItem)
```

Parameters

`tradeId` Guid

`blockIndex` long

`count` int

`outItem` List<Inventory.Item>

Returns

bool

UnlockInvalidSlot(OrderDigestListState, Address, Address)

```
public void UnlockInvalidSlot(OrderDigestListState digestListState, Address
agentAddress, Address avatarAddress)
```

Parameters

`digestListState` OrderDigestListState

`agentAddress` Address

`avatarAddress` Address

UpdateTradableItem(Guid, long, int, long)

```
public ITradableItem UpdateTradableItem(Guid tradableId, long blockIndex, int count,
long requiredBlockIndex)
```

Parameters

`tradableId` Guid

`blockIndex` long

`count` int

`requiredBlockIndex` long

Returns

ITradableItem

Class Inventory.Item

Namespace: Nekoyume.Model.Item

Assembly: Lib9c.dll

```
[Serializable]
public class Inventory.Item : IState, IComparer<Inventory.Item>,
IComparable<Inventory.Item>
```

Inheritance

object ← Inventory.Item

Implements

IState, IComparer<Inventory.Item>, IComparable<Inventory.Item>

Inherited Members

object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

Item(Dictionary)

```
public Item(Dictionary serialized)
```

Parameters

`serialized` Dictionary

Item(ItemBase, int)

```
public Item(ItemBase itemBase, int count = 1)
```

Parameters

`itemBase` ItemBase

count int

Fields

Lock

```
public ILock Lock
```

Field Value

ILock

count

```
public int count
```

Field Value

int

item

```
public ItemBase item
```

Field Value

ItemBase

Properties

Locked

```
public bool Locked { get; }
```

Property Value

bool

Methods

Compare(Item, Item)

Compares two objects and returns a value indicating whether one is less than, equal to, or greater than the other.

```
public int Compare(Inventory.Item x, Inventory.Item y)
```

Parameters

x Inventory.Item

The first object to compare.

y Inventory.Item

The second object to compare.

Returns

int

A signed integer that indicates the relative values of x and y, as shown in the following table.

Value	Meaning
Less than zero	x is less than y.
Zero	x equals y.
Greater than zero	x is greater than y.

CompareTo(Item)

Compares the current instance with another object of the same type and returns an integer that indicates whether the current instance precedes, follows, or occurs in the same position in the sort order as the other object.

```
public int CompareTo(Inventory.Item other)
```

Parameters

`other` Inventory.Item

An object to compare with this instance.

Returns

int

A value that indicates the relative order of the objects being compared. The return value has these meanings:

Value	Meaning
Less than zero	This instance precedes `other` in the sort order.
Zero	This instance occurs in the same position in the sort order as `other`.
Greater than zero	This instance follows `other` in the sort order.

Equals(Item)

```
protected bool Equals(Inventory.Item other)
```

Parameters

`other` Inventory.Item

Returns

bool

Equals(object)

Determines whether the specified object is equal to the current object.

```
public override bool Equals(object obj)
```

Parameters

`obj` object

The object to compare with the current object.

Returns

bool

true if the specified object is equal to the current object; otherwise, false.

GetHashCode()

Serves as the default hash function.

```
public override int GetHashCode()
```

Returns

int

A hash code for the current object.

LockUp(ILock)

```
public void LockUp(ILock iLock)
```

Parameters

`iLock` ILock

Serialize()

```
public IValue Serialize()
```

Returns

IValue

Unlock()

```
public void Unlock()
```

Class ItemBase

Namespace: Nekoyume.Model.Item

Assembly: Lib9c.dll

Base class for all items in the game. Supports both Dictionary and List serialization formats for backward compatibility.

Serialization Format: - Dictionary (Legacy): Uses key-value pairs for backward compatibility - List (New): Uses ordered list for better performance and smaller size

Field Order (List Format): 1. version - Serialization version number 2. id - Item ID (Binary format) 3. itemType - Item type as string 4. itemSubType - Item sub-type as string 5. grade - Item grade 6. elementalType - Elemental type as string

```
[Serializable]
public abstract class ItemBase : IItem, IState
```

Inheritance

object ← ItemBase

Implements

IItem, IState

Derived

Costume, ItemUsable, Material

Inherited Members

object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Remarks

This class implements dual serialization support to ensure smooth migration from the legacy Dictionary format to the new List format. The List format provides better performance and smaller serialized data size.

Example usage:

```
// Create item
var material = new Material(materialRow);
// Serialize to List format (new)
```

```
var serialized = material.Serialize(); // Returns List
// Deserialize from any format
var deserialized = new Material(serialized); // Supports both Dictionary and List
```

Constructors

ItemBase(IValue)

Constructor for deserialization that supports both Dictionary and List formats.

```
protected ItemBase(IValue serialized)
```

Parameters

`serialized` IValue

Serialized data in either Dictionary or List format

Exceptions

ArgumentNullException

Thrown when serialized is null

ArgumentException

Thrown when serialized format is not supported or has insufficient fields

ItemBase(ItemBase)

```
protected ItemBase(ItemBase other)
```

Parameters

`other` ItemBase

ItemBase(Row)

```
protected ItemBase(ItemSheet.Row data)
```

Parameters

`data` ItemSheet.Row

ItemBase(SerializationInfo, StreamingContext)

```
protected ItemBase(SerializationInfo info, StreamingContext _)
```

Parameters

`info` SerializationInfo

`_` StreamingContext

Fields

BaseFieldCount

```
protected const int BaseFieldCount = 6
```

Field Value

int

Codec

```
protected static readonly Codec Codec
```

Field Value

Codec

SerializationVersion

```
public const int SerializationVersion = 2
```

Field Value

int

Properties

ElementalType

```
public ElementalType ElementalType { get; set; }
```

Property Value

ElementalType

Grade

```
public int Grade { get; }
```

Property Value

int

Id

```
public int Id { get; }
```

Property Value

int

ItemSubType

```
public ItemSubType ItemSubType { get; }
```

Property Value

ItemSubType

ItemType

```
public ItemType ItemType { get; }
```

Property Value

ItemType

Methods

Equals(ItemBase)

```
protected bool Equals(ItemBase other)
```

Parameters

`other` ItemBase

Returns

bool

Equals(object)

Determines whether the specified object is equal to the current object.

```
public override bool Equals(object obj)
```

Parameters

`obj` object

The object to compare with the current object.

Returns

bool

true if the specified object is equal to the current object; otherwise, false.

GetFieldNames()

Gets the field names for serialization in order.

```
protected virtual string[] GetFieldNames()
```

Returns

string[]

Array of field names

GetHashCode()

Serves as the default hash function.

```
public override int GetHashCode()
```

Returns

int

A hash code for the current object.

GetObjectData(SerializationInfo, StreamingContext)

```
public void GetObjectData(SerializationInfo info, StreamingContext context)
```

Parameters

`info` SerializationInfo

`context` StreamingContext

Serialize()

Serializes the item to List format (new format). Order: [version, id, itemType, itemSubType, grade, elementalType]

```
public virtual IValue Serialize()
```

Returns

IValue

List containing serialized data

ToString()

Returns a string that represents the current object.

```
public override string ToString()
```

Returns

string

A string that represents the current object.

Class ItemFactory

Namespace: Nekoyume.Model.Item

Assembly: Lib9c.dll

Factory class for creating and deserializing items. Supports both Dictionary and List serialization formats for backward compatibility.

```
public static class ItemFactory
```

Inheritance

object ← ItemFactory

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Methods

CreateCostume(Row, Guid)

Creates a costume item.

```
public static Costume CreateCostume(CostumeItemSheet.Row row, Guid itemId)
```

Parameters

`row` CostumeItemSheet.Row

The costume item sheet row data

`itemId` Guid

The unique ID for the costume

Returns

Costume

The created costume

CreateItem(Row, IRandom)

Creates an item from the given row data.

```
public static ItemBase CreateItem(ItemSheet.Row row, IRandom random)
```

Parameters

`row` ItemSheet.Row

The item sheet row data

`random` IRandom

Random number generator

Returns

ItemBase

The created item

CreateItemUsable(Row, Guid, long, int, bool)

Creates a usable item (consumable or equipment) from the given row data.

```
public static ItemUsable CreateItemUsable(ItemSheet.Row itemRow, Guid id, long
requiredBlockIndex, int level = 0, bool madeWithMimisbrunnrRecipe = false)
```

Parameters

`itemRow` ItemSheet.Row

The item sheet row data

`id` Guid

The unique ID for the item

`requiredBlockIndex` long

The required block index for using the item

`level` int

The level of the equipment (default: 0)

`madeWithMimisbrunnrRecipe` bool

Whether the item was made with Mimisbrunnr recipe (default: false)

Returns

ItemUsable

The created usable item

CreateItemUsableV2(Row, Guid, long, int, IRandom, Row, bool, bool)

```
public static ItemUsable CreateItemUsableV2(ItemSheet.Row itemRow, Guid id, long
requiredBlockIndex, int level, IRandom random, EnhancementCostSheetV2.Row row, bool
isGreatSuccess, bool madeWithMimisbrunnrRecipe = false)
```

Parameters

`itemRow` ItemSheet.Row

`id` Guid

`requiredBlockIndex` long

`level` int

`random` IRandom

`row` EnhancementCostSheetV2.Row

`isGreatSuccess` bool

`madeWithMimisbrunnrRecipe` bool

Returns

ItemUsable

CreateMaterial(MaterialItemSheet, int)

Creates a material item from the sheet by ID.

```
public static Material CreateMaterial(MaterialItemSheet sheet, int itemId)
```

Parameters

`sheet` MaterialItemSheet

The material item sheet

`itemId` int

The material item ID

Returns

Material

The created material, or null if not found

CreateMaterial(Row)

Creates a material item from the given row data.

```
public static Material CreateMaterial(MaterialItemSheet.Row row)
```

Parameters

`row` MaterialItemSheet.Row

The material item sheet row data

Returns

Material

The created material

CreateTradableMaterial(Row)

Creates a tradable material item from the given row data.

```
public static TradableMaterial CreateTradableMaterial(MaterialItemSheet.Row row)
```

Parameters

`row` MaterialItemSheet.Row

The material item sheet row data

Returns

TradableMaterial

The created tradable material

Deserialize(IValue)

Deserializes an item from serialized data. Supports both Dictionary and List formats for backward compatibility.

```
public static ItemBase Deserialize(IValue serialized)
```

Parameters

`serialized` IValue

The serialized item data

Returns

ItemBase

The deserialized item

Exceptions

ArgumentException

Thrown when the serialization format is not supported

SelectIconId(int, bool, Row, int, CustomEquipmentCraftIconSheet, IRandom)

```
public static (int, bool) SelectIconId(int iconId, bool isRandom,
EquipmentItemSheet.Row equipmentRow, int relationship, CustomEquipmentCraftIconSheet
iconSheet, IRandom random)
```

Parameters

`iconId` int

`isRandom` bool

`equipmentRow` EquipmentItemSheet.Row

`relationship` int

`iconSheet` CustomEquipmentCraftIconSheet

`random` IRandom

Returns

(int, bool)

SelectOption(ItemSubType, CustomEquipmentCraftOptionSheet, IRandom)

```
public static CustomEquipmentCraftOptionSheet.Row SelectOption(ItemSubType
itemSubType, CustomEquipmentCraftOptionSheet optionSheet, IRandom random)
```

Parameters

`itemSubType` ItemSubType

`optionSheet` CustomEquipmentCraftOptionSheet

`random` IRandom

Returns

CustomEquipmentCraftOptionSheet.Row

Enum ItemSubType

Namespace: Nekoyume.Model.Item

Assembly: Lib9c.dll

```
public enum ItemSubType
```

Extension Methods
StateExtensions.Serialize(Enum)

Fields

ApStone = 16

Armor = 7

Aura = 19

Belt = 8

[Obsolete("ItemSubType.Chest has never been used outside the MaterialItemSheet. And we won't use it in the future until we have a specific reason.")] Chest = 17

Circle = 22

EarCostume = 3

EquipmentMaterial = 11

EyeCostume = 4

Food = 0

FoodMaterial = 12

FullCostume = 1

Grimoire = 20

HairCostume = 2

Hourglass = 15

MonsterPart = 13

Necklace = 9

NormalMaterial = 14

Ring = 10

Scroll = 21

TailCostume = 5

Title = 18

Weapon = 6

Class ItemSubTypeComparer

Namespace: Nekoyume.Model.Item

Assembly: Lib9c.dll

```
public class ItemSubTypeComparer : IEqualityComparer<ItemSubType>
```

Inheritance

object ← ItemSubTypeComparer

Implements

IEqualityComparer<ItemSubType>

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Fields

Instance

```
public static readonly ItemSubTypeComparer Instance
```

Field Value

ItemSubTypeComparer

Methods

Equals(ItemSubType, ItemSubType)

Determines whether the specified objects are equal.

```
public bool Equals(ItemSubType x, ItemSubType y)
```

Parameters

`x` ItemSubType

The first object of type `T` to compare.

`y` ItemSubType

The second object of type `T` to compare.

Returns

bool

[true] if the specified objects are equal; otherwise, [false].

GetHashCode(ItemSubType)

Returns a hash code for the specified object.

```
public int GetHashCode(ItemSubType obj)
```

Parameters

`obj` ItemSubType

The [object] for which a hash code is to be returned.

Returns

int

A hash code for the specified object.

Exceptions

ArgumentNullException

The type of `obj` is a reference type and `obj` is [null].

Enum ItemType

Namespace: Nekoyume.Model.Item

Assembly: Lib9c.dll

```
public enum ItemType
```

Extension Methods
StateExtensions.Serialize(Enum)

Fields

`Consumable = 0`

`Costume = 1`

`Equipment = 2`

`Material = 3`

Class ItemTypeComparer

Namespace: Nekoyume.Model.Item

Assembly: Lib9c.dll

```
public class ItemTypeComparer : IEqualityComparer<ItemType>
```

Inheritance

object ← ItemTypeComparer

Implements

IEqualityComparer<ItemType>

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Fields

Instance

```
public static readonly ItemTypeComparer Instance
```

Field Value

ItemTypeComparer

Methods

Equals(ItemType, ItemType)

Determines whether the specified objects are equal.

```
public bool Equals(ItemType x, ItemType y)
```

Parameters

`x` ItemType

The first object of type `T` to compare.

`y` ItemType

The second object of type `T` to compare.

Returns

bool

[true] if the specified objects are equal; otherwise, [false].

GetHashCode(ItemType)

Returns a hash code for the specified object.

```
public int GetHashCode(ItemType obj)
```

Parameters

`obj` ItemType

The [object] for which a hash code is to be returned.

Returns

int

A hash code for the specified object.

Exceptions

ArgumentNullException

The type of `obj` is a reference type and `obj` is [null].

Class ItemUsable

Namespace: Nekoyume.Model.Item

Assembly: Lib9c.dll

```
[Serializable]
public abstract class ItemUsable : ItemBase, INonFungibleItem, IItem, IState
```

Inheritance

object ← ItemBase ← ItemUsable

Implements

INonFungibleItem, IItem, IState

Derived

Consumable, Equipment

Inherited Members

ItemBase.SerializationVersion , ItemBase.BaseFieldCount , ItemBase.Codec , ItemBase.Id , ItemBase.Grade , ItemBase.ItemType , ItemBase.ItemSubType , ItemBase.ElementalType , ItemBase.GetObjectData(SerializationInfo, StreamingContext) , ItemBase.Equals(ItemBase) , ItemBase.ToString() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Constructors

ItemUsable(IValue)

Constructor for deserialization that supports both Dictionary and List formats.

```
protected ItemUsable(IValue serialized)
```

Parameters

`serialized` IValue

Serialized data in either Dictionary or List format

ItemUsable(Row, Guid, long)

```
protected ItemUsable(ItemSheet.Row data, Guid id, long requiredBlockIndex)
```

Parameters

`data` ItemSheet.Row

`id` Guid

`requiredBlockIndex` long

ItemUsable(SerializationInfo, StreamingContext)

```
protected ItemUsable(SerializationInfo info, StreamingContext _)
```

Parameters

`info` SerializationInfo

`_` StreamingContext

Fields

ITEM_USABLE_FIELD_COUNT

```
protected const int ITEM_USABLE_FIELD_COUNT = 11
```

Field Value

int

Properties

BuffSkills

```
public List<BuffSkill> BuffSkills { get; }
```

Property Value

List<BuffSkill>

ItemId

```
public Guid ItemId { get; }
```

Property Value

Guid

NonFungibleId

```
public Guid NonFungibleId { get; }
```

Property Value

Guid

RequiredBlockIndex

```
public long RequiredBlockIndex { get; set; }
```

Property Value

long

Skills

```
public List<Skill> Skills { get; }
```

Property Value

List<Skill>

StatsMap

```
public StatsMap StatsMap { get; }
```

Property Value

StatsMap

Methods

Equals(ItemUsable)

```
protected bool Equals(ItemUsable other)
```

Parameters

`other` ItemUsable

Returns

bool

Equals(object)

Determines whether the specified object is equal to the current object.

```
public override bool Equals(object obj)
```

Parameters

`obj` object

The object to compare with the current object.

Returns

bool

true if the specified object is equal to the current object; otherwise, false.

GetFieldNames()

Gets the field names for serialization in order.

```
protected override string[] GetFieldNames()
```

Returns

string[]

Array of field names

GetHashCode()

Serves as the default hash function.

```
public override int GetHashCode()
```

Returns

int

A hash code for the current object.

GetOptionCount()

```
public int GetOptionCount()
```

Returns

int

Serialize()

Serializes the item to List format (new format). Order: [baseData..., itemId, statsMap, skills, buffSkills, requiredBlockIndex]

```
public override IValue Serialize()
```

Returns

IValue

List containing serialized data

Update(long)

```
public void Update(long blockIndex)
```

Parameters

`blockIndex` long

Enum LockType

Namespace: Nekoyume.Model.Item

Assembly: Lib9c.dll

```
public enum LockType
```

Extension Methods
StateExtensions.Serialize(Enum)

Fields

`Order = 0`

Class Material

Namespace: Nekoyume.Model.Item

Assembly: Lib9c.dll

Represents a material item that can be used for crafting and other purposes. Supports both Dictionary and List serialization formats for backward compatibility.

Field Order (List Format): Base fields (0~5): version, id, itemType, itemSubType, grade, elementalType Material fields (6): itemId

Material Properties: - ItemId: Unique identifier for the material item

```
[Serializable]
public class Material : ItemBase, ISerializable, IFungibleItem, IItem, IState
```

Inheritance

object ← ItemBase ← Material

Implements

ISerializable, IFungibleItem, IItem, IState

Derived

TradableMaterial

Inherited Members

ItemBase.SerializationVersion , ItemBase.BaseFieldCount , ItemBase.Codec , ItemBase.Id , ItemBase.Grade , ItemBase.ItemType , ItemBase.ItemSubType , ItemBase.ElementalType , ItemBase.GetObjectData(SerializationInfo, StreamingContext) , ItemBase.Equals(ItemBase) , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Remarks

Material items are fungible items used primarily for crafting other items. They support both Binary and Text formats for ItemId during deserialization to maintain compatibility with different serialization formats.

Example usage:

```
// Create material
var material = new Material(materialRow);
// Use in crafting
var requiredMaterials = new Dictionary<HashDigest<SHA256>, int>
{
{ material.ItemId, 5 } // 5 of this material
};
// Check material properties
var grade = material.Grade;
var itemType = material.ItemType;
```

Constructors

Material(IValue)

Constructor for deserialization that supports both Dictionary and List formats.

```
public Material(IValue serialized)
```

Parameters

`serialized` IValue

Serialized data in either Dictionary or List format

Material(Material)

```
public Material(Material other)
```

Parameters

`other` Material

Material(Row)

```
public Material(MaterialItemSheet.Row data)
```

Parameters

`data` MaterialItemSheet.Row

Material(SerializationInfo, StreamingContext)

```
protected Material(SerializationInfo info, StreamingContext _)
```

Parameters

`info` SerializationInfo

`_` StreamingContext

Properties

FungibleId

```
public HashDigest<SHA256> FungibleId { get; }
```

Property Value

HashDigest<SHA256>

ItemId

```
public HashDigest<SHA256> ItemId { get; }
```

Property Value

HashDigest<SHA256>

Methods

Equals(Material)

```
protected bool Equals(Material other)
```

Parameters

`other` Material

Returns

bool

Equals(object)

Determines whether the specified object is equal to the current object.

```
public override bool Equals(object obj)
```

Parameters

`obj` object

The object to compare with the current object.

Returns

bool

true if the specified object is equal to the current object; otherwise, false.

GetFieldNames()

Gets the field names for serialization in order.

```
protected override string[] GetFieldNames()
```

Returns

string[]

Array of field names

GetHashCode()

Serves as the default hash function.

```
public override int GetHashCode()
```

Returns

int

A hash code for the current object.

Serialize()

Serializes the material to List format (new format). Order: [baseData..., itemId]

```
public override IValue Serialize()
```

Returns

IValue

List containing serialized data

ToString()

Returns a string that represents the current object.

```
public override string ToString()
```

Returns

string

A string that represents the current object.

Class Necklace

Namespace: Nekoyume.Model.Item

Assembly: Lib9c.dll

Represents necklace equipment items. Supports both Dictionary and List serialization formats for backward compatibility.

```
[Serializable]
public class Necklace : Equipment, INonFungibleItem, IEquippableItem, ITradableItem,
IItem, IState
```

Inheritance

object ← ItemBase ← ItemUsable ← Equipment ← Necklace

Implements

INonFungibleItem, IEquippableItem, ITradableItem, IItem, IState

Inherited Members

Equipment.equipped , Equipment.level , Equipment.Exp , Equipment.optionCountFromCombination , Equipment.IconId , Equipment.ByCustomCraft , Equipment.CraftWithRandom , Equipment.HasRandomOnlyIcon , Equipment.Stat , Equipment.SetId , Equipment.SpineResourcePath , Equipment.MadeWithMimisbrunnrRecipe , Equipment.UniqueStatType , Equipment.Equipped , Equipment.GetIncrementAmountOfEnhancement() , Equipment.Serialize() , Equipment.GetFieldNames() , Equipment.Equip() , Equipment.Unequip() , Equipment.LevelUpV1() , Equipment.LevelUp(IRandom, EnhancementCostSheetV2.Row, bool) , Equipment.SetLevel(IRandom, int, EnhancementCostSheetV3) , Equipment.GetRealExp(EquipmentItemSheet, EnhancementCostSheetV3) , Equipment.Equals(Equipment) , Equipment.Equals(object) , Equipment.GetHashCode() , ItemUsable.ITEM_USABLE_FIELD_COUNT , ItemUsable.ItemId , ItemUsable.NonFungibleId , ItemUsable.StatsMap , ItemUsable.Skills , ItemUsable.BuffSkills , ItemUsable.RequiredBlockIndex , ItemUsable.Equals(ItemUsable) , ItemUsable.GetOptionCount() , ItemUsable.Update(long) , ItemBase.SerializationVersion , ItemBase.BaseFieldCount , ItemBase.Codec , ItemBase.Id , ItemBase.Grade , ItemBase.ItemType , ItemBase.ItemSubType , ItemBase.ElementalType , ItemBase.GetObjectData(SerializationInfo, StreamingContext) , ItemBase.Equals(ItemBase) , ItemBase.ToString() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Extension Methods
EquipmentExtensions.GetRequirementLevel(Equipment, ItemRequirementSheet, EquipmentItemRecipeSheet, EquipmentItemSubRecipeSheetV2, EquipmentItemOptionSheet) , EquipmentExtensions.IsMadeWithMimisbrunnrRecipe(Equipment, EquipmentItemRecipeSheet, EquipmentItemSubRecipeSheetV2, EquipmentItemOptionSheet)

Constructors

Necklace(IValue)

Constructor for deserialization that supports both Dictionary and List formats.

```
public Necklace(IValue serialized)
```

Parameters

`serialized` IValue

Serialized data in either Dictionary or List format

Necklace(Row, Guid, long, bool)

```
public Necklace(EquipmentItemSheet.Row data, Guid id, long requiredBlockIndex, bool
madeWithMimisbrunnrRecipe = false)
```

Parameters

`data` EquipmentItemSheet.Row

`id` Guid

`requiredBlockIndex` long

`madeWithMimisbrunnrRecipe` bool

Necklace(SerializationInfo, StreamingContext)

```
protected Necklace(SerializationInfo info, StreamingContext _)
```

Parameters

`info` SerializationInfo

`_` StreamingContext

Properties

TradableId

```
public Guid TradableId { get; }
```

Property Value

Guid

Struct OrderLock

Namespace: Nekoyume.Model.Item

Assembly: Lib9c.dll

```
[Serializable]
public struct OrderLock : ILock
```

Implements
ILock

Inherited Members
ValueType.Equals(object) , ValueType.GetHashCode() , ValueType.ToString() , object.Equals(object, object) , object.GetType() , object.ReferenceEquals(object, object)

Constructors

OrderLock(List)

```
public OrderLock(List serialized)
```

Parameters

`serialized` List

OrderLock(Guid)

```
public OrderLock(Guid orderId)
```

Parameters

`orderId` Guid

Fields

OrderId

```
public readonly Guid OrderId
```

Field Value

Guid

Properties

Type

```
public LockType Type { get; }
```

Property Value

LockType

Methods

Serialize()

```
public IValue Serialize()
```

Returns

IValue

Class Ring

Namespace: Nekoyume.Model.Item

Assembly: Lib9c.dll

Represents ring equipment items. Supports both Dictionary and List serialization formats for backward compatibility.

```
[Serializable]
public class Ring : Equipment, INonFungibleItem, IEquippableItem, ITradableItem,
IItem, IState
```

Inheritance

object ← ItemBase ← ItemUsable ← Equipment ← Ring

Implements

INonFungibleItem, IEquippableItem, ITradableItem, IItem, IState

Inherited Members

Equipment.equipped , Equipment.level , Equipment.Exp , Equipment.optionCountFromCombination , Equipment.IconId , Equipment.ByCustomCraft , Equipment.CraftWithRandom , Equipment.HasRandomOnlyIcon , Equipment.Stat , Equipment.SetId , Equipment.SpineResourcePath , Equipment.MadeWithMimisbrunnrRecipe , Equipment.UniqueStatType , Equipment.Equipped , Equipment.GetIncrementAmountOfEnhancement() , Equipment.Serialize() , Equipment.GetFieldNames() , Equipment.Equip() , Equipment.Unequip() , Equipment.LevelUpV1() , Equipment.LevelUp(IRandom, EnhancementCostSheetV2.Row, bool) , Equipment.SetLevel(IRandom, int, EnhancementCostSheetV3) , Equipment.GetRealExp(EquipmentItemSheet, EnhancementCostSheetV3) , Equipment.Equals(Equipment) , Equipment.Equals(object) , Equipment.GetHashCode() , ItemUsable.ITEM_USABLE_FIELD_COUNT , ItemUsable.ItemId , ItemUsable.NonFungibleId , ItemUsable.StatsMap , ItemUsable.Skills , ItemUsable.BuffSkills , ItemUsable.RequiredBlockIndex , ItemUsable.Equals(ItemUsable) , ItemUsable.GetOptionCount() , ItemUsable.Update(long) , ItemBase.SerializationVersion , ItemBase.BaseFieldCount , ItemBase.Codec , ItemBase.Id , ItemBase.Grade , ItemBase.ItemType , ItemBase.ItemSubType , ItemBase.ElementalType , ItemBase.GetObjectData(SerializationInfo, StreamingContext) , ItemBase.Equals(ItemBase) , ItemBase.ToString() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Extension Methods
EquipmentExtensions.GetRequirementLevel(Equipment, ItemRequirementSheet, EquipmentItemRecipeSheet, EquipmentItemSubRecipeSheetV2, EquipmentItemOptionSheet) ,
EquipmentExtensions.IsMadeWithMimisbrunnrRecipe(Equipment, EquipmentItemRecipeSheet, EquipmentItemSubRecipeSheetV2, EquipmentItemOptionSheet)

Constructors

Ring(IValue)

Constructor for deserialization that supports both Dictionary and List formats.

```
public Ring(IValue serialized)
```

Parameters

`serialized` IValue

Serialized data in either Dictionary or List format

Ring(Row, Guid, long, bool)

```
public Ring(EquipmentItemSheet.Row data, Guid id, long requiredBlockIndex, bool
madeWithMimisbrunnrRecipe = false)
```

Parameters

`data` EquipmentItemSheet.Row

`id` Guid

`requiredBlockIndex` long

`madeWithMimisbrunnrRecipe` bool

Ring(SerializationInfo, StreamingContext)

```
protected Ring(SerializationInfo info, StreamingContext _)
```

Parameters

`info` SerializationInfo

`_` StreamingContext

Properties

TradableId

```
public Guid TradableId { get; }
```

Property Value

Guid

Class ShopItem

Namespace: Nekoyume.Model.Item

Assembly: Lib9c.dll

```
[Serializable]
public class ShopItem
```

Inheritance

object ← ShopItem

Inherited Members

object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

ShopItem(Dictionary)

```
public ShopItem(Dictionary serialized)
```

Parameters

`serialized` Dictionary

ShopItem(Address, Address, Guid, FungibleAssetValue, ITradableItem)

```
public ShopItem(Address sellerAgentAddress, Address sellerAvatarAddress, Guid
productId, FungibleAssetValue price, ITradableItem tradableItem)
```

Parameters

`sellerAgentAddress` Address

`sellerAvatarAddress` Address

`productId` Guid

`price` FungibleAssetValue

`tradableItem` ITradableItem

ShopItem(Address, Address, Guid, FungibleAssetValue, long, ITradableItem, int)

```
public ShopItem(Address sellerAgentAddress, Address sellerAvatarAddress, Guid
productId, FungibleAssetValue price, long expiredBlockIndex, ITradableItem
tradableItem, int tradableItemCount = 1)
```

Parameters

`sellerAgentAddress` Address

`sellerAvatarAddress` Address

`productId` Guid

`price` FungibleAssetValue

`expiredBlockIndex` long

`tradableItem` ITradableItem

`tradableItemCount` int

ShopItem(SerializationInfo, StreamingContext)

```
protected ShopItem(SerializationInfo info, StreamingContext _)
```

Parameters

`info` SerializationInfo

`_` StreamingContext

Fields

Codec

```
protected static readonly Codec Codec
```

Field Value

Codec

Costume

```
public readonly Costume Costume
```

Field Value

Costume

ExpiredBlockIndexKey

```
public const string ExpiredBlockIndexKey = "ebi"
```

Field Value

string

ItemUsable

```
public readonly ItemUsable ItemUsable
```

Field Value

ItemUsable

Price

```
public readonly FungibleAssetValue Price
```

Field Value

FungibleAssetValue

ProductId

```
public readonly Guid ProductId
```

Field Value

Guid

SellerAgentAddress

```
public readonly Address SellerAgentAddress
```

Field Value

Address

SellerAvatarAddress

```
public readonly Address SellerAvatarAddress
```

Field Value

Address

TradableFungibleItem

```
public readonly ITradableFungibleItem TradableFungibleItem
```

Field Value

ITradableFungibleItem

TradableFungibleItemCount

```
public readonly int TradableFungibleItemCount
```

Field Value

int

Properties

ExpiredBlockIndex

```
public long ExpiredBlockIndex { get; }
```

Property Value

long

Methods

Equals(ShopItem)

```
protected bool Equals(ShopItem other)
```

Parameters

`other` ShopItem

Returns

bool

Equals(object)

Determines whether the specified object is equal to the current object.

```
public override bool Equals(object obj)
```

Parameters

`obj` object

The object to compare with the current object.

Returns

bool

true if the specified object is equal to the current object; otherwise, false.

GetHashCode()

Serves as the default hash function.

```
public override int GetHashCode()
```

Returns

int

A hash code for the current object.

GetObjectData(SerializationInfo, StreamingContext)

```
public void GetObjectData(SerializationInfo info, StreamingContext context)
```

Parameters

`info` SerializationInfo

`context` StreamingContext

Serialize()

```
public IValue Serialize()
```

Returns

IValue

SerializeBackup1()

```
public IValue SerializeBackup1()
```

Returns

IValue

Class TradableMaterial

Namespace: Nekoyume.Model.Item

Assembly: Lib9c.dll

Represents a tradable material item that can be traded between players. Supports both Dictionary and List serialization formats for backward compatibility.

```
[Serializable]
public class TradableMaterial : Material, ISerializable, ITradableFungibleItem,
ITradableItem, IFungibleItem, IItem, IState, ICloneable
```

Inheritance

object ← ItemBase ← Material ← TradableMaterial

Implements

ISerializable, ITradableFungibleItem, ITradableItem, IFungibleItem, IItem, IState, ICloneable

Inherited Members

Material.ItemId , Material.FungibleId , Material.Equals(Material) , Material.GetFieldNames() , ItemBase.SerializationVersion , ItemBase.BaseFieldCount , ItemBase.Codec , ItemBase.Id , ItemBase.Grade , ItemBase.ItemType , ItemBase.ItemSubType , ItemBase.ElementalType , ItemBase.GetObjectData(SerializationInfo, StreamingContext) , ItemBase.Equals(ItemBase) , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Constructors

TradableMaterial(IValue)

Constructor for deserialization that supports both Dictionary and List formats.

```
public TradableMaterial(IValue serialized)
```

Parameters

`serialized` IValue

Serialized data in either Dictionary or List format

TradableMaterial(Row)

Initializes a new instance of the TradableMaterial class.

```
public TradableMaterial(MaterialItemSheet.Row data)
```

Parameters

`data` MaterialItemSheet.Row

The material item sheet row data

TradableMaterial(SerializationInfo, StreamingContext)

```
protected TradableMaterial(SerializationInfo info, StreamingContext _)
```

Parameters

`info` SerializationInfo

`_` StreamingContext

Properties

RequiredBlockIndex

Gets or sets the required block index for trading this material.

```
public long RequiredBlockIndex { get; set; }
```

Property Value

long

TradableId

Gets the tradable ID of this material.

```
public Guid TradableId { get; }
```

Property Value

Guid

Methods

Clone()

Creates a new object that is a copy of the current instance.

```
public object Clone()
```

Returns

object

A new object that is a copy of this instance.

DeriveTradableId(HashDigest<SHA256>)

Derives a tradable ID from a fungible ID.

```
public static Guid DeriveTradableId(HashDigest<SHA256> fungibleId)
```

Parameters

`fungibleId` HashDigest<SHA256>

The fungible ID to derive from

Returns

Guid

The derived tradable ID

Equals(TradableMaterial)

```
protected bool Equals(TradableMaterial other)
```

Parameters

`other` TradableMaterial

Returns

bool

Equals(object)

Determines whether the specified object is equal to the current object.

```
public override bool Equals(object obj)
```

Parameters

`obj` object

The object to compare with the current object.

Returns

bool

true if the specified object is equal to the current object; otherwise, false.

GetHashCode()

Serves as the default hash function.

```
public override int GetHashCode()
```

Returns

int

A hash code for the current object.

Serialize()

Serializes the tradable material to List format (new format). Order: [baseData..., requiredBlockIndex]

```
public override IValue Serialize()
```

Returns

IValue

List containing serialized data

ToString()

Returns a string that represents the current object.

```
public override string ToString()
```

Returns

string

A string that represents the current object.

Class Weapon

Namespace: Nekoyume.Model.Item

Assembly: Lib9c.dll

Represents weapon equipment items. Supports both Dictionary and List serialization formats for backward compatibility.

```
[Serializable]
public class Weapon : Equipment, INonFungibleItem, IEquippableItem, ITradableItem,
IItem, IState
```

Inheritance

object ← ItemBase ← ItemUsable ← Equipment ← Weapon

Implements

INonFungibleItem, IEquippableItem, ITradableItem, IItem, IState

Inherited Members

Equipment.equipped , Equipment.level , Equipment.Exp , Equipment.optionCountFromCombination , Equipment.IconId , Equipment.ByCustomCraft , Equipment.CraftWithRandom , Equipment.HasRandomOnlyIcon , Equipment.Stat , Equipment.SetId , Equipment.SpineResourcePath , Equipment.MadeWithMimisbrunnrRecipe , Equipment.UniqueStatType , Equipment.Equipped , Equipment.GetIncrementAmountOfEnhancement() , Equipment.Serialize() , Equipment.GetFieldNames() , Equipment.Equip() , Equipment.Unequip() , Equipment.LevelUpV1() , Equipment.LevelUp(IRandom, EnhancementCostSheetV2.Row, bool) , Equipment.SetLevel(IRandom, int, EnhancementCostSheetV3) , Equipment.GetRealExp(EquipmentItemSheet, EnhancementCostSheetV3) , Equipment.Equals(Equipment) , Equipment.Equals(object) , Equipment.GetHashCode() , ItemUsable.ITEM_USABLE_FIELD_COUNT , ItemUsable.ItemId , ItemUsable.NonFungibleId , ItemUsable.StatsMap , ItemUsable.Skills , ItemUsable.BuffSkills , ItemUsable.RequiredBlockIndex , ItemUsable.Equals(ItemUsable) , ItemUsable.GetOptionCount() , ItemUsable.Update(long) , ItemBase.SerializationVersion , ItemBase.BaseFieldCount , ItemBase.Codec , ItemBase.Id , ItemBase.Grade , ItemBase.ItemType , ItemBase.ItemSubType , ItemBase.ElementalType , ItemBase.GetObjectData(SerializationInfo, StreamingContext) , ItemBase.Equals(ItemBase) , ItemBase.ToString() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Extension Methods
EquipmentExtensions.GetRequirementLevel(Equipment, ItemRequirementSheet, EquipmentItemRecipeSheet, EquipmentItemSubRecipeSheetV2, EquipmentItemOptionSheet) ,
EquipmentExtensions.IsMadeWithMimisbrunnrRecipe(Equipment, EquipmentItemRecipeSheet, EquipmentItemSubRecipeSheetV2, EquipmentItemOptionSheet)

Constructors

Weapon(IValue)

Constructor for deserialization that supports both Dictionary and List formats.

```
public Weapon(IValue serialized)
```

Parameters

`serialized` IValue

Serialized data in either Dictionary or List format

Weapon(Row, Guid, long, bool)

```
public Weapon(EquipmentItemSheet.Row data, Guid id, long requiredBlockIndex, bool
madeWithMimisbrunnrRecipe = false)
```

Parameters

`data` EquipmentItemSheet.Row

`id` Guid

`requiredBlockIndex` long

`madeWithMimisbrunnrRecipe` bool

Weapon(SerializationInfo, StreamingContext)

```
protected Weapon(SerializationInfo info, StreamingContext _)
```

Parameters

`info` SerializationInfo

`_` StreamingContext

Properties

TradableId

```
public Guid TradableId { get; }
```

Property Value

Guid

Namespace Nekoyume.Model.Mail

Classes

AdventureBossRaffleWinnerMail

AttachmentMail

BuyerMail

CancelOrderMail

ClaimItemsMail

CombinationMail

CustomCraftMail

DailyRewardMail

GrindingMail

ItemEnhanceMail

Mail

MailBox

MaterialCraftMail

MonsterCollectionMail

OrderBuyerMail

OrderExpirationMail

OrderSellerMail

PatrolRewardMail

Represents a patrol reward mail that contains rewards such as fungible assets and items.

ProductBuyerMail

ProductCancelMail

ProductSellerMail

SellCancelMail

SellerMail

UnloadFromMyGaragesRecipientMail

WorldBossRewardMail

Represents a worldboss reward mail that contains rewards such as fungible assets and items.

Interfaces

IMail

Enums

MailType

Class AdventureBossRaffleWinnerMail

Namespace: Nekoyume.Model.Mail

Assembly: Lib9c.dll

```
public class AdventureBossRaffleWinnerMail : Mail, IState
```

Inheritance

object ← Mail ← AdventureBossRaffleWinnerMail

Implements

IState

Inherited Members

Mail.id , Mail.New , Mail.blockIndex , Mail.MailType , Mail.requiredBlockIndex , Mail.Deserialize(Dictionary) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

AdventureBossRaffleWinnerMail(Dictionary)

```
public AdventureBossRaffleWinnerMail(Dictionary serialized)
```

Parameters

`serialized` Dictionary

AdventureBossRaffleWinnerMail(long, Guid, long, long, FungibleAssetValue)

```
public AdventureBossRaffleWinnerMail(long blockIndex, Guid id, long
requiredBlockIndex, long season, FungibleAssetValue reward)
```

Parameters

`blockIndex` long

`id` Guid

`requiredBlockIndex` long

`season` long

`reward` FungibleAssetValue

Fields

Reward

```
public FungibleAssetValue Reward
```

Field Value

FungibleAssetValue

Season

```
public readonly long Season
```

Field Value

long

Properties

TypeId

```
protected override string TypeId { get; }
```

Property Value

string

Methods

Read(IMail)

```
public override void Read(IMail mail)
```

Parameters

`mail` IMail

Serialize()

```
public override IValue Serialize()
```

Returns

IValue

Class AttachmentMail

Namespace: Nekoyume.Model.Mail

Assembly: Lib9c.dll

```
[Serializable]
public abstract class AttachmentMail : Mail, IState
```

Inheritance

object ← Mail ← AttachmentMail

Implements

IState

Derived

BuyerMail, CombinationMail, DailyRewardMail, ItemEnhanceMail, MonsterCollectionMail, SellCancelMail, SellerMail

Inherited Members

Mail.id , Mail.New , Mail.blockIndex , Mail.MailType , Mail.requiredBlockIndex , Mail.Read(IMail) , Mail.TypeId , Mail.Deserialize(Dictionary) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

AttachmentMail(Dictionary)

```
public AttachmentMail(Dictionary serialized)
```

Parameters

`serialized` Dictionary

AttachmentMail(AttachmentActionResult, long, Guid, long)

```
protected AttachmentMail(AttachmentActionResult attachmentActionResult, long
blockIndex, Guid id, long requiredBlockIndex)
```

Parameters

`attachmentActionResult` AttachmentActionResult

`blockIndex` long

`id` Guid

`requiredBlockIndex` long

Fields

attachment

```
public AttachmentActionResult attachment
```

Field Value

AttachmentActionResult

Methods

Serialize()

```
public override IValue Serialize()
```

Returns

IValue

Class BuyerMail

Namespace: Nekoyume.Model.Mail

Assembly: Lib9c.dll

```
[Serializable]
public class BuyerMail : AttachmentMail, IState
```

Inheritance

object ← Mail ← AttachmentMail ← BuyerMail

Implements

IState

Inherited Members

AttachmentMail.attachment , AttachmentMail.Serialize() , Mail.id , Mail.New , Mail.blockIndex , Mail.requiredBlockIndex , Mail.Deserialize(Dictionary) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

BuyerMail(Dictionary)

```
public BuyerMail(Dictionary serialized)
```

Parameters

`serialized` Dictionary

BuyerMail(AttachmentActionResult, long, Guid, long)

```
public BuyerMail(AttachmentActionResult attachmentActionResult, long blockIndex,
Guid id, long requiredBlockIndex)
```

Parameters

`attachmentActionResult` AttachmentActionResult

`blockIndex` long

`id` Guid

`requiredBlockIndex` long

Properties

MailType

```
public override MailType MailType { get; }
```

Property Value

MailType

TypeId

```
protected override string TypeId { get; }
```

Property Value

string

Methods

Read(IMail)

```
public override void Read(IMail mail)
```

Parameters

`mail` IMail

Class CancelOrderMail

Namespace: Nekoyume.Model.Mail

Assembly: Lib9c.dll

```
[Serializable]
public class CancelOrderMail : Mail, IState
```

Inheritance

object ← Mail ← CancelOrderMail

Implements

IState

Inherited Members

Mail.id , Mail.New , Mail.blockIndex , Mail.requiredBlockIndex , Mail.Deserialize(Dictionary) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

CancelOrderMail(Dictionary)

```
public CancelOrderMail(Dictionary serialized)
```

Parameters

`serialized` Dictionary

CancelOrderMail(long, Guid, long, Guid)

```
public CancelOrderMail(long blockIndex, Guid id, long requiredBlockIndex,
Guid orderId)
```

Parameters

`blockIndex` long

`id` Guid

`requiredBlockIndex` long

`orderId` Guid

Fields

OrderId

```
public readonly Guid OrderId
```

Field Value

Guid

Properties

MailType

```
public override MailType MailType { get; }
```

Property Value

MailType

TypeId

```
protected override string TypeId { get; }
```

Property Value

string

Methods

Read(IMail)

```
public override void Read(IMail mail)
```

Parameters

`mail` IMail

Serialize()

```
public override IValue Serialize()
```

Returns

IValue

Class ClaimItemsMail

Namespace: Nekoyume.Model.Mail

Assembly: Lib9c.dll

```
public class ClaimItemsMail : Mail, IState
```

Inheritance

object ← Mail ← ClaimItemsMail

Implements

IState

Inherited Members

Mail.id , Mail.New , Mail.blockIndex , Mail.requiredBlockIndex , Mail.Deserialize(Dictionary) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

ClaimItemsMail(Dictionary)

```
public ClaimItemsMail(Dictionary serialized)
```

Parameters

`serialized` Dictionary

ClaimItemsMail(long, Guid, long, List<FungibleAssetValue>, List<(int id, int count)>, string)

```
public ClaimItemsMail(long blockIndex, Guid id, long requiredBlockIndex,
List<FungibleAssetValue> fungibleAssetValues, List<(int id, int count)> items,
```

```
    string memo)
```

Parameters

`blockIndex` long

`id` Guid

`requiredBlockIndex` long

`fungibleAssetValues` List<FungibleAssetValue>

`items` List<(int id, int count)>

`memo` string

Fields

FungibleAssetValues

```
public List<FungibleAssetValue> FungibleAssetValues
```

Field Value

List<FungibleAssetValue>

Items

```
public List<(int id, int count)> Items
```

Field Value

List<(int id, int count)>

Memo

```
public string Memo
```

Field Value

string

Properties

MailType

```
public override MailType MailType { get; }
```

Property Value

MailType

TypeId

```
protected override string TypeId { get; }
```

Property Value

string

Methods

Read(IMail)

```
public override void Read(IMail mail)
```

Parameters

`mail` IMail

Serialize()

```
public override IValue Serialize()
```

Returns

IValue

Class CombinationMail

Namespace: Nekoyume.Model.Mail

Assembly: Lib9c.dll

```
[Serializable]
public class CombinationMail : AttachmentMail, IState
```

Inheritance

object ← Mail ← AttachmentMail ← CombinationMail

Implements

IState

Inherited Members

AttachmentMail.attachment , AttachmentMail.Serialize() , Mail.id , Mail.New , Mail.blockIndex , Mail.requiredBlockIndex , Mail.Deserialize(Dictionary) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

CombinationMail(Dictionary)

```
public CombinationMail(Dictionary serialized)
```

Parameters

`serialized` Dictionary

CombinationMail(ResultModel, long, Guid, long)

```
public CombinationMail(CombinationConsumable5.ResultModel attachmentActionResult,
long blockIndex, Guid id, long requiredBlockIndex)
```

Parameters

`attachmentActionResult` CombinationConsumable5.ResultModel

`blockIndex` long

`id` Guid

`requiredBlockIndex` long

Properties

MailType

```
public override MailType MailType { get; }
```

Property Value

MailType

TypeId

```
protected override string TypeId { get; }
```

Property Value

string

Methods

Read(IMail)

```
public override void Read(IMail mail)
```

Parameters

`mail` IMail

Class CustomCraftMail

Namespace: Nekoyume.Model.Mail

Assembly: Lib9c.dll

```
public class CustomCraftMail : Mail, IState
```

Inheritance

object ← Mail ← CustomCraftMail

Implements

IState

Inherited Members

Mail.id , Mail.New , Mail.blockIndex , Mail.requiredBlockIndex , Mail.Deserialize(Dictionary) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

CustomCraftMail(Dictionary)

```
public CustomCraftMail(Dictionary serialized)
```

Parameters

`serialized` Dictionary

CustomCraftMail(long, Guid, long, Equipment)

```
public CustomCraftMail(long blockIndex, Guid id, long requiredBlockIndex,
Equipment equipment)
```

Parameters

`blockIndex` long

`id` Guid

`requiredBlockIndex` long

`equipment` Equipment

Fields

Equipment

```
public Equipment Equipment
```

Field Value

Equipment

Properties

MailType

```
public override MailType MailType { get; }
```

Property Value

MailType

TypeId

```
protected override string TypeId { get; }
```

Property Value

string

Methods

Read(IMail)

```
public override void Read(IMail mail)
```

Parameters

`mail` IMail

Serialize()

```
public override IValue Serialize()
```

Returns

IValue

Class DailyRewardMail

Namespace: Nekoyume.Model.Mail

Assembly: Lib9c.dll

```
[Serializable]
public class DailyRewardMail : AttachmentMail, IState
```

Inheritance

object ← Mail ← AttachmentMail ← DailyRewardMail

Implements

IState

Inherited Members

AttachmentMail.attachment , AttachmentMail.Serialize() , Mail.id , Mail.New , Mail.blockIndex , Mail.requiredBlockIndex , Mail.Deserialize(Dictionary) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

DailyRewardMail(Dictionary)

```
public DailyRewardMail(Dictionary serialized)
```

Parameters

`serialized` Dictionary

DailyRewardMail(AttachmentActionResult, long, Guid, long)

```
public DailyRewardMail(AttachmentActionResult attachmentActionResult, long
```

```
  blockIndex, Guid id, long requiredBlockIndex)
```

Parameters

`attachmentActionResult` AttachmentActionResult

`blockIndex` long

`id` Guid

`requiredBlockIndex` long

Properties

MailType

```
  public override MailType MailType { get; }
```

Property Value

MailType

TypeId

```
  protected override string TypeId { get; }
```

Property Value

string

Methods

Read(IMail)

```
  public override void Read(IMail mail)
```

Parameters

`mail` IMail

Class GrindingMail

Namespace: Nekoyume.Model.Mail

Assembly: Lib9c.dll

```
public class GrindingMail : Mail, IState
```

Inheritance

object ← Mail ← GrindingMail

Implements

IState

Inherited Members

Mail.id , Mail.New , Mail.blockIndex , Mail.requiredBlockIndex , Mail.Deserialize(Dictionary) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

GrindingMail(Dictionary)

```
public GrindingMail(Dictionary serialized)
```

Parameters

`serialized` Dictionary

GrindingMail(long, Guid, long, int, FungibleAssetValue, int)

```
public GrindingMail(long blockIndex, Guid id, long requiredBlockIndex, int
itemCount, FungibleAssetValue asset, int rewardMaterialCount)
```

Parameters

`blockIndex` long

`id` Guid

`requiredBlockIndex` long

`itemCount` int

`asset` FungibleAssetValue

`rewardMaterialCount` int

Fields

Asset

```
public FungibleAssetValue Asset
```

Field Value

FungibleAssetValue

ItemCount

```
public readonly int ItemCount
```

Field Value

int

RewardMaterialCount

```
public readonly int RewardMaterialCount
```

Field Value

int

Properties

MailType

```
public override MailType MailType { get; }
```

Property Value

MailType

TypeId

```
protected override string TypeId { get; }
```

Property Value

string

Methods

Read(IMail)

```
public override void Read(IMail mail)
```

Parameters

`mail` IMail

Serialize()

```
public override IValue Serialize()
```

Returns

IValue

Interface IMail

Namespace: Nekoyume.Model.Mail

Assembly: Lib9c.dll

```
public interface IMail
```

Methods

Read(AdventureBossRaffleWinnerMail)

```
void Read(AdventureBossRaffleWinnerMail adventureBossRaffleWinnerMail)
```

Parameters

`adventureBossRaffleWinnerMail` AdventureBossRaffleWinnerMail

Read(BuyerMail)

```
void Read(BuyerMail buyerMail)
```

Parameters

`buyerMail` BuyerMail

Read(CancelOrderMail)

```
void Read(CancelOrderMail cancelOrderMail)
```

Parameters

`cancelOrderMail` CancelOrderMail

Read(ClaimItemsMail)

```
void Read(ClaimItemsMail claimItemsMail)
```

Parameters

`claimItemsMail` ClaimItemsMail

Read(CombinationMail)

```
void Read(CombinationMail mail)
```

Parameters

`mail` CombinationMail

Read(CustomCraftMail)

```
void Read(CustomCraftMail customCraftMail)
```

Parameters

`customCraftMail` CustomCraftMail

Read(DailyRewardMail)

```
void Read(DailyRewardMail dailyRewardMail)
```

Parameters

`dailyRewardMail` DailyRewardMail

Read(GrindingMail)

```
void Read(GrindingMail grindingMail)
```

Parameters

`grindingMail` GrindingMail

Read(ItemEnhanceMail)

```
void Read(ItemEnhanceMail itemEnhanceMail)
```

Parameters

`itemEnhanceMail` ItemEnhanceMail

Read(MaterialCraftMail)

```
void Read(MaterialCraftMail materialCraftMail)
```

Parameters

`materialCraftMail` MaterialCraftMail

Read(MonsterCollectionMail)

```
void Read(MonsterCollectionMail monsterCollectionMail)
```

Parameters

`monsterCollectionMail` MonsterCollectionMail

Read(OrderBuyerMail)

```
void Read(OrderBuyerMail orderBuyerMail)
```

Parameters

`orderBuyerMail` OrderBuyerMail

Read(OrderExpirationMail)

```
void Read(OrderExpirationMail orderExpirationMail)
```

Parameters

`orderExpirationMail` OrderExpirationMail

Read(OrderSellerMail)

```
void Read(OrderSellerMail orderSellerMail)
```

Parameters

`orderSellerMail` OrderSellerMail

Read(PatrolRewardMail)

```
void Read(PatrolRewardMail patrolRewardMail)
```

Parameters

`patrolRewardMail` PatrolRewardMail

Read(ProductBuyerMail)

```
void Read(ProductBuyerMail productBuyerMail)
```

Parameters

`productBuyerMail` ProductBuyerMail

Read(ProductCancelMail)

```
void Read(ProductCancelMail productCancelMail)
```

Parameters

`productCancelMail` ProductCancelMail

Read(ProductSellerMail)

```
void Read(ProductSellerMail productSellerMail)
```

Parameters

`productSellerMail` ProductSellerMail

Read(SellCancelMail)

```
void Read(SellCancelMail mail)
```

Parameters

`mail` SellCancelMail

Read(SellerMail)

```
void Read(SellerMail sellerMail)
```

Parameters

`sellerMail` SellerMail

Read(UnloadFromMyGaragesRecipientMail)

```
void Read(UnloadFromMyGaragesRecipientMail unloadFromMyGaragesRecipientMail)
```

Parameters

`unloadFromMyGaragesRecipientMail` UnloadFromMyGaragesRecipientMail

Read(WorldBossRewardMail)

```
void Read(WorldBossRewardMail worldBossRewardMail)
```

Parameters

`worldBossRewardMail` WorldBossRewardMail

Class ItemEnhanceMail

Namespace: Nekoyume.Model.Mail

Assembly: Lib9c.dll

```
[Serializable]
public class ItemEnhanceMail : AttachmentMail, IState
```

Inheritance

object ← Mail ← AttachmentMail ← ItemEnhanceMail

Implements

IState

Inherited Members

AttachmentMail.attachment , AttachmentMail.Serialize() , Mail.id , Mail.New , Mail.blockIndex , Mail.requiredBlockIndex , Mail.Deserialize(Dictionary) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

ItemEnhanceMail(Dictionary)

```
public ItemEnhanceMail(Dictionary serialized)
```

Parameters

`serialized` Dictionary

ItemEnhanceMail(AttachmentActionResult, long, Guid, long)

```
public ItemEnhanceMail(AttachmentActionResult attachmentActionResult, long
```

```
  blockIndex, Guid id, long requiredBlockIndex)
```

Parameters

`attachmentActionResult` AttachmentActionResult

`blockIndex` long

`id` Guid

`requiredBlockIndex` long

Properties

MailType

```
  public override MailType MailType { get; }
```

Property Value

MailType

TypeId

```
  protected override string TypeId { get; }
```

Property Value

string

Methods

Read(IMail)

```
  public override void Read(IMail mail)
```

Parameters

`mail` IMail

Class Mail

Namespace: Nekoyume.Model.Mail

Assembly: Lib9c.dll

```
[Serializable]
public abstract class Mail : IState
```

Inheritance

object ← Mail

Implements

IState

Derived

AdventureBossRaffleWinnerMail, AttachmentMail, CancelOrderMail, ClaimItemsMail, CustomCraftMail, GrindingMail, MaterialCraftMail, OrderBuyerMail, OrderExpirationMail, OrderSellerMail, PatrolRewardMail, ProductBuyerMail, ProductCancelMail, ProductSellerMail, UnloadFromMyGaragesRecipientMail, WorldBossRewardMail

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

Mail(Dictionary)

```
protected Mail(Dictionary serialized)
```

Parameters

`serialized` Dictionary

Mail(long, Guid, long)

```
protected Mail(long blockIndex, Guid id, long requiredBlockIndex)
```

Parameters

`blockIndex` long

`id` Guid

`requiredBlockIndex` long

Fields

New

```
public bool New
```

Field Value

bool

blockIndex

```
public long blockIndex
```

Field Value

long

id

```
public Guid id
```

Field Value

Guid

requiredBlockIndex

```
public long requiredBlockIndex
```

Field Value

long

Properties

MailType

```
public virtual MailType MailType { get; }
```

Property Value

MailType

TypeId

```
protected abstract string TypeId { get; }
```

Property Value

string

Methods

Deserialize(Dictionary)

```
public static Mail Deserialize(Dictionary serialized)
```

Parameters

`serialized` Dictionary

Returns

Mail

Read(IMail)

```
public abstract void Read(IMail mail)
```

Parameters

`mail` IMail

Serialize()

```
public virtual IValue Serialize()
```

Returns

IValue

Class MailBox

Namespace: Nekoyume.Model.Mail

Assembly: Lib9c.dll

```
[Serializable]
public class MailBox : IEnumerable<Mail>, IEnumerable, IState
```

Inheritance

object ← MailBox

Implements

IEnumerable<Mail>, IEnumerable, IState

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

MailBox()

```
public MailBox()
```

MailBox(List)

```
public MailBox(List serialized)
```

Parameters

`serialized` List

Properties

Count

```
public int Count { get; }
```

Property Value

int

this[int]

```
public Mail this[int idx] { get; }
```

Parameters

`idx` int

Property Value

Mail

Methods

Add(Mail)

```
public void Add(Mail mail)
```

Parameters

`mail` Mail

CleanUp()

```
public void CleanUp()
```

CleanUpTemp(long)

```
[Obsolete("No longer in use.")]
public void CleanUpTemp(long blockIndex)
```

Parameters

`blockIndex` long

CleanUpV1()

```
[Obsolete("Use CleanUp")]
public void CleanUpV1()
```

GetEnumerator()

Returns an enumerator that iterates through the collection.

```
public IEnumerator<Mail> GetEnumerator()
```

Returns

IEnumerator<Mail>

An enumerator that can be used to iterate through the collection.

Remove(Mail)

```
public void Remove(Mail mail)
```

Parameters

`mail` Mail

Serialize()

```
public IValue Serialize()
```

Returns

IValue

Enum MailType

Namespace: Nekoyume.Model.Mail

Assembly: Lib9c.dll

```
public enum MailType
```

Extension Methods
StateExtensions.Serialize(Enum)

Fields

`Auction = 2`

`CustomCraft = 6`

`Grinding = 4`

`Summon = 5`

`System = 3`

`Workshop = 1`

Class MaterialCraftMail

Namespace: Nekoyume.Model.Mail

Assembly: Lib9c.dll

```
[Serializable]
public class MaterialCraftMail : Mail, IState
```

Inheritance

object ← Mail ← MaterialCraftMail

Implements

IState

Inherited Members

Mail.id , Mail.New , Mail.blockIndex , Mail.requiredBlockIndex , Mail.Deserialize(Dictionary) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

MaterialCraftMail(Dictionary)

```
public MaterialCraftMail(Dictionary serialized)
```

Parameters

`serialized` Dictionary

MaterialCraftMail(long, Guid, long, int, int)

```
public MaterialCraftMail(long blockIndex, Guid id, long requiredBlockIndex, int
itemCount, int itemId)
```

Parameters

`blockIndex` long

`id` Guid

`requiredBlockIndex` long

`itemCount` int

`itemId` int

Fields

ItemCount

```
public int ItemCount
```

Field Value

int

ItemId

```
public int ItemId
```

Field Value

int

Properties

MailType

```
public override MailType MailType { get; }
```

Property Value

MailType

TypeId

```
protected override string TypeId { get; }
```

Property Value

string

Methods

Read(IMail)

```
public override void Read(IMail mail)
```

Parameters

`mail` IMail

Serialize()

```
public override IValue Serialize()
```

Returns

IValue

Class MonsterCollectionMail

Namespace: Nekoyume.Model.Mail

Assembly: Lib9c.dll

```
[Serializable]
public class MonsterCollectionMail : AttachmentMail, IState
```

Inheritance

object ← Mail ← AttachmentMail ← MonsterCollectionMail

Implements

IState

Inherited Members

AttachmentMail.attachment , AttachmentMail.Serialize() , Mail.id , Mail.New , Mail.blockIndex , Mail.requiredBlockIndex , Mail.Deserialize(Dictionary) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

MonsterCollectionMail(Dictionary)

```
public MonsterCollectionMail(Dictionary serialized)
```

Parameters

`serialized` Dictionary

MonsterCollectionMail(AttachmentActionResult, long, Guid, long)

```
public MonsterCollectionMail(AttachmentActionResult attachmentActionResult, long
```

```
  blockIndex, Guid id, long requiredBlockIndex)
```

Parameters

`attachmentActionResult` AttachmentActionResult

`blockIndex` long

`id` Guid

`requiredBlockIndex` long

Properties

MailType

```
  public override MailType MailType { get; }
```

Property Value

MailType

TypeId

```
  protected override string TypeId { get; }
```

Property Value

string

Methods

Read(IMail)

```
  public override void Read(IMail mail)
```

Parameters

`mail` IMail

Class OrderBuyerMail

Namespace: Nekoyume.Model.Mail

Assembly: Lib9c.dll

```
[Serializable]
public class OrderBuyerMail : Mail, IState
```

Inheritance

object ← Mail ← OrderBuyerMail

Implements

IState

Inherited Members

Mail.id , Mail.New , Mail.blockIndex , Mail.requiredBlockIndex , Mail.Deserialize(Dictionary) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

OrderBuyerMail(Dictionary)

```
public OrderBuyerMail(Dictionary serialized)
```

Parameters

`serialized` Dictionary

OrderBuyerMail(long, Guid, long, Guid)

```
public OrderBuyerMail(long blockIndex, Guid id, long requiredBlockIndex,
Guid orderId)
```

Parameters

`blockIndex` long

`id` Guid

`requiredBlockIndex` long

`orderId` Guid

Fields

OrderId

```
public readonly Guid OrderId
```

Field Value

Guid

Properties

MailType

```
public override MailType MailType { get; }
```

Property Value

MailType

TypeId

```
protected override string TypeId { get; }
```

Property Value

string

Methods

Read(IMail)

```
public override void Read(IMail mail)
```

Parameters

`mail` IMail

Serialize()

```
public override IValue Serialize()
```

Returns

IValue

Class OrderExpirationMail

Namespace: Nekoyume.Model.Mail

Assembly: Lib9c.dll

```
[Serializable]
public class OrderExpirationMail : Mail, IState
```

Inheritance

object ← Mail ← OrderExpirationMail

Implements

IState

Inherited Members

Mail.id , Mail.New , Mail.blockIndex , Mail.requiredBlockIndex , Mail.Deserialize(Dictionary) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

OrderExpirationMail(Dictionary)

```
public OrderExpirationMail(Dictionary serialized)
```

Parameters

`serialized` Dictionary

OrderExpirationMail(long, Guid, long, Guid)

```
public OrderExpirationMail(long blockIndex, Guid id, long requiredBlockIndex,
Guid orderId)
```

Parameters

`blockIndex` long

`id` Guid

`requiredBlockIndex` long

`orderId` Guid

Fields

OrderId

```
public readonly Guid OrderId
```

Field Value

Guid

Properties

MailType

```
public override MailType MailType { get; }
```

Property Value

MailType

TypeId

```
protected override string TypeId { get; }
```

Property Value

string

Methods

Read(IMail)

```
public override void Read(IMail mail)
```

Parameters

`mail` IMail

Serialize()

```
public override IValue Serialize()
```

Returns

IValue

Class OrderSellerMail

Namespace: Nekoyume.Model.Mail

Assembly: Lib9c.dll

```
[Serializable]
public class OrderSellerMail : Mail, IState
```

Inheritance

object ← Mail ← OrderSellerMail

Implements

IState

Inherited Members

Mail.id , Mail.New , Mail.blockIndex , Mail.requiredBlockIndex , Mail.Deserialize(Dictionary) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

OrderSellerMail(Dictionary)

```
public OrderSellerMail(Dictionary serialized)
```

Parameters

`serialized` Dictionary

OrderSellerMail(long, Guid, long, Guid)

```
public OrderSellerMail(long blockIndex, Guid id, long requiredBlockIndex,
Guid orderId)
```

Parameters

`blockIndex` long

`id` Guid

`requiredBlockIndex` long

`orderId` Guid

Fields

OrderId

```
public readonly Guid OrderId
```

Field Value

Guid

Properties

MailType

```
public override MailType MailType { get; }
```

Property Value

MailType

TypeId

```
protected override string TypeId { get; }
```

Property Value

string

Methods

Read(IMail)

```
public override void Read(IMail mail)
```

Parameters

`mail` IMail

Serialize()

```
public override IValue Serialize()
```

Returns

IValue

Class PatrolRewardMail

Namespace: Nekoyume.Model.Mail

Assembly: Lib9c.dll

Represents a patrol reward mail that contains rewards such as fungible assets and items.

```
public class PatrolRewardMail : Mail, IState
```

Inheritance

object ← Mail ← PatrolRewardMail

Implements

IState

Inherited Members

Mail.id , Mail.New , Mail.blockIndex , Mail.requiredBlockIndex , Mail.Deserialize(Dictionary) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

PatrolRewardMail(Dictionary)

Constructor for deserializing a PatrolRewardMail from a Bencodex dictionary.

```
public PatrolRewardMail(Dictionary serialized)
```

Parameters

`serialized` Dictionary

The serialized dictionary representation of the mail.

PatrolRewardMail(long, Guid, long, List<FungibleAssetValue>, List<(int id, int count)>)

Constructor for creating a PatrolRewardMail with specified parameters.

```
public PatrolRewardMail(long blockIndex, Guid id, long requiredBlockIndex,
List<FungibleAssetValue> fungibleAssetValues, List<(int id, int count)> items)
```

Parameters

`blockIndex` long

The block index at which the mail is created.

`id` Guid

The unique identifier for the mail.

`requiredBlockIndex` long

The block index required to claim the mail.

`fungibleAssetValues` List<FungibleAssetValue>

The list of fungible asset rewards.

`items` List<(int id, int count)>

The list of item rewards.

Fields

FungibleAssetValues

A list of fungible asset values included in the reward.

```
public List<FungibleAssetValue> FungibleAssetValues
```

Field Value

List<FungibleAssetValue>

Items

A list of item rewards where each entry contains the item ID and count.

```
public List<(int id, int count)> Items
```

Field Value

List<(int id, int count)>

Properties

MailType

The type of mail. Defaults to Auction.

```
public override MailType MailType { get; }
```

Property Value

MailType

TypeId

The type identifier for the PatrolRewardMail class.

```
protected override string TypeId { get; }
```

Property Value

string

Methods

Read(IMail)

Reads the mail contents into the specified mail object.

```
public override void Read(IMail mail)
```

Parameters

`mail` IMail

The mail object to populate with this mail's data.

Serialize()

Serializes the PatrolRewardMail into a Bencodex-compatible format.

```
public override IValue Serialize()
```

Returns

IValue

A serialized representation of the mail.

Class ProductBuyerMail

Namespace: Nekoyume.Model.Mail

Assembly: Lib9c.dll

```
[Serializable]
public class ProductBuyerMail : Mail, IState
```

Inheritance

object ← Mail ← ProductBuyerMail

Implements

IState

Inherited Members

Mail.id , Mail.New , Mail.blockIndex , Mail.requiredBlockIndex , Mail.Deserialize(Dictionary) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

ProductBuyerMail(Dictionary)

```
public ProductBuyerMail(Dictionary serialized)
```

Parameters

`serialized` Dictionary

ProductBuyerMail(long, Guid, long, Guid, Product)

```
public ProductBuyerMail(long blockIndex, Guid id, long requiredBlockIndex, Guid
productId, Product product)
```

Parameters

`blockIndex` long

`id` Guid

`requiredBlockIndex` long

`productId` Guid

`product` Product

Fields

Product

```
public readonly Product Product
```

Field Value

Product

ProductId

```
public readonly Guid ProductId
```

Field Value

Guid

ProductKey

```
public const string ProductKey = "p"
```

Field Value

string

Properties

MailType

```
public override MailType MailType { get; }
```

Property Value

MailType

TypeId

```
protected override string TypeId { get; }
```

Property Value

string

Methods

Read(IMail)

```
public override void Read(IMail mail)
```

Parameters

`mail` IMail

Serialize()

```
public override IValue Serialize()
```

Returns

IValue

Class ProductCancelMail

Namespace: Nekoyume.Model.Mail

Assembly: Lib9c.dll

```
[Serializable]
public class ProductCancelMail : Mail, IState
```

Inheritance

object ← Mail ← ProductCancelMail

Implements

IState

Inherited Members

Mail.id , Mail.New , Mail.blockIndex , Mail.requiredBlockIndex , Mail.Deserialize(Dictionary) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

ProductCancelMail(Dictionary)

```
public ProductCancelMail(Dictionary serialized)
```

Parameters

`serialized` Dictionary

ProductCancelMail(long, Guid, long, Guid, Product)

```
public ProductCancelMail(long blockIndex, Guid id, long requiredBlockIndex, Guid
productId, Product product)
```

Parameters

`blockIndex` long

`id` Guid

`requiredBlockIndex` long

`productId` Guid

`product` Product

Fields

Product

```
public readonly Product Product
```

Field Value

Product

ProductId

```
public readonly Guid ProductId
```

Field Value

Guid

ProductKey

```
public const string ProductKey = "p"
```

Field Value

string

Properties

MailType

```
public override MailType MailType { get; }
```

Property Value

MailType

TypeId

```
protected override string TypeId { get; }
```

Property Value

string

Methods

Read(IMail)

```
public override void Read(IMail mail)
```

Parameters

`mail` IMail

Serialize()

```
public override IValue Serialize()
```

Returns

IValue

Class ProductSellerMail

Namespace: Nekoyume.Model.Mail

Assembly: Lib9c.dll

```
[Serializable]
public class ProductSellerMail : Mail, IState
```

Inheritance

object ← Mail ← ProductSellerMail

Implements

IState

Inherited Members

Mail.id , Mail.New , Mail.blockIndex , Mail.requiredBlockIndex , Mail.Deserialize(Dictionary) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

ProductSellerMail(Dictionary)

```
public ProductSellerMail(Dictionary serialized)
```

Parameters

`serialized` Dictionary

ProductSellerMail(long, Guid, long, Guid, Product)

```
public ProductSellerMail(long blockIndex, Guid id, long requiredBlockIndex, Guid
productId, Product product)
```

Parameters

`blockIndex` long

`id` Guid

`requiredBlockIndex` long

`productId` Guid

`product` Product

Fields

Product

```
public readonly Product Product
```

Field Value

Product

ProductId

```
public readonly Guid ProductId
```

Field Value

Guid

ProductKey

```
public const string ProductKey = "p"
```

Field Value

string

Properties

MailType

```
public override MailType MailType { get; }
```

Property Value

MailType

TypeId

```
protected override string TypeId { get; }
```

Property Value

string

Methods

Read(IMail)

```
public override void Read(IMail mail)
```

Parameters

`mail` IMail

Serialize()

```
public override IValue Serialize()
```

Returns

IValue

Class SellCancelMail

Namespace: Nekoyume.Model.Mail

Assembly: Lib9c.dll

```
[Serializable]
public class SellCancelMail : AttachmentMail, IState
```

Inheritance

object ← Mail ← AttachmentMail ← SellCancelMail

Implements

IState

Inherited Members

AttachmentMail.attachment , AttachmentMail.Serialize() , Mail.id , Mail.New , Mail.blockIndex , Mail.requiredBlockIndex , Mail.Deserialize(Dictionary) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

SellCancelMail(Dictionary)

```
public SellCancelMail(Dictionary serialized)
```

Parameters

`serialized` Dictionary

SellCancelMail(Result, long, Guid, long)

```
public SellCancelMail(SellCancellation.Result attachmentActionResult, long
blockIndex, Guid id, long requiredBlockIndex)
```

Parameters

`attachmentActionResult` SellCancellation.Result

`blockIndex` long

`id` Guid

`requiredBlockIndex` long

Properties

MailType

```
public override MailType MailType { get; }
```

Property Value

MailType

TypeId

```
protected override string TypeId { get; }
```

Property Value

string

Methods

Read(IMail)

```
public override void Read(IMail mail)
```

Parameters

`mail` IMail

Class SellerMail

Namespace: Nekoyume.Model.Mail

Assembly: Lib9c.dll

```
[Serializable]
public class SellerMail : AttachmentMail, IState
```

Inheritance

object ← Mail ← AttachmentMail ← SellerMail

Implements

IState

Inherited Members

AttachmentMail.attachment , AttachmentMail.Serialize() , Mail.id , Mail.New , Mail.blockIndex , Mail.requiredBlockIndex , Mail.Deserialize(Dictionary) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

SellerMail(Dictionary)

```
public SellerMail(Dictionary serialized)
```

Parameters

`serialized` Dictionary

SellerMail(AttachmentActionResult, long, Guid, long)

```
public SellerMail(AttachmentActionResult attachmentActionResult, long blockIndex,
Guid id, long requiredBlockIndex)
```

Parameters

`attachmentActionResult` AttachmentActionResult

`blockIndex` long

`id` Guid

`requiredBlockIndex` long

Properties

MailType

```
public override MailType MailType { get; }
```

Property Value

MailType

TypeId

```
protected override string TypeId { get; }
```

Property Value

string

Methods

Read(IMail)

```
public override void Read(IMail mail)
```

Parameters

`mail` IMail

Class UnloadFromMyGaragesRecipientMail

Namespace: Nekoyume.Model.Mail

Assembly: Lib9c.dll

```
public class UnloadFromMyGaragesRecipientMail : Mail, IState
```

Inheritance

object ← Mail ← UnloadFromMyGaragesRecipientMail

Implements

IState

Inherited Members

Mail.id , Mail.New , Mail.blockIndex , Mail.requiredBlockIndex , Mail.Deserialize(Dictionary) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

UnloadFromMyGaragesRecipientMail(Dictionary)

```
public UnloadFromMyGaragesRecipientMail(Dictionary serialized)
```

Parameters

`serialized` Dictionary

UnloadFromMyGaragesRecipientMail(long, Guid, long, IEnumerable<(Address balanceAddr, FungibleAssetValue value)>?, IEnumerable<(HashDigest<SHA256> fungibleId, int count)>?, string?)

```
public UnloadFromMyGaragesRecipientMail(long blockIndex, Guid id, long
requiredBlockIndex, IEnumerable<(Address balanceAddr, FungibleAssetValue value)>?
fungibleAssetValue, IEnumerable<(HashDigest<SHA256> fungibleId, int count)>?
fungibleIdAndCount, string? memo)
```

Parameters

`blockIndex` long

`id` Guid

`requiredBlockIndex` long

`fungibleAssetValue` IEnumerable<(Address balanceAddr, FungibleAssetValue value)>

`fungibleIdAndCount` IEnumerable<(HashDigest<SHA256> fungibleId, int count)>

`memo` string

Fields

FungibleAssetValues

```
public readonly IOrderedEnumerable<(Address balanceAddr, FungibleAssetValue
value)>? FungibleAssetValues
```

Field Value

IOrderedEnumerable<(Address balanceAddr, FungibleAssetValue value)>

FungibleIdAndCounts

```
public readonly IOrderedEnumerable<(HashDigest<SHA256> fungibleId, int
count)>? FungibleIdAndCounts
```

Field Value

IOrderedEnumerable<(HashDigest<SHA256> fungibleId, int count)>

Memo

```
public readonly string? Memo
```

Field Value

string

Properties

MailType

```
public override MailType MailType { get; }
```

Property Value

MailType

TypeId

```
protected override string TypeId { get; }
```

Property Value

string

Methods

Read(IMail)

```
public override void Read(IMail mail)
```

Parameters

`mail` IMail

Serialize()

```
public override IValue Serialize()
```

Returns

IValue

Class WorldBossRewardMail

Namespace: Nekoyume.Model.Mail

Assembly: Lib9c.dll

Represents a worldboss reward mail that contains rewards such as fungible assets and items.

```
public class WorldBossRewardMail : Mail, IState
```

Inheritance

object ← Mail ← WorldBossRewardMail

Implements

IState

Inherited Members

Mail.id , Mail.New , Mail.blockIndex , Mail.requiredBlockIndex , Mail.Deserialize(Dictionary) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

WorldBossRewardMail(Dictionary)

Constructor for deserializing a WorldBossRewardMail from a Bencodex dictionary.

```
public WorldBossRewardMail(Dictionary serialized)
```

Parameters

`serialized` Dictionary

The serialized dictionary representation of the mail.

WorldBossRewardMail(long, Guid, long, List<FungibleAssetValue>, List<(int id, int count)>)

Constructor for creating a WorldBossRewardMail with specified parameters.

```
public WorldBossRewardMail(long blockIndex, Guid id, long requiredBlockIndex,
List<FungibleAssetValue> fungibleAssetValues, List<(int id, int count)> items)
```

Parameters

`blockIndex` long

The block index at which the mail is created.

`id` Guid

The unique identifier for the mail.

`requiredBlockIndex` long

The block index required to claim the mail.

`fungibleAssetValues` List<FungibleAssetValue>

The list of fungible asset rewards.

`items` List<(int id, int count)>

The list of item rewards.

Fields

FungibleAssetValues

A list of fungible asset values included in the reward.

```
public List<FungibleAssetValue> FungibleAssetValues
```

Field Value

List<FungibleAssetValue>

Items

A list of item rewards where each entry contains the item ID and count.

```
public List<(int id, int count)> Items
```

Field Value

List<(int id, int count)>

Properties

MailType

The type of mail.

```
public override MailType MailType { get; }
```

Property Value

MailType

TypeId

The type identifier for the WorldBossRewardMail class.

```
protected override string TypeId { get; }
```

Property Value

string

Methods

Read(IMail)

Reads the mail contents into the specified mail object.

```
public override void Read(IMail mail)
```

Parameters

`mail` IMail

The mail object to populate with this mail's data.

Serialize()

Serializes the WorldBossRewardMail into a Bencodex-compatible format.

```
public override IValue Serialize()
```

Returns

IValue

A serialized representation of the mail.

Namespace Nekoyume.Model.Market

Classes

FavProduct

ItemProduct

Product

ProductFactory

ProductReceipt

ProductsState

Enums

ProductType

Class FavProduct

Namespace: Nekoyume.Model.Market

Assembly: Lib9c.dll

```
public class FavProduct : Product
```

Inheritance

object ← Product ← FavProduct

Inherited Members

Product.DeriveAddress(Guid) , Product.ProductId , Product.Type , Product.Price , Product.RegisteredBlockIndex , Product.SellerAvatarAddress , Product.SellerAgentAddress , Product.Validate(IProductInfo) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

FavProduct()

```
public FavProduct()
```

FavProduct(List)

```
public FavProduct(List serialized)
```

Parameters

`serialized` List

Fields

Asset

```
public FungibleAssetValue Asset
```

Field Value

FungibleAssetValue

Methods

Serialize()

```
public override IValue Serialize()
```

Returns

IValue

Class ItemProduct

Namespace: Nekoyume.Model.Market

Assembly: Lib9c.dll

```
public class ItemProduct : Product
```

Inheritance

object ← Product ← ItemProduct

Inherited Members

Product.DeriveAddress(Guid) , Product.ProductId , Product.Type , Product.Price , Product.RegisteredBlockIndex , Product.SellerAvatarAddress , Product.SellerAgentAddress , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

ItemProduct()

```
public ItemProduct()
```

ItemProduct(List)

```
public ItemProduct(List serialized)
```

Parameters

`serialized` List

Fields

ItemCount

```
public int ItemCount
```

Field Value

int

TradableItem

```
public ITradableItem TradableItem
```

Field Value

ITradableItem

Methods

Serialize()

```
public override IValue Serialize()
```

Returns

IValue

Validate(IProductInfo)

```
public override void Validate(IProductInfo productInfo)
```

Parameters

`productInfo` IProductInfo

Class Product

Namespace: Nekoyume.Model.Market

Assembly: Lib9c.dll

```
public class Product
```

Inheritance

object ← Product

Derived

FavProduct, ItemProduct

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

Product()

```
protected Product()
```

Product(List)

```
protected Product(List serialized)
```

Parameters

`serialized` List

Fields

Price

```
public FungibleAssetValue Price
```

Field Value

FungibleAssetValue

ProductId

```
public Guid ProductId
```

Field Value

Guid

RegisteredBlockIndex

```
public long RegisteredBlockIndex
```

Field Value

long

SellerAgentAddress

```
public Address SellerAgentAddress
```

Field Value

Address

SellerAvatarAddress

```
public Address SellerAvatarAddress
```

Field Value

Address

Type

```
public ProductType Type
```

Field Value

ProductType

Methods

DeriveAddress(Guid)

```
public static Address DeriveAddress(Guid productId)
```

Parameters

`productId` Guid

Returns

Address

Serialize()

```
public virtual IValue Serialize()
```

Returns

Validate(IProductInfo)

```
public virtual void Validate(IProductInfo productInfo)
```

Parameters

`productInfo` IProductInfo

Class ProductFactory

Namespace: Nekoyume.Model.Market

Assembly: Lib9c.dll

```
public static class ProductFactory
```

Inheritance

object ← ProductFactory

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Methods

DeserializeProduct(List)

```
public static Product DeserializeProduct(List serialized)
```

Parameters

`serialized` List

Returns

Product

DeserializeProductInfo(List)

```
public static IProductInfo DeserializeProductInfo(List serialized)
```

Parameters

`serialized` List

Returns

IProductInfo

DeserializeRegisterInfo(List)

```
public static IRegisterInfo DeserializeRegisterInfo(List serialized)
```

Parameters

`serialized` List

Returns

IRegisterInfo

Class ProductReceipt

Namespace: Nekoyume.Model.Market

Assembly: Lib9c.dll

```
public class ProductReceipt
```

Inheritance

object ← ProductReceipt

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

ProductReceipt(List)

```
public ProductReceipt(List serialized)
```

Parameters

`serialized` List

ProductReceipt(Guid, Address, Address, FungibleAssetValue, long)

```
public ProductReceipt(Guid productId, Address sellerAvatarAddress, Address
buyerAvatarAddress, FungibleAssetValue price, long purchasedBlockIndex)
```

Parameters

`productId` Guid

`sellerAvatarAddress` Address

`buyerAvatarAddress` Address

`price` FungibleAssetValue

`purchasedBlockIndex` long

Fields

BuyerAvatarAddress

```
public readonly Address BuyerAvatarAddress
```

Field Value

Address

Price

```
public FungibleAssetValue Price
```

Field Value

FungibleAssetValue

ProductId

```
public readonly Guid ProductId
```

Field Value

Guid

PurchasedBlockIndex

```
public long PurchasedBlockIndex
```

Field Value

long

SellerAvatarAddress

```
public readonly Address SellerAvatarAddress
```

Field Value

Address

Methods

DeriveAddress(Guid)

```
public static Address DeriveAddress(Guid productId)
```

Parameters

`productId` Guid

Returns

Address

Serialize()

```
public IValue Serialize()
```

Returns

IValue

Enum ProductType

Namespace: Nekoyume.Model.Market

Assembly: Lib9c.dll

```
public enum ProductType
```

Extension Methods
StateExtensions.Serialize(Enum)

Fields

`Fungible = 0`

`FungibleAssetValue = 1`

`NonFungible = 2`

Class ProductsState

Namespace: Nekoyume.Model.Market

Assembly: Lib9c.dll

```
public class ProductsState
```

Inheritance

object ← ProductsState

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

ProductsState()

```
public ProductsState()
```

ProductsState(List)

```
public ProductsState(List serialized)
```

Parameters

`serialized` List

Fields

ProductIds

```
public List<Guid> ProductIds
```

Field Value

List<Guid>

Methods

DeriveAddress(Address)

```
public static Address DeriveAddress(Address avatarAddress)
```

Parameters

`avatarAddress` Address

Returns

Address

Serialize()

```
public IValue Serialize()
```

Returns

IValue

Namespace Nekoyume.Model.Pet

Enums

PetOptionType

Enum PetOptionType

Namespace: Nekoyume.Model.Pet

Assembly: Lib9c.dll

```
public enum PetOptionType
```

Extension Methods
StateExtensions.Serialize(Enum)

Fields

AdditionalOptionRate = 2

AdditionalOptionRateByFixedValue = 3

DiscountMaterialCostCrystal = 5

IncreaseBlockPerHourglass = 4

ReduceRequiredBlock = 0

ReduceRequiredBlockByFixedValue = 1

Namespace Nekoyume.Model.Quest

Classes

CollectQuest

CombinationEquipmentQuest

CombinationQuest

GeneralQuest

GoldQuest

ItemEnhancementQuest

ItemGradeQuest

ItemTypeCollectQuest

MonsterQuest

Quest

QuestList

QuestReward

TradeQuest

UpdateListQuestsCountException

UpdateListVersionException

WorldQuest

Enums

QuestEventType

QuestType

Class CollectQuest

Namespace: Nekoyume.Model.Quest

Assembly: Lib9c.dll

```
[Serializable]
public class CollectQuest : Quest, IState
```

Inheritance

object ← Quest ← CollectQuest

Implements

IState

Inherited Members

Quest.isReceivable , Quest._current , Quest.Complete , Quest.Goal , Quest.Id , Quest.Reward , Quest.IsPaidInAction , Quest.Progress , Quest.GoalFormat , Quest.Deserialize(Dictionary) , Quest.DeserializeList(List) , Quest.Deserialize(IValue) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

CollectQuest(Dictionary)

```
public CollectQuest(Dictionary serialized)
```

Parameters

`serialized` Dictionary

CollectQuest(List)

```
public CollectQuest(List serialized)
```

Parameters

`serialized` List

CollectQuest(Row, QuestReward)

```
public CollectQuest(CollectQuestSheet.Row data, QuestReward reward)
```

Parameters

`data` CollectQuestSheet.Row

`reward` QuestReward

Fields

ItemId

```
public readonly int ItemId
```

Field Value

int

Properties

QuestType

```
public override QuestType QuestType { get; }
```

Property Value

QuestType

TypeId

```
protected override string TypeId { get; }
```

Property Value

string

Methods

Check()

```
public override void Check()
```

GetProgressText()

```
public override string GetProgressText()
```

Returns

string

Serialize()

```
public override IValue Serialize()
```

Returns

IValue

SerializeList()

```
public override IValue SerializeList()
```

Returns

IValue

Update(CollectionMap)

```
public void Update(CollectionMap itemMap)
```

Parameters

`itemMap` CollectionMap

Class CombinationEquipmentQuest

Namespace: Nekoyume.Model.Quest

Assembly: Lib9c.dll

```
public class CombinationEquipmentQuest : Quest, IState
```

Inheritance

object ← Quest ← CombinationEquipmentQuest

Implements

IState

Inherited Members

Quest.isReceivable , Quest._current , Quest.Complete , Quest.Goal , Quest.Id , Quest.Reward , Quest.IsPaidInAction , Quest.Progress , Quest.GoalFormat , Quest.Deserialize(Dictionary) , Quest.DeserializeList(List) , Quest.Deserialize(IValue) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

CombinationEquipmentQuest(Dictionary)

```
public CombinationEquipmentQuest(Dictionary serialized)
```

Parameters

`serialized` Dictionary

CombinationEquipmentQuest(List)

```
public CombinationEquipmentQuest(List serialized)
```

Parameters

`serialized` List

CombinationEquipmentQuest(Row, QuestReward, int)

```
public CombinationEquipmentQuest(QuestSheet.Row data, QuestReward reward,
int stageId)
```

Parameters

`data` QuestSheet.Row

`reward` QuestReward

`stageId` int

Properties

QuestType

```
public override QuestType QuestType { get; }
```

Property Value

QuestType

RecipeId

```
public int RecipeId { get; }
```

Property Value

int

StageId

```
public int StageId { get; }
```

Property Value

int

TypeId

```
protected override string TypeId { get; }
```

Property Value

string

Methods

Check()

```
public override void Check()
```

GetProgressText()

```
public override string GetProgressText()
```

Returns

string

Serialize()

```
public override IValue Serialize()
```

Returns

IValue

SerializeList()

```
public override IValue SerializeList()
```

Returns

IValue

Update(int)

```
public void Update(int recipeId)
```

Parameters

`recipeId` int

Class CombinationQuest

Namespace: Nekoyume.Model.Quest

Assembly: Lib9c.dll

```
[Serializable]
public class CombinationQuest : Quest, IState
```

Inheritance

object ← Quest ← CombinationQuest

Implements

IState

Inherited Members

Quest.isReceivable , Quest._current , Quest.Complete , Quest.Goal , Quest.Id , Quest.Reward , Quest.IsPaidInAction , Quest.Progress , Quest.GoalFormat , Quest.Deserialize(Dictionary) , Quest.DeserializeList(List) , Quest.Deserialize(IValue) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

CombinationQuest(Dictionary)

```
public CombinationQuest(Dictionary serialized)
```

Parameters

`serialized` Dictionary

CombinationQuest(List)

```
public CombinationQuest(List serialized)
```

Parameters

`serialized` List

CombinationQuest(Row, QuestReward)

```
public CombinationQuest(CombinationQuestSheet.Row data, QuestReward reward)
```

Parameters

`data` CombinationQuestSheet.Row

`reward` QuestReward

Properties

ItemSubType

```
public ItemSubType ItemSubType { get; }
```

Property Value

ItemSubType

ItemType

```
public ItemType ItemType { get; }
```

Property Value

ItemType

QuestType

```
public override QuestType QuestType { get; }
```

Property Value

QuestType

TypeId

```
protected override string TypeId { get; }
```

Property Value

string

Methods

Check()

```
public override void Check()
```

GetProgressText()

```
public override string GetProgressText()
```

Returns

string

Serialize()

```
public override IValue Serialize()
```

Returns

IValue

SerializeList()

```
public override IValue SerializeList()
```

Returns

IValue

Update(List<ItemBase>)

```
public void Update(List<ItemBase> items)
```

Parameters

`items` List<ItemBase>

Class GeneralQuest

Namespace: Nekoyume.Model.Quest

Assembly: Lib9c.dll

```
[Serializable]
public class GeneralQuest : Quest, IState
```

Inheritance

object ← Quest ← GeneralQuest

Implements

IState

Inherited Members

Quest.isReceivable , Quest._current , Quest.Complete , Quest.Goal , Quest.Id , Quest.Reward , Quest.IsPaidInAction , Quest.Progress , Quest.GoalFormat , Quest.Deserialize(Dictionary) , Quest.DeserializeList(List) , Quest.Deserialize(IValue) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

GeneralQuest(Dictionary)

```
public GeneralQuest(Dictionary serialized)
```

Parameters

`serialized` Dictionary

GeneralQuest(List)

```
public GeneralQuest(List serialized)
```

Parameters

`serialized` List

GeneralQuest(Row, QuestReward)

```
public GeneralQuest(GeneralQuestSheet.Row data, QuestReward reward)
```

Parameters

`data` GeneralQuestSheet.Row

`reward` QuestReward

Fields

Event

```
public readonly QuestEventType Event
```

Field Value

QuestEventType

Properties

QuestType

```
public override QuestType QuestType { get; }
```

Property Value

QuestType

TypeId

```
protected override string TypeId { get; }
```

Property Value

string

Methods

Check()

```
public override void Check()
```

GetProgressText()

```
public override string GetProgressText()
```

Returns

string

Serialize()

```
public override IValue Serialize()
```

Returns

IValue

SerializeList()

```
public override IValue SerializeList()
```

Returns

IValue

Update(CollectionMap)

```
public void Update(CollectionMap eventMap)
```

Parameters

`eventMap` CollectionMap

Class GoldQuest

Namespace: Nekoyume.Model.Quest

Assembly: Lib9c.dll

```
[Serializable]
public class GoldQuest : Quest, IState
```

Inheritance

object ← Quest ← GoldQuest

Implements

IState

Inherited Members

Quest.isReceivable , Quest._current , Quest.Complete , Quest.Goal , Quest.Id , Quest.Reward , Quest.IsPaidInAction , Quest.Progress , Quest.GoalFormat , Quest.Deserialize(Dictionary) , Quest.DeserializeList(List) , Quest.Deserialize(IValue) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

GoldQuest(Dictionary)

```
public GoldQuest(Dictionary serialized)
```

Parameters

`serialized` Dictionary

GoldQuest(List)

```
public GoldQuest(List serialized)
```

Parameters

`serialized` List

GoldQuest(Row, QuestReward)

```
public GoldQuest(GoldQuestSheet.Row data, QuestReward reward)
```

Parameters

`data` GoldQuestSheet.Row

`reward` QuestReward

Properties

QuestType

```
public override QuestType QuestType { get; }
```

Property Value

QuestType

Type

```
public TradeType Type { get; }
```

Property Value

TradeType

TypeId

```
protected override string TypeId { get; }
```

Property Value

string

Methods

Check()

```
public override void Check()
```

GetProgressText()

```
public override string GetProgressText()
```

Returns

string

Serialize()

```
public override IValue Serialize()
```

Returns

IValue

SerializeList()

```
public override IValue SerializeList()
```

Returns

IValue

Update(FungibleAssetValue)

```
public void Update(FungibleAssetValue gold)
```

Parameters

`gold` FungibleAssetValue

Class ItemEnhancementQuest

Namespace: Nekoyume.Model.Quest

Assembly: Lib9c.dll

```
[Serializable]
public class ItemEnhancementQuest : Quest, IState
```

Inheritance

object ← Quest ← ItemEnhancementQuest

Implements

IState

Inherited Members

Quest.isReceivable , Quest._current , Quest.Complete , Quest.Goal , Quest.Id , Quest.Reward , Quest.IsPaidInAction , Quest.GoalFormat , Quest.Deserialize(Dictionary) , Quest.DeserializeList(List) , Quest.Deserialize(IValue) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

ItemEnhancementQuest(Dictionary)

```
public ItemEnhancementQuest(Dictionary serialized)
```

Parameters

`serialized` Dictionary

ItemEnhancementQuest(List)

```
public ItemEnhancementQuest(List serialized)
```

Parameters

`serialized` List

ItemEnhancementQuest(Row, QuestReward)

```
public ItemEnhancementQuest(ItemEnhancementQuestSheet.Row data, QuestReward reward)
```

Parameters

`data` ItemEnhancementQuestSheet.Row

`reward` QuestReward

Properties

Count

```
public int Count { get; }
```

Property Value

int

Grade

```
public int Grade { get; }
```

Property Value

int

Progress

```
public override float Progress { get; }
```

Property Value

float

QuestType

```
public override QuestType QuestType { get; }
```

Property Value

QuestType

TypeId

```
protected override string TypeId { get; }
```

Property Value

string

Methods

Check()

```
public override void Check()
```

GetProgressText()

```
public override string GetProgressText()
```

Returns

string

Serialize()

```
public override IValue Serialize()
```

Returns

IValue

SerializeList()

```
public override IValue SerializeList()
```

Returns

IValue

Update(Equipment)

```
public void Update(Equipment equipment)
```

Parameters

`equipment` Equipment

Class ItemGradeQuest

Namespace: Nekoyume.Model.Quest

Assembly: Lib9c.dll

```
[Serializable]
public class ItemGradeQuest : Quest, IState
```

Inheritance

object ← Quest ← ItemGradeQuest

Implements

IState

Inherited Members

Quest.isReceivable , Quest._current , Quest.Complete , Quest.Goal , Quest.Id , Quest.Reward , Quest.IsPaidInAction , Quest.Progress , Quest.GoalFormat , Quest.Deserialize(Dictionary) , Quest.DeserializeList(List) , Quest.Deserialize(IValue) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

ItemGradeQuest(Dictionary)

```
public ItemGradeQuest(Dictionary serialized)
```

Parameters

`serialized` Dictionary

ItemGradeQuest(List)

```
public ItemGradeQuest(List serialized)
```

Parameters

`serialized` List

ItemGradeQuest(Row, QuestReward)

```
public ItemGradeQuest(ItemGradeQuestSheet.Row data, QuestReward reward)
```

Parameters

`data` ItemGradeQuestSheet.Row

`reward` QuestReward

Fields

Grade

```
public readonly int Grade
```

Field Value

int

ItemIds

```
public readonly List<int> ItemIds
```

Field Value

List<int>

Properties

QuestType

```
public override QuestType QuestType { get; }
```

Property Value

QuestType

TypeId

```
protected override string TypeId { get; }
```

Property Value

string

Methods

Check()

```
public override void Check()
```

GetProgressText()

```
public override string GetProgressText()
```

Returns

string

Serialize()

```
public override IValue Serialize()
```

Returns

IValue

SerializeList()

```
public override IValue SerializeList()
```

Returns

IValue

Update(ItemUsable)

```
public void Update(ItemUsable itemUsable)
```

Parameters

`itemUsable` ItemUsable

Class ItemTypeCollectQuest

Namespace: Nekoyume.Model.Quest

Assembly: Lib9c.dll

```
[Serializable]
public class ItemTypeCollectQuest : Quest, IState
```

Inheritance

object ← Quest ← ItemTypeCollectQuest

Implements

IState

Inherited Members

Quest.isReceivable , Quest._current , Quest.Complete , Quest.Goal , Quest.Id , Quest.Reward , Quest.IsPaidInAction , Quest.Progress , Quest.GoalFormat , Quest.Deserialize(Dictionary) , Quest.DeserializeList(List) , Quest.Deserialize(IValue) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

ItemTypeCollectQuest(Dictionary)

```
public ItemTypeCollectQuest(Dictionary serialized)
```

Parameters

`serialized` Dictionary

ItemTypeCollectQuest(List)

```
public ItemTypeCollectQuest(List serialized)
```

Parameters

`serialized` List

ItemTypeCollectQuest(Row, QuestReward)

```
public ItemTypeCollectQuest(ItemTypeCollectQuestSheet.Row data, QuestReward reward)
```

Parameters

`data` ItemTypeCollectQuestSheet.Row

`reward` QuestReward

Fields

ItemIds

```
public readonly List<int> ItemIds
```

Field Value

List<int>

ItemType

```
public readonly ItemType ItemType
```

Field Value

ItemType

Properties

QuestType

```
public override QuestType QuestType { get; }
```

Property Value

QuestType

TypeId

```
protected override string TypeId { get; }
```

Property Value

string

Methods

Check()

```
public override void Check()
```

GetProgressText()

```
public override string GetProgressText()
```

Returns

string

Serialize()

```
public override IValue Serialize()
```

Returns

IValue

SerializeList()

```
public override IValue SerializeList()
```

Returns

IValue

Update(ItemBase)

```
public void Update(ItemBase item)
```

Parameters

`item` ItemBase

Class MonsterQuest

Namespace: Nekoyume.Model.Quest

Assembly: Lib9c.dll

```
[Serializable]
public class MonsterQuest : Quest, IState
```

Inheritance

object ← Quest ← MonsterQuest

Implements

IState

Inherited Members

Quest.isReceivable , Quest._current , Quest.Complete , Quest.Goal , Quest.Id , Quest.Reward , Quest.IsPaidInAction , Quest.Progress , Quest.GoalFormat , Quest.Deserialize(Dictionary) , Quest.DeserializeList(List) , Quest.Deserialize(IValue) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

MonsterQuest(Dictionary)

```
public MonsterQuest(Dictionary serialized)
```

Parameters

`serialized` Dictionary

MonsterQuest(List)

```
public MonsterQuest(List serialized)
```

Parameters

`serialized` List

MonsterQuest(Row, QuestReward)

```
public MonsterQuest(MonsterQuestSheet.Row data, QuestReward reward)
```

Parameters

`data` MonsterQuestSheet.Row

`reward` QuestReward

Fields

MonsterId

```
public readonly int MonsterId
```

Field Value

int

Properties

QuestType

```
public override QuestType QuestType { get; }
```

Property Value

QuestType

TypeId

```
protected override string TypeId { get; }
```

Property Value

string

Methods

Check()

```
public override void Check()
```

GetProgressText()

```
public override string GetProgressText()
```

Returns

string

Serialize()

```
public override IValue Serialize()
```

Returns

IValue

SerializeList()

```
public override IValue SerializeList()
```

Returns

IValue

Update(CollectionMap)

```
public void Update(CollectionMap monsterMap)
```

Parameters

`monsterMap` CollectionMap

Class Quest

Namespace: Nekoyume.Model.Quest

Assembly: Lib9c.dll

```
[Serializable]
public abstract class Quest : IState
```

Inheritance

object ← Quest

Implements

IState

Derived

CollectQuest, CombinationEquipmentQuest, CombinationQuest, GeneralQuest, GoldQuest, ItemEnhancementQuest, ItemGradeQuest, ItemTypeCollectQuest, MonsterQuest, TradeQuest, WorldQuest

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

Quest(Dictionary)

```
protected Quest(Dictionary serialized)
```

Parameters

`serialized` Dictionary

Quest(List)

```
protected Quest(List serialized)
```

Parameters

`serialized` List

Quest(Row, QuestReward)

```
protected Quest(QuestSheet.Row data, QuestReward reward)
```

Parameters

`data` QuestSheet.Row

`reward` QuestReward

Fields

GoalFormat

```
public const string GoalFormat = "({0}/{1})"
```

Field Value

string

_current

```
protected int _current
```

Field Value

int

isReceivable

```
[NonSerialized]
public bool isReceivable
```

Field Value

bool

Properties

Complete

```
public bool Complete { get; protected set; }
```

Property Value

bool

Goal

```
public int Goal { get; }
```

Property Value

int

Id

```
public int Id { get; }
```

Property Value

int

IsPaidInAction

이미 퀘스트 보상이 액션에서 지급되었는가?

```
public bool IsPaidInAction { get; set; }
```

Property Value

bool

Progress

```
public virtual float Progress { get; }
```

Property Value

float

QuestType

```
public abstract QuestType QuestType { get; }
```

Property Value

QuestType

Reward

```
public QuestReward Reward { get; }
```

Property Value

QuestReward

TypeId

```
protected abstract string TypeId { get; }
```

Property Value

string

Methods

Check()

```
public abstract void Check()
```

Deserialize(Dictionary)

```
public static Quest Deserialize(Dictionary serialized)
```

Parameters

`serialized` Dictionary

Returns

Quest

Deserialize(IValue)

```
public static Quest Deserialize(IValue arg)
```

Parameters

`arg` IValue

Returns

Quest

DeserializeList(List)

```
public static Quest DeserializeList(List serialized)
```

Parameters

`serialized` List

Returns

Quest

GetProgressText()

```
public abstract string GetProgressText()
```

Returns

string

Serialize()

```
public virtual IValue Serialize()
```

Returns

IValue

SerializeList()

```
public virtual IValue SerializeList()
```

Returns

IValue

Enum QuestEventType

Namespace: Nekoyume.Model.Quest

Assembly: Lib9c.dll

```
public enum QuestEventType
```

Extension Methods
StateExtensions.Serialize(Enum)

Fields

`Complete = 4`

`Consumable = 6`

`Create = 0`

`Die = 3`

`Enhancement = 1`

`Equipment = 5`

`Level = 2`

Class QuestList

Namespace: Nekoyume.Model.Quest

Assembly: Lib9c.dll

```
[Serializable]
public class QuestList : IEnumerable<Quest>, IEnumerable, IState, ICloneable
```

Inheritance

object ← QuestList

Implements

IEnumerable<Quest>, IEnumerable, IState, ICloneable

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

QuestList(Dictionary)

```
public QuestList(Dictionary serialized)
```

Parameters

`serialized` Dictionary

QuestList(List)

```
public QuestList(List serialized)
```

Parameters

`serialized` List

QuestList(QuestSheet, QuestRewardSheet, QuestItemRewardSheet, EquipmentItemRecipeSheet, EquipmentItemSubRecipeSheet)

```
public QuestList(QuestSheet questSheet, QuestRewardSheet questRewardSheet,
QuestItemRewardSheet questItemRewardSheet, EquipmentItemRecipeSheet
equipmentItemRecipeSheet, EquipmentItemSubRecipeSheet equipmentItemSubRecipeSheet)
```

Parameters

`questSheet` QuestSheet

`questRewardSheet` QuestRewardSheet

`questItemRewardSheet` QuestItemRewardSheet

`equipmentItemRecipeSheet` EquipmentItemRecipeSheet

`equipmentItemSubRecipeSheet` EquipmentItemSubRecipeSheet

Fields

CompletedQuestIdsKey

```
public const string CompletedQuestIdsKey = "c"
```

Field Value

string

ListVersionKey

```
public const string ListVersionKey = "v"
```

Field Value

string

QuestsKey

```
public const string QuestsKey = "q"
```

Field Value

string

completedQuestIds

```
public List<int> completedQuestIds
```

Field Value

List<int>

Properties

ListVersion

```
public int ListVersion { get; }
```

Property Value

int

Methods

Clone()

Creates a new object that is a copy of the current instance.

```
public object Clone()
```

Returns

object

A new object that is a copy of this instance.

GetEnumerator()

Returns an enumerator that iterates through the collection.

```
public IEnumerator<Quest> GetEnumerator()
```

Returns

IEnumerator<Quest>

An enumerator that can be used to iterate through the collection.

Serialize()

```
public IValue Serialize()
```

Returns

IValue

SerializeDictionary()

```
[Obsolete("Dictionary type is obsolete.")]
public IValue SerializeDictionary()
```

Returns

IValue

SerializeList()

```
public IValue SerializeList()
```

Returns

IValue

UpdateCollectQuest(CollectionMap)

```
public void UpdateCollectQuest(CollectionMap itemMap)
```

Parameters

`itemMap` CollectionMap

UpdateCombinationEquipmentQuest(int)

```
public void UpdateCombinationEquipmentQuest(int recipeId)
```

Parameters

`recipeId` int

UpdateCombinationQuest(ItemUsable)

```
public void UpdateCombinationQuest(ItemUsable itemUsable)
```

Parameters

`itemUsable` ItemUsable

UpdateCompletedQuest(CollectionMap)

```
public CollectionMap UpdateCompletedQuest(CollectionMap eventMap)
```

Parameters

`eventMap` CollectionMap

Returns

CollectionMap

UpdateGeneralQuest(IEnumerable<QuestEventType>, CollectionMap)

```
public CollectionMap UpdateGeneralQuest(IEnumerable<QuestEventType> types,
CollectionMap eventMap)
```

Parameters

`types` IEnumerable<QuestEventType>

`eventMap` CollectionMap

Returns

CollectionMap

UpdateItemEnhancementQuest(Equipment)

```
public void UpdateItemEnhancementQuest(Equipment equipment)
```

Parameters

`equipment` Equipment

UpdateItemGradeQuest(ItemUsable)

```
public void UpdateItemGradeQuest(ItemUsable itemUsable)
```

Parameters

`itemUsable` ItemUsable

UpdateItemTypeCollectQuest(IEnumerable<ItemBase>)

```
public void UpdateItemTypeCollectQuest(IEnumerable<ItemBase> items)
```

Parameters

`items` IEnumerable<ItemBase>

UpdateList(QuestSheet, QuestRewardSheet, QuestItemRewardSheet, EquipmentItemRecipeSheet, ICollection<int>)

```
public void UpdateList(QuestSheet questSheet, QuestRewardSheet questRewardSheet,
QuestItemRewardSheet questItemRewardSheet, EquipmentItemRecipeSheet
equipmentItemRecipeSheet, ICollection<int> newIds)
```

Parameters

`questSheet` QuestSheet

`questRewardSheet` QuestRewardSheet

`questItemRewardSheet` QuestItemRewardSheet

`equipmentItemRecipeSheet` EquipmentItemRecipeSheet

`newIds` ICollection<int>

UpdateListV1(int, QuestSheet, QuestRewardSheet, QuestItemRewardSheet, EquipmentItemRecipeSheet)

```
[Obsolete("Use UpdateList()")]
public void UpdateListV1(int listVersion, QuestSheet questSheet, QuestRewardSheet
questRewardSheet, QuestItemRewardSheet questItemRewardSheet,
EquipmentItemRecipeSheet equipmentItemRecipeSheet)
```

Parameters

`listVersion` int

`questSheet` QuestSheet

`questRewardSheet` QuestRewardSheet

`questItemRewardSheet` QuestItemRewardSheet

`equipmentItemRecipeSheet` EquipmentItemRecipeSheet

Exceptions

UpdateListVersionException

UpdateListQuestsCountException

Exception

UpdateMonsterQuest(CollectionMap)

```
public void UpdateMonsterQuest(CollectionMap monsterMap)
```

Parameters

`monsterMap` CollectionMap

UpdateStageQuest(CollectionMap)

```
public void UpdateStageQuest(CollectionMap stageMap)
```

Parameters

`stageMap` CollectionMap

UpdateTradeQuest(TradeType, FungibleAssetValue)

```
public void UpdateTradeQuest(TradeType type, FungibleAssetValue price)
```

Parameters

`type` TradeType

`price` FungibleAssetValue

Class QuestReward

Namespace: Nekoyume.Model.Quest

Assembly: Lib9c.dll

```
[Serializable]
public class QuestReward : IState
```

Inheritance

object ← QuestReward

Implements

IState

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

QuestReward(Dictionary)

```
public QuestReward(Dictionary serialized)
```

Parameters

`serialized` Dictionary

QuestReward(Dictionary<int, int>)

```
public QuestReward(Dictionary<int, int> map)
```

Parameters

`map` Dictionary<int, int>

Fields

ItemMap

```
public readonly IEnumerable<Tuple<int, int>> ItemMap
```

Field Value

IEnumerable<Tuple<int, int>>

Methods

Serialize()

```
public IValue Serialize()
```

Returns

IValue

Enum QuestType

Namespace: Nekoyume.Model.Quest

Assembly: Lib9c.dll

```
public enum QuestType
```

Extension Methods
StateExtensions.Serialize(Enum)

Fields

`Adventure = 0`

`Craft = 2`

`Exchange = 3`

`Obtain = 1`

Class TradeQuest

Namespace: Nekoyume.Model.Quest

Assembly: Lib9c.dll

```
[Serializable]
public class TradeQuest : Quest, IState
```

Inheritance

object ← Quest ← TradeQuest

Implements

IState

Inherited Members

Quest.isReceivable , Quest._current , Quest.Complete , Quest.Goal , Quest.Id , Quest.Reward , Quest.IsPaidInAction , Quest.Progress , Quest.GoalFormat , Quest.Deserialize(Dictionary) , Quest.DeserializeList(List) , Quest.Deserialize(IValue) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

TradeQuest(Dictionary)

```
public TradeQuest(Dictionary serialized)
```

Parameters

`serialized` Dictionary

TradeQuest(List)

```
public TradeQuest(List serialized)
```

Parameters

`serialized` List

TradeQuest(Row, QuestReward)

```
public TradeQuest(TradeQuestSheet.Row data, QuestReward reward)
```

Parameters

`data` TradeQuestSheet.Row

`reward` QuestReward

Properties

QuestType

```
public override QuestType QuestType { get; }
```

Property Value

QuestType

Type

```
public TradeType Type { get; }
```

Property Value

TradeType

TypeId

```
protected override string TypeId { get; }
```

Property Value

string

Methods

Check()

```
public override void Check()
```

GetProgressText()

```
public override string GetProgressText()
```

Returns

string

Serialize()

```
public override IValue Serialize()
```

Returns

IValue

SerializeList()

```
public override IValue SerializeList()
```

Returns

IValue

Class UpdateListQuestsCountException

Namespace: Nekoyume.Model.Quest

Assembly: Lib9c.dll

```
[Serializable]
public class UpdateListQuestsCountException : ArgumentException, ISerializable
```

Inheritance

object ← Exception ← SystemException ← ArgumentException ← UpdateListQuestsCountException

Implements

ISerializable

Inherited Members

ArgumentException.GetObjectData(SerializationInfo, StreamingContext) , ArgumentException.Message , ArgumentException.ParamName , Exception.GetBaseException() , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Constructors

UpdateListQuestsCountException()

```
public UpdateListQuestsCountException()
```

UpdateListQuestsCountException(int, int)

```
public UpdateListQuestsCountException(int expected, int actual)
```

Parameters

`expected` int

`actual` int

UpdateListQuestsCountException(SerializationInfo, StreamingContext)

```
protected UpdateListQuestsCountException(SerializationInfo info,
StreamingContext context)
```

Parameters

`info` SerializationInfo

`context` StreamingContext

UpdateListQuestsCountException(string)

```
public UpdateListQuestsCountException(string s)
```

Parameters

`s` string

Class UpdateListVersionException

Namespace: Nekoyume.Model.Quest

Assembly: Lib9c.dll

```
[Serializable]
public class UpdateListVersionException : ArgumentOutOfRangeException, ISerializable
```

Inheritance

object ← Exception ← SystemException ← ArgumentException ← ArgumentOutOfRangeException ← UpdateListVersionException

Implements

ISerializable

Inherited Members

ArgumentOutOfRangeException.GetObjectData(SerializationInfo, StreamingContext) , ArgumentOutOfRangeException.ActualValue , ArgumentOutOfRangeException.Message , ArgumentException.ParamName , Exception.GetBaseException() , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Constructors

UpdateListVersionException()

```
public UpdateListVersionException()
```

UpdateListVersionException(int, int)

```
public UpdateListVersionException(int expected, int actual)
```

Parameters

`expected` int

`actual` int

UpdateListVersionException(SerializationInfo, StreamingContext)

```
protected UpdateListVersionException(SerializationInfo info,
StreamingContext context)
```

Parameters

`info` SerializationInfo

`context` StreamingContext

UpdateListVersionException(string)

```
public UpdateListVersionException(string s)
```

Parameters

`s` string

Class WorldQuest

Namespace: Nekoyume.Model.Quest

Assembly: Lib9c.dll

```
[Serializable]
public class WorldQuest : Quest, IState
```

Inheritance

object ← Quest ← WorldQuest

Implements

IState

Inherited Members

Quest.isReceivable , Quest._current , Quest.Complete , Quest.Goal , Quest.Id , Quest.Reward , Quest.IsPaidInAction , Quest.Progress , Quest.GoalFormat , Quest.Serialize() , Quest.SerializeList() , Quest.Deserialize(Dictionary) , Quest.DeserializeList(List) , Quest.Deserialize(IValue) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

WorldQuest(Dictionary)

```
public WorldQuest(Dictionary serialized)
```

Parameters

`serialized` Dictionary

WorldQuest(List)

```
public WorldQuest(List serialized)
```

Parameters

`serialized` List

WorldQuest(Row, QuestReward)

```
public WorldQuest(WorldQuestSheet.Row data, QuestReward reward)
```

Parameters

`data` WorldQuestSheet.Row

`reward` QuestReward

Properties

QuestType

```
public override QuestType QuestType { get; }
```

Property Value

QuestType

TypeId

```
protected override string TypeId { get; }
```

Property Value

string

Methods

Check()

```
public override void Check()
```

GetProgressText()

```
public override string GetProgressText()
```

Returns

string

Update(CollectionMap)

```
public void Update(CollectionMap stageMap)
```

Parameters

`stageMap` CollectionMap

Namespace Nekoyume.Model.Rune

Classes

DuplicatedRuneIdException

DuplicatedRuneSlotIndexException

ForbiddenRuneEquippedException

ForbiddenRuneTypeEquippedException

IsEquippableRuneException

IsUsableSlotException

MismatchRuneSlotTypeException

RuneCostDataNotFoundException

RuneCostNotFoundException

RuneInfosIsEmptyException

RuneListNotFoundException

RuneNotFoundException

RuneSlot

RuneStateNotFoundException

SlotIsAlreadyUnlockedException

SlotIsLockedException

SlotNotFoundException

SlotRuneTypeException

TryCountIsZeroException

Class DuplicatedRuneIdException

Namespace: Nekoyume.Model.Rune

Assembly: Lib9c.dll

```
[Serializable]
public class DuplicatedRuneIdException : Exception, ISerializable
```

Inheritance

object ← Exception ← DuplicatedRuneIdException

Implements

ISerializable

Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Constructors

DuplicatedRuneIdException(SerializationInfo, StreamingContext)

```
protected DuplicatedRuneIdException(SerializationInfo info,
StreamingContext context)
```

Parameters

`info` SerializationInfo

`context` StreamingContext

DuplicatedRuneIdException(string)

```
public DuplicatedRuneIdException(string message)
```

Parameters

`message` string

Class DuplicatedRuneSlotIndexException

Namespace: Nekoyume.Model.Rune

Assembly: Lib9c.dll

```
[Serializable]
public class DuplicatedRuneSlotIndexException : Exception, ISerializable
```

Inheritance

object ← Exception ← DuplicatedRuneSlotIndexException

Implements

ISerializable

Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Constructors

DuplicatedRuneSlotIndexException(SerializationInfo, StreamingContext)

```
protected DuplicatedRuneSlotIndexException(SerializationInfo info,
StreamingContext context)
```

Parameters

`info` SerializationInfo

`context` StreamingContext

DuplicatedRuneSlotIndexException(string)

```
public DuplicatedRuneSlotIndexException(string message)
```

Parameters

`message` string

Class ForbiddenRuneEquippedException

Namespace: Nekoyume.Model.Rune

Assembly: Lib9c.dll

```
[Serializable]
public class ForbiddenRuneEquippedException : Exception, ISerializable
```

Inheritance

object ← Exception ← ForbiddenRuneEquippedException

Implements

ISerializable

Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Constructors

ForbiddenRuneEquippedException(SerializationInfo, StreamingContext)

```
protected ForbiddenRuneEquippedException(SerializationInfo info,
StreamingContext context)
```

Parameters

`info` SerializationInfo

`context` StreamingContext

ForbiddenRuneEquippedException(string)

```
public ForbiddenRuneEquippedException(string message)
```

Parameters

`message` string

ForbiddenRuneEquippedException(string, Exception)

```
public ForbiddenRuneEquippedException(string message, Exception innerException)
```

Parameters

`message` string

`innerException` Exception

Class ForbiddenRuneTypeEquippedException

Namespace: Nekoyume.Model.Rune

Assembly: Lib9c.dll

```
[Serializable]
public class ForbiddenRuneTypeEquippedException : Exception, ISerializable
```

Inheritance

object ← Exception ← ForbiddenRuneTypeEquippedException

Implements

ISerializable

Inherited Members

Exception.GetBaseException() , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Constructors

ForbiddenRuneTypeEquippedException()

Initializes a new instance of the ForbiddenRuneTypeEquippedException class.

```
public ForbiddenRuneTypeEquippedException()
```

ForbiddenRuneTypeEquippedException(List<RuneType>, List<RuneType>)

Initializes a new instance of the ForbiddenRuneTypeEquippedException class.

```
public ForbiddenRuneTypeEquippedException(List<RuneType> forbiddenRuneTypes,
List<RuneType> equippedRuneTypes)
```

Parameters

`forbiddenRuneTypes` List<RuneType>

The list of forbidden rune types.

`equippedRuneTypes` List<RuneType>

The list of equipped rune types that are forbidden.

ForbiddenRuneTypeEquippedException(SerializationInfo, StreamingContext)

Initializes a new instance of the ForbiddenRuneTypeEquippedException class for serialization.

```
protected ForbiddenRuneTypeEquippedException(SerializationInfo info,
StreamingContext context)
```

Parameters

`info` SerializationInfo

The serialization info.

`context` StreamingContext

The streaming context.

ForbiddenRuneTypeEquippedException(string)

Initializes a new instance of the ForbiddenRuneTypeEquippedException class.

```
public ForbiddenRuneTypeEquippedException(string message)
```

Parameters

`message` string

The error message.

ForbiddenRuneTypeEquippedException(string, List<RuneType>, List<RuneType>)

Initializes a new instance of the ForbiddenRuneTypeEquippedException class.

```
public ForbiddenRuneTypeEquippedException(string message, List<RuneType>
forbiddenRuneTypes, List<RuneType> equippedRuneTypes)
```

Parameters

`message` string

The error message.

`forbiddenRuneTypes` List<RuneType>

The list of forbidden rune types.

`equippedRuneTypes` List<RuneType>

The list of equipped rune types that are forbidden.

ForbiddenRuneTypeEquippedException(string, Exception)

Initializes a new instance of the ForbiddenRuneTypeEquippedException class.

```
public ForbiddenRuneTypeEquippedException(string message, Exception innerException)
```

Parameters

`message` string

The error message.

`innerException` Exception

The inner exception.

Properties

EquippedRuneTypes

Gets the list of equipped rune types that are forbidden.

```
public List<RuneType> EquippedRuneTypes { get; }
```

Property Value

List<RuneType>

ForbiddenRuneTypes

Gets the list of forbidden rune types that were equipped.

```
public List<RuneType> ForbiddenRuneTypes { get; }
```

Property Value

List<RuneType>

Methods

GetObjectData(SerializationInfo, StreamingContext)

Gets object data for serialization.

```
public override void GetObjectData(SerializationInfo info, StreamingContext context)
```

Parameters

`info` SerializationInfo

The serialization info.

`context` StreamingContext

The streaming context.

Class IsEquippableRuneException

Namespace: Nekoyume.Model.Rune

Assembly: Lib9c.dll

```
[Serializable]
public class IsEquippableRuneException : Exception, ISerializable
```

Inheritance

object ← Exception ← IsEquippableRuneException

Implements

ISerializable

Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Constructors

IsEquippableRuneException(SerializationInfo, StreamingContext)

```
protected IsEquippableRuneException(SerializationInfo info,
StreamingContext context)
```

Parameters

`info` SerializationInfo

`context` StreamingContext

IsEquippableRuneException(string)

```
public IsEquippableRuneException(string message)
```

Parameters

`message` string

Class IsUsableSlotException

Namespace: Nekoyume.Model.Rune

Assembly: Lib9c.dll

```
[Serializable]
public class IsUsableSlotException : Exception, ISerializable
```

Inheritance

object ← Exception ← IsUsableSlotException

Implements

ISerializable

Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Constructors

IsUsableSlotException(SerializationInfo, StreamingContext)

```
protected IsUsableSlotException(SerializationInfo info, StreamingContext context)
```

Parameters

`info` SerializationInfo

`context` StreamingContext

IsUsableSlotException(string)

```
public IsUsableSlotException(string message)
```

Parameters

`message` string

Class MismatchRuneSlotTypeException

Namespace: Nekoyume.Model.Rune

Assembly: Lib9c.dll

```
[Serializable]
public class MismatchRuneSlotTypeException : Exception, ISerializable
```

Inheritance

object ← Exception ← MismatchRuneSlotTypeException

Implements

ISerializable

Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Constructors

MismatchRuneSlotTypeException(SerializationInfo, StreamingContext)

```
protected MismatchRuneSlotTypeException(SerializationInfo info,
StreamingContext context)
```

Parameters

`info` SerializationInfo

`context` StreamingContext

MismatchRuneSlotTypeException(string)

```
public MismatchRuneSlotTypeException(string message)
```

Parameters

`message` string

Class RuneCostDataNotFoundException

Namespace: Nekoyume.Model.Rune

Assembly: Lib9c.dll

```
[Serializable]
public class RuneCostDataNotFoundException : Exception, ISerializable
```

Inheritance

object ← Exception ← RuneCostDataNotFoundException

Implements

ISerializable

Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Constructors

RuneCostDataNotFoundException(SerializationInfo, StreamingContext)

```
protected RuneCostDataNotFoundException(SerializationInfo info,
StreamingContext context)
```

Parameters

`info` SerializationInfo

`context` StreamingContext

RuneCostDataNotFoundException(string)

```
public RuneCostDataNotFoundException(string message)
```

Parameters

`message` string

Class RuneCostNotFoundException

Namespace: Nekoyume.Model.Rune

Assembly: Lib9c.dll

```
[Serializable]
public class RuneCostNotFoundException : Exception, ISerializable
```

Inheritance

object ← Exception ← RuneCostNotFoundException

Implements

ISerializable

Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Constructors

RuneCostNotFoundException(SerializationInfo, StreamingContext)

```
protected RuneCostNotFoundException(SerializationInfo info,
StreamingContext context)
```

Parameters

`info` SerializationInfo

`context` StreamingContext

RuneCostNotFoundException(string)

```
public RuneCostNotFoundException(string message)
```

Parameters

`message` string

Class RuneInfosIsEmptyException

Namespace: Nekoyume.Model.Rune

Assembly: Lib9c.dll

```
[Serializable]
public class RuneInfosIsEmptyException : Exception, ISerializable
```

Inheritance

object ← Exception ← RuneInfosIsEmptyException

Implements

ISerializable

Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Constructors

RuneInfosIsEmptyException(SerializationInfo, StreamingContext)

```
protected RuneInfosIsEmptyException(SerializationInfo info,
StreamingContext context)
```

Parameters

`info` SerializationInfo

`context` StreamingContext

RuneInfosIsEmptyException(string)

```
public RuneInfosIsEmptyException(string message)
```

Parameters

`message` string

Class RuneListNotFoundException

Namespace: Nekoyume.Model.Rune

Assembly: Lib9c.dll

```
[Serializable]
public class RuneListNotFoundException : Exception, ISerializable
```

Inheritance

object ← Exception ← RuneListNotFoundException

Implements

ISerializable

Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Constructors

RuneListNotFoundException(SerializationInfo, StreamingContext)

```
protected RuneListNotFoundException(SerializationInfo info,
StreamingContext context)
```

Parameters

`info` SerializationInfo

`context` StreamingContext

RuneListNotFoundException(string)

```
public RuneListNotFoundException(string message)
```

Parameters

`message` string

Class RuneNotFoundException

Namespace: Nekoyume.Model.Rune

Assembly: Lib9c.dll

```
[Serializable]
public class RuneNotFoundException : Exception, ISerializable
```

Inheritance

object ← Exception ← RuneNotFoundException

Implements

ISerializable

Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Constructors

RuneNotFoundException(SerializationInfo, StreamingContext)

```
protected RuneNotFoundException(SerializationInfo info, StreamingContext context)
```

Parameters

`info` SerializationInfo

`context` StreamingContext

RuneNotFoundException(string)

```
public RuneNotFoundException(string message)
```

Parameters

`message` string

Class RuneSlot

Namespace: Nekoyume.Model.Rune

Assembly: Lib9c.dll

```
public class RuneSlot : IState
```

Inheritance

object ← RuneSlot

Implements

IState

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

RuneSlot(List)

```
public RuneSlot(List serialized)
```

Parameters

`serialized` List

RuneSlot(int, RuneSlotType, RuneType, bool)

```
public RuneSlot(int index, RuneSlotType runeSlotType, RuneType runeType,
bool isLock)
```

Parameters

`index` int

`runeSlotType` RuneSlotType

`runeType` RuneType

`isLock` bool

Properties

Index

```
public int Index { get; }
```

Property Value

int

IsLock

```
public bool IsLock { get; }
```

Property Value

bool

RuneId

```
public int? RuneId { get; }
```

Property Value

int?

RuneSlotType

```
public RuneSlotType RuneSlotType { get; }
```

Property Value

RuneSlotType

RuneType

```
public RuneType RuneType { get; }
```

Property Value

RuneType

Methods

Equip(int)

```
public void Equip(int runeId)
```

Parameters

`runeId` int

Serialize()

```
public IValue Serialize()
```

Returns

IValue

Unequip()

```
public void Unequip()
```

Unlock()

```
public void Unlock()
```

Class RuneStateNotFoundException

Namespace: Nekoyume.Model.Rune

Assembly: Lib9c.dll

```
[Serializable]
public class RuneStateNotFoundException : Exception, ISerializable
```

Inheritance

object ← Exception ← RuneStateNotFoundException

Implements

ISerializable

Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Constructors

RuneStateNotFoundException(SerializationInfo, StreamingContext)

```
protected RuneStateNotFoundException(SerializationInfo info,
StreamingContext context)
```

Parameters

`info` SerializationInfo

`context` StreamingContext

RuneStateNotFoundException(string)

```
public RuneStateNotFoundException(string message)
```

Parameters

`message` string

Class SlotIsAlreadyUnlockedException

Namespace: Nekoyume.Model.Rune

Assembly: Lib9c.dll

```
[Serializable]
public class SlotIsAlreadyUnlockedException : Exception, ISerializable
```

Inheritance

object ← Exception ← SlotIsAlreadyUnlockedException

Implements

ISerializable

Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Constructors

SlotIsAlreadyUnlockedException(SerializationInfo, StreamingContext)

```
protected SlotIsAlreadyUnlockedException(SerializationInfo info,
StreamingContext context)
```

Parameters

`info` SerializationInfo

`context` StreamingContext

SlotIsAlreadyUnlockedException(string)

```
public SlotIsAlreadyUnlockedException(string message)
```

Parameters

`message` string

Class SlotIsLockedException

Namespace: Nekoyume.Model.Rune

Assembly: Lib9c.dll

```
[Serializable]
public class SlotIsLockedException : Exception, ISerializable
```

Inheritance

object ← Exception ← SlotIsLockedException

Implements

ISerializable

Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Constructors

SlotIsLockedException(SerializationInfo, StreamingContext)

```
protected SlotIsLockedException(SerializationInfo info, StreamingContext context)
```

Parameters

`info` SerializationInfo

`context` StreamingContext

SlotIsLockedException(string)

```
public SlotIsLockedException(string message)
```

Parameters

`message` string

Class SlotNotFoundException

Namespace: Nekoyume.Model.Rune

Assembly: Lib9c.dll

```
[Serializable]
public class SlotNotFoundException : Exception, ISerializable
```

Inheritance

object ← Exception ← SlotNotFoundException

Implements

ISerializable

Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Constructors

SlotNotFoundException(SerializationInfo, StreamingContext)

```
protected SlotNotFoundException(SerializationInfo info, StreamingContext context)
```

Parameters

`info` SerializationInfo

`context` StreamingContext

SlotNotFoundException(string)

```
public SlotNotFoundException(string message)
```

Parameters

`message` string

Class SlotRuneTypeException

Namespace: Nekoyume.Model.Rune

Assembly: Lib9c.dll

```
[Serializable]
public class SlotRuneTypeException : Exception, ISerializable
```

Inheritance

object ← Exception ← SlotRuneTypeException

Implements

ISerializable

Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Constructors

SlotRuneTypeException(SerializationInfo, StreamingContext)

```
protected SlotRuneTypeException(SerializationInfo info, StreamingContext context)
```

Parameters

`info` SerializationInfo

`context` StreamingContext

SlotRuneTypeException(string)

```
public SlotRuneTypeException(string message)
```

Parameters

`message` string

Class TryCountIsZeroException

Namespace: Nekoyume.Model.Rune

Assembly: Lib9c.dll

```
[Serializable]
public class TryCountIsZeroException : Exception, ISerializable
```

Inheritance

object ← Exception ← TryCountIsZeroException

Implements

ISerializable

Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Constructors

TryCountIsZeroException(SerializationInfo, StreamingContext)

```
public TryCountIsZeroException(SerializationInfo info, StreamingContext context)
```

Parameters

`info` SerializationInfo

`context` StreamingContext

TryCountIsZeroException(string)

```
public TryCountIsZeroException(string msg)
```

Parameters

`msg` string

Namespace Nekoyume.Model.Skill

Classes

AreaAttack

AttackSkill

BlowAttack

BuffRemovalAttack

BuffSkill

DoubleAttack

FullBuffRemovalAttack
An attack skill that deals damage and removes all positive stat buffs from the target. Unlike BuffRemovalAttack which removes only the most recent buff, this skill removes every positive stat buff at once.

HealSkill

NormalAttack

ShatterStrike

Skill
Abstract base class for all skills in the game. Provides common functionality for skill execution and serialization.

SkillFactory

SkillTargetTypeExtension

UnexpectedOperationException

Structs

SkillCustomField

Interfaces

ISkill

Enums

ActionBuffType

SkillCategory

SkillTargetType

SkillType

Enum ActionBuffType

Namespace: Nekoyume.Model.Skill

Assembly: Lib9c.dll

```
public enum ActionBuffType
```

Extension Methods
StateExtensions.Serialize(Enum)

Fields

`Bleed = 0`

`Dispel = 4`

`Focus = 3`

`IceShield = 5`

`Stun = 1`

`Vampiric = 2`

Class AreaAttack

Namespace: Nekoyume.Model.Skill

Assembly: Lib9c.dll

```
[Serializable]
public class AreaAttack : AttackSkill, IState, ISkill
```

Inheritance

object ← Skill ← AttackSkill ← AreaAttack

Implements

IState, ISkill

Inherited Members

AttackSkill.ProcessDamage(CharacterBase, int, bool, bool) , AttackSkill.GetMultiplier(int, decimal) , Skill.SkillRow , Skill.Power , Skill.Chance , Skill.StatPowerRatio , Skill.ReferencedStatType , Skill.CustomField , Skill.Equals(Skill) , Skill.Equals(object) , Skill.GetHashCode() , Skill.ProcessBuff(CharacterBase, int, IEnumerable<Buff>, bool) , Skill.Update(int, long, int) , Skill.IsBuff() , Skill.IsDebuff() , Skill.Serialize() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

AreaAttack(Row, long, int, int, StatType)

```
public AreaAttack(SkillSheet.Row skillRow, long power, int chance, int
statPowerRatio, StatType referencedStatType)
```

Parameters

`skillRow` SkillSheet.Row

`power` long

`chance` int

`statPowerRatio` int

`referencedStatType` StatType

Methods

Use(CharacterBase, int, IEnumerable<Buff>, bool)

```
public override Skill Use(CharacterBase caster, int simulatorWaveTurn,
IEnumerable<Buff> buffs, bool copyCharacter)
```

Parameters

`caster` CharacterBase

`simulatorWaveTurn` int

`buffs` IEnumerable<Buff>

`copyCharacter` bool

Returns

Skill

Class AttackSkill

Namespace: Nekoyume.Model.Skill

Assembly: Lib9c.dll

```
[Serializable]
public abstract class AttackSkill : Skill, IState, ISkill
```

Inheritance

object ← Skill ← AttackSkill

Implements

IState, ISkill

Derived

AreaAttack, BlowAttack, BuffRemovalAttack, DoubleAttack, FullBuffRemovalAttack, NormalAttack, ShatterStrike

Inherited Members

Skill.SkillRow , Skill.Power , Skill.Chance , Skill.StatPowerRatio , Skill.ReferencedStatType , Skill.CustomField , Skill.Use(CharacterBase, int, IEnumerable<Buff>, bool) , Skill.Equals(Skill) , Skill.Equals(object) , Skill.GetHashCode() , Skill.ProcessBuff(CharacterBase, int, IEnumerable<Buff>, bool) , Skill.Update(int, long, int) , Skill.IsBuff() , Skill.IsDebuff() , Skill.Serialize() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

AttackSkill(Row, long, int, int, StatType)

```
protected AttackSkill(SkillSheet.Row skillRow, long power, int chance, int
statPowerRatio, StatType referencedStatType)
```

Parameters

`skillRow` SkillSheet.Row

`power` long

`chance` int

`statPowerRatio` int

`referencedStatType` StatType

Methods

GetMultiplier(int, decimal)

Calculates damage multipliers for multi-hit skills. For single hit skills, returns the total damage as is. For multi-hit skills, distributes the total damage across hits with the last hit dealing 30% more damage. This creates a pattern where the final hit is more impactful than the preceding hits.

```
protected static decimal[] GetMultiplier(int hitCount, decimal totalDamage)
```

Parameters

`hitCount` int

Number of hits for the skill

`totalDamage` decimal

Total damage to be distributed across hits

Returns

decimal[]

Array of damage multipliers for each hit

ProcessDamage(CharacterBase, int, bool, bool)

Processes damage calculation for attack skills with improved large value handling. Uses NumberConversionHelper.SafeDecimalToInt64 to prevent overflow when dealing with large

stat values. Implements DamageHelper.GetFinalDefense and DamageHelper.GetReducedDamage for consistent damage calculation.

todo: 캐릭터 스탯에 반영된 버프 효과가 스킬의 순수 데미지에는 영향을 주지 않는 로직. todo: 타겟의 회피 여부와 상관없이 버프의 확률로 발동되고 있음. 고민이 필요함.

```
protected IEnumerable<Skill.SkillInfo> ProcessDamage(CharacterBase caster, int
simulatorWaveTurn, bool isNormalAttack = false, bool copyCharacter = true)
```

Parameters

`caster` CharacterBase

The character casting the skill

`simulatorWaveTurn` int

Current wave turn in the battle simulator

`isNormalAttack` bool

Whether this is a normal attack

`copyCharacter` bool

Whether to copy the target character for skill info

Returns

IEnumerable<Skill.SkillInfo>

Collection of skill information including damage and effects

Class BlowAttack

Namespace: Nekoyume.Model.Skill

Assembly: Lib9c.dll

```
[Serializable]
public class BlowAttack : AttackSkill, IState, ISkill
```

Inheritance

object ← Skill ← AttackSkill ← BlowAttack

Implements

IState, ISkill

Inherited Members

AttackSkill.ProcessDamage(CharacterBase, int, bool, bool) , AttackSkill.GetMultiplier(int, decimal) , Skill.SkillRow , Skill.Power , Skill.Chance , Skill.StatPowerRatio , Skill.ReferencedStatType , Skill.CustomField , Skill.Equals(Skill) , Skill.Equals(object) , Skill.GetHashCode() , Skill.ProcessBuff(CharacterBase, int, IEnumerable<Buff>, bool) , Skill.Update(int, long, int) , Skill.IsBuff() , Skill.IsDebuff() , Skill.Serialize() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

BlowAttack(Row, long, int, int, StatType)

```
public BlowAttack(SkillSheet.Row skillRow, long power, int chance, int
statPowerRatio, StatType referencedStatType)
```

Parameters

`skillRow` SkillSheet.Row

`power` long

`chance` int

`statPowerRatio` int

`referencedStatType` StatType

Methods

Use(CharacterBase, int, IEnumerable<Buff>, bool)

```
public override Skill Use(CharacterBase caster, int simulatorWaveTurn,
IEnumerable<Buff> buffs, bool copyCharacter)
```

Parameters

`caster` CharacterBase

`simulatorWaveTurn` int

`buffs` IEnumerable<Buff>

`copyCharacter` bool

Returns

Skill

Class BuffRemovalAttack

Namespace: Nekoyume.Model.Skill

Assembly: Lib9c.dll

```
[Serializable]
public class BuffRemovalAttack : AttackSkill, IState, ISkill
```

Inheritance

object ← Skill ← AttackSkill ← BuffRemovalAttack

Implements

IState, ISkill

Inherited Members

AttackSkill.ProcessDamage(CharacterBase, int, bool, bool) , AttackSkill.GetMultiplier(int, decimal) , Skill.SkillRow , Skill.Power , Skill.Chance , Skill.StatPowerRatio , Skill.ReferencedStatType , Skill.CustomField , Skill.Equals(Skill) , Skill.Equals(object) , Skill.GetHashCode() , Skill.ProcessBuff(CharacterBase, int, IEnumerable<Buff>, bool) , Skill.Update(int, long, int) , Skill.IsBuff() , Skill.IsDebuff() , Skill.Serialize() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

BuffRemovalAttack(Row, long, int, int, StatType)

```
public BuffRemovalAttack(SkillSheet.Row skillRow, long power, int chance, int
statPowerRatio, StatType referencedStatType)
```

Parameters

`skillRow` SkillSheet.Row

`power` long

`chance` int

`statPowerRatio` int

`referencedStatType` StatType

Methods

Use(CharacterBase, int, IEnumerable<Buff>, bool)

```
public override Skill Use(CharacterBase caster, int simulatorWaveTurn,
IEnumerable<Buff> buffs, bool copyCharacter)
```

Parameters

`caster` CharacterBase

`simulatorWaveTurn` int

`buffs` IEnumerable<Buff>

`copyCharacter` bool

Returns

Skill

Class BuffSkill

Namespace: Nekoyume.Model.Skill

Assembly: Lib9c.dll

```
[Serializable]
public class BuffSkill : Skill, IState, ISkill
```

Inheritance

object ← Skill ← BuffSkill

Implements

IState, ISkill

Inherited Members

Skill.SkillRow , Skill.Power , Skill.Chance , Skill.StatPowerRatio , Skill.ReferencedStatType , Skill.CustomField , Skill.Equals(Skill) , Skill.Equals(object) , Skill.GetHashCode() , Skill.ProcessBuff(CharacterBase, int, IEnumerable<Buff>, bool) , Skill.Update(int, long, int) , Skill.IsBuff() , Skill.IsDebuff() , Skill.Serialize() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

BuffSkill(Row, long, int, int, StatType)

```
public BuffSkill(SkillSheet.Row skillRow, long power, int chance, int
statPowerRatio, StatType referencedStatType)
```

Parameters

`skillRow` SkillSheet.Row

`power` long

`chance` int

`statPowerRatio` int

`referencedStatType` StatType

Methods

Use(CharacterBase, int, IEnumerable<Buff>, bool)

```
public override Skill Use(CharacterBase caster, int simulatorWaveTurn,
IEnumerable<Buff> buffs, bool copyCharacter)
```

Parameters

`caster` CharacterBase

`simulatorWaveTurn` int

`buffs` IEnumerable<Buff>

`copyCharacter` bool

Returns

Skill

Class DoubleAttack

Namespace: Nekoyume.Model.Skill

Assembly: Lib9c.dll

```
[Serializable]
public class DoubleAttack : AttackSkill, IState, ISkill
```

Inheritance

object ← Skill ← AttackSkill ← DoubleAttack

Implements

IState, ISkill

Inherited Members

AttackSkill.ProcessDamage(CharacterBase, int, bool, bool) , AttackSkill.GetMultiplier(int, decimal) , Skill.SkillRow , Skill.Power , Skill.Chance , Skill.StatPowerRatio , Skill.ReferencedStatType , Skill.CustomField , Skill.Equals(Skill) , Skill.Equals(object) , Skill.GetHashCode() , Skill.ProcessBuff(CharacterBase, int, IEnumerable<Buff>, bool) , Skill.Update(int, long, int) , Skill.IsBuff() , Skill.IsDebuff() , Skill.Serialize() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

DoubleAttack(Row, long, int, int, StatType)

```
public DoubleAttack(SkillSheet.Row skillRow, long power, int chance, int
statPowerRatio, StatType referencedStatType)
```

Parameters

`skillRow` SkillSheet.Row

`power` long

`chance` int

`statPowerRatio` int

`referencedStatType` StatType

Methods

Use(CharacterBase, int, IEnumerable<Buff>, bool)

```
public override Skill Use(CharacterBase caster, int simulatorWaveTurn,
IEnumerable<Buff> buffs, bool copyCharacter)
```

Parameters

`caster` CharacterBase

`simulatorWaveTurn` int

`buffs` IEnumerable<Buff>

`copyCharacter` bool

Returns

Skill

Class FullBuffRemovalAttack

Namespace: Nekoyume.Model.Skill

Assembly: Lib9c.dll

An attack skill that deals damage and removes all positive stat buffs from the target. Unlike BuffRemovalAttack which removes only the most recent buff, this skill removes every positive stat buff at once.

```
[Serializable]
public class FullBuffRemovalAttack : AttackSkill, IState, ISkill
```

Inheritance

object ← Skill ← AttackSkill ← FullBuffRemovalAttack

Implements

IState, ISkill

Inherited Members

AttackSkill.ProcessDamage(CharacterBase, int, bool, bool) , AttackSkill.GetMultiplier(int, decimal) , Skill.SkillRow , Skill.Power , Skill.Chance , Skill.StatPowerRatio , Skill.ReferencedStatType , Skill.CustomField , Skill.Equals(Skill) , Skill.Equals(object) , Skill.GetHashCode() , Skill.ProcessBuff(CharacterBase, int, IEnumerable<Buff>, bool) , Skill.Update(int, long, int) , Skill.IsBuff() , Skill.IsDebuff() , Skill.Serialize() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

FullBuffRemovalAttack(Row, long, int, int, StatType)

```
public FullBuffRemovalAttack(SkillSheet.Row skillRow, long power, int chance, int
statPowerRatio, StatType referencedStatType)
```

Parameters

`skillRow` SkillSheet.Row

`power` long

`chance` int

`statPowerRatio` int

`referencedStatType` StatType

Methods

Use(CharacterBase, int, IEnumerable<Buff>, bool)

```
public override Skill Use(CharacterBase caster, int simulatorWaveTurn,
IEnumerable<Buff> buffs, bool copyCharacter)
```

Parameters

`caster` CharacterBase

`simulatorWaveTurn` int

`buffs` IEnumerable<Buff>

`copyCharacter` bool

Returns

Skill

Class HealSkill

Namespace: Nekoyume.Model.Skill

Assembly: Lib9c.dll

```
[Serializable]
public class HealSkill : Skill, IState, ISkill
```

Inheritance

object ← Skill ← HealSkill

Implements

IState, ISkill

Inherited Members

Skill.SkillRow , Skill.Power , Skill.Chance , Skill.StatPowerRatio , Skill.ReferencedStatType , Skill.CustomField , Skill.Equals(Skill) , Skill.Equals(object) , Skill.GetHashCode() , Skill.ProcessBuff(CharacterBase, int, IEnumerable<Buff>, bool) , Skill.Update(int, long, int) , Skill.IsBuff() , Skill.IsDebuff() , Skill.Serialize() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

HealSkill(Row, long, int, int, StatType)

```
public HealSkill(SkillSheet.Row skillRow, long power, int chance, int
statPowerRatio, StatType referencedStatType)
```

Parameters

`skillRow` SkillSheet.Row

`power` long

`chance` int

`statPowerRatio` int

`referencedStatType` StatType

Methods

ProcessHeal(CharacterBase, int, bool)

```
protected IEnumerable<Skill.SkillInfo> ProcessHeal(CharacterBase caster, int
simulatorWaveTurn, bool copyCharacter)
```

Parameters

`caster` CharacterBase

`simulatorWaveTurn` int

`copyCharacter` bool

Returns

IEnumerable<Skill.SkillInfo>

Use(CharacterBase, int, IEnumerable<Buff>, bool)

```
public override Skill Use(CharacterBase caster, int simulatorWaveTurn,
IEnumerable<Buff> buffs, bool copyCharacter)
```

Parameters

`caster` CharacterBase

`simulatorWaveTurn` int

`buffs` IEnumerable<Buff>

`copyCharacter` bool

Returns

Skill

Interface ISkill

Namespace: Nekoyume.Model.Skill

Assembly: Lib9c.dll

```
public interface ISkill
```

Properties

Chance

```
int Chance { get; }
```

Property Value

int

CustomField

```
SkillCustomField? CustomField { get; }
```

Property Value

SkillCustomField?

Power

Determines damage of `AttackSkill`. Determines effect of `BuffSkill`.

```
long Power { get; }
```

Property Value

long

ReferencedStatType

```
StatType ReferencedStatType { get; }
```

Property Value

StatType

SkillRow

```
SkillSheet.Row SkillRow { get; }
```

Property Value

SkillSheet.Row

StatPowerRatio

```
int StatPowerRatio { get; }
```

Property Value

int

Methods

IsBuff()

```
bool IsBuff()
```

Returns

bool

IsDebuff()

```
bool IsDebuff()
```

Returns

bool

Update(int, long, int)

```
void Update(int chance, long power, int statPowerRatio)
```

Parameters

`chance` int

`power` long

`statPowerRatio` int

Class NormalAttack

Namespace: Nekoyume.Model.Skill

Assembly: Lib9c.dll

```
[Serializable]
public class NormalAttack : AttackSkill, IState, ISkill
```

Inheritance

object ← Skill ← AttackSkill ← NormalAttack

Implements

IState, ISkill

Inherited Members

AttackSkill.ProcessDamage(CharacterBase, int, bool, bool) , AttackSkill.GetMultiplier(int, decimal) , Skill.SkillRow , Skill.Power , Skill.Chance , Skill.StatPowerRatio , Skill.ReferencedStatType , Skill.CustomField , Skill.Equals(Skill) , Skill.Equals(object) , Skill.GetHashCode() , Skill.ProcessBuff(CharacterBase, int, IEnumerable<Buff>, bool) , Skill.Update(int, long, int) , Skill.IsBuff() , Skill.IsDebuff() , Skill.Serialize() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

NormalAttack(Row, long, int, int, StatType)

```
public NormalAttack(SkillSheet.Row skillRow, long power, int chance, int
statPowerRatio, StatType referencedStatType)
```

Parameters

`skillRow` SkillSheet.Row

`power` long

`chance` int

`statPowerRatio` int

`referencedStatType` StatType

Methods

Use(CharacterBase, int, IEnumerable<Buff>, bool)

```
public override Skill Use(CharacterBase caster, int simulatorWaveTurn,
IEnumerable<Buff> buffs, bool copyCharacter)
```

Parameters

`caster` CharacterBase

`simulatorWaveTurn` int

`buffs` IEnumerable<Buff>

`copyCharacter` bool

Returns

Skill

Class ShatterStrike

Namespace: Nekoyume.Model.Skill

Assembly: Lib9c.dll

```
[Serializable]
public class ShatterStrike : AttackSkill, IState, ISkill
```

Inheritance

object ← Skill ← AttackSkill ← ShatterStrike

Implements

IState, ISkill

Inherited Members

AttackSkill.ProcessDamage(CharacterBase, int, bool, bool) , AttackSkill.GetMultiplier(int, decimal) , Skill.SkillRow , Skill.Power , Skill.Chance , Skill.StatPowerRatio , Skill.ReferencedStatType , Skill.CustomField , Skill.Equals(Skill) , Skill.Equals(object) , Skill.GetHashCode() , Skill.ProcessBuff(CharacterBase, int, IEnumerable<Buff>, bool) , Skill.Update(int, long, int) , Skill.IsBuff() , Skill.IsDebuff() , Skill.Serialize() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

ShatterStrike(Row, long, int, int, StatType)

```
public ShatterStrike(SkillSheet.Row skillRow, long power, int chance, int
statPowerRatio, StatType referencedStatType)
```

Parameters

`skillRow` SkillSheet.Row

`power` long

`chance` int

`statPowerRatio` int

`referencedStatType` StatType

Methods

Use(CharacterBase, int, IEnumerable<Buff>, bool)

```
public override Skill Use(CharacterBase caster, int simulatorWaveTurn,
IEnumerable<Buff> buffs, bool copyCharacter)
```

Parameters

`caster` CharacterBase

`simulatorWaveTurn` int

`buffs` IEnumerable<Buff>

`copyCharacter` bool

Returns

Skill

Class Skill

Namespace: Nekoyume.Model.Skill

Assembly: Lib9c.dll

Abstract base class for all skills in the game. Provides common functionality for skill execution and serialization.

```
[Serializable]
public abstract class Skill : IState, ISkill
```

Inheritance

object ← Skill

Implements

IState, ISkill

Derived

AttackSkill, BuffSkill, HealSkill

Inherited Members

object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Remarks

Skills are categorized by their type (Attack, Heal, Buff, Debuff) and category (NormalAttack, DoubleAttack, BlowAttack, etc.). Each skill has power, chance, and optional stat-based damage ratio.

Serialization format has been migrated from Dictionary to List for better performance and consistency. The new format stores data in the following order: [0] SkillRow (serialized SkillSheet.Row) [1] Power (long) [2] Chance (int) [3] StatPowerRatio (int, optional - only included if > 0) [4] ReferencedStatType (StatType, optional - only included if StatPowerRatio > 0)

Backward compatibility is maintained through SkillFactory.Deserialize() which automatically detects and handles both Dictionary and List formats.

Constructors

Skill(Row, long, int, int, StatType)

```
protected Skill(SkillSheet.Row skillRow, long power, int chance, int statPowerRatio,
StatType referencedStatType)
```

Parameters

`skillRow` SkillSheet.Row

`power` long

`chance` int

`statPowerRatio` int

`referencedStatType` StatType

Properties

Chance

```
public int Chance { get; }
```

Property Value

int

CustomField

```
public SkillCustomField? CustomField { get; set; }
```

Property Value

SkillCustomField?

Power

Determines damage of `AttackSkill`. Determines effect of `BuffSkill`.

```
public long Power { get; }
```

Property Value

long

ReferencedStatType

```
public StatType ReferencedStatType { get; }
```

Property Value

StatType

SkillRow

```
public SkillSheet.Row SkillRow { get; }
```

Property Value

SkillSheet.Row

StatPowerRatio

```
public int StatPowerRatio { get; }
```

Property Value

int

Methods

Equals(Skill)

```
protected bool Equals(Skill other)
```

Parameters

`other` Skill

Returns

bool

Equals(object)

Determines whether the specified object is equal to the current object.

```
public override bool Equals(object obj)
```

Parameters

`obj` object

The object to compare with the current object.

Returns

bool

true if the specified object is equal to the current object; otherwise, false.

GetHashCode()

Serves as the default hash function.

```
public override int GetHashCode()
```

Returns

int

A hash code for the current object.

IsBuff()

Determines if this skill is a buff skill.

```
public bool IsBuff()
```

Returns

bool

True if the skill type is Buff, false otherwise.

IsDebuff()

Determines if this skill is a debuff skill.

```
public bool IsDebuff()
```

Returns

bool

True if the skill type is Debuff, false otherwise.

ProcessBuff(CharacterBase, int, IEnumerable<Buff>, bool)

```
protected IEnumerable<Skill.SkillInfo> ProcessBuff(CharacterBase caster, int
simulatorWaveTurn, IEnumerable<Buff> buffs, bool copyCharacter)
```

Parameters

`caster` CharacterBase

`simulatorWaveTurn` int

`buffs` IEnumerable<Buff>

`copyCharacter` bool

Returns

IEnumerable<Skill.SkillInfo>

Serialize()

Serializes the skill to a List format for storage and transmission.

```
public IValue Serialize()
```

Returns

IValue

A List containing the serialized skill data in the following order: [0] SkillRow (serialized SkillSheet.Row) [1] Power (long) [2] Chance (int) [3] StatPowerRatio (int, optional - only included if > 0) [4] ReferencedStatType (StatType, optional - only included if StatPowerRatio > 0)

Remarks

This method implements the new List-based serialization format for better performance and consistency. The format is more compact than the previous Dictionary format and maintains all necessary data.

Optional fields (StatPowerRatio and ReferencedStatType) are only included when they have non-default values to minimize serialization overhead.

For backward compatibility, use SkillFactory.Deserialize() which can handle both the new List format and the legacy Dictionary format.

Update(int, long, int)

```
public void Update(int chance, long power, int statPowerRatio)
```

Parameters

`chance` int

`power` long

`statPowerRatio` int

Use(CharacterBase, int, IEnumerable<Buff>, bool)

```
public abstract Skill Use(CharacterBase caster, int simulatorWaveTurn,
IEnumerable<Buff> buffs, bool copyCharacter)
```

Parameters

`caster` CharacterBase

`simulatorWaveTurn` int

`buffs` IEnumerable<Buff>

`copyCharacter` bool

Returns

Skill

Enum SkillCategory

Namespace: Nekoyume.Model.Skill

Assembly: Lib9c.dll

```
public enum SkillCategory
```

Extension Methods
StateExtensions.Serialize(Enum)

Fields

`AreaAttack = 3`

`AttackBuff = 8`

`BlowAttack = 1`

`Buff = 15`

`BuffRemovalAttack = 4`

`CriticalBuff = 10`

`CriticalDamageBuff = 14`

`DamageReductionBuff = 13`

`Debuff = 16`

`DefenseBuff = 9`

`Dispel = 19`

`DoubleAttack = 2`

`Focus = 18`

`FullBuffRemovalAttack = 20`

Attack enemy and remove ALL positive stat buffs.

`HPBuff = 7`

```
Heal = 6

HitBuff = 11

NormalAttack = 0

ShatterStrike = 5

SpeedBuff = 12

TickDamage = 17
```

Struct SkillCustomField

Namespace: Nekoyume.Model.Skill

Assembly: Lib9c.dll

```
public struct SkillCustomField
```

Inherited Members
ValueType.Equals(object) , ValueType.GetHashCode() , ValueType.ToString() , object.Equals(object, object) , object.GetType() , object.ReferenceEquals(object, object)

Properties

BuffDuration

```
public int BuffDuration { readonly get; set; }
```

Property Value

int

BuffValue

```
public long BuffValue { readonly get; set; }
```

Property Value

long

Class SkillFactory

Namespace: Nekoyume.Model.Skill

Assembly: Lib9c.dll

```
public static class SkillFactory
```

Inheritance

object ← SkillFactory

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Methods

Deserialize(IValue)

Deserializes a skill from serialized data. Supports both Dictionary and List formats for backward compatibility.

```
public static Skill Deserialize(IValue serialized)
```

Parameters

`serialized` IValue

The serialized skill data

Returns

Skill

The deserialized skill

Remarks

This method automatically detects the serialization format and delegates to the appropriate deserialization method. It supports both the new List-based format and the legacy Dictionary format to ensure backward compatibility with existing data.

List format (new): [SkillRow, Power, Chance, StatPowerRatio?, ReferencedStatType?]
Dictionary format (legacy): {"skillRow": ..., "power": ..., "chance": ..., "stat_power_ratio": ..., "referenced_stat_type": ...}

Exceptions

ArgumentException

Thrown when the serialization format is not supported

DeserializeFromDictionary(Dictionary)

Deserializes a skill from the legacy Dictionary-based serialization format. This method is maintained for backward compatibility with existing data.

```
[Obsolete("Use Deserialize(IValue) instead.")]
public static Skill DeserializeFromDictionary(Dictionary serialized)
```

Parameters

`serialized` Dictionary

The serialized skill data in Dictionary format

Returns

Skill

The deserialized skill

Remarks

Expected Dictionary format: { "skillRow": SkillSheet.Row, "power": long, "chance": int, "stat_power_ratio": int (optional), "referenced_stat_type": StatType (optional) }

This method is marked as obsolete and should only be used for backward compatibility. New code should use the List-based format through Deserialize(IValue) or

DeserializeFromList(List).

DeserializeFromList(List)

Deserializes a skill from the new List-based serialization format.

```
public static Skill DeserializeFromList(List serialized)
```

Parameters

`serialized` List

The serialized skill data in List format

Returns

Skill

The deserialized skill

Remarks

Expected List format: [SkillRow, Power, Chance, StatPowerRatio?, ReferencedStatType?] Where StatPowerRatio and ReferencedStatType are optional and only included if StatPowerRatio > 0.

This method handles the new List-based serialization format which is more compact and performant than the previous Dictionary format.

Get(Row, long, int, int, StatType)

```
public static Skill Get(SkillSheet.Row skillRow, long power, int chance, int
statPowerRatio, StatType referencedStatType)
```

Parameters

`skillRow` SkillSheet.Row

`power` long

`chance` int

`statPowerRatio` int

`referencedStatType` StatType

Returns

Skill

GetForArena(Row, long, int, int, StatType)

```
public static ArenaSkill GetForArena(SkillSheet.Row skillRow, long power, int
chance, int statPowerRatio, StatType referencedStatType)
```

Parameters

`skillRow` SkillSheet.Row

`power` long

`chance` int

`statPowerRatio` int

`referencedStatType` StatType

Returns

ArenaSkill

GetV1(Row, long, int)

```
[Obsolete("Use Get() instead.")]
public static Skill GetV1(SkillSheet.Row skillRow, long power, int chance)
```

Parameters

`skillRow` SkillSheet.Row

`power` long

`chance` int

Returns

Skill

SelectSkill(ItemSubType, CustomEquipmentCraftRecipeSkillSheet, EquipmentItemOptionSheet, SkillSheet, IRandom)

```
public static Skill SelectSkill(ItemSubType itemSubType,
CustomEquipmentCraftRecipeSkillSheet recipeSkillSheet, EquipmentItemOptionSheet
itemOptionSheet, SkillSheet skillSheet, IRandom random)
```

Parameters

`itemSubType` ItemSubType

`recipeSkillSheet` CustomEquipmentCraftRecipeSkillSheet

`itemOptionSheet` EquipmentItemOptionSheet

`skillSheet` SkillSheet

`random` IRandom

Returns

Skill

Enum SkillTargetType

Namespace: Nekoyume.Model.Skill

Assembly: Lib9c.dll

```
public enum SkillTargetType
```

Extension Methods
SkillTargetTypeExtension.GetTarget(SkillTargetType, CharacterBase) ,
StateExtensions.Serialize(Enum)

Fields

`Ally = 3`

`Enemies = 1`

`Enemy = 0`

`Self = 2`

Class SkillTargetTypeExtension

Namespace: Nekoyume.Model.Skill

Assembly: Lib9c.dll

```
public static class SkillTargetTypeExtension
```

Inheritance

object ← SkillTargetTypeExtension

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Methods

GetTarget(SkillTargetType, CharacterBase)

```
public static IEnumerable<CharacterBase> GetTarget(this SkillTargetType value,
CharacterBase caster)
```

Parameters

`value` SkillTargetType

`caster` CharacterBase

Returns

IEnumerable<CharacterBase>

Enum SkillType

Namespace: Nekoyume.Model.Skill

Assembly: Lib9c.dll

```
public enum SkillType
```

Extension Methods
StateExtensions.Serialize(Enum)

Fields

`Attack = 0`

`Buff = 2`

`Debuff = 3`

`Heal = 1`

Class UnexpectedOperationException

Namespace: Nekoyume.Model.Skill

Assembly: Lib9c.dll

```
public class UnexpectedOperationException : Exception, ISerializable
```

Inheritance

object ← Exception ← UnexpectedOperationException

Implements

ISerializable

Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Constructors

UnexpectedOperationException(string)

```
public UnexpectedOperationException(string message)
```

Parameters

`message` string

Namespace Nekoyume.Model.Skill.Arena

Classes

ArenaAreaAttack

ArenaAttackSkill

ArenaBlowAttack

ArenaBuffRemovalAttack

ArenaBuffSkill

ArenaDoubleAttack

ArenaFullBuffRemovalAttack

Arena variant of FullBuffRemovalAttack. Deals damage and removes all positive stat buffs from the target.

ArenaHealSkill

ArenaNormalAttack

ArenaShatterStrike

ArenaSkill

Class ArenaAreaAttack

Namespace: Nekoyume.Model.Skill.Arena

Assembly: Lib9c.dll

```
[Serializable]
public class ArenaAreaAttack : ArenaAttackSkill, ISkill
```

Inheritance

object ← ArenaSkill ← ArenaAttackSkill ← ArenaAreaAttack

Implements

ISkill

Inherited Members

ArenaAttackSkill.ProcessDamage(ArenaCharacter, ArenaCharacter, int, bool) , ArenaAttackSkill.GetMultiplier(int, decimal) , ArenaSkill.SkillRow , ArenaSkill.Power , ArenaSkill.Chance , ArenaSkill.StatPowerRatio , ArenaSkill.ReferencedStatType , ArenaSkill.CustomField , ArenaSkill.Equals(Skill) , ArenaSkill.Equals(object) , ArenaSkill.GetHashCode() , ArenaSkill.ProcessBuff(ArenaCharacter, ArenaCharacter, int, IEnumerable<Buff>) , ArenaSkill.Update(int, long, int) , ArenaSkill.IsBuff() , ArenaSkill.IsDebuff() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

ArenaAreaAttack(Row, long, int, int, StatType)

```
public ArenaAreaAttack(SkillSheet.Row skillRow, long power, int chance, int
statPowerRatio, StatType referencedStatType)
```

Parameters

`skillRow` SkillSheet.Row

`power` long

`chance` int

`statPowerRatio` int

`referencedStatType` StatType

Methods

Use(ArenaCharacter, ArenaCharacter, int, IEnumerable<Buff>)

```
public override ArenaSkill Use(ArenaCharacter caster, ArenaCharacter target, int
turn, IEnumerable<Buff> buffs)
```

Parameters

`caster` ArenaCharacter

`target` ArenaCharacter

`turn` int

`buffs` IEnumerable<Buff>

Returns

ArenaSkill

Class ArenaAttackSkill

Namespace: Nekoyume.Model.Skill.Arena

Assembly: Lib9c.dll

```
[Serializable]
public abstract class ArenaAttackSkill : ArenaSkill, ISkill
```

Inheritance

object ← ArenaSkill ← ArenaAttackSkill

Implements

ISkill

Derived

ArenaAreaAttack, ArenaBlowAttack, ArenaBuffRemovalAttack, ArenaDoubleAttack, ArenaFullBuffRemovalAttack, ArenaNormalAttack, ArenaShatterStrike

Inherited Members

ArenaSkill.SkillRow , ArenaSkill.Power , ArenaSkill.Chance , ArenaSkill.StatPowerRatio , ArenaSkill.ReferencedStatType , ArenaSkill.CustomField , ArenaSkill.Use(ArenaCharacter, ArenaCharacter, int, IEnumerable<Buff>) , ArenaSkill.Equals(Skill) , ArenaSkill.Equals(object) , ArenaSkill.GetHashCode() , ArenaSkill.ProcessBuff(ArenaCharacter, ArenaCharacter, int, IEnumerable<Buff>) , ArenaSkill.Update(int, long, int) , ArenaSkill.IsBuff() , ArenaSkill.IsDebuff() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

ArenaAttackSkill(Row, long, int, int, StatType)

```
protected ArenaAttackSkill(SkillSheet.Row skillRow, long power, int chance, int
statPowerRatio, StatType referencedStatType)
```

Parameters

`skillRow` SkillSheet.Row

`power` long

`chance` int

`statPowerRatio` int

`referencedStatType` StatType

Methods

GetMultiplier(int, decimal)

Calculates damage multipliers for multi-hit arena skills. For single hit skills, returns the total damage as is. For multi-hit skills, distributes the total damage across hits with the last hit dealing 30% more damage. This creates a pattern where the final hit is more impactful than the preceding hits.

```
protected static decimal[] GetMultiplier(int hitCount, decimal totalDamage)
```

Parameters

`hitCount` int

Number of hits for the arena skill

`totalDamage` decimal

Total damage to be distributed across hits

Returns

decimal[]

Array of damage multipliers for each hit

ProcessDamage(ArenaCharacter, ArenaCharacter, int, bool)

Processes damage calculation for arena attack skills with improved large value handling. Uses NumberConversionHelper.SafeDecimalToInt64 to prevent overflow when dealing with

large stat values. Implements DamageHelper.GetFinalDefense and DamageHelper.GetReducedDamage for consistent damage calculation.

```
protected IEnumerable<ArenaSkill.ArenaSkillInfo> ProcessDamage(ArenaCharacter
caster, ArenaCharacter target, int simulatorWaveTurn, bool isNormalAttack = false)
```

Parameters

`caster` ArenaCharacter

The arena character casting the skill

`target` ArenaCharacter

The target arena character

`simulatorWaveTurn` int

Current wave turn in the arena simulator

`isNormalAttack` bool

Whether this is a normal attack

Returns

IEnumerable<ArenaSkill.ArenaSkillInfo>

Collection of arena skill information including damage and effects

Class ArenaBlowAttack

Namespace: Nekoyume.Model.Skill.Arena

Assembly: Lib9c.dll

```
[Serializable]
public class ArenaBlowAttack : ArenaAttackSkill, ISkill
```

Inheritance

object ← ArenaSkill ← ArenaAttackSkill ← ArenaBlowAttack

Implements

ISkill

Inherited Members

ArenaAttackSkill.ProcessDamage(ArenaCharacter, ArenaCharacter, int, bool) , ArenaAttackSkill.GetMultiplier(int, decimal) , ArenaSkill.SkillRow , ArenaSkill.Power , ArenaSkill.Chance , ArenaSkill.StatPowerRatio , ArenaSkill.ReferencedStatType , ArenaSkill.CustomField , ArenaSkill.Equals(Skill) , ArenaSkill.Equals(object) , ArenaSkill.GetHashCode() , ArenaSkill.ProcessBuff(ArenaCharacter, ArenaCharacter, int, IEnumerable<Buff>) , ArenaSkill.Update(int, long, int) , ArenaSkill.IsBuff() , ArenaSkill.IsDebuff() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

ArenaBlowAttack(Row, long, int, int, StatType)

```
public ArenaBlowAttack(SkillSheet.Row skillRow, long power, int chance, int
statPowerRatio, StatType referencedStatType)
```

Parameters

`skillRow` SkillSheet.Row

`power` long

`chance` int

`statPowerRatio` int

`referencedStatType` StatType

Methods

Use(ArenaCharacter, ArenaCharacter, int, IEnumerable<Buff>)

```
public override ArenaSkill Use(ArenaCharacter caster, ArenaCharacter target, int
turn, IEnumerable<Buff> buffs)
```

Parameters

`caster` ArenaCharacter

`target` ArenaCharacter

`turn` int

`buffs` IEnumerable<Buff>

Returns

ArenaSkill

Class ArenaBuffRemovalAttack

Namespace: Nekoyume.Model.Skill.Arena

Assembly: Lib9c.dll

```
[Serializable]
public class ArenaBuffRemovalAttack : ArenaAttackSkill, ISkill
```

Inheritance

object ← ArenaSkill ← ArenaAttackSkill ← ArenaBuffRemovalAttack

Implements

ISkill

Inherited Members

ArenaAttackSkill.ProcessDamage(ArenaCharacter, ArenaCharacter, int, bool) , ArenaAttackSkill.GetMultiplier(int, decimal) , ArenaSkill.SkillRow , ArenaSkill.Power , ArenaSkill.Chance , ArenaSkill.StatPowerRatio , ArenaSkill.ReferencedStatType , ArenaSkill.CustomField , ArenaSkill.Equals(Skill) , ArenaSkill.Equals(object) , ArenaSkill.GetHashCode() , ArenaSkill.ProcessBuff(ArenaCharacter, ArenaCharacter, int, IEnumerable<Buff>) , ArenaSkill.Update(int, long, int) , ArenaSkill.IsBuff() , ArenaSkill.IsDebuff() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

ArenaBuffRemovalAttack(Row, long, int, int, StatType)

```
public ArenaBuffRemovalAttack(SkillSheet.Row skillRow, long power, int chance, int
statPowerRatio, StatType referencedStatType)
```

Parameters

`skillRow` SkillSheet.Row

`power` long

`chance` int

`statPowerRatio` int

`referencedStatType` StatType

Methods

Use(ArenaCharacter, ArenaCharacter, int, IEnumerable<Buff>)

```
public override ArenaSkill Use(ArenaCharacter caster, ArenaCharacter target, int turn, IEnumerable<Buff> buffs)
```

Parameters

`caster` ArenaCharacter

`target` ArenaCharacter

`turn` int

`buffs` IEnumerable<Buff>

Returns

ArenaSkill

Class ArenaBuffSkill

Namespace: Nekoyume.Model.Skill.Arena

Assembly: Lib9c.dll

```
[Serializable]
public class ArenaBuffSkill : ArenaSkill, ISkill
```

Inheritance

object ← ArenaSkill ← ArenaBuffSkill

Implements

ISkill

Inherited Members

ArenaSkill.SkillRow , ArenaSkill.Power , ArenaSkill.Chance , ArenaSkill.StatPowerRatio , ArenaSkill.ReferencedStatType , ArenaSkill.CustomField , ArenaSkill.Equals(Skill) , ArenaSkill.Equals(object) , ArenaSkill.GetHashCode() , ArenaSkill.ProcessBuff(ArenaCharacter, ArenaCharacter, int, IEnumerable<Buff>) , ArenaSkill.Update(int, long, int) , ArenaSkill.IsBuff() , ArenaSkill.IsDebuff() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

ArenaBuffSkill(Row, long, int, int, StatType)

```
public ArenaBuffSkill(SkillSheet.Row skillRow, long power, int chance, int
statPowerRatio, StatType referencedStatType)
```

Parameters

`skillRow` SkillSheet.Row

`power` long

`chance` int

`statPowerRatio` int

`referencedStatType` StatType

Methods

Use(ArenaCharacter, ArenaCharacter, int, IEnumerable<Buff>)

```
public override ArenaSkill Use(ArenaCharacter caster, ArenaCharacter target, int
turn, IEnumerable<Buff> buffs)
```

Parameters

`caster` ArenaCharacter

`target` ArenaCharacter

`turn` int

`buffs` IEnumerable<Buff>

Returns

ArenaSkill

Class ArenaDoubleAttack

Namespace: Nekoyume.Model.Skill.Arena

Assembly: Lib9c.dll

```
[Serializable]
public class ArenaDoubleAttack : ArenaAttackSkill, ISkill
```

Inheritance

object ← ArenaSkill ← ArenaAttackSkill ← ArenaDoubleAttack

Implements

ISkill

Inherited Members

ArenaAttackSkill.ProcessDamage(ArenaCharacter, ArenaCharacter, int, bool) , ArenaAttackSkill.GetMultiplier(int, decimal) , ArenaSkill.SkillRow , ArenaSkill.Power , ArenaSkill.Chance , ArenaSkill.StatPowerRatio , ArenaSkill.ReferencedStatType , ArenaSkill.CustomField , ArenaSkill.Equals(Skill) , ArenaSkill.Equals(object) , ArenaSkill.GetHashCode() , ArenaSkill.ProcessBuff(ArenaCharacter, ArenaCharacter, int, IEnumerable<Buff>) , ArenaSkill.Update(int, long, int) , ArenaSkill.IsBuff() , ArenaSkill.IsDebuff() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

ArenaDoubleAttack(Row, long, int, int, StatType)

```
public ArenaDoubleAttack(SkillSheet.Row skillRow, long power, int chance, int
statPowerRatio, StatType referencedStatType)
```

Parameters

`skillRow` SkillSheet.Row

`power` long

`chance` int

`statPowerRatio` int

`referencedStatType` StatType

Methods

Use(ArenaCharacter, ArenaCharacter, int, IEnumerable<Buff>)

```
public override ArenaSkill Use(ArenaCharacter caster, ArenaCharacter target, int
turn, IEnumerable<Buff> buffs)
```

Parameters

`caster` ArenaCharacter

`target` ArenaCharacter

`turn` int

`buffs` IEnumerable<Buff>

Returns

ArenaSkill

Class ArenaFullBuffRemovalAttack

Namespace: Nekoyume.Model.Skill.Arena

Assembly: Lib9c.dll

Arena variant of FullBuffRemovalAttack. Deals damage and removes all positive stat buffs from the target.

```
[Serializable]
public class ArenaFullBuffRemovalAttack : ArenaAttackSkill, ISkill
```

Inheritance

object ← ArenaSkill ← ArenaAttackSkill ← ArenaFullBuffRemovalAttack

Implements

ISkill

Inherited Members

ArenaAttackSkill.ProcessDamage(ArenaCharacter, ArenaCharacter, int, bool) , ArenaAttackSkill.GetMultiplier(int, decimal) , ArenaSkill.SkillRow , ArenaSkill.Power , ArenaSkill.Chance , ArenaSkill.StatPowerRatio , ArenaSkill.ReferencedStatType , ArenaSkill.CustomField , ArenaSkill.Equals(Skill) , ArenaSkill.Equals(object) , ArenaSkill.GetHashCode() , ArenaSkill.ProcessBuff(ArenaCharacter, ArenaCharacter, int, IEnumerable<Buff>) , ArenaSkill.Update(int, long, int) , ArenaSkill.IsBuff() , ArenaSkill.IsDebuff() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

ArenaFullBuffRemovalAttack(Row, long, int, int, StatType)

```
public ArenaFullBuffRemovalAttack(SkillSheet.Row skillRow, long power, int chance,
int statPowerRatio, StatType referencedStatType)
```

Parameters

`skillRow` SkillSheet.Row

`power` long

`chance` int

`statPowerRatio` int

`referencedStatType` StatType

Methods

Use(ArenaCharacter, ArenaCharacter, int, IEnumerable<Buff>)

```
public override ArenaSkill Use(ArenaCharacter caster, ArenaCharacter target, int
turn, IEnumerable<Buff> buffs)
```

Parameters

`caster` ArenaCharacter

`target` ArenaCharacter

`turn` int

`buffs` IEnumerable<Buff>

Returns

ArenaSkill

Class ArenaHealSkill

Namespace: Nekoyume.Model.Skill.Arena

Assembly: Lib9c.dll

```
[Serializable]
public class ArenaHealSkill : ArenaSkill, ISkill
```

Inheritance

object ← ArenaSkill ← ArenaHealSkill

Implements

ISkill

Inherited Members

ArenaSkill.SkillRow , ArenaSkill.Power , ArenaSkill.Chance , ArenaSkill.StatPowerRatio , ArenaSkill.ReferencedStatType , ArenaSkill.CustomField , ArenaSkill.Equals(Skill) , ArenaSkill.Equals(object) , ArenaSkill.GetHashCode() , ArenaSkill.ProcessBuff(ArenaCharacter, ArenaCharacter, int, IEnumerable<Buff>) , ArenaSkill.Update(int, long, int) , ArenaSkill.IsBuff() , ArenaSkill.IsDebuff() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

ArenaHealSkill(Row, long, int, int, StatType)

```
public ArenaHealSkill(SkillSheet.Row skillRow, long power, int chance, int
statPowerRatio, StatType referencedStatType)
```

Parameters

`skillRow` SkillSheet.Row

`power` long

`chance` int

`statPowerRatio` int

`referencedStatType` StatType

Methods

Use(ArenaCharacter, ArenaCharacter, int, IEnumerable<Buff>)

```
public override ArenaSkill Use(ArenaCharacter caster, ArenaCharacter target, int turn, IEnumerable<Buff> buffs)
```

Parameters

`caster` ArenaCharacter

`target` ArenaCharacter

`turn` int

`buffs` IEnumerable<Buff>

Returns

ArenaSkill

Class ArenaNormalAttack

Namespace: Nekoyume.Model.Skill.Arena

Assembly: Lib9c.dll

```
[Serializable]
public class ArenaNormalAttack : ArenaAttackSkill, ISkill
```

Inheritance

object ← ArenaSkill ← ArenaAttackSkill ← ArenaNormalAttack

Implements

ISkill

Inherited Members

ArenaAttackSkill.ProcessDamage(ArenaCharacter, ArenaCharacter, int, bool) , ArenaAttackSkill.GetMultiplier(int, decimal) , ArenaSkill.SkillRow , ArenaSkill.Power , ArenaSkill.Chance , ArenaSkill.StatPowerRatio , ArenaSkill.ReferencedStatType , ArenaSkill.CustomField , ArenaSkill.Equals(Skill) , ArenaSkill.Equals(object) , ArenaSkill.GetHashCode() , ArenaSkill.ProcessBuff(ArenaCharacter, ArenaCharacter, int, IEnumerable<Buff>) , ArenaSkill.Update(int, long, int) , ArenaSkill.IsBuff() , ArenaSkill.IsDebuff() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

ArenaNormalAttack(Row, long, int, int, StatType)

```
public ArenaNormalAttack(SkillSheet.Row skillRow, long power, int chance, int
statPowerRatio, StatType referencedStatType)
```

Parameters

`skillRow` SkillSheet.Row

`power` long

`chance` int

`statPowerRatio` int

`referencedStatType` StatType

Methods

Use(ArenaCharacter, ArenaCharacter, int, IEnumerable<Buff>)

```
public override ArenaSkill Use(ArenaCharacter caster, ArenaCharacter target, int
turn, IEnumerable<Buff> buffs)
```

Parameters

`caster` ArenaCharacter

`target` ArenaCharacter

`turn` int

`buffs` IEnumerable<Buff>

Returns

ArenaSkill

Class ArenaShatterStrike

Namespace: Nekoyume.Model.Skill.Arena

Assembly: Lib9c.dll

```
[Serializable]
public class ArenaShatterStrike : ArenaAttackSkill, ISkill
```

Inheritance

object ← ArenaSkill ← ArenaAttackSkill ← ArenaShatterStrike

Implements

ISkill

Inherited Members

ArenaAttackSkill.ProcessDamage(ArenaCharacter, ArenaCharacter, int, bool) , ArenaAttackSkill.GetMultiplier(int, decimal) , ArenaSkill.SkillRow , ArenaSkill.Power , ArenaSkill.Chance , ArenaSkill.StatPowerRatio , ArenaSkill.ReferencedStatType , ArenaSkill.CustomField , ArenaSkill.Equals(Skill) , ArenaSkill.Equals(object) , ArenaSkill.GetHashCode() , ArenaSkill.ProcessBuff(ArenaCharacter, ArenaCharacter, int, IEnumerable<Buff>) , ArenaSkill.Update(int, long, int) , ArenaSkill.IsBuff() , ArenaSkill.IsDebuff() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

ArenaShatterStrike(Row, long, int, int, StatType)

```
public ArenaShatterStrike(SkillSheet.Row skillRow, long power, int chance, int
statPowerRatio, StatType referencedStatType)
```

Parameters

`skillRow` SkillSheet.Row

`power` long

`chance` int

`statPowerRatio` int

`referencedStatType` StatType

Methods

Use(ArenaCharacter, ArenaCharacter, int, IEnumerable<Buff>)

```
public override ArenaSkill Use(ArenaCharacter caster, ArenaCharacter target, int
turn, IEnumerable<Buff> buffs)
```

Parameters

`caster` ArenaCharacter

`target` ArenaCharacter

`turn` int

`buffs` IEnumerable<Buff>

Returns

ArenaSkill

Class ArenaSkill

Namespace: Nekoyume.Model.Skill.Arena

Assembly: Lib9c.dll

```
[Serializable]
public abstract class ArenaSkill : ISkill
```

Inheritance

object ← ArenaSkill

Implements

ISkill

Derived

ArenaAttackSkill, ArenaBuffSkill, ArenaHealSkill

Inherited Members

object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

ArenaSkill(Row, long, int, int, StatType)

```
protected ArenaSkill(SkillSheet.Row skillRow, long power, int chance, int
statPowerRatio, StatType referencedStatType)
```

Parameters

`skillRow` SkillSheet.Row

`power` long

`chance` int

`statPowerRatio` int

`referencedStatType` StatType

Properties

Chance

```
public int Chance { get; }
```

Property Value

int

CustomField

```
public SkillCustomField? CustomField { get; set; }
```

Property Value

SkillCustomField?

Power

Determines damage of `AttackSkill`. Determines effect of `BuffSkill`.

```
public long Power { get; }
```

Property Value

long

ReferencedStatType

```
public StatType ReferencedStatType { get; }
```

Property Value

StatType

SkillRow

```
public SkillSheet.Row SkillRow { get; }
```

Property Value

SkillSheet.Row

StatPowerRatio

```
public int StatPowerRatio { get; }
```

Property Value

int

Methods

Equals(Skill)

```
protected bool Equals(Skill other)
```

Parameters

`other` Skill

Returns

bool

Equals(object)

Determines whether the specified object is equal to the current object.

```
public override bool Equals(object obj)
```

Parameters

`obj` object

The object to compare with the current object.

Returns

bool

true if the specified object is equal to the current object; otherwise, false.

GetHashCode()

Serves as the default hash function.

```
public override int GetHashCode()
```

Returns

int

A hash code for the current object.

IsBuff()

```
public bool IsBuff()
```

Returns

bool

IsDebuff()

```
public bool IsDebuff()
```

Returns

bool

ProcessBuff(ArenaCharacter, ArenaCharacter, int, IEnumerable<Buff>)

```
protected IEnumerable<ArenaSkill.ArenaSkillInfo> ProcessBuff(ArenaCharacter caster,
ArenaCharacter target, int turn, IEnumerable<Buff> buffs)
```

Parameters

`caster` ArenaCharacter

`target` ArenaCharacter

`turn` int

`buffs` IEnumerable<Buff>

Returns

IEnumerable<ArenaSkill.ArenaSkillInfo>

Update(int, long, int)

```
public void Update(int chance, long power, int statPowerRatio)
```

Parameters

`chance` int

`power` long

`statPowerRatio` int

Use(ArenaCharacter, ArenaCharacter, int, IEnumerable<Buff>)

```
public abstract ArenaSkill Use(ArenaCharacter caster, ArenaCharacter target, int
turn, IEnumerable<Buff> buffs)
```

Parameters

`caster` ArenaCharacter

`target` ArenaCharacter

`turn` int

`buffs` IEnumerable<Buff>

Returns

ArenaSkill

Namespace Nekoyume.Model.Stake

Classes

Contract

StakeStateUtils

Structs

StakeState

Class Contract

Namespace: Nekoyume.Model.Stake

Assembly: Lib9c.dll

```
public class Contract
```

Inheritance

object ← Contract

Inherited Members

object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

Contract(IValue)

```
public Contract(IValue serialized)
```

Parameters

`serialized` IValue

Contract(StakePolicySheet)

```
public Contract(StakePolicySheet stakePolicySheet)
```

Parameters

`stakePolicySheet` StakePolicySheet

Contract(string, string, long, long)

```
public Contract(string stakeRegularFixedRewardSheetTableName, string
stakeRegularRewardSheetTableName, long rewardInterval, long lockupInterval)
```

Parameters

`stakeRegularFixedRewardSheetTableName` string

`stakeRegularRewardSheetTableName` string

`rewardInterval` long

`lockupInterval` long

Fields

StakeRegularFixedRewardSheetPrefix

```
public const string StakeRegularFixedRewardSheetPrefix =
"StakeRegularFixedRewardSheet_"
```

Field Value

string

StakeRegularRewardSheetPrefix

```
public const string StakeRegularRewardSheetPrefix = "StakeRegularRewardSheet_"
```

Field Value

string

StateTypeName

```
public const string StateTypeName = "stake_contract"
```

Field Value

string

StateTypeVersion

```
public const long StateTypeVersion = 1
```

Field Value

long

Properties

LockupInterval

```
[Obsolete("Not used because of guild system")]
public long LockupInterval { get; }
```

Property Value

long

RewardInterval

```
public long RewardInterval { get; }
```

Property Value

long

StakeRegularFixedRewardSheetTableName

```
public string StakeRegularFixedRewardSheetTableName { get; }
```

Property Value

string

StakeRegularRewardSheetTableName

```
public string StakeRegularRewardSheetTableName { get; }
```

Property Value

string

Methods

Equals(Contract)

```
protected bool Equals(Contract other)
```

Parameters

`other` Contract

Returns

bool

Equals(object)

Determines whether the specified object is equal to the current object.

```
public override bool Equals(object obj)
```

Parameters

`obj` object

The object to compare with the current object.

Returns

bool

true if the specified object is equal to the current object; otherwise, false.

GetHashCode()

Serves as the default hash function.

```
public override int GetHashCode()
```

Returns

int

A hash code for the current object.

Serialize()

```
public List Serialize()
```

Returns

List

Struct StakeState

Namespace: Nekoyume.Model.Stake

Assembly: Lib9c.dll

```
public readonly struct StakeState : IState
```

Implements
IState

Inherited Members
ValueType.ToString() , object.Equals(object, object) , object.GetType() , object.ReferenceEquals(object, object)

Constructors

StakeState(IValue)

```
public StakeState(IValue serialized)
```

Parameters

`serialized` IValue

StakeState(Contract, long, long, int)

```
public StakeState(Contract contract, long startedBlockIndex, long receivedBlockIndex
= 0, int stateVersion = 3)
```

Parameters

`contract` Contract

`startedBlockIndex` long

`receivedBlockIndex` long

`stateVersion` int

StakeState(LegacyStakeState, Contract)

```
public StakeState(LegacyStakeState legacyStakeState, Contract contract)
```

Parameters

`legacyStakeState` LegacyStakeState

`contract` Contract

Fields

Contract

```
public readonly Contract Contract
```

Field Value

Contract

LatestStateTypeVersion

```
public const int LatestStateTypeVersion = 3
```

Field Value

int

ReceivedBlockIndex

```
public readonly long ReceivedBlockIndex
```

Field Value

long

StartedBlockIndex

```
public readonly long StartedBlockIndex
```

Field Value

long

StateTypeName

```
public const string StateTypeName = "stake_state"
```

Field Value

string

StateVersion

```
public readonly int StateVersion
```

Field Value

int

Properties

CancellableBlockIndex

```
public long CancellableBlockIndex { get; }
```

Property Value

long

ClaimableBlockIndex

```
public long ClaimableBlockIndex { get; }
```

Property Value

long

ClaimedBlockIndex

```
public long ClaimedBlockIndex { get; }
```

Property Value

long

Methods

DeriveAddress(Address)

```
public static Address DeriveAddress(Address address)
```

Parameters

`address` Address

Returns

Address

Equals(StakeState)

```
public bool Equals(StakeState other)
```

Parameters

`other` StakeState

Returns

bool

Equals(object)

Indicates whether this instance and a specified object are equal.

```
public override bool Equals(object obj)
```

Parameters

`obj` object

The object to compare with the current instance.

Returns

bool

true if `obj` and this instance are the same type and represent the same value; otherwise, false.

GetHashCode()

Returns the hash code for this instance.

```
public override int GetHashCode()
```

Returns

int

A 32-bit signed integer that is the hash code for this instance.

Serialize()

```
public IValue Serialize()
```

Returns

IValue

Operators

operator ==(StakeState, StakeState)

```
public static bool operator ==(StakeState left, StakeState right)
```

Parameters

`left` StakeState

`right` StakeState

Returns

bool

operator !=(StakeState, StakeState)

```
public static bool operator !=(StakeState left, StakeState right)
```

Parameters

`left` StakeState

`right` StakeState

Returns

bool

Class StakeStateUtils

Namespace: Nekoyume.Model.Stake

Assembly: Lib9c.dll

```
public static class StakeStateUtils
```

Inheritance

object ← StakeStateUtils

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Methods

MigrateV1ToV2(IValue, GameConfigState)

```
public static StakeState? MigrateV1ToV2(IValue serialized,
GameConfigState gameConfigState)
```

Parameters

`serialized` IValue

`gameConfigState` GameConfigState

Returns

StakeState?

TryMigrateV1ToV2(IValue, GameConfigState, out StakeState)

```
public static bool TryMigrateV1ToV2(IValue serialized, GameConfigState
gameConfigState, out StakeState stakeState)
```

Parameters

`serialized` IValue

`gameConfigState` GameConfigState

`stakeState` StakeState

Returns

bool

TryMigrateV1ToV2(IWorldState, Address, out StakeState)

```
public static bool TryMigrateV1ToV2(IWorldState state, Address stakeStateAddr, out
StakeState stakeState)
```

Parameters

`state` IWorldState

`stakeStateAddr` Address

`stakeState` StakeState

Returns

bool

TryMigrateV2ToV3(IActionContext, IWorld, Address, StakeState, out (IWorld world, StakeState newStakeState)?)

```
public static bool TryMigrateV2ToV3(IActionContext context, IWorld world,
Address stakeStateAddr, StakeState stakeState, out (IWorld world, StakeState
```

```
newStakeState)? result)
```

Parameters

`context` IActionContext

`world` IWorld

`stakeStateAddr` Address

`stakeState` StakeState

`result` (IWorld world, StakeState newStakeState)?

Returns

bool

Namespace Nekoyume.Model.Stat

Classes

CharacterStats

Stat is built with _baseStats based on level, _equipmentStats based on equipments, _consumableStats based on consumables, _buffStats based on buffs and _optionalStats for runes, etc... Stat of character is built with total of these stats.

DecimalStat

Represents a single character statistic with base and additional values. Supports both Dictionary and List serialization formats for backward compatibility.

Serialization Format: - Dictionary (Legacy): Uses key-value pairs for backward compatibility - List (New): Uses ordered list for better performance and smaller size

Field Order (List Format): 1. version - Serialization version number 2. statType - The type of statistic 3. baseValue - Base value of the statistic 4. additionalValue - Additional value of the statistic

StatMap

Represents a collection of character statistics. Supports both Dictionary and List serialization formats for backward compatibility.

Serialization Format: - Dictionary (Legacy): Uses key-value pairs for backward compatibility - List (New): Uses ordered list for better performance and smaller size

Field Order (List Format): 1. version - Serialization version number 2. stats - List of DecimalStat objects ordered by StatType

StatModifier

StatTypeComparer

StatTypeExtension

Stats

StatsMap

Interfaces

IBaseAndAdditionalStats

IStats

Enums

StatModifier.OperationType

StatType

Class CharacterStats

Namespace: Nekoyume.Model.Stat

Assembly: Lib9c.dll

Stat is built with _baseStats based on level, _equipmentStats based on equipments, _consumableStats based on consumables, _buffStats based on buffs and _optionalStats for runes, etc... Stat of character is built with total of these stats.

```
[Serializable]
public class CharacterStats : Stats, IStats, IBaseAndAdditionalStats, ICloneable
```

Inheritance

object ← Stats ← CharacterStats

Implements

IStats, IBaseAndAdditionalStats, ICloneable

Inherited Members

Stats._statMap , Stats.HP , Stats.ATK , Stats.DEF , Stats.CRI , Stats.HIT , Stats.SPD , Stats.DRV , Stats.DRR , Stats.CDMG , Stats.ArmorPenetration , Stats.Thorn , Stats.LegacyDecimalStatTypes , Stats.Reset() , Stats.Set(StatMap, params Stats[]) , Stats.Set(StatsMap) , Stats.Modify(IEnumerable<StatModifier>) , Stats.Set(IEnumerable<StatModifier>, params Stats[]) , Stats.GetStatAsLong(StatType) , Stats.GetStat(StatType) , Stats.SetStatForTest(StatType, decimal) , Stats.GetStats(bool) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

CharacterStats(CharacterStats)

Initializes a new instance of the CharacterStats class by copying from another instance.

```
public CharacterStats(CharacterStats value)
```

Parameters

`value` CharacterStats

The character stats to copy from.

CharacterStats(Row, int, IReadOnlyList<StatModifier>)

Initializes a new instance of the CharacterStats class.

```
public CharacterStats(CharacterSheet.Row row, int level, IReadOnlyList<StatModifier>
initialStatModifiers = null)
```

Parameters

`row` CharacterSheet.Row

The character sheet row data.

`level` int

The character level.

`initialStatModifiers` IReadOnlyList<StatModifier>

Optional initial stat modifiers.

CharacterStats(WaveStatData)

Initializes a new instance of the CharacterStats class from world boss wave stat data.

```
public CharacterStats(WorldBossCharacterSheet.WaveStatData stat)
```

Parameters

`stat` WorldBossCharacterSheet.WaveStatData

The world boss wave stat data.

Fields

StatWithoutBuffs

Stat map without buff effects for arena calculations.

```
public readonly StatMap StatWithoutBuffs
```

Field Value

StatMap

Properties

AdditionalATK

Gets the additional ATK stat from equipment, buffs, etc.

```
public long AdditionalATK { get; }
```

Property Value

long

AdditionalArmorPenetration

Gets the additional ArmorPenetration stat from equipment, buffs, etc.

```
public long AdditionalArmorPenetration { get; }
```

Property Value

long

AdditionalCDMG

Gets the additional CDMG stat from equipment, buffs, etc.

```
public long AdditionalCDMG { get; }
```

Property Value

long

AdditionalCRI

Gets the additional CRI stat from equipment, buffs, etc.

```
public long AdditionalCRI { get; }
```

Property Value

long

AdditionalDEF

Gets the additional DEF stat from equipment, buffs, etc.

```
public long AdditionalDEF { get; }
```

Property Value

long

AdditionalDRR

Gets the additional DRR stat from equipment, buffs, etc.

```
public long AdditionalDRR { get; }
```

Property Value

long

AdditionalDRV

Gets the additional DRV stat from equipment, buffs, etc.

```
public long AdditionalDRV { get; }
```

Property Value

long

AdditionalHIT

Gets the additional HIT stat from equipment, buffs, etc.

```
public long AdditionalHIT { get; }
```

Property Value

long

AdditionalHP

Gets the additional HP stat from equipment, buffs, etc.

```
public long AdditionalHP { get; }
```

Property Value

long

AdditionalSPD

Gets the additional SPD stat from equipment, buffs, etc.

```
public long AdditionalSPD { get; }
```

Property Value

long

AdditionalThorn

Gets the additional Thorn stat from equipment, buffs, etc.

```
public long AdditionalThorn { get; }
```

Property Value

long

BaseATK

Gets the base ATK stat.

```
public long BaseATK { get; }
```

Property Value

long

BaseArmorPenetration

Gets the base ArmorPenetration stat.

```
public long BaseArmorPenetration { get; }
```

Property Value

long

BaseCDMG

Gets the base CDMG stat.

```
public long BaseCDMG { get; }
```

Property Value

long

BaseCRI

Gets the base CRI stat.

```
public long BaseCRI { get; }
```

Property Value

long

BaseDEF

Gets the base DEF stat.

```
public long BaseDEF { get; }
```

Property Value

long

BaseDRR

Gets the base DRR stat.

```
public long BaseDRR { get; }
```

Property Value

long

BaseDRV

Gets the base DRV stat.

```
public long BaseDRV { get; }
```

Property Value

long

BaseHIT

Gets the base HIT stat.

```
public long BaseHIT { get; }
```

Property Value

long

BaseHP

Gets the base HP stat.

```
public long BaseHP { get; }
```

Property Value

long

BaseSPD

Gets the base SPD stat.

```
public long BaseSPD { get; }
```

Property Value

long

BaseStats

Gets the base stats of the character.

```
public IStats BaseStats { get; }
```

Property Value

IStats

BaseThorn

Gets the base Thorn stat.

```
public long BaseThorn { get; }
```

Property Value

long

BuffStats

Gets the buff stats of the character.

```
public IStats BuffStats { get; }
```

Property Value

IStats

CollectionStats

Gets the collection stats of the character.

```
public IStats CollectionStats { get; }
```

Property Value

IStats

ConditionStats

Gets the condition stats of the character.

```
public IStats ConditionStats { get; }
```

Property Value

IStats

ConsumableStats

Gets the consumable stats of the character.

```
public IStats ConsumableStats { get; }
```

Property Value

IStats

CostumeStats

Gets the costume stats of the character.

```
public IStats CostumeStats { get; }
```

Property Value

IStats

EquipmentStats

Gets the equipment stats of the character.

```
public IStats EquipmentStats { get; }
```

Property Value

IStats

HpIncreasingModifier

Gets or sets the HP increasing modifier for arena battles.

```
public long HpIncreasingModifier { set; }
```

Property Value

long

IsArenaCharacter

Gets or sets whether this character is used in arena battles.

```
public bool IsArenaCharacter { set; }
```

Property Value

bool

Level

Gets the character level.

```
public int Level { get; }
```

Property Value

int

RuneStats

Gets the rune stats of the character.

```
public IStats RuneStats { get; }
```

Property Value

IStats

Methods

AddBuff(StatBuff, BuffLimitSheet, bool)

```
public void AddBuff(StatBuff buff, BuffLimitSheet buffLimitSheet, bool
updateImmediate = true)
```

Parameters

`buff` StatBuff

`buffLimitSheet` BuffLimitSheet

`updateImmediate` bool

AddCostume(IEnumerable<StatModifier>)

```
public void AddCostume(IEnumerable<StatModifier> statModifiers)
```

Parameters

`statModifiers` IEnumerable<StatModifier>

AddRune(IEnumerable<StatModifier>)

```
public void AddRune(IEnumerable<StatModifier> statModifiers)
```

Parameters

`statModifiers` IEnumerable<StatModifier>

AddRuneStat(RuneOptionInfo, int)

```
public void AddRuneStat(RuneOptionSheet.Row.RuneOptionInfo optionInfo,
int runeLevelBonus)
```

Parameters

`optionInfo` RuneOptionSheet.Row.RuneOptionInfo

`runeLevelBonus` int

ClearConditions(bool)

Clear all condition stat modifiers. This should be called after simulation ends to remove temporary conditions.

```
public CharacterStats ClearConditions(bool updateImmediate = true)
```

Parameters

`updateImmediate` bool

Whether to update stats immediately

Returns

CharacterStats

This instance for method chaining

Clone()

Creates a new object that is a copy of the current instance.

```
public override object Clone()
```

Returns

object

A new object that is a copy of this instance.

ConfigureStats(IReadOnlyCollection<Equipment>, IReadOnlyCollection<Costume>, IReadOnlyCollection<RuneOptionInfo>, CostumeStatSheet, List<StatModifier>, int)

```
public void ConfigureStats(IReadOnlyCollection<Equipment> equipments,
IReadOnlyCollection<Costume> costumes,
IReadOnlyCollection<RuneOptionSheet.Row.RuneOptionInfo> runeOptions,
CostumeStatSheet costumeStatSheet, List<StatModifier> collectionStatModifiers,
int runeLevelBonus)
```

Parameters

`equipments` IReadOnlyCollection<Equipment>

`costumes` IReadOnlyCollection<Costume>

`runeOptions` IReadOnlyCollection<RuneOptionSheet.Row.RuneOptionInfo>

`costumeStatSheet` CostumeStatSheet

`collectionStatModifiers` List<StatModifier>

`runeLevelBonus` int

GetAdditionalStats(bool)

```
public IEnumerable<(StatType statType, long additionalValue)>
GetAdditionalStats(bool ignoreZero = false)
```

Parameters

`ignoreZero` bool

Returns

IEnumerable<(StatType statType, long baseValue)>

GetBaseAndAdditionalStats(bool)

```
public IEnumerable<(StatType statType, long baseValue, long additionalValue)>
GetBaseAndAdditionalStats(bool ignoreZero = false)
```

Parameters

`ignoreZero` bool

Returns

IEnumerable<(StatType statType, long baseValue, long additionalValue)>

GetBaseStats(bool)

```
public IEnumerable<(StatType statType, long baseValue)> GetBaseStats(bool ignoreZero
= false)
```

Parameters

`ignoreZero` bool

Returns

IEnumerable<(StatType statType, long baseValue)>

IncreaseHpForArena()

Increases the HP of the character for the arena.

```
public void IncreaseHpForArena()
```

RemoveBuff(StatBuff, bool)

```
public void RemoveBuff(StatBuff buff, bool updateImmediate = true)
```

Parameters

`buff` StatBuff

`updateImmediate` bool

SetBuffs(IEnumerable<StatBuff>, BuffLimitSheet, bool)

Set stats based on buffs.

```
public CharacterStats SetBuffs(IEnumerable<StatBuff> value, BuffLimitSheet
buffLimitSheet, bool updateImmediate = true)
```

Parameters

`value` IEnumerable<StatBuff>

`buffLimitSheet` BuffLimitSheet

`updateImmediate` bool

Returns

CharacterStats

SetCollections(IEnumerable<StatModifier>, bool)

```
public CharacterStats SetCollections(IEnumerable<StatModifier> statModifiers, bool
updateImmediate = true)
```

Parameters

`statModifiers` IEnumerable<StatModifier>

`updateImmediate` bool

Returns

CharacterStats

SetConditions(IEnumerable<StatModifier>, bool)

Set stats based on infinite tower conditions. This is applied after buffs and is temporary for simulation only.

```
public CharacterStats SetConditions(IEnumerable<StatModifier> statModifiers, bool
updateImmediate = true)
```

Parameters

`statModifiers` IEnumerable<StatModifier>

Condition stat modifiers

`updateImmediate` bool

Whether to update stats immediately

Returns

CharacterStats

This instance for method chaining

SetConsumables(IEnumerable<Consumable>, bool)

Set stats based on consumables. Also recalculates stats from buffs.

```
public CharacterStats SetConsumables(IEnumerable<Consumable> consumables, bool
updateImmediate = true)
```

Parameters

`consumables` IEnumerable<Consumable>

`updateImmediate` bool

Returns

CharacterStats

SetCostume(IEnumerable<StatModifier>)

```
public void SetCostume(IEnumerable<StatModifier> statModifiers)
```

Parameters

`statModifiers` IEnumerable<StatModifier>

SetCostumeStat(IReadOnlyCollection<Costume>, CostumeStatSheet)

```
public void SetCostumeStat(IReadOnlyCollection<Costume> costumes,
CostumeStatSheet costumeStatSheet)
```

Parameters

`costumes` IReadOnlyCollection<Costume>

`costumeStatSheet` CostumeStatSheet

SetEquipments(IEnumerable<Equipment>, EquipmentItemSetEffectSheet, bool)

Set stats based on equipments. Also recalculates stats from consumables and buffs.

```
public CharacterStats SetEquipments(IEnumerable<Equipment> equipments,
EquipmentItemSetEffectSheet sheet, bool updateImmediate = true)
```

Parameters

`equipments` IEnumerable<Equipment>

`sheet` EquipmentItemSetEffectSheet

`updateImmediate` bool

Returns

CharacterStats

SetRunes(IEnumerable<StatModifier>, bool)

Set stats based on runes.

```
public CharacterStats SetRunes(IEnumerable<StatModifier> value, bool updateImmediate
= true)
```

Parameters

`value` IEnumerable<StatModifier>

`updateImmediate` bool

Returns

CharacterStats

SetStats(int, bool)

Set base stats based on character level.

```
public CharacterStats SetStats(int level, bool updateImmediate = true)
```

Parameters

`level` int

`updateImmediate` bool

Returns

CharacterStats

Class DecimalStat

Namespace: Nekoyume.Model.Stat

Assembly: Lib9c.dll

Represents a single character statistic with base and additional values. Supports both Dictionary and List serialization formats for backward compatibility.

Serialization Format: - Dictionary (Legacy): Uses key-value pairs for backward compatibility - List (New): Uses ordered list for better performance and smaller size

Field Order (List Format): 1. version - Serialization version number 2. statType - The type of statistic 3. baseValue - Base value of the statistic 4. additionalValue - Additional value of the statistic

```
[Serializable]
public class DecimalStat : ICloneable, IState
```

Inheritance

object ← DecimalStat

Implements

ICloneable, IState

Inherited Members

object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Extension Methods

StateExtensions.SerializeForLegacyEquipmentStat(DecimalStat)

Remarks

This class implements dual serialization support to ensure smooth migration from the legacy Dictionary format to the new List format. The List format provides better performance and smaller serialized data size.

Example usage:

```
// Create decimal stat
var stat = new DecimalStat(StatType.HP, 100, 50);
// Serialize to List format (new)
```

```
var serialized = stat.Serialize(); // Returns List
// Deserialize from any format
var deserialized = new DecimalStat(serialized); // Supports both Dictionary and List
```

Constructors

DecimalStat(Dictionary)

Constructor for backward compatibility with Dictionary format.

```
public DecimalStat(Dictionary serialized)
```

Parameters

`serialized` Dictionary

Dictionary containing serialized data

DecimalStat(IValue)

Constructor for deserialization that supports both Dictionary and List formats.

```
public DecimalStat(IValue serialized)
```

Parameters

`serialized` IValue

Serialized data in either Dictionary or List format

Exceptions

ArgumentNullException

Thrown when serialized is null

ArgumentException

Thrown when serialized format is not supported

DecimalStat(DecimalStat)

```
protected DecimalStat(DecimalStat value)
```

Parameters

`value` DecimalStat

DecimalStat(StatType, decimal, decimal)

```
public DecimalStat(StatType type, decimal value = 0, decimal additionalValue = 0)
```

Parameters

`type` StatType

`value` decimal

`additionalValue` decimal

Fields

StatType

```
public StatType StatType
```

Field Value

StatType

Properties

AdditionalValue

```
public decimal AdditionalValue { get; }
```

Property Value

decimal

AdditionalValueAsLong

```
public long AdditionalValueAsLong { get; }
```

Property Value

long

BaseValue

```
public decimal BaseValue { get; }
```

Property Value

decimal

BaseValueAsLong

```
public long BaseValueAsLong { get; }
```

Property Value

long

HasAdditionalValue

```
public bool HasAdditionalValue { get; }
```

Property Value

bool

HasAdditionalValueAsLong

```
public bool HasAdditionalValueAsLong { get; }
```

Property Value

bool

HasBaseValue

```
public bool HasBaseValue { get; }
```

Property Value

bool

HasBaseValueAsLong

```
public bool HasBaseValueAsLong { get; }
```

Property Value

bool

HasTotalValueAsLong

```
public bool HasTotalValueAsLong { get; }
```

Property Value

bool

TotalValue

```
public decimal TotalValue { get; }
```

Property Value

decimal

TotalValueAsLong

```
[Obsolete("For legacy equipments. (Before world 7 patch)")]
public long TotalValueAsLong { get; }
```

Property Value

long

Methods

AddAdditionalValue(decimal)

```
public void AddAdditionalValue(decimal value)
```

Parameters

`value` decimal

AddBaseValue(decimal)

```
public void AddBaseValue(decimal value)
```

Parameters

`value` decimal

Clone()

Creates a new object that is a copy of the current instance.

```
public virtual object Clone()
```

Returns

object

A new object that is a copy of this instance.

Deserialize(Dictionary)

Deserializes data from Dictionary format (legacy support).

```
public virtual void Deserialize(Dictionary serialized)
```

Parameters

`serialized` Dictionary

Dictionary containing serialized data

Equals(DecimalStat)

```
protected bool Equals(DecimalStat other)
```

Parameters

`other` DecimalStat

Returns

bool

Equals(object)

Determines whether the specified object is equal to the current object.

```
public override bool Equals(object obj)
```

Parameters

`obj` object

The object to compare with the current object.

Returns

bool

true if the specified object is equal to the current object; otherwise, false.

GetHashCode()

Serves as the default hash function.

```
public override int GetHashCode()
```

Returns

int

A hash code for the current object.

Reset()

```
public virtual void Reset()
```

Serialize()

Serializes the DecimalStat to List format (new format). Order: [version, statType, baseValue, additionalValue]

```
public virtual IValue Serialize()
```

Returns

IValue

List containing serialized data

SerializeWithoutAdditional()

```
public IValue SerializeWithoutAdditional()
```

Returns

IValue

SetAdditionalValue(decimal)

```
public void SetAdditionalValue(decimal value)
```

Parameters

`value` decimal

SetBaseValue(decimal)

```
public void SetBaseValue(decimal value)
```

Parameters

`value` decimal

Interface IBaseAndAdditionalStats

Namespace: Nekoyume.Model.Stat

Assembly: Lib9c.dll

```
public interface IBaseAndAdditionalStats
```

Properties

AdditionalATK

```
long AdditionalATK { get; }
```

Property Value

long

AdditionalArmorPenetration

```
long AdditionalArmorPenetration { get; }
```

Property Value

long

AdditionalCDMG

```
long AdditionalCDMG { get; }
```

Property Value

long

AdditionalCRI

```
long AdditionalCRI { get; }
```

Property Value

long

AdditionalDEF

```
long AdditionalDEF { get; }
```

Property Value

long

AdditionalDRR

```
long AdditionalDRR { get; }
```

Property Value

long

AdditionalDRV

```
long AdditionalDRV { get; }
```

Property Value

long

AdditionalHIT

```
long AdditionalHIT { get; }
```

Property Value

long

AdditionalHP

```
long AdditionalHP { get; }
```

Property Value

long

AdditionalSPD

```
long AdditionalSPD { get; }
```

Property Value

long

AdditionalThorn

```
long AdditionalThorn { get; }
```

Property Value

long

BaseATK

```
long BaseATK { get; }
```

Property Value

long

BaseArmorPenetration

```
long BaseArmorPenetration { get; }
```

Property Value

long

BaseCDMG

```
long BaseCDMG { get; }
```

Property Value

long

BaseCRI

```
long BaseCRI { get; }
```

Property Value

long

BaseDEF

```
long BaseDEF { get; }
```

Property Value

long

BaseDRR

```
long BaseDRR { get; }
```

Property Value

long

BaseDRV

```
long BaseDRV { get; }
```

Property Value

long

BaseHIT

```
long BaseHIT { get; }
```

Property Value

long

BaseHP

```
long BaseHP { get; }
```

Property Value

long

BaseSPD

```
long BaseSPD { get; }
```

Property Value

long

BaseThorn

```
long BaseThorn { get; }
```

Property Value

long

Methods

GetAdditionalStats(bool)

```
IEnumerable<(StatType statType, long additionalValue)> GetAdditionalStats(bool
ignoreZero = false)
```

Parameters

`ignoreZero` bool

Returns

IEnumerable<(StatType statType, long baseValue)>

GetBaseAndAdditionalStats(bool)

```
IEnumerable<(StatType statType, long baseValue, long additionalValue)>
GetBaseAndAdditionalStats(bool ignoreZero = false)
```

Parameters

`ignoreZero` bool

Returns

IEnumerable<(StatType statType, long baseValue, long additionalValue)>

GetBaseStats(bool)

```
IEnumerable<(StatType statType, long baseValue)> GetBaseStats(bool ignoreZero
= false)
```

Parameters

`ignoreZero` bool

Returns

IEnumerable<(StatType statType, long baseValue)>

Interface IStats

Namespace: Nekoyume.Model.Stat

Assembly: Lib9c.dll

```
public interface IStats
```

Properties

ATK

```
long ATK { get; }
```

Property Value

long

ArmorPenetration

```
long ArmorPenetration { get; }
```

Property Value

long

CDMG

```
long CDMG { get; }
```

Property Value

long

CRI

```
long CRI { get; }
```

Property Value

long

DEF

```
long DEF { get; }
```

Property Value

long

DRR

```
long DRR { get; }
```

Property Value

long

DRV

```
long DRV { get; }
```

Property Value

long

HIT

```
long HIT { get; }
```

Property Value

long

HP

```
long HP { get; }
```

Property Value

long

SPD

```
long SPD { get; }
```

Property Value

long

Thorn

```
long Thorn { get; }
```

Property Value

long

Methods

GetStats(bool)

```
IEnumerable<(StatType statType, long value)> GetStats(bool ignoreZero = false)
```

Parameters

`ignoreZero` bool

Returns

IEnumerable<(StatType statType, long baseValue)>

Class StatMap

Namespace: Nekoyume.Model.Stat

Assembly: Lib9c.dll

Represents a collection of character statistics. Supports both Dictionary and List serialization formats for backward compatibility.

Serialization Format: - Dictionary (Legacy): Uses key-value pairs for backward compatibility - List (New): Uses ordered list for better performance and smaller size

Field Order (List Format): 1. version - Serialization version number 2. stats - List of DecimalStat objects ordered by StatType

```
[Serializable]
public class StatMap : IStats, IBaseAndAdditionalStats, IState
```

Inheritance

object ← StatMap

Implements

IStats, IBaseAndAdditionalStats, IState

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Remarks

This class implements dual serialization support to ensure smooth migration from the legacy Dictionary format to the new List format. The List format provides better performance and smaller serialized data size.

Example usage:

```
// Create stat map
var statMap = new StatMap();
statMap[StatType.HP].SetBaseValue(100);
statMap[StatType.ATK].SetBaseValue(50);
// Serialize to List format (new)
var serialized = statMap.Serialize(); // Returns List
```

```
// Deserialize from any format
var deserialized = new StatMap(serialized); // Supports both Dictionary and List
```

Constructors

StatMap()

```
public StatMap()
```

StatMap(IValue)

Constructor for deserialization that supports both Dictionary and List formats.

```
public StatMap(IValue serialized)
```

Parameters

`serialized` IValue

Serialized data in either Dictionary or List format

Exceptions

ArgumentNullException

Thrown when serialized is null

ArgumentException

Thrown when serialized format is not supported

StatMap(StatMap)

```
public StatMap(StatMap statMap)
```

Parameters

`statMap` StatMap

Properties

ATK

```
public long ATK { get; }
```

Property Value

long

AdditionalATK

```
public long AdditionalATK { get; }
```

Property Value

long

AdditionalArmorPenetration

```
public long AdditionalArmorPenetration { get; }
```

Property Value

long

AdditionalCDMG

```
public long AdditionalCDMG { get; }
```

Property Value

long

AdditionalCRI

```
public long AdditionalCRI { get; }
```

Property Value

long

AdditionalDEF

```
public long AdditionalDEF { get; }
```

Property Value

long

AdditionalDRR

```
public long AdditionalDRR { get; }
```

Property Value

long

AdditionalDRV

```
public long AdditionalDRV { get; }
```

Property Value

long

AdditionalHIT

```
public long AdditionalHIT { get; }
```

Property Value

long

AdditionalHP

```
public long AdditionalHP { get; }
```

Property Value

long

AdditionalSPD

```
public long AdditionalSPD { get; }
```

Property Value

long

AdditionalThorn

```
public long AdditionalThorn { get; }
```

Property Value

long

ArmorPenetration

```
public long ArmorPenetration { get; }
```

Property Value

long

BaseATK

```
public long BaseATK { get; }
```

Property Value

long

BaseArmorPenetration

```
public long BaseArmorPenetration { get; }
```

Property Value

long

BaseCDMG

```
public long BaseCDMG { get; }
```

Property Value

long

BaseCRI

```
public long BaseCRI { get; }
```

Property Value

long

BaseDEF

```
public long BaseDEF { get; }
```

Property Value

long

BaseDRR

```
public long BaseDRR { get; }
```

Property Value

long

BaseDRV

```
public long BaseDRV { get; }
```

Property Value

long

BaseHIT

```
public long BaseHIT { get; }
```

Property Value

long

BaseHP

```
public long BaseHP { get; }
```

Property Value

long

BaseSPD

```
public long BaseSPD { get; }
```

Property Value

long

BaseThorn

```
public long BaseThorn { get; }
```

Property Value

long

CDMG

```
public long CDMG { get; }
```

Property Value

long

CRI

```
public long CRI { get; }
```

Property Value

long

DEF

```
public long DEF { get; }
```

Property Value

long

DRR

```
public long DRR { get; }
```

Property Value

long

DRV

```
public long DRV { get; }
```

Property Value

long

HIT

```
public long HIT { get; }
```

Property Value

long

HP

```
public long HP { get; }
```

Property Value

long

this[StatType]

```
public DecimalStat this[StatType type] { get; }
```

Parameters

`type` StatType

Property Value

DecimalStat

SPD

```
public long SPD { get; }
```

Property Value

long

Thorn

```
public long Thorn { get; }
```

Property Value

long

Methods

GetAdditionalStat(StatType)

```
public long GetAdditionalStat(StatType statType)
```

Parameters

`statType` StatType

Returns

long

GetAdditionalStats(bool)

```
public IEnumerable<(StatType statType, long additionalValue)>
GetAdditionalStats(bool ignoreZero = false)
```

Parameters

`ignoreZero` bool

Returns

IEnumerable<(StatType statType, long baseValue)>

GetBaseAndAdditionalStats(bool)

```
public IEnumerable<(StatType statType, long baseValue, long additionalValue)>
GetBaseAndAdditionalStats(bool ignoreZero = false)
```

Parameters

`ignoreZero` bool

Returns

IEnumerable<(StatType statType, long baseValue, long additionalValue)>

GetBaseStat(StatType)

```
public long GetBaseStat(StatType statType)
```

Parameters

`statType` StatType

Returns

long

GetBaseStats(bool)

```
public IEnumerable<(StatType statType, long baseValue)> GetBaseStats(bool ignoreZero
= false)
```

Parameters

`ignoreZero` bool

Returns

IEnumerable<(StatType statType, long baseValue)>

GetDecimalStats(bool)

```
public IEnumerable<DecimalStat> GetDecimalStats(bool ignoreZero)
```

Parameters

`ignoreZero` bool

Returns

IEnumerable<DecimalStat>

GetStat(StatType)

```
public decimal GetStat(StatType statType)
```

Parameters

`statType` StatType

Returns

decimal

GetStatAsLong(StatType)

```
public long GetStatAsLong(StatType statType)
```

Parameters

`statType` StatType

Returns

long

GetStats(bool)

```
public IEnumerable<(StatType statType, long value)> GetStats(bool ignoreZero
= false)
```

Parameters

`ignoreZero` bool

Returns

IEnumerable<(StatType statType, long baseValue)>

Reset()

```
public void Reset()
```

Serialize()

Serializes the StatMap to List format (new format). Order: [version, stats]

```
public IValue Serialize()
```

Returns

IValue

List containing serialized data

Class StatModifier

Namespace: Nekoyume.Model.Stat

Assembly: Lib9c.dll

```
[Serializable]
public class StatModifier
```

Inheritance

object ← StatModifier

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Constructors

StatModifier(DecimalStat)

```
public StatModifier(DecimalStat decimalStat)
```

Parameters

`decimalStat` DecimalStat

StatModifier(StatType, OperationType, long)

```
public StatModifier(StatType statType, StatModifier.OperationType operation,
long value)
```

Parameters

`statType` StatType

`operation` StatModifier.OperationType

value long

Properties

Operation

```
public StatModifier.OperationType Operation { get; }
```

Property Value

StatModifier.OperationType

StatType

```
public StatType StatType { get; }
```

Property Value

StatType

Value

```
public long Value { get; }
```

Property Value

long

Methods

GetModifiedAll(decimal)

value와 함께 value를 바탕으로 변경시킨 값의 합을 리턴한다.

```
public decimal GetModifiedAll(decimal value)
```

Parameters

`value` decimal

Returns

decimal

GetModifiedAll(long)

value와 함께 value를 바탕으로 변경시킨 값의 합을 리턴한다.

```
public long GetModifiedAll(long value)
```

Parameters

`value` long

Returns

long

GetModifiedValue(decimal)

value를 바탕으로 변경시킨 값만을 리턴한다.

```
public decimal GetModifiedValue(decimal value)
```

Parameters

`value` decimal

Returns

decimal

Exceptions

ArgumentOutOfRangeException

GetModifiedValue(int)

value를 바탕으로 변경시킨 값만을 리턴한다.

```
public long GetModifiedValue(int value)
```

Parameters

`value` int

Returns

long

Exceptions

ArgumentOutOfRangeException

Modify(DecimalStat)

value를 변경시킨다.

```
public void Modify(DecimalStat value)
```

Parameters

`value` DecimalStat

Exceptions

ArgumentOutOfRangeException

SetForTest(int)

```
public void SetForTest(int value)
```

Parameters

`value` int

ToString()

Returns a string that represents the current object.

```
public override string ToString()
```

Returns

string

A string that represents the current object.

Enum StatModifier.OperationType

Namespace: Nekoyume.Model.Stat

Assembly: Lib9c.dll

```
public enum StatModifier.OperationType
```

Extension Methods
StateExtensions.Serialize(Enum)

Fields

`Add = 0`

`Percentage = 1`

Enum StatType

Namespace: Nekoyume.Model.Stat

Assembly: Lib9c.dll

```
public enum StatType
```

Extension Methods
StatTypeExtension.Serialize(StatType) , StateExtensions.Serialize(Enum)

Fields

`ATK = 2`

Attack Power

`ArmorPenetration = 10`

Armor penetration

`CDMG = 9`

Crit Damage (Permyriad)

`CRI = 4`

Critical Chance

`DEF = 3`

Defence

`DRR = 8`

Damage Reduction Rate. Multiplied on original damage. (Permyriad)

`DRV = 7`

Damage Reduction Value. Subtracted from original damage.

`HIT = 5`

Hit Chance

`HP = 1`

Health Point

`NONE = 0`

Default, It's same with null.

`SPD = 6`

Speed

`Thorn = 11`

Apply {Thorn} damage to the attacker when an attack is attempted.

Class StatTypeComparer

Namespace: Nekoyume.Model.Stat

Assembly: Lib9c.dll

```
[Serializable]
public class StatTypeComparer : IEqualityComparer<StatType>
```

Inheritance

object ← StatTypeComparer

Implements

IEqualityComparer<StatType>

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Fields

Instance

```
public static readonly StatTypeComparer Instance
```

Field Value

StatTypeComparer

Methods

Equals(StatType, StatType)

Determines whether the specified objects are equal.

```
public bool Equals(StatType x, StatType y)
```

Parameters

`x` StatType

The first object of type `T` to compare.

`y` StatType

The second object of type `T` to compare.

Returns

bool

[true] if the specified objects are equal; otherwise, [false].

GetHashCode(StatType)

Returns a hash code for the specified object.

```
public int GetHashCode(StatType obj)
```

Parameters

`obj` StatType

The [object] for which a hash code is to be returned.

Returns

int

A hash code for the specified object.

Exceptions

ArgumentNullException

The type of `obj` is a reference type and `obj` is [null].

Class StatTypeExtension

Namespace: Nekoyume.Model.Stat

Assembly: Lib9c.dll

```
public static class StatTypeExtension
```

Inheritance

object ← StatTypeExtension

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Methods

Deserialize(Binary)

```
public static StatType Deserialize(Binary serialized)
```

Parameters

`serialized` Binary

Returns

StatType

Serialize(StatType)

```
public static IKey Serialize(this StatType statType)
```

Parameters

`statType` StatType

Returns

IKey

Class Stats

Namespace: Nekoyume.Model.Stat

Assembly: Lib9c.dll

```
[Serializable]
public class Stats : IStats, ICloneable
```

Inheritance

object ← Stats

Implements

IStats, ICloneable

Derived

CharacterStats

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

Stats()

```
public Stats()
```

Stats(Stats)

```
public Stats(Stats value)
```

Parameters

`value` Stats

Fields

LegacyDecimalStatTypes

```
protected readonly HashSet<StatType> LegacyDecimalStatTypes
```

Field Value

HashSet<StatType>

_statMap

```
protected readonly StatMap _statMap
```

Field Value

StatMap

Properties

ATK

```
public long ATK { get; }
```

Property Value

long

ArmorPenetration

```
public long ArmorPenetration { get; }
```

Property Value

long

CDMG

```
public long CDMG { get; }
```

Property Value

long

CRI

```
public long CRI { get; }
```

Property Value

long

DEF

```
public long DEF { get; }
```

Property Value

long

DRR

```
public long DRR { get; }
```

Property Value

long

DRV

```
public long DRV { get; }
```

Property Value

long

HIT

```
public long HIT { get; }
```

Property Value

long

HP

```
public long HP { get; }
```

Property Value

long

SPD

```
public long SPD { get; }
```

Property Value

long

Thorn

```
public long Thorn { get; }
```

Property Value

long

Methods

Clone()

Creates a new object that is a copy of the current instance.

```
public virtual object Clone()
```

Returns

object

A new object that is a copy of this instance.

GetStat(StatType)

```
public decimal GetStat(StatType statType)
```

Parameters

`statType` StatType

Returns

decimal

GetStatAsLong(StatType)

```
public long GetStatAsLong(StatType statType)
```

Parameters

`statType` StatType

Returns

long

GetStats(bool)

```
public IEnumerable<(StatType statType, long value)> GetStats(bool ignoreZero
= false)
```

Parameters

`ignoreZero` bool

Returns

IEnumerable<(StatType statType, long baseValue)>

Modify(IEnumerable<StatModifier>)

```
public void Modify(IEnumerable<StatModifier> statModifiers)
```

Parameters

`statModifiers` IEnumerable<StatModifier>

Reset()

```
public void Reset()
```

Set(StatMap, params Stats[])

```
public void Set(StatMap statMap, params Stats[] statsArray)
```

Parameters

`statMap` StatMap

`statsArray` Stats[]

Set(StatsMap)

```
public void Set(StatsMap value)
```

Parameters

`value` StatsMap

Set(IEnumerable<StatModifier>, params Stats[])

```
public void Set(IEnumerable<StatModifier> statModifiers, params Stats[] baseStats)
```

Parameters

`statModifiers` IEnumerable<StatModifier>

`baseStats` Stats[]

SetStatForTest(StatType, decimal)

Use this only for testing.

```
public void SetStatForTest(StatType statType, decimal value)
```

Parameters

`statType` StatType

value decimal

Class StatsMap

Namespace: Nekoyume.Model.Stat

Assembly: Lib9c.dll

```
[Serializable]
public class StatsMap : IStats, IBaseAndAdditionalStats, IState
```

Inheritance

object ← StatsMap

Implements

IStats, IBaseAndAdditionalStats, IState

Inherited Members

object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Properties

ATK

```
public long ATK { get; }
```

Property Value

long

AdditionalATK

```
public long AdditionalATK { get; }
```

Property Value

long

AdditionalArmorPenetration

```
public long AdditionalArmorPenetration { get; }
```

Property Value

long

AdditionalCDMG

```
public long AdditionalCDMG { get; }
```

Property Value

long

AdditionalCRI

```
public long AdditionalCRI { get; }
```

Property Value

long

AdditionalDEF

```
public long AdditionalDEF { get; }
```

Property Value

long

AdditionalDRR

```
public long AdditionalDRR { get; }
```

Property Value

long

AdditionalDRV

```
public long AdditionalDRV { get; }
```

Property Value

long

AdditionalHIT

```
public long AdditionalHIT { get; }
```

Property Value

long

AdditionalHP

```
public long AdditionalHP { get; }
```

Property Value

long

AdditionalSPD

```
public long AdditionalSPD { get; }
```

Property Value

long

AdditionalThorn

```
public long AdditionalThorn { get; }
```

Property Value

long

ArmorPenetration

```
public long ArmorPenetration { get; }
```

Property Value

long

BaseATK

```
public long BaseATK { get; }
```

Property Value

long

BaseArmorPenetration

```
public long BaseArmorPenetration { get; }
```

Property Value

long

BaseCDMG

```
public long BaseCDMG { get; }
```

Property Value

long

BaseCRI

```
public long BaseCRI { get; }
```

Property Value

long

BaseDEF

```
public long BaseDEF { get; }
```

Property Value

long

BaseDRR

```
public long BaseDRR { get; }
```

Property Value

long

BaseDRV

```
public long BaseDRV { get; }
```

Property Value

long

BaseHIT

```
public long BaseHIT { get; }
```

Property Value

long

BaseHP

```
public long BaseHP { get; }
```

Property Value

long

BaseSPD

```
public long BaseSPD { get; }
```

Property Value

long

BaseThorn

```
public long BaseThorn { get; }
```

Property Value

long

CDMG

```
public long CDMG { get; }
```

Property Value

long

CRI

```
public long CRI { get; }
```

Property Value

long

DEF

```
public long DEF { get; }
```

Property Value

long

DRR

```
public long DRR { get; }
```

Property Value

long

DRV

```
public long DRV { get; }
```

Property Value

long

HIT

```
public long HIT { get; }
```

Property Value

long

HP

```
public long HP { get; }
```

Property Value

long

SPD

```
public long SPD { get; }
```

Property Value

long

Thorn

```
public long Thorn { get; }
```

Property Value

long

Methods

AddStatAdditionalValue(StatModifier)

```
public void AddStatAdditionalValue(StatModifier statModifier)
```

Parameters

`statModifier` StatModifier

AddStatAdditionalValue(StatType, decimal)

```
public void AddStatAdditionalValue(StatType key, decimal additionalValue)
```

Parameters

`key` StatType

`additionalValue` decimal

AddStatValue(StatType, decimal)

```
public void AddStatValue(StatType key, decimal value)
```

Parameters

`key` StatType

`value` decimal

Deserialize(IValue)

```
public void Deserialize(IValue serialized)
```

Parameters

`serialized` IValue

Equals(StatsMap)

```
protected bool Equals(StatsMap other)
```

Parameters

`other` StatsMap

Returns

bool

Equals(object)

Determines whether the specified object is equal to the current object.

```
public override bool Equals(object obj)
```

Parameters

`obj` object

The object to compare with the current object.

Returns

bool

true if the specified object is equal to the current object; otherwise, false.

GetAdditionalStat(StatType)

```
public long GetAdditionalStat(StatType statType)
```

Parameters

`statType` StatType

Returns

long

GetAdditionalStats()

```
public IEnumerable<DecimalStat> GetAdditionalStats()
```

Returns

IEnumerable<DecimalStat>

GetAdditionalStats(bool)

```
public IEnumerable<(StatType statType, long additionalValue)>
GetAdditionalStats(bool ignoreZero = false)
```

Parameters

`ignoreZero` bool

Returns

IEnumerable<(StatType statType, long baseValue)>

GetBaseAndAdditionalStats(bool)

```
public IEnumerable<(StatType statType, long baseValue, long additionalValue)>
GetBaseAndAdditionalStats(bool ignoreZero = false)
```

Parameters

`ignoreZero` bool

Returns

IEnumerable<(StatType statType, long baseValue, long additionalValue)>

GetBaseStat(StatType)

```
public long GetBaseStat(StatType statType)
```

Parameters

`statType` StatType

Returns

long

GetBaseStats(bool)

```
public IEnumerable<(StatType statType, long baseValue)> GetBaseStats(bool ignoreZero
= false)
```

Parameters

`ignoreZero` bool

Returns

IEnumerable<(StatType statType, long baseValue)>

GetDecimalStats(bool)

```
public IEnumerable<DecimalStat> GetDecimalStats(bool ignoreZero)
```

Parameters

`ignoreZero` bool

Returns

IEnumerable<DecimalStat>

GetHashCode()

Serves as the default hash function.

```
public override int GetHashCode()
```

Returns

int

A hash code for the current object.

GetStat(StatType)

```
public long GetStat(StatType statType)
```

Parameters

`statType` StatType

Returns

long

GetStats(bool)

```
public IEnumerable<(StatType statType, long value)> GetStats(bool ignoreZero
= false)
```

Parameters

`ignoreZero` bool

Returns

IEnumerable<(StatType statType, long baseValue)>

Serialize()

```
public IValue Serialize()
```

Returns

SetStatAdditionalValue(StatType, decimal)

```
public void SetStatAdditionalValue(StatType key, decimal additionalValue)
```

Parameters

`key` StatType

`additionalValue` decimal

Namespace Nekoyume.Model.State

Classes

ActivatedAccountsState

AdminState

AgentState

Agent□ □□ □□□□.

AllCombinationSlotState

This is new version of combination slot State. This state stores all combinationSlot states of an avatar. AllCombinationSlotState has all combinationSlotStates as dictionary and has methods to get/set/update each combinationSlotState. Use this with CombinationSlotStateModule.

AllRuneState

This is new version of rune state. This state stores all rune states of an avatar. AllRuneState has all RuneStates as dictionary and has methods to get/set/update each RuneState. Use this with RuneStateModule.

ArenaAvatarState

Introduced at https://github.com/planetarium/lib9c/pull/1156

ArenaInfo

ArenaInfo.Record

AuthorizedMinersState

AvatarState

Agent□ □□□□ □ Avatar□ □□ □□□□.

CollectionState

Represents the state of a collection.

CombinationSlotState

CpState

CpState is a state that stores the cp of the player.

CreditsState

CrystalCostState

CrystalRandomSkillState

DuplicateRedeemException

FailedToUnregisterInShopStateException

GameConfigState

GoldBalanceState

GoldCurrencyState

HammerPointState

InfiniteTowerBoardState

Represents the board state for infinite tower, tracking clear counts per floor.

InvalidRedeemCodeException

ItemSlotState

LazyState<TState, TEncoding>

LegacyStakeState

LegacyStakeState.StakeAchievements

MarketState

MonsterCollectionResult

MonsterCollectionState

MonsterCollectionState0

PendingActivationState

PetState

RaiderState

RankingInfo

RankingMapState

RankingState

RankingState0

RankingState1

RedeemCodeState

RedeemCodeState.Reward

RuneSlotState

RuneState

ShardedShopState

ShardedShopStateV2

ShopState
This is a model class of shop state.

ShopStateAlreadyContainsException

StakeRewardCalculator
Provides methods for calculating staking rewards from reward sheets.

State

StateExtensions

WeeklyArenaState

WorldBossKillRewardRecord

WorldBossState
Represents the state of a World Boss in the game.

Interfaces

IState

Delegates

StateExtensions.IValueTryParseDelegate<T>

Class ActivatedAccountsState

Namespace: Nekoyume.Model.State

Assembly: Lib9c.dll

```
[Serializable]
public class ActivatedAccountsState : State, IState, ISerializable
```

Inheritance

object ← State ← ActivatedAccountsState

Implements

IState, ISerializable

Inherited Members

State.address , State.SerializeBase() , State.SerializeV2Base() , State.SerializeListBase() , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

ActivatedAccountsState()

```
public ActivatedAccountsState()
```

ActivatedAccountsState(Dictionary)

```
public ActivatedAccountsState(Dictionary serialized)
```

Parameters

`serialized` Dictionary

ActivatedAccountsState(IImmutableSet<Address>)

```
public ActivatedAccountsState(IImmutableSet<Address> accounts)
```

Parameters

`accounts` IImmutableSet<Address>

ActivatedAccountsState(SerializationInfo, StreamingContext)

```
protected ActivatedAccountsState(SerializationInfo info, StreamingContext context)
```

Parameters

`info` SerializationInfo

`context` StreamingContext

Fields

Address

```
public static readonly Address Address
```

Field Value

Address

Properties

Accounts

```
public IImmutableSet<Address> Accounts { get; }
```

Property Value

IImmutableSet<Address>

Methods

AddAccount(Address)

```
public ActivatedAccountsState AddAccount(Address account)
```

Parameters

`account` Address

Returns

ActivatedAccountsState

GetObjectData(SerializationInfo, StreamingContext)

Populates a SerializationInfo with the data needed to serialize the target object.

```
public void GetObjectData(SerializationInfo info, StreamingContext context)
```

Parameters

`info` SerializationInfo

The SerializationInfo to populate with data.

`context` StreamingContext

The destination (see StreamingContext) for this serialization.

Exceptions

SecurityException

The caller does not have the required permission.

Remove(Address)

```
public void Remove(Address account)
```

Parameters

`account` Address

Serialize()

```
public override IValue Serialize()
```

Returns

IValue

Class AdminState

Namespace: Nekoyume.Model.State

Assembly: Lib9c.dll

```
[Serializable]
public class AdminState : State, IState
```

Inheritance

object ← State ← AdminState

Implements

IState

Inherited Members

State.address , State.SerializeBase() , State.SerializeV2Base() , State.SerializeListBase() , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

AdminState(Dictionary)

```
public AdminState(Dictionary serialized)
```

Parameters

`serialized` Dictionary

AdminState(Address, long)

```
public AdminState(Address adminAddress, long validUntil)
```

Parameters

`adminAddress` Address

`validUntil` long

Fields

Address

```
public static readonly Address Address
```

Field Value

Address

Properties

AdminAddress

```
public Address AdminAddress { get; }
```

Property Value

Address

ValidUntil

```
public long ValidUntil { get; }
```

Property Value

long

Methods

Serialize()

```
public override IValue Serialize()
```

Returns

IValue

Class AgentState

Namespace: Nekoyume.Model.State

Assembly: Lib9c.dll

Agent□ □□ □□□□.

```
[Serializable]
public class AgentState : State, IState, ICloneable
```

Inheritance

object ← State ← AgentState

Implements

IState, ICloneable

Inherited Members

State.address , State.SerializeBase() , State.SerializeV2Base() , State.SerializeListBase() , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

AgentState(Dictionary)

```
public AgentState(Dictionary serialized)
```

Parameters

`serialized` Dictionary

AgentState(List)

```
public AgentState(List serialized)
```

Parameters

`serialized` List

AgentState(Address)

```
public AgentState(Address address)
```

Parameters

`address` Address

Fields

CurrentVersion

```
public const int CurrentVersion = 1
```

Field Value

int

avatarAddresses

```
public readonly Dictionary<int, Address> avatarAddresses
```

Field Value

Dictionary<int, Address>

Properties

MonsterCollectionRound

```
public int MonsterCollectionRound { get; }
```

Property Value

int

Version

```
public int Version { get; }
```

Property Value

int

Methods

Clone()

Creates a new object that is a copy of the current instance.

```
public object Clone()
```

Returns

object

A new object that is a copy of this instance.

IncreaseCollectionRound()

```
public void IncreaseCollectionRound()
```

Serialize()

```
public override IValue Serialize()
```

Returns

IValue

SerializeList()

```
public IValue SerializeList()
```

Returns

IValue

Class AllCombinationSlotState

Namespace: Nekoyume.Model.State

Assembly: Lib9c.dll

This is new version of combination slot State. This state stores all combinationSlot states of an avatar. AllCombinationSlotState has all combinationSlotStates as dictionary and has methods to get/set/update each combinationSlotState. Use this with CombinationSlotState Module.

```
public class AllCombinationSlotState : IState, IEnumerable<CombinationSlotState>,
IEnumerable
```

Inheritance

object ← AllCombinationSlotState

Implements

IState, IEnumerable<CombinationSlotState>, IEnumerable

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

AllCombinationSlotState()

```
public AllCombinationSlotState()
```

AllCombinationSlotState(List)

```
public AllCombinationSlotState(List serialized)
```

Parameters

serialized List

Properties

CombinationSlots

```
public Dictionary<int, CombinationSlotState> CombinationSlots { get; }
```

Property Value

Dictionary<int, CombinationSlotState>

Methods

AddSlot(Address, int)

```
public void AddSlot(Address address, int index = 0)
```

Parameters

address Address

index int

AddSlot(CombinationSlotState)

```
public void AddSlot(CombinationSlotState combinationSlotState)
```

Parameters

combinationSlotState CombinationSlotState

GetEnumerator()

Returns an enumerator that iterates through the collection.

```
public IEnumerator<CombinationSlotState> GetEnumerator()
```

Returns

IEnumerator<CombinationSlotState>

An enumerator that can be used to iterate through the collection.

GetSlot(int)

```
public CombinationSlotState GetSlot(int slotStateIndex)
```

Parameters

`slotStateIndex` int

Returns

CombinationSlotState

MigrationLegacySlotState(IWorldState, Address)

만약 AllCombinationSlotState가 없다면, 슬롯 확장 업데이트 전 4개의 슬롯을 가져와서 채워넣는다.

```
public static AllCombinationSlotState MigrationLegacySlotState(IWorldState
worldState, Address avatarAddress)
```

Parameters

`worldState` IWorldState

Slot 상태를 가져올 world state

`avatarAddress` Address

Migration을 진행할 아바타

Returns

AllCombinationSlotState

Migration AllCombinationSlotState

MigrationLegacySlotState(Func<int, CombinationSlotState?>, Address)

AllCombinationSlotState , , 4 .

```
public static AllCombinationSlotState MigrationLegacySlotState(Func<int,
CombinationSlotState?> stateFactory, Address avatarAddress)
```

Parameters

`stateFactory` Func<int, CombinationSlotState>

CombinationSlotState

`avatarAddress` Address

Migration

Returns

AllCombinationSlotState

Migration AllCombinationSlotState

Serialize()

```
public IValue Serialize()
```

Returns

IValue

SetSlot(CombinationSlotState)

```
public void SetSlot(CombinationSlotState combinationSlotState)
```

Parameters

`combinationSlotState` CombinationSlotState

TryGetSlot(int, out CombinationSlotState?)

```
public bool TryGetSlot(int slotStateIndex, out CombinationSlotState?
combinationSlotState)
```

Parameters

`slotStateIndex` int

`combinationSlotState` CombinationSlotState

Returns

bool

UnlockSlot(Address, int)

```
public void UnlockSlot(Address avatarAddress, int index)
```

Parameters

`avatarAddress` Address

`index` int

Class AllRuneState

Namespace: Nekoyume.Model.State

Assembly: Lib9c.dll

This is new version of rune state. This state stores all rune states of an avatar. AllRuneState has all RuneStates as dictionary and has methods to get/set/update each RuneState. Use this with RuneStateModule.

```
public class AllRuneState : IState
```

Inheritance

object ← AllRuneState

Implements

IState

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

AllRuneState()

```
public AllRuneState()
```

AllRuneState(List)

```
public AllRuneState(List serialized)
```

Parameters

`serialized` List

AllRuneState(int, int)

```
public AllRuneState(int runeId, int level = 0)
```

Parameters

`runeId` int

`level` int

Properties

Runes

```
public Dictionary<int, RuneState> Runes { get; }
```

Property Value

Dictionary<int, RuneState>

Methods

AddRuneState(RuneState)

```
public void AddRuneState(RuneState runeState)
```

Parameters

`runeState` RuneState

AddRuneState(int, int)

```
public void AddRuneState(int runeId, int level = 0)
```

Parameters

`runeId` int

`level` int

GetRuneState(int)

```
public RuneState GetRuneState(int runeId)
```

Parameters

`runeId` int

Returns

RuneState

Serialize()

```
public IValue Serialize()
```

Returns

IValue

SetRuneState(RuneState)

```
public void SetRuneState(RuneState runeState)
```

Parameters

`runeState` RuneState

SetRuneState(int, int)

```
public void SetRuneState(int runeId, int level)
```

Parameters

`runeId` int

`level` int

TryGetRuneState(int, out RuneState)

```
public bool TryGetRuneState(int runeId, out RuneState runeState)
```

Parameters

`runeId` int

`runeState` RuneState

Returns

bool

Class ArenaAvatarState

Namespace: Nekoyume.Model.State

Assembly: Lib9c.dll

Introduced at https://github.com/planetarium/lib9c/pull/1156

```
public class ArenaAvatarState : IState
```

Inheritance

object ← ArenaAvatarState

Implements

IState

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

ArenaAvatarState(List)

```
public ArenaAvatarState(List serialized)
```

Parameters

`serialized` List

ArenaAvatarState(AvatarState)

```
public ArenaAvatarState(AvatarState avatarState)
```

Parameters

`avatarState` AvatarState

Fields

Address

```
public Address Address
```

Field Value

Address

LastBattleBlockIndex

```
public long LastBattleBlockIndex
```

Field Value

long

Properties

Costumes

```
public List<Guid> Costumes { get; }
```

Property Value

List<Guid>

Equipments

```
public List<Guid> Equipments { get; }
```

Property Value

List<Guid>

Methods

DeriveAddress(Address)

```
public static Address DeriveAddress(Address avatarAddress)
```

Parameters

`avatarAddress` Address

Returns

Address

Serialize()

```
public IValue Serialize()
```

Returns

IValue

UpdateCostumes(List<Guid>)

```
public void UpdateCostumes(List<Guid> costumes)
```

Parameters

`costumes` List<Guid>

UpdateEquipment(List<Guid>)

```
public void UpdateEquipment(List<Guid> equipments)
```

Parameters

`equipments` List<Guid>

Class ArenaInfo

Namespace: Nekoyume.Model.State

Assembly: Lib9c.dll

```
public class ArenaInfo : IState
```

Inheritance

object ← ArenaInfo

Implements

IState

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

ArenaInfo(Dictionary)

```
public ArenaInfo(Dictionary serialized)
```

Parameters

`serialized` Dictionary

ArenaInfo(ArenaInfo)

```
public ArenaInfo(ArenaInfo prevInfo)
```

Parameters

`prevInfo` ArenaInfo

ArenaInfo(AvatarState, CharacterSheet, CostumeStatSheet, bool)

```
public ArenaInfo(AvatarState avatarState, CharacterSheet characterSheet,
CostumeStatSheet costumeStatSheet, bool active)
```

Parameters

`avatarState` AvatarState

`characterSheet` CharacterSheet

`costumeStatSheet` CostumeStatSheet

`active` bool

ArenaInfo(AvatarState, CharacterSheet, bool)

```
public ArenaInfo(AvatarState avatarState, CharacterSheet characterSheet,
bool active)
```

Parameters

`avatarState` AvatarState

`characterSheet` CharacterSheet

`active` bool

Fields

AgentAddress

```
public readonly Address AgentAddress
```

Field Value

Address

ArenaRecord

```
public readonly ArenaInfo.Record ArenaRecord
```

Field Value

ArenaInfo.Record

ArmorId

```
[Obsolete("Not used anymore since v100070")]
public int ArmorId
```

Field Value

int

AvatarAddress

```
public readonly Address AvatarAddress
```

Field Value

Address

AvatarName

```
public readonly string AvatarName
```

Field Value

string

CombatPoint

```
[Obsolete("Not used anymore since v100070")]
public long CombatPoint
```

Field Value

long

Level

```
[Obsolete("Not used anymore since v100070")]
public int Level
```

Field Value

int

Receive

```
[Obsolete("Not used anymore since v100070")]
public bool Receive
```

Field Value

bool

Properties

Active

```
public bool Active { get; }
```

Property Value

bool

DailyChallengeCount

```
public int DailyChallengeCount { get; }
```

Property Value

int

Score

```
public int Score { get; }
```

Property Value

int

Methods

Activate()

```
public void Activate()
```

Clone()

```
public ArenaInfo Clone()
```

Returns

ArenaInfo

GetRewardCount()

```
public int GetRewardCount()
```

Returns

int

IsActive(Dictionary)

```
public static bool IsActive(Dictionary serialized)
```

Parameters

`serialized` Dictionary

Returns

bool

ResetCount()

```
public void ResetCount()
```

Serialize()

```
public IValue Serialize()
```

Returns

IValue

Update(ArenaInfo, Result, Func<int, int, Result, (int challengerScore, int defenderScore)>)

```
public int Update(ArenaInfo enemyInfo, BattleLog.Result result, Func<int, int,
BattleLog.Result, (int challengerScore, int defenderScore)> scoreGetter)
```

Parameters

`enemyInfo` ArenaInfo

`result` BattleLog.Result

`scoreGetter` Func<int, int, BattleLog.Result, (int challengerScore, int defenderScore)>

Returns

int

UpdateV1(AvatarState, CharacterSheet)

```
[Obsolete("Use Update()")]
public void UpdateV1(AvatarState state, CharacterSheet characterSheet)
```

Parameters

`state` AvatarState

`characterSheet` CharacterSheet

UpdateV2(AvatarState, CharacterSheet, CostumeStatSheet)

```
[Obsolete("Use Update()")]
public void UpdateV2(AvatarState state, CharacterSheet characterSheet,
CostumeStatSheet costumeStatSheet)
```

Parameters

`state` AvatarState

`characterSheet` CharacterSheet

`costumeStatSheet` CostumeStatSheet

UpdateV3(AvatarState, ArenaInfo, Result)

```
[Obsolete("Use Update()")]
public int UpdateV3(AvatarState avatarState, ArenaInfo enemyInfo,
BattleLog.Result result)
```

Parameters

`avatarState` AvatarState

`enemyInfo` ArenaInfo

`result` BattleLog.Result

Returns

int

UpdateV4(ArenaInfo, Result)

```
[Obsolete("Use Update()")]
public int UpdateV4(ArenaInfo enemyInfo, BattleLog.Result result)
```

Parameters

`enemyInfo` ArenaInfo

`result` BattleLog.Result

Returns

int

UpdateV5(ArenaInfo, Result)

```
[Obsolete("Use Update()")]
public int UpdateV5(ArenaInfo enemyInfo, BattleLog.Result result)
```

Parameters

`enemyInfo` ArenaInfo

`result` BattleLog.Result

Returns

int

Class ArenaInfo.Record

Namespace: Nekoyume.Model.State

Assembly: Lib9c.dll

```
public class ArenaInfo.Record : IState
```

Inheritance

object ← ArenaInfo.Record

Implements

IState

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

Record()

```
public Record()
```

Record(Dictionary)

```
public Record(Dictionary serialized)
```

Parameters

`serialized` Dictionary

Fields

Draw

```
public int Draw
```

Field Value

int

Lose

```
public int Lose
```

Field Value

int

Win

```
public int Win
```

Field Value

int

Methods

Serialize()

```
public IValue Serialize()
```

Returns

IValue

Class AuthorizedMinersState

Namespace: Nekoyume.Model.State

Assembly: Lib9c.dll

```
public class AuthorizedMinersState : State, IState
```

Inheritance

object ← State ← AuthorizedMinersState

Implements

IState

Inherited Members

State.address , State.SerializeBase() , State.SerializeV2Base() , State.SerializeListBase() , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

AuthorizedMinersState(Dictionary)

```
public AuthorizedMinersState(Dictionary serialized)
```

Parameters

`serialized` Dictionary

AuthorizedMinersState(IEnumerable<Address>, long, long)

```
public AuthorizedMinersState(IEnumerable<Address> miners, long interval,
long validUntil)
```

Parameters

`miners` IEnumerable<Address>

`interval` long

`validUntil` long

Fields

Address

```
public static readonly Address Address
```

Field Value

Address

Properties

Interval

```
public long Interval { get; }
```

Property Value

long

Miners

```
public ImmutableHashSet<Address> Miners { get; }
```

Property Value

ImmutableHashSet<Address>

ValidUntil

```
public long ValidUntil { get; }
```

Property Value

long

Methods

Serialize()

```
public override IValue Serialize()
```

Returns

IValue

Class AvatarState

Namespace: Nekoyume.Model.State

Assembly: Lib9c.dll

Agent□ □□□□ □ Avatar□ □□ □□□□.

```
[Serializable]
public class AvatarState : State, IState, ICloneable
```

Inheritance

object ← State ← AvatarState

Implements

IState, ICloneable

Inherited Members

State.address , State.SerializeBase() , State.SerializeV2Base() , State.SerializeListBase() , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Extension Methods

AvatarStateExtensions.GetLevelAndExp(AvatarState, CharacterLevelSheet, int, int) , AvatarStateExtensions.GetLevelAndExpForEventDungeon(AvatarState, CharacterLevelSheet, EventScheduleSheet.Row, int, int) , AvatarStateExtensions.GetLevelAndExpV1(AvatarState, CharacterLevelSheet, int, int) , AvatarStateExtensions.UpdateExp(AvatarState, int, long) , AvatarStateExtensions.UpdateInventory(AvatarState, List<ItemBase>) , AvatarStateExtensions.UpdateMonsterMap(AvatarState, StageWaveSheet, int) , AvatarStateExtensions.ValidEquipmentAndCostume(AvatarState, IEnumerable<Guid>, List<Guid>, ItemRequirementSheet, EquipmentItemRecipeSheet, EquipmentItemSubRecipeSheetV2, EquipmentItemOptionSheet, long, string) , AvatarStateExtensions.ValidEquipmentAndCostumeV2(AvatarState, IEnumerable<Guid>, List<Guid>, ItemRequirementSheet, EquipmentItemRecipeSheet, EquipmentItemSubRecipeSheetV2, EquipmentItemOptionSheet, long, string, GameConfigState)

Constructors

AvatarState(Dictionary)

```
public AvatarState(Dictionary serialized)
```

Parameters

`serialized` Dictionary

AvatarState(List)

```
public AvatarState(List serialized)
```

Parameters

`serialized` List

AvatarState(Address, Address, long, QuestList, WorldInformation, Address, string)

```
public AvatarState(Address address, Address agentAddress, long blockIndex, QuestList
questList, WorldInformation worldInformation, Address rankingMapAddress, string name
= null)
```

Parameters

`address` Address

`agentAddress` Address

`blockIndex` long

`questList` QuestList

`worldInformation` WorldInformation

`rankingMapAddress` Address

`name` string

AvatarState(AvatarState)

```
public AvatarState(AvatarState avatarState)
```

Parameters

`avatarState` AvatarState

Fields

CombinationSlotCapacity

```
public const int CombinationSlotCapacity = 8
```

Field Value

int

CurrentVersion

```
public const int CurrentVersion = 2
```

Field Value

int

DefaultCombinationSlotCount

```
public const int DefaultCombinationSlotCount = 4
```

Field Value

int

RankingMapAddress

```
[Obsolete("don't use this field.")]
public readonly Address RankingMapAddress
```

Field Value

Address

actionPoint

```
public int actionPoint
```

Field Value

int

agentAddress

```
public Address agentAddress
```

Field Value

Address

blockIndex

```
public long blockIndex
```

Field Value

long

characterId

```
public int characterId
```

Field Value

int

combinationSlotAddresses

```
[Obsolete("don't use this field, use AllCombinationSlotState instead.")]
public List<Address> combinationSlotAddresses
```

Field Value

List<Address>

dailyRewardReceivedIndex

```
public long dailyRewardReceivedIndex
```

Field Value

long

ear

```
public int ear
```

Field Value

int

eventMap

```
public CollectionMap eventMap
```

Field Value

CollectionMap

exp

```
public long exp
```

Field Value

long

hair

```
public int hair
```

Field Value

int

inventory

```
public Inventory inventory
```

Field Value

Inventory

itemMap

```
public CollectionMap itemMap
```

Field Value

CollectionMap

lens

```
public int lens
```

Field Value

int

level

```
public int level
```

Field Value

int

mailBox

```
public MailBox mailBox
```

Field Value

MailBox

monsterMap

```
public CollectionMap monsterMap
```

Field Value

CollectionMap

name

```
public string name
```

Field Value

string

questList

```
public QuestList questList
```

Field Value

QuestList

stageMap

```
public CollectionMap stageMap
```

Field Value

CollectionMap

tail

```
public int tail
```

Field Value

int

updatedAt

```
public long updatedAt
```

Field Value

long

worldInformation

```
public WorldInformation worldInformation
```

Field Value

WorldInformation

Properties

NameWithHash

```
public string NameWithHash { get; }
```

Property Value

string

Nonce

```
public int Nonce { get; }
```

Property Value

int

Version

```
public int Version { get; }
```

Property Value

int

Methods

Apply(Player, long)

```
public void Apply(Player player, long index)
```

Parameters

`player` Player

`index` long

Clone()

Creates a new object that is a copy of the current instance.

```
public object Clone()
```

Returns

object

A new object that is a copy of this instance.

Create(Address, Address, long, AvatarSheets, Address, string)

```
public static AvatarState Create(Address address, Address agentAddress, long
blockIndex, AvatarSheets avatarSheets, Address rankingMapAddress, string name
= null)
```

Parameters

`address` Address

`agentAddress` Address

`blockIndex` long

`avatarSheets` AvatarSheets

`rankingMapAddress` Address

`name` string

Returns

AvatarState

CreateAvatarAddress()

```
public static Address CreateAvatarAddress()
```

Returns

Address

Customize(int, int, int, int)

```
public void Customize(int hair, int lens, int ear, int tail)
```

Parameters

`hair` int

`lens` int

`ear` int

`tail` int

EquipItems(IEnumerable<Guid>)

```
public void EquipItems(IEnumerable<Guid> itemIds)
```

Parameters

`itemIds` IEnumerable<Guid>

GetArmorId()

```
public int GetArmorId()
```

Returns

int

GetNonFungibleItems<T>(List<Guid>)

```
public List<T> GetNonFungibleItems<T>(List<Guid> itemIds)
```

Parameters

`itemIds` List<Guid>

Returns

List<T>

Type Parameters

T

GetPortraitId()

```
public int GetPortraitId()
```

Returns

int

Serialize()

```
public override IValue Serialize()
```

Returns

IValue

SerializeList()

```
public IValue SerializeList()
```

Returns

IValue

Update(IStageSimulator)

```
public void Update(IStageSimulator stageSimulator)
```

Parameters

`stageSimulator` IStageSimulator

Update(Mail)

```
public void Update(Mail mail)
```

Parameters

`mail` Mail

Update2(Mail)

```
[Obsolete("Use Update")]
public void Update2(Mail mail)
```

Parameters

`mail` Mail

Update3(Mail)

```
[Obsolete("Use Update")]
public void Update3(Mail mail)
```

Parameters

`mail` Mail

UpdateFromAddCostume(Costume)

```
public void UpdateFromAddCostume(Costume costume)
```

Parameters

`costume` Costume

UpdateFromAddItem(ItemBase, int, bool)

```
public void UpdateFromAddItem(ItemBase itemUsable, int count, bool canceled)
```

Parameters

`itemUsable` ItemBase

`count` int

`canceled` bool

UpdateFromAddItem(ItemUsable, bool)

```
public void UpdateFromAddItem(ItemUsable itemUsable, bool canceled)
```

Parameters

`itemUsable` ItemUsable

`canceled` bool

UpdateFromCombination(ItemUsable)

```
public void UpdateFromCombination(ItemUsable itemUsable)
```

Parameters

`itemUsable` ItemUsable

UpdateFromItemEnhancement(Equipment)

```
public void UpdateFromItemEnhancement(Equipment equipment)
```

Parameters

`equipment` Equipment

UpdateFromQuestReward(Quest, MaterialItemSheet)

```
public void UpdateFromQuestReward(Quest quest, MaterialItemSheet materialItemSheet)
```

Parameters

`quest` Quest

`materialItemSheet` MaterialItemSheet

UpdateFromRapidCombination(ResultModel, long)

```
public void UpdateFromRapidCombination(CombinationConsumable5.ResultModel result,
long requiredIndex)
```

Parameters

`result` CombinationConsumable5.ResultModel

`requiredIndex` long

UpdateFromRapidCombinationV2(ResultModel, long)

```
public void UpdateFromRapidCombinationV2(RapidCombination5.ResultModel result,
long requiredIndex)
```

Parameters

`result` RapidCombination5.ResultModel

`requiredIndex` long

UpdateGeneralQuest(IEnumerable<QuestEventType>)

```
public void UpdateGeneralQuest(IEnumerable<QuestEventType> types)
```

Parameters

`types` IEnumerable<QuestEventType>

UpdateQuestRewards(MaterialItemSheet)

```
public void UpdateQuestRewards(MaterialItemSheet materialItemSheet)
```

Parameters

`materialItemSheet` MaterialItemSheet

UpdateTemp(Mail, long)

```
[Obsolete("No longer in use.")]
public void UpdateTemp(Mail mail, long currentBlockIndex)
```

Parameters

`mail` Mail

`currentBlockIndex` long

ValidateConsumable(List<Guid>, long)

```
public List<int> ValidateConsumable(List<Guid> consumableIds,
long currentBlockIndex)
```

Parameters

`consumableIds` List<Guid>

`currentBlockIndex` long

Returns

List<int>

ValidateConsumableV2(List<Guid>, long, GameConfigState)

```
public List<int> ValidateConsumableV2(List<Guid> consumableIds, long
currentBlockIndex, GameConfigState gameConfigState)
```

Parameters

`consumableIds` List<Guid>

`currentBlockIndex` long

`gameConfigState` GameConfigState

Returns

List<int>

ValidateCostume(HashSet<int>)

```
public void ValidateCostume(HashSet<int> costumeIds)
```

Parameters

`costumeIds` HashSet<int>

ValidateCostume(IEnumerable<Guid>)

```
public List<int> ValidateCostume(IEnumerable<Guid> costumeIds)
```

Parameters

`costumeIds` IEnumerable<Guid>

Returns

List<int>

ValidateCostumeV2(IEnumerable<Guid>, GameConfigState)

```
public List<Costume> ValidateCostumeV2(IEnumerable<Guid> costumeIds,
GameConfigState gameConfigState)
```

Parameters

`costumeIds` IEnumerable<Guid>

`gameConfigState` GameConfigState

Returns

List<Costume>

ValidateEquipments(List<Guid>, long)

```
public void ValidateEquipments(List<Guid> equipmentIds, long blockIndex)
```

Parameters

`equipmentIds` List<Guid>

`blockIndex` long

ValidateEquipmentsV2(List<Guid>, long)

```
public List<Equipment> ValidateEquipmentsV2(List<Guid> equipmentIds,
long blockIndex)
```

Parameters

`equipmentIds` List<Guid>

`blockIndex` long

Returns

List<Equipment>

ValidateEquipmentsV3(List<Guid>, long, GameConfigState)

```
public List<Equipment> ValidateEquipmentsV3(List<Guid> equipmentIds, long
blockIndex, GameConfigState gameConfigState)
```

Parameters

`equipmentIds` List<Guid>

`blockIndex` long

`gameConfigState` GameConfigState

Returns

List<Equipment>

ValidateItemRequirement(List<int>, List<Equipment>, ItemRequirementSheet, EquipmentItemRecipeSheet, EquipmentItemSubRecipeSheetV2, EquipmentItemOptionSheet, string)

```
public void ValidateItemRequirement(List<int> itemIds, List<Equipment> equipments,
ItemRequirementSheet requirementSheet, EquipmentItemRecipeSheet recipeSheet,
EquipmentItemSubRecipeSheetV2 subRecipeSheet, EquipmentItemOptionSheet
itemOptionSheet, string addressesHex)
```

Parameters

`itemIds` List<int>

`equipments` List<Equipment>

`requirementSheet` ItemRequirementSheet

`recipeSheet` EquipmentItemRecipeSheet

`subRecipeSheet` EquipmentItemSubRecipeSheetV2

`itemOptionSheet` EquipmentItemOptionSheet

`addressesHex` string

Class CollectionState

Namespace: Nekoyume.Model.State

Assembly: Lib9c.dll

Represents the state of a collection.

```
public class CollectionState : IBencodable
```

Inheritance

object ← CollectionState

Implements

IBencodable

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

CollectionState()

```
public CollectionState()
```

CollectionState(IValue)

```
public CollectionState(IValue bencoded)
```

Parameters

`bencoded` IValue

CollectionState(List)

```
public CollectionState(List serialized)
```

Parameters

`serialized` List

Fields

Ids

```
public SortedSet<int> Ids
```

Field Value

SortedSet<int>

Properties

Bencoded

An Bencodex.Types.IValue representation of this object that can be decoded back to instantiate an equal object. The decoded object must be equal to the original in the sense that Equals(T) should be true.

```
public IValue Bencoded { get; }
```

Property Value

IValue

Remarks

Note that the only requirement is that the produced Bencodex.Types.IValue can be decoded back to an equal object. This representation may not be canonical in the sense that

additional junk data may be present in an Bencodex.Types.IValue that one may wish to decode and this may be discarded while decoding.

A specific implemnetation may decide to only allow the canonical representation to be decoded.

Methods

GetModifiers(CollectionSheet)

```
public List<StatModifier> GetModifiers(CollectionSheet collectionSheet)
```

Parameters

`collectionSheet` CollectionSheet

Returns

List<StatModifier>

Class CombinationSlotState

Namespace: Nekoyume.Model.State

Assembly: Lib9c.dll

```
public class CombinationSlotState : State, IState
```

Inheritance

object ← State ← CombinationSlotState

Implements

IState

Inherited Members

State.address , State.SerializeBase() , State.SerializeV2Base() , State.SerializeListBase() , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Extension Methods

CombinationSlotStateExtensions.TryGetMail(CombinationSlotState, long, long, out CombinationMail, out ItemEnhanceMail) , CombinationSlotStateExtensions.TryGetResultId(CombinationSlotState, out Guid) , CombinationSlotStateExtensions.ValidateFromAction(CombinationSlotState, long, AvatarState, int, string, string)

Constructors

CombinationSlotState(Dictionary)

```
public CombinationSlotState(Dictionary serialized)
```

Parameters

`serialized` Dictionary

CombinationSlotState(Address, int)

```
public CombinationSlotState(Address address, int index = 0)
```

Parameters

`address` Address

`index` int

Fields

DeriveFormat

```
public const string DeriveFormat = "combination-slot-{0}"
```

Field Value

string

Properties

Index

It is a CombinationSlot index. start from 0.

```
public int Index { get; set; }
```

Property Value

int

IsUnlocked

```
public bool IsUnlocked { get; }
```

Property Value

bool

PetId

```
public int? PetId { get; }
```

Property Value

int?

RequiredBlockIndex

```
public long RequiredBlockIndex { get; }
```

Property Value

long

Result

```
public AttachmentActionResult Result { get; }
```

Property Value

AttachmentActionResult

WorkCompleteBlockIndex

Serialize key is "unlockBlockIndex".

```
public long WorkCompleteBlockIndex { get; }
```

Property Value

long

WorkStartBlockIndex

Serialize key is "startBlockIndex".

```
public long WorkStartBlockIndex { get; }
```

Property Value

long

Methods

DeriveAddress(Address, int)

Is only used for migration legacy state.

```
public static Address DeriveAddress(Address address, int slotIndex)
```

Parameters

`address` Address

`slotIndex` int

Returns

Address

Serialize()

```
public override IValue Serialize()
```

Returns

IValue

Unlock()

```
public bool Unlock()
```

Returns

bool

Update(AttachmentActionResult, long, long, int?)

```
public void Update(AttachmentActionResult result, long blockIndex, long
workCompleteBlockIndex, int? petId = null)
```

Parameters

`result` AttachmentActionResult

`blockIndex` long

`workCompleteBlockIndex` long

`petId` int?

Update(long)

```
public void Update(long blockIndex)
```

Parameters

`blockIndex` long

Update(long, Material, int)

```
public void Update(long blockIndex, Material material, int count)
```

Parameters

`blockIndex` long

`material` Material

`count` int

UpdateV2(long, Material, int)

```
public void UpdateV2(long blockIndex, Material material, int count)
```

Parameters

`blockIndex` long

`material` Material

`count` int

Validate(AvatarState, long)

```
[Obsolete("Use ValidateV2")]
public bool Validate(AvatarState avatarState, long blockIndex)
```

Parameters

`avatarState` AvatarState

`blockIndex` long

Returns

bool

ValidateSlotIndex(int)

```
public static bool ValidateSlotIndex(int index)
```

Parameters

`index` int

Returns

bool

ValidateV2(long)

```
public bool ValidateV2(long blockIndex)
```

Parameters

`blockIndex` long

Returns

bool

Class CpState

Namespace: Nekoyume.Model.State

Assembly: Lib9c.dll

CpState is a state that stores the cp of the player.

```
public class CpState : IBencodable, IState
```

Inheritance

object ← CpState

Implements

IBencodable, IState

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

CpState(IValue)

```
public CpState(IValue bencoded)
```

Parameters

`bencoded` IValue

CpState(long)

```
public CpState(long cp)
```

Parameters

cp long

Fields

Cp

```
public long Cp
```

Field Value

long

Properties

Bencoded

An Bencodex.Types.IValue representation of this object that can be decoded back to instantiate an equal object. The decoded object must be equal to the original in the sense that Equals(T) should be true.

```
public IValue Bencoded { get; }
```

Property Value

IValue

Remarks

Note that the only requirement is that the produced Bencodex.Types.IValue can be decoded back to an equal object. This representation may not be canonical in the sense that additional junk data may be present in an Bencodex.Types.IValue that one may wish to decode and this may be discarded while decoding.

A specific implemnetation may decide to only allow the canonical representation to be decoded.

Methods

Serialize()

```
public IValue Serialize()
```

Returns

IValue

Class CreditsState

Namespace: Nekoyume.Model.State

Assembly: Lib9c.dll

```
[Serializable]
public class CreditsState : State, IState
```

Inheritance

object ← State ← CreditsState

Implements

IState

Inherited Members

State.address , State.SerializeBase() , State.SerializeV2Base() , State.SerializeListBase() , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

CreditsState(Dictionary)

```
public CreditsState(Dictionary serialized)
```

Parameters

`serialized` Dictionary

CreditsState(IEnumerable<string>)

```
public CreditsState(IEnumerable<string> names)
```

Parameters

names IEnumerable<string>

Fields

Address

```
public static readonly Address Address
```

Field Value

Address

Properties

Names

```
public IImmutableList<string> Names { get; }
```

Property Value

IImmutableList<string>

Methods

Serialize()

```
public override IValue Serialize()
```

Returns

IValue

Class CrystalCostState

Namespace: Nekoyume.Model.State

Assembly: Lib9c.dll

```
[Serializable]
public class CrystalCostState : IState
```

Inheritance

object ← CrystalCostState

Implements

IState

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

CrystalCostState(Address, List)

```
public CrystalCostState(Address address, List serialized)
```

Parameters

`address` Address

`serialized` List

CrystalCostState(Address, FungibleAssetValue)

```
public CrystalCostState(Address address, FungibleAssetValue crystal)
```

Parameters

`address` Address

`crystal` FungibleAssetValue

Fields

Address

```
public Address Address
```

Field Value

Address

CRYSTAL

```
public FungibleAssetValue CRYSTAL
```

Field Value

FungibleAssetValue

Count

```
public int Count
```

Field Value

int

DailyIntervalIndex

```
public const long DailyIntervalIndex = 7200
```

Field Value

long

Methods

Serialize()

```
public IValue Serialize()
```

Returns

IValue

Class CrystalRandomSkillState

Namespace: Nekoyume.Model.State

Assembly: Lib9c.dll

```
public class CrystalRandomSkillState : IState
```

Inheritance

object ← CrystalRandomSkillState

Implements

IState

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

CrystalRandomSkillState(Address, List)

```
public CrystalRandomSkillState(Address address, List serialized)
```

Parameters

`address` Address

`serialized` List

CrystalRandomSkillState(Address, int)

```
public CrystalRandomSkillState(Address address, int stageId)
```

Parameters

`address` Address

`stageId` int

Properties

Address

```
public Address Address { get; }
```

Property Value

Address

SkillIds

```
public List<int> SkillIds { get; }
```

Property Value

List<int>

StageId

```
public int StageId { get; }
```

Property Value

int

StarCount

```
public int StarCount { get; }
```

Property Value

int

Methods

GetHighestRankSkill(CrystalRandomBuffSheet)

```
public int GetHighestRankSkill(CrystalRandomBuffSheet crystalRandomBuffSheet)
```

Parameters

`crystalRandomBuffSheet` CrystalRandomBuffSheet

Returns

int

GetSkill(int, CrystalRandomBuffSheet, SkillSheet)

```
public static Skill GetSkill(int skillId, CrystalRandomBuffSheet
crystalRandomBuffSheet, SkillSheet skillSheet)
```

Parameters

`skillId` int

`crystalRandomBuffSheet` CrystalRandomBuffSheet

`skillSheet` SkillSheet

Returns

Skill

Serialize()

```
public IValue Serialize()
```

Returns

IValue

Update(List<int>)

```
public void Update(List<int> skillIds)
```

Parameters

`skillIds` List<int>

Update(int, CrystalStageBuffGachaSheet)

```
public void Update(int gotStarCount, CrystalStageBuffGachaSheet sheet)
```

Parameters

`gotStarCount` int

`sheet` CrystalStageBuffGachaSheet

Class DuplicateRedeemException

Namespace: Nekoyume.Model.State

Assembly: Lib9c.dll

```
[Serializable]
public class DuplicateRedeemException : InvalidOperationException, ISerializable
```

Inheritance

object ← Exception ← SystemException ← InvalidOperationException ← DuplicateRedeemException

Implements

ISerializable

Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Constructors

DuplicateRedeemException(SerializationInfo, StreamingContext)

```
public DuplicateRedeemException(SerializationInfo info, StreamingContext context)
```

Parameters

`info` SerializationInfo

`context` StreamingContext

DuplicateRedeemException(string)

```
public DuplicateRedeemException(string s)
```

Parameters

s [string]

Class FailedToUnregisterInShopStateException

Namespace: Nekoyume.Model.State

Assembly: Lib9c.dll

```
[Serializable]
public class FailedToUnregisterInShopStateException : Exception, ISerializable
```

Inheritance

object ← Exception ← FailedToUnregisterInShopStateException

Implements

ISerializable

Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Constructors

FailedToUnregisterInShopStateException(SerializationInfo, StreamingContext)

```
protected FailedToUnregisterInShopStateException(SerializationInfo info,
StreamingContext context)
```

Parameters

`info` SerializationInfo

`context` StreamingContext

FailedToUnregisterInShopStateException(string)

```
public FailedToUnregisterInShopStateException(string message)
```

Parameters

`message` string

Class GameConfigState

Namespace: Nekoyume.Model.State

Assembly: Lib9c.dll

```
[Serializable]
public class GameConfigState : State, IState
```

Inheritance

object ← State ← GameConfigState

Implements

IState

Inherited Members

State.address , State.SerializeBase() , State.SerializeV2Base() , State.SerializeListBase() , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

GameConfigState()

```
public GameConfigState()
```

GameConfigState(Dictionary)

```
public GameConfigState(Dictionary serialized)
```

Parameters

`serialized` Dictionary

GameConfigState(string)

```
public GameConfigState(string csv)
```

Parameters

`csv` string

Fields

Address

```
public static readonly Address Address
```

Field Value

Address

Properties

ActionPointMax

```
public int ActionPointMax { get; }
```

Property Value

int

AdventureBossActiveInterval

```
public long AdventureBossActiveInterval { get; }
```

Property Value

long

AdventureBossClaimInterval

```
public long AdventureBossClaimInterval { get; }
```

Property Value

long

AdventureBossInactiveInterval

```
public long AdventureBossInactiveInterval { get; }
```

Property Value

long

AdventureBossMinBounty

```
public BigInteger AdventureBossMinBounty { get; }
```

Property Value

BigInteger

AdventureBossNcgApRatio

```
public decimal AdventureBossNcgApRatio { get; }
```

Property Value

decimal

AdventureBossNcgRuneRatio

```
public decimal AdventureBossNcgRuneRatio { get; }
```

Property Value

decimal

AdventureBossWantedRequiredStakingLevel

```
public int AdventureBossWantedRequiredStakingLevel { get; }
```

Property Value

int

BattleArenaInterval

```
public int BattleArenaInterval { get; }
```

Property Value

int

CustomEquipmentCraftIconCostMultiplier

```
public long CustomEquipmentCraftIconCostMultiplier { get; }
```

Property Value

long

DailyArenaInterval

```
public int DailyArenaInterval { get; }
```

Property Value

int

DailyRewardInterval

```
public int DailyRewardInterval { get; }
```

Property Value

int

DailyRuneRewardAmount

```
public int DailyRuneRewardAmount { get; }
```

Property Value

int

DailyWorldBossInterval

```
public int DailyWorldBossInterval { get; }
```

Property Value

int

HourglassPerBlock

```
public int HourglassPerBlock { get; }
```

Property Value

int

RequireCharacterLevel_ConsumableSlot1

```
public int RequireCharacterLevel_ConsumableSlot1 { get; }
```

Property Value

int

RequireCharacterLevel_ConsumableSlot2

```
public int RequireCharacterLevel_ConsumableSlot2 { get; }
```

Property Value

int

RequireCharacterLevel_ConsumableSlot3

```
public int RequireCharacterLevel_ConsumableSlot3 { get; }
```

Property Value

int

RequireCharacterLevel_ConsumableSlot4

```
public int RequireCharacterLevel_ConsumableSlot4 { get; }
```

Property Value

int

RequireCharacterLevel_ConsumableSlot5

```
public int RequireCharacterLevel_ConsumableSlot5 { get; }
```

Property Value

int

RequireCharacterLevel_EarCostumeSlot

```
public int RequireCharacterLevel_EarCostumeSlot { get; }
```

Property Value

int

RequireCharacterLevel_EquipmentSlotArmor

```
public int RequireCharacterLevel_EquipmentSlotArmor { get; }
```

Property Value

int

RequireCharacterLevel_EquipmentSlotAura

```
public int RequireCharacterLevel_EquipmentSlotAura { get; }
```

Property Value

int

RequireCharacterLevel_EquipmentSlotBelt

```
public int RequireCharacterLevel_EquipmentSlotBelt { get; }
```

Property Value

int

RequireCharacterLevel_EquipmentSlotGrimoire

```
public int RequireCharacterLevel_EquipmentSlotGrimoire { get; }
```

Property Value

int

RequireCharacterLevel_EquipmentSlotNecklace

```
public int RequireCharacterLevel_EquipmentSlotNecklace { get; }
```

Property Value

int

RequireCharacterLevel_EquipmentSlotRing1

```
public int RequireCharacterLevel_EquipmentSlotRing1 { get; }
```

Property Value

int

RequireCharacterLevel_EquipmentSlotRing2

```
public int RequireCharacterLevel_EquipmentSlotRing2 { get; }
```

Property Value

int

RequireCharacterLevel_EquipmentSlotWeapon

```
public int RequireCharacterLevel_EquipmentSlotWeapon { get; }
```

Property Value

int

RequireCharacterLevel_EyeCostumeSlot

```
public int RequireCharacterLevel_EyeCostumeSlot { get; }
```

Property Value

int

RequireCharacterLevel_FullCostumeSlot

```
public int RequireCharacterLevel_FullCostumeSlot { get; }
```

Property Value

int

RequireCharacterLevel_HairCostumeSlot

```
public int RequireCharacterLevel_HairCostumeSlot { get; }
```

Property Value

int

RequireCharacterLevel_TailCostumeSlot

```
public int RequireCharacterLevel_TailCostumeSlot { get; }
```

Property Value

int

RequireCharacterLevel_TitleSlot

```
public int RequireCharacterLevel_TitleSlot { get; }
```

Property Value

int

RuneSkillSlotCrystalUnlockCost

```
public int RuneSkillSlotCrystalUnlockCost { get; }
```

Property Value

int

RuneSkillSlotUnlockCost

```
public int RuneSkillSlotUnlockCost { get; }
```

Property Value

int

RuneStatSlotCrystalUnlockCost

```
public int RuneStatSlotCrystalUnlockCost { get; }
```

Property Value

int

RuneStatSlotUnlockCost

```
public int RuneStatSlotUnlockCost { get; }
```

Property Value

int

ShatterStrikeMaxDamage

```
public long ShatterStrikeMaxDamage { get; }
```

Property Value

long

StakeRegularFixedRewardSheet_V2_StartBlockIndex

```
public long StakeRegularFixedRewardSheet_V2_StartBlockIndex { get; }
```

Property Value

long

StakeRegularRewardSheet_V2_StartBlockIndex

```
public long StakeRegularRewardSheet_V2_StartBlockIndex { get; }
```

Property Value

long

StakeRegularRewardSheet_V3_StartBlockIndex

```
public long StakeRegularRewardSheet_V3_StartBlockIndex { get; }
```

Property Value

long

StakeRegularRewardSheet_V4_StartBlockIndex

```
public long StakeRegularRewardSheet_V4_StartBlockIndex { get; }
```

Property Value

long

StakeRegularRewardSheet_V5_StartBlockIndex

```
public long StakeRegularRewardSheet_V5_StartBlockIndex { get; }
```

Property Value

long

WeeklyArenaInterval

```
public int WeeklyArenaInterval { get; }
```

Property Value

int

WorldBossRequiredInterval

```
public int WorldBossRequiredInterval { get; }
```

Property Value

int

Methods

Serialize()

```
public override IValue Serialize()
```

Returns

IValue

Set(GameConfigSheet)

```
public void Set(GameConfigSheet sheet)
```

Parameters

`sheet` GameConfigSheet

Update(Row)

```
public void Update(GameConfigSheet.Row row)
```

Parameters

`row` GameConfigSheet.Row

Class GoldBalanceState

Namespace: Nekoyume.Model.State

Assembly: Lib9c.dll

```
[Serializable]
public class GoldBalanceState : State, IState, ICloneable
```

Inheritance

object ← State ← GoldBalanceState

Implements

IState, ICloneable

Inherited Members

State.address , State.SerializeBase() , State.SerializeV2Base() , State.SerializeListBase() , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

GoldBalanceState(Dictionary)

```
public GoldBalanceState(Dictionary serialized)
```

Parameters

`serialized` Dictionary

GoldBalanceState(IValue)

```
public GoldBalanceState(IValue serialized)
```

Parameters

`serialized` IValue

GoldBalanceState(Address, FungibleAssetValue)

```
public GoldBalanceState(Address address, FungibleAssetValue gold)
```

Parameters

`address` Address

`gold` FungibleAssetValue

Fields

Gold

```
public readonly FungibleAssetValue Gold
```

Field Value

FungibleAssetValue

Methods

Add(FungibleAssetValue)

```
public GoldBalanceState Add(FungibleAssetValue adder)
```

Parameters

`adder` FungibleAssetValue

Returns

GoldBalanceState

Clone()

Creates a new object that is a copy of the current instance.

```
public object Clone()
```

Returns

object

A new object that is a copy of this instance.

Serialize()

```
public override IValue Serialize()
```

Returns

IValue

Class GoldCurrencyState

Namespace: Nekoyume.Model.State

Assembly: Lib9c.dll

```
[Serializable]
public class GoldCurrencyState : State, IState, ISerializable
```

Inheritance

object ← State ← GoldCurrencyState

Implements

IState, ISerializable

Inherited Members

State.address , State.SerializeBase() , State.SerializeV2Base() , State.SerializeListBase() , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

GoldCurrencyState(Dictionary)

```
public GoldCurrencyState(Dictionary serialized)
```

Parameters

`serialized` Dictionary

GoldCurrencyState(Currency)

```
public GoldCurrencyState(Currency currency)
```

Parameters

`currency` Currency

GoldCurrencyState(Currency, long)

```
public GoldCurrencyState(Currency currency, long initialSupply)
```

Parameters

`currency` Currency

`initialSupply` long

GoldCurrencyState(SerializationInfo, StreamingContext)

```
protected GoldCurrencyState(SerializationInfo info, StreamingContext context)
```

Parameters

`info` SerializationInfo

`context` StreamingContext

Fields

Address

```
public static readonly Address Address
```

Field Value

Address

DEFAULT_INITIAL_SUPPLY

```
public const long DEFAULT_INITIAL_SUPPLY = 1000000000
```

Field Value

long

Properties

Currency

```
public Currency Currency { get; }
```

Property Value

Currency

InitialSupply

```
public long InitialSupply { get; }
```

Property Value

long

Methods

GetObjectData(SerializationInfo, StreamingContext)

Populates a SerializationInfo with the data needed to serialize the target object.

```
public void GetObjectData(SerializationInfo info, StreamingContext context)
```

Parameters

`info` SerializationInfo

The SerializationInfo to populate with data.

`context` StreamingContext

The destination (see StreamingContext) for this serialization.

Exceptions

SecurityException

The caller does not have the required permission.

Serialize()

```
public override IValue Serialize()
```

Returns

IValue

Class HammerPointState

Namespace: Nekoyume.Model.State

Assembly: Lib9c.dll

```
public class HammerPointState : IState
```

Inheritance

object ← HammerPointState

Implements

IState

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

HammerPointState(Address, List)

```
public HammerPointState(Address address, List serialized)
```

Parameters

`address` Address

`serialized` List

HammerPointState(Address, int)

```
public HammerPointState(Address address, int recipeId)
```

Parameters

`address` Address

`recipeId` int

Properties

Address

```
public Address Address { get; }
```

Property Value

Address

HammerPoint

```
public int HammerPoint { get; }
```

Property Value

int

RecipeId

```
public int RecipeId { get; }
```

Property Value

int

Methods

AddHammerPoint(int, CrystalHammerPointSheet)

```
public void AddHammerPoint(int point, CrystalHammerPointSheet sheet = null)
```

Parameters

`point` int

`sheet` CrystalHammerPointSheet

ResetHammerPoint()

```
public void ResetHammerPoint()
```

Serialize()

```
public IValue Serialize()
```

Returns

IValue

Interface IState

Namespace: Nekoyume.Model.State

Assembly: Lib9c.dll

```
public interface IState
```

Methods

Serialize()

```
IValue Serialize()
```

Returns

IValue

Class InfiniteTowerBoardState

Namespace: Nekoyume.Model.State

Assembly: Lib9c.dll

Represents the board state for infinite tower, tracking clear counts per floor.

```
[Serializable]
public class InfiniteTowerBoardState : IState
```

Inheritance

object ← InfiniteTowerBoardState

Implements

IState

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

InfiniteTowerBoardState(List)

Creates an infinite tower board state from serialized data.

```
public InfiniteTowerBoardState(List serialized)
```

Parameters

`serialized` List

The serialized data.

InfiniteTowerBoardState(int)

Creates a new infinite tower board state.

```
public InfiniteTowerBoardState(int infiniteTowerId)
```

Parameters

`infiniteTowerId` int

The infinite tower ID.

Properties

FloorClearCounts

Dictionary mapping floor ID to the number of players who cleared it.

```
public Dictionary<int, int> FloorClearCounts { get; }
```

Property Value

Dictionary<int, int>

InfiniteTowerId

The infinite tower ID this board belongs to.

```
public int InfiniteTowerId { get; }
```

Property Value

int

LastUpdatedBlockIndex

The block index when this board was last updated.

```
public long LastUpdatedBlockIndex { get; }
```

Property Value

long

Methods

GetFloorClearCount(int)

Gets the clear count for a specific floor.

```
public int GetFloorClearCount(int floorId)
```

Parameters

`floorId` int

The floor ID.

Returns

int

The number of players who cleared this floor.

GetStatistics()

Gets statistics about the board state.

```
public Dictionary<string, object> GetStatistics()
```

Returns

Dictionary<string, object>

A dictionary containing various statistics.

GetTopFloorsByClearCount(int)

Gets the top floors by clear count.

```
public List<int> GetTopFloorsByClearCount(int limit = 10)
```

Parameters

`limit` int

The maximum number of floors to return.

Returns

List<int>

A list of floor IDs sorted by clear count (descending).

GetTotalClearedFloors()

Gets the total number of floors that have been cleared by at least one player.

```
public int GetTotalClearedFloors()
```

Returns

int

The number of floors with at least one clear.

RecordFloorClear(int, long)

Records a floor clear for a player.

```
public void RecordFloorClear(int floorId, long blockIndex)
```

Parameters

`floorId` int

The floor ID that was cleared.

`blockIndex` long

The current block index.

Serialize()

Serializes the state to Bencodex format.

```
public IValue Serialize()
```

Returns

IValue

The serialized state.

Class InvalidRedeemCodeException

Namespace: Nekoyume.Model.State

Assembly: Lib9c.dll

```
[Serializable]
public class InvalidRedeemCodeException : KeyNotFoundException, ISerializable
```

Inheritance

object ← Exception ← SystemException ← KeyNotFoundException ← InvalidRedeemCodeException

Implements

ISerializable

Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Constructors

InvalidRedeemCodeException()

```
public InvalidRedeemCodeException()
```

InvalidRedeemCodeException(SerializationInfo, StreamingContext)

```
public InvalidRedeemCodeException(SerializationInfo info, StreamingContext context)
```

Parameters

`info` SerializationInfo

`context` StreamingContext

InvalidRedeemCodeException(string)

```
public InvalidRedeemCodeException(string s)
```

Parameters

`s` string

Class ItemSlotState

Namespace: Nekoyume.Model.State

Assembly: Lib9c.dll

```
public class ItemSlotState : IState
```

Inheritance

object ← ItemSlotState

Implements

IState

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

ItemSlotState(List)

```
public ItemSlotState(List serialized)
```

Parameters

`serialized` List

ItemSlotState(BattleType)

```
public ItemSlotState(BattleType battleType)
```

Parameters

`battleType` BattleType

Properties

BattleType

```
public BattleType BattleType { get; }
```

Property Value

BattleType

Costumes

```
public List<Guid> Costumes { get; }
```

Property Value

List<Guid>

Equipments

```
public List<Guid> Equipments { get; }
```

Property Value

List<Guid>

Methods

DeriveAddress(Address, BattleType)

```
public static Address DeriveAddress(Address avatarAddress, BattleType battleType)
```

Parameters

`avatarAddress` Address

`battleType` BattleType

Returns

Address

Serialize()

```
public IValue Serialize()
```

Returns

IValue

UpdateCostumes(List<Guid>)

```
public void UpdateCostumes(List<Guid> costumes)
```

Parameters

`costumes` List<Guid>

UpdateEquipment(List<Guid>)

```
public void UpdateEquipment(List<Guid> equipments)
```

Parameters

`equipments` List<Guid>

Class LazyState<TState, TEncoding>

Namespace: Nekoyume.Model.State

Assembly: Lib9c.dll

```
[Serializable]
public sealed class LazyState<TState, TEncoding> : IState, ISerializable where
TState : IState where TEncoding : IValue
```

Type Parameters

`TState`

`TEncoding`

Inheritance

object ← LazyState<TState, TEncoding>

Implements

IState, ISerializable

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

LazyState(TState)

```
public LazyState(TState loadedValue)
```

Parameters

`loadedValue` TState

LazyState(TEncoding, Func<TEncoding, TState>)

```
public LazyState(TEncoding serialized, Func<TEncoding, TState> loader)
```

Parameters

`serialized` TEncoding

`loader` Func<TEncoding, TState>

Properties

State

```
public TState State { get; }
```

Property Value

TState

Methods

GetObjectData(SerializationInfo, StreamingContext)

Populates a SerializationInfo with the data needed to serialize the target object.

```
public void GetObjectData(SerializationInfo info, StreamingContext context)
```

Parameters

`info` SerializationInfo

The SerializationInfo to populate with data.

`context` StreamingContext

The destination (see StreamingContext) for this serialization.

Exceptions

SecurityException

The caller does not have the required permission.

GetStateOrSerializedEncoding(out TState, out TEncoding)

```
public bool GetStateOrSerializedEncoding(out TState loadedState, out
TEncoding serialized)
```

Parameters

`loadedState` TState

`serialized` TEncoding

Returns

bool

LoadState(LazyState<TState, TEncoding>)

```
public static TState LoadState(LazyState<TState, TEncoding> lazyState)
```

Parameters

`lazyState` LazyState<TState, TEncoding>

Returns

TState

LoadStatePair<T>(KeyValuePair<T, LazyState<TState, TEncoding>>)

```
public static KeyValuePair<T, TState> LoadStatePair<T>(KeyValuePair<T,
LazyState<TState, TEncoding>> lazyPair)
```

Parameters

`lazyPair` KeyValuePair<T, LazyState<TState, TEncoding>>

Returns

KeyValuePair<T, TState>

Type Parameters

`T`

Serialize()

```
public IValue Serialize()
```

Returns

IValue

Class LegacyStakeState

Namespace: Nekoyume.Model.State

Assembly: Lib9c.dll

```
public class LegacyStakeState : State, IState
```

Inheritance

object ← State ← LegacyStakeState

Implements

IState

Inherited Members

State.address , State.SerializeBase() , State.SerializeV2Base() , State.SerializeListBase() , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

LegacyStakeState(Dictionary)

```
public LegacyStakeState(Dictionary serialized)
```

Parameters

`serialized` Dictionary

LegacyStakeState(Address, long)

```
public LegacyStakeState(Address address, long startedBlockIndex)
```

Parameters

`address` Address

`startedBlockIndex` long

LegacyStakeState(Address, long, long, long, StakeAchievements)

```
public LegacyStakeState(Address address, long startedBlockIndex, long
receivedBlockIndex, long cancellableBlockIndex, LegacyStakeState.StakeAchievements
achievements)
```

Parameters

`address` Address

`startedBlockIndex` long

`receivedBlockIndex` long

`cancellableBlockIndex` long

`achievements` LegacyStakeState.StakeAchievements

Fields

CurrencyAsRewardStartIndex

```
public const long CurrencyAsRewardStartIndex = 6910000
```

Field Value

long

LockupInterval

```
public const long LockupInterval = 201600
```

Field Value

long

RewardInterval

```
public const long RewardInterval = 50400
```

Field Value

long

StakeRewardSheetV2Index

```
public const long StakeRewardSheetV2Index = 6700000
```

Field Value

long

StakeRewardSheetV3Index

```
public const long StakeRewardSheetV3Index = 7650000
```

Field Value

long

Properties

Achievements

```
public LegacyStakeState.StakeAchievements Achievements { get; }
```

Property Value

LegacyStakeState.StakeAchievements

CancellableBlockIndex

```
public long CancellableBlockIndex { get; }
```

Property Value

long

ReceivedBlockIndex

```
public long ReceivedBlockIndex { get; }
```

Property Value

long

StartedBlockIndex

```
public long StartedBlockIndex { get; }
```

Property Value

long

Methods

CalculateAccumulatedCurrencyCrystalRewards(long, out int, out int, out int)

```
public int CalculateAccumulatedCurrencyCrystalRewards(long blockIndex, out int
v1Step, out int v2Step, out int v3Step)
```

Parameters

`blockIndex` long

`v1Step` int

`v2Step` int

`v3Step` int

Returns

int

CalculateAccumulatedCurrencyRewards(long, out int, out int, out int)

```
public int CalculateAccumulatedCurrencyRewards(long blockIndex, out int v1Step, out int v2Step, out int v3Step)
```

Parameters

`blockIndex` long

`v1Step` int

`v2Step` int

`v3Step` int

Returns

int

CalculateAccumulatedCurrencyRewardsV1(long, out int, out int)

```
[Obsolete("Use CalculateAccumulatedCurrencyRewards() instead.")]
public int CalculateAccumulatedCurrencyRewardsV1(long blockIndex, out int v1Step,
```

```
out int v2Step)
```

Parameters

`blockIndex` long

`v1Step` int

`v2Step` int

Returns

int

CalculateAccumulatedCurrencyRewardsV2(long, out int, out int)

```
[Obsolete("Use CalculateAccumulatedCurrencyRewards() instead.")]
public int CalculateAccumulatedCurrencyRewardsV2(long blockIndex, out int v1Step,
out int v2Step)
```

Parameters

`blockIndex` long

`v1Step` int

`v2Step` int

Returns

int

CalculateAccumulatedItemRewards(long, out int, out int, out int)

```
public int CalculateAccumulatedItemRewards(long blockIndex, out int v1Step, out int
v2Step, out int v3Step)
```

Parameters

`blockIndex` long

`v1Step` int

`v2Step` int

`v3Step` int

Returns

int

CalculateAccumulatedItemRewardsV1(long)

```
[Obsolete("Use CalculateAccumulatedItemRewards() instead.")]
public int CalculateAccumulatedItemRewardsV1(long blockIndex)
```

Parameters

`blockIndex` long

Returns

int

CalculateAccumulatedItemRewardsV1(long, out int, out int)

```
[Obsolete("Use CalculateAccumulatedItemRewards() instead.")]
public int CalculateAccumulatedItemRewardsV1(long blockIndex, out int v1Step, out
int v2Step)
```

Parameters

`blockIndex` long

`v1Step` int

`v2Step` int

Returns

int

CalculateAccumulatedItemRewardsV2(long, out int, out int)

```
[Obsolete("Use CalculateAccumulatedItemRewards() instead.")]
public int CalculateAccumulatedItemRewardsV2(long blockIndex, out int v1Step, out int v2Step)
```

Parameters

`blockIndex` long

`v1Step` int

`v2Step` int

Returns

int

CalculateAccumulatedItemRewardsV3(long, out int, out int)

```
[Obsolete("Use CalculateAccumulatedItemRewards() instead.")]
public int CalculateAccumulatedItemRewardsV3(long blockIndex, out int v1Step, out int v2Step)
```

Parameters

`blockIndex` long

`v1Step` int

`v2Step` int

Returns

int

CalculateAccumulatedRuneRewards(long)

```
public int CalculateAccumulatedRuneRewards(long blockIndex)
```

Parameters

`blockIndex` long

Returns

int

CalculateAccumulatedRuneRewards(long, out int, out int, out int)

```
public int CalculateAccumulatedRuneRewards(long blockIndex, out int v1Step, out int
v2Step, out int v3Step)
```

Parameters

`blockIndex` long

`v1Step` int

`v2Step` int

`v3Step` int

Returns

int

CalculateAccumulatedRuneRewardsV1(long)

```
[Obsolete("Use CalculateAccumulatedRuneRewards() instead.")]
public int CalculateAccumulatedRuneRewardsV1(long blockIndex)
```

Parameters

`blockIndex` long

Returns

int

CalculateAccumulatedRuneRewardsV1(long, out int, out int)

```
[Obsolete("Use CalculateAccumulatedRuneRewards() instead.")]
public int CalculateAccumulatedRuneRewardsV1(long blockIndex, out int v1Step, out
int v2Step)
```

Parameters

`blockIndex` long

`v1Step` int

`v2Step` int

Returns

int

CalculateAccumulatedRuneRewardsV2(long, out int, out int)

```
[Obsolete("Use CalculateAccumulatedRuneRewards() instead.")]
public int CalculateAccumulatedRuneRewardsV2(long blockIndex, out int v1Step, out
```

```
int v2Step)
```

Parameters

`blockIndex` long

`v1Step` int

`v2Step` int

Returns

int

CalculateAccumulatedRuneRewardsV3(long, out int, out int)

```
[Obsolete("Use CalculateAccumulatedRuneRewards() instead.")]
public int CalculateAccumulatedRuneRewardsV3(long blockIndex, out int v1Step, out
int v2Step)
```

Parameters

`blockIndex` long

`v1Step` int

`v2Step` int

Returns

int

Claim(long)

```
public void Claim(long blockIndex)
```

Parameters

`blockIndex` long

DeriveAddress(Address)

```
public static Address DeriveAddress(Address agentAddress)
```

Parameters

`agentAddress` Address

Returns

Address

GetClaimableBlockIndex(long)

```
public long GetClaimableBlockIndex(long blockIndex)
```

Parameters

`blockIndex` long

Returns

long

GetRewardStep(long, long?)

Calculate accumulated rewards step.

```
public int GetRewardStep(long currentBlockIndex, long? rewardStartBlockIndex)
```

Parameters

`currentBlockIndex` long

The block index of the current block.

`rewardStartBlockIndex` long?

The block index of the reward start block. If not null, the return value is calculated differently. StakeStateTest.GetRewardStep()

Returns

int

The accumulated rewards step.

IsCancellable(long)

```
public bool IsCancellable(long blockIndex)
```

Parameters

`blockIndex` long

Returns

bool

IsClaimable(long)

```
public bool IsClaimable(long blockIndex)
```

Parameters

`blockIndex` long

Returns

bool

IsClaimable(long, out int, out int)

```
public bool IsClaimable(long blockIndex, out int v1Step, out int v2Step)
```

Parameters

`blockIndex` long

`v1Step` int

`v2Step` int

Returns

bool

Serialize()

```
public override IValue Serialize()
```

Returns

IValue

SerializeV2()

```
public IValue SerializeV2()
```

Returns

IValue

Class LegacyStakeState.StakeAchievements

Namespace: Nekoyume.Model.State

Assembly: Lib9c.dll

```
public class LegacyStakeState.StakeAchievements
```

Inheritance

object ← LegacyStakeState.StakeAchievements

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

StakeAchievements(Dictionary)

```
public StakeAchievements(Dictionary serialized)
```

Parameters

`serialized` Dictionary

StakeAchievements(Dictionary<int, int>)

```
public StakeAchievements(Dictionary<int, int> achievements = null)
```

Parameters

`achievements` Dictionary<int, int>

Methods

Achieve(int, int)

```
public void Achieve(int level, int step)
```

Parameters

`level` int

`step` int

Check(int, int)

```
public bool Check(int level, int step)
```

Parameters

`level` int

`step` int

Returns

bool

Serialize()

```
public IValue Serialize()
```

Returns

IValue

Class MarketState

Namespace: Nekoyume.Model.State

Assembly: Lib9c.dll

```
public class MarketState
```

Inheritance

object ← MarketState

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

MarketState()

```
public MarketState()
```

MarketState(IValue)

```
public MarketState(IValue rawList)
```

Parameters

`rawList` IValue

Fields

AvatarAddresses

```
public List<Address> AvatarAddresses
```

Field Value

List<Address>

Methods

Serialize()

```
public IValue Serialize()
```

Returns

IValue

Class MonsterCollectionResult

Namespace: Nekoyume.Model.State

Assembly: Lib9c.dll

```
[Serializable]
public class MonsterCollectionResult : AttachmentActionResult, IState
```

Inheritance

object ← AttachmentActionResult ← MonsterCollectionResult

Implements

IState

Inherited Members

AttachmentActionResult.itemUsable , AttachmentActionResult.costume , AttachmentActionResult.tradableFungibleItem , AttachmentActionResult.tradableFungibleItemCount , AttachmentActionResult.SerializeBackup1() , AttachmentActionResult.Deserialize(Dictionary) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

MonsterCollectionResult(Dictionary)

```
public MonsterCollectionResult(Dictionary serialized)
```

Parameters

`serialized` Dictionary

MonsterCollectionResult(Guid, Address, List<RewardInfo>)

```
public MonsterCollectionResult(Guid guid, Address address,
List<MonsterCollectionRewardSheet.RewardInfo> rewardInfos)
```

Parameters

`guid` Guid

`address` Address

`rewardInfos` List<MonsterCollectionRewardSheet.RewardInfo>

Fields

avatarAddress

```
public Address avatarAddress
```

Field Value

Address

id

```
public Guid id
```

Field Value

Guid

rewards

```
public List<MonsterCollectionRewardSheet.RewardInfo> rewards
```

Field Value

List<MonsterCollectionRewardSheet.RewardInfo>

Properties

TypeId

```
protected override string TypeId { get; }
```

Property Value

string

Methods

Serialize()

```
public override IValue Serialize()
```

Returns

IValue

Class MonsterCollectionState

Namespace: Nekoyume.Model.State

Assembly: Lib9c.dll

```
[Serializable]
public class MonsterCollectionState : State, IState
```

Inheritance

object ← State ← MonsterCollectionState

Implements

IState

Inherited Members

State.address , State.SerializeBase() , State.SerializeV2Base() , State.SerializeListBase() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

MonsterCollectionState(Dictionary)

```
public MonsterCollectionState(Dictionary serialized)
```

Parameters

`serialized` Dictionary

MonsterCollectionState(Address, int, long)

```
public MonsterCollectionState(Address address, int level, long blockIndex)
```

Parameters

`address` Address

`level` int

`blockIndex` long

MonsterCollectionState(Address, int, long, MonsterCollectionRewardSheet)

```
public MonsterCollectionState(Address address, int level, long blockIndex,
MonsterCollectionRewardSheet monsterCollectionRewardSheet)
```

Parameters

`address` Address

`level` int

`blockIndex` long

`monsterCollectionRewardSheet` MonsterCollectionRewardSheet

Fields

DeriveFormat

```
public const string DeriveFormat = "monster-collection-{0}"
```

Field Value

string

LockUpInterval

```
public const long LockUpInterval = 201600
```

Field Value

long

RewardInterval

```
public const long RewardInterval = 50400
```

Field Value

long

Properties

ExpiredBlockIndex

```
public long ExpiredBlockIndex { get; }
```

Property Value

long

Level

```
public int Level { get; }
```

Property Value

int

ReceivedBlockIndex

```
public long ReceivedBlockIndex { get; }
```

Property Value

long

RewardLevel

```
public long RewardLevel { get; }
```

Property Value

long

StartedBlockIndex

```
public long StartedBlockIndex { get; }
```

Property Value

long

Methods

CalculateRewards(MonsterCollectionRewardSheet, long)

```
public List<MonsterCollectionRewardSheet.RewardInfo>
CalculateRewards(MonsterCollectionRewardSheet sheet, long blockIndex)
```

Parameters

`sheet` MonsterCollectionRewardSheet

`blockIndex` long

Returns

List<MonsterCollectionRewardSheet.RewardInfo>

CalculateStep(long)

```
public int CalculateStep(long blockIndex)
```

Parameters

`blockIndex` long

Returns

int

Claim(long)

```
public void Claim(long blockIndex)
```

Parameters

`blockIndex` long

DeriveAddress(Address, int)

```
public static Address DeriveAddress(Address baseAddress, int round)
```

Parameters

`baseAddress` Address

`round` int

Returns

Address

Equals(MonsterCollectionState)

```
protected bool Equals(MonsterCollectionState other)
```

Parameters

`other` MonsterCollectionState

Returns

bool

Equals(object)

Determines whether the specified object is equal to the current object.

```
public override bool Equals(object obj)
```

Parameters

`obj` object

The object to compare with the current object.

Returns

bool

true if the specified object is equal to the current object; otherwise, false.

GetHashCode()

Serves as the default hash function.

```
public override int GetHashCode()
```

Returns

int

A hash code for the current object.

IsLocked(long)

```
public bool IsLocked(long blockIndex)
```

Parameters

`blockIndex` long

Returns

bool

Serialize()

```
public override IValue Serialize()
```

Returns

IValue

Class MonsterCollectionState0

Namespace: Nekoyume.Model.State

Assembly: Lib9c.dll

```
[Obsolete("This class is reserved for backwards compatibility with
MonsterCollection0 only.", false)]
[Serializable]
public class MonsterCollectionState0 : State, IState
```

Inheritance

object ← State ← MonsterCollectionState0

Implements

IState

Inherited Members

State.address , State.SerializeBase() , State.SerializeV2Base() , State.SerializeListBase() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

MonsterCollectionState0(Dictionary)

```
public MonsterCollectionState0(Dictionary serialized)
```

Parameters

`serialized` Dictionary

MonsterCollectionState0(Address, int, long)

```
public MonsterCollectionState0(Address address, int level, long blockIndex)
```

Parameters

`address` Address

`level` int

`blockIndex` long

MonsterCollectionState0(Address, int, long, MonsterCollectionRewardSheet)

```
public MonsterCollectionState0(Address address, int level, long blockIndex,
MonsterCollectionRewardSheet monsterCollectionRewardSheet)
```

Parameters

`address` Address

`level` int

`blockIndex` long

`monsterCollectionRewardSheet` MonsterCollectionRewardSheet

Fields

DeriveFormat

```
public const string DeriveFormat = "monster-collection-{0}"
```

Field Value

string

ExpirationIndex

```
public const long ExpirationIndex = 201600
```

Field Value

long

LockUpInterval

```
public const long LockUpInterval = 201600
```

Field Value

long

RewardCapacity

```
public const int RewardCapacity = 4
```

Field Value

int

RewardInterval

```
public const long RewardInterval = 50400
```

Field Value

long

Properties

End

```
public bool End { get; }
```

Property Value

bool

ExpiredBlockIndex

```
public long ExpiredBlockIndex { get; }
```

Property Value

long

Level

```
public int Level { get; }
```

Property Value

int

ReceivedBlockIndex

```
public long ReceivedBlockIndex { get; }
```

Property Value

long

RewardLevel

```
public long RewardLevel { get; }
```

Property Value

long

RewardLevelMap

```
public Dictionary<long, List<MonsterCollectionRewardSheet.RewardInfo>>
RewardLevelMap { get; }
```

Property Value

Dictionary<long, List<MonsterCollectionRewardSheet.RewardInfo>>

RewardMap

```
public Dictionary<long, MonsterCollectionResult> RewardMap { get; }
```

Property Value

Dictionary<long, MonsterCollectionResult>

StartedBlockIndex

```
public long StartedBlockIndex { get; }
```

Property Value

long

Methods

CanReceive(long)

```
public bool CanReceive(long blockIndex)
```

Parameters

`blockIndex` long

Returns

bool

DeriveAddress(Address, int)

```
public static Address DeriveAddress(Address baseAddress, int collectRound)
```

Parameters

`baseAddress` Address

`collectRound` int

Returns

Address

Equals(MonsterCollectionState0)

```
protected bool Equals(MonsterCollectionState0 other)
```

Parameters

`other` MonsterCollectionState0

Returns

bool

Equals(object)

Determines whether the specified object is equal to the current object.

```
public override bool Equals(object obj)
```

Parameters

`obj` object

The object to compare with the current object.

Returns

bool

true if the specified object is equal to the current object; otherwise, false.

GetHashCode()

Serves as the default hash function.

```
public override int GetHashCode()
```

Returns

int

A hash code for the current object.

GetRewardLevel(long)

```
public long GetRewardLevel(long blockIndex)
```

Parameters

`blockIndex` long

Returns

long

IsLock(long)

```
public bool IsLock(long blockIndex)
```

Parameters

`blockIndex` long

Returns

bool

Receive(long)

```
public void Receive(long blockIndex)
```

Parameters

`blockIndex` long

Serialize()

```
public override IValue Serialize()
```

Returns

IValue

SerializeV2()

```
public IValue SerializeV2()
```

Returns

IValue

Update(int, long, MonsterCollectionRewardSheet)

```
public void Update(int level, long rewardLevel, MonsterCollectionRewardSheet
monsterCollectionRewardSheet)
```

Parameters

`level` int

`rewardLevel` long

`monsterCollectionRewardSheet` MonsterCollectionRewardSheet

UpdateRewardMap(long, MonsterCollectionResult, long)

```
public void UpdateRewardMap(long rewardLevel, MonsterCollectionResult avatarAddress,
long blockIndex)
```

Parameters

`rewardLevel` long

`avatarAddress` MonsterCollectionResult

`blockIndex` long

Class PendingActivationState

Namespace: Nekoyume.Model.State

Assembly: Lib9c.dll

```
[Serializable]
public class PendingActivationState : State, IState, ISerializable
```

Inheritance

object ← State ← PendingActivationState

Implements

IState, ISerializable

Inherited Members

State.address , State.SerializeBase() , State.SerializeV2Base() , State.SerializeListBase() , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

PendingActivationState(Dictionary)

```
public PendingActivationState(Dictionary serialized)
```

Parameters

`serialized` Dictionary

PendingActivationState(byte[], PublicKey)

```
public PendingActivationState(byte[] nonce, PublicKey publicKey)
```

Parameters

`nonce` byte[]

`publicKey` PublicKey

PendingActivationState(SerializationInfo, StreamingContext)

```
protected PendingActivationState(SerializationInfo info, StreamingContext context)
```

Parameters

`info` SerializationInfo

`context` StreamingContext

Properties

Nonce

```
public byte[] Nonce { get; }
```

Property Value

byte[]

PublicKey

```
public PublicKey PublicKey { get; }
```

Property Value

PublicKey

Methods

GetObjectData(SerializationInfo, StreamingContext)

Populates a SerializationInfo with the data needed to serialize the target object.

```
public void GetObjectData(SerializationInfo info, StreamingContext context)
```

Parameters

`info` SerializationInfo

The SerializationInfo to populate with data.

`context` StreamingContext

The destination (see StreamingContext) for this serialization.

Exceptions

SecurityException

The caller does not have the required permission.

Serialize()

```
public override IValue Serialize()
```

Returns

IValue

Verify(byte[])

```
public bool Verify(byte[] signature)
```

Parameters

`signature` byte[]

Returns

bool

Class PetState

Namespace: Nekoyume.Model.State

Assembly: Lib9c.dll

```
public class PetState : IState
```

Inheritance

object ← PetState

Implements

IState

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

PetState(List)

```
public PetState(List serialized)
```

Parameters

`serialized` List

PetState(int)

```
public PetState(int petId)
```

Parameters

`petId` int

Properties

Level

```
public int Level { get; }
```

Property Value

int

PetId

```
public int PetId { get; }
```

Property Value

int

UnlockedBlockIndex

```
public long UnlockedBlockIndex { get; }
```

Property Value

long

Methods

DeriveAddress(Address, int)

```
public static Address DeriveAddress(Address avatarAddress, int petId)
```

Parameters

`avatarAddress` Address

`petId` int

Returns

Address

LevelUp()

```
public void LevelUp()
```

Serialize()

```
public IValue Serialize()
```

Returns

IValue

Update(long)

```
public void Update(long unlockedIndex)
```

Parameters

`unlockedIndex` long

Validate(long)

```
public bool Validate(long blockIndex)
```

Parameters

`blockIndex` long

Returns

bool

Class RaiderState

Namespace: Nekoyume.Model.State

Assembly: Lib9c.dll

```
[Serializable]
public class RaiderState : IState
```

Inheritance

object ← RaiderState

Implements

IState

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

RaiderState()

```
public RaiderState()
```

RaiderState(List)

```
public RaiderState(List rawState)
```

Parameters

`rawState` List

Fields

AvatarAddress

```
public Address AvatarAddress
```

Field Value

Address

AvatarName

```
public string AvatarName
```

Field Value

string

ClaimedBlockIndex

```
public long ClaimedBlockIndex
```

Field Value

long

Cp

```
public long Cp
```

Field Value

long

HasClaimedReward

```
public bool HasClaimedReward
```

Field Value

bool

HighScore

```
public long HighScore
```

Field Value

long

IconId

```
public int IconId
```

Field Value

int

LatestBossLevel

```
public int LatestBossLevel
```

Field Value

int

LatestRewardRank

```
public int LatestRewardRank
```

Field Value

int

Level

```
public int Level
```

Field Value

int

PurchaseCount

```
public int PurchaseCount
```

Field Value

int

RefillBlockIndex

```
public long RefillBlockIndex
```

Field Value

long

RemainChallengeCount

```
public int RemainChallengeCount
```

Field Value

int

TotalChallengeCount

```
public int TotalChallengeCount
```

Field Value

int

TotalScore

```
public long TotalScore
```

Field Value

long

UpdatedBlockIndex

```
public long UpdatedBlockIndex
```

Field Value

long

Methods

Serialize()

```
public IValue Serialize()
```

Returns

IValue

Update(AvatarState, long, long, bool, long)

```
public void Update(AvatarState avatarState, long cp, long score, bool payNcg,
long blockIndex)
```

Parameters

`avatarState` AvatarState

`cp` long

`score` long

`payNcg` bool

`blockIndex` long

Class RankingInfo

Namespace: Nekoyume.Model.State

Assembly: Lib9c.dll

```
public class RankingInfo : IState
```

Inheritance

object ← RankingInfo

Implements

IState

Inherited Members

object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

RankingInfo(Dictionary)

```
public RankingInfo(Dictionary serialized)
```

Parameters

`serialized` Dictionary

RankingInfo(AvatarState)

```
public RankingInfo(AvatarState avatarState)
```

Parameters

`avatarState` AvatarState

Fields

AgentAddress

```
public readonly Address AgentAddress
```

Field Value

Address

ArmorId

```
public readonly int ArmorId
```

Field Value

int

AvatarAddress

```
public readonly Address AvatarAddress
```

Field Value

Address

AvatarName

```
public readonly string AvatarName
```

Field Value

string

Exp

```
public readonly long Exp
```

Field Value

long

Level

```
public readonly int Level
```

Field Value

int

StageClearedBlockIndex

```
public readonly long StageClearedBlockIndex
```

Field Value

long

UpdatedAt

```
public readonly long UpdatedAt
```

Field Value

long

Methods

Equals(RankingInfo)

```
protected bool Equals(RankingInfo other)
```

Parameters

`other` RankingInfo

Returns

bool

Equals(object)

Determines whether the specified object is equal to the current object.

```
public override bool Equals(object obj)
```

Parameters

`obj` object

The object to compare with the current object.

Returns

bool

true if the specified object is equal to the current object; otherwise, false.

GetHashCode()

Serves as the default hash function.

```
public override int GetHashCode()
```

Returns

int

A hash code for the current object.

Serialize()

```
public IValue Serialize()
```

Returns

IValue

Class RankingMapState

Namespace: Nekoyume.Model.State

Assembly: Lib9c.dll

```
public class RankingMapState : State, IState
```

Inheritance

object ← State ← RankingMapState

Implements

IState

Inherited Members

State.address , State.SerializeBase() , State.SerializeV2Base() , State.SerializeListBase() , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

RankingMapState(Dictionary)

```
public RankingMapState(Dictionary serialized)
```

Parameters

`serialized` Dictionary

RankingMapState(Address)

```
public RankingMapState(Address address)
```

Parameters

`address` Address

Fields

Capacity

```
public const int Capacity = 500
```

Field Value

int

Methods

GetRankingInfos(long?)

```
public List<RankingInfo> GetRankingInfos(long? blockOffset)
```

Parameters

`blockOffset` long?

Returns

List<RankingInfo>

Serialize()

```
public override IValue Serialize()
```

Returns

IValue

Update(AvatarState)

```
public void Update(AvatarState state)
```

Parameters

`state` AvatarState

Class RankingState

Namespace: Nekoyume.Model.State

Assembly: Lib9c.dll

```
[Serializable]
public class RankingState : State, IState, ISerializable
```

Inheritance

object ← State ← RankingState

Implements

IState, ISerializable

Inherited Members

State.address , State.SerializeBase() , State.SerializeV2Base() , State.SerializeListBase() , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

RankingState()

```
public RankingState()
```

RankingState(Dictionary)

```
public RankingState(Dictionary serialized)
```

Parameters

`serialized` Dictionary

RankingState(SerializationInfo, StreamingContext)

```
public RankingState(SerializationInfo info, StreamingContext context)
```

Parameters

`info` SerializationInfo

`context` StreamingContext

Fields

Address

```
public static readonly Address Address
```

Field Value

Address

RankingMapCapacity

```
public const int RankingMapCapacity = 150
```

Field Value

int

Properties

RankingMap

```
public Dictionary<Address, ImmutableHashSet<Address>> RankingMap { get; }
```

Property Value

Dictionary<Address, ImmutableHashSet<Address>>

Methods

Derive(int)

```
public static Address Derive(int index)
```

Parameters

`index` int

Returns

Address

GetObjectData(SerializationInfo, StreamingContext)

Populates a SerializationInfo with the data needed to serialize the target object.

```
public void GetObjectData(SerializationInfo info, StreamingContext context)
```

Parameters

`info` SerializationInfo

The SerializationInfo to populate with data.

`context` StreamingContext

The destination (see StreamingContext) for this serialization.

Exceptions

SecurityException

The caller does not have the required permission.

Serialize()

```
public override IValue Serialize()
```

Returns

IValue

UpdateRankingMap(Address)

```
public Address UpdateRankingMap(Address avatarAddress)
```

Parameters

`avatarAddress` Address

Returns

Address

Class RankingState0

Namespace: Nekoyume.Model.State

Assembly: Lib9c.dll

```
[Serializable]
public class RankingState0 : State, IState, ISerializable
```

Inheritance

object ← State ← RankingState0

Implements

IState, ISerializable

Inherited Members

State.address , State.SerializeBase() , State.SerializeV2Base() , State.SerializeListBase() , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

RankingState0()

```
public RankingState0()
```

RankingState0(Dictionary)

```
public RankingState0(Dictionary serialized)
```

Parameters

`serialized` Dictionary

RankingState0(SerializationInfo, StreamingContext)

```
public RankingState0(SerializationInfo info, StreamingContext context)
```

Parameters

`info` SerializationInfo

`context` StreamingContext

Fields

Address

```
public static readonly Address Address
```

Field Value

Address

RankingMapCapacity

```
public const int RankingMapCapacity = 100
```

Field Value

int

Properties

RankingMap

```
public Dictionary<Address, ImmutableHashSet<Address>> RankingMap { get; }
```

Property Value

Dictionary<Address, ImmutableHashSet<Address>>

Methods

Derive(int)

```
public static Address Derive(int index)
```

Parameters

`index` int

Returns

Address

GetObjectData(SerializationInfo, StreamingContext)

Populates a SerializationInfo with the data needed to serialize the target object.

```
public void GetObjectData(SerializationInfo info, StreamingContext context)
```

Parameters

`info` SerializationInfo

The SerializationInfo to populate with data.

`context` StreamingContext

The destination (see StreamingContext) for this serialization.

Exceptions

SecurityException

The caller does not have the required permission.

Serialize()

```
public override IValue Serialize()
```

Returns

IValue

UpdateRankingMap(Address)

```
public Address UpdateRankingMap(Address avatarAddress)
```

Parameters

`avatarAddress` Address

Returns

Address

Class RankingState1

Namespace: Nekoyume.Model.State

Assembly: Lib9c.dll

```
[Serializable]
public class RankingState1 : State, IState, ISerializable
```

Inheritance

object ← State ← RankingState1

Implements

IState, ISerializable

Inherited Members

State.address , State.SerializeBase() , State.SerializeV2Base() , State.SerializeListBase() , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

RankingState1()

```
public RankingState1()
```

RankingState1(Dictionary)

```
public RankingState1(Dictionary serialized)
```

Parameters

`serialized` Dictionary

RankingState1(SerializationInfo, StreamingContext)

```
public RankingState1(SerializationInfo info, StreamingContext context)
```

Parameters

`info` SerializationInfo

`context` StreamingContext

Fields

Address

```
public static readonly Address Address
```

Field Value

Address

RankingMapCapacity

```
public const int RankingMapCapacity = 150
```

Field Value

int

Properties

RankingMap

```
public Dictionary<Address, ImmutableHashSet<Address>> RankingMap { get; }
```

Property Value

Dictionary<Address, ImmutableHashSet<Address>>

Methods

Derive(int)

```
public static Address Derive(int index)
```

Parameters

`index` int

Returns

Address

GetObjectData(SerializationInfo, StreamingContext)

Populates a SerializationInfo with the data needed to serialize the target object.

```
public void GetObjectData(SerializationInfo info, StreamingContext context)
```

Parameters

`info` SerializationInfo

The SerializationInfo to populate with data.

`context` StreamingContext

The destination (see StreamingContext) for this serialization.

Exceptions

SecurityException

The caller does not have the required permission.

Serialize()

```
public override IValue Serialize()
```

Returns

IValue

UpdateRankingMap(Address)

```
public Address UpdateRankingMap(Address avatarAddress)
```

Parameters

`avatarAddress` Address

Returns

Address

Class RedeemCodeState

Namespace: Nekoyume.Model.State

Assembly: Lib9c.dll

```
[Serializable]
public class RedeemCodeState : State, IState
```

Inheritance

object ← State ← RedeemCodeState

Implements

IState

Inherited Members

State.address , State.SerializeBase() , State.SerializeV2Base() , State.SerializeListBase() , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

RedeemCodeState(Dictionary)

```
public RedeemCodeState(Dictionary serialized)
```

Parameters

`serialized` Dictionary

RedeemCodeState(RedeemCodeListSheet)

```
public RedeemCodeState(RedeemCodeListSheet sheet)
```

Parameters

`sheet` RedeemCodeListSheet

RedeemCodeState(Dictionary<PublicKey, Reward>)

```
public RedeemCodeState(Dictionary<PublicKey, RedeemCodeState.Reward> rewardMap)
```

Parameters

`rewardMap` Dictionary<PublicKey, RedeemCodeState.Reward>

Fields

Address

```
public static readonly Address Address
```

Field Value

Address

Properties

Map

```
public IReadOnlyDictionary<PublicKey, RedeemCodeState.Reward> Map { get; }
```

Property Value

IReadOnlyDictionary<PublicKey, RedeemCodeState.Reward>

Methods

Redeem(string, Address)

```
public int Redeem(string code, Address userAddress)
```

Parameters

`code` string

`userAddress` Address

Returns

int

Serialize()

```
public override IValue Serialize()
```

Returns

IValue

Update(RedeemCodeListSheet)

```
public void Update(RedeemCodeListSheet sheet)
```

Parameters

`sheet` RedeemCodeListSheet

Class RedeemCodeState.Reward

Namespace: Nekoyume.Model.State

Assembly: Lib9c.dll

```
public class RedeemCodeState.Reward
```

Inheritance

object ← RedeemCodeState.Reward

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

Reward(Dictionary)

```
public Reward(Dictionary serialized)
```

Parameters

`serialized` Dictionary

Reward(int)

```
public Reward(int rewardId)
```

Parameters

`rewardId` int

Fields

RewardId

```
public readonly int RewardId
```

Field Value

int

UserAddress

```
public Address? UserAddress
```

Field Value

Address?

Methods

Serialize()

```
public IValue Serialize()
```

Returns

IValue

Class RuneSlotState

Namespace: Nekoyume.Model.State

Assembly: Lib9c.dll

```
public class RuneSlotState : IState
```

Inheritance

object ← RuneSlotState

Implements

IState

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

RuneSlotState(List)

```
public RuneSlotState(List serialized)
```

Parameters

`serialized` List

RuneSlotState(BattleType)

```
public RuneSlotState(BattleType battleType)
```

Parameters

`battleType` BattleType

Properties

BattleType

```
public BattleType BattleType { get; }
```

Property Value

BattleType

Methods

DeriveAddress(Address, BattleType)

```
public static Address DeriveAddress(Address avatarAddress, BattleType battleType)
```

Parameters

`avatarAddress` Address

`battleType` BattleType

Returns

Address

GetEquippedRuneSlotInfos()

```
public List<RuneSlotInfo> GetEquippedRuneSlotInfos()
```

Returns

List<RuneSlotInfo>

GetRuneSlot()

```
public List<RuneSlot> GetRuneSlot()
```

Returns

List<RuneSlot>

Serialize()

```
public IValue Serialize()
```

Returns

IValue

Unlock(int)

```
public void Unlock(int index)
```

Parameters

`index` int

UpdateSlot(List<RuneSlotInfo>, RuneListSheet)

```
public void UpdateSlot(List<RuneSlotInfo> runeInfos, RuneListSheet runeListSheet)
```

Parameters

`runeInfos` List<RuneSlotInfo>

`runeListSheet` RuneListSheet

UpdateSlotV2(List<RuneSlotInfo>, RuneListSheet)

```
[Obsolete("UpdateSlotV2")]
public void UpdateSlotV2(List<RuneSlotInfo> runeInfos, RuneListSheet runeListSheet)
```

Parameters

`runeInfos` List<RuneSlotInfo>

`runeListSheet` RuneListSheet

Class RuneState

Namespace: Nekoyume.Model.State

Assembly: Lib9c.dll

```
public class RuneState : IState
```

Inheritance

object ← RuneState

Implements

IState

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

RuneState(List)

```
public RuneState(List serialized)
```

Parameters

`serialized` List

RuneState(int)

```
public RuneState(int runeId)
```

Parameters

`runeId` int

RuneState(int, int)

```
public RuneState(int runeId, int level)
```

Parameters

`runeId` int

`level` int

Properties

Level

```
public int Level { get; }
```

Property Value

int

RuneId

```
public int RuneId { get; }
```

Property Value

int

Methods

DeriveAddress(Address, int)

```
public static Address DeriveAddress(Address avatarAddress, int runeId)
```

Parameters

`avatarAddress` Address

`runeId` int

Returns

Address

LevelUp(int)

```
public void LevelUp(int level = 1)
```

Parameters

`level` int

Serialize()

```
public IValue Serialize()
```

Returns

IValue

Class ShardedShopState

Namespace: Nekoyume.Model.State

Assembly: Lib9c.dll

```
public class ShardedShopState : State, IState
```

Inheritance

object ← State ← ShardedShopState

Implements

IState

Inherited Members

State.address , State.SerializeBase() , State.SerializeV2Base() , State.SerializeListBase() , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

ShardedShopState(Dictionary)

```
public ShardedShopState(Dictionary serialized)
```

Parameters

`serialized` Dictionary

ShardedShopState(Address)

```
public ShardedShopState(Address address)
```

Parameters

`address` Address

Fields

AddressKeys

```
public static readonly IReadOnlyList<string> AddressKeys
```

Field Value

IReadOnlyList<string>

Products

```
public readonly Dictionary<Guid, ShopItem> Products
```

Field Value

Dictionary<Guid, ShopItem>

Methods

DeriveAddress(ItemSubType, Guid)

```
public static Address DeriveAddress(ItemSubType itemSubType, Guid productId)
```

Parameters

`itemSubType` ItemSubType

`productId` Guid

Returns

Address

DeriveAddress(ItemSubType, string)

```
public static Address DeriveAddress(ItemSubType itemSubType, string nonce)
```

Parameters

`itemSubType` ItemSubType

`nonce` string

Returns

Address

Register(ShopItem)

```
public void Register(ShopItem shopItem)
```

Parameters

`shopItem` ShopItem

Serialize()

```
public override IValue Serialize()
```

Returns

IValue

Class ShardedShopStateV2

Namespace: Nekoyume.Model.State

Assembly: Lib9c.dll

```
[Serializable]
public class ShardedShopStateV2 : State, IState
```

Inheritance

object ← State ← ShardedShopStateV2

Implements

IState

Inherited Members

State.address , State.SerializeBase() , State.SerializeV2Base() , State.SerializeListBase() , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

ShardedShopStateV2(Dictionary)

```
public ShardedShopStateV2(Dictionary serialized)
```

Parameters

`serialized` Dictionary

ShardedShopStateV2(Address)

```
public ShardedShopStateV2(Address address)
```

Parameters

`address` Address

Properties

OrderDigestList

```
public IReadOnlyList<OrderDigest> OrderDigestList { get; }
```

Property Value

IReadOnlyList<OrderDigest>

Methods

Add(OrderDigest, long)

```
public void Add(OrderDigest orderDigest, long blockIndex)
```

Parameters

`orderDigest` OrderDigest

`blockIndex` long

DeriveAddress(ItemSubType, Guid)

```
public static Address DeriveAddress(ItemSubType itemSubType, Guid orderId)
```

Parameters

`itemSubType` ItemSubType

`orderId` Guid

Returns

Address

DeriveAddress(ItemSubType, string)

```
public static Address DeriveAddress(ItemSubType itemSubType, string nonce)
```

Parameters

`itemSubType` ItemSubType

`nonce` string

Returns

Address

Remove(Order, long)

```
public void Remove(Order order, long blockIndex)
```

Parameters

`order` Order

`blockIndex` long

Serialize()

```
public override IValue Serialize()
```

Returns

IValue

Class ShopState

Namespace: Nekoyume.Model.State

Assembly: Lib9c.dll

This is a model class of shop state.

```
[Serializable]
public class ShopState : State, IState
```

Inheritance

object ← State ← ShopState

Implements

IState

Inherited Members

State.address , State.SerializeBase() , State.SerializeV2Base() , State.SerializeListBase() , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

ShopState()

```
public ShopState()
```

ShopState(Dictionary)

```
public ShopState(Dictionary serialized)
```

Parameters

`serialized` Dictionary

Fields

Address

```
public static readonly Address Address
```

Field Value

Address

Properties

Products

```
public IReadOnlyDictionary<Guid, ShopItem> Products { get; }
```

Property Value

IReadOnlyDictionary<Guid, ShopItem>

Methods

Register(ShopItem)

```
public void Register(ShopItem shopItem)
```

Parameters

`shopItem` ShopItem

Serialize()

```
public override IValue Serialize()
```

Returns

IValue

TryGet(Address, Guid, out ShopItem)

```
public bool TryGet(Address sellerAgentAddress, Guid productId, out
ShopItem shopItem)
```

Parameters

`sellerAgentAddress` Address

`productId` Guid

`shopItem` ShopItem

Returns

bool

TryUnregister(Guid, out ShopItem)

```
public bool TryUnregister(Guid productId, out ShopItem unregisteredItem)
```

Parameters

`productId` Guid

`unregisteredItem` ShopItem

Returns

bool

Unregister(ShopItem)

```
public void Unregister(ShopItem shopItem)
```

Parameters

`shopItem` ShopItem

Unregister(Guid)

```
public void Unregister(Guid productId)
```

Parameters

`productId` Guid

Class ShopStateAlreadyContainsException

Namespace: Nekoyume.Model.State

Assembly: Lib9c.dll

```
[Serializable]
public class ShopStateAlreadyContainsException : Exception, ISerializable
```

Inheritance

object ← Exception ← ShopStateAlreadyContainsException

Implements

ISerializable

Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Constructors

ShopStateAlreadyContainsException(SerializationInfo, StreamingContext)

```
protected ShopStateAlreadyContainsException(SerializationInfo info,
StreamingContext context)
```

Parameters

`info` SerializationInfo

`context` StreamingContext

ShopStateAlreadyContainsException(string)

```
public ShopStateAlreadyContainsException(string message)
```

Parameters

`message` string

Class StakeRewardCalculator

Namespace: Nekoyume.Model.State

Assembly: Lib9c.dll

Provides methods for calculating staking rewards from reward sheets.

```
public static class StakeRewardCalculator
```

Inheritance

object ← StakeRewardCalculator

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Methods

CalculateFixedRewards(int, IRandom, StakeRegularFixedRewardSheet, ItemSheet, int)

Calculates fixed item rewards for a given stake level. Fixed rewards have a predetermined item count (unlike rate-based rewards in CalculateRewards(Currency, FungibleAssetValue, int, int, StakeRegularRewardSheet, ItemSheet, IRandom)), and the `tradable` flag on each StakeRegularFixedRewardSheet.RewardInfo controls whether material items are created as tradable or non-tradable.

```
public static Dictionary<ItemBase, int> CalculateFixedRewards(int stakeLevel,
IRandom random, StakeRegularFixedRewardSheet stakeRegularFixedRewardSheet, ItemSheet
itemSheet, int rewardSteps)
```

Parameters

`stakeLevel` int

The current stake level used to look up the reward row.

`random` IRandom

Random source used for non-material item creation.

`stakeRegularFixedRewardSheet` StakeRegularFixedRewardSheet

Sheet containing fixed reward definitions.

`itemSheet` ItemSheet

Sheet used to look up item metadata by ID.

`rewardSteps` int

The number of reward steps accumulated since the last claim. Each reward's count is multiplied by this value.

Returns

Dictionary<ItemBase, int>

A dictionary mapping each rewarded ItemBase to its total count.

CalculateRewards(Currency, FungibleAssetValue, int, int, StakeRegularRewardSheet, ItemSheet, IRandom)

Calculates rate-based item and currency rewards for a given staking level. Unlike CalculateFixedRewards(int, IRandom, StakeRegularFixedRewardSheet, ItemSheet, int), reward quantities are derived from the ratio of the staked NCG amount to each reward's Decimal Rate. Rewards whose computed quantity is zero or negative are skipped.

```
public static (Dictionary<ItemBase, int> itemResult, List<FungibleAssetValue>
favResult) CalculateRewards(Currency ncg, FungibleAssetValue stakedNcg, int
stakingLevel, int rewardSteps, StakeRegularRewardSheet stakeRegularRewardSheet,
ItemSheet itemSheet, IRandom random)
```

Parameters

`ncg` Currency

The NCG currency definition.

`stakedNcg` FungibleAssetValue

The total amount of staked NCG.

`stakingLevel` int

The current staking level used to look up the reward row.

`rewardSteps` int

The number of reward steps accumulated since the last claim. Each reward quantity is multiplied by this value.

`stakeRegularRewardSheet` StakeRegularRewardSheet

Sheet containing rate-based reward definitions.

`itemSheet` ItemSheet

Sheet used to look up item metadata by ID.

`random` IRandom

Random source used for non-material item creation.

Returns

(Dictionary<ItemBase, int> itemResult, List<FungibleAssetValue> favResult)

A tuple of:

- `itemResult`: a dictionary mapping each rewarded ItemBase to its total count.
- `favResult`: a list of Libplanet.Types.Assets.FungibleAssetValue for rune and currency rewards.

Class State

Namespace: Nekoyume.Model.State

Assembly: Lib9c.dll

```
[Serializable]
public abstract class State : IState
```

Inheritance

object ← State

Implements

IState

Derived

ActivatedAccountsState, AdminState, AgentState, AuthorizedMinersState, AvatarState, CombinationSlotState, CreditsState, GameConfigState, GoldBalanceState, GoldCurrencyState, LegacyStakeState, MonsterCollectionState, MonsterCollectionState0, PendingActivationState, RankingMapState, RankingState, RankingState0, RankingState1, RedeemCodeState, ShardedShopState, ShardedShopStateV2, ShopState, WeeklyArenaState

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

State(IValue)

```
protected State(IValue iValue)
```

Parameters

`iValue` IValue

State(Address)

```
protected State(Address address)
```

Parameters

`address` Address

Fields

address

```
public Address address
```

Field Value

Address

Methods

Serialize()

```
public abstract IValue Serialize()
```

Returns

IValue

SerializeBase()

```
protected IValue SerializeBase()
```

Returns

IValue

SerializeListBase()

```
protected IValue SerializeListBase()
```

Returns

IValue

SerializeV2Base()

```
protected IValue SerializeV2Base()
```

Returns

IValue

Class StateExtensions

Namespace: Nekoyume.Model.State

Assembly: Lib9c.dll

```
public static class StateExtensions
```

Inheritance

object ← StateExtensions

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Methods

Deserialize<T>(Func<IValue, T>, IValue)

```
public static T? Deserialize<T>(Func<IValue, T> deserializer, IValue serialized)
where T : struct
```

Parameters

`deserializer` Func<IValue, T>

`serialized` IValue

Returns

T?

Type Parameters

`T`

GetAddress(Dictionary, string, Address)

```
public static Address GetAddress(this Dictionary serialized, string key, Address
defaultValue = default)
```

Parameters

`serialized` Dictionary

`key` string

`defaultValue` Address

Returns

Address

GetBoolean(Dictionary, string, bool)

```
public static bool GetBoolean(this Dictionary serialized, string key, bool
defaultValue = false)
```

Parameters

`serialized` Dictionary

`key` string

`defaultValue` bool

Returns

bool

GetDateTimeOffset(Dictionary, string, DateTimeOffset)

```
public static DateTimeOffset GetDateTimeOffset(this Dictionary serialized, string
key, DateTimeOffset defaultValue = default)
```

Parameters

`serialized` Dictionary

`key` string

`defaultValue` DateTimeOffset

Returns

DateTimeOffset

GetDecimal(Dictionary, string, decimal)

```
public static decimal GetDecimal(this Dictionary serialized, string key, decimal
defaultValue = 0)
```

Parameters

`serialized` Dictionary

`key` string

`defaultValue` decimal

Returns

decimal

GetDecimalStat(Dictionary, string, DecimalStat)

```
public static DecimalStat GetDecimalStat(this Dictionary serialized, string key,
DecimalStat defaultValue = null)
```

Parameters

`serialized` Dictionary

`key` string

`defaultValue` DecimalStat

Returns

DecimalStat

GetGuid(Dictionary, string, Guid)

```
public static Guid GetGuid(this Dictionary serialized, string key, Guid defaultValue
= default)
```

Parameters

`serialized` Dictionary

`key` string

`defaultValue` Guid

Returns

Guid

GetInteger(Dictionary, string, int)

```
public static int GetInteger(this Dictionary serialized, string key, int
defaultValue = 0)
```

Parameters

`serialized` Dictionary

`key` string

`defaultValue` int

Returns

int

GetLong(Dictionary, string, long)

```
public static long GetLong(this Dictionary serialized, string key, long defaultValue
= 0)
```

Parameters

`serialized` Dictionary

`key` string

`defaultValue` long

Returns

long

GetStat(Dictionary)

```
public static (StatType statType, decimal baseValue, decimal additionalValue)
GetStat(this Dictionary serialized)
```

Parameters

`serialized` Dictionary

Returns

(StatType statType, decimal baseValue, decimal additionalValue)

GetString(Dictionary, string, string)

```
public static string GetString(this Dictionary serialized, string key, string
defaultValue = "")
```

Parameters

`serialized` Dictionary

`key` string

`defaultValue` string

Returns

string

GetValue<T>(Dictionary, string, T, IValueTryParseDelegate<T>)

```
public static T GetValue<T>(this Dictionary serialized, string key, T defaultValue,
StateExtensions.IValueTryParseDelegate<T> tryParser)
```

Parameters

`serialized` Dictionary

`key` string

`defaultValue` T

`tryParser` StateExtensions.IValueTryParseDelegate<T>

Returns

T

Type Parameters

`T`

Serialize(HashDigest<SHA256>)

```
public static IValue Serialize(this HashDigest<SHA256> hashDigest)
```

Parameters

`hashDigest` HashDigest<SHA256>

Returns

IValue

Serialize(Address)

```
public static IValue Serialize(this Address address)
```

Parameters

`address` Address

Returns

IValue

Serialize(PublicKey)

```
public static IValue Serialize(this PublicKey key)
```

Parameters

`key` PublicKey

Returns

IValue

Serialize(FungibleAssetValue)

```
public static IValue Serialize(this FungibleAssetValue value)
```

Parameters

`value` FungibleAssetValue

Returns

IValue

Serialize(bool)

```
public static IValue Serialize(this bool boolean)
```

Parameters

`boolean` bool

Returns

IValue

Serialize(Dictionary<Material, int>)

```
public static IValue Serialize(this Dictionary<Material, int> value)
```

Parameters

`value` Dictionary<Material, int>

Returns

IValue

Serialize(IEnumerable<(Address, Address, IValue)>)

```
public static IValue Serialize(this IEnumerable<(Address, Address, IValue)> value)
```

Parameters

`value` IEnumerable<(Address, Address, IValue)>

Returns

IValue

Serialize(DateTimeOffset)

```
public static IValue Serialize(this DateTimeOffset dateTime)
```

Parameters

`dateTime` DateTimeOffset

Returns

IValue

Serialize(decimal)

```
public static IValue Serialize(this decimal number)
```

Parameters

`number` decimal

Returns

IValue

Serialize(Enum)

```
public static IValue Serialize(this Enum type)
```

Parameters

`type` Enum

Returns

IValue

Serialize(Guid)

```
public static IValue Serialize(this Guid number)
```

Parameters

`number` Guid

Returns

IValue

Serialize(int)

```
public static IValue Serialize(this int number)
```

Parameters

`number` int

Returns

IValue

Serialize(long)

```
public static IValue Serialize(this long number)
```

Parameters

`number` long

Returns

IValue

Serialize(Address?)

```
public static IValue Serialize(this Address? address)
```

Parameters

`address` Address?

Returns

IValue

Serialize(FungibleAssetValue?)

```
public static IValue Serialize(this FungibleAssetValue? value)
```

Parameters

`value` FungibleAssetValue?

Returns

IValue

Serialize(bool?)

```
public static IValue Serialize(this bool? boolean)
```

Parameters

`boolean` bool?

Returns

IValue

Serialize(DateTimeOffset?)

```
public static IValue Serialize(this DateTimeOffset? dateTime)
```

Parameters

`dateTime` DateTimeOffset?

Returns

IValue

Serialize(decimal?)

```
public static IValue Serialize(this decimal? number)
```

Parameters

`number` decimal?

Returns

IValue

Serialize(Guid?)

```
public static IValue Serialize(this Guid? number)
```

Parameters

`number` Guid?

Returns

IValue

Serialize(int?)

```
public static IValue Serialize(this int? number)
```

Parameters

`number` int?

Returns

IValue

Serialize(long?)

```
public static IValue Serialize(this long? number)
```

Parameters

`number` long?

Returns

IValue

Serialize(BigInteger?)

```
public static IValue Serialize(this BigInteger? number)
```

Parameters

`number` BigInteger?

Returns

IValue

Serialize(BigInteger)

```
public static IValue Serialize(this BigInteger number)
```

Parameters

`number` BigInteger

Returns

IValue

Serialize(string)

```
public static IValue Serialize(this string text)
```

Parameters

`text` string

Returns

IValue

SerializeForLegacyEquipmentStat(DecimalStat)

```
public static IValue SerializeForLegacyEquipmentStat(this DecimalStat decimalStat)
```

Parameters

`decimalStat` DecimalStat

Returns

IValue

Serialize<T>(IEnumerable<T>)

```
public static IValue Serialize<T>(this IEnumerable<T> values) where T : IValue
```

Parameters

`values` IEnumerable<T>

Returns

IValue

Type Parameters

`T`

Serialize<T>(Func<T, IValue>, T?)

```
public static IValue Serialize<T>(Func<T, IValue> serializer, T? value) where T
: struct
```

Parameters

`serializer` Func<T, IValue>

`value` T?

Returns

IValue

Type Parameters

`T`

ToAddress(IValue)

```
public static Address ToAddress(this IValue serialized)
```

Parameters

`serialized` IValue

Returns

Address

ToArray<T>(IValue, Func<IValue, T>)

```
public static T[] ToArray<T>(this IValue serialized, Func<IValue, T> deserializer)
```

Parameters

`serialized` IValue

`deserializer` Func<IValue, T>

Returns

T[]

Type Parameters

`T`

ToBigInteger(IValue)

```
public static BigInteger ToBigInteger(this IValue serialized)
```

Parameters

`serialized` IValue

Returns

BigInteger

ToBoolean(IValue)

```
public static bool ToBoolean(this IValue serialized)
```

Parameters

`serialized` IValue

Returns

bool

ToDateTimeOffset(IValue)

```
public static DateTimeOffset ToDateTimeOffset(this IValue serialized)
```

Parameters

`serialized` IValue

Returns

DateTimeOffset

ToDecimal(IValue)

```
public static decimal ToDecimal(this IValue serialized)
```

Parameters

`serialized` IValue

Returns

decimal

ToDecimalStat(Dictionary)

```
public static DecimalStat ToDecimalStat(this Dictionary serialized)
```

Parameters

`serialized` Dictionary

Returns

DecimalStat

ToDecimalStat(IValue)

```
public static DecimalStat ToDecimalStat(this IValue serialized)
```

Parameters

`serialized` IValue

Returns

DecimalStat

ToDictionary_Material_int(IValue)

```
public static Dictionary<Material, int> ToDictionary_Material_int(this
IValue serialized)
```

Parameters

`serialized` IValue

Returns

Dictionary<Material, int>

ToDotnetString(IValue)

```
public static string ToDotnetString(this IValue serialized)
```

Parameters

`serialized` IValue

Returns

string

ToEnum<T>(IValue)

```
public static T ToEnum<T>(this IValue serialized) where T : struct
```

Parameters

`serialized` IValue

Returns

T

Type Parameters

T

ToEnumerable<T>(IValue, Func<IValue, T>)

```
public static IEnumerable<T> ToEnumerable<T>(this IValue serialized, Func<IValue,
T> deserializer)
```

Parameters

serialized IValue

deserializer Func<IValue, T>

Returns

IEnumerable<T>

Type Parameters

T

ToFungibleAssetValue(IValue)

```
public static FungibleAssetValue ToFungibleAssetValue(this IValue serialized)
```

Parameters

serialized IValue

Returns

FungibleAssetValue

ToGuid(IValue)

```
public static Guid ToGuid(this IValue serialized)
```

Parameters

`serialized` IValue

Returns

Guid

ToHashSet<T>(IValue, Func<IValue, T>)

```
public static HashSet<T> ToHashSet<T>(this IValue serialized, Func<IValue,
T> deserializer)
```

Parameters

`serialized` IValue

`deserializer` Func<IValue, T>

Returns

HashSet<T>

Type Parameters

`T`

ToImmutableHashSet<T>(IValue, Func<IValue, T>)

```
public static ImmutableHashSet<T> ToImmutableHashSet<T>(this IValue serialized,
Func<IValue, T> deserializer)
```

Parameters

`serialized` IValue

`deserializer` Func<IValue, T>

Returns

ImmutableHashSet<T>

Type Parameters

T

ToInteger(IValue)

```
public static int ToInteger(this IValue serialized)
```

Parameters

serialized IValue

Returns

int

ToItemId(IValue)

```
public static HashDigest<SHA256> ToItemId(this IValue serialized)
```

Parameters

serialized IValue

Returns

HashDigest<SHA256>

ToList<T>(IValue, Func<IValue, T>)

```
public static List<T> ToList<T>(this IValue serialized, Func<IValue,
T> deserializer)
```

Parameters

`serialized` IValue

`deserializer` Func<IValue, T>

Returns

List<T>

Type Parameters

`T`

ToLock(IValue)

```
public static ILock ToLock(this IValue serialized)
```

Parameters

`serialized` IValue

Returns

ILock

ToLong(IValue)

```
public static long ToLong(this IValue serialized)
```

Parameters

`serialized` IValue

Returns

long

ToNullableAddress(IValue)

```
public static Address? ToNullableAddress(this IValue serialized)
```

Parameters

`serialized` IValue

Returns

Address?

ToNullableBigInteger(IValue)

```
public static BigInteger? ToNullableBigInteger(this IValue serialized)
```

Parameters

`serialized` IValue

Returns

BigInteger?

ToNullableBoolean(IValue)

```
public static bool? ToNullableBoolean(this IValue serialized)
```

Parameters

`serialized` IValue

Returns

bool?

ToNullableDateTimeOffset(IValue)

```
public static DateTimeOffset? ToNullableDateTimeOffset(this IValue serialized)
```

Parameters

`serialized` IValue

Returns

DateTimeOffset?

ToNullableDecimal(IValue)

```
public static decimal? ToNullableDecimal(this IValue serialized)
```

Parameters

`serialized` IValue

Returns

decimal?

ToNullableFungibleAssetValue(IValue)

```
public static FungibleAssetValue? ToNullableFungibleAssetValue(this
IValue serialized)
```

Parameters

`serialized` IValue

Returns

FungibleAssetValue?

ToNullableGuid(IValue)

```
public static Guid? ToNullableGuid(this IValue serialized)
```

Parameters

`serialized` IValue

Returns

Guid?

ToNullableInteger(IValue)

```
public static int? ToNullableInteger(this IValue serialized)
```

Parameters

`serialized` IValue

Returns

int?

ToNullableLong(IValue)

```
public static long? ToNullableLong(this IValue serialized)
```

Parameters

`serialized` IValue

Returns

long?

ToPublicKey(IValue)

```
public static PublicKey ToPublicKey(this IValue serialized)
```

Parameters

`serialized` IValue

Returns

PublicKey

ToPurchaseInfo(IValue)

```
public static PurchaseInfo0 ToPurchaseInfo(this IValue serialized)
```

Parameters

`serialized` IValue

Returns

PurchaseInfo0

ToPurchaseInfoLegacy(IValue)

```
public static BuyMultiple.PurchaseInfo ToPurchaseInfoLegacy(this IValue serialized)
```

Parameters

`serialized` IValue

Returns

BuyMultiple.PurchaseInfo

ToPurchaseResult(IValue)

```
public static Buy7.PurchaseResult ToPurchaseResult(this IValue serialized)
```

Parameters

`serialized` IValue

Returns

Buy7.PurchaseResult

ToPurchaseResultLegacy(IValue)

```
public static BuyMultiple.PurchaseResult ToPurchaseResultLegacy(this
IValue serialized)
```

Parameters

`serialized` IValue

Returns

BuyMultiple.PurchaseResult

ToSellerResult(IValue)

```
public static Buy7.SellerResult ToSellerResult(this IValue serialized)
```

Parameters

`serialized` IValue

Returns

Buy7.SellerResult

ToStateList(IValue)

```
public static List<(Address, Address, IValue)> ToStateList(this IValue serialized)
```

Parameters

`serialized` IValue

Returns

List<(Address, Address, IValue)>

Delegate StateExtensions.IValueTryParseDelegate<T>

Namespace: Nekoyume.Model.State

Assembly: Lib9c.dll

```
public delegate bool StateExtensions.IValueTryParseDelegate<T>(IValue input, out T output)
```

Parameters

`input` IValue

`output` T

Returns

bool

Type Parameters

`T`

Class WeeklyArenaState

Namespace: Nekoyume.Model.State

Assembly: Lib9c.dll

```
[Serializable]
public class WeeklyArenaState : State, IState, IDictionary<Address, ArenaInfo>,
ICollection<KeyValuePair<Address, ArenaInfo>>, IEnumerable<KeyValuePair<Address,
ArenaInfo>>, IEnumerable, ISerializable
```

Inheritance

object ← State ← WeeklyArenaState

Implements

IState, IDictionary<Address, ArenaInfo>, ICollection<KeyValuePair<Address, ArenaInfo>>, IEnumerable<KeyValuePair<Address, ArenaInfo>>, IEnumerable, ISerializable

Inherited Members

State.address , State.SerializeBase() , State.SerializeV2Base() , State.SerializeListBase() , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

WeeklyArenaState(Dictionary)

```
public WeeklyArenaState(Dictionary serialized)
```

Parameters

`serialized` Dictionary

WeeklyArenaState(IValue)

```
public WeeklyArenaState(IValue iValue)
```

Parameters

`iValue` IValue

WeeklyArenaState(Address)

```
public WeeklyArenaState(Address address)
```

Parameters

`address` Address

WeeklyArenaState(int)

```
public WeeklyArenaState(int index)
```

Parameters

`index` int

WeeklyArenaState(SerializationInfo, StreamingContext)

```
protected WeeklyArenaState(SerializationInfo info, StreamingContext context)
```

Parameters

`info` SerializationInfo

`context` StreamingContext

Fields

Ended

```
public bool Ended
```

Field Value

bool

ResetIndex

```
public long ResetIndex
```

Field Value

long

Properties

Count

Gets the number of elements contained in the ICollection<T>.

```
public int Count { get; }
```

Property Value

int

The number of elements contained in the ICollection<T>.

IsReadOnly

Gets a value indicating whether the ICollection<T> is read-only.

```
public bool IsReadOnly { get; }
```

Property Value

bool

true if the ICollection<T> is read-only; otherwise, false.

this[Address]

Gets or sets the element with the specified key.

```
public ArenaInfo this[Address key] { get; set; }
```

Parameters

`key` Address

The key of the element to get or set.

Property Value

ArenaInfo

The element with the specified key.

Exceptions

ArgumentNullException

`key` is null.

KeyNotFoundException

The property is retrieved and `key` is not found.

NotSupportedException

The property is set and the IDictionary<TKey, TValue> is read-only.

Keys

Gets an ICollection<T> containing the keys of the IDictionary<TKey, TValue>.

```
public ICollection<Address> Keys { get; }
```

Property Value

ICollection<Address>

An ICollection<T> containing the keys of the object that implements IDictionary<TKey, TValue>.

Map

```
public IReadOnlyDictionary<Address, LazyState<ArenaInfo, Dictionary>> Map { get; }
```

Property Value

IReadOnlyDictionary<Address, LazyState<ArenaInfo, Dictionary>>

OrderedArenaInfos

```
public List<ArenaInfo> OrderedArenaInfos { get; }
```

Property Value

List<ArenaInfo>

Values

Gets an ICollection<T> containing the values in the IDictionary<TKey, TValue>.

```
public ICollection<ArenaInfo> Values { get; }
```

Property Value

ICollection<ArenaInfo>

An ICollection<T> containing the values in the object that implements IDictionary<TKey, TValue>.

Methods

Add(Address, ArenaInfo)

Adds an element with the provided key and value to the IDictionary<TKey, TValue>.

```
public void Add(Address key, ArenaInfo value)
```

Parameters

`key` Address

The object to use as the key of the element to add.

`value` ArenaInfo

The object to use as the value of the element to add.

Exceptions

ArgumentNullException

`key` is null.

ArgumentException

An element with the same key already exists in the IDictionary<TKey, TValue>.

NotSupportedException

The IDictionary<TKey, TValue> is read-only.

Add(KeyValuePair<Address, ArenaInfo>)

Adds an item to the ICollection<T>.

```
public void Add(KeyValuePair<Address, ArenaInfo> item)
```

Parameters

`item` KeyValuePair<Address, ArenaInfo>

The object to add to the ICollection<T>.

Exceptions

NotSupportedException

The ICollection<T> is read-only.

Clear()

Removes all items from the ICollection<T>.

```
public void Clear()
```

Exceptions

NotSupportedException

The ICollection<T> is read-only.

Contains(KeyValuePair<Address, ArenaInfo>)

Determines whether the ICollection<T> contains a specific value.

```
public bool Contains(KeyValuePair<Address, ArenaInfo> item)
```

Parameters

`item` KeyValuePair<Address, ArenaInfo>

The object to locate in the ICollection<T>.

Returns

bool

true if `item` is found in the ICollection<T>; otherwise, false.

ContainsKey(Address)

Determines whether the IDictionary<TKey, TValue> contains an element with the specified key.

```
public bool ContainsKey(Address key)
```

Parameters

`key` Address

The key to locate in the IDictionary<TKey, TValue>.

Returns

bool

true if the IDictionary<TKey, TValue> contains an element with the key; otherwise, false.

Exceptions

ArgumentNullException

`key` is null.

CopyTo(KeyValuePair<Address, ArenaInfo>[], int)

Copies the elements of the ICollection<T> to an Array, starting at a particular Array index.

```
public void CopyTo(KeyValuePair<Address, ArenaInfo>[] array, int arrayIndex)
```

Parameters

`array` KeyValuePair<Address, ArenaInfo>[]

The one-dimensional Array that is the destination of the elements copied from ICollection<T>. The Array must have zero-based indexing.

`arrayIndex` int

The zero-based index in `array` at which copying begins.

Exceptions

ArgumentNullException

`array` is null.

ArgumentOutOfRangeException

`arrayIndex` is less than 0.

ArgumentException

The number of elements in the source ICollection<T> is greater than the available space from `arrayIndex` to the end of the destination `array`.

DeriveAddress(int)

```
public static Address DeriveAddress(int index)
```

Parameters

`index` int

Returns

Address

End()

```
public void End()
```

GetArenaInfo(Address)

```
public ArenaInfo GetArenaInfo(Address avatarAddress)
```

Parameters

`avatarAddress` Address

Returns

ArenaInfo

GetArenaInfos(Address, int, int)

Get arena rank information.

```
public List<(int rank, ArenaInfo arenaInfo)> GetArenaInfos(Address avatarAddress,
int upperRange = 10, int lowerRange = 10)
```

Parameters

`avatarAddress` Address

The base value of the range that want to get.

`upperRange` int

The upper range than base value in the ranges that want to get.

`lowerRange` int

The lower range than base value in the ranges that want to get.

Returns

List<(int rank, ArenaInfo arenaInfo)>

A list of tuples that contains `int` and `ArenaInfo`.

GetArenaInfos(int, int?)

Get arena rank information.

```
public List<(int rank, ArenaInfo arenaInfo)> GetArenaInfos(int firstRank = 1, int?
count = null)
```

Parameters

`firstRank` int

The first rank in the range that want to get.

`count` int?

The count of the range that want to get.

Returns

List<(int rank, ArenaInfo arenaInfo)>

A list of tuples that contains `int` and `ArenaInfo`.

GetEnumerator()

Returns an enumerator that iterates through the collection.

```
public IEnumerator<KeyValuePair<Address, ArenaInfo>> GetEnumerator()
```

Returns

IEnumerator<KeyValuePair<Address, ArenaInfo>>

An enumerator that can be used to iterate through the collection.

GetObjectData(SerializationInfo, StreamingContext)

Populates a SerializationInfo with the data needed to serialize the target object.

```
public void GetObjectData(SerializationInfo info, StreamingContext context)
```

Parameters

`info` SerializationInfo

The SerializationInfo to populate with data.

`context` StreamingContext

The destination (see StreamingContext) for this serialization.

Exceptions

SecurityException

The caller does not have the required permission.

Remove(Address)

Removes the element with the specified key from the IDictionary<TKey, TValue>.

```
public bool Remove(Address key)
```

Parameters

`key` Address

The key of the element to remove.

Returns

bool

true if the element is successfully removed; otherwise, false. This method also returns false if `key` was not found in the original IDictionary<TKey, TValue>.

Exceptions

ArgumentNullException

`key` is null.

NotSupportedException

The IDictionary<TKey, TValue> is read-only.

Remove(KeyValuePair<Address, ArenaInfo>)

Removes the first occurrence of a specific object from the ICollection<T>.

```
public bool Remove(KeyValuePair<Address, ArenaInfo> item)
```

Parameters

`item` KeyValuePair<Address, ArenaInfo>

The object to remove from the ICollection<T>.

Returns

bool

true if `item` was successfully removed from the ICollection<T>; otherwise, false. This method also returns false if `item` is not found in the original ICollection<T>.

Exceptions

NotSupportedException

The ICollection<T> is read-only.

ResetCount(long)

```
public void ResetCount(long ctxBlockIndex)
```

Parameters

`ctxBlockIndex` long

Serialize()

```
public override IValue Serialize()
```

Returns

IValue

Set(AvatarState, CharacterSheet)

```
public void Set(AvatarState avatarState, CharacterSheet characterSheet)
```

Parameters

`avatarState` AvatarState

`characterSheet` CharacterSheet

SetReceive(Address)

```
public void SetReceive(Address avatarAddress)
```

Parameters

`avatarAddress` Address

SetV2(AvatarState, CharacterSheet, CostumeStatSheet)

```
public void SetV2(AvatarState avatarState, CharacterSheet characterSheet,
CostumeStatSheet costumeStatSheet)
```

Parameters

`avatarState` AvatarState

`characterSheet` CharacterSheet

`costumeStatSheet` CostumeStatSheet

TryGetValue(Address, out ArenaInfo)

Gets the value associated with the specified key.

```
public bool TryGetValue(Address key, out ArenaInfo value)
```

Parameters

`key` Address

The key whose value to get.

`value` ArenaInfo

When this method returns, the value associated with the specified key, if the key is found; otherwise, the default value for the type of the `value` parameter. This parameter is passed uninitialized.

Returns

bool

true if the object that implements IDictionary<TKey, TValue> contains an element with the specified key; otherwise, false.

Exceptions

ArgumentNullException

`key` is null.

Update(ArenaInfo)

```
public void Update(ArenaInfo info)
```

Parameters

`info` ArenaInfo

Update(WeeklyArenaState, long)

```
public void Update(WeeklyArenaState prevState, long index)
```

Parameters

`prevState` WeeklyArenaState

`index` long

Class WorldBossKillRewardRecord

Namespace: Nekoyume.Model.State

Assembly: Lib9c.dll

```
public class WorldBossKillRewardRecord : IDictionary<int, bool>,
ICollection<KeyValuePair<int, bool>>, IEnumerable<KeyValuePair<int, bool>>,
IEnumerable, IState
```

Inheritance

object ← WorldBossKillRewardRecord

Implements

IDictionary<int, bool>, ICollection<KeyValuePair<int, bool>>, IEnumerable<KeyValuePair<int, bool>>, IEnumerable, IState

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

WorldBossKillRewardRecord()

```
public WorldBossKillRewardRecord()
```

WorldBossKillRewardRecord(List)

```
public WorldBossKillRewardRecord(List serialized)
```

Parameters

`serialized` List

Properties

Count

Gets the number of elements contained in the ICollection<T>.

```
public int Count { get; }
```

Property Value

int

The number of elements contained in the ICollection<T>.

IsReadOnly

Gets a value indicating whether the ICollection<T> is read-only.

```
public bool IsReadOnly { get; }
```

Property Value

bool

true if the ICollection<T> is read-only; otherwise, false.

this[int]

Gets or sets the element with the specified key.

```
public bool this[int key] { get; set; }
```

Parameters

`key` int

The key of the element to get or set.

Property Value

bool

The element with the specified key.

Exceptions

ArgumentNullException

`key` is null.

KeyNotFoundException

The property is retrieved and `key` is not found.

NotSupportedException

The property is set and the IDictionary<TKey, TValue> is read-only.

Keys

Gets an ICollection<T> containing the keys of the IDictionary<TKey, TValue>.

```
public ICollection<int> Keys { get; }
```

Property Value

ICollection<int>

An ICollection<T> containing the keys of the object that implements IDictionary<TKey, TValue>.

Values

Gets an ICollection<T> containing the values in the IDictionary<TKey, TValue>.

```
public ICollection<bool> Values { get; }
```

Property Value

ICollection<bool>

An ICollection<T> containing the values in the object that implements IDictionary<TKey, TValue>.

Methods

Add(KeyValuePair<int, bool>)

Adds an item to the ICollection<T>.

```
public void Add(KeyValuePair<int, bool> item)
```

Parameters

`item` KeyValuePair<int, bool>

The object to add to the ICollection<T>.

Exceptions

NotSupportedException

The ICollection<T> is read-only.

Add(int, bool)

Adds an element with the provided key and value to the IDictionary<TKey, TValue>.

```
public void Add(int key, bool value)
```

Parameters

`key` int

The object to use as the key of the element to add.

`value` bool

The object to use as the value of the element to add.

Exceptions

ArgumentNullException

`key` is null.

ArgumentException

An element with the same key already exists in the IDictionary<TKey, TValue>.

NotSupportedException

The IDictionary<TKey, TValue> is read-only.

Clear()

Removes all items from the ICollection<T>.

```
public void Clear()
```

Exceptions

NotSupportedException

The ICollection<T> is read-only.

Contains(KeyValuePair<int, bool>)

Determines whether the ICollection<T> contains a specific value.

```
public bool Contains(KeyValuePair<int, bool> item)
```

Parameters

`item` KeyValuePair<int, bool>

The object to locate in the ICollection<T>.

Returns

bool

true if `item` is found in the ICollection<T>; otherwise, false.

ContainsKey(int)

Determines whether the IDictionary<TKey, TValue> contains an element with the specified key.

```
public bool ContainsKey(int key)
```

Parameters

`key` int

The key to locate in the IDictionary<TKey, TValue>.

Returns

bool

true if the IDictionary<TKey, TValue> contains an element with the key; otherwise, false.

Exceptions

ArgumentNullException

`key` is null.

CopyTo(KeyValuePair<int, bool>[], int)

Copies the elements of the ICollection<T> to an Array, starting at a particular Array index.

```
public void CopyTo(KeyValuePair<int, bool>[] array, int arrayIndex)
```

Parameters

`array` KeyValuePair<int, bool>[]

The one-dimensional Array that is the destination of the elements copied from ICollection<T>. The Array must have zero-based indexing.

`arrayIndex` int

The zero-based index in `array` at which copying begins.

Exceptions

ArgumentNullException

`array` is null.

ArgumentOutOfRangeException

`arrayIndex` is less than 0.

ArgumentException

The number of elements in the source ICollection<T> is greater than the available space from `arrayIndex` to the end of the destination `array`.

GetEnumerator()

Returns an enumerator that iterates through the collection.

```
public IEnumerator<KeyValuePair<int, bool>> GetEnumerator()
```

Returns

IEnumerator<KeyValuePair<int, bool>>

An enumerator that can be used to iterate through the collection.

IsClaimable(int)

```
public bool IsClaimable(int level)
```

Parameters

`level` int

Returns

bool

Remove(KeyValuePair<int, bool>)

Removes the first occurrence of a specific object from the ICollection<T>.

```
public bool Remove(KeyValuePair<int, bool> item)
```

Parameters

`item` KeyValuePair<int, bool>

The object to remove from the ICollection<T>.

Returns

bool

true if `item` was successfully removed from the ICollection<T>; otherwise, false. This method also returns false if `item` is not found in the original ICollection<T>.

Exceptions

NotSupportedException

The ICollection<T> is read-only.

Remove(int)

Removes the element with the specified key from the IDictionary<TKey, TValue>.

```
public bool Remove(int key)
```

Parameters

`key` int

The key of the element to remove.

Returns

bool

true if the element is successfully removed; otherwise, false. This method also returns false if `key` was not found in the original IDictionary<TKey, TValue>.

Exceptions

ArgumentNullException

`key` is null.

NotSupportedException

The IDictionary<TKey, TValue> is read-only.

Serialize()

```
public IValue Serialize()
```

Returns

IValue

TryGetValue(int, out bool)

Gets the value associated with the specified key.

```
public bool TryGetValue(int key, out bool value)
```

Parameters

`key` int

The key whose value to get.

`value` bool

When this method returns, the value associated with the specified key, if the key is found; otherwise, the default value for the type of the `value` parameter. This parameter is passed uninitialized.

Returns

bool

true if the object that implements IDictionary<TKey, TValue> contains an element with the specified key; otherwise, false.

Exceptions

ArgumentNullException

`key` is null.

Class WorldBossState

Namespace: Nekoyume.Model.State

Assembly: Lib9c.dll

Represents the state of a World Boss in the game.

```
[Serializable]
public class WorldBossState : IState
```

Inheritance

object ← WorldBossState

Implements

IState

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

WorldBossState(List)

Initializes a new instance of the WorldBossState class from serialized data.

```
public WorldBossState(List serialized)
```

Parameters

`serialized` List

The serialized data for the World Boss state.

WorldBossState(Row, Row)

Initializes a new instance of the WorldBossState class using the specified row data.

```
public WorldBossState(WorldBossListSheet.Row row, WorldBossGlobalHpSheet.Row hpRow)
```

Parameters

`row` WorldBossListSheet.Row

The row containing the World Boss data.

`hpRow` WorldBossGlobalHpSheet.Row

The row containing the World Boss HP data.

Fields

CurrentHp

Current hit points of the World Boss.

```
public BigInteger CurrentHp
```

Field Value

BigInteger

EndedBlockIndex

Block index when the World Boss season ended.

```
public long EndedBlockIndex
```

Field Value

long

Id

The boss id of the World Boss from WorldBossListSheet.

```
public int Id
```

Field Value

int

Level

Current boss level.

```
public int Level
```

Field Value

int

StartedBlockIndex

Block index when the World Boss season started.

```
public long StartedBlockIndex
```

Field Value

long

TotalDamage

Total damage dealt to the World Boss in season.

```
public BigInteger TotalDamage
```

Field Value

BigInteger

Methods

Serialize()

Serializes the current state of the World Boss into a serializable format.

```
public IValue Serialize()
```

Returns

IValue

A serialized representation of the World Boss state.

Namespace Nekoyume.Model.Swap

Classes

SwapPool

Swap pool for swap between currencies.

Class SwapPool

Namespace: Nekoyume.Model.Swap

Assembly: Lib9c.dll

Swap pool for swap between currencies.

```
public class SwapPool
```

Inheritance

object ← SwapPool

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

SwapPool(SwapRateSheet)

Constructor for SwapPool.

```
public SwapPool(SwapRateSheet swapRateSheet)
```

Parameters

`swapRateSheet` SwapRateSheet

Swap rate sheet to refer.

Properties

Address

Address of the swap pool.

```
public static Address Address { get; }
```

Property Value

Address

SwapRateSheet

Swap rate sheet to refer.

```
public SwapRateSheet SwapRateSheet { get; }
```

Property Value

SwapRateSheet

Methods

ConvertToSwapFAV(FungibleAssetValue, Currency, out FungibleAssetValue)

Convert `from` to `to` with swap rate.

```
public FungibleAssetValue ConvertToSwapFAV(FungibleAssetValue from, Currency to, out
FungibleAssetValue remainder)
```

Parameters

`from` FungibleAssetValue

Libplanet.Types.Assets.FungibleAssetValue to swap from.

`to` Currency

Libplanet.Types.Assets.Currency to swap to.

`remainder` FungibleAssetValue

Remainder of the swap. This will be returned to the Address who requested swap.

Returns

FungibleAssetValue

Libplanet.Types.Assets.FungibleAssetValue after swapping.

Exceptions

SheetRowNotFoundException

Thrown when the swap rate sheet does not contain the rate for `from` and `to`.

ConvertToSwapFAV(FungibleAssetValue, Currency, BigInteger, BigInteger, out FungibleAssetValue)

Convert `from` to `to` with swap rate.

```
public static FungibleAssetValue ConvertToSwapFAV(FungibleAssetValue from,
Currency to, BigInteger rateNumerator, BigInteger rateDenominator, out
FungibleAssetValue remainder)
```

Parameters

`from` FungibleAssetValue

Libplanet.Types.Assets.FungibleAssetValue to swap from.

`to` Currency

Libplanet.Types.Assets.Currency to swap to.

`rateNumerator` BigInteger

Numerator of the swap rate.

`rateDenominator` BigInteger

Denominator of the swap rate.

`remainder` FungibleAssetValue

Remainder of the swap. This will be returned to the Address who requested swap.

Returns

FungibleAssetValue

Libplanet.Types.Assets.FungibleAssetValue after swapping.

Exceptions

ArgumentException

Thrown when `from` and `to` are the same currency.

Swap(IWorld, IActionContext, FungibleAssetValue, Currency)

Swap the currency from `from` to `to`.

```
public IWorld Swap(IWorld world, IActionContext context, FungibleAssetValue from,
Currency to)
```

Parameters

`world` IWorld

Libplanet.Action.State.IWorld to swap the currency.

`context` IActionContext

Libplanet.Action.IActionContext to swap the currency.

`from` FungibleAssetValue

Libplanet.Types.Assets.FungibleAssetValue to swap from.

`to` Currency

Libplanet.Types.Assets.Currency to swap to.

Returns

IWorld

New Libplanet.Action.State.IWorld after swapping the currency.

Exceptions

ArgumentException

Thrown when `from` and `to` are the same currency.

Namespace Nekoyume.Module

Classes

ActionPointModule

AdventureBossModule

AgentModule

ArenaModule

AvatarModule

CollectionModule
Provides utility methods for working with collection states in the world state.

CombinationSlotStateModule
CombinationSlotStateModule is the module to use CombinationSlotState/AllCombinationSlotState with account.

DailyRewardModule

GiftModule

GlobalCounterModule
Reads and writes the global counters that drive hard world openings.

InfiniteTowerModule

LegacyModule

PatrolRewardModule
Provides utility methods for handling patrol reward states.

RelationshipModule

RuneStateModule
RuneStateModule is the module to use RuneState/AllRuneState with account.

StateResolver

WorldOpenModule
Records when hard world openings were confirmed and decides whether they are in effect.

Class ActionPointModule

Namespace: Nekoyume.Module

Assembly: Lib9c.dll

```
public static class ActionPointModule
```

Inheritance

object ← ActionPointModule

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Methods

GetActionPoint(IWorldState, Address)

```
public static long GetActionPoint(this IWorldState worldState,
Address avatarAddress)
```

Parameters

`worldState` IWorldState

`avatarAddress` Address

Returns

long

SetActionPoint(IWorld, Address, long)

```
public static IWorld SetActionPoint(this IWorld world, Address avatarAddress,
long actionPoint)
```

Parameters

`world` IWorld

`avatarAddress` Address

`actionPoint` long

Returns

IWorld

TryGetActionPoint(IWorldState, Address, out long)

```
public static bool TryGetActionPoint(this IWorldState worldState, Address
avatarAddress, out long actionPoint)
```

Parameters

`worldState` IWorldState

`avatarAddress` Address

`actionPoint` long

Returns

bool

Class AdventureBossModule

Namespace: Nekoyume.Module

Assembly: Lib9c.dll

```
public static class AdventureBossModule
```

Inheritance

object ← AdventureBossModule

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Fields

LatestSeasonAddress

```
public static readonly Address LatestSeasonAddress
```

Field Value

Address

Methods

GetBountyBoard(IWorldState, long)

```
public static BountyBoard GetBountyBoard(this IWorldState worldState, long season)
```

Parameters

`worldState` IWorldState

`season` long

Returns

BountyBoard

GetExploreBoard(IWorldState, long)

```
public static ExploreBoard GetExploreBoard(this IWorldState worldState, long season)
```

Parameters

`worldState` IWorldState

`season` long

Returns

ExploreBoard

GetExplorer(IWorldState, long, Address)

```
public static Explorer GetExplorer(this IWorldState worldState, long season,
Address avatarAddress)
```

Parameters

`worldState` IWorldState

`season` long

`avatarAddress` Address

Returns

Explorer

GetExplorerList(IWorldState, long)

```
public static ExplorerList GetExplorerList(this IWorldState worldState, long season)
```

Parameters

`worldState` IWorldState

`season` long

Returns

ExplorerList

GetLatestAdventureBossSeason(IWorldState)

Get brief adventure boss season info of latest season. This only has brief, readonly data, so we must use SeasonInfo to handle season itself.

```
public static SeasonInfo GetLatestAdventureBossSeason(this IWorldState worldState)
```

Parameters

`worldState` IWorldState

Returns

SeasonInfo

LatestSeason state. If no season is started, return a state with default value(0)

GetSeasonInfo(IWorldState, long)

```
public static SeasonInfo GetSeasonInfo(this IWorldState worldState, long season)
```

Parameters

`worldState` IWorldState

`season` long

Returns

SeasonInfo

SetBountyBoard(IWorld, long, BountyBoard)

```
public static IWorld SetBountyBoard(this IWorld world, long season,
BountyBoard bountyBoard)
```

Parameters

`world` IWorld

`season` long

`bountyBoard` BountyBoard

Returns

IWorld

SetExploreBoard(IWorld, long, ExploreBoard)

```
public static IWorld SetExploreBoard(this IWorld world, long season,
ExploreBoard exploreBoard)
```

Parameters

`world` IWorld

`season` long

`exploreBoard` ExploreBoard

Returns

IWorld

SetExplorer(IWorld, long, Explorer)

```
public static IWorld SetExplorer(this IWorld world, long season, Explorer explorer)
```

Parameters

`world` IWorld

`season` long

`explorer` Explorer

Returns

IWorld

SetExplorerList(IWorld, long, ExplorerList)

```
public static IWorld SetExplorerList(this IWorld world, long season,
ExplorerList explorerList)
```

Parameters

`world` IWorld

`season` long

`explorerList` ExplorerList

Returns

IWorld

SetLatestAdventureBossSeason(IWorld, SeasonInfo)

```
public static IWorld SetLatestAdventureBossSeason(this IWorld world,
SeasonInfo latestSeasonInfo)
```

Parameters

`world` IWorld

`latestSeasonInfo` SeasonInfo

Returns

IWorld

SetSeasonInfo(IWorld, SeasonInfo)

```
public static IWorld SetSeasonInfo(this IWorld world, SeasonInfo seasonInfo)
```

Parameters

`world` IWorld

`seasonInfo` SeasonInfo

Returns

IWorld

TryGetExploreBoard(IWorldState, long, out ExploreBoard)

```
public static bool TryGetExploreBoard(this IWorldState worldState, long season, out
ExploreBoard exploreBoard)
```

Parameters

`worldState` IWorldState

`season` long

`exploreBoard` ExploreBoard

Returns

bool

TryGetExplorer(IWorldState, long, Address, out Explorer)

```
public static bool TryGetExplorer(this IWorldState worldState, long season, Address
avatarAddress, out Explorer explorer)
```

Parameters

`worldState` IWorldState

`season` long

`avatarAddress` Address

`explorer` Explorer

Returns

bool

Class AgentModule

Namespace: Nekoyume.Module

Assembly: Lib9c.dll

```
public static class AgentModule
```

Inheritance

object ← AgentModule

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Methods

GetAgentState(IWorldState, Address)

```
public static AgentState? GetAgentState(this IWorldState worldState,
Address address)
```

Parameters

`worldState` IWorldState

`address` Address

Returns

AgentState

SetAgentState(IWorld, Address, AgentState)

```
public static IWorld SetAgentState(this IWorld world, Address agent,
AgentState state)
```

Parameters

`world` IWorld

`agent` Address

`state` AgentState

Returns

IWorld

Class ArenaModule

Namespace: Nekoyume.Module

Assembly: Lib9c.dll

```
public static class ArenaModule
```

Inheritance

object ← ArenaModule

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Methods

GetArenaParticipant(IWorldState, int, int, Address)

```
public static ArenaParticipant GetArenaParticipant(this IWorldState worldState, int
championshipId, int round, Address avatarAddress)
```

Parameters

`worldState` IWorldState

`championshipId` int

`round` int

`avatarAddress` Address

Returns

ArenaParticipant

GetArenaParticipantState(IWorldState, Address, Address)

```
public static IValue GetArenaParticipantState(this IWorldState worldState, Address
accountAddress, Address stateAddress)
```

Parameters

`worldState` IWorldState

`accountAddress` Address

`stateAddress` Address

Returns

IValue

GetArenaParticipantState(IWorldState, int, int, Address)

```
public static IValue GetArenaParticipantState(this IWorldState worldState, int
championshipId, int round, Address avatarAddress)
```

Parameters

`worldState` IWorldState

`championshipId` int

`round` int

`avatarAddress` Address

Returns

IValue

SetArenaParticipant(IWorld, Address, Address, IValue)

```
public static IWorld SetArenaParticipant(this IWorld world, Address accountAddress,
Address stateAddress, IValue arenaParticipantState)
```

Parameters

`world` IWorld

`accountAddress` Address

`stateAddress` Address

`arenaParticipantState` IValue

Returns

IWorld

SetArenaParticipant(IWorld, int, int, Address, IValue)

```
public static IWorld SetArenaParticipant(this IWorld world, int championshipId, int
round, Address avatarAddress, IValue arenaParticipantState)
```

Parameters

`world` IWorld

`championshipId` int

`round` int

`avatarAddress` Address

`arenaParticipantState` IValue

Returns

IWorld

SetArenaParticipant(IWorld, int, int, Address, ArenaParticipant)

```
public static IWorld SetArenaParticipant(this IWorld world, int championshipId, int
round, Address avatarAddress, ArenaParticipant arenaParticipant)
```

Parameters

`world` IWorld

`championshipId` int

`round` int

`avatarAddress` Address

`arenaParticipant` ArenaParticipant

Returns

IWorld

Class AvatarModule

Namespace: Nekoyume.Module

Assembly: Lib9c.dll

```
public static class AvatarModule
```

Inheritance

object ← AvatarModule

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Methods

GetAvatarState(IWorldState, Address, bool, bool, bool)

```
public static AvatarState GetAvatarState(this IWorldState worldState, Address
address, bool getInventory = true, bool getWorldInformation = true, bool
getQuestList = true)
```

Parameters

`worldState` IWorldState

`address` Address

`getInventory` bool

`getWorldInformation` bool

`getQuestList` bool

Returns

AvatarState

GetEnemyAvatarState(IWorldState, Address)

```
public static AvatarState GetEnemyAvatarState(this IWorldState worldState,
Address avatarAddress)
```

Parameters

`worldState` IWorldState

`avatarAddress` Address

Returns

AvatarState

SetAvatarState(IWorld, Address, AvatarState, bool, bool, bool, bool)

```
public static IWorld SetAvatarState(this IWorld world, Address avatarAddress,
AvatarState state, bool setAvatar = true, bool setInventory = true, bool
setWorldInformation = true, bool setQuestList = true)
```

Parameters

`world` IWorld

`avatarAddress` Address

`state` AvatarState

`setAvatar` bool

`setInventory` bool

`setWorldInformation` bool

`setQuestList` bool

Returns

TryGetAvatarState(IWorldState, Address, Address, out AvatarState)

```
public static bool TryGetAvatarState(this IWorldState worldState, Address
agentAddress, Address avatarAddress, out AvatarState avatarState)
```

Parameters

`worldState` IWorldState

`agentAddress` Address

`avatarAddress` Address

`avatarState` AvatarState

Returns

bool

Class CollectionModule

Namespace: Nekoyume.Module

Assembly: Lib9c.dll

Provides utility methods for working with collection states in the world state.

```
public static class CollectionModule
```

Inheritance

object ← CollectionModule

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Methods

GetCollectionState(IWorldState, Address)

Get the CollectionState for a given address from the world state.

```
public static CollectionState GetCollectionState(this IWorldState worldState,
Address address)
```

Parameters

`worldState` IWorldState

The world state object.

`address` Address

The address to retrieve the CollectionState.

Returns

CollectionState

The CollectionState corresponding to the address.

Exceptions

FailedLoadStateException

Thrown when the Collection state for the given address is not found in the world state.

InvalidCastException

Thrown when the serialized Collection state is not in the correct format.

GetCollectionStates(IWorldState, IReadOnlyList<Address>)

Retrieves the collection states for the given addresses from the world state.

```
public static Dictionary<Address, CollectionState> GetCollectionStates(this
IWorldState worldState, IReadOnlyList<Address> addresses)
```

Parameters

`worldState` IWorldState

The world state used to retrieve the collection states.

`addresses` IReadOnlyList<Address>

The list of addresses to retrieve the collection states for.

Returns

Dictionary<Address, CollectionState>

A dictionary of Address and CollectionState pairs representing the collection states for the given addresses, or an empty dictionary for addresses that do not have a collection state.

SetCollectionState(IWorld, Address, CollectionState)

Sets the state of a collection in the world.

```
public static IWorld SetCollectionState(this IWorld world, Address collection,
CollectionState state)
```

Parameters

`world` IWorld

`collection` Address

The address of the collection.

`state` CollectionState

The state of the collection.

Returns

IWorld

The updated world with the updated collection state.

TryGetCollectionState(IWorldState, Address, out CollectionState)

Tries to get the collection state for a specific address from the given world state.

```
public static bool TryGetCollectionState(this IWorldState worldState, Address
address, out CollectionState collectionState)
```

Parameters

`worldState` IWorldState

The world state from which to get the collection state.

`address` Address

The address for which to retrieve the collection state.

`collectionState` CollectionState

The resulting collection state, if found.

Returns

bool

True if the collection state is found, otherwise false.

Class CombinationSlotStateModule

Namespace: Nekoyume.Module

Assembly: Lib9c.dll

CombinationSlotStateModule is the module to use CombinationSlotState/AllCombinationSlotState with account.

```
public static class CombinationSlotStateModule
```

Inheritance

object ← CombinationSlotStateModule

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Methods

GetAllCombinationSlotState(IWorldState, Address)

```
public static AllCombinationSlotState GetAllCombinationSlotState(this IWorldState worldState, Address avatarAddress)
```

Parameters

`worldState` IWorldState

`avatarAddress` Address

Returns

AllCombinationSlotState

SetCombinationSlotState(IWorld, Address, AllCombinationSlotState)

```
public static IWorld SetCombinationSlotState(this IWorld world, Address
avatarAddress, AllCombinationSlotState combinationSlotState)
```

Parameters

`world` IWorld

`avatarAddress` Address

`combinationSlotState` AllCombinationSlotState

Returns

IWorld

Class DailyRewardModule

Namespace: Nekoyume.Module

Assembly: Lib9c.dll

```
public static class DailyRewardModule
```

Inheritance

object ← DailyRewardModule

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Methods

GetDailyRewardReceivedBlockIndex(IWorldState, Address)

```
public static long GetDailyRewardReceivedBlockIndex(this IWorldState worldState,
Address avatarAddress)
```

Parameters

`worldState` IWorldState

`avatarAddress` Address

Returns

long

SetDailyRewardReceivedBlockIndex(IWorld, Address, long)

```
public static IWorld SetDailyRewardReceivedBlockIndex(this IWorld world, Address
avatarAddress, long blockIndex)
```

Parameters

`world` IWorld

`avatarAddress` Address

`blockIndex` long

Returns

IWorld

TryGetDailyRewardReceivedBlockIndex(IWorldState, Address, out long)

```
public static bool TryGetDailyRewardReceivedBlockIndex(this IWorldState worldState,
Address avatarAddress, out long receivedBlockIndex)
```

Parameters

`worldState` IWorldState

`avatarAddress` Address

`receivedBlockIndex` long

Returns

bool

Class GiftModule

Namespace: Nekoyume.Module

Assembly: Lib9c.dll

```
public static class GiftModule
```

Inheritance

object ← GiftModule

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Methods

GetClaimedGifts(IWorldState, Address)

```
public static List<int> GetClaimedGifts(this IWorldState state,
Address avatarAddress)
```

Parameters

`state` IWorldState

`avatarAddress` Address

Returns

List<int>

SetClaimedGifts(IWorld, Address, List<int>)

```
public static IWorld SetClaimedGifts(this IWorld world, Address avatarAddress,
List<int> claimedGiftIds)
```

Parameters

`world` IWorld

`avatarAddress` Address

`claimedGiftIds` List<int>

Returns

IWorld

Class GlobalCounterModule

Namespace: Nekoyume.Module

Assembly: Lib9c.dll

Reads and writes the global counters that drive hard world openings.

```
public static class GlobalCounterModule
```

Inheritance

object ← GlobalCounterModule

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Methods

AddCounter(IWorld, int, long)

Adds `delta` to counter `id`.

```
public static IWorld AddCounter(this IWorld world, int id, long delta)
```

Parameters

`world` IWorld

`id` int

`delta` long

Returns

IWorld

Remarks

Does not check `delta`; it must be positive, since counters only grow. Tick(IWorld, Global CounterType, int, long, int, long) guards this for every action path.

CounterKey(int)

The state key of counter `id`.

```
public static Address CounterKey(int id)
```

Parameters

`id` int

Returns

Address

GetCounter(IWorldState, int)

The value of counter `id`, or 0 if it was never written.

```
public static long GetCounter(this IWorldState worldState, int id)
```

Parameters

`worldState` IWorldState

`id` int

Returns

long

Class InfiniteTowerModule

Namespace: Nekoyume.Module

Assembly: Lib9c.dll

```
public static class InfiniteTowerModule
```

Inheritance

object ← InfiniteTowerModule

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Methods

GetFloorClearCount(IWorld, int, int)

Get the clear count for a specific floor in a specific tower.

```
public static int GetFloorClearCount(this IWorld worldState, int infiniteTowerId,
int floorId)
```

Parameters

`worldState` IWorld

The world state.

`infiniteTowerId` int

The infinite tower ID.

`floorId` int

The floor ID.

Returns

int

The number of times the floor has been cleared.

GetInfiniteTowerBoardState(IWorld, int)

Get infinite tower board state for a specific tower.

```
public static InfiniteTowerBoardState GetInfiniteTowerBoardState(this IWorld
worldState, int infiniteTowerId)
```

Parameters

`worldState` IWorld

The world state.

`infiniteTowerId` int

The infinite tower ID.

Returns

InfiniteTowerBoardState

InfiniteTowerBoardState for the tower.

GetInfiniteTowerInfo(IWorld, Address, int)

Get infinite tower info for a specific avatar and tower.

```
public static InfiniteTowerInfo GetInfiniteTowerInfo(this IWorld worldState, Address
avatarAddress, int infiniteTowerId)
```

Parameters

`worldState` IWorld

The world state.

`avatarAddress` Address

The avatar address.

`infiniteTowerId` int

The infinite tower ID.

Returns

InfiniteTowerInfo

InfiniteTowerInfo for the avatar and tower.

RecordFloorClear(IWorld, int, int, long)

Update infinite tower board state by recording a floor clear.

```
public static IWorld RecordFloorClear(this IWorld world, int infiniteTowerId, int
floorId, long blockIndex)
```

Parameters

`world` IWorld

The world.

`infiniteTowerId` int

The infinite tower ID.

`floorId` int

The floor ID that was cleared.

`blockIndex` long

The block index when the floor was cleared.

Returns

IWorld

Updated world with the recorded floor clear.

SetInfiniteTowerBoardState(IWorld, InfiniteTowerBoardState)

Set infinite tower board state for a specific tower.

```
public static IWorld SetInfiniteTowerBoardState(this IWorld world,
InfiniteTowerBoardState infiniteTowerBoardState)
```

Parameters

`world` IWorld

The world.

`infiniteTowerBoardState` InfiniteTowerBoardState

The infinite tower board state to set.

Returns

IWorld

Updated world with the infinite tower board state.

SetInfiniteTowerInfo(IWorld, Address, InfiniteTowerInfo)

Set infinite tower info for a specific avatar and tower.

```
public static IWorld SetInfiniteTowerInfo(this IWorld world, Address avatarAddress,
InfiniteTowerInfo infiniteTowerInfo)
```

Parameters

`world` IWorld

The world.

`avatarAddress` Address

The avatar address.

`infiniteTowerInfo` InfiniteTowerInfo

The infinite tower info to set.

Returns

IWorld

Updated world with the infinite tower info.

TryGetInfiniteTowerBoardState(IWorld, int, out InfiniteTowerBoardState)

Try to get infinite tower board state for a specific tower.

```
public static bool TryGetInfiniteTowerBoardState(this IWorld worldState, int
infiniteTowerId, out InfiniteTowerBoardState infiniteTowerBoardState)
```

Parameters

`worldState` IWorld

The world state.

`infiniteTowerId` int

The infinite tower ID.

`infiniteTowerBoardState` InfiniteTowerBoardState

The infinite tower board state if found.

Returns

bool

True if the infinite tower board state was found, false otherwise.

TryGetInfiniteTowerInfo(IWorld, Address, int, out InfiniteTowerInfo)

Try to get infinite tower info for a specific avatar and tower.

```
public static bool TryGetInfiniteTowerInfo(this IWorld worldState, Address
avatarAddress, int infiniteTowerId, out InfiniteTowerInfo infiniteTowerInfo)
```

Parameters

`worldState` IWorld

The world state.

`avatarAddress` Address

The avatar address.

`infiniteTowerId` int

The infinite tower ID.

`infiniteTowerInfo` InfiniteTowerInfo

The infinite tower info if found.

Returns

bool

True if the infinite tower info was found, false otherwise.

Class LegacyModule

Namespace: Nekoyume.Module

Assembly: Lib9c.dll

```
public static class LegacyModule
```

Inheritance

object ← LegacyModule

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Methods

GetArenaAvatarState(IWorldState, Address, AvatarState)

```
public static ArenaAvatarState GetArenaAvatarState(this IWorldState worldState,
Address arenaAvatarStateAddress, AvatarState avatarState)
```

Parameters

`worldState` IWorldState

`arenaAvatarStateAddress` Address

`avatarState` AvatarState

Returns

ArenaAvatarState

GetArenaInfo(IWorldState, Address, AvatarState, CharacterSheet, CostumeStatSheet)

```
public static (Address arenaInfoAddress, ArenaInfo arenaInfo, bool isNewArenaInfo)
GetArenaInfo(this IWorldState worldState, Address weeklyArenaAddress, AvatarState
avatarState, CharacterSheet characterSheet, CostumeStatSheet costumeStatSheet)
```

Parameters

`worldState` IWorldState

`weeklyArenaAddress` Address

`avatarState` AvatarState

`characterSheet` CharacterSheet

`costumeStatSheet` CostumeStatSheet

Returns

(Address arenaInfoAddress, ArenaInfo arenaInfo, bool isNewArenaInfo)

GetArenaParticipants(IWorldState, Address, int, int)

```
public static ArenaParticipants GetArenaParticipants(this IWorldState worldState,
Address arenaParticipantsAddress, int id, int round)
```

Parameters

`worldState` IWorldState

`arenaParticipantsAddress` Address

`id` int

`round` int

Returns

ArenaParticipants

GetAvatarSheets(IWorldState)

```
public static AvatarSheets GetAvatarSheets(this IWorldState worldState)
```

Parameters

`worldState` IWorldState

Returns

AvatarSheets

GetCombinationSlotStateLegacy(IWorldState, Address, int)

```
[Obsolete("Use AllCombinationSlotState.GetRuneState() instead.")]
public static CombinationSlotState GetCombinationSlotStateLegacy(this IWorldState
worldState, Address avatarAddress, int index)
```

Parameters

`worldState` IWorldState

`avatarAddress` Address

`index` int

Returns

CombinationSlotState

GetCouponWallet(IWorldState, Address)

```
public static IImmutableDictionary<Guid, Coupon> GetCouponWallet(this IWorldState
```

```
worldState, Address agentAddress)
```

Parameters

`worldState` IWorldState

`agentAddress` Address

Returns

IImmutableDictionary<Guid, Coupon>

GetCrystalCostState(IWorldState, Address)

```
public static CrystalCostState GetCrystalCostState(this IWorldState worldState,
Address address)
```

Parameters

`worldState` IWorldState

`address` Address

Returns

CrystalCostState

GetCrystalCostStates(IWorldState, long, long)

```
public static (CrystalCostState DailyCostState, CrystalCostState WeeklyCostState,
CrystalCostState PrevWeeklyCostState, CrystalCostState BeforePrevWeeklyCostState)
GetCrystalCostStates(this IWorldState worldState, long blockIndex, long interval)
```

Parameters

`worldState` IWorldState

`blockIndex` long

`interval` long

Returns

(CrystalCostState DailyCostState, CrystalCostState WeeklyCostState, CrystalCostState PrevWeeklyCostState, CrystalCostState BeforePrevWeeklyCostState)

GetGameConfigState(IWorldState)

```
public static GameConfigState GetGameConfigState(this IWorldState worldState)
```

Parameters

`worldState` IWorldState

Returns

GameConfigState

GetGoldBalanceState(IWorldState, Address, Currency)

```
public static GoldBalanceState GetGoldBalanceState(this IWorldState worldState,
Address address, Currency currency)
```

Parameters

`worldState` IWorldState

`address` Address

`currency` Currency

Returns

GoldBalanceState

GetGoldCurrency(IWorldState)

```
public static Currency GetGoldCurrency(this IWorldState worldState)
```

Parameters

`worldState` IWorldState

Returns

Currency

GetGoldDistribution(IWorldState)

```
public static IEnumerable<GoldDistribution> GetGoldDistribution(this
IWorldState worldState)
```

Parameters

`worldState` IWorldState

Returns

IEnumerable<GoldDistribution>

GetItemSheet(IWorldState)

```
public static ItemSheet GetItemSheet(this IWorldState worldState)
```

Parameters

`worldState` IWorldState

Returns

ItemSheet

GetLegacyState(IWorldState, Address)

```
public static IValue GetLegacyState(this IWorldState worldState, Address address)
```

Parameters

`worldState` IWorldState

`address` Address

Returns

IValue

GetLegacyStates(IWorldState, IReadOnlyList<Address>)

```
public static IReadOnlyList<IValue?> GetLegacyStates(this IWorldState worldState,
IReadOnlyList<Address> addresses)
```

Parameters

`worldState` IWorldState

`addresses` IReadOnlyList<Address>

Returns

IReadOnlyList<IValue>

GetLegacyStatesAsDict(IWorldState, params Address[])

```
public static Dictionary<Address, IValue> GetLegacyStatesAsDict(this IWorldState
worldState, params Address[] addresses)
```

Parameters

`worldState` IWorldState

`addresses` Address[]

Returns

Dictionary<Address, IValue>

GetQuestSheet(IWorldState)

```
public static QuestSheet GetQuestSheet(this IWorldState worldState)
```

Parameters

`worldState` IWorldState

Returns

QuestSheet

GetRaiderState(IWorldState, Address)

```
public static RaiderState GetRaiderState(this IWorldState worldState,
Address raiderAddress)
```

Parameters

`worldState` IWorldState

`raiderAddress` Address

Returns

RaiderState

GetRaiderState(IWorldState, Address, int)

```
public static RaiderState GetRaiderState(this IWorldState worldState, Address
avatarAddress, int raidId)
```

Parameters

`worldState` IWorldState

`avatarAddress` Address

`raidId` int

Returns

RaiderState

GetRankingSimulatorSheets(IWorldState)

```
public static RankingSimulatorSheets GetRankingSimulatorSheets(this
IWorldState worldState)
```

Parameters

`worldState` IWorldState

Returns

RankingSimulatorSheets

GetRankingSimulatorSheetsV1(IWorldState)

```
public static RankingSimulatorSheetsV1 GetRankingSimulatorSheetsV1(this
IWorldState worldState)
```

Parameters

`worldState` IWorldState

Returns

RankingSimulatorSheetsV1

GetRankingState(IWorldState)

```
public static RankingState GetRankingState(this IWorldState worldState)
```

Parameters

`worldState` IWorldState

Returns

RankingState

GetRankingState0(IWorldState)

```
public static RankingState0 GetRankingState0(this IWorldState worldState)
```

Parameters

`worldState` IWorldState

Returns

RankingState0

GetRankingState1(IWorldState)

```
public static RankingState1 GetRankingState1(this IWorldState worldState)
```

Parameters

`worldState` IWorldState

Returns

RankingState1

GetRedeemCodeState(IWorldState)

```
public static RedeemCodeState GetRedeemCodeState(this IWorldState worldState)
```

Parameters

`worldState` IWorldState

Returns

RedeemCodeState

GetSheetCsv(IWorldState, Address)

```
public static string GetSheetCsv(this IWorldState worldState, Address address)
```

Parameters

`worldState` IWorldState

`address` Address

Returns

string

GetSheetCsv<T>(IWorldState)

```
public static string GetSheetCsv<T>(this IWorldState worldState) where T :
ISheet, new()
```

Parameters

`worldState` IWorldState

Returns

string

Type Parameters

`T`

GetSheet<T>(IWorldState)

```
public static T GetSheet<T>(this IWorldState worldState) where T : ISheet, new()
```

Parameters

`worldState` IWorldState

Returns

T

Type Parameters

`T`

GetSheet<T>(IWorldState, Address)

```
public static T GetSheet<T>(this IWorldState worldState, Address sheetAddr) where T
: ISheet, new()
```

Parameters

`worldState` IWorldState

`sheetAddr` Address

Returns

T

Type Parameters

T

GetSheets(IWorldState, bool, bool, bool, bool, bool, bool, bool, bool, bool, IEnumerable<Type>)

```
public static Dictionary<Type, (Address address, ISheet sheet)> GetSheets(this
IWorldState worldState, bool containAvatarSheets = false, bool containItemSheet =
false, bool containQuestSheet = false, bool containSimulatorSheets = false, bool
containStageSimulatorSheets = false, bool containRankingSimulatorSheets = false,
bool containArenaSimulatorSheets = false, bool containValidateItemRequirementSheets
= false, bool containRaidSimulatorSheets = false, IEnumerable<Type> sheetTypes
= null)
```

Parameters

worldState IWorldState

containAvatarSheets bool

containItemSheet bool

containQuestSheet bool

containSimulatorSheets bool

containStageSimulatorSheets bool

containRankingSimulatorSheets bool

containArenaSimulatorSheets bool

containValidateItemRequirementSheets bool

containRaidSimulatorSheets bool

sheetTypes IEnumerable<Type>

Returns

Dictionary<Type, (Address address, ISheet sheet)>

GetSheets(IWorldState, params Type[])

```
public static Dictionary<Type, (Address address, ISheet sheet)> GetSheets(this
IWorldState worldState, params Type[] sheetTypes)
```

Parameters

`worldState` IWorldState

`sheetTypes` Type[]

Returns

Dictionary<Type, (Address address, ISheet sheet)>

GetSheets(IWorldState, params (Type sheetType, string sheetName)[])

```
public static Dictionary<Type, (Address address, ISheet sheet)> GetSheets(this
IWorldState worldState, params (Type sheetType, string sheetName)[] sheetTuples)
```

Parameters

`worldState` IWorldState

`sheetTuples` (Type sheetType, string sheetName)[]

Returns

Dictionary<Type, (Address address, ISheet sheet)>

GetSheetsV1(IWorldState, bool, bool, bool, bool, bool, bool, bool, bool, bool, IEnumerable<Type>)

```
public static Dictionary<Type, (Address address, ISheet sheet)> GetSheetsV1(this
IWorldState worldState, bool containAvatarSheets = false, bool containItemSheet =
false, bool containQuestSheet = false, bool containSimulatorSheets = false, bool
containStageSimulatorSheets = false, bool containRankingSimulatorSheets = false,
bool containArenaSimulatorSheets = false, bool containValidateItemRequirementSheets
= false, bool containRaidSimulatorSheets = false, IEnumerable<Type> sheetTypes
= null)
```

Parameters

`worldState` IWorldState

`containAvatarSheets` bool

`containItemSheet` bool

`containQuestSheet` bool

`containSimulatorSheets` bool

`containStageSimulatorSheets` bool

`containRankingSimulatorSheets` bool

`containArenaSimulatorSheets` bool

`containValidateItemRequirementSheets` bool

`containRaidSimulatorSheets` bool

`sheetTypes` IEnumerable<Type>

Returns

Dictionary<Type, (Address address, ISheet sheet)>

GetSheetsV100291(IWorldState, bool, bool, bool, bool, bool, bool, bool, bool, bool, IEnumerable<Type>)

```
public static Dictionary<Type, (Address address, ISheet sheet)>
GetSheetsV100291(this IWorldState worldState, bool containAvatarSheets = false, bool
containItemSheet = false, bool containQuestSheet = false, bool
containSimulatorSheets = false, bool containStageSimulatorSheets = false, bool
containRankingSimulatorSheets = false, bool containArenaSimulatorSheets = false,
bool containValidateItemRequirementSheets = false, IEnumerable<Type> sheetTypes
= null)
```

Parameters

`worldState` IWorldState

`containAvatarSheets` bool

`containItemSheet` bool

`containQuestSheet` bool

`containSimulatorSheets` bool

`containStageSimulatorSheets` bool

`containRankingSimulatorSheets` bool

`containArenaSimulatorSheets` bool

`containValidateItemRequirementSheets` bool

`sheetTypes` IEnumerable<Type>

Returns

Dictionary<Type, (Address address, ISheet sheet)>

GetShopState(IWorldState)

```
public static ShopState GetShopState(this IWorldState worldState)
```

Parameters

`worldState` IWorldState

Returns

ShopState

GetStageSimulatorSheets(IWorldState)

```
public static StageSimulatorSheets GetStageSimulatorSheets(this
IWorldState worldState)
```

Parameters

`worldState` IWorldState

Returns

StageSimulatorSheets

GetStageSimulatorSheetsV1(IWorldState)

```
public static StageSimulatorSheetsV1 GetStageSimulatorSheetsV1(this
IWorldState worldState)
```

Parameters

`worldState` IWorldState

Returns

StageSimulatorSheetsV1

GetWeeklyArenaState(IWorldState, Address)

```
public static WeeklyArenaState GetWeeklyArenaState(this IWorldState worldState,
Address address)
```

Parameters

`worldState` IWorldState

`address` Address

Returns

WeeklyArenaState

GetWeeklyArenaState(IWorldState, int)

```
public static WeeklyArenaState GetWeeklyArenaState(this IWorldState worldState,
int index)
```

Parameters

`worldState` IWorldState

`index` int

Returns

WeeklyArenaState

MarkBalanceChanged(IWorld, IActionContext, Currency, params Address[])

```
public static IWorld MarkBalanceChanged(this IWorld world, IActionContext context,
Currency currency, params Address[] accounts)
```

Parameters

`world` IWorld

`context` IActionContext

`currency` Currency

`accounts` Address[]

Returns

IWorld

RemoveLegacyState(IWorld, Address)

```
public static IWorld RemoveLegacyState(this IWorld world, Address address)
```

Parameters

`world` IWorld

`address` Address

Returns

IWorld

SetCouponWallet(IWorld, Address, IImmutableDictionary<Guid, Coupon>)

```
public static IWorld SetCouponWallet(this IWorld world, Address agentAddress,
IImmutableDictionary<Guid, Coupon> couponWallet)
```

Parameters

`world` IWorld

`agentAddress` Address

`couponWallet` IImmutableDictionary<Guid, Coupon>

Returns

IWorld

SetLegacyState(IWorld, Address, IValue)

```
public static IWorld SetLegacyState(this IWorld world, Address address,
IValue state)
```

Parameters

`world` IWorld

`address` Address

`state` IValue

Returns

IWorld

SetWorldBossKillReward(IWorld, IActionContext, Address, WorldBossKillRewardRecord, int, WorldBossState, RuneWeightSheet, WorldBossKillRewardSheet, RuneSheet, MaterialItemSheet, IRandom, Inventory, Address, Address, Row)

```
public static IWorld SetWorldBossKillReward(this IWorld world, IActionContext
context, Address rewardInfoAddress, WorldBossKillRewardRecord rewardRecord, int
rank, WorldBossState bossState, RuneWeightSheet runeWeightSheet,
WorldBossKillRewardSheet worldBossKillRewardSheet, RuneSheet runeSheet,
MaterialItemSheet materialItemSheet, IRandom random, Inventory inventory, Address
avatarAddress, Address agentAddress, BoostScheduleSheet.Row killRewardBoost = null)
```

Parameters

`world` IWorld

`context` IActionContext

`rewardInfoAddress` Address

`rewardRecord` WorldBossKillRewardRecord

`rank` int

`bossState` WorldBossState

`runeWeightSheet` RuneWeightSheet

`worldBossKillRewardSheet` WorldBossKillRewardSheet

`runeSheet` RuneSheet

`materialItemSheet` MaterialItemSheet

`random` IRandom

`inventory` Inventory

`avatarAddress` Address

`agentAddress` Address

`killRewardBoost` BoostScheduleSheet.Row

Returns

IWorld

TryGetArenaAvatarState(IWorldState, Address, out ArenaAvatarState)

```
public static bool TryGetArenaAvatarState(this IWorldState worldState, Address
arenaAvatarStateAddress, out ArenaAvatarState arenaAvatarState)
```

Parameters

`worldState` IWorldState

`arenaAvatarStateAddress` Address

`arenaAvatarState` ArenaAvatarState

Returns

bool

TryGetArenaInformation(IWorldState, Address, out ArenaInformation)

```
public static bool TryGetArenaInformation(this IWorldState worldState, Address arenaInformationAddress, out ArenaInformation arenaInformation)
```

Parameters

`worldState` IWorldState

`arenaInformationAddress` Address

`arenaInformation` ArenaInformation

Returns

bool

TryGetArenaParticipants(IWorldState, Address, out ArenaParticipants)

```
public static bool TryGetArenaParticipants(this IWorldState worldState, Address arenaParticipantsAddress, out ArenaParticipants arenaParticipants)
```

Parameters

`worldState` IWorldState

`arenaParticipantsAddress` Address

`arenaParticipants` ArenaParticipants

Returns

bool

TryGetArenaScore(IWorldState, Address, out ArenaScore)

```
public static bool TryGetArenaScore(this IWorldState worldState, Address
arenaScoreAddress, out ArenaScore arenaScore)
```

Parameters

`worldState` IWorldState

`arenaScoreAddress` Address

`arenaScore` ArenaScore

Returns

bool

TryGetGoldBalance(IWorldState, Address, Currency, out FungibleAssetValue)

```
public static bool TryGetGoldBalance(this IWorldState worldState, Address address,
Currency currency, out FungibleAssetValue balance)
```

Parameters

`worldState` IWorldState

`address` Address

`currency` Currency

`balance` FungibleAssetValue

Returns

bool

TryGetLegacyStakeState(IWorldState, Address, out LegacyStakeState)

```
public static bool TryGetLegacyStakeState(this IWorldState worldState, Address agentAddress, out LegacyStakeState legacyStakeState)
```

Parameters

`worldState` IWorldState

`agentAddress` Address

`legacyStakeState` LegacyStakeState

Returns

bool

TryGetLegacyState<T>(IWorldState, Address, out T)

```
public static bool TryGetLegacyState<T>(this IWorldState worldState, Address address, out T result) where T : IValue
```

Parameters

`worldState` IWorldState

`address` Address

`result` T

Returns

bool

Type Parameters

`T`

TryGetPatchedSheet<T>(IWorldState, out T)

Loads `T` only when the chain has already been patched with it, and lets a malformed sheet throw instead of reporting it as missing.

```
public static bool TryGetPatchedSheet<T>(this IWorldState worldState, out T sheet)
where T : ISheet, new()
```

Parameters

`worldState` IWorldState

The world to read the sheet from.

`sheet` T

The parsed sheet, or null when absent.

Returns

bool

false when the sheet has never been patched onto this chain.

Type Parameters

`T`

The sheet to load.

Remarks

Unlike TryGetSheet<T>(IWorldState, out T) this does not swallow a parse failure. A sheet that carries restrictions must not be able to switch itself off by becoming unreadable; only genuine absence may fall back, and that is also what keeps re-evaluation of blocks older than the patch identical.

TryGetSheet<T>(IWorldState, Address, out T)

```
public static bool TryGetSheet<T>(this IWorldState worldState, Address address, out
T sheet) where T : ISheet, new()
```

Parameters

`worldState` IWorldState

`address` Address

`sheet` T

Returns

bool

Type Parameters

`T`

TryGetSheet<T>(IWorldState, out T)

```
public static bool TryGetSheet<T>(this IWorldState worldState, out T sheet) where T
: ISheet, new()
```

Parameters

`worldState` IWorldState

`sheet` T

Returns

bool

Type Parameters

`T`

TryGetStakeState(IWorldState, Address, out StakeState)

```
public static bool TryGetStakeState(this IWorldState worldState, Address agentAddr,
out StakeState stakeState)
```

Parameters

`worldState` IWorldState

`agentAddr` Address

`stakeState` StakeState

Returns

bool

ValidateWorldId(IWorldState, Address, int)

```
public static void ValidateWorldId(this IWorldState worldState, Address
avatarAddress, int worldId)
```

Parameters

`worldState` IWorldState

`avatarAddress` Address

`worldId` int

Class PatrolRewardModule

Namespace: Nekoyume.Module

Assembly: Lib9c.dll

Provides utility methods for handling patrol reward states.

```
public static class PatrolRewardModule
```

Inheritance

object ← PatrolRewardModule

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Methods

GetPatrolRewardClaimedBlockIndex(IWorldState, Address)

Retrieves the block index at which the patrol reward was last claimed for a given avatar.

```
public static long GetPatrolRewardClaimedBlockIndex(this IWorldState worldState,
Address avatarAddress)
```

Parameters

`worldState` IWorldState

The current world state.

`avatarAddress` Address

The address of the avatar.

Returns

long

The block index when the patrol reward was last claimed.

Exceptions

FailedLoadStateException

Thrown if the state could not be loaded.

SetPatrolRewardClaimedBlockIndex(IWorld, Address, long)

Sets the block index at which the patrol reward was last claimed for a given avatar.

```
public static IWorld SetPatrolRewardClaimedBlockIndex(this IWorld world, Address
avatarAddress, long blockIndex)
```

Parameters

`world` IWorld

The current world instance.

`avatarAddress` Address

The address of the avatar.

`blockIndex` long

The block index to set.

Returns

IWorld

The updated world instance.

TryGetPatrolRewardClaimedBlockIndex(IWorldState, Address, out long)

Tries to retrieve the block index at which the patrol reward was last claimed for a given avatar.

```
public static bool TryGetPatrolRewardClaimedBlockIndex(this IWorldState worldState,
Address avatarAddress, out long blockIndex)
```

Parameters

`worldState` IWorldState

The current world state.

`avatarAddress` Address

The address of the avatar.

`blockIndex` long

Outputs the block index when the patrol reward was last claimed.

Returns

bool

`true` if the block index was successfully retrieved; otherwise, `false`.

Class RelationshipModule

Namespace: Nekoyume.Module

Assembly: Lib9c.dll

```
public static class RelationshipModule
```

Inheritance

object ← RelationshipModule

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Methods

GetRelationship(IWorldState, Address)

```
public static int GetRelationship(this IWorldState state, Address avatarAddress)
```

Parameters

`state` IWorldState

`avatarAddress` Address

Returns

int

SetRelationship(IWorld, Address, int)

```
public static IWorld SetRelationship(this IWorld world, Address avatarAddress,
int relationship)
```

Parameters

`world` IWorld

`avatarAddress` Address

`relationship` int

Returns

IWorld

Class RuneStateModule

Namespace: Nekoyume.Module

Assembly: Lib9c.dll

RuneStateModule is the module to use RuneState/AllRuneState with account.

```
public static class RuneStateModule
```

Inheritance

object ← RuneStateModule

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Methods

GetRuneState(IWorldState, Address, out bool)

```
public static AllRuneState GetRuneState(this IWorldState worldState, Address
avatarAddress, out bool migrateRequired)
```

Parameters

`worldState` IWorldState

`avatarAddress` Address

`migrateRequired` bool

Returns

AllRuneState

SetRuneState(IWorld, Address, AllRuneState)

```
public static IWorld SetRuneState(this IWorld world, Address avatarAddress,
AllRuneState allRuneState)
```

Parameters

`world` IWorld

`avatarAddress` Address

`allRuneState` AllRuneState

Returns

IWorld

Class StateResolver

Namespace: Nekoyume.Module

Assembly: Lib9c.dll

```
public static class StateResolver
```

Inheritance

object ← StateResolver

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Methods

GetResolvedState(IWorldState, Address, Address)

Gets the account of address `address` if it exists. If not, returns from the legacy account.

```
public static IValue? GetResolvedState(this IWorldState world, Address address,
Address accountAddress)
```

Parameters

`world` IWorldState

An Libplanet.Action.State.IWorld instance of to get state from.

`address` Address

The Libplanet.Crypto.Address of the state to get.

`accountAddress` Address

The Libplanet.Crypto.Address of the account to get.

Returns

IValue

An Libplanet.Action.State.IAccountState instance of address `address` if exits. If not, legacy account.

GetResolvedState(IWorldState, Address, Address, Address)

Gets the account of address `address` if it exists. If not, returns from the legacy account with `legacyAddress`.

```
public static IValue? GetResolvedState(this IWorldState world, Address address,
Address accountAddress, Address legacyAddress)
```

Parameters

`world` IWorldState

An Libplanet.Action.State.IWorld instance of to get state from.

`address` Address

The Libplanet.Crypto.Address of the state to get.

`accountAddress` Address

The Libplanet.Crypto.Address of the account to get.

`legacyAddress` Address

The Libplanet.Crypto.Address of the state to get from the legacy account.

Returns

IValue

An Libplanet.Action.State.IAccountState instance of address `address` if exits. If not, legacy account.

Class WorldOpenModule

Namespace: Nekoyume.Module

Assembly: Lib9c.dll

Records when hard world openings were confirmed and decides whether they are in effect.

```
public static class WorldOpenModule
```

Inheritance

object ← WorldOpenModule

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Methods

ConfirmOpen(IWorld, int, long)

Confirms the opening at `blockIndex`. Never reverts: an already confirmed opening is left as it is.

```
public static IWorld ConfirmOpen(this IWorld world, int openId, long blockIndex)
```

Parameters

`world` IWorld

`openId` int

`blockIndex` long

Returns

IWorld

IsApplied(IWorldState, WorldOpenSheet, Row, long)

Whether `open` and all its prerequisites are in effect at `block`. Iterative with a visited set (a cycle must not overflow the stack); a cycle, an unknown prerequisite or an invalid row (see IsValid) anywhere in the chain means not applied.

```
public static bool IsApplied(this IWorldState state, WorldOpenSheet sheet,
WorldOpenSheet.Row open, long block)
```

Parameters

`state` IWorldState

`sheet` WorldOpenSheet

`open` WorldOpenSheet.Row

`block` long

Returns

bool

IsPending(IWorldState, Row, long)

Whether the opening is valid, unconfirmed and before its fallback, i.e. counters for it may still be written.

```
public static bool IsPending(this IWorldState state, WorldOpenSheet.Row r,
long block)
```

Parameters

`state` IWorldState

`r` WorldOpenSheet.Row

`block` long

Returns

bool

OpenKey(int)

The state key of opening `openId`.

```
public static Address OpenKey(int openId)
```

Parameters

`openId` int

Returns

Address

TryGetAppliedBlock(IWorldState, WorldOpenSheet, Row, out long)

The first block at which IsApplied(IWorldState, WorldOpenSheet, Row, long) becomes true for `open`, given the current state: the latest over the prerequisite chain of each row's own effective block (its fallback when unconfirmed, otherwise `min(fallback, max(confirmed + delay, min))`). Confirming an opening later can only bring this block earlier, never later.

```
public static bool TryGetAppliedBlock(this IWorldState state, WorldOpenSheet sheet,
WorldOpenSheet.Row open, out long appliedBlock)
```

Parameters

`state` IWorldState

`sheet` WorldOpenSheet

`open` WorldOpenSheet.Row

`appliedBlock` long

Returns

bool

false when the chain has a cycle, an unknown prerequisite or an invalid row, i.e. when IsApplied(IWorldState, WorldOpenSheet, Row, long) is false at every block.

TryGetConfirmedBlock(IWorldState, int, out long)

Gets the block at which opening `openId` was confirmed.

```
public static bool TryGetConfirmedBlock(this IWorldState state, int openId, out
long block)
```

Parameters

`state` IWorldState

`openId` int

`block` long

Returns

bool

Namespace Nekoyume.Module.CombinationSlot

Classes

CombinationSlotNotFoundException

DuplicatedCombinationSlotIndexException

Class CombinationSlotNotFoundException

Namespace: Nekoyume.Module.CombinationSlot

Assembly: Lib9c.dll

```
[Serializable]
public class CombinationSlotNotFoundException : Exception, ISerializable
```

Inheritance

object ← Exception ← CombinationSlotNotFoundException

Implements

ISerializable

Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Constructors

CombinationSlotNotFoundException(SerializationInfo, StreamingContext)

```
protected CombinationSlotNotFoundException(SerializationInfo info,
StreamingContext context)
```

Parameters

`info` SerializationInfo

`context` StreamingContext

CombinationSlotNotFoundException(string)

```
public CombinationSlotNotFoundException(string message)
```

Parameters

`message` string

Class DuplicatedCombinationSlotIndexException

Namespace: Nekoyume.Module.CombinationSlot

Assembly: Lib9c.dll

```
[Serializable]
public class DuplicatedCombinationSlotIndexException : Exception, ISerializable
```

Inheritance

object ← Exception ← DuplicatedCombinationSlotIndexException

Implements

ISerializable

Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Constructors

DuplicatedCombinationSlotIndexException(SerializationInfo, StreamingContext)

```
protected DuplicatedCombinationSlotIndexException(SerializationInfo info,
StreamingContext context)
```

Parameters

`info` SerializationInfo

`context` StreamingContext

DuplicatedCombinationSlotIndexException(string)

```
public DuplicatedCombinationSlotIndexException(string message)
```

Parameters

`message` string

Namespace Nekoyume.Module.Guild

Classes

GuildBanModule

GuildDelegateeModule

GuildMemberCounterModule

GuildModule

GuildParticipantModule

Class GuildBanModule

Namespace: Nekoyume.Module.Guild

Assembly: Lib9c.dll

```
public static class GuildBanModule
```

Inheritance

object ← GuildBanModule

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Methods

Ban(GuildRepository, AgentAddress, AgentAddress)

```
public static void Ban(this GuildRepository repository, AgentAddress signer,
AgentAddress target)
```

Parameters

`repository` GuildRepository

`signer` AgentAddress

`target` AgentAddress

IsBanned(GuildRepository, GuildAddress, Address)

```
public static bool IsBanned(this GuildRepository repository, GuildAddress
guildAddress, Address agentAddress)
```

Parameters

`repository` GuildRepository

`guildAddress` GuildAddress

`agentAddress` Address

Returns

bool

RemoveBanList(GuildRepository, GuildAddress)

```
public static void RemoveBanList(this GuildRepository repository,
GuildAddress guildAddress)
```

Parameters

`repository` GuildRepository

`guildAddress` GuildAddress

Unban(GuildRepository, AgentAddress, Address)

```
public static void Unban(this GuildRepository repository, AgentAddress signer,
Address target)
```

Parameters

`repository` GuildRepository

`signer` AgentAddress

`target` Address

Class GuildDelegateeModule

Namespace: Nekoyume.Module.Guild

Assembly: Lib9c.dll

```
public static class GuildDelegateeModule
```

Inheritance

object ← GuildDelegateeModule

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Methods

CreateDelegatee(GuildRepository, Address)

```
public static GuildDelegatee CreateDelegatee(this GuildRepository repository,
Address address)
```

Parameters

`repository` GuildRepository

`address` Address

Returns

GuildDelegatee

TryGetDelegatee(GuildRepository, Address, out GuildDelegatee?)

```
public static bool TryGetDelegatee(this GuildRepository repository, Address address,
out GuildDelegatee? guildDelegatee)
```

Parameters

`repository` GuildRepository

`address` Address

`guildDelegatee` GuildDelegatee

Returns

bool

Class GuildMemberCounterModule

Namespace: Nekoyume.Module.Guild

Assembly: Lib9c.dll

```
public static class GuildMemberCounterModule
```

Inheritance

object ← GuildMemberCounterModule

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Methods

DecreaseGuildMemberCount(GuildRepository, GuildAddress)

```
public static GuildRepository DecreaseGuildMemberCount(this GuildRepository
repository, GuildAddress guildAddress)
```

Parameters

`repository` GuildRepository

`guildAddress` GuildAddress

Returns

GuildRepository

GetGuildMemberCount(GuildRepository, GuildAddress)

```
public static BigInteger GetGuildMemberCount(this GuildRepository repository,
GuildAddress guildAddress)
```

Parameters

`repository` GuildRepository

`guildAddress` GuildAddress

Returns

BigInteger

IncreaseGuildMemberCount(GuildRepository, GuildAddress)

```
public static GuildRepository IncreaseGuildMemberCount(this GuildRepository
repository, GuildAddress guildAddress)
```

Parameters

`repository` GuildRepository

`guildAddress` GuildAddress

Returns

GuildRepository

Class GuildModule

Namespace: Nekoyume.Module.Guild

Assembly: Lib9c.dll

```
public static class GuildModule
```

Inheritance

object ← GuildModule

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Methods

ConvertCurrency(FungibleAssetValue, Currency)

```
public static (FungibleAssetValue TargetFAV, FungibleAssetValue Remainder)
ConvertCurrency(FungibleAssetValue sourceFAV, Currency targetCurrency)
```

Parameters

`sourceFAV` FungibleAssetValue

`targetCurrency` Currency

Returns

(FungibleAssetValue TargetFAV, FungibleAssetValue Remainder)

FAVFromShare(BigInteger, BigInteger, FungibleAssetValue)

```
public static FungibleAssetValue FAVFromShare(BigInteger share, BigInteger
totalShares, FungibleAssetValue totalDelegated)
```

Parameters

`share` BigInteger

`totalShares` BigInteger

`totalDelegated` FungibleAssetValue

Returns

FungibleAssetValue

GetDelegated(IWorld, Address)

```
public static FungibleAssetValue GetDelegated(this IWorld world,
Address agentAddress)
```

Parameters

`world` IWorld

`agentAddress` Address

Returns

FungibleAssetValue

GetDelegationInfo(IWorld, Address)

```
public static (BigInteger Share, BigInteger TotalShare, FungibleAssetValue
TotalDelegated) GetDelegationInfo(this IWorld world, Address agentAddress)
```

Parameters

`world` IWorld

`agentAddress` Address

Returns

(BigInteger Share, BigInteger TotalShare, FungibleAssetValue TotalDelegated)

GetDelegationInfo(IWorldState, Address)

```
public static (BigInteger Share, BigInteger TotalShare, FungibleAssetValue
TotalDelegated) GetDelegationInfo(this IWorldState worldState, Address agentAddress)
```

Parameters

`worldState` IWorldState

`agentAddress` Address

Returns

(BigInteger Share, BigInteger TotalShare, FungibleAssetValue TotalDelegated)

GetDelegationInfo(GuildRepository, Address)

```
public static (BigInteger Share, BigInteger TotalShare, FungibleAssetValue
TotalDelegated) GetDelegationInfo(this GuildRepository repository,
Address agentAddress)
```

Parameters

`repository` GuildRepository

`agentAddress` Address

Returns

(BigInteger Share, BigInteger TotalShare, FungibleAssetValue TotalDelegated)

GetGuildRepository(IWorld, IActionContext)

```
public static GuildRepository GetGuildRepository(this IWorld world,
IActionContext context)
```

Parameters

`world` IWorld

`context` IActionContext

Returns

GuildRepository

GetStaked(IWorld, Address)

```
public static FungibleAssetValue GetStaked(this IWorld world, Address agentAddress)
```

Parameters

`world` IWorld

`agentAddress` Address

Returns

FungibleAssetValue

GetStaked(IWorldState, Address)

```
public static FungibleAssetValue GetStaked(this IWorldState worldState,
Address agentAddress)
```

Parameters

`worldState` IWorldState

`agentAddress` Address

Returns

FungibleAssetValue

MakeGuild(GuildRepository, GuildAddress, Address)

```
public static GuildRepository MakeGuild(this GuildRepository repository,
GuildAddress guildAddress, Address validatorAddress)
```

Parameters

`repository` GuildRepository

`guildAddress` GuildAddress

`validatorAddress` Address

Returns

GuildRepository

RemoveGuild(GuildRepository)

```
public static GuildRepository RemoveGuild(this GuildRepository repository)
```

Parameters

`repository` GuildRepository

Returns

GuildRepository

TryGetGuild(GuildRepository, GuildAddress, out Guild?)

```
public static bool TryGetGuild(this GuildRepository repository, GuildAddress
guildAddress, out Guild? guild)
```

Parameters

`repository` GuildRepository

`guildAddress` GuildAddress

`guild` Guild

Returns

bool

Class GuildParticipantModule

Namespace: Nekoyume.Module.Guild

Assembly: Lib9c.dll

```
public static class GuildParticipantModule
```

Inheritance

object ← GuildParticipantModule

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Methods

GetJoinedGuild(GuildRepository, AgentAddress)

```
public static GuildAddress? GetJoinedGuild(this GuildRepository repository,
AgentAddress agentAddress)
```

Parameters

`repository` GuildRepository

`agentAddress` AgentAddress

Returns

GuildAddress?

JoinGuild(GuildRepository, GuildAddress, AgentAddress)

```
public static GuildRepository JoinGuild(this GuildRepository repository,
GuildAddress guildAddress, AgentAddress target)
```

Parameters

`repository` GuildRepository

`guildAddress` GuildAddress

`target` AgentAddress

Returns

GuildRepository

LeaveGuild(GuildRepository, AgentAddress)

```
public static GuildRepository LeaveGuild(this GuildRepository repository,
AgentAddress agentAddress)
```

Parameters

`repository` GuildRepository

`agentAddress` AgentAddress

Returns

GuildRepository

MoveGuild(GuildRepository, AgentAddress, GuildAddress)

```
public static GuildRepository MoveGuild(this GuildRepository repository,
AgentAddress guildParticipantAddress, GuildAddress dstGuildAddress)
```

Parameters

`repository` GuildRepository

`guildParticipantAddress` AgentAddress

`dstGuildAddress` GuildAddress

Returns

GuildRepository

TryGetGuildParticipant(GuildRepository, AgentAddress, out GuildParticipant?)

```
public static bool TryGetGuildParticipant(this GuildRepository repository,
AgentAddress agentAddress, out GuildParticipant? guildParticipant)
```

Parameters

`repository` GuildRepository

`agentAddress` AgentAddress

`guildParticipant` GuildParticipant

Returns

bool

Namespace Nekoyume.Module.Validator Delegation

Classes

AbstainHistoryModule

ValidatorDelegateeModule

ValidatorDelegatorModule

ValidatorListModule

Class AbstainHistoryModule

Namespace: Nekoyume.Module.ValidatorDelegation

Assembly: Lib9c.dll

```
public static class AbstainHistoryModule
```

Inheritance

object ← AbstainHistoryModule

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Methods

GetAbstainHistory(ValidatorRepository)

```
public static AbstainHistory GetAbstainHistory(this ValidatorRepository repository)
```

Parameters

`repository` ValidatorRepository

Returns

AbstainHistory

SetAbstainHistory(ValidatorRepository, AbstainHistory)

```
public static ValidatorRepository SetAbstainHistory(this ValidatorRepository
repository, AbstainHistory abstainHistory)
```

Parameters

`repository` ValidatorRepository

`abstainHistory` AbstainHistory

Returns

ValidatorRepository

Class ValidatorDelegateeModule

Namespace: Nekoyume.Module.ValidatorDelegation

Assembly: Lib9c.dll

```
public static class ValidatorDelegateeModule
```

Inheritance

object ← ValidatorDelegateeModule

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Methods

CreateDelegatee(ValidatorRepository, PublicKey, BigInteger)

```
public static ValidatorDelegatee CreateDelegatee(this ValidatorRepository
repository, PublicKey publicKey, BigInteger commissionPercentage)
```

Parameters

`repository` ValidatorRepository

`publicKey` PublicKey

`commissionPercentage` BigInteger

Returns

ValidatorDelegatee

TryGetDelegatee(ValidatorRepository, Address, out ValidatorDelegatee?)

```
public static bool TryGetDelegatee(this ValidatorRepository repository, Address address, out ValidatorDelegatee? validatorDelegatee)
```

Parameters

`repository` ValidatorRepository

`address` Address

`validatorDelegatee` ValidatorDelegatee

Returns

bool

Class ValidatorDelegatorModule

Namespace: Nekoyume.Module.ValidatorDelegation

Assembly: Lib9c.dll

```
public static class ValidatorDelegatorModule
```

Inheritance

object ← ValidatorDelegatorModule

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Methods

TryGetDelegator(ValidatorRepository, Address, out ValidatorDelegator?)

```
public static bool TryGetDelegator(this ValidatorRepository repository, Address
address, out ValidatorDelegator? validatorDelegator)
```

Parameters

`repository` ValidatorRepository

`address` Address

`validatorDelegator` ValidatorDelegator

Returns

bool

Class ValidatorListModule

Namespace: Nekoyume.Module.ValidatorDelegation

Assembly: Lib9c.dll

```
public static class ValidatorListModule
```

Inheritance

object ← ValidatorListModule

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Methods

GetProposerInfo(ValidatorRepository)

```
public static ProposerInfo GetProposerInfo(this ValidatorRepository repository)
```

Parameters

`repository` ValidatorRepository

Returns

ProposerInfo

GetValidatorList(ValidatorRepository)

```
public static ValidatorList GetValidatorList(this ValidatorRepository repository)
```

Parameters

`repository` ValidatorRepository

Returns

ValidatorList

SetProposerInfo(ValidatorRepository, ProposerInfo)

```
public static ValidatorRepository SetProposerInfo(this ValidatorRepository
repository, ProposerInfo proposerInfo)
```

Parameters

`repository` ValidatorRepository

`proposerInfo` ProposerInfo

Returns

ValidatorRepository

TryGetProposerInfo(ValidatorRepository, Address, out ProposerInfo?)

```
public static bool TryGetProposerInfo(this ValidatorRepository repository, Address
address, out ProposerInfo? proposerInfo)
```

Parameters

`repository` ValidatorRepository

`address` Address

`proposerInfo` ProposerInfo

Returns

bool

TryGetValidatorList(ValidatorRepository, out ValidatorList?)

```
public static bool TryGetValidatorList(this ValidatorRepository repository, out
ValidatorList? validatorList)
```

Parameters

`repository` ValidatorRepository

`validatorList` ValidatorList

Returns

bool

Namespace Nekoyume.TableData

Classes

ActionBuffSheet

ActionBuffSheet.Row

ArenaSheet

ArenaSheet.RoundData

ArenaSheet.Row

ArenaSimulatorSheets

ArenaSimulatorSheetsV1

ArenaSimulatorSheetsV100291

AvatarSheets

BoostScheduleSheet

Schedules a temporary adjustment of a game value — a reward amount, a cost, a guarantee count — for a block range, so that an event can be run with `patch_table_sheet` instead of rewriting the value's own sheet and patching it back afterwards.

Columns: `id`, `target`, `target_id`, `op`, `value`, `start_block`, `end_block`. A `_memo` column, like any `_` prefixed column, is dropped on load and may be used for notes.

`target_id` is a single id (`50`), an inclusive range (`1~50`) or `*` for every id. `~` rather than `-` keeps a spreadsheet from reading a range as a date on its way to the CSV.

`target` names what is adjusted and is matched as a plain string by the call site that owns it, so the sheet itself never needs to know which targets exist. A node that predates a target simply finds no match for it instead of failing to read the sheet. BoostScheduleSheet.Targets lists the targets this version applies.

A row applies in `[start_block, end_block)`. When several rows match, the one with the lowest `id` wins; rows are never combined, so overlapping rows cannot stack into a larger adjustment than any single row declares.

Reading is total: a row that does not parse is kept but marked invalid and never matches, so a typo fails toward "no adjustment" instead of aborting every action that reads the

sheet. ValidateCsv(string) rejects such a row at patch time so it should not reach the chain in the first place.

Callers must tolerate the sheet being absent — it reaches an existing chain only by `patch_table_sheet`. Absence means no adjustment, which is also what keeps re-evaluation of pre-patch blocks identical.

BoostScheduleSheet.Row
: One scheduled adjustment.

BoostScheduleSheet.TargetInfo
: How to write a row for one target: reference text for people and tools, never read by an action. See Infos.

BoostScheduleSheet.Targets
: The targets this version applies. A row naming any other target is valid data but has no effect until a version that applies it is running.

BuffLimitSheet
: Upper limits of stat buffs and lower limits of stat debuffs, per buff group.

BuffLimitSheet.Row
: The limit of one buff group.

BuffLinkSheet

BuffLinkSheet.Row

BuffSheet

BuffSheet.Row

CharacterLevelSheet

CharacterLevelSheet.Row

CharacterSheet

CharacterSheet.Row

CharacterSheetExtension

ClaimableGiftsSheet
: Sheet for claimable gifts that can be claimed by players during specific block index ranges. Each row contains gift ID, block index range, and a list of items to be given.

ClaimableGiftsSheet.Row

Represents a single row in the ClaimableGiftsSheet. Contains gift ID, block index range, and items to be given.

CollectQuestSheet

CollectQuestSheet.Row

CollectionSheet

CollectionSheet.RequiredMaterial

CollectionSheet.Row

CombinationEquipmentQuestSheet

CombinationEquipmentQuestSheet.Row

CombinationQuestSheet

CombinationQuestSheet.Row

ConsumableItemRecipeSheet

ConsumableItemRecipeSheet.Row

ConsumableItemSheet

ConsumableItemSheet.Row

CostumeItemSheet

CostumeItemSheet.Row

CostumeStatSheet

CostumeStatSheet.Row

CreateAvatarFavSheet

CreateAvatarFavSheet.Row

CreateAvatarItemSheet

CreateAvatarItemSheet.Row

EnemySkillSheet

EnemySkillSheet.Row

EnhancementCostSheet

EnhancementCostSheet.Row

EnhancementCostSheetV2

EnhancementCostSheetV2.Row

EnhancementCostSheetV3

EnhancementCostSheetV3.Row

EquipmentItemOptionSheet
Stat and skill options that crafting can add to equipment.

EquipmentItemOptionSheet.Row
One option and the ranges its values are drawn from.

EquipmentItemRecipeSheet
Equipment crafting recipes.

EquipmentItemRecipeSheet.Row
One equipment recipe.

EquipmentItemSetEffectSheet

EquipmentItemSetEffectSheet.Row

EquipmentItemSheet

EquipmentItemSheet.Row

EquipmentItemSubRecipeSheet

EquipmentItemSubRecipeSheet.Row

EquipmentItemSubRecipeSheetV2

EquipmentItemSubRecipeSheetV2.Row

GameConfigSheet

GameConfigSheet.Row

GeneralQuestSheet

GeneralQuestSheet.Row

GlobalCounterSheet

Defines the global counters that world-open conditions (WorldOpenConditionSheet) are measured with: what each counter counts, over which targets, and the minimum avatar level that contributes.

Columns: `id`, `type`, `target_begin`, `target_end`, `min_avatar_level`. The target is a stage id or a recipe id depending on `type`; `[target_begin, target_end]` is inclusive.

Reading is total, like BoostScheduleSheet: a row that does not parse is kept but marked invalid and never matches. ValidateCsv(string) rejects such a row at patch time.

A patch cannot change what an id means once recorded; see WorldOpenIdentity.

GlobalCounterSheet.Row

One counter definition.

GoldQuestSheet

GoldQuestSheet.Row

InfiniteTowerConditionSheet

Infinite tower condition sheet for managing battle conditions and restrictions. Uses StatType, SkillTargetType, and StatModifier.OperationType for flexible stat modifications.

InfiniteTowerConditionSheet.Row

Represents a row in the InfiniteTowerConditionSheet containing condition data.

InfiniteTowerFloorSheet

Infinite tower floor sheet for managing tower floors and their properties.

InfiniteTowerFloorSheet.Row

Represents a row in the InfiniteTowerFloorSheet containing floor data.

InfiniteTowerFloorWaveSheet

Infinite tower floor wave sheet for managing wave configurations. Supports multiple monster types per wave, similar to StageWaveSheet.

InfiniteTowerFloorWaveSheet.MonsterData

Represents a single monster type within a wave.

InfiniteTowerFloorWaveSheet.Row

Represents a row in the InfiniteTowerFloorWaveSheet containing wave data for a specific floor.

InfiniteTowerFloorWaveSheet.WaveData

Represents a single wave within a floor, containing multiple monster types.

InfiniteTowerScheduleSheet

Infinite tower schedule sheet for managing season schedules and ticket policies.

InfiniteTowerScheduleSheet.Row

Represents a row in the InfiniteTowerScheduleSheet containing schedule data.

ItemConfigForGradeSheet

ItemConfigForGradeSheet.Row

ItemEnhancementQuestSheet

ItemEnhancementQuestSheet.Row

ItemGradeQuestSheet

ItemGradeQuestSheet.Row

ItemRequirementSheet

ItemRequirementSheet.Row

ItemSheet

ItemSheet.Row

ItemSheet.RowBase

ItemTypeCollectQuestSheet

ItemTypeCollectQuestSheet.Row

MaterialItemSheet

MaterialItemSheet.Row

MimisbrunnrSheet

MimisbrunnrSheet.Row

MonsterCollectionRewardSheet

MonsterCollectionRewardSheet.RewardInfo

MonsterCollectionRewardSheet.Row

MonsterCollectionSheet

MonsterCollectionSheet.Row

MonsterQuestSheet

MonsterQuestSheet.Row

PetCostSheet

PetCostSheet.PetCostData

PetCostSheet.Row

PetSheet

PetSheet.Row

QuestItemRewardSheet

QuestItemRewardSheet.Row

QuestRewardSheet

QuestRewardSheet.Row

QuestSheet

QuestSheet.Row

RaidSimulatorSheets

RaidSimulatorSheetsV1

RankingSimulatorSheets

RankingSimulatorSheetsV1

RankingSimulatorSheetsV100291

RedeemCodeListSheet

RedeemCodeListSheet.Row

RedeemRewardSheet

RedeemRewardSheet.RewardInfo

RedeemRewardSheet.Row

RestrictionSheet

Declares what a given item or currency may not be used for, so that a restriction can be adjusted with `patch_table_sheet` instead of a headless and client release.

A row targets either an item id (numeric key) or a currency ticker (non-numeric key). Which one is looked up is decided by the call site — a market action already holds an item id or a ticker — so the sheet needs no discriminator column.

Columns: `key`, `market_registrable`, `synthesize_material`. The first governs the market registration path only — a fungible item can still reach another account through a garage or by being wrapped into a currency, so a row here does not make its target account bound by itself. The second closes the conversion path that would otherwise turn a restricted item into an unrestricted one.

An empty cell means "unspecified" and leaves the behavior that predates this sheet untouched. An unparsable cell is read as `false`(= restricted) on purpose: a typo then fails toward blocking, which is recoverable, rather than toward leaking. ValidateCsv(string) rejects such a cell at patch time so it should not reach the chain in the first place.

A key is an item id, not an item instance, so a row applies to every copy that exists — give a restricted item a new id rather than reusing one players already hold. A false `market_registrable` also stops `ReRegisterProduct`, leaving an existing seller able to cancel but not to reprice, and it does not retract listings that are already up: `BuyProduct` does not consult this sheet, so a restriction added after the fact still needs those products cancelled. A false `synthesize_material` only stops the item from being consumed as a material — an item that must not be handed out by synthesis is kept out of the result pool by leaving it out of `SynthesizeWeightSheet`, which is a separate sheet from this one.

Callers must tolerate the sheet being absent — it reaches an existing chain only by `patch_table_sheet`, so loading it as a required sheet would fail every affected action until each chain is patched. Absence has to fall back to the restrictions that are still hardcoded, which is also what keeps re-evaluation of pre-patch blocks identical.

RestrictionSheet.Row

One target and what it may not be used for.

RuneCostSheet

Represents a sheet containing rune cost data organized by level ranges. This sheet stores cost information for rune enhancement across different level intervals, allowing for efficient data compression and retrieval.

RuneCostSheet.Row

Represents a row in the RuneCostSheet containing cost data for a specific rune. Each row can contain multiple cost data entries for different level ranges.

RuneCostSheet.RuneCostData

Represents cost data for a specific level range of rune enhancement. Contains information about required materials and success rates for a given level interval.

RuneOptionSheet

RuneOptionSheet.Row

RuneOptionSheet.Row.RuneOptionInfo

RuneSheet

RuneSheet.Row

RuneWeightSheet

RuneWeightSheet.Row

RuneWeightSheet.RuneInfo

SetEffectExtension

SheetRowColumnException

SheetRowNotFoundException

SheetRowValidateException

SheetRow<T>

Sheet<TKey, TValue>

SimulatorSheets

SimulatorSheetsV1

SimulatorSheetsV100291

SkillActionBuffSheet

SkillActionBuffSheet.Row

SkillBuffSheet

SkillBuffSheet.Row

SkillSheet

Sheet containing skill data loaded from CSV files. Provides access to skill definitions and their properties.

SkillSheet.Row

Represents a single skill row from the skill sheet. Contains skill metadata and properties used for skill creation and execution.

StageDialogSheet

StageDialogSheet.Row

StageSheet

StageSheet.FavRewardData

StageSheet.RewardData

StageSheet.Row

StageSimulatorSheets

StageSimulatorSheetsV1

StageSimulatorSheetsV100291

StageWaveSheet

StageWaveSheet.MonsterData

StageWaveSheet.Row

StageWaveSheet.WaveData

StakeAchievementRewardSheet

StakeAchievementRewardSheet.RewardInfo

StakeAchievementRewardSheet.Row

StakeAchievementRewardSheet.Step

StakeActionPointCoefficientSheet

StakeActionPointCoefficientSheet.Row

StakeRegularFixedRewardSheet

This sheet is used for setting the regular rewards for staking. The difference between this sheet and StakeRegularRewardSheet is that the StakeRegularFixedRewardSheet.RewardInfo of this sheet has a fixed count of reward.

CSV column order: level, required_gold, item_id, count[, tradable]

The optional `tradable` column controls whether material items are created as tradable (TradableMaterial) or non-tradable (Material). Omitting the column defaults to `true`, preserving backward compatibility with existing CSV data.

StakeRegularFixedRewardSheet.RewardInfo

StakeRegularFixedRewardSheet.Row

StakeRegularRewardSheet

This sheet is used for setting the regular rewards for staking. The difference between this sheet and StakeRegularFixedRewardSheet is that the StakeRegularRewardSheet.RewardInfo of this sheet has a rate and a reward type. The count of reward is calculated by the rate and the amount of staked currency.

StakeRegularRewardSheet.RewardInfo

StakeRegularRewardSheet.Row

StatBuffSheet

StatBuffSheet.Row

SweepRequiredCPSheet

SweepRequiredCPSheet.Row

SynthesizeSheet

Represents a SynthesizeSheet.

SynthesizeSheet.Row

SynthesizeWeightSheet

Represents a SynthesizeWeightSheet.

SynthesizeWeightSheet.Row

TableExtensions

TradeQuestSheet

TradeQuestSheet.Row

UnlockCombinationSlotCostSheet

UnlockCombinationSlotCostSheet.Row

WeeklyArenaRewardSheet

WeeklyArenaRewardSheet.RewardData

WeeklyArenaRewardSheet.Row

WorldBossActionPatternSheet

WorldBossActionPatternSheet.ActionPatternData

WorldBossActionPatternSheet.Row

WorldBossBattleRewardSheet

WorldBossBattleRewardSheet.Row

WorldBossCharacterSheet

WorldBossCharacterSheet.Row

WorldBossCharacterSheet.WaveStatData

WorldBossContributionRewardSheet
Represents the reward sheet for world boss contributions.

WorldBossContributionRewardSheet.RewardModel

WorldBossContributionRewardSheet.Row
Represents each row in the sheet.

WorldBossGlobalHpSheet

WorldBossGlobalHpSheet.Row

WorldBossKillRewardSheet

WorldBossKillRewardSheet.Row

WorldBossListSheet

WorldBossListSheet.Row

WorldBossRankRewardSheet

WorldBossRankRewardSheet.Row

WorldBossRankingRewardSheet

WorldBossRankingRewardSheet.Row

WorldBossStatSheetExtension

WorldOpenConditionSheet

The conditions that open a WorldOpenSheet opening: within one `condition_group` every counter must reach its threshold, and any one group being satisfied opens it (AND within a group, OR across groups).

Columns: `open_id`, `condition_group`, `counter_id`, `threshold`. Cross-sheet references (`open_id`, `counter_id`) are not validated here.

`open_id` is not unique, so a row's key is a synthetic sequential number assigned in the order rows are added (CSV order, starting at 1). Look rows up with GroupsOf(int) or OpenIdsReferencing(int) rather than by key. A blank line is kept as an invalid row (open_id 0); both lookups skip invalid rows, so code reading `OrderedList` directly must check IsValid too.

A patch cannot change what an id means once recorded; see WorldOpenIdentity.

WorldOpenConditionSheet.Row

One counter requirement of a condition group.

WorldOpenSheet

Defines when each hard world segment opens: up to which stage, which opening it follows, and the block window (earliest, fallback, activation delay) around it. The conditions that open it are in WorldOpenConditionSheet.

Columns: `id`, `world_id`, `stage_end`, `prerequisite_open_id` (empty for none), `min_block_index`, `fallback_block_index`, `effective_delay_block`.

Set(IReadOnlyList<string>) does not validate across rows, so a test can build an invalid sheet directly; ValidateCsv(string) does, at patch time.

A patch cannot change what an id means once recorded; see WorldOpenIdentity.

WorldOpenSheet.Row

One opening.

WorldQuestSheet

WorldQuestSheet.Row

WorldSheet

WorldSheet.Row

WorldUnlockSheet

WorldUnlockSheet.Row

Structs

EquipmentItemSubRecipeSheet.MaterialInfo

EquipmentItemSubRecipeSheet.OptionInfo

EquipmentItemSubRecipeSheetV2.OptionInfo

SynthesizeSheet.SynthesizeData
: synthesize data for each subtype

WorldBossRankingRewardSheet.Row.RuneInfo

Interfaces

ISheet

IStakeRewardRow

IStakeRewardSheet

IWorldBossRewardRow

IWorldBossRewardSheet

Enums

BoostScheduleSheet.Operation
: How a row's Value is combined with the value it adjusts.

CreateAvatarFavSheet.Target

GlobalCounterType
: What a GlobalCounterSheet counter counts.

RewardType

StakeRegularRewardSheet.StakeRewardType

The reward type of stake. Do not use the whole element of this enum at once in actions. (e.g., count, loop) Do versioning when you change or remove the elements of this enum.

Class ActionBuffSheet

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class ActionBuffSheet : Sheet<int, ActionBuffSheet.Row>, IDictionary<int,
ActionBuffSheet.Row>, ICollection<KeyValuePair<int, ActionBuffSheet.Row>>,
IEnumerable<KeyValuePair<int, ActionBuffSheet.Row>>, IEnumerable, ISheet
```

Inheritance

object ← Sheet<int, ActionBuffSheet.Row> ← ActionBuffSheet

Implements

IDictionary<int, ActionBuffSheet.Row>,
ICollection<KeyValuePair<int, ActionBuffSheet.Row>>,
IEnumerable<KeyValuePair<int, ActionBuffSheet.Row>>, IEnumerable, ISheet

Inherited Members

Sheet<int, ActionBuffSheet.Row>.Name , Sheet<int, ActionBuffSheet.Row>.OrderedList ,
Sheet<int, ActionBuffSheet.Row>.First , Sheet<int, ActionBuffSheet.Row>.Last ,
Sheet<int, ActionBuffSheet.Row>.Keys , Sheet<int, ActionBuffSheet.Row>.Values ,
Sheet<int, ActionBuffSheet.Row>.Count , Sheet<int, ActionBuffSheet.Row>.IsReadOnly ,
Sheet<int, ActionBuffSheet.Row>.this[int] ,
Sheet<int, ActionBuffSheet.Row>.Set(string, bool) ,
Sheet<int, ActionBuffSheet.Row>.Set<T>(Sheet<int, T>, bool) ,
Sheet<int, ActionBuffSheet.Row>.GetEnumerator() ,
Sheet<int, ActionBuffSheet.Row>.TryGetValue(int, out ActionBuffSheet.Row, bool) ,
Sheet<int, ActionBuffSheet.Row>.AddRow(int, ActionBuffSheet.Row) ,
Sheet<int, ActionBuffSheet.Row>.Add(int, ActionBuffSheet.Row) ,
Sheet<int, ActionBuffSheet.Row>.ContainsKey(int) ,
Sheet<int, ActionBuffSheet.Row>.Remove(int) ,
Sheet<int, ActionBuffSheet.Row>.TryGetValue(int, out ActionBuffSheet.Row) ,
Sheet<int, ActionBuffSheet.Row>.Add(KeyValuePair<int, ActionBuffSheet.Row>) ,
Sheet<int, ActionBuffSheet.Row>.Clear() ,
Sheet<int, ActionBuffSheet.Row>.Contains(KeyValuePair<int, ActionBuffSheet.Row>) ,
Sheet<int, ActionBuffSheet.Row>.CopyTo(KeyValuePair<int, ActionBuffSheet.Row>[], int) ,
Sheet<int, ActionBuffSheet.Row>.Remove(KeyValuePair<int, ActionBuffSheet.Row>) ,
Sheet<int, ActionBuffSheet.Row>.Serialize() , object.Equals(object) ,

object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

ActionBuffSheet()

```
public ActionBuffSheet()
```

Class ActionBuffSheet.Row

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class ActionBuffSheet.Row : SheetRow<int>
```

Inheritance

object ← SheetRow<int> ← ActionBuffSheet.Row

Inherited Members

SheetRow<int>.Validate() , SheetRow<int>.EndOfSheetInitialize() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Properties

ATKPowerRatio

```
public decimal ATKPowerRatio { get; }
```

Property Value

decimal

ActionBuffType

```
public ActionBuffType ActionBuffType { get; }
```

Property Value

ActionBuffType

Chance

```
public int Chance { get; }
```

Property Value

int

Duration

```
public int Duration { get; }
```

Property Value

int

ElementalType

```
public ElementalType ElementalType { get; }
```

Property Value

ElementalType

GroupId

```
public int GroupId { get; }
```

Property Value

int

Id

```
public int Id { get; }
```

Property Value

int

Key

```
public override int Key { get; }
```

Property Value

int

TargetType

```
public SkillTargetType TargetType { get; }
```

Property Value

SkillTargetType

Methods

Set(IReadOnlyList<string>)

```
public override void Set(IReadOnlyList<string> fields)
```

Parameters

`fields` IReadOnlyList<string>

Class ArenaSheet

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class ArenaSheet : Sheet<int, ArenaSheet.Row>, IDictionary<int,
ArenaSheet.Row>, ICollection<KeyValuePair<int, ArenaSheet.Row>>,
IEnumerable<KeyValuePair<int, ArenaSheet.Row>>, IEnumerable, ISheet
```

Inheritance

object ← Sheet<int, ArenaSheet.Row> ← ArenaSheet

Implements

IDictionary<int, ArenaSheet.Row>,
ICollection<KeyValuePair<int, ArenaSheet.Row>>,
IEnumerable<KeyValuePair<int, ArenaSheet.Row>>, IEnumerable, ISheet

Inherited Members

Sheet<int, ArenaSheet.Row>.Name , Sheet<int, ArenaSheet.Row>.OrderedList ,
Sheet<int, ArenaSheet.Row>.First , Sheet<int, ArenaSheet.Row>.Last ,
Sheet<int, ArenaSheet.Row>.Keys , Sheet<int, ArenaSheet.Row>.Values ,
Sheet<int, ArenaSheet.Row>.Count , Sheet<int, ArenaSheet.Row>.IsReadOnly ,
Sheet<int, ArenaSheet.Row>.this[int] , Sheet<int, ArenaSheet.Row>.Set(string, bool) ,
Sheet<int, ArenaSheet.Row>.Set<T>(Sheet<int, T>, bool) ,
Sheet<int, ArenaSheet.Row>.GetEnumerator() ,
Sheet<int, ArenaSheet.Row>.TryGetValue(int, out ArenaSheet.Row, bool) ,
Sheet<int, ArenaSheet.Row>.AddRow(int, ArenaSheet.Row) ,
Sheet<int, ArenaSheet.Row>.Add(int, ArenaSheet.Row) ,
Sheet<int, ArenaSheet.Row>.ContainsKey(int) , Sheet<int, ArenaSheet.Row>.Remove(int) ,
Sheet<int, ArenaSheet.Row>.TryGetValue(int, out ArenaSheet.Row) ,
Sheet<int, ArenaSheet.Row>.Add(KeyValuePair<int, ArenaSheet.Row>) ,
Sheet<int, ArenaSheet.Row>.Clear() ,
Sheet<int, ArenaSheet.Row>.Contains(KeyValuePair<int, ArenaSheet.Row>) ,
Sheet<int, ArenaSheet.Row>.CopyTo(KeyValuePair<int, ArenaSheet.Row>[], int) ,
Sheet<int, ArenaSheet.Row>.Remove(KeyValuePair<int, ArenaSheet.Row>) ,
Sheet<int, ArenaSheet.Row>.Serialize() , object.Equals(object) ,
object.Equals(object, object) , object.GetHashCode() , object.GetType() ,
object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

ArenaSheet()

```
public ArenaSheet()
```

Methods

AddRow(int, Row)

```
protected override void AddRow(int key, ArenaSheet.Row value)
```

Parameters

`key` int

`value` ArenaSheet.Row

GetRoundByBlockIndex(long)

```
public ArenaSheet.RoundData GetRoundByBlockIndex(long blockIndex)
```

Parameters

`blockIndex` long

Returns

ArenaSheet.RoundData

GetRowByBlockIndex(long)

```
public ArenaSheet.Row GetRowByBlockIndex(long blockIndex)
```

Parameters

`blockIndex` long

Returns

ArenaSheet.Row

Class ArenaSheet.RoundData

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class ArenaSheet.RoundData
```

Inheritance

object ← ArenaSheet.RoundData

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

RoundData(int, int, ArenaType, long, long, int, long, long, long, int, int, int)

```
public RoundData(int championshipId, int round, ArenaType arenaType, long
startBlockIndex, long endBlockIndex, int requiredMedalCount, long entranceFee, long
ticketPrice, long additionalTicketPrice, int maxPurchaseCount, int
maxPurchaseCountWithInterval, int medalId = 0)
```

Parameters

`championshipId` int

`round` int

`arenaType` ArenaType

`startBlockIndex` long

`endBlockIndex` long

`requiredMedalCount` int

`entranceFee` long

`ticketPrice` long

`additionalTicketPrice` long

`maxPurchaseCount` int

`maxPurchaseCountWithInterval` int

`medalId` int

Properties

AdditionalTicketPrice

```
public long AdditionalTicketPrice { get; }
```

Property Value

long

ArenaType

```
public ArenaType ArenaType { get; }
```

Property Value

ArenaType

ChampionshipId

```
public int ChampionshipId { get; }
```

Property Value

int

EndBlockIndex

```
public long EndBlockIndex { get; }
```

Property Value

long

EntranceFee

```
public long EntranceFee { get; }
```

Property Value

long

MaxPurchaseCount

```
public int MaxPurchaseCount { get; }
```

Property Value

int

MaxPurchaseCountWithInterval

```
public int MaxPurchaseCountWithInterval { get; }
```

Property Value

int

MedalId

```
public int MedalId { get; }
```

Property Value

int

RequiredMedalCount

```
public int RequiredMedalCount { get; }
```

Property Value

int

Round

```
public int Round { get; }
```

Property Value

int

StartBlockIndex

```
public long StartBlockIndex { get; }
```

Property Value

long

TicketPrice

```
public long TicketPrice { get; }
```

Property Value

long

Methods

IsTheRoundOpened(long)

```
public bool IsTheRoundOpened(long blockIndex)
```

Parameters

`blockIndex` long

Returns

bool

Class ArenaSheet.Row

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class ArenaSheet.Row : SheetRow<int>
```

Inheritance

object ← SheetRow<int> ← ArenaSheet.Row

Inherited Members

SheetRow<int>.Validate() , SheetRow<int>.EndOfSheetInitialize() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Properties

ChampionshipId

```
public int ChampionshipId { get; }
```

Property Value

int

Key

```
public override int Key { get; }
```

Property Value

int

Round

```
public List<ArenaSheet.RoundData> Round { get; }
```

Property Value

List<ArenaSheet.RoundData>

Methods

Set(IReadOnlyList<string>)

```
public override void Set(IReadOnlyList<string> fields)
```

Parameters

`fields` IReadOnlyList<string>

TryGetChampionshipRound(out RoundData)

```
public bool TryGetChampionshipRound(out ArenaSheet.RoundData roundData)
```

Parameters

`roundData` ArenaSheet.RoundData

Returns

bool

TryGetRound(int, out RoundData)

```
public bool TryGetRound(int round, out ArenaSheet.RoundData roundData)
```

Parameters

`round` int

`roundData` ArenaSheet.RoundData

Returns

bool

Class ArenaSimulatorSheets

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
public class ArenaSimulatorSheets : SimulatorSheets
```

Inheritance

object ← SimulatorSheetsV1 ← SimulatorSheets ← ArenaSimulatorSheets

Inherited Members

SimulatorSheets.RuneOptionSheet , SimulatorSheets.RuneListSheet , SimulatorSheets.RuneLevelBonusSheet , SimulatorSheetsV1.MaterialItemSheet , SimulatorSheetsV1.SkillSheet , SimulatorSheetsV1.SkillBuffSheet , SimulatorSheetsV1.SkillActionBuffSheet , SimulatorSheetsV1.ActionBuffSheet , SimulatorSheetsV1.StatBuffSheet , SimulatorSheetsV1.CharacterSheet , SimulatorSheetsV1.CharacterLevelSheet , SimulatorSheetsV1.EquipmentItemSetEffectSheet , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

ArenaSimulatorSheets(MaterialItemSheet, SkillSheet, SkillBuffSheet, StatBuffSheet, SkillActionBuffSheet, ActionBuffSheet, CharacterSheet, CharacterLevelSheet, EquipmentItemSetEffectSheet, CostumeStatSheet, WeeklyArenaRewardSheet, RuneOptionSheet, RuneListSheet, RuneLevelBonusSheet)

```
public ArenaSimulatorSheets(MaterialItemSheet materialItemSheet, SkillSheet
skillSheet, SkillBuffSheet skillBuffSheet, StatBuffSheet statBuffSheet,
SkillActionBuffSheet skillActionBuffSheet, ActionBuffSheet actionBuffSheet,
CharacterSheet characterSheet, CharacterLevelSheet characterLevelSheet,
EquipmentItemSetEffectSheet equipmentItemSetEffectSheet, CostumeStatSheet
costumeStatSheet, WeeklyArenaRewardSheet weeklyArenaRewardSheet,
```

```
RuneOptionSheet runeOptionSheet, RuneListSheet runeListSheet,
RuneLevelBonusSheet runeLevelBonusSheet)
```

Parameters

`materialItemSheet` MaterialItemSheet

`skillSheet` SkillSheet

`skillBuffSheet` SkillBuffSheet

`statBuffSheet` StatBuffSheet

`skillActionBuffSheet` SkillActionBuffSheet

`actionBuffSheet` ActionBuffSheet

`characterSheet` CharacterSheet

`characterLevelSheet` CharacterLevelSheet

`equipmentItemSetEffectSheet` EquipmentItemSetEffectSheet

`costumeStatSheet` CostumeStatSheet

`weeklyArenaRewardSheet` WeeklyArenaRewardSheet

`runeOptionSheet` RuneOptionSheet

`runeListSheet` RuneListSheet

`runeLevelBonusSheet` RuneLevelBonusSheet

Properties

CostumeStatSheet

```
public CostumeStatSheet CostumeStatSheet { get; }
```

Property Value

CostumeStatSheet

WeeklyArenaRewardSheet

```
public WeeklyArenaRewardSheet WeeklyArenaRewardSheet { get; }
```

Property Value

WeeklyArenaRewardSheet

Class ArenaSimulatorSheetsV1

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
public class ArenaSimulatorSheetsV1 : SimulatorSheetsV1
```

Inheritance

object ← SimulatorSheetsV1 ← ArenaSimulatorSheetsV1

Inherited Members

SimulatorSheetsV1.MaterialItemSheet , SimulatorSheetsV1.SkillSheet , SimulatorSheetsV1.SkillBuffSheet , SimulatorSheetsV1.SkillActionBuffSheet , SimulatorSheetsV1.ActionBuffSheet , SimulatorSheetsV1.StatBuffSheet , SimulatorSheetsV1.CharacterSheet , SimulatorSheetsV1.CharacterLevelSheet , SimulatorSheetsV1.EquipmentItemSetEffectSheet , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

ArenaSimulatorSheetsV1(MaterialItemSheet, SkillSheet, SkillBuffSheet, StatBuffSheet, SkillActionBuffSheet, ActionBuffSheet, CharacterSheet, CharacterLevelSheet, EquipmentItemSetEffectSheet, CostumeStatSheet, WeeklyArenaRewardSheet)

```
public ArenaSimulatorSheetsV1(MaterialItemSheet materialItemSheet, SkillSheet
skillSheet, SkillBuffSheet skillBuffSheet, StatBuffSheet statBuffSheet,
SkillActionBuffSheet skillActionBuffSheet, ActionBuffSheet actionBuffSheet,
CharacterSheet characterSheet, CharacterLevelSheet characterLevelSheet,
EquipmentItemSetEffectSheet equipmentItemSetEffectSheet, CostumeStatSheet
costumeStatSheet, WeeklyArenaRewardSheet weeklyArenaRewardSheet)
```

Parameters

`materialItemSheet` MaterialItemSheet

`skillSheet` SkillSheet

`skillBuffSheet` SkillBuffSheet

`statBuffSheet` StatBuffSheet

`skillActionBuffSheet` SkillActionBuffSheet

`actionBuffSheet` ActionBuffSheet

`characterSheet` CharacterSheet

`characterLevelSheet` CharacterLevelSheet

`equipmentItemSetEffectSheet` EquipmentItemSetEffectSheet

`costumeStatSheet` CostumeStatSheet

`weeklyArenaRewardSheet` WeeklyArenaRewardSheet

Properties

CostumeStatSheet

```
public CostumeStatSheet CostumeStatSheet { get; }
```

Property Value

CostumeStatSheet

WeeklyArenaRewardSheet

```
public WeeklyArenaRewardSheet WeeklyArenaRewardSheet { get; }
```

Property Value

WeeklyArenaRewardSheet

Class ArenaSimulatorSheetsV100291

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
public class ArenaSimulatorSheetsV100291 : SimulatorSheetsV100291
```

Inheritance

object ← SimulatorSheetsV100291 ← ArenaSimulatorSheetsV100291

Inherited Members

SimulatorSheetsV100291.MaterialItemSheet , SimulatorSheetsV100291.SkillSheet , SimulatorSheetsV100291.SkillBuffSheet , SimulatorSheetsV100291.BuffSheet , SimulatorSheetsV100291.CharacterSheet , SimulatorSheetsV100291.CharacterLevelSheet , SimulatorSheetsV100291.EquipmentItemSetEffectSheet , SimulatorSheetsV100291.ToSimulatorSheetsV1() , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

ArenaSimulatorSheetsV100291(MaterialItemSheet, SkillSheet, SkillBuffSheet, BuffSheet, CharacterSheet, CharacterLevelSheet, EquipmentItemSetEffectSheet, CostumeStatSheet, WeeklyArenaRewardSheet)

```
public ArenaSimulatorSheetsV100291(MaterialItemSheet materialItemSheet, SkillSheet
skillSheet, SkillBuffSheet skillBuffSheet, BuffSheet buffSheet, CharacterSheet
characterSheet, CharacterLevelSheet characterLevelSheet, EquipmentItemSetEffectSheet
equipmentItemSetEffectSheet, CostumeStatSheet costumeStatSheet,
WeeklyArenaRewardSheet weeklyArenaRewardSheet)
```

Parameters

`materialItemSheet` MaterialItemSheet

`skillSheet` SkillSheet

`skillBuffSheet` SkillBuffSheet

`buffSheet` BuffSheet

`characterSheet` CharacterSheet

`characterLevelSheet` CharacterLevelSheet

`equipmentItemSetEffectSheet` EquipmentItemSetEffectSheet

`costumeStatSheet` CostumeStatSheet

`weeklyArenaRewardSheet` WeeklyArenaRewardSheet

Properties

CostumeStatSheet

```
public CostumeStatSheet CostumeStatSheet { get; }
```

Property Value

CostumeStatSheet

WeeklyArenaRewardSheet

```
public WeeklyArenaRewardSheet WeeklyArenaRewardSheet { get; }
```

Property Value

WeeklyArenaRewardSheet

Methods

ToArenaSimulatorSheetsV1()

```
public ArenaSimulatorSheetsV1 ToArenaSimulatorSheetsV1()
```

Returns

ArenaSimulatorSheetsV1

Class AvatarSheets

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
public class AvatarSheets
```

Inheritance

object ← AvatarSheets

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

AvatarSheets(WorldSheet, QuestSheet, QuestRewardSheet, QuestItemRewardSheet, EquipmentItemRecipeSheet, EquipmentItemSubRecipeSheet)

```
public AvatarSheets(WorldSheet worldSheet, QuestSheet questSheet,
QuestRewardSheet questRewardSheet, QuestItemRewardSheet questItemRewardSheet,
EquipmentItemRecipeSheet equipmentItemRecipeSheet, EquipmentItemSubRecipeSheet
equipmentItemSubRecipeSheet)
```

Parameters

`worldSheet` WorldSheet

`questSheet` QuestSheet

`questRewardSheet` QuestRewardSheet

`questItemRewardSheet` QuestItemRewardSheet

`equipmentItemRecipeSheet` EquipmentItemRecipeSheet

`equipmentItemSubRecipeSheet` EquipmentItemSubRecipeSheet

Fields

EquipmentItemRecipeSheet

```
public readonly EquipmentItemRecipeSheet EquipmentItemRecipeSheet
```

Field Value

EquipmentItemRecipeSheet

EquipmentItemSubRecipeSheet

```
public readonly EquipmentItemSubRecipeSheet EquipmentItemSubRecipeSheet
```

Field Value

EquipmentItemSubRecipeSheet

QuestItemRewardSheet

```
public readonly QuestItemRewardSheet QuestItemRewardSheet
```

Field Value

QuestItemRewardSheet

QuestRewardSheet

```
public readonly QuestRewardSheet QuestRewardSheet
```

Field Value

QuestRewardSheet

QuestSheet

```
public readonly QuestSheet QuestSheet
```

Field Value

QuestSheet

WorldSheet

```
public readonly WorldSheet WorldSheet
```

Field Value

WorldSheet

Class BoostScheduleSheet

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

Schedules a temporary adjustment of a game value — a reward amount, a cost, a guarantee count — for a block range, so that an event can be run with `patch_table_sheet` instead of rewriting the value's own sheet and patching it back afterwards.

Columns: `id`, `target`, `target_id`, `op`, `value`, `start_block`, `end_block`. A `_memo` column, like any `_` prefixed column, is dropped on load and may be used for notes.

`target_id` is a single id (`50`), an inclusive range (`1~50`) or `*` for every id. `~` rather than `-` keeps a spreadsheet from reading a range as a date on its way to the CSV.

`target` names what is adjusted and is matched as a plain string by the call site that owns it, so the sheet itself never needs to know which targets exist. A node that predates a target simply finds no match for it instead of failing to read the sheet. BoostScheduleSheet.Targets lists the targets this version applies.

A row applies in `[start_block, end_block)`. When several rows match, the one with the lowest `id` wins; rows are never combined, so overlapping rows cannot stack into a larger adjustment than any single row declares.

Reading is total: a row that does not parse is kept but marked invalid and never matches, so a typo fails toward "no adjustment" instead of aborting every action that reads the sheet. ValidateCsv(string) rejects such a row at patch time so it should not reach the chain in the first place.

Callers must tolerate the sheet being absent — it reaches an existing chain only by `patch_table_sheet`. Absence means no adjustment, which is also what keeps re-evaluation of pre-patch blocks identical.

```
[Serializable]
public class BoostScheduleSheet : Sheet<int, BoostScheduleSheet.Row>,
IDictionary<int, BoostScheduleSheet.Row>, ICollection<KeyValuePair<int,
BoostScheduleSheet.Row>>, IEnumerable<KeyValuePair<int, BoostScheduleSheet.Row>>,
IEnumerable, ISheet
```

Inheritance

object ← Sheet<int, BoostScheduleSheet.Row> ← BoostScheduleSheet

Implements
IDictionary<int, BoostScheduleSheet.Row>,
ICollection<KeyValuePair<int, BoostScheduleSheet.Row>>,
IEnumerable<KeyValuePair<int, BoostScheduleSheet.Row>>, IEnumerable, ISheet

Inherited Members
Sheet<int, BoostScheduleSheet.Row>.Name ,
Sheet<int, BoostScheduleSheet.Row>.OrderedList ,
Sheet<int, BoostScheduleSheet.Row>.First , Sheet<int, BoostScheduleSheet.Row>.Last ,
Sheet<int, BoostScheduleSheet.Row>.Keys ,
Sheet<int, BoostScheduleSheet.Row>.Values ,
Sheet<int, BoostScheduleSheet.Row>.Count ,
Sheet<int, BoostScheduleSheet.Row>.IsReadOnly ,
Sheet<int, BoostScheduleSheet.Row>.this[int] ,
Sheet<int, BoostScheduleSheet.Row>.Set(string, bool) ,
Sheet<int, BoostScheduleSheet.Row>.Set<T>(Sheet<int, T>, bool) ,
Sheet<int, BoostScheduleSheet.Row>.GetEnumerator() ,
Sheet<int, BoostScheduleSheet.Row>.TryGetValue(int, out BoostScheduleSheet.Row, bool)
,
Sheet<int, BoostScheduleSheet.Row>.AddRow(int, BoostScheduleSheet.Row) ,
Sheet<int, BoostScheduleSheet.Row>.Add(int, BoostScheduleSheet.Row) ,
Sheet<int, BoostScheduleSheet.Row>.ContainsKey(int) ,
Sheet<int, BoostScheduleSheet.Row>.Remove(int) ,
Sheet<int, BoostScheduleSheet.Row>.TryGetValue(int, out BoostScheduleSheet.Row) ,
Sheet<int, BoostScheduleSheet.Row>.Add(KeyValuePair<int, BoostScheduleSheet.Row>) ,
Sheet<int, BoostScheduleSheet.Row>.Clear() ,
Sheet<int, BoostScheduleSheet.Row>.Contains(KeyValuePair<int,
BoostScheduleSheet.Row>) ,
Sheet<int, BoostScheduleSheet.Row>.CopyTo(KeyValuePair<int, BoostScheduleSheet.Row>
[], int) ,
Sheet<int, BoostScheduleSheet.Row>.Remove(KeyValuePair<int,
BoostScheduleSheet.Row>) ,
Sheet<int, BoostScheduleSheet.Row>.Serialize() , object.Equals(object) ,
object.Equals(object, object) , object.GetHashCode() , object.GetType() ,
object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

BoostScheduleSheet()

Creates an empty sheet.

```
public BoostScheduleSheet()
```

Fields

AnyTargetId

The `target_id` cell that matches every id of a target.

```
public const string AnyTargetId = "*"
```

Field Value

string

ColumnNames

Column names in their expected order, after `_` prefixed columns are dropped.

```
public static readonly string[] ColumnNames
```

Field Value

string[]

MaxAbsoluteValue

Upper bound of `|value|`. Keeps every Apply(int) and Apply(long) inside decimal range so that applying a row can never throw: the worst case, MaxValue × `int.MaxValue` ≈ 2e28, stays under MaxValue ≈ 7.9e28. Raising this bound must keep that true, or a row would make every action it applies to fail.

```
public const decimal MaxAbsoluteValue = 2147483647
```

Field Value

decimal

TargetIdRangeSeparator

Separates the two ends of a `target_id` range, as in `1~50`.

```
public const char TargetIdRangeSeparator = '~'
```

Field Value

char

Methods

AddRow(int, Row)

Adds a parsed row to the sheet.

```
protected override void AddRow(int key, BoostScheduleSheet.Row value)
```

Parameters

`key` int

The row's key.

`value` BoostScheduleSheet.Row

The row to add.

Remarks

A row without a usable id is dropped and a duplicated id keeps the first row instead of throwing, because a throw here would fail every action that reads the sheet.

FindActive(string, int, long)

Finds the row that applies to `targetId` of `target` at `blockIndex`.

```
public BoostScheduleSheet.Row FindActive(string target, int targetId,
long blockIndex)
```

Parameters

`target` string

The target being adjusted. See BoostScheduleSheet.Targets.

`targetId` int

The id within the target.

`blockIndex` long

The block being evaluated.

Returns

BoostScheduleSheet.Row

The matching row with the lowest id, or `null` when none applies.

ValidateCsv(string)

Parses `csv` strictly so that a malformed schedule fails the `patch_table_sheet` transaction instead of landing on the chain as a row that silently never applies.

```
public static void ValidateCsv(string csv)
```

Parameters

`csv` string

The sheet's CSV, as the patch would write it.

Remarks

Whether a target is applied by any running code is deliberately not checked here: the set of targets grows with releases, and a patch must not be accepted by one node version and rejected by another. Compare against Applied off-chain.

Exceptions

SheetRowValidateException

- The header does not match ColumnNames, or a row has an unusable or duplicated id, an empty target, an unparsable cell, a value out of range for its operation, or a block range that is empty.

Enum BoostScheduleSheet.Operation

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

How a row's Value is combined with the value it adjusts.

```
public enum BoostScheduleSheet.Operation
```

Extension Methods
StateExtensions.Serialize(Enum)

Fields

`Add = 1`

`base + value`. `value` must be an integer.

`Mul = 0`

`floor(base × value)`.

Class BoostScheduleSheet.Row

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

One scheduled adjustment.

```
[Serializable]
public class BoostScheduleSheet.Row : SheetRow<int>
```

Inheritance

object ← SheetRow<int> ← BoostScheduleSheet.Row

Inherited Members

SheetRow<int>.Validate() , SheetRow<int>.EndOfSheetInitialize() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Properties

EndBlockIndex

First block index this row no longer applies to.

```
public long EndBlockIndex { get; }
```

Property Value

long

Id

The row id. Also the precedence among overlapping rows: lower wins.

```
public int Id { get; }
```

Property Value

int

IsValid

Whether every cell parsed. An invalid row never matches.

```
public bool IsValid { get; }
```

Property Value

bool

Key

```
public override int Key { get; }
```

Property Value

int

Op

How Value is combined with the adjusted value.

```
public BoostScheduleSheet.Operation Op { get; }
```

Property Value

BoostScheduleSheet.Operation

StartBlockIndex

First block index this row applies to.

```
public long StartBlockIndex { get; }
```

Property Value

long

Target

What this row adjusts. See BoostScheduleSheet.Targets.

```
public string Target { get; }
```

Property Value

string

TargetIdBegin

First id within Target this row applies to, or `null` for all.

```
public int? TargetIdBegin { get; }
```

Property Value

int?

TargetIdEnd

Last id within Target this row applies to, inclusive, or `null` for all. Equals TargetIdBegin for a single id.

```
public int? TargetIdEnd { get; }
```

Property Value

int?

Value

The operand of Op.

```
public decimal Value { get; }
```

Property Value

decimal

Methods

Apply(int)

Applies this row to `baseValue`, rounding down and clamping the result to `[0, int.MaxValue]`.

```
public int Apply(int baseValue)
```

Parameters

`baseValue` int

The value before adjustment.

Returns

int

The adjusted value.

Remarks

Rounding down means a small base can be left unchanged by a multiplier — e.g. `1 × 1.5` stays `1`. Use Add when every target must actually move.

Apply(long)

Apply(int) for 64-bit values: same rounding (down) and lower bound (`0`), clamped to Max Value instead of MaxValue so that large bases such as hammer experience are not cut.

```
public long Apply(long baseValue)
```

Parameters

`baseValue` long

The value before adjustment.

Returns

long

The adjusted value.

Matches(string, int, long)

Whether this row applies to `targetId` of `target` at `blockIndex`.

```
public bool Matches(string target, int targetId, long blockIndex)
```

Parameters

`target` string

The target being adjusted.

`targetId` int

The id within the target.

`blockIndex` long

The block being evaluated.

Returns

bool

`true` when the row is valid and covers all three.

Set(IReadOnlyList<string>)

Reads one row.

```
public override void Set(IReadOnlyList<string> fields)
```

Parameters

`fields` IReadOnlyList<string>

The row's cells, with "_" prefixed columns already dropped.

Remarks

Total by design: Sheet<TKey, TValue> does not guard this call, so a throw here would fail every action that reads the sheet.

Class BoostScheduleSheet.TargetInfo

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

How to write a row for one target: reference text for people and tools, never read by an action. See Infos.

```
public sealed class BoostScheduleSheet.TargetInfo
```

Inheritance

object ← BoostScheduleSheet.TargetInfo

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

TargetInfo(string, string, string, string, string, string, string, string)

Creates a description.

```
public TargetInfo(string target, string adjusts, string targetIdMeaning, string
targetIdSheet, string usualOp, string valueRule, string notes, string example)
```

Parameters

`target` string

`adjusts` string

`targetIdMeaning` string

`targetIdSheet` string

`usualOp` string

`valueRule` string

`notes` string

`example` string

Properties

Adjusts

What a row changes.

```
public string Adjusts { get; }
```

Property Value

string

Example

An example row, with the block range left as placeholders.

```
public string Example { get; }
```

Property Value

string

Notes

What the row does not cover, and what to release first.

```
public string Notes { get; }
```

Property Value

string

Target

The `target` cell.

```
public string Target { get; }
```

Property Value

string

TargetIdMeaning

What the `target_id` cell names (a description, not a value).

```
public string TargetIdMeaning { get; }
```

Property Value

string

TargetIdSheet

The sheet whose ids `target_id` takes.

```
public string TargetIdSheet { get; }
```

Property Value

string

UsualOp

The `op` and value direction an event normally uses.

```
public string UsualOp { get; }
```

Property Value

string

ValueRule

How the value is applied: rounding and bounds.

```
public string ValueRule { get; }
```

Property Value

string

Class BoostScheduleSheet.Targets

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

The targets this version applies. A row naming any other target is valid data but has no effect until a version that applies it is running.

```
public static class BoostScheduleSheet.Targets
```

Inheritance

object ← BoostScheduleSheet.Targets

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Fields

AdventureBossFirstClearReward

Amount of each `AdventureBossFloorFirstRewardSheet` reward, granted by `ExploreAdventureBoss` when a floor is cleared for the first time in a season. `target_id` is the floor number, as for AdventureBossFloorReward. Bounded as AdventureBossFloorReward (at most 5 times each amount).

```
public const string AdventureBossFirstClearReward =
"ADVENTURE_BOSS_FIRST_CLEAR_REWARD"
```

Field Value

string

AdventureBossFloorReward

Amount of the reward an adventure boss floor drops on every clear — the one entry drawn from `AdventureBossFloorSheet` rewards per floor, by both `ExploreAdventureBoss` and `SweepAdventureBoss`. `target_id` is the floor number (`AdventureBossFloorSheetfloor`, 1 for the first floor of every boss), not the floor row id.

The adjusted amount is kept between 1 and `AdventureBossHelper.MaxAdventureBossRewardBoostFactor` (5) times the drawn amount, `ADD` included. See `AdventureBossHelper.ApplyRewardBoost`.

```
public const string AdventureBossFloorReward = "ADVENTURE_BOSS_FLOOR_REWARD"
```

Field Value

string

Applied

Every target above, for tooling that warns about rows no running code applies.

```
public static readonly IReadOnlyCollection<string> Applied
```

Field Value

IReadOnlyCollection<string>

CostumeSummonGuarantee

Guarantee count of a `CostumeSummonSheet` group. `target_id` is the group id.

```
public const string CostumeSummonGuarantee = "COSTUME_SUMMON_GUARANTEE"
```

Field Value

string

EnhancementHammerExp

Experience one enhancement hammer gives an equipment in `ItemEnhancement` (`EnhancementCostSheetV3` hammer rows). `target_id` is the hammer item id (`600301~600308`), so grades can get different multipliers.

The adjusted experience is stored on the equipment like any other experience: the equipment ends up exactly as if it had gathered that much normally, so it neither owes the boost back on its next enhancement nor is worth less as a material. The result is kept between the configured experience and `ItemEnhancement.MaxHammerExpBoostFactor` times it — a row can only raise it. See `ItemEnhancement.GetHammerExp`.

Patch a row only after a client that previews with the adjusted experience is released: an old client shows a lower target level and a lower NCG cost than the action charges (cost follows the level reached), so an avatar holding only the shown NCG fails, and it lets players add more hammers than the max level needs.

```
public const string EnhancementHammerExp = "ENHANCEMENT_HAMMER_EXP"
```

Field Value

string

EquipmentSummonGuarantee

Guarantee count of an `EquipmentSummonSheet` group. `target_id` is the group id.

```
public const string EquipmentSummonGuarantee = "EQUIPMENT_SUMMON_GUARANTEE"
```

Field Value

string

Infos

How to write a row for each target above, for tooling that helps people fill the sheet (the backoffice serves this as is). It is reference text only: no action reads it, so changing it never changes what a row does. Rules shared by every row are in RowRules.

```
public static readonly IReadOnlyList<BoostScheduleSheet.TargetInfo> Infos
```

Field Value

IReadOnlyList<BoostScheduleSheet.TargetInfo>

RowRules

Rules every row follows whatever its target, for the same tooling as Infos. Reference text only.

```
public static readonly IReadOnlyList<string> RowRules
```

Field Value

IReadOnlyList<string>

RuneSummonGuarantee

Guarantee count of a `RuneSummonSheet` group. `target_id` is the group id.

```
public const string RuneSummonGuarantee = "RUNE_SUMMON_GUARANTEE"
```

Field Value

string

StageFavReward

Fungible asset rewards (e.g. crystal, rune stones) of an adventure stage clear (`HackAndSlash`, `HackAndSlashSweep`). `target_id` is the `StageSheet` stage id. Event dungeon stages are not covered.

Applied per clear to the amount of each drawn ticker after it is drawn, so the random draws are the same as without the row. See `StageSimulator.ApplyFavRewardBoost`.

```
public const string StageFavReward = "STAGE_FAV_REWARD"
```

Field Value

string

StageItemReward

Material item rewards of an adventure stage clear (`HackAndSlash`, `HackAndSlashSweep`). `target_id` is the `StageSheet` stage id, so `1~300` covers stages 1 through 300. Event dungeon stages are not covered.

Applied per clear to the count of each distinct item after the items are drawn, so the random draws are the same as without the row: `MUL 2` grants exactly twice each drawn item and `ADD 1` one more of each. See `StageSimulator.GetBoostedItemRewardCount`.

Boosted items are ordinary rewards everywhere else too: they count toward collect quest progress, and a circle stays tradable.

Rounding is down, per item id per clear, with one exception: a `MUL` above 1 turns an item drawn once in a clear (how most stages drop) into at least 2, so `MUL 1.5` grants 2 of a single drop. A count of 2 or more keeps the plain floor (`MUL 1.2` on 2 stays 2, `MUL 1.5` on 2 becomes 3); this is not a general "at least one more". `ADD` and a `MUL` of 1 or below are unaffected. Beware that any `MUL` just above 1 (e.g. `MUL 1.01`) also turns a single drop into 2 — effectively a 100% boost for those items, however small the value. StageFavReward amounts are large and always plainly floored.

```
public const string StageItemReward = "STAGE_ITEM_REWARD"
```

Field Value

string

SummonRuneEnhancementCrystalCost

Crystal cost of one `RuneEnhancement` try, per `RuneCostSheet` level, for **summonable runes only**: a rune whose id is a recipe id of any `RuneSummonSheet` row, whatever its ratio and whether or not that group is currently sold. Other runes are never adjusted. The summon pool is read from the chain when the try is evaluated, so a rune added to or removed from `RuneSummonSheet` gains or loses the adjustment with it.

`target_id` is the rune level the try attempts to reach — the current level plus one — so `1~50` adjusts the tries from level 0 up to level 50. It cannot narrow the row to particular rune ids. The adjusted cost never exceeds the sheet's cost and never drops below 1 while the sheet's cost is positive. NCG cost and success rate are not adjustable.

```
public const string SummonRuneEnhancementCrystalCost =
"SUMMON_RUNE_ENHANCEMENT_CRYSTAL_COST"
```

Field Value

string

SummonRuneEnhancementRuneStoneCost

Rune stone cost of one `RuneEnhancement` try, per `RuneCostSheet` level, for summonable runes only. Scope, `target_id` and bounds as in SummonRuneEnhancementCrystalCost.

```
public const string SummonRuneEnhancementRuneStoneCost =
"SUMMON_RUNE_ENHANCEMENT_RUNE_STONE_COST"
```

Field Value

string

SynthesizeRequiredCount

`required_count` of a `SynthesizeSheet` grade — how many materials one synthesis consumes. `target_id` is the material grade id (`grade_id`, e.g. `1~4`), and a row applies to every `item_sub_type` of that grade alike. The adjusted count never drops below 1 — which keeps synthesis from being free, not from being cheap: a typo such as `MUL 0.06` brings every grade down to 1, so keep values within the range planning intended.

Patch a row only after a client that picks materials by the adjusted count is released: the action still requires the material count to be a multiple of it, so a client choosing by the configured count fails every synthesis while the row is active.

```
public const string SynthesizeRequiredCount = "SYNTHESIZE_REQUIRED_COUNT"
```

Field Value

string

WorldBossBattleReward

Every amount — each drawn rune, crystal, circle — of the `WorldBossBattleRewardSheet` reward a `Raid` grants for its battle. `target_id` is the boss id (`WorldBossListSheet.boss_id`, e.g. `900001`), and the block is the one the raid is played in.

Each adjusted amount is capped at `WorldBossHelper.MaxWorldBossRewardBoostFactor` (5) times the drawn amount, `ADD` included; the same cap holds for WorldBossKillReward and WorldBossRankReward. See `WorldBossHelper.ApplyRewardBoost`.

```
public const string WorldBossBattleReward = "WORLD_BOSS_BATTLE_REWARD"
```

Field Value

string

WorldBossKillReward

Every amount — each drawn rune, crystal, circle — of each `WorldBossKillRewardSheet` reward, granted either by the `Raid` that raises the boss level or by `ClaimWordBossKillReward`. `target_id` is the boss id (`WorldBossListSheet.boss_id`), and the block is the one the reward is granted in, not the one the boss was killed in: a kill left unclaimed until the event is boosted when it is claimed during the event. Capped at 5 times each amount, as WorldBossBattle Reward.

```
public const string WorldBossKillReward = "WORLD_BOSS_KILL_REWARD"
```

Field Value

string

WorldBossRankReward

Every amount — each drawn rune, crystal, circle — of each `WorldBossRankRewardSheet` reward `ClaimRaidReward` grants. `target_id` is the boss id (`WorldBossListSheet.boss_id`), and the block is the one the reward is claimed in, not the one the rank was reached in: a rank left unclaimed until the event is boosted when it is claimed during the event. Capped at 5 times each amount, as WorldBossBattleReward.

```
public const string WorldBossRankReward = "WORLD_BOSS_RANK_REWARD"
```

Field Value

string

Class BuffLimitSheet

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

Upper limits of stat buffs and lower limits of stat debuffs, per buff group.

```
public class BuffLimitSheet : Sheet<int, BuffLimitSheet.Row>, IDictionary<int,
BuffLimitSheet.Row>, ICollection<KeyValuePair<int, BuffLimitSheet.Row>>,
IEnumerable<KeyValuePair<int, BuffLimitSheet.Row>>, IEnumerable, ISheet
```

Inheritance

object ← Sheet<int, BuffLimitSheet.Row> ← BuffLimitSheet

Implements

IDictionary<int, BuffLimitSheet.Row>,
ICollection<KeyValuePair<int, BuffLimitSheet.Row>>,
IEnumerable<KeyValuePair<int, BuffLimitSheet.Row>>, IEnumerable, ISheet

Inherited Members

Sheet<int, BuffLimitSheet.Row>.Name , Sheet<int, BuffLimitSheet.Row>.OrderedList ,
Sheet<int, BuffLimitSheet.Row>.First , Sheet<int, BuffLimitSheet.Row>.Last ,
Sheet<int, BuffLimitSheet.Row>.Keys , Sheet<int, BuffLimitSheet.Row>.Values ,
Sheet<int, BuffLimitSheet.Row>.Count , Sheet<int, BuffLimitSheet.Row>.IsReadOnly ,
Sheet<int, BuffLimitSheet.Row>.this[int] ,
Sheet<int, BuffLimitSheet.Row>.Set(string, bool) ,
Sheet<int, BuffLimitSheet.Row>.Set<T>(Sheet<int, T>, bool) ,
Sheet<int, BuffLimitSheet.Row>.GetEnumerator() ,
Sheet<int, BuffLimitSheet.Row>.TryGetValue(int, out BuffLimitSheet.Row, bool) ,
Sheet<int, BuffLimitSheet.Row>.AddRow(int, BuffLimitSheet.Row) ,
Sheet<int, BuffLimitSheet.Row>.Add(int, BuffLimitSheet.Row) ,
Sheet<int, BuffLimitSheet.Row>.ContainsKey(int) ,
Sheet<int, BuffLimitSheet.Row>.Remove(int) ,
Sheet<int, BuffLimitSheet.Row>.TryGetValue(int, out BuffLimitSheet.Row) ,
Sheet<int, BuffLimitSheet.Row>.Add(KeyValuePair<int, BuffLimitSheet.Row>) ,
Sheet<int, BuffLimitSheet.Row>.Clear() ,
Sheet<int, BuffLimitSheet.Row>.Contains(KeyValuePair<int, BuffLimitSheet.Row>) ,
Sheet<int, BuffLimitSheet.Row>.CopyTo(KeyValuePair<int, BuffLimitSheet.Row>[], int) ,
Sheet<int, BuffLimitSheet.Row>.Remove(KeyValuePair<int, BuffLimitSheet.Row>) ,
Sheet<int, BuffLimitSheet.Row>.Serialize() , object.Equals(object) ,

object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

BuffLimitSheet()

Creates an empty sheet.

```
public BuffLimitSheet()
```

Class BuffLimitSheet.Row

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

The limit of one buff group.

```
public class BuffLimitSheet.Row : SheetRow<int>
```

Inheritance

object ← SheetRow<int> ← BuffLimitSheet.Row

Inherited Members

SheetRow<int>.Validate() , SheetRow<int>.EndOfSheetInitialize() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Properties

GroupId

The GroupId this limit applies to.

```
public int GroupId { get; set; }
```

Property Value

int

Key

```
public override int Key { get; }
```

Property Value

int

StatModifier

How Value limits the buff: a percentage of the stat or an absolute amount.

```
public StatModifier.OperationType StatModifier { get; }
```

Property Value

StatModifier.OperationType

Value

The limit. An absolute (Add) limit follows stat magnitudes, which pass MaxValue in hard worlds, so it is a long.

```
public long Value { get; set; }
```

Property Value

long

Methods

GetModifier(StatType)

The limit as a StatModifier on `statType`.

```
public StatModifier GetModifier(StatType statType)
```

Parameters

`statType` StatType

The stat the limited buff modifies.

Returns

StatModifier

The modifier that applies the limit.

Set(IReadOnlyList<string>)

```
public override void Set(IReadOnlyList<string> fields)
```

Parameters

`fields` IReadOnlyList<string>

Class BuffLinkSheet

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
public class BuffLinkSheet : Sheet<int, BuffLinkSheet.Row>, IDictionary<int,
BuffLinkSheet.Row>, ICollection<KeyValuePair<int, BuffLinkSheet.Row>>,
IEnumerable<KeyValuePair<int, BuffLinkSheet.Row>>, IEnumerable, ISheet
```

Inheritance

object ← Sheet<int, BuffLinkSheet.Row> ← BuffLinkSheet

Implements

IDictionary<int, BuffLinkSheet.Row>,
ICollection<KeyValuePair<int, BuffLinkSheet.Row>>,
IEnumerable<KeyValuePair<int, BuffLinkSheet.Row>>, IEnumerable, ISheet

Inherited Members

Sheet<int, BuffLinkSheet.Row>.Name , Sheet<int, BuffLinkSheet.Row>.OrderedList ,
Sheet<int, BuffLinkSheet.Row>.First , Sheet<int, BuffLinkSheet.Row>.Last ,
Sheet<int, BuffLinkSheet.Row>.Keys , Sheet<int, BuffLinkSheet.Row>.Values ,
Sheet<int, BuffLinkSheet.Row>.Count , Sheet<int, BuffLinkSheet.Row>.IsReadOnly ,
Sheet<int, BuffLinkSheet.Row>.this[int] , Sheet<int, BuffLinkSheet.Row>.Set(string, bool) ,
Sheet<int, BuffLinkSheet.Row>.Set<T>(Sheet<int, T>, bool) ,
Sheet<int, BuffLinkSheet.Row>.GetEnumerator() ,
Sheet<int, BuffLinkSheet.Row>.TryGetValue(int, out BuffLinkSheet.Row, bool) ,
Sheet<int, BuffLinkSheet.Row>.AddRow(int, BuffLinkSheet.Row) ,
Sheet<int, BuffLinkSheet.Row>.Add(int, BuffLinkSheet.Row) ,
Sheet<int, BuffLinkSheet.Row>.ContainsKey(int) ,
Sheet<int, BuffLinkSheet.Row>.Remove(int) ,
Sheet<int, BuffLinkSheet.Row>.TryGetValue(int, out BuffLinkSheet.Row) ,
Sheet<int, BuffLinkSheet.Row>.Add(KeyValuePair<int, BuffLinkSheet.Row>) ,
Sheet<int, BuffLinkSheet.Row>.Clear() ,
Sheet<int, BuffLinkSheet.Row>.Contains(KeyValuePair<int, BuffLinkSheet.Row>) ,
Sheet<int, BuffLinkSheet.Row>.CopyTo(KeyValuePair<int, BuffLinkSheet.Row>[], int) ,
Sheet<int, BuffLinkSheet.Row>.Remove(KeyValuePair<int, BuffLinkSheet.Row>) ,
Sheet<int, BuffLinkSheet.Row>.Serialize() , object.Equals(object) ,
object.Equals(object, object) , object.GetHashCode() , object.GetType() ,
object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

BuffLinkSheet()

```
public BuffLinkSheet()
```

Class BuffLinkSheet.Row

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class BuffLinkSheet.Row : SheetRow<int>
```

Inheritance

object ← SheetRow<int> ← BuffLinkSheet.Row

Inherited Members

SheetRow<int>.Validate() , SheetRow<int>.EndOfSheetInitialize() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Properties

BuffId

```
public int BuffId { get; }
```

Property Value

int

Key

```
public override int Key { get; }
```

Property Value

int

LinkedBuffId

```
public int LinkedBuffId { get; }
```

Property Value

int

Methods

Set(IReadOnlyList<string>)

```
public override void Set(IReadOnlyList<string> fields)
```

Parameters

`fields` IReadOnlyList<string>

Class BuffSheet

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class BuffSheet : Sheet<int, BuffSheet.Row>, IDictionary<int, BuffSheet.Row>,
ICollection<KeyValuePair<int, BuffSheet.Row>>, IEnumerable<KeyValuePair<int,
BuffSheet.Row>>, IEnumerable, ISheet
```

Inheritance

object ← Sheet<int, BuffSheet.Row> ← BuffSheet

Implements

IDictionary<int, BuffSheet.Row>, ICollection<KeyValuePair<int, BuffSheet.Row>>, IEnumerable<KeyValuePair<int, BuffSheet.Row>>, IEnumerable, ISheet

Inherited Members

Sheet<int, BuffSheet.Row>.Name , Sheet<int, BuffSheet.Row>.OrderedList , Sheet<int, BuffSheet.Row>.First , Sheet<int, BuffSheet.Row>.Last , Sheet<int, BuffSheet.Row>.Keys , Sheet<int, BuffSheet.Row>.Values , Sheet<int, BuffSheet.Row>.Count , Sheet<int, BuffSheet.Row>.IsReadOnly , Sheet<int, BuffSheet.Row>.this[int] , Sheet<int, BuffSheet.Row>.Set(string, bool) , Sheet<int, BuffSheet.Row>.Set<T>(Sheet<int, T>, bool) , Sheet<int, BuffSheet.Row>.GetEnumerator() , Sheet<int, BuffSheet.Row>.TryGetValue(int, out BuffSheet.Row, bool) , Sheet<int, BuffSheet.Row>.AddRow(int, BuffSheet.Row) , Sheet<int, BuffSheet.Row>.Add(int, BuffSheet.Row) , Sheet<int, BuffSheet.Row>.ContainsKey(int) , Sheet<int, BuffSheet.Row>.Remove(int) , Sheet<int, BuffSheet.Row>.TryGetValue(int, out BuffSheet.Row) , Sheet<int, BuffSheet.Row>.Add(KeyValuePair<int, BuffSheet.Row>) , Sheet<int, BuffSheet.Row>.Clear() , Sheet<int, BuffSheet.Row>.Contains(KeyValuePair<int, BuffSheet.Row>) , Sheet<int, BuffSheet.Row>.CopyTo(KeyValuePair<int, BuffSheet.Row>[], int) , Sheet<int, BuffSheet.Row>.Remove(KeyValuePair<int, BuffSheet.Row>) , Sheet<int, BuffSheet.Row>.Serialize() , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

BuffSheet()

```
public BuffSheet()
```

Class BuffSheet.Row

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class BuffSheet.Row : SheetRow<int>
```

Inheritance

object ← SheetRow<int> ← BuffSheet.Row

Inherited Members

SheetRow<int>.Validate() , SheetRow<int>.EndOfSheetInitialize() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Properties

Chance

100: 100%

```
public int Chance { get; }
```

Property Value

int

Duration

Turn count.

```
public int Duration { get; }
```

Property Value

int

GroupId

GroupId는 같은 GroupId를 공유하는 버프 중 가장 작은 값으로 함. Id가 다르더라도 GroupId가 같으면, 서로 덮어씀. 이때의 우선순위는 Id가 큰 것이 남는 것으로 구현함.

```
public int GroupId { get; }
```

Property Value

int

IconResource

```
public string IconResource { get; }
```

Property Value

string

Id

```
public int Id { get; }
```

Property Value

int

Key

```
public override int Key { get; }
```

Property Value

int

StatModifier

```
public StatModifier StatModifier { get; }
```

Property Value

StatModifier

TargetType

```
public SkillTargetType TargetType { get; }
```

Property Value

SkillTargetType

Methods

Set(IReadOnlyList<string>)

```
public override void Set(IReadOnlyList<string> fields)
```

Parameters

`fields` IReadOnlyList<string>

Class CharacterLevelSheet

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class CharacterLevelSheet : Sheet<int, CharacterLevelSheet.Row>,
IDictionary<int, CharacterLevelSheet.Row>, ICollection<KeyValuePair<int,
CharacterLevelSheet.Row>>, IEnumerable<KeyValuePair<int, CharacterLevelSheet.Row>>,
IEnumerable, ISheet
```

Inheritance

object ← Sheet<int, CharacterLevelSheet.Row> ← CharacterLevelSheet

Implements

IDictionary<int, CharacterLevelSheet.Row>,
ICollection<KeyValuePair<int, CharacterLevelSheet.Row>>,
IEnumerable<KeyValuePair<int, CharacterLevelSheet.Row>>, IEnumerable, ISheet

Inherited Members

Sheet<int, CharacterLevelSheet.Row>.Name ,
Sheet<int, CharacterLevelSheet.Row>.OrderedList ,
Sheet<int, CharacterLevelSheet.Row>.First , Sheet<int, CharacterLevelSheet.Row>.Last ,
Sheet<int, CharacterLevelSheet.Row>.Keys ,
Sheet<int, CharacterLevelSheet.Row>.Values ,
Sheet<int, CharacterLevelSheet.Row>.Count ,
Sheet<int, CharacterLevelSheet.Row>.IsReadOnly ,
Sheet<int, CharacterLevelSheet.Row>.this[int] ,
Sheet<int, CharacterLevelSheet.Row>.Set(string, bool) ,
Sheet<int, CharacterLevelSheet.Row>.Set<T>(Sheet<int, T>, bool) ,
Sheet<int, CharacterLevelSheet.Row>.GetEnumerator() ,
Sheet<int, CharacterLevelSheet.Row>.TryGetValue(int, out CharacterLevelSheet.Row, bool)
,
Sheet<int, CharacterLevelSheet.Row>.AddRow(int, CharacterLevelSheet.Row) ,
Sheet<int, CharacterLevelSheet.Row>.Add(int, CharacterLevelSheet.Row) ,
Sheet<int, CharacterLevelSheet.Row>.ContainsKey(int) ,
Sheet<int, CharacterLevelSheet.Row>.Remove(int) ,
Sheet<int, CharacterLevelSheet.Row>.TryGetValue(int, out CharacterLevelSheet.Row) ,
Sheet<int, CharacterLevelSheet.Row>.Add(KeyValuePair<int, CharacterLevelSheet.Row>) ,
Sheet<int, CharacterLevelSheet.Row>.Clear() ,

Sheet<int, CharacterLevelSheet.Row>.Contains(KeyValuePair<int, CharacterLevelSheet.Row>) , Sheet<int, CharacterLevelSheet.Row>.CopyTo(KeyValuePair<int, CharacterLevelSheet.Row>[], int) , Sheet<int, CharacterLevelSheet.Row>.Remove(KeyValuePair<int, CharacterLevelSheet.Row>) , Sheet<int, CharacterLevelSheet.Row>.Serialize() , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

CharacterLevelSheet()

```
public CharacterLevelSheet()
```

Methods

GetLevel(long)

```
public int GetLevel(long exp)
```

Parameters

exp long

Returns

int

TryGetLevel(long, out int)

```
public bool TryGetLevel(long exp, out int level)
```

Parameters

`exp` long

`level` int

Returns

bool

Class CharacterLevelSheet.Row

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class CharacterLevelSheet.Row : SheetRow<int>
```

Inheritance

object ← SheetRow<int> ← CharacterLevelSheet.Row

Inherited Members

SheetRow<int>.Validate() , SheetRow<int>.EndOfSheetInitialize() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Properties

Exp

```
public long Exp { get; }
```

Property Value

long

ExpNeed

```
public long ExpNeed { get; }
```

Property Value

long

Key

```
public override int Key { get; }
```

Property Value

int

Level

```
public int Level { get; }
```

Property Value

int

Methods

Set(IReadOnlyList<string>)

```
public override void Set(IReadOnlyList<string> fields)
```

Parameters

`fields` IReadOnlyList<string>

Class CharacterSheet

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class CharacterSheet : Sheet<int, CharacterSheet.Row>, IDictionary<int,
CharacterSheet.Row>, ICollection<KeyValuePair<int, CharacterSheet.Row>>,
IEnumerable<KeyValuePair<int, CharacterSheet.Row>>, IEnumerable, ISheet
```

Inheritance

object ← Sheet<int, CharacterSheet.Row> ← CharacterSheet

Implements

IDictionary<int, CharacterSheet.Row>,
ICollection<KeyValuePair<int, CharacterSheet.Row>>,
IEnumerable<KeyValuePair<int, CharacterSheet.Row>>, IEnumerable, ISheet

Inherited Members

Sheet<int, CharacterSheet.Row>.Name , Sheet<int, CharacterSheet.Row>.OrderedList ,
Sheet<int, CharacterSheet.Row>.First , Sheet<int, CharacterSheet.Row>.Last ,
Sheet<int, CharacterSheet.Row>.Keys , Sheet<int, CharacterSheet.Row>.Values ,
Sheet<int, CharacterSheet.Row>.Count , Sheet<int, CharacterSheet.Row>.IsReadOnly ,
Sheet<int, CharacterSheet.Row>.this[int] ,
Sheet<int, CharacterSheet.Row>.Set(string, bool) ,
Sheet<int, CharacterSheet.Row>.Set<T>(Sheet<int, T>, bool) ,
Sheet<int, CharacterSheet.Row>.GetEnumerator() ,
Sheet<int, CharacterSheet.Row>.TryGetValue(int, out CharacterSheet.Row, bool) ,
Sheet<int, CharacterSheet.Row>.AddRow(int, CharacterSheet.Row) ,
Sheet<int, CharacterSheet.Row>.Add(int, CharacterSheet.Row) ,
Sheet<int, CharacterSheet.Row>.ContainsKey(int) ,
Sheet<int, CharacterSheet.Row>.Remove(int) ,
Sheet<int, CharacterSheet.Row>.TryGetValue(int, out CharacterSheet.Row) ,
Sheet<int, CharacterSheet.Row>.Add(KeyValuePair<int, CharacterSheet.Row>) ,
Sheet<int, CharacterSheet.Row>.Clear() ,
Sheet<int, CharacterSheet.Row>.Contains(KeyValuePair<int, CharacterSheet.Row>) ,
Sheet<int, CharacterSheet.Row>.CopyTo(KeyValuePair<int, CharacterSheet.Row>[], int) ,
Sheet<int, CharacterSheet.Row>.Remove(KeyValuePair<int, CharacterSheet.Row>) ,
Sheet<int, CharacterSheet.Row>.Serialize() , object.Equals(object) ,

object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

CharacterSheet()

```
public CharacterSheet()
```

Class CharacterSheet.Row

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class CharacterSheet.Row : SheetRow<int>
```

Inheritance

object ← SheetRow<int> ← CharacterSheet.Row

Inherited Members

SheetRow<int>.Validate() , SheetRow<int>.EndOfSheetInitialize() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Extension Methods

CharacterSheetExtension.ToStats(CharacterSheet.Row, int)

Properties

ATK

```
public decimal ATK { get; }
```

Property Value

decimal

AttackRange

```
public float AttackRange { get; }
```

Property Value

float

CRI

```
public decimal CRI { get; }
```

Property Value

decimal

DEF

```
public decimal DEF { get; }
```

Property Value

decimal

ElementalType

```
public ElementalType ElementalType { get; }
```

Property Value

ElementalType

HIT

```
public decimal HIT { get; }
```

Property Value

decimal

HP

```
public decimal HP { get; }
```

Property Value

decimal

Id

```
public int Id { get; }
```

Property Value

int

Key

```
public override int Key { get; }
```

Property Value

int

LvATK

```
public decimal LvATK { get; }
```

Property Value

decimal

LvCRI

```
public decimal LvCRI { get; }
```

Property Value

decimal

LvDEF

```
public decimal LvDEF { get; }
```

Property Value

decimal

LvHIT

```
public decimal LvHIT { get; }
```

Property Value

decimal

LvHP

```
public decimal LvHP { get; }
```

Property Value

decimal

LvSPD

```
public decimal LvSPD { get; }
```

Property Value

decimal

RunSpeed

```
public float RunSpeed { get; }
```

Property Value

float

SPD

```
public decimal SPD { get; }
```

Property Value

decimal

SizeType

```
public SizeType SizeType { get; }
```

Property Value

SizeType

Methods

Set(IReadOnlyList<string>)

```
public override void Set(IReadOnlyList<string> fields)
```

Parameters

`fields` IReadOnlyList<string>

Class CharacterSheetExtension

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
public static class CharacterSheetExtension
```

Inheritance

object ← CharacterSheetExtension

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Methods

ToStats(Row, int)

```
public static StatsMap ToStats(this CharacterSheet.Row row, int level)
```

Parameters

`row` CharacterSheet.Row

`level` int

Returns

StatsMap

Class ClaimableGiftsSheet

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

Sheet for claimable gifts that can be claimed by players during specific block index ranges. Each row contains gift ID, block index range, and a list of items to be given.

```
public class ClaimableGiftsSheet : Sheet<int, ClaimableGiftsSheet.Row>,
IDictionary<int, ClaimableGiftsSheet.Row>, ICollection<KeyValuePair<int,
ClaimableGiftsSheet.Row>>, IEnumerable<KeyValuePair<int, ClaimableGiftsSheet.Row>>,
IEnumerable, ISheet
```

Inheritance

object ← Sheet<int, ClaimableGiftsSheet.Row> ← ClaimableGiftsSheet

Implements

IDictionary<int, ClaimableGiftsSheet.Row>,
ICollection<KeyValuePair<int, ClaimableGiftsSheet.Row>>,
IEnumerable<KeyValuePair<int, ClaimableGiftsSheet.Row>>, IEnumerable, ISheet

Inherited Members

Sheet<int, ClaimableGiftsSheet.Row>.Name ,
Sheet<int, ClaimableGiftsSheet.Row>.OrderedList ,
Sheet<int, ClaimableGiftsSheet.Row>.First , Sheet<int, ClaimableGiftsSheet.Row>.Last ,
Sheet<int, ClaimableGiftsSheet.Row>.Keys , Sheet<int, ClaimableGiftsSheet.Row>.Values ,
Sheet<int, ClaimableGiftsSheet.Row>.Count ,
Sheet<int, ClaimableGiftsSheet.Row>.IsReadOnly ,
Sheet<int, ClaimableGiftsSheet.Row>.this[int] ,
Sheet<int, ClaimableGiftsSheet.Row>.Set(string, bool) ,
Sheet<int, ClaimableGiftsSheet.Row>.Set<T>(Sheet<int, T>, bool) ,
Sheet<int, ClaimableGiftsSheet.Row>.GetEnumerator() ,
Sheet<int, ClaimableGiftsSheet.Row>.TryGetValue(int, out ClaimableGiftsSheet.Row, bool) ,
Sheet<int, ClaimableGiftsSheet.Row>.AddRow(int, ClaimableGiftsSheet.Row) ,
Sheet<int, ClaimableGiftsSheet.Row>.Add(int, ClaimableGiftsSheet.Row) ,
Sheet<int, ClaimableGiftsSheet.Row>.ContainsKey(int) ,
Sheet<int, ClaimableGiftsSheet.Row>.Remove(int) ,
Sheet<int, ClaimableGiftsSheet.Row>.TryGetValue(int, out ClaimableGiftsSheet.Row) ,
Sheet<int, ClaimableGiftsSheet.Row>.Add(KeyValuePair<int, ClaimableGiftsSheet.Row>) ,
Sheet<int, ClaimableGiftsSheet.Row>.Clear() ,

Sheet<int, ClaimableGiftsSheet.Row>.Contains(KeyValuePair<int, ClaimableGiftsSheet.Row>) , Sheet<int, ClaimableGiftsSheet.Row>.CopyTo(KeyValuePair<int, ClaimableGiftsSheet.Row>[], int) , Sheet<int, ClaimableGiftsSheet.Row>.Remove(KeyValuePair<int, ClaimableGiftsSheet.Row>) , Sheet<int, ClaimableGiftsSheet.Row>.Serialize() , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

ClaimableGiftsSheet()

Initializes a new instance of the ClaimableGiftsSheet class.

```
public ClaimableGiftsSheet()
```

Methods

TryFindRowByBlockIndex(long, out Row)

Tries to find a row that is claimable at the given block index.

```
public bool TryFindRowByBlockIndex(long blockIndex, out ClaimableGiftsSheet.Row row)
```

Parameters

`blockIndex` long

Block index to search for

`row` ClaimableGiftsSheet.Row

When this method returns, contains the row if found; otherwise, null

Returns

bool

True if a claimable row is found; otherwise, false

Class ClaimableGiftsSheet.Row

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

Represents a single row in the ClaimableGiftsSheet. Contains gift ID, block index range, and items to be given.

```
public class ClaimableGiftsSheet.Row : SheetRow<int>
```

Inheritance

object ← SheetRow<int> ← ClaimableGiftsSheet.Row

Inherited Members

SheetRow<int>.Validate() , SheetRow<int>.EndOfSheetInitialize() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Fields

EndedBlockIndex

Block index when this gift becomes unclaimable (inclusive).

```
public long EndedBlockIndex
```

Field Value

long

Id

Gift ID that uniquely identifies this gift.

```
public int Id
```

Field Value

int

Items

List of items to be given when this gift is claimed. Each tuple contains (itemId, quantity, tradable).

```
public List<(int itemId, int quantity, bool tradable)> Items
```

Field Value

List<(int id, int count, bool tradable)>

StartedBlockIndex

Block index when this gift becomes claimable (inclusive).

```
public long StartedBlockIndex
```

Field Value

long

Properties

Key

Gets the key for this row, which is the gift ID.

```
public override int Key { get; }
```

Property Value

int

Methods

Set(IReadOnlyList<string>)

Sets the row data from CSV fields. Expected format:
id,started_block_index,ended_block_index,item_1_id,item_1_quantity,item_1_tradable,...,item_5_id,item_5_quantity,item_5_tradable
Supports up to 5 items. Empty or zero values for item_id or quantity will be skipped.

```
public override void Set(IReadOnlyList<string> fields)
```

Parameters

`fields` IReadOnlyList<string>

CSV field values

Validate(long)

Validates whether the given block index is within the claimable range.

```
public bool Validate(long blockIndex)
```

Parameters

`blockIndex` long

Block index to validate

Returns

bool

True if the block index is within [StartedBlockIndex, EndedBlockIndex], false otherwise

Class CollectQuestSheet

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class CollectQuestSheet : Sheet<int, CollectQuestSheet.Row>, IDictionary<int,
CollectQuestSheet.Row>, ICollection<KeyValuePair<int, CollectQuestSheet.Row>>,
IEnumerable<KeyValuePair<int, CollectQuestSheet.Row>>, IEnumerable, ISheet
```

Inheritance

object ← Sheet<int, CollectQuestSheet.Row> ← CollectQuestSheet

Implements

IDictionary<int, CollectQuestSheet.Row>,
ICollection<KeyValuePair<int, CollectQuestSheet.Row>>,
IEnumerable<KeyValuePair<int, CollectQuestSheet.Row>>, IEnumerable, ISheet

Inherited Members

Sheet<int, CollectQuestSheet.Row>.Name ,
Sheet<int, CollectQuestSheet.Row>.OrderedList ,
Sheet<int, CollectQuestSheet.Row>.First , Sheet<int, CollectQuestSheet.Row>.Last ,
Sheet<int, CollectQuestSheet.Row>.Keys , Sheet<int, CollectQuestSheet.Row>.Values ,
Sheet<int, CollectQuestSheet.Row>.Count ,
Sheet<int, CollectQuestSheet.Row>.IsReadOnly ,
Sheet<int, CollectQuestSheet.Row>.this[int] ,
Sheet<int, CollectQuestSheet.Row>.Set(string, bool) ,
Sheet<int, CollectQuestSheet.Row>.Set<T>(Sheet<int, T>, bool) ,
Sheet<int, CollectQuestSheet.Row>.GetEnumerator() ,
Sheet<int, CollectQuestSheet.Row>.TryGetValue(int, out CollectQuestSheet.Row, bool) ,
Sheet<int, CollectQuestSheet.Row>.AddRow(int, CollectQuestSheet.Row) ,
Sheet<int, CollectQuestSheet.Row>.Add(int, CollectQuestSheet.Row) ,
Sheet<int, CollectQuestSheet.Row>.ContainsKey(int) ,
Sheet<int, CollectQuestSheet.Row>.Remove(int) ,
Sheet<int, CollectQuestSheet.Row>.TryGetValue(int, out CollectQuestSheet.Row) ,
Sheet<int, CollectQuestSheet.Row>.Add(KeyValuePair<int, CollectQuestSheet.Row>) ,
Sheet<int, CollectQuestSheet.Row>.Clear() ,
Sheet<int, CollectQuestSheet.Row>.Contains(KeyValuePair<int, CollectQuestSheet.Row>) ,

Sheet<int, CollectQuestSheet.Row>.CopyTo(KeyValuePair<int, CollectQuestSheet.Row>[], int) ,
Sheet<int, CollectQuestSheet.Row>.Remove(KeyValuePair<int, CollectQuestSheet.Row>) ,
Sheet<int, CollectQuestSheet.Row>.Serialize() , object.Equals(object) ,
object.Equals(object, object) , object.GetHashCode() , object.GetType() ,
object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

CollectQuestSheet()

```
public CollectQuestSheet()
```

Class CollectQuestSheet.Row

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class CollectQuestSheet.Row : QuestSheet.Row, IState
```

Inheritance

object ← SheetRow<int> ← QuestSheet.Row ← CollectQuestSheet.Row

Implements

IState

Inherited Members

QuestSheet.Row.Key , QuestSheet.Row.Id , QuestSheet.Row.Goal , QuestSheet.Row.QuestRewardId , QuestSheet.Row.Serialize() , QuestSheet.Row.Deserialize(Dictionary) , SheetRow<int>.Validate() , SheetRow<int>.EndOfSheetInitialize() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Properties

ItemId

```
public int ItemId { get; }
```

Property Value

int

Methods

Set(IReadOnlyList<string>)

```
public override void Set(IReadOnlyList<string> fields)
```

Parameters

`fields` IReadOnlyList<string>

Class CollectionSheet

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
public class CollectionSheet : Sheet<int, CollectionSheet.Row>, IDictionary<int,
CollectionSheet.Row>, ICollection<KeyValuePair<int, CollectionSheet.Row>>,
IEnumerable<KeyValuePair<int, CollectionSheet.Row>>, IEnumerable, ISheet
```

Inheritance

object ← Sheet<int, CollectionSheet.Row> ← CollectionSheet

Implements

IDictionary<int, CollectionSheet.Row>,
ICollection<KeyValuePair<int, CollectionSheet.Row>>,
IEnumerable<KeyValuePair<int, CollectionSheet.Row>>, IEnumerable, ISheet

Inherited Members

Sheet<int, CollectionSheet.Row>.Name , Sheet<int, CollectionSheet.Row>.OrderedList ,
Sheet<int, CollectionSheet.Row>.First , Sheet<int, CollectionSheet.Row>.Last ,
Sheet<int, CollectionSheet.Row>.Keys , Sheet<int, CollectionSheet.Row>.Values ,
Sheet<int, CollectionSheet.Row>.Count , Sheet<int, CollectionSheet.Row>.IsReadOnly ,
Sheet<int, CollectionSheet.Row>.this[int] ,
Sheet<int, CollectionSheet.Row>.Set(string, bool) ,
Sheet<int, CollectionSheet.Row>.Set<T>(Sheet<int, T>, bool) ,
Sheet<int, CollectionSheet.Row>.GetEnumerator() ,
Sheet<int, CollectionSheet.Row>.TryGetValue(int, out CollectionSheet.Row, bool) ,
Sheet<int, CollectionSheet.Row>.AddRow(int, CollectionSheet.Row) ,
Sheet<int, CollectionSheet.Row>.Add(int, CollectionSheet.Row) ,
Sheet<int, CollectionSheet.Row>.ContainsKey(int) ,
Sheet<int, CollectionSheet.Row>.Remove(int) ,
Sheet<int, CollectionSheet.Row>.TryGetValue(int, out CollectionSheet.Row) ,
Sheet<int, CollectionSheet.Row>.Add(KeyValuePair<int, CollectionSheet.Row>) ,
Sheet<int, CollectionSheet.Row>.Clear() ,
Sheet<int, CollectionSheet.Row>.Contains(KeyValuePair<int, CollectionSheet.Row>) ,
Sheet<int, CollectionSheet.Row>.CopyTo(KeyValuePair<int, CollectionSheet.Row>[], int) ,
Sheet<int, CollectionSheet.Row>.Remove(KeyValuePair<int, CollectionSheet.Row>) ,
Sheet<int, CollectionSheet.Row>.Serialize() , object.Equals(object) ,

object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

CollectionSheet()

```
public CollectionSheet()
```

Class CollectionSheet.RequiredMaterial

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
public class CollectionSheet.RequiredMaterial
```

Inheritance

object ← CollectionSheet.RequiredMaterial

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Fields

Count

```
public int Count
```

Field Value

int

ItemId

```
public int ItemId
```

Field Value

int

Level

```
public int Level
```

Field Value

int

SkillContains

```
public bool SkillContains
```

Field Value

bool

Methods

GetMaterial(IEnumerable<ICollectionMaterial>)

Retrieves the ICollectionMaterial object from the given collection of materials based on the item ID and count.

```
public ICollectionMaterial GetMaterial(IEnumerable<ICollectionMaterial> materials)
```

Parameters

`materials` IEnumerable<ICollectionMaterial>

The collection of materials to search.

Returns

ICollectionMaterial

The ICollectionMaterial object if found; otherwise, an exception is thrown.

Validate(INonFungibleItem)

```
public bool Validate(INonFungibleItem nonFungibleItem)
```

Parameters

`nonFungibleItem` INonFungibleItem

Returns

bool

Class CollectionSheet.Row

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
public class CollectionSheet.Row : SheetRow<int>
```

Inheritance

object ← SheetRow<int> ← CollectionSheet.Row

Inherited Members

SheetRow<int>.Validate() , SheetRow<int>.EndOfSheetInitialize() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Fields

Materials

```
public List<CollectionSheet.RequiredMaterial> Materials
```

Field Value

List<CollectionSheet.RequiredMaterial>

StatModifiers

```
public List<StatModifier> StatModifiers
```

Field Value

List<StatModifier>

Properties

Id

```
public int Id { get; }
```

Property Value

int

Key

```
public override int Key { get; }
```

Property Value

int

Methods

Set(IReadOnlyList<string>)

```
public override void Set(IReadOnlyList<string> fields)
```

Parameters

`fields` IReadOnlyList<string>

Class CombinationEquipmentQuestSheet

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class CombinationEquipmentQuestSheet : Sheet<int,
CombinationEquipmentQuestSheet.Row>, IDictionary<int,
CombinationEquipmentQuestSheet.Row>, ICollection<KeyValuePair<int,
CombinationEquipmentQuestSheet.Row>>, IEnumerable<KeyValuePair<int,
CombinationEquipmentQuestSheet.Row>>, IEnumerable, ISheet
```

Inheritance

object ← Sheet<int, CombinationEquipmentQuestSheet.Row> ← CombinationEquipmentQuestSheet

Implements

IDictionary<int, CombinationEquipmentQuestSheet.Row>, ICollection<KeyValuePair<int, CombinationEquipmentQuestSheet.Row>>, IEnumerable<KeyValuePair<int, CombinationEquipmentQuestSheet.Row>>, IEnumerable, ISheet

Inherited Members

Sheet<int, CombinationEquipmentQuestSheet.Row>.Name ,
Sheet<int, CombinationEquipmentQuestSheet.Row>.OrderedList ,
Sheet<int, CombinationEquipmentQuestSheet.Row>.First ,
Sheet<int, CombinationEquipmentQuestSheet.Row>.Last ,
Sheet<int, CombinationEquipmentQuestSheet.Row>.Keys ,
Sheet<int, CombinationEquipmentQuestSheet.Row>.Values ,
Sheet<int, CombinationEquipmentQuestSheet.Row>.Count ,
Sheet<int, CombinationEquipmentQuestSheet.Row>.IsReadOnly ,
Sheet<int, CombinationEquipmentQuestSheet.Row>.this[int] ,
Sheet<int, CombinationEquipmentQuestSheet.Row>.Set(string, bool) ,
Sheet<int, CombinationEquipmentQuestSheet.Row>.Set<T>(Sheet<int, T>, bool) ,
Sheet<int, CombinationEquipmentQuestSheet.Row>.GetEnumerator() ,
Sheet<int, CombinationEquipmentQuestSheet.Row>.TryGetValue(int, out CombinationEquipmentQuestSheet.Row, bool) ,
Sheet<int, CombinationEquipmentQuestSheet.Row>.AddRow(int, CombinationEquipmentQuestSheet.Row) ,

Sheet<int, CombinationEquipmentQuestSheet.Row>.Add(int, CombinationEquipmentQuestSheet.Row) , Sheet<int, CombinationEquipmentQuestSheet.Row>.ContainsKey(int) , Sheet<int, CombinationEquipmentQuestSheet.Row>.Remove(int) , Sheet<int, CombinationEquipmentQuestSheet.Row>.TryGetValue(int, out CombinationEquipmentQuestSheet.Row) , Sheet<int, CombinationEquipmentQuestSheet.Row>.Add(KeyValuePair<int, CombinationEquipmentQuestSheet.Row>) , Sheet<int, CombinationEquipmentQuestSheet.Row>.Clear() , Sheet<int, CombinationEquipmentQuestSheet.Row>.Contains(KeyValuePair<int, CombinationEquipmentQuestSheet.Row>) , Sheet<int, CombinationEquipmentQuestSheet.Row>.CopyTo(KeyValuePair<int, CombinationEquipmentQuestSheet.Row>[], int) , Sheet<int, CombinationEquipmentQuestSheet.Row>.Remove(KeyValuePair<int, CombinationEquipmentQuestSheet.Row>) , Sheet<int, CombinationEquipmentQuestSheet.Row>.Serialize() , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

CombinationEquipmentQuestSheet()

```
public CombinationEquipmentQuestSheet()
```

Class CombinationEquipmentQuestSheet.Row

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class CombinationEquipmentQuestSheet.Row : QuestSheet.Row, IState
```

Inheritance

object ← SheetRow<int> ← QuestSheet.Row ← CombinationEquipmentQuestSheet.Row

Implements

IState

Inherited Members

QuestSheet.Row.Key , QuestSheet.Row.Id , QuestSheet.Row.Goal , QuestSheet.Row.QuestRewardId , QuestSheet.Row.Serialize() , QuestSheet.Row.Deserialize(Dictionary) , SheetRow<int>.Validate() , SheetRow<int>.EndOfSheetInitialize() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Properties

RecipeId

```
public int RecipeId { get; }
```

Property Value

int

Methods

Set(IReadOnlyList<string>)

```
public override void Set(IReadOnlyList<string> fields)
```

Parameters

`fields` IReadOnlyList<string>

Class CombinationQuestSheet

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class CombinationQuestSheet : Sheet<int, CombinationQuestSheet.Row>,
IDictionary<int, CombinationQuestSheet.Row>, ICollection<KeyValuePair<int,
CombinationQuestSheet.Row>>, IEnumerable<KeyValuePair<int,
CombinationQuestSheet.Row>>, IEnumerable, ISheet
```

Inheritance

object ← Sheet<int, CombinationQuestSheet.Row> ← CombinationQuestSheet

Implements

IDictionary<int, CombinationQuestSheet.Row>,
ICollection<KeyValuePair<int, CombinationQuestSheet.Row>>,
IEnumerable<KeyValuePair<int, CombinationQuestSheet.Row>>, IEnumerable,
ISheet

Inherited Members

Sheet<int, CombinationQuestSheet.Row>.Name ,
Sheet<int, CombinationQuestSheet.Row>.OrderedList ,
Sheet<int, CombinationQuestSheet.Row>.First ,
Sheet<int, CombinationQuestSheet.Row>.Last ,
Sheet<int, CombinationQuestSheet.Row>.Keys ,
Sheet<int, CombinationQuestSheet.Row>.Values ,
Sheet<int, CombinationQuestSheet.Row>.Count ,
Sheet<int, CombinationQuestSheet.Row>.IsReadOnly ,
Sheet<int, CombinationQuestSheet.Row>.this[int] ,
Sheet<int, CombinationQuestSheet.Row>.Set(string, bool) ,
Sheet<int, CombinationQuestSheet.Row>.Set<T>(Sheet<int, T>, bool) ,
Sheet<int, CombinationQuestSheet.Row>.GetEnumerator() ,
Sheet<int, CombinationQuestSheet.Row>.TryGetValue(int, out
CombinationQuestSheet.Row, bool) ,
Sheet<int, CombinationQuestSheet.Row>.AddRow(int, CombinationQuestSheet.Row) ,
Sheet<int, CombinationQuestSheet.Row>.Add(int, CombinationQuestSheet.Row) ,
Sheet<int, CombinationQuestSheet.Row>.ContainsKey(int) ,
Sheet<int, CombinationQuestSheet.Row>.Remove(int) ,

Sheet<int, CombinationQuestSheet.Row>.TryGetValue(int, out CombinationQuestSheet.Row) , Sheet<int, CombinationQuestSheet.Row>.Add(KeyValuePair<int, CombinationQuestSheet.Row>) , Sheet<int, CombinationQuestSheet.Row>.Clear() , Sheet<int, CombinationQuestSheet.Row>.Contains(KeyValuePair<int, CombinationQuestSheet.Row>) , Sheet<int, CombinationQuestSheet.Row>.CopyTo(KeyValuePair<int, CombinationQuestSheet.Row>[], int) , Sheet<int, CombinationQuestSheet.Row>.Remove(KeyValuePair<int, CombinationQuestSheet.Row>) , Sheet<int, CombinationQuestSheet.Row>.Serialize() , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

CombinationQuestSheet()

```
public CombinationQuestSheet()
```

Class CombinationQuestSheet.Row

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class CombinationQuestSheet.Row : QuestSheet.Row, IState
```

Inheritance

object ← SheetRow<int> ← QuestSheet.Row ← CombinationQuestSheet.Row

Implements

IState

Inherited Members

QuestSheet.Row.Key , QuestSheet.Row.Id , QuestSheet.Row.Goal , QuestSheet.Row.QuestRewardId , QuestSheet.Row.Serialize() , QuestSheet.Row.Deserialize(Dictionary) , SheetRow<int>.Validate() , SheetRow<int>.EndOfSheetInitialize() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Properties

ItemSubType

```
public ItemSubType ItemSubType { get; }
```

Property Value

ItemSubType

ItemType

```
public ItemType ItemType { get; }
```

Property Value

ItemType

Methods

Set(IReadOnlyList<string>)

```
public override void Set(IReadOnlyList<string> fields)
```

Parameters

`fields` IReadOnlyList<string>

Class ConsumableItemRecipeSheet

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class ConsumableItemRecipeSheet : Sheet<int, ConsumableItemRecipeSheet.Row>,
IDictionary<int, ConsumableItemRecipeSheet.Row>, ICollection<KeyValuePair<int,
ConsumableItemRecipeSheet.Row>>, IEnumerable<KeyValuePair<int,
ConsumableItemRecipeSheet.Row>>, IEnumerable, ISheet
```

Inheritance

object ← Sheet<int, ConsumableItemRecipeSheet.Row> ← ConsumableItemRecipeSheet

Implements

IDictionary<int, ConsumableItemRecipeSheet.Row>,
ICollection<KeyValuePair<int, ConsumableItemRecipeSheet.Row>>,
IEnumerable<KeyValuePair<int, ConsumableItemRecipeSheet.Row>>, IEnumerable,
ISheet

Inherited Members

Sheet<int, ConsumableItemRecipeSheet.Row>.Name ,
Sheet<int, ConsumableItemRecipeSheet.Row>.OrderedList ,
Sheet<int, ConsumableItemRecipeSheet.Row>.First ,
Sheet<int, ConsumableItemRecipeSheet.Row>.Last ,
Sheet<int, ConsumableItemRecipeSheet.Row>.Keys ,
Sheet<int, ConsumableItemRecipeSheet.Row>.Values ,
Sheet<int, ConsumableItemRecipeSheet.Row>.Count ,
Sheet<int, ConsumableItemRecipeSheet.Row>.IsReadOnly ,
Sheet<int, ConsumableItemRecipeSheet.Row>.this[int] ,
Sheet<int, ConsumableItemRecipeSheet.Row>.Set(string, bool) ,
Sheet<int, ConsumableItemRecipeSheet.Row>.Set<T>(Sheet<int, T>, bool) ,
Sheet<int, ConsumableItemRecipeSheet.Row>.GetEnumerator() ,
Sheet<int, ConsumableItemRecipeSheet.Row>.TryGetValue(int, out
ConsumableItemRecipeSheet.Row, bool) ,
Sheet<int, ConsumableItemRecipeSheet.Row>.AddRow(int,
ConsumableItemRecipeSheet.Row) ,
Sheet<int, ConsumableItemRecipeSheet.Row>.Add(int, ConsumableItemRecipeSheet.Row)
,
Sheet<int, ConsumableItemRecipeSheet.Row>.ContainsKey(int) ,

Sheet<int, ConsumableItemRecipeSheet.Row>.Remove(int) ,
Sheet<int, ConsumableItemRecipeSheet.Row>.TryGetValue(int, out
ConsumableItemRecipeSheet.Row) ,
Sheet<int, ConsumableItemRecipeSheet.Row>.Add(KeyValuePair<int,
ConsumableItemRecipeSheet.Row>) ,
Sheet<int, ConsumableItemRecipeSheet.Row>.Clear() ,
Sheet<int, ConsumableItemRecipeSheet.Row>.Contains(KeyValuePair<int,
ConsumableItemRecipeSheet.Row>) ,
Sheet<int, ConsumableItemRecipeSheet.Row>.CopyTo(KeyValuePair<int,
ConsumableItemRecipeSheet.Row>[], int) ,
Sheet<int, ConsumableItemRecipeSheet.Row>.Remove(KeyValuePair<int,
ConsumableItemRecipeSheet.Row>) ,
Sheet<int, ConsumableItemRecipeSheet.Row>.Serialize() , object.Equals(object) ,
object.Equals(object, object) , object.GetHashCode() , object.GetType() ,
object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

ConsumableItemRecipeSheet()

```
public ConsumableItemRecipeSheet()
```

Class ConsumableItemRecipeSheet.Row

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class ConsumableItemRecipeSheet.Row : SheetRow<int>
```

Inheritance

object ← SheetRow<int> ← ConsumableItemRecipeSheet.Row

Derived

EventConsumableItemRecipeSheet.Row

Inherited Members

SheetRow<int>.Validate() , SheetRow<int>.EndOfSheetInitialize() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Properties

Id

```
public int Id { get; }
```

Property Value

int

Key

```
public override int Key { get; }
```

Property Value

int

Materials

```
public List<EquipmentItemSubRecipeSheet.MaterialInfo> Materials { get; }
```

Property Value

List<EquipmentItemSubRecipeSheet.MaterialInfo>

RequiredActionPoint

```
public int RequiredActionPoint { get; }
```

Property Value

int

RequiredBlockIndex

```
public long RequiredBlockIndex { get; }
```

Property Value

long

RequiredGold

```
public decimal RequiredGold { get; }
```

Property Value

decimal

ResultConsumableItemId

```
public int ResultConsumableItemId { get; }
```

Property Value

int

Methods

GetAllMaterials()

```
public IEnumerable<EquipmentItemSubRecipeSheet.MaterialInfo> GetAllMaterials()
```

Returns

IEnumerable<EquipmentItemSubRecipeSheet.MaterialInfo>

Set(IReadOnlyList<string>)

```
public override void Set(IReadOnlyList<string> fields)
```

Parameters

`fields` IReadOnlyList<string>

Class ConsumableItemSheet

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class ConsumableItemSheet : Sheet<int, ConsumableItemSheet.Row>,
IDictionary<int, ConsumableItemSheet.Row>, ICollection<KeyValuePair<int,
ConsumableItemSheet.Row>>, IEnumerable<KeyValuePair<int, ConsumableItemSheet.Row>>,
IEnumerable, ISheet
```

Inheritance

object ← Sheet<int, ConsumableItemSheet.Row> ← ConsumableItemSheet

Implements

IDictionary<int, ConsumableItemSheet.Row>,
ICollection<KeyValuePair<int, ConsumableItemSheet.Row>>,
IEnumerable<KeyValuePair<int, ConsumableItemSheet.Row>>, IEnumerable, ISheet

Inherited Members

Sheet<int, ConsumableItemSheet.Row>.Name ,
Sheet<int, ConsumableItemSheet.Row>.OrderedList ,
Sheet<int, ConsumableItemSheet.Row>.First ,
Sheet<int, ConsumableItemSheet.Row>.Last ,
Sheet<int, ConsumableItemSheet.Row>.Keys ,
Sheet<int, ConsumableItemSheet.Row>.Values ,
Sheet<int, ConsumableItemSheet.Row>.Count ,
Sheet<int, ConsumableItemSheet.Row>.IsReadOnly ,
Sheet<int, ConsumableItemSheet.Row>.this[int] ,
Sheet<int, ConsumableItemSheet.Row>.Set(string, bool) ,
Sheet<int, ConsumableItemSheet.Row>.Set<T>(Sheet<int, T>, bool) ,
Sheet<int, ConsumableItemSheet.Row>.GetEnumerator() ,
Sheet<int, ConsumableItemSheet.Row>.TryGetValue(int, out ConsumableItemSheet.Row, bool) ,
Sheet<int, ConsumableItemSheet.Row>.AddRow(int, ConsumableItemSheet.Row) ,
Sheet<int, ConsumableItemSheet.Row>.Add(int, ConsumableItemSheet.Row) ,
Sheet<int, ConsumableItemSheet.Row>.ContainsKey(int) ,
Sheet<int, ConsumableItemSheet.Row>.Remove(int) ,
Sheet<int, ConsumableItemSheet.Row>.TryGetValue(int, out ConsumableItemSheet.Row) ,

Sheet<int, ConsumableItemSheet.Row>.Add(KeyValuePair<int, ConsumableItemSheet.Row>) , Sheet<int, ConsumableItemSheet.Row>.Clear() , Sheet<int, ConsumableItemSheet.Row>.Contains(KeyValuePair<int, ConsumableItemSheet.Row>) , Sheet<int, ConsumableItemSheet.Row>.CopyTo(KeyValuePair<int, ConsumableItemSheet.Row>[], int) , Sheet<int, ConsumableItemSheet.Row>.Remove(KeyValuePair<int, ConsumableItemSheet.Row>) , Sheet<int, ConsumableItemSheet.Row>.Serialize() , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

ConsumableItemSheet()

```
public ConsumableItemSheet()
```

Methods

AddRow(int, Row)

```
protected override void AddRow(int key, ConsumableItemSheet.Row value)
```

Parameters

`key` int

`value` ConsumableItemSheet.Row

Class ConsumableItemSheet.Row

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class ConsumableItemSheet.Row : ItemSheet.Row, IState
```

Inheritance

object ← SheetRow<int> ← ItemSheet.RowBase ← ItemSheet.Row ← ConsumableItemSheet.Row

Implements

IState

Inherited Members

ItemSheet.Row.Key , ItemSheet.Row.Id , ItemSheet.Row.ItemSubType , ItemSheet.Row.Grade , ItemSheet.Row.ElementalType , SheetRow<int>.Validate() , SheetRow<int>.EndOfSheetInitialize() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

Row()

```
public Row()
```

Row(Dictionary)

```
public Row(Dictionary serialized)
```

Parameters

`serialized` Dictionary

Properties

ItemType

```
public override ItemType ItemType { get; }
```

Property Value

ItemType

Stats

```
public List<DecimalStat> Stats { get; }
```

Property Value

List<DecimalStat>

Methods

Deserialize(Dictionary)

```
public static ConsumableItemSheet.Row Deserialize(Dictionary serialized)
```

Parameters

`serialized` Dictionary

Returns

ConsumableItemSheet.Row

Serialize()

```
public override IValue Serialize()
```

Returns

IValue

Set(IReadOnlyList<string>)

```
public override void Set(IReadOnlyList<string> fields)
```

Parameters

`fields` IReadOnlyList<string>

Class CostumeItemSheet

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class CostumeItemSheet : Sheet<int, CostumeItemSheet.Row>, IDictionary<int,
CostumeItemSheet.Row>, ICollection<KeyValuePair<int, CostumeItemSheet.Row>>,
IEnumerable<KeyValuePair<int, CostumeItemSheet.Row>>, IEnumerable, ISheet
```

Inheritance

object ← Sheet<int, CostumeItemSheet.Row> ← CostumeItemSheet

Implements

IDictionary<int, CostumeItemSheet.Row>,
ICollection<KeyValuePair<int, CostumeItemSheet.Row>>,
IEnumerable<KeyValuePair<int, CostumeItemSheet.Row>>, IEnumerable, ISheet

Inherited Members

Sheet<int, CostumeItemSheet.Row>.Name ,
Sheet<int, CostumeItemSheet.Row>.OrderedList ,
Sheet<int, CostumeItemSheet.Row>.First , Sheet<int, CostumeItemSheet.Row>.Last ,
Sheet<int, CostumeItemSheet.Row>.Keys , Sheet<int, CostumeItemSheet.Row>.Values ,
Sheet<int, CostumeItemSheet.Row>.Count ,
Sheet<int, CostumeItemSheet.Row>.IsReadOnly ,
Sheet<int, CostumeItemSheet.Row>.this[int] ,
Sheet<int, CostumeItemSheet.Row>.Set(string, bool) ,
Sheet<int, CostumeItemSheet.Row>.Set<T>(Sheet<int, T>, bool) ,
Sheet<int, CostumeItemSheet.Row>.GetEnumerator() ,
Sheet<int, CostumeItemSheet.Row>.TryGetValue(int, out CostumeItemSheet.Row, bool) ,
Sheet<int, CostumeItemSheet.Row>.AddRow(int, CostumeItemSheet.Row) ,
Sheet<int, CostumeItemSheet.Row>.Add(int, CostumeItemSheet.Row) ,
Sheet<int, CostumeItemSheet.Row>.ContainsKey(int) ,
Sheet<int, CostumeItemSheet.Row>.Remove(int) ,
Sheet<int, CostumeItemSheet.Row>.TryGetValue(int, out CostumeItemSheet.Row) ,
Sheet<int, CostumeItemSheet.Row>.Add(KeyValuePair<int, CostumeItemSheet.Row>) ,
Sheet<int, CostumeItemSheet.Row>.Clear() ,
Sheet<int, CostumeItemSheet.Row>.Contains(KeyValuePair<int, CostumeItemSheet.Row>)
,

Sheet<int, CostumeItemSheet.Row>.CopyTo(KeyValuePair<int, CostumeItemSheet.Row>[], int) ,
Sheet<int, CostumeItemSheet.Row>.Remove(KeyValuePair<int, CostumeItemSheet.Row>) ,
Sheet<int, CostumeItemSheet.Row>.Serialize() , object.Equals(object) ,
object.Equals(object, object) , object.GetHashCode() , object.GetType() ,
object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

CostumeItemSheet()

```
public CostumeItemSheet()
```

Class CostumeItemSheet.Row

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class CostumeItemSheet.Row : ItemSheet.Row
```

Inheritance

object ← SheetRow<int> ← ItemSheet.RowBase ← ItemSheet.Row ← CostumeItemSheet.Row

Inherited Members

ItemSheet.Row.Key , ItemSheet.Row.Id , ItemSheet.Row.ItemSubType , ItemSheet.Row.Grade , ItemSheet.Row.ElementalType , SheetRow<int>.Validate() , SheetRow<int>.EndOfSheetInitialize() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

Row()

```
public Row()
```

Row(Dictionary)

```
public Row(Dictionary serialized)
```

Parameters

`serialized` Dictionary

Properties

ItemType

```
public override ItemType ItemType { get; }
```

Property Value

ItemType

SpineResourcePath

```
public string SpineResourcePath { get; }
```

Property Value

string

Methods

Deserialize(Dictionary)

```
public static CostumeItemSheet.Row Deserialize(Dictionary serialized)
```

Parameters

`serialized` Dictionary

Returns

CostumeItemSheet.Row

Serialize()

```
public override IValue Serialize()
```

Returns

IValue

Set(IReadOnlyList<string>)

```
public override void Set(IReadOnlyList<string> fields)
```

Parameters

`fields` IReadOnlyList<string>

Class CostumeStatSheet

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class CostumeStatSheet : Sheet<int, CostumeStatSheet.Row>, IDictionary<int,
CostumeStatSheet.Row>, ICollection<KeyValuePair<int, CostumeStatSheet.Row>>,
IEnumerable<KeyValuePair<int, CostumeStatSheet.Row>>, IEnumerable, ISheet
```

Inheritance

object ← Sheet<int, CostumeStatSheet.Row> ← CostumeStatSheet

Implements

IDictionary<int, CostumeStatSheet.Row>,
ICollection<KeyValuePair<int, CostumeStatSheet.Row>>,
IEnumerable<KeyValuePair<int, CostumeStatSheet.Row>>, IEnumerable, ISheet

Inherited Members

Sheet<int, CostumeStatSheet.Row>.Name ,
Sheet<int, CostumeStatSheet.Row>.OrderedList ,
Sheet<int, CostumeStatSheet.Row>.First , Sheet<int, CostumeStatSheet.Row>.Last ,
Sheet<int, CostumeStatSheet.Row>.Keys , Sheet<int, CostumeStatSheet.Row>.Values ,
Sheet<int, CostumeStatSheet.Row>.Count ,
Sheet<int, CostumeStatSheet.Row>.IsReadOnly ,
Sheet<int, CostumeStatSheet.Row>.this[int] ,
Sheet<int, CostumeStatSheet.Row>.Set(string, bool) ,
Sheet<int, CostumeStatSheet.Row>.Set<T>(Sheet<int, T>, bool) ,
Sheet<int, CostumeStatSheet.Row>.GetEnumerator() ,
Sheet<int, CostumeStatSheet.Row>.TryGetValue(int, out CostumeStatSheet.Row, bool) ,
Sheet<int, CostumeStatSheet.Row>.AddRow(int, CostumeStatSheet.Row) ,
Sheet<int, CostumeStatSheet.Row>.Add(int, CostumeStatSheet.Row) ,
Sheet<int, CostumeStatSheet.Row>.ContainsKey(int) ,
Sheet<int, CostumeStatSheet.Row>.Remove(int) ,
Sheet<int, CostumeStatSheet.Row>.TryGetValue(int, out CostumeStatSheet.Row) ,
Sheet<int, CostumeStatSheet.Row>.Add(KeyValuePair<int, CostumeStatSheet.Row>) ,
Sheet<int, CostumeStatSheet.Row>.Clear() ,
Sheet<int, CostumeStatSheet.Row>.Contains(KeyValuePair<int, CostumeStatSheet.Row>) ,

Sheet<int, CostumeStatSheet.Row>.CopyTo(KeyValuePair<int, CostumeStatSheet.Row>[], int) ,
Sheet<int, CostumeStatSheet.Row>.Remove(KeyValuePair<int, CostumeStatSheet.Row>) ,
Sheet<int, CostumeStatSheet.Row>.Serialize() , object.Equals(object) ,
object.Equals(object, object) , object.GetHashCode() , object.GetType() ,
object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

CostumeStatSheet()

```
public CostumeStatSheet()
```

Class CostumeStatSheet.Row

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class CostumeStatSheet.Row : SheetRow<int>
```

Inheritance

object ← SheetRow<int> ← CostumeStatSheet.Row

Inherited Members

SheetRow<int>.Validate() , SheetRow<int>.EndOfSheetInitialize() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Properties

CostumeId

```
public int CostumeId { get; }
```

Property Value

int

Id

```
public int Id { get; }
```

Property Value

int

Key

```
public override int Key { get; }
```

Property Value

int

Stat

```
public decimal Stat { get; }
```

Property Value

decimal

StatType

```
public StatType StatType { get; }
```

Property Value

StatType

Methods

Set(IReadOnlyList<string>)

```
public override void Set(IReadOnlyList<string> fields)
```

Parameters

`fields` IReadOnlyList<string>

Class CreateAvatarFavSheet

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
public class CreateAvatarFavSheet : Sheet<string, CreateAvatarFavSheet.Row>,
IDictionary<string, CreateAvatarFavSheet.Row>, ICollection<KeyValuePair<string,
CreateAvatarFavSheet.Row>>, IEnumerable<KeyValuePair<string,
CreateAvatarFavSheet.Row>>, IEnumerable, ISheet
```

Inheritance

object ← Sheet<string, CreateAvatarFavSheet.Row> ← CreateAvatarFavSheet

Implements

IDictionary<string, CreateAvatarFavSheet.Row>,
ICollection<KeyValuePair<string, CreateAvatarFavSheet.Row>>,
IEnumerable<KeyValuePair<string, CreateAvatarFavSheet.Row>>, IEnumerable,
ISheet

Inherited Members

Sheet<string, CreateAvatarFavSheet.Row>.Name ,
Sheet<string, CreateAvatarFavSheet.Row>.OrderedList ,
Sheet<string, CreateAvatarFavSheet.Row>.First ,
Sheet<string, CreateAvatarFavSheet.Row>.Last ,
Sheet<string, CreateAvatarFavSheet.Row>.Keys ,
Sheet<string, CreateAvatarFavSheet.Row>.Values ,
Sheet<string, CreateAvatarFavSheet.Row>.Count ,
Sheet<string, CreateAvatarFavSheet.Row>.IsReadOnly ,
Sheet<string, CreateAvatarFavSheet.Row>.this[string] ,
Sheet<string, CreateAvatarFavSheet.Row>.Set(string, bool) ,
Sheet<string, CreateAvatarFavSheet.Row>.Set<T>(Sheet<string, T>, bool) ,
Sheet<string, CreateAvatarFavSheet.Row>.GetEnumerator() ,
Sheet<string, CreateAvatarFavSheet.Row>.TryGetValue(string, out
CreateAvatarFavSheet.Row, bool) ,
Sheet<string, CreateAvatarFavSheet.Row>.AddRow(string, CreateAvatarFavSheet.Row) ,
Sheet<string, CreateAvatarFavSheet.Row>.Add(string, CreateAvatarFavSheet.Row) ,
Sheet<string, CreateAvatarFavSheet.Row>.ContainsKey(string) ,
Sheet<string, CreateAvatarFavSheet.Row>.Remove(string) ,

Sheet<string, CreateAvatarFavSheet.Row>.TryGetValue(string, out CreateAvatarFavSheet.Row) , Sheet<string, CreateAvatarFavSheet.Row>.Add(KeyValuePair<string, CreateAvatarFavSheet.Row>) , Sheet<string, CreateAvatarFavSheet.Row>.Clear() , Sheet<string, CreateAvatarFavSheet.Row>.Contains(KeyValuePair<string, CreateAvatarFavSheet.Row>) , Sheet<string, CreateAvatarFavSheet.Row>.CopyTo(KeyValuePair<string, CreateAvatarFavSheet.Row>[], int) , Sheet<string, CreateAvatarFavSheet.Row>.Remove(KeyValuePair<string, CreateAvatarFavSheet.Row>) , Sheet<string, CreateAvatarFavSheet.Row>.Serialize() , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

CreateAvatarFavSheet()

```
public CreateAvatarFavSheet()
```

Class CreateAvatarFavSheet.Row

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
public class CreateAvatarFavSheet.Row : SheetRow<string>
```

Inheritance

object ← SheetRow<string> ← CreateAvatarFavSheet.Row

Inherited Members

SheetRow<string>.Validate() , SheetRow<string>.EndOfSheetInitialize() , SheetRow<string>.Equals(object) , SheetRow<string>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Properties

Currency

```
public Currency Currency { get; }
```

Property Value

Currency

Key

```
public override string Key { get; }
```

Property Value

string

Quantity

```
public int Quantity { get; }
```

Property Value

int

Target

```
public CreateAvatarFavSheet.Target Target { get; }
```

Property Value

CreateAvatarFavSheet.Target

Methods

Set(IReadOnlyList<string>)

```
public override void Set(IReadOnlyList<string> fields)
```

Parameters

`fields` IReadOnlyList<string>

Enum CreateAvatarFavSheet.Target

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
public enum CreateAvatarFavSheet.Target
```

Extension Methods
StateExtensions.Serialize(Enum)

Fields

`Agent = 0`

`Avatar = 1`

Class CreateAvatarItemSheet

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
public class CreateAvatarItemSheet : Sheet<int, CreateAvatarItemSheet.Row>,
IDictionary<int, CreateAvatarItemSheet.Row>, ICollection<KeyValuePair<int,
CreateAvatarItemSheet.Row>>, IEnumerable<KeyValuePair<int,
CreateAvatarItemSheet.Row>>, IEnumerable, ISheet
```

Inheritance

object ← Sheet<int, CreateAvatarItemSheet.Row> ← CreateAvatarItemSheet

Implements

IDictionary<int, CreateAvatarItemSheet.Row>,
ICollection<KeyValuePair<int, CreateAvatarItemSheet.Row>>,
IEnumerable<KeyValuePair<int, CreateAvatarItemSheet.Row>>, IEnumerable,
ISheet

Inherited Members

Sheet<int, CreateAvatarItemSheet.Row>.Name ,
Sheet<int, CreateAvatarItemSheet.Row>.OrderedList ,
Sheet<int, CreateAvatarItemSheet.Row>.First ,
Sheet<int, CreateAvatarItemSheet.Row>.Last ,
Sheet<int, CreateAvatarItemSheet.Row>.Keys ,
Sheet<int, CreateAvatarItemSheet.Row>.Values ,
Sheet<int, CreateAvatarItemSheet.Row>.Count ,
Sheet<int, CreateAvatarItemSheet.Row>.IsReadOnly ,
Sheet<int, CreateAvatarItemSheet.Row>.this[int] ,
Sheet<int, CreateAvatarItemSheet.Row>.Set(string, bool) ,
Sheet<int, CreateAvatarItemSheet.Row>.Set<T>(Sheet<int, T>, bool) ,
Sheet<int, CreateAvatarItemSheet.Row>.GetEnumerator() ,
Sheet<int, CreateAvatarItemSheet.Row>.TryGetValue(int, out CreateAvatarItemSheet.Row,
bool) ,
Sheet<int, CreateAvatarItemSheet.Row>.AddRow(int, CreateAvatarItemSheet.Row) ,
Sheet<int, CreateAvatarItemSheet.Row>.Add(int, CreateAvatarItemSheet.Row) ,
Sheet<int, CreateAvatarItemSheet.Row>.ContainsKey(int) ,
Sheet<int, CreateAvatarItemSheet.Row>.Remove(int) ,

Sheet<int, CreateAvatarItemSheet.Row>.TryGetValue(int, out CreateAvatarItemSheet.Row) , Sheet<int, CreateAvatarItemSheet.Row>.Add(KeyValuePair<int, CreateAvatarItemSheet.Row>) , Sheet<int, CreateAvatarItemSheet.Row>.Clear() , Sheet<int, CreateAvatarItemSheet.Row>.Contains(KeyValuePair<int, CreateAvatarItemSheet.Row>) , Sheet<int, CreateAvatarItemSheet.Row>.CopyTo(KeyValuePair<int, CreateAvatarItemSheet.Row>[], int) , Sheet<int, CreateAvatarItemSheet.Row>.Remove(KeyValuePair<int, CreateAvatarItemSheet.Row>) , Sheet<int, CreateAvatarItemSheet.Row>.Serialize() , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

CreateAvatarItemSheet()

```
public CreateAvatarItemSheet()
```

Class CreateAvatarItemSheet.Row

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class CreateAvatarItemSheet.Row : SheetRow<int>
```

Inheritance

object ← SheetRow<int> ← CreateAvatarItemSheet.Row

Inherited Members

SheetRow<int>.Validate() , SheetRow<int>.EndOfSheetInitialize() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Properties

Count

```
public int Count { get; }
```

Property Value

int

ItemId

```
public int ItemId { get; }
```

Property Value

int

Key

```
public override int Key { get; }
```

Property Value

int

Methods

Set(IReadOnlyList<string>)

```
public override void Set(IReadOnlyList<string> fields)
```

Parameters

`fields` IReadOnlyList<string>

Class EnemySkillSheet

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class EnemySkillSheet : Sheet<int, EnemySkillSheet.Row>, IDictionary<int,
EnemySkillSheet.Row>, ICollection<KeyValuePair<int, EnemySkillSheet.Row>>,
IEnumerable<KeyValuePair<int, EnemySkillSheet.Row>>, IEnumerable, ISheet
```

Inheritance

object ← Sheet<int, EnemySkillSheet.Row> ← EnemySkillSheet

Implements

IDictionary<int, EnemySkillSheet.Row>,
ICollection<KeyValuePair<int, EnemySkillSheet.Row>>,
IEnumerable<KeyValuePair<int, EnemySkillSheet.Row>>, IEnumerable, ISheet

Inherited Members

Sheet<int, EnemySkillSheet.Row>.Name , Sheet<int, EnemySkillSheet.Row>.OrderedList ,
Sheet<int, EnemySkillSheet.Row>.First , Sheet<int, EnemySkillSheet.Row>.Last ,
Sheet<int, EnemySkillSheet.Row>.Keys , Sheet<int, EnemySkillSheet.Row>.Values ,
Sheet<int, EnemySkillSheet.Row>.Count , Sheet<int, EnemySkillSheet.Row>.IsReadOnly ,
Sheet<int, EnemySkillSheet.Row>.this[int] ,
Sheet<int, EnemySkillSheet.Row>.Set(string, bool) ,
Sheet<int, EnemySkillSheet.Row>.Set<T>(Sheet<int, T>, bool) ,
Sheet<int, EnemySkillSheet.Row>.GetEnumerator() ,
Sheet<int, EnemySkillSheet.Row>.TryGetValue(int, out EnemySkillSheet.Row, bool) ,
Sheet<int, EnemySkillSheet.Row>.AddRow(int, EnemySkillSheet.Row) ,
Sheet<int, EnemySkillSheet.Row>.Add(int, EnemySkillSheet.Row) ,
Sheet<int, EnemySkillSheet.Row>.ContainsKey(int) ,
Sheet<int, EnemySkillSheet.Row>.Remove(int) ,
Sheet<int, EnemySkillSheet.Row>.TryGetValue(int, out EnemySkillSheet.Row) ,
Sheet<int, EnemySkillSheet.Row>.Add(KeyValuePair<int, EnemySkillSheet.Row>) ,
Sheet<int, EnemySkillSheet.Row>.Clear() ,
Sheet<int, EnemySkillSheet.Row>.Contains(KeyValuePair<int, EnemySkillSheet.Row>) ,
Sheet<int, EnemySkillSheet.Row>.CopyTo(KeyValuePair<int, EnemySkillSheet.Row>[], int) ,
Sheet<int, EnemySkillSheet.Row>.Remove(KeyValuePair<int, EnemySkillSheet.Row>) ,
Sheet<int, EnemySkillSheet.Row>.Serialize() , object.Equals(object) ,

object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

EnemySkillSheet()

```
public EnemySkillSheet()
```

Class EnemySkillSheet.Row

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class EnemySkillSheet.Row : SheetRow<int>
```

Inheritance

object ← SheetRow<int> ← EnemySkillSheet.Row

Inherited Members

SheetRow<int>.Validate() , SheetRow<int>.EndOfSheetInitialize() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Fields

characterId

```
public int characterId
```

Field Value

int

id

```
public int id
```

Field Value

int

skillId

```
public int skillId
```

Field Value

int

Properties

Key

```
public override int Key { get; }
```

Property Value

int

Methods

Set(IReadOnlyList<string>)

```
public override void Set(IReadOnlyList<string> fields)
```

Parameters

`fields` IReadOnlyList<string>

Class EnhancementCostSheet

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
public class EnhancementCostSheet : Sheet<int, EnhancementCostSheet.Row>,
IDictionary<int, EnhancementCostSheet.Row>, ICollection<KeyValuePair<int,
EnhancementCostSheet.Row>>, IEnumerable<KeyValuePair<int,
EnhancementCostSheet.Row>>, IEnumerable, ISheet
```

Inheritance

object ← Sheet<int, EnhancementCostSheet.Row> ← EnhancementCostSheet

Implements

IDictionary<int, EnhancementCostSheet.Row>,
ICollection<KeyValuePair<int, EnhancementCostSheet.Row>>,
IEnumerable<KeyValuePair<int, EnhancementCostSheet.Row>>, IEnumerable,
ISheet

Inherited Members

Sheet<int, EnhancementCostSheet.Row>.Name ,
Sheet<int, EnhancementCostSheet.Row>.OrderedList ,
Sheet<int, EnhancementCostSheet.Row>.First ,
Sheet<int, EnhancementCostSheet.Row>.Last ,
Sheet<int, EnhancementCostSheet.Row>.Keys ,
Sheet<int, EnhancementCostSheet.Row>.Values ,
Sheet<int, EnhancementCostSheet.Row>.Count ,
Sheet<int, EnhancementCostSheet.Row>.IsReadOnly ,
Sheet<int, EnhancementCostSheet.Row>.this[int] ,
Sheet<int, EnhancementCostSheet.Row>.Set(string, bool) ,
Sheet<int, EnhancementCostSheet.Row>.Set<T>(Sheet<int, T>, bool) ,
Sheet<int, EnhancementCostSheet.Row>.GetEnumerator() ,
Sheet<int, EnhancementCostSheet.Row>.TryGetValue(int, out
EnhancementCostSheet.Row, bool) ,
Sheet<int, EnhancementCostSheet.Row>.AddRow(int, EnhancementCostSheet.Row) ,
Sheet<int, EnhancementCostSheet.Row>.Add(int, EnhancementCostSheet.Row) ,
Sheet<int, EnhancementCostSheet.Row>.ContainsKey(int) ,
Sheet<int, EnhancementCostSheet.Row>.Remove(int) ,

Sheet<int, EnhancementCostSheet.Row>.TryGetValue(int, out EnhancementCostSheet.Row) , Sheet<int, EnhancementCostSheet.Row>.Add(KeyValuePair<int, EnhancementCostSheet.Row>) , Sheet<int, EnhancementCostSheet.Row>.Clear() , Sheet<int, EnhancementCostSheet.Row>.Contains(KeyValuePair<int, EnhancementCostSheet.Row>) , Sheet<int, EnhancementCostSheet.Row>.CopyTo(KeyValuePair<int, EnhancementCostSheet.Row>[], int) , Sheet<int, EnhancementCostSheet.Row>.Remove(KeyValuePair<int, EnhancementCostSheet.Row>) , Sheet<int, EnhancementCostSheet.Row>.Serialize() , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

EnhancementCostSheet()

```
public EnhancementCostSheet()
```

Class EnhancementCostSheet.Row

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
public class EnhancementCostSheet.Row : SheetRow<int>
```

Inheritance

object ← SheetRow<int> ← EnhancementCostSheet.Row

Derived

EnhancementCostSheetV2.Row, EnhancementCostSheetV3.Row

Inherited Members

SheetRow<int>.Validate() , SheetRow<int>.EndOfSheetInitialize() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Properties

Cost

```
public BigInteger Cost { get; }
```

Property Value

BigInteger

Grade

```
public int Grade { get; }
```

Property Value

int

Id

```
public int Id { get; }
```

Property Value

int

ItemSubType

```
public ItemSubType ItemSubType { get; }
```

Property Value

ItemSubType

Key

```
public override int Key { get; }
```

Property Value

int

Level

```
public int Level { get; }
```

Property Value

int

Methods

Set(IReadOnlyList<string>)

```
public override void Set(IReadOnlyList<string> fields)
```

Parameters

`fields` IReadOnlyList<string>

Class EnhancementCostSheetV2

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
public class EnhancementCostSheetV2 : Sheet<int, EnhancementCostSheetV2.Row>,
IDictionary<int, EnhancementCostSheetV2.Row>, ICollection<KeyValuePair<int,
EnhancementCostSheetV2.Row>>, IEnumerable<KeyValuePair<int,
EnhancementCostSheetV2.Row>>, IEnumerable, ISheet
```

Inheritance

object ← Sheet<int, EnhancementCostSheetV2.Row> ← EnhancementCostSheetV2

Implements

IDictionary<int, EnhancementCostSheetV2.Row>,
ICollection<KeyValuePair<int, EnhancementCostSheetV2.Row>>,
IEnumerable<KeyValuePair<int, EnhancementCostSheetV2.Row>>, IEnumerable,
ISheet

Inherited Members

Sheet<int, EnhancementCostSheetV2.Row>.Name ,
Sheet<int, EnhancementCostSheetV2.Row>.OrderedList ,
Sheet<int, EnhancementCostSheetV2.Row>.First ,
Sheet<int, EnhancementCostSheetV2.Row>.Last ,
Sheet<int, EnhancementCostSheetV2.Row>.Keys ,
Sheet<int, EnhancementCostSheetV2.Row>.Values ,
Sheet<int, EnhancementCostSheetV2.Row>.Count ,
Sheet<int, EnhancementCostSheetV2.Row>.IsReadOnly ,
Sheet<int, EnhancementCostSheetV2.Row>.this[int] ,
Sheet<int, EnhancementCostSheetV2.Row>.Set(string, bool) ,
Sheet<int, EnhancementCostSheetV2.Row>.Set<T>(Sheet<int, T>, bool) ,
Sheet<int, EnhancementCostSheetV2.Row>.GetEnumerator() ,
Sheet<int, EnhancementCostSheetV2.Row>.TryGetValue(int, out
EnhancementCostSheetV2.Row, bool) ,
Sheet<int, EnhancementCostSheetV2.Row>.AddRow(int, EnhancementCostSheetV2.Row) ,
Sheet<int, EnhancementCostSheetV2.Row>.Add(int, EnhancementCostSheetV2.Row) ,
Sheet<int, EnhancementCostSheetV2.Row>.ContainsKey(int) ,
Sheet<int, EnhancementCostSheetV2.Row>.Remove(int) ,

Sheet<int, EnhancementCostSheetV2.Row>.TryGetValue(int, out EnhancementCostSheetV2.Row) , Sheet<int, EnhancementCostSheetV2.Row>.Add(KeyValuePair<int, EnhancementCostSheetV2.Row>) , Sheet<int, EnhancementCostSheetV2.Row>.Clear() , Sheet<int, EnhancementCostSheetV2.Row>.Contains(KeyValuePair<int, EnhancementCostSheetV2.Row>) , Sheet<int, EnhancementCostSheetV2.Row>.CopyTo(KeyValuePair<int, EnhancementCostSheetV2.Row>[], int) , Sheet<int, EnhancementCostSheetV2.Row>.Remove(KeyValuePair<int, EnhancementCostSheetV2.Row>) , Sheet<int, EnhancementCostSheetV2.Row>.Serialize() , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

EnhancementCostSheetV2()

```
public EnhancementCostSheetV2()
```

Class EnhancementCostSheetV2.Row

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
public class EnhancementCostSheetV2.Row : EnhancementCostSheet.Row
```

Inheritance

object ← SheetRow<int> ← EnhancementCostSheet.Row ← EnhancementCostSheetV2.Row

Inherited Members

EnhancementCostSheet.Row.Key , EnhancementCostSheet.Row.Id , EnhancementCostSheet.Row.ItemSubType , EnhancementCostSheet.Row.Grade , EnhancementCostSheet.Row.Level , EnhancementCostSheet.Row.Cost , SheetRow<int>.Validate() , SheetRow<int>.EndOfSheetInitialize() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Properties

BaseStatGrowthMax

```
public int BaseStatGrowthMax { get; }
```

Property Value

int

BaseStatGrowthMin

```
public int BaseStatGrowthMin { get; }
```

Property Value

int

ExtraSkillChanceGrowthMax

```
public int ExtraSkillChanceGrowthMax { get; }
```

Property Value

int

ExtraSkillChanceGrowthMin

```
public int ExtraSkillChanceGrowthMin { get; }
```

Property Value

int

ExtraSkillDamageGrowthMax

```
public int ExtraSkillDamageGrowthMax { get; }
```

Property Value

int

ExtraSkillDamageGrowthMin

```
public int ExtraSkillDamageGrowthMin { get; }
```

Property Value

int

ExtraStatGrowthMax

```
public int ExtraStatGrowthMax { get; }
```

Property Value

int

ExtraStatGrowthMin

```
public int ExtraStatGrowthMin { get; }
```

Property Value

int

FailRatio

```
public int FailRatio { get; }
```

Property Value

int

FailRequiredBlockIndex

```
public int FailRequiredBlockIndex { get; }
```

Property Value

int

GreatSuccessRatio

```
public int GreatSuccessRatio { get; }
```

Property Value

int

GreatSuccessRequiredBlockIndex

```
public int GreatSuccessRequiredBlockIndex { get; }
```

Property Value

int

SuccessRatio

```
public int SuccessRatio { get; }
```

Property Value

int

SuccessRequiredBlockIndex

```
public int SuccessRequiredBlockIndex { get; }
```

Property Value

int

Methods

Set(IReadOnlyList<string>)

```
public override void Set(IReadOnlyList<string> fields)
```

Parameters

`fields` IReadOnlyList<string>

Class EnhancementCostSheetV3

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
public class EnhancementCostSheetV3 : Sheet<int, EnhancementCostSheetV3.Row>,
IDictionary<int, EnhancementCostSheetV3.Row>, ICollection<KeyValuePair<int,
EnhancementCostSheetV3.Row>>, IEnumerable<KeyValuePair<int,
EnhancementCostSheetV3.Row>>, IEnumerable, ISheet
```

Inheritance

object ← Sheet<int, EnhancementCostSheetV3.Row> ← EnhancementCostSheetV3

Implements

IDictionary<int, EnhancementCostSheetV3.Row>,
ICollection<KeyValuePair<int, EnhancementCostSheetV3.Row>>,
IEnumerable<KeyValuePair<int, EnhancementCostSheetV3.Row>>, IEnumerable,
ISheet

Inherited Members

Sheet<int, EnhancementCostSheetV3.Row>.Name ,
Sheet<int, EnhancementCostSheetV3.Row>.OrderedList ,
Sheet<int, EnhancementCostSheetV3.Row>.First ,
Sheet<int, EnhancementCostSheetV3.Row>.Last ,
Sheet<int, EnhancementCostSheetV3.Row>.Keys ,
Sheet<int, EnhancementCostSheetV3.Row>.Values ,
Sheet<int, EnhancementCostSheetV3.Row>.Count ,
Sheet<int, EnhancementCostSheetV3.Row>.IsReadOnly ,
Sheet<int, EnhancementCostSheetV3.Row>.this[int] ,
Sheet<int, EnhancementCostSheetV3.Row>.Set(string, bool) ,
Sheet<int, EnhancementCostSheetV3.Row>.Set<T>(Sheet<int, T>, bool) ,
Sheet<int, EnhancementCostSheetV3.Row>.GetEnumerator() ,
Sheet<int, EnhancementCostSheetV3.Row>.TryGetValue(int, out
EnhancementCostSheetV3.Row, bool) ,
Sheet<int, EnhancementCostSheetV3.Row>.AddRow(int, EnhancementCostSheetV3.Row) ,
Sheet<int, EnhancementCostSheetV3.Row>.Add(int, EnhancementCostSheetV3.Row) ,
Sheet<int, EnhancementCostSheetV3.Row>.ContainsKey(int) ,
Sheet<int, EnhancementCostSheetV3.Row>.Remove(int) ,

Sheet<int, EnhancementCostSheetV3.Row>.TryGetValue(int, out EnhancementCostSheetV3.Row) , Sheet<int, EnhancementCostSheetV3.Row>.Add(KeyValuePair<int, EnhancementCostSheetV3.Row>) , Sheet<int, EnhancementCostSheetV3.Row>.Clear() , Sheet<int, EnhancementCostSheetV3.Row>.Contains(KeyValuePair<int, EnhancementCostSheetV3.Row>) , Sheet<int, EnhancementCostSheetV3.Row>.CopyTo(KeyValuePair<int, EnhancementCostSheetV3.Row>[], int) , Sheet<int, EnhancementCostSheetV3.Row>.Remove(KeyValuePair<int, EnhancementCostSheetV3.Row>) , Sheet<int, EnhancementCostSheetV3.Row>.Serialize() , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

EnhancementCostSheetV3()

```
public EnhancementCostSheetV3()
```

Methods

GetHammerExp(int)

```
public long GetHammerExp(int hammerId)
```

Parameters

`hammerId` int

Returns

long

Class EnhancementCostSheetV3.Row

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
public class EnhancementCostSheetV3.Row : EnhancementCostSheet.Row
```

Inheritance

object ← SheetRow<int> ← EnhancementCostSheet.Row ← EnhancementCostSheetV3.Row

Inherited Members

EnhancementCostSheet.Row.Key , EnhancementCostSheet.Row.Id , EnhancementCostSheet.Row.ItemSubType , EnhancementCostSheet.Row.Grade , EnhancementCostSheet.Row.Level , EnhancementCostSheet.Row.Cost , SheetRow<int>.Validate() , SheetRow<int>.EndOfSheetInitialize() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Properties

BaseStatGrowthMax

```
public int BaseStatGrowthMax { get; }
```

Property Value

int

BaseStatGrowthMin

```
public int BaseStatGrowthMin { get; }
```

Property Value

int

Exp

```
public long Exp { get; }
```

Property Value

long

ExtraSkillChanceGrowthMax

```
public int ExtraSkillChanceGrowthMax { get; }
```

Property Value

int

ExtraSkillChanceGrowthMin

```
public int ExtraSkillChanceGrowthMin { get; }
```

Property Value

int

ExtraSkillDamageGrowthMax

```
public int ExtraSkillDamageGrowthMax { get; }
```

Property Value

int

ExtraSkillDamageGrowthMin

```
public int ExtraSkillDamageGrowthMin { get; }
```

Property Value

int

ExtraStatGrowthMax

```
public int ExtraStatGrowthMax { get; }
```

Property Value

int

ExtraStatGrowthMin

```
public int ExtraStatGrowthMin { get; }
```

Property Value

int

RequiredBlockIndex

```
public int RequiredBlockIndex { get; }
```

Property Value

int

Methods

Set(IReadOnlyList<string>)

```
public override void Set(IReadOnlyList<string> fields)
```

Parameters

`fields` IReadOnlyList<string>

Class EquipmentItemOptionSheet

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

Stat and skill options that crafting can add to equipment.

```
public class EquipmentItemOptionSheet : Sheet<int, EquipmentItemOptionSheet.Row>,
IDictionary<int, EquipmentItemOptionSheet.Row>, ICollection<KeyValuePair<int,
EquipmentItemOptionSheet.Row>>, IEnumerable<KeyValuePair<int,
EquipmentItemOptionSheet.Row>>, IEnumerable, ISheet
```

Inheritance

object ← Sheet<int, EquipmentItemOptionSheet.Row> ← EquipmentItemOptionSheet

Implements

IDictionary<int, EquipmentItemOptionSheet.Row>, ICollection<KeyValuePair<int, EquipmentItemOptionSheet.Row>>, IEnumerable<KeyValuePair<int, EquipmentItemOptionSheet.Row>>, IEnumerable, ISheet

Inherited Members

Sheet<int, EquipmentItemOptionSheet.Row>.Name ,
Sheet<int, EquipmentItemOptionSheet.Row>.OrderedList ,
Sheet<int, EquipmentItemOptionSheet.Row>.First ,
Sheet<int, EquipmentItemOptionSheet.Row>.Last ,
Sheet<int, EquipmentItemOptionSheet.Row>.Keys ,
Sheet<int, EquipmentItemOptionSheet.Row>.Values ,
Sheet<int, EquipmentItemOptionSheet.Row>.Count ,
Sheet<int, EquipmentItemOptionSheet.Row>.IsReadOnly ,
Sheet<int, EquipmentItemOptionSheet.Row>.this[int] ,
Sheet<int, EquipmentItemOptionSheet.Row>.Set(string, bool) ,
Sheet<int, EquipmentItemOptionSheet.Row>.Set<T>(Sheet<int, T>, bool) ,
Sheet<int, EquipmentItemOptionSheet.Row>.GetEnumerator() ,
Sheet<int, EquipmentItemOptionSheet.Row>.TryGetValue(int, out EquipmentItemOptionSheet.Row, bool) ,
Sheet<int, EquipmentItemOptionSheet.Row>.AddRow(int, EquipmentItemOptionSheet.Row) ,
Sheet<int, EquipmentItemOptionSheet.Row>.Add(int, EquipmentItemOptionSheet.Row) ,
Sheet<int, EquipmentItemOptionSheet.Row>.ContainsKey(int) ,

Sheet<int, EquipmentItemOptionSheet.Row>.Remove(int) ,
Sheet<int, EquipmentItemOptionSheet.Row>.TryGetValue(int, out
EquipmentItemOptionSheet.Row) ,
Sheet<int, EquipmentItemOptionSheet.Row>.Add(KeyValuePair<int,
EquipmentItemOptionSheet.Row>) ,
Sheet<int, EquipmentItemOptionSheet.Row>.Clear() ,
Sheet<int, EquipmentItemOptionSheet.Row>.Contains(KeyValuePair<int,
EquipmentItemOptionSheet.Row>) ,
Sheet<int, EquipmentItemOptionSheet.Row>.CopyTo(KeyValuePair<int,
EquipmentItemOptionSheet.Row>[], int) ,
Sheet<int, EquipmentItemOptionSheet.Row>.Remove(KeyValuePair<int,
EquipmentItemOptionSheet.Row>) ,
Sheet<int, EquipmentItemOptionSheet.Row>.Serialize() , object.Equals(object) ,
object.Equals(object, object) , object.GetHashCode() , object.GetType() ,
object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

EquipmentItemOptionSheet()

Creates an empty sheet.

```
public EquipmentItemOptionSheet()
```

Class EquipmentItemOptionSheet.Row

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

One option and the ranges its values are drawn from.

```
public class EquipmentItemOptionSheet.Row : SheetRow<int>
```

Inheritance

object ← SheetRow<int> ← EquipmentItemOptionSheet.Row

Inherited Members

SheetRow<int>.Validate() , SheetRow<int>.EndOfSheetInitialize() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Properties

Id

The option id.

```
public int Id { get; }
```

Property Value

int

Key

```
public override int Key { get; }
```

Property Value

int

ReferencedStatType

The stat the stat damage ratio refers to.

```
public StatType ReferencedStatType { get; }
```

Property Value

StatType

SkillChanceMax

The inclusive upper bound of the skill chance.

```
public int SkillChanceMax { get; }
```

Property Value

int

SkillChanceMin

The inclusive lower bound of the skill chance.

```
public int SkillChanceMin { get; }
```

Property Value

int

SkillDamageMax

The inclusive upper bound of the skill power.

```
public long SkillDamageMax { get; }
```

Property Value

long

SkillDamageMin

```
public long SkillDamageMin { get; }
```

Property Value

long

SkillId

The skill this option grants, or 0 for a stat option.

```
public int SkillId { get; }
```

Property Value

int

StatDamageRatioMax

The inclusive upper bound of the stat damage ratio.

```
public int StatDamageRatioMax { get; }
```

Property Value

int

StatDamageRatioMin

The inclusive lower bound of the stat damage ratio.

```
public int StatDamageRatioMin { get; }
```

Property Value

int

StatMax

The inclusive upper bound of the stat value.

```
public long StatMax { get; }
```

Property Value

long

StatMin

The inclusive lower bound of the stat value.

```
public long StatMin { get; }
```

Property Value

long

StatType

The stat this option adds, or NONE for a skill option.

```
public StatType StatType { get; }
```

Property Value

StatType

Methods

Set(IReadOnlyList<string>)

```
public override void Set(IReadOnlyList<string> fields)
```

Parameters

`fields` IReadOnlyList<string>

Class EquipmentItemRecipeSheet

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

Equipment crafting recipes.

```
public class EquipmentItemRecipeSheet : Sheet<int, EquipmentItemRecipeSheet.Row>,
IDictionary<int, EquipmentItemRecipeSheet.Row>, ICollection<KeyValuePair<int,
EquipmentItemRecipeSheet.Row>>, IEnumerable<KeyValuePair<int,
EquipmentItemRecipeSheet.Row>>, IEnumerable, ISheet
```

Inheritance

object ← Sheet<int, EquipmentItemRecipeSheet.Row> ← EquipmentItemRecipeSheet

Implements

IDictionary<int, EquipmentItemRecipeSheet.Row>,
ICollection<KeyValuePair<int, EquipmentItemRecipeSheet.Row>>,
IEnumerable<KeyValuePair<int, EquipmentItemRecipeSheet.Row>>, IEnumerable,
ISheet

Inherited Members

Sheet<int, EquipmentItemRecipeSheet.Row>.Name ,
Sheet<int, EquipmentItemRecipeSheet.Row>.OrderedList ,
Sheet<int, EquipmentItemRecipeSheet.Row>.First ,
Sheet<int, EquipmentItemRecipeSheet.Row>.Last ,
Sheet<int, EquipmentItemRecipeSheet.Row>.Keys ,
Sheet<int, EquipmentItemRecipeSheet.Row>.Values ,
Sheet<int, EquipmentItemRecipeSheet.Row>.Count ,
Sheet<int, EquipmentItemRecipeSheet.Row>.IsReadOnly ,
Sheet<int, EquipmentItemRecipeSheet.Row>.this[int] ,
Sheet<int, EquipmentItemRecipeSheet.Row>.Set(string, bool) ,
Sheet<int, EquipmentItemRecipeSheet.Row>.Set<T>(Sheet<int, T>, bool) ,
Sheet<int, EquipmentItemRecipeSheet.Row>.GetEnumerator() ,
Sheet<int, EquipmentItemRecipeSheet.Row>.TryGetValue(int, out
EquipmentItemRecipeSheet.Row, bool) ,
Sheet<int, EquipmentItemRecipeSheet.Row>.AddRow(int, EquipmentItemRecipeSheet.Row)
,
Sheet<int, EquipmentItemRecipeSheet.Row>.Add(int, EquipmentItemRecipeSheet.Row) ,
Sheet<int, EquipmentItemRecipeSheet.Row>.ContainsKey(int) ,

Sheet<int, EquipmentItemRecipeSheet.Row>.Remove(int) ,
Sheet<int, EquipmentItemRecipeSheet.Row>.TryGetValue(int, out
EquipmentItemRecipeSheet.Row) ,
Sheet<int, EquipmentItemRecipeSheet.Row>.Add(KeyValuePair<int,
EquipmentItemRecipeSheet.Row>) ,
Sheet<int, EquipmentItemRecipeSheet.Row>.Clear() ,
Sheet<int, EquipmentItemRecipeSheet.Row>.Contains(KeyValuePair<int,
EquipmentItemRecipeSheet.Row>) ,
Sheet<int, EquipmentItemRecipeSheet.Row>.CopyTo(KeyValuePair<int,
EquipmentItemRecipeSheet.Row>[], int) ,
Sheet<int, EquipmentItemRecipeSheet.Row>.Remove(KeyValuePair<int,
EquipmentItemRecipeSheet.Row>) ,
Sheet<int, EquipmentItemRecipeSheet.Row>.Serialize() , object.Equals(object) ,
object.Equals(object, object) , object.GetHashCode() , object.GetType() ,
object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

EquipmentItemRecipeSheet()

Creates an empty sheet.

```
public EquipmentItemRecipeSheet()
```

Class EquipmentItemRecipeSheet.Row

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

One equipment recipe.

```
public class EquipmentItemRecipeSheet.Row : SheetRow<int>
```

Inheritance

object ← SheetRow<int> ← EquipmentItemRecipeSheet.Row

Inherited Members

SheetRow<int>.Validate() , SheetRow<int>.EndOfSheetInitialize() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Properties

CRYSTAL

Crystal to unlock the recipe (column `required_crystal`), 0 for a recipe open from the start.

```
public long CRYSTAL { get; }
```

Property Value

long

Id

The recipe id.

```
public int Id { get; }
```

Property Value

int

ItemSubType

The result equipment's sub type, when given.

```
public ItemSubType? ItemSubType { get; }
```

Property Value

ItemSubType?

Key

```
public override int Key { get; }
```

Property Value

int

MaterialCount

How many of the main material are used.

```
public int MaterialCount { get; }
```

Property Value

int

MaterialId

The main material.

```
public int MaterialId { get; }
```

Property Value

int

RequiredActionPoint

Action points the craft costs.

```
public int RequiredActionPoint { get; }
```

Property Value

int

RequiredBlockIndex

Blocks the craft takes.

```
public long RequiredBlockIndex { get; }
```

Property Value

long

RequiredGold

NCG the craft costs.

```
public long RequiredGold { get; }
```

Property Value

long

ResultEquipmentId

The equipment the recipe makes.

```
public int ResultEquipmentId { get; }
```

Property Value

int

SubRecipeIds

The sub recipes of this recipe.

```
public List<int> SubRecipeIds { get; }
```

Property Value

List<int>

UnlockStage

The stage that must be cleared to unlock the recipe.

```
public int UnlockStage { get; }
```

Property Value

int

Methods

GetAllMaterials(EquipmentItemSubRecipeSheetV2, CraftType)

```
public IEnumerable<EquipmentItemSubRecipeSheet.MaterialInfo>
GetAllMaterials(EquipmentItemSubRecipeSheetV2 sheet, CraftType subRecipeType
= CraftType.Normal)
```

Parameters

`sheet` EquipmentItemSubRecipeSheetV2

`subRecipeType` CraftType

Returns

IEnumerable<EquipmentItemSubRecipeSheet.MaterialInfo>

IsMimisBrunnrSubRecipe(int?)

```
[Obsolete("It is obsoleted. Check
EquipmentItemSubRecipeSheetV2.IsMimisbrunnrSubRecipe.")]
public bool IsMimisBrunnrSubRecipe(int? subRecipeId)
```

Parameters

`subRecipeId` int?

Returns

bool

Set(IReadOnlyList<string>)

```
public override void Set(IReadOnlyList<string> fields)
```

Parameters

`fields` IReadOnlyList<string>

Class EquipmentItemSetEffectSheet

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class EquipmentItemSetEffectSheet : Sheet<int,
EquipmentItemSetEffectSheet.Row>, IDictionary<int, EquipmentItemSetEffectSheet.Row>,
ICollection<KeyValuePair<int, EquipmentItemSetEffectSheet.Row>>,
IEnumerable<KeyValuePair<int, EquipmentItemSetEffectSheet.Row>>, IEnumerable, ISheet
```

Inheritance

object ← Sheet<int, EquipmentItemSetEffectSheet.Row> ← EquipmentItemSetEffectSheet

Implements

IDictionary<int, EquipmentItemSetEffectSheet.Row>, ICollection<KeyValuePair<int, EquipmentItemSetEffectSheet.Row>>, IEnumerable<KeyValuePair<int, EquipmentItemSetEffectSheet.Row>>, IEnumerable, ISheet

Inherited Members

Sheet<int, EquipmentItemSetEffectSheet.Row>.Name ,
Sheet<int, EquipmentItemSetEffectSheet.Row>.OrderedList ,
Sheet<int, EquipmentItemSetEffectSheet.Row>.First ,
Sheet<int, EquipmentItemSetEffectSheet.Row>.Last ,
Sheet<int, EquipmentItemSetEffectSheet.Row>.Keys ,
Sheet<int, EquipmentItemSetEffectSheet.Row>.Values ,
Sheet<int, EquipmentItemSetEffectSheet.Row>.Count ,
Sheet<int, EquipmentItemSetEffectSheet.Row>.IsReadOnly ,
Sheet<int, EquipmentItemSetEffectSheet.Row>.this[int] ,
Sheet<int, EquipmentItemSetEffectSheet.Row>.Set(string, bool) ,
Sheet<int, EquipmentItemSetEffectSheet.Row>.Set<T>(Sheet<int, T>, bool) ,
Sheet<int, EquipmentItemSetEffectSheet.Row>.GetEnumerator() ,
Sheet<int, EquipmentItemSetEffectSheet.Row>.TryGetValue(int, out EquipmentItemSetEffectSheet.Row, bool) ,
Sheet<int, EquipmentItemSetEffectSheet.Row>.AddRow(int, EquipmentItemSetEffectSheet.Row) ,
Sheet<int, EquipmentItemSetEffectSheet.Row>.Add(int, EquipmentItemSetEffectSheet.Row) ,

Sheet<int, EquipmentItemSetEffectSheet.Row>.ContainsKey(int) ,
Sheet<int, EquipmentItemSetEffectSheet.Row>.Remove(int) ,
Sheet<int, EquipmentItemSetEffectSheet.Row>.TryGetValue(int, out EquipmentItemSetEffectSheet.Row) ,
Sheet<int, EquipmentItemSetEffectSheet.Row>.Add(KeyValuePair<int, EquipmentItemSetEffectSheet.Row>) ,
Sheet<int, EquipmentItemSetEffectSheet.Row>.Clear() ,
Sheet<int, EquipmentItemSetEffectSheet.Row>.Contains(KeyValuePair<int, EquipmentItemSetEffectSheet.Row>) ,
Sheet<int, EquipmentItemSetEffectSheet.Row>.CopyTo(KeyValuePair<int, EquipmentItemSetEffectSheet.Row>[], int) ,
Sheet<int, EquipmentItemSetEffectSheet.Row>.Remove(KeyValuePair<int, EquipmentItemSetEffectSheet.Row>) ,
Sheet<int, EquipmentItemSetEffectSheet.Row>.Serialize() , object.Equals(object) ,
object.Equals(object, object) , object.GetHashCode() , object.GetType() ,
object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Extension Methods
SetEffectExtension.GetSetEffectRows(EquipmentItemSetEffectSheet, IEnumerable<Equipment>)

Constructors

EquipmentItemSetEffectSheet()

```
public EquipmentItemSetEffectSheet()
```

Methods

AddRow(int, Row)

```
protected override void AddRow(int key, EquipmentItemSetEffectSheet.Row value)
```

Parameters

`key` int

`value` EquipmentItemSetEffectSheet.Row

Class EquipmentItemSetEffectSheet.Row

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class EquipmentItemSetEffectSheet.Row : SheetRow<int>
```

Inheritance

object ← SheetRow<int> ← EquipmentItemSetEffectSheet.Row

Inherited Members

SheetRow<int>.Validate() , SheetRow<int>.EndOfSheetInitialize() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Properties

Id

```
public int Id { get; }
```

Property Value

int

Key

```
public override int Key { get; }
```

Property Value

int

StatModifiers

```
public Dictionary<int, StatModifier> StatModifiers { get; }
```

Property Value

Dictionary<int, StatModifier>

Methods

Set(IReadOnlyList<string>)

```
public override void Set(IReadOnlyList<string> fields)
```

Parameters

`fields` IReadOnlyList<string>

Class EquipmentItemSheet

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class EquipmentItemSheet : Sheet<int, EquipmentItemSheet.Row>,
IDictionary<int, EquipmentItemSheet.Row>, ICollection<KeyValuePair<int,
EquipmentItemSheet.Row>>, IEnumerable<KeyValuePair<int, EquipmentItemSheet.Row>>,
IEnumerable, ISheet
```

Inheritance

object ← Sheet<int, EquipmentItemSheet.Row> ← EquipmentItemSheet

Implements

IDictionary<int, EquipmentItemSheet.Row>,
ICollection<KeyValuePair<int, EquipmentItemSheet.Row>>,
IEnumerable<KeyValuePair<int, EquipmentItemSheet.Row>>, IEnumerable, ISheet

Inherited Members

Sheet<int, EquipmentItemSheet.Row>.Name ,
Sheet<int, EquipmentItemSheet.Row>.OrderedList ,
Sheet<int, EquipmentItemSheet.Row>.First , Sheet<int, EquipmentItemSheet.Row>.Last ,
Sheet<int, EquipmentItemSheet.Row>.Keys ,
Sheet<int, EquipmentItemSheet.Row>.Values ,
Sheet<int, EquipmentItemSheet.Row>.Count ,
Sheet<int, EquipmentItemSheet.Row>.IsReadOnly ,
Sheet<int, EquipmentItemSheet.Row>.this[int] ,
Sheet<int, EquipmentItemSheet.Row>.Set(string, bool) ,
Sheet<int, EquipmentItemSheet.Row>.Set<T>(Sheet<int, T>, bool) ,
Sheet<int, EquipmentItemSheet.Row>.GetEnumerator() ,
Sheet<int, EquipmentItemSheet.Row>.TryGetValue(int, out EquipmentItemSheet.Row,
bool) ,
Sheet<int, EquipmentItemSheet.Row>.AddRow(int, EquipmentItemSheet.Row) ,
Sheet<int, EquipmentItemSheet.Row>.Add(int, EquipmentItemSheet.Row) ,
Sheet<int, EquipmentItemSheet.Row>.ContainsKey(int) ,
Sheet<int, EquipmentItemSheet.Row>.Remove(int) ,
Sheet<int, EquipmentItemSheet.Row>.TryGetValue(int, out EquipmentItemSheet.Row) ,
Sheet<int, EquipmentItemSheet.Row>.Add(KeyValuePair<int, EquipmentItemSheet.Row>) ,
Sheet<int, EquipmentItemSheet.Row>.Clear() ,

Sheet<int, EquipmentItemSheet.Row>.Contains(KeyValuePair<int, EquipmentItemSheet.Row>) , Sheet<int, EquipmentItemSheet.Row>.CopyTo(KeyValuePair<int, EquipmentItemSheet.Row>[], int) , Sheet<int, EquipmentItemSheet.Row>.Remove(KeyValuePair<int, EquipmentItemSheet.Row>) , Sheet<int, EquipmentItemSheet.Row>.Serialize() , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

EquipmentItemSheet()

```
public EquipmentItemSheet()
```

Class EquipmentItemSheet.Row

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class EquipmentItemSheet.Row : ItemSheet.Row
```

Inheritance

object ← SheetRow<int> ← ItemSheet.RowBase ← ItemSheet.Row ← EquipmentItemSheet.Row

Inherited Members

ItemSheet.Row.Key , ItemSheet.Row.Id , ItemSheet.Row.ItemSubType , ItemSheet.Row.Grade , ItemSheet.Row.ElementalType , ItemSheet.Row.Deserialize(Dictionary) , SheetRow<int>.Validate() , SheetRow<int>.EndOfSheetInitialize() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

Row()

```
public Row()
```

Row(Dictionary)

```
public Row(Dictionary serialized)
```

Parameters

`serialized` Dictionary

Properties

AttackRange

```
public decimal AttackRange { get; }
```

Property Value

decimal

Exp

```
public long? Exp { get; }
```

Property Value

long?

ItemType

```
public override ItemType ItemType { get; }
```

Property Value

ItemType

SetId

```
public int SetId { get; }
```

Property Value

int

SpineResourcePath

```
public string SpineResourcePath { get; }
```

Property Value

string

Stat

```
public DecimalStat Stat { get; }
```

Property Value

DecimalStat

Methods

Serialize()

```
public override IValue Serialize()
```

Returns

IValue

Set(IReadOnlyList<string>)

```
public override void Set(IReadOnlyList<string> fields)
```

Parameters

`fields` IReadOnlyList<string>

Class EquipmentItemSubRecipeSheet

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
public class EquipmentItemSubRecipeSheet : Sheet<int,
EquipmentItemSubRecipeSheet.Row>, IDictionary<int, EquipmentItemSubRecipeSheet.Row>,
ICollection<KeyValuePair<int, EquipmentItemSubRecipeSheet.Row>>,
IEnumerable<KeyValuePair<int, EquipmentItemSubRecipeSheet.Row>>, IEnumerable, ISheet
```

Inheritance

object ← Sheet<int, EquipmentItemSubRecipeSheet.Row> ← EquipmentItemSubRecipeSheet

Implements

IDictionary<int, EquipmentItemSubRecipeSheet.Row>, ICollection<KeyValuePair<int, EquipmentItemSubRecipeSheet.Row>>, IEnumerable<KeyValuePair<int, EquipmentItemSubRecipeSheet.Row>>, IEnumerable, ISheet

Inherited Members

Sheet<int, EquipmentItemSubRecipeSheet.Row>.Name ,
Sheet<int, EquipmentItemSubRecipeSheet.Row>.OrderedList ,
Sheet<int, EquipmentItemSubRecipeSheet.Row>.First ,
Sheet<int, EquipmentItemSubRecipeSheet.Row>.Last ,
Sheet<int, EquipmentItemSubRecipeSheet.Row>.Keys ,
Sheet<int, EquipmentItemSubRecipeSheet.Row>.Values ,
Sheet<int, EquipmentItemSubRecipeSheet.Row>.Count ,
Sheet<int, EquipmentItemSubRecipeSheet.Row>.IsReadOnly ,
Sheet<int, EquipmentItemSubRecipeSheet.Row>.this[int] ,
Sheet<int, EquipmentItemSubRecipeSheet.Row>.Set(string, bool) ,
Sheet<int, EquipmentItemSubRecipeSheet.Row>.Set<T>(Sheet<int, T>, bool) ,
Sheet<int, EquipmentItemSubRecipeSheet.Row>.GetEnumerator() ,
Sheet<int, EquipmentItemSubRecipeSheet.Row>.TryGetValue(int, out EquipmentItemSubRecipeSheet.Row, bool) ,
Sheet<int, EquipmentItemSubRecipeSheet.Row>.AddRow(int, EquipmentItemSubRecipeSheet.Row) ,
Sheet<int, EquipmentItemSubRecipeSheet.Row>.Add(int, EquipmentItemSubRecipeSheet.Row) ,

Sheet<int, EquipmentItemSubRecipeSheet.Row>.ContainsKey(int) ,
Sheet<int, EquipmentItemSubRecipeSheet.Row>.Remove(int) ,
Sheet<int, EquipmentItemSubRecipeSheet.Row>.TryGetValue(int, out
EquipmentItemSubRecipeSheet.Row) ,
Sheet<int, EquipmentItemSubRecipeSheet.Row>.Add(KeyValuePair<int,
EquipmentItemSubRecipeSheet.Row>) ,
Sheet<int, EquipmentItemSubRecipeSheet.Row>.Clear() ,
Sheet<int, EquipmentItemSubRecipeSheet.Row>.Contains(KeyValuePair<int,
EquipmentItemSubRecipeSheet.Row>) ,
Sheet<int, EquipmentItemSubRecipeSheet.Row>.CopyTo(KeyValuePair<int,
EquipmentItemSubRecipeSheet.Row>[], int) ,
Sheet<int, EquipmentItemSubRecipeSheet.Row>.Remove(KeyValuePair<int,
EquipmentItemSubRecipeSheet.Row>) ,
Sheet<int, EquipmentItemSubRecipeSheet.Row>.Serialize() , object.Equals(object) ,
object.Equals(object, object) , object.GetHashCode() , object.GetType() ,
object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

EquipmentItemSubRecipeSheet()

```
public EquipmentItemSubRecipeSheet()
```

Struct EquipmentItemSubRecipeSheet.MaterialInfo

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
public readonly struct EquipmentItemSubRecipeSheet.MaterialInfo
```

Inherited Members
ValueType.Equals(object) , ValueType.GetHashCode() , ValueType.ToString() , object.Equals(object, object) , object.GetType() , object.ReferenceEquals(object, object)

Constructors

MaterialInfo(int, int)

```
public MaterialInfo(int id, int count)
```

Parameters

`id` int

`count` int

Fields

Count

```
public readonly int Count
```

Field Value

int

Id

```
public readonly int Id
```

Field Value

int

Struct EquipmentItemSubRecipeSheet.OptionInfo

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
public readonly struct EquipmentItemSubRecipeSheet.OptionInfo
```

Inherited Members
ValueType.Equals(object) , ValueType.GetHashCode() , ValueType.ToString() , object.Equals(object, object) , object.GetType() , object.ReferenceEquals(object, object)

Constructors

OptionInfo(int, decimal)

```
public OptionInfo(int id, decimal ratio)
```

Parameters

`id` int

`ratio` decimal

Fields

Id

```
public readonly int Id
```

Field Value

int

Ratio

```
public readonly decimal Ratio
```

Field Value

decimal

Class EquipmentItemSubRecipeSheet.Row

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
public class EquipmentItemSubRecipeSheet.Row : SheetRow<int>
```

Inheritance

object ← SheetRow<int> ← EquipmentItemSubRecipeSheet.Row

Inherited Members

SheetRow<int>.Validate() , SheetRow<int>.EndOfSheetInitialize() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Properties

Id

```
public int Id { get; }
```

Property Value

int

Key

```
public override int Key { get; }
```

Property Value

int

Materials

```
public List<EquipmentItemSubRecipeSheet.MaterialInfo> Materials { get; }
```

Property Value

List<EquipmentItemSubRecipeSheet.MaterialInfo>

MaxOptionLimit

```
public int MaxOptionLimit { get; }
```

Property Value

int

Options

```
public List<EquipmentItemSubRecipeSheet.OptionInfo> Options { get; }
```

Property Value

List<EquipmentItemSubRecipeSheet.OptionInfo>

RequiredActionPoint

```
public int RequiredActionPoint { get; }
```

Property Value

int

RequiredBlockIndex

```
public long RequiredBlockIndex { get; }
```

Property Value

long

RequiredGold

```
public long RequiredGold { get; }
```

Property Value

long

Methods

Set(IReadOnlyList<string>)

```
public override void Set(IReadOnlyList<string> fields)
```

Parameters

`fields` IReadOnlyList<string>

Class EquipmentItemSubRecipeSheetV2

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
public class EquipmentItemSubRecipeSheetV2 : Sheet<int,
EquipmentItemSubRecipeSheetV2.Row>, IDictionary<int,
EquipmentItemSubRecipeSheetV2.Row>, ICollection<KeyValuePair<int,
EquipmentItemSubRecipeSheetV2.Row>>, IEnumerable<KeyValuePair<int,
EquipmentItemSubRecipeSheetV2.Row>>, IEnumerable, ISheet
```

Inheritance

object ← Sheet<int, EquipmentItemSubRecipeSheetV2.Row> ← EquipmentItemSubRecipeSheetV2

Implements

IDictionary<int, EquipmentItemSubRecipeSheetV2.Row>, ICollection<KeyValuePair<int, EquipmentItemSubRecipeSheetV2.Row>>, IEnumerable<KeyValuePair<int, EquipmentItemSubRecipeSheetV2.Row>>, IEnumerable, ISheet

Inherited Members

Sheet<int, EquipmentItemSubRecipeSheetV2.Row>.Name ,
Sheet<int, EquipmentItemSubRecipeSheetV2.Row>.OrderedList ,
Sheet<int, EquipmentItemSubRecipeSheetV2.Row>.First ,
Sheet<int, EquipmentItemSubRecipeSheetV2.Row>.Last ,
Sheet<int, EquipmentItemSubRecipeSheetV2.Row>.Keys ,
Sheet<int, EquipmentItemSubRecipeSheetV2.Row>.Values ,
Sheet<int, EquipmentItemSubRecipeSheetV2.Row>.Count ,
Sheet<int, EquipmentItemSubRecipeSheetV2.Row>.IsReadOnly ,
Sheet<int, EquipmentItemSubRecipeSheetV2.Row>.this[int] ,
Sheet<int, EquipmentItemSubRecipeSheetV2.Row>.Set(string, bool) ,
Sheet<int, EquipmentItemSubRecipeSheetV2.Row>.Set<T>(Sheet<int, T>, bool) ,
Sheet<int, EquipmentItemSubRecipeSheetV2.Row>.GetEnumerator() ,
Sheet<int, EquipmentItemSubRecipeSheetV2.Row>.TryGetValue(int, out EquipmentItemSubRecipeSheetV2.Row, bool) ,
Sheet<int, EquipmentItemSubRecipeSheetV2.Row>.AddRow(int, EquipmentItemSubRecipeSheetV2.Row) ,
Sheet<int, EquipmentItemSubRecipeSheetV2.Row>.Add(int, EquipmentItemSubRecipeSheetV2.Row) ,

Sheet<int, EquipmentItemSubRecipeSheetV2.Row>.ContainsKey(int) ,
Sheet<int, EquipmentItemSubRecipeSheetV2.Row>.Remove(int) ,
Sheet<int, EquipmentItemSubRecipeSheetV2.Row>.TryGetValue(int, out
EquipmentItemSubRecipeSheetV2.Row) ,
Sheet<int, EquipmentItemSubRecipeSheetV2.Row>.Add(KeyValuePair<int,
EquipmentItemSubRecipeSheetV2.Row>) ,
Sheet<int, EquipmentItemSubRecipeSheetV2.Row>.Clear() ,
Sheet<int, EquipmentItemSubRecipeSheetV2.Row>.Contains(KeyValuePair<int,
EquipmentItemSubRecipeSheetV2.Row>) ,
Sheet<int, EquipmentItemSubRecipeSheetV2.Row>.CopyTo(KeyValuePair<int,
EquipmentItemSubRecipeSheetV2.Row>[], int) ,
Sheet<int, EquipmentItemSubRecipeSheetV2.Row>.Remove(KeyValuePair<int,
EquipmentItemSubRecipeSheetV2.Row>) ,
Sheet<int, EquipmentItemSubRecipeSheetV2.Row>.Serialize() , object.Equals(object) ,
object.Equals(object, object) , object.GetHashCode() , object.GetType() ,
object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

EquipmentItemSubRecipeSheetV2()

```
public EquipmentItemSubRecipeSheetV2()
```

Struct EquipmentItemSubRecipeSheetV2.OptionInfo

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
public readonly struct EquipmentItemSubRecipeSheetV2.OptionInfo
```

Inherited Members

ValueType.Equals(object) , ValueType.GetHashCode() , ValueType.ToString() , object.Equals(object, object) , object.GetType() , object.ReferenceEquals(object, object)

Constructors

OptionInfo(int, int, int)

```
public OptionInfo(int id, int ratio, int requiredBlockIndex = 0)
```

Parameters

`id` int

`ratio` int

`requiredBlockIndex` int

Fields

Id

```
public readonly int Id
```

Field Value

int

Ratio

```
public readonly int Ratio
```

Field Value

int

RequiredBlockIndex

```
public readonly int RequiredBlockIndex
```

Field Value

int

Class EquipmentItemSubRecipeSheetV2.Row

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
public class EquipmentItemSubRecipeSheetV2.Row : SheetRow<int>
```

Inheritance

object ← SheetRow<int> ← EquipmentItemSubRecipeSheetV2.Row

Inherited Members

SheetRow<int>.Validate() , SheetRow<int>.EndOfSheetInitialize() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Properties

Id

```
public int Id { get; }
```

Property Value

int

IsMimisbrunnrSubRecipe

```
public bool? IsMimisbrunnrSubRecipe { get; }
```

Property Value

bool?

Key

```
public override int Key { get; }
```

Property Value

int

Materials

```
public List<EquipmentItemSubRecipeSheet.MaterialInfo> Materials { get; }
```

Property Value

List<EquipmentItemSubRecipeSheet.MaterialInfo>

Options

```
public List<EquipmentItemSubRecipeSheetV2.OptionInfo> Options { get; }
```

Property Value

List<EquipmentItemSubRecipeSheetV2.OptionInfo>

RequiredActionPoint

```
public int RequiredActionPoint { get; }
```

Property Value

int

RequiredBlockIndex

```
public long RequiredBlockIndex { get; }
```

Property Value

long

RequiredGold

```
public long RequiredGold { get; }
```

Property Value

long

RewardHammerPoint

```
public int? RewardHammerPoint { get; }
```

Property Value

int?

Methods

Set(IReadOnlyList<string>)

```
public override void Set(IReadOnlyList<string> fields)
```

Parameters

`fields` IReadOnlyList<string>

Class GameConfigSheet

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class GameConfigSheet : Sheet<string, GameConfigSheet.Row>,
IDictionary<string, GameConfigSheet.Row>, ICollection<KeyValuePair<string,
GameConfigSheet.Row>>, IEnumerable<KeyValuePair<string, GameConfigSheet.Row>>,
IEnumerable, ISheet
```

Inheritance

object ← Sheet<string, GameConfigSheet.Row> ← GameConfigSheet

Implements

IDictionary<string, GameConfigSheet.Row>,
ICollection<KeyValuePair<string, GameConfigSheet.Row>>,
IEnumerable<KeyValuePair<string, GameConfigSheet.Row>>, IEnumerable, ISheet

Inherited Members

Sheet<string, GameConfigSheet.Row>.Name ,
Sheet<string, GameConfigSheet.Row>.OrderedList ,
Sheet<string, GameConfigSheet.Row>.First , Sheet<string, GameConfigSheet.Row>.Last ,
Sheet<string, GameConfigSheet.Row>.Keys ,
Sheet<string, GameConfigSheet.Row>.Values ,
Sheet<string, GameConfigSheet.Row>.Count ,
Sheet<string, GameConfigSheet.Row>.IsReadOnly ,
Sheet<string, GameConfigSheet.Row>.this[string] ,
Sheet<string, GameConfigSheet.Row>.Set(string, bool) ,
Sheet<string, GameConfigSheet.Row>.Set<T>(Sheet<string, T>, bool) ,
Sheet<string, GameConfigSheet.Row>.GetEnumerator() ,
Sheet<string, GameConfigSheet.Row>.TryGetValue(string, out GameConfigSheet.Row, bool) ,
Sheet<string, GameConfigSheet.Row>.AddRow(string, GameConfigSheet.Row) ,
Sheet<string, GameConfigSheet.Row>.Add(string, GameConfigSheet.Row) ,
Sheet<string, GameConfigSheet.Row>.ContainsKey(string) ,
Sheet<string, GameConfigSheet.Row>.Remove(string) ,
Sheet<string, GameConfigSheet.Row>.TryGetValue(string, out GameConfigSheet.Row) ,
Sheet<string, GameConfigSheet.Row>.Add(KeyValuePair<string, GameConfigSheet.Row>)
,

Sheet<string, GameConfigSheet.Row>.Clear() ,
Sheet<string, GameConfigSheet.Row>.Contains(KeyValuePair<string, GameConfigSheet.Row>) ,
Sheet<string, GameConfigSheet.Row>.CopyTo(KeyValuePair<string, GameConfigSheet.Row>[], int) ,
Sheet<string, GameConfigSheet.Row>.Remove(KeyValuePair<string, GameConfigSheet.Row>) ,
Sheet<string, GameConfigSheet.Row>.Serialize() , object.Equals(object) ,
object.Equals(object, object) , object.GetHashCode() , object.GetType() ,
object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

GameConfigSheet()

```
public GameConfigSheet()
```

Class GameConfigSheet.Row

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class GameConfigSheet.Row : SheetRow<string>
```

Inheritance

object ← SheetRow<string> ← GameConfigSheet.Row

Inherited Members

SheetRow<string>.Validate() , SheetRow<string>.EndOfSheetInitialize() , SheetRow<string>.Equals(object) , SheetRow<string>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Properties

Key

```
public override string Key { get; }
```

Property Value

string

Value

```
public string Value { get; }
```

Property Value

string

Methods

Set(IReadOnlyList<string>)

```
public override void Set(IReadOnlyList<string> fields)
```

Parameters

`fields` IReadOnlyList<string>

Class GeneralQuestSheet

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class GeneralQuestSheet : Sheet<int, GeneralQuestSheet.Row>, IDictionary<int,
GeneralQuestSheet.Row>, ICollection<KeyValuePair<int, GeneralQuestSheet.Row>>,
IEnumerable<KeyValuePair<int, GeneralQuestSheet.Row>>, IEnumerable, ISheet
```

Inheritance

object ← Sheet<int, GeneralQuestSheet.Row> ← GeneralQuestSheet

Implements

IDictionary<int, GeneralQuestSheet.Row>,
ICollection<KeyValuePair<int, GeneralQuestSheet.Row>>,
IEnumerable<KeyValuePair<int, GeneralQuestSheet.Row>>, IEnumerable, ISheet

Inherited Members

Sheet<int, GeneralQuestSheet.Row>.Name ,
Sheet<int, GeneralQuestSheet.Row>.OrderedList ,
Sheet<int, GeneralQuestSheet.Row>.First , Sheet<int, GeneralQuestSheet.Row>.Last ,
Sheet<int, GeneralQuestSheet.Row>.Keys , Sheet<int, GeneralQuestSheet.Row>.Values ,
Sheet<int, GeneralQuestSheet.Row>.Count ,
Sheet<int, GeneralQuestSheet.Row>.IsReadOnly ,
Sheet<int, GeneralQuestSheet.Row>.this[int] ,
Sheet<int, GeneralQuestSheet.Row>.Set(string, bool) ,
Sheet<int, GeneralQuestSheet.Row>.Set<T>(Sheet<int, T>, bool) ,
Sheet<int, GeneralQuestSheet.Row>.GetEnumerator() ,
Sheet<int, GeneralQuestSheet.Row>.TryGetValue(int, out GeneralQuestSheet.Row, bool) ,
Sheet<int, GeneralQuestSheet.Row>.AddRow(int, GeneralQuestSheet.Row) ,
Sheet<int, GeneralQuestSheet.Row>.Add(int, GeneralQuestSheet.Row) ,
Sheet<int, GeneralQuestSheet.Row>.ContainsKey(int) ,
Sheet<int, GeneralQuestSheet.Row>.Remove(int) ,
Sheet<int, GeneralQuestSheet.Row>.TryGetValue(int, out GeneralQuestSheet.Row) ,
Sheet<int, GeneralQuestSheet.Row>.Add(KeyValuePair<int, GeneralQuestSheet.Row>) ,
Sheet<int, GeneralQuestSheet.Row>.Clear() ,
Sheet<int, GeneralQuestSheet.Row>.Contains(KeyValuePair<int,
GeneralQuestSheet.Row>) ,

Sheet<int, GeneralQuestSheet.Row>.CopyTo(KeyValuePair<int, GeneralQuestSheet.Row>[], int) ,
Sheet<int, GeneralQuestSheet.Row>.Remove(KeyValuePair<int, GeneralQuestSheet.Row>) ,
Sheet<int, GeneralQuestSheet.Row>.Serialize() , object.Equals(object) ,
object.Equals(object, object) , object.GetHashCode() , object.GetType() ,
object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

GeneralQuestSheet()

```
public GeneralQuestSheet()
```

Class GeneralQuestSheet.Row

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class GeneralQuestSheet.Row : QuestSheet.Row, IState
```

Inheritance

object ← SheetRow<int> ← QuestSheet.Row ← GeneralQuestSheet.Row

Implements

IState

Inherited Members

QuestSheet.Row.Key , QuestSheet.Row.Id , QuestSheet.Row.Goal , QuestSheet.Row.QuestRewardId , QuestSheet.Row.Serialize() , QuestSheet.Row.Deserialize(Dictionary) , SheetRow<int>.Validate() , SheetRow<int>.EndOfSheetInitialize() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Properties

Event

```
public QuestEventType Event { get; }
```

Property Value

QuestEventType

Methods

Set(IReadOnlyList<string>)

```
public override void Set(IReadOnlyList<string> fields)
```

Parameters

`fields` IReadOnlyList<string>

Class GlobalCounterSheet

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

Defines the global counters that world-open conditions (WorldOpenConditionSheet) are measured with: what each counter counts, over which targets, and the minimum avatar level that contributes.

Columns: `id`, `type`, `target_begin`, `target_end`, `min_avatar_level`. The target is a stage id or a recipe id depending on `type`; `[target_begin, target_end]` is inclusive.

Reading is total, like BoostScheduleSheet: a row that does not parse is kept but marked invalid and never matches. ValidateCsv(string) rejects such a row at patch time.

A patch cannot change what an id means once recorded; see WorldOpenIdentity.

```
[Serializable]
public class GlobalCounterSheet : Sheet<int, GlobalCounterSheet.Row>,
IDictionary<int, GlobalCounterSheet.Row>, ICollection<KeyValuePair<int,
GlobalCounterSheet.Row>>, IEnumerable<KeyValuePair<int, GlobalCounterSheet.Row>>,
IEnumerable, ISheet
```

Inheritance

object ← Sheet<int, GlobalCounterSheet.Row> ← GlobalCounterSheet

Implements

IDictionary<int, GlobalCounterSheet.Row>,
ICollection<KeyValuePair<int, GlobalCounterSheet.Row>>,
IEnumerable<KeyValuePair<int, GlobalCounterSheet.Row>>, IEnumerable, ISheet

Inherited Members

Sheet<int, GlobalCounterSheet.Row>.Name ,
Sheet<int, GlobalCounterSheet.Row>.OrderedList ,
Sheet<int, GlobalCounterSheet.Row>.First , Sheet<int, GlobalCounterSheet.Row>.Last ,
Sheet<int, GlobalCounterSheet.Row>.Keys , Sheet<int, GlobalCounterSheet.Row>.Values ,
Sheet<int, GlobalCounterSheet.Row>.Count ,
Sheet<int, GlobalCounterSheet.Row>.IsReadOnly ,
Sheet<int, GlobalCounterSheet.Row>.this[int] ,
Sheet<int, GlobalCounterSheet.Row>.Set(string, bool) ,
Sheet<int, GlobalCounterSheet.Row>.Set<T>(Sheet<int, T>, bool) ,

Sheet<int, GlobalCounterSheet.Row>.GetEnumerator() ,
Sheet<int, GlobalCounterSheet.Row>.TryGetValue(int, out GlobalCounterSheet.Row, bool) ,
Sheet<int, GlobalCounterSheet.Row>.AddRow(int, GlobalCounterSheet.Row) ,
Sheet<int, GlobalCounterSheet.Row>.Add(int, GlobalCounterSheet.Row) ,
Sheet<int, GlobalCounterSheet.Row>.ContainsKey(int) ,
Sheet<int, GlobalCounterSheet.Row>.Remove(int) ,
Sheet<int, GlobalCounterSheet.Row>.TryGetValue(int, out GlobalCounterSheet.Row) ,
Sheet<int, GlobalCounterSheet.Row>.Add(KeyValuePair<int, GlobalCounterSheet.Row>) ,
Sheet<int, GlobalCounterSheet.Row>.Clear() ,
Sheet<int, GlobalCounterSheet.Row>.Contains(KeyValuePair<int, GlobalCounterSheet.Row>) ,
Sheet<int, GlobalCounterSheet.Row>.CopyTo(KeyValuePair<int, GlobalCounterSheet.Row>[], int) ,
Sheet<int, GlobalCounterSheet.Row>.Remove(KeyValuePair<int, GlobalCounterSheet.Row>) ,
Sheet<int, GlobalCounterSheet.Row>.Serialize() , object.Equals(object) ,
object.Equals(object, object) , object.GetHashCode() , object.GetType() ,
object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

GlobalCounterSheet()

Creates an empty sheet.

```
public GlobalCounterSheet()
```

Fields

ColumnNames

Column names in their expected order, after _ prefixed columns are dropped.

```
public static readonly string[] ColumnNames
```

Field Value

string[]

Methods

AddRow(int, Row)

Adds a parsed row. A row without a usable id is dropped and a duplicated id keeps the first row instead of throwing, because a throw would fail every action that reads the sheet.

```
protected override void AddRow(int key, GlobalCounterSheet.Row value)
```

Parameters

`key` int

`value` GlobalCounterSheet.Row

Matching(GlobalCounterType, int)

The valid counters of `type` whose target range contains `target`.

```
public IEnumerable<GlobalCounterSheet.Row> Matching(GlobalCounterType type,
int target)
```

Parameters

`type` GlobalCounterType

`target` int

Returns

IEnumerable<GlobalCounterSheet.Row>

ValidateCsv(string)

Parses `csv` strictly so that a malformed sheet fails the `patch_table_sheet` transaction instead of landing on the chain.

```
public static void ValidateCsv(string csv)
```

Parameters

`csv` string

Exceptions

SheetRowValidateException

The header does not match ColumnNames, an id is not a positive unique integer, a cell does not parse, the type is unknown, `target_begin` is greater than `target_end`, or a `STAGE_FIRST_CLEAR` counter spans more than one stage.

Class GlobalCounterSheet.Row

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

One counter definition.

```
[Serializable]
public class GlobalCounterSheet.Row : SheetRow<int>
```

Inheritance

object ← SheetRow<int> ← GlobalCounterSheet.Row

Inherited Members

SheetRow<int>.Validate() , SheetRow<int>.EndOfSheetInitialize() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Properties

Id

The counter id.

```
public int Id { get; }
```

Property Value

int

IsValid

Whether every cell parsed. An invalid row never matches.

```
public bool IsValid { get; }
```

Property Value

bool

Key

```
public override int Key { get; }
```

Property Value

int

MinAvatarLevel

Minimum avatar level whose action counts.

```
public int MinAvatarLevel { get; }
```

Property Value

int

TargetBegin

First target id counted, inclusive.

```
public int TargetBegin { get; }
```

Property Value

int

TargetEnd

Last target id counted, inclusive.

```
public int TargetEnd { get; }
```

Property Value

int

Type

What the counter counts.

```
public GlobalCounterType Type { get; }
```

Property Value

GlobalCounterType

Methods

Set(IReadOnlyList<string>)

```
public override void Set(IReadOnlyList<string> fields)
```

Parameters

`fields` IReadOnlyList<string>

Enum GlobalCounterType

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

What a GlobalCounterSheet counter counts.

```
public enum GlobalCounterType
```

Extension Methods
StateExtensions.Serialize(Enum)

Fields

`RECIPE_CRAFT = 2`

An avatar crafts a recipe.

`STAGE_FIRST_CLEAR = 0`

An avatar clears a stage for the first time.

`STAGE_PLAY = 1`

An avatar clears a stage in a `HackAndSlash` battle (every wave cleared, the same signal as a stage clear), counted per such battle: a multi-play action adds its cleared battles; a lost or partly cleared battle adds nothing. Sweeping does not count.

Class GoldQuestSheet

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class GoldQuestSheet : Sheet<int, GoldQuestSheet.Row>, IDictionary<int,
GoldQuestSheet.Row>, ICollection<KeyValuePair<int, GoldQuestSheet.Row>>,
IEnumerable<KeyValuePair<int, GoldQuestSheet.Row>>, IEnumerable, ISheet
```

Inheritance

object ← Sheet<int, GoldQuestSheet.Row> ← GoldQuestSheet

Implements

IDictionary<int, GoldQuestSheet.Row>,
ICollection<KeyValuePair<int, GoldQuestSheet.Row>>,
IEnumerable<KeyValuePair<int, GoldQuestSheet.Row>>, IEnumerable, ISheet

Inherited Members

Sheet<int, GoldQuestSheet.Row>.Name , Sheet<int, GoldQuestSheet.Row>.OrderedList ,
Sheet<int, GoldQuestSheet.Row>.First , Sheet<int, GoldQuestSheet.Row>.Last ,
Sheet<int, GoldQuestSheet.Row>.Keys , Sheet<int, GoldQuestSheet.Row>.Values ,
Sheet<int, GoldQuestSheet.Row>.Count , Sheet<int, GoldQuestSheet.Row>.IsReadOnly ,
Sheet<int, GoldQuestSheet.Row>.this[int] ,
Sheet<int, GoldQuestSheet.Row>.Set(string, bool) ,
Sheet<int, GoldQuestSheet.Row>.Set<T>(Sheet<int, T>, bool) ,
Sheet<int, GoldQuestSheet.Row>.GetEnumerator() ,
Sheet<int, GoldQuestSheet.Row>.TryGetValue(int, out GoldQuestSheet.Row, bool) ,
Sheet<int, GoldQuestSheet.Row>.AddRow(int, GoldQuestSheet.Row) ,
Sheet<int, GoldQuestSheet.Row>.Add(int, GoldQuestSheet.Row) ,
Sheet<int, GoldQuestSheet.Row>.ContainsKey(int) ,
Sheet<int, GoldQuestSheet.Row>.Remove(int) ,
Sheet<int, GoldQuestSheet.Row>.TryGetValue(int, out GoldQuestSheet.Row) ,
Sheet<int, GoldQuestSheet.Row>.Add(KeyValuePair<int, GoldQuestSheet.Row>) ,
Sheet<int, GoldQuestSheet.Row>.Clear() ,
Sheet<int, GoldQuestSheet.Row>.Contains(KeyValuePair<int, GoldQuestSheet.Row>) ,
Sheet<int, GoldQuestSheet.Row>.CopyTo(KeyValuePair<int, GoldQuestSheet.Row>[], int) ,
Sheet<int, GoldQuestSheet.Row>.Remove(KeyValuePair<int, GoldQuestSheet.Row>) ,
Sheet<int, GoldQuestSheet.Row>.Serialize() , object.Equals(object) ,

object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

GoldQuestSheet()

```
public GoldQuestSheet()
```

Class GoldQuestSheet.Row

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class GoldQuestSheet.Row : QuestSheet.Row, IState
```

Inheritance

object ← SheetRow<int> ← QuestSheet.Row ← GoldQuestSheet.Row

Implements

IState

Inherited Members

QuestSheet.Row.Key , QuestSheet.Row.Id , QuestSheet.Row.Goal , QuestSheet.Row.QuestRewardId , QuestSheet.Row.Serialize() , QuestSheet.Row.Deserialize(Dictionary) , SheetRow<int>.Validate() , SheetRow<int>.EndOfSheetInitialize() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Properties

Type

```
public TradeType Type { get; }
```

Property Value

TradeType

Methods

Set(IReadOnlyList<string>)

```
public override void Set(IReadOnlyList<string> fields)
```

Parameters

`fields` IReadOnlyList<string>

Interface ISheet

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
public interface ISheet
```

Methods

Set(string, bool)

```
void Set(string csv, bool isReversed = false)
```

Parameters

`csv` string

`isReversed` bool

Interface IStakeRewardRow

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
public interface IStakeRewardRow
```

Properties

Level

```
int Level { get; }
```

Property Value

int

RequiredGold

```
long RequiredGold { get; }
```

Property Value

long

Interface IStakeRewardSheet

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
public interface IStakeRewardSheet : ISheet
```

Inherited Members
ISheet.Set(string, bool)

Extension Methods
SheetsExtensions.FindLevelByStakedAmount(IStakeRewardSheet, Address, FungibleAssetValue)

Properties

OrderedRows

```
IReadOnlyList<IStakeRewardRow> OrderedRows { get; }
```

Property Value

IReadOnlyList<IStakeRewardRow>

Interface IWorldBossRewardRow

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
public interface IWorldBossRewardRow
```

Properties

BossId

```
int BossId { get; }
```

Property Value

int

Circle

```
int Circle { get; }
```

Property Value

int

Crystal

```
int Crystal { get; }
```

Property Value

int

Rank

```
int Rank { get; }
```

Property Value

int

Rune

```
int Rune { get; }
```

Property Value

int

Interface IWorldBossRewardSheet

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
public interface IWorldBossRewardSheet : ISheet
```

Inherited Members
ISheet.Set(string, bool)

Properties

OrderedRows

```
IReadOnlyList<IWorldBossRewardRow> OrderedRows { get; }
```

Property Value

IReadOnlyList<IWorldBossRewardRow>

Class InfiniteTowerConditionSheet

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

Infinite tower condition sheet for managing battle conditions and restrictions. Uses StatType, SkillTargetType, and StatModifier.OperationType for flexible stat modifications.

```
[Serializable]
public class InfiniteTowerConditionSheet : Sheet<int,
InfiniteTowerConditionSheet.Row>, IDictionary<int, InfiniteTowerConditionSheet.Row>,
ICollection<KeyValuePair<int, InfiniteTowerConditionSheet.Row>>,
IEnumerable<KeyValuePair<int, InfiniteTowerConditionSheet.Row>>, IEnumerable, ISheet
```

Inheritance

object ← Sheet<int, InfiniteTowerConditionSheet.Row> ← InfiniteTowerConditionSheet

Implements

IDictionary<int, InfiniteTowerConditionSheet.Row>,
ICollection<KeyValuePair<int, InfiniteTowerConditionSheet.Row>>,
IEnumerable<KeyValuePair<int, InfiniteTowerConditionSheet.Row>>, IEnumerable,
ISheet

Inherited Members

Sheet<int, InfiniteTowerConditionSheet.Row>.Name ,
Sheet<int, InfiniteTowerConditionSheet.Row>.OrderedList ,
Sheet<int, InfiniteTowerConditionSheet.Row>.First ,
Sheet<int, InfiniteTowerConditionSheet.Row>.Last ,
Sheet<int, InfiniteTowerConditionSheet.Row>.Keys ,
Sheet<int, InfiniteTowerConditionSheet.Row>.Values ,
Sheet<int, InfiniteTowerConditionSheet.Row>.Count ,
Sheet<int, InfiniteTowerConditionSheet.Row>.IsReadOnly ,
Sheet<int, InfiniteTowerConditionSheet.Row>.this[int] ,
Sheet<int, InfiniteTowerConditionSheet.Row>.Set(string, bool) ,
Sheet<int, InfiniteTowerConditionSheet.Row>.Set<T>(Sheet<int, T>, bool) ,
Sheet<int, InfiniteTowerConditionSheet.Row>.GetEnumerator() ,
Sheet<int, InfiniteTowerConditionSheet.Row>.TryGetValue(int, out
InfiniteTowerConditionSheet.Row, bool) ,
Sheet<int, InfiniteTowerConditionSheet.Row>.AddRow(int,
InfiniteTowerConditionSheet.Row) ,

Sheet<int, InfiniteTowerConditionSheet.Row>.Add(int, InfiniteTowerConditionSheet.Row) ,
Sheet<int, InfiniteTowerConditionSheet.Row>.ContainsKey(int) ,
Sheet<int, InfiniteTowerConditionSheet.Row>.Remove(int) ,
Sheet<int, InfiniteTowerConditionSheet.Row>.TryGetValue(int, out
InfiniteTowerConditionSheet.Row) ,
Sheet<int, InfiniteTowerConditionSheet.Row>.Add(KeyValuePair<int,
InfiniteTowerConditionSheet.Row>) ,
Sheet<int, InfiniteTowerConditionSheet.Row>.Clear() ,
Sheet<int, InfiniteTowerConditionSheet.Row>.Contains(KeyValuePair<int,
InfiniteTowerConditionSheet.Row>) ,
Sheet<int, InfiniteTowerConditionSheet.Row>.CopyTo(KeyValuePair<int,
InfiniteTowerConditionSheet.Row>[], int) ,
Sheet<int, InfiniteTowerConditionSheet.Row>.Remove(KeyValuePair<int,
InfiniteTowerConditionSheet.Row>) ,
Sheet<int, InfiniteTowerConditionSheet.Row>.Serialize() , object.Equals(object) ,
object.Equals(object, object) , object.GetHashCode() , object.GetType() ,
object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

InfiniteTowerConditionSheet()

```
public InfiniteTowerConditionSheet()
```

Class InfiniteTowerConditionSheet.Row

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

Represents a row in the InfiniteTowerConditionSheet containing condition data.

```
[Serializable]
public class InfiniteTowerConditionSheet.Row : SheetRow<int>
```

Inheritance

object ← SheetRow<int> ← InfiniteTowerConditionSheet.Row

Inherited Members

SheetRow<int>.Validate() , SheetRow<int>.EndOfSheetInitialize() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Properties

Id

Gets the unique identifier for this condition.

```
public int Id { get; }
```

Property Value

int

Key

Gets the condition ID as the key for this row.

```
public override int Key { get; }
```

Property Value

int

OperationType

Gets the operation type for the stat modification.

```
public StatModifier.OperationType OperationType { get; }
```

Property Value

StatModifier.OperationType

StatType

Gets the stat type that this condition affects.

```
public StatType StatType { get; }
```

Property Value

StatType

TargetType

Gets the list of target types for this condition (Self, Enemy, etc.).

```
public List<SkillTargetType> TargetType { get; }
```

Property Value

List<SkillTargetType>

Value

Gets the value for the stat modification.

```
public int Value { get; }
```

Property Value

int

Methods

GetStatModifier()

Creates a StatModifier from this condition row.

```
public StatModifier GetStatModifier()
```

Returns

StatModifier

Set(IReadOnlyList<string>)

```
public override void Set(IReadOnlyList<string> fields)
```

Parameters

`fields` IReadOnlyList<string>

Class InfiniteTowerFloorSheet

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

Infinite tower floor sheet for managing tower floors and their properties.

```
[Serializable]
public class InfiniteTowerFloorSheet : Sheet<int, InfiniteTowerFloorSheet.Row>,
IDictionary<int, InfiniteTowerFloorSheet.Row>, ICollection<KeyValuePair<int,
InfiniteTowerFloorSheet.Row>>, IEnumerable<KeyValuePair<int,
InfiniteTowerFloorSheet.Row>>, IEnumerable, ISheet
```

Inheritance

object ← Sheet<int, InfiniteTowerFloorSheet.Row> ← InfiniteTowerFloorSheet

Implements

IDictionary<int, InfiniteTowerFloorSheet.Row>,
ICollection<KeyValuePair<int, InfiniteTowerFloorSheet.Row>>,
IEnumerable<KeyValuePair<int, InfiniteTowerFloorSheet.Row>>, IEnumerable,
ISheet

Inherited Members

Sheet<int, InfiniteTowerFloorSheet.Row>.Name ,
Sheet<int, InfiniteTowerFloorSheet.Row>.OrderedList ,
Sheet<int, InfiniteTowerFloorSheet.Row>.First ,
Sheet<int, InfiniteTowerFloorSheet.Row>.Last ,
Sheet<int, InfiniteTowerFloorSheet.Row>.Keys ,
Sheet<int, InfiniteTowerFloorSheet.Row>.Values ,
Sheet<int, InfiniteTowerFloorSheet.Row>.Count ,
Sheet<int, InfiniteTowerFloorSheet.Row>.IsReadOnly ,
Sheet<int, InfiniteTowerFloorSheet.Row>.this[int] ,
Sheet<int, InfiniteTowerFloorSheet.Row>.Set(string, bool) ,
Sheet<int, InfiniteTowerFloorSheet.Row>.Set<T>(Sheet<int, T>, bool) ,
Sheet<int, InfiniteTowerFloorSheet.Row>.GetEnumerator() ,
Sheet<int, InfiniteTowerFloorSheet.Row>.TryGetValue(int, out InfiniteTowerFloorSheet.Row,
bool) ,
Sheet<int, InfiniteTowerFloorSheet.Row>.AddRow(int, InfiniteTowerFloorSheet.Row) ,
Sheet<int, InfiniteTowerFloorSheet.Row>.Add(int, InfiniteTowerFloorSheet.Row) ,
Sheet<int, InfiniteTowerFloorSheet.Row>.ContainsKey(int) ,

Sheet<int, InfiniteTowerFloorSheet.Row>.Remove(int) ,
Sheet<int, InfiniteTowerFloorSheet.Row>.TryGetValue(int, out InfiniteTowerFloorSheet.Row)
,
Sheet<int, InfiniteTowerFloorSheet.Row>.Add(KeyValuePair<int,
InfiniteTowerFloorSheet.Row>) ,
Sheet<int, InfiniteTowerFloorSheet.Row>.Clear() ,
Sheet<int, InfiniteTowerFloorSheet.Row>.Contains(KeyValuePair<int,
InfiniteTowerFloorSheet.Row>) ,
Sheet<int, InfiniteTowerFloorSheet.Row>.CopyTo(KeyValuePair<int,
InfiniteTowerFloorSheet.Row>[], int) ,
Sheet<int, InfiniteTowerFloorSheet.Row>.Remove(KeyValuePair<int,
InfiniteTowerFloorSheet.Row>) ,
Sheet<int, InfiniteTowerFloorSheet.Row>.Serialize() , object.Equals(object) ,
object.Equals(object, object) , object.GetHashCode() , object.GetType() ,
object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

InfiniteTowerFloorSheet()

```
public InfiniteTowerFloorSheet()
```

Class InfiniteTowerFloorSheet.Row

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

Represents a row in the InfiniteTowerFloorSheet containing floor data.

```
[Serializable]
public class InfiniteTowerFloorSheet.Row : SheetRow<int>
```

Inheritance

object ← SheetRow<int> ← InfiniteTowerFloorSheet.Row

Inherited Members

SheetRow<int>.Validate() , SheetRow<int>.EndOfSheetInitialize() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Properties

EnemyInitialStatModifiers

Gets the list of initial stat modifiers applied to enemies on this floor.

```
public List<StatModifier> EnemyInitialStatModifiers { get; }
```

Property Value

List<StatModifier>

Floor

Gets the floor number.

```
public int Floor { get; }
```

Property Value

int

ForbiddenItemSubTypes

Gets the list of forbidden item subtypes for this floor.

```
public List<ItemSubType> ForbiddenItemSubTypes { get; }
```

Property Value

List<ItemSubType>

ForbiddenRuneTypes

Gets the list of forbidden rune types for this floor.

```
public List<RuneType> ForbiddenRuneTypes { get; }
```

Property Value

List<RuneType>

FungibleAssetRewardAmount1

Gets the amount of the first fungible asset reward.

```
public int? FungibleAssetRewardAmount1 { get; }
```

Property Value

int?

FungibleAssetRewardAmount2

Gets the amount of the second fungible asset reward.

```
public int? FungibleAssetRewardAmount2 { get; }
```

Property Value

int?

FungibleAssetRewardAmount3

Gets the amount of the third fungible asset reward.

```
public int? FungibleAssetRewardAmount3 { get; }
```

Property Value

int?

FungibleAssetRewardAmount4

Gets the amount of the fourth fungible asset reward.

```
public int? FungibleAssetRewardAmount4 { get; }
```

Property Value

int?

FungibleAssetRewardAmount5

Gets the amount of the fifth fungible asset reward.

```
public int? FungibleAssetRewardAmount5 { get; }
```

Property Value

int?

FungibleAssetRewardTicker1

Gets the first fungible asset reward ticker for this floor.

```
public string? FungibleAssetRewardTicker1 { get; }
```

Property Value

string

FungibleAssetRewardTicker2

Gets the second fungible asset reward ticker for this floor.

```
public string? FungibleAssetRewardTicker2 { get; }
```

Property Value

string

FungibleAssetRewardTicker3

Gets the third fungible asset reward ticker for this floor.

```
public string? FungibleAssetRewardTicker3 { get; }
```

Property Value

string

FungibleAssetRewardTicker4

Gets the fourth fungible asset reward ticker for this floor.

```
public string? FungibleAssetRewardTicker4 { get; }
```

Property Value

string

FungibleAssetRewardTicker5

Gets the fifth fungible asset reward ticker for this floor.

```
public string? FungibleAssetRewardTicker5 { get; }
```

Property Value

string

GuaranteedConditionId

Gets the guaranteed condition ID that will always be applied to this floor.

```
public int GuaranteedConditionId { get; }
```

Property Value

int

Id

Gets the unique identifier for this floor.

```
public int Id { get; }
```

Property Value

int

ItemRewardCount1

Gets the count of the first item reward.

```
public int? ItemRewardCount1 { get; }
```

Property Value

int?

ItemRewardCount2

Gets the count of the second item reward.

```
public int? ItemRewardCount2 { get; }
```

Property Value

int?

ItemRewardCount3

Gets the count of the third item reward.

```
public int? ItemRewardCount3 { get; }
```

Property Value

int?

ItemRewardCount4

Gets the count of the fourth item reward.

```
public int? ItemRewardCount4 { get; }
```

Property Value

int?

ItemRewardCount5

Gets the count of the fifth item reward.

```
public int? ItemRewardCount5 { get; }
```

Property Value

int?

ItemRewardId1

Gets the first item reward ID for this floor.

```
public int? ItemRewardId1 { get; }
```

Property Value

int?

ItemRewardId2

Gets the second item reward ID for this floor.

```
public int? ItemRewardId2 { get; }
```

Property Value

int?

ItemRewardId3

Gets the third item reward ID for this floor.

```
public int? ItemRewardId3 { get; }
```

Property Value

int?

ItemRewardId4

Gets the fourth item reward ID for this floor.

```
public int? ItemRewardId4 { get; }
```

Property Value

int?

ItemRewardId5

Gets the fifth item reward ID for this floor.

```
public int? ItemRewardId5 { get; }
```

Property Value

int?

Key

Gets the floor ID as the key for this row.

```
public override int Key { get; }
```

Property Value

int

MaterialCostCount

Gets the amount of material required for purchasing tickets.

```
public int? MaterialCostCount { get; }
```

Property Value

int?

MaterialCostId

Gets the material item ID required for purchasing tickets.

```
public int? MaterialCostId { get; }
```

Property Value

int?

MaxCp

Gets the maximum allowed combat power for this floor.

```
public long? MaxCp { get; }
```

Property Value

long?

MaxItemGrade

Gets the maximum item grade allowed for this floor.

```
public int? MaxItemGrade { get; }
```

Property Value

int?

MaxItemLevel

Gets the maximum item level allowed for this floor.

```
public int? MaxItemLevel { get; }
```

Property Value

int?

MaxRandomConditions

Gets the maximum number of random conditions to apply.

```
public int MaxRandomConditions { get; }
```

Property Value

int

MinItemGrade

Gets the minimum item grade allowed for this floor.

```
public int? MinItemGrade { get; }
```

Property Value

int?

MinItemLevel

Gets the minimum item level allowed for this floor.

```
public int? MinItemLevel { get; }
```

Property Value

int?

MinRandomConditions

Gets the minimum number of random conditions to apply.

```
public int MinRandomConditions { get; }
```

Property Value

int

NcgCost

Gets the NCG cost for purchasing tickets for this floor.

```
public int? NcgCost { get; }
```

Property Value

int?

RandomConditionId1

Gets the first random condition ID for weighted selection.

```
public int? RandomConditionId1 { get; }
```

Property Value

int?

RandomConditionId2

Gets the second random condition ID for weighted selection.

```
public int? RandomConditionId2 { get; }
```

Property Value

int?

RandomConditionId3

Gets the third random condition ID for weighted selection.

```
public int? RandomConditionId3 { get; }
```

Property Value

int?

RandomConditionId4

Gets the fourth random condition ID for weighted selection.

```
public int? RandomConditionId4 { get; }
```

Property Value

int?

RandomConditionId5

Gets the fifth random condition ID for weighted selection.

```
public int? RandomConditionId5 { get; }
```

Property Value

int?

RandomConditionWeight1

Gets the weight for the first random condition.

```
public int? RandomConditionWeight1 { get; }
```

Property Value

int?

RandomConditionWeight2

Gets the weight for the second random condition.

```
public int? RandomConditionWeight2 { get; }
```

Property Value

int?

RandomConditionWeight3

Gets the weight for the third random condition.

```
public int? RandomConditionWeight3 { get; }
```

Property Value

int?

RandomConditionWeight4

Gets the weight for the fourth random condition.

```
public int? RandomConditionWeight4 { get; }
```

Property Value

int?

RandomConditionWeight5

Gets the weight for the fifth random condition.

```
public int? RandomConditionWeight5 { get; }
```

Property Value

int?

RequiredCp

Gets the minimum required combat power for this floor.

```
public long? RequiredCp { get; }
```

Property Value

long?

RequiredElementalTypes

Gets the list of required elemental types for equipment on this floor.

```
public List<ElementalType> RequiredElementalTypes { get; }
```

Property Value

List<ElementalType>

Methods

GetBattleCondition(BattleConditionType)

Gets battle conditions of a specific type from this floor's restrictions. This method can be used by clients for simulation and replay purposes.

```
public InfiniteTowerBattleCondition? GetBattleCondition(BattleConditionType
conditionType)
```

Parameters

`conditionType` BattleConditionType

The type of battle condition to retrieve

Returns

InfiniteTowerBattleCondition

Battle condition of the specified type, or null if not found

GetBattleConditions()

Creates a list of battle conditions from this floor's restrictions. This method can be used by clients for simulation and replay purposes.

```
public List<InfiniteTowerBattleCondition> GetBattleConditions()
```

Returns

List<InfiniteTowerBattleCondition>

List of battle conditions representing this floor's restrictions

GetFungibleAssetRewards()

Gets all fungible asset rewards as a list of tuples (ticker, amount).

```
public List<(string ticker, int amount)> GetFungibleAssetRewards()
```

Returns

List<(string ticker, int amount)>

GetItemRewards()

Gets all item rewards as a list of tuples (itemId, count).

```
public List<(int itemId, int count)> GetItemRewards()
```

Returns

List<(int id, int count)>

GetRandomConditions(InfiniteTowerConditionSheet, IRandom, int?)

Selects random conditions from all available conditions using uniform random selection. Uses Fisher-Yates shuffle algorithm to avoid index issues when removing items. This method can be used by clients for simulation and replay purposes.

```
public List<InfiniteTowerCondition> GetRandomConditions(InfiniteTowerConditionSheet
conditionSheet, IRandom random, int? guaranteedConditionId = null)
```

Parameters

`conditionSheet` InfiniteTowerConditionSheet

The condition sheet containing all available conditions

`random` IRandom

Random number generator

`guaranteedConditionId` int?

Optional guaranteed condition ID to exclude from random selection

Returns

List<InfiniteTowerCondition>

List of selected random conditions

Exceptions

InvalidOperationException

Thrown when available conditions are insufficient for minimum count requirement

GetRandomConditionsWithWeights()

Gets random conditions with weights for weighted selection.

```
public List<(int conditionId, int weight)> GetRandomConditionsWithWeights()
```

Returns

List<(int id, int count)>

GetRandomConditionsWithWeights(InfiniteTowerConditionSheet, IRandom, int?)

Selects random conditions using weighted selection for this floor. This method can be used by clients for simulation and replay purposes.

```
public List<InfiniteTowerCondition>
GetRandomConditionsWithWeights(InfiniteTowerConditionSheet conditionSheet, IRandom
```

```
  random, int? guaranteedConditionId = null)
```

Parameters

`conditionSheet` InfiniteTowerConditionSheet

The condition sheet containing all available conditions

`random` IRandom

Random number generator

`guaranteedConditionId` int?

Optional guaranteed condition ID to exclude from random selection

Returns

List<InfiniteTowerCondition>

List of selected random conditions

ProcessRewards(IActionContext, IWorld, AvatarState, Address, EquipmentItemSheet, MaterialItemSheet, ConsumableItemSheet, CostumeItemSheet, IRandom)

Processes rewards for successful floor completion.

```
  public IWorld ProcessRewards(IActionContext context, IWorld states, AvatarState
  avatarState, Address avatarAddress, EquipmentItemSheet equipmentSheet,
  MaterialItemSheet materialSheet, ConsumableItemSheet consumableSheet,
  CostumeItemSheet costumeSheet, IRandom random)
```

Parameters

`context` IActionContext

The action context

`states` IWorld

The world state

`avatarState` AvatarState

The avatar state to update

`avatarAddress` Address

The avatar address

`equipmentSheet` EquipmentItemSheet

The equipment item sheet

`materialSheet` MaterialItemSheet

The material item sheet

`consumableSheet` ConsumableItemSheet

The consumable item sheet

`costumeSheet` CostumeItemSheet

The costume item sheet

`random` IRandom

Random number generator for creating items

Returns

IWorld

Updated world state

Exceptions

InvalidItemIdException

Thrown when item ID is not found in any item sheet

Set(IReadOnlyList<string>)

```
public override void Set(IReadOnlyList<string> fields)
```

Parameters

`fields` IReadOnlyList<string>

ValidateCpRequirements(long)

Validates CP requirements (minimum and maximum) for the floor.

```
public void ValidateCpRequirements(long currentCp)
```

Parameters

`currentCp` long

ValidateCurrencyForTicketPurchase(IActionContext, IWorld, Address, bool, string)

Validates that the player has enough currency to purchase a ticket.

```
public void ValidateCurrencyForTicketPurchase(IActionContext context, IWorld states,
Address avatarAddress, bool useNcgForTicket, string addressesHex)
```

Parameters

`context` IActionContext

The action context

`states` IWorld

The world state

`avatarAddress` Address

The avatar address

`useNcgForTicket` bool

Whether to use NCG for ticket purchase

`addressesHex` string

Addresses hex for logging

Exceptions

InvalidOperationException

Thrown when cost configuration is missing

InsufficientBalanceException

Thrown when NCG balance is insufficient

NotEnoughMaterialException

Thrown when material count is insufficient

ValidateEquipmentElementalType(List<Equipment>?)

Validates equipment elemental types against the floor's requirements. This method can be used by clients for simulation and replay purposes.

```
public void ValidateEquipmentElementalType(List<Equipment>? equipments)
```

Parameters

`equipments` List<Equipment>

List of equipment to validate

Exceptions

InvalidElementalException

Thrown when equipment has invalid elemental type

ValidateEquipmentElementalType<T>(List<T>)

Validates equipment elemental type restrictions for the floor.

```
public void ValidateEquipmentElementalType<T>(List<T> equipmentList) where T
: ItemBase
```

Parameters

`equipmentList` List<T>

Type Parameters

`T`

ValidateFloorRestrictions(List<Equipment>, List<Costume>)

Validates floor-specific restrictions for equipment and costumes. This method can be used by clients for simulation and replay purposes.

```
public void ValidateFloorRestrictions(List<Equipment> equipmentList,
List<Costume> costumeList)
```

Parameters

`equipmentList` List<Equipment>

List of equipment to validate

`costumeList` List<Costume>

List of costumes to validate

Exceptions

Exception

Thrown when equipment or costume restrictions are violated

ValidateItemGradeRestrictions<T>(List<T>)

Validates item grade restrictions for the floor (applies to both equipment and costumes).

```
public void ValidateItemGradeRestrictions<T>(List<T> itemList) where T : ItemBase
```

Parameters

`itemList` List<T>

Type Parameters

`T`

ValidateItemLevelRestrictions<T>(List<T>)

Validates item level restrictions for the floor (applies to both equipment and costumes).

```
public void ValidateItemLevelRestrictions<T>(List<T> itemList) where T : ItemBase
```

Parameters

`itemList` List<T>

Type Parameters

`T`

ValidateItemTypeRestrictions<T>(List<T>)

Validates item subtype restrictions for the floor (applies to both equipment and costumes).

```
public void ValidateItemTypeRestrictions<T>(List<T> itemList) where T : ItemBase
```

Parameters

`itemList` List<T>

Type Parameters

`T`

ValidateRuneTypes(List<RuneSlotInfo>?, RuneListSheet)

Validates equipped rune types against the floor's forbidden list. This method can be used by clients for simulation and replay purposes.

```
public void ValidateRuneTypes(List<RuneSlotInfo>? runeInfos,
RuneListSheet runeListSheet)
```

Parameters

`runeInfos` List<RuneSlotInfo>

List of rune slot information

`runeListSheet` RuneListSheet

Rune list sheet for type lookup

Exceptions

ForbiddenRuneTypeEquippedException

Thrown when forbidden rune types are equipped

Class InfiniteTowerFloorWaveSheet

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

Infinite tower floor wave sheet for managing wave configurations. Supports multiple monster types per wave, similar to StageWaveSheet.

```
[Serializable]
public class InfiniteTowerFloorWaveSheet : Sheet<int,
InfiniteTowerFloorWaveSheet.Row>, IDictionary<int, InfiniteTowerFloorWaveSheet.Row>,
ICollection<KeyValuePair<int, InfiniteTowerFloorWaveSheet.Row>>,
IEnumerable<KeyValuePair<int, InfiniteTowerFloorWaveSheet.Row>>, IEnumerable, ISheet
```

Inheritance

object ← Sheet<int, InfiniteTowerFloorWaveSheet.Row> ← InfiniteTowerFloorWaveSheet

Implements

IDictionary<int, InfiniteTowerFloorWaveSheet.Row>,
ICollection<KeyValuePair<int, InfiniteTowerFloorWaveSheet.Row>>,
IEnumerable<KeyValuePair<int, InfiniteTowerFloorWaveSheet.Row>>, IEnumerable,
ISheet

Inherited Members

Sheet<int, InfiniteTowerFloorWaveSheet.Row>.Name ,
Sheet<int, InfiniteTowerFloorWaveSheet.Row>.OrderedList ,
Sheet<int, InfiniteTowerFloorWaveSheet.Row>.First ,
Sheet<int, InfiniteTowerFloorWaveSheet.Row>.Last ,
Sheet<int, InfiniteTowerFloorWaveSheet.Row>.Keys ,
Sheet<int, InfiniteTowerFloorWaveSheet.Row>.Values ,
Sheet<int, InfiniteTowerFloorWaveSheet.Row>.Count ,
Sheet<int, InfiniteTowerFloorWaveSheet.Row>.IsReadOnly ,
Sheet<int, InfiniteTowerFloorWaveSheet.Row>.this[int] ,
Sheet<int, InfiniteTowerFloorWaveSheet.Row>.Set(string, bool) ,
Sheet<int, InfiniteTowerFloorWaveSheet.Row>.Set<T>(Sheet<int, T>, bool) ,
Sheet<int, InfiniteTowerFloorWaveSheet.Row>.GetEnumerator() ,
Sheet<int, InfiniteTowerFloorWaveSheet.Row>.TryGetValue(int, out
InfiniteTowerFloorWaveSheet.Row, bool) ,
Sheet<int, InfiniteTowerFloorWaveSheet.Row>.AddRow(int,
InfiniteTowerFloorWaveSheet.Row) ,

Sheet<int, InfiniteTowerFloorWaveSheet.Row>.Add(int, InfiniteTowerFloorWaveSheet.Row) ,
Sheet<int, InfiniteTowerFloorWaveSheet.Row>.ContainsKey(int) ,
Sheet<int, InfiniteTowerFloorWaveSheet.Row>.Remove(int) ,
Sheet<int, InfiniteTowerFloorWaveSheet.Row>.TryGetValue(int, out InfiniteTowerFloorWaveSheet.Row) ,
Sheet<int, InfiniteTowerFloorWaveSheet.Row>.Add(KeyValuePair<int, InfiniteTowerFloorWaveSheet.Row>) ,
Sheet<int, InfiniteTowerFloorWaveSheet.Row>.Clear() ,
Sheet<int, InfiniteTowerFloorWaveSheet.Row>.Contains(KeyValuePair<int, InfiniteTowerFloorWaveSheet.Row>) ,
Sheet<int, InfiniteTowerFloorWaveSheet.Row>.CopyTo(KeyValuePair<int, InfiniteTowerFloorWaveSheet.Row>[], int) ,
Sheet<int, InfiniteTowerFloorWaveSheet.Row>.Remove(KeyValuePair<int, InfiniteTowerFloorWaveSheet.Row>) ,
Sheet<int, InfiniteTowerFloorWaveSheet.Row>.Serialize() , object.Equals(object) ,
object.Equals(object, object) , object.GetHashCode() , object.GetType() ,
object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

InfiniteTowerFloorWaveSheet()

Initializes a new instance of the InfiniteTowerFloorWaveSheet class.

```
public InfiniteTowerFloorWaveSheet()
```

Methods

AddRow(int, Row)

Adds a row to the sheet, merging waves if the floor already exists.

```
protected override void AddRow(int key, InfiniteTowerFloorWaveSheet.Row value)
```

Parameters

`key` int

The floor ID key.

`value` InfiniteTowerFloorWaveSheet.Row

The row data to add.

Class InfiniteTowerFloorWaveSheet.MonsterData

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

Represents a single monster type within a wave.

```
[Serializable]
public class InfiniteTowerFloorWaveSheet.MonsterData
```

Inheritance

object ← InfiniteTowerFloorWaveSheet.MonsterData

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

MonsterData(int, int, int)

Initializes a new instance of the MonsterData class.

```
public MonsterData(int characterId, int level, int count)
```

Parameters

`characterId` int

The character ID.

`level` int

The monster level.

`count` int

The number of monsters.

Properties

CharacterId

Gets the character ID of the monster.

```
public int CharacterId { get; }
```

Property Value

int

Count

Gets the number of monsters of this type in the wave.

```
public int Count { get; }
```

Property Value

int

Level

Gets the level of the monster.

```
public int Level { get; }
```

Property Value

int

Class InfiniteTowerFloorWaveSheet.Row

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

Represents a row in the InfiniteTowerFloorWaveSheet containing wave data for a specific floor.

```
[Serializable]
public class InfiniteTowerFloorWaveSheet.Row : SheetRow<int>
```

Inheritance

object ← SheetRow<int> ← InfiniteTowerFloorWaveSheet.Row

Inherited Members

SheetRow<int>.Validate() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Properties

FloorId

Gets the floor ID this row represents.

```
public int FloorId { get; }
```

Property Value

int

HasBoss

Gets whether any wave in this floor contains a boss monster.

```
public bool HasBoss { get; }
```

Property Value

bool

Key

Gets the floor ID as the key for this row.

```
public override int Key { get; }
```

Property Value

int

TotalMonsterIds

Gets the list of all unique monster character IDs across all waves in this floor.

```
public List<int> TotalMonsterIds { get; }
```

Property Value

List<int>

Waves

Gets the list of waves for this floor.

```
public List<InfiniteTowerFloorWaveSheet.WaveData> Waves { get; }
```

Property Value

List<InfiniteTowerFloorWaveSheet.WaveData>

Methods

EndOfSheetInitialize()

Performs post-initialization tasks after all rows have been loaded. Sorts waves by number and populates derived properties.

```
public override void EndOfSheetInitialize()
```

Set(IReadOnlyList<string>)

Sets the row data from CSV fields. Parses multiple monster types per wave from the CSV data.

```
public override void Set(IReadOnlyList<string> fields)
```

Parameters

`fields` IReadOnlyList<string>

CSV field data.

Class InfiniteTowerFloorWaveSheet.WaveData

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

Represents a single wave within a floor, containing multiple monster types.

```
[Serializable]
public class InfiniteTowerFloorWaveSheet.WaveData
```

Inheritance

object ← InfiniteTowerFloorWaveSheet.WaveData

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

WaveData(int, List<MonsterData>, bool)

Initializes a new instance of the WaveData class.

```
public WaveData(int number, List<InfiniteTowerFloorWaveSheet.MonsterData> monsters,
bool hasBoss)
```

Parameters

`number` int

The wave number.

`monsters` List<InfiniteTowerFloorWaveSheet.MonsterData>

List of monsters in this wave.

`hasBoss` bool

Whether this wave has a boss.

Properties

HasBoss

Gets whether this wave contains a boss monster.

```
public bool HasBoss { get; }
```

Property Value

bool

Monsters

Gets the list of monster types in this wave.

```
public List<InfiniteTowerFloorWaveSheet.MonsterData> Monsters { get; }
```

Property Value

List<InfiniteTowerFloorWaveSheet.MonsterData>

Number

Gets the wave number within the floor.

```
public int Number { get; }
```

Property Value

int

Class InfiniteTowerScheduleSheet

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

Infinite tower schedule sheet for managing season schedules and ticket policies.

```
[Serializable]
public class InfiniteTowerScheduleSheet : Sheet<int,
InfiniteTowerScheduleSheet.Row>, IDictionary<int, InfiniteTowerScheduleSheet.Row>,
ICollection<KeyValuePair<int, InfiniteTowerScheduleSheet.Row>>,
IEnumerable<KeyValuePair<int, InfiniteTowerScheduleSheet.Row>>, IEnumerable, ISheet
```

Inheritance

object ← Sheet<int, InfiniteTowerScheduleSheet.Row> ← InfiniteTowerScheduleSheet

Implements

IDictionary<int, InfiniteTowerScheduleSheet.Row>,
ICollection<KeyValuePair<int, InfiniteTowerScheduleSheet.Row>>,
IEnumerable<KeyValuePair<int, InfiniteTowerScheduleSheet.Row>>, IEnumerable,
ISheet

Inherited Members

Sheet<int, InfiniteTowerScheduleSheet.Row>.Name ,
Sheet<int, InfiniteTowerScheduleSheet.Row>.OrderedList ,
Sheet<int, InfiniteTowerScheduleSheet.Row>.First ,
Sheet<int, InfiniteTowerScheduleSheet.Row>.Last ,
Sheet<int, InfiniteTowerScheduleSheet.Row>.Keys ,
Sheet<int, InfiniteTowerScheduleSheet.Row>.Values ,
Sheet<int, InfiniteTowerScheduleSheet.Row>.Count ,
Sheet<int, InfiniteTowerScheduleSheet.Row>.IsReadOnly ,
Sheet<int, InfiniteTowerScheduleSheet.Row>.this[int] ,
Sheet<int, InfiniteTowerScheduleSheet.Row>.Set(string, bool) ,
Sheet<int, InfiniteTowerScheduleSheet.Row>.Set<T>(Sheet<int, T>, bool) ,
Sheet<int, InfiniteTowerScheduleSheet.Row>.GetEnumerator() ,
Sheet<int, InfiniteTowerScheduleSheet.Row>.TryGetValue(int, out
InfiniteTowerScheduleSheet.Row, bool) ,
Sheet<int, InfiniteTowerScheduleSheet.Row>.AddRow(int, InfiniteTowerScheduleSheet.Row)
,
Sheet<int, InfiniteTowerScheduleSheet.Row>.Add(int, InfiniteTowerScheduleSheet.Row) ,

Sheet<int, InfiniteTowerScheduleSheet.Row>.ContainsKey(int) ,
Sheet<int, InfiniteTowerScheduleSheet.Row>.Remove(int) ,
Sheet<int, InfiniteTowerScheduleSheet.Row>.TryGetValue(int, out
InfiniteTowerScheduleSheet.Row) ,
Sheet<int, InfiniteTowerScheduleSheet.Row>.Add(KeyValuePair<int,
InfiniteTowerScheduleSheet.Row>) ,
Sheet<int, InfiniteTowerScheduleSheet.Row>.Clear() ,
Sheet<int, InfiniteTowerScheduleSheet.Row>.Contains(KeyValuePair<int,
InfiniteTowerScheduleSheet.Row>) ,
Sheet<int, InfiniteTowerScheduleSheet.Row>.CopyTo(KeyValuePair<int,
InfiniteTowerScheduleSheet.Row>[], int) ,
Sheet<int, InfiniteTowerScheduleSheet.Row>.Remove(KeyValuePair<int,
InfiniteTowerScheduleSheet.Row>) ,
Sheet<int, InfiniteTowerScheduleSheet.Row>.Serialize() , object.Equals(object) ,
object.Equals(object, object) , object.GetHashCode() , object.GetType() ,
object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

InfiniteTowerScheduleSheet()

```
public InfiniteTowerScheduleSheet()
```

Class InfiniteTowerScheduleSheet.Row

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

Represents a row in the InfiniteTowerScheduleSheet containing schedule data.

```
[Serializable]
public class InfiniteTowerScheduleSheet.Row : SheetRow<int>
```

Inheritance

object ← SheetRow<int> ← InfiniteTowerScheduleSheet.Row

Inherited Members

SheetRow<int>.Validate() , SheetRow<int>.EndOfSheetInitialize() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Properties

DailyFreeTickets

Gets the number of free tickets given daily.

```
public int DailyFreeTickets { get; }
```

Property Value

int

EndBlockIndex

Gets the block index when the schedule ends.

```
public long EndBlockIndex { get; }
```

Property Value

long

FloorBegin

Gets the starting floor number for this schedule.

```
public int FloorBegin { get; }
```

Property Value

int

FloorEnd

Gets the ending floor number for this schedule.

```
public int FloorEnd { get; }
```

Property Value

int

FloorsCount

Gets the total number of floors in this schedule.

```
public int FloorsCount { get; }
```

Property Value

int

Id

Gets the unique identifier for this schedule.

```
public int Id { get; }
```

Property Value

int

InfiniteTowerId

Gets the infinite tower ID this schedule belongs to.

```
public int InfiniteTowerId { get; }
```

Property Value

int

Key

Gets the schedule ID as the key for this row.

```
public override int Key { get; }
```

Property Value

int

MaxTickets

Gets the maximum number of tickets a player can hold.

```
public int MaxTickets { get; }
```

Property Value

int

ResetIntervalBlocks

Gets the number of blocks between ticket resets.

```
public int ResetIntervalBlocks { get; }
```

Property Value

int

StartBlockIndex

Gets the block index when the schedule starts.

```
public long StartBlockIndex { get; }
```

Property Value

long

Methods

GetRemainingBlocks(long)

Gets the remaining blocks until the schedule ends.

```
public long GetRemainingBlocks(long currentBlockIndex)
```

Parameters

`currentBlockIndex` long

Current block index

Returns

long

Number of blocks remaining, or 0 if already ended

HasEnded(long)

Checks if the infinite tower schedule has ended.

```
public bool HasEnded(long currentBlockIndex)
```

Parameters

`currentBlockIndex` long

Current block index

Returns

bool

True if the schedule has ended, false otherwise

HasStarted(long)

Checks if the infinite tower schedule has started.

```
public bool HasStarted(long currentBlockIndex)
```

Parameters

`currentBlockIndex` long

Current block index

Returns

bool

True if the schedule has started, false otherwise

IsActive(long)

Checks if the infinite tower schedule is currently active.

```
public bool IsActive(long currentBlockIndex)
```

Parameters

`currentBlockIndex` long

Current block index

Returns

bool

True if the schedule is active, false otherwise

Set(IReadOnlyList<string>)

```
public override void Set(IReadOnlyList<string> fields)
```

Parameters

`fields` IReadOnlyList<string>

ValidateFloorRange(int, string)

Validates that the floor ID is within this schedule's floor range.

```
public void ValidateFloorRange(int floorId, string addressesHex)
```

Parameters

`floorId` int

Floor ID to validate

`addressesHex` string

Addresses hex for logging

Exceptions

InvalidOperationException

Thrown when floor ID is out of range

ValidateInfiniteTowerId(int, string)

Validates that the infinite tower ID matches this schedule's configuration.

```
public void ValidateInfiniteTowerId(int expectedInfiniteTowerId,
string addressesHex)
```

Parameters

`expectedInfiniteTowerId` int

The expected infinite tower ID

`addressesHex` string

Addresses hex for logging

Exceptions

SheetRowNotFoundException

Thrown when infinite tower ID mismatch

ValidateScheduleTiming(long, int, string)

Validates schedule timing for the infinite tower.

```
public void ValidateScheduleTiming(long currentBlockIndex, int infiniteTowerId,
```

```
  string addressesHex)
```

Parameters

`currentBlockIndex` long

Current block index

`infiniteTowerId` int

The infinite tower ID for error messages

`addressesHex` string

Addresses hex for logging

Exceptions

InfiniteTowerScheduleNotStartedException

Thrown when schedule has not started

InfiniteTowerScheduleEndedException

Thrown when schedule has ended

InfiniteTowerScheduleNotActiveException

Thrown when schedule is not active

Class ItemConfigForGradeSheet

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class ItemConfigForGradeSheet : Sheet<int, ItemConfigForGradeSheet.Row>,
IDictionary<int, ItemConfigForGradeSheet.Row>, ICollection<KeyValuePair<int,
ItemConfigForGradeSheet.Row>>, IEnumerable<KeyValuePair<int,
ItemConfigForGradeSheet.Row>>, IEnumerable, ISheet
```

Inheritance

object ← Sheet<int, ItemConfigForGradeSheet.Row> ← ItemConfigForGradeSheet

Implements

IDictionary<int, ItemConfigForGradeSheet.Row>,
ICollection<KeyValuePair<int, ItemConfigForGradeSheet.Row>>,
IEnumerable<KeyValuePair<int, ItemConfigForGradeSheet.Row>>, IEnumerable,
ISheet

Inherited Members

Sheet<int, ItemConfigForGradeSheet.Row>.Name ,
Sheet<int, ItemConfigForGradeSheet.Row>.OrderedList ,
Sheet<int, ItemConfigForGradeSheet.Row>.First ,
Sheet<int, ItemConfigForGradeSheet.Row>.Last ,
Sheet<int, ItemConfigForGradeSheet.Row>.Keys ,
Sheet<int, ItemConfigForGradeSheet.Row>.Values ,
Sheet<int, ItemConfigForGradeSheet.Row>.Count ,
Sheet<int, ItemConfigForGradeSheet.Row>.IsReadOnly ,
Sheet<int, ItemConfigForGradeSheet.Row>.this[int] ,
Sheet<int, ItemConfigForGradeSheet.Row>.Set(string, bool) ,
Sheet<int, ItemConfigForGradeSheet.Row>.Set<T>(Sheet<int, T>, bool) ,
Sheet<int, ItemConfigForGradeSheet.Row>.GetEnumerator() ,
Sheet<int, ItemConfigForGradeSheet.Row>.TryGetValue(int, out
ItemConfigForGradeSheet.Row, bool) ,
Sheet<int, ItemConfigForGradeSheet.Row>.AddRow(int, ItemConfigForGradeSheet.Row) ,
Sheet<int, ItemConfigForGradeSheet.Row>.Add(int, ItemConfigForGradeSheet.Row) ,
Sheet<int, ItemConfigForGradeSheet.Row>.ContainsKey(int) ,
Sheet<int, ItemConfigForGradeSheet.Row>.Remove(int) ,

Sheet<int, ItemConfigForGradeSheet.Row>.TryGetValue(int, out ItemConfigForGradeSheet.Row) , Sheet<int, ItemConfigForGradeSheet.Row>.Add(KeyValuePair<int, ItemConfigForGradeSheet.Row>) , Sheet<int, ItemConfigForGradeSheet.Row>.Clear() , Sheet<int, ItemConfigForGradeSheet.Row>.Contains(KeyValuePair<int, ItemConfigForGradeSheet.Row>) , Sheet<int, ItemConfigForGradeSheet.Row>.CopyTo(KeyValuePair<int, ItemConfigForGradeSheet.Row>[], int) , Sheet<int, ItemConfigForGradeSheet.Row>.Remove(KeyValuePair<int, ItemConfigForGradeSheet.Row>) , Sheet<int, ItemConfigForGradeSheet.Row>.Serialize() , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

ItemConfigForGradeSheet()

```
public ItemConfigForGradeSheet()
```

Class ItemConfigForGradeSheet.Row

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class ItemConfigForGradeSheet.Row : SheetRow<int>
```

Inheritance

object ← SheetRow<int> ← ItemConfigForGradeSheet.Row

Inherited Members

SheetRow<int>.Validate() , SheetRow<int>.EndOfSheetInitialize() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Properties

EnhancementLimit

```
public int EnhancementLimit { get; }
```

Property Value

int

Id

```
public int Id { get; }
```

Property Value

int

Key

```
public override int Key { get; }
```

Property Value

int

MonsterPartsCountForCombination

```
public int MonsterPartsCountForCombination { get; }
```

Property Value

int

MonsterPartsCountForCombinationWithNCG

```
public int MonsterPartsCountForCombinationWithNCG { get; }
```

Property Value

int

RandomBuffSkillMaxCountForCombination

```
public int RandomBuffSkillMaxCountForCombination { get; }
```

Property Value

int

RandomBuffSkillMinCountForCombination

```
public int RandomBuffSkillMinCountForCombination { get; }
```

Property Value

int

Methods

Set(IReadOnlyList<string>)

```
public override void Set(IReadOnlyList<string> fields)
```

Parameters

`fields` IReadOnlyList<string>

Class ItemEnhancementQuestSheet

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class ItemEnhancementQuestSheet : Sheet<int, ItemEnhancementQuestSheet.Row>,
IDictionary<int, ItemEnhancementQuestSheet.Row>, ICollection<KeyValuePair<int,
ItemEnhancementQuestSheet.Row>>, IEnumerable<KeyValuePair<int,
ItemEnhancementQuestSheet.Row>>, IEnumerable, ISheet
```

Inheritance

object ← Sheet<int, ItemEnhancementQuestSheet.Row> ← ItemEnhancementQuestSheet

Implements

IDictionary<int, ItemEnhancementQuestSheet.Row>, ICollection<KeyValuePair<int, ItemEnhancementQuestSheet.Row>>, IEnumerable<KeyValuePair<int, ItemEnhancementQuestSheet.Row>>, IEnumerable, ISheet

Inherited Members

Sheet<int, ItemEnhancementQuestSheet.Row>.Name ,
Sheet<int, ItemEnhancementQuestSheet.Row>.OrderedList ,
Sheet<int, ItemEnhancementQuestSheet.Row>.First ,
Sheet<int, ItemEnhancementQuestSheet.Row>.Last ,
Sheet<int, ItemEnhancementQuestSheet.Row>.Keys ,
Sheet<int, ItemEnhancementQuestSheet.Row>.Values ,
Sheet<int, ItemEnhancementQuestSheet.Row>.Count ,
Sheet<int, ItemEnhancementQuestSheet.Row>.IsReadOnly ,
Sheet<int, ItemEnhancementQuestSheet.Row>.this[int] ,
Sheet<int, ItemEnhancementQuestSheet.Row>.Set(string, bool) ,
Sheet<int, ItemEnhancementQuestSheet.Row>.Set<T>(Sheet<int, T>, bool) ,
Sheet<int, ItemEnhancementQuestSheet.Row>.GetEnumerator() ,
Sheet<int, ItemEnhancementQuestSheet.Row>.TryGetValue(int, out ItemEnhancementQuestSheet.Row, bool) ,
Sheet<int, ItemEnhancementQuestSheet.Row>.AddRow(int, ItemEnhancementQuestSheet.Row) ,
Sheet<int, ItemEnhancementQuestSheet.Row>.Add(int, ItemEnhancementQuestSheet.Row) ,

Sheet<int, ItemEnhancementQuestSheet.Row>.ContainsKey(int) ,
Sheet<int, ItemEnhancementQuestSheet.Row>.Remove(int) ,
Sheet<int, ItemEnhancementQuestSheet.Row>.TryGetValue(int, out
ItemEnhancementQuestSheet.Row) ,
Sheet<int, ItemEnhancementQuestSheet.Row>.Add(KeyValuePair<int,
ItemEnhancementQuestSheet.Row>) ,
Sheet<int, ItemEnhancementQuestSheet.Row>.Clear() ,
Sheet<int, ItemEnhancementQuestSheet.Row>.Contains(KeyValuePair<int,
ItemEnhancementQuestSheet.Row>) ,
Sheet<int, ItemEnhancementQuestSheet.Row>.CopyTo(KeyValuePair<int,
ItemEnhancementQuestSheet.Row>[], int) ,
Sheet<int, ItemEnhancementQuestSheet.Row>.Remove(KeyValuePair<int,
ItemEnhancementQuestSheet.Row>) ,
Sheet<int, ItemEnhancementQuestSheet.Row>.Serialize() , object.Equals(object) ,
object.Equals(object, object) , object.GetHashCode() , object.GetType() ,
object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

ItemEnhancementQuestSheet()

```
public ItemEnhancementQuestSheet()
```

Class ItemEnhancementQuestSheet.Row

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class ItemEnhancementQuestSheet.Row : QuestSheet.Row, IState
```

Inheritance

object ← SheetRow<int> ← QuestSheet.Row ← ItemEnhancementQuestSheet.Row

Implements

IState

Inherited Members

QuestSheet.Row.Key , QuestSheet.Row.Id , QuestSheet.Row.Goal , QuestSheet.Row.QuestRewardId , QuestSheet.Row.Serialize() , QuestSheet.Row.Deserialize(Dictionary) , SheetRow<int>.Validate() , SheetRow<int>.EndOfSheetInitialize() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Properties

Count

```
public int Count { get; }
```

Property Value

int

Grade

```
public int Grade { get; }
```

Property Value

int

Methods

Set(IReadOnlyList<string>)

```
public override void Set(IReadOnlyList<string> fields)
```

Parameters

`fields` IReadOnlyList<string>

Class ItemGradeQuestSheet

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class ItemGradeQuestSheet : Sheet<int, ItemGradeQuestSheet.Row>,
IDictionary<int, ItemGradeQuestSheet.Row>, ICollection<KeyValuePair<int,
ItemGradeQuestSheet.Row>>, IEnumerable<KeyValuePair<int, ItemGradeQuestSheet.Row>>,
IEnumerable, ISheet
```

Inheritance

object ← Sheet<int, ItemGradeQuestSheet.Row> ← ItemGradeQuestSheet

Implements

IDictionary<int, ItemGradeQuestSheet.Row>,
ICollection<KeyValuePair<int, ItemGradeQuestSheet.Row>>,
IEnumerable<KeyValuePair<int, ItemGradeQuestSheet.Row>>, IEnumerable, ISheet

Inherited Members

Sheet<int, ItemGradeQuestSheet.Row>.Name ,
Sheet<int, ItemGradeQuestSheet.Row>.OrderedList ,
Sheet<int, ItemGradeQuestSheet.Row>.First ,
Sheet<int, ItemGradeQuestSheet.Row>.Last ,
Sheet<int, ItemGradeQuestSheet.Row>.Keys ,
Sheet<int, ItemGradeQuestSheet.Row>.Values ,
Sheet<int, ItemGradeQuestSheet.Row>.Count ,
Sheet<int, ItemGradeQuestSheet.Row>.IsReadOnly ,
Sheet<int, ItemGradeQuestSheet.Row>.this[int] ,
Sheet<int, ItemGradeQuestSheet.Row>.Set(string, bool) ,
Sheet<int, ItemGradeQuestSheet.Row>.Set<T>(Sheet<int, T>, bool) ,
Sheet<int, ItemGradeQuestSheet.Row>.GetEnumerator() ,
Sheet<int, ItemGradeQuestSheet.Row>.TryGetValue(int, out ItemGradeQuestSheet.Row, bool) ,
Sheet<int, ItemGradeQuestSheet.Row>.AddRow(int, ItemGradeQuestSheet.Row) ,
Sheet<int, ItemGradeQuestSheet.Row>.Add(int, ItemGradeQuestSheet.Row) ,
Sheet<int, ItemGradeQuestSheet.Row>.ContainsKey(int) ,
Sheet<int, ItemGradeQuestSheet.Row>.Remove(int) ,
Sheet<int, ItemGradeQuestSheet.Row>.TryGetValue(int, out ItemGradeQuestSheet.Row) ,

Sheet<int, ItemGradeQuestSheet.Row>.Add(KeyValuePair<int, ItemGradeQuestSheet.Row>) ,
Sheet<int, ItemGradeQuestSheet.Row>.Clear() ,
Sheet<int, ItemGradeQuestSheet.Row>.Contains(KeyValuePair<int, ItemGradeQuestSheet.Row>) ,
Sheet<int, ItemGradeQuestSheet.Row>.CopyTo(KeyValuePair<int, ItemGradeQuestSheet.Row>[], int) ,
Sheet<int, ItemGradeQuestSheet.Row>.Remove(KeyValuePair<int, ItemGradeQuestSheet.Row>) ,
Sheet<int, ItemGradeQuestSheet.Row>.Serialize() , object.Equals(object) ,
object.Equals(object, object) , object.GetHashCode() , object.GetType() ,
object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

ItemGradeQuestSheet()

```
public ItemGradeQuestSheet()
```

Class ItemGradeQuestSheet.Row

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class ItemGradeQuestSheet.Row : QuestSheet.Row, IState
```

Inheritance

object ← SheetRow<int> ← QuestSheet.Row ← ItemGradeQuestSheet.Row

Implements

IState

Inherited Members

QuestSheet.Row.Key , QuestSheet.Row.Id , QuestSheet.Row.Goal , QuestSheet.Row.QuestRewardId , QuestSheet.Row.Serialize() , QuestSheet.Row.Deserialize(Dictionary) , SheetRow<int>.Validate() , SheetRow<int>.EndOfSheetInitialize() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Properties

Grade

```
public int Grade { get; }
```

Property Value

int

Methods

Set(IReadOnlyList<string>)

```
public override void Set(IReadOnlyList<string> fields)
```

Parameters

`fields` IReadOnlyList<string>

Class ItemRequirementSheet

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
public class ItemRequirementSheet : Sheet<int, ItemRequirementSheet.Row>,
IDictionary<int, ItemRequirementSheet.Row>, ICollection<KeyValuePair<int,
ItemRequirementSheet.Row>>, IEnumerable<KeyValuePair<int,
ItemRequirementSheet.Row>>, IEnumerable, ISheet
```

Inheritance

object ← Sheet<int, ItemRequirementSheet.Row> ← ItemRequirementSheet

Implements

IDictionary<int, ItemRequirementSheet.Row>,
ICollection<KeyValuePair<int, ItemRequirementSheet.Row>>,
IEnumerable<KeyValuePair<int, ItemRequirementSheet.Row>>, IEnumerable,
ISheet

Inherited Members

Sheet<int, ItemRequirementSheet.Row>.Name ,
Sheet<int, ItemRequirementSheet.Row>.OrderedList ,
Sheet<int, ItemRequirementSheet.Row>.First ,
Sheet<int, ItemRequirementSheet.Row>.Last ,
Sheet<int, ItemRequirementSheet.Row>.Keys ,
Sheet<int, ItemRequirementSheet.Row>.Values ,
Sheet<int, ItemRequirementSheet.Row>.Count ,
Sheet<int, ItemRequirementSheet.Row>.IsReadOnly ,
Sheet<int, ItemRequirementSheet.Row>.this[int] ,
Sheet<int, ItemRequirementSheet.Row>.Set(string, bool) ,
Sheet<int, ItemRequirementSheet.Row>.Set<T>(Sheet<int, T>, bool) ,
Sheet<int, ItemRequirementSheet.Row>.GetEnumerator() ,
Sheet<int, ItemRequirementSheet.Row>.TryGetValue(int, out ItemRequirementSheet.Row,
bool) ,
Sheet<int, ItemRequirementSheet.Row>.AddRow(int, ItemRequirementSheet.Row) ,
Sheet<int, ItemRequirementSheet.Row>.Add(int, ItemRequirementSheet.Row) ,
Sheet<int, ItemRequirementSheet.Row>.ContainsKey(int) ,
Sheet<int, ItemRequirementSheet.Row>.Remove(int) ,
Sheet<int, ItemRequirementSheet.Row>.TryGetValue(int, out ItemRequirementSheet.Row) ,

Sheet<int, ItemRequirementSheet.Row>.Add(KeyValuePair<int, ItemRequirementSheet.Row>) ,
Sheet<int, ItemRequirementSheet.Row>.Clear() ,
Sheet<int, ItemRequirementSheet.Row>.Contains(KeyValuePair<int, ItemRequirementSheet.Row>) ,
Sheet<int, ItemRequirementSheet.Row>.CopyTo(KeyValuePair<int, ItemRequirementSheet.Row>[], int) ,
Sheet<int, ItemRequirementSheet.Row>.Remove(KeyValuePair<int, ItemRequirementSheet.Row>) ,
Sheet<int, ItemRequirementSheet.Row>.Serialize() , object.Equals(object) ,
object.Equals(object, object) , object.GetHashCode() , object.GetType() ,
object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

ItemRequirementSheet()

```
public ItemRequirementSheet()
```

Class ItemRequirementSheet.Row

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
public class ItemRequirementSheet.Row : SheetRow<int>
```

Inheritance

object ← SheetRow<int> ← ItemRequirementSheet.Row

Inherited Members

SheetRow<int>.Validate() , SheetRow<int>.EndOfSheetInitialize() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Properties

ItemId

```
public int ItemId { get; }
```

Property Value

int

Key

```
public override int Key { get; }
```

Property Value

int

Level

```
public int Level { get; }
```

Property Value

int

MimisLevel

```
public int MimisLevel { get; }
```

Property Value

int

Methods

Set(IReadOnlyList<string>)

```
public override void Set(IReadOnlyList<string> fields)
```

Parameters

`fields` IReadOnlyList<string>

Class ItemSheet

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class ItemSheet : Sheet<int, ItemSheet.Row>, IDictionary<int, ItemSheet.Row>,
ICollection<KeyValuePair<int, ItemSheet.Row>>, IEnumerable<KeyValuePair<int,
ItemSheet.Row>>, IEnumerable, ISheet
```

Inheritance

object ← Sheet<int, ItemSheet.Row> ← ItemSheet

Implements

IDictionary<int, ItemSheet.Row>,
ICollection<KeyValuePair<int, ItemSheet.Row>>,
IEnumerable<KeyValuePair<int, ItemSheet.Row>>, IEnumerable, ISheet

Inherited Members

Sheet<int, ItemSheet.Row>.Name , Sheet<int, ItemSheet.Row>.OrderedList ,
Sheet<int, ItemSheet.Row>.First , Sheet<int, ItemSheet.Row>.Last ,
Sheet<int, ItemSheet.Row>.Keys , Sheet<int, ItemSheet.Row>.Values ,
Sheet<int, ItemSheet.Row>.Count , Sheet<int, ItemSheet.Row>.IsReadOnly ,
Sheet<int, ItemSheet.Row>.this[int] , Sheet<int, ItemSheet.Row>.Set(string, bool) ,
Sheet<int, ItemSheet.Row>.Set<T>(Sheet<int, T>, bool) ,
Sheet<int, ItemSheet.Row>.GetEnumerator() ,
Sheet<int, ItemSheet.Row>.TryGetValue(int, out ItemSheet.Row, bool) ,
Sheet<int, ItemSheet.Row>.AddRow(int, ItemSheet.Row) ,
Sheet<int, ItemSheet.Row>.Add(int, ItemSheet.Row) ,
Sheet<int, ItemSheet.Row>.ContainsKey(int) , Sheet<int, ItemSheet.Row>.Remove(int) ,
Sheet<int, ItemSheet.Row>.TryGetValue(int, out ItemSheet.Row) ,
Sheet<int, ItemSheet.Row>.Add(KeyValuePair<int, ItemSheet.Row>) ,
Sheet<int, ItemSheet.Row>.Clear() ,
Sheet<int, ItemSheet.Row>.Contains(KeyValuePair<int, ItemSheet.Row>) ,
Sheet<int, ItemSheet.Row>.CopyTo(KeyValuePair<int, ItemSheet.Row>[], int) ,
Sheet<int, ItemSheet.Row>.Remove(KeyValuePair<int, ItemSheet.Row>) ,
Sheet<int, ItemSheet.Row>.Serialize() , object.Equals(object) ,
object.Equals(object, object) , object.GetHashCode() , object.GetType() ,
object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

ItemSheet()

```
public ItemSheet()
```

Class ItemSheet.Row

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class ItemSheet.Row : ItemSheet.RowBase
```

Inheritance

object ← SheetRow<int> ← ItemSheet.RowBase ← ItemSheet.Row

Derived

ConsumableItemSheet.Row, CostumeItemSheet.Row, EquipmentItemSheet.Row, MaterialItemSheet.Row

Inherited Members

SheetRow<int>.Validate() , SheetRow<int>.EndOfSheetInitialize() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

Row()

```
public Row()
```

Row(Dictionary)

```
public Row(Dictionary serialized)
```

Parameters

`serialized` Dictionary

Properties

ElementalType

```
public ElementalType ElementalType { get; }
```

Property Value

ElementalType

Grade

```
public int Grade { get; }
```

Property Value

int

Id

```
public int Id { get; }
```

Property Value

int

ItemSubType

```
public ItemSubType ItemSubType { get; protected set; }
```

Property Value

ItemSubType

ItemType

```
public virtual ItemType ItemType { get; }
```

Property Value

ItemType

Key

```
public override int Key { get; }
```

Property Value

int

Methods

Deserialize(Dictionary)

```
public static ItemSheet.Row Deserialize(Dictionary serialized)
```

Parameters

`serialized` Dictionary

Returns

ItemSheet.Row

Serialize()

```
public override IValue Serialize()
```

Returns

IValue

Set(IReadOnlyList<string>)

```
public override void Set(IReadOnlyList<string> fields)
```

Parameters

`fields` IReadOnlyList<string>

Class ItemSheet.RowBase

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public abstract class ItemSheet.RowBase : SheetRow<int>
```

Inheritance

object ← SheetRow<int> ← ItemSheet.RowBase

Derived

ItemSheet.Row

Inherited Members

SheetRow<int>.Key , SheetRow<int>.Set(IReadOnlyList<string>) , SheetRow<int>.Validate() , SheetRow<int>.EndOfSheetInitialize() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Methods

Serialize()

```
public abstract IValue Serialize()
```

Returns

IValue

Class ItemTypeCollectQuestSheet

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class ItemTypeCollectQuestSheet : Sheet<int, ItemTypeCollectQuestSheet.Row>,
IDictionary<int, ItemTypeCollectQuestSheet.Row>, ICollection<KeyValuePair<int,
ItemTypeCollectQuestSheet.Row>>, IEnumerable<KeyValuePair<int,
ItemTypeCollectQuestSheet.Row>>, IEnumerable, ISheet
```

Inheritance

object ← Sheet<int, ItemTypeCollectQuestSheet.Row> ← ItemTypeCollectQuestSheet

Implements

IDictionary<int, ItemTypeCollectQuestSheet.Row>,
ICollection<KeyValuePair<int, ItemTypeCollectQuestSheet.Row>>,
IEnumerable<KeyValuePair<int, ItemTypeCollectQuestSheet.Row>>, IEnumerable,
ISheet

Inherited Members

Sheet<int, ItemTypeCollectQuestSheet.Row>.Name ,
Sheet<int, ItemTypeCollectQuestSheet.Row>.OrderedList ,
Sheet<int, ItemTypeCollectQuestSheet.Row>.First ,
Sheet<int, ItemTypeCollectQuestSheet.Row>.Last ,
Sheet<int, ItemTypeCollectQuestSheet.Row>.Keys ,
Sheet<int, ItemTypeCollectQuestSheet.Row>.Values ,
Sheet<int, ItemTypeCollectQuestSheet.Row>.Count ,
Sheet<int, ItemTypeCollectQuestSheet.Row>.IsReadOnly ,
Sheet<int, ItemTypeCollectQuestSheet.Row>.this[int] ,
Sheet<int, ItemTypeCollectQuestSheet.Row>.Set(string, bool) ,
Sheet<int, ItemTypeCollectQuestSheet.Row>.Set<T>(Sheet<int, T>, bool) ,
Sheet<int, ItemTypeCollectQuestSheet.Row>.GetEnumerator() ,
Sheet<int, ItemTypeCollectQuestSheet.Row>.TryGetValue(int, out
ItemTypeCollectQuestSheet.Row, bool) ,
Sheet<int, ItemTypeCollectQuestSheet.Row>.AddRow(int, ItemTypeCollectQuestSheet.Row)
,
Sheet<int, ItemTypeCollectQuestSheet.Row>.Add(int, ItemTypeCollectQuestSheet.Row) ,
Sheet<int, ItemTypeCollectQuestSheet.Row>.ContainsKey(int) ,
Sheet<int, ItemTypeCollectQuestSheet.Row>.Remove(int) ,

Sheet<int, ItemTypeCollectQuestSheet.Row>.TryGetValue(int, out ItemTypeCollectQuestSheet.Row) , Sheet<int, ItemTypeCollectQuestSheet.Row>.Add(KeyValuePair<int, ItemTypeCollectQuestSheet.Row>) , Sheet<int, ItemTypeCollectQuestSheet.Row>.Clear() , Sheet<int, ItemTypeCollectQuestSheet.Row>.Contains(KeyValuePair<int, ItemTypeCollectQuestSheet.Row>) , Sheet<int, ItemTypeCollectQuestSheet.Row>.CopyTo(KeyValuePair<int, ItemTypeCollectQuestSheet.Row>[], int) , Sheet<int, ItemTypeCollectQuestSheet.Row>.Remove(KeyValuePair<int, ItemTypeCollectQuestSheet.Row>) , Sheet<int, ItemTypeCollectQuestSheet.Row>.Serialize() , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

ItemTypeCollectQuestSheet()

```
public ItemTypeCollectQuestSheet()
```

Class ItemTypeCollectQuestSheet.Row

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class ItemTypeCollectQuestSheet.Row : QuestSheet.Row, IState
```

Inheritance

object ← SheetRow<int> ← QuestSheet.Row ← ItemTypeCollectQuestSheet.Row

Implements

IState

Inherited Members

QuestSheet.Row.Key , QuestSheet.Row.Id , QuestSheet.Row.Goal , QuestSheet.Row.QuestRewardId , QuestSheet.Row.Serialize() , QuestSheet.Row.Deserialize(Dictionary) , SheetRow<int>.Validate() , SheetRow<int>.EndOfSheetInitialize() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Properties

ItemType

```
public ItemType ItemType { get; }
```

Property Value

ItemType

Methods

Set(IReadOnlyList<string>)

```
public override void Set(IReadOnlyList<string> fields)
```

Parameters

`fields` IReadOnlyList<string>

Class MaterialItemSheet

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class MaterialItemSheet : Sheet<int, MaterialItemSheet.Row>, IDictionary<int,
MaterialItemSheet.Row>, ICollection<KeyValuePair<int, MaterialItemSheet.Row>>,
IEnumerable<KeyValuePair<int, MaterialItemSheet.Row>>, IEnumerable, ISheet
```

Inheritance

object ← Sheet<int, MaterialItemSheet.Row> ← MaterialItemSheet

Implements

IDictionary<int, MaterialItemSheet.Row>,
ICollection<KeyValuePair<int, MaterialItemSheet.Row>>,
IEnumerable<KeyValuePair<int, MaterialItemSheet.Row>>, IEnumerable, ISheet

Inherited Members

Sheet<int, MaterialItemSheet.Row>.Name ,
Sheet<int, MaterialItemSheet.Row>.OrderedList ,
Sheet<int, MaterialItemSheet.Row>.First , Sheet<int, MaterialItemSheet.Row>.Last ,
Sheet<int, MaterialItemSheet.Row>.Keys , Sheet<int, MaterialItemSheet.Row>.Values ,
Sheet<int, MaterialItemSheet.Row>.Count ,
Sheet<int, MaterialItemSheet.Row>.IsReadOnly ,
Sheet<int, MaterialItemSheet.Row>.this[int] ,
Sheet<int, MaterialItemSheet.Row>.Set(string, bool) ,
Sheet<int, MaterialItemSheet.Row>.Set<T>(Sheet<int, T>, bool) ,
Sheet<int, MaterialItemSheet.Row>.GetEnumerator() ,
Sheet<int, MaterialItemSheet.Row>.TryGetValue(int, out MaterialItemSheet.Row, bool) ,
Sheet<int, MaterialItemSheet.Row>.AddRow(int, MaterialItemSheet.Row) ,
Sheet<int, MaterialItemSheet.Row>.Add(int, MaterialItemSheet.Row) ,
Sheet<int, MaterialItemSheet.Row>.ContainsKey(int) ,
Sheet<int, MaterialItemSheet.Row>.Remove(int) ,
Sheet<int, MaterialItemSheet.Row>.TryGetValue(int, out MaterialItemSheet.Row) ,
Sheet<int, MaterialItemSheet.Row>.Add(KeyValuePair<int, MaterialItemSheet.Row>) ,
Sheet<int, MaterialItemSheet.Row>.Clear() ,
Sheet<int, MaterialItemSheet.Row>.Contains(KeyValuePair<int, MaterialItemSheet.Row>) ,

Sheet<int, MaterialItemSheet.Row>.CopyTo(KeyValuePair<int, MaterialItemSheet.Row>[], int) ,
Sheet<int, MaterialItemSheet.Row>.Remove(KeyValuePair<int, MaterialItemSheet.Row>) ,
Sheet<int, MaterialItemSheet.Row>.Serialize() , object.Equals(object) ,
object.Equals(object, object) , object.GetHashCode() , object.GetType() ,
object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Extension Methods
MaterialItemSheetExtensions.ValidateFromAction(MaterialItemSheet, List<EquipmentItemSubRecipeSheet.MaterialInfo>, Dictionary<int, int>, string)

Constructors

MaterialItemSheet()

```
public MaterialItemSheet()
```

Class MaterialItemSheet.Row

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class MaterialItemSheet.Row : ItemSheet.Row, ISerializable
```

Inheritance

object ← SheetRow<int> ← ItemSheet.RowBase ← ItemSheet.Row ← MaterialItemSheet.Row

Implements

ISerializable

Inherited Members

ItemSheet.Row.Key , ItemSheet.Row.Id , ItemSheet.Row.ItemSubType , ItemSheet.Row.Grade , ItemSheet.Row.ElementalType , ItemSheet.Row.Serialize() , ItemSheet.Row.Deserialize(Dictionary) , SheetRow<int>.Validate() , SheetRow<int>.EndOfSheetInitialize() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

Row()

```
public Row()
```

Row(Dictionary)

```
public Row(Dictionary serialized)
```

Parameters

`serialized` Dictionary

Row(SerializationInfo, StreamingContext)

```
protected Row(SerializationInfo info, StreamingContext context)
```

Parameters

`info` SerializationInfo

`context` StreamingContext

Properties

ItemId

```
public HashDigest<SHA256> ItemId { get; }
```

Property Value

HashDigest<SHA256>

ItemType

```
public override ItemType ItemType { get; }
```

Property Value

ItemType

Methods

GetObjectData(SerializationInfo, StreamingContext)

Populates a SerializationInfo with the data needed to serialize the target object.

```
public void GetObjectData(SerializationInfo info, StreamingContext context)
```

Parameters

`info` SerializationInfo

The SerializationInfo to populate with data.

`context` StreamingContext

The destination (see StreamingContext) for this serialization.

Exceptions

SecurityException

The caller does not have the required permission.

Set(IReadOnlyList<string>)

```
public override void Set(IReadOnlyList<string> fields)
```

Parameters

`fields` IReadOnlyList<string>

Class MimisbrunnrSheet

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class MimisbrunnrSheet : Sheet<int, MimisbrunnrSheet.Row>, IDictionary<int,
MimisbrunnrSheet.Row>, ICollection<KeyValuePair<int, MimisbrunnrSheet.Row>>,
IEnumerable<KeyValuePair<int, MimisbrunnrSheet.Row>>, IEnumerable, ISheet
```

Inheritance

object ← Sheet<int, MimisbrunnrSheet.Row> ← MimisbrunnrSheet

Implements

IDictionary<int, MimisbrunnrSheet.Row>,
ICollection<KeyValuePair<int, MimisbrunnrSheet.Row>>,
IEnumerable<KeyValuePair<int, MimisbrunnrSheet.Row>>, IEnumerable, ISheet

Inherited Members

Sheet<int, MimisbrunnrSheet.Row>.Name ,
Sheet<int, MimisbrunnrSheet.Row>.OrderedList ,
Sheet<int, MimisbrunnrSheet.Row>.First , Sheet<int, MimisbrunnrSheet.Row>.Last ,
Sheet<int, MimisbrunnrSheet.Row>.Keys , Sheet<int, MimisbrunnrSheet.Row>.Values ,
Sheet<int, MimisbrunnrSheet.Row>.Count ,
Sheet<int, MimisbrunnrSheet.Row>.IsReadOnly ,
Sheet<int, MimisbrunnrSheet.Row>.this[int] ,
Sheet<int, MimisbrunnrSheet.Row>.Set(string, bool) ,
Sheet<int, MimisbrunnrSheet.Row>.Set<T>(Sheet<int, T>, bool) ,
Sheet<int, MimisbrunnrSheet.Row>.GetEnumerator() ,
Sheet<int, MimisbrunnrSheet.Row>.TryGetValue(int, out MimisbrunnrSheet.Row, bool) ,
Sheet<int, MimisbrunnrSheet.Row>.AddRow(int, MimisbrunnrSheet.Row) ,
Sheet<int, MimisbrunnrSheet.Row>.Add(int, MimisbrunnrSheet.Row) ,
Sheet<int, MimisbrunnrSheet.Row>.ContainsKey(int) ,
Sheet<int, MimisbrunnrSheet.Row>.Remove(int) ,
Sheet<int, MimisbrunnrSheet.Row>.TryGetValue(int, out MimisbrunnrSheet.Row) ,
Sheet<int, MimisbrunnrSheet.Row>.Add(KeyValuePair<int, MimisbrunnrSheet.Row>) ,
Sheet<int, MimisbrunnrSheet.Row>.Clear() ,
Sheet<int, MimisbrunnrSheet.Row>.Contains(KeyValuePair<int, MimisbrunnrSheet.Row>) ,

Sheet<int, MimisbrunnrSheet.Row>.CopyTo(KeyValuePair<int, MimisbrunnrSheet.Row>[], int) ,
Sheet<int, MimisbrunnrSheet.Row>.Remove(KeyValuePair<int, MimisbrunnrSheet.Row>) ,
Sheet<int, MimisbrunnrSheet.Row>.Serialize() , object.Equals(object) ,
object.Equals(object, object) , object.GetHashCode() , object.GetType() ,
object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

MimisbrunnrSheet()

```
public MimisbrunnrSheet()
```

Class MimisbrunnrSheet.Row

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class MimisbrunnrSheet.Row : SheetRow<int>
```

Inheritance

object ← SheetRow<int> ← MimisbrunnrSheet.Row

Inherited Members

SheetRow<int>.Validate() , SheetRow<int>.EndOfSheetInitialize() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Properties

ElementalTypes

```
public List<ElementalType> ElementalTypes { get; }
```

Property Value

List<ElementalType>

Id

```
public int Id { get; }
```

Property Value

int

Key

```
public override int Key { get; }
```

Property Value

int

Methods

Set(IReadOnlyList<string>)

```
public override void Set(IReadOnlyList<string> fields)
```

Parameters

`fields` IReadOnlyList<string>

Class MonsterCollectionRewardSheet

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class MonsterCollectionRewardSheet : Sheet<int,
MonsterCollectionRewardSheet.Row>, IDictionary<int,
MonsterCollectionRewardSheet.Row>, ICollection<KeyValuePair<int,
MonsterCollectionRewardSheet.Row>>, IEnumerable<KeyValuePair<int,
MonsterCollectionRewardSheet.Row>>, IEnumerable, ISheet
```

Inheritance

object ← Sheet<int, MonsterCollectionRewardSheet.Row> ← MonsterCollectionRewardSheet

Implements

IDictionary<int, MonsterCollectionRewardSheet.Row>, ICollection<KeyValuePair<int, MonsterCollectionRewardSheet.Row>>, IEnumerable<KeyValuePair<int, MonsterCollectionRewardSheet.Row>>, IEnumerable, ISheet

Inherited Members

Sheet<int, MonsterCollectionRewardSheet.Row>.Name ,
Sheet<int, MonsterCollectionRewardSheet.Row>.OrderedList ,
Sheet<int, MonsterCollectionRewardSheet.Row>.First ,
Sheet<int, MonsterCollectionRewardSheet.Row>.Last ,
Sheet<int, MonsterCollectionRewardSheet.Row>.Keys ,
Sheet<int, MonsterCollectionRewardSheet.Row>.Values ,
Sheet<int, MonsterCollectionRewardSheet.Row>.Count ,
Sheet<int, MonsterCollectionRewardSheet.Row>.IsReadOnly ,
Sheet<int, MonsterCollectionRewardSheet.Row>.this[int] ,
Sheet<int, MonsterCollectionRewardSheet.Row>.Set(string, bool) ,
Sheet<int, MonsterCollectionRewardSheet.Row>.Set<T>(Sheet<int, T>, bool) ,
Sheet<int, MonsterCollectionRewardSheet.Row>.GetEnumerator() ,
Sheet<int, MonsterCollectionRewardSheet.Row>.TryGetValue(int, out MonsterCollectionRewardSheet.Row, bool) ,
Sheet<int, MonsterCollectionRewardSheet.Row>.AddRow(int, MonsterCollectionRewardSheet.Row) ,

Sheet<int, MonsterCollectionRewardSheet.Row>.Add(int, MonsterCollectionRewardSheet.Row) , Sheet<int, MonsterCollectionRewardSheet.Row>.ContainsKey(int) , Sheet<int, MonsterCollectionRewardSheet.Row>.Remove(int) , Sheet<int, MonsterCollectionRewardSheet.Row>.TryGetValue(int, out MonsterCollectionRewardSheet.Row) , Sheet<int, MonsterCollectionRewardSheet.Row>.Add(KeyValuePair<int, MonsterCollectionRewardSheet.Row>) , Sheet<int, MonsterCollectionRewardSheet.Row>.Clear() , Sheet<int, MonsterCollectionRewardSheet.Row>.Contains(KeyValuePair<int, MonsterCollectionRewardSheet.Row>) , Sheet<int, MonsterCollectionRewardSheet.Row>.CopyTo(KeyValuePair<int, MonsterCollectionRewardSheet.Row>[], int) , Sheet<int, MonsterCollectionRewardSheet.Row>.Remove(KeyValuePair<int, MonsterCollectionRewardSheet.Row>) , Sheet<int, MonsterCollectionRewardSheet.Row>.Serialize() , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

MonsterCollectionRewardSheet()

```
public MonsterCollectionRewardSheet()
```

Methods

AddRow(int, Row)

```
protected override void AddRow(int key, MonsterCollectionRewardSheet.Row value)
```

Parameters

`key` int

`value` MonsterCollectionRewardSheet.Row

Class MonsterCollectionRewardSheet.RewardInfo

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class MonsterCollectionRewardSheet.RewardInfo
```

Inheritance

object ← MonsterCollectionRewardSheet.RewardInfo

Inherited Members

object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

RewardInfo(Dictionary)

```
public RewardInfo(Dictionary dictionary)
```

Parameters

`dictionary` Dictionary

RewardInfo(int, int)

```
public RewardInfo(int itemId, int quantity)
```

Parameters

`itemId` int

`quantity` int

RewardInfo(params string[])

```
public RewardInfo(params string[] fields)
```

Parameters

`fields` string[]

Fields

ItemId

```
public readonly int ItemId
```

Field Value

int

Quantity

```
public readonly int Quantity
```

Field Value

int

Methods

Equals(RewardInfo)

```
protected bool Equals(MonsterCollectionRewardSheet.RewardInfo other)
```

Parameters

`other` MonsterCollectionRewardSheet.RewardInfo

Returns

bool

Equals(object)

Determines whether the specified object is equal to the current object.

```
public override bool Equals(object obj)
```

Parameters

`obj` object

The object to compare with the current object.

Returns

bool

true if the specified object is equal to the current object; otherwise, false.

GetHashCode()

Serves as the default hash function.

```
public override int GetHashCode()
```

Returns

int

A hash code for the current object.

Serialize()

```
public IValue Serialize()
```

Returns

IValue

Class MonsterCollectionRewardSheet.Row

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class MonsterCollectionRewardSheet.Row : SheetRow<int>
```

Inheritance

object ← SheetRow<int> ← MonsterCollectionRewardSheet.Row

Inherited Members

SheetRow<int>.Validate() , SheetRow<int>.EndOfSheetInitialize() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Properties

Key

```
public override int Key { get; }
```

Property Value

int

MonsterCollectionLevel

```
public int MonsterCollectionLevel { get; }
```

Property Value

int

Rewards

```
public List<MonsterCollectionRewardSheet.RewardInfo> Rewards { get; }
```

Property Value

List<MonsterCollectionRewardSheet.RewardInfo>

Methods

Set(IReadOnlyList<string>)

```
public override void Set(IReadOnlyList<string> fields)
```

Parameters

`fields` IReadOnlyList<string>

Class MonsterCollectionSheet

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class MonsterCollectionSheet : Sheet<int, MonsterCollectionSheet.Row>,
IDictionary<int, MonsterCollectionSheet.Row>, ICollection<KeyValuePair<int,
MonsterCollectionSheet.Row>>, IEnumerable<KeyValuePair<int,
MonsterCollectionSheet.Row>>, IEnumerable, ISheet
```

Inheritance

object ← Sheet<int, MonsterCollectionSheet.Row> ← MonsterCollectionSheet

Implements

IDictionary<int, MonsterCollectionSheet.Row>,
ICollection<KeyValuePair<int, MonsterCollectionSheet.Row>>,
IEnumerable<KeyValuePair<int, MonsterCollectionSheet.Row>>, IEnumerable,
ISheet

Inherited Members

Sheet<int, MonsterCollectionSheet.Row>.Name ,
Sheet<int, MonsterCollectionSheet.Row>.OrderedList ,
Sheet<int, MonsterCollectionSheet.Row>.First ,
Sheet<int, MonsterCollectionSheet.Row>.Last ,
Sheet<int, MonsterCollectionSheet.Row>.Keys ,
Sheet<int, MonsterCollectionSheet.Row>.Values ,
Sheet<int, MonsterCollectionSheet.Row>.Count ,
Sheet<int, MonsterCollectionSheet.Row>.IsReadOnly ,
Sheet<int, MonsterCollectionSheet.Row>.this[int] ,
Sheet<int, MonsterCollectionSheet.Row>.Set(string, bool) ,
Sheet<int, MonsterCollectionSheet.Row>.Set<T>(Sheet<int, T>, bool) ,
Sheet<int, MonsterCollectionSheet.Row>.GetEnumerator() ,
Sheet<int, MonsterCollectionSheet.Row>.TryGetValue(int, out MonsterCollectionSheet.Row,
bool) ,
Sheet<int, MonsterCollectionSheet.Row>.AddRow(int, MonsterCollectionSheet.Row) ,
Sheet<int, MonsterCollectionSheet.Row>.Add(int, MonsterCollectionSheet.Row) ,
Sheet<int, MonsterCollectionSheet.Row>.ContainsKey(int) ,
Sheet<int, MonsterCollectionSheet.Row>.Remove(int) ,

Sheet<int, MonsterCollectionSheet.Row>.TryGetValue(int, out MonsterCollectionSheet.Row) , Sheet<int, MonsterCollectionSheet.Row>.Add(KeyValuePair<int, MonsterCollectionSheet.Row>) , Sheet<int, MonsterCollectionSheet.Row>.Clear() , Sheet<int, MonsterCollectionSheet.Row>.Contains(KeyValuePair<int, MonsterCollectionSheet.Row>) , Sheet<int, MonsterCollectionSheet.Row>.CopyTo(KeyValuePair<int, MonsterCollectionSheet.Row>[], int) , Sheet<int, MonsterCollectionSheet.Row>.Remove(KeyValuePair<int, MonsterCollectionSheet.Row>) , Sheet<int, MonsterCollectionSheet.Row>.Serialize() , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

MonsterCollectionSheet()

```
public MonsterCollectionSheet()
```

Class MonsterCollectionSheet.Row

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class MonsterCollectionSheet.Row : SheetRow<int>
```

Inheritance

object ← SheetRow<int> ← MonsterCollectionSheet.Row

Inherited Members

SheetRow<int>.Validate() , SheetRow<int>.EndOfSheetInitialize() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Properties

Key

```
public override int Key { get; }
```

Property Value

int

Level

```
public int Level { get; }
```

Property Value

int

RequiredGold

```
public int RequiredGold { get; }
```

Property Value

int

RewardId

```
public int RewardId { get; }
```

Property Value

int

Methods

Set(IReadOnlyList<string>)

```
public override void Set(IReadOnlyList<string> fields)
```

Parameters

`fields` IReadOnlyList<string>

Class MonsterQuestSheet

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class MonsterQuestSheet : Sheet<int, MonsterQuestSheet.Row>, IDictionary<int,
MonsterQuestSheet.Row>, ICollection<KeyValuePair<int, MonsterQuestSheet.Row>>,
IEnumerable<KeyValuePair<int, MonsterQuestSheet.Row>>, IEnumerable, ISheet
```

Inheritance

object ← Sheet<int, MonsterQuestSheet.Row> ← MonsterQuestSheet

Implements

IDictionary<int, MonsterQuestSheet.Row>,
ICollection<KeyValuePair<int, MonsterQuestSheet.Row>>,
IEnumerable<KeyValuePair<int, MonsterQuestSheet.Row>>, IEnumerable, ISheet

Inherited Members

Sheet<int, MonsterQuestSheet.Row>.Name ,
Sheet<int, MonsterQuestSheet.Row>.OrderedList ,
Sheet<int, MonsterQuestSheet.Row>.First , Sheet<int, MonsterQuestSheet.Row>.Last ,
Sheet<int, MonsterQuestSheet.Row>.Keys , Sheet<int, MonsterQuestSheet.Row>.Values ,
Sheet<int, MonsterQuestSheet.Row>.Count ,
Sheet<int, MonsterQuestSheet.Row>.IsReadOnly ,
Sheet<int, MonsterQuestSheet.Row>.this[int] ,
Sheet<int, MonsterQuestSheet.Row>.Set(string, bool) ,
Sheet<int, MonsterQuestSheet.Row>.Set<T>(Sheet<int, T>, bool) ,
Sheet<int, MonsterQuestSheet.Row>.GetEnumerator() ,
Sheet<int, MonsterQuestSheet.Row>.TryGetValue(int, out MonsterQuestSheet.Row, bool) ,
Sheet<int, MonsterQuestSheet.Row>.AddRow(int, MonsterQuestSheet.Row) ,
Sheet<int, MonsterQuestSheet.Row>.Add(int, MonsterQuestSheet.Row) ,
Sheet<int, MonsterQuestSheet.Row>.ContainsKey(int) ,
Sheet<int, MonsterQuestSheet.Row>.Remove(int) ,
Sheet<int, MonsterQuestSheet.Row>.TryGetValue(int, out MonsterQuestSheet.Row) ,
Sheet<int, MonsterQuestSheet.Row>.Add(KeyValuePair<int, MonsterQuestSheet.Row>) ,
Sheet<int, MonsterQuestSheet.Row>.Clear() ,
Sheet<int, MonsterQuestSheet.Row>.Contains(KeyValuePair<int,
MonsterQuestSheet.Row>) ,

Sheet<int, MonsterQuestSheet.Row>.CopyTo(KeyValuePair<int, MonsterQuestSheet.Row>[], int) ,
Sheet<int, MonsterQuestSheet.Row>.Remove(KeyValuePair<int, MonsterQuestSheet.Row>) ,
Sheet<int, MonsterQuestSheet.Row>.Serialize() , object.Equals(object) ,
object.Equals(object, object) , object.GetHashCode() , object.GetType() ,
object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

MonsterQuestSheet()

```
public MonsterQuestSheet()
```

Class MonsterQuestSheet.Row

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class MonsterQuestSheet.Row : QuestSheet.Row, IState
```

Inheritance

object ← SheetRow<int> ← QuestSheet.Row ← MonsterQuestSheet.Row

Implements

IState

Inherited Members

QuestSheet.Row.Key , QuestSheet.Row.Id , QuestSheet.Row.Goal , QuestSheet.Row.QuestRewardId , QuestSheet.Row.Serialize() , QuestSheet.Row.Deserialize(Dictionary) , SheetRow<int>.Validate() , SheetRow<int>.EndOfSheetInitialize() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Properties

MonsterId

```
public int MonsterId { get; }
```

Property Value

int

Methods

Set(IReadOnlyList<string>)

```
public override void Set(IReadOnlyList<string> fields)
```

Parameters

`fields` IReadOnlyList<string>

Class PetCostSheet

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class PetCostSheet : Sheet<int, PetCostSheet.Row>, IDictionary<int,
PetCostSheet.Row>, ICollection<KeyValuePair<int, PetCostSheet.Row>>,
IEnumerable<KeyValuePair<int, PetCostSheet.Row>>, IEnumerable, ISheet
```

Inheritance

object ← Sheet<int, PetCostSheet.Row> ← PetCostSheet

Implements

IDictionary<int, PetCostSheet.Row>,
ICollection<KeyValuePair<int, PetCostSheet.Row>>,
IEnumerable<KeyValuePair<int, PetCostSheet.Row>>, IEnumerable, ISheet

Inherited Members

Sheet<int, PetCostSheet.Row>.Name , Sheet<int, PetCostSheet.Row>.OrderedList ,
Sheet<int, PetCostSheet.Row>.First , Sheet<int, PetCostSheet.Row>.Last ,
Sheet<int, PetCostSheet.Row>.Keys , Sheet<int, PetCostSheet.Row>.Values ,
Sheet<int, PetCostSheet.Row>.Count , Sheet<int, PetCostSheet.Row>.IsReadOnly ,
Sheet<int, PetCostSheet.Row>.this[int] , Sheet<int, PetCostSheet.Row>.Set(string, bool) ,
Sheet<int, PetCostSheet.Row>.Set<T>(Sheet<int, T>, bool) ,
Sheet<int, PetCostSheet.Row>.GetEnumerator() ,
Sheet<int, PetCostSheet.Row>.TryGetValue(int, out PetCostSheet.Row, bool) ,
Sheet<int, PetCostSheet.Row>.AddRow(int, PetCostSheet.Row) ,
Sheet<int, PetCostSheet.Row>.Add(int, PetCostSheet.Row) ,
Sheet<int, PetCostSheet.Row>.ContainsKey(int) ,
Sheet<int, PetCostSheet.Row>.Remove(int) ,
Sheet<int, PetCostSheet.Row>.TryGetValue(int, out PetCostSheet.Row) ,
Sheet<int, PetCostSheet.Row>.Add(KeyValuePair<int, PetCostSheet.Row>) ,
Sheet<int, PetCostSheet.Row>.Clear() ,
Sheet<int, PetCostSheet.Row>.Contains(KeyValuePair<int, PetCostSheet.Row>) ,
Sheet<int, PetCostSheet.Row>.CopyTo(KeyValuePair<int, PetCostSheet.Row>[], int) ,
Sheet<int, PetCostSheet.Row>.Remove(KeyValuePair<int, PetCostSheet.Row>) ,
Sheet<int, PetCostSheet.Row>.Serialize() , object.Equals(object) ,

object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

PetCostSheet()

```
public PetCostSheet()
```

Methods

AddRow(int, Row)

```
protected override void AddRow(int key, PetCostSheet.Row value)
```

Parameters

`key` int

`value` PetCostSheet.Row

Class PetCostSheet.PetCostData

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class PetCostSheet.PetCostData
```

Inheritance

object ← PetCostSheet.PetCostData

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

PetCostData(int, int, int)

```
public PetCostData(int level, int soulStoneQuantity, int ncgQuantity)
```

Parameters

`level` int

`soulStoneQuantity` int

`ncgQuantity` int

Properties

Level

```
public int Level { get; }
```

Property Value

int

NcgQuantity

```
public int NcgQuantity { get; }
```

Property Value

int

SoulStoneQuantity

```
public int SoulStoneQuantity { get; }
```

Property Value

int

Class PetCostSheet.Row

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class PetCostSheet.Row : SheetRow<int>
```

Inheritance

object ← SheetRow<int> ← PetCostSheet.Row

Inherited Members

SheetRow<int>.Validate() , SheetRow<int>.EndOfSheetInitialize() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Properties

Cost

```
public List<PetCostSheet.PetCostData> Cost { get; }
```

Property Value

List<PetCostSheet.PetCostData>

Key

```
public override int Key { get; }
```

Property Value

int

PetId

```
public int PetId { get; }
```

Property Value

int

Methods

Set(IReadOnlyList<string>)

```
public override void Set(IReadOnlyList<string> fields)
```

Parameters

`fields` IReadOnlyList<string>

TryGetCost(int, out PetCostData)

```
public bool TryGetCost(int level, out PetCostSheet.PetCostData costData)
```

Parameters

`level` int

`costData` PetCostSheet.PetCostData

Returns

bool

Class PetSheet

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class PetSheet : Sheet<int, PetSheet.Row>, IDictionary<int, PetSheet.Row>,
ICollection<KeyValuePair<int, PetSheet.Row>>, IEnumerable<KeyValuePair<int,
PetSheet.Row>>, IEnumerable, ISheet
```

Inheritance

object ← Sheet<int, PetSheet.Row> ← PetSheet

Implements

IDictionary<int, PetSheet.Row>, ICollection<KeyValuePair<int, PetSheet.Row>>, IEnumerable<KeyValuePair<int, PetSheet.Row>>, IEnumerable, ISheet

Inherited Members

Sheet<int, PetSheet.Row>.Name , Sheet<int, PetSheet.Row>.OrderedList , Sheet<int, PetSheet.Row>.First , Sheet<int, PetSheet.Row>.Last , Sheet<int, PetSheet.Row>.Keys , Sheet<int, PetSheet.Row>.Values , Sheet<int, PetSheet.Row>.Count , Sheet<int, PetSheet.Row>.IsReadOnly , Sheet<int, PetSheet.Row>.this[int] , Sheet<int, PetSheet.Row>.Set(string, bool) , Sheet<int, PetSheet.Row>.Set<T>(Sheet<int, T>, bool) , Sheet<int, PetSheet.Row>.GetEnumerator() , Sheet<int, PetSheet.Row>.TryGetValue(int, out PetSheet.Row, bool) , Sheet<int, PetSheet.Row>.AddRow(int, PetSheet.Row) , Sheet<int, PetSheet.Row>.Add(int, PetSheet.Row) , Sheet<int, PetSheet.Row>.ContainsKey(int) , Sheet<int, PetSheet.Row>.Remove(int) , Sheet<int, PetSheet.Row>.TryGetValue(int, out PetSheet.Row) , Sheet<int, PetSheet.Row>.Add(KeyValuePair<int, PetSheet.Row>) , Sheet<int, PetSheet.Row>.Clear() , Sheet<int, PetSheet.Row>.Contains(KeyValuePair<int, PetSheet.Row>) , Sheet<int, PetSheet.Row>.CopyTo(KeyValuePair<int, PetSheet.Row>[], int) , Sheet<int, PetSheet.Row>.Remove(KeyValuePair<int, PetSheet.Row>) , Sheet<int, PetSheet.Row>.Serialize() , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

PetSheet()

```
public PetSheet()
```

Class PetSheet.Row

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
public class PetSheet.Row : SheetRow<int>
```

Inheritance

object ← SheetRow<int> ← PetSheet.Row

Inherited Members

SheetRow<int>.Validate() , SheetRow<int>.EndOfSheetInitialize() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Properties

Grade

```
public int Grade { get; }
```

Property Value

int

Id

```
public int Id { get; }
```

Property Value

int

Key

```
public override int Key { get; }
```

Property Value

int

SoulStoneTicker

```
public string SoulStoneTicker { get; }
```

Property Value

string

Methods

Set(IReadOnlyList<string>)

```
public override void Set(IReadOnlyList<string> fields)
```

Parameters

`fields` IReadOnlyList<string>

Class QuestItemRewardSheet

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class QuestItemRewardSheet : Sheet<int, QuestItemRewardSheet.Row>,
IDictionary<int, QuestItemRewardSheet.Row>, ICollection<KeyValuePair<int,
QuestItemRewardSheet.Row>>, IEnumerable<KeyValuePair<int,
QuestItemRewardSheet.Row>>, IEnumerable, ISheet
```

Inheritance

object ← Sheet<int, QuestItemRewardSheet.Row> ← QuestItemRewardSheet

Implements

IDictionary<int, QuestItemRewardSheet.Row>,
ICollection<KeyValuePair<int, QuestItemRewardSheet.Row>>,
IEnumerable<KeyValuePair<int, QuestItemRewardSheet.Row>>, IEnumerable,
ISheet

Inherited Members

Sheet<int, QuestItemRewardSheet.Row>.Name ,
Sheet<int, QuestItemRewardSheet.Row>.OrderedList ,
Sheet<int, QuestItemRewardSheet.Row>.First ,
Sheet<int, QuestItemRewardSheet.Row>.Last ,
Sheet<int, QuestItemRewardSheet.Row>.Keys ,
Sheet<int, QuestItemRewardSheet.Row>.Values ,
Sheet<int, QuestItemRewardSheet.Row>.Count ,
Sheet<int, QuestItemRewardSheet.Row>.IsReadOnly ,
Sheet<int, QuestItemRewardSheet.Row>.this[int] ,
Sheet<int, QuestItemRewardSheet.Row>.Set(string, bool) ,
Sheet<int, QuestItemRewardSheet.Row>.Set<T>(Sheet<int, T>, bool) ,
Sheet<int, QuestItemRewardSheet.Row>.GetEnumerator() ,
Sheet<int, QuestItemRewardSheet.Row>.TryGetValue(int, out QuestItemRewardSheet.Row,
bool) ,
Sheet<int, QuestItemRewardSheet.Row>.AddRow(int, QuestItemRewardSheet.Row) ,
Sheet<int, QuestItemRewardSheet.Row>.Add(int, QuestItemRewardSheet.Row) ,
Sheet<int, QuestItemRewardSheet.Row>.ContainsKey(int) ,
Sheet<int, QuestItemRewardSheet.Row>.Remove(int) ,

Sheet<int, QuestItemRewardSheet.Row>.TryGetValue(int, out QuestItemRewardSheet.Row) , Sheet<int, QuestItemRewardSheet.Row>.Add(KeyValuePair<int, QuestItemRewardSheet.Row>) , Sheet<int, QuestItemRewardSheet.Row>.Clear() , Sheet<int, QuestItemRewardSheet.Row>.Contains(KeyValuePair<int, QuestItemRewardSheet.Row>) , Sheet<int, QuestItemRewardSheet.Row>.CopyTo(KeyValuePair<int, QuestItemRewardSheet.Row>[], int) , Sheet<int, QuestItemRewardSheet.Row>.Remove(KeyValuePair<int, QuestItemRewardSheet.Row>) , Sheet<int, QuestItemRewardSheet.Row>.Serialize() , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

QuestItemRewardSheet()

```
public QuestItemRewardSheet()
```

Class QuestItemRewardSheet.Row

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class QuestItemRewardSheet.Row : SheetRow<int>
```

Inheritance

object ← SheetRow<int> ← QuestItemRewardSheet.Row

Inherited Members

SheetRow<int>.Validate() , SheetRow<int>.EndOfSheetInitialize() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Properties

Count

```
public int Count { get; }
```

Property Value

int

Id

```
public int Id { get; }
```

Property Value

int

ItemId

```
public int ItemId { get; }
```

Property Value

int

Key

```
public override int Key { get; }
```

Property Value

int

Methods

Set(IReadOnlyList<string>)

```
public override void Set(IReadOnlyList<string> fields)
```

Parameters

`fields` IReadOnlyList<string>

Class QuestRewardSheet

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class QuestRewardSheet : Sheet<int, QuestRewardSheet.Row>, IDictionary<int,
QuestRewardSheet.Row>, ICollection<KeyValuePair<int, QuestRewardSheet.Row>>,
IEnumerable<KeyValuePair<int, QuestRewardSheet.Row>>, IEnumerable, ISheet
```

Inheritance

object ← Sheet<int, QuestRewardSheet.Row> ← QuestRewardSheet

Implements

IDictionary<int, QuestRewardSheet.Row>,
ICollection<KeyValuePair<int, QuestRewardSheet.Row>>,
IEnumerable<KeyValuePair<int, QuestRewardSheet.Row>>, IEnumerable, ISheet

Inherited Members

Sheet<int, QuestRewardSheet.Row>.Name ,
Sheet<int, QuestRewardSheet.Row>.OrderedList ,
Sheet<int, QuestRewardSheet.Row>.First , Sheet<int, QuestRewardSheet.Row>.Last ,
Sheet<int, QuestRewardSheet.Row>.Keys , Sheet<int, QuestRewardSheet.Row>.Values ,
Sheet<int, QuestRewardSheet.Row>.Count ,
Sheet<int, QuestRewardSheet.Row>.IsReadOnly ,
Sheet<int, QuestRewardSheet.Row>.this[int] ,
Sheet<int, QuestRewardSheet.Row>.Set(string, bool) ,
Sheet<int, QuestRewardSheet.Row>.Set<T>(Sheet<int, T>, bool) ,
Sheet<int, QuestRewardSheet.Row>.GetEnumerator() ,
Sheet<int, QuestRewardSheet.Row>.TryGetValue(int, out QuestRewardSheet.Row, bool) ,
Sheet<int, QuestRewardSheet.Row>.AddRow(int, QuestRewardSheet.Row) ,
Sheet<int, QuestRewardSheet.Row>.Add(int, QuestRewardSheet.Row) ,
Sheet<int, QuestRewardSheet.Row>.ContainsKey(int) ,
Sheet<int, QuestRewardSheet.Row>.Remove(int) ,
Sheet<int, QuestRewardSheet.Row>.TryGetValue(int, out QuestRewardSheet.Row) ,
Sheet<int, QuestRewardSheet.Row>.Add(KeyValuePair<int, QuestRewardSheet.Row>) ,
Sheet<int, QuestRewardSheet.Row>.Clear() ,
Sheet<int, QuestRewardSheet.Row>.Contains(KeyValuePair<int, QuestRewardSheet.Row>)
,

Sheet<int, QuestRewardSheet.Row>.CopyTo(KeyValuePair<int, QuestRewardSheet.Row>[], int) ,
Sheet<int, QuestRewardSheet.Row>.Remove(KeyValuePair<int, QuestRewardSheet.Row>) ,
Sheet<int, QuestRewardSheet.Row>.Serialize() , object.Equals(object) ,
object.Equals(object, object) , object.GetHashCode() , object.GetType() ,
object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

QuestRewardSheet()

```
public QuestRewardSheet()
```

Methods

AddRow(int, Row)

```
protected override void AddRow(int key, QuestRewardSheet.Row value)
```

Parameters

`key` int

`value` QuestRewardSheet.Row

Class QuestRewardSheet.Row

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class QuestRewardSheet.Row : SheetRow<int>
```

Inheritance

object ← SheetRow<int> ← QuestRewardSheet.Row

Inherited Members

SheetRow<int>.Validate() , SheetRow<int>.EndOfSheetInitialize() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Properties

Id

```
public int Id { get; }
```

Property Value

int

Key

```
public override int Key { get; }
```

Property Value

int

RewardIds

```
public List<int> RewardIds { get; }
```

Property Value

List<int>

Methods

Set(IReadOnlyList<string>)

```
public override void Set(IReadOnlyList<string> fields)
```

Parameters

`fields` IReadOnlyList<string>

Class QuestSheet

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class QuestSheet : Sheet<int, QuestSheet.Row>, IDictionary<int,
QuestSheet.Row>, ICollection<KeyValuePair<int, QuestSheet.Row>>,
IEnumerable<KeyValuePair<int, QuestSheet.Row>>, IEnumerable, ISheet
```

Inheritance

object ← Sheet<int, QuestSheet.Row> ← QuestSheet

Implements

IDictionary<int, QuestSheet.Row>,
ICollection<KeyValuePair<int, QuestSheet.Row>>,
IEnumerable<KeyValuePair<int, QuestSheet.Row>>, IEnumerable, ISheet

Inherited Members

Sheet<int, QuestSheet.Row>.Name , Sheet<int, QuestSheet.Row>.OrderedList ,
Sheet<int, QuestSheet.Row>.First , Sheet<int, QuestSheet.Row>.Last ,
Sheet<int, QuestSheet.Row>.Keys , Sheet<int, QuestSheet.Row>.Values ,
Sheet<int, QuestSheet.Row>.Count , Sheet<int, QuestSheet.Row>.IsReadOnly ,
Sheet<int, QuestSheet.Row>.this[int] , Sheet<int, QuestSheet.Row>.Set(string, bool) ,
Sheet<int, QuestSheet.Row>.Set<T>(Sheet<int, T>, bool) ,
Sheet<int, QuestSheet.Row>.GetEnumerator() ,
Sheet<int, QuestSheet.Row>.TryGetValue(int, out QuestSheet.Row, bool) ,
Sheet<int, QuestSheet.Row>.AddRow(int, QuestSheet.Row) ,
Sheet<int, QuestSheet.Row>.Add(int, QuestSheet.Row) ,
Sheet<int, QuestSheet.Row>.ContainsKey(int) , Sheet<int, QuestSheet.Row>.Remove(int) ,
Sheet<int, QuestSheet.Row>.TryGetValue(int, out QuestSheet.Row) ,
Sheet<int, QuestSheet.Row>.Add(KeyValuePair<int, QuestSheet.Row>) ,
Sheet<int, QuestSheet.Row>.Clear() ,
Sheet<int, QuestSheet.Row>.Contains(KeyValuePair<int, QuestSheet.Row>) ,
Sheet<int, QuestSheet.Row>.CopyTo(KeyValuePair<int, QuestSheet.Row>[], int) ,
Sheet<int, QuestSheet.Row>.Remove(KeyValuePair<int, QuestSheet.Row>) ,
Sheet<int, QuestSheet.Row>.Serialize() , object.Equals(object) ,
object.Equals(object, object) , object.GetHashCode() , object.GetType() ,
object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

QuestSheet()

```
public QuestSheet()
```

Class QuestSheet.Row

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class QuestSheet.Row : SheetRow<int>, IState
```

Inheritance

object ← SheetRow<int> ← QuestSheet.Row

Implements

IState

Derived

CollectQuestSheet.Row, CombinationEquipmentQuestSheet.Row, CombinationQuestSheet.Row, GeneralQuestSheet.Row, GoldQuestSheet.Row, ItemEnhancementQuestSheet.Row, ItemGradeQuestSheet.Row, ItemTypeCollectQuestSheet.Row, MonsterQuestSheet.Row, TradeQuestSheet.Row, WorldQuestSheet.Row

Inherited Members

SheetRow<int>.Validate() , SheetRow<int>.EndOfSheetInitialize() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Properties

Goal

```
public int Goal { get; }
```

Property Value

int

Id

```
public int Id { get; }
```

Property Value

int

Key

```
public override int Key { get; }
```

Property Value

int

QuestRewardId

```
public int QuestRewardId { get; }
```

Property Value

int

Methods

Deserialize(Dictionary)

```
public static QuestSheet.Row Deserialize(Dictionary serialized)
```

Parameters

`serialized` Dictionary

Returns

QuestSheet.Row

Serialize()

```
public IValue Serialize()
```

Returns

IValue

Set(IReadOnlyList<string>)

```
public override void Set(IReadOnlyList<string> fields)
```

Parameters

`fields` IReadOnlyList<string>

Class RaidSimulatorSheets

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
public class RaidSimulatorSheets : SimulatorSheets
```

Inheritance

object ← SimulatorSheetsV1 ← SimulatorSheets ← RaidSimulatorSheets

Inherited Members

SimulatorSheets.RuneOptionSheet , SimulatorSheets.RuneListSheet , SimulatorSheets.RuneLevelBonusSheet , SimulatorSheetsV1.MaterialItemSheet , SimulatorSheetsV1.SkillSheet , SimulatorSheetsV1.SkillBuffSheet , SimulatorSheetsV1.SkillActionBuffSheet , SimulatorSheetsV1.ActionBuffSheet , SimulatorSheetsV1.StatBuffSheet , SimulatorSheetsV1.CharacterSheet , SimulatorSheetsV1.CharacterLevelSheet , SimulatorSheetsV1.EquipmentItemSetEffectSheet , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

RaidSimulatorSheets(MaterialItemSheet, SkillSheet, SkillBuffSheet, StatBuffSheet, SkillActionBuffSheet, ActionBuffSheet, CharacterSheet, CharacterLevelSheet, EquipmentItemSetEffectSheet, WorldBossCharacterSheet, WorldBossActionPatternSheet, WorldBossBattleRewardSheet, RuneWeightSheet, RuneSheet, RuneOptionSheet, RuneListSheet, RuneLevelBonusSheet)

```
public RaidSimulatorSheets(MaterialItemSheet materialItemSheet, SkillSheet
skillSheet, SkillBuffSheet skillBuffSheet, StatBuffSheet statBuffSheet,
SkillActionBuffSheet skillActionBuffSheet, ActionBuffSheet actionBuffSheet,
```

```
CharacterSheet characterSheet, CharacterLevelSheet characterLevelSheet,
EquipmentItemSetEffectSheet equipmentItemSetEffectSheet, WorldBossCharacterSheet
worldBossCharacterSheet, WorldBossActionPatternSheet worldBossActionPatternSheet,
WorldBossBattleRewardSheet worldBossBattleRewardSheet, RuneWeightSheet
runeWeightSheet, RuneSheet runeSheet, RuneOptionSheet runeOptionSheet, RuneListSheet
runeListSheet, RuneLevelBonusSheet runeLevelBonusSheet)
```

Parameters

`materialItemSheet` MaterialItemSheet

`skillSheet` SkillSheet

`skillBuffSheet` SkillBuffSheet

`statBuffSheet` StatBuffSheet

`skillActionBuffSheet` SkillActionBuffSheet

`actionBuffSheet` ActionBuffSheet

`characterSheet` CharacterSheet

`characterLevelSheet` CharacterLevelSheet

`equipmentItemSetEffectSheet` EquipmentItemSetEffectSheet

`worldBossCharacterSheet` WorldBossCharacterSheet

`worldBossActionPatternSheet` WorldBossActionPatternSheet

`worldBossBattleRewardSheet` WorldBossBattleRewardSheet

`runeWeightSheet` RuneWeightSheet

`runeSheet` RuneSheet

`runeOptionSheet` RuneOptionSheet

`runeListSheet` RuneListSheet

`runeLevelBonusSheet` RuneLevelBonusSheet

Properties

RuneSheet

```
public RuneSheet RuneSheet { get; }
```

Property Value

RuneSheet

RuneWeightSheet

```
public RuneWeightSheet RuneWeightSheet { get; }
```

Property Value

RuneWeightSheet

WorldBossActionPatternSheet

```
public WorldBossActionPatternSheet WorldBossActionPatternSheet { get; }
```

Property Value

WorldBossActionPatternSheet

WorldBossBattleRewardSheet

```
public WorldBossBattleRewardSheet WorldBossBattleRewardSheet { get; }
```

Property Value

WorldBossBattleRewardSheet

WorldBossCharacterSheet

```
public WorldBossCharacterSheet WorldBossCharacterSheet { get; }
```

Property Value

WorldBossCharacterSheet

Class RaidSimulatorSheetsV1

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
public class RaidSimulatorSheetsV1 : SimulatorSheetsV1
```

Inheritance

object ← SimulatorSheetsV1 ← RaidSimulatorSheetsV1

Inherited Members

SimulatorSheetsV1.MaterialItemSheet , SimulatorSheetsV1.SkillSheet , SimulatorSheetsV1.SkillBuffSheet , SimulatorSheetsV1.SkillActionBuffSheet , SimulatorSheetsV1.ActionBuffSheet , SimulatorSheetsV1.StatBuffSheet , SimulatorSheetsV1.CharacterSheet , SimulatorSheetsV1.CharacterLevelSheet , SimulatorSheetsV1.EquipmentItemSetEffectSheet , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

RaidSimulatorSheetsV1(MaterialItemSheet, SkillSheet, SkillBuffSheet, StatBuffSheet, SkillActionBuffSheet, ActionBuffSheet, CharacterSheet, CharacterLevelSheet, EquipmentItemSetEffectSheet, WorldBossCharacterSheet, WorldBossActionPatternSheet, WorldBossBattleRewardSheet, RuneWeightSheet, RuneSheet)

```
public RaidSimulatorSheetsV1(MaterialItemSheet materialItemSheet, SkillSheet
skillSheet, SkillBuffSheet skillBuffSheet, StatBuffSheet statBuffSheet,
SkillActionBuffSheet skillActionBuffSheet, ActionBuffSheet actionBuffSheet,
CharacterSheet characterSheet, CharacterLevelSheet characterLevelSheet,
EquipmentItemSetEffectSheet equipmentItemSetEffectSheet, WorldBossCharacterSheet
worldBossCharacterSheet, WorldBossActionPatternSheet worldBossActionPatternSheet,
```

```
WorldBossBattleRewardSheet worldBossBattleRewardSheet, RuneWeightSheet
runeWeightSheet, RuneSheet runeSheet)
```

Parameters

`materialItemSheet` MaterialItemSheet

`skillSheet` SkillSheet

`skillBuffSheet` SkillBuffSheet

`statBuffSheet` StatBuffSheet

`skillActionBuffSheet` SkillActionBuffSheet

`actionBuffSheet` ActionBuffSheet

`characterSheet` CharacterSheet

`characterLevelSheet` CharacterLevelSheet

`equipmentItemSetEffectSheet` EquipmentItemSetEffectSheet

`worldBossCharacterSheet` WorldBossCharacterSheet

`worldBossActionPatternSheet` WorldBossActionPatternSheet

`worldBossBattleRewardSheet` WorldBossBattleRewardSheet

`runeWeightSheet` RuneWeightSheet

`runeSheet` RuneSheet

Properties

RuneSheet

```
public RuneSheet RuneSheet { get; }
```

Property Value

RuneSheet

RuneWeightSheet

```
public RuneWeightSheet RuneWeightSheet { get; }
```

Property Value

RuneWeightSheet

WorldBossActionPatternSheet

```
public WorldBossActionPatternSheet WorldBossActionPatternSheet { get; }
```

Property Value

WorldBossActionPatternSheet

WorldBossBattleRewardSheet

```
public WorldBossBattleRewardSheet WorldBossBattleRewardSheet { get; }
```

Property Value

WorldBossBattleRewardSheet

WorldBossCharacterSheet

```
public WorldBossCharacterSheet WorldBossCharacterSheet { get; }
```

Property Value

WorldBossCharacterSheet

Class RankingSimulatorSheets

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
public class RankingSimulatorSheets : SimulatorSheets
```

Inheritance

object ← SimulatorSheetsV1 ← SimulatorSheets ← RankingSimulatorSheets

Inherited Members

SimulatorSheets.RuneOptionSheet , SimulatorSheets.RuneListSheet , SimulatorSheets.RuneLevelBonusSheet , SimulatorSheetsV1.MaterialItemSheet , SimulatorSheetsV1.SkillSheet , SimulatorSheetsV1.SkillBuffSheet , SimulatorSheetsV1.SkillActionBuffSheet , SimulatorSheetsV1.ActionBuffSheet , SimulatorSheetsV1.StatBuffSheet , SimulatorSheetsV1.CharacterSheet , SimulatorSheetsV1.CharacterLevelSheet , SimulatorSheetsV1.EquipmentItemSetEffectSheet , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

RankingSimulatorSheets(MaterialItemSheet, SkillSheet, SkillBuffSheet, StatBuffSheet, SkillActionBuffSheet, ActionBuffSheet, CharacterSheet, CharacterLevelSheet, EquipmentItemSetEffectSheet, WeeklyArenaRewardSheet, RuneOptionSheet, RuneListSheet, RuneLevelBonusSheet)

```
public RankingSimulatorSheets(MaterialItemSheet materialItemSheet, SkillSheet
skillSheet, SkillBuffSheet skillBuffSheet, StatBuffSheet statBuffSheet,
SkillActionBuffSheet skillActionBuffSheet, ActionBuffSheet actionBuffSheet,
CharacterSheet characterSheet, CharacterLevelSheet characterLevelSheet,
EquipmentItemSetEffectSheet equipmentItemSetEffectSheet, WeeklyArenaRewardSheet
```

```
  weeklyArenaRewardSheet, RuneOptionSheet runeOptionSheet, RuneListSheet
  runeListSheet, RuneLevelBonusSheet runeLevelBonusSheet)
```

Parameters

`materialItemSheet` MaterialItemSheet

`skillSheet` SkillSheet

`skillBuffSheet` SkillBuffSheet

`statBuffSheet` StatBuffSheet

`skillActionBuffSheet` SkillActionBuffSheet

`actionBuffSheet` ActionBuffSheet

`characterSheet` CharacterSheet

`characterLevelSheet` CharacterLevelSheet

`equipmentItemSetEffectSheet` EquipmentItemSetEffectSheet

`weeklyArenaRewardSheet` WeeklyArenaRewardSheet

`runeOptionSheet` RuneOptionSheet

`runeListSheet` RuneListSheet

`runeLevelBonusSheet` RuneLevelBonusSheet

Fields

WeeklyArenaRewardSheet

```
  public readonly WeeklyArenaRewardSheet WeeklyArenaRewardSheet
```

Field Value

WeeklyArenaRewardSheet

Class RankingSimulatorSheetsV1

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
public class RankingSimulatorSheetsV1 : SimulatorSheetsV1
```

Inheritance

object ← SimulatorSheetsV1 ← RankingSimulatorSheetsV1

Inherited Members

SimulatorSheetsV1.MaterialItemSheet , SimulatorSheetsV1.SkillSheet , SimulatorSheetsV1.SkillBuffSheet , SimulatorSheetsV1.SkillActionBuffSheet , SimulatorSheetsV1.ActionBuffSheet , SimulatorSheetsV1.StatBuffSheet , SimulatorSheetsV1.CharacterSheet , SimulatorSheetsV1.CharacterLevelSheet , SimulatorSheetsV1.EquipmentItemSetEffectSheet , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

RankingSimulatorSheetsV1(MaterialItemSheet, SkillSheet, SkillBuffSheet, StatBuffSheet, SkillActionBuffSheet, ActionBuffSheet, CharacterSheet, CharacterLevelSheet, EquipmentItemSetEffectSheet, WeeklyArenaRewardSheet)

```
public RankingSimulatorSheetsV1(MaterialItemSheet materialItemSheet, SkillSheet
skillSheet, SkillBuffSheet skillBuffSheet, StatBuffSheet statBuffSheet,
SkillActionBuffSheet skillActionBuffSheet, ActionBuffSheet actionBuffSheet,
CharacterSheet characterSheet, CharacterLevelSheet characterLevelSheet,
EquipmentItemSetEffectSheet equipmentItemSetEffectSheet, WeeklyArenaRewardSheet
weeklyArenaRewardSheet)
```

Parameters

`materialItemSheet` MaterialItemSheet

`skillSheet` SkillSheet

`skillBuffSheet` SkillBuffSheet

`statBuffSheet` StatBuffSheet

`skillActionBuffSheet` SkillActionBuffSheet

`actionBuffSheet` ActionBuffSheet

`characterSheet` CharacterSheet

`characterLevelSheet` CharacterLevelSheet

`equipmentItemSetEffectSheet` EquipmentItemSetEffectSheet

`weeklyArenaRewardSheet` WeeklyArenaRewardSheet

Fields

WeeklyArenaRewardSheet

```
public readonly WeeklyArenaRewardSheet WeeklyArenaRewardSheet
```

Field Value

WeeklyArenaRewardSheet

Class RankingSimulatorSheetsV100291

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
public class RankingSimulatorSheetsV100291 : SimulatorSheetsV100291
```

Inheritance

object ← SimulatorSheetsV100291 ← RankingSimulatorSheetsV100291

Inherited Members

SimulatorSheetsV100291.MaterialItemSheet , SimulatorSheetsV100291.SkillSheet , SimulatorSheetsV100291.SkillBuffSheet , SimulatorSheetsV100291.BuffSheet , SimulatorSheetsV100291.CharacterSheet , SimulatorSheetsV100291.CharacterLevelSheet , SimulatorSheetsV100291.EquipmentItemSetEffectSheet , SimulatorSheetsV100291.ToSimulatorSheetsV1() , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

RankingSimulatorSheetsV100291(MaterialItemSheet, SkillSheet, SkillBuffSheet, BuffSheet, CharacterSheet, CharacterLevelSheet, EquipmentItemSetEffectSheet, WeeklyArenaRewardSheet)

```
public RankingSimulatorSheetsV100291(MaterialItemSheet materialItemSheet, SkillSheet
skillSheet, SkillBuffSheet skillBuffSheet, BuffSheet buffSheet, CharacterSheet
characterSheet, CharacterLevelSheet characterLevelSheet, EquipmentItemSetEffectSheet
equipmentItemSetEffectSheet, WeeklyArenaRewardSheet weeklyArenaRewardSheet)
```

Parameters

`materialItemSheet` MaterialItemSheet

`skillSheet` SkillSheet

`skillBuffSheet` SkillBuffSheet

`buffSheet` BuffSheet

`characterSheet` CharacterSheet

`characterLevelSheet` CharacterLevelSheet

`equipmentItemSetEffectSheet` EquipmentItemSetEffectSheet

`weeklyArenaRewardSheet` WeeklyArenaRewardSheet

Fields

WeeklyArenaRewardSheet

```
public readonly WeeklyArenaRewardSheet WeeklyArenaRewardSheet
```

Field Value

WeeklyArenaRewardSheet

Methods

ToRankingSimulatorSheetsV1()

```
public RankingSimulatorSheetsV1 ToRankingSimulatorSheetsV1()
```

Returns

RankingSimulatorSheetsV1

Class RedeemCodeListSheet

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class RedeemCodeListSheet : Sheet<int, RedeemCodeListSheet.Row>,
IDictionary<int, RedeemCodeListSheet.Row>, ICollection<KeyValuePair<int,
RedeemCodeListSheet.Row>>, IEnumerable<KeyValuePair<int, RedeemCodeListSheet.Row>>,
IEnumerable, ISheet
```

Inheritance

object ← Sheet<int, RedeemCodeListSheet.Row> ← RedeemCodeListSheet

Implements

IDictionary<int, RedeemCodeListSheet.Row>,
ICollection<KeyValuePair<int, RedeemCodeListSheet.Row>>,
IEnumerable<KeyValuePair<int, RedeemCodeListSheet.Row>>, IEnumerable, ISheet

Inherited Members

Sheet<int, RedeemCodeListSheet.Row>.Name ,
Sheet<int, RedeemCodeListSheet.Row>.OrderedList ,
Sheet<int, RedeemCodeListSheet.Row>.First ,
Sheet<int, RedeemCodeListSheet.Row>.Last ,
Sheet<int, RedeemCodeListSheet.Row>.Keys ,
Sheet<int, RedeemCodeListSheet.Row>.Values ,
Sheet<int, RedeemCodeListSheet.Row>.Count ,
Sheet<int, RedeemCodeListSheet.Row>.IsReadOnly ,
Sheet<int, RedeemCodeListSheet.Row>.this[int] ,
Sheet<int, RedeemCodeListSheet.Row>.Set(string, bool) ,
Sheet<int, RedeemCodeListSheet.Row>.Set<T>(Sheet<int, T>, bool) ,
Sheet<int, RedeemCodeListSheet.Row>.GetEnumerator() ,
Sheet<int, RedeemCodeListSheet.Row>.TryGetValue(int, out RedeemCodeListSheet.Row, bool) ,
Sheet<int, RedeemCodeListSheet.Row>.AddRow(int, RedeemCodeListSheet.Row) ,
Sheet<int, RedeemCodeListSheet.Row>.Add(int, RedeemCodeListSheet.Row) ,
Sheet<int, RedeemCodeListSheet.Row>.ContainsKey(int) ,
Sheet<int, RedeemCodeListSheet.Row>.Remove(int) ,
Sheet<int, RedeemCodeListSheet.Row>.TryGetValue(int, out RedeemCodeListSheet.Row) ,

Sheet<int, RedeemCodeListSheet.Row>.Add(KeyValuePair<int, RedeemCodeListSheet.Row>) , Sheet<int, RedeemCodeListSheet.Row>.Clear() , Sheet<int, RedeemCodeListSheet.Row>.Contains(KeyValuePair<int, RedeemCodeListSheet.Row>) , Sheet<int, RedeemCodeListSheet.Row>.CopyTo(KeyValuePair<int, RedeemCodeListSheet.Row>[], int) , Sheet<int, RedeemCodeListSheet.Row>.Remove(KeyValuePair<int, RedeemCodeListSheet.Row>) , Sheet<int, RedeemCodeListSheet.Row>.Serialize() , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

RedeemCodeListSheet()

```
public RedeemCodeListSheet()
```

Class RedeemCodeListSheet.Row

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class RedeemCodeListSheet.Row : SheetRow<int>
```

Inheritance

object ← SheetRow<int> ← RedeemCodeListSheet.Row

Inherited Members

SheetRow<int>.Validate() , SheetRow<int>.EndOfSheetInitialize() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Properties

Id

```
public int Id { get; }
```

Property Value

int

Key

```
public override int Key { get; }
```

Property Value

int

PublicKey

```
public PublicKey PublicKey { get; }
```

Property Value

PublicKey

PublicKeyBinary

```
public Binary PublicKeyBinary { get; }
```

Property Value

Binary

RewardId

```
public int RewardId { get; }
```

Property Value

int

Methods

Set(IReadOnlyList<string>)

```
public override void Set(IReadOnlyList<string> fields)
```

Parameters

`fields` IReadOnlyList<string>

Class RedeemRewardSheet

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class RedeemRewardSheet : Sheet<int, RedeemRewardSheet.Row>, IDictionary<int,
RedeemRewardSheet.Row>, ICollection<KeyValuePair<int, RedeemRewardSheet.Row>>,
IEnumerable<KeyValuePair<int, RedeemRewardSheet.Row>>, IEnumerable, ISheet
```

Inheritance

object ← Sheet<int, RedeemRewardSheet.Row> ← RedeemRewardSheet

Implements

IDictionary<int, RedeemRewardSheet.Row>,
ICollection<KeyValuePair<int, RedeemRewardSheet.Row>>,
IEnumerable<KeyValuePair<int, RedeemRewardSheet.Row>>, IEnumerable, ISheet

Inherited Members

Sheet<int, RedeemRewardSheet.Row>.Name ,
Sheet<int, RedeemRewardSheet.Row>.OrderedList ,
Sheet<int, RedeemRewardSheet.Row>.First , Sheet<int, RedeemRewardSheet.Row>.Last ,
Sheet<int, RedeemRewardSheet.Row>.Keys ,
Sheet<int, RedeemRewardSheet.Row>.Values ,
Sheet<int, RedeemRewardSheet.Row>.Count ,
Sheet<int, RedeemRewardSheet.Row>.IsReadOnly ,
Sheet<int, RedeemRewardSheet.Row>.this[int] ,
Sheet<int, RedeemRewardSheet.Row>.Set(string, bool) ,
Sheet<int, RedeemRewardSheet.Row>.Set<T>(Sheet<int, T>, bool) ,
Sheet<int, RedeemRewardSheet.Row>.GetEnumerator() ,
Sheet<int, RedeemRewardSheet.Row>.TryGetValue(int, out RedeemRewardSheet.Row,
bool) ,
Sheet<int, RedeemRewardSheet.Row>.AddRow(int, RedeemRewardSheet.Row) ,
Sheet<int, RedeemRewardSheet.Row>.Add(int, RedeemRewardSheet.Row) ,
Sheet<int, RedeemRewardSheet.Row>.ContainsKey(int) ,
Sheet<int, RedeemRewardSheet.Row>.Remove(int) ,
Sheet<int, RedeemRewardSheet.Row>.TryGetValue(int, out RedeemRewardSheet.Row) ,
Sheet<int, RedeemRewardSheet.Row>.Add(KeyValuePair<int, RedeemRewardSheet.Row>)
,

Sheet<int, RedeemRewardSheet.Row>.Clear() ,
Sheet<int, RedeemRewardSheet.Row>.Contains(KeyValuePair<int,
RedeemRewardSheet.Row>) ,
Sheet<int, RedeemRewardSheet.Row>.CopyTo(KeyValuePair<int,
RedeemRewardSheet.Row>[], int) ,
Sheet<int, RedeemRewardSheet.Row>.Remove(KeyValuePair<int,
RedeemRewardSheet.Row>) ,
Sheet<int, RedeemRewardSheet.Row>.Serialize() , object.Equals(object) ,
object.Equals(object, object) , object.GetHashCode() , object.GetType() ,
object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

RedeemRewardSheet()

```
public RedeemRewardSheet()
```

Methods

AddRow(int, Row)

```
protected override void AddRow(int key, RedeemRewardSheet.Row value)
```

Parameters

`key` int

`value` RedeemRewardSheet.Row

Class RedeemRewardSheet.RewardInfo

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class RedeemRewardSheet.RewardInfo : IState
```

Inheritance

object ← RedeemRewardSheet.RewardInfo

Implements

IState

Inherited Members

object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

RewardInfo(Dictionary)

```
public RewardInfo(Dictionary serialized)
```

Parameters

`serialized` Dictionary

RewardInfo(params string[])

```
public RewardInfo(params string[] fields)
```

Parameters

`fields` string[]

Fields

ItemId

```
public readonly int? ItemId
```

Field Value

int?

Quantity

```
public readonly int Quantity
```

Field Value

int

Type

```
public readonly RewardType Type
```

Field Value

RewardType

Methods

Equals(RewardInfo)

```
protected bool Equals(RedeemRewardSheet.RewardInfo other)
```

Parameters

`other` RedeemRewardSheet.RewardInfo

Returns

bool

Equals(object)

Determines whether the specified object is equal to the current object.

```
public override bool Equals(object obj)
```

Parameters

`obj` object

The object to compare with the current object.

Returns

bool

true if the specified object is equal to the current object; otherwise, false.

GetHashCode()

Serves as the default hash function.

```
public override int GetHashCode()
```

Returns

int

A hash code for the current object.

Serialize()

```
public IValue Serialize()
```

Returns

IValue

Class RedeemRewardSheet.Row

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class RedeemRewardSheet.Row : SheetRow<int>
```

Inheritance

object ← SheetRow<int> ← RedeemRewardSheet.Row

Inherited Members

SheetRow<int>.Validate() , SheetRow<int>.EndOfSheetInitialize() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Properties

Id

```
public int Id { get; }
```

Property Value

int

Key

```
public override int Key { get; }
```

Property Value

int

Rewards

```
public List<RedeemRewardSheet.RewardInfo> Rewards { get; }
```

Property Value

List<RedeemRewardSheet.RewardInfo>

Methods

Set(IReadOnlyList<string>)

```
public override void Set(IReadOnlyList<string> fields)
```

Parameters

`fields` IReadOnlyList<string>

Class RestrictionSheet

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

Declares what a given item or currency may not be used for, so that a restriction can be adjusted with `patch_table_sheet` instead of a headless and client release.

A row targets either an item id (numeric key) or a currency ticker (non-numeric key). Which one is looked up is decided by the call site — a market action already holds an item id or a ticker — so the sheet needs no discriminator column.

Columns: `key`, `market_registrable`, `synthesize_material`. The first governs the market registration path only — a fungible item can still reach another account through a garage or by being wrapped into a currency, so a row here does not make its target account bound by itself. The second closes the conversion path that would otherwise turn a restricted item into an unrestricted one.

An empty cell means "unspecified" and leaves the behavior that predates this sheet untouched. An unparsable cell is read as `false`(= restricted) on purpose: a typo then fails toward blocking, which is recoverable, rather than toward leaking. ValidateCsv(string) rejects such a cell at patch time so it should not reach the chain in the first place.

A key is an item id, not an item instance, so a row applies to every copy that exists — give a restricted item a new id rather than reusing one players already hold. A false `market_registrable` also stops `ReRegisterProduct`, leaving an existing seller able to cancel but not to reprice, and it does not retract listings that are already up: `BuyProduct` does not consult this sheet, so a restriction added after the fact still needs those products cancelled. A false `synthesize_material` only stops the item from being consumed as a material — an item that must not be handed out by synthesis is kept out of the result pool by leaving it out of `SynthesizeWeightSheet`, which is a separate sheet from this one.

Callers must tolerate the sheet being absent — it reaches an existing chain only by `patch_table_sheet`, so loading it as a required sheet would fail every affected action until each chain is patched. Absence has to fall back to the restrictions that are still hardcoded, which is also what keeps re-evaluation of pre-patch blocks identical.

```
[Serializable]
public class RestrictionSheet : Sheet<string, RestrictionSheet.Row>,
IDictionary<string, RestrictionSheet.Row>, ICollection<KeyValuePair<string,
```

```
RestrictionSheet.Row>>, IEnumerable<KeyValuePair<string, RestrictionSheet.Row>>,
IEnumerable, ISheet
```

Inheritance

object ← Sheet<string, RestrictionSheet.Row> ← RestrictionSheet

Implements

IDictionary<string, RestrictionSheet.Row>,
ICollection<KeyValuePair<string, RestrictionSheet.Row>>,
IEnumerable<KeyValuePair<string, RestrictionSheet.Row>>, IEnumerable, ISheet

Inherited Members

Sheet<string, RestrictionSheet.Row>.Name ,
Sheet<string, RestrictionSheet.Row>.OrderedList ,
Sheet<string, RestrictionSheet.Row>.First , Sheet<string, RestrictionSheet.Row>.Last ,
Sheet<string, RestrictionSheet.Row>.Keys , Sheet<string, RestrictionSheet.Row>.Values ,
Sheet<string, RestrictionSheet.Row>.Count ,
Sheet<string, RestrictionSheet.Row>.IsReadOnly ,
Sheet<string, RestrictionSheet.Row>.this[string] ,
Sheet<string, RestrictionSheet.Row>.Set(string, bool) ,
Sheet<string, RestrictionSheet.Row>.Set<T>(Sheet<string, T>, bool) ,
Sheet<string, RestrictionSheet.Row>.GetEnumerator() ,
Sheet<string, RestrictionSheet.Row>.TryGetValue(string, out RestrictionSheet.Row, bool) ,
Sheet<string, RestrictionSheet.Row>.AddRow(string, RestrictionSheet.Row) ,
Sheet<string, RestrictionSheet.Row>.Add(string, RestrictionSheet.Row) ,
Sheet<string, RestrictionSheet.Row>.ContainsKey(string) ,
Sheet<string, RestrictionSheet.Row>.Remove(string) ,
Sheet<string, RestrictionSheet.Row>.TryGetValue(string, out RestrictionSheet.Row) ,
Sheet<string, RestrictionSheet.Row>.Add(KeyValuePair<string, RestrictionSheet.Row>) ,
Sheet<string, RestrictionSheet.Row>.Clear() ,
Sheet<string, RestrictionSheet.Row>.Contains(KeyValuePair<string,
RestrictionSheet.Row>) ,
Sheet<string, RestrictionSheet.Row>.CopyTo(KeyValuePair<string, RestrictionSheet.Row>
[], int) ,
Sheet<string, RestrictionSheet.Row>.Remove(KeyValuePair<string, RestrictionSheet.Row>)
,
Sheet<string, RestrictionSheet.Row>.Serialize() , object.Equals(object) ,
object.Equals(object, object) , object.GetHashCode() , object.GetType() ,
object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

RestrictionSheet()

Creates an empty sheet.

```
public RestrictionSheet()
```

Fields

ColumnNames

Column names in their expected order, after _ prefixed columns are dropped.

```
public static readonly string[] ColumnNames
```

Field Value

string[]

Methods

AddRow(string, Row)

Adds a parsed row to the sheet.

```
protected override void AddRow(string key, RestrictionSheet.Row value)
```

Parameters

`key` string

The row's key.

`value` RestrictionSheet.Row

The row to add.

Remarks

Blank keys are dropped and duplicate keys are merged toward the restrictive value instead of throwing, because a throw here would take the entire policy offline and with it every restriction the sheet carries.

IsCurrencyMarketRegistrable(string)

Whether the currency may be registered on the market as a fungible asset product. True when unlisted or unspecified.

```
public bool IsCurrencyMarketRegistrable(string ticker)
```

Parameters

`ticker` string

Returns

bool

IsItemMarketRegistrable(int)

Whether the item may be registered on the market. True when unlisted or unspecified.

```
public bool IsItemMarketRegistrable(int itemId)
```

Parameters

`itemId` int

Returns

bool

IsItemSynthesizeMaterial(int)

Whether the item may be consumed as a synthesis material. True when unlisted or unspecified.

```
public bool IsItemSynthesizeMaterial(int itemId)
```

Parameters

`itemId` int

Returns

bool

ToKey(int)

Renders an item id the way a row key spells it.

```
public static string ToKey(int itemId)
```

Parameters

`itemId` int

Returns

string

ValidateCsv(string)

Parses `csv` strictly so that a malformed policy fails the `patch_table_sheet` transaction instead of silently landing on the chain, where a row that no lookup can match reads as "unrestricted".

```
public static void ValidateCsv(string csv)
```

Parameters

csv string

The sheet's CSV, as the patch would write it.

Exceptions

SheetRowValidateException

The header does not match ColumnNames, or a row has no key, a key no lookup can produce, an unparsable policy cell, no policy at all, a synthesis policy on a currency, or a key an earlier row already claimed.

Class RestrictionSheet.Row

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

One target and what it may not be used for.

```
[Serializable]
public class RestrictionSheet.Row : SheetRow<string>
```

Inheritance

object ← SheetRow<string> ← RestrictionSheet.Row

Inherited Members

SheetRow<string>.Validate() , SheetRow<string>.EndOfSheetInitialize() , SheetRow<string>.Equals(object) , SheetRow<string>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Properties

Key

The item id or ticker this row restricts, which is also how it is looked up.

```
public override string Key { get; }
```

Property Value

string

MarketRegistrable

Whether the target may be registered on the market. `null` when the cell is empty(= unspecified).

```
public bool? MarketRegistrable { get; }
```

Property Value

bool?

SynthesizeMaterial

Whether the target may be consumed as a synthesis material. `null` when the cell is empty(= unspecified).

```
public bool? SynthesizeMaterial { get; }
```

Property Value

bool?

Target

An item id in its string form, or a currency ticker.

```
public string Target { get; }
```

Property Value

string

Methods

Set(IReadOnlyList<string>)

Reads one row.

```
public override void Set(IReadOnlyList<string> fields)
```

Parameters

`fields` IReadOnlyList<string>

The row's cells, with "_" prefixed columns already dropped.

Remarks

Total by design: Sheet<TKey, TValue> does not guard this call, so a throw here would abort the whole sheet and take every restriction with it.

Enum RewardType

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
public enum RewardType
```

Extension Methods
StateExtensions.Serialize(Enum)

Fields

`Gold = 1`

`Item = 0`

Class RuneCostSheet

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

Represents a sheet containing rune cost data organized by level ranges. This sheet stores cost information for rune enhancement across different level intervals, allowing for efficient data compression and retrieval.

```
[Serializable]
public class RuneCostSheet : Sheet<int, RuneCostSheet.Row>, IDictionary<int,
RuneCostSheet.Row>, ICollection<KeyValuePair<int, RuneCostSheet.Row>>,
IEnumerable<KeyValuePair<int, RuneCostSheet.Row>>, IEnumerable, ISheet
```

Inheritance

object ← Sheet<int, RuneCostSheet.Row> ← RuneCostSheet

Implements

IDictionary<int, RuneCostSheet.Row>,
ICollection<KeyValuePair<int, RuneCostSheet.Row>>,
IEnumerable<KeyValuePair<int, RuneCostSheet.Row>>, IEnumerable, ISheet

Inherited Members

Sheet<int, RuneCostSheet.Row>.Name , Sheet<int, RuneCostSheet.Row>.OrderedList ,
Sheet<int, RuneCostSheet.Row>.First , Sheet<int, RuneCostSheet.Row>.Last ,
Sheet<int, RuneCostSheet.Row>.Keys , Sheet<int, RuneCostSheet.Row>.Values ,
Sheet<int, RuneCostSheet.Row>.Count , Sheet<int, RuneCostSheet.Row>.IsReadOnly ,
Sheet<int, RuneCostSheet.Row>.this[int] ,
Sheet<int, RuneCostSheet.Row>.Set(string, bool) ,
Sheet<int, RuneCostSheet.Row>.Set<T>(Sheet<int, T>, bool) ,
Sheet<int, RuneCostSheet.Row>.GetEnumerator() ,
Sheet<int, RuneCostSheet.Row>.TryGetValue(int, out RuneCostSheet.Row, bool) ,
Sheet<int, RuneCostSheet.Row>.AddRow(int, RuneCostSheet.Row) ,
Sheet<int, RuneCostSheet.Row>.Add(int, RuneCostSheet.Row) ,
Sheet<int, RuneCostSheet.Row>.ContainsKey(int) ,
Sheet<int, RuneCostSheet.Row>.Remove(int) ,
Sheet<int, RuneCostSheet.Row>.TryGetValue(int, out RuneCostSheet.Row) ,
Sheet<int, RuneCostSheet.Row>.Add(KeyValuePair<int, RuneCostSheet.Row>) ,
Sheet<int, RuneCostSheet.Row>.Clear() ,
Sheet<int, RuneCostSheet.Row>.Contains(KeyValuePair<int, RuneCostSheet.Row>) ,

Sheet<int, RuneCostSheet.Row>.CopyTo(KeyValuePair<int, RuneCostSheet.Row>[], int) , Sheet<int, RuneCostSheet.Row>.Remove(KeyValuePair<int, RuneCostSheet.Row>) , Sheet<int, RuneCostSheet.Row>.Serialize() , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

RuneCostSheet()

Initializes a new instance of the RuneCostSheet class.

```
public RuneCostSheet()
```

Methods

AddRow(int, Row)

Adds a row to the sheet, handling cases where the key already exists. If a row with the same key exists, the cost data is merged.

```
protected override void AddRow(int key, RuneCostSheet.Row value)
```

Parameters

`key` int

The unique identifier for the row

`value` RuneCostSheet.Row

The row data to add

Class RuneCostSheet.Row

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

Represents a row in the RuneCostSheet containing cost data for a specific rune. Each row can contain multiple cost data entries for different level ranges.

```
[Serializable]
public class RuneCostSheet.Row : SheetRow<int>
```

Inheritance

object ← SheetRow<int> ← RuneCostSheet.Row

Inherited Members

SheetRow<int>.Validate() , SheetRow<int>.EndOfSheetInitialize() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Properties

Cost

Gets the collection of cost data entries for different level ranges.

```
public List<RuneCostSheet.RuneCostData> Cost { get; }
```

Property Value

List<RuneCostSheet.RuneCostData>

Key

Gets the unique identifier for this row, which is the rune ID.

```
public override int Key { get; }
```

Property Value

int

RuneId

Gets the rune identifier associated with this cost data.

```
public int RuneId { get; }
```

Property Value

int

Methods

GetCostsInRange(int, int)

Retrieves all cost data entries that overlap with the specified level range.

```
public IEnumerable<RuneCostSheet.RuneCostData> GetCostsInRange(int startLevel,
int endLevel)
```

Parameters

`startLevel` int

The starting level of the range (inclusive)

`endLevel` int

The ending level of the range (inclusive)

Returns

IEnumerable<RuneCostSheet.RuneCostData>

An enumerable collection of cost data entries that overlap with the specified range

Set(IReadOnlyList<string>)

Sets the row data from the provided CSV fields.

```
public override void Set(IReadOnlyList<string> fields)
```

Parameters

`fields` IReadOnlyList<string>

The CSV fields containing rune cost information

TryGetCost(int, out RuneCostData)

Attempts to find cost data for a specific level.

```
public bool TryGetCost(int level, out RuneCostSheet.RuneCostData costData)
```

Parameters

`level` int

The level to search for

`costData` RuneCostSheet.RuneCostData

When this method returns, contains the cost data for the specified level, if found; otherwise, null

Returns

bool

true if cost data was found for the specified level; otherwise, false

Class RuneCostSheet.RuneCostData

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

Represents cost data for a specific level range of rune enhancement. Contains information about required materials and success rates for a given level interval.

```
[Serializable]
public class RuneCostSheet.RuneCostData
```

Inheritance

object ← RuneCostSheet.RuneCostData

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

RuneCostData(int, int, int, long, int, int)

Initializes a new instance of the RuneCostData class with specified parameters.

```
public RuneCostData(int levelStart, int levelEnd, int runeStoneQuantity, long
crystalQuantity, int ncgQuantity, int levelUpSuccessRate)
```

Parameters

`levelStart` int

The starting level of the cost range

`levelEnd` int

The ending level of the cost range

`runeStoneQuantity` int

Required rune stone quantity

`crystalQuantity` long

Required crystal quantity

`ncgQuantity` int

Required NCG quantity

`levelUpSuccessRate` int

Success rate for enhancement

Properties

CrystalQuantity

The quantity of crystals required for enhancement

```
public long CrystalQuantity { get; }
```

Property Value

long

LevelEnd

The ending level of this cost range (inclusive)

```
public int LevelEnd { get; }
```

Property Value

int

LevelStart

The starting level of this cost range (inclusive)

```
public int LevelStart { get; }
```

Property Value

int

LevelUpSuccessRate

The success rate percentage for rune enhancement (in basis points, e.g., 10000 = 100%)

```
public int LevelUpSuccessRate { get; }
```

Property Value

int

NcgQuantity

The quantity of NCG (Nine Chronicles Gold) required for enhancement

```
public int NcgQuantity { get; }
```

Property Value

int

RuneStoneQuantity

The quantity of rune stones required for enhancement

```
public int RuneStoneQuantity { get; }
```

Property Value

int

Methods

ContainsLevel(int)

Determines whether the specified level falls within this cost range.

```
public bool ContainsLevel(int level)
```

Parameters

`level` int

The level to check

Returns

bool

true if the level is within the range (inclusive); otherwise, false

Class RuneOptionSheet

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class RuneOptionSheet : Sheet<int, RuneOptionSheet.Row>, IDictionary<int,
RuneOptionSheet.Row>, ICollection<KeyValuePair<int, RuneOptionSheet.Row>>,
IEnumerable<KeyValuePair<int, RuneOptionSheet.Row>>, IEnumerable, ISheet
```

Inheritance

object ← Sheet<int, RuneOptionSheet.Row> ← RuneOptionSheet

Implements

IDictionary<int, RuneOptionSheet.Row>,
ICollection<KeyValuePair<int, RuneOptionSheet.Row>>,
IEnumerable<KeyValuePair<int, RuneOptionSheet.Row>>, IEnumerable, ISheet

Inherited Members

Sheet<int, RuneOptionSheet.Row>.Name ,
Sheet<int, RuneOptionSheet.Row>.OrderedList , Sheet<int, RuneOptionSheet.Row>.First ,
Sheet<int, RuneOptionSheet.Row>.Last , Sheet<int, RuneOptionSheet.Row>.Keys ,
Sheet<int, RuneOptionSheet.Row>.Values , Sheet<int, RuneOptionSheet.Row>.Count ,
Sheet<int, RuneOptionSheet.Row>.IsReadOnly ,
Sheet<int, RuneOptionSheet.Row>.this[int] ,
Sheet<int, RuneOptionSheet.Row>.Set(string, bool) ,
Sheet<int, RuneOptionSheet.Row>.Set<T>(Sheet<int, T>, bool) ,
Sheet<int, RuneOptionSheet.Row>.GetEnumerator() ,
Sheet<int, RuneOptionSheet.Row>.TryGetValue(int, out RuneOptionSheet.Row, bool) ,
Sheet<int, RuneOptionSheet.Row>.AddRow(int, RuneOptionSheet.Row) ,
Sheet<int, RuneOptionSheet.Row>.Add(int, RuneOptionSheet.Row) ,
Sheet<int, RuneOptionSheet.Row>.ContainsKey(int) ,
Sheet<int, RuneOptionSheet.Row>.Remove(int) ,
Sheet<int, RuneOptionSheet.Row>.TryGetValue(int, out RuneOptionSheet.Row) ,
Sheet<int, RuneOptionSheet.Row>.Add(KeyValuePair<int, RuneOptionSheet.Row>) ,
Sheet<int, RuneOptionSheet.Row>.Clear() ,
Sheet<int, RuneOptionSheet.Row>.Contains(KeyValuePair<int, RuneOptionSheet.Row>) ,
Sheet<int, RuneOptionSheet.Row>.CopyTo(KeyValuePair<int, RuneOptionSheet.Row>[],
int) ,

Sheet<int, RuneOptionSheet.Row>.Remove(KeyValuePair<int, RuneOptionSheet.Row>) , Sheet<int, RuneOptionSheet.Row>.Serialize() , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

RuneOptionSheet()

```
public RuneOptionSheet()
```

Methods

AddRow(int, Row)

```
protected override void AddRow(int key, RuneOptionSheet.Row value)
```

Parameters

`key` int

`value` RuneOptionSheet.Row

TryGetOptionInfo(int, int, out RuneOptionInfo)

```
public bool TryGetOptionInfo(int runeId, int level, out
RuneOptionSheet.Row.RuneOptionInfo optionInfo)
```

Parameters

`runeId` int

`level` int

`optionInfo` RuneOptionSheet.Row.RuneOptionInfo

Returns

bool

Class RuneOptionSheet.Row

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class RuneOptionSheet.Row : SheetRow<int>
```

Inheritance

object ← SheetRow<int> ← RuneOptionSheet.Row

Inherited Members

SheetRow<int>.Validate() , SheetRow<int>.EndOfSheetInitialize() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Properties

Key

```
public override int Key { get; }
```

Property Value

int

LevelOptionMap

```
public Dictionary<int, RuneOptionSheet.Row.RuneOptionInfo> LevelOptionMap { get; }
```

Property Value

Dictionary<int, RuneOptionSheet.Row.RuneOptionInfo>

RuneId

```
public int RuneId { get; }
```

Property Value

int

Methods

Set(IReadOnlyList<string>)

```
public override void Set(IReadOnlyList<string> fields)
```

Parameters

`fields` IReadOnlyList<string>

Class RuneOptionSheet.Row.RuneOptionInfo

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
public class RuneOptionSheet.Row.RuneOptionInfo
```

Inheritance

object ← RuneOptionSheet.Row.RuneOptionInfo

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

RuneOptionInfo(long, List<(DecimalStat, OperationType)>)

```
public RuneOptionInfo(long cp, List<(DecimalStat, StatModifier.OperationType)>
stats)
```

Parameters

`cp` long

`stats` List<(DecimalStat stat, StatModifier.OperationType operationType)>

RuneOptionInfo(long, List<(DecimalStat, OperationType)>, int, int, int, decimal, OperationType, StatType, StatReferenceType, int)

```
public RuneOptionInfo(long cp, List<(DecimalStat, StatModifier.OperationType)>
stats, int skillId, int skillCooldown, int skillChance, decimal skillValue,
StatModifier.OperationType skillValueType, StatType skillStatType, StatReferenceType
statReferenceType, int buffDuration)
```

Parameters

`cp` long

`stats` List<(DecimalStat stat, StatModifier.OperationType operationType)>

`skillId` int

`skillCooldown` int

`skillChance` int

`skillValue` decimal

`skillValueType` StatModifier.OperationType

`skillStatType` StatType

`statReferenceType` StatReferenceType

`buffDuration` int

Properties

BuffDuration

```
public int BuffDuration { get; set; }
```

Property Value

int

Cp

```
public long Cp { get; }
```

Property Value

long

SkillChance

```
public int SkillChance { get; set; }
```

Property Value

int

SkillCooldown

```
public int SkillCooldown { get; set; }
```

Property Value

int

SkillId

```
public int SkillId { get; set; }
```

Property Value

int

SkillStatType

```
public StatType SkillStatType { get; set; }
```

Property Value

StatType

SkillValue

```
public decimal SkillValue { get; set; }
```

Property Value

decimal

SkillValueType

```
public StatModifier.OperationType SkillValueType { get; set; }
```

Property Value

StatModifier.OperationType

StatReferenceType

```
public StatReferenceType StatReferenceType { get; set; }
```

Property Value

StatReferenceType

Stats

```
public List<(DecimalStat stat, StatModifier.OperationType operationType)> Stats {
get; set; }
```

Property Value

List<(DecimalStat stat, StatModifier.OperationType operationType)>

Class RuneSheet

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
public class RuneSheet : Sheet<int, RuneSheet.Row>, IDictionary<int, RuneSheet.Row>,
ICollection<KeyValuePair<int, RuneSheet.Row>>, IEnumerable<KeyValuePair<int,
RuneSheet.Row>>, IEnumerable, ISheet
```

Inheritance

object ← Sheet<int, RuneSheet.Row> ← RuneSheet

Implements

IDictionary<int, RuneSheet.Row>,
ICollection<KeyValuePair<int, RuneSheet.Row>>,
IEnumerable<KeyValuePair<int, RuneSheet.Row>>, IEnumerable, ISheet

Inherited Members

Sheet<int, RuneSheet.Row>.Name , Sheet<int, RuneSheet.Row>.OrderedList ,
Sheet<int, RuneSheet.Row>.First , Sheet<int, RuneSheet.Row>.Last ,
Sheet<int, RuneSheet.Row>.Keys , Sheet<int, RuneSheet.Row>.Values ,
Sheet<int, RuneSheet.Row>.Count , Sheet<int, RuneSheet.Row>.IsReadOnly ,
Sheet<int, RuneSheet.Row>.this[int] , Sheet<int, RuneSheet.Row>.Set(string, bool) ,
Sheet<int, RuneSheet.Row>.Set<T>(Sheet<int, T>, bool) ,
Sheet<int, RuneSheet.Row>.GetEnumerator() ,
Sheet<int, RuneSheet.Row>.TryGetValue(int, out RuneSheet.Row, bool) ,
Sheet<int, RuneSheet.Row>.AddRow(int, RuneSheet.Row) ,
Sheet<int, RuneSheet.Row>.Add(int, RuneSheet.Row) ,
Sheet<int, RuneSheet.Row>.ContainsKey(int) , Sheet<int, RuneSheet.Row>.Remove(int) ,
Sheet<int, RuneSheet.Row>.TryGetValue(int, out RuneSheet.Row) ,
Sheet<int, RuneSheet.Row>.Add(KeyValuePair<int, RuneSheet.Row>) ,
Sheet<int, RuneSheet.Row>.Clear() ,
Sheet<int, RuneSheet.Row>.Contains(KeyValuePair<int, RuneSheet.Row>) ,
Sheet<int, RuneSheet.Row>.CopyTo(KeyValuePair<int, RuneSheet.Row>[], int) ,
Sheet<int, RuneSheet.Row>.Remove(KeyValuePair<int, RuneSheet.Row>) ,
Sheet<int, RuneSheet.Row>.Serialize() , object.Equals(object) ,
object.Equals(object, object) , object.GetHashCode() , object.GetType() ,
object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

RuneSheet()

```
public RuneSheet()
```

Class RuneSheet.Row

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
public class RuneSheet.Row : SheetRow<int>
```

Inheritance

object ← SheetRow<int> ← RuneSheet.Row

Inherited Members

SheetRow<int>.Validate() , SheetRow<int>.EndOfSheetInitialize() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Fields

Id

```
public int Id
```

Field Value

int

Ticker

```
public string Ticker
```

Field Value

string

Properties

Key

```
public override int Key { get; }
```

Property Value

int

Methods

Set(IReadOnlyList<string>)

```
public override void Set(IReadOnlyList<string> fields)
```

Parameters

`fields` IReadOnlyList<string>

Class RuneWeightSheet

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
public class RuneWeightSheet : Sheet<int, RuneWeightSheet.Row>, IDictionary<int,
RuneWeightSheet.Row>, ICollection<KeyValuePair<int, RuneWeightSheet.Row>>,
IEnumerable<KeyValuePair<int, RuneWeightSheet.Row>>, IEnumerable, ISheet
```

Inheritance

object ← Sheet<int, RuneWeightSheet.Row> ← RuneWeightSheet

Implements

IDictionary<int, RuneWeightSheet.Row>,
ICollection<KeyValuePair<int, RuneWeightSheet.Row>>,
IEnumerable<KeyValuePair<int, RuneWeightSheet.Row>>, IEnumerable, ISheet

Inherited Members

Sheet<int, RuneWeightSheet.Row>.Name ,
Sheet<int, RuneWeightSheet.Row>.OrderedList , Sheet<int, RuneWeightSheet.Row>.First ,
Sheet<int, RuneWeightSheet.Row>.Last , Sheet<int, RuneWeightSheet.Row>.Keys ,
Sheet<int, RuneWeightSheet.Row>.Values , Sheet<int, RuneWeightSheet.Row>.Count ,
Sheet<int, RuneWeightSheet.Row>.IsReadOnly ,
Sheet<int, RuneWeightSheet.Row>.this[int] ,
Sheet<int, RuneWeightSheet.Row>.Set(string, bool) ,
Sheet<int, RuneWeightSheet.Row>.Set<T>(Sheet<int, T>, bool) ,
Sheet<int, RuneWeightSheet.Row>.GetEnumerator() ,
Sheet<int, RuneWeightSheet.Row>.TryGetValue(int, out RuneWeightSheet.Row, bool) ,
Sheet<int, RuneWeightSheet.Row>.AddRow(int, RuneWeightSheet.Row) ,
Sheet<int, RuneWeightSheet.Row>.Add(int, RuneWeightSheet.Row) ,
Sheet<int, RuneWeightSheet.Row>.ContainsKey(int) ,
Sheet<int, RuneWeightSheet.Row>.Remove(int) ,
Sheet<int, RuneWeightSheet.Row>.TryGetValue(int, out RuneWeightSheet.Row) ,
Sheet<int, RuneWeightSheet.Row>.Add(KeyValuePair<int, RuneWeightSheet.Row>) ,
Sheet<int, RuneWeightSheet.Row>.Clear() ,
Sheet<int, RuneWeightSheet.Row>.Contains(KeyValuePair<int, RuneWeightSheet.Row>) ,
Sheet<int, RuneWeightSheet.Row>.CopyTo(KeyValuePair<int, RuneWeightSheet.Row>[],
int) ,
Sheet<int, RuneWeightSheet.Row>.Remove(KeyValuePair<int, RuneWeightSheet.Row>) ,

Sheet<int, RuneWeightSheet.Row>.Serialize() , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

RuneWeightSheet()

```
public RuneWeightSheet()
```

Methods

AddRow(int, Row)

```
protected override void AddRow(int key, RuneWeightSheet.Row value)
```

Parameters

`key` int

`value` RuneWeightSheet.Row

Class RuneWeightSheet.Row

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
public class RuneWeightSheet.Row : SheetRow<int>
```

Inheritance

object ← SheetRow<int> ← RuneWeightSheet.Row

Inherited Members

SheetRow<int>.Validate() , SheetRow<int>.EndOfSheetInitialize() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Fields

BossId

```
public int BossId
```

Field Value

int

Id

```
public int Id
```

Field Value

int

Rank

```
public int Rank
```

Field Value

int

RuneInfos

```
public List<RuneWeightSheet.RuneInfo> RuneInfos
```

Field Value

List<RuneWeightSheet.RuneInfo>

Properties

Key

```
public override int Key { get; }
```

Property Value

int

Methods

Set(IReadOnlyList<string>)

```
public override void Set(IReadOnlyList<string> fields)
```

Parameters

`fields` IReadOnlyList<string>

Class RuneWeightSheet.RuneInfo

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
public class RuneWeightSheet.RuneInfo
```

Inheritance

object ← RuneWeightSheet.RuneInfo

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

RuneInfo(int, decimal)

```
public RuneInfo(int runeId, decimal weight)
```

Parameters

`runeId` int

`weight` decimal

Fields

RuneId

```
public int RuneId
```

Field Value

int

Weight

```
public decimal Weight
```

Field Value

decimal

Class SetEffectExtension

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
public static class SetEffectExtension
```

Inheritance

object ← SetEffectExtension

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Methods

GetSetEffectRows(EquipmentItemSetEffectSheet, IEnumerable<Equipment>)

```
public static List<EquipmentItemSetEffectSheet.Row> GetSetEffectRows(this
EquipmentItemSetEffectSheet sheet, IEnumerable<Equipment> equipments)
```

Parameters

`sheet` EquipmentItemSetEffectSheet

`equipments` IEnumerable<Equipment>

Returns

List<EquipmentItemSetEffectSheet.Row>

Class SheetRowColumnException

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class SheetRowColumnException : Exception, ISerializable
```

Inheritance

object ← Exception ← SheetRowColumnException

Implements

ISerializable

Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Constructors

SheetRowColumnException(SerializationInfo, StreamingContext)

```
protected SheetRowColumnException(SerializationInfo info, StreamingContext context)
```

Parameters

`info` SerializationInfo

`context` StreamingContext

SheetRowColumnException(string)

```
public SheetRowColumnException(string message)
```

Parameters

`message` string

Class SheetRowNotFoundException

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class SheetRowNotFoundException : Exception, ISerializable
```

Inheritance

object ← Exception ← SheetRowNotFoundException

Implements

ISerializable

Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Constructors

SheetRowNotFoundException(SerializationInfo, StreamingContext)

```
protected SheetRowNotFoundException(SerializationInfo info,
StreamingContext context)
```

Parameters

`info` SerializationInfo

`context` StreamingContext

SheetRowNotFoundException(string)

```
public SheetRowNotFoundException(string message)
```

Parameters

`message` string

SheetRowNotFoundException(string, int)

```
public SheetRowNotFoundException(string sheetName, int intKey)
```

Parameters

`sheetName` string

`intKey` int

SheetRowNotFoundException(string, long)

```
public SheetRowNotFoundException(string sheetName, long longKey)
```

Parameters

`sheetName` string

`longKey` long

SheetRowNotFoundException(string, string)

```
public SheetRowNotFoundException(string sheetName, string key)
```

Parameters

`sheetName` string

key string

SheetRowNotFoundException(string, string, int)

```
public SheetRowNotFoundException(string addressesHex, string sheetName, int intKey)
```

Parameters

addressesHex string

sheetName string

intKey int

SheetRowNotFoundException(string, string, string)

```
public SheetRowNotFoundException(string sheetName, string condition, string value)
```

Parameters

sheetName string

condition string

value string

SheetRowNotFoundException(string, string, string, int)

```
public SheetRowNotFoundException(string actionType, string addressesHex, string
sheetName, int intKey)
```

Parameters

actionType string

addressesHex string

`sheetName` string

`intKey` int

Class SheetRowValidateException

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class SheetRowValidateException : Exception, ISerializable
```

Inheritance

object ← Exception ← SheetRowValidateException

Implements

ISerializable

Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , Exception.SerializeObjectState , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Constructors

SheetRowValidateException(SerializationInfo, StreamingContext)

```
protected SheetRowValidateException(SerializationInfo info,
StreamingContext context)
```

Parameters

`info` SerializationInfo

`context` StreamingContext

SheetRowValidateException(string)

```
public SheetRowValidateException(string message)
```

Parameters

`message` string

Class SheetRow<T>

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public abstract class SheetRow<T>
```

Type Parameters

T

Inheritance

object ← SheetRow<T>

Derived

ActionBuffSheet.Row, AdventureBossContributionRewardSheet.Row, AdventureBossFloorFirstRewardSheet.Row, AdventureBossFloorPointSheet.Row, AdventureBossFloorSheet.Row, AdventureBossFloorWaveSheet.Row, AdventureBossNcgRewardRatioSheet.Row, AdventureBossSheet.Row, AdventureBossUnlockFloorCostSheet.Row, AdventureBossWantedRewardSheet.Row, ArenaSheet.Row, BoostScheduleSheet.Row, BuffLimitSheet.Row, BuffLinkSheet.Row, BuffSheet.Row, CharacterLevelSheet.Row, CharacterSheet.Row, ClaimableGiftsSheet.Row, CollectionSheet.Row, ConsumableItemRecipeSheet.Row, CostumeStatSheet.Row, CreateAvatarFavSheet.Row, CreateAvatarItemSheet.Row, CrystalEquipmentGrindingSheet.Row, CrystalFluctuationSheet.Row, CrystalHammerPointSheet.Row, CrystalMaterialCostSheet.Row, CrystalMonsterCollectionMultiplierSheet.Row, CrystalRandomBuffSheet.Row, CrystalStageBuffGachaSheet.Row, CustomEquipmentCraftIconSheet.Row, CustomEquipmentCraftOptionSheet.Row, CustomEquipmentCraftRecipeSheet.Row, CustomEquipmentCraftRecipeSkillSheet.Row, CustomEquipmentCraftRelationshipSheet.Row, EnemySkillSheet.Row, EnhancementCostSheet.Row, EquipmentItemOptionSheet.Row, EquipmentItemRecipeSheet.Row, EquipmentItemSetEffectSheet.Row, EquipmentItemSubRecipeSheet.Row, EquipmentItemSubRecipeSheetV2.Row, EventMaterialItemRecipeSheet.Row, EventScheduleSheet.Row, PatrolRewardSheet.Row, GameConfigSheet.Row, LoadIntoMyGaragesCostSheet.Row, GlobalCounterSheet.Row, GrandFinaleParticipantsSheet.Row, GrandFinaleScheduleSheet.Row, InfiniteTowerConditionSheet.Row, InfiniteTowerFloorSheet.Row,

InfiniteTowerFloorWaveSheet.Row, InfiniteTowerScheduleSheet.Row, ItemConfigForGradeSheet.Row, ItemRequirementSheet.Row, ItemSheet.RowBase, MimisbrunnrSheet.Row, MonsterCollectionRewardSheet.Row, MonsterCollectionSheet.Row, PetOptionSheet.Row, PetCostSheet.Row, PetSheet.Row, QuestItemRewardSheet.Row, QuestRewardSheet.Row, QuestSheet.Row, RedeemCodeListSheet.Row, RedeemRewardSheet.Row, RestrictionSheet.Row, RuneLevelBonusSheet.Row, RuneListSheet.Row, RuneCostSheet.Row, RuneOptionSheet.Row, RuneSheet.Row, RuneWeightSheet.Row, SkillActionBuffSheet.Row, SkillBuffSheet.Row, SkillSheet.Row, StageDialogSheet.Row, StageSheet.Row, StageWaveSheet.Row, StakePolicySheet.Row, StakeAchievementRewardSheet.Row, StakeActionPointCoefficientSheet.Row, StakeRegularFixedRewardSheet.Row, StakeRegularRewardSheet.Row, StatBuffSheet.Row, SummonSheet.Row, SwapRateSheet.Row, SweepRequiredCPSheet.Row, SynthesizeSheet.Row, SynthesizeWeightSheet.Row, UnlockCombinationSlotCostSheet.Row, WeeklyArenaRewardSheet.Row, WorldBossActionPatternSheet.Row, WorldBossBattleRewardSheet.Row, WorldBossCharacterSheet.Row, WorldBossContributionRewardSheet.Row, WorldBossGlobalHpSheet.Row, WorldBossKillRewardSheet.Row, WorldBossListSheet.Row, WorldBossRankRewardSheet.Row, WorldBossRankingRewardSheet.Row, WorldOpenConditionSheet.Row, WorldOpenSheet.Row, WorldSheet.Row, WorldUnlockSheet.Row

Inherited Members

object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Properties

Key

```
public abstract T Key { get; }
```

Property Value

T

Methods

EndOfSheetInitialize()

```
public virtual void EndOfSheetInitialize()
```

Equals(object)

Determines whether the specified object is equal to the current object.

```
public override bool Equals(object obj)
```

Parameters

`obj` object

The object to compare with the current object.

Returns

bool

true if the specified object is equal to the current object; otherwise, false.

GetHashCode()

Serves as the default hash function.

```
public override int GetHashCode()
```

Returns

int

A hash code for the current object.

Set(IReadOnlyList<string>)

```
public abstract void Set(IReadOnlyList<string> fields)
```

Parameters

`fields` IReadOnlyList<string>

Validate()

```
public virtual void Validate()
```

Class Sheet<TKey, TValue>

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public abstract class Sheet<TKey, TValue> : IDictionary<TKey, TValue>,
ICollection<KeyValuePair<TKey, TValue>>, IEnumerable<KeyValuePair<TKey, TValue>>,
IEnumerable, ISheet where TKey : notnull where TValue : SheetRow<TKey>, new()
```

Type Parameters

TKey

TValue

Inheritance

object ← Sheet<TKey, TValue>

Implements

IDictionary<TKey, TValue>, ICollection<KeyValuePair<TKey, TValue>>, IEnumerable<KeyValuePair<TKey, TValue>>, IEnumerable, ISheet

Derived

ActionBuffSheet, AdventureBossContributionRewardSheet, AdventureBossFloorFirstRewardSheet, AdventureBossFloorPointSheet, AdventureBossFloorSheet, AdventureBossFloorWaveSheet, AdventureBossNcgRewardRatioSheet, AdventureBossSheet, AdventureBossUnlockFloorCostSheet, AdventureBossWantedRewardSheet, ArenaSheet, BoostScheduleSheet, BuffLimitSheet, BuffLinkSheet, BuffSheet, CharacterLevelSheet, CharacterSheet, ClaimableGiftsSheet, CollectQuestSheet, CollectionSheet, CombinationEquipmentQuestSheet, CombinationQuestSheet, ConsumableItemRecipeSheet, ConsumableItemSheet, CostumeItemSheet, CostumeStatSheet, CreateAvatarFavSheet, CreateAvatarItemSheet, CrystalEquipmentGrindingSheet, CrystalFluctuationSheet, CrystalHammerPointSheet, CrystalMaterialCostSheet, CrystalMonsterCollectionMultiplierSheet, CrystalRandomBuffSheet, CrystalStageBuffGachaSheet, CustomEquipmentCraftIconSheet, CustomEquipmentCraftOptionSheet, CustomEquipmentCraftRecipeSheet, CustomEquipmentCraftRecipeSkillSheet, CustomEquipmentCraftRelationshipSheet, EnemySkillSheet, EnhancementCostSheet, EnhancementCostSheetV2,

EnhancementCostSheetV3, EquipmentItemOptionSheet, EquipmentItemRecipeSheet, EquipmentItemSetEffectSheet, EquipmentItemSheet, EquipmentItemSubRecipeSheet, EquipmentItemSubRecipeSheetV2, EventConsumableItemRecipeSheet, EventDungeonSheet, EventDungeonStageSheet, EventDungeonStageWaveSheet, EventMaterialItemRecipeSheet, EventScheduleSheet, PatrolRewardSheet, GameConfigSheet, LoadIntoMyGaragesCostSheet, GeneralQuestSheet, GlobalCounterSheet, GoldQuestSheet, GrandFinaleParticipantsSheet, GrandFinaleScheduleSheet, InfiniteTowerConditionSheet, InfiniteTowerFloorSheet, InfiniteTowerFloorWaveSheet, InfiniteTowerScheduleSheet, ItemConfigForGradeSheet, ItemEnhancementQuestSheet, ItemGradeQuestSheet, ItemRequirementSheet, ItemSheet, ItemTypeCollectQuestSheet, MaterialItemSheet, MimisbrunnrSheet, MonsterCollectionRewardSheet, MonsterCollectionSheet, MonsterQuestSheet, PetOptionSheet, PetCostSheet, PetSheet, QuestItemRewardSheet, QuestRewardSheet, QuestSheet, RedeemCodeListSheet, RedeemRewardSheet, RestrictionSheet, RuneLevelBonusSheet, RuneListSheet, RuneCostSheet, RuneOptionSheet, RuneSheet, RuneWeightSheet, SkillActionBuffSheet, SkillBuffSheet, SkillSheet, StageDialogSheet, StageSheet, StageWaveSheet, StakePolicySheet, StakeAchievementRewardSheet, StakeActionPointCoefficientSheet, StakeRegularFixedRewardSheet, StakeRegularRewardSheet, StatBuffSheet, SummonSheet, SwapRateSheet, SweepRequiredCPSheet, SynthesizeSheet, SynthesizeWeightSheet, TradeQuestSheet, UnlockCombinationSlotCostSheet, WeeklyArenaRewardSheet, WorldBossActionPatternSheet, WorldBossBattleRewardSheet, WorldBossCharacterSheet, WorldBossContributionRewardSheet, WorldBossGlobalHpSheet, WorldBossKillRewardSheet, WorldBossListSheet, WorldBossRankRewardSheet, WorldBossRankingRewardSheet, WorldOpenConditionSheet, WorldOpenSheet, WorldQuestSheet, WorldSheet, WorldUnlockSheet

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

Sheet(string)

```
protected Sheet(string name)
```

Parameters

`name` string

Properties

Count

Gets the number of elements contained in the ICollection<T>.

```
public int Count { get; }
```

Property Value

int

The number of elements contained in the ICollection<T>.

First

```
public TValue? First { get; }
```

Property Value

TValue

IsReadOnly

Gets a value indicating whether the ICollection<T> is read-only.

```
public bool IsReadOnly { get; }
```

Property Value

bool

true if the ICollection<T> is read-only; otherwise, false.

this[TKey]

Gets or sets the element with the specified key.

```
public TValue this[TKey key] { get; set; }
```

Parameters

`key` TKey

The key of the element to get or set.

Property Value

TValue

The element with the specified key.

Exceptions

ArgumentNullException

`key` is null.

KeyNotFoundException

The property is retrieved and `key` is not found.

NotSupportedException

The property is set and the IDictionary<TKey, TValue> is read-only.

Keys

Gets an ICollection<T> containing the keys of the IDictionary<TKey, TValue>.

```
public ICollection<TKey> Keys { get; }
```

Property Value

ICollection<TKey>

An ICollection<T> containing the keys of the object that implements IDictionary<TKey, TValue>.

Last

```
public TValue? Last { get; }
```

Property Value

TValue

Name

```
public string Name { get; }
```

Property Value

string

OrderedList

```
public IReadOnlyList<TValue>? OrderedList { get; }
```

Property Value

IReadOnlyList<TValue>

Values

Gets an ICollection<T> containing the values in the IDictionary<TKey, TValue>.

```
public ICollection<TValue> Values { get; }
```

Property Value

ICollection<TValue>

An ICollection<T> containing the values in the object that implements IDictionary<TKey, TValue>.

Methods

Add(KeyValuePair<TKey, TValue>)

Adds an item to the ICollection<T>.

```
public void Add(KeyValuePair<TKey, TValue> item)
```

Parameters

`item` KeyValuePair<TKey, TValue>

The object to add to the ICollection<T>.

Exceptions

NotSupportedException

The ICollection<T> is read-only.

Add(TKey, TValue)

Adds an element with the provided key and value to the IDictionary<TKey, TValue>.

```
public void Add(TKey key, TValue value)
```

Parameters

`key` TKey

The object to use as the key of the element to add.

`value` TValue

The object to use as the value of the element to add.

Exceptions

ArgumentNullException

`key` is null.

ArgumentException

An element with the same key already exists in the IDictionary<TKey, TValue>.

NotSupportedException

The IDictionary<TKey, TValue> is read-only.

AddRow(TKey, TValue)

```
protected virtual void AddRow(TKey key, TValue value)
```

Parameters

`key` TKey

`value` TValue

Clear()

Removes all items from the ICollection<T>.

```
public void Clear()
```

Exceptions

NotSupportedException

The ICollection<T> is read-only.

Contains(KeyValuePair<TKey, TValue>)

Determines whether the ICollection<T> contains a specific value.

```
public bool Contains(KeyValuePair<TKey, TValue> item)
```

Parameters

`item` KeyValuePair<TKey, TValue>

The object to locate in the ICollection<T>.

Returns

bool

true if `item` is found in the ICollection<T>; otherwise, false.

ContainsKey(TKey)

Determines whether the IDictionary<TKey, TValue> contains an element with the specified key.

```
public bool ContainsKey(TKey key)
```

Parameters

`key` TKey

The key to locate in the IDictionary<TKey, TValue>.

Returns

bool

true if the IDictionary<TKey, TValue> contains an element with the key; otherwise, false.

Exceptions

ArgumentNullException

`key` is null.

CopyTo(KeyValuePair<TKey, TValue>[], int)

Copies the elements of the ICollection<T> to an Array, starting at a particular Array index.

```
public void CopyTo(KeyValuePair<TKey, TValue>[] array, int arrayIndex)
```

Parameters

`array` KeyValuePair<TKey, TValue>[]

The one-dimensional Array that is the destination of the elements copied from ICollection<T>. The Array must have zero-based indexing.

`arrayIndex` int

The zero-based index in `array` at which copying begins.

Exceptions

ArgumentNullException

`array` is null.

ArgumentOutOfRangeException

`arrayIndex` is less than 0.

ArgumentException

The number of elements in the source ICollection<T> is greater than the available space from `arrayIndex` to the end of the destination `array`.

GetEnumerator()

```
public IEnumerator<TValue> GetEnumerator()
```

Returns

IEnumerator<TValue>

Remove(KeyValuePair<TKey, TValue>)

Removes the first occurrence of a specific object from the ICollection<T>.

```
public bool Remove(KeyValuePair<TKey, TValue> item)
```

Parameters

`item` KeyValuePair<TKey, TValue>

The object to remove from the ICollection<T>.

Returns

bool

true if `item` was successfully removed from the ICollection<T>; otherwise, false. This method also returns false if `item` is not found in the original ICollection<T>.

Exceptions

NotSupportedException

The ICollection<T> is read-only.

Remove(TKey)

Removes the element with the specified key from the IDictionary<TKey, TValue>.

```
public bool Remove(TKey key)
```

Parameters

`key` TKey

The key of the element to remove.

Returns

bool

true if the element is successfully removed; otherwise, false. This method also returns false if `key` was not found in the original IDictionary<TKey, TValue>.

Exceptions

ArgumentNullException

`key` is null.

NotSupportedException

The IDictionary<TKey, TValue> is read-only.

Serialize()

```
public IValue Serialize()
```

Returns

IValue

Set(string, bool)

```
public virtual void Set(string csv, bool isReversed = false)
```

Parameters

`csv` string

`isReversed` bool

true: csv값의 column과 row가 뒤집혀서 작성되어 있다고 판단합니다.

Exceptions

ArgumentNullException

InvalidDataException

Set<T>(Sheet<TKey, T>, bool)

```
public void Set<T>(Sheet<TKey, T> sheet, bool executePostSet = true) where T :
TValue, new()
```

Parameters

`sheet` Sheet<TKey, T>

`executePostSet` bool

Type Parameters

`T`

TryGetValue(TKey, out TValue)

Gets the value associated with the specified key.

```
public bool TryGetValue(TKey key, out TValue value)
```

Parameters

`key` TKey

The key whose value to get.

`value` TValue

When this method returns, the value associated with the specified key, if the key is found; otherwise, the default value for the type of the `value` parameter. This parameter is passed uninitialized.

Returns

bool

true if the object that implements IDictionary<TKey, TValue> contains an element with the specified key; otherwise, false.

Exceptions

ArgumentNullException

`key` is null.

TryGetValue(TKey, out TValue, bool)

```
public bool TryGetValue(TKey key, out TValue value, bool throwException)
```

Parameters

`key` TKey

`value` TValue

`throwException` bool

Returns

bool

Class SimulatorSheets

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
public class SimulatorSheets : SimulatorSheetsV1
```

Inheritance

object ← SimulatorSheetsV1 ← SimulatorSheets

Derived

ArenaSimulatorSheets, RaidSimulatorSheets, RankingSimulatorSheets, StageSimulatorSheets

Inherited Members

SimulatorSheetsV1.MaterialItemSheet , SimulatorSheetsV1.SkillSheet , SimulatorSheetsV1.SkillBuffSheet , SimulatorSheetsV1.SkillActionBuffSheet , SimulatorSheetsV1.ActionBuffSheet , SimulatorSheetsV1.StatBuffSheet , SimulatorSheetsV1.CharacterSheet , SimulatorSheetsV1.CharacterLevelSheet , SimulatorSheetsV1.EquipmentItemSetEffectSheet , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

SimulatorSheets(MaterialItemSheet, SkillSheet, SkillBuffSheet, StatBuffSheet, SkillActionBuffSheet, ActionBuffSheet, CharacterSheet, CharacterLevelSheet, EquipmentItemSetEffectSheet, RuneOptionSheet, RuneListSheet, RuneLevelBonusSheet)

```
public SimulatorSheets(MaterialItemSheet materialItemSheet, SkillSheet skillSheet,
SkillBuffSheet skillBuffSheet, StatBuffSheet statBuffSheet, SkillActionBuffSheet
skillActionBuffSheet, ActionBuffSheet actionBuffSheet, CharacterSheet
characterSheet, CharacterLevelSheet characterLevelSheet, EquipmentItemSetEffectSheet
equipmentItemSetEffectSheet, RuneOptionSheet runeOptionSheet, RuneListSheet
runeListSheet, RuneLevelBonusSheet runeLevelBonusSheet)
```

Parameters

`materialItemSheet` MaterialItemSheet

`skillSheet` SkillSheet

`skillBuffSheet` SkillBuffSheet

`statBuffSheet` StatBuffSheet

`skillActionBuffSheet` SkillActionBuffSheet

`actionBuffSheet` ActionBuffSheet

`characterSheet` CharacterSheet

`characterLevelSheet` CharacterLevelSheet

`equipmentItemSetEffectSheet` EquipmentItemSetEffectSheet

`runeOptionSheet` RuneOptionSheet

`runeListSheet` RuneListSheet

`runeLevelBonusSheet` RuneLevelBonusSheet

Fields

RuneLevelBonusSheet

```
public readonly RuneLevelBonusSheet RuneLevelBonusSheet
```

Field Value

RuneLevelBonusSheet

RuneListSheet

```
public readonly RuneListSheet RuneListSheet
```

Field Value

RuneListSheet

RuneOptionSheet

```
public readonly RuneOptionSheet RuneOptionSheet
```

Field Value

RuneOptionSheet

Class SimulatorSheetsV1

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
public class SimulatorSheetsV1
```

Inheritance

object ← SimulatorSheetsV1

Derived

ArenaSimulatorSheetsV1, RaidSimulatorSheetsV1, RankingSimulatorSheetsV1, SimulatorSheets, StageSimulatorSheetsV1

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

SimulatorSheetsV1(MaterialItemSheet, SkillSheet, SkillBuffSheet, StatBuffSheet, SkillActionBuffSheet, ActionBuffSheet, CharacterSheet, CharacterLevelSheet, EquipmentItemSetEffectSheet)

```
public SimulatorSheetsV1(MaterialItemSheet materialItemSheet, SkillSheet skillSheet,
SkillBuffSheet skillBuffSheet, StatBuffSheet statBuffSheet,
SkillActionBuffSheet skillActionBuffSheet, ActionBuffSheet actionBuffSheet,
CharacterSheet characterSheet, CharacterLevelSheet characterLevelSheet,
EquipmentItemSetEffectSheet equipmentItemSetEffectSheet)
```

Parameters

`materialItemSheet` MaterialItemSheet

`skillSheet` SkillSheet

`skillBuffSheet` SkillBuffSheet

`statBuffSheet` StatBuffSheet

`skillActionBuffSheet` SkillActionBuffSheet

`actionBuffSheet` ActionBuffSheet

`characterSheet` CharacterSheet

`characterLevelSheet` CharacterLevelSheet

`equipmentItemSetEffectSheet` EquipmentItemSetEffectSheet

Fields

ActionBuffSheet

```
public readonly ActionBuffSheet ActionBuffSheet
```

Field Value

ActionBuffSheet

CharacterLevelSheet

```
public readonly CharacterLevelSheet CharacterLevelSheet
```

Field Value

CharacterLevelSheet

CharacterSheet

```
public readonly CharacterSheet CharacterSheet
```

Field Value

CharacterSheet

EquipmentItemSetEffectSheet

```
public readonly EquipmentItemSetEffectSheet EquipmentItemSetEffectSheet
```

Field Value

EquipmentItemSetEffectSheet

MaterialItemSheet

```
public readonly MaterialItemSheet MaterialItemSheet
```

Field Value

MaterialItemSheet

SkillActionBuffSheet

```
public readonly SkillActionBuffSheet SkillActionBuffSheet
```

Field Value

SkillActionBuffSheet

SkillBuffSheet

```
public readonly SkillBuffSheet SkillBuffSheet
```

Field Value

SkillBuffSheet

SkillSheet

```
public readonly SkillSheet SkillSheet
```

Field Value

SkillSheet

StatBuffSheet

```
public readonly StatBuffSheet StatBuffSheet
```

Field Value

StatBuffSheet

Class SimulatorSheetsV100291

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
public class SimulatorSheetsV100291
```

Inheritance

object ← SimulatorSheetsV100291

Derived

ArenaSimulatorSheetsV100291, RankingSimulatorSheetsV100291, StageSimulatorSheetsV100291

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

SimulatorSheetsV100291(MaterialItemSheet, SkillSheet, SkillBuffSheet, BuffSheet, CharacterSheet, CharacterLevelSheet, EquipmentItemSetEffectSheet)

```
public SimulatorSheetsV100291(MaterialItemSheet materialItemSheet,
SkillSheet skillSheet, SkillBuffSheet skillBuffSheet, BuffSheet buffSheet,
CharacterSheet characterSheet, CharacterLevelSheet characterLevelSheet,
EquipmentItemSetEffectSheet equipmentItemSetEffectSheet)
```

Parameters

`materialItemSheet` MaterialItemSheet

`skillSheet` SkillSheet

`skillBuffSheet` SkillBuffSheet

`buffSheet` BuffSheet

`characterSheet` CharacterSheet

`characterLevelSheet` CharacterLevelSheet

`equipmentItemSetEffectSheet` EquipmentItemSetEffectSheet

Fields

BuffSheet

```
public readonly BuffSheet BuffSheet
```

Field Value

BuffSheet

CharacterLevelSheet

```
public readonly CharacterLevelSheet CharacterLevelSheet
```

Field Value

CharacterLevelSheet

CharacterSheet

```
public readonly CharacterSheet CharacterSheet
```

Field Value

CharacterSheet

EquipmentItemSetEffectSheet

```
public readonly EquipmentItemSetEffectSheet EquipmentItemSetEffectSheet
```

Field Value

EquipmentItemSetEffectSheet

MaterialItemSheet

```
public readonly MaterialItemSheet MaterialItemSheet
```

Field Value

MaterialItemSheet

SkillBuffSheet

```
public readonly SkillBuffSheet SkillBuffSheet
```

Field Value

SkillBuffSheet

SkillSheet

```
public readonly SkillSheet SkillSheet
```

Field Value

SkillSheet

Methods

ToSimulatorSheetsV1()

```
public SimulatorSheetsV1 ToSimulatorSheetsV1()
```

Returns

SimulatorSheetsV1

Class SkillActionBuffSheet

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class SkillActionBuffSheet : Sheet<int, SkillActionBuffSheet.Row>,
IDictionary<int, SkillActionBuffSheet.Row>, ICollection<KeyValuePair<int,
SkillActionBuffSheet.Row>>, IEnumerable<KeyValuePair<int,
SkillActionBuffSheet.Row>>, IEnumerable, ISheet
```

Inheritance

object ← Sheet<int, SkillActionBuffSheet.Row> ← SkillActionBuffSheet

Implements

IDictionary<int, SkillActionBuffSheet.Row>,
ICollection<KeyValuePair<int, SkillActionBuffSheet.Row>>,
IEnumerable<KeyValuePair<int, SkillActionBuffSheet.Row>>, IEnumerable, ISheet

Inherited Members

Sheet<int, SkillActionBuffSheet.Row>.Name ,
Sheet<int, SkillActionBuffSheet.Row>.OrderedList ,
Sheet<int, SkillActionBuffSheet.Row>.First , Sheet<int, SkillActionBuffSheet.Row>.Last ,
Sheet<int, SkillActionBuffSheet.Row>.Keys , Sheet<int, SkillActionBuffSheet.Row>.Values ,
Sheet<int, SkillActionBuffSheet.Row>.Count ,
Sheet<int, SkillActionBuffSheet.Row>.IsReadOnly ,
Sheet<int, SkillActionBuffSheet.Row>.this[int] ,
Sheet<int, SkillActionBuffSheet.Row>.Set(string, bool) ,
Sheet<int, SkillActionBuffSheet.Row>.Set<T>(Sheet<int, T>, bool) ,
Sheet<int, SkillActionBuffSheet.Row>.GetEnumerator() ,
Sheet<int, SkillActionBuffSheet.Row>.TryGetValue(int, out SkillActionBuffSheet.Row, bool) ,
Sheet<int, SkillActionBuffSheet.Row>.AddRow(int, SkillActionBuffSheet.Row) ,
Sheet<int, SkillActionBuffSheet.Row>.Add(int, SkillActionBuffSheet.Row) ,
Sheet<int, SkillActionBuffSheet.Row>.ContainsKey(int) ,
Sheet<int, SkillActionBuffSheet.Row>.Remove(int) ,
Sheet<int, SkillActionBuffSheet.Row>.TryGetValue(int, out SkillActionBuffSheet.Row) ,
Sheet<int, SkillActionBuffSheet.Row>.Add(KeyValuePair<int, SkillActionBuffSheet.Row>) ,
Sheet<int, SkillActionBuffSheet.Row>.Clear() ,
Sheet<int, SkillActionBuffSheet.Row>.Contains(KeyValuePair<int,
SkillActionBuffSheet.Row>) ,

Sheet<int, SkillActionBuffSheet.Row>.CopyTo(KeyValuePair<int, SkillActionBuffSheet.Row>[], int) ,
Sheet<int, SkillActionBuffSheet.Row>.Remove(KeyValuePair<int, SkillActionBuffSheet.Row>) ,
Sheet<int, SkillActionBuffSheet.Row>.Serialize() , object.Equals(object) ,
object.Equals(object, object) , object.GetHashCode() , object.GetType() ,
object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

SkillActionBuffSheet()

```
public SkillActionBuffSheet()
```

Methods

AddRow(int, Row)

```
protected override void AddRow(int key, SkillActionBuffSheet.Row value)
```

Parameters

`key` int

`value` SkillActionBuffSheet.Row

Class SkillActionBuffSheet.Row

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class SkillActionBuffSheet.Row : SheetRow<int>
```

Inheritance

object ← SheetRow<int> ← SkillActionBuffSheet.Row

Inherited Members

SheetRow<int>.Validate() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Properties

BuffIds

```
public List<int> BuffIds { get; }
```

Property Value

List<int>

Key

```
public override int Key { get; }
```

Property Value

int

SkillId

```
public int SkillId { get; }
```

Property Value

int

Methods

EndOfSheetInitialize()

```
public override void EndOfSheetInitialize()
```

Set(IReadOnlyList<string>)

```
public override void Set(IReadOnlyList<string> fields)
```

Parameters

`fields` IReadOnlyList<string>

Class SkillBuffSheet

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class SkillBuffSheet : Sheet<int, SkillBuffSheet.Row>, IDictionary<int,
SkillBuffSheet.Row>, ICollection<KeyValuePair<int, SkillBuffSheet.Row>>,
IEnumerable<KeyValuePair<int, SkillBuffSheet.Row>>, IEnumerable, ISheet
```

Inheritance

object ← Sheet<int, SkillBuffSheet.Row> ← SkillBuffSheet

Implements

IDictionary<int, SkillBuffSheet.Row>,
ICollection<KeyValuePair<int, SkillBuffSheet.Row>>,
IEnumerable<KeyValuePair<int, SkillBuffSheet.Row>>, IEnumerable, ISheet

Inherited Members

Sheet<int, SkillBuffSheet.Row>.Name , Sheet<int, SkillBuffSheet.Row>.OrderedList ,
Sheet<int, SkillBuffSheet.Row>.First , Sheet<int, SkillBuffSheet.Row>.Last ,
Sheet<int, SkillBuffSheet.Row>.Keys , Sheet<int, SkillBuffSheet.Row>.Values ,
Sheet<int, SkillBuffSheet.Row>.Count , Sheet<int, SkillBuffSheet.Row>.IsReadOnly ,
Sheet<int, SkillBuffSheet.Row>.this[int] , Sheet<int, SkillBuffSheet.Row>.Set(string, bool) ,
Sheet<int, SkillBuffSheet.Row>.Set<T>(Sheet<int, T>, bool) ,
Sheet<int, SkillBuffSheet.Row>.GetEnumerator() ,
Sheet<int, SkillBuffSheet.Row>.TryGetValue(int, out SkillBuffSheet.Row, bool) ,
Sheet<int, SkillBuffSheet.Row>.AddRow(int, SkillBuffSheet.Row) ,
Sheet<int, SkillBuffSheet.Row>.Add(int, SkillBuffSheet.Row) ,
Sheet<int, SkillBuffSheet.Row>.ContainsKey(int) ,
Sheet<int, SkillBuffSheet.Row>.Remove(int) ,
Sheet<int, SkillBuffSheet.Row>.TryGetValue(int, out SkillBuffSheet.Row) ,
Sheet<int, SkillBuffSheet.Row>.Add(KeyValuePair<int, SkillBuffSheet.Row>) ,
Sheet<int, SkillBuffSheet.Row>.Clear() ,
Sheet<int, SkillBuffSheet.Row>.Contains(KeyValuePair<int, SkillBuffSheet.Row>) ,
Sheet<int, SkillBuffSheet.Row>.CopyTo(KeyValuePair<int, SkillBuffSheet.Row>[], int) ,
Sheet<int, SkillBuffSheet.Row>.Remove(KeyValuePair<int, SkillBuffSheet.Row>) ,
Sheet<int, SkillBuffSheet.Row>.Serialize() , object.Equals(object) ,

object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

SkillBuffSheet()

```
public SkillBuffSheet()
```

Methods

AddRow(int, Row)

```
protected override void AddRow(int key, SkillBuffSheet.Row value)
```

Parameters

`key` int

`value` SkillBuffSheet.Row

Class SkillBuffSheet.Row

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class SkillBuffSheet.Row : SheetRow<int>
```

Inheritance

object ← SheetRow<int> ← SkillBuffSheet.Row

Inherited Members

SheetRow<int>.Validate() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Properties

BuffIds

```
public List<int> BuffIds { get; }
```

Property Value

List<int>

Key

```
public override int Key { get; }
```

Property Value

int

SkillId

```
public int SkillId { get; }
```

Property Value

int

Methods

EndOfSheetInitialize()

```
public override void EndOfSheetInitialize()
```

Set(IReadOnlyList<string>)

```
public override void Set(IReadOnlyList<string> fields)
```

Parameters

`fields` IReadOnlyList<string>

Class SkillSheet

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

Sheet containing skill data loaded from CSV files. Provides access to skill definitions and their properties.

```
[Serializable]
public class SkillSheet : Sheet<int, SkillSheet.Row>, IDictionary<int,
SkillSheet.Row>, ICollection<KeyValuePair<int, SkillSheet.Row>>,
IEnumerable<KeyValuePair<int, SkillSheet.Row>>, IEnumerable, ISheet
```

Inheritance

object ← Sheet<int, SkillSheet.Row> ← SkillSheet

Implements

IDictionary<int, SkillSheet.Row>, ICollection<KeyValuePair<int, SkillSheet.Row>>, IEnumerable<KeyValuePair<int, SkillSheet.Row>>, IEnumerable, ISheet

Inherited Members

Sheet<int, SkillSheet.Row>.Name , Sheet<int, SkillSheet.Row>.OrderedList , Sheet<int, SkillSheet.Row>.First , Sheet<int, SkillSheet.Row>.Last , Sheet<int, SkillSheet.Row>.Keys , Sheet<int, SkillSheet.Row>.Values , Sheet<int, SkillSheet.Row>.Count , Sheet<int, SkillSheet.Row>.IsReadOnly , Sheet<int, SkillSheet.Row>.this[int] , Sheet<int, SkillSheet.Row>.Set(string, bool) , Sheet<int, SkillSheet.Row>.Set<T>(Sheet<int, T>, bool) , Sheet<int, SkillSheet.Row>.GetEnumerator() , Sheet<int, SkillSheet.Row>.TryGetValue(int, out SkillSheet.Row, bool) , Sheet<int, SkillSheet.Row>.AddRow(int, SkillSheet.Row) , Sheet<int, SkillSheet.Row>.Add(int, SkillSheet.Row) , Sheet<int, SkillSheet.Row>.ContainsKey(int) , Sheet<int, SkillSheet.Row>.Remove(int) , Sheet<int, SkillSheet.Row>.TryGetValue(int, out SkillSheet.Row) , Sheet<int, SkillSheet.Row>.Add(KeyValuePair<int, SkillSheet.Row>) , Sheet<int, SkillSheet.Row>.Clear() , Sheet<int, SkillSheet.Row>.Contains(KeyValuePair<int, SkillSheet.Row>) , Sheet<int, SkillSheet.Row>.CopyTo(KeyValuePair<int, SkillSheet.Row>[], int) , Sheet<int, SkillSheet.Row>.Remove(KeyValuePair<int, SkillSheet.Row>) , Sheet<int, SkillSheet.Row>.Serialize() , object.Equals(object) ,

object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

SkillSheet()

```
public SkillSheet()
```

Class SkillSheet.Row

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

Represents a single skill row from the skill sheet. Contains skill metadata and properties used for skill creation and execution.

```
[Serializable]
public class SkillSheet.Row : SheetRow<int>, IState
```

Inheritance

object ← SheetRow<int> ← SkillSheet.Row

Implements

IState

Inherited Members

SheetRow<int>.Validate() , SheetRow<int>.EndOfSheetInitialize() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Remarks

SkillSheet.Row maps 1:1 to the following 9 fields in SkillSheet.csv: [0] id [1] _name [2] elemental_type [3] skill_type [4] skill_category [5] skill_target_type [6] hit_count [7] cooldown [8] combo

Serialization and deserialization are performed as a List in this order. (Dictionary is also supported for backward compatibility.)

Constructors

Row()

```
public Row()
```

Row(Dictionary)

```
public Row(Dictionary serialized)
```

Parameters

`serialized` Dictionary

Row(List)

```
public Row(List serialized)
```

Parameters

`serialized` List

Properties

Combo

```
public bool Combo { get; }
```

Property Value

bool

Cooldown

```
public int Cooldown { get; }
```

Property Value

int

ElementalType

```
public ElementalType ElementalType { get; }
```

Property Value

ElementalType

HitCount

```
public int HitCount { get; }
```

Property Value

int

Id

```
public int Id { get; }
```

Property Value

int

Key

```
public override int Key { get; }
```

Property Value

int

SkillCategory

```
public SkillCategory SkillCategory { get; }
```

Property Value

SkillCategory

SkillTargetType

```
public SkillTargetType SkillTargetType { get; }
```

Property Value

SkillTargetType

SkillType

```
public SkillType SkillType { get; }
```

Property Value

SkillType

Methods

Deserialize(Dictionary)

```
[Obsolete("Use Deserialize(IValue) instead.")]
public static SkillSheet.Row Deserialize(Dictionary serialized)
```

Parameters

`serialized` Dictionary

Returns

SkillSheet.Row

Deserialize(IValue)

Deserializes a skill row from serialized data. Supports both Dictionary and List formats for backward compatibility.

```
public static SkillSheet.Row Deserialize(IValue serialized)
```

Parameters

`serialized` IValue

The serialized skill row data

Returns

SkillSheet.Row

The deserialized skill row

Exceptions

ArgumentException

Thrown when the serialization format is not supported

DeserializeFromDictionary(Dictionary)

```
public static SkillSheet.Row DeserializeFromDictionary(Dictionary serialized)
```

Parameters

`serialized` Dictionary

Returns

SkillSheet.Row

DeserializeFromList(List)

```
public static SkillSheet.Row DeserializeFromList(List serialized)
```

Parameters

`serialized` List

Returns

SkillSheet.Row

Serialize()

```
public IValue Serialize()
```

Returns

IValue

Set(IReadOnlyList<string>)

```
public override void Set(IReadOnlyList<string> fields)
```

Parameters

`fields` IReadOnlyList<string>

Class StageDialogSheet

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class StageDialogSheet : Sheet<int, StageDialogSheet.Row>, IDictionary<int,
StageDialogSheet.Row>, ICollection<KeyValuePair<int, StageDialogSheet.Row>>,
IEnumerable<KeyValuePair<int, StageDialogSheet.Row>>, IEnumerable, ISheet
```

Inheritance

object ← Sheet<int, StageDialogSheet.Row> ← StageDialogSheet

Implements

IDictionary<int, StageDialogSheet.Row>,
ICollection<KeyValuePair<int, StageDialogSheet.Row>>,
IEnumerable<KeyValuePair<int, StageDialogSheet.Row>>, IEnumerable, ISheet

Inherited Members

Sheet<int, StageDialogSheet.Row>.Name ,
Sheet<int, StageDialogSheet.Row>.OrderedList , Sheet<int, StageDialogSheet.Row>.First ,
Sheet<int, StageDialogSheet.Row>.Last , Sheet<int, StageDialogSheet.Row>.Keys ,
Sheet<int, StageDialogSheet.Row>.Values , Sheet<int, StageDialogSheet.Row>.Count ,
Sheet<int, StageDialogSheet.Row>.IsReadOnly ,
Sheet<int, StageDialogSheet.Row>.this[int] ,
Sheet<int, StageDialogSheet.Row>.Set(string, bool) ,
Sheet<int, StageDialogSheet.Row>.Set<T>(Sheet<int, T>, bool) ,
Sheet<int, StageDialogSheet.Row>.GetEnumerator() ,
Sheet<int, StageDialogSheet.Row>.TryGetValue(int, out StageDialogSheet.Row, bool) ,
Sheet<int, StageDialogSheet.Row>.AddRow(int, StageDialogSheet.Row) ,
Sheet<int, StageDialogSheet.Row>.Add(int, StageDialogSheet.Row) ,
Sheet<int, StageDialogSheet.Row>.ContainsKey(int) ,
Sheet<int, StageDialogSheet.Row>.Remove(int) ,
Sheet<int, StageDialogSheet.Row>.TryGetValue(int, out StageDialogSheet.Row) ,
Sheet<int, StageDialogSheet.Row>.Add(KeyValuePair<int, StageDialogSheet.Row>) ,
Sheet<int, StageDialogSheet.Row>.Clear() ,
Sheet<int, StageDialogSheet.Row>.Contains(KeyValuePair<int, StageDialogSheet.Row>) ,
Sheet<int, StageDialogSheet.Row>.CopyTo(KeyValuePair<int, StageDialogSheet.Row>[],
int) ,

Sheet<int, StageDialogSheet.Row>.Remove(KeyValuePair<int, StageDialogSheet.Row>) , Sheet<int, StageDialogSheet.Row>.Serialize() , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

StageDialogSheet()

```
public StageDialogSheet()
```

Class StageDialogSheet.Row

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class StageDialogSheet.Row : SheetRow<int>
```

Inheritance

object ← SheetRow<int> ← StageDialogSheet.Row

Inherited Members

SheetRow<int>.Validate() , SheetRow<int>.EndOfSheetInitialize() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Properties

DialogId

```
public int DialogId { get; }
```

Property Value

int

Id

```
public int Id { get; }
```

Property Value

int

Key

```
public override int Key { get; }
```

Property Value

int

StageId

```
public int StageId { get; }
```

Property Value

int

Methods

Set(IReadOnlyList<string>)

```
public override void Set(IReadOnlyList<string> fields)
```

Parameters

`fields` IReadOnlyList<string>

Class StageSheet

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class StageSheet : Sheet<int, StageSheet.Row>, IDictionary<int,
StageSheet.Row>, ICollection<KeyValuePair<int, StageSheet.Row>>,
IEnumerable<KeyValuePair<int, StageSheet.Row>>, IEnumerable, ISheet
```

Inheritance

object ← Sheet<int, StageSheet.Row> ← StageSheet

Implements

IDictionary<int, StageSheet.Row>,
ICollection<KeyValuePair<int, StageSheet.Row>>,
IEnumerable<KeyValuePair<int, StageSheet.Row>>, IEnumerable, ISheet

Inherited Members

Sheet<int, StageSheet.Row>.Name , Sheet<int, StageSheet.Row>.OrderedList ,
Sheet<int, StageSheet.Row>.First , Sheet<int, StageSheet.Row>.Last ,
Sheet<int, StageSheet.Row>.Keys , Sheet<int, StageSheet.Row>.Values ,
Sheet<int, StageSheet.Row>.Count , Sheet<int, StageSheet.Row>.IsReadOnly ,
Sheet<int, StageSheet.Row>.this[int] , Sheet<int, StageSheet.Row>.Set(string, bool) ,
Sheet<int, StageSheet.Row>.Set<T>(Sheet<int, T>, bool) ,
Sheet<int, StageSheet.Row>.GetEnumerator() ,
Sheet<int, StageSheet.Row>.TryGetValue(int, out StageSheet.Row, bool) ,
Sheet<int, StageSheet.Row>.AddRow(int, StageSheet.Row) ,
Sheet<int, StageSheet.Row>.Add(int, StageSheet.Row) ,
Sheet<int, StageSheet.Row>.ContainsKey(int) , Sheet<int, StageSheet.Row>.Remove(int) ,
Sheet<int, StageSheet.Row>.TryGetValue(int, out StageSheet.Row) ,
Sheet<int, StageSheet.Row>.Add(KeyValuePair<int, StageSheet.Row>) ,
Sheet<int, StageSheet.Row>.Clear() ,
Sheet<int, StageSheet.Row>.Contains(KeyValuePair<int, StageSheet.Row>) ,
Sheet<int, StageSheet.Row>.CopyTo(KeyValuePair<int, StageSheet.Row>[], int) ,
Sheet<int, StageSheet.Row>.Remove(KeyValuePair<int, StageSheet.Row>) ,
Sheet<int, StageSheet.Row>.Serialize() , object.Equals(object) ,
object.Equals(object, object) , object.GetHashCode() , object.GetType() ,
object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

StageSheet()

```
public StageSheet()
```

Methods

AddRow(int, Row)

```
protected override void AddRow(int key, StageSheet.Row value)
```

Parameters

`key` int

`value` StageSheet.Row

Class StageSheet.FavRewardData

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class StageSheet.FavRewardData
```

Inheritance

object ← StageSheet.FavRewardData

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

FavRewardData(string, decimal, int, int)

Initializes a new instance of StageSheet.FavRewardData.

```
public FavRewardData(string ticker, decimal ratio, int min, int max)
```

Parameters

`ticker` string

The fungible asset ticker.

`ratio` decimal

The selection weight.

`min` int

Minimum granted amount (inclusive).

`max` int

Maximum granted amount (inclusive).

Properties

Max

Gets the maximum amount granted when selected.

```
public int Max { get; }
```

Property Value

int

Min

Gets the minimum amount granted when selected.

```
public int Min { get; }
```

Property Value

int

Ratio

Gets the weight used for random selection.

```
public decimal Ratio { get; }
```

Property Value

decimal

Ticker

Gets the currency ticker of the fungible asset reward (e.g., `CRYSTAL`).

```
public string Ticker { get; }
```

Property Value

string

Class StageSheet.RewardData

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class StageSheet.RewardData
```

Inheritance

object ← StageSheet.RewardData

Derived

WeeklyArenaRewardSheet.RewardData

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

RewardData(int, decimal, int, int)

```
public RewardData(int itemId, decimal ratio, int min, int max)
```

Parameters

`itemId` int

`ratio` decimal

`min` int

`max` int

Properties

ItemId

```
public int ItemId { get; }
```

Property Value

int

Max

```
public int Max { get; }
```

Property Value

int

Min

```
public int Min { get; }
```

Property Value

int

Ratio

```
public decimal Ratio { get; }
```

Property Value

decimal

Class StageSheet.Row

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class StageSheet.Row : SheetRow<int>
```

Inheritance

object ← SheetRow<int> ← StageSheet.Row

Derived

EventDungeonStageSheet.Row

Inherited Members

SheetRow<int>.Validate() , SheetRow<int>.EndOfSheetInitialize() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Properties

BGM

```
public string BGM { get; }
```

Property Value

string

Background

```
public string Background { get; }
```

Property Value

string

CostAP

```
public int CostAP { get; }
```

Property Value

int

DropItemMax

```
public int DropItemMax { get; }
```

Property Value

int

DropItemMin

```
public int DropItemMin { get; }
```

Property Value

int

EnemyInitialStatModifiers

```
public List<StatModifier> EnemyInitialStatModifiers { get; }
```

Property Value

List<StatModifier>

EntryCostItemCount

Gets the additional entry cost material item count required per play for this stage. A value of 0 means no additional entry material cost is required.

```
public int EntryCostItemCount { get; }
```

Property Value

int

EntryCostItemId

Gets the additional entry cost material item ID required to play this stage. A value of 0 means no additional entry material cost is required.

```
public int EntryCostItemId { get; }
```

Property Value

int

FavDropMax

Gets the maximum number of draws from FavRewards (inclusive).

```
public int FavDropMax { get; }
```

Property Value

int

FavDropMin

Gets the minimum number of draws from FavRewards (inclusive).

```
public int FavDropMin { get; }
```

Property Value

int

FavRewards

Gets the pool of fungible asset reward candidates for this stage. Rewards are selected by weight (Ratio) and granted as (ticker, amount) pairs.

```
public List<StageSheet.FavRewardData> FavRewards { get; }
```

Property Value

List<StageSheet.FavRewardData>

Id

```
public int Id { get; }
```

Property Value

int

Key

```
public override int Key { get; }
```

Property Value

int

Rewards

```
public List<StageSheet.RewardData> Rewards { get; }
```

Property Value

List<StageSheet.RewardData>

TurnLimit

```
public int TurnLimit { get; }
```

Property Value

int

Methods

CloneWithId(int)

Creates a shallow clone of this row with a new stage ID.

```
public StageSheet.Row CloneWithId(int newId)
```

Parameters

`newId` int

The stage ID to assign to the cloned row.

Returns

StageSheet.Row

A new StageSheet.Row with copied values and the specified ID.

Set(IReadOnlyList<string>)

```
public override void Set(IReadOnlyList<string> fields)
```

Parameters

`fields` IReadOnlyList<string>

Class StageSimulatorSheets

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
public class StageSimulatorSheets : SimulatorSheets
```

Inheritance

object ← SimulatorSheetsV1 ← SimulatorSheets ← StageSimulatorSheets

Inherited Members

SimulatorSheets.RuneOptionSheet , SimulatorSheets.RuneListSheet , SimulatorSheets.RuneLevelBonusSheet , SimulatorSheetsV1.MaterialItemSheet , SimulatorSheetsV1.SkillSheet , SimulatorSheetsV1.SkillBuffSheet , SimulatorSheetsV1.SkillActionBuffSheet , SimulatorSheetsV1.ActionBuffSheet , SimulatorSheetsV1.StatBuffSheet , SimulatorSheetsV1.CharacterSheet , SimulatorSheetsV1.CharacterLevelSheet , SimulatorSheetsV1.EquipmentItemSetEffectSheet , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

StageSimulatorSheets(MaterialItemSheet, SkillSheet, SkillBuffSheet, StatBuffSheet, SkillActionBuffSheet, ActionBuffSheet, CharacterSheet, CharacterLevelSheet, EquipmentItemSetEffectSheet, StageSheet, StageWaveSheet, EnemySkillSheet, RuneOptionSheet, RuneListSheet, RuneLevelBonusSheet)

```
public StageSimulatorSheets(MaterialItemSheet materialItemSheet, SkillSheet
skillSheet, SkillBuffSheet skillBuffSheet, StatBuffSheet statBuffSheet,
SkillActionBuffSheet skillActionBuffSheet, ActionBuffSheet actionBuffSheet,
CharacterSheet characterSheet, CharacterLevelSheet characterLevelSheet,
EquipmentItemSetEffectSheet equipmentItemSetEffectSheet, StageSheet
stageSheet, StageWaveSheet stageWaveSheet, EnemySkillSheet enemySkillSheet,
```

```
RuneOptionSheet runeOptionSheet, RuneListSheet runeListSheet,
RuneLevelBonusSheet runeLevelBonusSheet)
```

Parameters

`materialItemSheet` MaterialItemSheet

`skillSheet` SkillSheet

`skillBuffSheet` SkillBuffSheet

`statBuffSheet` StatBuffSheet

`skillActionBuffSheet` SkillActionBuffSheet

`actionBuffSheet` ActionBuffSheet

`characterSheet` CharacterSheet

`characterLevelSheet` CharacterLevelSheet

`equipmentItemSetEffectSheet` EquipmentItemSetEffectSheet

`stageSheet` StageSheet

`stageWaveSheet` StageWaveSheet

`enemySkillSheet` EnemySkillSheet

`runeOptionSheet` RuneOptionSheet

`runeListSheet` RuneListSheet

`runeLevelBonusSheet` RuneLevelBonusSheet

Fields

EnemySkillSheet

```
public readonly EnemySkillSheet EnemySkillSheet
```

Field Value

EnemySkillSheet

StageSheet

```
public readonly StageSheet StageSheet
```

Field Value

StageSheet

StageWaveSheet

```
public readonly StageWaveSheet StageWaveSheet
```

Field Value

StageWaveSheet

Class StageSimulatorSheetsV1

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
public class StageSimulatorSheetsV1 : SimulatorSheetsV1
```

Inheritance

object ← SimulatorSheetsV1 ← StageSimulatorSheetsV1

Inherited Members

SimulatorSheetsV1.MaterialItemSheet , SimulatorSheetsV1.SkillSheet , SimulatorSheetsV1.SkillBuffSheet , SimulatorSheetsV1.SkillActionBuffSheet , SimulatorSheetsV1.ActionBuffSheet , SimulatorSheetsV1.StatBuffSheet , SimulatorSheetsV1.CharacterSheet , SimulatorSheetsV1.CharacterLevelSheet , SimulatorSheetsV1.EquipmentItemSetEffectSheet , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

StageSimulatorSheetsV1(MaterialItemSheet, SkillSheet, SkillBuffSheet, StatBuffSheet, SkillActionBuffSheet, ActionBuffSheet, CharacterSheet, CharacterLevelSheet, EquipmentItemSetEffectSheet, StageSheet, StageWaveSheet, EnemySkillSheet)

```
public StageSimulatorSheetsV1(MaterialItemSheet materialItemSheet, SkillSheet
skillSheet, SkillBuffSheet skillBuffSheet, StatBuffSheet statBuffSheet,
SkillActionBuffSheet skillActionBuffSheet, ActionBuffSheet actionBuffSheet,
CharacterSheet characterSheet, CharacterLevelSheet characterLevelSheet,
EquipmentItemSetEffectSheet equipmentItemSetEffectSheet, StageSheet stageSheet,
StageWaveSheet stageWaveSheet, EnemySkillSheet enemySkillSheet)
```

Parameters

`materialItemSheet` MaterialItemSheet

`skillSheet` SkillSheet

`skillBuffSheet` SkillBuffSheet

`statBuffSheet` StatBuffSheet

`skillActionBuffSheet` SkillActionBuffSheet

`actionBuffSheet` ActionBuffSheet

`characterSheet` CharacterSheet

`characterLevelSheet` CharacterLevelSheet

`equipmentItemSetEffectSheet` EquipmentItemSetEffectSheet

`stageSheet` StageSheet

`stageWaveSheet` StageWaveSheet

`enemySkillSheet` EnemySkillSheet

Fields

EnemySkillSheet

```
public readonly EnemySkillSheet EnemySkillSheet
```

Field Value

EnemySkillSheet

StageSheet

```
public readonly StageSheet StageSheet
```

Field Value

StageSheet

StageWaveSheet

```
public readonly StageWaveSheet StageWaveSheet
```

Field Value

StageWaveSheet

Class StageSimulatorSheetsV100291

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
public class StageSimulatorSheetsV100291 : SimulatorSheetsV100291
```

Inheritance

object ← SimulatorSheetsV100291 ← StageSimulatorSheetsV100291

Inherited Members

SimulatorSheetsV100291.MaterialItemSheet , SimulatorSheetsV100291.SkillSheet , SimulatorSheetsV100291.SkillBuffSheet , SimulatorSheetsV100291.BuffSheet , SimulatorSheetsV100291.CharacterSheet , SimulatorSheetsV100291.CharacterLevelSheet , SimulatorSheetsV100291.EquipmentItemSetEffectSheet , SimulatorSheetsV100291.ToSimulatorSheetsV1() , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

StageSimulatorSheetsV100291(MaterialItemSheet, SkillSheet, SkillBuffSheet, BuffSheet, CharacterSheet, CharacterLevelSheet, EquipmentItemSetEffectSheet, StageSheet, StageWaveSheet, EnemySkillSheet)

```
public StageSimulatorSheetsV100291(MaterialItemSheet materialItemSheet, SkillSheet
skillSheet, SkillBuffSheet skillBuffSheet, BuffSheet buffSheet, CharacterSheet
characterSheet, CharacterLevelSheet characterLevelSheet, EquipmentItemSetEffectSheet
equipmentItemSetEffectSheet, StageSheet stageSheet, StageWaveSheet stageWaveSheet,
EnemySkillSheet enemySkillSheet)
```

Parameters

`materialItemSheet` MaterialItemSheet

`skillSheet` SkillSheet

`skillBuffSheet` SkillBuffSheet

`buffSheet` BuffSheet

`characterSheet` CharacterSheet

`characterLevelSheet` CharacterLevelSheet

`equipmentItemSetEffectSheet` EquipmentItemSetEffectSheet

`stageSheet` StageSheet

`stageWaveSheet` StageWaveSheet

`enemySkillSheet` EnemySkillSheet

Fields

EnemySkillSheet

```
public readonly EnemySkillSheet EnemySkillSheet
```

Field Value

EnemySkillSheet

StageSheet

```
public readonly StageSheet StageSheet
```

Field Value

StageSheet

StageWaveSheet

```
public readonly StageWaveSheet StageWaveSheet
```

Field Value

StageWaveSheet

Methods

ToStageSimulatorSheetsV1()

```
public StageSimulatorSheetsV1 ToStageSimulatorSheetsV1()
```

Returns

StageSimulatorSheetsV1

Class StageWaveSheet

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class StageWaveSheet : Sheet<int, StageWaveSheet.Row>, IDictionary<int,
StageWaveSheet.Row>, ICollection<KeyValuePair<int, StageWaveSheet.Row>>,
IEnumerable<KeyValuePair<int, StageWaveSheet.Row>>, IEnumerable, ISheet
```

Inheritance

object ← Sheet<int, StageWaveSheet.Row> ← StageWaveSheet

Implements

IDictionary<int, StageWaveSheet.Row>,
ICollection<KeyValuePair<int, StageWaveSheet.Row>>,
IEnumerable<KeyValuePair<int, StageWaveSheet.Row>>, IEnumerable, ISheet

Inherited Members

Sheet<int, StageWaveSheet.Row>.Name , Sheet<int, StageWaveSheet.Row>.OrderedList ,
Sheet<int, StageWaveSheet.Row>.First , Sheet<int, StageWaveSheet.Row>.Last ,
Sheet<int, StageWaveSheet.Row>.Keys , Sheet<int, StageWaveSheet.Row>.Values ,
Sheet<int, StageWaveSheet.Row>.Count , Sheet<int, StageWaveSheet.Row>.IsReadOnly ,
Sheet<int, StageWaveSheet.Row>.this[int] ,
Sheet<int, StageWaveSheet.Row>.Set(string, bool) ,
Sheet<int, StageWaveSheet.Row>.Set<T>(Sheet<int, T>, bool) ,
Sheet<int, StageWaveSheet.Row>.GetEnumerator() ,
Sheet<int, StageWaveSheet.Row>.TryGetValue(int, out StageWaveSheet.Row, bool) ,
Sheet<int, StageWaveSheet.Row>.AddRow(int, StageWaveSheet.Row) ,
Sheet<int, StageWaveSheet.Row>.Add(int, StageWaveSheet.Row) ,
Sheet<int, StageWaveSheet.Row>.ContainsKey(int) ,
Sheet<int, StageWaveSheet.Row>.Remove(int) ,
Sheet<int, StageWaveSheet.Row>.TryGetValue(int, out StageWaveSheet.Row) ,
Sheet<int, StageWaveSheet.Row>.Add(KeyValuePair<int, StageWaveSheet.Row>) ,
Sheet<int, StageWaveSheet.Row>.Clear() ,
Sheet<int, StageWaveSheet.Row>.Contains(KeyValuePair<int, StageWaveSheet.Row>) ,
Sheet<int, StageWaveSheet.Row>.CopyTo(KeyValuePair<int, StageWaveSheet.Row>[], int)
,
Sheet<int, StageWaveSheet.Row>.Remove(KeyValuePair<int, StageWaveSheet.Row>) ,

Sheet<int, StageWaveSheet.Row>.Serialize() , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

StageWaveSheet()

```
public StageWaveSheet()
```

Methods

AddRow(int, Row)

```
protected override void AddRow(int key, StageWaveSheet.Row value)
```

Parameters

`key` int

`value` StageWaveSheet.Row

Class StageWaveSheet.MonsterData

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class StageWaveSheet.MonsterData
```

Inheritance

object ← StageWaveSheet.MonsterData

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

MonsterData(int, int, int)

```
public MonsterData(int characterId, int level, int count)
```

Parameters

`characterId` int

`level` int

`count` int

Properties

CharacterId

```
public int CharacterId { get; }
```

Property Value

int

Count

```
public int Count { get; }
```

Property Value

int

Level

```
public int Level { get; }
```

Property Value

int

Class StageWaveSheet.Row

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class StageWaveSheet.Row : SheetRow<int>
```

Inheritance

object ← SheetRow<int> ← StageWaveSheet.Row

Derived

EventDungeonStageWaveSheet.Row

Inherited Members

SheetRow<int>.Validate() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Properties

HasBoss

```
public bool HasBoss { get; }
```

Property Value

bool

Key

```
public override int Key { get; }
```

Property Value

int

StageId

```
public int StageId { get; }
```

Property Value

int

TotalMonsterIds

```
public List<int> TotalMonsterIds { get; }
```

Property Value

List<int>

Waves

```
public List<StageWaveSheet.WaveData> Waves { get; }
```

Property Value

List<StageWaveSheet.WaveData>

Methods

CloneWithStageId(int)

Creates a shallow clone of this row with a new stage ID.

```
public StageWaveSheet.Row CloneWithStageId(int newStageId)
```

Parameters

`newStageId` int

The stage ID to assign to the cloned row.

Returns

StageWaveSheet.Row

A new StageWaveSheet.Row with copied wave data and the specified stage ID.

EndOfSheetInitialize()

```
public override void EndOfSheetInitialize()
```

Set(IReadOnlyList<string>)

```
public override void Set(IReadOnlyList<string> fields)
```

Parameters

`fields` IReadOnlyList<string>

Class StageWaveSheet.WaveData

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class StageWaveSheet.WaveData
```

Inheritance

object ← StageWaveSheet.WaveData

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

WaveData(int, List<MonsterData>, bool)

```
public WaveData(int number, List<StageWaveSheet.MonsterData> monsters, bool hasBoss)
```

Parameters

`number` int

`monsters` List<StageWaveSheet.MonsterData>

`hasBoss` bool

Properties

HasBoss

```
public bool HasBoss { get; }
```

Property Value

bool

Monsters

```
public List<StageWaveSheet.MonsterData> Monsters { get; }
```

Property Value

List<StageWaveSheet.MonsterData>

Number

```
public int Number { get; }
```

Property Value

int

Class StakeAchievementRewardSheet

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class StakeAchievementRewardSheet : Sheet<int,
StakeAchievementRewardSheet.Row>, IDictionary<int, StakeAchievementRewardSheet.Row>,
ICollection<KeyValuePair<int, StakeAchievementRewardSheet.Row>>,
IEnumerable<KeyValuePair<int, StakeAchievementRewardSheet.Row>>, IEnumerable,
IStakeRewardSheet, ISheet
```

Inheritance

object ← Sheet<int, StakeAchievementRewardSheet.Row> ← StakeAchievementRewardSheet

Implements

IDictionary<int, StakeAchievementRewardSheet.Row>,
ICollection<KeyValuePair<int, StakeAchievementRewardSheet.Row>>,
IEnumerable<KeyValuePair<int, StakeAchievementRewardSheet.Row>>,
IEnumerable, IStakeRewardSheet, ISheet

Inherited Members

Sheet<int, StakeAchievementRewardSheet.Row>.Name ,
Sheet<int, StakeAchievementRewardSheet.Row>.OrderedList ,
Sheet<int, StakeAchievementRewardSheet.Row>.First ,
Sheet<int, StakeAchievementRewardSheet.Row>.Last ,
Sheet<int, StakeAchievementRewardSheet.Row>.Keys ,
Sheet<int, StakeAchievementRewardSheet.Row>.Values ,
Sheet<int, StakeAchievementRewardSheet.Row>.Count ,
Sheet<int, StakeAchievementRewardSheet.Row>.IsReadOnly ,
Sheet<int, StakeAchievementRewardSheet.Row>.this[int] ,
Sheet<int, StakeAchievementRewardSheet.Row>.Set(string, bool) ,
Sheet<int, StakeAchievementRewardSheet.Row>.Set<T>(Sheet<int, T>, bool) ,
Sheet<int, StakeAchievementRewardSheet.Row>.GetEnumerator() ,
Sheet<int, StakeAchievementRewardSheet.Row>.TryGetValue(int, out StakeAchievementRewardSheet.Row, bool) ,
Sheet<int, StakeAchievementRewardSheet.Row>.AddRow(int, StakeAchievementRewardSheet.Row) ,

Sheet<int, StakeAchievementRewardSheet.Row>.Add(int, StakeAchievementRewardSheet.Row) , Sheet<int, StakeAchievementRewardSheet.Row>.ContainsKey(int) , Sheet<int, StakeAchievementRewardSheet.Row>.Remove(int) , Sheet<int, StakeAchievementRewardSheet.Row>.TryGetValue(int, out StakeAchievementRewardSheet.Row) , Sheet<int, StakeAchievementRewardSheet.Row>.Add(KeyValuePair<int, StakeAchievementRewardSheet.Row>) , Sheet<int, StakeAchievementRewardSheet.Row>.Clear() , Sheet<int, StakeAchievementRewardSheet.Row>.Contains(KeyValuePair<int, StakeAchievementRewardSheet.Row>) , Sheet<int, StakeAchievementRewardSheet.Row>.CopyTo(KeyValuePair<int, StakeAchievementRewardSheet.Row>[], int) , Sheet<int, StakeAchievementRewardSheet.Row>.Remove(KeyValuePair<int, StakeAchievementRewardSheet.Row>) , Sheet<int, StakeAchievementRewardSheet.Row>.Serialize() , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Extension Methods

SheetsExtensions.FindLevelByStakedAmount(IStakeRewardSheet, Address, FungibleAssetValue)

Constructors

StakeAchievementRewardSheet()

```
public StakeAchievementRewardSheet()
```

Properties

OrderedRows

```
public IReadOnlyList<IStakeRewardRow> OrderedRows { get; }
```

Property Value

IReadOnlyList<IStakeRewardRow>

Methods

AddRow(int, Row)

```
protected override void AddRow(int key, StakeAchievementRewardSheet.Row value)
```

Parameters

`key` int

`value` StakeAchievementRewardSheet.Row

FindStep(int, long)

```
public int FindStep(int level, long stakedBlockPeriod)
```

Parameters

`level` int

`stakedBlockPeriod` long

Returns

int

Class StakeAchievementRewardSheet.RewardInfo

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class StakeAchievementRewardSheet.RewardInfo
```

Inheritance

object ← StakeAchievementRewardSheet.RewardInfo

Inherited Members

object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

RewardInfo(Dictionary)

```
public RewardInfo(Dictionary dictionary)
```

Parameters

`dictionary` Dictionary

RewardInfo(int, int)

```
public RewardInfo(int itemId, int quantity)
```

Parameters

`itemId` int

quantity int

RewardInfo(params string[])

```
public RewardInfo(params string[] fields)
```

Parameters

fields string[]

Fields

ItemId

```
public readonly int ItemId
```

Field Value

int

Quantity

```
public readonly int Quantity
```

Field Value

int

Methods

Equals(RewardInfo)

```
protected bool Equals(StakeAchievementRewardSheet.RewardInfo other)
```

Parameters

`other` StakeAchievementRewardSheet.RewardInfo

Returns

bool

Equals(object)

Determines whether the specified object is equal to the current object.

```
public override bool Equals(object obj)
```

Parameters

`obj` object

The object to compare with the current object.

Returns

bool

true if the specified object is equal to the current object; otherwise, false.

GetHashCode()

Serves as the default hash function.

```
public override int GetHashCode()
```

Returns

int

A hash code for the current object.

Serialize()

```
public IValue Serialize()
```

Returns

IValue

Class StakeAchievementRewardSheet.Row

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class StakeAchievementRewardSheet.Row : SheetRow<int>, IStakeRewardRow
```

Inheritance

object ← SheetRow<int> ← StakeAchievementRewardSheet.Row

Implements

IStakeRewardRow

Inherited Members

SheetRow<int>.Validate() , SheetRow<int>.EndOfSheetInitialize() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Properties

Key

```
public override int Key { get; }
```

Property Value

int

Level

```
public int Level { get; }
```

Property Value

int

RequiredGold

```
public long RequiredGold { get; }
```

Property Value

long

Steps

```
public List<StakeAchievementRewardSheet.Step> Steps { get; }
```

Property Value

List<StakeAchievementRewardSheet.Step>

Methods

Set(IReadOnlyList<string>)

```
public override void Set(IReadOnlyList<string> fields)
```

Parameters

`fields` IReadOnlyList<string>

Class StakeAchievementRewardSheet.Step

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
public class StakeAchievementRewardSheet.Step
```

Inheritance

object ← StakeAchievementRewardSheet.Step

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

Step(long, long, List<RewardInfo>)

```
public Step(long requiredGold, long requiredBlockIndex,
List<StakeAchievementRewardSheet.RewardInfo> rewards)
```

Parameters

`requiredGold` long

`requiredBlockIndex` long

`rewards` List<StakeAchievementRewardSheet.RewardInfo>

Properties

RequiredBlockIndex

```
public long RequiredBlockIndex { get; }
```

Property Value

long

RequiredGold

```
public long RequiredGold { get; }
```

Property Value

long

Rewards

```
public List<StakeAchievementRewardSheet.RewardInfo> Rewards { get; }
```

Property Value

List<StakeAchievementRewardSheet.RewardInfo>

Class StakeActionPointCoefficientSheet

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class StakeActionPointCoefficientSheet : Sheet<int,
StakeActionPointCoefficientSheet.Row>, IDictionary<int,
StakeActionPointCoefficientSheet.Row>, ICollection<KeyValuePair<int,
StakeActionPointCoefficientSheet.Row>>, IEnumerable<KeyValuePair<int,
StakeActionPointCoefficientSheet.Row>>, IEnumerable, IStakeRewardSheet, ISheet
```

Inheritance

object ← Sheet<int, StakeActionPointCoefficientSheet.Row> ← StakeActionPointCoefficientSheet

Implements

IDictionary<int, StakeActionPointCoefficientSheet.Row>, ICollection<KeyValuePair<int, StakeActionPointCoefficientSheet.Row>>, IEnumerable<KeyValuePair<int, StakeActionPointCoefficientSheet.Row>>, IEnumerable, IStakeRewardSheet, ISheet

Inherited Members

Sheet<int, StakeActionPointCoefficientSheet.Row>.Name ,
Sheet<int, StakeActionPointCoefficientSheet.Row>.OrderedList ,
Sheet<int, StakeActionPointCoefficientSheet.Row>.First ,
Sheet<int, StakeActionPointCoefficientSheet.Row>.Last ,
Sheet<int, StakeActionPointCoefficientSheet.Row>.Keys ,
Sheet<int, StakeActionPointCoefficientSheet.Row>.Values ,
Sheet<int, StakeActionPointCoefficientSheet.Row>.Count ,
Sheet<int, StakeActionPointCoefficientSheet.Row>.IsReadOnly ,
Sheet<int, StakeActionPointCoefficientSheet.Row>.this[int] ,
Sheet<int, StakeActionPointCoefficientSheet.Row>.Set(string, bool) ,
Sheet<int, StakeActionPointCoefficientSheet.Row>.Set<T>(Sheet<int, T>, bool) ,
Sheet<int, StakeActionPointCoefficientSheet.Row>.GetEnumerator() ,
Sheet<int, StakeActionPointCoefficientSheet.Row>.TryGetValue(int, out StakeActionPointCoefficientSheet.Row, bool) ,
Sheet<int, StakeActionPointCoefficientSheet.Row>.AddRow(int, StakeActionPointCoefficientSheet.Row) ,

Sheet<int, StakeActionPointCoefficientSheet.Row>.Add(int, StakeActionPointCoefficientSheet.Row) , Sheet<int, StakeActionPointCoefficientSheet.Row>.ContainsKey(int) , Sheet<int, StakeActionPointCoefficientSheet.Row>.Remove(int) , Sheet<int, StakeActionPointCoefficientSheet.Row>.TryGetValue(int, out StakeActionPointCoefficientSheet.Row) , Sheet<int, StakeActionPointCoefficientSheet.Row>.Add(KeyValuePair<int, StakeActionPointCoefficientSheet.Row>) , Sheet<int, StakeActionPointCoefficientSheet.Row>.Clear() , Sheet<int, StakeActionPointCoefficientSheet.Row>.Contains(KeyValuePair<int, StakeActionPointCoefficientSheet.Row>) , Sheet<int, StakeActionPointCoefficientSheet.Row>.CopyTo(KeyValuePair<int, StakeActionPointCoefficientSheet.Row>[], int) , Sheet<int, StakeActionPointCoefficientSheet.Row>.Remove(KeyValuePair<int, StakeActionPointCoefficientSheet.Row>) , Sheet<int, StakeActionPointCoefficientSheet.Row>.Serialize() , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Extension Methods

SheetsExtensions.FindLevelByStakedAmount(IStakeRewardSheet, Address, FungibleAssetValue) , SheetsExtensions.GetActionPointByStaking(StakeActionPointCoefficientSheet, int, int, int)

Constructors

StakeActionPointCoefficientSheet()

```
public StakeActionPointCoefficientSheet()
```

Properties

OrderedRows

```
public IReadOnlyList<IStakeRewardRow> OrderedRows { get; }
```

Property Value

IReadOnlyList<IStakeRewardRow>

Methods

AddRow(int, Row)

```
protected override void AddRow(int key, StakeActionPointCoefficientSheet.Row value)
```

Parameters

`key` int

`value` StakeActionPointCoefficientSheet.Row

Class StakeActionPointCoefficientSheet.Row

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
public class StakeActionPointCoefficientSheet.Row : SheetRow<int>, IStakeRewardRow
```

Inheritance

object ← SheetRow<int> ← StakeActionPointCoefficientSheet.Row

Implements

IStakeRewardRow

Inherited Members

SheetRow<int>.Validate() , SheetRow<int>.EndOfSheetInitialize() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Properties

Coefficient

```
public int Coefficient { get; }
```

Property Value

int

Key

```
public override int Key { get; }
```

Property Value

int

Level

```
public int Level { get; }
```

Property Value

int

RequiredGold

```
public long RequiredGold { get; }
```

Property Value

long

Methods

Set(IReadOnlyList<string>)

```
public override void Set(IReadOnlyList<string> fields)
```

Parameters

`fields` IReadOnlyList<string>

Class StakeRegularFixedRewardSheet

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

This sheet is used for setting the regular rewards for staking. The difference between this sheet and StakeRegularRewardSheet is that the StakeRegularFixedRewardSheet.RewardInfo of this sheet has a fixed count of reward.

CSV column order: level, required_gold, item_id, count[, tradable]

The optional `tradable` column controls whether material items are created as tradable (TradableMaterial) or non-tradable (Material). Omitting the column defaults to `true`, preserving backward compatibility with existing CSV data.

```
[Serializable]
public class StakeRegularFixedRewardSheet : Sheet<int,
StakeRegularFixedRewardSheet.Row>, IDictionary<int,
StakeRegularFixedRewardSheet.Row>, ICollection<KeyValuePair<int,
StakeRegularFixedRewardSheet.Row>>, IEnumerable<KeyValuePair<int,
StakeRegularFixedRewardSheet.Row>>, IEnumerable, IStakeRewardSheet, ISheet
```

Inheritance

object ← Sheet<int, StakeRegularFixedRewardSheet.Row> ← StakeRegularFixedRewardSheet

Implements

IDictionary<int, StakeRegularFixedRewardSheet.Row>, ICollection<KeyValuePair<int, StakeRegularFixedRewardSheet.Row>>, IEnumerable<KeyValuePair<int, StakeRegularFixedRewardSheet.Row>>, IEnumerable, IStakeRewardSheet, ISheet

Inherited Members

Sheet<int, StakeRegularFixedRewardSheet.Row>.Name ,
Sheet<int, StakeRegularFixedRewardSheet.Row>.OrderedList ,
Sheet<int, StakeRegularFixedRewardSheet.Row>.First ,
Sheet<int, StakeRegularFixedRewardSheet.Row>.Last ,
Sheet<int, StakeRegularFixedRewardSheet.Row>.Keys ,
Sheet<int, StakeRegularFixedRewardSheet.Row>.Values ,
Sheet<int, StakeRegularFixedRewardSheet.Row>.Count ,
Sheet<int, StakeRegularFixedRewardSheet.Row>.IsReadOnly ,

Sheet<int, StakeRegularFixedRewardSheet.Row>.this[int] ,
Sheet<int, StakeRegularFixedRewardSheet.Row>.Set(string, bool) ,
Sheet<int, StakeRegularFixedRewardSheet.Row>.Set<T>(Sheet<int, T>, bool) ,
Sheet<int, StakeRegularFixedRewardSheet.Row>.GetEnumerator() ,
Sheet<int, StakeRegularFixedRewardSheet.Row>.TryGetValue(int, out StakeRegularFixedRewardSheet.Row, bool) ,
Sheet<int, StakeRegularFixedRewardSheet.Row>.AddRow(int, StakeRegularFixedRewardSheet.Row) ,
Sheet<int, StakeRegularFixedRewardSheet.Row>.Add(int, StakeRegularFixedRewardSheet.Row) ,
Sheet<int, StakeRegularFixedRewardSheet.Row>.ContainsKey(int) ,
Sheet<int, StakeRegularFixedRewardSheet.Row>.Remove(int) ,
Sheet<int, StakeRegularFixedRewardSheet.Row>.TryGetValue(int, out StakeRegularFixedRewardSheet.Row) ,
Sheet<int, StakeRegularFixedRewardSheet.Row>.Add(KeyValuePair<int, StakeRegularFixedRewardSheet.Row>) ,
Sheet<int, StakeRegularFixedRewardSheet.Row>.Clear() ,
Sheet<int, StakeRegularFixedRewardSheet.Row>.Contains(KeyValuePair<int, StakeRegularFixedRewardSheet.Row>) ,
Sheet<int, StakeRegularFixedRewardSheet.Row>.CopyTo(KeyValuePair<int, StakeRegularFixedRewardSheet.Row>[], int) ,
Sheet<int, StakeRegularFixedRewardSheet.Row>.Remove(KeyValuePair<int, StakeRegularFixedRewardSheet.Row>) ,
Sheet<int, StakeRegularFixedRewardSheet.Row>.Serialize() , object.Equals(object) ,
object.Equals(object, object) , object.GetHashCode() , object.GetType() ,
object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Extension Methods
SheetsExtensions.FindLevelByStakedAmount(IStakeRewardSheet, Address, FungibleAssetValue)

Constructors

StakeRegularFixedRewardSheet()

```
public StakeRegularFixedRewardSheet()
```

Properties

OrderedRows

```
public IReadOnlyList<IStakeRewardRow> OrderedRows { get; }
```

Property Value

IReadOnlyList<IStakeRewardRow>

Methods

AddRow(int, Row)

```
protected override void AddRow(int key, StakeRegularFixedRewardSheet.Row value)
```

Parameters

`key` int

`value` StakeRegularFixedRewardSheet.Row

Class StakeRegularFixedRewardSheet.RewardInfo

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class StakeRegularFixedRewardSheet.RewardInfo
```

Inheritance

object ← StakeRegularFixedRewardSheet.RewardInfo

Inherited Members

object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

RewardInfo(int, int, bool)

```
public RewardInfo(int itemId, int count, bool tradable = true)
```

Parameters

`itemId` int

The item ID to reward.

`count` int

The fixed count of items to reward.

`tradable` bool

Whether the rewarded material item should be tradable. Defaults to `true`.

RewardInfo(params string[])

Initializes a StakeRegularFixedRewardSheet.RewardInfo from CSV field tokens. Expected order: item_id, count[, tradable].

```
public RewardInfo(params string[] fields)
```

Parameters

`fields` string[]

Fields

Count

```
public readonly int Count
```

Field Value

int

ItemId

```
public readonly int ItemId
```

Field Value

int

Tradable

Whether the rewarded item is tradable. If `true`, materials are created as TradableMaterial; otherwise they are created as non-tradable Material. Defaults to `true` when the column is omitted in the CSV.

```
public readonly bool Tradable
```

Field Value

bool

Methods

Equals(RewardInfo)

```
protected bool Equals(StakeRegularFixedRewardSheet.RewardInfo other)
```

Parameters

`other` StakeRegularFixedRewardSheet.RewardInfo

Returns

bool

Equals(object)

Determines whether the specified object is equal to the current object.

```
public override bool Equals(object obj)
```

Parameters

`obj` object

The object to compare with the current object.

Returns

bool

true if the specified object is equal to the current object; otherwise, false.

GetHashCode()

Serves as the default hash function.

```
public override int GetHashCode()
```

Returns

int

A hash code for the current object.

Class StakeRegularFixedRewardSheet.Row

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class StakeRegularFixedRewardSheet.Row : SheetRow<int>, IStakeRewardRow
```

Inheritance

object ← SheetRow<int> ← StakeRegularFixedRewardSheet.Row

Implements

IStakeRewardRow

Inherited Members

SheetRow<int>.Validate() , SheetRow<int>.EndOfSheetInitialize() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Properties

Key

```
public override int Key { get; }
```

Property Value

int

Level

```
public int Level { get; }
```

Property Value

int

RequiredGold

```
public long RequiredGold { get; }
```

Property Value

long

Rewards

```
public List<StakeRegularFixedRewardSheet.RewardInfo> Rewards { get; }
```

Property Value

List<StakeRegularFixedRewardSheet.RewardInfo>

Methods

Set(IReadOnlyList<string>)

```
public override void Set(IReadOnlyList<string> fields)
```

Parameters

`fields` IReadOnlyList<string>

Class StakeRegularRewardSheet

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

This sheet is used for setting the regular rewards for staking. The difference between this sheet and StakeRegularFixedRewardSheet is that the StakeRegularRewardSheet.RewardInfo of this sheet has a rate and a reward type. The count of reward is calculated by the rate and the amount of staked currency.

```
[Serializable]
public class StakeRegularRewardSheet : Sheet<int, StakeRegularRewardSheet.Row>,
IDictionary<int, StakeRegularRewardSheet.Row>, ICollection<KeyValuePair<int,
StakeRegularRewardSheet.Row>>, IEnumerable<KeyValuePair<int,
StakeRegularRewardSheet.Row>>, IEnumerable, IStakeRewardSheet, ISheet
```

Inheritance

object ← Sheet<int, StakeRegularRewardSheet.Row> ← StakeRegularRewardSheet

Implements

IDictionary<int, StakeRegularRewardSheet.Row>,
ICollection<KeyValuePair<int, StakeRegularRewardSheet.Row>>,
IEnumerable<KeyValuePair<int, StakeRegularRewardSheet.Row>>, IEnumerable,
IStakeRewardSheet, ISheet

Inherited Members

Sheet<int, StakeRegularRewardSheet.Row>.Name ,
Sheet<int, StakeRegularRewardSheet.Row>.OrderedList ,
Sheet<int, StakeRegularRewardSheet.Row>.First ,
Sheet<int, StakeRegularRewardSheet.Row>.Last ,
Sheet<int, StakeRegularRewardSheet.Row>.Keys ,
Sheet<int, StakeRegularRewardSheet.Row>.Values ,
Sheet<int, StakeRegularRewardSheet.Row>.Count ,
Sheet<int, StakeRegularRewardSheet.Row>.IsReadOnly ,
Sheet<int, StakeRegularRewardSheet.Row>.this[int] ,
Sheet<int, StakeRegularRewardSheet.Row>.Set(string, bool) ,
Sheet<int, StakeRegularRewardSheet.Row>.Set<T>(Sheet<int, T>, bool) ,
Sheet<int, StakeRegularRewardSheet.Row>.GetEnumerator() ,
Sheet<int, StakeRegularRewardSheet.Row>.TryGetValue(int, out
StakeRegularRewardSheet.Row, bool) ,

Sheet<int, StakeRegularRewardSheet.Row>.AddRow(int, StakeRegularRewardSheet.Row) ,
Sheet<int, StakeRegularRewardSheet.Row>.Add(int, StakeRegularRewardSheet.Row) ,
Sheet<int, StakeRegularRewardSheet.Row>.ContainsKey(int) ,
Sheet<int, StakeRegularRewardSheet.Row>.Remove(int) ,
Sheet<int, StakeRegularRewardSheet.Row>.TryGetValue(int, out StakeRegularRewardSheet.Row) ,
Sheet<int, StakeRegularRewardSheet.Row>.Add(KeyValuePair<int, StakeRegularRewardSheet.Row>) ,
Sheet<int, StakeRegularRewardSheet.Row>.Clear() ,
Sheet<int, StakeRegularRewardSheet.Row>.Contains(KeyValuePair<int, StakeRegularRewardSheet.Row>) ,
Sheet<int, StakeRegularRewardSheet.Row>.CopyTo(KeyValuePair<int, StakeRegularRewardSheet.Row>[], int) ,
Sheet<int, StakeRegularRewardSheet.Row>.Remove(KeyValuePair<int, StakeRegularRewardSheet.Row>) ,
Sheet<int, StakeRegularRewardSheet.Row>.Serialize() , object.Equals(object) ,
object.Equals(object, object) , object.GetHashCode() , object.GetType() ,
object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Extension Methods
SheetsExtensions.FindLevelByStakedAmount(IStakeRewardSheet, Address, FungibleAssetValue)

Constructors

StakeRegularRewardSheet()

```
public StakeRegularRewardSheet()
```

Properties

OrderedRows

```
public IReadOnlyList<IStakeRewardRow> OrderedRows { get; }
```

Property Value

IReadOnlyList<IStakeRewardRow>

Methods

AddRow(int, Row)

```
protected override void AddRow(int key, StakeRegularRewardSheet.Row value)
```

Parameters

`key` int

`value` StakeRegularRewardSheet.Row

Class StakeRegularRewardSheet.RewardInfo

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class StakeRegularRewardSheet.RewardInfo
```

Inheritance

object ← StakeRegularRewardSheet.RewardInfo

Inherited Members

object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

RewardInfo(int, int, StakeRewardType, string, int?, decimal)

```
public RewardInfo(int itemId, int rate = 0, StakeRegularRewardSheet.StakeRewardType
type = StakeRewardType.Item, string currencyTicker = null, int?
currencyDecimalPlaces = null, decimal decimalRate = 0)
```

Parameters

`itemId` int

`rate` int

`type` StakeRegularRewardSheet.StakeRewardType

`currencyTicker` string

`currencyDecimalPlaces` int?

`decimalRate` decimal

RewardInfo(params string[])

```
public RewardInfo(params string[] fields)
```

Parameters

`fields` string[]

Fields

CurrencyDecimalPlaces

The decimal places of currency. This field is only used when Type is Currency.

```
public readonly int? CurrencyDecimalPlaces
```

Field Value

int?

CurrencyTicker

The ticker of currency. This field is only used when Type is Currency.

```
public readonly string CurrencyTicker
```

Field Value

string

DecimalRate

The decimal rate of reward. This field is used from `V3`(in blockchain state) or later. This field is set to `Rate` when `ClaimStakeReward.V2` or earlier.

```
public readonly decimal DecimalRate
```

Field Value

decimal

ItemId

```
public readonly int ItemId
```

Field Value

int

Rate

The rate of reward.

```
[Obsolete("This field is used from `ClaimStakeReward.V2` or earlier. Use
`DecimalRate` instead. Deprecated this field because we need to support
decimal rate.")]
public readonly int Rate
```

Field Value

int

Tradable

```
public readonly bool Tradable
```

Field Value

bool

Type

```
public readonly StakeRegularRewardSheet.StakeRewardType Type
```

Field Value

StakeRegularRewardSheet.StakeRewardType

Methods

Equals(RewardInfo)

```
protected bool Equals(StakeRegularRewardSheet.RewardInfo other)
```

Parameters

`other` StakeRegularRewardSheet.RewardInfo

Returns

bool

Equals(object)

Determines whether the specified object is equal to the current object.

```
public override bool Equals(object obj)
```

Parameters

`obj` object

The object to compare with the current object.

Returns

bool

true if the specified object is equal to the current object; otherwise, false.

GetHashCode()

Serves as the default hash function.

```
public override int GetHashCode()
```

Returns

int

A hash code for the current object.

Class StakeRegularRewardSheet.Row

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class StakeRegularRewardSheet.Row : SheetRow<int>, IStakeRewardRow
```

Inheritance

object ← SheetRow<int> ← StakeRegularRewardSheet.Row

Implements

IStakeRewardRow

Inherited Members

SheetRow<int>.Validate() , SheetRow<int>.EndOfSheetInitialize() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Properties

Key

```
public override int Key { get; }
```

Property Value

int

Level

```
public int Level { get; }
```

Property Value

int

RequiredGold

```
public long RequiredGold { get; }
```

Property Value

long

Rewards

```
public List<StakeRegularRewardSheet.RewardInfo> Rewards { get; }
```

Property Value

List<StakeRegularRewardSheet.RewardInfo>

Methods

Set(IReadOnlyList<string>)

```
public override void Set(IReadOnlyList<string> fields)
```

Parameters

`fields` IReadOnlyList<string>

Enum StakeRegularRewardSheet.StakeRewardType

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

The reward type of stake. Do not use the whole element of this enum at once in actions. (e.g., count, loop) Do versioning when you change or remove the elements of this enum.

```
public enum StakeRegularRewardSheet.StakeRewardType
```

Extension Methods
StateExtensions.Serialize(Enum)

Fields

`Currency = 2`

`Item = 0`

`Rune = 1`

Class StatBuffSheet

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class StatBuffSheet : Sheet<int, StatBuffSheet.Row>, IDictionary<int,
StatBuffSheet.Row>, ICollection<KeyValuePair<int, StatBuffSheet.Row>>,
IEnumerable<KeyValuePair<int, StatBuffSheet.Row>>, IEnumerable, ISheet
```

Inheritance

object ← Sheet<int, StatBuffSheet.Row> ← StatBuffSheet

Implements

IDictionary<int, StatBuffSheet.Row>,
ICollection<KeyValuePair<int, StatBuffSheet.Row>>,
IEnumerable<KeyValuePair<int, StatBuffSheet.Row>>, IEnumerable, ISheet

Inherited Members

Sheet<int, StatBuffSheet.Row>.Name , Sheet<int, StatBuffSheet.Row>.OrderedList ,
Sheet<int, StatBuffSheet.Row>.First , Sheet<int, StatBuffSheet.Row>.Last ,
Sheet<int, StatBuffSheet.Row>.Keys , Sheet<int, StatBuffSheet.Row>.Values ,
Sheet<int, StatBuffSheet.Row>.Count , Sheet<int, StatBuffSheet.Row>.IsReadOnly ,
Sheet<int, StatBuffSheet.Row>.this[int] , Sheet<int, StatBuffSheet.Row>.Set(string, bool) ,
Sheet<int, StatBuffSheet.Row>.Set<T>(Sheet<int, T>, bool) ,
Sheet<int, StatBuffSheet.Row>.GetEnumerator() ,
Sheet<int, StatBuffSheet.Row>.TryGetValue(int, out StatBuffSheet.Row, bool) ,
Sheet<int, StatBuffSheet.Row>.AddRow(int, StatBuffSheet.Row) ,
Sheet<int, StatBuffSheet.Row>.Add(int, StatBuffSheet.Row) ,
Sheet<int, StatBuffSheet.Row>.ContainsKey(int) ,
Sheet<int, StatBuffSheet.Row>.Remove(int) ,
Sheet<int, StatBuffSheet.Row>.TryGetValue(int, out StatBuffSheet.Row) ,
Sheet<int, StatBuffSheet.Row>.Add(KeyValuePair<int, StatBuffSheet.Row>) ,
Sheet<int, StatBuffSheet.Row>.Clear() ,
Sheet<int, StatBuffSheet.Row>.Contains(KeyValuePair<int, StatBuffSheet.Row>) ,
Sheet<int, StatBuffSheet.Row>.CopyTo(KeyValuePair<int, StatBuffSheet.Row>[], int) ,
Sheet<int, StatBuffSheet.Row>.Remove(KeyValuePair<int, StatBuffSheet.Row>) ,
Sheet<int, StatBuffSheet.Row>.Serialize() , object.Equals(object) ,

object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

StatBuffSheet()

```
public StatBuffSheet()
```

Class StatBuffSheet.Row

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class StatBuffSheet.Row : SheetRow<int>
```

Inheritance

object ← SheetRow<int> ← StatBuffSheet.Row

Inherited Members

SheetRow<int>.Validate() , SheetRow<int>.EndOfSheetInitialize() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Properties

Chance

100: 100%

```
public int Chance { get; }
```

Property Value

int

Duration

Turn count.

```
public int Duration { get; }
```

Property Value

int

GroupId

GroupId는 같은 GroupId를 공유하는 버프 중 가장 작은 값으로 함. Id가 다르더라도 GroupId가 같으면, 서로 덮어씀. 이때의 우선순위는 Id가 큰 것이 남는 것으로 구현함.

```
public int GroupId { get; }
```

Property Value

int

Id

```
public int Id { get; }
```

Property Value

int

IsEnhanceable

```
public bool IsEnhanceable { get; }
```

Property Value

bool

Key

```
public override int Key { get; }
```

Property Value

int

MaxStack

```
public int MaxStack { get; }
```

Property Value

int

OperationType

```
public StatModifier.OperationType OperationType { get; }
```

Property Value

StatModifier.OperationType

StatType

```
public StatType StatType { get; }
```

Property Value

StatType

TargetType

```
public SkillTargetType TargetType { get; }
```

Property Value

SkillTargetType

Value

```
public long Value { get; }
```

Property Value

long

Methods

Set(IReadOnlyList<string>)

```
public override void Set(IReadOnlyList<string> fields)
```

Parameters

`fields` IReadOnlyList<string>

Class SweepRequiredCPSheet

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
public class SweepRequiredCPSheet : Sheet<int, SweepRequiredCPSheet.Row>,
IDictionary<int, SweepRequiredCPSheet.Row>, ICollection<KeyValuePair<int,
SweepRequiredCPSheet.Row>>, IEnumerable<KeyValuePair<int,
SweepRequiredCPSheet.Row>>, IEnumerable, ISheet
```

Inheritance

object ← Sheet<int, SweepRequiredCPSheet.Row> ← SweepRequiredCPSheet

Implements

IDictionary<int, SweepRequiredCPSheet.Row>,
ICollection<KeyValuePair<int, SweepRequiredCPSheet.Row>>,
IEnumerable<KeyValuePair<int, SweepRequiredCPSheet.Row>>, IEnumerable,
ISheet

Inherited Members

Sheet<int, SweepRequiredCPSheet.Row>.Name ,
Sheet<int, SweepRequiredCPSheet.Row>.OrderedList ,
Sheet<int, SweepRequiredCPSheet.Row>.First ,
Sheet<int, SweepRequiredCPSheet.Row>.Last ,
Sheet<int, SweepRequiredCPSheet.Row>.Keys ,
Sheet<int, SweepRequiredCPSheet.Row>.Values ,
Sheet<int, SweepRequiredCPSheet.Row>.Count ,
Sheet<int, SweepRequiredCPSheet.Row>.IsReadOnly ,
Sheet<int, SweepRequiredCPSheet.Row>.this[int] ,
Sheet<int, SweepRequiredCPSheet.Row>.Set(string, bool) ,
Sheet<int, SweepRequiredCPSheet.Row>.Set<T>(Sheet<int, T>, bool) ,
Sheet<int, SweepRequiredCPSheet.Row>.GetEnumerator() ,
Sheet<int, SweepRequiredCPSheet.Row>.TryGetValue(int, out SweepRequiredCPSheet.Row,
bool) ,
Sheet<int, SweepRequiredCPSheet.Row>.AddRow(int, SweepRequiredCPSheet.Row) ,
Sheet<int, SweepRequiredCPSheet.Row>.Add(int, SweepRequiredCPSheet.Row) ,
Sheet<int, SweepRequiredCPSheet.Row>.ContainsKey(int) ,
Sheet<int, SweepRequiredCPSheet.Row>.Remove(int) ,

Sheet<int, SweepRequiredCPSheet.Row>.TryGetValue(int, out SweepRequiredCPSheet.Row) , Sheet<int, SweepRequiredCPSheet.Row>.Add(KeyValuePair<int, SweepRequiredCPSheet.Row>) , Sheet<int, SweepRequiredCPSheet.Row>.Clear() , Sheet<int, SweepRequiredCPSheet.Row>.Contains(KeyValuePair<int, SweepRequiredCPSheet.Row>) , Sheet<int, SweepRequiredCPSheet.Row>.CopyTo(KeyValuePair<int, SweepRequiredCPSheet.Row>[], int) , Sheet<int, SweepRequiredCPSheet.Row>.Remove(KeyValuePair<int, SweepRequiredCPSheet.Row>) , Sheet<int, SweepRequiredCPSheet.Row>.Serialize() , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

SweepRequiredCPSheet()

```
public SweepRequiredCPSheet()
```

Class SweepRequiredCPSheet.Row

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
public class SweepRequiredCPSheet.Row : SheetRow<int>
```

Inheritance

object ← SheetRow<int> ← SweepRequiredCPSheet.Row

Inherited Members

SheetRow<int>.Validate() , SheetRow<int>.EndOfSheetInitialize() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Properties

Key

```
public override int Key { get; }
```

Property Value

int

RequiredCP

```
public long RequiredCP { get; }
```

Property Value

long

StageId

```
public int StageId { get; }
```

Property Value

int

Methods

Set(IReadOnlyList<string>)

```
public override void Set(IReadOnlyList<string> fields)
```

Parameters

`fields` IReadOnlyList<string>

Class SynthesizeSheet

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

Represents a SynthesizeSheet.

```
[Serializable]
public class SynthesizeSheet : Sheet<int, SynthesizeSheet.Row>, IDictionary<int,
SynthesizeSheet.Row>, ICollection<KeyValuePair<int, SynthesizeSheet.Row>>,
IEnumerable<KeyValuePair<int, SynthesizeSheet.Row>>, IEnumerable, ISheet
```

Inheritance

object ← Sheet<int, SynthesizeSheet.Row> ← SynthesizeSheet

Implements

IDictionary<int, SynthesizeSheet.Row>,
ICollection<KeyValuePair<int, SynthesizeSheet.Row>>,
IEnumerable<KeyValuePair<int, SynthesizeSheet.Row>>, IEnumerable, ISheet

Inherited Members

Sheet<int, SynthesizeSheet.Row>.Name , Sheet<int, SynthesizeSheet.Row>.OrderedList ,
Sheet<int, SynthesizeSheet.Row>.First , Sheet<int, SynthesizeSheet.Row>.Last ,
Sheet<int, SynthesizeSheet.Row>.Keys , Sheet<int, SynthesizeSheet.Row>.Values ,
Sheet<int, SynthesizeSheet.Row>.Count , Sheet<int, SynthesizeSheet.Row>.IsReadOnly ,
Sheet<int, SynthesizeSheet.Row>.this[int] ,
Sheet<int, SynthesizeSheet.Row>.Set(string, bool) ,
Sheet<int, SynthesizeSheet.Row>.Set<T>(Sheet<int, T>, bool) ,
Sheet<int, SynthesizeSheet.Row>.GetEnumerator() ,
Sheet<int, SynthesizeSheet.Row>.TryGetValue(int, out SynthesizeSheet.Row, bool) ,
Sheet<int, SynthesizeSheet.Row>.AddRow(int, SynthesizeSheet.Row) ,
Sheet<int, SynthesizeSheet.Row>.Add(int, SynthesizeSheet.Row) ,
Sheet<int, SynthesizeSheet.Row>.ContainsKey(int) ,
Sheet<int, SynthesizeSheet.Row>.Remove(int) ,
Sheet<int, SynthesizeSheet.Row>.TryGetValue(int, out SynthesizeSheet.Row) ,
Sheet<int, SynthesizeSheet.Row>.Add(KeyValuePair<int, SynthesizeSheet.Row>) ,
Sheet<int, SynthesizeSheet.Row>.Clear() ,
Sheet<int, SynthesizeSheet.Row>.Contains(KeyValuePair<int, SynthesizeSheet.Row>) ,
Sheet<int, SynthesizeSheet.Row>.CopyTo(KeyValuePair<int, SynthesizeSheet.Row>[], int) ,
Sheet<int, SynthesizeSheet.Row>.Remove(KeyValuePair<int, SynthesizeSheet.Row>) ,

Sheet<int, SynthesizeSheet.Row>.Serialize() , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

SynthesizeSheet()

```
public SynthesizeSheet()
```

Fields

SucceedRateMax

```
public const int SucceedRateMax = 10000
```

Field Value

int

SucceedRateMin

```
public const int SucceedRateMin = 0
```

Field Value

int

Methods

AddRow(int, Row)

```
protected override void AddRow(int key, SynthesizeSheet.Row value)
```

Parameters

`key` int

`value` SynthesizeSheet.Row

Class SynthesizeSheet.Row

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class SynthesizeSheet.Row : SheetRow<int>
```

Inheritance

object ← SheetRow<int> ← SynthesizeSheet.Row

Inherited Members

SheetRow<int>.Validate() , SheetRow<int>.EndOfSheetInitialize() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Properties

GradeId

```
public int GradeId { get; }
```

Property Value

int

Key

```
public override int Key { get; }
```

Property Value

int

RequiredCountDict

```
public Dictionary<ItemSubType, SynthesizeSheet.SynthesizeData> RequiredCountDict {
get; }
```

Property Value

Dictionary<ItemSubType, SynthesizeSheet.SynthesizeData>

Methods

Set(IReadOnlyList<string>)

```
public override void Set(IReadOnlyList<string> fields)
```

Parameters

`fields` IReadOnlyList<string>

Struct SynthesizeSheet.SynthesizeData

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

synthesize data for each subtype

```
public struct SynthesizeSheet.SynthesizeData
```

Inherited Members
ValueType.Equals(object) , ValueType.GetHashCode() , ValueType.ToString() , object.Equals(object, object) , object.GetType() , object.ReferenceEquals(object, object)

Fields

RequiredCount

```
public int RequiredCount
```

Field Value

int

SucceedRate

```
public int SucceedRate
```

Field Value

int

Class SynthesizeWeightSheet

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

Represents a SynthesizeWeightSheet.

```
[Serializable]
public class SynthesizeWeightSheet : Sheet<int, SynthesizeWeightSheet.Row>,
IDictionary<int, SynthesizeWeightSheet.Row>, ICollection<KeyValuePair<int,
SynthesizeWeightSheet.Row>>, IEnumerable<KeyValuePair<int,
SynthesizeWeightSheet.Row>>, IEnumerable, ISheet
```

Inheritance

object ← Sheet<int, SynthesizeWeightSheet.Row> ← SynthesizeWeightSheet

Implements

IDictionary<int, SynthesizeWeightSheet.Row>,
ICollection<KeyValuePair<int, SynthesizeWeightSheet.Row>>,
IEnumerable<KeyValuePair<int, SynthesizeWeightSheet.Row>>, IEnumerable,
ISheet

Inherited Members

Sheet<int, SynthesizeWeightSheet.Row>.Name ,
Sheet<int, SynthesizeWeightSheet.Row>.OrderedList ,
Sheet<int, SynthesizeWeightSheet.Row>.First ,
Sheet<int, SynthesizeWeightSheet.Row>.Last ,
Sheet<int, SynthesizeWeightSheet.Row>.Keys ,
Sheet<int, SynthesizeWeightSheet.Row>.Values ,
Sheet<int, SynthesizeWeightSheet.Row>.Count ,
Sheet<int, SynthesizeWeightSheet.Row>.IsReadOnly ,
Sheet<int, SynthesizeWeightSheet.Row>.this[int] ,
Sheet<int, SynthesizeWeightSheet.Row>.Set(string, bool) ,
Sheet<int, SynthesizeWeightSheet.Row>.Set<T>(Sheet<int, T>, bool) ,
Sheet<int, SynthesizeWeightSheet.Row>.GetEnumerator() ,
Sheet<int, SynthesizeWeightSheet.Row>.TryGetValue(int, out SynthesizeWeightSheet.Row,
bool) ,
Sheet<int, SynthesizeWeightSheet.Row>.AddRow(int, SynthesizeWeightSheet.Row) ,
Sheet<int, SynthesizeWeightSheet.Row>.Add(int, SynthesizeWeightSheet.Row) ,
Sheet<int, SynthesizeWeightSheet.Row>.ContainsKey(int) ,

Sheet<int, SynthesizeWeightSheet.Row>.Remove(int) ,
Sheet<int, SynthesizeWeightSheet.Row>.TryGetValue(int, out SynthesizeWeightSheet.Row)
,
Sheet<int, SynthesizeWeightSheet.Row>.Add(KeyValuePair<int,
SynthesizeWeightSheet.Row>) ,
Sheet<int, SynthesizeWeightSheet.Row>.Clear() ,
Sheet<int, SynthesizeWeightSheet.Row>.Contains(KeyValuePair<int,
SynthesizeWeightSheet.Row>) ,
Sheet<int, SynthesizeWeightSheet.Row>.CopyTo(KeyValuePair<int,
SynthesizeWeightSheet.Row>[], int) ,
Sheet<int, SynthesizeWeightSheet.Row>.Remove(KeyValuePair<int,
SynthesizeWeightSheet.Row>) ,
Sheet<int, SynthesizeWeightSheet.Row>.Serialize() , object.Equals(object) ,
object.Equals(object, object) , object.GetHashCode() , object.GetType() ,
object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

SynthesizeWeightSheet()

Creates an empty sheet.

```
public SynthesizeWeightSheet()
```

Fields

DefaultWeight

Weight of an item this sheet does not list, or lists with an unreadable weight.

```
public const int DefaultWeight = 0
```

Field Value

int

Remarks

This used to be 10000, which meant a result pool held every item of its grade and sub type until someone remembered to write a 0 here. An item sheet row was therefore enough to put a costume into circulation, which is how four grade 8 costumes were handed out in 2026-08. A pool is spelled out now instead: what this sheet does not list is not drawn.

Class SynthesizeWeightSheet.Row

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class SynthesizeWeightSheet.Row : SheetRow<int>
```

Inheritance

object ← SheetRow<int> ← SynthesizeWeightSheet.Row

Inherited Members

SheetRow<int>.Validate() , SheetRow<int>.EndOfSheetInitialize() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Properties

ItemId

```
public int ItemId { get; }
```

Property Value

int

Key

```
public override int Key { get; }
```

Property Value

int

Weight

```
public int Weight { get; }
```

Property Value

int

Methods

Set(IReadOnlyList<string>)

```
public override void Set(IReadOnlyList<string> fields)
```

Parameters

`fields` IReadOnlyList<string>

Class TableExtensions

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
public static class TableExtensions
```

Inheritance

object ← TableExtensions

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Methods

ParseBigInteger(string)

```
public static BigInteger ParseBigInteger(string value)
```

Parameters

`value` string

Returns

BigInteger

ParseBool(string, bool)

```
public static bool ParseBool(string value, bool defaultValue)
```

Parameters

`value` string

`defaultValue` bool

Returns

bool

ParseCurrency(string)

Parse a currency from a string.

```
public static Currency ParseCurrency(string value)
```

Parameters

`value` string

The string to parse. Currency should be in the form of "ticker;decimalPlaces;minters;totalSupplyTractable;maximumSupply".

Returns

Currency

The parsed currency.

Exceptions

ArgumentException

The string is not a valid currency.

ParseDecimal(string)

```
public static decimal ParseDecimal(string value)
```

Parameters

`value` string

Returns

decimal

ParseDecimal(string, decimal)

```
public static decimal ParseDecimal(string value, decimal defaultValue)
```

Parameters

`value` string

`defaultValue` decimal

Returns

decimal

ParseElementalTypes(string)

Parses a colon-separated list of elemental types. Accepts numeric values matching ElementalType enum. Returns an empty list if input is null/empty or no valid items.

```
public static List<ElementalType> ParseElementalTypes(string value)
```

Parameters

`value` string

Returns

List<ElementalType>

ParseFraction(string)

Parse a fraction from a string.

```
public static SwapRateSheet.Fraction ParseFraction(string value)
```

Parameters

`value` string

The string to parse. Fraction should be in the form of "numerator/denominator".

Returns

SwapRateSheet.Fraction

The parsed fraction.

Exceptions

ArgumentException

The string is not a valid fraction.

ParseInt(string)

```
public static int ParseInt(string value)
```

Parameters

`value` string

Returns

int

ParseInt(string, int)

```
public static int ParseInt(string value, int defaultValue)
```

Parameters

`value` string

`defaultValue` int

Returns

int

ParseIntList(string)

Parses a colon-separated list of integers. Returns an empty list for null/empty input. Invalid tokens are ignored.

```
public static List<int> ParseIntList(string value)
```

Parameters

`value` string

Returns

List<int>

ParseIntOrNull(string)

Parses an integer or returns null for empty input.

```
public static int? ParseIntOrNull(string value)
```

Parameters

`value` string

Returns

int?

ParseItemSubTypes(string)

Parses a colon-separated list of item subtypes. Accepts numeric values matching ItemSubType enum. Returns an empty list if input is null/empty or no valid items.

```
public static List<ItemSubType> ParseItemSubTypes(string value)
```

Parameters

`value` string

Returns

List<ItemSubType>

ParseLong(string)

```
public static long ParseLong(string value)
```

Parameters

`value` string

Returns

long

ParseLong(string, long)

```
public static long ParseLong(string value, long defaultValue)
```

Parameters

`value` string

`defaultValue` long

Returns

long

ParseRuneTypes(string)

Parses a comma-separated list of rune types from a string. Returns an empty list if input is null/empty or no valid items.

```
public static List<RuneType> ParseRuneTypes(string value)
```

Parameters

`value` string

Returns

List<RuneType>

ParseSkillTargetTypes(string)

Parses a colon-separated list of skill target types. Accepts numeric values matching SkillTargetType enum. Returns an empty list if input is null/empty or no valid items.

```
public static List<SkillTargetType> ParseSkillTargetTypes(string value)
```

Parameters

`value` string

Returns

List<SkillTargetType>

ParseStringOrNull(string)

Parses a trimmed string or returns null for empty input.

```
public static string? ParseStringOrNull(string value)
```

Parameters

`value` string

Returns

string

TryParseBigInteger(string, out BigInteger)

```
public static bool TryParseBigInteger(string value, out BigInteger result)
```

Parameters

`value` string

`result` BigInteger

Returns

bool

TryParseCurrency(string, out Currency)

Try to parse a currency from a string.

```
public static bool TryParseCurrency(string value, out Currency currency)
```

Parameters

`value` string

> The string to parse. Currency should be in the form of "ticker;decimalPlaces;minters;totalSupplyTractable;maximumSupply".

`currency` Currency

The parsed currency.

Returns

bool

Whether the parsing is successful.

TryParseDecimal(string, out decimal)

```
public static bool TryParseDecimal(string value, out decimal result)
```

Parameters

`value` string

`result` decimal

Returns

bool

TryParseFloat(string, out float)

```
public static bool TryParseFloat(string value, out float result)
```

Parameters

`value` string

`result` float

Returns

bool

TryParseFraction(string, out Fraction)

Try to parse a fraction from a string.

```
public static bool TryParseFraction(string value, out SwapRateSheet.Fraction result)
```

Parameters

`value` string

The string to parse. Fraction should be in the form of "numerator/denominator".

`result` SwapRateSheet.Fraction

The parsed fraction.

Returns

bool

TryParseInt(string, out int)

```
public static bool TryParseInt(string value, out int result)
```

Parameters

`value` string

`result` int

Returns

bool

TryParseLong(string, out long)

```
public static bool TryParseLong(string value, out long result)
```

Parameters

`value` string

`result` long

Returns

bool

Class TradeQuestSheet

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class TradeQuestSheet : Sheet<int, TradeQuestSheet.Row>, IDictionary<int,
TradeQuestSheet.Row>, ICollection<KeyValuePair<int, TradeQuestSheet.Row>>,
IEnumerable<KeyValuePair<int, TradeQuestSheet.Row>>, IEnumerable, ISheet
```

Inheritance

object ← Sheet<int, TradeQuestSheet.Row> ← TradeQuestSheet

Implements

IDictionary<int, TradeQuestSheet.Row>,
ICollection<KeyValuePair<int, TradeQuestSheet.Row>>,
IEnumerable<KeyValuePair<int, TradeQuestSheet.Row>>, IEnumerable, ISheet

Inherited Members

Sheet<int, TradeQuestSheet.Row>.Name , Sheet<int, TradeQuestSheet.Row>.OrderedList ,
Sheet<int, TradeQuestSheet.Row>.First , Sheet<int, TradeQuestSheet.Row>.Last ,
Sheet<int, TradeQuestSheet.Row>.Keys , Sheet<int, TradeQuestSheet.Row>.Values ,
Sheet<int, TradeQuestSheet.Row>.Count , Sheet<int, TradeQuestSheet.Row>.IsReadOnly ,
Sheet<int, TradeQuestSheet.Row>.this[int] ,
Sheet<int, TradeQuestSheet.Row>.Set(string, bool) ,
Sheet<int, TradeQuestSheet.Row>.Set<T>(Sheet<int, T>, bool) ,
Sheet<int, TradeQuestSheet.Row>.GetEnumerator() ,
Sheet<int, TradeQuestSheet.Row>.TryGetValue(int, out TradeQuestSheet.Row, bool) ,
Sheet<int, TradeQuestSheet.Row>.AddRow(int, TradeQuestSheet.Row) ,
Sheet<int, TradeQuestSheet.Row>.Add(int, TradeQuestSheet.Row) ,
Sheet<int, TradeQuestSheet.Row>.ContainsKey(int) ,
Sheet<int, TradeQuestSheet.Row>.Remove(int) ,
Sheet<int, TradeQuestSheet.Row>.TryGetValue(int, out TradeQuestSheet.Row) ,
Sheet<int, TradeQuestSheet.Row>.Add(KeyValuePair<int, TradeQuestSheet.Row>) ,
Sheet<int, TradeQuestSheet.Row>.Clear() ,
Sheet<int, TradeQuestSheet.Row>.Contains(KeyValuePair<int, TradeQuestSheet.Row>) ,
Sheet<int, TradeQuestSheet.Row>.CopyTo(KeyValuePair<int, TradeQuestSheet.Row>[], int)
,
Sheet<int, TradeQuestSheet.Row>.Remove(KeyValuePair<int, TradeQuestSheet.Row>) ,

Sheet<int, TradeQuestSheet.Row>.Serialize() , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

TradeQuestSheet()

```
public TradeQuestSheet()
```

Class TradeQuestSheet.Row

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class TradeQuestSheet.Row : QuestSheet.Row, IState
```

Inheritance

object ← SheetRow<int> ← QuestSheet.Row ← TradeQuestSheet.Row

Implements

IState

Inherited Members

QuestSheet.Row.Key , QuestSheet.Row.Id , QuestSheet.Row.Goal , QuestSheet.Row.QuestRewardId , QuestSheet.Row.Serialize() , QuestSheet.Row.Deserialize(Dictionary) , SheetRow<int>.Validate() , SheetRow<int>.EndOfSheetInitialize() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Properties

Type

```
public TradeType Type { get; }
```

Property Value

TradeType

Methods

Set(IReadOnlyList<string>)

```
public override void Set(IReadOnlyList<string> fields)
```

Parameters

`fields` IReadOnlyList<string>

Class UnlockCombinationSlotCostSheet

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class UnlockCombinationSlotCostSheet : Sheet<int,
UnlockCombinationSlotCostSheet.Row>, IDictionary<int,
UnlockCombinationSlotCostSheet.Row>, ICollection<KeyValuePair<int,
UnlockCombinationSlotCostSheet.Row>>, IEnumerable<KeyValuePair<int,
UnlockCombinationSlotCostSheet.Row>>, IEnumerable, ISheet
```

Inheritance

object ← Sheet<int, UnlockCombinationSlotCostSheet.Row> ← UnlockCombinationSlotCostSheet

Implements

IDictionary<int, UnlockCombinationSlotCostSheet.Row>, ICollection<KeyValuePair<int, UnlockCombinationSlotCostSheet.Row>>, IEnumerable<KeyValuePair<int, UnlockCombinationSlotCostSheet.Row>>, IEnumerable, ISheet

Inherited Members

Sheet<int, UnlockCombinationSlotCostSheet.Row>.Name ,
Sheet<int, UnlockCombinationSlotCostSheet.Row>.OrderedList ,
Sheet<int, UnlockCombinationSlotCostSheet.Row>.First ,
Sheet<int, UnlockCombinationSlotCostSheet.Row>.Last ,
Sheet<int, UnlockCombinationSlotCostSheet.Row>.Keys ,
Sheet<int, UnlockCombinationSlotCostSheet.Row>.Values ,
Sheet<int, UnlockCombinationSlotCostSheet.Row>.Count ,
Sheet<int, UnlockCombinationSlotCostSheet.Row>.IsReadOnly ,
Sheet<int, UnlockCombinationSlotCostSheet.Row>.this[int] ,
Sheet<int, UnlockCombinationSlotCostSheet.Row>.Set(string, bool) ,
Sheet<int, UnlockCombinationSlotCostSheet.Row>.Set<T>(Sheet<int, T>, bool) ,
Sheet<int, UnlockCombinationSlotCostSheet.Row>.GetEnumerator() ,
Sheet<int, UnlockCombinationSlotCostSheet.Row>.TryGetValue(int, out UnlockCombinationSlotCostSheet.Row, bool) ,
Sheet<int, UnlockCombinationSlotCostSheet.Row>.AddRow(int, UnlockCombinationSlotCostSheet.Row) ,

Sheet<int, UnlockCombinationSlotCostSheet.Row>.Add(int, UnlockCombinationSlotCostSheet.Row) ,
Sheet<int, UnlockCombinationSlotCostSheet.Row>.ContainsKey(int) ,
Sheet<int, UnlockCombinationSlotCostSheet.Row>.Remove(int) ,
Sheet<int, UnlockCombinationSlotCostSheet.Row>.TryGetValue(int, out UnlockCombinationSlotCostSheet.Row) ,
Sheet<int, UnlockCombinationSlotCostSheet.Row>.Add(KeyValuePair<int, UnlockCombinationSlotCostSheet.Row>) ,
Sheet<int, UnlockCombinationSlotCostSheet.Row>.Clear() ,
Sheet<int, UnlockCombinationSlotCostSheet.Row>.Contains(KeyValuePair<int, UnlockCombinationSlotCostSheet.Row>) ,
Sheet<int, UnlockCombinationSlotCostSheet.Row>.CopyTo(KeyValuePair<int, UnlockCombinationSlotCostSheet.Row>[], int) ,
Sheet<int, UnlockCombinationSlotCostSheet.Row>.Remove(KeyValuePair<int, UnlockCombinationSlotCostSheet.Row>) ,
Sheet<int, UnlockCombinationSlotCostSheet.Row>.Serialize() , object.Equals(object) ,
object.Equals(object, object) , object.GetHashCode() , object.GetType() ,
object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

UnlockCombinationSlotCostSheet()

```
public UnlockCombinationSlotCostSheet()
```

Class UnlockCombinationSlotCostSheet.Row

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class UnlockCombinationSlotCostSheet.Row : SheetRow<int>
```

Inheritance

object ← SheetRow<int> ← UnlockCombinationSlotCostSheet.Row

Inherited Members

SheetRow<int>.Validate() , SheetRow<int>.EndOfSheetInitialize() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Fields

CrystalPrice

```
public int CrystalPrice
```

Field Value

int

GoldenDustPrice

```
public int GoldenDustPrice
```

Field Value

int

NcgPrice

```
public BigInteger NcgPrice
```

Field Value

BigInteger

RubyDustPrice

```
public int RubyDustPrice
```

Field Value

int

SlotId

```
public int SlotId
```

Field Value

int

Properties

Key

```
public override int Key { get; }
```

Property Value

int

Methods

Set(IReadOnlyList<string>)

```
public override void Set(IReadOnlyList<string> fields)
```

Parameters

`fields` IReadOnlyList<string>

Class WeeklyArenaRewardSheet

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class WeeklyArenaRewardSheet : Sheet<int, WeeklyArenaRewardSheet.Row>,
IDictionary<int, WeeklyArenaRewardSheet.Row>, ICollection<KeyValuePair<int,
WeeklyArenaRewardSheet.Row>>, IEnumerable<KeyValuePair<int,
WeeklyArenaRewardSheet.Row>>, IEnumerable, ISheet
```

Inheritance

object ← Sheet<int, WeeklyArenaRewardSheet.Row> ← WeeklyArenaRewardSheet

Implements

IDictionary<int, WeeklyArenaRewardSheet.Row>,
ICollection<KeyValuePair<int, WeeklyArenaRewardSheet.Row>>,
IEnumerable<KeyValuePair<int, WeeklyArenaRewardSheet.Row>>, IEnumerable,
ISheet

Inherited Members

Sheet<int, WeeklyArenaRewardSheet.Row>.Name ,
Sheet<int, WeeklyArenaRewardSheet.Row>.OrderedList ,
Sheet<int, WeeklyArenaRewardSheet.Row>.First ,
Sheet<int, WeeklyArenaRewardSheet.Row>.Last ,
Sheet<int, WeeklyArenaRewardSheet.Row>.Keys ,
Sheet<int, WeeklyArenaRewardSheet.Row>.Values ,
Sheet<int, WeeklyArenaRewardSheet.Row>.Count ,
Sheet<int, WeeklyArenaRewardSheet.Row>.IsReadOnly ,
Sheet<int, WeeklyArenaRewardSheet.Row>.this[int] ,
Sheet<int, WeeklyArenaRewardSheet.Row>.Set(string, bool) ,
Sheet<int, WeeklyArenaRewardSheet.Row>.Set<T>(Sheet<int, T>, bool) ,
Sheet<int, WeeklyArenaRewardSheet.Row>.GetEnumerator() ,
Sheet<int, WeeklyArenaRewardSheet.Row>.TryGetValue(int, out
WeeklyArenaRewardSheet.Row, bool) ,
Sheet<int, WeeklyArenaRewardSheet.Row>.AddRow(int, WeeklyArenaRewardSheet.Row) ,
Sheet<int, WeeklyArenaRewardSheet.Row>.Add(int, WeeklyArenaRewardSheet.Row) ,
Sheet<int, WeeklyArenaRewardSheet.Row>.ContainsKey(int) ,
Sheet<int, WeeklyArenaRewardSheet.Row>.Remove(int) ,

Sheet<int, WeeklyArenaRewardSheet.Row>.TryGetValue(int, out WeeklyArenaRewardSheet.Row) , Sheet<int, WeeklyArenaRewardSheet.Row>.Add(KeyValuePair<int, WeeklyArenaRewardSheet.Row>) , Sheet<int, WeeklyArenaRewardSheet.Row>.Clear() , Sheet<int, WeeklyArenaRewardSheet.Row>.Contains(KeyValuePair<int, WeeklyArenaRewardSheet.Row>) , Sheet<int, WeeklyArenaRewardSheet.Row>.CopyTo(KeyValuePair<int, WeeklyArenaRewardSheet.Row>[], int) , Sheet<int, WeeklyArenaRewardSheet.Row>.Remove(KeyValuePair<int, WeeklyArenaRewardSheet.Row>) , Sheet<int, WeeklyArenaRewardSheet.Row>.Serialize() , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

WeeklyArenaRewardSheet()

```
public WeeklyArenaRewardSheet()
```

Class WeeklyArenaRewardSheet.RewardData

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class WeeklyArenaRewardSheet.RewardData : StageSheet.RewardData
```

Inheritance

object ← StageSheet.RewardData ← WeeklyArenaRewardSheet.RewardData

Inherited Members

StageSheet.RewardData.ItemId , StageSheet.RewardData.Ratio , StageSheet.RewardData.Min , StageSheet.RewardData.Max , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

RewardData(int, decimal, int, int, int)

```
public RewardData(int itemId, decimal ratio, int min, int max, int requiredLevel)
```

Parameters

`itemId` int

`ratio` decimal

`min` int

`max` int

`requiredLevel` int

Properties

RequiredLevel

```
public int RequiredLevel { get; }
```

Property Value

int

Class WeeklyArenaRewardSheet.Row

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class WeeklyArenaRewardSheet.Row : SheetRow<int>
```

Inheritance

object ← SheetRow<int> ← WeeklyArenaRewardSheet.Row

Inherited Members

SheetRow<int>.Validate() , SheetRow<int>.EndOfSheetInitialize() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Properties

Id

```
public int Id { get; }
```

Property Value

int

Key

```
public override int Key { get; }
```

Property Value

int

Reward

```
public WeeklyArenaRewardSheet.RewardData Reward { get; }
```

Property Value

WeeklyArenaRewardSheet.RewardData

Methods

Set(IReadOnlyList<string>)

```
public override void Set(IReadOnlyList<string> fields)
```

Parameters

`fields` IReadOnlyList<string>

Class WorldBossActionPatternSheet

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class WorldBossActionPatternSheet : Sheet<int,
WorldBossActionPatternSheet.Row>, IDictionary<int, WorldBossActionPatternSheet.Row>,
ICollection<KeyValuePair<int, WorldBossActionPatternSheet.Row>>,
IEnumerable<KeyValuePair<int, WorldBossActionPatternSheet.Row>>, IEnumerable, ISheet
```

Inheritance

object ← Sheet<int, WorldBossActionPatternSheet.Row> ← WorldBossActionPatternSheet

Implements

IDictionary<int, WorldBossActionPatternSheet.Row>,
ICollection<KeyValuePair<int, WorldBossActionPatternSheet.Row>>,
IEnumerable<KeyValuePair<int, WorldBossActionPatternSheet.Row>>, IEnumerable,
ISheet

Inherited Members

Sheet<int, WorldBossActionPatternSheet.Row>.Name ,
Sheet<int, WorldBossActionPatternSheet.Row>.OrderedList ,
Sheet<int, WorldBossActionPatternSheet.Row>.First ,
Sheet<int, WorldBossActionPatternSheet.Row>.Last ,
Sheet<int, WorldBossActionPatternSheet.Row>.Keys ,
Sheet<int, WorldBossActionPatternSheet.Row>.Values ,
Sheet<int, WorldBossActionPatternSheet.Row>.Count ,
Sheet<int, WorldBossActionPatternSheet.Row>.IsReadOnly ,
Sheet<int, WorldBossActionPatternSheet.Row>.this[int] ,
Sheet<int, WorldBossActionPatternSheet.Row>.Set(string, bool) ,
Sheet<int, WorldBossActionPatternSheet.Row>.Set<T>(Sheet<int, T>, bool) ,
Sheet<int, WorldBossActionPatternSheet.Row>.GetEnumerator() ,
Sheet<int, WorldBossActionPatternSheet.Row>.TryGetValue(int, out
WorldBossActionPatternSheet.Row, bool) ,
Sheet<int, WorldBossActionPatternSheet.Row>.AddRow(int,
WorldBossActionPatternSheet.Row) ,
Sheet<int, WorldBossActionPatternSheet.Row>.Add(int, WorldBossActionPatternSheet.Row)
,
Sheet<int, WorldBossActionPatternSheet.Row>.ContainsKey(int) ,

Sheet<int, WorldBossActionPatternSheet.Row>.Remove(int) ,
Sheet<int, WorldBossActionPatternSheet.Row>.TryGetValue(int, out
WorldBossActionPatternSheet.Row) ,
Sheet<int, WorldBossActionPatternSheet.Row>.Add(KeyValuePair<int,
WorldBossActionPatternSheet.Row>) ,
Sheet<int, WorldBossActionPatternSheet.Row>.Clear() ,
Sheet<int, WorldBossActionPatternSheet.Row>.Contains(KeyValuePair<int,
WorldBossActionPatternSheet.Row>) ,
Sheet<int, WorldBossActionPatternSheet.Row>.CopyTo(KeyValuePair<int,
WorldBossActionPatternSheet.Row>[], int) ,
Sheet<int, WorldBossActionPatternSheet.Row>.Remove(KeyValuePair<int,
WorldBossActionPatternSheet.Row>) ,
Sheet<int, WorldBossActionPatternSheet.Row>.Serialize() , object.Equals(object) ,
object.Equals(object, object) , object.GetHashCode() , object.GetType() ,
object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

WorldBossActionPatternSheet()

```
public WorldBossActionPatternSheet()
```

Methods

AddRow(int, Row)

```
protected override void AddRow(int key, WorldBossActionPatternSheet.Row value)
```

Parameters

`key` int

`value` WorldBossActionPatternSheet.Row

Class WorldBossActionPatternSheet.ActionPatternData

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class WorldBossActionPatternSheet.ActionPatternData
```

Inheritance

object ← WorldBossActionPatternSheet.ActionPatternData

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Fields

SkillIds

```
public readonly List<int> SkillIds
```

Field Value

List<int>

Wave

```
public int Wave
```

Field Value

int

Class WorldBossActionPatternSheet.Row

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class WorldBossActionPatternSheet.Row : SheetRow<int>
```

Inheritance

object ← SheetRow<int> ← WorldBossActionPatternSheet.Row

Inherited Members

SheetRow<int>.Validate() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Fields

BossId

```
public int BossId
```

Field Value

int

Patterns

```
public List<WorldBossActionPatternSheet.ActionPatternData> Patterns
```

Field Value

List<WorldBossActionPatternSheet.ActionPatternData>

Properties

Key

```
public override int Key { get; }
```

Property Value

int

Methods

EndOfSheetInitialize()

```
public override void EndOfSheetInitialize()
```

Set(IReadOnlyList<string>)

```
public override void Set(IReadOnlyList<string> fields)
```

Parameters

`fields` IReadOnlyList<string>

Class WorldBossBattleRewardSheet

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
public class WorldBossBattleRewardSheet : Sheet<int,
WorldBossBattleRewardSheet.Row>, IDictionary<int, WorldBossBattleRewardSheet.Row>,
ICollection<KeyValuePair<int, WorldBossBattleRewardSheet.Row>>,
IEnumerable<KeyValuePair<int, WorldBossBattleRewardSheet.Row>>, IEnumerable,
IWorldBossRewardSheet, ISheet
```

Inheritance

object ← Sheet<int, WorldBossBattleRewardSheet.Row> ← WorldBossBattleRewardSheet

Implements

IDictionary<int, WorldBossBattleRewardSheet.Row>,
ICollection<KeyValuePair<int, WorldBossBattleRewardSheet.Row>>,
IEnumerable<KeyValuePair<int, WorldBossBattleRewardSheet.Row>>, IEnumerable,
IWorldBossRewardSheet, ISheet

Inherited Members

Sheet<int, WorldBossBattleRewardSheet.Row>.Name ,
Sheet<int, WorldBossBattleRewardSheet.Row>.OrderedList ,
Sheet<int, WorldBossBattleRewardSheet.Row>.First ,
Sheet<int, WorldBossBattleRewardSheet.Row>.Last ,
Sheet<int, WorldBossBattleRewardSheet.Row>.Keys ,
Sheet<int, WorldBossBattleRewardSheet.Row>.Values ,
Sheet<int, WorldBossBattleRewardSheet.Row>.Count ,
Sheet<int, WorldBossBattleRewardSheet.Row>.IsReadOnly ,
Sheet<int, WorldBossBattleRewardSheet.Row>.this[int] ,
Sheet<int, WorldBossBattleRewardSheet.Row>.Set(string, bool) ,
Sheet<int, WorldBossBattleRewardSheet.Row>.Set<T>(Sheet<int, T>, bool) ,
Sheet<int, WorldBossBattleRewardSheet.Row>.GetEnumerator() ,
Sheet<int, WorldBossBattleRewardSheet.Row>.TryGetValue(int, out
WorldBossBattleRewardSheet.Row, bool) ,
Sheet<int, WorldBossBattleRewardSheet.Row>.AddRow(int,
WorldBossBattleRewardSheet.Row) ,
Sheet<int, WorldBossBattleRewardSheet.Row>.Add(int, WorldBossBattleRewardSheet.Row)
,
Sheet<int, WorldBossBattleRewardSheet.Row>.ContainsKey(int) ,

Sheet<int, WorldBossBattleRewardSheet.Row>.Remove(int) ,
Sheet<int, WorldBossBattleRewardSheet.Row>.TryGetValue(int, out
WorldBossBattleRewardSheet.Row) ,
Sheet<int, WorldBossBattleRewardSheet.Row>.Add(KeyValuePair<int,
WorldBossBattleRewardSheet.Row>) ,
Sheet<int, WorldBossBattleRewardSheet.Row>.Clear() ,
Sheet<int, WorldBossBattleRewardSheet.Row>.Contains(KeyValuePair<int,
WorldBossBattleRewardSheet.Row>) ,
Sheet<int, WorldBossBattleRewardSheet.Row>.CopyTo(KeyValuePair<int,
WorldBossBattleRewardSheet.Row>[], int) ,
Sheet<int, WorldBossBattleRewardSheet.Row>.Remove(KeyValuePair<int,
WorldBossBattleRewardSheet.Row>) ,
Sheet<int, WorldBossBattleRewardSheet.Row>.Serialize() , object.Equals(object) ,
object.Equals(object, object) , object.GetHashCode() , object.GetType() ,
object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

WorldBossBattleRewardSheet()

```
public WorldBossBattleRewardSheet()
```

Properties

OrderedRows

```
public IReadOnlyList<IWorldBossRewardRow> OrderedRows { get; }
```

Property Value

IReadOnlyList<IWorldBossRewardRow>

Class WorldBossBattleRewardSheet.Row

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
public class WorldBossBattleRewardSheet.Row : SheetRow<int>, IWorldBossRewardRow
```

Inheritance

object ← SheetRow<int> ← WorldBossBattleRewardSheet.Row

Implements

IWorldBossRewardRow

Inherited Members

SheetRow<int>.Validate() , SheetRow<int>.EndOfSheetInitialize() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Fields

Id

```
public int Id
```

Field Value

int

RuneMax

```
public int RuneMax
```

Field Value

int

RuneMin

```
public int RuneMin
```

Field Value

int

Properties

BossId

```
public int BossId { get; }
```

Property Value

int

Circle

```
public int Circle { get; }
```

Property Value

int

Crystal

```
public int Crystal { get; }
```

Property Value

int

Key

```
public override int Key { get; }
```

Property Value

int

Rank

```
public int Rank { get; }
```

Property Value

int

Rune

```
public int Rune { get; }
```

Property Value

int

Methods

Set(IReadOnlyList<string>)

```
public override void Set(IReadOnlyList<string> fields)
```

Parameters

`fields` IReadOnlyList<string>

SetRune(IRandom)

```
public void SetRune(IRandom random)
```

Parameters

`random` IRandom

Class WorldBossCharacterSheet

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class WorldBossCharacterSheet : Sheet<int, WorldBossCharacterSheet.Row>,
IDictionary<int, WorldBossCharacterSheet.Row>, ICollection<KeyValuePair<int,
WorldBossCharacterSheet.Row>>, IEnumerable<KeyValuePair<int,
WorldBossCharacterSheet.Row>>, IEnumerable, ISheet
```

Inheritance

object ← Sheet<int, WorldBossCharacterSheet.Row> ← WorldBossCharacterSheet

Implements

IDictionary<int, WorldBossCharacterSheet.Row>,
ICollection<KeyValuePair<int, WorldBossCharacterSheet.Row>>,
IEnumerable<KeyValuePair<int, WorldBossCharacterSheet.Row>>, IEnumerable,
ISheet

Inherited Members

Sheet<int, WorldBossCharacterSheet.Row>.Name ,
Sheet<int, WorldBossCharacterSheet.Row>.OrderedList ,
Sheet<int, WorldBossCharacterSheet.Row>.First ,
Sheet<int, WorldBossCharacterSheet.Row>.Last ,
Sheet<int, WorldBossCharacterSheet.Row>.Keys ,
Sheet<int, WorldBossCharacterSheet.Row>.Values ,
Sheet<int, WorldBossCharacterSheet.Row>.Count ,
Sheet<int, WorldBossCharacterSheet.Row>.IsReadOnly ,
Sheet<int, WorldBossCharacterSheet.Row>.this[int] ,
Sheet<int, WorldBossCharacterSheet.Row>.Set(string, bool) ,
Sheet<int, WorldBossCharacterSheet.Row>.Set<T>(Sheet<int, T>, bool) ,
Sheet<int, WorldBossCharacterSheet.Row>.GetEnumerator() ,
Sheet<int, WorldBossCharacterSheet.Row>.TryGetValue(int, out
WorldBossCharacterSheet.Row, bool) ,
Sheet<int, WorldBossCharacterSheet.Row>.AddRow(int, WorldBossCharacterSheet.Row) ,
Sheet<int, WorldBossCharacterSheet.Row>.Add(int, WorldBossCharacterSheet.Row) ,
Sheet<int, WorldBossCharacterSheet.Row>.ContainsKey(int) ,
Sheet<int, WorldBossCharacterSheet.Row>.Remove(int) ,

Sheet<int, WorldBossCharacterSheet.Row>.TryGetValue(int, out WorldBossCharacterSheet.Row) , Sheet<int, WorldBossCharacterSheet.Row>.Add(KeyValuePair<int, WorldBossCharacterSheet.Row>) , Sheet<int, WorldBossCharacterSheet.Row>.Clear() , Sheet<int, WorldBossCharacterSheet.Row>.Contains(KeyValuePair<int, WorldBossCharacterSheet.Row>) , Sheet<int, WorldBossCharacterSheet.Row>.CopyTo(KeyValuePair<int, WorldBossCharacterSheet.Row>[], int) , Sheet<int, WorldBossCharacterSheet.Row>.Remove(KeyValuePair<int, WorldBossCharacterSheet.Row>) , Sheet<int, WorldBossCharacterSheet.Row>.Serialize() , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

WorldBossCharacterSheet()

```
public WorldBossCharacterSheet()
```

Methods

AddRow(int, Row)

```
protected override void AddRow(int key, WorldBossCharacterSheet.Row value)
```

Parameters

`key` int

`value` WorldBossCharacterSheet.Row

Class WorldBossCharacterSheet.Row

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class WorldBossCharacterSheet.Row : SheetRow<int>
```

Inheritance

object ← SheetRow<int> ← WorldBossCharacterSheet.Row

Inherited Members

SheetRow<int>.Validate() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Properties

BossId

```
public int BossId { get; }
```

Property Value

int

Key

```
public override int Key { get; }
```

Property Value

int

WaveStats

```
public List<WorldBossCharacterSheet.WaveStatData> WaveStats { get; }
```

Property Value

List<WorldBossCharacterSheet.WaveStatData>

Methods

EndOfSheetInitialize()

```
public override void EndOfSheetInitialize()
```

Set(IReadOnlyList<string>)

```
public override void Set(IReadOnlyList<string> fields)
```

Parameters

`fields` IReadOnlyList<string>

Class WorldBossCharacterSheet.WaveStatData

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class WorldBossCharacterSheet.WaveStatData
```

Inheritance

object ← WorldBossCharacterSheet.WaveStatData

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Extension Methods

WorldBossStatSheetExtension.ToStats(WorldBossCharacterSheet.WaveStatData)

Properties

ATK

```
public decimal ATK { get; set; }
```

Property Value

decimal

CRI

```
public decimal CRI { get; set; }
```

Property Value

decimal

DEF

```
public decimal DEF { get; set; }
```

Property Value

decimal

ElementalType

```
public ElementalType ElementalType { get; set; }
```

Property Value

ElementalType

EnrageSkillId

```
public int EnrageSkillId { get; set; }
```

Property Value

int

EnrageTurn

```
public int EnrageTurn { get; set; }
```

Property Value

int

HIT

```
public decimal HIT { get; set; }
```

Property Value

decimal

HP

```
public decimal HP { get; set; }
```

Property Value

decimal

Level

```
public int Level { get; set; }
```

Property Value

int

SPD

```
public decimal SPD { get; set; }
```

Property Value

decimal

TurnLimit

```
public int TurnLimit { get; set; }
```

Property Value

int

Wave

```
public int Wave { get; set; }
```

Property Value

int

Class WorldBossContributionRewardSheet

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

Represents the reward sheet for world boss contributions.

```
public class WorldBossContributionRewardSheet : Sheet<int,
WorldBossContributionRewardSheet.Row>, IDictionary<int,
WorldBossContributionRewardSheet.Row>, ICollection<KeyValuePair<int,
WorldBossContributionRewardSheet.Row>>, IEnumerable<KeyValuePair<int,
WorldBossContributionRewardSheet.Row>>, IEnumerable, ISheet
```

Inheritance

object ← Sheet<int, WorldBossContributionRewardSheet.Row> ← WorldBossContributionRewardSheet

Implements

IDictionary<int, WorldBossContributionRewardSheet.Row>, ICollection<KeyValuePair<int, WorldBossContributionRewardSheet.Row>>, IEnumerable<KeyValuePair<int, WorldBossContributionRewardSheet.Row>>, IEnumerable, ISheet

Inherited Members

Sheet<int, WorldBossContributionRewardSheet.Row>.Name ,
Sheet<int, WorldBossContributionRewardSheet.Row>.OrderedList ,
Sheet<int, WorldBossContributionRewardSheet.Row>.First ,
Sheet<int, WorldBossContributionRewardSheet.Row>.Last ,
Sheet<int, WorldBossContributionRewardSheet.Row>.Keys ,
Sheet<int, WorldBossContributionRewardSheet.Row>.Values ,
Sheet<int, WorldBossContributionRewardSheet.Row>.Count ,
Sheet<int, WorldBossContributionRewardSheet.Row>.IsReadOnly ,
Sheet<int, WorldBossContributionRewardSheet.Row>.this[int] ,
Sheet<int, WorldBossContributionRewardSheet.Row>.Set(string, bool) ,
Sheet<int, WorldBossContributionRewardSheet.Row>.Set<T>(Sheet<int, T>, bool) ,
Sheet<int, WorldBossContributionRewardSheet.Row>.GetEnumerator() ,
Sheet<int, WorldBossContributionRewardSheet.Row>.TryGetValue(int, out WorldBossContributionRewardSheet.Row, bool) ,
Sheet<int, WorldBossContributionRewardSheet.Row>.AddRow(int, WorldBossContributionRewardSheet.Row) ,

Sheet<int, WorldBossContributionRewardSheet.Row>.Add(int, WorldBossContributionRewardSheet.Row) , Sheet<int, WorldBossContributionRewardSheet.Row>.ContainsKey(int) , Sheet<int, WorldBossContributionRewardSheet.Row>.Remove(int) , Sheet<int, WorldBossContributionRewardSheet.Row>.TryGetValue(int, out WorldBossContributionRewardSheet.Row) , Sheet<int, WorldBossContributionRewardSheet.Row>.Add(KeyValuePair<int, WorldBossContributionRewardSheet.Row>) , Sheet<int, WorldBossContributionRewardSheet.Row>.Clear() , Sheet<int, WorldBossContributionRewardSheet.Row>.Contains(KeyValuePair<int, WorldBossContributionRewardSheet.Row>) , Sheet<int, WorldBossContributionRewardSheet.Row>.CopyTo(KeyValuePair<int, WorldBossContributionRewardSheet.Row>[], int) , Sheet<int, WorldBossContributionRewardSheet.Row>.Remove(KeyValuePair<int, WorldBossContributionRewardSheet.Row>) , Sheet<int, WorldBossContributionRewardSheet.Row>.Serialize() , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

WorldBossContributionRewardSheet()

Initializes a new instance of the WorldBossContributionRewardSheet class.

```
public WorldBossContributionRewardSheet()
```

Class WorldBossContributionRewardSheet.RewardModel

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class WorldBossContributionRewardSheet.RewardModel
```

Inheritance

object ← WorldBossContributionRewardSheet.RewardModel

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

RewardModel(BigInteger, int, string)

Initializes a new instance of the WorldBossContributionRewardSheet.RewardModel class.

```
public RewardModel(BigInteger count, int itemId, string ticker)
```

Parameters

`count` BigInteger

The quantity of the reward.

`itemId` int

The ID of the item if reward is ItemBase.

`ticker` string

The ticker of the item if reward is FungibleAssetValue.

Fields

Count

Gets or sets the quantity of the reward.

```
public BigInteger Count
```

Field Value

BigInteger

ItemId

Gets or sets the ID of the item.

```
public int ItemId
```

Field Value

int

Ticker

Gets or sets the ticker of the item.

```
public string Ticker
```

Field Value

string

Class WorldBossContributionRewardSheet.Row

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

Represents each row in the sheet.

```
public class WorldBossContributionRewardSheet.Row : SheetRow<int>
```

Inheritance

object ← SheetRow<int> ← WorldBossContributionRewardSheet.Row

Inherited Members

SheetRow<int>.Validate() , SheetRow<int>.EndOfSheetInitialize() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Fields

BossId

Gets or sets the ID of the boss.

```
public int BossId
```

Field Value

int

Rewards

Gets the list of rewards.

```
public readonly List<WorldBossContributionRewardSheet.RewardModel> Rewards
```

Field Value

List<WorldBossContributionRewardSheet.RewardModel>

Properties

Key

Gets the key value of the row.

```
public override int Key { get; }
```

Property Value

int

Methods

Set(IReadOnlyList<string>)

Sets the row based on given fields.

```
public override void Set(IReadOnlyList<string> fields)
```

Parameters

`fields` IReadOnlyList<string>

The fields to set the row.

Class WorldBossGlobalHpSheet

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class WorldBossGlobalHpSheet : Sheet<int, WorldBossGlobalHpSheet.Row>,
IDictionary<int, WorldBossGlobalHpSheet.Row>, ICollection<KeyValuePair<int,
WorldBossGlobalHpSheet.Row>>, IEnumerable<KeyValuePair<int,
WorldBossGlobalHpSheet.Row>>, IEnumerable, ISheet
```

Inheritance

object ← Sheet<int, WorldBossGlobalHpSheet.Row> ← WorldBossGlobalHpSheet

Implements

IDictionary<int, WorldBossGlobalHpSheet.Row>,
ICollection<KeyValuePair<int, WorldBossGlobalHpSheet.Row>>,
IEnumerable<KeyValuePair<int, WorldBossGlobalHpSheet.Row>>, IEnumerable,
ISheet

Inherited Members

Sheet<int, WorldBossGlobalHpSheet.Row>.Name ,
Sheet<int, WorldBossGlobalHpSheet.Row>.OrderedList ,
Sheet<int, WorldBossGlobalHpSheet.Row>.First ,
Sheet<int, WorldBossGlobalHpSheet.Row>.Last ,
Sheet<int, WorldBossGlobalHpSheet.Row>.Keys ,
Sheet<int, WorldBossGlobalHpSheet.Row>.Values ,
Sheet<int, WorldBossGlobalHpSheet.Row>.Count ,
Sheet<int, WorldBossGlobalHpSheet.Row>.IsReadOnly ,
Sheet<int, WorldBossGlobalHpSheet.Row>.this[int] ,
Sheet<int, WorldBossGlobalHpSheet.Row>.Set(string, bool) ,
Sheet<int, WorldBossGlobalHpSheet.Row>.Set<T>(Sheet<int, T>, bool) ,
Sheet<int, WorldBossGlobalHpSheet.Row>.GetEnumerator() ,
Sheet<int, WorldBossGlobalHpSheet.Row>.TryGetValue(int, out
WorldBossGlobalHpSheet.Row, bool) ,
Sheet<int, WorldBossGlobalHpSheet.Row>.AddRow(int, WorldBossGlobalHpSheet.Row) ,
Sheet<int, WorldBossGlobalHpSheet.Row>.Add(int, WorldBossGlobalHpSheet.Row) ,
Sheet<int, WorldBossGlobalHpSheet.Row>.ContainsKey(int) ,
Sheet<int, WorldBossGlobalHpSheet.Row>.Remove(int) ,

Sheet<int, WorldBossGlobalHpSheet.Row>.TryGetValue(int, out WorldBossGlobalHpSheet.Row) , Sheet<int, WorldBossGlobalHpSheet.Row>.Add(KeyValuePair<int, WorldBossGlobalHpSheet.Row>) , Sheet<int, WorldBossGlobalHpSheet.Row>.Clear() , Sheet<int, WorldBossGlobalHpSheet.Row>.Contains(KeyValuePair<int, WorldBossGlobalHpSheet.Row>) , Sheet<int, WorldBossGlobalHpSheet.Row>.CopyTo(KeyValuePair<int, WorldBossGlobalHpSheet.Row>[], int) , Sheet<int, WorldBossGlobalHpSheet.Row>.Remove(KeyValuePair<int, WorldBossGlobalHpSheet.Row>) , Sheet<int, WorldBossGlobalHpSheet.Row>.Serialize() , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

WorldBossGlobalHpSheet()

```
public WorldBossGlobalHpSheet()
```

Class WorldBossGlobalHpSheet.Row

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
public class WorldBossGlobalHpSheet.Row : SheetRow<int>
```

Inheritance

object ← SheetRow<int> ← WorldBossGlobalHpSheet.Row

Inherited Members

SheetRow<int>.Validate() , SheetRow<int>.EndOfSheetInitialize() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Fields

Hp

```
public BigInteger Hp
```

Field Value

BigInteger

Level

```
public int Level
```

Field Value

int

Properties

Key

```
public override int Key { get; }
```

Property Value

int

Methods

Set(IReadOnlyList<string>)

```
public override void Set(IReadOnlyList<string> fields)
```

Parameters

`fields` IReadOnlyList<string>

Class WorldBossKillRewardSheet

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
public class WorldBossKillRewardSheet : Sheet<int, WorldBossKillRewardSheet.Row>,
IDictionary<int, WorldBossKillRewardSheet.Row>, ICollection<KeyValuePair<int,
WorldBossKillRewardSheet.Row>>, IEnumerable<KeyValuePair<int,
WorldBossKillRewardSheet.Row>>, IEnumerable, IWorldBossRewardSheet, ISheet
```

Inheritance

object ← Sheet<int, WorldBossKillRewardSheet.Row> ← WorldBossKillRewardSheet

Implements

IDictionary<int, WorldBossKillRewardSheet.Row>,
ICollection<KeyValuePair<int, WorldBossKillRewardSheet.Row>>,
IEnumerable<KeyValuePair<int, WorldBossKillRewardSheet.Row>>, IEnumerable,
IWorldBossRewardSheet, ISheet

Inherited Members

Sheet<int, WorldBossKillRewardSheet.Row>.Name ,
Sheet<int, WorldBossKillRewardSheet.Row>.OrderedList ,
Sheet<int, WorldBossKillRewardSheet.Row>.First ,
Sheet<int, WorldBossKillRewardSheet.Row>.Last ,
Sheet<int, WorldBossKillRewardSheet.Row>.Keys ,
Sheet<int, WorldBossKillRewardSheet.Row>.Values ,
Sheet<int, WorldBossKillRewardSheet.Row>.Count ,
Sheet<int, WorldBossKillRewardSheet.Row>.IsReadOnly ,
Sheet<int, WorldBossKillRewardSheet.Row>.this[int] ,
Sheet<int, WorldBossKillRewardSheet.Row>.Set(string, bool) ,
Sheet<int, WorldBossKillRewardSheet.Row>.Set<T>(Sheet<int, T>, bool) ,
Sheet<int, WorldBossKillRewardSheet.Row>.GetEnumerator() ,
Sheet<int, WorldBossKillRewardSheet.Row>.TryGetValue(int, out
WorldBossKillRewardSheet.Row, bool) ,
Sheet<int, WorldBossKillRewardSheet.Row>.AddRow(int, WorldBossKillRewardSheet.Row) ,
Sheet<int, WorldBossKillRewardSheet.Row>.Add(int, WorldBossKillRewardSheet.Row) ,
Sheet<int, WorldBossKillRewardSheet.Row>.ContainsKey(int) ,
Sheet<int, WorldBossKillRewardSheet.Row>.Remove(int) ,

Sheet<int, WorldBossKillRewardSheet.Row>.TryGetValue(int, out WorldBossKillRewardSheet.Row) ,
Sheet<int, WorldBossKillRewardSheet.Row>.Add(KeyValuePair<int, WorldBossKillRewardSheet.Row>) ,
Sheet<int, WorldBossKillRewardSheet.Row>.Clear() ,
Sheet<int, WorldBossKillRewardSheet.Row>.Contains(KeyValuePair<int, WorldBossKillRewardSheet.Row>) ,
Sheet<int, WorldBossKillRewardSheet.Row>.CopyTo(KeyValuePair<int, WorldBossKillRewardSheet.Row>[], int) ,
Sheet<int, WorldBossKillRewardSheet.Row>.Remove(KeyValuePair<int, WorldBossKillRewardSheet.Row>) ,
Sheet<int, WorldBossKillRewardSheet.Row>.Serialize() , object.Equals(object) ,
object.Equals(object, object) , object.GetHashCode() , object.GetType() ,
object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

WorldBossKillRewardSheet()

```
public WorldBossKillRewardSheet()
```

Properties

OrderedRows

```
public IReadOnlyList<IWorldBossRewardRow> OrderedRows { get; }
```

Property Value

IReadOnlyList<IWorldBossRewardRow>

Class WorldBossKillRewardSheet.Row

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
public class WorldBossKillRewardSheet.Row : SheetRow<int>, IWorldBossRewardRow
```

Inheritance

object ← SheetRow<int> ← WorldBossKillRewardSheet.Row

Implements

IWorldBossRewardRow

Inherited Members

SheetRow<int>.Validate() , SheetRow<int>.EndOfSheetInitialize() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Fields

Id

```
public int Id
```

Field Value

int

RuneMax

```
public int RuneMax
```

Field Value

int

RuneMin

```
public int RuneMin
```

Field Value

int

Properties

BossId

```
public int BossId { get; }
```

Property Value

int

Circle

```
public int Circle { get; }
```

Property Value

int

Crystal

```
public int Crystal { get; }
```

Property Value

int

Key

```
public override int Key { get; }
```

Property Value

int

Rank

```
public int Rank { get; }
```

Property Value

int

Rune

```
public int Rune { get; }
```

Property Value

int

Methods

Set(IReadOnlyList<string>)

```
public override void Set(IReadOnlyList<string> fields)
```

Parameters

`fields` IReadOnlyList<string>

SetRune(IRandom)

```
public void SetRune(IRandom random)
```

Parameters

`random` IRandom

Class WorldBossListSheet

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
public class WorldBossListSheet : Sheet<int, WorldBossListSheet.Row>,
IDictionary<int, WorldBossListSheet.Row>, ICollection<KeyValuePair<int,
WorldBossListSheet.Row>>, IEnumerable<KeyValuePair<int, WorldBossListSheet.Row>>,
IEnumerable, ISheet
```

Inheritance

object ← Sheet<int, WorldBossListSheet.Row> ← WorldBossListSheet

Implements

IDictionary<int, WorldBossListSheet.Row>,
ICollection<KeyValuePair<int, WorldBossListSheet.Row>>,
IEnumerable<KeyValuePair<int, WorldBossListSheet.Row>>, IEnumerable, ISheet

Inherited Members

Sheet<int, WorldBossListSheet.Row>.Name ,
Sheet<int, WorldBossListSheet.Row>.OrderedList ,
Sheet<int, WorldBossListSheet.Row>.First , Sheet<int, WorldBossListSheet.Row>.Last ,
Sheet<int, WorldBossListSheet.Row>.Keys , Sheet<int, WorldBossListSheet.Row>.Values ,
Sheet<int, WorldBossListSheet.Row>.Count ,
Sheet<int, WorldBossListSheet.Row>.IsReadOnly ,
Sheet<int, WorldBossListSheet.Row>.this[int] ,
Sheet<int, WorldBossListSheet.Row>.Set(string, bool) ,
Sheet<int, WorldBossListSheet.Row>.Set<T>(Sheet<int, T>, bool) ,
Sheet<int, WorldBossListSheet.Row>.GetEnumerator() ,
Sheet<int, WorldBossListSheet.Row>.TryGetValue(int, out WorldBossListSheet.Row, bool) ,
Sheet<int, WorldBossListSheet.Row>.AddRow(int, WorldBossListSheet.Row) ,
Sheet<int, WorldBossListSheet.Row>.Add(int, WorldBossListSheet.Row) ,
Sheet<int, WorldBossListSheet.Row>.ContainsKey(int) ,
Sheet<int, WorldBossListSheet.Row>.Remove(int) ,
Sheet<int, WorldBossListSheet.Row>.TryGetValue(int, out WorldBossListSheet.Row) ,
Sheet<int, WorldBossListSheet.Row>.Add(KeyValuePair<int, WorldBossListSheet.Row>) ,
Sheet<int, WorldBossListSheet.Row>.Clear() ,
Sheet<int, WorldBossListSheet.Row>.Contains(KeyValuePair<int,
WorldBossListSheet.Row>) ,

Sheet<int, WorldBossListSheet.Row>.CopyTo(KeyValuePair<int, WorldBossListSheet.Row>[], int) ,
Sheet<int, WorldBossListSheet.Row>.Remove(KeyValuePair<int, WorldBossListSheet.Row>) ,
Sheet<int, WorldBossListSheet.Row>.Serialize() , object.Equals(object) ,
object.Equals(object, object) , object.GetHashCode() , object.GetType() ,
object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Extension Methods
SheetsExtensions.FindPreviousRaidIdByBlockIndex(WorldBossListSheet, long) ,
SheetsExtensions.FindPreviousRowByBlockIndex(WorldBossListSheet, long) ,
SheetsExtensions.FindRaidIdByBlockIndex(WorldBossListSheet, long) ,
SheetsExtensions.FindRowByBlockIndex(WorldBossListSheet, long)

Constructors

WorldBossListSheet()

```
public WorldBossListSheet()
```

Class WorldBossListSheet.Row

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
public class WorldBossListSheet.Row : SheetRow<int>
```

Inheritance

object ← SheetRow<int> ← WorldBossListSheet.Row

Inherited Members

SheetRow<int>.Validate() , SheetRow<int>.EndOfSheetInitialize() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Fields

AdditionalTicketPrice

```
public int AdditionalTicketPrice
```

Field Value

int

BossId

```
public int BossId
```

Field Value

int

EndedBlockIndex

```
public long EndedBlockIndex
```

Field Value

long

EntranceFee

```
public int EntranceFee
```

Field Value

int

Id

```
public int Id
```

Field Value

int

MaxPurchaseCount

```
public int MaxPurchaseCount
```

Field Value

int

StartedBlockIndex

```
public long StartedBlockIndex
```

Field Value

long

TicketPrice

```
public int TicketPrice
```

Field Value

int

Properties

Key

```
public override int Key { get; }
```

Property Value

int

Methods

Set(IReadOnlyList<string>)

```
public override void Set(IReadOnlyList<string> fields)
```

Parameters

`fields` IReadOnlyList<string>

Class WorldBossRankRewardSheet

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
public class WorldBossRankRewardSheet : Sheet<int, WorldBossRankRewardSheet.Row>,
IDictionary<int, WorldBossRankRewardSheet.Row>, ICollection<KeyValuePair<int,
WorldBossRankRewardSheet.Row>>, IEnumerable<KeyValuePair<int,
WorldBossRankRewardSheet.Row>>, IEnumerable, IWorldBossRewardSheet, ISheet
```

Inheritance

object ← Sheet<int, WorldBossRankRewardSheet.Row> ← WorldBossRankRewardSheet

Implements

IDictionary<int, WorldBossRankRewardSheet.Row>,
ICollection<KeyValuePair<int, WorldBossRankRewardSheet.Row>>,
IEnumerable<KeyValuePair<int, WorldBossRankRewardSheet.Row>>, IEnumerable,
IWorldBossRewardSheet, ISheet

Inherited Members

Sheet<int, WorldBossRankRewardSheet.Row>.Name ,
Sheet<int, WorldBossRankRewardSheet.Row>.OrderedList ,
Sheet<int, WorldBossRankRewardSheet.Row>.First ,
Sheet<int, WorldBossRankRewardSheet.Row>.Last ,
Sheet<int, WorldBossRankRewardSheet.Row>.Keys ,
Sheet<int, WorldBossRankRewardSheet.Row>.Values ,
Sheet<int, WorldBossRankRewardSheet.Row>.Count ,
Sheet<int, WorldBossRankRewardSheet.Row>.IsReadOnly ,
Sheet<int, WorldBossRankRewardSheet.Row>.this[int] ,
Sheet<int, WorldBossRankRewardSheet.Row>.Set(string, bool) ,
Sheet<int, WorldBossRankRewardSheet.Row>.Set<T>(Sheet<int, T>, bool) ,
Sheet<int, WorldBossRankRewardSheet.Row>.GetEnumerator() ,
Sheet<int, WorldBossRankRewardSheet.Row>.TryGetValue(int, out
WorldBossRankRewardSheet.Row, bool) ,
Sheet<int, WorldBossRankRewardSheet.Row>.AddRow(int,
WorldBossRankRewardSheet.Row) ,
Sheet<int, WorldBossRankRewardSheet.Row>.Add(int, WorldBossRankRewardSheet.Row) ,
Sheet<int, WorldBossRankRewardSheet.Row>.ContainsKey(int) ,
Sheet<int, WorldBossRankRewardSheet.Row>.Remove(int) ,

Sheet<int, WorldBossRankRewardSheet.Row>.TryGetValue(int, out WorldBossRankRewardSheet.Row) , Sheet<int, WorldBossRankRewardSheet.Row>.Add(KeyValuePair<int, WorldBossRankRewardSheet.Row>) , Sheet<int, WorldBossRankRewardSheet.Row>.Clear() , Sheet<int, WorldBossRankRewardSheet.Row>.Contains(KeyValuePair<int, WorldBossRankRewardSheet.Row>) , Sheet<int, WorldBossRankRewardSheet.Row>.CopyTo(KeyValuePair<int, WorldBossRankRewardSheet.Row>[], int) , Sheet<int, WorldBossRankRewardSheet.Row>.Remove(KeyValuePair<int, WorldBossRankRewardSheet.Row>) , Sheet<int, WorldBossRankRewardSheet.Row>.Serialize() , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

WorldBossRankRewardSheet()

```
public WorldBossRankRewardSheet()
```

Properties

OrderedRows

```
public IReadOnlyList<IWorldBossRewardRow> OrderedRows { get; }
```

Property Value

IReadOnlyList<IWorldBossRewardRow>

Class WorldBossRankRewardSheet.Row

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
public class WorldBossRankRewardSheet.Row : SheetRow<int>, IWorldBossRewardRow
```

Inheritance

object ← SheetRow<int> ← WorldBossRankRewardSheet.Row

Implements

IWorldBossRewardRow

Inherited Members

SheetRow<int>.Validate() , SheetRow<int>.EndOfSheetInitialize() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Fields

Id

```
public int Id
```

Field Value

int

Properties

BossId

```
public int BossId { get; }
```

Property Value

int

Circle

```
public int Circle { get; }
```

Property Value

int

Crystal

```
public int Crystal { get; }
```

Property Value

int

Key

```
public override int Key { get; }
```

Property Value

int

Rank

```
public int Rank { get; }
```

Property Value

int

Rune

```
public int Rune { get; }
```

Property Value

int

Methods

Set(IReadOnlyList<string>)

```
public override void Set(IReadOnlyList<string> fields)
```

Parameters

`fields` IReadOnlyList<string>

Class WorldBossRankingRewardSheet

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
public class WorldBossRankingRewardSheet : Sheet<int,
WorldBossRankingRewardSheet.Row>, IDictionary<int, WorldBossRankingRewardSheet.Row>,
ICollection<KeyValuePair<int, WorldBossRankingRewardSheet.Row>>,
IEnumerable<KeyValuePair<int, WorldBossRankingRewardSheet.Row>>, IEnumerable, ISheet
```

Inheritance

object ← Sheet<int, WorldBossRankingRewardSheet.Row> ← WorldBossRankingRewardSheet

Implements

IDictionary<int, WorldBossRankingRewardSheet.Row>, ICollection<KeyValuePair<int, WorldBossRankingRewardSheet.Row>>, IEnumerable<KeyValuePair<int, WorldBossRankingRewardSheet.Row>>, IEnumerable, ISheet

Inherited Members

Sheet<int, WorldBossRankingRewardSheet.Row>.Name ,
Sheet<int, WorldBossRankingRewardSheet.Row>.OrderedList ,
Sheet<int, WorldBossRankingRewardSheet.Row>.First ,
Sheet<int, WorldBossRankingRewardSheet.Row>.Last ,
Sheet<int, WorldBossRankingRewardSheet.Row>.Keys ,
Sheet<int, WorldBossRankingRewardSheet.Row>.Values ,
Sheet<int, WorldBossRankingRewardSheet.Row>.Count ,
Sheet<int, WorldBossRankingRewardSheet.Row>.IsReadOnly ,
Sheet<int, WorldBossRankingRewardSheet.Row>.this[int] ,
Sheet<int, WorldBossRankingRewardSheet.Row>.Set(string, bool) ,
Sheet<int, WorldBossRankingRewardSheet.Row>.Set<T>(Sheet<int, T>, bool) ,
Sheet<int, WorldBossRankingRewardSheet.Row>.GetEnumerator() ,
Sheet<int, WorldBossRankingRewardSheet.Row>.TryGetValue(int, out WorldBossRankingRewardSheet.Row, bool) ,
Sheet<int, WorldBossRankingRewardSheet.Row>.AddRow(int, WorldBossRankingRewardSheet.Row) ,
Sheet<int, WorldBossRankingRewardSheet.Row>.Add(int, WorldBossRankingRewardSheet.Row) ,

Sheet<int, WorldBossRankingRewardSheet.Row>.ContainsKey(int) ,
Sheet<int, WorldBossRankingRewardSheet.Row>.Remove(int) ,
Sheet<int, WorldBossRankingRewardSheet.Row>.TryGetValue(int, out
WorldBossRankingRewardSheet.Row) ,
Sheet<int, WorldBossRankingRewardSheet.Row>.Add(KeyValuePair<int,
WorldBossRankingRewardSheet.Row>) ,
Sheet<int, WorldBossRankingRewardSheet.Row>.Clear() ,
Sheet<int, WorldBossRankingRewardSheet.Row>.Contains(KeyValuePair<int,
WorldBossRankingRewardSheet.Row>) ,
Sheet<int, WorldBossRankingRewardSheet.Row>.CopyTo(KeyValuePair<int,
WorldBossRankingRewardSheet.Row>[], int) ,
Sheet<int, WorldBossRankingRewardSheet.Row>.Remove(KeyValuePair<int,
WorldBossRankingRewardSheet.Row>) ,
Sheet<int, WorldBossRankingRewardSheet.Row>.Serialize() , object.Equals(object) ,
object.Equals(object, object) , object.GetHashCode() , object.GetType() ,
object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

WorldBossRankingRewardSheet()

```
public WorldBossRankingRewardSheet()
```

Methods

FindRow(int, int, int)

```
public WorldBossRankingRewardSheet.Row FindRow(int bossId, int ranking, int rate)
```

Parameters

`bossId` int

`ranking` int

`rate` int

Returns

WorldBossRankingRewardSheet.Row

Class WorldBossRankingRewardSheet.Row

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
public class WorldBossRankingRewardSheet.Row : SheetRow<int>
```

Inheritance

object ← SheetRow<int> ← WorldBossRankingRewardSheet.Row

Inherited Members

SheetRow<int>.Validate() , SheetRow<int>.EndOfSheetInitialize() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Fields

BossId

```
public int BossId
```

Field Value

int

Crystal

```
public int Crystal
```

Field Value

int

Id

```
public int Id
```

Field Value

int

Materials

```
public List<(int itemId, int quantity)> Materials
```

Field Value

List<(int id, int count)>

RankingMax

```
public int RankingMax
```

Field Value

int

RankingMin

```
public int RankingMin
```

Field Value

int

RateMax

```
public int RateMax
```

Field Value

int

RateMin

```
public int RateMin
```

Field Value

int

Runes

```
public List<WorldBossRankingRewardSheet.Row.RuneInfo> Runes
```

Field Value

List<WorldBossRankingRewardSheet.Row.RuneInfo>

Properties

Key

```
public override int Key { get; }
```

Property Value

int

Methods

GetRewards(RuneSheet, MaterialItemSheet)

```
public List<FungibleAssetValue> GetRewards(RuneSheet runeSheet,
MaterialItemSheet materialSheet)
```

Parameters

`runeSheet` RuneSheet

`materialSheet` MaterialItemSheet

Returns

List<FungibleAssetValue>

Set(IReadOnlyList<string>)

```
public override void Set(IReadOnlyList<string> fields)
```

Parameters

`fields` IReadOnlyList<string>

Struct WorldBossRankingRewardSheet.Row.RuneInfo

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
public struct WorldBossRankingRewardSheet.Row.RuneInfo
```

Inherited Members

ValueType.Equals(object) , ValueType.GetHashCode() , ValueType.ToString() , object.Equals(object, object) , object.GetType() , object.ReferenceEquals(object, object)

Constructors

RuneInfo(int, int)

```
public RuneInfo(int id, int qty)
```

Parameters

`id` int

`qty` int

Fields

RuneId

```
public int RuneId
```

Field Value

int

RuneQty

```
public int RuneQty
```

Field Value

int

Class WorldBossStatSheetExtension

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
public static class WorldBossStatSheetExtension
```

Inheritance

object ← WorldBossStatSheetExtension

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Methods

ToStats(WaveStatData)

```
public static StatsMap ToStats(this WorldBossCharacterSheet.WaveStatData statData)
```

Parameters

`statData` WorldBossCharacterSheet.WaveStatData

Returns

StatsMap

Class WorldOpenConditionSheet

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

The conditions that open a WorldOpenSheet opening: within one `condition_group` every counter must reach its threshold, and any one group being satisfied opens it (AND within a group, OR across groups).

Columns: `open_id`, `condition_group`, `counter_id`, `threshold`. Cross-sheet references (`open_id`, `counter_id`) are not validated here.

`open_id` is not unique, so a row's key is a synthetic sequential number assigned in the order rows are added (CSV order, starting at 1). Look rows up with GroupsOf(int) or OpenIds Referencing(int) rather than by key. A blank line is kept as an invalid row (open_id 0); both lookups skip invalid rows, so code reading `OrderedList` directly must check IsValid too.

A patch cannot change what an id means once recorded; see WorldOpenIdentity.

```
[Serializable]
public class WorldOpenConditionSheet : Sheet<int, WorldOpenConditionSheet.Row>,
IDictionary<int, WorldOpenConditionSheet.Row>, ICollection<KeyValuePair<int,
WorldOpenConditionSheet.Row>>, IEnumerable<KeyValuePair<int,
WorldOpenConditionSheet.Row>>, IEnumerable, ISheet
```

Inheritance

object ← Sheet<int, WorldOpenConditionSheet.Row> ← WorldOpenConditionSheet

Implements

IDictionary<int, WorldOpenConditionSheet.Row>,
ICollection<KeyValuePair<int, WorldOpenConditionSheet.Row>>,
IEnumerable<KeyValuePair<int, WorldOpenConditionSheet.Row>>, IEnumerable,
ISheet

Inherited Members

Sheet<int, WorldOpenConditionSheet.Row>.Name ,
Sheet<int, WorldOpenConditionSheet.Row>.OrderedList ,
Sheet<int, WorldOpenConditionSheet.Row>.First ,
Sheet<int, WorldOpenConditionSheet.Row>.Last ,
Sheet<int, WorldOpenConditionSheet.Row>.Keys ,
Sheet<int, WorldOpenConditionSheet.Row>.Values ,

Sheet<int, WorldOpenConditionSheet.Row>.Count ,
Sheet<int, WorldOpenConditionSheet.Row>.IsReadOnly ,
Sheet<int, WorldOpenConditionSheet.Row>.this[int] ,
Sheet<int, WorldOpenConditionSheet.Row>.Set(string, bool) ,
Sheet<int, WorldOpenConditionSheet.Row>.Set<T>(Sheet<int, T>, bool) ,
Sheet<int, WorldOpenConditionSheet.Row>.GetEnumerator() ,
Sheet<int, WorldOpenConditionSheet.Row>.TryGetValue(int, out WorldOpenConditionSheet.Row, bool) ,
Sheet<int, WorldOpenConditionSheet.Row>.AddRow(int, WorldOpenConditionSheet.Row) ,
Sheet<int, WorldOpenConditionSheet.Row>.Add(int, WorldOpenConditionSheet.Row) ,
Sheet<int, WorldOpenConditionSheet.Row>.ContainsKey(int) ,
Sheet<int, WorldOpenConditionSheet.Row>.Remove(int) ,
Sheet<int, WorldOpenConditionSheet.Row>.TryGetValue(int, out WorldOpenConditionSheet.Row) ,
Sheet<int, WorldOpenConditionSheet.Row>.Add(KeyValuePair<int, WorldOpenConditionSheet.Row>) ,
Sheet<int, WorldOpenConditionSheet.Row>.Clear() ,
Sheet<int, WorldOpenConditionSheet.Row>.Contains(KeyValuePair<int, WorldOpenConditionSheet.Row>) ,
Sheet<int, WorldOpenConditionSheet.Row>.CopyTo(KeyValuePair<int, WorldOpenConditionSheet.Row>[], int) ,
Sheet<int, WorldOpenConditionSheet.Row>.Remove(KeyValuePair<int, WorldOpenConditionSheet.Row>) ,
Sheet<int, WorldOpenConditionSheet.Row>.Serialize() , object.Equals(object) ,
object.Equals(object, object) , object.GetHashCode() , object.GetType() ,
object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

WorldOpenConditionSheet()

Creates an empty sheet.

```
public WorldOpenConditionSheet()
```

Fields

ColumnNames

Column names in their expected order, after _ prefixed columns are dropped.

```
public static readonly string[] ColumnNames
```

Field Value

string[]

Methods

AddRow(int, Row)

Assigns the synthetic key. Rows are numbered in the order they are added.

```
protected override void AddRow(int key, WorldOpenConditionSheet.Row value)
```

Parameters

`key` int

`value` WorldOpenConditionSheet.Row

GroupsOf(int)

The condition groups of `openId`, each as the rows (counters) of that group, ordered by group then by row order.

```
public IEnumerable<IReadOnlyList<WorldOpenConditionSheet.Row>> GroupsOf(int openId)
```

Parameters

`openId` int

Returns

IEnumerable<IReadOnlyList<WorldOpenConditionSheet.Row>>

OpenIdsReferencing(int)

The ids of the openings that have a condition on `counterId`.

```
public IEnumerable<int> OpenIdsReferencing(int counterId)
```

Parameters

`counterId` int

Returns

IEnumerable<int>

ValidateCsv(string)

Parses `csv` strictly so that a malformed sheet fails the `patch_table_sheet` transaction instead of landing on the chain.

```
public static void ValidateCsv(string csv)
```

Parameters

`csv` string

Remarks

References to other sheets are not checked here; the backoffice does that.

Exceptions

SheetRowValidateException

> The header does not match ColumnNames, a cell does not parse, or `open_id`, `condition_group`, `counter_id` or `threshold` is not positive.

Class WorldOpenConditionSheet.Row

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

One counter requirement of a condition group.

```
[Serializable]
public class WorldOpenConditionSheet.Row : SheetRow<int>
```

Inheritance

object ← SheetRow<int> ← WorldOpenConditionSheet.Row

Inherited Members

SheetRow<int>.Validate() , SheetRow<int>.EndOfSheetInitialize() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Properties

ConditionGroup

The group; counters of one group are AND-ed, groups are OR-ed.

```
public int ConditionGroup { get; }
```

Property Value

int

CounterId

The GlobalCounterSheet counter that is measured.

```
public int CounterId { get; }
```

Property Value

int

IsValid

Whether every cell parsed and is in range.

```
public bool IsValid { get; }
```

Property Value

bool

Key

The synthetic row number, assigned when the row is added to the sheet.

```
public override int Key { get; }
```

Property Value

int

OpenId

The opening this condition belongs to.

```
public int OpenId { get; }
```

Property Value

int

Threshold

The count the counter must reach.

```
public long Threshold { get; }
```

Property Value

long

Methods

Set(IReadOnlyList<string>)

```
public override void Set(IReadOnlyList<string> fields)
```

Parameters

`fields` IReadOnlyList<string>

Class WorldOpenSheet

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

Defines when each hard world segment opens: up to which stage, which opening it follows, and the block window (earliest, fallback, activation delay) around it. The conditions that open it are in WorldOpenConditionSheet.

Columns: `id`, `world_id`, `stage_end`, `prerequisite_open_id` (empty for none), `min_block_index`, `fallback_block_index`, `effective_delay_block`.

Set(IReadOnlyList<string>) does not validate across rows, so a test can build an invalid sheet directly; ValidateCsv(string) does, at patch time.

A patch cannot change what an id means once recorded; see WorldOpenIdentity.

```
[Serializable]
public class WorldOpenSheet : Sheet<int, WorldOpenSheet.Row>, IDictionary<int,
WorldOpenSheet.Row>, ICollection<KeyValuePair<int, WorldOpenSheet.Row>>,
IEnumerable<KeyValuePair<int, WorldOpenSheet.Row>>, IEnumerable, ISheet
```

Inheritance

object ← Sheet<int, WorldOpenSheet.Row> ← WorldOpenSheet

Implements

IDictionary<int, WorldOpenSheet.Row>,
ICollection<KeyValuePair<int, WorldOpenSheet.Row>>,
IEnumerable<KeyValuePair<int, WorldOpenSheet.Row>>, IEnumerable, ISheet

Inherited Members

Sheet<int, WorldOpenSheet.Row>.Name , Sheet<int, WorldOpenSheet.Row>.OrderedList ,
Sheet<int, WorldOpenSheet.Row>.First , Sheet<int, WorldOpenSheet.Row>.Last ,
Sheet<int, WorldOpenSheet.Row>.Keys , Sheet<int, WorldOpenSheet.Row>.Values ,
Sheet<int, WorldOpenSheet.Row>.Count , Sheet<int, WorldOpenSheet.Row>.IsReadOnly ,
Sheet<int, WorldOpenSheet.Row>.this[int] ,
Sheet<int, WorldOpenSheet.Row>.Set(string, bool) ,
Sheet<int, WorldOpenSheet.Row>.Set<T>(Sheet<int, T>, bool) ,
Sheet<int, WorldOpenSheet.Row>.GetEnumerator() ,
Sheet<int, WorldOpenSheet.Row>.TryGetValue(int, out WorldOpenSheet.Row, bool) ,
Sheet<int, WorldOpenSheet.Row>.AddRow(int, WorldOpenSheet.Row) ,

Sheet<int, WorldOpenSheet.Row>.Add(int, WorldOpenSheet.Row) ,
Sheet<int, WorldOpenSheet.Row>.ContainsKey(int) ,
Sheet<int, WorldOpenSheet.Row>.Remove(int) ,
Sheet<int, WorldOpenSheet.Row>.TryGetValue(int, out WorldOpenSheet.Row) ,
Sheet<int, WorldOpenSheet.Row>.Add(KeyValuePair<int, WorldOpenSheet.Row>) ,
Sheet<int, WorldOpenSheet.Row>.Clear() ,
Sheet<int, WorldOpenSheet.Row>.Contains(KeyValuePair<int, WorldOpenSheet.Row>) ,
Sheet<int, WorldOpenSheet.Row>.CopyTo(KeyValuePair<int, WorldOpenSheet.Row>[], int) ,
Sheet<int, WorldOpenSheet.Row>.Remove(KeyValuePair<int, WorldOpenSheet.Row>) ,
Sheet<int, WorldOpenSheet.Row>.Serialize() , object.Equals(object) ,
object.Equals(object, object) , object.GetHashCode() , object.GetType() ,
object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

WorldOpenSheet()

Creates an empty sheet.

```
public WorldOpenSheet()
```

Fields

ColumnNames

Column names in their expected order, after _ prefixed columns are dropped.

```
public static readonly string[] ColumnNames
```

Field Value

string[]

Methods

AddRow(int, Row)

Adds a parsed row. A row without a usable id is dropped and a duplicated id keeps the first row instead of throwing.

```
protected override void AddRow(int key, WorldOpenSheet.Row value)
```

Parameters

`key` int

`value` WorldOpenSheet.Row

ByWorld(int)

The openings of `worldId`.

```
public IEnumerable<WorldOpenSheet.Row> ByWorld(int worldId)
```

Parameters

`worldId` int

Returns

IEnumerable<WorldOpenSheet.Row>

ValidateCsv(string)

Parses `csv` strictly so that a malformed sheet fails the `patch_table_sheet` transaction instead of landing on the chain.

```
public static void ValidateCsv(string csv)
```

Parameters

`csv` string

Remarks

References to other sheets (conditions, counters) are not checked here.

Exceptions

SheetRowValidateException

The header does not match ColumnNames; an id is not a positive unique integer; a cell does not parse; `fallback_block_index` is not positive; `min_block_index` is negative or greater than the fallback; `effective_delay_block` is negative or greater than the fallback; a prerequisite is unknown, refers to itself or is part of a cycle; `stage_end` does not increase along a prerequisite; or a row other than the one with the lowest `stage_end` does not take the row just below it (by `stage_end`, across all worlds) as its prerequisite.

Class WorldOpenSheet.Row

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

One opening.

```
[Serializable]
public class WorldOpenSheet.Row : SheetRow<int>
```

Inheritance

object ← SheetRow<int> ← WorldOpenSheet.Row

Inherited Members

SheetRow<int>.Validate() , SheetRow<int>.EndOfSheetInitialize() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Properties

EffectiveDelayBlock

Blocks between the opening being decided and taking effect.

```
public long EffectiveDelayBlock { get; }
```

Property Value

long

FallbackBlockIndex

Block at which the opening happens regardless of conditions.

```
public long FallbackBlockIndex { get; }
```

Property Value

long

Id

The opening id.

```
public int Id { get; }
```

Property Value

int

IsValid

Whether every cell parsed.

```
public bool IsValid { get; }
```

Property Value

bool

Key

```
public override int Key { get; }
```

Property Value

int

MinBlockIndex

Earliest block the opening may happen at.

```
public long MinBlockIndex { get; }
```

Property Value

long

PrerequisiteOpenId

The opening that must happen first, or `null`.

```
public int? PrerequisiteOpenId { get; }
```

Property Value

int?

StageEnd

The last stage this opening makes playable.

```
public int StageEnd { get; }
```

Property Value

int

WorldId

The world this opening belongs to.

```
public int WorldId { get; }
```

Property Value

int

Methods

Set(IReadOnlyList<string>)

```
public override void Set(IReadOnlyList<string> fields)
```

Parameters

`fields` IReadOnlyList<string>

Class WorldQuestSheet

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class WorldQuestSheet : Sheet<int, WorldQuestSheet.Row>, IDictionary<int,
WorldQuestSheet.Row>, ICollection<KeyValuePair<int, WorldQuestSheet.Row>>,
IEnumerable<KeyValuePair<int, WorldQuestSheet.Row>>, IEnumerable, ISheet
```

Inheritance

object ← Sheet<int, WorldQuestSheet.Row> ← WorldQuestSheet

Implements

IDictionary<int, WorldQuestSheet.Row>,
ICollection<KeyValuePair<int, WorldQuestSheet.Row>>,
IEnumerable<KeyValuePair<int, WorldQuestSheet.Row>>, IEnumerable, ISheet

Inherited Members

Sheet<int, WorldQuestSheet.Row>.Name , Sheet<int, WorldQuestSheet.Row>.OrderedList ,
Sheet<int, WorldQuestSheet.Row>.First , Sheet<int, WorldQuestSheet.Row>.Last ,
Sheet<int, WorldQuestSheet.Row>.Keys , Sheet<int, WorldQuestSheet.Row>.Values ,
Sheet<int, WorldQuestSheet.Row>.Count , Sheet<int, WorldQuestSheet.Row>.IsReadOnly ,
Sheet<int, WorldQuestSheet.Row>.this[int] ,
Sheet<int, WorldQuestSheet.Row>.Set(string, bool) ,
Sheet<int, WorldQuestSheet.Row>.Set<T>(Sheet<int, T>, bool) ,
Sheet<int, WorldQuestSheet.Row>.GetEnumerator() ,
Sheet<int, WorldQuestSheet.Row>.TryGetValue(int, out WorldQuestSheet.Row, bool) ,
Sheet<int, WorldQuestSheet.Row>.AddRow(int, WorldQuestSheet.Row) ,
Sheet<int, WorldQuestSheet.Row>.Add(int, WorldQuestSheet.Row) ,
Sheet<int, WorldQuestSheet.Row>.ContainsKey(int) ,
Sheet<int, WorldQuestSheet.Row>.Remove(int) ,
Sheet<int, WorldQuestSheet.Row>.TryGetValue(int, out WorldQuestSheet.Row) ,
Sheet<int, WorldQuestSheet.Row>.Add(KeyValuePair<int, WorldQuestSheet.Row>) ,
Sheet<int, WorldQuestSheet.Row>.Clear() ,
Sheet<int, WorldQuestSheet.Row>.Contains(KeyValuePair<int, WorldQuestSheet.Row>) ,
Sheet<int, WorldQuestSheet.Row>.CopyTo(KeyValuePair<int, WorldQuestSheet.Row>[], int)
,
Sheet<int, WorldQuestSheet.Row>.Remove(KeyValuePair<int, WorldQuestSheet.Row>) ,

Sheet<int, WorldQuestSheet.Row>.Serialize() , object.Equals(object) ,
object.Equals(object, object) , object.GetHashCode() , object.GetType() ,
object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

WorldQuestSheet()

```
public WorldQuestSheet()
```

Class WorldQuestSheet.Row

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class WorldQuestSheet.Row : QuestSheet.Row, IState
```

Inheritance

object ← SheetRow<int> ← QuestSheet.Row ← WorldQuestSheet.Row

Implements

IState

Inherited Members

QuestSheet.Row.Key , QuestSheet.Row.Id , QuestSheet.Row.Goal , QuestSheet.Row.QuestRewardId , QuestSheet.Row.Set(IReadOnlyList<string>) , QuestSheet.Row.Serialize() , QuestSheet.Row.Deserialize(Dictionary) , SheetRow<int>.Validate() , SheetRow<int>.EndOfSheetInitialize() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Class WorldSheet

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class WorldSheet : Sheet<int, WorldSheet.Row>, IDictionary<int,
WorldSheet.Row>, ICollection<KeyValuePair<int, WorldSheet.Row>>,
IEnumerable<KeyValuePair<int, WorldSheet.Row>>, IEnumerable, ISheet
```

Inheritance

object ← Sheet<int, WorldSheet.Row> ← WorldSheet

Implements

IDictionary<int, WorldSheet.Row>,
ICollection<KeyValuePair<int, WorldSheet.Row>>,
IEnumerable<KeyValuePair<int, WorldSheet.Row>>, IEnumerable, ISheet

Inherited Members

Sheet<int, WorldSheet.Row>.Name , Sheet<int, WorldSheet.Row>.OrderedList ,
Sheet<int, WorldSheet.Row>.First , Sheet<int, WorldSheet.Row>.Last ,
Sheet<int, WorldSheet.Row>.Keys , Sheet<int, WorldSheet.Row>.Values ,
Sheet<int, WorldSheet.Row>.Count , Sheet<int, WorldSheet.Row>.IsReadOnly ,
Sheet<int, WorldSheet.Row>.this[int] , Sheet<int, WorldSheet.Row>.Set(string, bool) ,
Sheet<int, WorldSheet.Row>.Set<T>(Sheet<int, T>, bool) ,
Sheet<int, WorldSheet.Row>.GetEnumerator() ,
Sheet<int, WorldSheet.Row>.TryGetValue(int, out WorldSheet.Row, bool) ,
Sheet<int, WorldSheet.Row>.AddRow(int, WorldSheet.Row) ,
Sheet<int, WorldSheet.Row>.Add(int, WorldSheet.Row) ,
Sheet<int, WorldSheet.Row>.ContainsKey(int) , Sheet<int, WorldSheet.Row>.Remove(int) ,
Sheet<int, WorldSheet.Row>.TryGetValue(int, out WorldSheet.Row) ,
Sheet<int, WorldSheet.Row>.Add(KeyValuePair<int, WorldSheet.Row>) ,
Sheet<int, WorldSheet.Row>.Clear() ,
Sheet<int, WorldSheet.Row>.Contains(KeyValuePair<int, WorldSheet.Row>) ,
Sheet<int, WorldSheet.Row>.CopyTo(KeyValuePair<int, WorldSheet.Row>[], int) ,
Sheet<int, WorldSheet.Row>.Remove(KeyValuePair<int, WorldSheet.Row>) ,
Sheet<int, WorldSheet.Row>.Serialize() , object.Equals(object) ,
object.Equals(object, object) , object.GetHashCode() , object.GetType() ,
object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

WorldSheet()

```
public WorldSheet()
```

Class WorldSheet.Row

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class WorldSheet.Row : SheetRow<int>
```

Inheritance

object ← SheetRow<int> ← WorldSheet.Row

Derived

EventDungeonSheet.Row

Inherited Members

SheetRow<int>.Validate() , SheetRow<int>.EndOfSheetInitialize() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Properties

Id

```
public int Id { get; }
```

Property Value

int

Key

```
public override int Key { get; }
```

Property Value

int

Name

```
public string Name { get; }
```

Property Value

string

StageBegin

```
public int StageBegin { get; }
```

Property Value

int

StageEnd

```
public int StageEnd { get; }
```

Property Value

int

StagesCount

```
public int StagesCount { get; }
```

Property Value

int

Methods

Set(IReadOnlyList<string>)

```
public override void Set(IReadOnlyList<string> fields)
```

Parameters

`fields` IReadOnlyList<string>

Class WorldUnlockSheet

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class WorldUnlockSheet : Sheet<int, WorldUnlockSheet.Row>, IDictionary<int,
WorldUnlockSheet.Row>, ICollection<KeyValuePair<int, WorldUnlockSheet.Row>>,
IEnumerable<KeyValuePair<int, WorldUnlockSheet.Row>>, IEnumerable, ISheet
```

Inheritance

object ← Sheet<int, WorldUnlockSheet.Row> ← WorldUnlockSheet

Implements

IDictionary<int, WorldUnlockSheet.Row>,
ICollection<KeyValuePair<int, WorldUnlockSheet.Row>>,
IEnumerable<KeyValuePair<int, WorldUnlockSheet.Row>>, IEnumerable, ISheet

Inherited Members

Sheet<int, WorldUnlockSheet.Row>.Name ,
Sheet<int, WorldUnlockSheet.Row>.OrderedList , Sheet<int, WorldUnlockSheet.Row>.First ,
Sheet<int, WorldUnlockSheet.Row>.Last , Sheet<int, WorldUnlockSheet.Row>.Keys ,
Sheet<int, WorldUnlockSheet.Row>.Values , Sheet<int, WorldUnlockSheet.Row>.Count ,
Sheet<int, WorldUnlockSheet.Row>.IsReadOnly ,
Sheet<int, WorldUnlockSheet.Row>.this[int] ,
Sheet<int, WorldUnlockSheet.Row>.Set(string, bool) ,
Sheet<int, WorldUnlockSheet.Row>.Set<T>(Sheet<int, T>, bool) ,
Sheet<int, WorldUnlockSheet.Row>.GetEnumerator() ,
Sheet<int, WorldUnlockSheet.Row>.TryGetValue(int, out WorldUnlockSheet.Row, bool) ,
Sheet<int, WorldUnlockSheet.Row>.AddRow(int, WorldUnlockSheet.Row) ,
Sheet<int, WorldUnlockSheet.Row>.Add(int, WorldUnlockSheet.Row) ,
Sheet<int, WorldUnlockSheet.Row>.ContainsKey(int) ,
Sheet<int, WorldUnlockSheet.Row>.Remove(int) ,
Sheet<int, WorldUnlockSheet.Row>.TryGetValue(int, out WorldUnlockSheet.Row) ,
Sheet<int, WorldUnlockSheet.Row>.Add(KeyValuePair<int, WorldUnlockSheet.Row>) ,
Sheet<int, WorldUnlockSheet.Row>.Clear() ,
Sheet<int, WorldUnlockSheet.Row>.Contains(KeyValuePair<int, WorldUnlockSheet.Row>) ,
Sheet<int, WorldUnlockSheet.Row>.CopyTo(KeyValuePair<int, WorldUnlockSheet.Row>[],
int) ,

Sheet<int, WorldUnlockSheet.Row>.Remove(KeyValuePair<int, WorldUnlockSheet.Row>) , Sheet<int, WorldUnlockSheet.Row>.Serialize() , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

WorldUnlockSheet()

```
public WorldUnlockSheet()
```

Methods

TryGetUnlockedInformation(int, int, out List<int>)

```
public bool TryGetUnlockedInformation(int clearedWorldId, int clearedStageId, out
List<int> worldIdsToUnlock)
```

Parameters

`clearedWorldId` int

`clearedStageId` int

`worldIdsToUnlock` List<int>

Returns

bool

Class WorldUnlockSheet.Row

Namespace: Nekoyume.TableData

Assembly: Lib9c.dll

```
[Serializable]
public class WorldUnlockSheet.Row : SheetRow<int>
```

Inheritance

object ← SheetRow<int> ← WorldUnlockSheet.Row

Inherited Members

SheetRow<int>.Validate() , SheetRow<int>.EndOfSheetInitialize() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Properties

CRYSTAL

```
public int CRYSTAL { get; }
```

Property Value

int

Id

```
public int Id { get; }
```

Property Value

int

Key

```
public override int Key { get; }
```

Property Value

int

StageId

```
public int StageId { get; }
```

Property Value

int

WorldId

```
public int WorldId { get; }
```

Property Value

int

WorldIdToUnlock

```
public int WorldIdToUnlock { get; }
```

Property Value

int

Methods

Set(IReadOnlyList<string>)

```
public override void Set(IReadOnlyList<string> fields)
```

Parameters

`fields` IReadOnlyList<string>

Namespace Nekoyume.TableData.AdventureBoss

Classes

AdventureBossContributionRewardSheet

AdventureBossContributionRewardSheet.Row

AdventureBossFloorFirstRewardSheet

AdventureBossFloorFirstRewardSheet.Row

AdventureBossFloorPointSheet

AdventureBossFloorPointSheet.Row

AdventureBossFloorSheet

AdventureBossFloorSheet.RewardData

AdventureBossFloorSheet.Row

AdventureBossFloorWaveSheet

AdventureBossFloorWaveSheet.MonsterData

AdventureBossFloorWaveSheet.Row

AdventureBossFloorWaveSheet.WaveData

AdventureBossNcgRewardRatioSheet

AdventureBossNcgRewardRatioSheet.Row

AdventureBossSheet

AdventureBossSheet.RewardAmountData

AdventureBossSheet.RewardRatioData

AdventureBossSheet.Row

AdventureBossUnlockFloorCostSheet

AdventureBossUnlockFloorCostSheet.Row

AdventureBossWantedRewardSheet

AdventureBossWantedRewardSheet.Row

Class AdventureBossContributionRewardSheet

Namespace: Nekoyume.TableData.AdventureBoss

Assembly: Lib9c.dll

```
[Serializable]
public class AdventureBossContributionRewardSheet : Sheet<int,
AdventureBossContributionRewardSheet.Row>, IDictionary<int,
AdventureBossContributionRewardSheet.Row>, ICollection<KeyValuePair<int,
AdventureBossContributionRewardSheet.Row>>, IEnumerable<KeyValuePair<int,
AdventureBossContributionRewardSheet.Row>>, IEnumerable, ISheet
```

Inheritance

object ← Sheet<int, AdventureBossContributionRewardSheet.Row> ← AdventureBossContributionRewardSheet

Implements

IDictionary<int, AdventureBossContributionRewardSheet.Row>, ICollection<KeyValuePair<int, AdventureBossContributionRewardSheet.Row>>, IEnumerable<KeyValuePair<int, AdventureBossContributionRewardSheet.Row>>, IEnumerable, ISheet

Inherited Members

Sheet<int, AdventureBossContributionRewardSheet.Row>.Name ,
Sheet<int, AdventureBossContributionRewardSheet.Row>.OrderedList ,
Sheet<int, AdventureBossContributionRewardSheet.Row>.First ,
Sheet<int, AdventureBossContributionRewardSheet.Row>.Last ,
Sheet<int, AdventureBossContributionRewardSheet.Row>.Keys ,
Sheet<int, AdventureBossContributionRewardSheet.Row>.Values ,
Sheet<int, AdventureBossContributionRewardSheet.Row>.Count ,
Sheet<int, AdventureBossContributionRewardSheet.Row>.IsReadOnly ,
Sheet<int, AdventureBossContributionRewardSheet.Row>.this[int] ,
Sheet<int, AdventureBossContributionRewardSheet.Row>.Set(string, bool) ,
Sheet<int, AdventureBossContributionRewardSheet.Row>.Set<T>(Sheet<int, T>, bool) ,
Sheet<int, AdventureBossContributionRewardSheet.Row>.GetEnumerator() ,
Sheet<int, AdventureBossContributionRewardSheet.Row>.TryGetValue(int, out AdventureBossContributionRewardSheet.Row, bool) ,

Sheet<int, AdventureBossContributionRewardSheet.Row>.AddRow(int, AdventureBossContributionRewardSheet.Row) , Sheet<int, AdventureBossContributionRewardSheet.Row>.Add(int, AdventureBossContributionRewardSheet.Row) , Sheet<int, AdventureBossContributionRewardSheet.Row>.ContainsKey(int) , Sheet<int, AdventureBossContributionRewardSheet.Row>.Remove(int) , Sheet<int, AdventureBossContributionRewardSheet.Row>.TryGetValue(int, out AdventureBossContributionRewardSheet.Row) , Sheet<int, AdventureBossContributionRewardSheet.Row>.Add(KeyValuePair<int, AdventureBossContributionRewardSheet.Row>) , Sheet<int, AdventureBossContributionRewardSheet.Row>.Clear() , Sheet<int, AdventureBossContributionRewardSheet.Row>.Contains(KeyValuePair<int, AdventureBossContributionRewardSheet.Row>) , Sheet<int, AdventureBossContributionRewardSheet.Row>.CopyTo(KeyValuePair<int, AdventureBossContributionRewardSheet.Row>[], int) , Sheet<int, AdventureBossContributionRewardSheet.Row>.Remove(KeyValuePair<int, AdventureBossContributionRewardSheet.Row>) , Sheet<int, AdventureBossContributionRewardSheet.Row>.Serialize() , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

AdventureBossContributionRewardSheet()

```
public AdventureBossContributionRewardSheet()
```

Class AdventureBossContributionRewardSheet.Row

Namespace: Nekoyume.TableData.AdventureBoss

Assembly: Lib9c.dll

```
[Serializable]
public class AdventureBossContributionRewardSheet.Row : SheetRow<int>
```

Inheritance

object ← SheetRow<int> ← AdventureBossContributionRewardSheet.Row

Inherited Members

SheetRow<int>.Validate() , SheetRow<int>.EndOfSheetInitialize() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Fields

AdventureBossId

```
public int AdventureBossId
```

Field Value

int

Id

```
public int Id
```

Field Value

int

Rewards

```
public List<AdventureBossSheet.RewardRatioData> Rewards
```

Field Value

List<AdventureBossSheet.RewardRatioData>

Properties

Key

```
public override int Key { get; }
```

Property Value

int

Methods

Set(IReadOnlyList<string>)

```
public override void Set(IReadOnlyList<string> fields)
```

Parameters

`fields` IReadOnlyList<string>

Class AdventureBossFloorFirstRewardSheet

Namespace: Nekoyume.TableData.AdventureBoss

Assembly: Lib9c.dll

```
[Serializable]
public class AdventureBossFloorFirstRewardSheet : Sheet<int,
AdventureBossFloorFirstRewardSheet.Row>, IDictionary<int,
AdventureBossFloorFirstRewardSheet.Row>, ICollection<KeyValuePair<int,
AdventureBossFloorFirstRewardSheet.Row>>, IEnumerable<KeyValuePair<int,
AdventureBossFloorFirstRewardSheet.Row>>, IEnumerable, ISheet
```

Inheritance

object ← Sheet<int, AdventureBossFloorFirstRewardSheet.Row> ← AdventureBossFloorFirstRewardSheet

Implements

IDictionary<int, AdventureBossFloorFirstRewardSheet.Row>, ICollection<KeyValuePair<int, AdventureBossFloorFirstRewardSheet.Row>>, IEnumerable<KeyValuePair<int, AdventureBossFloorFirstRewardSheet.Row>>, IEnumerable, ISheet

Inherited Members

Sheet<int, AdventureBossFloorFirstRewardSheet.Row>.Name ,
Sheet<int, AdventureBossFloorFirstRewardSheet.Row>.OrderedList ,
Sheet<int, AdventureBossFloorFirstRewardSheet.Row>.First ,
Sheet<int, AdventureBossFloorFirstRewardSheet.Row>.Last ,
Sheet<int, AdventureBossFloorFirstRewardSheet.Row>.Keys ,
Sheet<int, AdventureBossFloorFirstRewardSheet.Row>.Values ,
Sheet<int, AdventureBossFloorFirstRewardSheet.Row>.Count ,
Sheet<int, AdventureBossFloorFirstRewardSheet.Row>.IsReadOnly ,
Sheet<int, AdventureBossFloorFirstRewardSheet.Row>.this[int] ,
Sheet<int, AdventureBossFloorFirstRewardSheet.Row>.Set(string, bool) ,
Sheet<int, AdventureBossFloorFirstRewardSheet.Row>.Set<T>(Sheet<int, T>, bool) ,
Sheet<int, AdventureBossFloorFirstRewardSheet.Row>.GetEnumerator() ,
Sheet<int, AdventureBossFloorFirstRewardSheet.Row>.TryGetValue(int, out AdventureBossFloorFirstRewardSheet.Row, bool) ,

Sheet<int, AdventureBossFloorFirstRewardSheet.Row>.AddRow(int, AdventureBossFloorFirstRewardSheet.Row) , Sheet<int, AdventureBossFloorFirstRewardSheet.Row>.Add(int, AdventureBossFloorFirstRewardSheet.Row) , Sheet<int, AdventureBossFloorFirstRewardSheet.Row>.ContainsKey(int) , Sheet<int, AdventureBossFloorFirstRewardSheet.Row>.Remove(int) , Sheet<int, AdventureBossFloorFirstRewardSheet.Row>.TryGetValue(int, out AdventureBossFloorFirstRewardSheet.Row) , Sheet<int, AdventureBossFloorFirstRewardSheet.Row>.Add(KeyValuePair<int, AdventureBossFloorFirstRewardSheet.Row>) , Sheet<int, AdventureBossFloorFirstRewardSheet.Row>.Clear() , Sheet<int, AdventureBossFloorFirstRewardSheet.Row>.Contains(KeyValuePair<int, AdventureBossFloorFirstRewardSheet.Row>) , Sheet<int, AdventureBossFloorFirstRewardSheet.Row>.CopyTo(KeyValuePair<int, AdventureBossFloorFirstRewardSheet.Row>[], int) , Sheet<int, AdventureBossFloorFirstRewardSheet.Row>.Remove(KeyValuePair<int, AdventureBossFloorFirstRewardSheet.Row>) , Sheet<int, AdventureBossFloorFirstRewardSheet.Row>.Serialize() , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

AdventureBossFloorFirstRewardSheet()

```
public AdventureBossFloorFirstRewardSheet()
```

Class AdventureBossFloorFirstRewardSheet.Row

Namespace: Nekoyume.TableData.AdventureBoss

Assembly: Lib9c.dll

```
[Serializable]
public class AdventureBossFloorFirstRewardSheet.Row : SheetRow<int>
```

Inheritance

object ← SheetRow<int> ← AdventureBossFloorFirstRewardSheet.Row

Inherited Members

SheetRow<int>.Validate() , SheetRow<int>.EndOfSheetInitialize() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Fields

FloorId

```
public int FloorId
```

Field Value

int

Rewards

```
public List<AdventureBossSheet.RewardAmountData> Rewards
```

Field Value

List<AdventureBossSheet.RewardAmountData>

Properties

Key

```
public override int Key { get; }
```

Property Value

int

Methods

Set(IReadOnlyList<string>)

```
public override void Set(IReadOnlyList<string> fields)
```

Parameters

`fields` IReadOnlyList<string>

Class AdventureBossFloorPointSheet

Namespace: Nekoyume.TableData.AdventureBoss

Assembly: Lib9c.dll

```
[Serializable]
public class AdventureBossFloorPointSheet : Sheet<int,
AdventureBossFloorPointSheet.Row>, IDictionary<int,
AdventureBossFloorPointSheet.Row>, ICollection<KeyValuePair<int,
AdventureBossFloorPointSheet.Row>>, IEnumerable<KeyValuePair<int,
AdventureBossFloorPointSheet.Row>>, IEnumerable, ISheet
```

Inheritance

object ← Sheet<int, AdventureBossFloorPointSheet.Row> ← AdventureBossFloorPointSheet

Implements

IDictionary<int, AdventureBossFloorPointSheet.Row>, ICollection<KeyValuePair<int, AdventureBossFloorPointSheet.Row>>, IEnumerable<KeyValuePair<int, AdventureBossFloorPointSheet.Row>>, IEnumerable, ISheet

Inherited Members

Sheet<int, AdventureBossFloorPointSheet.Row>.Name ,
Sheet<int, AdventureBossFloorPointSheet.Row>.OrderedList ,
Sheet<int, AdventureBossFloorPointSheet.Row>.First ,
Sheet<int, AdventureBossFloorPointSheet.Row>.Last ,
Sheet<int, AdventureBossFloorPointSheet.Row>.Keys ,
Sheet<int, AdventureBossFloorPointSheet.Row>.Values ,
Sheet<int, AdventureBossFloorPointSheet.Row>.Count ,
Sheet<int, AdventureBossFloorPointSheet.Row>.IsReadOnly ,
Sheet<int, AdventureBossFloorPointSheet.Row>.this[int] ,
Sheet<int, AdventureBossFloorPointSheet.Row>.Set(string, bool) ,
Sheet<int, AdventureBossFloorPointSheet.Row>.Set<T>(Sheet<int, T>, bool) ,
Sheet<int, AdventureBossFloorPointSheet.Row>.GetEnumerator() ,
Sheet<int, AdventureBossFloorPointSheet.Row>.TryGetValue(int, out AdventureBossFloorPointSheet.Row, bool) ,
Sheet<int, AdventureBossFloorPointSheet.Row>.AddRow(int, AdventureBossFloorPointSheet.Row) ,

Sheet<int, AdventureBossFloorPointSheet.Row>.Add(int, AdventureBossFloorPointSheet.Row) , Sheet<int, AdventureBossFloorPointSheet.Row>.ContainsKey(int) , Sheet<int, AdventureBossFloorPointSheet.Row>.Remove(int) , Sheet<int, AdventureBossFloorPointSheet.Row>.TryGetValue(int, out AdventureBossFloorPointSheet.Row) , Sheet<int, AdventureBossFloorPointSheet.Row>.Add(KeyValuePair<int, AdventureBossFloorPointSheet.Row>) , Sheet<int, AdventureBossFloorPointSheet.Row>.Clear() , Sheet<int, AdventureBossFloorPointSheet.Row>.Contains(KeyValuePair<int, AdventureBossFloorPointSheet.Row>) , Sheet<int, AdventureBossFloorPointSheet.Row>.CopyTo(KeyValuePair<int, AdventureBossFloorPointSheet.Row>[], int) , Sheet<int, AdventureBossFloorPointSheet.Row>.Remove(KeyValuePair<int, AdventureBossFloorPointSheet.Row>) , Sheet<int, AdventureBossFloorPointSheet.Row>.Serialize() , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

AdventureBossFloorPointSheet()

```
public AdventureBossFloorPointSheet()
```

Class AdventureBossFloorPointSheet.Row

Namespace: Nekoyume.TableData.AdventureBoss

Assembly: Lib9c.dll

```
[Serializable]
public class AdventureBossFloorPointSheet.Row : SheetRow<int>
```

Inheritance

object ← SheetRow<int> ← AdventureBossFloorPointSheet.Row

Inherited Members

SheetRow<int>.Validate() , SheetRow<int>.EndOfSheetInitialize() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Fields

FloorId

```
public int FloorId
```

Field Value

int

MaxPoint

```
public int MaxPoint
```

Field Value

int

MinPoint

```
public int MinPoint
```

Field Value

int

Properties

Key

```
public override int Key { get; }
```

Property Value

int

Methods

Set(IReadOnlyList<string>)

```
public override void Set(IReadOnlyList<string> fields)
```

Parameters

`fields` IReadOnlyList<string>

Class AdventureBossFloorSheet

Namespace: Nekoyume.TableData.AdventureBoss

Assembly: Lib9c.dll

```
[Serializable]
public class AdventureBossFloorSheet : Sheet<int, AdventureBossFloorSheet.Row>,
IDictionary<int, AdventureBossFloorSheet.Row>, ICollection<KeyValuePair<int,
AdventureBossFloorSheet.Row>>, IEnumerable<KeyValuePair<int,
AdventureBossFloorSheet.Row>>, IEnumerable, ISheet
```

Inheritance

object ← Sheet<int, AdventureBossFloorSheet.Row> ← AdventureBossFloorSheet

Implements

IDictionary<int, AdventureBossFloorSheet.Row>,
ICollection<KeyValuePair<int, AdventureBossFloorSheet.Row>>,
IEnumerable<KeyValuePair<int, AdventureBossFloorSheet.Row>>, IEnumerable,
ISheet

Inherited Members

Sheet<int, AdventureBossFloorSheet.Row>.Name ,
Sheet<int, AdventureBossFloorSheet.Row>.OrderedList ,
Sheet<int, AdventureBossFloorSheet.Row>.First ,
Sheet<int, AdventureBossFloorSheet.Row>.Last ,
Sheet<int, AdventureBossFloorSheet.Row>.Keys ,
Sheet<int, AdventureBossFloorSheet.Row>.Values ,
Sheet<int, AdventureBossFloorSheet.Row>.Count ,
Sheet<int, AdventureBossFloorSheet.Row>.IsReadOnly ,
Sheet<int, AdventureBossFloorSheet.Row>.this[int] ,
Sheet<int, AdventureBossFloorSheet.Row>.Set(string, bool) ,
Sheet<int, AdventureBossFloorSheet.Row>.Set<T>(Sheet<int, T>, bool) ,
Sheet<int, AdventureBossFloorSheet.Row>.GetEnumerator() ,
Sheet<int, AdventureBossFloorSheet.Row>.TryGetValue(int, out
AdventureBossFloorSheet.Row, bool) ,
Sheet<int, AdventureBossFloorSheet.Row>.AddRow(int, AdventureBossFloorSheet.Row) ,
Sheet<int, AdventureBossFloorSheet.Row>.Add(int, AdventureBossFloorSheet.Row) ,
Sheet<int, AdventureBossFloorSheet.Row>.ContainsKey(int) ,
Sheet<int, AdventureBossFloorSheet.Row>.Remove(int) ,

Sheet<int, AdventureBossFloorSheet.Row>.TryGetValue(int, out AdventureBossFloorSheet.Row) , Sheet<int, AdventureBossFloorSheet.Row>.Add(KeyValuePair<int, AdventureBossFloorSheet.Row>) , Sheet<int, AdventureBossFloorSheet.Row>.Clear() , Sheet<int, AdventureBossFloorSheet.Row>.Contains(KeyValuePair<int, AdventureBossFloorSheet.Row>) , Sheet<int, AdventureBossFloorSheet.Row>.CopyTo(KeyValuePair<int, AdventureBossFloorSheet.Row>[], int) , Sheet<int, AdventureBossFloorSheet.Row>.Remove(KeyValuePair<int, AdventureBossFloorSheet.Row>) , Sheet<int, AdventureBossFloorSheet.Row>.Serialize() , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

AdventureBossFloorSheet()

```
public AdventureBossFloorSheet()
```

Class AdventureBossFloorSheet.RewardData

Namespace: Nekoyume.TableData.AdventureBoss

Assembly: Lib9c.dll

```
[Serializable]
public class AdventureBossFloorSheet.RewardData
```

Inheritance

object ← AdventureBossFloorSheet.RewardData

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

RewardData(string, int, int, int, int)

```
public RewardData(string itemType, int itemId, int min, int max, int ratio)
```

Parameters

`itemType` string

`itemId` int

`min` int

`max` int

`ratio` int

Properties

ItemId

```
public int ItemId { get; }
```

Property Value

int

ItemType

```
public string ItemType { get; }
```

Property Value

string

Max

```
public int Max { get; }
```

Property Value

int

Min

```
public int Min { get; }
```

Property Value

int

Ratio

```
public int Ratio { get; }
```

Property Value

int

Class AdventureBossFloorSheet.Row

Namespace: Nekoyume.TableData.AdventureBoss

Assembly: Lib9c.dll

```
[Serializable]
public class AdventureBossFloorSheet.Row : SheetRow<int>
```

Inheritance

object ← SheetRow<int> ← AdventureBossFloorSheet.Row

Inherited Members

SheetRow<int>.Validate() , SheetRow<int>.EndOfSheetInitialize() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Properties

AdventureBossId

```
public int AdventureBossId { get; }
```

Property Value

int

BGM

```
public string BGM { get; }
```

Property Value

string

Background

```
public string Background { get; }
```

Property Value

string

EnemyInitialStatModifiers

```
public List<StatModifier> EnemyInitialStatModifiers { get; }
```

Property Value

List<StatModifier>

Floor

```
public int Floor { get; }
```

Property Value

int

Id

```
public int Id { get; }
```

Property Value

int

Key

```
public override int Key { get; }
```

Property Value

int

MaxDropItem

```
public int MaxDropItem { get; }
```

Property Value

int

MinDropItem

```
public int MinDropItem { get; }
```

Property Value

int

Rewards

```
public List<AdventureBossFloorSheet.RewardData> Rewards { get; }
```

Property Value

List<AdventureBossFloorSheet.RewardData>

StageBuffSkillId

```
public int StageBuffSkillId { get; }
```

Property Value

int

TurnLimit

```
public int TurnLimit { get; }
```

Property Value

int

Methods

Set(IReadOnlyList<string>)

```
public override void Set(IReadOnlyList<string> fields)
```

Parameters

`fields` IReadOnlyList<string>

Class AdventureBossFloorWaveSheet

Namespace: Nekoyume.TableData.AdventureBoss

Assembly: Lib9c.dll

```
[Serializable]
public class AdventureBossFloorWaveSheet : Sheet<int,
AdventureBossFloorWaveSheet.Row>, IDictionary<int, AdventureBossFloorWaveSheet.Row>,
ICollection<KeyValuePair<int, AdventureBossFloorWaveSheet.Row>>,
IEnumerable<KeyValuePair<int, AdventureBossFloorWaveSheet.Row>>, IEnumerable, ISheet
```

Inheritance

object ← Sheet<int, AdventureBossFloorWaveSheet.Row> ← AdventureBossFloorWaveSheet

Implements

IDictionary<int, AdventureBossFloorWaveSheet.Row>, ICollection<KeyValuePair<int, AdventureBossFloorWaveSheet.Row>>, IEnumerable<KeyValuePair<int, AdventureBossFloorWaveSheet.Row>>, IEnumerable, ISheet

Inherited Members

Sheet<int, AdventureBossFloorWaveSheet.Row>.Name ,
Sheet<int, AdventureBossFloorWaveSheet.Row>.OrderedList ,
Sheet<int, AdventureBossFloorWaveSheet.Row>.First ,
Sheet<int, AdventureBossFloorWaveSheet.Row>.Last ,
Sheet<int, AdventureBossFloorWaveSheet.Row>.Keys ,
Sheet<int, AdventureBossFloorWaveSheet.Row>.Values ,
Sheet<int, AdventureBossFloorWaveSheet.Row>.Count ,
Sheet<int, AdventureBossFloorWaveSheet.Row>.IsReadOnly ,
Sheet<int, AdventureBossFloorWaveSheet.Row>.this[int] ,
Sheet<int, AdventureBossFloorWaveSheet.Row>.Set(string, bool) ,
Sheet<int, AdventureBossFloorWaveSheet.Row>.Set<T>(Sheet<int, T>, bool) ,
Sheet<int, AdventureBossFloorWaveSheet.Row>.GetEnumerator() ,
Sheet<int, AdventureBossFloorWaveSheet.Row>.TryGetValue(int, out AdventureBossFloorWaveSheet.Row, bool) ,
Sheet<int, AdventureBossFloorWaveSheet.Row>.AddRow(int, AdventureBossFloorWaveSheet.Row) ,
Sheet<int, AdventureBossFloorWaveSheet.Row>.Add(int, AdventureBossFloorWaveSheet.Row) ,

Sheet<int, AdventureBossFloorWaveSheet.Row>.ContainsKey(int) ,
Sheet<int, AdventureBossFloorWaveSheet.Row>.Remove(int) ,
Sheet<int, AdventureBossFloorWaveSheet.Row>.TryGetValue(int, out
AdventureBossFloorWaveSheet.Row) ,
Sheet<int, AdventureBossFloorWaveSheet.Row>.Add(KeyValuePair<int,
AdventureBossFloorWaveSheet.Row>) ,
Sheet<int, AdventureBossFloorWaveSheet.Row>.Clear() ,
Sheet<int, AdventureBossFloorWaveSheet.Row>.Contains(KeyValuePair<int,
AdventureBossFloorWaveSheet.Row>) ,
Sheet<int, AdventureBossFloorWaveSheet.Row>.CopyTo(KeyValuePair<int,
AdventureBossFloorWaveSheet.Row>[], int) ,
Sheet<int, AdventureBossFloorWaveSheet.Row>.Remove(KeyValuePair<int,
AdventureBossFloorWaveSheet.Row>) ,
Sheet<int, AdventureBossFloorWaveSheet.Row>.Serialize() , object.Equals(object) ,
object.Equals(object, object) , object.GetHashCode() , object.GetType() ,
object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

AdventureBossFloorWaveSheet()

```
public AdventureBossFloorWaveSheet()
```

Class AdventureBossFloorWaveSheet.MonsterData

Namespace: Nekoyume.TableData.AdventureBoss

Assembly: Lib9c.dll

```
[Serializable]
public class AdventureBossFloorWaveSheet.MonsterData
```

Inheritance

object ← AdventureBossFloorWaveSheet.MonsterData

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

MonsterData(int, int, int)

```
public MonsterData(int characterId, int level, int count)
```

Parameters

`characterId` int

`level` int

`count` int

Properties

CharacterId

```
public int CharacterId { get; }
```

Property Value

int

Count

```
public int Count { get; }
```

Property Value

int

Level

```
public int Level { get; }
```

Property Value

int

Class AdventureBossFloorWaveSheet.Row

Namespace: Nekoyume.TableData.AdventureBoss

Assembly: Lib9c.dll

```
[Serializable]
public class AdventureBossFloorWaveSheet.Row : SheetRow<int>
```

Inheritance

object ← SheetRow<int> ← AdventureBossFloorWaveSheet.Row

Inherited Members

SheetRow<int>.Validate() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Properties

FloorId

```
public int FloorId { get; }
```

Property Value

int

HasBoss

```
public bool HasBoss { get; }
```

Property Value

bool

Key

```
public override int Key { get; }
```

Property Value

int

TotalMonsterIds

```
public List<int> TotalMonsterIds { get; }
```

Property Value

List<int>

Waves

```
public List<AdventureBossFloorWaveSheet.WaveData> Waves { get; }
```

Property Value

List<AdventureBossFloorWaveSheet.WaveData>

Methods

EndOfSheetInitialize()

```
public override void EndOfSheetInitialize()
```

Set(IReadOnlyList<string>)

```
public override void Set(IReadOnlyList<string> fields)
```

Parameters

`fields` IReadOnlyList<string>

Class AdventureBossFloorWaveSheet.WaveData

Namespace: Nekoyume.TableData.AdventureBoss

Assembly: Lib9c.dll

```
[Serializable]
public class AdventureBossFloorWaveSheet.WaveData
```

Inheritance

object ← AdventureBossFloorWaveSheet.WaveData

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

WaveData(int, List<MonsterData>, bool)

```
public WaveData(int number, List<AdventureBossFloorWaveSheet.MonsterData> monsters,
bool hasBoss)
```

Parameters

`number` int

`monsters` List<AdventureBossFloorWaveSheet.MonsterData>

`hasBoss` bool

Properties

HasBoss

```
public bool HasBoss { get; }
```

Property Value

bool

Monsters

```
public List<AdventureBossFloorWaveSheet.MonsterData> Monsters { get; }
```

Property Value

List<AdventureBossFloorWaveSheet.MonsterData>

Number

```
public int Number { get; }
```

Property Value

int

Class AdventureBossNcgRewardRatioSheet

Namespace: Nekoyume.TableData.AdventureBoss

Assembly: Lib9c.dll

```
[Serializable]
public class AdventureBossNcgRewardRatioSheet : Sheet<int,
AdventureBossNcgRewardRatioSheet.Row>, IDictionary<int,
AdventureBossNcgRewardRatioSheet.Row>, ICollection<KeyValuePair<int,
AdventureBossNcgRewardRatioSheet.Row>>, IEnumerable<KeyValuePair<int,
AdventureBossNcgRewardRatioSheet.Row>>, IEnumerable, ISheet
```

Inheritance

object ← Sheet<int, AdventureBossNcgRewardRatioSheet.Row> ← AdventureBossNcgRewardRatioSheet

Implements

IDictionary<int, AdventureBossNcgRewardRatioSheet.Row>, ICollection<KeyValuePair<int, AdventureBossNcgRewardRatioSheet.Row>>, IEnumerable<KeyValuePair<int, AdventureBossNcgRewardRatioSheet.Row>>, IEnumerable, ISheet

Inherited Members

Sheet<int, AdventureBossNcgRewardRatioSheet.Row>.Name ,
Sheet<int, AdventureBossNcgRewardRatioSheet.Row>.OrderedList ,
Sheet<int, AdventureBossNcgRewardRatioSheet.Row>.First ,
Sheet<int, AdventureBossNcgRewardRatioSheet.Row>.Last ,
Sheet<int, AdventureBossNcgRewardRatioSheet.Row>.Keys ,
Sheet<int, AdventureBossNcgRewardRatioSheet.Row>.Values ,
Sheet<int, AdventureBossNcgRewardRatioSheet.Row>.Count ,
Sheet<int, AdventureBossNcgRewardRatioSheet.Row>.IsReadOnly ,
Sheet<int, AdventureBossNcgRewardRatioSheet.Row>.this[int] ,
Sheet<int, AdventureBossNcgRewardRatioSheet.Row>.Set(string, bool) ,
Sheet<int, AdventureBossNcgRewardRatioSheet.Row>.Set<T>(Sheet<int, T>, bool) ,
Sheet<int, AdventureBossNcgRewardRatioSheet.Row>.GetEnumerator() ,
Sheet<int, AdventureBossNcgRewardRatioSheet.Row>.TryGetValue(int, out AdventureBossNcgRewardRatioSheet.Row, bool) ,
Sheet<int, AdventureBossNcgRewardRatioSheet.Row>.AddRow(int, AdventureBossNcgRewardRatioSheet.Row) ,

Sheet<int, AdventureBossNcgRewardRatioSheet.Row>.Add(int, AdventureBossNcgRewardRatioSheet.Row) , Sheet<int, AdventureBossNcgRewardRatioSheet.Row>.ContainsKey(int) , Sheet<int, AdventureBossNcgRewardRatioSheet.Row>.Remove(int) , Sheet<int, AdventureBossNcgRewardRatioSheet.Row>.TryGetValue(int, out AdventureBossNcgRewardRatioSheet.Row) , Sheet<int, AdventureBossNcgRewardRatioSheet.Row>.Add(KeyValuePair<int, AdventureBossNcgRewardRatioSheet.Row>) , Sheet<int, AdventureBossNcgRewardRatioSheet.Row>.Clear() , Sheet<int, AdventureBossNcgRewardRatioSheet.Row>.Contains(KeyValuePair<int, AdventureBossNcgRewardRatioSheet.Row>) , Sheet<int, AdventureBossNcgRewardRatioSheet.Row>.CopyTo(KeyValuePair<int, AdventureBossNcgRewardRatioSheet.Row>[], int) , Sheet<int, AdventureBossNcgRewardRatioSheet.Row>.Remove(KeyValuePair<int, AdventureBossNcgRewardRatioSheet.Row>) , Sheet<int, AdventureBossNcgRewardRatioSheet.Row>.Serialize() , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

AdventureBossNcgRewardRatioSheet()

```
public AdventureBossNcgRewardRatioSheet()
```

Class AdventureBossNcgRewardRatioSheet.Row

Namespace: Nekoyume.TableData.AdventureBoss

Assembly: Lib9c.dll

```
[Serializable]
public class AdventureBossNcgRewardRatioSheet.Row : SheetRow<int>
```

Inheritance

object ← SheetRow<int> ← AdventureBossNcgRewardRatioSheet.Row

Inherited Members

SheetRow<int>.Validate() , SheetRow<int>.EndOfSheetInitialize() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Fields

ItemId

```
public int ItemId
```

Field Value

int

Ratio

```
public decimal Ratio
```

Field Value

decimal

Properties

Key

```
public override int Key { get; }
```

Property Value

int

Methods

Set(IReadOnlyList<string>)

```
public override void Set(IReadOnlyList<string> fields)
```

Parameters

`fields` IReadOnlyList<string>

Class AdventureBossSheet

Namespace: Nekoyume.TableData.AdventureBoss

Assembly: Lib9c.dll

```
[Serializable]
public class AdventureBossSheet : Sheet<int, AdventureBossSheet.Row>,
IDictionary<int, AdventureBossSheet.Row>, ICollection<KeyValuePair<int,
AdventureBossSheet.Row>>, IEnumerable<KeyValuePair<int, AdventureBossSheet.Row>>,
IEnumerable, ISheet
```

Inheritance

object ← Sheet<int, AdventureBossSheet.Row> ← AdventureBossSheet

Implements

IDictionary<int, AdventureBossSheet.Row>,
ICollection<KeyValuePair<int, AdventureBossSheet.Row>>,
IEnumerable<KeyValuePair<int, AdventureBossSheet.Row>>, IEnumerable, ISheet

Inherited Members

Sheet<int, AdventureBossSheet.Row>.Name ,
Sheet<int, AdventureBossSheet.Row>.OrderedList ,
Sheet<int, AdventureBossSheet.Row>.First , Sheet<int, AdventureBossSheet.Row>.Last ,
Sheet<int, AdventureBossSheet.Row>.Keys ,
Sheet<int, AdventureBossSheet.Row>.Values ,
Sheet<int, AdventureBossSheet.Row>.Count ,
Sheet<int, AdventureBossSheet.Row>.IsReadOnly ,
Sheet<int, AdventureBossSheet.Row>.this[int] ,
Sheet<int, AdventureBossSheet.Row>.Set(string, bool) ,
Sheet<int, AdventureBossSheet.Row>.Set<T>(Sheet<int, T>, bool) ,
Sheet<int, AdventureBossSheet.Row>.GetEnumerator() ,
Sheet<int, AdventureBossSheet.Row>.TryGetValue(int, out AdventureBossSheet.Row, bool)
,
Sheet<int, AdventureBossSheet.Row>.AddRow(int, AdventureBossSheet.Row) ,
Sheet<int, AdventureBossSheet.Row>.Add(int, AdventureBossSheet.Row) ,
Sheet<int, AdventureBossSheet.Row>.ContainsKey(int) ,
Sheet<int, AdventureBossSheet.Row>.Remove(int) ,
Sheet<int, AdventureBossSheet.Row>.TryGetValue(int, out AdventureBossSheet.Row) ,
Sheet<int, AdventureBossSheet.Row>.Add(KeyValuePair<int, AdventureBossSheet.Row>) ,
Sheet<int, AdventureBossSheet.Row>.Clear() ,

Sheet<int, AdventureBossSheet.Row>.Contains(KeyValuePair<int, AdventureBossSheet.Row>) , Sheet<int, AdventureBossSheet.Row>.CopyTo(KeyValuePair<int, AdventureBossSheet.Row>[], int) , Sheet<int, AdventureBossSheet.Row>.Remove(KeyValuePair<int, AdventureBossSheet.Row>) , Sheet<int, AdventureBossSheet.Row>.Serialize() , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

AdventureBossSheet()

```
public AdventureBossSheet()
```

Class AdventureBossSheet.RewardAmountData

Namespace: Nekoyume.TableData.AdventureBoss

Assembly: Lib9c.dll

```
[Serializable]
public class AdventureBossSheet.RewardAmountData
```

Inheritance

object ← AdventureBossSheet.RewardAmountData

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

RewardAmountData(string, int, int)

```
public RewardAmountData(string itemType, int itemId, int amount)
```

Parameters

`itemType` string

`itemId` int

`amount` int

Properties

Amount

```
public int Amount { get; }
```

Property Value

int

ItemId

```
public int ItemId { get; }
```

Property Value

int

ItemType

```
public string ItemType { get; }
```

Property Value

string

Class AdventureBossSheet.RewardRatioData

Namespace: Nekoyume.TableData.AdventureBoss

Assembly: Lib9c.dll

```
[Serializable]
public class AdventureBossSheet.RewardRatioData
```

Inheritance

object ← AdventureBossSheet.RewardRatioData

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

RewardRatioData(string, int, int)

```
public RewardRatioData(string itemType, int itemId, int ratio)
```

Parameters

`itemType` string

`itemId` int

`ratio` int

Properties

ItemId

```
public int ItemId { get; }
```

Property Value

int

ItemType

```
public string ItemType { get; }
```

Property Value

string

Ratio

```
public int Ratio { get; }
```

Property Value

int

Class AdventureBossSheet.Row

Namespace: Nekoyume.TableData.AdventureBoss

Assembly: Lib9c.dll

```
[Serializable]
public class AdventureBossSheet.Row : SheetRow<int>
```

Inheritance

object ← SheetRow<int> ← AdventureBossSheet.Row

Inherited Members

SheetRow<int>.Validate() , SheetRow<int>.EndOfSheetInitialize() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Properties

BossId

```
public int BossId { get; }
```

Property Value

int

ExploreAp

```
public int ExploreAp { get; }
```

Property Value

int

Id

```
public int Id { get; }
```

Property Value

int

Key

```
public override int Key { get; }
```

Property Value

int

SweepAp

```
public int SweepAp { get; }
```

Property Value

int

Methods

Set(IReadOnlyList<string>)

```
public override void Set(IReadOnlyList<string> fields)
```

Parameters

`fields` IReadOnlyList<string>

Class AdventureBossUnlockFloorCostSheet

Namespace: Nekoyume.TableData.AdventureBoss

Assembly: Lib9c.dll

```
[Serializable]
public class AdventureBossUnlockFloorCostSheet : Sheet<int,
AdventureBossUnlockFloorCostSheet.Row>, IDictionary<int,
AdventureBossUnlockFloorCostSheet.Row>, ICollection<KeyValuePair<int,
AdventureBossUnlockFloorCostSheet.Row>>, IEnumerable<KeyValuePair<int,
AdventureBossUnlockFloorCostSheet.Row>>, IEnumerable, ISheet
```

Inheritance

object ← Sheet<int, AdventureBossUnlockFloorCostSheet.Row> ← AdventureBossUnlockFloorCostSheet

Implements

IDictionary<int, AdventureBossUnlockFloorCostSheet.Row>, ICollection<KeyValuePair<int, AdventureBossUnlockFloorCostSheet.Row>>, IEnumerable<KeyValuePair<int, AdventureBossUnlockFloorCostSheet.Row>>, IEnumerable, ISheet

Inherited Members

Sheet<int, AdventureBossUnlockFloorCostSheet.Row>.Name ,
Sheet<int, AdventureBossUnlockFloorCostSheet.Row>.OrderedList ,
Sheet<int, AdventureBossUnlockFloorCostSheet.Row>.First ,
Sheet<int, AdventureBossUnlockFloorCostSheet.Row>.Last ,
Sheet<int, AdventureBossUnlockFloorCostSheet.Row>.Keys ,
Sheet<int, AdventureBossUnlockFloorCostSheet.Row>.Values ,
Sheet<int, AdventureBossUnlockFloorCostSheet.Row>.Count ,
Sheet<int, AdventureBossUnlockFloorCostSheet.Row>.IsReadOnly ,
Sheet<int, AdventureBossUnlockFloorCostSheet.Row>.this[int] ,
Sheet<int, AdventureBossUnlockFloorCostSheet.Row>.Set(string, bool) ,
Sheet<int, AdventureBossUnlockFloorCostSheet.Row>.Set<T>(Sheet<int, T>, bool) ,
Sheet<int, AdventureBossUnlockFloorCostSheet.Row>.GetEnumerator() ,
Sheet<int, AdventureBossUnlockFloorCostSheet.Row>.TryGetValue(int, out AdventureBossUnlockFloorCostSheet.Row, bool) ,
Sheet<int, AdventureBossUnlockFloorCostSheet.Row>.AddRow(int, AdventureBossUnlockFloorCostSheet.Row) ,

Sheet<int, AdventureBossUnlockFloorCostSheet.Row>.Add(int, AdventureBossUnlockFloorCostSheet.Row) , Sheet<int, AdventureBossUnlockFloorCostSheet.Row>.ContainsKey(int) , Sheet<int, AdventureBossUnlockFloorCostSheet.Row>.Remove(int) , Sheet<int, AdventureBossUnlockFloorCostSheet.Row>.TryGetValue(int, out AdventureBossUnlockFloorCostSheet.Row) , Sheet<int, AdventureBossUnlockFloorCostSheet.Row>.Add(KeyValuePair<int, AdventureBossUnlockFloorCostSheet.Row>) , Sheet<int, AdventureBossUnlockFloorCostSheet.Row>.Clear() , Sheet<int, AdventureBossUnlockFloorCostSheet.Row>.Contains(KeyValuePair<int, AdventureBossUnlockFloorCostSheet.Row>) , Sheet<int, AdventureBossUnlockFloorCostSheet.Row>.CopyTo(KeyValuePair<int, AdventureBossUnlockFloorCostSheet.Row>[], int) , Sheet<int, AdventureBossUnlockFloorCostSheet.Row>.Remove(KeyValuePair<int, AdventureBossUnlockFloorCostSheet.Row>) , Sheet<int, AdventureBossUnlockFloorCostSheet.Row>.Serialize() , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

AdventureBossUnlockFloorCostSheet()

```
public AdventureBossUnlockFloorCostSheet()
```

Class AdventureBossUnlockFloorCostSheet.Row

Namespace: Nekoyume.TableData.AdventureBoss

Assembly: Lib9c.dll

```
[Serializable]
public class AdventureBossUnlockFloorCostSheet.Row : SheetRow<int>
```

Inheritance

object ← SheetRow<int> ← AdventureBossUnlockFloorCostSheet.Row

Inherited Members

SheetRow<int>.Validate() , SheetRow<int>.EndOfSheetInitialize() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Fields

FloorId

```
public int FloorId
```

Field Value

int

GoldenDustPrice

```
public int GoldenDustPrice
```

Field Value

int

NcgPrice

```
public BigInteger NcgPrice
```

Field Value

BigInteger

Properties

Key

```
public override int Key { get; }
```

Property Value

int

Methods

Set(IReadOnlyList<string>)

```
public override void Set(IReadOnlyList<string> fields)
```

Parameters

`fields` IReadOnlyList<string>

Class AdventureBossWantedRewardSheet

Namespace: Nekoyume.TableData.AdventureBoss

Assembly: Lib9c.dll

```
[Serializable]
public class AdventureBossWantedRewardSheet : Sheet<int,
AdventureBossWantedRewardSheet.Row>, IDictionary<int,
AdventureBossWantedRewardSheet.Row>, ICollection<KeyValuePair<int,
AdventureBossWantedRewardSheet.Row>>, IEnumerable<KeyValuePair<int,
AdventureBossWantedRewardSheet.Row>>, IEnumerable, ISheet
```

Inheritance

object ← Sheet<int, AdventureBossWantedRewardSheet.Row> ← AdventureBossWantedRewardSheet

Implements

IDictionary<int, AdventureBossWantedRewardSheet.Row>, ICollection<KeyValuePair<int, AdventureBossWantedRewardSheet.Row>>, IEnumerable<KeyValuePair<int, AdventureBossWantedRewardSheet.Row>>, IEnumerable, ISheet

Inherited Members

Sheet<int, AdventureBossWantedRewardSheet.Row>.Name ,
Sheet<int, AdventureBossWantedRewardSheet.Row>.OrderedList ,
Sheet<int, AdventureBossWantedRewardSheet.Row>.First ,
Sheet<int, AdventureBossWantedRewardSheet.Row>.Last ,
Sheet<int, AdventureBossWantedRewardSheet.Row>.Keys ,
Sheet<int, AdventureBossWantedRewardSheet.Row>.Values ,
Sheet<int, AdventureBossWantedRewardSheet.Row>.Count ,
Sheet<int, AdventureBossWantedRewardSheet.Row>.IsReadOnly ,
Sheet<int, AdventureBossWantedRewardSheet.Row>.this[int] ,
Sheet<int, AdventureBossWantedRewardSheet.Row>.Set(string, bool) ,
Sheet<int, AdventureBossWantedRewardSheet.Row>.Set<T>(Sheet<int, T>, bool) ,
Sheet<int, AdventureBossWantedRewardSheet.Row>.GetEnumerator() ,
Sheet<int, AdventureBossWantedRewardSheet.Row>.TryGetValue(int, out AdventureBossWantedRewardSheet.Row, bool) ,
Sheet<int, AdventureBossWantedRewardSheet.Row>.AddRow(int, AdventureBossWantedRewardSheet.Row) ,

Sheet<int, AdventureBossWantedRewardSheet.Row>.Add(int, AdventureBossWantedRewardSheet.Row) , Sheet<int, AdventureBossWantedRewardSheet.Row>.ContainsKey(int) , Sheet<int, AdventureBossWantedRewardSheet.Row>.Remove(int) , Sheet<int, AdventureBossWantedRewardSheet.Row>.TryGetValue(int, out AdventureBossWantedRewardSheet.Row) , Sheet<int, AdventureBossWantedRewardSheet.Row>.Add(KeyValuePair<int, AdventureBossWantedRewardSheet.Row>) , Sheet<int, AdventureBossWantedRewardSheet.Row>.Clear() , Sheet<int, AdventureBossWantedRewardSheet.Row>.Contains(KeyValuePair<int, AdventureBossWantedRewardSheet.Row>) , Sheet<int, AdventureBossWantedRewardSheet.Row>.CopyTo(KeyValuePair<int, AdventureBossWantedRewardSheet.Row>[], int) , Sheet<int, AdventureBossWantedRewardSheet.Row>.Remove(KeyValuePair<int, AdventureBossWantedRewardSheet.Row>) , Sheet<int, AdventureBossWantedRewardSheet.Row>.Serialize() , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

AdventureBossWantedRewardSheet()

```
public AdventureBossWantedRewardSheet()
```

Class AdventureBossWantedRewardSheet.Row

Namespace: Nekoyume.TableData.AdventureBoss

Assembly: Lib9c.dll

```
[Serializable]
public class AdventureBossWantedRewardSheet.Row : SheetRow<int>
```

Inheritance

object ← SheetRow<int> ← AdventureBossWantedRewardSheet.Row

Inherited Members

SheetRow<int>.Validate() , SheetRow<int>.EndOfSheetInitialize() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Fields

AdventureBossId

```
public int AdventureBossId
```

Field Value

int

FixedReward

```
public AdventureBossSheet.RewardRatioData FixedReward
```

Field Value

AdventureBossSheet.RewardRatioData

Id

```
public int Id
```

Field Value

int

RandomRewards

```
public List<AdventureBossSheet.RewardRatioData> RandomRewards
```

Field Value

List<AdventureBossSheet.RewardRatioData>

Properties

Key

```
public override int Key { get; }
```

Property Value

int

Methods

Set(IReadOnlyList<string>)

```
public override void Set(IReadOnlyList<string> fields)
```

Parameters

`fields` IReadOnlyList<string>

Namespace Nekoyume.TableData.Crystal

Classes

CrystalEquipmentGrindingSheet
: Crystal and materials paid for grinding each equipment.

CrystalEquipmentGrindingSheet.Row
: One equipment's grinding reward.

CrystalFluctuationSheet

CrystalFluctuationSheet.Row

CrystalHammerPointSheet
: Hammer points and their crystal price per equipment recipe.

CrystalHammerPointSheet.Row
: One recipe's hammer point settings.

CrystalMaterialCostSheet

CrystalMaterialCostSheet.Row
: The crystal price of one material.

CrystalMonsterCollectionMultiplierSheet

CrystalMonsterCollectionMultiplierSheet.Row

CrystalRandomBuffSheet

CrystalRandomBuffSheet.Row

CrystalStageBuffGachaSheet

CrystalStageBuffGachaSheet.Row

Enums

CrystalFluctuationSheet.ServiceType

CrystalRandomBuffSheet.Row.BuffRank

Class CrystalEquipmentGrindingSheet

Namespace: Nekoyume.TableData.Crystal

Assembly: Lib9c.dll

Crystal and materials paid for grinding each equipment.

```
public class CrystalEquipmentGrindingSheet : Sheet<int,
CrystalEquipmentGrindingSheet.Row>, IDictionary<int,
CrystalEquipmentGrindingSheet.Row>, ICollection<KeyValuePair<int,
CrystalEquipmentGrindingSheet.Row>>, IEnumerable<KeyValuePair<int,
CrystalEquipmentGrindingSheet.Row>>, IEnumerable, ISheet
```

Inheritance

object ← Sheet<int, CrystalEquipmentGrindingSheet.Row> ← CrystalEquipmentGrindingSheet

Implements

IDictionary<int, CrystalEquipmentGrindingSheet.Row>, ICollection<KeyValuePair<int, CrystalEquipmentGrindingSheet.Row>>, IEnumerable<KeyValuePair<int, CrystalEquipmentGrindingSheet.Row>>, IEnumerable, ISheet

Inherited Members

Sheet<int, CrystalEquipmentGrindingSheet.Row>.Name ,
Sheet<int, CrystalEquipmentGrindingSheet.Row>.OrderedList ,
Sheet<int, CrystalEquipmentGrindingSheet.Row>.First ,
Sheet<int, CrystalEquipmentGrindingSheet.Row>.Last ,
Sheet<int, CrystalEquipmentGrindingSheet.Row>.Keys ,
Sheet<int, CrystalEquipmentGrindingSheet.Row>.Values ,
Sheet<int, CrystalEquipmentGrindingSheet.Row>.Count ,
Sheet<int, CrystalEquipmentGrindingSheet.Row>.IsReadOnly ,
Sheet<int, CrystalEquipmentGrindingSheet.Row>.this[int] ,
Sheet<int, CrystalEquipmentGrindingSheet.Row>.Set(string, bool) ,
Sheet<int, CrystalEquipmentGrindingSheet.Row>.Set<T>(Sheet<int, T>, bool) ,
Sheet<int, CrystalEquipmentGrindingSheet.Row>.GetEnumerator() ,
Sheet<int, CrystalEquipmentGrindingSheet.Row>.TryGetValue(int, out CrystalEquipmentGrindingSheet.Row, bool) ,
Sheet<int, CrystalEquipmentGrindingSheet.Row>.AddRow(int, CrystalEquipmentGrindingSheet.Row) ,

Sheet<int, CrystalEquipmentGrindingSheet.Row>.Add(int, CrystalEquipmentGrindingSheet.Row) , Sheet<int, CrystalEquipmentGrindingSheet.Row>.ContainsKey(int) , Sheet<int, CrystalEquipmentGrindingSheet.Row>.Remove(int) , Sheet<int, CrystalEquipmentGrindingSheet.Row>.TryGetValue(int, out CrystalEquipmentGrindingSheet.Row) , Sheet<int, CrystalEquipmentGrindingSheet.Row>.Add(KeyValuePair<int, CrystalEquipmentGrindingSheet.Row>) , Sheet<int, CrystalEquipmentGrindingSheet.Row>.Clear() , Sheet<int, CrystalEquipmentGrindingSheet.Row>.Contains(KeyValuePair<int, CrystalEquipmentGrindingSheet.Row>) , Sheet<int, CrystalEquipmentGrindingSheet.Row>.CopyTo(KeyValuePair<int, CrystalEquipmentGrindingSheet.Row>[], int) , Sheet<int, CrystalEquipmentGrindingSheet.Row>.Remove(KeyValuePair<int, CrystalEquipmentGrindingSheet.Row>) , Sheet<int, CrystalEquipmentGrindingSheet.Row>.Serialize() , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

CrystalEquipmentGrindingSheet()

Creates an empty sheet.

```
public CrystalEquipmentGrindingSheet()
```

Class CrystalEquipmentGrindingSheet.Row

Namespace: Nekoyume.TableData.Crystal

Assembly: Lib9c.dll

One equipment's grinding reward.

```
public class CrystalEquipmentGrindingSheet.Row : SheetRow<int>
```

Inheritance

object ← SheetRow<int> ← CrystalEquipmentGrindingSheet.Row

Inherited Members

SheetRow<int>.Validate() , SheetRow<int>.EndOfSheetInitialize() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Fields

CRYSTAL

Crystal paid for grinding this equipment (column `gain_crystal`).

```
public long CRYSTAL
```

Field Value

long

EnchantBaseId

The row whose crystal is added per enhancement level.

```
public int EnchantBaseId
```

Field Value

int

ItemId

The equipment id this row applies to.

```
public int ItemId
```

Field Value

int

RewardMaterials

Materials also given for grinding.

```
public List<(int materialId, int count)> RewardMaterials
```

Field Value

List<(int id, int count)>

Properties

Key

```
public override int Key { get; }
```

Property Value

int

Methods

Set(IReadOnlyList<string>)

```
public override void Set(IReadOnlyList<string> fields)
```

Parameters

`fields` IReadOnlyList<string>

Class CrystalFluctuationSheet

Namespace: Nekoyume.TableData.Crystal

Assembly: Lib9c.dll

```
public class CrystalFluctuationSheet : Sheet<int, CrystalFluctuationSheet.Row>,
IDictionary<int, CrystalFluctuationSheet.Row>, ICollection<KeyValuePair<int,
CrystalFluctuationSheet.Row>>, IEnumerable<KeyValuePair<int,
CrystalFluctuationSheet.Row>>, IEnumerable, ISheet
```

Inheritance

object ← Sheet<int, CrystalFluctuationSheet.Row> ← CrystalFluctuationSheet

Implements

IDictionary<int, CrystalFluctuationSheet.Row>,
ICollection<KeyValuePair<int, CrystalFluctuationSheet.Row>>,
IEnumerable<KeyValuePair<int, CrystalFluctuationSheet.Row>>, IEnumerable,
ISheet

Inherited Members

Sheet<int, CrystalFluctuationSheet.Row>.Name ,
Sheet<int, CrystalFluctuationSheet.Row>.OrderedList ,
Sheet<int, CrystalFluctuationSheet.Row>.First ,
Sheet<int, CrystalFluctuationSheet.Row>.Last ,
Sheet<int, CrystalFluctuationSheet.Row>.Keys ,
Sheet<int, CrystalFluctuationSheet.Row>.Values ,
Sheet<int, CrystalFluctuationSheet.Row>.Count ,
Sheet<int, CrystalFluctuationSheet.Row>.IsReadOnly ,
Sheet<int, CrystalFluctuationSheet.Row>.this[int] ,
Sheet<int, CrystalFluctuationSheet.Row>.Set(string, bool) ,
Sheet<int, CrystalFluctuationSheet.Row>.Set<T>(Sheet<int, T>, bool) ,
Sheet<int, CrystalFluctuationSheet.Row>.GetEnumerator() ,
Sheet<int, CrystalFluctuationSheet.Row>.TryGetValue(int, out CrystalFluctuationSheet.Row,
bool) ,
Sheet<int, CrystalFluctuationSheet.Row>.AddRow(int, CrystalFluctuationSheet.Row) ,
Sheet<int, CrystalFluctuationSheet.Row>.Add(int, CrystalFluctuationSheet.Row) ,
Sheet<int, CrystalFluctuationSheet.Row>.ContainsKey(int) ,
Sheet<int, CrystalFluctuationSheet.Row>.Remove(int) ,

Sheet<int, CrystalFluctuationSheet.Row>.TryGetValue(int, out CrystalFluctuationSheet.Row) , Sheet<int, CrystalFluctuationSheet.Row>.Add(KeyValuePair<int, CrystalFluctuationSheet.Row>) , Sheet<int, CrystalFluctuationSheet.Row>.Clear() , Sheet<int, CrystalFluctuationSheet.Row>.Contains(KeyValuePair<int, CrystalFluctuationSheet.Row>) , Sheet<int, CrystalFluctuationSheet.Row>.CopyTo(KeyValuePair<int, CrystalFluctuationSheet.Row>[], int) , Sheet<int, CrystalFluctuationSheet.Row>.Remove(KeyValuePair<int, CrystalFluctuationSheet.Row>) , Sheet<int, CrystalFluctuationSheet.Row>.Serialize() , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

CrystalFluctuationSheet()

```
public CrystalFluctuationSheet()
```

Class CrystalFluctuationSheet.Row

Namespace: Nekoyume.TableData.Crystal

Assembly: Lib9c.dll

```
public class CrystalFluctuationSheet.Row : SheetRow<int>
```

Inheritance

object ← SheetRow<int> ← CrystalFluctuationSheet.Row

Inherited Members

SheetRow<int>.Validate() , SheetRow<int>.EndOfSheetInitialize() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Fields

BlockInterval

```
public long BlockInterval
```

Field Value

long

Id

```
public int Id
```

Field Value

int

MaximumRate

```
public int MaximumRate
```

Field Value

int

MinimumRate

```
public int MinimumRate
```

Field Value

int

Type

```
public CrystalFluctuationSheet.ServiceType Type
```

Field Value

CrystalFluctuationSheet.ServiceType

Properties

Key

```
public override int Key { get; }
```

Property Value

int

Methods

Set(IReadOnlyList<string>)

```
public override void Set(IReadOnlyList<string> fields)
```

Parameters

`fields` IReadOnlyList<string>

Enum CrystalFluctuationSheet.ServiceType

Namespace: Nekoyume.TableData.Crystal

Assembly: Lib9c.dll

```
public enum CrystalFluctuationSheet.ServiceType
```

Extension Methods
StateExtensions.Serialize(Enum)

Fields

`Combination = 0`

Class CrystalHammerPointSheet

Namespace: Nekoyume.TableData.Crystal

Assembly: Lib9c.dll

Hammer points and their crystal price per equipment recipe.

```
public class CrystalHammerPointSheet : Sheet<int, CrystalHammerPointSheet.Row>,
IDictionary<int, CrystalHammerPointSheet.Row>, ICollection<KeyValuePair<int,
CrystalHammerPointSheet.Row>>, IEnumerable<KeyValuePair<int,
CrystalHammerPointSheet.Row>>, IEnumerable, ISheet
```

Inheritance

object ← Sheet<int, CrystalHammerPointSheet.Row> ← CrystalHammerPointSheet

Implements

IDictionary<int, CrystalHammerPointSheet.Row>, ICollection<KeyValuePair<int, CrystalHammerPointSheet.Row>>, IEnumerable<KeyValuePair<int, CrystalHammerPointSheet.Row>>, IEnumerable, ISheet

Inherited Members

Sheet<int, CrystalHammerPointSheet.Row>.Name ,
Sheet<int, CrystalHammerPointSheet.Row>.OrderedList ,
Sheet<int, CrystalHammerPointSheet.Row>.First ,
Sheet<int, CrystalHammerPointSheet.Row>.Last ,
Sheet<int, CrystalHammerPointSheet.Row>.Keys ,
Sheet<int, CrystalHammerPointSheet.Row>.Values ,
Sheet<int, CrystalHammerPointSheet.Row>.Count ,
Sheet<int, CrystalHammerPointSheet.Row>.IsReadOnly ,
Sheet<int, CrystalHammerPointSheet.Row>.this[int] ,
Sheet<int, CrystalHammerPointSheet.Row>.Set(string, bool) ,
Sheet<int, CrystalHammerPointSheet.Row>.Set<T>(Sheet<int, T>, bool) ,
Sheet<int, CrystalHammerPointSheet.Row>.GetEnumerator() ,
Sheet<int, CrystalHammerPointSheet.Row>.TryGetValue(int, out CrystalHammerPointSheet.Row, bool) ,
Sheet<int, CrystalHammerPointSheet.Row>.AddRow(int, CrystalHammerPointSheet.Row) ,
Sheet<int, CrystalHammerPointSheet.Row>.Add(int, CrystalHammerPointSheet.Row) ,
Sheet<int, CrystalHammerPointSheet.Row>.ContainsKey(int) ,
Sheet<int, CrystalHammerPointSheet.Row>.Remove(int) ,

Sheet<int, CrystalHammerPointSheet.Row>.TryGetValue(int, out CrystalHammerPointSheet.Row) , Sheet<int, CrystalHammerPointSheet.Row>.Add(KeyValuePair<int, CrystalHammerPointSheet.Row>) , Sheet<int, CrystalHammerPointSheet.Row>.Clear() , Sheet<int, CrystalHammerPointSheet.Row>.Contains(KeyValuePair<int, CrystalHammerPointSheet.Row>) , Sheet<int, CrystalHammerPointSheet.Row>.CopyTo(KeyValuePair<int, CrystalHammerPointSheet.Row>[], int) , Sheet<int, CrystalHammerPointSheet.Row>.Remove(KeyValuePair<int, CrystalHammerPointSheet.Row>) , Sheet<int, CrystalHammerPointSheet.Row>.Serialize() , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

CrystalHammerPointSheet()

Creates an empty sheet.

```
public CrystalHammerPointSheet()
```

Class CrystalHammerPointSheet.Row

Namespace: Nekoyume.TableData.Crystal

Assembly: Lib9c.dll

One recipe's hammer point settings.

```
public class CrystalHammerPointSheet.Row : SheetRow<int>
```

Inheritance

object ← SheetRow<int> ← CrystalHammerPointSheet.Row

Inherited Members

SheetRow<int>.Validate() , SheetRow<int>.EndOfSheetInitialize() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Properties

CRYSTAL

Crystal cost of one hammer point (column `crystal_cost`).

```
public long CRYSTAL { get; }
```

Property Value

long

Key

```
public override int Key { get; }
```

Property Value

int

MaxPoint

The hammer points needed for a super craft.

```
public int MaxPoint { get; }
```

Property Value

int

RecipeId

The equipment recipe id.

```
public int RecipeId { get; }
```

Property Value

int

Methods

Set(IReadOnlyList<string>)

```
public override void Set(IReadOnlyList<string> fields)
```

Parameters

`fields` IReadOnlyList<string>

Class CrystalMaterialCostSheet

Namespace: Nekoyume.TableData.Crystal

Assembly: Lib9c.dll

```
[Serializable]
public class CrystalMaterialCostSheet : Sheet<int, CrystalMaterialCostSheet.Row>,
IDictionary<int, CrystalMaterialCostSheet.Row>, ICollection<KeyValuePair<int,
CrystalMaterialCostSheet.Row>>, IEnumerable<KeyValuePair<int,
CrystalMaterialCostSheet.Row>>, IEnumerable, ISheet
```

Inheritance

object ← Sheet<int, CrystalMaterialCostSheet.Row> ← CrystalMaterialCostSheet

Implements

IDictionary<int, CrystalMaterialCostSheet.Row>,
ICollection<KeyValuePair<int, CrystalMaterialCostSheet.Row>>,
IEnumerable<KeyValuePair<int, CrystalMaterialCostSheet.Row>>, IEnumerable,
ISheet

Inherited Members

Sheet<int, CrystalMaterialCostSheet.Row>.Name ,
Sheet<int, CrystalMaterialCostSheet.Row>.OrderedList ,
Sheet<int, CrystalMaterialCostSheet.Row>.First ,
Sheet<int, CrystalMaterialCostSheet.Row>.Last ,
Sheet<int, CrystalMaterialCostSheet.Row>.Keys ,
Sheet<int, CrystalMaterialCostSheet.Row>.Values ,
Sheet<int, CrystalMaterialCostSheet.Row>.Count ,
Sheet<int, CrystalMaterialCostSheet.Row>.IsReadOnly ,
Sheet<int, CrystalMaterialCostSheet.Row>.this[int] ,
Sheet<int, CrystalMaterialCostSheet.Row>.Set(string, bool) ,
Sheet<int, CrystalMaterialCostSheet.Row>.Set<T>(Sheet<int, T>, bool) ,
Sheet<int, CrystalMaterialCostSheet.Row>.GetEnumerator() ,
Sheet<int, CrystalMaterialCostSheet.Row>.TryGetValue(int, out
CrystalMaterialCostSheet.Row, bool) ,
Sheet<int, CrystalMaterialCostSheet.Row>.AddRow(int, CrystalMaterialCostSheet.Row) ,
Sheet<int, CrystalMaterialCostSheet.Row>.Add(int, CrystalMaterialCostSheet.Row) ,
Sheet<int, CrystalMaterialCostSheet.Row>.ContainsKey(int) ,
Sheet<int, CrystalMaterialCostSheet.Row>.Remove(int) ,

Sheet<int, CrystalMaterialCostSheet.Row>.TryGetValue(int, out CrystalMaterialCostSheet.Row) , Sheet<int, CrystalMaterialCostSheet.Row>.Add(KeyValuePair<int, CrystalMaterialCostSheet.Row>) , Sheet<int, CrystalMaterialCostSheet.Row>.Clear() , Sheet<int, CrystalMaterialCostSheet.Row>.Contains(KeyValuePair<int, CrystalMaterialCostSheet.Row>) , Sheet<int, CrystalMaterialCostSheet.Row>.CopyTo(KeyValuePair<int, CrystalMaterialCostSheet.Row>[], int) , Sheet<int, CrystalMaterialCostSheet.Row>.Remove(KeyValuePair<int, CrystalMaterialCostSheet.Row>) , Sheet<int, CrystalMaterialCostSheet.Row>.Serialize() , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

CrystalMaterialCostSheet()

Creates an empty sheet.

```
public CrystalMaterialCostSheet()
```

Class CrystalMaterialCostSheet.Row

Namespace: Nekoyume.TableData.Crystal

Assembly: Lib9c.dll

The crystal price of one material.

```
public class CrystalMaterialCostSheet.Row : SheetRow<int>
```

Inheritance

object ← SheetRow<int> ← CrystalMaterialCostSheet.Row

Inherited Members

SheetRow<int>.Validate() , SheetRow<int>.EndOfSheetInitialize() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Properties

CRYSTAL

Crystal that replaces one of the material.

```
public long CRYSTAL { get; }
```

Property Value

long

ItemId

The material id.

```
public int ItemId { get; }
```

Property Value

int

Key

```
public override int Key { get; }
```

Property Value

int

Methods

Set(IReadOnlyList<string>)

```
public override void Set(IReadOnlyList<string> fields)
```

Parameters

`fields` IReadOnlyList<string>

Class CrystalMonsterCollectionMultiplierSheet

Namespace: Nekoyume.TableData.Crystal

Assembly: Lib9c.dll

```
[Serializable]
public class CrystalMonsterCollectionMultiplierSheet : Sheet<int,
CrystalMonsterCollectionMultiplierSheet.Row>, IDictionary<int,
CrystalMonsterCollectionMultiplierSheet.Row>, ICollection<KeyValuePair<int,
CrystalMonsterCollectionMultiplierSheet.Row>>, IEnumerable<KeyValuePair<int,
CrystalMonsterCollectionMultiplierSheet.Row>>, IEnumerable, ISheet
```

Inheritance

object ← Sheet<int, CrystalMonsterCollectionMultiplierSheet.Row> ← CrystalMonsterCollectionMultiplierSheet

Implements

IDictionary<int, CrystalMonsterCollectionMultiplierSheet.Row>, ICollection<KeyValuePair<int, CrystalMonsterCollectionMultiplierSheet.Row>>, IEnumerable<KeyValuePair<int, CrystalMonsterCollectionMultiplierSheet.Row>>, IEnumerable, ISheet

Inherited Members

Sheet<int, CrystalMonsterCollectionMultiplierSheet.Row>.Name ,
Sheet<int, CrystalMonsterCollectionMultiplierSheet.Row>.OrderedList ,
Sheet<int, CrystalMonsterCollectionMultiplierSheet.Row>.First ,
Sheet<int, CrystalMonsterCollectionMultiplierSheet.Row>.Last ,
Sheet<int, CrystalMonsterCollectionMultiplierSheet.Row>.Keys ,
Sheet<int, CrystalMonsterCollectionMultiplierSheet.Row>.Values ,
Sheet<int, CrystalMonsterCollectionMultiplierSheet.Row>.Count ,
Sheet<int, CrystalMonsterCollectionMultiplierSheet.Row>.IsReadOnly ,
Sheet<int, CrystalMonsterCollectionMultiplierSheet.Row>.this[int] ,
Sheet<int, CrystalMonsterCollectionMultiplierSheet.Row>.Set(string, bool) ,
Sheet<int, CrystalMonsterCollectionMultiplierSheet.Row>.Set<T>(Sheet<int, T>, bool) ,
Sheet<int, CrystalMonsterCollectionMultiplierSheet.Row>.GetEnumerator() ,
Sheet<int, CrystalMonsterCollectionMultiplierSheet.Row>.TryGetValue(int, out CrystalMonsterCollectionMultiplierSheet.Row, bool) ,

Sheet<int, CrystalMonsterCollectionMultiplierSheet.Row>.AddRow(int, CrystalMonsterCollectionMultiplierSheet.Row) , Sheet<int, CrystalMonsterCollectionMultiplierSheet.Row>.Add(int, CrystalMonsterCollectionMultiplierSheet.Row) , Sheet<int, CrystalMonsterCollectionMultiplierSheet.Row>.ContainsKey(int) , Sheet<int, CrystalMonsterCollectionMultiplierSheet.Row>.Remove(int) , Sheet<int, CrystalMonsterCollectionMultiplierSheet.Row>.TryGetValue(int, out CrystalMonsterCollectionMultiplierSheet.Row) , Sheet<int, CrystalMonsterCollectionMultiplierSheet.Row>.Add(KeyValuePair<int, CrystalMonsterCollectionMultiplierSheet.Row>) , Sheet<int, CrystalMonsterCollectionMultiplierSheet.Row>.Clear() , Sheet<int, CrystalMonsterCollectionMultiplierSheet.Row>.Contains(KeyValuePair<int, CrystalMonsterCollectionMultiplierSheet.Row>) , Sheet<int, CrystalMonsterCollectionMultiplierSheet.Row>.CopyTo(KeyValuePair<int, CrystalMonsterCollectionMultiplierSheet.Row>[], int) , Sheet<int, CrystalMonsterCollectionMultiplierSheet.Row>.Remove(KeyValuePair<int, CrystalMonsterCollectionMultiplierSheet.Row>) , Sheet<int, CrystalMonsterCollectionMultiplierSheet.Row>.Serialize() , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

CrystalMonsterCollectionMultiplierSheet()

```
public CrystalMonsterCollectionMultiplierSheet()
```

Class CrystalMonsterCollectionMultiplierSheet.Row

Namespace: Nekoyume.TableData.Crystal

Assembly: Lib9c.dll

```
[Serializable]
public class CrystalMonsterCollectionMultiplierSheet.Row : SheetRow<int>
```

Inheritance

object ← SheetRow<int> ← CrystalMonsterCollectionMultiplierSheet.Row

Inherited Members

SheetRow<int>.Validate() , SheetRow<int>.EndOfSheetInitialize() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Fields

Level

```
public int Level
```

Field Value

int

Multiplier

```
public int Multiplier
```

Field Value

int

Properties

Key

```
public override int Key { get; }
```

Property Value

int

Methods

Set(IReadOnlyList<string>)

```
public override void Set(IReadOnlyList<string> fields)
```

Parameters

`fields` IReadOnlyList<string>

Class CrystalRandomBuffSheet

Namespace: Nekoyume.TableData.Crystal

Assembly: Lib9c.dll

```
[Serializable]
public class CrystalRandomBuffSheet : Sheet<int, CrystalRandomBuffSheet.Row>,
IDictionary<int, CrystalRandomBuffSheet.Row>, ICollection<KeyValuePair<int,
CrystalRandomBuffSheet.Row>>, IEnumerable<KeyValuePair<int,
CrystalRandomBuffSheet.Row>>, IEnumerable, ISheet
```

Inheritance

object ← Sheet<int, CrystalRandomBuffSheet.Row> ← CrystalRandomBuffSheet

Implements

IDictionary<int, CrystalRandomBuffSheet.Row>,
ICollection<KeyValuePair<int, CrystalRandomBuffSheet.Row>>,
IEnumerable<KeyValuePair<int, CrystalRandomBuffSheet.Row>>, IEnumerable,
ISheet

Inherited Members

Sheet<int, CrystalRandomBuffSheet.Row>.Name ,
Sheet<int, CrystalRandomBuffSheet.Row>.OrderedList ,
Sheet<int, CrystalRandomBuffSheet.Row>.First ,
Sheet<int, CrystalRandomBuffSheet.Row>.Last ,
Sheet<int, CrystalRandomBuffSheet.Row>.Keys ,
Sheet<int, CrystalRandomBuffSheet.Row>.Values ,
Sheet<int, CrystalRandomBuffSheet.Row>.Count ,
Sheet<int, CrystalRandomBuffSheet.Row>.IsReadOnly ,
Sheet<int, CrystalRandomBuffSheet.Row>.this[int] ,
Sheet<int, CrystalRandomBuffSheet.Row>.Set(string, bool) ,
Sheet<int, CrystalRandomBuffSheet.Row>.Set<T>(Sheet<int, T>, bool) ,
Sheet<int, CrystalRandomBuffSheet.Row>.GetEnumerator() ,
Sheet<int, CrystalRandomBuffSheet.Row>.TryGetValue(int, out
CrystalRandomBuffSheet.Row, bool) ,
Sheet<int, CrystalRandomBuffSheet.Row>.AddRow(int, CrystalRandomBuffSheet.Row) ,
Sheet<int, CrystalRandomBuffSheet.Row>.Add(int, CrystalRandomBuffSheet.Row) ,
Sheet<int, CrystalRandomBuffSheet.Row>.ContainsKey(int) ,
Sheet<int, CrystalRandomBuffSheet.Row>.Remove(int) ,

Sheet<int, CrystalRandomBuffSheet.Row>.TryGetValue(int, out CrystalRandomBuffSheet.Row) , Sheet<int, CrystalRandomBuffSheet.Row>.Add(KeyValuePair<int, CrystalRandomBuffSheet.Row>) , Sheet<int, CrystalRandomBuffSheet.Row>.Clear() , Sheet<int, CrystalRandomBuffSheet.Row>.Contains(KeyValuePair<int, CrystalRandomBuffSheet.Row>) , Sheet<int, CrystalRandomBuffSheet.Row>.CopyTo(KeyValuePair<int, CrystalRandomBuffSheet.Row>[], int) , Sheet<int, CrystalRandomBuffSheet.Row>.Remove(KeyValuePair<int, CrystalRandomBuffSheet.Row>) , Sheet<int, CrystalRandomBuffSheet.Row>.Serialize() , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

CrystalRandomBuffSheet()

```
public CrystalRandomBuffSheet()
```

Class CrystalRandomBuffSheet.Row

Namespace: Nekoyume.TableData.Crystal

Assembly: Lib9c.dll

```
public class CrystalRandomBuffSheet.Row : SheetRow<int>
```

Inheritance

object ← SheetRow<int> ← CrystalRandomBuffSheet.Row

Inherited Members

SheetRow<int>.Validate() , SheetRow<int>.EndOfSheetInitialize() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Fields

Id

```
public int Id
```

Field Value

int

Rank

```
public CrystalRandomBuffSheet.Row.BuffRank Rank
```

Field Value

CrystalRandomBuffSheet.Row.BuffRank

Ratio

```
public decimal Ratio
```

Field Value

decimal

SkillId

```
public int SkillId
```

Field Value

int

Properties

Key

```
public override int Key { get; }
```

Property Value

int

Methods

Set(IReadOnlyList<string>)

```
public override void Set(IReadOnlyList<string> fields)
```

Parameters

`fields` IReadOnlyList<string>

Enum CrystalRandomBuffSheet.Row.BuffRank

Namespace: Nekoyume.TableData.Crystal

Assembly: Lib9c.dll

```
public enum CrystalRandomBuffSheet.Row.BuffRank
```

Extension Methods

StateExtensions.Serialize(Enum)

Fields

A = 3

B = 4

S = 2

SS = 1

Class CrystalStageBuffGachaSheet

Namespace: Nekoyume.TableData.Crystal

Assembly: Lib9c.dll

```
[Serializable]
public class CrystalStageBuffGachaSheet : Sheet<int,
CrystalStageBuffGachaSheet.Row>, IDictionary<int, CrystalStageBuffGachaSheet.Row>,
ICollection<KeyValuePair<int, CrystalStageBuffGachaSheet.Row>>,
IEnumerable<KeyValuePair<int, CrystalStageBuffGachaSheet.Row>>, IEnumerable, ISheet
```

Inheritance

object ← Sheet<int, CrystalStageBuffGachaSheet.Row> ← CrystalStageBuffGachaSheet

Implements

IDictionary<int, CrystalStageBuffGachaSheet.Row>,
ICollection<KeyValuePair<int, CrystalStageBuffGachaSheet.Row>>,
IEnumerable<KeyValuePair<int, CrystalStageBuffGachaSheet.Row>>, IEnumerable,
ISheet

Inherited Members

Sheet<int, CrystalStageBuffGachaSheet.Row>.Name ,
Sheet<int, CrystalStageBuffGachaSheet.Row>.OrderedList ,
Sheet<int, CrystalStageBuffGachaSheet.Row>.First ,
Sheet<int, CrystalStageBuffGachaSheet.Row>.Last ,
Sheet<int, CrystalStageBuffGachaSheet.Row>.Keys ,
Sheet<int, CrystalStageBuffGachaSheet.Row>.Values ,
Sheet<int, CrystalStageBuffGachaSheet.Row>.Count ,
Sheet<int, CrystalStageBuffGachaSheet.Row>.IsReadOnly ,
Sheet<int, CrystalStageBuffGachaSheet.Row>.this[int] ,
Sheet<int, CrystalStageBuffGachaSheet.Row>.Set(string, bool) ,
Sheet<int, CrystalStageBuffGachaSheet.Row>.Set<T>(Sheet<int, T>, bool) ,
Sheet<int, CrystalStageBuffGachaSheet.Row>.GetEnumerator() ,
Sheet<int, CrystalStageBuffGachaSheet.Row>.TryGetValue(int, out
CrystalStageBuffGachaSheet.Row, bool) ,
Sheet<int, CrystalStageBuffGachaSheet.Row>.AddRow(int,
CrystalStageBuffGachaSheet.Row) ,
Sheet<int, CrystalStageBuffGachaSheet.Row>.Add(int, CrystalStageBuffGachaSheet.Row) ,
Sheet<int, CrystalStageBuffGachaSheet.Row>.ContainsKey(int) ,
Sheet<int, CrystalStageBuffGachaSheet.Row>.Remove(int) ,

Sheet<int, CrystalStageBuffGachaSheet.Row>.TryGetValue(int, out CrystalStageBuffGachaSheet.Row) , Sheet<int, CrystalStageBuffGachaSheet.Row>.Add(KeyValuePair<int, CrystalStageBuffGachaSheet.Row>) , Sheet<int, CrystalStageBuffGachaSheet.Row>.Clear() , Sheet<int, CrystalStageBuffGachaSheet.Row>.Contains(KeyValuePair<int, CrystalStageBuffGachaSheet.Row>) , Sheet<int, CrystalStageBuffGachaSheet.Row>.CopyTo(KeyValuePair<int, CrystalStageBuffGachaSheet.Row>[], int) , Sheet<int, CrystalStageBuffGachaSheet.Row>.Remove(KeyValuePair<int, CrystalStageBuffGachaSheet.Row>) , Sheet<int, CrystalStageBuffGachaSheet.Row>.Serialize() , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

CrystalStageBuffGachaSheet()

```
public CrystalStageBuffGachaSheet()
```

Methods

AddRow(int, Row)

```
protected override void AddRow(int key, CrystalStageBuffGachaSheet.Row value)
```

Parameters

`key` int

`value` CrystalStageBuffGachaSheet.Row

Class CrystalStageBuffGachaSheet.Row

Namespace: Nekoyume.TableData.Crystal

Assembly: Lib9c.dll

```
public class CrystalStageBuffGachaSheet.Row : SheetRow<int>
```

Inheritance

object ← SheetRow<int> ← CrystalStageBuffGachaSheet.Row

Inherited Members

SheetRow<int>.Validate() , SheetRow<int>.EndOfSheetInitialize() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Fields

AdvancedCost

```
public int AdvancedCost
```

Field Value

int

MaxStar

```
public int MaxStar
```

Field Value

int

NormalCost

```
public int NormalCost
```

Field Value

int

StageId

```
public int StageId
```

Field Value

int

Properties

Key

```
public override int Key { get; }
```

Property Value

int

Methods

Set(IReadOnlyList<string>)

```
public override void Set(IReadOnlyList<string> fields)
```

Parameters

`fields` IReadOnlyList<string>

Namespace Nekoyume.TableData.Custom EquipmentCraft

Classes

CustomEquipmentCraftIconSheet

CustomEquipmentCraftIconSheet.Row

CustomEquipmentCraftOptionSheet

CustomEquipmentCraftOptionSheet.Row

CustomEquipmentCraftRecipeSheet

CustomEquipmentCraftRecipeSheet.Row

CustomEquipmentCraftRecipeSkillSheet

CustomEquipmentCraftRecipeSkillSheet.Row

CustomEquipmentCraftRelationshipSheet

CustomEquipmentCraftRelationshipSheet.Row

Structs

CustomEquipmentCraftOptionSheet.SubStat

CustomEquipmentCraftRelationshipSheet.CpGroup
A total CP range and its selection weight.

CustomEquipmentCraftRelationshipSheet.MaterialCost

Class CustomEquipmentCraftIconSheet

Namespace: Nekoyume.TableData.CustomEquipmentCraft

Assembly: Lib9c.dll

```
[Serializable]
public class CustomEquipmentCraftIconSheet : Sheet<int,
CustomEquipmentCraftIconSheet.Row>, IDictionary<int,
CustomEquipmentCraftIconSheet.Row>, ICollection<KeyValuePair<int,
CustomEquipmentCraftIconSheet.Row>>, IEnumerable<KeyValuePair<int,
CustomEquipmentCraftIconSheet.Row>>, IEnumerable, ISheet
```

Inheritance

object ← Sheet<int, CustomEquipmentCraftIconSheet.Row> ← CustomEquipmentCraftIconSheet

Implements

IDictionary<int, CustomEquipmentCraftIconSheet.Row>, ICollection<KeyValuePair<int, CustomEquipmentCraftIconSheet.Row>>, IEnumerable<KeyValuePair<int, CustomEquipmentCraftIconSheet.Row>>, IEnumerable, ISheet

Inherited Members

Sheet<int, CustomEquipmentCraftIconSheet.Row>.Name ,
Sheet<int, CustomEquipmentCraftIconSheet.Row>.OrderedList ,
Sheet<int, CustomEquipmentCraftIconSheet.Row>.First ,
Sheet<int, CustomEquipmentCraftIconSheet.Row>.Last ,
Sheet<int, CustomEquipmentCraftIconSheet.Row>.Keys ,
Sheet<int, CustomEquipmentCraftIconSheet.Row>.Values ,
Sheet<int, CustomEquipmentCraftIconSheet.Row>.Count ,
Sheet<int, CustomEquipmentCraftIconSheet.Row>.IsReadOnly ,
Sheet<int, CustomEquipmentCraftIconSheet.Row>.this[int] ,
Sheet<int, CustomEquipmentCraftIconSheet.Row>.Set(string, bool) ,
Sheet<int, CustomEquipmentCraftIconSheet.Row>.Set<T>(Sheet<int, T>, bool) ,
Sheet<int, CustomEquipmentCraftIconSheet.Row>.GetEnumerator() ,
Sheet<int, CustomEquipmentCraftIconSheet.Row>.TryGetValue(int, out CustomEquipmentCraftIconSheet.Row, bool) ,
Sheet<int, CustomEquipmentCraftIconSheet.Row>.AddRow(int, CustomEquipmentCraftIconSheet.Row) ,

Sheet<int, CustomEquipmentCraftIconSheet.Row>.Add(int, CustomEquipmentCraftIconSheet.Row) , Sheet<int, CustomEquipmentCraftIconSheet.Row>.ContainsKey(int) , Sheet<int, CustomEquipmentCraftIconSheet.Row>.Remove(int) , Sheet<int, CustomEquipmentCraftIconSheet.Row>.TryGetValue(int, out CustomEquipmentCraftIconSheet.Row) , Sheet<int, CustomEquipmentCraftIconSheet.Row>.Add(KeyValuePair<int, CustomEquipmentCraftIconSheet.Row>) , Sheet<int, CustomEquipmentCraftIconSheet.Row>.Clear() , Sheet<int, CustomEquipmentCraftIconSheet.Row>.Contains(KeyValuePair<int, CustomEquipmentCraftIconSheet.Row>) , Sheet<int, CustomEquipmentCraftIconSheet.Row>.CopyTo(KeyValuePair<int, CustomEquipmentCraftIconSheet.Row>[], int) , Sheet<int, CustomEquipmentCraftIconSheet.Row>.Remove(KeyValuePair<int, CustomEquipmentCraftIconSheet.Row>) , Sheet<int, CustomEquipmentCraftIconSheet.Row>.Serialize() , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

CustomEquipmentCraftIconSheet()

```
public CustomEquipmentCraftIconSheet()
```

Class CustomEquipmentCraftIconSheet.Row

Namespace: Nekoyume.TableData.CustomEquipmentCraft

Assembly: Lib9c.dll

```
[Serializable]
public class CustomEquipmentCraftIconSheet.Row : SheetRow<int>
```

Inheritance

object ← SheetRow<int> ← CustomEquipmentCraftIconSheet.Row

Inherited Members

SheetRow<int>.Validate() , SheetRow<int>.EndOfSheetInitialize() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Properties

IconId

```
public int IconId { get; }
```

Property Value

int

Id

```
public int Id { get; }
```

Property Value

int

ItemSubType

```
public ItemSubType ItemSubType { get; }
```

Property Value

ItemSubType

Key

```
public override int Key { get; }
```

Property Value

int

RandomOnly

```
public bool RandomOnly { get; }
```

Property Value

bool

Ratio

```
public int Ratio { get; }
```

Property Value

int

RequiredRelationship

```
public int RequiredRelationship { get; }
```

Property Value

int

Methods

Set(IReadOnlyList<string>)

```
public override void Set(IReadOnlyList<string> fields)
```

Parameters

`fields` IReadOnlyList<string>

Class CustomEquipmentCraftOptionSheet

Namespace: Nekoyume.TableData.CustomEquipmentCraft

Assembly: Lib9c.dll

```
[Serializable]
public class CustomEquipmentCraftOptionSheet : Sheet<int,
CustomEquipmentCraftOptionSheet.Row>, IDictionary<int,
CustomEquipmentCraftOptionSheet.Row>, ICollection<KeyValuePair<int,
CustomEquipmentCraftOptionSheet.Row>>, IEnumerable<KeyValuePair<int,
CustomEquipmentCraftOptionSheet.Row>>, IEnumerable, ISheet
```

Inheritance

object ← Sheet<int, CustomEquipmentCraftOptionSheet.Row> ← CustomEquipmentCraftOptionSheet

Implements

IDictionary<int, CustomEquipmentCraftOptionSheet.Row>, ICollection<KeyValuePair<int, CustomEquipmentCraftOptionSheet.Row>>, IEnumerable<KeyValuePair<int, CustomEquipmentCraftOptionSheet.Row>>, IEnumerable, ISheet

Inherited Members

Sheet<int, CustomEquipmentCraftOptionSheet.Row>.Name ,
Sheet<int, CustomEquipmentCraftOptionSheet.Row>.OrderedList ,
Sheet<int, CustomEquipmentCraftOptionSheet.Row>.First ,
Sheet<int, CustomEquipmentCraftOptionSheet.Row>.Last ,
Sheet<int, CustomEquipmentCraftOptionSheet.Row>.Keys ,
Sheet<int, CustomEquipmentCraftOptionSheet.Row>.Values ,
Sheet<int, CustomEquipmentCraftOptionSheet.Row>.Count ,
Sheet<int, CustomEquipmentCraftOptionSheet.Row>.IsReadOnly ,
Sheet<int, CustomEquipmentCraftOptionSheet.Row>.this[int] ,
Sheet<int, CustomEquipmentCraftOptionSheet.Row>.Set(string, bool) ,
Sheet<int, CustomEquipmentCraftOptionSheet.Row>.Set<T>(Sheet<int, T>, bool) ,
Sheet<int, CustomEquipmentCraftOptionSheet.Row>.GetEnumerator() ,
Sheet<int, CustomEquipmentCraftOptionSheet.Row>.TryGetValue(int, out CustomEquipmentCraftOptionSheet.Row, bool) ,
Sheet<int, CustomEquipmentCraftOptionSheet.Row>.AddRow(int, CustomEquipmentCraftOptionSheet.Row) ,

Sheet<int, CustomEquipmentCraftOptionSheet.Row>.Add(int, CustomEquipmentCraftOptionSheet.Row) , Sheet<int, CustomEquipmentCraftOptionSheet.Row>.ContainsKey(int) , Sheet<int, CustomEquipmentCraftOptionSheet.Row>.Remove(int) , Sheet<int, CustomEquipmentCraftOptionSheet.Row>.TryGetValue(int, out CustomEquipmentCraftOptionSheet.Row) , Sheet<int, CustomEquipmentCraftOptionSheet.Row>.Add(KeyValuePair<int, CustomEquipmentCraftOptionSheet.Row>) , Sheet<int, CustomEquipmentCraftOptionSheet.Row>.Clear() , Sheet<int, CustomEquipmentCraftOptionSheet.Row>.Contains(KeyValuePair<int, CustomEquipmentCraftOptionSheet.Row>) , Sheet<int, CustomEquipmentCraftOptionSheet.Row>.CopyTo(KeyValuePair<int, CustomEquipmentCraftOptionSheet.Row>[], int) , Sheet<int, CustomEquipmentCraftOptionSheet.Row>.Remove(KeyValuePair<int, CustomEquipmentCraftOptionSheet.Row>) , Sheet<int, CustomEquipmentCraftOptionSheet.Row>.Serialize() , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

CustomEquipmentCraftOptionSheet()

```
public CustomEquipmentCraftOptionSheet()
```

Class CustomEquipmentCraftOptionSheet.Row

Namespace: Nekoyume.TableData.CustomEquipmentCraft

Assembly: Lib9c.dll

```
[Serializable]
public class CustomEquipmentCraftOptionSheet.Row : SheetRow<int>
```

Inheritance

object ← SheetRow<int> ← CustomEquipmentCraftOptionSheet.Row

Inherited Members

SheetRow<int>.Validate() , SheetRow<int>.EndOfSheetInitialize() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Properties

Id

```
public int Id { get; }
```

Property Value

int

ItemSubType

```
public ItemSubType ItemSubType { get; }
```

Property Value

ItemSubType

Key

```
public override int Key { get; }
```

Property Value

int

Ratio

```
public int Ratio { get; }
```

Property Value

int

SubStatData

```
public List<CustomEquipmentCraftOptionSheet.SubStat> SubStatData { get; }
```

Property Value

List<CustomEquipmentCraftOptionSheet.SubStat>

TotalOptionRatio

```
public int TotalOptionRatio { get; }
```

Property Value

int

Methods

Set(IReadOnlyList<string>)

```
public override void Set(IReadOnlyList<string> fields)
```

Parameters

`fields` IReadOnlyList<string>

Struct CustomEquipmentCraftOptionSheet.SubStat

Namespace: Nekoyume.TableData.CustomEquipmentCraft

Assembly: Lib9c.dll

```
public struct CustomEquipmentCraftOptionSheet.SubStat
```

Inherited Members

ValueType.Equals(object) , ValueType.GetHashCode() , ValueType.ToString() , object.Equals(object, object) , object.GetType() , object.ReferenceEquals(object, object)

Fields

Ratio

```
public int Ratio
```

Field Value

int

StatType

```
public StatType StatType
```

Field Value

StatType

Class CustomEquipmentCraftRecipeSheet

Namespace: Nekoyume.TableData.CustomEquipmentCraft

Assembly: Lib9c.dll

```
[Serializable]
public class CustomEquipmentCraftRecipeSheet : Sheet<int,
CustomEquipmentCraftRecipeSheet.Row>, IDictionary<int,
CustomEquipmentCraftRecipeSheet.Row>, ICollection<KeyValuePair<int,
CustomEquipmentCraftRecipeSheet.Row>>, IEnumerable<KeyValuePair<int,
CustomEquipmentCraftRecipeSheet.Row>>, IEnumerable, ISheet
```

Inheritance

object ← Sheet<int, CustomEquipmentCraftRecipeSheet.Row> ← CustomEquipmentCraftRecipeSheet

Implements

IDictionary<int, CustomEquipmentCraftRecipeSheet.Row>, ICollection<KeyValuePair<int, CustomEquipmentCraftRecipeSheet.Row>>, IEnumerable<KeyValuePair<int, CustomEquipmentCraftRecipeSheet.Row>>, IEnumerable, ISheet

Inherited Members

Sheet<int, CustomEquipmentCraftRecipeSheet.Row>.Name ,
Sheet<int, CustomEquipmentCraftRecipeSheet.Row>.OrderedList ,
Sheet<int, CustomEquipmentCraftRecipeSheet.Row>.First ,
Sheet<int, CustomEquipmentCraftRecipeSheet.Row>.Last ,
Sheet<int, CustomEquipmentCraftRecipeSheet.Row>.Keys ,
Sheet<int, CustomEquipmentCraftRecipeSheet.Row>.Values ,
Sheet<int, CustomEquipmentCraftRecipeSheet.Row>.Count ,
Sheet<int, CustomEquipmentCraftRecipeSheet.Row>.IsReadOnly ,
Sheet<int, CustomEquipmentCraftRecipeSheet.Row>.this[int] ,
Sheet<int, CustomEquipmentCraftRecipeSheet.Row>.Set(string, bool) ,
Sheet<int, CustomEquipmentCraftRecipeSheet.Row>.Set<T>(Sheet<int, T>, bool) ,
Sheet<int, CustomEquipmentCraftRecipeSheet.Row>.GetEnumerator() ,
Sheet<int, CustomEquipmentCraftRecipeSheet.Row>.TryGetValue(int, out CustomEquipmentCraftRecipeSheet.Row, bool) ,
Sheet<int, CustomEquipmentCraftRecipeSheet.Row>.AddRow(int, CustomEquipmentCraftRecipeSheet.Row) ,

Sheet<int, CustomEquipmentCraftRecipeSheet.Row>.Add(int, CustomEquipmentCraftRecipeSheet.Row) , Sheet<int, CustomEquipmentCraftRecipeSheet.Row>.ContainsKey(int) , Sheet<int, CustomEquipmentCraftRecipeSheet.Row>.Remove(int) , Sheet<int, CustomEquipmentCraftRecipeSheet.Row>.TryGetValue(int, out CustomEquipmentCraftRecipeSheet.Row) , Sheet<int, CustomEquipmentCraftRecipeSheet.Row>.Add(KeyValuePair<int, CustomEquipmentCraftRecipeSheet.Row>) , Sheet<int, CustomEquipmentCraftRecipeSheet.Row>.Clear() , Sheet<int, CustomEquipmentCraftRecipeSheet.Row>.Contains(KeyValuePair<int, CustomEquipmentCraftRecipeSheet.Row>) , Sheet<int, CustomEquipmentCraftRecipeSheet.Row>.CopyTo(KeyValuePair<int, CustomEquipmentCraftRecipeSheet.Row>[], int) , Sheet<int, CustomEquipmentCraftRecipeSheet.Row>.Remove(KeyValuePair<int, CustomEquipmentCraftRecipeSheet.Row>) , Sheet<int, CustomEquipmentCraftRecipeSheet.Row>.Serialize() , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

CustomEquipmentCraftRecipeSheet()

```
public CustomEquipmentCraftRecipeSheet()
```

Class CustomEquipmentCraftRecipeSheet.Row

Namespace: Nekoyume.TableData.CustomEquipmentCraft

Assembly: Lib9c.dll

```
[Serializable]
public class CustomEquipmentCraftRecipeSheet.Row : SheetRow<int>
```

Inheritance

object ← SheetRow<int> ← CustomEquipmentCraftRecipeSheet.Row

Inherited Members

SheetRow<int>.Validate() , SheetRow<int>.EndOfSheetInitialize() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Properties

CircleAmount

```
public int CircleAmount { get; }
```

Property Value

int

Id

```
public int Id { get; }
```

Property Value

int

ItemSubType

```
public ItemSubType ItemSubType { get; }
```

Property Value

ItemSubType

Key

```
public override int Key { get; }
```

Property Value

int

RequiredBlock

```
public long RequiredBlock { get; }
```

Property Value

long

ScrollAmount

```
public int ScrollAmount { get; }
```

Property Value

int

Methods

Set(IReadOnlyList<string>)

```
public override void Set(IReadOnlyList<string> fields)
```

Parameters

`fields` IReadOnlyList<string>

Class CustomEquipmentCraftRecipeSkillSheet

Namespace: Nekoyume.TableData.CustomEquipmentCraft

Assembly: Lib9c.dll

```
[Serializable]
public class CustomEquipmentCraftRecipeSkillSheet : Sheet<int,
CustomEquipmentCraftRecipeSkillSheet.Row>, IDictionary<int,
CustomEquipmentCraftRecipeSkillSheet.Row>, ICollection<KeyValuePair<int,
CustomEquipmentCraftRecipeSkillSheet.Row>>, IEnumerable<KeyValuePair<int,
CustomEquipmentCraftRecipeSkillSheet.Row>>, IEnumerable, ISheet
```

Inheritance

object ← Sheet<int, CustomEquipmentCraftRecipeSkillSheet.Row> ← CustomEquipmentCraftRecipeSkillSheet

Implements

IDictionary<int, CustomEquipmentCraftRecipeSkillSheet.Row>, ICollection<KeyValuePair<int, CustomEquipmentCraftRecipeSkillSheet.Row>>, IEnumerable<KeyValuePair<int, CustomEquipmentCraftRecipeSkillSheet.Row>>, IEnumerable, ISheet

Inherited Members

Sheet<int, CustomEquipmentCraftRecipeSkillSheet.Row>.Name ,
Sheet<int, CustomEquipmentCraftRecipeSkillSheet.Row>.OrderedList ,
Sheet<int, CustomEquipmentCraftRecipeSkillSheet.Row>.First ,
Sheet<int, CustomEquipmentCraftRecipeSkillSheet.Row>.Last ,
Sheet<int, CustomEquipmentCraftRecipeSkillSheet.Row>.Keys ,
Sheet<int, CustomEquipmentCraftRecipeSkillSheet.Row>.Values ,
Sheet<int, CustomEquipmentCraftRecipeSkillSheet.Row>.Count ,
Sheet<int, CustomEquipmentCraftRecipeSkillSheet.Row>.IsReadOnly ,
Sheet<int, CustomEquipmentCraftRecipeSkillSheet.Row>.this[int] ,
Sheet<int, CustomEquipmentCraftRecipeSkillSheet.Row>.Set(string, bool) ,
Sheet<int, CustomEquipmentCraftRecipeSkillSheet.Row>.Set<T>(Sheet<int, T>, bool) ,
Sheet<int, CustomEquipmentCraftRecipeSkillSheet.Row>.GetEnumerator() ,
Sheet<int, CustomEquipmentCraftRecipeSkillSheet.Row>.TryGetValue(int, out CustomEquipmentCraftRecipeSkillSheet.Row, bool) ,

Sheet<int, CustomEquipmentCraftRecipeSkillSheet.Row>.AddRow(int, CustomEquipmentCraftRecipeSkillSheet.Row) , Sheet<int, CustomEquipmentCraftRecipeSkillSheet.Row>.Add(int, CustomEquipmentCraftRecipeSkillSheet.Row) , Sheet<int, CustomEquipmentCraftRecipeSkillSheet.Row>.ContainsKey(int) , Sheet<int, CustomEquipmentCraftRecipeSkillSheet.Row>.Remove(int) , Sheet<int, CustomEquipmentCraftRecipeSkillSheet.Row>.TryGetValue(int, out CustomEquipmentCraftRecipeSkillSheet.Row) , Sheet<int, CustomEquipmentCraftRecipeSkillSheet.Row>.Add(KeyValuePair<int, CustomEquipmentCraftRecipeSkillSheet.Row>) , Sheet<int, CustomEquipmentCraftRecipeSkillSheet.Row>.Clear() , Sheet<int, CustomEquipmentCraftRecipeSkillSheet.Row>.Contains(KeyValuePair<int, CustomEquipmentCraftRecipeSkillSheet.Row>) , Sheet<int, CustomEquipmentCraftRecipeSkillSheet.Row>.CopyTo(KeyValuePair<int, CustomEquipmentCraftRecipeSkillSheet.Row>[], int) , Sheet<int, CustomEquipmentCraftRecipeSkillSheet.Row>.Remove(KeyValuePair<int, CustomEquipmentCraftRecipeSkillSheet.Row>) , Sheet<int, CustomEquipmentCraftRecipeSkillSheet.Row>.Serialize() , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

CustomEquipmentCraftRecipeSkillSheet()

```
public CustomEquipmentCraftRecipeSkillSheet()
```

Class CustomEquipmentCraftRecipeSkillSheet.Row

Namespace: Nekoyume.TableData.CustomEquipmentCraft

Assembly: Lib9c.dll

```
[Serializable]
public class CustomEquipmentCraftRecipeSkillSheet.Row : SheetRow<int>
```

Inheritance

object ← SheetRow<int> ← CustomEquipmentCraftRecipeSkillSheet.Row

Inherited Members

SheetRow<int>.Validate() , SheetRow<int>.EndOfSheetInitialize() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Properties

Id

```
public int Id { get; }
```

Property Value

int

ItemOptionId

```
public int ItemOptionId { get; }
```

Property Value

int

ItemSubType

```
public ItemSubType ItemSubType { get; }
```

Property Value

ItemSubType

Key

```
public override int Key { get; }
```

Property Value

int

Ratio

```
public int Ratio { get; }
```

Property Value

int

Methods

Set(IReadOnlyList<string>)

```
public override void Set(IReadOnlyList<string> fields)
```

Parameters

`fields` IReadOnlyList<string>

Class CustomEquipmentCraftRelationshipSheet

Namespace: Nekoyume.TableData.CustomEquipmentCraft

Assembly: Lib9c.dll

```
[Serializable]
public class CustomEquipmentCraftRelationshipSheet : Sheet<int,
CustomEquipmentCraftRelationshipSheet.Row>, IDictionary<int,
CustomEquipmentCraftRelationshipSheet.Row>, ICollection<KeyValuePair<int,
CustomEquipmentCraftRelationshipSheet.Row>>, IEnumerable<KeyValuePair<int,
CustomEquipmentCraftRelationshipSheet.Row>>, IEnumerable, ISheet
```

Inheritance

object ← Sheet<int, CustomEquipmentCraftRelationshipSheet.Row> ← CustomEquipmentCraftRelationshipSheet

Implements

IDictionary<int, CustomEquipmentCraftRelationshipSheet.Row>, ICollection<KeyValuePair<int, CustomEquipmentCraftRelationshipSheet.Row>>, IEnumerable<KeyValuePair<int, CustomEquipmentCraftRelationshipSheet.Row>>, IEnumerable, ISheet

Inherited Members

Sheet<int, CustomEquipmentCraftRelationshipSheet.Row>.Name ,
Sheet<int, CustomEquipmentCraftRelationshipSheet.Row>.OrderedList ,
Sheet<int, CustomEquipmentCraftRelationshipSheet.Row>.First ,
Sheet<int, CustomEquipmentCraftRelationshipSheet.Row>.Last ,
Sheet<int, CustomEquipmentCraftRelationshipSheet.Row>.Keys ,
Sheet<int, CustomEquipmentCraftRelationshipSheet.Row>.Values ,
Sheet<int, CustomEquipmentCraftRelationshipSheet.Row>.Count ,
Sheet<int, CustomEquipmentCraftRelationshipSheet.Row>.IsReadOnly ,
Sheet<int, CustomEquipmentCraftRelationshipSheet.Row>.this[int] ,
Sheet<int, CustomEquipmentCraftRelationshipSheet.Row>.Set(string, bool) ,
Sheet<int, CustomEquipmentCraftRelationshipSheet.Row>.Set<T>(Sheet<int, T>, bool) ,
Sheet<int, CustomEquipmentCraftRelationshipSheet.Row>.GetEnumerator() ,
Sheet<int, CustomEquipmentCraftRelationshipSheet.Row>.TryGetValue(int, out CustomEquipmentCraftRelationshipSheet.Row, bool) ,

Sheet<int, CustomEquipmentCraftRelationshipSheet.Row>.AddRow(int, CustomEquipmentCraftRelationshipSheet.Row) , Sheet<int, CustomEquipmentCraftRelationshipSheet.Row>.Add(int, CustomEquipmentCraftRelationshipSheet.Row) , Sheet<int, CustomEquipmentCraftRelationshipSheet.Row>.ContainsKey(int) , Sheet<int, CustomEquipmentCraftRelationshipSheet.Row>.Remove(int) , Sheet<int, CustomEquipmentCraftRelationshipSheet.Row>.TryGetValue(int, out CustomEquipmentCraftRelationshipSheet.Row) , Sheet<int, CustomEquipmentCraftRelationshipSheet.Row>.Add(KeyValuePair<int, CustomEquipmentCraftRelationshipSheet.Row>) , Sheet<int, CustomEquipmentCraftRelationshipSheet.Row>.Clear() , Sheet<int, CustomEquipmentCraftRelationshipSheet.Row>.Contains(KeyValuePair<int, CustomEquipmentCraftRelationshipSheet.Row>) , Sheet<int, CustomEquipmentCraftRelationshipSheet.Row>.CopyTo(KeyValuePair<int, CustomEquipmentCraftRelationshipSheet.Row>[], int) , Sheet<int, CustomEquipmentCraftRelationshipSheet.Row>.Remove(KeyValuePair<int, CustomEquipmentCraftRelationshipSheet.Row>) , Sheet<int, CustomEquipmentCraftRelationshipSheet.Row>.Serialize() , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

CustomEquipmentCraftRelationshipSheet()

```
public CustomEquipmentCraftRelationshipSheet()
```

Struct CustomEquipmentCraftRelationshipSheet.CpGroup

Namespace: Nekoyume.TableData.CustomEquipmentCraft

Assembly: Lib9c.dll

A total CP range and its selection weight.

```
public struct CustomEquipmentCraftRelationshipSheet.CpGroup
```

Inherited Members
ValueType.Equals(object) , ValueType.GetHashCode() , ValueType.ToString() , object.Equals(object, object) , object.GetType() , object.ReferenceEquals(object, object)

Fields

MaxCp

The inclusive upper bound of the total CP.

```
public long MaxCp
```

Field Value

long

MinCp

The inclusive lower bound of the total CP.

```
public long MinCp
```

Field Value

long

Ratio

The weight of this group among the row's groups.

```
public int Ratio
```

Field Value

int

Methods

SelectCp(IRandom)

```
public long SelectCp(IRandom random)
```

Parameters

`random` IRandom

Returns

long

Struct CustomEquipmentCraftRelationshipSheet.MaterialCost

Namespace: Nekoyume.TableData.CustomEquipmentCraft

Assembly: Lib9c.dll

```
public struct CustomEquipmentCraftRelationshipSheet.MaterialCost
```

Inherited Members

ValueType.Equals(object) , ValueType.GetHashCode() , ValueType.ToString() , object.Equals(object, object) , object.GetType() , object.ReferenceEquals(object, object)

Fields

Amount

```
public int Amount
```

Field Value

int

ItemId

```
public int ItemId
```

Field Value

int

Class CustomEquipmentCraftRelationshipSheet.Row

Namespace: Nekoyume.TableData.CustomEquipmentCraft

Assembly: Lib9c.dll

```
[Serializable]
public class CustomEquipmentCraftRelationshipSheet.Row : SheetRow<int>
```

Inheritance

object ← SheetRow<int> ← CustomEquipmentCraftRelationshipSheet.Row

Inherited Members

SheetRow<int>.Validate() , SheetRow<int>.EndOfSheetInitialize() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Properties

ArmorItemId

```
public int ArmorItemId { get; }
```

Property Value

int

BeltItemId

```
public int BeltItemId { get; }
```

Property Value

int

CostMultiplier

```
public long CostMultiplier { get; }
```

Property Value

long

CpGroups

```
public List<CustomEquipmentCraftRelationshipSheet.CpGroup> CpGroups { get; }
```

Property Value

List<CustomEquipmentCraftRelationshipSheet.CpGroup>

GoldAmount

```
public BigInteger GoldAmount { get; }
```

Property Value

BigInteger

Key

```
public override int Key { get; }
```

Property Value

int

MaterialCosts

```
public List<CustomEquipmentCraftRelationshipSheet.MaterialCost> MaterialCosts {
get; }
```

Property Value

List<CustomEquipmentCraftRelationshipSheet.MaterialCost>

NecklaceItemId

```
public int NecklaceItemId { get; }
```

Property Value

int

Relationship

```
public int Relationship { get; }
```

Property Value

int

RequiredBlockMultiplier

```
public long RequiredBlockMultiplier { get; }
```

Property Value

long

RingItemId

```
public int RingItemId { get; }
```

Property Value

int

WeaponItemId

```
public int WeaponItemId { get; }
```

Property Value

int

Methods

GetItemId(ItemSubType)

```
public int GetItemId(ItemSubType itemSubType)
```

Parameters

`itemSubType` ItemSubType

Returns

int

Set(IReadOnlyList<string>)

```
public override void Set(IReadOnlyList<string> fields)
```

Parameters

`fields` IReadOnlyList<string>

Namespace Nekoyume.TableData.Event

Classes

EventConsumableItemRecipeSheet

EventConsumableItemRecipeSheet.Row

EventDungeonSheet

EventDungeonSheet.Row

EventDungeonStageSheet

EventDungeonStageSheet.Row

EventDungeonStageWaveSheet

EventDungeonStageWaveSheet.Row

EventMaterialItemRecipeSheet

EventMaterialItemRecipeSheet.Row

EventScheduleSheet

EventScheduleSheet.Row

PatrolRewardSheet
: Represents the PatrolRewardSheet, which defines patrol reward policies and rules.

PatrolRewardSheet.RewardModel
: Represents a reward model consisting of item counts, IDs, and tickers.

PatrolRewardSheet.Row
: Represents a row in the PatrolRewardSheet, defining patrol reward policies for specific levels and intervals.

Class EventConsumableItemRecipeSheet

Namespace: Nekoyume.TableData.Event

Assembly: Lib9c.dll

```
[Serializable]
public class EventConsumableItemRecipeSheet : Sheet<int,
EventConsumableItemRecipeSheet.Row>, IDictionary<int,
EventConsumableItemRecipeSheet.Row>, ICollection<KeyValuePair<int,
EventConsumableItemRecipeSheet.Row>>, IEnumerable<KeyValuePair<int,
EventConsumableItemRecipeSheet.Row>>, IEnumerable, ISheet
```

Inheritance

object ← Sheet<int, EventConsumableItemRecipeSheet.Row> ← EventConsumableItemRecipeSheet

Implements

IDictionary<int, EventConsumableItemRecipeSheet.Row>, ICollection<KeyValuePair<int, EventConsumableItemRecipeSheet.Row>>, IEnumerable<KeyValuePair<int, EventConsumableItemRecipeSheet.Row>>, IEnumerable, ISheet

Inherited Members

Sheet<int, EventConsumableItemRecipeSheet.Row>.Name ,
Sheet<int, EventConsumableItemRecipeSheet.Row>.OrderedList ,
Sheet<int, EventConsumableItemRecipeSheet.Row>.First ,
Sheet<int, EventConsumableItemRecipeSheet.Row>.Last ,
Sheet<int, EventConsumableItemRecipeSheet.Row>.Keys ,
Sheet<int, EventConsumableItemRecipeSheet.Row>.Values ,
Sheet<int, EventConsumableItemRecipeSheet.Row>.Count ,
Sheet<int, EventConsumableItemRecipeSheet.Row>.IsReadOnly ,
Sheet<int, EventConsumableItemRecipeSheet.Row>.this[int] ,
Sheet<int, EventConsumableItemRecipeSheet.Row>.Set(string, bool) ,
Sheet<int, EventConsumableItemRecipeSheet.Row>.Set<T>(Sheet<int, T>, bool) ,
Sheet<int, EventConsumableItemRecipeSheet.Row>.GetEnumerator() ,
Sheet<int, EventConsumableItemRecipeSheet.Row>.TryGetValue(int, out EventConsumableItemRecipeSheet.Row, bool) ,
Sheet<int, EventConsumableItemRecipeSheet.Row>.AddRow(int, EventConsumableItemRecipeSheet.Row) ,

Sheet<int, EventConsumableItemRecipeSheet.Row>.Add(int, EventConsumableItemRecipeSheet.Row) , Sheet<int, EventConsumableItemRecipeSheet.Row>.ContainsKey(int) , Sheet<int, EventConsumableItemRecipeSheet.Row>.Remove(int) , Sheet<int, EventConsumableItemRecipeSheet.Row>.TryGetValue(int, out EventConsumableItemRecipeSheet.Row) , Sheet<int, EventConsumableItemRecipeSheet.Row>.Add(KeyValuePair<int, EventConsumableItemRecipeSheet.Row>) , Sheet<int, EventConsumableItemRecipeSheet.Row>.Clear() , Sheet<int, EventConsumableItemRecipeSheet.Row>.Contains(KeyValuePair<int, EventConsumableItemRecipeSheet.Row>) , Sheet<int, EventConsumableItemRecipeSheet.Row>.CopyTo(KeyValuePair<int, EventConsumableItemRecipeSheet.Row>[], int) , Sheet<int, EventConsumableItemRecipeSheet.Row>.Remove(KeyValuePair<int, EventConsumableItemRecipeSheet.Row>) , Sheet<int, EventConsumableItemRecipeSheet.Row>.Serialize() , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Extension Methods

EventRecipeExtensions.GetRecipeRows(EventConsumableItemRecipeSheet, int) , EventRecipeExtensions.ValidateFromAction(EventConsumableItemRecipeSheet, int, string, string)

Constructors

EventConsumableItemRecipeSheet()

```
public EventConsumableItemRecipeSheet()
```

Class EventConsumableItemRecipeSheet.Row

Namespace: Nekoyume.TableData.Event

Assembly: Lib9c.dll

```
[Serializable]
public class EventConsumableItemRecipeSheet.Row : ConsumableItemRecipeSheet.Row
```

Inheritance

object ← SheetRow<int> ← ConsumableItemRecipeSheet.Row ← EventConsumableItemRecipeSheet.Row

Inherited Members

ConsumableItemRecipeSheet.Row.Key , ConsumableItemRecipeSheet.Row.Id , ConsumableItemRecipeSheet.Row.RequiredBlockIndex , ConsumableItemRecipeSheet.Row.RequiredActionPoint , ConsumableItemRecipeSheet.Row.RequiredGold , ConsumableItemRecipeSheet.Row.Materials , ConsumableItemRecipeSheet.Row.ResultConsumableItemId , ConsumableItemRecipeSheet.Row.Set(IReadOnlyList<string>) , ConsumableItemRecipeSheet.Row.GetAllMaterials() , SheetRow<int>.Validate() , SheetRow<int>.EndOfSheetInitialize() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Class EventDungeonSheet

Namespace: Nekoyume.TableData.Event

Assembly: Lib9c.dll

```
[Serializable]
public class EventDungeonSheet : Sheet<int, EventDungeonSheet.Row>, IDictionary<int,
EventDungeonSheet.Row>, ICollection<KeyValuePair<int, EventDungeonSheet.Row>>,
IEnumerable<KeyValuePair<int, EventDungeonSheet.Row>>, IEnumerable, ISheet
```

Inheritance

object ← Sheet<int, EventDungeonSheet.Row> ← EventDungeonSheet

Implements

IDictionary<int, EventDungeonSheet.Row>,
ICollection<KeyValuePair<int, EventDungeonSheet.Row>>,
IEnumerable<KeyValuePair<int, EventDungeonSheet.Row>>, IEnumerable, ISheet

Inherited Members

Sheet<int, EventDungeonSheet.Row>.Name ,
Sheet<int, EventDungeonSheet.Row>.OrderedList ,
Sheet<int, EventDungeonSheet.Row>.First , Sheet<int, EventDungeonSheet.Row>.Last ,
Sheet<int, EventDungeonSheet.Row>.Keys , Sheet<int, EventDungeonSheet.Row>.Values ,
Sheet<int, EventDungeonSheet.Row>.Count ,
Sheet<int, EventDungeonSheet.Row>.IsReadOnly ,
Sheet<int, EventDungeonSheet.Row>.this[int] ,
Sheet<int, EventDungeonSheet.Row>.Set(string, bool) ,
Sheet<int, EventDungeonSheet.Row>.Set<T>(Sheet<int, T>, bool) ,
Sheet<int, EventDungeonSheet.Row>.GetEnumerator() ,
Sheet<int, EventDungeonSheet.Row>.TryGetValue(int, out EventDungeonSheet.Row, bool)
,
Sheet<int, EventDungeonSheet.Row>.AddRow(int, EventDungeonSheet.Row) ,
Sheet<int, EventDungeonSheet.Row>.Add(int, EventDungeonSheet.Row) ,
Sheet<int, EventDungeonSheet.Row>.ContainsKey(int) ,
Sheet<int, EventDungeonSheet.Row>.Remove(int) ,
Sheet<int, EventDungeonSheet.Row>.TryGetValue(int, out EventDungeonSheet.Row) ,
Sheet<int, EventDungeonSheet.Row>.Add(KeyValuePair<int, EventDungeonSheet.Row>) ,
Sheet<int, EventDungeonSheet.Row>.Clear() ,

Sheet<int, EventDungeonSheet.Row>.Contains(KeyValuePair<int, EventDungeonSheet.Row>) ,
Sheet<int, EventDungeonSheet.Row>.CopyTo(KeyValuePair<int, EventDungeonSheet.Row>[], int) ,
Sheet<int, EventDungeonSheet.Row>.Remove(KeyValuePair<int, EventDungeonSheet.Row>) ,
Sheet<int, EventDungeonSheet.Row>.Serialize() , object.Equals(object) ,
object.Equals(object, object) , object.GetHashCode() , object.GetType() ,
object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Extension Methods
EventDungeonExtensions.TryGetRowByEventDungeonStageId(EventDungeonSheet, int, out EventDungeonSheet.Row) ,
EventDungeonExtensions.TryGetRowByEventScheduleId(EventDungeonSheet, int, out EventDungeonSheet.Row) ,
EventDungeonExtensions.ValidateFromAction(EventDungeonSheet, int, int, string, string)

Constructors

EventDungeonSheet()

```
public EventDungeonSheet()
```

Class EventDungeonSheet.Row

Namespace: Nekoyume.TableData.Event

Assembly: Lib9c.dll

```
[Serializable]
public class EventDungeonSheet.Row : WorldSheet.Row
```

Inheritance

object ← SheetRow<int> ← WorldSheet.Row ← EventDungeonSheet.Row

Inherited Members

WorldSheet.Row.Key , WorldSheet.Row.Id , WorldSheet.Row.Name , WorldSheet.Row.StageBegin , WorldSheet.Row.StageEnd , WorldSheet.Row.StagesCount , WorldSheet.Row.Set(IReadOnlyList<string>) , SheetRow<int>.Validate() , SheetRow<int>.EndOfSheetInitialize() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Class EventDungeonStageSheet

Namespace: Nekoyume.TableData.Event

Assembly: Lib9c.dll

```
[Serializable]
public class EventDungeonStageSheet : Sheet<int, EventDungeonStageSheet.Row>,
IDictionary<int, EventDungeonStageSheet.Row>, ICollection<KeyValuePair<int,
EventDungeonStageSheet.Row>>, IEnumerable<KeyValuePair<int,
EventDungeonStageSheet.Row>>, IEnumerable, ISheet
```

Inheritance

object ← Sheet<int, EventDungeonStageSheet.Row> ← EventDungeonStageSheet

Implements

IDictionary<int, EventDungeonStageSheet.Row>,
ICollection<KeyValuePair<int, EventDungeonStageSheet.Row>>,
IEnumerable<KeyValuePair<int, EventDungeonStageSheet.Row>>, IEnumerable,
ISheet

Inherited Members

Sheet<int, EventDungeonStageSheet.Row>.Name ,
Sheet<int, EventDungeonStageSheet.Row>.OrderedList ,
Sheet<int, EventDungeonStageSheet.Row>.First ,
Sheet<int, EventDungeonStageSheet.Row>.Last ,
Sheet<int, EventDungeonStageSheet.Row>.Keys ,
Sheet<int, EventDungeonStageSheet.Row>.Values ,
Sheet<int, EventDungeonStageSheet.Row>.Count ,
Sheet<int, EventDungeonStageSheet.Row>.IsReadOnly ,
Sheet<int, EventDungeonStageSheet.Row>.this[int] ,
Sheet<int, EventDungeonStageSheet.Row>.Set(string, bool) ,
Sheet<int, EventDungeonStageSheet.Row>.Set<T>(Sheet<int, T>, bool) ,
Sheet<int, EventDungeonStageSheet.Row>.GetEnumerator() ,
Sheet<int, EventDungeonStageSheet.Row>.TryGetValue(int, out
EventDungeonStageSheet.Row, bool) ,
Sheet<int, EventDungeonStageSheet.Row>.AddRow(int, EventDungeonStageSheet.Row) ,
Sheet<int, EventDungeonStageSheet.Row>.Add(int, EventDungeonStageSheet.Row) ,
Sheet<int, EventDungeonStageSheet.Row>.ContainsKey(int) ,
Sheet<int, EventDungeonStageSheet.Row>.Remove(int) ,

Sheet<int, EventDungeonStageSheet.Row>.TryGetValue(int, out EventDungeonStageSheet.Row) , Sheet<int, EventDungeonStageSheet.Row>.Add(KeyValuePair<int, EventDungeonStageSheet.Row>) , Sheet<int, EventDungeonStageSheet.Row>.Clear() , Sheet<int, EventDungeonStageSheet.Row>.Contains(KeyValuePair<int, EventDungeonStageSheet.Row>) , Sheet<int, EventDungeonStageSheet.Row>.CopyTo(KeyValuePair<int, EventDungeonStageSheet.Row>[], int) , Sheet<int, EventDungeonStageSheet.Row>.Remove(KeyValuePair<int, EventDungeonStageSheet.Row>) , Sheet<int, EventDungeonStageSheet.Row>.Serialize() , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Extension Methods

EventDungeonExtensions.GetStageRows(EventDungeonStageSheet, int, int) , EventDungeonExtensions.ValidateFromAction(EventDungeonStageSheet, int, string, string)

Constructors

EventDungeonStageSheet()

```
public EventDungeonStageSheet()
```

Class EventDungeonStageSheet.Row

Namespace: Nekoyume.TableData.Event

Assembly: Lib9c.dll

```
[Serializable]
public class EventDungeonStageSheet.Row : StageSheet.Row
```

Inheritance

object ← SheetRow<int> ← StageSheet.Row ← EventDungeonStageSheet.Row

Inherited Members

StageSheet.Row.Key , StageSheet.Row.Id , StageSheet.Row.CostAP , StageSheet.Row.EntryCostItemId , StageSheet.Row.EntryCostItemCount , StageSheet.Row.TurnLimit , StageSheet.Row.EnemyInitialStatModifiers , StageSheet.Row.Background , StageSheet.Row.BGM , StageSheet.Row.Rewards , StageSheet.Row.DropItemMin , StageSheet.Row.DropItemMax , StageSheet.Row.FavRewards , StageSheet.Row.FavDropMin , StageSheet.Row.FavDropMax , StageSheet.Row.Set(IReadOnlyList<string>) , StageSheet.Row.CloneWithId(int) , SheetRow<int>.Validate() , SheetRow<int>.EndOfSheetInitialize() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Extension Methods

EventDungeonExtensions.GetStageNumber(EventDungeonStageSheet.Row)

Class EventDungeonStageWaveSheet

Namespace: Nekoyume.TableData.Event

Assembly: Lib9c.dll

```
[Serializable]
public class EventDungeonStageWaveSheet : Sheet<int,
EventDungeonStageWaveSheet.Row>, IDictionary<int, EventDungeonStageWaveSheet.Row>,
ICollection<KeyValuePair<int, EventDungeonStageWaveSheet.Row>>,
IEnumerable<KeyValuePair<int, EventDungeonStageWaveSheet.Row>>, IEnumerable, ISheet
```

Inheritance

object ← Sheet<int, EventDungeonStageWaveSheet.Row> ← EventDungeonStageWaveSheet

Implements

IDictionary<int, EventDungeonStageWaveSheet.Row>, ICollection<KeyValuePair<int, EventDungeonStageWaveSheet.Row>>, IEnumerable<KeyValuePair<int, EventDungeonStageWaveSheet.Row>>, IEnumerable, ISheet

Inherited Members

Sheet<int, EventDungeonStageWaveSheet.Row>.Name ,
Sheet<int, EventDungeonStageWaveSheet.Row>.OrderedList ,
Sheet<int, EventDungeonStageWaveSheet.Row>.First ,
Sheet<int, EventDungeonStageWaveSheet.Row>.Last ,
Sheet<int, EventDungeonStageWaveSheet.Row>.Keys ,
Sheet<int, EventDungeonStageWaveSheet.Row>.Values ,
Sheet<int, EventDungeonStageWaveSheet.Row>.Count ,
Sheet<int, EventDungeonStageWaveSheet.Row>.IsReadOnly ,
Sheet<int, EventDungeonStageWaveSheet.Row>.this[int] ,
Sheet<int, EventDungeonStageWaveSheet.Row>.Set(string, bool) ,
Sheet<int, EventDungeonStageWaveSheet.Row>.Set<T>(Sheet<int, T>, bool) ,
Sheet<int, EventDungeonStageWaveSheet.Row>.GetEnumerator() ,
Sheet<int, EventDungeonStageWaveSheet.Row>.TryGetValue(int, out EventDungeonStageWaveSheet.Row, bool) ,
Sheet<int, EventDungeonStageWaveSheet.Row>.AddRow(int, EventDungeonStageWaveSheet.Row) ,
Sheet<int, EventDungeonStageWaveSheet.Row>.Add(int, EventDungeonStageWaveSheet.Row) ,

Sheet<int, EventDungeonStageWaveSheet.Row>.ContainsKey(int) ,
Sheet<int, EventDungeonStageWaveSheet.Row>.Remove(int) ,
Sheet<int, EventDungeonStageWaveSheet.Row>.TryGetValue(int, out
EventDungeonStageWaveSheet.Row) ,
Sheet<int, EventDungeonStageWaveSheet.Row>.Add(KeyValuePair<int,
EventDungeonStageWaveSheet.Row>) ,
Sheet<int, EventDungeonStageWaveSheet.Row>.Clear() ,
Sheet<int, EventDungeonStageWaveSheet.Row>.Contains(KeyValuePair<int,
EventDungeonStageWaveSheet.Row>) ,
Sheet<int, EventDungeonStageWaveSheet.Row>.CopyTo(KeyValuePair<int,
EventDungeonStageWaveSheet.Row>[], int) ,
Sheet<int, EventDungeonStageWaveSheet.Row>.Remove(KeyValuePair<int,
EventDungeonStageWaveSheet.Row>) ,
Sheet<int, EventDungeonStageWaveSheet.Row>.Serialize() , object.Equals(object) ,
object.Equals(object, object) , object.GetHashCode() , object.GetType() ,
object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Extension Methods
EventDungeonExtensions.GetStageWaveRows(EventDungeonStageWaveSheet, int, int)

Constructors

EventDungeonStageWaveSheet()

```
public EventDungeonStageWaveSheet()
```

Methods

AddRow(int, Row)

```
protected override void AddRow(int key, EventDungeonStageWaveSheet.Row value)
```

Parameters

`key` int

`value` EventDungeonStageWaveSheet.Row

Class EventDungeonStageWaveSheet.Row

Namespace: Nekoyume.TableData.Event

Assembly: Lib9c.dll

```
[Serializable]
public class EventDungeonStageWaveSheet.Row : StageWaveSheet.Row
```

Inheritance

object ← SheetRow<int> ← StageWaveSheet.Row ← EventDungeonStageWaveSheet.Row

Inherited Members

StageWaveSheet.Row.Key , StageWaveSheet.Row.StageId , StageWaveSheet.Row.Waves , StageWaveSheet.Row.HasBoss , StageWaveSheet.Row.TotalMonsterIds , StageWaveSheet.Row.Set(IReadOnlyList<string>) , StageWaveSheet.Row.CloneWithStageId(int) , StageWaveSheet.Row.EndOfSheetInitialize() , SheetRow<int>.Validate() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Class EventMaterialItemRecipeSheet

Namespace: Nekoyume.TableData.Event

Assembly: Lib9c.dll

```
[Serializable]
public class EventMaterialItemRecipeSheet : Sheet<int,
EventMaterialItemRecipeSheet.Row>, IDictionary<int,
EventMaterialItemRecipeSheet.Row>, ICollection<KeyValuePair<int,
EventMaterialItemRecipeSheet.Row>>, IEnumerable<KeyValuePair<int,
EventMaterialItemRecipeSheet.Row>>, IEnumerable, ISheet
```

Inheritance

object ← Sheet<int, EventMaterialItemRecipeSheet.Row> ← EventMaterialItemRecipeSheet

Implements

IDictionary<int, EventMaterialItemRecipeSheet.Row>, ICollection<KeyValuePair<int, EventMaterialItemRecipeSheet.Row>>, IEnumerable<KeyValuePair<int, EventMaterialItemRecipeSheet.Row>>, IEnumerable, ISheet

Inherited Members

Sheet<int, EventMaterialItemRecipeSheet.Row>.Name ,
Sheet<int, EventMaterialItemRecipeSheet.Row>.OrderedList ,
Sheet<int, EventMaterialItemRecipeSheet.Row>.First ,
Sheet<int, EventMaterialItemRecipeSheet.Row>.Last ,
Sheet<int, EventMaterialItemRecipeSheet.Row>.Keys ,
Sheet<int, EventMaterialItemRecipeSheet.Row>.Values ,
Sheet<int, EventMaterialItemRecipeSheet.Row>.Count ,
Sheet<int, EventMaterialItemRecipeSheet.Row>.IsReadOnly ,
Sheet<int, EventMaterialItemRecipeSheet.Row>.this[int] ,
Sheet<int, EventMaterialItemRecipeSheet.Row>.Set(string, bool) ,
Sheet<int, EventMaterialItemRecipeSheet.Row>.Set<T>(Sheet<int, T>, bool) ,
Sheet<int, EventMaterialItemRecipeSheet.Row>.GetEnumerator() ,
Sheet<int, EventMaterialItemRecipeSheet.Row>.TryGetValue(int, out EventMaterialItemRecipeSheet.Row, bool) ,
Sheet<int, EventMaterialItemRecipeSheet.Row>.AddRow(int, EventMaterialItemRecipeSheet.Row) ,

Sheet<int, EventMaterialItemRecipeSheet.Row>.Add(int, EventMaterialItemRecipeSheet.Row) ,
Sheet<int, EventMaterialItemRecipeSheet.Row>.ContainsKey(int) ,
Sheet<int, EventMaterialItemRecipeSheet.Row>.Remove(int) ,
Sheet<int, EventMaterialItemRecipeSheet.Row>.TryGetValue(int, out EventMaterialItemRecipeSheet.Row) ,
Sheet<int, EventMaterialItemRecipeSheet.Row>.Add(KeyValuePair<int, EventMaterialItemRecipeSheet.Row>) ,
Sheet<int, EventMaterialItemRecipeSheet.Row>.Clear() ,
Sheet<int, EventMaterialItemRecipeSheet.Row>.Contains(KeyValuePair<int, EventMaterialItemRecipeSheet.Row>) ,
Sheet<int, EventMaterialItemRecipeSheet.Row>.CopyTo(KeyValuePair<int, EventMaterialItemRecipeSheet.Row>[], int) ,
Sheet<int, EventMaterialItemRecipeSheet.Row>.Remove(KeyValuePair<int, EventMaterialItemRecipeSheet.Row>) ,
Sheet<int, EventMaterialItemRecipeSheet.Row>.Serialize() , object.Equals(object) ,
object.Equals(object, object) , object.GetHashCode() , object.GetType() ,
object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Extension Methods
EventRecipeExtensions.GetRecipeRows(EventMaterialItemRecipeSheet, int) ,
EventRecipeExtensions.ValidateFromAction(EventMaterialItemRecipeSheet, int, string, string)

Constructors

EventMaterialItemRecipeSheet()

```
public EventMaterialItemRecipeSheet()
```

Class EventMaterialItemRecipeSheet.Row

Namespace: Nekoyume.TableData.Event

Assembly: Lib9c.dll

```
[Serializable]
public class EventMaterialItemRecipeSheet.Row : SheetRow<int>
```

Inheritance

object ← SheetRow<int> ← EventMaterialItemRecipeSheet.Row

Inherited Members

SheetRow<int>.Validate() , SheetRow<int>.EndOfSheetInitialize() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Extension Methods

EventRecipeExtensions.ValidateFromAction(EventMaterialItemRecipeSheet.Row, MaterialItemSheet, Dictionary<int, int>, string, string)

Properties

Id

```
public int Id { get; }
```

Property Value

int

Key

```
public override int Key { get; }
```

Property Value

int

RequiredMaterialsCount

```
public int RequiredMaterialsCount { get; }
```

Property Value

int

RequiredMaterialsId

```
public List<int> RequiredMaterialsId { get; }
```

Property Value

List<int>

ResultMaterialItemCount

```
public int ResultMaterialItemCount { get; }
```

Property Value

int

ResultMaterialItemId

```
public int ResultMaterialItemId { get; }
```

Property Value

int

Methods

Set(IReadOnlyList<string>)

```
public override void Set(IReadOnlyList<string> fields)
```

Parameters

`fields` IReadOnlyList<string>

Class EventScheduleSheet

Namespace: Nekoyume.TableData.Event

Assembly: Lib9c.dll

```
public class EventScheduleSheet : Sheet<int, EventScheduleSheet.Row>,
IDictionary<int, EventScheduleSheet.Row>, ICollection<KeyValuePair<int,
EventScheduleSheet.Row>>, IEnumerable<KeyValuePair<int, EventScheduleSheet.Row>>,
IEnumerable, ISheet
```

Inheritance

object ← Sheet<int, EventScheduleSheet.Row> ← EventScheduleSheet

Implements

IDictionary<int, EventScheduleSheet.Row>,
ICollection<KeyValuePair<int, EventScheduleSheet.Row>>,
IEnumerable<KeyValuePair<int, EventScheduleSheet.Row>>, IEnumerable, ISheet

Inherited Members

Sheet<int, EventScheduleSheet.Row>.Name ,
Sheet<int, EventScheduleSheet.Row>.OrderedList ,
Sheet<int, EventScheduleSheet.Row>.First , Sheet<int, EventScheduleSheet.Row>.Last ,
Sheet<int, EventScheduleSheet.Row>.Keys ,
Sheet<int, EventScheduleSheet.Row>.Values ,
Sheet<int, EventScheduleSheet.Row>.Count ,
Sheet<int, EventScheduleSheet.Row>.IsReadOnly ,
Sheet<int, EventScheduleSheet.Row>.this[int] ,
Sheet<int, EventScheduleSheet.Row>.Set(string, bool) ,
Sheet<int, EventScheduleSheet.Row>.Set<T>(Sheet<int, T>, bool) ,
Sheet<int, EventScheduleSheet.Row>.GetEnumerator() ,
Sheet<int, EventScheduleSheet.Row>.TryGetValue(int, out EventScheduleSheet.Row, bool)
,
Sheet<int, EventScheduleSheet.Row>.AddRow(int, EventScheduleSheet.Row) ,
Sheet<int, EventScheduleSheet.Row>.Add(int, EventScheduleSheet.Row) ,
Sheet<int, EventScheduleSheet.Row>.ContainsKey(int) ,
Sheet<int, EventScheduleSheet.Row>.Remove(int) ,
Sheet<int, EventScheduleSheet.Row>.TryGetValue(int, out EventScheduleSheet.Row) ,
Sheet<int, EventScheduleSheet.Row>.Add(KeyValuePair<int, EventScheduleSheet.Row>) ,
Sheet<int, EventScheduleSheet.Row>.Clear() ,

Sheet<int, EventScheduleSheet.Row>.Contains(KeyValuePair<int, EventScheduleSheet.Row>) , Sheet<int, EventScheduleSheet.Row>.CopyTo(KeyValuePair<int, EventScheduleSheet.Row>[], int) , Sheet<int, EventScheduleSheet.Row>.Remove(KeyValuePair<int, EventScheduleSheet.Row>) , Sheet<int, EventScheduleSheet.Row>.Serialize() , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Extension Methods

EventScheduleExtensions.TryGetRowForDungeon(EventScheduleSheet, long, out EventScheduleSheet.Row) , EventScheduleExtensions.TryGetRowForRecipe(EventScheduleSheet, long, out EventScheduleSheet.Row) , EventScheduleExtensions.ValidateFromActionForDungeon(EventScheduleSheet, long, int, int, string, string) , EventScheduleExtensions.ValidateFromActionForRecipe(EventScheduleSheet, long, int, int, string, string)

Constructors

EventScheduleSheet()

```
public EventScheduleSheet()
```

Class EventScheduleSheet.Row

Namespace: Nekoyume.TableData.Event

Assembly: Lib9c.dll

```
[Serializable]
public class EventScheduleSheet.Row : SheetRow<int>
```

Inheritance

object ← SheetRow<int> ← EventScheduleSheet.Row

Inherited Members

SheetRow<int>.Validate() , SheetRow<int>.EndOfSheetInitialize() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Extension Methods

EventScheduleExtensions.GetDungeonTicketCost(EventScheduleSheet.Row, int, Currency) , EventScheduleExtensions.GetDungeonTicketCostV1(EventScheduleSheet.Row, int) , EventScheduleExtensions.GetStageExp(EventScheduleSheet.Row, int, int)

Properties

DungeonEndBlockIndex

```
public long DungeonEndBlockIndex { get; }
```

Property Value

long

DungeonExpSeedValue

```
public int DungeonExpSeedValue { get; }
```

Property Value

int

DungeonTicketAdditionalPrice

This value is divided by 10 and used. e.g., DecimalMath.DecimalEx.Floor(row.DungeonTicketAdditionalPrice * .1m, 2);

```
public int DungeonTicketAdditionalPrice { get; }
```

Property Value

int

DungeonTicketPrice

This value is divided by 10 and used. e.g., DecimalMath.DecimalEx.Floor(row.DungeonTicketPrice * .1m, 2);

```
public int DungeonTicketPrice { get; }
```

Property Value

int

DungeonTicketsMax

```
public int DungeonTicketsMax { get; }
```

Property Value

int

DungeonTicketsResetIntervalBlockRange

```
public int DungeonTicketsResetIntervalBlockRange { get; }
```

Property Value

int

Id

```
public int Id { get; }
```

Property Value

int

Key

```
public override int Key { get; }
```

Property Value

int

RecipeEndBlockIndex

```
public long RecipeEndBlockIndex { get; }
```

Property Value

long

StartBlockIndex

```
public long StartBlockIndex { get; }
```

Property Value

long

Methods

Set(IReadOnlyList<string>)

```
public override void Set(IReadOnlyList<string> fields)
```

Parameters

`fields` IReadOnlyList<string>

Class PatrolRewardSheet

Namespace: Nekoyume.TableData.Event

Assembly: Lib9c.dll

Represents the PatrolRewardSheet, which defines patrol reward policies and rules.

```
[Serializable]
public class PatrolRewardSheet : Sheet<int, PatrolRewardSheet.Row>, IDictionary<int,
PatrolRewardSheet.Row>, ICollection<KeyValuePair<int, PatrolRewardSheet.Row>>,
IEnumerable<KeyValuePair<int, PatrolRewardSheet.Row>>, IEnumerable, ISheet
```

Inheritance

object ← Sheet<int, PatrolRewardSheet.Row> ← PatrolRewardSheet

Implements

IDictionary<int, PatrolRewardSheet.Row>,
ICollection<KeyValuePair<int, PatrolRewardSheet.Row>>,
IEnumerable<KeyValuePair<int, PatrolRewardSheet.Row>>, IEnumerable, ISheet

Inherited Members

Sheet<int, PatrolRewardSheet.Row>.Name ,
Sheet<int, PatrolRewardSheet.Row>.OrderedList ,
Sheet<int, PatrolRewardSheet.Row>.First , Sheet<int, PatrolRewardSheet.Row>.Last ,
Sheet<int, PatrolRewardSheet.Row>.Keys , Sheet<int, PatrolRewardSheet.Row>.Values ,
Sheet<int, PatrolRewardSheet.Row>.Count ,
Sheet<int, PatrolRewardSheet.Row>.IsReadOnly ,
Sheet<int, PatrolRewardSheet.Row>.this[int] ,
Sheet<int, PatrolRewardSheet.Row>.Set(string, bool) ,
Sheet<int, PatrolRewardSheet.Row>.Set<T>(Sheet<int, T>, bool) ,
Sheet<int, PatrolRewardSheet.Row>.GetEnumerator() ,
Sheet<int, PatrolRewardSheet.Row>.TryGetValue(int, out PatrolRewardSheet.Row, bool) ,
Sheet<int, PatrolRewardSheet.Row>.AddRow(int, PatrolRewardSheet.Row) ,
Sheet<int, PatrolRewardSheet.Row>.Add(int, PatrolRewardSheet.Row) ,
Sheet<int, PatrolRewardSheet.Row>.ContainsKey(int) ,
Sheet<int, PatrolRewardSheet.Row>.Remove(int) ,
Sheet<int, PatrolRewardSheet.Row>.TryGetValue(int, out PatrolRewardSheet.Row) ,
Sheet<int, PatrolRewardSheet.Row>.Add(KeyValuePair<int, PatrolRewardSheet.Row>) ,
Sheet<int, PatrolRewardSheet.Row>.Clear() ,

Sheet<int, PatrolRewardSheet.Row>.Contains(KeyValuePair<int, PatrolRewardSheet.Row>) , Sheet<int, PatrolRewardSheet.Row>.CopyTo(KeyValuePair<int, PatrolRewardSheet.Row>[], int) , Sheet<int, PatrolRewardSheet.Row>.Remove(KeyValuePair<int, PatrolRewardSheet.Row>) , Sheet<int, PatrolRewardSheet.Row>.Serialize() , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

PatrolRewardSheet()

Initializes a new instance of the PatrolRewardSheet class.

```
public PatrolRewardSheet()
```

Methods

FindByLevel(int, long)

Finds the reward policy row based on avatar level and block index.

```
public PatrolRewardSheet.Row FindByLevel(int level, long blockIndex)
```

Parameters

`level` int

The avatar's level.

`blockIndex` long

The current block index.

Returns

PatrolRewardSheet.Row

The corresponding reward policy row.

Exceptions

InvalidOperationException

Thrown if no activated policy matches the criteria.

Class PatrolRewardSheet.RewardModel

Namespace: Nekoyume.TableData.Event

Assembly: Lib9c.dll

Represents a reward model consisting of item counts, IDs, and tickers.

```
[Serializable]
public class PatrolRewardSheet.RewardModel
```

Inheritance

object ← PatrolRewardSheet.RewardModel

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

RewardModel(int, int, string)

```
public RewardModel(int count, int itemId, string ticker)
```

Parameters

`count` int

`itemId` int

`ticker` string

Fields

Count

```
public int Count
```

Field Value

int

ItemId

```
public int ItemId
```

Field Value

int

Ticker

```
public string Ticker
```

Field Value

string

Class PatrolRewardSheet.Row

Namespace: Nekoyume.TableData.Event

Assembly: Lib9c.dll

Represents a row in the PatrolRewardSheet, defining patrol reward policies for specific levels and intervals.

```
[Serializable]
public class PatrolRewardSheet.Row : SheetRow<int>
```

Inheritance

object ← SheetRow<int> ← PatrolRewardSheet.Row

Inherited Members

SheetRow<int>.Validate() , SheetRow<int>.EndOfSheetInitialize() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Fields

MaxRewardCount

```
public const int MaxRewardCount = 100
```

Field Value

int

Properties

EndedBlockIndex

```
public long EndedBlockIndex { get; set; }
```

Property Value

long

Id

```
public int Id { get; set; }
```

Property Value

int

Interval

```
public long Interval { get; set; }
```

Property Value

long

Key

```
public override int Key { get; }
```

Property Value

int

MaxLevel

```
public int? MaxLevel { get; set; }
```

Property Value

int?

MinimumLevel

```
public int MinimumLevel { get; set; }
```

Property Value

int

Rewards

```
public List<PatrolRewardSheet.RewardModel> Rewards { get; set; }
```

Property Value

List<PatrolRewardSheet.RewardModel>

StartedBlockIndex

```
public long StartedBlockIndex { get; set; }
```

Property Value

long

Methods

Set(IReadOnlyList<string>)

Sets the row data using a list of string fields.

```
public override void Set(IReadOnlyList<string> fields)
```

Parameters

`fields` IReadOnlyList<string>

The fields to parse into the row.

Namespace Nekoyume.TableData.Garages

Classes

LoadIntoMyGaragesCostSheet

LoadIntoMyGaragesCostSheet.Row

Class LoadIntoMyGaragesCostSheet

Namespace: Nekoyume.TableData.Garages

Assembly: Lib9c.dll

```
[Serializable]
public class LoadIntoMyGaragesCostSheet : Sheet<int,
LoadIntoMyGaragesCostSheet.Row>, IDictionary<int, LoadIntoMyGaragesCostSheet.Row>,
ICollection<KeyValuePair<int, LoadIntoMyGaragesCostSheet.Row>>,
IEnumerable<KeyValuePair<int, LoadIntoMyGaragesCostSheet.Row>>, IEnumerable, ISheet
```

Inheritance

object ← Sheet<int, LoadIntoMyGaragesCostSheet.Row> ← LoadIntoMyGaragesCostSheet

Implements

IDictionary<int, LoadIntoMyGaragesCostSheet.Row>, ICollection<KeyValuePair<int, LoadIntoMyGaragesCostSheet.Row>>, IEnumerable<KeyValuePair<int, LoadIntoMyGaragesCostSheet.Row>>, IEnumerable, ISheet

Inherited Members

Sheet<int, LoadIntoMyGaragesCostSheet.Row>.Name ,
Sheet<int, LoadIntoMyGaragesCostSheet.Row>.OrderedList ,
Sheet<int, LoadIntoMyGaragesCostSheet.Row>.First ,
Sheet<int, LoadIntoMyGaragesCostSheet.Row>.Last ,
Sheet<int, LoadIntoMyGaragesCostSheet.Row>.Keys ,
Sheet<int, LoadIntoMyGaragesCostSheet.Row>.Values ,
Sheet<int, LoadIntoMyGaragesCostSheet.Row>.Count ,
Sheet<int, LoadIntoMyGaragesCostSheet.Row>.IsReadOnly ,
Sheet<int, LoadIntoMyGaragesCostSheet.Row>.this[int] ,
Sheet<int, LoadIntoMyGaragesCostSheet.Row>.Set(string, bool) ,
Sheet<int, LoadIntoMyGaragesCostSheet.Row>.Set<T>(Sheet<int, T>, bool) ,
Sheet<int, LoadIntoMyGaragesCostSheet.Row>.GetEnumerator() ,
Sheet<int, LoadIntoMyGaragesCostSheet.Row>.TryGetValue(int, out LoadIntoMyGaragesCostSheet.Row, bool) ,
Sheet<int, LoadIntoMyGaragesCostSheet.Row>.AddRow(int, LoadIntoMyGaragesCostSheet.Row) ,
Sheet<int, LoadIntoMyGaragesCostSheet.Row>.Add(int, LoadIntoMyGaragesCostSheet.Row) ,

Sheet<int, LoadIntoMyGaragesCostSheet.Row>.ContainsKey(int) ,
Sheet<int, LoadIntoMyGaragesCostSheet.Row>.Remove(int) ,
Sheet<int, LoadIntoMyGaragesCostSheet.Row>.TryGetValue(int, out
LoadIntoMyGaragesCostSheet.Row) ,
Sheet<int, LoadIntoMyGaragesCostSheet.Row>.Add(KeyValuePair<int,
LoadIntoMyGaragesCostSheet.Row>) ,
Sheet<int, LoadIntoMyGaragesCostSheet.Row>.Clear() ,
Sheet<int, LoadIntoMyGaragesCostSheet.Row>.Contains(KeyValuePair<int,
LoadIntoMyGaragesCostSheet.Row>) ,
Sheet<int, LoadIntoMyGaragesCostSheet.Row>.CopyTo(KeyValuePair<int,
LoadIntoMyGaragesCostSheet.Row>[], int) ,
Sheet<int, LoadIntoMyGaragesCostSheet.Row>.Remove(KeyValuePair<int,
LoadIntoMyGaragesCostSheet.Row>) ,
Sheet<int, LoadIntoMyGaragesCostSheet.Row>.Serialize() , object.Equals(object) ,
object.Equals(object, object) , object.GetHashCode() , object.GetType() ,
object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

LoadIntoMyGaragesCostSheet()

```
public LoadIntoMyGaragesCostSheet()
```

Methods

GetGarageCost(HashDigest<SHA256>, int)

```
public FungibleAssetValue GetGarageCost(HashDigest<SHA256> fungibleId, int count)
```

Parameters

`fungibleId` HashDigest<SHA256>

`count` int

Returns

GetGarageCost(FungibleAssetValue)

```
public FungibleAssetValue GetGarageCost(FungibleAssetValue fav)
```

Parameters

`fav` FungibleAssetValue

Returns

FungibleAssetValue

GetGarageCost(IEnumerable<FungibleAssetValue>, IEnumerable<(HashDigest<SHA256> fungibleId, int count)>)

```
public FungibleAssetValue GetGarageCost(IEnumerable<FungibleAssetValue>
fungibleAssetValues, IEnumerable<(HashDigest<SHA256> fungibleId, int
count)> fungibleIdAndCounts)
```

Parameters

`fungibleAssetValues` IEnumerable<FungibleAssetValue>

`fungibleIdAndCounts` IEnumerable<(HashDigest<SHA256> fungibleId, int count)>

Returns

FungibleAssetValue

GetGarageCost(IEnumerable<FungibleAssetValue>, IEnumerable<(int itemId, int count, bool tradable)>)

```
public FungibleAssetValue GetGarageCost(IEnumerable<FungibleAssetValue>
fungibleAssetValues, IEnumerable<(int itemId, int count, bool
tradable)> itemIdAndCounts)
```

Parameters

`fungibleAssetValues` IEnumerable<FungibleAssetValue>

`itemIdAndCounts` IEnumerable<(int itemId, int count, bool tradable)>

Returns

FungibleAssetValue

GetGarageCost(int, int)

```
public FungibleAssetValue GetGarageCost(int itemId, int count)
```

Parameters

`itemId` int

`count` int

Returns

FungibleAssetValue

GetGarageCostPerUnit(HashDigest<SHA256>)

```
public decimal GetGarageCostPerUnit(HashDigest<SHA256> fungibleId)
```

Parameters

`fungibleId` HashDigest<SHA256>

Returns

decimal

GetGarageCostPerUnit(int)

```
public decimal GetGarageCostPerUnit(int itemId)
```

Parameters

`itemId` int

Returns

decimal

GetGarageCostPerUnit(string)

```
public decimal GetGarageCostPerUnit(string currencyTicker)
```

Parameters

`currencyTicker` string

Returns

decimal

HasCost(HashDigest<SHA256>)

```
public bool HasCost(HashDigest<SHA256> fungibleId)
```

Parameters

`fungibleId` HashDigest<SHA256>

Returns

bool

HasCost(string)

```
public bool HasCost(string currencyTicker)
```

Parameters

`currencyTicker` string

Returns

bool

Class LoadIntoMyGaragesCostSheet.Row

Namespace: Nekoyume.TableData.Garages

Assembly: Lib9c.dll

```
[Serializable]
public class LoadIntoMyGaragesCostSheet.Row : SheetRow<int>
```

Inheritance

object ← SheetRow<int> ← LoadIntoMyGaragesCostSheet.Row

Inherited Members

SheetRow<int>.Validate() , SheetRow<int>.EndOfSheetInitialize() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Properties

CurrencyTicker

```
public string CurrencyTicker { get; }
```

Property Value

string

FungibleId

```
public HashDigest<SHA256>? FungibleId { get; }
```

Property Value

HashDigest<SHA256>?

GarageCostPerUnit

```
public decimal GarageCostPerUnit { get; }
```

Property Value

decimal

Id

```
public int Id { get; }
```

Property Value

int

ItemId

```
public int ItemId { get; }
```

Property Value

int

Key

```
public override int Key { get; }
```

Property Value

int

Methods

Set(IReadOnlyList<string>)

```
public override void Set(IReadOnlyList<string> fields)
```

Parameters

`fields` IReadOnlyList<string>

Namespace Nekoyume.TableData.GrandFinale

Classes

GrandFinaleParticipantsSheet

GrandFinaleParticipantsSheet.Row

GrandFinaleScheduleSheet

GrandFinaleScheduleSheet.Row

Class GrandFinaleParticipantsSheet

Namespace: Nekoyume.TableData.GrandFinale

Assembly: Lib9c.dll

```
[Serializable]
public class GrandFinaleParticipantsSheet : Sheet<int,
GrandFinaleParticipantsSheet.Row>, IDictionary<int,
GrandFinaleParticipantsSheet.Row>, ICollection<KeyValuePair<int,
GrandFinaleParticipantsSheet.Row>>, IEnumerable<KeyValuePair<int,
GrandFinaleParticipantsSheet.Row>>, IEnumerable, ISheet
```

Inheritance

object ← Sheet<int, GrandFinaleParticipantsSheet.Row> ← GrandFinaleParticipantsSheet

Implements

IDictionary<int, GrandFinaleParticipantsSheet.Row>,
ICollection<KeyValuePair<int, GrandFinaleParticipantsSheet.Row>>,
IEnumerable<KeyValuePair<int, GrandFinaleParticipantsSheet.Row>>, IEnumerable,
ISheet

Inherited Members

Sheet<int, GrandFinaleParticipantsSheet.Row>.Name ,
Sheet<int, GrandFinaleParticipantsSheet.Row>.OrderedList ,
Sheet<int, GrandFinaleParticipantsSheet.Row>.First ,
Sheet<int, GrandFinaleParticipantsSheet.Row>.Last ,
Sheet<int, GrandFinaleParticipantsSheet.Row>.Keys ,
Sheet<int, GrandFinaleParticipantsSheet.Row>.Values ,
Sheet<int, GrandFinaleParticipantsSheet.Row>.Count ,
Sheet<int, GrandFinaleParticipantsSheet.Row>.IsReadOnly ,
Sheet<int, GrandFinaleParticipantsSheet.Row>.this[int] ,
Sheet<int, GrandFinaleParticipantsSheet.Row>.Set(string, bool) ,
Sheet<int, GrandFinaleParticipantsSheet.Row>.Set<T>(Sheet<int, T>, bool) ,
Sheet<int, GrandFinaleParticipantsSheet.Row>.GetEnumerator() ,
Sheet<int, GrandFinaleParticipantsSheet.Row>.TryGetValue(int, out
GrandFinaleParticipantsSheet.Row, bool) ,
Sheet<int, GrandFinaleParticipantsSheet.Row>.AddRow(int,
GrandFinaleParticipantsSheet.Row) ,
Sheet<int, GrandFinaleParticipantsSheet.Row>.Add(int, GrandFinaleParticipantsSheet.Row)
,

Sheet<int, GrandFinaleParticipantsSheet.Row>.ContainsKey(int) ,
Sheet<int, GrandFinaleParticipantsSheet.Row>.Remove(int) ,
Sheet<int, GrandFinaleParticipantsSheet.Row>.TryGetValue(int, out
GrandFinaleParticipantsSheet.Row) ,
Sheet<int, GrandFinaleParticipantsSheet.Row>.Add(KeyValuePair<int,
GrandFinaleParticipantsSheet.Row>) ,
Sheet<int, GrandFinaleParticipantsSheet.Row>.Clear() ,
Sheet<int, GrandFinaleParticipantsSheet.Row>.Contains(KeyValuePair<int,
GrandFinaleParticipantsSheet.Row>) ,
Sheet<int, GrandFinaleParticipantsSheet.Row>.CopyTo(KeyValuePair<int,
GrandFinaleParticipantsSheet.Row>[], int) ,
Sheet<int, GrandFinaleParticipantsSheet.Row>.Remove(KeyValuePair<int,
GrandFinaleParticipantsSheet.Row>) ,
Sheet<int, GrandFinaleParticipantsSheet.Row>.Serialize() , object.Equals(object) ,
object.Equals(object, object) , object.GetHashCode() , object.GetType() ,
object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

GrandFinaleParticipantsSheet()

```
public GrandFinaleParticipantsSheet()
```

Methods

AddRow(int, Row)

```
protected override void AddRow(int key, GrandFinaleParticipantsSheet.Row value)
```

Parameters

`key` int

`value` GrandFinaleParticipantsSheet.Row

Class GrandFinaleParticipantsSheet.Row

Namespace: Nekoyume.TableData.GrandFinale

Assembly: Lib9c.dll

```
[Serializable]
public class GrandFinaleParticipantsSheet.Row : SheetRow<int>
```

Inheritance

object ← SheetRow<int> ← GrandFinaleParticipantsSheet.Row

Inherited Members

SheetRow<int>.Validate() , SheetRow<int>.EndOfSheetInitialize() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Properties

GrandFinaleId

```
public int GrandFinaleId { get; }
```

Property Value

int

Key

```
public override int Key { get; }
```

Property Value

int

Participants

```
public List<Address> Participants { get; }
```

Property Value

List<Address>

Methods

Set(IReadOnlyList<string>)

```
public override void Set(IReadOnlyList<string> fields)
```

Parameters

`fields` IReadOnlyList<string>

Class GrandFinaleScheduleSheet

Namespace: Nekoyume.TableData.GrandFinale

Assembly: Lib9c.dll

```
[Serializable]
public class GrandFinaleScheduleSheet : Sheet<int, GrandFinaleScheduleSheet.Row>,
IDictionary<int, GrandFinaleScheduleSheet.Row>, ICollection<KeyValuePair<int,
GrandFinaleScheduleSheet.Row>>, IEnumerable<KeyValuePair<int,
GrandFinaleScheduleSheet.Row>>, IEnumerable, ISheet
```

Inheritance

object ← Sheet<int, GrandFinaleScheduleSheet.Row> ← GrandFinaleScheduleSheet

Implements

IDictionary<int, GrandFinaleScheduleSheet.Row>,
ICollection<KeyValuePair<int, GrandFinaleScheduleSheet.Row>>,
IEnumerable<KeyValuePair<int, GrandFinaleScheduleSheet.Row>>, IEnumerable,
ISheet

Inherited Members

Sheet<int, GrandFinaleScheduleSheet.Row>.Name ,
Sheet<int, GrandFinaleScheduleSheet.Row>.OrderedList ,
Sheet<int, GrandFinaleScheduleSheet.Row>.First ,
Sheet<int, GrandFinaleScheduleSheet.Row>.Last ,
Sheet<int, GrandFinaleScheduleSheet.Row>.Keys ,
Sheet<int, GrandFinaleScheduleSheet.Row>.Values ,
Sheet<int, GrandFinaleScheduleSheet.Row>.Count ,
Sheet<int, GrandFinaleScheduleSheet.Row>.IsReadOnly ,
Sheet<int, GrandFinaleScheduleSheet.Row>.this[int] ,
Sheet<int, GrandFinaleScheduleSheet.Row>.Set(string, bool) ,
Sheet<int, GrandFinaleScheduleSheet.Row>.Set<T>(Sheet<int, T>, bool) ,
Sheet<int, GrandFinaleScheduleSheet.Row>.GetEnumerator() ,
Sheet<int, GrandFinaleScheduleSheet.Row>.TryGetValue(int, out
GrandFinaleScheduleSheet.Row, bool) ,
Sheet<int, GrandFinaleScheduleSheet.Row>.AddRow(int, GrandFinaleScheduleSheet.Row) ,
Sheet<int, GrandFinaleScheduleSheet.Row>.Add(int, GrandFinaleScheduleSheet.Row) ,
Sheet<int, GrandFinaleScheduleSheet.Row>.ContainsKey(int) ,
Sheet<int, GrandFinaleScheduleSheet.Row>.Remove(int) ,

Sheet<int, GrandFinaleScheduleSheet.Row>.TryGetValue(int, out GrandFinaleScheduleSheet.Row) , Sheet<int, GrandFinaleScheduleSheet.Row>.Add(KeyValuePair<int, GrandFinaleScheduleSheet.Row>) , Sheet<int, GrandFinaleScheduleSheet.Row>.Clear() , Sheet<int, GrandFinaleScheduleSheet.Row>.Contains(KeyValuePair<int, GrandFinaleScheduleSheet.Row>) , Sheet<int, GrandFinaleScheduleSheet.Row>.CopyTo(KeyValuePair<int, GrandFinaleScheduleSheet.Row>[], int) , Sheet<int, GrandFinaleScheduleSheet.Row>.Remove(KeyValuePair<int, GrandFinaleScheduleSheet.Row>) , Sheet<int, GrandFinaleScheduleSheet.Row>.Serialize() , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

GrandFinaleScheduleSheet()

```
public GrandFinaleScheduleSheet()
```

Methods

GetRowByBlockIndex(long)

```
public GrandFinaleScheduleSheet.Row GetRowByBlockIndex(long blockIndex)
```

Parameters

`blockIndex` long

Returns

GrandFinaleScheduleSheet.Row

Class GrandFinaleScheduleSheet.Row

Namespace: Nekoyume.TableData.GrandFinale

Assembly: Lib9c.dll

```
[Serializable]
public class GrandFinaleScheduleSheet.Row : SheetRow<int>
```

Inheritance

object ← SheetRow<int> ← GrandFinaleScheduleSheet.Row

Inherited Members

SheetRow<int>.Validate() , SheetRow<int>.EndOfSheetInitialize() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Properties

EndBlockIndex

```
public long EndBlockIndex { get; }
```

Property Value

long

Id

```
public int Id { get; }
```

Property Value

int

Key

```
public override int Key { get; }
```

Property Value

int

StartBlockIndex

```
public long StartBlockIndex { get; }
```

Property Value

long

Methods

IsOpened(long)

```
public bool IsOpened(long blockIndex)
```

Parameters

`blockIndex` long

Returns

bool

Set(IReadOnlyList<string>)

```
public override void Set(IReadOnlyList<string> fields)
```

Parameters

`fields` IReadOnlyList<string>

Namespace Nekoyume.TableData.Pet

Classes

PetOptionSheet

PetOptionSheet.Row

PetOptionSheet.Row.PetOptionInfo

Class PetOptionSheet

Namespace: Nekoyume.TableData.Pet

Assembly: Lib9c.dll

```
[Serializable]
public class PetOptionSheet : Sheet<int, PetOptionSheet.Row>, IDictionary<int,
PetOptionSheet.Row>, ICollection<KeyValuePair<int, PetOptionSheet.Row>>,
IEnumerable<KeyValuePair<int, PetOptionSheet.Row>>, IEnumerable, ISheet
```

Inheritance

object ← Sheet<int, PetOptionSheet.Row> ← PetOptionSheet

Implements

IDictionary<int, PetOptionSheet.Row>,
ICollection<KeyValuePair<int, PetOptionSheet.Row>>,
IEnumerable<KeyValuePair<int, PetOptionSheet.Row>>, IEnumerable, ISheet

Inherited Members

Sheet<int, PetOptionSheet.Row>.Name , Sheet<int, PetOptionSheet.Row>.OrderedList ,
Sheet<int, PetOptionSheet.Row>.First , Sheet<int, PetOptionSheet.Row>.Last ,
Sheet<int, PetOptionSheet.Row>.Keys , Sheet<int, PetOptionSheet.Row>.Values ,
Sheet<int, PetOptionSheet.Row>.Count , Sheet<int, PetOptionSheet.Row>.IsReadOnly ,
Sheet<int, PetOptionSheet.Row>.this[int] ,
Sheet<int, PetOptionSheet.Row>.Set(string, bool) ,
Sheet<int, PetOptionSheet.Row>.Set<T>(Sheet<int, T>, bool) ,
Sheet<int, PetOptionSheet.Row>.GetEnumerator() ,
Sheet<int, PetOptionSheet.Row>.TryGetValue(int, out PetOptionSheet.Row, bool) ,
Sheet<int, PetOptionSheet.Row>.AddRow(int, PetOptionSheet.Row) ,
Sheet<int, PetOptionSheet.Row>.Add(int, PetOptionSheet.Row) ,
Sheet<int, PetOptionSheet.Row>.ContainsKey(int) ,
Sheet<int, PetOptionSheet.Row>.Remove(int) ,
Sheet<int, PetOptionSheet.Row>.TryGetValue(int, out PetOptionSheet.Row) ,
Sheet<int, PetOptionSheet.Row>.Add(KeyValuePair<int, PetOptionSheet.Row>) ,
Sheet<int, PetOptionSheet.Row>.Clear() ,
Sheet<int, PetOptionSheet.Row>.Contains(KeyValuePair<int, PetOptionSheet.Row>) ,
Sheet<int, PetOptionSheet.Row>.CopyTo(KeyValuePair<int, PetOptionSheet.Row>[], int) ,
Sheet<int, PetOptionSheet.Row>.Remove(KeyValuePair<int, PetOptionSheet.Row>) ,
Sheet<int, PetOptionSheet.Row>.Serialize() , object.Equals(object) ,

object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

PetOptionSheet()

```
public PetOptionSheet()
```

Methods

AddRow(int, Row)

```
protected override void AddRow(int key, PetOptionSheet.Row value)
```

Parameters

`key` int

`value` PetOptionSheet.Row

Class PetOptionSheet.Row

Namespace: Nekoyume.TableData.Pet

Assembly: Lib9c.dll

```
[Serializable]
public class PetOptionSheet.Row : SheetRow<int>
```

Inheritance

object ← SheetRow<int> ← PetOptionSheet.Row

Inherited Members

SheetRow<int>.Validate() , SheetRow<int>.EndOfSheetInitialize() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Properties

Key

```
public override int Key { get; }
```

Property Value

int

LevelOptionMap

```
public Dictionary<int, PetOptionSheet.Row.PetOptionInfo> LevelOptionMap { get; }
```

Property Value

Dictionary<int, PetOptionSheet.Row.PetOptionInfo>

PetId

```
public int PetId { get; }
```

Property Value

int

Methods

Set(IReadOnlyList<string>)

```
public override void Set(IReadOnlyList<string> fields)
```

Parameters

`fields` IReadOnlyList<string>

Class PetOptionSheet.Row.PetOptionInfo

Namespace: Nekoyume.TableData.Pet

Assembly: Lib9c.dll

```
public class PetOptionSheet.Row.PetOptionInfo
```

Inheritance

object ← PetOptionSheet.Row.PetOptionInfo

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

PetOptionInfo(PetOptionType, decimal)

```
public PetOptionInfo(PetOptionType optionType, decimal optionValue)
```

Parameters

`optionType` PetOptionType

`optionValue` decimal

Properties

OptionType

```
public PetOptionType OptionType { get; }
```

Property Value

PetOptionType

OptionValue

```
public decimal OptionValue { get; }
```

Property Value

decimal

Namespace Nekoyume.TableData.Rune

Classes

RuneLevelBonusSheet

RuneLevelBonusSheet.Row

RuneListSheet

RuneListSheet.Row

Class RuneLevelBonusSheet

Namespace: Nekoyume.TableData.Rune

Assembly: Lib9c.dll

```
public class RuneLevelBonusSheet : Sheet<int, RuneLevelBonusSheet.Row>,
IDictionary<int, RuneLevelBonusSheet.Row>, ICollection<KeyValuePair<int,
RuneLevelBonusSheet.Row>>, IEnumerable<KeyValuePair<int, RuneLevelBonusSheet.Row>>,
IEnumerable, ISheet
```

Inheritance

object ← Sheet<int, RuneLevelBonusSheet.Row> ← RuneLevelBonusSheet

Implements

IDictionary<int, RuneLevelBonusSheet.Row>,
ICollection<KeyValuePair<int, RuneLevelBonusSheet.Row>>,
IEnumerable<KeyValuePair<int, RuneLevelBonusSheet.Row>>, IEnumerable, ISheet

Inherited Members

Sheet<int, RuneLevelBonusSheet.Row>.Name ,
Sheet<int, RuneLevelBonusSheet.Row>.OrderedList ,
Sheet<int, RuneLevelBonusSheet.Row>.First ,
Sheet<int, RuneLevelBonusSheet.Row>.Last ,
Sheet<int, RuneLevelBonusSheet.Row>.Keys ,
Sheet<int, RuneLevelBonusSheet.Row>.Values ,
Sheet<int, RuneLevelBonusSheet.Row>.Count ,
Sheet<int, RuneLevelBonusSheet.Row>.IsReadOnly ,
Sheet<int, RuneLevelBonusSheet.Row>.this[int] ,
Sheet<int, RuneLevelBonusSheet.Row>.Set(string, bool) ,
Sheet<int, RuneLevelBonusSheet.Row>.Set<T>(Sheet<int, T>, bool) ,
Sheet<int, RuneLevelBonusSheet.Row>.GetEnumerator() ,
Sheet<int, RuneLevelBonusSheet.Row>.TryGetValue(int, out RuneLevelBonusSheet.Row, bool) ,
Sheet<int, RuneLevelBonusSheet.Row>.AddRow(int, RuneLevelBonusSheet.Row) ,
Sheet<int, RuneLevelBonusSheet.Row>.Add(int, RuneLevelBonusSheet.Row) ,
Sheet<int, RuneLevelBonusSheet.Row>.ContainsKey(int) ,
Sheet<int, RuneLevelBonusSheet.Row>.Remove(int) ,
Sheet<int, RuneLevelBonusSheet.Row>.TryGetValue(int, out RuneLevelBonusSheet.Row) ,

Sheet<int, RuneLevelBonusSheet.Row>.Add(KeyValuePair<int, RuneLevelBonusSheet.Row>) , Sheet<int, RuneLevelBonusSheet.Row>.Clear() , Sheet<int, RuneLevelBonusSheet.Row>.Contains(KeyValuePair<int, RuneLevelBonusSheet.Row>) , Sheet<int, RuneLevelBonusSheet.Row>.CopyTo(KeyValuePair<int, RuneLevelBonusSheet.Row>[], int) , Sheet<int, RuneLevelBonusSheet.Row>.Remove(KeyValuePair<int, RuneLevelBonusSheet.Row>) , Sheet<int, RuneLevelBonusSheet.Row>.Serialize() , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

RuneLevelBonusSheet()

```
public RuneLevelBonusSheet()
```

Class RuneLevelBonusSheet.Row

Namespace: Nekoyume.TableData.Rune

Assembly: Lib9c.dll

```
public class RuneLevelBonusSheet.Row : SheetRow<int>
```

Inheritance

object ← SheetRow<int> ← RuneLevelBonusSheet.Row

Inherited Members

SheetRow<int>.Validate() , SheetRow<int>.EndOfSheetInitialize() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Fields

Bonus

```
public int Bonus
```

Field Value

int

Id

```
public int Id
```

Field Value

int

RuneLevel

```
public int RuneLevel
```

Field Value

int

Properties

Key

```
public override int Key { get; }
```

Property Value

int

Methods

Set(IReadOnlyList<string>)

```
public override void Set(IReadOnlyList<string> fields)
```

Parameters

`fields` IReadOnlyList<string>

Class RuneListSheet

Namespace: Nekoyume.TableData.Rune

Assembly: Lib9c.dll

```
public class RuneListSheet : Sheet<int, RuneListSheet.Row>, IDictionary<int,
RuneListSheet.Row>, ICollection<KeyValuePair<int, RuneListSheet.Row>>,
IEnumerable<KeyValuePair<int, RuneListSheet.Row>>, IEnumerable, ISheet
```

Inheritance

object ← Sheet<int, RuneListSheet.Row> ← RuneListSheet

Implements

IDictionary<int, RuneListSheet.Row>,
ICollection<KeyValuePair<int, RuneListSheet.Row>>,
IEnumerable<KeyValuePair<int, RuneListSheet.Row>>, IEnumerable, ISheet

Inherited Members

Sheet<int, RuneListSheet.Row>.Name , Sheet<int, RuneListSheet.Row>.OrderedList ,
Sheet<int, RuneListSheet.Row>.First , Sheet<int, RuneListSheet.Row>.Last ,
Sheet<int, RuneListSheet.Row>.Keys , Sheet<int, RuneListSheet.Row>.Values ,
Sheet<int, RuneListSheet.Row>.Count , Sheet<int, RuneListSheet.Row>.IsReadOnly ,
Sheet<int, RuneListSheet.Row>.this[int] ,
Sheet<int, RuneListSheet.Row>.Set(string, bool) ,
Sheet<int, RuneListSheet.Row>.Set<T>(Sheet<int, T>, bool) ,
Sheet<int, RuneListSheet.Row>.GetEnumerator() ,
Sheet<int, RuneListSheet.Row>.TryGetValue(int, out RuneListSheet.Row, bool) ,
Sheet<int, RuneListSheet.Row>.AddRow(int, RuneListSheet.Row) ,
Sheet<int, RuneListSheet.Row>.Add(int, RuneListSheet.Row) ,
Sheet<int, RuneListSheet.Row>.ContainsKey(int) ,
Sheet<int, RuneListSheet.Row>.Remove(int) ,
Sheet<int, RuneListSheet.Row>.TryGetValue(int, out RuneListSheet.Row) ,
Sheet<int, RuneListSheet.Row>.Add(KeyValuePair<int, RuneListSheet.Row>) ,
Sheet<int, RuneListSheet.Row>.Clear() ,
Sheet<int, RuneListSheet.Row>.Contains(KeyValuePair<int, RuneListSheet.Row>) ,
Sheet<int, RuneListSheet.Row>.CopyTo(KeyValuePair<int, RuneListSheet.Row>[], int) ,
Sheet<int, RuneListSheet.Row>.Remove(KeyValuePair<int, RuneListSheet.Row>) ,
Sheet<int, RuneListSheet.Row>.Serialize() , object.Equals(object) ,

object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

RuneListSheet()

```
public RuneListSheet()
```

Class RuneListSheet.Row

Namespace: Nekoyume.TableData.Rune

Assembly: Lib9c.dll

```
public class RuneListSheet.Row : SheetRow<int>
```

Inheritance

object ← SheetRow<int> ← RuneListSheet.Row

Inherited Members

SheetRow<int>.Validate() , SheetRow<int>.EndOfSheetInitialize() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Fields

BonusCoef

```
public int BonusCoef
```

Field Value

int

Grade

```
public int Grade
```

Field Value

int

Id

```
public int Id
```

Field Value

int

RequiredLevel

```
public int RequiredLevel
```

Field Value

int

RuneType

```
public int RuneType
```

Field Value

int

UsePlace

```
public int UsePlace
```

Field Value

int

Properties

Key

```
public override int Key { get; }
```

Property Value

int

Methods

Set(IReadOnlyList<string>)

```
public override void Set(IReadOnlyList<string> fields)
```

Parameters

`fields` IReadOnlyList<string>

Namespace Nekoyume.TableData.Stake

Classes

StakePolicySheet

StakePolicySheet.Row

Class StakePolicySheet

Namespace: Nekoyume.TableData.Stake

Assembly: Lib9c.dll

```
public class StakePolicySheet : Sheet<string, StakePolicySheet.Row>,
IDictionary<string, StakePolicySheet.Row>, ICollection<KeyValuePair<string,
StakePolicySheet.Row>>, IEnumerable<KeyValuePair<string, StakePolicySheet.Row>>,
IEnumerable, ISheet
```

Inheritance

object ← Sheet<string, StakePolicySheet.Row> ← StakePolicySheet

Implements

IDictionary<string, StakePolicySheet.Row>,
ICollection<KeyValuePair<string, StakePolicySheet.Row>>,
IEnumerable<KeyValuePair<string, StakePolicySheet.Row>>, IEnumerable, ISheet

Inherited Members

Sheet<string, StakePolicySheet.Row>.Name ,
Sheet<string, StakePolicySheet.Row>.OrderedList ,
Sheet<string, StakePolicySheet.Row>.First , Sheet<string, StakePolicySheet.Row>.Last ,
Sheet<string, StakePolicySheet.Row>.Keys , Sheet<string, StakePolicySheet.Row>.Values ,
Sheet<string, StakePolicySheet.Row>.Count ,
Sheet<string, StakePolicySheet.Row>.IsReadOnly ,
Sheet<string, StakePolicySheet.Row>.this[string] ,
Sheet<string, StakePolicySheet.Row>.Set(string, bool) ,
Sheet<string, StakePolicySheet.Row>.Set<T>(Sheet<string, T>, bool) ,
Sheet<string, StakePolicySheet.Row>.GetEnumerator() ,
Sheet<string, StakePolicySheet.Row>.TryGetValue(string, out StakePolicySheet.Row, bool) ,
Sheet<string, StakePolicySheet.Row>.AddRow(string, StakePolicySheet.Row) ,
Sheet<string, StakePolicySheet.Row>.Add(string, StakePolicySheet.Row) ,
Sheet<string, StakePolicySheet.Row>.ContainsKey(string) ,
Sheet<string, StakePolicySheet.Row>.Remove(string) ,
Sheet<string, StakePolicySheet.Row>.TryGetValue(string, out StakePolicySheet.Row) ,
Sheet<string, StakePolicySheet.Row>.Add(KeyValuePair<string, StakePolicySheet.Row>) ,
Sheet<string, StakePolicySheet.Row>.Clear() ,
Sheet<string, StakePolicySheet.Row>.Contains(KeyValuePair<string,
StakePolicySheet.Row>) ,

Sheet<string, StakePolicySheet.Row>.CopyTo(KeyValuePair<string, StakePolicySheet.Row>[], int) ,
Sheet<string, StakePolicySheet.Row>.Remove(KeyValuePair<string, StakePolicySheet.Row>) ,
Sheet<string, StakePolicySheet.Row>.Serialize() , object.Equals(object) ,
object.Equals(object, object) , object.GetHashCode() , object.GetType() ,
object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

StakePolicySheet()

```
public StakePolicySheet()
```

Fields

RequiredAttrNames

```
public static readonly string[] RequiredAttrNames
```

Field Value

string[]

SheetPrefixRules

```
public static readonly (string attrName, string value)[] SheetPrefixRules
```

Field Value

(string attrName, string value)[]

Properties

LockupIntervalValue

```
public long LockupIntervalValue { get; }
```

Property Value

long

RewardIntervalValue

```
public long RewardIntervalValue { get; }
```

Property Value

long

StakeRegularFixedRewardSheetValue

```
public string StakeRegularFixedRewardSheetValue { get; }
```

Property Value

string

StakeRegularRewardSheetValue

```
public string StakeRegularRewardSheetValue { get; }
```

Property Value

string

Methods

Set(string, bool)

```
public override void Set(string csv, bool isReversed = false)
```

Parameters

`csv` string

`isReversed` bool

true: csv값의 column과 row가 뒤집혀서 작성되어 있다고 판단합니다.

Exceptions

ArgumentNullException

InvalidDataException

Class StakePolicySheet.Row

Namespace: Nekoyume.TableData.Stake

Assembly: Lib9c.dll

```
[Serializable]
public class StakePolicySheet.Row : SheetRow<string>
```

Inheritance

object ← SheetRow<string> ← StakePolicySheet.Row

Inherited Members

SheetRow<string>.EndOfSheetInitialize() , SheetRow<string>.Equals(object) , SheetRow<string>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Properties

AttrName

```
public string AttrName { get; }
```

Property Value

string

Key

```
public override string Key { get; }
```

Property Value

string

Value

```
public string Value { get; }
```

Property Value

string

Methods

Set(IReadOnlyList<string>)

```
public override void Set(IReadOnlyList<string> fields)
```

Parameters

`fields` IReadOnlyList<string>

Validate()

```
public override void Validate()
```

Namespace Nekoyume.TableData.Summon

Classes

CostumeSummonSheet

EquipmentSummonSheet

RuneSummonSheet

SummonSheet

SummonSheet.Row

Class CostumeSummonSheet

Namespace: Nekoyume.TableData.Summon

Assembly: Lib9c.dll

```
public class CostumeSummonSheet : SummonSheet, IDictionary<int, SummonSheet.Row>,
ICollection<KeyValuePair<int, SummonSheet.Row>>, IEnumerable<KeyValuePair<int,
SummonSheet.Row>>, IEnumerable, ISheet
```

Inheritance

object ← Sheet<int, SummonSheet.Row> ← SummonSheet ← CostumeSummonSheet

Implements

IDictionary<int, SummonSheet.Row>,
ICollection<KeyValuePair<int, SummonSheet.Row>>,
IEnumerable<KeyValuePair<int, SummonSheet.Row>>, IEnumerable, ISheet

Inherited Members

Sheet<int, SummonSheet.Row>.Name , Sheet<int, SummonSheet.Row>.OrderedList ,
Sheet<int, SummonSheet.Row>.First , Sheet<int, SummonSheet.Row>.Last ,
Sheet<int, SummonSheet.Row>.Keys , Sheet<int, SummonSheet.Row>.Values ,
Sheet<int, SummonSheet.Row>.Count , Sheet<int, SummonSheet.Row>.IsReadOnly ,
Sheet<int, SummonSheet.Row>.this[int] ,
Sheet<int, SummonSheet.Row>.Set(string, bool) ,
Sheet<int, SummonSheet.Row>.Set<T>(Sheet<int, T>, bool) ,
Sheet<int, SummonSheet.Row>.GetEnumerator() ,
Sheet<int, SummonSheet.Row>.TryGetValue(int, out SummonSheet.Row, bool) ,
Sheet<int, SummonSheet.Row>.AddRow(int, SummonSheet.Row) ,
Sheet<int, SummonSheet.Row>.Add(int, SummonSheet.Row) ,
Sheet<int, SummonSheet.Row>.ContainsKey(int) ,
Sheet<int, SummonSheet.Row>.Remove(int) ,
Sheet<int, SummonSheet.Row>.TryGetValue(int, out SummonSheet.Row) ,
Sheet<int, SummonSheet.Row>.Add(KeyValuePair<int, SummonSheet.Row>) ,
Sheet<int, SummonSheet.Row>.Clear() ,
Sheet<int, SummonSheet.Row>.Contains(KeyValuePair<int, SummonSheet.Row>) ,
Sheet<int, SummonSheet.Row>.CopyTo(KeyValuePair<int, SummonSheet.Row>[], int) ,
Sheet<int, SummonSheet.Row>.Remove(KeyValuePair<int, SummonSheet.Row>) ,
Sheet<int, SummonSheet.Row>.Serialize() , object.Equals(object) ,

object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Class EquipmentSummonSheet

Namespace: Nekoyume.TableData.Summon

Assembly: Lib9c.dll

```
public class EquipmentSummonSheet : SummonSheet, IDictionary<int, SummonSheet.Row>,
ICollection<KeyValuePair<int, SummonSheet.Row>>, IEnumerable<KeyValuePair<int,
SummonSheet.Row>>, IEnumerable, ISheet
```

Inheritance

object ← Sheet<int, SummonSheet.Row> ← SummonSheet ← EquipmentSummonSheet

Implements

IDictionary<int, SummonSheet.Row>,
ICollection<KeyValuePair<int, SummonSheet.Row>>,
IEnumerable<KeyValuePair<int, SummonSheet.Row>>, IEnumerable, ISheet

Inherited Members

Sheet<int, SummonSheet.Row>.Name , Sheet<int, SummonSheet.Row>.OrderedList ,
Sheet<int, SummonSheet.Row>.First , Sheet<int, SummonSheet.Row>.Last ,
Sheet<int, SummonSheet.Row>.Keys , Sheet<int, SummonSheet.Row>.Values ,
Sheet<int, SummonSheet.Row>.Count , Sheet<int, SummonSheet.Row>.IsReadOnly ,
Sheet<int, SummonSheet.Row>.this[int] ,
Sheet<int, SummonSheet.Row>.Set(string, bool) ,
Sheet<int, SummonSheet.Row>.Set<T>(Sheet<int, T>, bool) ,
Sheet<int, SummonSheet.Row>.GetEnumerator() ,
Sheet<int, SummonSheet.Row>.TryGetValue(int, out SummonSheet.Row, bool) ,
Sheet<int, SummonSheet.Row>.AddRow(int, SummonSheet.Row) ,
Sheet<int, SummonSheet.Row>.Add(int, SummonSheet.Row) ,
Sheet<int, SummonSheet.Row>.ContainsKey(int) ,
Sheet<int, SummonSheet.Row>.Remove(int) ,
Sheet<int, SummonSheet.Row>.TryGetValue(int, out SummonSheet.Row) ,
Sheet<int, SummonSheet.Row>.Add(KeyValuePair<int, SummonSheet.Row>) ,
Sheet<int, SummonSheet.Row>.Clear() ,
Sheet<int, SummonSheet.Row>.Contains(KeyValuePair<int, SummonSheet.Row>) ,
Sheet<int, SummonSheet.Row>.CopyTo(KeyValuePair<int, SummonSheet.Row>[], int) ,
Sheet<int, SummonSheet.Row>.Remove(KeyValuePair<int, SummonSheet.Row>) ,
Sheet<int, SummonSheet.Row>.Serialize() , object.Equals(object) ,

object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Class RuneSummonSheet

Namespace: Nekoyume.TableData.Summon

Assembly: Lib9c.dll

```
public class RuneSummonSheet : SummonSheet, IDictionary<int, SummonSheet.Row>,
ICollection<KeyValuePair<int, SummonSheet.Row>>, IEnumerable<KeyValuePair<int,
SummonSheet.Row>>, IEnumerable, ISheet
```

Inheritance

object ← Sheet<int, SummonSheet.Row> ← SummonSheet ← RuneSummonSheet

Implements

IDictionary<int, SummonSheet.Row>,
ICollection<KeyValuePair<int, SummonSheet.Row>>,
IEnumerable<KeyValuePair<int, SummonSheet.Row>>, IEnumerable, ISheet

Inherited Members

Sheet<int, SummonSheet.Row>.Name , Sheet<int, SummonSheet.Row>.OrderedList ,
Sheet<int, SummonSheet.Row>.First , Sheet<int, SummonSheet.Row>.Last ,
Sheet<int, SummonSheet.Row>.Keys , Sheet<int, SummonSheet.Row>.Values ,
Sheet<int, SummonSheet.Row>.Count , Sheet<int, SummonSheet.Row>.IsReadOnly ,
Sheet<int, SummonSheet.Row>.this[int] ,
Sheet<int, SummonSheet.Row>.Set(string, bool) ,
Sheet<int, SummonSheet.Row>.Set<T>(Sheet<int, T>, bool) ,
Sheet<int, SummonSheet.Row>.GetEnumerator() ,
Sheet<int, SummonSheet.Row>.TryGetValue(int, out SummonSheet.Row, bool) ,
Sheet<int, SummonSheet.Row>.AddRow(int, SummonSheet.Row) ,
Sheet<int, SummonSheet.Row>.Add(int, SummonSheet.Row) ,
Sheet<int, SummonSheet.Row>.ContainsKey(int) ,
Sheet<int, SummonSheet.Row>.Remove(int) ,
Sheet<int, SummonSheet.Row>.TryGetValue(int, out SummonSheet.Row) ,
Sheet<int, SummonSheet.Row>.Add(KeyValuePair<int, SummonSheet.Row>) ,
Sheet<int, SummonSheet.Row>.Clear() ,
Sheet<int, SummonSheet.Row>.Contains(KeyValuePair<int, SummonSheet.Row>) ,
Sheet<int, SummonSheet.Row>.CopyTo(KeyValuePair<int, SummonSheet.Row>[], int) ,
Sheet<int, SummonSheet.Row>.Remove(KeyValuePair<int, SummonSheet.Row>) ,
Sheet<int, SummonSheet.Row>.Serialize() , object.Equals(object) ,

object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Class SummonSheet

Namespace: Nekoyume.TableData.Summon

Assembly: Lib9c.dll

```
public class SummonSheet : Sheet<int, SummonSheet.Row>, IDictionary<int,
SummonSheet.Row>, ICollection<KeyValuePair<int, SummonSheet.Row>>,
IEnumerable<KeyValuePair<int, SummonSheet.Row>>, IEnumerable, ISheet
```

Inheritance

object ← Sheet<int, SummonSheet.Row> ← SummonSheet

Implements

IDictionary<int, SummonSheet.Row>,
ICollection<KeyValuePair<int, SummonSheet.Row>>,
IEnumerable<KeyValuePair<int, SummonSheet.Row>>, IEnumerable, ISheet

Derived

CostumeSummonSheet, EquipmentSummonSheet, RuneSummonSheet

Inherited Members

Sheet<int, SummonSheet.Row>.Name , Sheet<int, SummonSheet.Row>.OrderedList ,
Sheet<int, SummonSheet.Row>.First , Sheet<int, SummonSheet.Row>.Last ,
Sheet<int, SummonSheet.Row>.Keys , Sheet<int, SummonSheet.Row>.Values ,
Sheet<int, SummonSheet.Row>.Count , Sheet<int, SummonSheet.Row>.IsReadOnly ,
Sheet<int, SummonSheet.Row>.this[int] ,
Sheet<int, SummonSheet.Row>.Set(string, bool) ,
Sheet<int, SummonSheet.Row>.Set<T>(Sheet<int, T>, bool) ,
Sheet<int, SummonSheet.Row>.GetEnumerator() ,
Sheet<int, SummonSheet.Row>.TryGetValue(int, out SummonSheet.Row, bool) ,
Sheet<int, SummonSheet.Row>.AddRow(int, SummonSheet.Row) ,
Sheet<int, SummonSheet.Row>.Add(int, SummonSheet.Row) ,
Sheet<int, SummonSheet.Row>.ContainsKey(int) ,
Sheet<int, SummonSheet.Row>.Remove(int) ,
Sheet<int, SummonSheet.Row>.TryGetValue(int, out SummonSheet.Row) ,
Sheet<int, SummonSheet.Row>.Add(KeyValuePair<int, SummonSheet.Row>) ,
Sheet<int, SummonSheet.Row>.Clear() ,
Sheet<int, SummonSheet.Row>.Contains(KeyValuePair<int, SummonSheet.Row>) ,
Sheet<int, SummonSheet.Row>.CopyTo(KeyValuePair<int, SummonSheet.Row>[], int) ,
Sheet<int, SummonSheet.Row>.Remove(KeyValuePair<int, SummonSheet.Row>) ,

Sheet<int, SummonSheet.Row>.Serialize() , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

SummonSheet()

```
public SummonSheet()
```

Class SummonSheet.Row

Namespace: Nekoyume.TableData.Summon

Assembly: Lib9c.dll

```
public class SummonSheet.Row : SheetRow<int>
```

Inheritance

object ← SheetRow<int> ← SummonSheet.Row

Inherited Members

SheetRow<int>.Validate() , SheetRow<int>.EndOfSheetInitialize() , SheetRow<int>.Equals(object) , SheetRow<int>.GetHashCode() , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Fields

MaxRecipeCount

Maximum number of recipes allowed per summon group.

```
public const int MaxRecipeCount = 100
```

Field Value

int

Recipes

List of recipes available for this summon group. Each tuple contains (recipeId, ratio) where ratio determines the probability of selection.

```
public readonly List<(int, int)> Recipes
```

Field Value

List<(int id, int count)>

Properties

CostMaterial

Material ID required for summoning.

```
public int CostMaterial { get; }
```

Property Value

int

CostMaterialCount

Quantity of material required for summoning.

```
public int CostMaterialCount { get; }
```

Property Value

int

CostNcg

NCG (Nine Chronicles Gold) cost for summoning.

```
public int CostNcg { get; }
```

Property Value

int

GroupId

Unique identifier for this summon group.

```
public int GroupId { get; }
```

Property Value

int

GuaranteeCount11

Number of items to guarantee for 11 summons. Null if not configured.

```
public int? GuaranteeCount11 { get; }
```

Property Value

int?

GuaranteeCount110

Number of items to guarantee for 110 summons. Null if not configured.

```
public int? GuaranteeCount110 { get; }
```

Property Value

int?

Key

```
public override int Key { get; }
```

Property Value

int

MinimumGrade11

Minimum grade to guarantee for 11 summons. Null if not configured.

```
public int? MinimumGrade11 { get; }
```

Property Value

int?

MinimumGrade110

Minimum grade to guarantee for 110 summons. Null if not configured.

```
public int? MinimumGrade110 { get; }
```

Property Value

int?

Methods

CumulativeRatio(int)

Calculates the cumulative ratio up to the specified recipe index. Used for weighted random selection of recipes.

```
public int CumulativeRatio(int index)
```

Parameters

`index` int

Recipe index (1-based)

Returns

int

Cumulative ratio up to the specified index

Exceptions

IndexOutOfRangeException

Thrown when index is out of valid range

Set(IReadOnlyList<string>)

Sets the row data from CSV fields, supporting both legacy and new formats with grade guarantee settings. Legacy format: groupID,cost_material,cost_material_count,cost_ncg,recipe1ID,recipe1ratio,... New format: groupID,cost_material,cost_material_count,cost_ncg,GUARANTEE,min_grade_11,count_11,min_grade_110,count_110,recipe1ID,recipe1rati

```
public override void Set(IReadOnlyList<string> fields)
```

Parameters

`fields` IReadOnlyList<string>

CSV field values

TotalRatio()

Calculates the total ratio of all recipes in this summon group.

```
public int TotalRatio()
```

Returns

int

Sum of all recipe ratios

UseGradeGuarantee(int)

Determines if grade guarantee is enabled for this summon group based on summon count. Returns true if guarantee settings are configured for the specified summon count. For 11 summons: checks if MinimumGrade11 and GuaranteeCount11 are configured. For 110 summons: checks if MinimumGrade110 and GuaranteeCount110 are configured. For other counts: returns false.

```
public bool UseGradeGuarantee(int summonCount)
```

Parameters

`summonCount` int

Number of summons to check guarantee for

Returns

bool

True if grade guarantee is configured for the specified summon count, false otherwise

Namespace Nekoyume.TableData.Swap

Classes

SwapRateSheet

Represents a sheet of swap rates.

SwapRateSheet.CurrencyPair

Represents a currency pair for swap.

SwapRateSheet.Fraction

Represents a fraction.

SwapRateSheet.Row

Represents a row of SwapRateSheet.

Class SwapRateSheet

Namespace: Nekoyume.TableData.Swap

Assembly: Lib9c.dll

Represents a sheet of swap rates.

```
public class SwapRateSheet : Sheet<SwapRateSheet.CurrencyPair, SwapRateSheet.Row>,
IDictionary<SwapRateSheet.CurrencyPair, SwapRateSheet.Row>,
ICollection<KeyValuePair<SwapRateSheet.CurrencyPair, SwapRateSheet.Row>>,
IEnumerable<KeyValuePair<SwapRateSheet.CurrencyPair, SwapRateSheet.Row>>,
IEnumerable, ISheet
```

Inheritance

object ← Sheet<SwapRateSheet.CurrencyPair, SwapRateSheet.Row> ← SwapRateSheet

Implements

IDictionary<SwapRateSheet.CurrencyPair, SwapRateSheet.Row>,
ICollection<KeyValuePair<SwapRateSheet.CurrencyPair, SwapRateSheet.Row>>,
IEnumerable<KeyValuePair<SwapRateSheet.CurrencyPair, SwapRateSheet.Row>>,
IEnumerable, ISheet

Inherited Members

Sheet<SwapRateSheet.CurrencyPair, SwapRateSheet.Row>.Name ,
Sheet<SwapRateSheet.CurrencyPair, SwapRateSheet.Row>.OrderedList ,
Sheet<SwapRateSheet.CurrencyPair, SwapRateSheet.Row>.First ,
Sheet<SwapRateSheet.CurrencyPair, SwapRateSheet.Row>.Last ,
Sheet<SwapRateSheet.CurrencyPair, SwapRateSheet.Row>.Keys ,
Sheet<SwapRateSheet.CurrencyPair, SwapRateSheet.Row>.Values ,
Sheet<SwapRateSheet.CurrencyPair, SwapRateSheet.Row>.Count ,
Sheet<SwapRateSheet.CurrencyPair, SwapRateSheet.Row>.IsReadOnly ,
Sheet<SwapRateSheet.CurrencyPair,
SwapRateSheet.Row>.this[SwapRateSheet.CurrencyPair] ,
Sheet<SwapRateSheet.CurrencyPair, SwapRateSheet.Row>.Set(string, bool) ,
Sheet<SwapRateSheet.CurrencyPair, SwapRateSheet.Row>.Set<T>
(Sheet<SwapRateSheet.CurrencyPair, T>, bool) ,
Sheet<SwapRateSheet.CurrencyPair, SwapRateSheet.Row>.GetEnumerator() ,
Sheet<SwapRateSheet.CurrencyPair,
SwapRateSheet.Row>.TryGetValue(SwapRateSheet.CurrencyPair, out SwapRateSheet.Row,
bool) ,

Sheet<SwapRateSheet.CurrencyPair, SwapRateSheet.Row>.AddRow(SwapRateSheet.CurrencyPair, SwapRateSheet.Row) , Sheet<SwapRateSheet.CurrencyPair, SwapRateSheet.Row>.Add(SwapRateSheet.CurrencyPair, SwapRateSheet.Row) , Sheet<SwapRateSheet.CurrencyPair, SwapRateSheet.Row>.ContainsKey(SwapRateSheet.CurrencyPair) , Sheet<SwapRateSheet.CurrencyPair, SwapRateSheet.Row>.Remove(SwapRateSheet.CurrencyPair) , Sheet<SwapRateSheet.CurrencyPair, SwapRateSheet.Row>.TryGetValue(SwapRateSheet.CurrencyPair, out SwapRateSheet.Row) , Sheet<SwapRateSheet.CurrencyPair, SwapRateSheet.Row>.Add(KeyValuePair<SwapRateSheet.CurrencyPair, SwapRateSheet.Row>) , Sheet<SwapRateSheet.CurrencyPair, SwapRateSheet.Row>.Clear() , Sheet<SwapRateSheet.CurrencyPair, SwapRateSheet.Row>.Contains(KeyValuePair<SwapRateSheet.CurrencyPair, SwapRateSheet.Row>) , Sheet<SwapRateSheet.CurrencyPair, SwapRateSheet.Row>.CopyTo(KeyValuePair<SwapRateSheet.CurrencyPair, SwapRateSheet.Row>[], int) , Sheet<SwapRateSheet.CurrencyPair, SwapRateSheet.Row>.Remove(KeyValuePair<SwapRateSheet.CurrencyPair, SwapRateSheet.Row>) , Sheet<SwapRateSheet.CurrencyPair, SwapRateSheet.Row>.Serialize() , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

SwapRateSheet()

Initializes a new instance of the SwapRateSheet class.

```
public SwapRateSheet()
```

Class SwapRateSheet.CurrencyPair

Namespace: Nekoyume.TableData.Swap

Assembly: Lib9c.dll

Represents a currency pair for swap.

```
public class SwapRateSheet.CurrencyPair : IEquatable<SwapRateSheet.CurrencyPair>,
IComparable<SwapRateSheet.CurrencyPair>
```

Inheritance

object ← SwapRateSheet.CurrencyPair

Implements

IEquatable<SwapRateSheet.CurrencyPair>,
IComparable<SwapRateSheet.CurrencyPair>

Inherited Members

object.Equals(object, object) , object.GetType() , object.MemberwiseClone() ,
object.ReferenceEquals(object, object) , object.ToString()

Constructors

CurrencyPair(Currency, Currency)

Initializes a new instance of the SwapRateSheet.CurrencyPair class.

```
public CurrencyPair(Currency from, Currency to)
```

Parameters

`from` Currency

The currency to swap from.

`to` Currency

The currency to swap to.

Properties

From

The currency to swap from.

```
public Currency From { get; }
```

Property Value

Currency

To

The currency to swap to.

```
public Currency To { get; }
```

Property Value

Currency

Methods

CompareTo(CurrencyPair?)

Compares the current instance with another object of the same type and returns an integer that indicates whether the current instance precedes, follows, or occurs in the same position in the sort order as the other object.

```
public int CompareTo(SwapRateSheet.CurrencyPair? other)
```

Parameters

`other` SwapRateSheet.CurrencyPair

An object to compare with this instance.

Returns

int

A value that indicates the relative order of the objects being compared. The return value has these meanings:

Value	Meaning
Less than zero	This instance precedes `other` in the sort order.
Zero	This instance occurs in the same position in the sort order as `other`.
Greater than zero	This instance follows `other` in the sort order.

Equals(CurrencyPair?)

Indicates whether the current object is equal to another object of the same type.

```
public bool Equals(SwapRateSheet.CurrencyPair? other)
```

Parameters

`other` SwapRateSheet.CurrencyPair

An object to compare with this object.

Returns

bool

true if the current object is equal to the `other` parameter; otherwise, false.

Equals(object?)

Determines whether the specified object is equal to the current object.

```
public override bool Equals(object? obj)
```

Parameters

`obj` object

The object to compare with the current object.

Returns

bool

true if the specified object is equal to the current object; otherwise, false.

GetHashCode()

Serves as the default hash function.

```
public override int GetHashCode()
```

Returns

int

A hash code for the current object.

Operators

operator ==(CurrencyPair, CurrencyPair)

Determines whether two specified instances of SwapRateSheet.CurrencyPair are equal.

```
public static bool operator ==(SwapRateSheet.CurrencyPair left,
SwapRateSheet.CurrencyPair right)
```

Parameters

`left` SwapRateSheet.CurrencyPair

The first SwapRateSheet.CurrencyPair to compare.

`right` SwapRateSheet.CurrencyPair

The second SwapRateSheet.CurrencyPair to compare.

Returns

bool

true if `left` and `right` represent the same value;

operator >(CurrencyPair, CurrencyPair)

Determines whether one specified SwapRateSheet.CurrencyPair is greater than another specified SwapRateSheet.CurrencyPair.

```
public static bool operator >(SwapRateSheet.CurrencyPair left,
SwapRateSheet.CurrencyPair right)
```

Parameters

`left` SwapRateSheet.CurrencyPair

The first SwapRateSheet.CurrencyPair to compare.

`right` SwapRateSheet.CurrencyPair

The second SwapRateSheet.CurrencyPair to compare.

Returns

bool

true if `left` is greater than `right`;

operator >=(CurrencyPair, CurrencyPair)

Determines whether one specified SwapRateSheet.CurrencyPair is greater than or equal to another specified SwapRateSheet.CurrencyPair.

```
public static bool operator >=(SwapRateSheet.CurrencyPair left,
SwapRateSheet.CurrencyPair right)
```

Parameters

`left` SwapRateSheet.CurrencyPair

The first SwapRateSheet.CurrencyPair to compare.

`right` SwapRateSheet.CurrencyPair

The second SwapRateSheet.CurrencyPair to compare.

Returns

bool

true if `left` is greater than or equal to `right`;

operator !=(CurrencyPair, CurrencyPair)

Determines whether two specified instances of SwapRateSheet.CurrencyPair are not equal.

```
public static bool operator !=(SwapRateSheet.CurrencyPair left,
SwapRateSheet.CurrencyPair right)
```

Parameters

`left` SwapRateSheet.CurrencyPair

`right` SwapRateSheet.CurrencyPair

Returns

bool

operator <(CurrencyPair, CurrencyPair)

Determines whether one specified SwapRateSheet.CurrencyPair is less than another specified SwapRateSheet.CurrencyPair.

```
public static bool operator <(SwapRateSheet.CurrencyPair left,
SwapRateSheet.CurrencyPair right)
```

Parameters

`left` SwapRateSheet.CurrencyPair

The first SwapRateSheet.CurrencyPair to compare.

`right` SwapRateSheet.CurrencyPair

Returns

bool

operator <=(CurrencyPair, CurrencyPair)

Determines whether one specified SwapRateSheet.CurrencyPair is less than or equal to another specified SwapRateSheet.CurrencyPair.

```
public static bool operator <=(SwapRateSheet.CurrencyPair left,
SwapRateSheet.CurrencyPair right)
```

Parameters

`left` SwapRateSheet.CurrencyPair

The first SwapRateSheet.CurrencyPair to compare.

`right` SwapRateSheet.CurrencyPair

The second SwapRateSheet.CurrencyPair to compare.

Returns

bool

true if `left` is less than or equal to `right`;

Class SwapRateSheet.Fraction

Namespace: Nekoyume.TableData.Swap

Assembly: Lib9c.dll

Represents a fraction.

```
public class SwapRateSheet.Fraction : IEquatable<SwapRateSheet.Fraction>
```

Inheritance

object ← SwapRateSheet.Fraction

Implements

IEquatable<SwapRateSheet.Fraction>

Inherited Members

object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

Fraction(BigInteger, BigInteger)

Initializes a new instance of the SwapRateSheet.Fraction class.

```
public Fraction(BigInteger numerator, BigInteger denominator)
```

Parameters

`numerator` BigInteger

The numerator of the fraction.

`denominator` BigInteger

The denominator of the fraction.

Properties

Denominator

The denominator of the fraction.

```
public BigInteger Denominator { get; }
```

Property Value

BigInteger

Numerator

The numerator of the fraction.

```
public BigInteger Numerator { get; }
```

Property Value

BigInteger

Methods

Equals(Fraction?)

Indicates whether the current object is equal to another object of the same type.

```
public bool Equals(SwapRateSheet.Fraction? other)
```

Parameters

`other` SwapRateSheet.Fraction

An object to compare with this object.

Returns

bool

true if the current object is equal to the `other` parameter; otherwise, false.

Equals(object?)

Determines whether the specified object is equal to the current object.

```
public override bool Equals(object? obj)
```

Parameters

`obj` object

The object to compare with the current object.

Returns

bool

true if the specified object is equal to the current object; otherwise, false.

GetHashCode()

Serves as the default hash function.

```
public override int GetHashCode()
```

Returns

int

A hash code for the current object.

Class SwapRateSheet.Row

Namespace: Nekoyume.TableData.Swap

Assembly: Lib9c.dll

Represents a row of SwapRateSheet.

```
public class SwapRateSheet.Row : SheetRow<SwapRateSheet.CurrencyPair>
```

Inheritance

object ← SheetRow<SwapRateSheet.CurrencyPair> ← SwapRateSheet.Row

Inherited Members

SheetRow<SwapRateSheet.CurrencyPair>.Validate() ,
SheetRow<SwapRateSheet.CurrencyPair>.EndOfSheetInitialize() ,
SheetRow<SwapRateSheet.CurrencyPair>.Equals(object) ,
SheetRow<SwapRateSheet.CurrencyPair>.GetHashCode() , object.Equals(object, object) ,
object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) ,
object.ToString()

Properties

Key

```
public override SwapRateSheet.CurrencyPair Key { get; }
```

Property Value

SwapRateSheet.CurrencyPair

Pair

The currency pair for swap.

```
public SwapRateSheet.CurrencyPair? Pair { get; }
```

Property Value

SwapRateSheet.CurrencyPair

Rate

The swap rate represented by SwapRateSheet.Fraction.

```
public SwapRateSheet.Fraction? Rate { get; }
```

Property Value

SwapRateSheet.Fraction

Methods

Set(IReadOnlyList<string>)

```
public override void Set(IReadOnlyList<string> fields)
```

Parameters

`fields` IReadOnlyList<string>

Namespace Nekoyume.TypedAddress

Structs

AgentAddress

GuildAddress

PledgeAddress

Struct AgentAddress

Namespace: Nekoyume.TypedAddress

Assembly: Lib9c.dll

```
public readonly struct AgentAddress : IBencodable
```

Implements
IBencodable

Inherited Members
object.Equals(object, object) , object.GetType() , object.ReferenceEquals(object, object)

Extension Methods
AgentAddressExtensions.GetPledgeAddress(AgentAddress)

Constructors

AgentAddress(IValue)

```
public AgentAddress(IValue value)
```

Parameters

`value` IValue

AgentAddress(Address)

```
public AgentAddress(Address address)
```

Parameters

`address` Address

AgentAddress(byte[])

```
public AgentAddress(byte[] bytes)
```

Parameters

`bytes` byte[]

AgentAddress(string)

```
public AgentAddress(string hex)
```

Parameters

`hex` string

Properties

Bencoded

An Bencodex.Types.IValue representation of this object that can be decoded back to instantiate an equal object. The decoded object must be equal to the original in the sense that Equals(T) should be true.

```
public IValue Bencoded { get; }
```

Property Value

IValue

Remarks

Note that the only requirement is that the produced Bencodex.Types.IValue can be decoded back to an equal object. This representation may not be canonical in the sense that additional junk data may be present in an Bencodex.Types.IValue that one may wish to decode and this may be discarded while decoding.

A specific implemnetation may decide to only allow the canonical representation to be decoded.

Methods

Equals(AgentAddress)

```
public bool Equals(AgentAddress other)
```

Parameters

`other` AgentAddress

Returns

bool

Equals(object)

Indicates whether this instance and a specified object are equal.

```
public override bool Equals(object obj)
```

Parameters

`obj` object

The object to compare with the current instance.

Returns

bool

true if `obj` and this instance are the same type and represent the same value; otherwise, false.

GetHashCode()

Returns the hash code for this instance.

```
public override int GetHashCode()
```

Returns

int

A 32-bit signed integer that is the hash code for this instance.

ToString()

Returns the fully qualified type name of this instance.

```
public override string ToString()
```

Returns

string

The fully qualified type name.

Operators

operator ==(AgentAddress, AgentAddress)

```
public static bool operator ==(AgentAddress left, AgentAddress right)
```

Parameters

`left` AgentAddress

`right` AgentAddress

Returns

bool

implicit operator Address(AgentAddress)

```
public static implicit operator Address(AgentAddress agentAddress)
```

Parameters

`agentAddress` AgentAddress

Returns

Address

operator !=(AgentAddress, AgentAddress)

```
public static bool operator !=(AgentAddress left, AgentAddress right)
```

Parameters

`left` AgentAddress

`right` AgentAddress

Returns

bool

Struct GuildAddress

Namespace: Nekoyume.TypedAddress

Assembly: Lib9c.dll

```
public readonly struct GuildAddress : IBencodable, IEquatable<GuildAddress>
```

Implements
IBencodable, IEquatable<GuildAddress>

Inherited Members
object.Equals(object, object) , object.GetType() , object.ReferenceEquals(object, object)

Constructors

GuildAddress(IValue)

```
public GuildAddress(IValue value)
```

Parameters

`value` IValue

GuildAddress(Address)

```
public GuildAddress(Address address)
```

Parameters

`address` Address

GuildAddress(byte[])

```
public GuildAddress(byte[] bytes)
```

Parameters

`bytes` byte[]

GuildAddress(string)

```
public GuildAddress(string hex)
```

Parameters

`hex` string

Properties

Bencoded

An Bencodex.Types.IValue representation of this object that can be decoded back to instantiate an equal object. The decoded object must be equal to the original in the sense that Equals(T) should be true.

```
public IValue Bencoded { get; }
```

Property Value

IValue

Remarks

Note that the only requirement is that the produced Bencodex.Types.IValue can be decoded back to an equal object. This representation may not be canonical in the sense that additional junk data may be present in an Bencodex.Types.IValue that one may wish to decode and this may be discarded while decoding.

A specific implemnetation may decide to only allow the canonical representation to be decoded.

Methods

Equals(GuildAddress)

Indicates whether the current object is equal to another object of the same type.

```
public bool Equals(GuildAddress other)
```

Parameters

`other` GuildAddress

An object to compare with this object.

Returns

bool

true if the current object is equal to the `other` parameter; otherwise, false.

Equals(object)

Indicates whether this instance and a specified object are equal.

```
public override bool Equals(object obj)
```

Parameters

`obj` object

The object to compare with the current instance.

Returns

bool

true if `obj` and this instance are the same type and represent the same value; otherwise, false.

GetHashCode()

Returns the hash code for this instance.

```
public override int GetHashCode()
```

Returns

int

A 32-bit signed integer that is the hash code for this instance.

ToString()

Returns the fully qualified type name of this instance.

```
public override string ToString()
```

Returns

string

The fully qualified type name.

Operators

operator ==(GuildAddress, GuildAddress)

```
public static bool operator ==(GuildAddress left, GuildAddress right)
```

Parameters

`left` GuildAddress

`right` GuildAddress

Returns

bool

implicit operator Address(GuildAddress)

```
public static implicit operator Address(GuildAddress guildAddress)
```

Parameters

`guildAddress` GuildAddress

Returns

Address

operator !=(GuildAddress, GuildAddress)

```
public static bool operator !=(GuildAddress left, GuildAddress right)
```

Parameters

`left` GuildAddress

`right` GuildAddress

Returns

bool

Struct PledgeAddress

Namespace: Nekoyume.TypedAddress

Assembly: Lib9c.dll

```
public readonly struct PledgeAddress : IBencodable
```

Implements
IBencodable

Inherited Members
object.Equals(object, object) , object.GetType() , object.ReferenceEquals(object, object)

Constructors

PledgeAddress(IValue)

```
public PledgeAddress(IValue value)
```

Parameters

`value` IValue

PledgeAddress(Address)

```
public PledgeAddress(Address address)
```

Parameters

`address` Address

PledgeAddress(byte[])

```
public PledgeAddress(byte[] bytes)
```

Parameters

`bytes` byte[]

PledgeAddress(string)

```
public PledgeAddress(string hex)
```

Parameters

`hex` string

Properties

Bencoded

An Bencodex.Types.IValue representation of this object that can be decoded back to instantiate an equal object. The decoded object must be equal to the original in the sense that Equals(T) should be true.

```
public IValue Bencoded { get; }
```

Property Value

IValue

Remarks

Note that the only requirement is that the produced Bencodex.Types.IValue can be decoded back to an equal object. This representation may not be canonical in the sense that additional junk data may be present in an Bencodex.Types.IValue that one may wish to decode and this may be discarded while decoding.

A specific implemnetation may decide to only allow the canonical representation to be decoded.

Methods

Equals(PledgeAddress)

```
public bool Equals(PledgeAddress other)
```

Parameters

`other` PledgeAddress

Returns

bool

Equals(object)

Indicates whether this instance and a specified object are equal.

```
public override bool Equals(object obj)
```

Parameters

`obj` object

The object to compare with the current instance.

Returns

bool

true if `obj` and this instance are the same type and represent the same value; otherwise, false.

GetHashCode()

Returns the hash code for this instance.

```
public override int GetHashCode()
```

Returns

int

A 32-bit signed integer that is the hash code for this instance.

ToString()

Returns the fully qualified type name of this instance.

```
public override string ToString()
```

Returns

string

The fully qualified type name.

Operators

operator ==(PledgeAddress, PledgeAddress)

```
public static bool operator ==(PledgeAddress left, PledgeAddress right)
```

Parameters

`left` PledgeAddress

`right` PledgeAddress

Returns

bool

implicit operator Address(PledgeAddress)

```
public static implicit operator Address(PledgeAddress agentAddress)
```

Parameters

`agentAddress` PledgeAddress

Returns

Address

operator !=(PledgeAddress, PledgeAddress)

```
public static bool operator !=(PledgeAddress left, PledgeAddress right)
```

Parameters

`left` PledgeAddress

`right` PledgeAddress

Returns

bool

Namespace Nekoyume.Validator Delegation

Classes

AbstainHistory

PayMaster

PayMaster is a contract that manages the proxy of mortgage and refund of gas.

ProposerInfo

ValidatorComparer

ValidatorDelegatee

ValidatorDelegator

ValidatorList

ValidatorRepository

Class AbstainHistory

Namespace: Nekoyume.ValidatorDelegation

Assembly: Lib9c.dll

```
public sealed class AbstainHistory
```

Inheritance

object ← AbstainHistory

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

AbstainHistory()

```
public AbstainHistory()
```

AbstainHistory(IValue)

```
public AbstainHistory(IValue bencoded)
```

Parameters

`bencoded` IValue

AbstainHistory(List)

```
public AbstainHistory(List bencoded)
```

Parameters

`bencoded` List

Properties

Address

```
public static Address Address { get; }
```

Property Value

Address

Bencoded

```
public IValue Bencoded { get; }
```

Property Value

IValue

History

```
public SortedDictionary<PublicKey, List<long>> History { get; }
```

Property Value

SortedDictionary<PublicKey, List<long>>

MaxAbstainAllowance

Maximum abstain allowance to slash and jail.

```
public static int MaxAbstainAllowance { get; }
```

Property Value

int

WindowSize

```
public static int WindowSize { get; }
```

Property Value

int

Methods

FindToSlashAndAdd(IEnumerable<PublicKey>, long)

```
public List<PublicKey> FindToSlashAndAdd(IEnumerable<PublicKey> abstainList,
long height)
```

Parameters

`abstainList` IEnumerable<PublicKey>

`height` long

Returns

List<PublicKey>

Class PayMaster

Namespace: Nekoyume.ValidatorDelegation

Assembly: Lib9c.dll

PayMaster is a contract that manages the proxy of mortgage and refund of gas.

```
public static class PayMaster
```

Inheritance

object ← PayMaster

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Methods

Mortgage(IWorld, IActionContext, Address, Address, FungibleAssetValue)

Mortgage gas with the PayMaster.

```
public static IWorld Mortgage(this IWorld world, IActionContext context, Address
address, Address payMaster, FungibleAssetValue gasToMortgage)
```

Parameters

`world` IWorld

The Libplanet.Action.State.IWorld where the mortgage happen.

`context` IActionContext

The Libplanet.Action.IActionContext where the mortgage happen.

`address` Address

The Libplanet.Crypto.Address of the user to mortgage gas.

`payMaster` Address

The Libplanet.Crypto.Address of the paymaster.

`gasToMortgage` FungibleAssetValue

The Libplanet.Types.Assets.FungibleAssetValue of gas to mortgage.

Returns

IWorld

The updated Libplanet.Action.State.IWorld.

Refund(IWorld, IActionContext, Address, FungibleAssetValue)

Refund gas with the PayMaster.

```
public static IWorld Refund(this IWorld world, IActionContext context, Address
address, FungibleAssetValue gasToRefund)
```

Parameters

`world` IWorld

The Libplanet.Action.State.IWorld where the refund happen.

`context` IActionContext

The Libplanet.Action.IActionContext where the refund happen.

`address` Address

The Libplanet.Crypto.Address of the user to refund gas.

`gasToRefund` FungibleAssetValue

The Libplanet.Types.Assets.FungibleAssetValue of gas to refund.

Returns

IWorld

The updated Libplanet.Action.State.IWorld.

Class ProposerInfo

Namespace: Nekoyume.ValidatorDelegation

Assembly: Lib9c.dll

```
public class ProposerInfo
```

Inheritance

object ← ProposerInfo

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

ProposerInfo(IValue)

```
public ProposerInfo(IValue bencoded)
```

Parameters

`bencoded` IValue

ProposerInfo(List)

```
public ProposerInfo(List bencoded)
```

Parameters

`bencoded` List

ProposerInfo(long, Address)

```
public ProposerInfo(long blockIndex, Address proposer)
```

Parameters

`blockIndex` long

`proposer` Address

Properties

Address

```
public static Address Address { get; }
```

Property Value

Address

Bencoded

```
public IValue Bencoded { get; }
```

Property Value

IValue

BlockIndex

```
public long BlockIndex { get; }
```

Property Value

long

Proposer

```
public Address Proposer { get; }
```

Property Value

Address

Class ValidatorComparer

Namespace: Nekoyume.ValidatorDelegation

Assembly: Lib9c.dll

```
public class ValidatorComparer : IComparer<Validator>
```

Inheritance

object ← ValidatorComparer

Implements

IComparer<Validator>

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Methods

Compare(Validator?, Validator?)

Compares two objects and returns a value indicating whether one is less than, equal to, or greater than the other.

```
public int Compare(Validator? x, Validator? y)
```

Parameters

x Validator

The first object to compare.

y Validator

The second object to compare.

Returns

int

A signed integer that indicates the relative values of x and y, as shown in the following table.

Value	Meaning
Less than zero	x is less than y.
Zero	x equals y.
Greater than zero	x is greater than y.

Class ValidatorDelegatee

Namespace: Nekoyume.ValidatorDelegation

Assembly: Lib9c.dll

```
public sealed class ValidatorDelegatee : Delegatee<ValidatorRepository,
ValidatorDelegatee, ValidatorDelegator>, IDelegatee, IEquatable<ValidatorDelegatee>,
IBencodable
```

Inheritance

object ← Delegatee<ValidatorRepository, ValidatorDelegatee, ValidatorDelegator> ← ValidatorDelegatee

Implements

IDelegatee, IEquatable<ValidatorDelegatee>, IBencodable

Inherited Members

Delegatee<ValidatorRepository, ValidatorDelegatee, ValidatorDelegator>.Metadata ,
Delegatee<ValidatorRepository, ValidatorDelegatee, ValidatorDelegator>.Repository ,
Delegatee<ValidatorRepository, ValidatorDelegatee, ValidatorDelegator>.Address ,
Delegatee<ValidatorRepository, ValidatorDelegatee, ValidatorDelegator>.AccountAddress ,
Delegatee<ValidatorRepository, ValidatorDelegatee, ValidatorDelegator>.MetadataAddress
,
Delegatee<ValidatorRepository, ValidatorDelegatee,
ValidatorDelegator>.DelegationCurrency ,
Delegatee<ValidatorRepository, ValidatorDelegatee, ValidatorDelegator>.RewardCurrencies
,
Delegatee<ValidatorRepository, ValidatorDelegatee,
ValidatorDelegator>.DelegationPoolAddress ,
Delegatee<ValidatorRepository, ValidatorDelegatee,
ValidatorDelegator>.RewardPoolAddress ,
Delegatee<ValidatorRepository, ValidatorDelegatee,
ValidatorDelegator>.RewardRemainderPoolAddress ,
Delegatee<ValidatorRepository, ValidatorDelegatee,
ValidatorDelegator>.SlashedPoolAddress ,
Delegatee<ValidatorRepository, ValidatorDelegatee, ValidatorDelegator>.UnbondingPeriod
,
Delegatee<ValidatorRepository, ValidatorDelegatee,
ValidatorDelegator>.MaxUnbondLockInEntries ,

Delegatee<ValidatorRepository, ValidatorDelegatee, ValidatorDelegator>.MaxRebondGraceEntries ,
Delegatee<ValidatorRepository, ValidatorDelegatee, ValidatorDelegator>.TotalDelegated ,
Delegatee<ValidatorRepository, ValidatorDelegatee, ValidatorDelegator>.TotalShares ,
Delegatee<ValidatorRepository, ValidatorDelegatee, ValidatorDelegator>.Jailed ,
Delegatee<ValidatorRepository, ValidatorDelegatee, ValidatorDelegator>.JailedUntil ,
Delegatee<ValidatorRepository, ValidatorDelegatee, ValidatorDelegator>.Tombstoned ,
Delegatee<ValidatorRepository, ValidatorDelegatee, ValidatorDelegator>.MetadataBencoded ,
Delegatee<ValidatorRepository, ValidatorDelegatee, ValidatorDelegator>.ShareFromFAV(FungibleAssetValue) ,
Delegatee<ValidatorRepository, ValidatorDelegatee, ValidatorDelegator>.FAVFromShare(BigInteger) ,
Delegatee<ValidatorRepository, ValidatorDelegatee, ValidatorDelegator>.Jail(long) ,
Delegatee<ValidatorRepository, ValidatorDelegatee, ValidatorDelegator>.Unjail(long) ,
Delegatee<ValidatorRepository, ValidatorDelegatee, ValidatorDelegator>.Tombstone() ,
Delegatee<ValidatorRepository, ValidatorDelegatee, ValidatorDelegator>.BondAddress(Address) ,
Delegatee<ValidatorRepository, ValidatorDelegatee, ValidatorDelegator>.UnbondLockInAddress(Address) ,
Delegatee<ValidatorRepository, ValidatorDelegatee, ValidatorDelegator>.RebondGraceAddress(Address) ,
Delegatee<ValidatorRepository, ValidatorDelegatee, ValidatorDelegator>.DistributionPoolAddress() ,
Delegatee<ValidatorRepository, ValidatorDelegatee, ValidatorDelegator>.CurrentRewardBaseAddress() ,
Delegatee<ValidatorRepository, ValidatorDelegatee, ValidatorDelegator>.RewardBaseAddress(long) ,
Delegatee<ValidatorRepository, ValidatorDelegatee, ValidatorDelegator>.CurrentLumpSumRewardsRecordAddress() ,
Delegatee<ValidatorRepository, ValidatorDelegatee, ValidatorDelegator>.LumpSumRewardsRecordAddress(long) ,
Delegatee<ValidatorRepository, ValidatorDelegatee, ValidatorDelegator>.Bond(ValidatorDelegator, FungibleAssetValue, long) ,
Delegatee<ValidatorRepository, ValidatorDelegatee, ValidatorDelegator>.Unbond(ValidatorDelegator, BigInteger, long) ,
Delegatee<ValidatorRepository, ValidatorDelegatee, ValidatorDelegator>.DistributeReward(ValidatorDelegator, long) ,
Delegatee<ValidatorRepository, ValidatorDelegatee, ValidatorDelegator>.CollectRewards(long) ,

Delegatee<ValidatorRepository, ValidatorDelegatee, ValidatorDelegator>.Slash(BigInteger, long, long) ,
Delegatee<ValidatorRepository, ValidatorDelegatee, ValidatorDelegator>.StartNewRewardPeriod(long) ,
Delegatee<ValidatorRepository, ValidatorDelegatee, ValidatorDelegator>.CalculateRewards(BigInteger, RewardBase, RewardBase) ,
object.Equals(object, object) , object.GetType() ,
object.ReferenceEquals(object, object) , object.ToString()

Constructors

ValidatorDelegatee(Address, IValue, ValidatorRepository)

```
public ValidatorDelegatee(Address address, IValue bencoded,
ValidatorRepository repository)
```

Parameters

`address` Address

`bencoded` IValue

`repository` ValidatorRepository

ValidatorDelegatee(Address, List, ValidatorRepository)

```
public ValidatorDelegatee(Address address, List bencoded, ValidatorRepository
repository)
```

Parameters

`address` Address

`bencoded` List

`repository` ValidatorRepository

ValidatorDelegatee(Address, PublicKey, BigInteger, long, IEnumerable<Currency>, ValidatorRepository)

```
public ValidatorDelegatee(Address address, PublicKey publicKey, BigInteger
commissionPercentage, long creationHeight, IEnumerable<Currency> rewardCurrencies,
ValidatorRepository repository)
```

Parameters

`address` Address

`publicKey` PublicKey

`commissionPercentage` BigInteger

`creationHeight` long

`rewardCurrencies` IEnumerable<Currency>

`repository` ValidatorRepository

Properties

ActiveDelegationPoolAddress

```
public static Address ActiveDelegationPoolAddress { get; }
```

Property Value

Address

BaseProposerRewardPercentage

```
public static BigInteger BaseProposerRewardPercentage { get; }
```

Property Value

BigInteger

Bencoded

```
public List Bencoded { get; }
```

Property Value

List

BonusProposerRewardPercentage

```
public static BigInteger BonusProposerRewardPercentage { get; }
```

Property Value

BigInteger

CommissionPercentage

```
public BigInteger CommissionPercentage { get; }
```

Property Value

BigInteger

CommissionPercentageLastUpdateHeight

```
public long CommissionPercentageLastUpdateHeight { get; }
```

Property Value

long

CommissionPercentageMaxChange

```
public static BigInteger CommissionPercentageMaxChange { get; }
```

Property Value

BigInteger

CommissionPercentageUpdateCooldown

```
public static long CommissionPercentageUpdateCooldown { get; }
```

Property Value

long

DefaultCommissionPercentage

```
public static BigInteger DefaultCommissionPercentage { get; }
```

Property Value

BigInteger

DefaultUnbondingPeriod

```
public static long DefaultUnbondingPeriod { get; }
```

Property Value

long

InactiveDelegationPoolAddress

```
public static Address InactiveDelegationPoolAddress { get; }
```

Property Value

Address

IsActive

```
public bool IsActive { get; }
```

Property Value

bool

MaxCommissionPercentage

```
public static BigInteger MaxCommissionPercentage { get; }
```

Property Value

BigInteger

MinCommissionPercentage

```
public static BigInteger MinCommissionPercentage { get; }
```

Property Value

BigInteger

MinSelfDelegation

```
public FungibleAssetValue MinSelfDelegation { get; }
```

Property Value

FungibleAssetValue

Power

```
public BigInteger Power { get; }
```

Property Value

BigInteger

PublicKey

```
public PublicKey PublicKey { get; }
```

Property Value

PublicKey

Validator

```
public Validator Validator { get; }
```

Property Value

Validator

ValidatorDelegationCurrency

```
public static Currency ValidatorDelegationCurrency { get; }
```

Property Value

Currency

ValidatorMaxRebondGraceEntries

```
public static int ValidatorMaxRebondGraceEntries { get; }
```

Property Value

int

ValidatorMaxUnbondLockInEntries

```
public static int ValidatorMaxUnbondLockInEntries { get; }
```

Property Value

int

Methods

Activate()

```
public void Activate()
```

AllocateReward(FungibleAssetValue, BigInteger, BigInteger, Address, long)

```
public void AllocateReward(FungibleAssetValue rewardToAllocate, BigInteger
```

```
validatorPower, BigInteger validatorSetPower, Address RewardSource, long height)
```

Parameters

`rewardToAllocate` FungibleAssetValue

`validatorPower` BigInteger

`validatorSetPower` BigInteger

`RewardSource` Address

`height` long

Deactivate()

```
public void Deactivate()
```

Equals(IDelegatee?)

```
public bool Equals(IDelegatee? other)
```

Parameters

`other` IDelegatee

Returns

bool

Equals(ValidatorDelegatee?)

Indicates whether the current object is equal to another object of the same type.

```
public bool Equals(ValidatorDelegatee? other)
```

Parameters

`other` ValidatorDelegatee

An object to compare with this object.

Returns

bool

true if the current object is equal to the `other` parameter; otherwise, false.

Equals(object?)

Determines whether the specified object is equal to the current object.

```
public override bool Equals(object? obj)
```

Parameters

`obj` object

The object to compare with the current object.

Returns

bool

true if the specified object is equal to the current object; otherwise, false.

GetHashCode()

Serves as the default hash function.

```
public override int GetHashCode()
```

Returns

int

A hash code for the current object.

OnDelegationChanged(long)

```
protected override void OnDelegationChanged(long height)
```

Parameters

`height` long

OnEnjailed()

```
protected override void OnEnjailed()
```

OnUnjailed()

```
protected override void OnUnjailed()
```

SetCommissionPercentage(BigInteger, long)

```
public void SetCommissionPercentage(BigInteger percentage, long height)
```

Parameters

`percentage` BigInteger

`height` long

Unjail(long)

```
public void Unjail(long height)
```

Parameters

`height` long

Class ValidatorDelegator

Namespace: Nekoyume.ValidatorDelegation

Assembly: Lib9c.dll

```
public sealed class ValidatorDelegator : Delegator<ValidatorRepository,
ValidatorDelegatee, ValidatorDelegator>, IDelegator, IEquatable<ValidatorDelegator>
```

Inheritance

object ← Delegator<ValidatorRepository, ValidatorDelegatee, ValidatorDelegator> ← ValidatorDelegator

Implements

IDelegator, IEquatable<ValidatorDelegator>

Inherited Members

Delegator<ValidatorRepository, ValidatorDelegatee, ValidatorDelegator>.Metadata ,
Delegator<ValidatorRepository, ValidatorDelegatee, ValidatorDelegator>.Repository ,
Delegator<ValidatorRepository, ValidatorDelegatee, ValidatorDelegator>.Address ,
Delegator<ValidatorRepository, ValidatorDelegatee, ValidatorDelegator>.AccountAddress ,
Delegator<ValidatorRepository, ValidatorDelegatee, ValidatorDelegator>.MetadataAddress
,
Delegator<ValidatorRepository, ValidatorDelegatee,
ValidatorDelegator>.DelegationPoolAddress ,
Delegator<ValidatorRepository, ValidatorDelegatee, ValidatorDelegator>.RewardAddress ,
Delegator<ValidatorRepository, ValidatorDelegatee, ValidatorDelegator>.Delegatees ,
Delegator<ValidatorRepository, ValidatorDelegatee, ValidatorDelegator>.UnbondingRefs ,
Delegator<ValidatorRepository, ValidatorDelegatee,
ValidatorDelegator>.MetadataBencoded ,
Delegator<ValidatorRepository, ValidatorDelegatee,
ValidatorDelegator>.Delegate(ValidatorDelegatee, FungibleAssetValue, long) ,
Delegator<ValidatorRepository, ValidatorDelegatee,
ValidatorDelegator>.Undelegate(ValidatorDelegatee, BigInteger, long) ,
Delegator<ValidatorRepository, ValidatorDelegatee,
ValidatorDelegator>.Redelegate(ValidatorDelegatee, ValidatorDelegatee, BigInteger, long) ,
Delegator<ValidatorRepository, ValidatorDelegatee,
ValidatorDelegator>.CancelUndelegate(ValidatorDelegatee, FungibleAssetValue, long) ,
Delegator<ValidatorRepository, ValidatorDelegatee,
ValidatorDelegator>.ClaimReward(ValidatorDelegatee, long) ,

Delegator<ValidatorRepository, ValidatorDelegatee, ValidatorDelegator>.ReleaseUnbondings(long) , object.Equals(object, object) , object.GetType() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

ValidatorDelegator(Address, Address, ValidatorRepository)

```
public ValidatorDelegator(Address address, Address delegationPoolAddress,
ValidatorRepository repository)
```

Parameters

`address` Address

`delegationPoolAddress` Address

`repository` ValidatorRepository

ValidatorDelegator(Address, ValidatorRepository)

```
public ValidatorDelegator(Address address, ValidatorRepository repository)
```

Parameters

`address` Address

`repository` ValidatorRepository

Methods

Equals(ValidatorDelegator?)

Indicates whether the current object is equal to another object of the same type.

```
public bool Equals(ValidatorDelegator? other)
```

Parameters

`other` ValidatorDelegator

An object to compare with this object.

Returns

bool

true if the current object is equal to the `other` parameter; otherwise, false.

Equals(object?)

Determines whether the specified object is equal to the current object.

```
public override bool Equals(object? obj)
```

Parameters

`obj` object

The object to compare with the current object.

Returns

bool

true if the specified object is equal to the current object; otherwise, false.

GetHashCode()

Serves as the default hash function.

```
public override int GetHashCode()
```

Returns

int

A hash code for the current object.

Class ValidatorList

Namespace: Nekoyume.ValidatorDelegation

Assembly: Lib9c.dll

```
public sealed class ValidatorList : IBencodable
```

Inheritance

object ← ValidatorList

Implements

IBencodable

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

ValidatorList()

```
public ValidatorList()
```

ValidatorList(IValue)

```
public ValidatorList(IValue bencoded)
```

Parameters

`bencoded` IValue

ValidatorList(List)

```
public ValidatorList(List bencoded)
```

Parameters

`bencoded` List

Properties

Address

```
public static Address Address { get; }
```

Property Value

Address

Bencoded

```
public List Bencoded { get; }
```

Property Value

List

MaxActiveSetSize

```
public static int MaxActiveSetSize { get; }
```

Property Value

int

Validators

```
public ImmutableList<Validator> Validators { get; }
```

Property Value

ImmutableList<Validator>

Methods

ActiveSet()

```
public List<Validator> ActiveSet()
```

Returns

List<Validator>

InActiveSet()

```
public List<Validator> InActiveSet()
```

Returns

List<Validator>

RemoveValidator(PublicKey)

```
public ValidatorList RemoveValidator(PublicKey publicKey)
```

Parameters

`publicKey` PublicKey

Returns

ValidatorList

SetValidator(Validator)

```
public ValidatorList SetValidator(Validator validator)
```

Parameters

`validator` Validator

Returns

ValidatorList

Class ValidatorRepository

Namespace: Nekoyume.ValidatorDelegation

Assembly: Lib9c.dll

```
public sealed class ValidatorRepository : DelegationRepository<ValidatorRepository,
ValidatorDelegatee, ValidatorDelegator>, IDelegationRepository
```

Inheritance

object ←

DelegationRepository<ValidatorRepository, ValidatorDelegatee, ValidatorDelegator> ← ValidatorRepository

Implements

IDelegationRepository

Inherited Members

DelegationRepository<ValidatorRepository, ValidatorDelegatee, ValidatorDelegator>.World ,
DelegationRepository<ValidatorRepository, ValidatorDelegatee, ValidatorDelegator>.ActionContext ,
DelegationRepository<ValidatorRepository, ValidatorDelegatee, ValidatorDelegator>.DelegateeAccountAddress ,
DelegationRepository<ValidatorRepository, ValidatorDelegatee, ValidatorDelegator>.DelegatorAccountAddress ,
DelegationRepository<ValidatorRepository, ValidatorDelegatee, ValidatorDelegator>.DelegateeMetadataAccountAddress ,
DelegationRepository<ValidatorRepository, ValidatorDelegatee, ValidatorDelegator>.DelegatorMetadataAccountAddress ,
DelegationRepository<ValidatorRepository, ValidatorDelegatee, ValidatorDelegator>.BondAccountAddress ,
DelegationRepository<ValidatorRepository, ValidatorDelegatee, ValidatorDelegator>.UnbondLockInAccountAddress ,
DelegationRepository<ValidatorRepository, ValidatorDelegatee, ValidatorDelegator>.RebondGraceAccountAddress ,
DelegationRepository<ValidatorRepository, ValidatorDelegatee, ValidatorDelegator>.UnbondingSetAccountAddress ,
DelegationRepository<ValidatorRepository, ValidatorDelegatee, ValidatorDelegator>.RewardBaseAccountAddress ,

DelegationRepository<ValidatorRepository, ValidatorDelegatee, ValidatorDelegator>.LumpSumRewardsRecordAccountAddress ,
DelegationRepository<ValidatorRepository, ValidatorDelegatee, ValidatorDelegator>.GetDelegatee(Address) ,
DelegationRepository<ValidatorRepository, ValidatorDelegatee, ValidatorDelegator>.GetDelegator(Address) ,
DelegationRepository<ValidatorRepository, ValidatorDelegatee, ValidatorDelegator>.SetDelegatee(ValidatorDelegatee) ,
DelegationRepository<ValidatorRepository, ValidatorDelegatee, ValidatorDelegator>.SetDelegator(ValidatorDelegator) ,
DelegationRepository<ValidatorRepository, ValidatorDelegatee, ValidatorDelegator>.GetDelegateeMetadata(Address) ,
DelegationRepository<ValidatorRepository, ValidatorDelegatee, ValidatorDelegator>.GetDelegatorMetadata(Address) ,
DelegationRepository<ValidatorRepository, ValidatorDelegatee, ValidatorDelegator>.GetBond(ValidatorDelegatee, Address) ,
DelegationRepository<ValidatorRepository, ValidatorDelegatee, ValidatorDelegator>.GetUnbondLockIn(ValidatorDelegatee, Address) ,
DelegationRepository<ValidatorRepository, ValidatorDelegatee, ValidatorDelegator>.GetUnlimitedUnbondLockIn(Address) ,
DelegationRepository<ValidatorRepository, ValidatorDelegatee, ValidatorDelegator>.GetRebondGrace(ValidatorDelegatee, Address) ,
DelegationRepository<ValidatorRepository, ValidatorDelegatee, ValidatorDelegator>.GetUnlimitedRebondGrace(Address) ,
DelegationRepository<ValidatorRepository, ValidatorDelegatee, ValidatorDelegator>.GetCurrentRewardBase(ValidatorDelegatee) ,
DelegationRepository<ValidatorRepository, ValidatorDelegatee, ValidatorDelegator>.GetRewardBase(ValidatorDelegatee, long) ,
DelegationRepository<ValidatorRepository, ValidatorDelegatee, ValidatorDelegator>.GetLumpSumRewardsRecord(ValidatorDelegatee, long) ,
DelegationRepository<ValidatorRepository, ValidatorDelegatee, ValidatorDelegator>.GetCurrentLumpSumRewardsRecord(ValidatorDelegatee) ,
DelegationRepository<ValidatorRepository, ValidatorDelegatee, ValidatorDelegator>.GetBalance(Address, Currency) ,
DelegationRepository<ValidatorRepository, ValidatorDelegatee, ValidatorDelegator>.SetDelegateeMetadata(DelegateeMetadata) ,
DelegationRepository<ValidatorRepository, ValidatorDelegatee, ValidatorDelegator>.SetDelegatorMetadata(DelegatorMetadata) ,
DelegationRepository<ValidatorRepository, ValidatorDelegatee, ValidatorDelegator>.SetBond(Bond) ,

DelegationRepository<ValidatorRepository, ValidatorDelegatee, ValidatorDelegator>.SetUnbondLockIn(UnbondLockIn) , DelegationRepository<ValidatorRepository, ValidatorDelegatee, ValidatorDelegator>.SetRebondGrace(RebondGrace) , DelegationRepository<ValidatorRepository, ValidatorDelegatee, ValidatorDelegator>.SetRewardBase(RewardBase) , DelegationRepository<ValidatorRepository, ValidatorDelegatee, ValidatorDelegator>.SetLumpSumRewardsRecord(LumpSumRewardsRecord) , DelegationRepository<ValidatorRepository, ValidatorDelegatee, ValidatorDelegator>.RemoveLumpSumRewardsRecord(LumpSumRewardsRecord) , DelegationRepository<ValidatorRepository, ValidatorDelegatee, ValidatorDelegator>.TransferAsset(Address, Address, FungibleAssetValue) , DelegationRepository<ValidatorRepository, ValidatorDelegatee, ValidatorDelegator>.UpdateWorld(IWorld) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.ReferenceEquals(object, object) , object.ToString()

Extension Methods

AbstainHistoryModule.GetAbstainHistory(ValidatorRepository) , AbstainHistoryModule.SetAbstainHistory(ValidatorRepository, AbstainHistory) , ValidatorDelegateeModule.CreateDelegatee(ValidatorRepository, PublicKey, BigInteger) , ValidatorDelegateeModule.TryGetDelegatee(ValidatorRepository, Address, out ValidatorDelegatee?) , ValidatorDelegatorModule.TryGetDelegator(ValidatorRepository, Address, out ValidatorDelegator?) , ValidatorListModule.GetProposerInfo(ValidatorRepository) , ValidatorListModule.GetValidatorList(ValidatorRepository) , ValidatorListModule.SetProposerInfo(ValidatorRepository, ProposerInfo) , ValidatorListModule.TryGetProposerInfo(ValidatorRepository, Address, out ProposerInfo?) , ValidatorListModule.TryGetValidatorList(ValidatorRepository, out ValidatorList?)

Constructors

ValidatorRepository(IWorld, IActionContext)

```
public ValidatorRepository(IWorld world, IActionContext actionContext)
```

Parameters

`world` IWorld

`actionContext` IActionContext

ValidatorRepository(IDelegationRepository)

```
public ValidatorRepository(IDelegationRepository repository)
```

Parameters

`repository` IDelegationRepository

Properties

World

```
public override IWorld World { get; }
```

Property Value

IWorld

Methods

GetDelegatee(Address)

```
public override ValidatorDelegatee GetDelegatee(Address address)
```

Parameters

`address` Address

Returns

ValidatorDelegatee

GetDelegator(Address)

```
public override ValidatorDelegator GetDelegator(Address address)
```

Parameters

`address` Address

Returns

ValidatorDelegator

GetValidatorList()

```
public ValidatorList GetValidatorList()
```

Returns

ValidatorList

SetCommissionPercentage(Address, BigInteger, long)

```
public void SetCommissionPercentage(Address address, BigInteger
commissionPercentage, long height)
```

Parameters

`address` Address

`commissionPercentage` BigInteger

`height` long

SetDelegatee(ValidatorDelegatee)

```
public override void SetDelegatee(ValidatorDelegatee delegatee)
```

Parameters

`delegatee` ValidatorDelegatee

SetDelegator(ValidatorDelegator)

```
public override void SetDelegator(ValidatorDelegator delegator)
```

Parameters

`delegator` ValidatorDelegator

SetValidatorList(ValidatorList)

```
public void SetValidatorList(ValidatorList validatorList)
```

Parameters

`validatorList` ValidatorList

UpdateWorld(IWorld)

```
public override void UpdateWorld(IWorld world)
```

Parameters

`world` IWorld